# ETSI TS 137 571-1 V19.1.0 (2026-09)



TECHNICAL SPECIFICATION

**Universal Mobile Telecommunications System (UMTS);**
**LTE;**
**5G;**
**User Equipment (UE) conformance specification for UE positioning;**
**Part 1: Conformance test specification**
**(3GPP TS 37.571-1 version 19.1.0 Release 19)**



Reference
RTS/TSGR-0537571-1vj10

Keywords
5G,LTE,UMTS

***ETSI***

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

***Important notice***

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

***Notice of disclaimer & limitation of liability***

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

***Copyright Notification***

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

# Intellectual Property Rights

## Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

## Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

**DECT™**, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

# Legal Notice

This Technical Specification (TS) has been produced by the ETSI 3rd Generation Partnership Project (3GPP).

The present document may refer to technical specifications or reports using their 3GPP identities. These shall be interpreted as being references to the corresponding ETSI deliverables.

The cross reference between 3GPP and ETSI identities can be found at 3GPP to ETSI numbering cross-referencing.

# Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

# Contents

Intellectual Property Rights ..... 2
Legal Notice ..... 2
Modal verbs terminology ..... 2
Foreword ..... 55
Introduction ..... 55
1 Scope ..... 56
2 References ..... 56
3 Definitions, symbols and abbreviations ..... 59
3.1 Definitions ..... 59
3.2 Symbols ..... 59
3.3 Abbreviations ..... 60
4 General test conditions ..... 62
4.1 Introduction ..... 62
4.2 GNSS test conditions ..... 62
4.2.0 General ..... 62
4.2.1 GNSS signals ..... 62
4.2.2 GNSS frequency ..... 62
4.2.3 GNSS static propagation conditions ..... 62
4.2.4 GNSS multi-path conditions ..... 62
4.2.5 UEs supporting multiple satellite signals ..... 63
4.2.6 GNSS multi System Time Offsets ..... 64
4.3 UTRA test conditions ..... 64
4.3.1 UTRA frequency band and frequency range ..... 64
4.3.2 UTRA frequency ..... 64
4.3.3 Sensors ..... 64
4.4 E-UTRA test conditions ..... 64
4.4.1 E-UTRA frequency band and frequency range ..... 64
4.4.2 Groups of bands ..... 64
4.4.3 Sensors ..... 65
4.5 A-GNSS test conditions ..... 65
4.5.1 General ..... 65
4.5.2 UTRAN measurement parameters ..... 65
4.5.2.1 UE based A-GNSS measurement parameters ..... 65
4.5.2.2 UE assisted A-GNSS measurement parameters ..... 65
4.5.2.3 2D position error ..... 65
4.5.2.4 Response time ..... 65
4.5.3 E-UTRAN and NR measurement parameters ..... 66
4.5.3.1 UE based A-GNSS measurement parameters ..... 66
4.5.3.2 UE assisted A-GNSS measurement parameters ..... 66
4.5.3.3 2D Error definition ..... 66
4.5.3.4 Response time ..... 66
4.5.4 Converting A-GNSS UE-assisted measurement reports into position estimates ..... 66
4.6 ECID test conditions ..... 66
4.6.1 Simulated cells ..... 66
4.6.2 Propagation conditions ..... 66
4.6.2.1 Static ..... 66
4.6.2.2 Multi-path fading ..... 67
4.6.3 UE Rx - Tx time difference reporting range ..... 67
4.7 OTDOA test conditions ..... 68
4.7.1 Simulated cells ..... 68
4.7.2 Propagation conditions ..... 68
4.7.2.1 Static ..... 68
4.7.2.2 Multi-path fading ..... 68

4.7.3 Response time ..... 69
4.7.4 RSTD reporting range ..... 69
4.7.5 RSTD Carrier Aggregation Test Cases with Different Channel Bandwidth Combinations ..... 69
4.8 MBS test conditions ..... 70
4.8.1 MBS signals ..... 70
4.8.2 Propagation conditions ..... 70
4.8.2.1 Static ..... 70
4.8.2.2 Multi-path fading ..... 70
4.8.3 Response time ..... 70
4.9 WLAN test conditions ..... 70
4.9.1 Simulated WLAN Access Points ..... 70
4.9.2 Propagation conditions ..... 70
4.9.2.1 Static ..... 70
4.9.3 Response time ..... 71
4.9.4 Void ..... 71
4.10 BLE test conditions ..... 71
4.10.1 Simulated BLE ..... 71
4.10.2 Propagation conditions ..... 71
4.10.2.1 Static ..... 71
4.10.3 Response time ..... 71
4.11 NB-IOT test conditions ..... 71
4.11.1 Groups of bands ..... 71
4.11.2 NB-IOT inband mode ..... 71
4.12 NR test conditions ..... 72
4.12.1 NR terminology ..... 72
4.12.2 NR frequency band and frequency range ..... 72
4.12.3 Groups of bands ..... 72
4.12.4 Sensors ..... 72
4.12.5 Default signal conditions for FR2 ..... 72
4.12.6 Frequency Bands for Testing ..... 73
4.12.6.1 EN-DC band combination groups ..... 73
4.12.6.2 Applicable EN-DC band combinations for performing A-GNSS Sensitivity Requirements ..... 73
4.12.6.3 Test frequencies for EN-DC band combinations ..... 73
4.14 Multi-RTT test conditions ..... 73
4.14.1 Simulated cells ..... 73
4.14.2 Propagation conditions ..... 73
4.14.3 Measurement Reporting Requirements ..... 74
4.14.3.1 Measurement Reporting Requirements in RRC_CONNECTED state ..... 74
4.14.3.2 Measurement Reporting Requirements in RRC_INACTIVE state ..... 74
4.14.4 Measurement Period Requirements ..... 74
4.14.4.1 Measurement Period Requirements in RRC_CONNECTED state within Measurement Gaps ..... 74
4.14.4.2 Measurement Period Requirements in RRC_CONNECTED state without Measurement Gaps ..... 76
4.14.4.2A Measurement Period Requirements in RRC_CONNECTED state with Bandwidth Aggregation ..... 76
4.14.4.3 Measurement Period Requirements in RRC_INACTIVE state ..... 76
4.14.4.4 Measurement Period Requirements in RRC_INACTIVE state with Bandwidth Aggregation ..... 76
4.14.5 Measurement Accuracy Requirements ..... 76
4.14.5B Measurement Accuracy Requirements with Bandwidth Aggregation ..... 85
4.14.6 Reporting mapping ..... 85
4.14.6.1 Absolute UE Rx-Tx Measurement Report Mapping ..... 85
4.14.6.2 Differential UE Rx-Tx Measurement Report Mapping ..... 87
4.14.6.3 Additional Path Report Mapping for UE Rx-Tx Time Difference ..... 89
4.14A Multi-RTT test conditions for Redcap UE ..... 91
4.14A.1 Simulated cells ..... 91
4.14A.2 Propagation conditions ..... 91
4.14A.3 Measurement Reporting Requirements ..... 91
4.14A.3.1 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH with Measurement Gaps ..... 91
4.14A.3.2 Measurement Reporting Requirements in RRC_INACTIVE state ..... 91
4.14A.4 Measurement Period Requirements ..... 91
4.14A.4.1 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH within Measurement Gaps ..... 91

4.14A.4.2 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH without Measurement Gaps ..... 91
4.14A.4.3 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH within both Measurement Gaps and PPW ..... 91
4.14A.4.4 Measurement Reporting Requirements in RRC_CONNECTED state with RX FH ..... 91
4.14A.4.5 Measurement Period Requirements in RRC_INACTIVE state without RX FH ..... 91
4.14A.4.6 Measurement Period Requirements in RRC_INACTIVE state with RX FH ..... 91
4.14A.5 Measurement Accuracy Requirements ..... 92
4.14A.5.1 Measurement Accuracy Requirements for 2RX RedCap UE without RX FH ..... 92
4.14A.5.2 Measurement Accuracy Requirements for 1RX RedCap UE without RX FH ..... 92
4.14A.5.3 Measurement Accuracy Requirements for 2RX RedCap UE with RX FH ..... 92
4.14A.5.4 Measurement Accuracy Requirements for 1RX RedCap UE with RX FH ..... 92
4.14A.6 Reporting mapping ..... 92
4.14A.6.1 Absolute UE Rx-Tx Measurement Report Mapping ..... 92
4.14A.6.2 Differential UE Rx-Tx Measurement Report Mapping ..... 92
4.14A.6.3 Additional Path Report Mapping for UE Rx-Tx Time Difference ..... 92
4.15 DL-TDOA test conditions ..... 92
4.15.1 Simulated cells ..... 92
4.15.2 Propagation conditions ..... 92
4.15.3 Measurement Reporting Requirements ..... 92
4.15.3.1 Measurement Reporting Requirements in RRC_CONNECTED state ..... 92
4.15.3.2 Measurement Reporting Requirements in RRC_INACTIVE state ..... 93
4.15.4 Measurement Period Requirements ..... 93
4.15.4.1 Measurement Period Requirements within Measurement Gaps ..... 93
4.15.4.2 Measurement Period Requirements without Measurement Gaps ..... 95
4.15.4.3 Measurement Period Requirements in RRC_INACTIVE state ..... 95
4.15.4.4 Measurement Period Requirements in RRC_INACTIVE state with Bandwidth Aggregation ..... 95
4.15.5 Measurement Accuracy Requirements ..... 99
4.15.5A Measurement Accuracy Requirements with Bandwidth Aggregation ..... 105
4.15.6 Reporting mapping ..... 111
4.15.6.1 Absolute DL RSTD Measurement Reporting ..... 111
4.15.6.2 Differential Reporting for DL RSTD Measurement ..... 112
4.15.6.3 Additional Path Report Mapping for DL RSTD ..... 114
4.16 DL-AoD test conditions ..... 116
4.16.1 Simulated cells ..... 116
4.16.2 Propagation conditions ..... 116
4.16.3 Measurement Reporting Requirements ..... 116
4.16.3.1 Measurement Reporting Requirements in RRC_CONNECTED state ..... 116
4.16.3.2 Measurement Reporting Requirements in RRC_INACTIVE state ..... 116
4.16.4 PRS-RSRP Measurement Period Requirements ..... 117
4.16.4.1 Measurement Period Requirements within Measurement Gaps ..... 117
4.16.4.2 Measurement Period Requirements without Measurement Gaps ..... 118
4.16.4.3 Measurement Period Requirements in RRC_INACTIVE state ..... 118
4.16.5 PRS-RSRP Measurement Accuracy Requirements ..... 118
4.16.6 PRS-RSRP Reporting mapping ..... 125
4.16.6.1 Absolute PRS-RSRP Measurement Report Mapping ..... 125
4.16.6.2 Differential Report Mapping for PRS-RSRP Measurement ..... 126
4.16.7 PRS-RSRPP Measurement Period Requirements ..... 127
4.16.8 PRS-RSRPP Measurement Accuracy Requirements ..... 127
4.16.9 PRS-RSRPP Reporting mapping ..... 127
4.16.9.1 Absolute PRS-RSRPP Measurement Report Mapping ..... 127
4.16.9.2 Differential Report Mapping for PRS-RSRPP Measurement ..... 128

4A Support of 4 Rx capable UEs ..... 129
4A.0 Introduction ..... 129
4A.1 RAT Independent Tests ..... 130
4A.2 RAT Dependent Tests ..... 130
4A.2.1 Principle of testing ..... 130
4A.2.1.1 Single carrier tests ..... 130
4A.2.1.2 Carrier Aggregation tests ..... 130
4A.2.1.3 Antenna connection for bands where 2RX is supported ..... 130
4A.2.1.4 Antenna connection for bands where 4RX is supported ..... 130

4B Applicability of tests for types and Categories of UE ..... 130
4B.1 Introduction ..... 130
4B.2 Applicability of requirements and tests ..... 130
5 UTRA A-GPS Minimum Performance requirements ..... 133
5.1 General ..... 133
5.2 Sensitivity ..... 133
5.2.1 Sensitivity Coarse Time Assistance ..... 133
5.2.1.1 Definition and applicability ..... 133
5.2.1.2 Minimum requirements ..... 133
5.2.1.3 Test purpose ..... 133
5.2.1.4 Method of test ..... 134
5.2.1.4.1 Initial conditions ..... 134
5.2.1.4.2 Procedure ..... 134
5.2.1.5 Test Requirements ..... 134
5.2.2 Sensitivity Fine Time Assistance ..... 135
5.2.2.1 Definition and applicability ..... 135
5.2.2.2 Minimum requirements ..... 135
5.2.2.3 Test purpose ..... 136
5.2.2.4 Method of test ..... 136
5.2.2.4.1 Initial conditions ..... 136
5.2.2.4.2 Procedure ..... 136
5.2.2.5 Test Requirements ..... 136
5.3 Nominal Accuracy ..... 137
5.3.1 Definition and applicability ..... 137
5.3.2 Minimum requirements ..... 137
5.3.3 Test purpose ..... 137
5.3.4 Method of test ..... 138
5.3.4.1 Initial conditions ..... 138
5.3.4.2 Procedure ..... 138
5.3.5 Test Requirements ..... 139
5.4 Dynamic Range ..... 139
5.4.1 Definition and applicability ..... 139
5.4.2 Minimum requirements ..... 139
5.4.3 Test purpose ..... 140
5.4.4 Method of test ..... 140
5.4.4.1 Initial conditions ..... 140
5.4.4.2 Procedure ..... 140
5.4.5 Test Requirements ..... 141
5.5 Multi-path Performance ..... 141
5.5.1 Definition and applicability ..... 141
5.5.2 Minimum requirements ..... 141
5.5.3 Test purpose ..... 142
5.5.4 Method of test ..... 142
5.5.4.1 Initial conditions ..... 142
5.5.4.2 Procedure ..... 142
5.5.5 Test Requirements ..... 143
5.6 Moving Scenario and Periodic Update Performance ..... 143
5.6.1 Definition and applicability ..... 143
5.6.2 Minimum requirements ..... 143
5.6.3 Test purpose ..... 144
5.6.4 Method of test ..... 144
5.6.4.1 Initial conditions ..... 144
5.6.4.2 Procedure ..... 145
5.6.5 Test Requirements ..... 145
6 UTRA A-GNSS Minimum Performance requirements ..... 147
6.1 General ..... 147
6.2 Sensitivity ..... 147
6.2.1 Sensitivity Coarse Time Assistance ..... 147
6.2.1.1 Definition and applicability ..... 147
6.2.1.2 Minimum requirements ..... 147

6.2.1.3 Test purpose ..... 148
6.2.1.4 Method of test ..... 148
6.2.1.4.1 Initial conditions ..... 148
6.2.1.4.2 Procedure ..... 148
6.2.1.5 Test Requirements ..... 149
6.2.2 Sensitivity Fine Time Assistance ..... 150
6.2.2.1 Definition and applicability ..... 150
6.2.2.2 Minimum requirements ..... 150
6.2.2.3 Test purpose ..... 151
6.2.2.4 Method of test ..... 151
6.2.2.4.1 Initial conditions ..... 151
6.2.2.4.2 Procedure ..... 151
6.2.2.5 Test Requirements ..... 152
6.3 Nominal Accuracy ..... 153
6.3.1 Definition and applicability ..... 153
6.3.2 Minimum requirements ..... 153
6.3.3 Test purpose ..... 154
6.3.4 Method of test ..... 154
6.3.4.1 Initial conditions ..... 154
6.3.4.2 Procedure ..... 154
6.3.5 Test Requirements ..... 155
6.4 Dynamic Range ..... 156
6.4.1 Definition and applicability ..... 156
6.4.2 Minimum requirements ..... 156
6.4.3 Test purpose ..... 157
6.4.4 Method of test ..... 157
6.4.4.1 Initial conditions ..... 157
6.4.4.2 Procedure ..... 157
6.4.5 Test Requirements ..... 158
6.5 Multi-path Performance ..... 159
6.5.1 Definition and applicability ..... 159
6.5.2 Minimum requirements ..... 159
6.5.3 Test purpose ..... 160
6.5.4 Method of test ..... 160
6.5.4.1 Initial conditions ..... 160
6.5.4.2 Procedure ..... 160
6.5.5 Test Requirements ..... 161
6.6 Moving Scenario and Periodic Update Performance ..... 162
6.6.1 Definition and applicability ..... 162
6.6.2 Minimum requirements ..... 162
6.6.3 Test purpose ..... 163
6.6.4 Method of test ..... 163
6.6.4.1 Initial conditions ..... 163
6.6.4.2 Procedure ..... 164
6.6.5 Test Requirements ..... 165

7 E-UTRA A-GNSS minimum performance requirements ..... 167
7.0 General ..... 167
7.1 Sensitivity ..... 167
7.1.1 Sensitivity Coarse time assistance ..... 167
7.1.1.1 Sub-tests ..... 167
7.1.1.2 Test purpose ..... 167
7.1.1.3 Test applicability ..... 167
7.1.1.4 Minimum conformance requirements ..... 167
7.1.1.5 Test description ..... 168
7.1.1.5.1 Initial conditions ..... 168
7.1.1.5.2 Test procedure ..... 169
7.1.1.5.3 Message contents ..... 170
7.1.1.6 Test requirement ..... 171
7.1.2 Sensitivity Fine time assistance ..... 172
7.1.2.1 Sub-tests ..... 172
7.1.2.2 Test purpose ..... 172

7.1.2.3 Test applicability ..... 172
7.1.2.4 Minimum conformance requirements ..... 172
7.1.2.5 Test description ..... 173
7.1.2.5.1 Initial conditions ..... 173
7.1.2.5.2 Test procedure ..... 174
7.1.2.5.3 Message contents ..... 174
7.1.2.6 Test requirement ..... 175
7.2 Nominal Accuracy ..... 176
7.2.1 Sub-tests ..... 176
7.2.2 Test purpose ..... 177
7.2.3 Test applicability ..... 177
7.2.4 Minimum conformance requirements ..... 177
7.2.5 Test description ..... 178
7.2.5.1 Initial conditions ..... 178
7.2.5.2 Test procedure ..... 178
7.2.5.3 Message contents ..... 179
7.2.6 Test requirement ..... 180
7.3 Dynamic Range ..... 181
7.3.1 Sub-tests ..... 181
7.3.2 Test purpose ..... 182
7.3.3 Test applicability ..... 182
7.3.4 Minimum conformance requirements ..... 182
7.3.5 Test description ..... 183
7.3.5.1 Initial conditions ..... 183
7.3.5.2 Test procedure ..... 183
7.3.5.3 Message contents ..... 184
7.3.6 Test requirement ..... 185
7.4 Multi-Path scenario ..... 186
7.4.1 Sub-tests ..... 186
7.4.2 Test purpose ..... 187
7.4.3 Test applicability ..... 187
7.4.4 Minimum conformance requirements ..... 187
7.4.5 Test description ..... 188
7.4.5.1 Initial conditions ..... 188
7.4.5.2 Test procedure ..... 188
7.4.5.3 Message contents ..... 189
7.4.6 Test requirement ..... 190
7.5 Moving scenario and periodic update (Rel-9 to Rel-13) ..... 191
7.5.1 Sub-tests ..... 191
7.5.2 Test purpose ..... 192
7.5.3 Test applicability ..... 192
7.5.4 Minimum conformance requirements ..... 192
7.5.5 Test description ..... 193
7.5.5.1 Initial conditions ..... 193
7.5.5.2 Test procedure ..... 194
7.5.5.3 Message contents ..... 195
7.5.6 Test requirement ..... 196
7.5A Moving scenario and periodic update (Rel-14 onwards) ..... 197
7.5A.1 Sub-tests ..... 197
7.5A.2 Test purpose ..... 197
7.5A.3 Test applicability ..... 197
7.5A.4 Minimum conformance requirements ..... 197
7.5A.5 Test description ..... 197
7.5A.6 Test requirement ..... 197

8 E-UTRA ECID measurement requirements ..... 197
8.0 General ..... 197
8.1 UE Rx – Tx Time Difference ..... 197
8.1.1 E-UTRAN FDD UE Rx – Tx time difference case (Rel-9 to Rel-11) ..... 197
8.1.1.1 Test purpose ..... 197
8.1.1.2 Test applicability ..... 197
8.1.1.3 Minimum conformance requirements ..... 198

8.1.1.4 Test description ..... 198
8.1.1.4.1 Initial conditions ..... 198
8.1.1.4.2 Test procedure ..... 199
8.1.1.4.3 Message contents ..... 200
8.1.1.5 Test requirement ..... 202
8.1.1A E-UTRAN FDD UE Rx – Tx time difference case (Rel-12 onwards) ..... 204
8.1.1A.1 Test purpose ..... 204
8.1.1A.2 Test applicability ..... 204
8.1.1A.3 Minimum conformance requirements ..... 204
8.1.1A.4 Test description ..... 204
8.1.1A.4.1 Initial conditions ..... 205
8.1.1A.4.2 Test procedure ..... 205
8.1.1A.4.3 Message contents ..... 205
8.1.1A.5 Test requirement ..... 205
8.1.1B E-UTRAN FDD UE Rx – Tx time difference case for UE Category 1bis ..... 205
8.1.1B.1 Test purpose ..... 205
8.1.1B.2 Test applicability ..... 205
8.1.1B.3 Minimum conformance requirements ..... 205
8.1.1B.4 Test description ..... 205
8.1.1B.4.1 Initial conditions ..... 205
8.1.1B.4.2 Test procedure ..... 205
8.1.1B.4.3 Message contents ..... 205
8.1.1B.5 Test requirement ..... 206
8.1.2 E-UTRAN TDD UE Rx – Tx time difference case (Rel-9 to Rel-11) ..... 206
8.1.2.1 Test purpose ..... 206
8.1.2.2 Test applicability ..... 206
8.1.2.3 Minimum conformance requirements ..... 206
8.1.2.4 Test description ..... 206
8.1.2.4.1 Initial conditions ..... 206
8.1.2.4.2 Test procedure ..... 207
8.1.2.4.3 Message contents ..... 207
8.1.2.5 Test requirement ..... 210
8.1.2A E-UTRAN TDD UE Rx – Tx time difference case (Rel-12 onwards) ..... 212
8.1.2A.1 Test purpose ..... 212
8.1.2A.2 Test applicability ..... 212
8.1.2A.3 Minimum conformance requirements ..... 212
8.1.2A.4 Test description ..... 212
8.1.2A.4.1 Initial conditions ..... 212
8.1.2A.4.2 Test procedure ..... 212
8.1.2A.4.3 Message contents ..... 212
8.1.2A.5 Test requirement ..... 212
8.1.2B E-UTRAN TDD UE Rx – Tx time difference case for UE Category 1bis ..... 213
8.1.2B.1 Test purpose ..... 213
8.1.2B.2 Test applicability ..... 213
8.1.2B.3 Minimum conformance requirements ..... 213
8.1.2B.4 Test description ..... 213
8.1.2B.4.1 Initial conditions ..... 213
8.1.2B.4.2 Test procedure ..... 213
8.1.2B.4.3 Message contents ..... 213
8.1.2B.5 Test requirement ..... 213
8.1.3 E-UTRAN FDD UE Rx–Tx Time Difference under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) ..... 213
8.1.3.1 Test purpose ..... 213
8.1.3.2 Test applicability ..... 213
8.1.3.3 Minimum conformance requirements ..... 213
8.1.3.4 Test description ..... 214
8.1.3.4.1 Initial conditions ..... 215
8.1.3.4.2 Test procedure ..... 215
8.1.3.4.3 Message contents ..... 216
8.1.3.5 Test requirement ..... 219
8.1.4 E-UTRAN TDD UE Rx–Tx Time Difference under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) ..... 221

8.1.4.1 Test purpose ..... 221
8.1.4.2 Test applicability ..... 221
8.1.4.3 Minimum conformance requirements ..... 221
8.1.4.4 Test description ..... 222
8.1.4.4.1 Initial conditions ..... 222
8.1.4.4.2 Test procedure ..... 223
8.1.4.4.3 Message contents ..... 224
8.1.4.5 Test requirement ..... 226
8.1.5 E-UTRAN FDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) ..... 228
8.1.5.1 Test purpose ..... 228
8.1.5.2 Test applicability ..... 228
8.1.5.3 Minimum conformance requirements ..... 228
8.1.5.4 Test description ..... 229
8.1.5.4.1 Initial conditions ..... 230
8.1.5.4.2 Test procedure ..... 231
8.1.5.4.3 Message contents ..... 231
8.1.5.5 Test requirement ..... 235
8.1.6 E-UTRAN TDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) ..... 237
8.1.6.1 Test purpose ..... 237
8.1.6.2 Test applicability ..... 237
8.1.6.3 Minimum conformance requirements ..... 237
8.1.6.4 Test description ..... 238
8.1.6.4.1 Initial conditions ..... 239
8.1.6.4.2 Test procedure ..... 240
8.1.6.4.3 Message contents ..... 240
8.1.6.5 Test requirement ..... 244
8.1.7 E-UTRAN FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA ..... 246
8.1.7.1 Test purpose ..... 246
8.1.7.2 Test applicability ..... 246
8.1.7.3 Minimum conformance requirements ..... 246
8.1.7.4 Test description ..... 247
8.1.7.4.1 Initial conditions ..... 247
8.1.7.4.2 Test procedure ..... 247
8.1.7.4.3 Message contents ..... 248
8.1.7.5 Test requirement ..... 250
8.1.8 E-UTRAN HD-FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA ..... 251
8.1.8.1 Test purpose ..... 251
8.1.8.2 Test applicability ..... 251
8.1.8.3 Minimum conformance requirements ..... 251
8.1.8.4 Test description ..... 251
8.1.8.4.1 Initial conditions ..... 251
8.1.8.4.2 Test procedure ..... 251
8.1.8.4.3 Message contents ..... 252
8.1.8.5 Test requirement ..... 252
8.1.9 E-UTRAN TDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA ..... 253
8.1.9.1 Test purpose ..... 253
8.1.9.2 Test applicability ..... 253
8.1.9.3 Minimum conformance requirements ..... 253
8.1.9.4 Test description ..... 253
8.1.9.4.1 Initial conditions ..... 253
8.1.9.4.2 Test procedure ..... 253
8.1.9.4.3 Message contents ..... 254
8.1.9.5 Test requirement ..... 254

9 E-UTRA OTDOA measurement requirements ..... 256
9.0 General ..... 256
9.1 RSTD Intra-Frequency Measurements ..... 256
9.1.1 FDD RSTD Measurement Reporting Delay ..... 256
9.1.1.1 Test purpose ..... 256
9.1.1.2 Test applicability ..... 256

9.1.1.3 Minimum conformance requirements....256
9.1.1.4 Test description....257
9.1.1.4.1 Initial conditions....257
9.1.1.4.2 Test procedure....259
9.1.1.4.3 Message contents....260
9.1.1.5 Test requirement....263
9.1.1A FDD RSTD Measurement Reporting Delay for UE Category 1bis....265
9.1.1A.1 Test purpose....265
9.1.1A.2 Test applicability....265
9.1.1A.3 Minimum conformance requirements....265
9.1.1A.4 Test description....266
9.1.1A.4.1 Initial conditions....266
9.1.1A.4.2 Test procedure....267
9.1.1A.4.3 Message contents....267
9.1.1A.5 Test requirement....268
9.1.2 TDD RSTD Measurement Reporting Delay....268
9.1.2.1 Test purpose....268
9.1.2.2 Test applicability....268
9.1.2.3 Minimum conformance requirements....268
9.1.2.4 Test description....269
9.1.2.4.1 Initial conditions....269
9.1.2.4.2 Test procedure....271
9.1.2.4.3 Message contents....272
9.1.2.5 Test requirement....275
9.1.2A TDD RSTD Measurement Reporting Delay for UE Category 1bis....277
9.1.2A.1 Test purpose....277
9.1.2A.2 Test applicability....277
9.1.2A.3 Minimum conformance requirements....277
9.1.2A.4 Test description....278
9.1.2A.4.1 Initial conditions....278
9.1.2A.4.2 Test procedure....279
9.1.2A.4.3 Message contents....279
9.1.2A.5 Test requirement....280
9.1.3 FDD RSTD Measurement Accuracy....280
9.1.3.1 Test purpose....280
9.1.3.2 Test applicability....280
9.1.3.3 Minimum conformance requirements....280
9.1.3.4 Test description....281
9.1.3.4.1 Initial conditions....281
9.1.3.4.2 Test procedure....283
9.1.3.4.3 Message contents....284
9.1.3.5 Test requirement....287
9.1.3A FDD RSTD Measurement Accuracy for UE Category 1bis....288
9.1.3A.1 Test purpose....288
9.1.3A.2 Test applicability....288
9.1.3A.3 Minimum conformance requirements....288
9.1.3A.4 Test description....288
9.1.3A.4.1 Initial conditions....288
9.1.3A.4.2 Test procedure....289
9.1.3A.4.3 Message contents....289
9.1.3A.5 Test requirement....290
9.1.4 TDD RSTD Measurement Accuracy....290
9.1.4.1 Test purpose....290
9.1.4.2 Test applicability....290
9.1.4.3 Minimum conformance requirements....290
9.1.4.4 Test description....290
9.1.4.4.1 Initial conditions....290
9.1.4.4.2 Test procedure....292
9.1.4.4.3 Message contents....293
9.1.4.5 Test requirement....296
9.1.4A TDD RSTD Measurement Accuracy for UE Category 1bis....297
9.1.4A.1 Test purpose....297

9.1.4A.2 Test applicability ..... 297
9.1.4A.3 Minimum conformance requirements ..... 297
9.1.4A.4 Test description ..... 297
9.1.4A.4.1 Initial conditions ..... 297
9.1.4A.4.2 Test procedure ..... 299
9.1.4A.4.3 Message contents ..... 299
9.1.4A.5 Test requirement ..... 299
9.2 RSTD Inter-Frequency Measurements ..... 300
9.2.1 FDD-FDD inter-frequency RSTD measurement reporting delay ..... 300
9.2.1.1 Test purpose ..... 300
9.2.1.2 Test applicability ..... 300
9.2.1.3 Minimum conformance requirements ..... 300
9.2.1.4 Test description ..... 301
9.2.1.4.1 Initial conditions ..... 301
9.2.1.4.2 Test procedure ..... 303
9.2.1.4.3 Message contents ..... 304
9.2.1.5 Test requirement ..... 307
9.2.1A FDD-FDD inter-frequency RSTD Measurement Reporting Delay for UE Category 1bis ..... 309
9.2.1A.1 Test purpose ..... 309
9.2.1A.2 Test applicability ..... 309
9.2.1A.3 Minimum conformance requirements ..... 310
9.2.1A.4 Test description ..... 310
9.2.1A.4.1 Initial conditions ..... 310
9.2.1A.4.2 Test procedure ..... 312
9.2.1A.4.3 Message contents ..... 312
9.2.1A.5 Test requirement ..... 312
9.2.2 TDD-TDD inter-frequency RSTD measurement reporting delay ..... 313
9.2.2.1 Test purpose ..... 313
9.2.2.2 Test applicability ..... 313
9.2.2.3 Minimum conformance requirements ..... 313
9.2.2.4 Test description ..... 314
9.2.2.4.1 Initial conditions ..... 314
9.2.2.4.2 Test procedure ..... 316
9.2.2.4.3 Message contents ..... 317
9.2.2.5 Test requirement ..... 320
9.2.2A TDD-TDD inter-frequency RSTD Measurement Reporting Delay for UE Category 1bis ..... 322
9.2.2A.1 Test purpose ..... 322
9.2.2A.2 Test applicability ..... 323
9.2.2A.3 Minimum conformance requirements ..... 323
9.2.2A.4 Test description ..... 323
9.2.2A.4.1 Initial conditions ..... 323
9.2.2A.4.2 Test procedure ..... 325
9.2.2A.4.3 Message contents ..... 325
9.2.2A.5 Test requirement ..... 325
9.2.3 Void ..... 326
9.2.4 FDD-FDD inter-frequency RSTD Accuracy ..... 326
9.2.4.1 Test purpose ..... 326
9.2.4.2 Test applicability ..... 326
9.2.4.3 Minimum conformance requirements ..... 326
9.2.4.4 Test description ..... 327
9.2.4.4.1 Initial conditions ..... 327
9.2.4.4.2 Test procedure ..... 329
9.2.4.4.3 Message contents ..... 330
9.2.4.5 Test requirement ..... 333
9.2.4A FDD-FDD inter-frequency RSTD Accuracy for UE Category 1bis ..... 334
9.2.4A.1 Test purpose ..... 334
9.2.4A.2 Test applicability ..... 334
9.2.4A.3 Minimum conformance requirements ..... 334
9.2.4A.4 Test description ..... 334
9.2.4A.4.1 Initial conditions ..... 334
9.2.4A.4.2 Test procedure ..... 335
9.2.4A.4.3 Message contents ..... 335

9.2.4A.5 Test requirement ....336
9.2.5 TDD-TDD inter-frequency RSTD Accuracy ....336
9.2.5.1 Test purpose ....336
9.2.5.2 Test applicability ....336
9.2.5.3 Minimum conformance requirements ....336
9.2.5.4 Test description ....336
9.2.5.4.1 Initial conditions ....336
9.2.5.4.2 Test procedure ....338
9.2.5.4.3 Message contents ....339
9.2.5.5 Test requirement ....342
9.2.5A TDD-TDD inter-frequency RSTD Accuracy for UE Category 1bis ....343
9.2.5A.1 Test purpose ....343
9.2.5A.2 Test applicability ....343
9.2.5A.3 Minimum conformance requirements ....343
9.2.5A.4 Test description ....343
9.2.5A.4.1 Initial conditions ....343
9.2.5A.4.2 Test procedure ....344
9.2.5A.4.3 Message contents ....344
9.2.5A.5 Test requirement ....345
9.3 RSTD Intra-Frequency Measurements for UE Category M1/M2 ....346
9.3.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A ....346
9.3.1.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 ....346
9.3.1.1.1 Test purpose ....346
9.3.1.1.2 Test applicability ....346
9.3.1.1.3 Minimum conformance requirements ....346
9.3.1.1.4 Test description ....348
9.3.1.1.4.1 Initial conditions ....348
9.3.1.1.4.2 Test procedure ....350
9.3.1.1.4.3 Message contents ....351
9.3.1.1.5 Test requirement ....353
9.3.1.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 ....355
9.3.1.2.1 Test purpose ....355
9.3.1.2.2 Test applicability ....355
9.3.1.2.3 Minimum conformance requirements ....355
9.3.1.2.4 Test description ....355
9.3.1.2.4.1 Initial conditions ....355
9.3.1.2.4.2 Test procedure ....357
9.3.1.2.4.3 Message contents ....357
9.3.1.2.5 Test requirement ....357
9.3.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A ....357
9.3.2.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 ....357
9.3.2.1.1 Test purpose ....357
9.3.2.1.2 Test applicability ....357
9.3.2.1.3 Minimum conformance requirements ....357
9.3.2.1.4 Test description ....357
9.3.2.1.4.1 Initial conditions ....357
9.3.2.1.4.2 Test procedure ....359
9.3.2.1.4.3 Message contents ....359
9.3.2.1.5 Test requirement ....359
9.3.2.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 ....359
9.3.2.2.1 Test purpose ....359
9.3.2.2.2 Test applicability ....359
9.3.2.2.3 Minimum conformance requirements ....359
9.3.2.2.4 Test description ....359
9.3.2.2.4.1 Initial conditions ....359
9.3.2.2.4.2 Test procedure ....361
9.3.2.2.4.3 Message contents ....361
9.3.2.2.5 Test requirement ....361
9.3.3 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A ....361
9.3.3.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 ....361
9.3.3.1.1 Test purpose ....361
9.3.3.1.2 Test applicability ....361

9.3.3.1.3 Minimum conformance requirements ..... 361
9.3.3.1.4 Test description ..... 362
9.3.3.1.4.1 Initial conditions ..... 362
9.3.3.1.4.2 Test procedure ..... 363
9.3.3.1.4.3 Message contents ..... 363
9.3.3.1.5 Test requirement ..... 363
9.3.3.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 ..... 365
9.3.3.2.1 Test purpose ..... 365
9.3.3.2.2 Test applicability ..... 365
9.3.3.2.3 Minimum conformance requirements ..... 365
9.3.3.2.4 Test description ..... 366
9.3.3.2.4.1 Initial conditions ..... 366
9.3.3.2.4.2 Test procedure ..... 367
9.3.3.2.4.3 Message contents ..... 367
9.3.3.2.5 Test requirement ..... 367
9.3.4 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B ..... 368
9.3.4.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 ..... 368
9.3.4.1.1 Test purpose ..... 368
9.3.4.1.2 Test applicability ..... 368
9.3.4.1.3 Minimum conformance requirements ..... 368
9.3.4.1.4 Test description ..... 368
9.3.4.1.4.1 Initial conditions ..... 368
9.3.4.1.4.2 Test procedure ..... 370
9.3.4.1.4.3 Message contents ..... 370
9.3.4.1.5 Test requirement ..... 370
9.3.4.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 ..... 372
9.3.4.2.1 Test purpose ..... 372
9.3.4.2.2 Test applicability ..... 372
9.3.4.2.3 Minimum conformance requirements ..... 372
9.3.4.2.4 Test description ..... 372
9.3.4.2.4.1 Initial conditions ..... 372
9.3.4.2.4.2 Test procedure ..... 374
9.3.4.2.4.3 Message contents ..... 374
9.3.4.2.5 Test requirement ..... 374
9.3.5 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B ..... 375
9.3.5.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 ..... 375
9.3.5.1.1 Test purpose ..... 375
9.3.5.1.2 Test applicability ..... 375
9.3.5.1.3 Minimum conformance requirements ..... 375
9.3.5.1.4 Test description ..... 375
9.3.5.1.4.1 Initial conditions ..... 375
9.3.5.1.4.2 Test procedure ..... 377
9.3.5.1.4.3 Message contents ..... 377
9.3.5.1.5 Test requirement ..... 377
9.3.5.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 ..... 377
9.3.5.2.1 Test purpose ..... 377
9.3.5.2.2 Test applicability ..... 377
9.3.5.2.3 Minimum conformance requirements ..... 377
9.3.5.2.4 Test description ..... 377
9.3.5.2.4.1 Initial conditions ..... 377
9.3.5.2.4.2 Test procedure ..... 378
9.3.5.2.4.3 Message contents ..... 379
9.3.5.2.5 Test requirement ..... 379
9.3.6 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B ..... 379
9.3.6.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 ..... 379
9.3.6.1.1 Test purpose ..... 379
9.3.6.1.2 Test applicability ..... 379
9.3.6.1.3 Minimum conformance requirements ..... 379
9.3.6.1.4 Test description ..... 379
9.3.6.1.4.1 Initial conditions ..... 379
9.3.6.1.4.2 Test procedure ..... 381
9.3.6.1.4.3 Message contents ..... 381

9.3.6.1.5 Test requirement ..... 381
9.3.6.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 ..... 383
9.3.6.2.1 Test purpose ..... 383
9.3.6.2.2 Test applicability ..... 383
9.3.6.2.3 Minimum conformance requirements ..... 383
9.3.6.2.4 Test description ..... 383
9.3.6.2.4.1 Initial conditions ..... 383
9.3.6.2.4.2 Test procedure ..... 385
9.3.6.2.4.3 Message contents ..... 385
9.3.6.2.5 Test requirement ..... 385
9.3.7 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A ..... 386
9.3.7.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 ..... 386
9.3.7.1.1 Test purpose ..... 386
9.3.7.1.2 Test applicability ..... 386
9.3.7.1.3 Minimum conformance requirements ..... 386
9.3.7.1.4 Test description ..... 387
9.3.7.1.4.1 Initial conditions ..... 387
9.3.7.1.4.2 Test procedure ..... 389
9.3.7.1.4.3 Message contents ..... 390
9.3.7.1.5 Test requirement ..... 391
9.3.7.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 ..... 392
9.3.7.2.1 Test purpose ..... 392
9.3.7.2.2 Test applicability ..... 392
9.3.7.2.3 Minimum conformance requirements ..... 392
9.3.7.2.4 Test description ..... 394
9.3.7.2.4.1 Initial conditions ..... 394
9.3.7.2.4.2 Test procedure ..... 395
9.3.7.2.4.3 Message contents ..... 395
9.3.7.2.5 Test requirement ..... 395
9.3.8 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A ..... 396
9.3.8.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 ..... 396
9.3.8.1.1 Test purpose ..... 396
9.3.8.1.2 Test applicability ..... 396
9.3.8.1.3 Minimum conformance requirements ..... 396
9.3.8.1.4 Test description ..... 396
9.3.8.1.4.1 Initial conditions ..... 396
9.3.8.1.4.2 Test procedure ..... 397
9.3.8.1.4.3 Message contents ..... 397
9.3.8.1.5 Test requirement ..... 397
9.3.8.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 ..... 398
9.3.8.2.1 Test purpose ..... 398
9.3.8.2.2 Test applicability ..... 398
9.3.8.2.3 Minimum conformance requirements ..... 398
9.3.8.2.4 Test description ..... 398
9.3.8.2.4.1 Initial conditions ..... 398
9.3.8.2.4.2 Test procedure ..... 399
9.3.8.2.4.3 Message contents ..... 399
9.3.8.2.5 Test requirement ..... 399
9.3.9 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A ..... 400
9.3.9.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 ..... 400
9.3.9.1.1 Test purpose ..... 400
9.3.9.1.2 Test applicability ..... 400
9.3.9.1.3 Minimum conformance requirements ..... 400
9.3.9.1.4 Test description ..... 400
9.3.9.1.4.1 Initial conditions ..... 400
9.3.9.1.4.2 Test procedure ..... 401
9.3.9.1.4.3 Message contents ..... 402
9.3.9.1.5 Test requirement ..... 402
9.3.9.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 ..... 402
9.3.9.2.1 Test purpose ..... 402
9.3.9.2.2 Test applicability ..... 402
9.3.9.2.3 Minimum conformance requirements ..... 402

9.3.9.2.4 Test description ..... 402
9.3.9.2.4.1 Initial conditions ..... 402
9.3.9.2.4.2 Test procedure ..... 404
9.3.9.2.4.3 Message contents ..... 404
9.3.9.2.5 Test requirement ..... 404
9.3.10 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B ..... 404
9.3.10.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 ..... 404
9.3.10.1.1 Test purpose ..... 404
9.3.10.1.2 Test applicability ..... 404
9.3.10.1.3 Minimum conformance requirements ..... 404
9.3.10.1.4 Test description ..... 405
9.3.10.1.4.1 Initial conditions ..... 405
9.3.10.1.4.2 Test procedure ..... 407
9.3.10.1.4.3 Message contents ..... 407
9.3.10.1.5 Test requirement ..... 407
9.3.10.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 ..... 408
9.3.10.2.1 Test purpose ..... 408
9.3.10.2.2 Test applicability ..... 408
9.3.10.2.3 Minimum conformance requirements ..... 409
9.3.10.2.4 Test description ..... 410
9.3.10.2.4.1 Initial conditions ..... 410
9.3.10.2.4.2 Test procedure ..... 411
9.3.10.2.4.3 Message contents ..... 411
9.3.10.2.5 Test requirement ..... 412
9.3.11 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B ..... 412
9.3.11.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 ..... 412
9.3.11.1.1 Test purpose ..... 412
9.3.11.1.2 Test applicability ..... 412
9.3.11.1.3 Minimum conformance requirements ..... 412
9.3.11.1.4 Test description ..... 413
9.3.11.1.4.1 Initial conditions ..... 413
9.3.11.1.4.2 Test procedure ..... 414
9.3.11.1.4.3 Message contents ..... 414
9.3.11.1.5 Test requirement ..... 414
9.3.11.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 ..... 414
9.3.11.2.1 Test purpose ..... 414
9.3.11.2.2 Test applicability ..... 414
9.3.11.2.3 Minimum conformance requirements ..... 414
9.3.11.2.4 Test description ..... 415
9.3.11.2.4.1 Initial conditions ..... 415
9.3.11.2.4.2 Test procedure ..... 416
9.3.11.2.4.3 Message contents ..... 416
9.3.11.2.5 Test requirement ..... 416
9.3.12 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B ..... 416
9.3.12.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 ..... 416
9.3.12.1.1 Test purpose ..... 416
9.3.12.1.2 Test applicability ..... 416
9.3.12.1.3 Minimum conformance requirements ..... 417
9.3.12.1.4 Test description ..... 417
9.3.12.1.4.1 Initial conditions ..... 417
9.3.12.1.4.2 Test procedure ..... 418
9.3.12.1.4.3 Message contents ..... 418
9.3.12.1.5 Test requirement ..... 418
9.3.12.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 ..... 418
9.3.12.2.1 Test purpose ..... 418
9.3.12.2.2 Test applicability ..... 419
9.3.12.2.3 Minimum conformance requirements ..... 419
9.3.12.2.4 Test description ..... 419
9.3.12.2.4.1 Initial conditions ..... 419
9.3.12.2.4.2 Test procedure ..... 420
9.3.12.2.4.3 Message contents ..... 420
9.3.12.2.5 Test requirement ..... 420

9.3.13 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions....421
9.3.13.1 Test purpose....421
9.3.13.2 Test applicability....421
9.3.13.3 Minimum conformance requirements....421
9.3.13.4 Test description....421
9.3.13.4.1 Initial conditions....421
9.3.13.4.2 Test procedure....423
9.3.13.4.3 Message contents....423
9.3.13.5 Test requirement....424
9.3.14 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions....424
9.3.14.1 Test purpose....424
9.3.14.2 Test applicability....424
9.3.14.3 Minimum conformance requirements....424
9.3.14.4 Test description....424
9.3.14.4.1 Initial conditions....424
9.3.14.4.2 Test procedure....426
9.3.14.4.3 Message contents....426
9.3.14.5 Test requirement....426
9.3.15 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions....426
9.3.15.1 Test purpose....426
9.3.15.2 Test applicability....427
9.3.15.3 Minimum conformance requirements....427
9.3.15.4 Test description....427
9.3.15.4.1 Initial conditions....427
9.3.15.4.2 Test procedure....429
9.3.15.4.3 Message contents....429
9.3.15.5 Test requirement....429
9.3.16 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions....429
9.3.16.1 Test purpose....429
9.3.16.2 Test applicability....430
9.3.16.3 Minimum conformance requirements....430
9.3.16.4 Test description....430
9.3.16.4.1 Initial conditions....430
9.3.16.4.2 Test procedure....432
9.3.16.4.3 Message contents....432
9.3.16.5 Test requirement....432
9.3.17 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions....432
9.3.17.1 Test purpose....432
9.3.17.2 Test applicability....432
9.3.17.3 Minimum conformance requirements....432
9.3.17.4 Test description....432
9.3.17.4.1 Initial conditions....432
9.3.17.4.2 Test procedure....434
9.3.17.4.3 Message contents....434
9.3.17.5 Test requirement....434
9.3.18 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions....434
9.3.18.1 Test purpose....434
9.3.18.2 Test applicability....435
9.3.18.3 Minimum conformance requirements....435
9.3.18.4 Test description....435
9.3.18.4.1 Initial conditions....435
9.3.18.4.2 Test procedure....437
9.3.18.4.3 Message contents....437
9.3.18.5 Test requirement....437
9.4 RSTD Inter-Frequency Measurements for UE Category M1/M2....438
9.4.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A....438

9.4.1.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 ..... 438
9.4.1.1.1 Test purpose ..... 438
9.4.1.1.2 Test applicability ..... 438
9.4.1.1.3 Minimum conformance requirements ..... 438
9.4.1.1.4 Test description ..... 439
9.4.1.1.4.1 Initial conditions ..... 439
9.4.1.1.4.2 Test procedure ..... 442
9.4.1.1.4.3 Message contents ..... 443
9.4.1.1.5 Test requirement ..... 443
9.4.1.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 ..... 445
9.4.1.2.1 Test purpose ..... 445
9.4.1.2.2 Test applicability ..... 445
9.4.1.2.3 Minimum conformance requirements ..... 446
9.4.1.2.4 Test description ..... 446
9.4.1.2.4.1 Initial conditions ..... 446
9.4.1.2.4.2 Test procedure ..... 447
9.4.1.2.4.3 Message contents ..... 447
9.4.1.2.5 Test requirement ..... 448
9.4.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A ..... 448
9.4.2.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 ..... 448
9.4.2.1.1 Test purpose ..... 448
9.4.2.1.2 Test applicability ..... 449
9.4.2.1.3 Minimum conformance requirements ..... 449
9.4.2.1.4 Test description ..... 449
9.4.2.1.4.1 Initial conditions ..... 449
9.4.2.1.4.2 Test procedure ..... 450
9.4.2.1.4.3 Message contents ..... 450
9.4.2.1.5 Test requirement ..... 450
9.4.2.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 ..... 451
9.4.2.2.1 Test purpose ..... 451
9.4.2.2.2 Test applicability ..... 451
9.4.2.2.3 Minimum conformance requirements ..... 451
9.4.2.2.4 Test description ..... 451
9.4.2.2.4.1 Initial conditions ..... 451
9.4.2.2.4.2 Test procedure ..... 453
9.4.2.2.4.3 Message contents ..... 453
9.4.2.2.5 Test requirement ..... 453
9.4.3 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A ..... 453
9.4.3.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 ..... 453
9.4.3.1.1 Test purpose ..... 453
9.4.3.1.2 Test applicability ..... 453
9.4.3.1.3 Minimum conformance requirements ..... 453
9.4.3.1.4 Test description ..... 453
9.4.3.1.4.1 Initial conditions ..... 453
9.4.3.1.4.2 Test procedure ..... 455
9.4.3.1.4.3 Message contents ..... 455
9.4.3.1.5 Test requirement ..... 455
9.4.3.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 ..... 457
9.4.3.2.1 Test purpose ..... 457
9.4.3.2.2 Test applicability ..... 457
9.4.3.2.3 Minimum conformance requirements ..... 457
9.4.3.2.4 Test description ..... 458
9.4.3.2.4.1 Initial conditions ..... 458
9.4.3.2.4.2 Test procedure ..... 460
9.4.3.2.4.3 Message contents ..... 460
9.4.3.2.5 Test requirement ..... 460
9.4.4 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B ..... 460
9.4.4.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 ..... 460
9.4.4.1.1 Test purpose ..... 460
9.4.4.1.2 Test applicability ..... 460
9.4.4.1.3 Minimum conformance requirements ..... 460
9.4.4.1.4 Test description ..... 461

9.4.4.1.4.1 Initial conditions....461
9.4.4.1.4.2 Test procedure....462
9.4.4.1.4.3 Message contents....462
9.4.4.1.5 Test requirement....463
9.4.4.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2....465
9.4.4.2.1 Test purpose....465
9.4.4.2.2 Test applicability....465
9.4.4.2.3 Minimum conformance requirements....465
9.4.4.2.4 Test description....465
9.4.4.2.4.1 Initial conditions....465
9.4.4.2.4.2 Test procedure....466
9.4.4.2.4.3 Message contents....467
9.4.4.2.5 Test requirement....467
9.4.5 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B....468
9.4.5.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1....468
9.4.5.1.1 Test purpose....468
9.4.5.1.2 Test applicability....468
9.4.5.1.3 Minimum conformance requirements....468
9.4.5.1.4 Test description....468
9.4.5.1.4.1 Initial conditions....468
9.4.5.1.4.2 Test procedure....469
9.4.5.1.4.3 Message contents....469
9.4.5.1.5 Test requirement....470
9.4.5.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2....470
9.4.5.2.1 Test purpose....470
9.4.5.2.2 Test applicability....470
9.4.5.2.3 Minimum conformance requirements....470
9.4.5.2.4 Test description....470
9.4.5.2.4.1 Initial conditions....470
9.4.5.2.4.2 Test procedure....472
9.4.5.2.4.3 Message contents....472
9.4.5.2.5 Test requirement....472
9.4.6 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B....472
9.4.6.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1....472
9.4.6.1.1 Test purpose....472
9.4.6.1.2 Test applicability....472
9.4.6.1.3 Minimum conformance requirements....472
9.4.6.1.4 Test description....472
9.4.6.1.4.1 Initial conditions....472
9.4.6.1.4.2 Test procedure....474
9.4.6.1.4.3 Message contents....474
9.4.6.1.5 Test requirement....474
9.4.6.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2....476
9.4.6.2.1 Test purpose....476
9.4.6.2.2 Test applicability....476
9.4.6.2.3 Minimum conformance requirements....476
9.4.6.2.4 Test description....476
9.4.6.2.4.1 Initial conditions....476
9.4.6.2.4.2 Test procedure....478
9.4.6.2.4.3 Message contents....478
9.4.6.2.5 Test requirement....478
9.4.7 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A....479
9.4.7.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1....479
9.4.7.1.1 Test purpose....479
9.4.7.1.2 Test applicability....479
9.4.7.1.3 Minimum conformance requirements....479
9.4.7.1.4 Test description....480
9.4.7.1.4.1 Initial conditions....480
9.4.7.1.4.2 Test procedure....482
9.4.7.1.4.3 Message contents....482
9.4.7.1.5 Test requirement....483
9.4.7.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2....484

9.4.7.2.1 Test purpose ..... 484
9.4.7.2.2 Test applicability ..... 484
9.4.7.2.3 Minimum conformance requirements ..... 484
9.4.7.2.4 Test description ..... 485
9.4.7.2.4.1 Initial conditions ..... 485
9.4.7.2.4.2 Test procedure ..... 487
9.4.7.2.4.3 Message contents ..... 487
9.4.7.2.5 Test requirement ..... 487
9.4.8 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A ..... 488
9.4.8.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 ..... 488
9.4.8.1.1 Test purpose ..... 488
9.4.8.1.2 Test applicability ..... 488
9.4.8.1.3 Minimum conformance requirements ..... 488
9.4.8.1.4 Test description ..... 488
9.4.8.1.4.1 Initial conditions ..... 488
9.4.8.1.4.2 Test procedure ..... 489
9.4.8.1.4.3 Message contents ..... 490
9.4.8.1.5 Test requirement ..... 490
9.4.8.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 ..... 490
9.4.8.2.1 Test purpose ..... 490
9.4.8.2.2 Test applicability ..... 490
9.4.8.2.3 Minimum conformance requirements ..... 490
9.4.8.2.4 Test description ..... 490
9.4.8.2.4.1 Initial conditions ..... 490
9.4.8.2.4.2 Test procedure ..... 491
9.4.8.2.4.3 Message contents ..... 491
9.4.8.2.5 Test requirement ..... 491
9.4.9 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A ..... 492
9.4.9.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 ..... 492
9.4.9.1.1 Test purpose ..... 492
9.4.9.1.2 Test applicability ..... 492
9.4.9.1.3 Minimum conformance requirements ..... 492
9.4.9.1.4 Test description ..... 492
9.4.9.1.4.1 Initial conditions ..... 492
9.4.9.1.4.2 Test procedure ..... 493
9.4.9.1.4.3 Message contents ..... 494
9.4.9.1.5 Test requirement ..... 494
9.4.9.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 ..... 495
9.4.9.2.1 Test purpose ..... 495
9.4.9.2.2 Test applicability ..... 495
9.4.9.2.3 Minimum conformance requirements ..... 495
9.4.9.2.4 Test description ..... 495
9.4.9.2.4.1 Initial conditions ..... 495
9.4.9.2.4.2 Test procedure ..... 496
9.4.9.2.4.3 Message contents ..... 496
9.4.9.2.5 Test requirement ..... 496
9.4.10 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B ..... 497
9.4.10.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 ..... 497
9.4.10.1.1 Test purpose ..... 497
9.4.10.1.2 Test applicability ..... 497
9.4.10.1.3 Minimum conformance requirements ..... 497
9.4.10.1.4 Test description ..... 498
9.4.10.1.4.1 Initial conditions ..... 498
9.4.10.1.4.2 Test procedure ..... 499
9.4.10.1.4.3 Message contents ..... 499
9.4.10.1.5 Test requirement ..... 500
9.4.10.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 ..... 501
9.4.10.2.1 Test purpose ..... 501
9.4.10.2.2 Test applicability ..... 501
9.4.10.2.3 Minimum conformance requirements ..... 501
9.4.10.2.4 Test description ..... 502
9.4.10.2.4.1 Initial conditions ..... 502

9.4.10.2.4.2 Test procedure ..... 504
9.4.10.2.4.3 Message contents ..... 504
9.4.10.2.5 Test requirement ..... 504
9.4.11 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B ..... 505
9.4.11.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 ..... 505
9.4.11.1.1 Test purpose ..... 505
9.4.11.1.2 Test applicability ..... 505
9.4.11.1.3 Minimum conformance requirements ..... 505
9.4.11.1.4 Test description ..... 505
9.4.11.1.4.1 Initial conditions ..... 505
9.4.11.1.4.2 Test procedure ..... 507
9.4.11.1.4.3 Message contents ..... 507
9.4.11.1.5 Test requirement ..... 507
9.4.11.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 ..... 507
9.4.11.2.1 Test purpose ..... 507
9.4.11.2.2 Test applicability ..... 507
9.4.11.2.3 Minimum conformance requirements ..... 507
9.4.11.2.4 Test description ..... 507
9.4.11.2.4.1 Initial conditions ..... 507
9.4.11.2.4.2 Test procedure ..... 508
9.4.11.2.4.3 Message contents ..... 508
9.4.11.2.5 Test requirement ..... 508
9.4.12 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B ..... 509
9.4.12.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 ..... 509
9.4.12.1.1 Test purpose ..... 509
9.4.12.1.2 Test applicability ..... 509
9.4.12.1.3 Minimum conformance requirements ..... 509
9.4.12.1.4 Test description ..... 509
9.4.12.1.4.1 Initial conditions ..... 509
9.4.12.1.4.2 Test procedure ..... 510
9.4.12.1.4.3 Message contents ..... 511
9.4.12.1.5 Test requirement ..... 511
9.4.12.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 ..... 512
9.4.12.2.1 Test purpose ..... 512
9.4.12.2.2 Test applicability ..... 512
9.4.12.2.3 Minimum conformance requirements ..... 512
9.4.12.2.4 Test description ..... 512
9.4.12.2.4.1 Initial conditions ..... 512
9.4.12.2.4.2 Test procedure ..... 513
9.4.12.2.4.3 Message contents ..... 513
9.4.12.2.5 Test requirement ..... 513
9.4.13 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions ..... 514
9.4.13.1 Test purpose ..... 514
9.4.13.2 Test applicability ..... 514
9.4.13.3 Minimum conformance requirements ..... 514
9.4.13.4 Test description ..... 514
9.4.13.4.1 Initial conditions ..... 514
9.4.13.4.2 Test procedure ..... 516
9.4.13.4.3 Message contents ..... 516
9.4.13.5 Test requirement ..... 517
9.4.14 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions ..... 517
9.4.14.1 Test purpose ..... 517
9.4.14.2 Test applicability ..... 517
9.4.14.3 Minimum conformance requirements ..... 517
9.4.14.4 Test description ..... 518
9.4.14.4.1 Initial conditions ..... 518
9.4.14.4.2 Test procedure ..... 519
9.4.14.4.3 Message contents ..... 519
9.4.14.5 Test requirement ..... 519

9.4.15 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions....520
9.4.15.1 Test purpose....520
9.4.15.2 Test applicability....520
9.4.15.3 Minimum conformance requirements....520
9.4.15.4 Test description....520
9.4.15.4.1 Initial conditions....520
9.4.15.4.2 Test procedure....522
9.4.15.4.3 Message contents....522
9.4.15.5 Test requirement....522
9.4.16 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions....522
9.4.16.1 Test purpose....522
9.4.16.2 Test applicability....522
9.4.16.3 Minimum conformance requirements....522
9.4.16.4 Test description....522
9.4.16.4.1 Initial conditions....522
9.4.16.4.2 Test procedure....524
9.4.16.4.3 Message contents....524
9.4.16.5 Test requirement....524
9.4.17 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions....525
9.4.17.1 Test purpose....525
9.4.17.2 Test applicability....525
9.4.17.3 Minimum conformance requirements....525
9.4.17.4 Test description....525
9.4.17.4.1 Initial conditions....525
9.4.17.4.2 Test procedure....527
9.4.17.4.3 Message contents....527
9.4.17.5 Test requirement....527
9.4.18 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions....527
9.4.18.1 Test purpose....527
9.4.18.2 Test applicability....527
9.4.18.3 Minimum conformance requirements....527
9.4.18.4 Test description....527
9.4.18.4.1 Initial conditions....527
9.4.18.4.2 Test procedure....529
9.4.18.4.3 Message contents....529
9.4.18.5 Test requirement....530
9.5 HD-FDD RSTD Intra-Frequency Measurements for NB-IOT....531
9.5.1 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage....531
9.5.1.1 Test purpose....531
9.5.1.2 Test applicability....531
9.5.1.3 Minimum conformance requirements....531
9.5.1.4 Test description....533
9.5.1.4.1 Initial conditions....533
9.5.1.4.2 Test procedure....534
9.5.1.4.3 Message contents....535
9.5.1.5 Test requirement....537
9.5.2 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage....539
9.5.2.1 Test purpose....539
9.5.2.2 Test applicability....540
9.5.2.3 Minimum conformance requirements....540
9.5.2.4 Test description....540
9.5.2.4.1 Initial conditions....540
9.5.2.4.2 Test procedure....542
9.5.2.4.3 Message contents....542
9.5.2.5 Test requirement....543

9.5.3 HD-FDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Standalone Mode in enhanced coverage ..... 544
9.5.3.1 Test purpose ..... 544
9.5.3.2 Test applicability ..... 545
9.5.3.3 Minimum conformance requirements ..... 545
9.5.3.4 Test description ..... 546
9.5.3.4.1 Initial conditions ..... 546
9.5.3.4.2 Test procedure ..... 548
9.5.3.4.3 Message contents ..... 549
9.5.3.5 Test requirement ..... 549
9.5.4 Void ..... 552
9.6 HD-FDD RSTD Inter-Frequency Measurements for NB-IOT ..... 552
9.6.1 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage ..... 552
9.6.1.1 Test purpose ..... 552
9.6.1.2 Test applicability ..... 552
9.6.1.3 Minimum conformance requirements ..... 552
9.6.1.4 Test description ..... 554
9.6.1.4.1 Initial conditions ..... 554
9.6.1.4.2 Test procedure ..... 556
9.6.1.4.3 Message contents ..... 556
9.6.1.5 Test requirement ..... 557
9.6.2 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage ..... 559
9.6.2.1 Test purpose ..... 559
9.6.2.2 Test applicability ..... 560
9.6.2.3 Minimum conformance requirements ..... 560
9.6.2.4 Test description ..... 560
9.6.2.4.1 Initial conditions ..... 560
9.6.2.4.2 Test procedure ..... 562
9.6.2.4.3 Message contents ..... 563
9.6.2.5 Test requirement ..... 563
9.6.3 HD-FDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Standalone Mode in enhanced coverage ..... 564
9.6.3.1 Test purpose ..... 564
9.6.3.2 Test applicability ..... 565
9.6.3.3 Minimum conformance requirements ..... 565
9.6.3.4 Test description ..... 566
9.6.3.4.1 Initial conditions ..... 566
9.6.3.4.2 Test procedure ..... 568
9.6.3.4.3 Message contents ..... 569
9.6.3.5 Test requirement ..... 569
9.6.4 Void ..... 572
9.7 TDD RSTD Intra-Frequency Measurements for NB-IOT ..... 572
9.7.1 TDD Intra Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage ..... 572
9.7.2 TDD Intra Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage ..... 577
9.7.3 TDD Intra Frequency RSTD Measurement Reporting Delay for NB-IOT Standalone Mode in enhanced coverage ..... 583
9.8 TDD RSTD Inter-Frequency Measurements for NB-IOT ..... 589
9.8.1 TDD Inter Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage ..... 589
9.8.2 TDD Inter Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage ..... 595

10 E-UTRA OTDOA measurement requirements for Carrier Aggregation ..... 607
10.0 General ..... 607
10.1 FDD RSTD Measurement Reporting Delay for Carrier Aggregation ..... 607
10.1.1 Test purpose ..... 607
10.1.2 Test applicability ..... 607
10.1.3 Minimum conformance requirements ..... 608
10.1.3.1 Measurements on the secondary component carrier ..... 608
10.1.3.2 Measurements on both primary component carrier and secondary component carrier ..... 608

10.1.4 Test description....608
10.1.4.1 Initial conditions....608
10.1.4.2 Test procedure....610
10.1.4.3 Message contents....612
10.1.5 Test requirement....615
10.1A FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz Bandwidth....617
10.1A.1 Test purpose....617
10.1A.2 Test applicability....617
10.1A.3 Minimum conformance requirements....617
10.1A.4 Test description....617
10.1A.4.1 Initial conditions....617
10.1A.4.2 Test procedure....618
10.1A.4.3 Message contents....618
10.1A.5 Test requirement....618
10.1B FDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz + 5 MHz Bandwidth....618
10.1B.1 Test purpose....618
10.1B.2 Test applicability....619
10.1B.3 Minimum conformance requirements....619
10.1B.4 Test description....619
10.1B.4.1 Initial conditions....619
10.1B.4.2 Test procedure....619
10.1B.4.3 Message contents....619
10.1B.5 Test requirement....619
10.1C FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth....620
10.1C.1 Test purpose....620
10.1C.2 Test applicability....620
10.1C.3 Minimum conformance requirements....620
10.1C.4 Test description....620
10.1C.4.1 Initial conditions....620
10.1C.4.2 Test procedure....621
10.1C.4.3 Message contents....621
10.1C.5 Test requirement....621
10.2 TDD RSTD Measurement Reporting Delay for Carrier Aggregation....622
10.2.1 Test purpose....622
10.2.2 Test applicability....622
10.2.3 Minimum conformance requirements....622
10.2.3.1 Measurements on the secondary component carrier....622
10.2.3.2 Measurements on both primary component carrier and secondary component carrier....622
10.2.4 Test description....622
10.2.4.1 Initial conditions....622
10.2.4.2 Test procedure....625
10.2.4.3 Message contents....626
10.2.5 Test requirement....629
10.2A TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz Bandwidth....631
10.2A.1 Test purpose....631
10.2A.2 Test applicability....632
10.2A.3 Minimum conformance requirements....632
10.2A.4 Test description....632
10.2A.4.1 Initial conditions....632
10.2A.4.2 Test procedure....632
10.2A.4.3 Message contents....632
10.2A.5 Test requirement....632
10.2B TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 5 MHz + 5 MHz Bandwidth....633
10.2B.1 Test purpose....633
10.2B.2 Test applicability....633
10.2B.3 Minimum conformance requirements....633
10.2B.4 Test description....633
10.2B.4.1 Initial conditions....633
10.2B.4.2 Test procedure....634
10.2B.4.3 Message contents....634
10.2B.5 Test requirement....634
10.2C TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz + 5 MHz Bandwidth....635

10.2C.1 Test purpose ..... 635
10.2C.2 Test applicability ..... 635
10.2C.3 Minimum conformance requirements ..... 635
10.2C.4 Test description ..... 635
10.2C.4.1 Initial conditions ..... 635
10.2C.4.2 Test procedure ..... 635
10.2C.4.3 Message contents ..... 635
10.2C.5 Test requirement ..... 636
10.2D TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz +10 MHz Bandwidth ..... 636
10.2D.1 Test purpose ..... 636
10.2D.2 Test applicability ..... 636
10.2D.3 Minimum conformance requirements ..... 636
10.2D.4 Test description ..... 637
10.2D.4.1 Initial conditions ..... 637
10.2D.4.2 Test procedure ..... 637
10.2D.4.3 Message contents ..... 637
10.2D.5 Test requirement ..... 637
10.3 FDD RSTD Measurement Accuracy for Carrier Aggregation ..... 638
10.3.1 Test purpose ..... 638
10.3.2 Test applicability ..... 638
10.3.3 Minimum conformance requirements ..... 638
10.3.4 Test description ..... 638
10.3.4.1 Initial conditions ..... 638
10.3.4.2 Test procedure ..... 640
10.3.4.3 Message contents ..... 641
10.3.5 Test requirement ..... 643
10.3A FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-10 and Rel-11) ..... 644
10.3A.1 Test purpose ..... 644
10.3A.2 Test applicability ..... 644
10.3A.3 Minimum conformance requirements ..... 644
10.3A.4 Test description ..... 644
10.3A.4.1 Initial conditions ..... 644
10.3A.4.2 Test procedure ..... 645
10.3A.4.3 Message contents ..... 645
10.3A.5 Test requirement ..... 645
10.3A_1 FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-12 onwards) ..... 645
10.3A_1.1 Test purpose ..... 645
10.3A_1.2 Test applicability ..... 645
10.3A_1.3 Minimum conformance requirements ..... 646
10.3A_1.4 Test description ..... 646
10.3A_1.4.1 Initial conditions ..... 646
10.3A_1.4.2 Test procedure ..... 646
10.3A_1.4.3 Message contents ..... 646
10.3A_1.5 Test requirement ..... 646
10.3B FDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz + 5 MHz Bandwidth ..... 646
10.3B.1 Test purpose ..... 646
10.3B.2 Test applicability ..... 646
10.3B.3 Minimum conformance requirements ..... 646
10.3B.4 Test description ..... 646
10.3B.4.1 Initial conditions ..... 646
10.3B.4.2 Test procedure ..... 647
10.3B.4.3 Message contents ..... 647
10.3B.5 Test requirement ..... 647
10.3C FDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz + 5 MHz Bandwidth ..... 648
10.3C.1 Test purpose ..... 648
10.3C.2 Test applicability ..... 648
10.3C.3 Minimum conformance requirements ..... 648
10.3C.4 Test description ..... 648
10.3C.4.1 Initial conditions ..... 648
10.3C.4.2 Test procedure ..... 648
10.3C.4.3 Message contents ..... 649

10.3C.5 Test requirement ..... 649
10.4 TDD RSTD Measurement Accuracy for Carrier Aggregation ..... 649
10.4.1 Test purpose ..... 649
10.4.2 Test applicability ..... 649
10.4.3 Minimum conformance requirements ..... 649
10.4.4 Test description ..... 650
10.4.4.1 Initial conditions ..... 650
10.4.4.2 Test procedure ..... 651
10.4.4.3 Message contents ..... 652
10.4.5 Test requirement ..... 655
10.4A TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-10 and Rel-11) ..... 656
10.4A.1 Test purpose ..... 656
10.4A.2 Test applicability ..... 656
10.4A.3 Minimum conformance requirements ..... 656
10.4A.4 Test description ..... 656
10.4A.4.1 Initial conditions ..... 656
10.4A.4.2 Test procedure ..... 656
10.4A.4.3 Message contents ..... 656
10.4A.5 Test requirement ..... 656
10.4A_1 TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-12 onwards) ..... 657
10.4A_1.1 Test purpose ..... 657
10.4A_1.2 Test applicability ..... 657
10.4A_1.3 Minimum conformance requirements ..... 657
10.4A_1.4 Test description ..... 657
10.4A_1.4.1 Initial conditions ..... 657
10.4A_1.4.2 Test procedure ..... 657
10.4A_1.4.3 Message contents ..... 657
10.4A_1.5 Test requirement ..... 657
10.4B TDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz + 5 MHz bandwidth ..... 658
10.4B.1 Test purpose ..... 658
10.4B.2 Test applicability ..... 658
10.4B.3 Minimum conformance requirements ..... 658
10.4B.4 Test description ..... 658
10.4B.4.1 Initial conditions ..... 658
10.4B.4.2 Test procedure ..... 658
10.4B.4.3 Message contents ..... 658
10.4B.5 Test requirement ..... 659
10.4C TDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz + 5 MHz Bandwidth ..... 659
10.4C.1 Test purpose ..... 659
10.4C.2 Test applicability ..... 659
10.4C.3 Minimum conformance requirements ..... 659
10.4C.4 Test description ..... 659
10.4C.4.1 Initial conditions ..... 659
10.4C.4.2 Test procedure ..... 660
10.4C.4.3 Message contents ..... 660
10.4C.5 Test requirement ..... 660
10.4D TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz+10 MHz Bandwidth ..... 661
10.4D.1 Test purpose ..... 661
10.4D.2 Test applicability ..... 661
10.4D.3 Minimum conformance requirements ..... 661
10.4D.4 Test description ..... 661
10.4D.4.1 Initial conditions ..... 661
10.4D.4.2 Test procedure ..... 661
10.4D.4.3 Message contents ..... 661
10.4D.5 Test requirement ..... 662
10.5 FDD 3 DL CA RSTD Measurement Reporting Delay ..... 662
10.5.1 Test Purpose ..... 662
10.5.2 Test applicability ..... 662
10.5.3 Minimum conformance requirements ..... 662
10.5.3.1 Measurements on the secondary component carrier ..... 662
10.5.3.2 Measurements on both primary component carrier and secondary component carrier ..... 662

10.5.3.3 Measurements on different secondary component carriers ..... 663
10.5.4 Test description ..... 663
10.5.4.1 Initial conditions ..... 663
10.5.4.2 Test procedure ..... 666
10.5.4.3 Message contents ..... 667
10.5.5 Test Requirements ..... 670
10.6 TDD 3 DL CA RSTD Measurement Reporting Delay ..... 672
10.6.1 Test Purpose ..... 672
10.6.2 Test applicability ..... 673
10.6.3 Minimum conformance requirements ..... 673
10.6.3.1 Measurements on the secondary component carrier ..... 673
10.6.3.2 Measurements on both primary component carrier and secondary component carrier ..... 673
10.6.3.3 Measurements on different secondary component carriers ..... 673
10.6.4 Test description ..... 674
10.6.4.1 Initial conditions ..... 674
10.6.4.2 Test procedure ..... 676
10.6.4.3 Message contents ..... 678
10.6.5 Test Requirements ..... 681
10.7 FDD RSTD Measurement Accuracy for 3DL Carrier Aggregation ..... 683
10.7.1 Test purpose ..... 683
10.7.2 Test applicability ..... 683
10.7.3 Minimum conformance requirements ..... 683
10.7.4 Test description ..... 684
10.7.4.1 Initial conditions ..... 684
10.7.4.2 Test procedure ..... 686
10.7.4.3 Message contents ..... 687
10.7.5 Test requirement ..... 690
10.8 TDD RSTD Measurement Accuracy for 3DL Carrier Aggregation ..... 691
10.8.1 Test purpose ..... 691
10.8.2 Test applicability ..... 691
10.8.3 Minimum conformance requirements ..... 691
10.8.4 Test description ..... 692
10.8.4.1 Initial conditions ..... 692
10.8.4.2 Test procedure ..... 694
10.8.4.3 Message contents ..... 695
10.8.5 Test requirement ..... 698

11 E-UTRA and NR MBS measurement requirements ..... 699
11.0 General ..... 699
11.1 MBS Measurement Reporting Delay (Release 13 only) ..... 699
11.1.1 Test purpose ..... 699
11.1.2 Test applicability ..... 699
11.1.3 Minimum conformance requirements ..... 699
11.1.4 Test description ..... 700
11.1.4.1 Initial conditions ..... 700
11.1.4.2 Test procedure ..... 700
11.1.4.3 Message contents ..... 700
11.1.5 Test requirement ..... 702
11.1A MBS Measurement Reporting Delay (Release 14 Onwards) ..... 703
11.1A.1 Test purpose ..... 703
11.1A.2 Test applicability ..... 703
11.1A.3 Minimum conformance requirements ..... 703
11.1A.4 Test description ..... 703
11.1A.4.1 Initial conditions ..... 703
11.1A.4.2 Test procedure ..... 703
11.1A.4.3 Message contents ..... 703
11.1A.5 Test requirement ..... 704
11.1B MBS Measurement Reporting Delay (NR) ..... 704
11.1B.1 Test purpose ..... 704
11.1B.2 Test applicability ..... 705
11.1B.3 Minimum conformance requirements ..... 705
11.1B.4 Test description ..... 705

11.1B.4.1 Initial conditions ..... 705
11.1B.4.2 Test procedure ..... 705
11.1B.4.3 Message contents ..... 705
11.1B.5 Test requirement ..... 705
11.2 MBS Sensitivity Measurement Accuracy (Release 13 only) ..... 705
11.2.1 Test purpose ..... 705
11.2.2 Test applicability ..... 705
11.2.3 Minimum conformance requirements ..... 705
11.2.4 Test description ..... 706
11.2.4.1 Initial conditions ..... 706
11.2.4.2 Test procedure ..... 706
11.2.4.3 Message contents ..... 706
11.2.5 Test requirement ..... 708
11.2A MBS Sensitivity Measurement Accuracy (Release 14 Onwards) ..... 709
11.2A.1 Test purpose ..... 709
11.2A.2 Test applicability ..... 709
11.2A.3 Minimum conformance requirements ..... 709
11.2A.4 Test description ..... 710
11.2A.4.1 Initial conditions ..... 710
11.2A.4.2 Test procedure ..... 710
11.2A.4.3 Message contents ..... 710
11.2A.5 Test requirement ..... 711
11.2B MBS Sensitivity Measurement Accuracy (NR) ..... 711
11.2B.1 Test purpose ..... 711
11.2B.2 Test applicability ..... 711
11.2B.3 Minimum conformance requirements ..... 712
11.2B.4 Test description ..... 712
11.2B.4.1 Initial conditions ..... 712
11.2B.4.2 Test procedure ..... 712
11.2B.4.3 Message contents ..... 712
11.2B.5 Test requirement ..... 712
11.3 MBS Nominal Measurement Accuracy (Release 13 only) ..... 712
11.3.1 Test purpose ..... 712
11.3.2 Test applicability ..... 712
11.3.3 Minimum conformance requirements ..... 712
11.3.4 Test description ..... 713
11.3.4.1 Initial conditions ..... 713
11.3.4.2 Test procedure ..... 713
11.3.4.3 Message contents ..... 713
11.3.5 Test requirement ..... 715
11.3A MBS Nominal Measurement Accuracy (Release 14 Onwards) ..... 716
11.3A.1 Test purpose ..... 716
11.3A.2 Test applicability ..... 716
11.3A.3 Minimum conformance requirements ..... 716
11.3A.4 Test description ..... 717
11.3A.4.1 Initial conditions ..... 717
11.3A.4.2 Test procedure ..... 717
11.3A.4.3 Message contents ..... 717
11.3A.5 Test requirement ..... 718
11.3B MBS Nominal Measurement Accuracy (NR) ..... 718
11.3B.1 Test purpose ..... 718
11.3B.2 Test applicability ..... 718
11.3B.3 Minimum conformance requirements ..... 718
11.3B.4 Test description ..... 719
11.3B.4.1 Initial conditions ..... 719
11.3B.4.2 Test procedure ..... 719
11.3B.4.3 Message contents ..... 719
11.3B.5 Test requirement ..... 719
11.4 MBS Dynamic Range Measurement Accuracy (Release 13 only) ..... 719
11.4.1 Test purpose ..... 719
11.4.2 Test applicability ..... 719
11.4.3 Minimum conformance requirements ..... 719

11.4.4 Test description....720
11.4.4.1 Initial conditions....720
11.4.4.2 Test procedure....720
11.4.4.3 Message contents....720
11.4.5 Test requirement....722
11.4A MBS Dynamic Range Measurement Accuracy (Release 14 Onwards)....724
11.4A.1 Test purpose....724
11.4A.2 Test applicability....724
11.4A.3 Minimum conformance requirements....724
11.4A.4 Test description....724
11.4A.4.1 Initial conditions....724
11.4A.4.2 Test procedure....724
11.4A.4.3 Message contents....724
11.4A.5 Test requirement....726
11.4B MBS Dynamic Range Measurement Accuracy (NR)....726
11.4B.1 Test purpose....726
11.4B.2 Test applicability....726
11.4B.3 Minimum conformance requirements....726
11.4B.4 Test description....726
11.4B.4.1 Initial conditions....726
11.4B.4.2 Test procedure....727
11.4B.4.3 Message contents....727
11.4B.5 Test requirement....727
11.5 MBS Measurement Accuracy in Multipath (Release 13 only)....727
11.5.1 Test purpose....727
11.5.2 Test applicability....727
11.5.3 Minimum conformance requirements....727
11.5.4 Test description....727
11.5.4.1 Initial conditions....727
11.5.4.2 Test procedure....728
11.5.4.3 Message contents....728
11.5.5 Test requirement....730
11.5A MBS Measurement Accuracy in Multipath (Release 14 Onwards)....731
11.5A.1 Test purpose....731
11.5A.2 Test applicability....731
11.5A.3 Minimum conformance requirements....731
11.5A.4 Test description....731
11.5A.4.1 Initial conditions....731
11.5A.4.2 Test procedure....731
11.5A.4.3 Message contents....732
11.5A.5 Test requirement....733
11.5B MBS Measurement Accuracy in Multipath (NR)....733
11.5B.1 Test purpose....733
11.5B.2 Test applicability....733
11.5B.3 Minimum conformance requirements....733
11.5B.4 Test description....733
11.5B.4.1 Initial conditions....733
11.5B.4.2 Test procedure....733
11.5B.4.3 Message contents....733
11.5B.5 Test requirement....733
12 E-UTRA WLAN and BLE measurement requirements....733
12.0 General....733
12.1 WLAN Access Point Identification and Reporting Delay....734
12.1.1 WLAN AP Identification and reporting delay under nominal conditions....734
12.1.1.1 Test purpose....734
12.1.1.2 Test applicability....734
12.1.1.3 Minimum conformance requirements....734
12.1.1.4 Test description....735
12.1.1.4.1 Initial conditions....735
12.1.1.4.2 Test procedure....735
12.1.1.4.3 Message contents....736

12.1.1.5 Test requirement ..... 737
12.1.2 WLAN AP Identification and reporting delay under dynamic range conditions ..... 739
12.1.2.1 Test purpose ..... 739
12.1.2.2 Test applicability ..... 739
12.1.2.3 Minimum conformance requirements ..... 739
12.1.2.4 Test description ..... 740
12.1.2.4.1 Initial conditions ..... 740
12.1.2.4.2 Test procedure ..... 740
12.1.2.4.3 Message contents ..... 740
12.1.2.5 Test requirement ..... 740
12.2 BLE Identification and Reporting Delay ..... 741
12.2.1 Bluetooth identification ..... 741
12.2.1.1 Test purpose ..... 741
12.2.1.2 Test applicability ..... 741
12.2.1.3 Minimum conformance requirements ..... 742
12.2.1.4 Test description ..... 742
12.2.1.4.1 Initial conditions ..... 742
12.2.1.4.2 Test procedure ..... 742
12.2.1.4.3 Message contents ..... 743
12.2.1.5 Test requirement ..... 744
13 NR A-GNSS minimum performance requirements ..... 746
13.1 General ..... 746
13.2 Sensitivity ..... 746
13.2.1 Sensitivity Coarse time assistance ..... 746
13.2.1.1 Sub-tests ..... 746
13.2.1.2 Test purpose ..... 746
13.2.1.3 Test applicability ..... 746
13.2.1.4 Minimum conformance requirements ..... 747
13.2.1.5 Test description ..... 748
13.2.1.5.1 Initial conditions ..... 748
13.2.1.5.2 Test procedure ..... 748
13.2.1.5.3 Message contents ..... 749
13.2.1.6 Test requirement ..... 750
13.2.2 Sensitivity Fine time assistance ..... 751
13.2.2.1 Sub-tests ..... 751
13.2.2.2 Test purpose ..... 751
13.2.2.3 Test applicability ..... 752
13.2.2.4 Minimum conformance requirements ..... 752
13.2.2.5 Test description ..... 753
13.2.2.5.1 Initial conditions ..... 753
13.2.2.5.2 Test procedure ..... 753
13.2.2.5.3 Message contents ..... 754
13.2.2.6 Test requirement ..... 755
13.3 Nominal Accuracy ..... 756
13.3.1 Sub-tests ..... 756
13.3.2 Test purpose ..... 756
13.3.3 Test applicability ..... 756
13.3.4 Minimum conformance requirements ..... 756
13.3.5 Test description ..... 757
13.3.5.1 Initial conditions ..... 757
13.3.5.2 Test procedure ..... 758
13.3.5.3 Message contents ..... 759
13.3.6 Test requirement ..... 760
13.4 Dynamic Range ..... 761
13.4.1 Sub-tests ..... 761
13.4.2 Test purpose ..... 761
13.4.3 Test applicability ..... 761
13.4.4 Minimum conformance requirements ..... 761
13.4.5 Test description ..... 762
13.4.5.1 Initial conditions ..... 762
13.4.5.2 Test procedure ..... 763

13.4.5.3 Message contents ..... 764
13.4.6 Test requirement ..... 764
13.5 Multi-Path scenario ..... 766
13.5.1 Sub-tests ..... 766
13.5.2 Test purpose ..... 766
13.5.3 Test applicability ..... 766
13.5.4 Minimum conformance requirements ..... 766
13.5.5 Test description ..... 767
13.5.5.1 Initial conditions ..... 767
13.5.5.2 Test procedure ..... 768
13.5.5.3 Message contents ..... 769
13.5.6 Test requirement ..... 769
13.6 Moving scenario and periodic update (Rel-9 to Rel-13) ..... 771
13.6.1 Sub-tests ..... 771
13.6.2 Test purpose ..... 771
13.6.3 Test applicability ..... 771
13.6.4 Minimum conformance requirements ..... 771
13.6.5 Test description ..... 772
13.6.5.1 Initial conditions ..... 772
13.6.5.2 Test procedure ..... 773
13.6.5.3 Message contents ..... 774
13.6.6 Test requirement ..... 775
13.7 Moving scenario and periodic update (Rel-14 onwards) ..... 776
13.7.1 Sub-tests ..... 776
13.7.2 Test purpose ..... 776
13.7.3 Test applicability ..... 777
13.7.4 Minimum conformance requirements ..... 777
13.7.5 Test description ..... 777
13.7.6 Test requirement ..... 777

14 NR RSTD measurement requirements ..... 777
14.1 General ..... 777
14.2 NR RSTD measurement period test cases in RRC_CONNECTED ..... 777
14.2.1 NR RSTD measurement period test case for single positioning frequency layer in FR1 SA ..... 777
14.2.1.1 Test purpose ..... 777
14.2.1.2 Test applicability ..... 777
14.2.1.3 Minimum conformance requirements ..... 777
14.2.1.4 Test description ..... 779
14.2.1.4.1 Initial conditions ..... 779
14.2.1.4.2 Test procedure ..... 779
14.2.1.4.3 Message contents ..... 780
14.2.1.5 Test requirement ..... 786
14.2.2 NR RSTD measurement period test case for dual positioning frequency layer in FR1 SA ..... 792
14.2.2.1 Test purpose ..... 792
14.2.2.2 Test applicability ..... 792
14.2.2.3 Minimum conformance requirements ..... 792
14.2.2.4 Test description ..... 792
14.2.2.4.1 Initial conditions ..... 792
14.2.2.4.2 Test procedure ..... 793
14.2.2.4.3 Message contents ..... 793
14.2.2.5 Test requirement ..... 795
14.2.3 NR RSTD measurement period test case for single positioning frequency layer in FR2 SA ..... 801
14.2.3.1 Test purpose ..... 801
14.2.3.2 Test applicability ..... 801
14.2.3.3 Minimum conformance requirements ..... 801
14.2.3.4 Test description ..... 801
14.2.3.4.1 Initial conditions ..... 801
14.2.3.4.2 Test procedure ..... 802
14.2.3.4.3 Message contents ..... 803
14.2.3.5 Test requirement ..... 808
14.2.4 NR RSTD measurement period test case for dual positioning frequency layers in FR2 SA ..... 813
14.2.4.1 Test purpose ..... 813

14.2.4.2 Test applicability ..... 813
14.2.4.3 Minimum conformance requirements ..... 813
14.2.4.4 Test description ..... 814
14.2.4.4.1 Initial conditions ..... 814
14.2.4.4.2 Test procedure ..... 814
14.2.4.4.3 Message contents ..... 814
14.2.4.5 Test requirement ..... 820
14.2.5 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 824
14.2.5.1 Test purpose ..... 824
14.2.5.2 Test applicability ..... 825
14.2.5.3 Minimum conformance requirements ..... 825
14.2.5.4 Test description ..... 827
14.2.5.4.1 Initial conditions ..... 827
14.2.5.4.2 Test procedure ..... 828
14.2.5.4.3 Message contents ..... 828
14.2.5.5 Test requirement ..... 835
14.2.6 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA ..... 839
14.2.6.1 Test purpose ..... 839
14.2.6.2 Test applicability ..... 839
14.2.6.3 Minimum conformance requirements ..... 840
14.2.6.4 Test description ..... 842
14.2.6.4.1 Initial conditions ..... 842
14.2.6.4.2 Test procedure ..... 842
14.2.6.4.3 Message contents ..... 843
14.2.6.5 Test requirement ..... 851
14.2.7 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR1 SA ..... 856
14.2.7.1 Test purpose ..... 856
14.2.7.2 Test applicability ..... 856
14.2.7.3 Minimum conformance requirements ..... 856
14.2.7.4 Test description ..... 857
14.2.7.4.1 Initial conditions ..... 857
14.2.7.4.2 Test procedure ..... 857
14.2.7.4.3 Message contents ..... 857
14.2.7.5 Test requirement ..... 864
14.2.8 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 869
14.2.8.1 Test purpose ..... 869
14.2.8.2 Test applicability ..... 869
14.2.8.3 Minimum conformance requirements ..... 869
14.2.8.4 Test description ..... 871
14.2.8.4.1 Initial conditions ..... 871
14.2.8.4.2 Test procedure ..... 872
14.2.8.4.3 Message contents ..... 873
14.2.8.5 Test requirement ..... 879
14.2.9 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA ..... 883
14.2.9.1 Test purpose ..... 883
14.2.9.2 Test applicability ..... 883
14.2.9.3 Minimum conformance requirements ..... 883
14.2.9.4 Test description ..... 886
14.2.9.4.1 Initial conditions ..... 886
14.2.9.4.2 Test procedure ..... 886
14.2.9.4.3 Message contents ..... 887
14.2.9.5 Test requirement ..... 895
14.2.10 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR2 SA ..... 899
14.2.10.1 Test purpose ..... 899
14.2.10.2 Test applicability ..... 899
14.2.10.3 Minimum conformance requirements ..... 899

14.2.10.4 Test description ..... 899
14.2.10.4.1 Initial conditions ..... 899
14.2.10.4.2 Test procedure ..... 900
14.2.10.4.3 Message contents ..... 900
14.2.10.5 Test requirement ..... 906
14.2.11 NR RSTD measurement reporting delay test case for PRS aggregation in FR1 ..... 910
14.2.11.1 Test purpose ..... 910
14.2.11.2 Test applicability ..... 910
14.2.11.4 Test description ..... 914
14.2.11.4.1 Initial conditions ..... 914
14.2.11.4.2 Test procedure ..... 914
14.2.11.4.3 Message contents ..... 915
14.2.11.5 Test requirement ..... 923
14.2.12 NR RSTD measurement reporting delay test case for PRS aggregation in FR2 ..... 929
14.2.12.1 Test purpose ..... 929
14.2.12.4 Test description ..... 929
14.2.12.4.1 Initial conditions ..... 929
14.2.12.4.2 Test procedure ..... 930
14.2.12.4.3 Message contents ..... 930
14.2.12.5 Test requirement ..... 934
14.2.13 NR RSTD measurement reporting delay test case in FR1 for Redcap UE without Rx FH ..... 937
14.2.13.1 Test purpose ..... 938
14.2.13.2 Test applicability ..... 938
14.2.13.3 Minimum conformance requirements ..... 938
14.2.13.4 Test description ..... 945
14.2.13.4.1 Initial conditions ..... 945
14.2.13.4.2 Test procedure ..... 945
14.2.13.4.3 Message contents ..... 945
14.2.13.5 Test requirement ..... 945
14.2.14 RSTD measurement reporting delay test case in FR2 for Redcap UE without Rx FH ..... 949
14.2.14.1 Test purpose ..... 950
14.2.14.2 Test applicability ..... 950
14.2.14.3 Minimum conformance requirements ..... 950
14.2.14.4 Test description ..... 950
14.2.14.4.1 Initial conditions ..... 950
14.2.14.4.2 Test procedure ..... 950
14.2.14.4.3 Message contents ..... 950
14.2.14.5 Test requirement ..... 950
14.2.15 NR RSTD measurement reporting delay test case in FR1 for Redcap UE with Rx FH ..... 954
14.2.15.1 Test purpose ..... 954
14.2.15.2 Test applicability ..... 954
14.2.15.3 Minimum conformance requirements ..... 955
14.2.15.4 Test description ..... 957
14.2.15.4.1 Initial conditions ..... 957
14.2.15.4.2 Test procedure ..... 957
14.2.15.4.3 Message contents ..... 957
14.2.15.5 Test requirement ..... 957
14.2.16 NR RSTD measurement reporting delay test case in FR2 for Redcap UE with Rx FH ..... 961
14.2.16.1 Test purpose ..... 961
14.2.16.2 Test applicability ..... 961
14.2.16.3 Minimum conformance requirements ..... 961
14.2.16.4 Test description ..... 961
14.2.16.4.1 Initial conditions ..... 961
14.2.16.4.2 Test procedure ..... 961
14.2.16.4.3 Message contents ..... 961
14.2.16.5 Test requirement ..... 961
14.3 NR RSTD measurement accuracy test case in RRC_CONNECTED ..... 964
14.3.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA ..... 964
14.3.1.1 Test purpose ..... 964
14.3.1.2 Test applicability ..... 964
14.3.1.3 Minimum conformance requirements ..... 965
14.3.1.4 Test description ..... 967

14.3.1.4.1 Initial conditions ..... 967
14.3.1.4.2 Test procedure ..... 967
14.3.1.4.3 Message contents ..... 968
14.3.1.5 Test requirement ..... 974
14.3.2 NR RSTD measurement accuracy test case for dual positioning frequency layer in FR1 SA ..... 976
14.3.2.1 Test purpose ..... 976
14.3.2.2 Test applicability ..... 976
14.3.2.3 Minimum conformance requirements ..... 976
14.3.2.4 Test description ..... 976
14.3.2.4.1 Initial conditions ..... 976
14.3.2.4.2 Test procedure ..... 977
14.3.2.4.3 Message contents ..... 977
14.3.2.5 Test requirement ..... 979
14.3.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA ..... 981
14.3.3.1 Test purpose ..... 981
14.3.3.2 Test applicability ..... 981
14.3.3.3 Minimum conformance requirements ..... 982
14.3.3.4 Test description ..... 982
14.3.3.4.1 Initial conditions ..... 982
14.3.3.4.2 Test procedure ..... 982
14.3.3.4.3 Message contents ..... 983
14.3.3.5 Test requirement ..... 988
14.3.4 NR RSTD measurement accuracy test case for dual positioning frequency layer in FR2 SA ..... 990
14.3.4.1 Test purpose ..... 990
14.3.4.2 Test applicability ..... 990
14.3.4.3 Minimum conformance requirements ..... 990
14.3.4.4 Test description ..... 990
14.3.4.4.1 Initial conditions ..... 990
14.3.4.4.2 Test procedure ..... 991
14.3.4.4.3 Message contents ..... 991
14.3.4.5 Test requirement ..... 996
14.3.5 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 998
14.3.5.1 Test purpose ..... 998
14.3.5.2 Test applicability ..... 998
14.3.5.3 Minimum conformance requirements ..... 998
14.3.5.4 Test description ..... 1000
14.3.5.4.1 Initial conditions ..... 1000
14.3.5.4.2 Test procedure ..... 1001
14.3.5.4.3 Message contents ..... 1002
14.3.5.5 Test requirement ..... 1008
14.3.6 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR1 SA ..... 1010
14.3.6.1 Test purpose ..... 1010
14.3.6.2 Test applicability ..... 1010
14.3.6.3 Minimum conformance requirements ..... 1010
14.3.6.4 Test description ..... 1010
14.3.6.4.1 Initial conditions ..... 1010
14.3.6.4.2 Test procedure ..... 1011
14.3.6.4.3 Message contents ..... 1012
14.3.6.5 Test requirement ..... 1018
14.3.7 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 1020
14.3.7.1 Test purpose ..... 1020
14.3.7.2 Test applicability ..... 1020
14.3.7.3 Minimum conformance requirements ..... 1020
14.3.7.4 Test description ..... 1022
14.3.7.4.1 Initial conditions ..... 1022
14.3.7.4.2 Test procedure ..... 1022
14.3.7.4.3 Message contents ..... 1023
14.3.7.5 Test requirement ..... 1029

14.3.8 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR2 SA ..... 1031
14.3.8.1 Test purpose ..... 1031
14.3.8.2 Test applicability ..... 1031
14.3.8.3 Minimum conformance requirements ..... 1031
14.3.8.4 Test description ..... 1031
14.3.8.4.1 Initial conditions ..... 1031
14.3.8.4.2 Test procedure ..... 1032
14.3.8.4.3 Message contents ..... 1032
14.3.8.5 Test requirement ..... 1038
14.3.9 NR RSTD measurement accuracy test case for PRS aggregation in FR1 SA ..... 1040
14.3.9.1 Test purpose ..... 1040
14.3.9.2 Test applicability ..... 1040
14.3.9.3 Minimum conformance requirements ..... 1040
14.3.9.4 Test description ..... 1046
14.3.9.4.1 Initial conditions ..... 1046
14.3.9.4.2 Test procedure ..... 1046
14.3.9.4.3 Message contents ..... 1047
14.3.9.5 Test requirement ..... 1054
The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.14.3.10 NR RSTD measurement accuracy test case for PRS aggregation in FR2 SA ..... 1056
14.3.10.1 Test purpose ..... 1056
14.3.10.2 Test applicability ..... 1056
14.3.10.3 Minimum conformance requirements ..... 1056
14.3.10.4 Test description ..... 1056
14.3.10.4.1 Initial conditions ..... 1056
14.3.10.4.2 Test procedure ..... 1056
14.3.10.4.3 Message contents ..... 1057
14.3.10.5 Test requirements ..... 1060
14.3.11 NR RSTD measurement accuracy test case in FR1 for Redcap UE without Rx FH ..... 1061
14.3.11.1 Test purpose ..... 1061
14.3.11.2 Test applicability ..... 1062
14.3.11.3 Minimum conformance requirements ..... 1062
14.3.11.4 Test description ..... 1062
14.3.11.4.1 Initial conditions ..... 1062
14.3.11.4.2 Test procedure ..... 1062
14.3.11.4.3 Message contents ..... 1062
14.3.11.5 Test requirement ..... 1062
14.3.12 NR RSTD measurement accuracy test case in FR2 for Redcap UE without Rx FH ..... 1064
14.3.12.1 Test purpose ..... 1065
14.3.12.2 Test applicability ..... 1065
14.3.12.3 Minimum conformance requirements ..... 1065
14.3.12.4 Test description ..... 1065
14.3.12.4.1 Initial conditions ..... 1065
14.3.12.4.2 Test procedure ..... 1065
14.3.12.4.3 Message contents ..... 1065
14.3.12.5 Test requirement ..... 1065
14.3.13 NR RSTD measurement accuracy test case in FR1 for Redcap UE with Rx FH ..... 1067
14.3.13.1 Test purpose ..... 1067
14.3.13.2 Test applicability ..... 1067
14.3.13.3 Minimum conformance requirements ..... 1067
14.3.13.4 Test description ..... 1068
14.3.13.4.1 Initial conditions ..... 1068
14.3.13.4.2 Test procedure ..... 1068
14.3.13.4.3 Message contents ..... 1068
14.3.13.5 Test requirement ..... 1068
14.3.14 NR RSTD measurement accuracy test case in FR2 for Redcap UE with Rx FH ..... 1070
14.3.14.1 Test purpose ..... 1070
14.3.14.2 Test applicability ..... 1070
14.3.14.3 Minimum conformance requirements ..... 1070
14.3.14.4 Test description ..... 1071
14.3.14.4.1 Initial conditions ..... 1071

14.3.14.4.2 Test procedure ..... 1071
14.3.14.4.3 Message contents ..... 1071
14.3.14.5 Test requirement ..... 1071
14.4 NR RSTD measurement period test cases in RRC_INACTIVE ..... 1073
14.4.1 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR1 SA ..... 1073
14.4.1.1 Test purpose ..... 1073
14.4.1.2 Test applicability ..... 1073
14.4.1.3 Minimum conformance requirements ..... 1073
14.4.1.4 Test description ..... 1075
14.4.1.4.1 Initial conditions ..... 1075
14.4.1.4.2 Test procedure ..... 1076
14.4.1.4.3 Message contents ..... 1077
14.4.1.5 Test requirement ..... 1085
14.4.2 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 1089
14.4.2.1 Test purpose ..... 1089
14.4.2.2 Test applicability ..... 1089
14.4.2.3 Minimum conformance requirements ..... 1089
14.4.2.4 Test description ..... 1090
14.4.2.4.1 Initial conditions ..... 1090
14.4.2.4.2 Test procedure ..... 1090
14.4.2.4.3 Message contents ..... 1090
14.4.2.5 Test requirement ..... 1094
14.4.3 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR2 SA ..... 1098
14.4.3.1 Test purpose ..... 1098
14.4.3.2 Test applicability ..... 1098
14.4.3.3 Minimum conformance requirements ..... 1098
14.4.3.4 Test description ..... 1100
14.4.3.4.1 Initial conditions ..... 1100
14.4.3.4.2 Test procedure ..... 1101
14.4.3.4.3 Message contents ..... 1102
14.4.3.5 Test requirement ..... 1110
14.4.4 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 1114
14.4.4.1 Test purpose ..... 1114
14.4.4.2 Test applicability ..... 1114
14.4.4.3 Minimum conformance requirements ..... 1114
14.4.4.4 Test description ..... 1114
14.4.4.4.1 Initial conditions ..... 1114
14.4.4.4.2 Test procedure ..... 1115
14.4.4.4.3 Message contents ..... 1115
14.4.4.5 Test requirement ..... 1117
14.4.5 NR RSTD measurement reporting delay test case for PRS aggregation in FR1 SA in RRC_INACTIVE state ..... 1121
14.4.5.1 Test purpose ..... 1121
14.4.5.2 Test applicability ..... 1121
14.4.5.3 Minimum conformance requirements ..... 1121
14.4.5.4 Test description ..... 1124
14.4.5.4.1 Initial conditions ..... 1124
14.4.5.4.2 Test procedure ..... 1125
14.4.5.4.3 Message contents ..... 1126
14.4.5.5 Test requirement ..... 1139
14.4.6 NR RSTD measurement reporting delay test case for PRS aggregation in FR2 SA in RRC_INACTIVE state ..... 1142
14.4.6.1 Test purpose ..... 1142
14.4.6.2 Test applicability ..... 1142
14.4.6.3 Minimum conformance requirements ..... 1142
14.4.6.4 Test description ..... 1142
14.4.6.4.1 Initial conditions ..... 1142
14.4.6.4.2 Test procedure ..... 1143
14.4.6.4.3 Message contents ..... 1143
14.4.6.5 Test requirement ..... 1149

14.4.7 NR RSTD measurement reporting delay test case in FR1 for Redcap UE without Rx FH ..... 1152
14.4.7.1 Test purpose ..... 1152
14.4.7.2 Test applicability ..... 1152
14.4.7.3 Minimum conformance requirements ..... 1152
14.4.7.4 Test description ..... 1155
14.4.7.4.1 Initial conditions ..... 1155
14.4.7.4.2 Test procedure ..... 1156
14.4.7.4.3 Message contents ..... 1157
14.4.7.5 Test requirement ..... 1165
14.4.8 NR RSTD measurement reporting delay test case in FR2 for Redcap UE without Rx FH ..... 1168
14.4.8.1 Test purpose ..... 1168
14.4.8.2 Test applicability ..... 1169
14.4.8.3 Minimum conformance requirements ..... 1169
14.4.8.4 Test description ..... 1169
14.4.8.4.1 Initial conditions ..... 1169
14.4.8.4.2 Test procedure ..... 1169
14.4.8.4.3 Message contents ..... 1169
14.4.8.5 Test requirement ..... 1172
14.5 NR RSTD measurement accuracy test case in RRC_INACTIVE ..... 1174
14.5.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA ..... 1174
14.5.1.1 Test purpose ..... 1174
14.5.1.2 Test applicability ..... 1174
14.5.1.3 Minimum conformance requirements ..... 1175
14.5.1.4 Test description ..... 1176
14.5.1.4.1 Initial conditions ..... 1177
14.5.1.4.2 Test procedure ..... 1177
14.5.1.4.3 Message contents ..... 1178
14.5.1.5 Test requirement ..... 1182
14.5.2 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA ..... 1184
14.5.2.1 Test purpose ..... 1184
14.5.2.2 Test applicability ..... 1184
14.5.2.3 Minimum conformance requirements ..... 1184
14.5.2.4 Test description ..... 1186
14.5.2.4.1 Initial conditions ..... 1186
14.5.2.4.2 Test procedure ..... 1186
14.5.2.4.3 Message contents ..... 1186
14.5.2.5 Test requirement ..... 1190
14.5.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA ..... 1192
14.5.3.1 Test purpose ..... 1192
14.5.3.2 Test applicability ..... 1192
14.5.3.3 Minimum conformance requirements ..... 1192
14.5.3.4 Test description ..... 1193
14.5.3.4.1 Initial conditions ..... 1194
14.5.3.4.2 Test procedure ..... 1194
14.5.3.4.3 Message contents ..... 1195
14.5.3.5 Test requirement ..... 1198
14.5.4 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA ..... 1200
14.5.4.1 Test purpose ..... 1200
14.5.4.2 Test applicability ..... 1200
14.5.4.3 Minimum conformance requirements ..... 1200
14.5.4.4 Test description ..... 1202
14.5.4.4.1 Initial conditions ..... 1202
14.5.4.4.2 Test procedure ..... 1202
14.5.4.4.3 Message contents ..... 1202
14.5.4.5 Test requirement ..... 1206
14.5.5 RSTD measurement accuracy for PRS aggregation in FR1 in RRC_INACTIVE state ..... 1207
14.5.5.1 Test purpose ..... 1207
14.5.5.2 Test applicability ..... 1207
14.5.5.3 Minimum conformance requirements ..... 1208
14.5.5.4 Test description ..... 1213

14.5.5.4.1 Initial conditions ..... 1213
14.5.5.4.2 Test procedure ..... 1213
14.5.5.4.3 Message contents ..... 1214
14.5.5.5 Test requirement ..... 1221
14.5.6 RSTD measurement accuracy for PRS aggregation in FR2 in RRC_INACTIVE state ..... 1222
14.5.6.1 Test purpose ..... 1222
14.5.6.2 Test applicability ..... 1223
14.5.6.3 Minimum conformance requirements ..... 1223
14.5.6.4 Test description ..... 1223
14.5.6.4.1 Initial conditions ..... 1223
14.5.6.4.2 Test procedure ..... 1223
14.5.6.4.3 Message contents ..... 1224
14.5.6.5 Test requirement ..... 1230

15 UE Rx-Tx time difference measurement requirements ..... 1232
15.1 General ..... 1232
15.2 UE Rx-Tx time difference measurement period test cases in RRC_CONNECTED ..... 1232
15.2.1 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR1 SA ..... 1232
15.2.1.1 Test purpose ..... 1232
15.2.1.2 Test applicability ..... 1232
15.2.1.3 Minimum conformance requirements ..... 1232
15.2.1.4 Test description ..... 1234
15.2.1.4.1 Initial conditions ..... 1234
15.2.1.4.2 Test procedure ..... 1234
15.2.1.4.3 Message contents ..... 1235
15.2.1.5 Test requirement ..... 1242
15.2.2 UE Rx-Tx time difference measurement period for dual positioning frequency layer in FR1 SA ..... 1244
15.2.2.1 Test purpose ..... 1244
15.2.2.2 Test applicability ..... 1245
15.2.2.3 Minimum conformance requirements ..... 1245
15.2.2.4 Test description ..... 1245
15.2.2.4.1 Initial conditions ..... 1245
15.2.2.4.2 Test procedure ..... 1245
15.2.2.4.3 Message contents ..... 1245
15.2.2.5 Test requirement ..... 1253
15.2.3 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR2 SA ..... 1255
15.2.3.1 Test purpose ..... 1255
15.2.3.2 Test applicability ..... 1256
15.2.3.3 Minimum conformance requirements ..... 1256
15.2.3.4 Test description ..... 1256
15.2.3.4.1 Initial conditions ..... 1256
15.2.3.4.2 Test procedure ..... 1256
15.2.3.4.3 Message contents ..... 1256
15.2.3.5 Test requirement ..... 1263
15.2.4 UE Rx-Tx time difference measurement period for dual positioning frequency layer in FR2 SA ..... 1266
15.2.4.1 Test purpose ..... 1266
15.2.4.2 Test applicability ..... 1266
15.2.4.3 Minimum conformance requirements ..... 1266
15.2.4.4 Test description ..... 1266
15.2.4.4.1 Initial conditions ..... 1266
15.2.4.4.2 Test procedure ..... 1266
15.2.4.4.3 Message contents ..... 1267
15.2.4.5 Test requirement ..... 1273
15.2.5 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 1276
15.2.5.1 Test purpose ..... 1276
15.2.5.2 Test applicability ..... 1276
15.2.5.3 Minimum conformance requirements ..... 1276
15.2.5.4 Test description ..... 1278
15.2.5.4.1 Initial conditions ..... 1278
15.2.5.4.2 Test procedure ..... 1279
15.2.5.4.3 Message contents ..... 1280

15.2.5.5 Test requirement ..... 1287
15.2.6 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA ..... 1290
15.2.6.1 Test purpose ..... 1290
15.2.6.2 Test applicability ..... 1291
15.2.6.3 Minimum conformance requirements ..... 1291
15.2.6.4 Test description ..... 1293
15.2.6.4.1 Initial conditions ..... 1293
15.2.6.4.2 Test procedure ..... 1294
15.2.6.4.3 Message contents ..... 1295
15.2.6.5 Test requirement ..... 1302
15.2.7 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with RxTx TEG in FR1 SA ..... 1306
15.2.7.1 Test purpose ..... 1306
15.2.7.2 Test applicability ..... 1306
15.2.7.3 Minimum conformance requirements ..... 1306
15.2.7.4 Test description ..... 1306
15.2.7.4.1 Initial conditions ..... 1306
15.2.7.4.2 Test procedure ..... 1307
15.2.7.4.3 Message contents ..... 1307
15.2.7.5 Test requirement ..... 1314
15.2.8 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 1318
15.2.8.1 Test purpose ..... 1318
15.2.8.2 Test applicability ..... 1318
15.2.8.3 Minimum conformance requirements ..... 1318
15.2.8.4 Test description ..... 1320
15.2.8.4.1 Initial conditions ..... 1320
15.2.8.4.2 Test procedure ..... 1321
15.2.8.4.3 Message contents ..... 1322
15.2.8.5 Test requirement ..... 1329
15.2.9 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA ..... 1331
15.2.9.1 Test purpose ..... 1331
15.2.9.2 Test applicability ..... 1332
15.2.9.3 Minimum conformance requirements ..... 1332
15.2.9.4 Test description ..... 1334
15.2.9.4.1 Initial conditions ..... 1334
15.2.9.4.2 Test procedure ..... 1335
15.2.9.4.3 Message contents ..... 1336
15.2.9.5 Test requirement ..... 1343
15.2.10 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with RxTx TEG in FR2 SA ..... 1346
15.2.10.1 Test purpose ..... 1346
15.2.10.2 Test applicability ..... 1346
15.2.10.3 Minimum conformance requirements ..... 1346
15.2.10.4 Test description ..... 1346
15.2.10.4.1 Initial conditions ..... 1346
15.2.10.4.2 Test procedure ..... 1347
15.2.10.4.3 Message contents ..... 1347
15.2.10.5 Test requirement ..... 1354
15.2.11 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR1 SA ..... 1356
15.2.11.1 Test purpose ..... 1356
15.2.11.2 Test applicability ..... 1357
15.2.11.3 Minimum conformance requirements ..... 1357
15.2.11.4 Test description ..... 1360
15.2.11.4.1 Initial conditions ..... 1360
15.2.11.4.2 Test procedure ..... 1360
15.2.11.4.3 Message contents ..... 1361
15.2.11.5 Test requirement ..... 1370
15.2.12 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR2 SA ..... 1374

This test case is incomplete for test frequencies > 40.8 GHz15.2.12.1 Test purpose ..... 1374
15.2.12.2 Test applicability ..... 1374
15.2.12.3 Minimum conformance requirements ..... 1374
15.2.12.4 Test description ..... 1374
15.2.12.4.1 Initial conditions ..... 1374
15.2.12.4.2 Test procedure ..... 1375
15.2.12.4.3 Message contents ..... 1375
15.2.12.5 Test requirement ..... 1379
15.2.13 UE Rx-Tx time difference measurement reporting delay test case in FR1 for Redcap UE without Rx FH ..... 1382
15.2.13.1 Test purpose ..... 1382
15.2.13.2 Test applicability ..... 1382
15.2.13.3 Minimum conformance requirements ..... 1382
15.2.13.4 Test description ..... 1382
15.2.13.4.1 Initial conditions ..... 1382
15.2.13.4.2 Test procedure ..... 1382
15.2.13.4.3 Message contents ..... 1383
15.2.13.5 Test requirement ..... 1385
15.2.14 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE without Rx FH ..... 1389
15.2.14.1 Test purpose ..... 1389
15.2.14.2 Test applicability ..... 1389
15.2.14.3 Minimum conformance requirements ..... 1389
15.2.14.4 Test description ..... 1389
15.2.14.4.1 Initial conditions ..... 1389
15.2.14.4.2 Test procedure ..... 1389
15.2.14.4.3 Message contents ..... 1390
15.2.14.5 Test requirement ..... 1395
15.2.15 UE Rx-Tx time difference measurement reporting delay test case in FR1 for Redcap UE with Rx FH 1398
15.2.15.1 Test purpose ..... 1398
15.2.15.2 Test applicability ..... 1398
15.2.15.3 Minimum conformance requirements ..... 1398
15.2.15.4 Test description ..... 1400
15.2.15.4.1 Initial conditions ..... 1400
15.2.15.4.2 Test procedure ..... 1401
15.2.15.4.3 Message contents ..... 1401
15.2.15.5 Test requirement ..... 1407
15.2.16 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE with Rx FH 1407
15.2.16.1 Test purpose ..... 1408
15.2.16.2 Test applicability ..... 1408
15.2.16.3 Minimum conformance requirements ..... 1408
15.2.16.4 Test description ..... 1408
15.2.16.4.1 Initial conditions ..... 1408
15.2.16.4.2 Test procedure ..... 1408
15.2.16.4.3 Message contents ..... 1408
15.2.16.5 Test requirement ..... 1414
15.3 UE Rx-Tx time difference measurement accuracy test cases in RRC_CONNECTED ..... 1418
15.3.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA ..1418
15.3.1.1 Test purpose ..... 1418
15.3.1.2 Test applicability ..... 1418
15.3.1.3 Minimum conformance requirements ..... 1418
15.3.1.4 Test description ..... 1420
15.3.1.4.1 Initial conditions ..... 1420
15.3.1.4.2 Test procedure ..... 1420
15.3.1.4.3 Message contents ..... 1421
15.3.1.5 Test requirement ..... 1429
15.3.2 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA ..1431
15.3.2.1 Test purpose ..... 1431
15.3.2.2 Test applicability ..... 1431
15.3.2.3 Minimum conformance requirements ..... 1432
15.3.2.4 Test description ..... 1432

15.3.2.4.1 Initial conditions ..... 1432
15.3.2.4.2 Test procedure ..... 1432
15.3.2.4.3 Message contents ..... 1432
15.3.2.5 Test requirement ..... 1439
15.3.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA ..... 1441
15.3.3.1 Test purpose ..... 1441
15.3.3.2 Test applicability ..... 1441
15.3.3.3 Minimum conformance requirements ..... 1441
15.3.3.4 Test description ..... 1442
15.3.3.4.1 Initial conditions ..... 1442
15.3.3.4.2 Test procedure ..... 1443
15.3.3.4.3 Message contents ..... 1444
15.3.3.5 Test requirement ..... 1451
15.3.4 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR1 SA ..... 1453
15.3.4.1 Test purpose ..... 1453
15.3.4.2 Test applicability ..... 1453
15.3.4.3 Minimum conformance requirements ..... 1453
15.3.4.4 Test description ..... 1454
15.3.4.4.1 Initial conditions ..... 1454
15.3.4.4.2 Test procedure ..... 1455
15.3.4.4.3 Message contents ..... 1456
15.3.4.5 Test requirement ..... 1463
15.3.5 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA ..... 1466
15.3.5.1 Test purpose ..... 1466
15.3.5.2 Test applicability ..... 1466
15.3.5.3 Minimum conformance requirements ..... 1466
15.3.5.4 Test description ..... 1467
15.3.5.4.1 Initial conditions ..... 1467
15.3.5.4.2 Test procedure ..... 1467
15.3.5.4.3 Message contents ..... 1468
15.3.5.5 Test requirement ..... 1475
15.3.6 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR2 SA ..... 1477
15.3.6.1 Test purpose ..... 1477
15.3.6.2 Test applicability ..... 1477
15.3.6.3 Minimum conformance requirements ..... 1477
15.3.6.4 Test description ..... 1478
15.3.6.4.1 Initial conditions ..... 1478
15.3.6.4.2 Test procedure ..... 1479
15.3.6.4.3 Message contents ..... 1479
15.3.6.5 Test requirement ..... 1487
15.3.7 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR1 SA ..... 1489
15.3.7.1 Test purpose ..... 1489
15.3.7.2 Test applicability ..... 1489
15.3.7.3 Minimum conformance requirements ..... 1489
15.3.7.4 Test description ..... 1492
15.3.7.4.1 Initial conditions ..... 1492
15.3.7.4.2 Test procedure ..... 1493
15.3.7.4.3 Message contents ..... 1494
15.3.7.5 Test requirement ..... 1503
15.3.8 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR2 SA ..... 1505
This test case is incomplete for test frequencies > 40.8 GHz15.3.8.1 ..... Test purpose ..... 1505
15.3.8.2 Test applicability ..... 1506
15.3.8.3 Minimum conformance requirements ..... 1506
15.3.8.4 Test description ..... 1506
15.3.8.4.1 Initial conditions ..... 1506
15.3.8.4.2 Test procedure ..... 1506
15.3.8.4.3 Message contents ..... 1506

15.3.8.5 Test requirement ....1511
15.3.9 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH ....1513
15.3.9.1 Test purpose ....1513
15.3.9.2 Test applicability ....1513
15.3.9.3 Minimum conformance requirements ....1513
15.3.9.4 Test description ....1515
15.3.9.4.1 Initial conditions ....1515
15.3.9.4.2 Test procedure ....1515
15.3.9.4.3 Message contents ....1515
15.3.9.5 Test requirement ....1520
15.3.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH ....1522
15.3.10.1 Test purpose ....1522
15.3.10.2 Test applicability ....1523
15.3.10.3 Minimum conformance requirements ....1523
15.3.10.4 Test description ....1523
15.3.10.4.1 Initial conditions ....1523
15.3.10.4.2 Test procedure ....1523
15.3.10.4.3 Message contents ....1523
15.3.10.5 Test requirement ....1528
15.3.11 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE with Rx FH ....1530
15.3.11.1 Test purpose ....1530
15.3.11.2 Test applicability ....1530
15.3.11.3 Minimum conformance requirements ....1530
15.3.11.4 Test description ....1534
15.3.11.4.1 Initial conditions ....1534
15.3.11.4.2 Test procedure ....1534
15.3.11.4.3 Message contents ....1534
15.3.11.5 Test requirement ....1539
15.3.12 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH ....1540
15.3.12.1 Test purpose ....1540
15.3.12.2 Test applicability ....1540
15.3.12.3 Minimum conformance requirements ....1540
15.3.12.4 Test description ....1540
15.3.12.4.1 Initial conditions ....1540
15.3.12.4.2 Test procedure ....1540
15.3.12.4.3 Message contents ....1541
15.3.12.5 Test requirement ....1545
15.4 UE Rx-Tx time difference measurement period test cases in RRC_INACTIVE ....1547
15.4.1 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR1 SA ....1547
15.4.1.1 Test purpose ....1547
15.4.1.2 Test applicability ....1547
15.4.1.3 Minimum conformance requirements ....1547
15.4.1.4 Test description ....1549
15.4.1.4.1 Initial conditions ....1549
15.4.1.4.2 Test procedure ....1550
15.4.1.4.3 Message contents ....1551
15.4.1.5 Test requirement ....1559
15.4.2 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA ....1562
15.4.2.1 Test purpose ....1562
15.4.2.2 Test applicability ....1562
15.4.2.3 Minimum conformance requirements ....1563
15.4.2.4 Test description ....1563
15.4.2.4.1 Initial conditions ....1563
15.4.2.4.2 Test procedure ....1563
15.4.2.4.3 Message contents ....1563
15.4.2.5 Test requirement ....1565
15.4.3 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR2 SA ....1568
15.4.3.1 Test purpose ....1568
15.4.3.2 Test applicability ....1568

15.4.3.3 Minimum conformance requirements ..... 1568
15.4.3.4 Test description ..... 1568
15.4.3.4.1 Initial conditions ..... 1568
15.4.3.4.2 Test procedure ..... 1569
15.4.3.4.3 Message contents ..... 1569
15.4.3.5 Test requirement ..... 1572
15.4.4 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 1575
15.4.4.1 Test purpose ..... 1575
15.4.4.2 Test applicability ..... 1575
15.4.4.3 Minimum conformance requirements ..... 1575
15.4.4.4 Test description ..... 1575
15.4.4.4.1 Initial conditions ..... 1575
15.4.4.4.2 Test procedure ..... 1576
15.4.4.4.3 Message contents ..... 1576
15.4.4.5 Test requirement ..... 1580
15.4.5 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR1 SA ..... 1583
15.4.5.1 Test purpose ..... 1583
15.4.5.2 Test applicability ..... 1583
15.4.5.3 Minimum conformance requirements ..... 1583
15.4.5.4 Test description ..... 1586
15.4.5.4.1 Initial conditions ..... 1586
15.4.5.4.2 Test procedure ..... 1586
15.4.5.4.3 Message contents ..... 1588
15.4.5.5 Test requirement ..... 1594
15.4.6 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR2 SA ..... 1598
15.4.6.1 Test purpose ..... 1598
15.4.6.2 Test applicability ..... 1598
15.4.6.3 Minimum conformance requirements ..... 1598
15.4.6.4 Test description ..... 1598
15.4.6.4.2 Test procedure ..... 1599
15.4.6.4.3 Message contents ..... 1599
15.4.6.5 Test requirement ..... 1606
15.4.7 UE Rx-Tx time difference measurement reporting delay in FR1 for Redcap UE without Rx FH ..... 1609
15.4.7.1 Test purpose ..... 1609
15.4.7.2 Test applicability ..... 1609
15.4.7.3 Minimum conformance requirements ..... 1609
15.4.7.4 Test description ..... 1612
15.4.7.4.1 Initial conditions ..... 1612
15.4.7.4.2 Test procedure ..... 1612
15.4.7.4.3 Message contents ..... 1612
15.4.7.5 Test requirement ..... 1617
15.4.8 UE Rx-Tx time difference measurement reporting delay in FR2 for Redcap UE without Rx FH ..... 1620
15.4.8.1 Test purpose ..... 1621
15.4.8.2 Test applicability ..... 1621
15.4.8.3 Minimum conformance requirements ..... 1621
15.4.8.4 Test description ..... 1621
15.4.8.4.1 Initial conditions ..... 1621
15.4.8.4.2 Test procedure ..... 1621
15.4.8.4.3 Message contents ..... 1621
15.4.8.5 Test requirement ..... 1627
15.4.9 UE Rx-Tx time difference measurement reporting delay in FR1 for Redcap UE with Rx FH ..... 1630
15.4.9.1 Test purpose ..... 1630
15.4.9.2 Test applicability ..... 1630
15.4.9.3 Minimum conformance requirements ..... 1630
15.4.9.4 Test description ..... 1632
15.4.9.4.1 Initial conditions ..... 1632
15.4.9.4.2 Test procedure ..... 1633
15.4.9.4.3 Message contents ..... 1633
15.4.9.5 Test requirement ..... 1638
15.4.10 UE Rx-Tx time difference measurement reporting delay in FR2 for Redcap UE with Rx FH ..... 1638
15.4.10.1 Test purpose ..... 1638

15.4.10.2 Test applicability ..... 1638
15.4.10.3 Minimum conformance requirements ..... 1638
15.4.10.4 Test description ..... 1638
15.4.10.4.1 Initial conditions ..... 1638
15.4.10.4.2 Test procedure ..... 1639
15.4.10.4.3 Message contents ..... 1639
15.4.10.5 Test requirement ..... 1644
15.4.11 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for normal UE in FR1 SA ..... 1644
15.4.11.1 Test purpose ..... 1645
15.4.11.2 Test applicability ..... 1645
15.4.11.3 Minimum conformance requirements ..... 1645
15.4.11.4 Test description ..... 1647
15.4.11.4.1 Initial conditions ..... 1647
15.4.11.4.2 Test procedure ..... 1648
15.4.11.4.3 Message contents ..... 1648
15.4.11.5 Test requirement ..... 1651
15.4.12 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for RedCap UE in FR1 SA ..... 1651
15.4.12.1 Test purpose ..... 1651
15.4.12.2 Test applicability ..... 1651
15.4.12.3 Minimum conformance requirements ..... 1652
15.4.12.4 Test description ..... 1652
15.4.12.4.1 Initial conditions ..... 1652
15.4.12.4.2 Test procedure ..... 1652
15.4.12.4.3 Message contents ..... 1652
15.4.12.5 Test requirement ..... 1658
15.4.13 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for normal UE in FR2 SA ..... 1658
15.4.13.1 Test purpose ..... 1658
15.4.13.2 Test applicability ..... 1658
15.4.13.3 Minimum conformance requirements ..... 1658
15.4.13.4 Test description ..... 1658
15.4.13.4.1 Initial conditions ..... 1658
15.4.13.4.2 Test procedure ..... 1659
15.4.13.4.3 Message contents ..... 1659
15.4.13.5 Test requirement ..... 1665
15.4.14 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for RedCap UE in FR2 SA ..... 1665
15.4.14.1 Test purpose ..... 1665
15.4.14.2 Test applicability ..... 1665
15.4.14.3 Minimum conformance requirements ..... 1666
15.4.14.4 Test description ..... 1666
15.4.14.4.1 Initial conditions ..... 1666
15.4.14.4.2 Test procedure ..... 1666
15.4.14.4.3 Message contents ..... 1666
15.4.14.5 Test requirement ..... 1673
15.5 UE Rx-Tx time difference measurement accuracy test cases in RRC_INACTIVE ..... 1673
15.5.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA ..... 1673
15.5.1.1 Test purpose ..... 1673
15.5.1.2 Test applicability ..... 1673
15.5.1.3 Minimum conformance requirements ..... 1673
15.5.1.4 Test description ..... 1675
15.5.1.4.1 Initial conditions ..... 1675
15.5.1.4.2 Test procedure ..... 1676
15.5.1.4.3 Message contents ..... 1677
15.5.1.5 Test requirement ..... 1680
15.5.2 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR1 SA ..... 1682
15.5.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA ..... 1691
15.5.3.1 Test purpose ..... 1691
15.5.3.2 Test applicability ..... 1691
15.5.3.3 Minimum conformance requirements ..... 1691

15.5.3.4 Test description ..... 1693
15.5.3.4.1 Initial conditions ..... 1693
15.5.3.4.2 Test procedure ..... 1693
15.5.3.4.3 Message contents ..... 1694
15.5.3.5 Test requirement ..... 1698
15.5.5 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR1 SA ..... 1707
15.5.5.1 Test purpose ..... 1707
15.5.5.2 Test applicability ..... 1708
15.5.5.3 Minimum conformance requirements ..... 1708
15.5.5.4 Test description ..... 1708
15.5.5.4.1 Initial conditions ..... 1708
15.5.5.4.2 Test procedure ..... 1708
15.5.5.4.3 Message contents ..... 1709
15.5.5.5 Test requirement ..... 1716
15.5.6 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR2 SA ..... 1716
15.5.6.1 Test purpose ..... 1716
15.5.6.2 Test applicability ..... 1716
15.5.6.3 Minimum conformance requirements ..... 1717
15.5.6.4 Test description ..... 1717
15.5.6.4.1 Initial conditions ..... 1717
15.5.6.4.2 Test procedure ..... 1717
15.5.6.4.3 Message contents ..... 1717
15.5.6.5 Test requirement ..... 1725
15.5.7 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH ..... 1725
15.5.7.1 Test purpose ..... 1725
15.5.7.2 Test applicability ..... 1725
15.5.7.3 Minimum conformance requirements ..... 1725
15.5.7.4 Test description ..... 1725
15.5.7.4.1 Initial conditions ..... 1726
15.5.7.4.2 Test procedure ..... 1726
15.5.7.4.3 Message contents ..... 1726
15.5.7.5 Test requirement ..... 1731
15.5.8 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH ..... 1732
15.5.8.1 Test purpose ..... 1733
15.5.8.2 Test applicability ..... 1733
15.5.8.3 Minimum conformance requirements ..... 1733
15.5.8.4 Test description ..... 1733
15.5.8.4.1 Initial conditions ..... 1733
15.5.8.4.2 Test procedure ..... 1733
15.5.8.4.3 Message contents ..... 1733
15.5.8.5 Test requirement ..... 1739
15.5.9 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE with Rx FH ..... 1741
15.5.9.1 Test purpose ..... 1741
15.5.9.2 Test applicability ..... 1741
15.5.9.3 Minimum conformance requirements ..... 1741
15.5.9.4 Test description ..... 1741
15.5.9.4.1 Initial conditions ..... 1741
15.5.9.4.2 Test procedure ..... 1742
15.5.9.4.3 Message contents ..... 1742
15.5.9.5 Test requirement ..... 1747
15.5.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH ..... 1747
15.5.10.1 Test purpose ..... 1747
15.5.10.2 Test applicability ..... 1747
15.5.10.3 Minimum conformance requirements ..... 1747
15.5.10.4 Test description ..... 1747
15.5.10.4.1 Initial conditions ..... 1748
15.5.10.4.2 Test procedure ..... 1748
15.5.10.4.3 Message contents ..... 1748
15.5.10.5 Test requirement ..... 1754

16 NR PRS-RSRP measurement requirements ..... 1755
16.1 General ..... 1755

16.2 NR PRS-RSRP measurement period test cases in RRC_CONNECTED ..... 1755
16.2.1 PRS-RSRP measurement period test case for single positioning frequency layer in FR1 SA ..... 1755
16.2.1.1 Test purpose ..... 1755
16.2.1.2 Test applicability ..... 1755
16.2.1.3 Minimum conformance requirements ..... 1755
16.2.1.4 Test description ..... 1758
16.2.1.4.1 Initial conditions ..... 1758
16.2.1.4.2 Test procedure ..... 1758
16.2.1.4.3 Message contents ..... 1759
16.2.1.5 Test requirement ..... 1765
16.2.2 PRS-RSRP measurement period test case for dual positioning frequency layer in FR1 SA ..... 1768
16.2.2.1 Test purpose ..... 1768
16.2.2.2 Test applicability ..... 1768
16.2.2.3 Minimum conformance requirements ..... 1768
16.2.2.4 Test description ..... 1768
16.2.2.4.1 Initial conditions ..... 1768
16.2.2.4.2 Test procedure ..... 1769
16.2.2.4.3 Message contents ..... 1769
16.2.2.5 Test requirement ..... 1775
16.2.3 PRS-RSRP measurement period test case for single positioning frequency layer in FR2 SA ..... 1778
16.2.3.1 Test purpose ..... 1778
16.2.3.2 Test applicability ..... 1778
16.2.3.3 Minimum conformance requirements ..... 1778
16.2.3.4 Test description ..... 1780
16.2.3.4.1 Initial conditions ..... 1780
16.2.3.4.2 Test procedure ..... 1780
16.2.3.4.3 Message contents ..... 1781
16.2.3.5 Test requirement ..... 1786
16.2.4 PRS-RSRP measurement period test case for dual positioning frequency layer in FR2 SA ..... 1789
16.2.4.1 Test purpose ..... 1789
16.2.4.2 Test applicability ..... 1789
16.2.4.3 Minimum conformance requirements ..... 1790
16.2.4.4 Test description ..... 1790
16.2.4.4.1 Initial conditions ..... 1790
16.2.4.4.2 Test procedure ..... 1790
16.2.4.4.3 Message contents ..... 1790
16.2.4.5 Test requirement ..... 1796
16.2.5 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR1 SA ..... 1799
16.2.5.1 Test purpose ..... 1799
16.2.5.2 Test applicability ..... 1799
16.2.5.3 Minimum conformance requirements ..... 1799
16.2.5.4 Test description ..... 1801
16.2.5.4.1 Initial conditions ..... 1801
16.2.5.4.2 Test procedure ..... 1802
16.2.5.4.3 Message contents ..... 1802
16.2.5.5 Test requirement ..... 1808
16.2.6 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA ..... 1812
16.2.6.1 Test purpose ..... 1812
16.2.6.2 Test applicability ..... 1812
16.2.6.3 Minimum conformance requirements ..... 1812
16.2.6.4 Test description ..... 1814
16.2.6.4.1 Initial conditions ..... 1814
16.2.6.4.2 Test procedure ..... 1814
16.2.6.4.3 Message contents ..... 1815
16.2.6.5 Test requirement ..... 1822
16.2.7 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR2 SA ..... 1826
16.2.7.1 Test purpose ..... 1826
16.2.7.2 Test applicability ..... 1826
16.2.7.3 Minimum conformance requirements ..... 1826

16.2.7.4 Test description ..... 1828
16.2.7.4.1 Initial conditions ..... 1828
16.2.7.4.2 Test procedure ..... 1828
16.2.7.4.3 Message contents ..... 1829
16.2.7.5 Test requirement ..... 1835
16.2.8 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA ..... 1838
16.2.8.1 Test purpose ..... 1838
16.2.8.2 Test applicability ..... 1838
16.2.8.3 Minimum conformance requirements ..... 1838
16.2.8.4 Test description ..... 1840
16.2.8.4.1 Initial conditions ..... 1840
16.2.8.4.2 Test procedure ..... 1841
16.2.8.4.3 Message contents ..... 1842
16.2.8.5 Test requirement ..... 1848
16.2.9 PRS-RSRP measurement delay in FR1 for Redcap UE without Rx FH ..... 1851
16.2.9.1 Test purpose ..... 1851
16.2.9.2 Test applicability ..... 1851
16.2.9.3 Minimum conformance requirements ..... 1851
16.2.9.4 Test description ..... 1854
16.2.9.4.1 Initial conditions ..... 1854
16.2.9.4.2 Test procedures ..... 1854
16.2.9.4.3 Message contents ..... 1855
16.2.9.5 Test requirement ..... 1862
16.2.10 PRS-RSRP measurement delay in FR2 for Redcap UE without Rx FH for single positioning frequency layer ..... 1864
16.2.10.1 Test purpose ..... 1864
16.2.10.2 Test applicability ..... 1864
16.2.10.3 Minimum conformance requirements ..... 1864
16.2.10.4 Test description ..... 1864
16.2.10.4.1 Initial conditions ..... 1864
16.2.10.4.2 Test procedures ..... 1864
16.2.10.4.3 Message contents ..... 1865
16.2.10.5 Test requirement ..... 1866
16.2.11 PRS-RSRP measurement delay in FR2 for Redcap UE without Rx FH for dual positioning frequency layer ..... 1869
16.2.11.1 Test purpose ..... 1869
16.2.11.2 Test applicability ..... 1869
16.2.11.3 Minimum conformance requirements ..... 1869
16.2.11.4.1 Initial conditions ..... 1869
16.2.11.4.2 Test procedures ..... 1869
16.2.11.4.3 Message contents ..... 1870
16.2.11.5 Test requirement ..... 1872
16.2.12 PRS-RSRP measurement delay in FR1 for Redcap UE with Rx FH ..... 1874
16.2.12.1 Test purpose ..... 1874
16.2.12.2 Test applicability ..... 1874
16.2.12.3 Minimum conformance requirements ..... 1874
16.2.12.4 Test description ..... 1876
16.2.12.4.1 Initial conditions ..... 1876
16.2.12.4.2 Test procedures ..... 1877
16.2.12.4.3 Message contents ..... 1877
16.2.12.5 Test requirement ..... 1883
16.2.13 PRS-RSRP measurement delay in FR2 for Redcap UE with Rx FH ..... 1885
16.2.13.1 Test purpose ..... 1886
16.2.13.2 Test applicability ..... 1886
16.2.13.3 Minimum conformance requirements ..... 1886
16.2.13.4 Test description ..... 1886
16.2.13.4.1 Initial conditions ..... 1886
16.2.13.4.2 Test procedures ..... 1886
16.2.13.4.3 Message contents ..... 1886
16.2.13.5 Test requirement ..... 1890
16.3 NR PRS-RSRP measurement accuracy test case in RRC_CONNECTED ..... 1892

16.3.1 PRS-RSRP measurement accuracy with PRS in FR1 ..... 1892
16.3.1.1 Test purpose ..... 1892
16.3.1.2 Test applicability ..... 1892
16.3.1.3 Minimum conformance requirements ..... 1892
16.3.1.4 Test description ..... 1894
16.3.1.4.1 Initial conditions ..... 1894
16.3.1.4.2 Test procedure ..... 1895
16.3.1.4.3 Message contents ..... 1896
16.3.1.5 Test requirement ..... 1903
16.3.2 PRS-RSRP measurement accuracy with PRS in FR2 ..... 1905
16.3.2.1 Test purpose ..... 1905
16.3.2.2 Test applicability ..... 1905
16.3.2.3 Minimum conformance requirements ..... 1906
16.3.2.4 Test description ..... 1907
16.3.2.4.1 Initial conditions ..... 1907
16.3.2.4.2 Test procedure ..... 1907
16.3.2.4.3 Message contents ..... 1908
16.3.2.5 Test requirement ..... 1914
16.3.3 PRS-RSRP measurement accuracy with PRS in FR1 with reduced sample number ..... 1916
16.3.3.1 Test purpose ..... 1916
16.3.3.2 Test applicability ..... 1917
16.3.3.3 Minimum conformance requirements ..... 1917
16.3.3.4 Test description ..... 1918
16.3.3.4.1 Initial conditions ..... 1918
16.3.3.4.2 Test procedure ..... 1919
16.3.3.4.3 Message contents ..... 1919
16.3.3.5 Test requirement ..... 1925
16.3.4 PRS-RSRP measurement accuracy with PRS in FR2 with reduced sample number ..... 1928
16.3.4.1 Test purpose ..... 1928
16.3.4.2 Test applicability ..... 1928
16.3.4.3 Minimum conformance requirements ..... 1928
16.3.4.4 Test description ..... 1929
16.3.4.4.1 Initial conditions ..... 1929
16.3.4.4.2 Test procedure ..... 1929
16.3.4.4.3 Message contents ..... 1930
16.3.4.5 Test requirement ..... 1935
16.4 NR PRS-RSRP measurement period test cases in RRC_INACTIVE ..... 1937
16.4.1 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR1 SA ..... 1937
16.4.1.1 Test purpose ..... 1937
16.4.1.2 Test applicability ..... 1937
16.4.1.3 Minimum conformance requirements ..... 1938
16.4.1.4 Test description ..... 1939
16.4.1.4.1 Initial conditions ..... 1939
16.4.1.4.2 Test procedure ..... 1940
16.4.1.4.3 Message contents ..... 1941
16.4.1.5 Test requirement ..... 1948
16.4.2 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 1951
16.4.2.1 Test purpose ..... 1951
16.4.2.2 Test applicability ..... 1951
16.4.2.3 Minimum conformance requirements ..... 1951
16.4.2.4 Test description ..... 1951
16.4.2.4.1 Initial conditions ..... 1951
16.4.2.4.2 Test procedure ..... 1952
16.4.2.4.3 Message contents ..... 1952
16.4.2.5 Test requirement ..... 1955
16.4.3 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR2 SA ..... 1958
16.4.3.1 Test purpose ..... 1959
16.4.3.2 Test applicability ..... 1959
16.4.3.3 Minimum conformance requirements ..... 1959
16.4.3.4 Test description ..... 1960
16.4.3.4.1 Initial conditions ..... 1960

16.4.3.4.2 Test procedure ..... 1961
16.4.3.4.3 Message contents ..... 1962
16.4.3.5 Test requirement ..... 1969
16.4.4 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 1972
16.4.4.1 Test purpose ..... 1972
16.4.4.2 Test applicability ..... 1972
16.4.4.3 Minimum conformance requirements ..... 1973
16.4.4.4 Test description ..... 1973
16.4.4.4.1 Initial conditions ..... 1973
16.4.4.4.2 Test procedure ..... 1973
16.4.4.4.3 Message contents ..... 1973
16.4.4.5 Test requirement ..... 1976
16.5 NR PRS-RSRP measurement accuracy test case in RRC_INACTIVE ..... 1979
16.5.1 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 SA ..... 1979
16.5.1.1 Test purpose ..... 1979
16.5.1.2 Test applicability ..... 1979
16.5.1.3 Minimum conformance requirements ..... 1979
16.5.1.4 Test description ..... 1981
16.5.1.4.1 Initial conditions ..... 1981
16.5.1.4.2 Test procedure ..... 1982
16.5.1.4.3 Message contents ..... 1983
16.5.1.5 Test requirement ..... 1985
16.5.2 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 with reduced sample number ..... 1988
16.5.2.1 Test purpose ..... 1988
16.5.2.2 Test applicability ..... 1988
16.5.2.3 Minimum conformance requirements ..... 1988
16.5.2.4 Test description ..... 1990
16.5.2.4.1 Initial conditions ..... 1990
16.5.2.4.2 Test procedure ..... 1990
16.5.2.4.3 Message contents ..... 1990
16.5.2.5 Test requirement ..... 1992
16.5.3 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 SA ..... 1995
16.5.3.1 Test purpose ..... 1995
16.5.3.2 Test applicability ..... 1995
16.5.3.3 Minimum conformance requirements ..... 1995
16.5.3.4 Test description ..... 1995
16.5.3.4.1 Initial conditions ..... 1995
16.5.3.4.2 Test procedure ..... 1995
16.5.3.4.3 Message contents ..... 1997
16.5.3.5 Test requirement ..... 1999
16.5.4 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 with reduced sample number ..... 2001
16.5.4.1 Test purpose ..... 2001
16.5.4.2 Test applicability ..... 2001
16.5.4.3 Minimum conformance requirements ..... 2001
16.5.4.4 Test description ..... 2001
16.5.4.4.1 Initial conditions ..... 2001
16.5.4.4.2 Test procedure ..... 2002
16.5.4.4.3 Message contents ..... 2002
16.5.4.5 Test requirement ..... 2005

17 NR PRS-RSRPP measurement requirements ..... 2007
17.1 General ..... 2007
17.2 NR PRS-RSRPP measurement period test cases in RRC_CONNECTED ..... 2007
17.2.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA ..... 2007
17.2.1.1 Test purpose ..... 2007
17.2.1.2 Test applicability ..... 2007

17.2.1.3 Minimum conformance requirements ....2008
17.2.1.4 Test description ....2009
17.2.1.4.1 Initial conditions ....2009
17.2.1.4.2 Test procedure ....2010
17.2.1.4.3 Message contents ....2011
17.2.1.5 Test requirement ....2017
17.2.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA ....2019
17.2.2.1 Test purpose ....2019
17.2.2.2 Test applicability ....2019
17.2.2.3 Minimum conformance requirements ....2020
17.2.2.4 Test description ....2020
17.2.2.4.1 Initial conditions ....2020
17.2.2.4.2 Test procedure ....2020
17.2.2.4.3 Message contents ....2020
17.2.2.5 Test requirement ....2026
17.2.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA ....2029
17.2.3.1 Test purpose ....2029
17.2.3.2 Test applicability ....2029
17.2.3.3 Minimum conformance requirements ....2029
17.2.3.4 Test description ....2029
17.2.3.4.1 Initial conditions ....2029
17.2.3.4.2 Test procedure ....2030
17.2.3.4.3 Message contents ....2031
17.2.3.5 Test requirement ....2038
17.2.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA ....2041
17.2.4.1 Test purpose ....2041
17.2.4.2 Test applicability ....2041
17.2.4.3 Minimum conformance requirements ....2041
17.2.4.4 Test description ....2041
17.2.4.4.1 Initial conditions ....2041
17.2.4.4.2 Test procedure ....2042
17.2.4.4.3 Message contents ....2042
17.2.4.5 Test requirement ....2048
17.2.5 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA ....2051
17.2.5.1 Test purpose ....2051
17.2.5.2 Test applicability ....2051
17.2.5.3 Minimum conformance requirements ....2051
17.2.5.4 Test description ....2052
17.2.5.4.1 Initial conditions ....2052
17.2.5.4.2 Test procedure ....2052
17.2.5.4.3 Message contents ....2052
17.2.5.5 Test requirement ....2058
17.2.6 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA ....2061
17.2.6.1 Test purpose ....2062
17.2.6.2 Test applicability ....2062
17.2.6.3 Minimum conformance requirements ....2062
17.2.6.4 Test description ....2062
17.2.6.4.1 Initial conditions ....2062
17.2.6.4.2 Test procedure ....2062
17.2.6.4.3 Message contents ....2062
17.2.6.5 Test requirement ....2068
17.3 NR PRS-RSRPP measurement accuracy test case in RRC_CONNECTED ....2072
17.3.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA ....2072
17.3.1.1 Test purpose ....2072
17.3.1.2 Test applicability ....2072
17.3.1.3 Minimum conformance requirements ....2072
17.3.1.4 Test description ....2073

17.3.1.4.1 Initial conditions ..... 2073
17.3.1.4.2 Test procedure ..... 2074
17.3.1.4.3 Message contents ..... 2074
17.3.1.5 Test requirement ..... 2081
17.3.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 2083
17.3.2.1 Test purpose ..... 2083
17.3.2.2 Test applicability ..... 2083
17.3.2.3 Minimum conformance requirements ..... 2083
17.3.2.4 Test description ..... 2084
17.3.2.4.1 Initial conditions ..... 2084
17.3.2.4.2 Test procedure ..... 2085
17.3.2.4.3 Message contents ..... 2085
17.3.2.5 Test requirement ..... 2088
17.3.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA ..... 2090
17.3.3.1 Test purpose ..... 2090
17.3.3.2 Test applicability ..... 2090
17.3.3.3 Minimum conformance requirements ..... 2090
17.3.3.4 Test description ..... 2091
17.3.3.4.1 Initial conditions ..... 2091
17.3.3.4.2 Test procedure ..... 2092
17.3.3.4.3 Message contents ..... 2092
17.3.3.5 Test requirement ..... 2094
17.3.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 2096
17.3.4.1 Test purpose ..... 2096
17.3.4.2 Test applicability ..... 2096
17.3.4.3 Minimum conformance requirements ..... 2096
17.3.4.4 Test description ..... 2097
17.3.4.4.1 Initial conditions ..... 2097
17.3.4.4.2 Test procedure ..... 2097
17.3.4.4.3 Message contents ..... 2097
17.3.4.5 Test requirement ..... 2101
17.4 NR PRS-RSRPP measurement period test cases in RRC_INACTIVE ..... 2102
17.4.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA ..... 2102
17.4.1.1 Test purpose ..... 2102
17.4.1.2 Test applicability ..... 2103
17.4.1.3 Minimum conformance requirements ..... 2103
17.4.1.4 Test description ..... 2103
17.4.1.4.1 Initial conditions ..... 2103
17.4.1.4.2 Test procedure ..... 2103
17.4.1.4.3 Message contents ..... 2104
17.4.1.5 Test requirement ..... 2112
17.4.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 2115
17.4.2.1 Test purpose ..... 2115
17.4.2.2 Test applicability ..... 2115
17.4.2.3 Minimum conformance requirements ..... 2115
17.4.2.4 Test description ..... 2115
17.4.2.4.1 Initial conditions ..... 2115
17.4.2.4.2 Test procedure ..... 2116
17.4.2.4.3 Message contents ..... 2116
17.4.2.5 Test requirement ..... 2118
17.4.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA ..... 2121
17.4.3.1 Test purpose ..... 2121
17.4.3.2 Test applicability ..... 2121
17.4.3.3 Minimum conformance requirements ..... 2121
17.4.3.4 Test description ..... 2121
17.4.3.4.1 Initial conditions ..... 2121
17.4.3.4.2 Test procedure ..... 2122

17.4.3.4.3 Message contents ..... 2122
17.4.3.5 Test requirement ..... 2124
17.4.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 2128
17.4.4.1 Test purpose ..... 2128
17.4.4.2 Test applicability ..... 2128
17.4.4.3 Minimum conformance requirements ..... 2128
17.4.4.4 Test description ..... 2128
17.4.4.4.1 Initial conditions ..... 2128
17.4.4.4.2 Test procedure ..... 2128
17.4.4.4.3 Message contents ..... 2129
17.4.4.5 Test requirement ..... 2132
17.5 NR PRS-RSRPP measurement accuracy test case in RRC_INACTIVE ..... 2136
17.5.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA ..... 2136
17.5.1.1 Test purpose ..... 2136
17.5.1.2 Test applicability ..... 2136
17.5.1.3 Minimum conformance requirements ..... 2136
17.5.1.4 Test description ..... 2136
17.5.1.4.1 Initial conditions ..... 2136
17.5.1.4.2 Test procedure ..... 2136
17.5.1.4.3 Message contents ..... 2138
17.5.1.5 Test requirement ..... 2140
17.5.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA ..... 2142
17.5.2.1 Test purpose ..... 2142
17.5.2.2 Test applicability ..... 2142
17.5.2.3 Minimum conformance requirements ..... 2142
17.5.2.4 Test description ..... 2142
17.5.2.4.1 Initial conditions ..... 2142
17.5.2.4.2 Test procedure ..... 2143
17.5.2.4.3 Message contents ..... 2143
17.5.2.5 Test requirement ..... 2146
17.5.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA ..... 2148
17.5.3.1 Test purpose ..... 2149
17.5.3.2 Test applicability ..... 2149
17.5.3.3 Minimum conformance requirements ..... 2149
17.5.3.4 Test description ..... 2149
17.5.3.4.1 Initial conditions ..... 2149
17.5.3.4.2 Test procedure ..... 2149
17.5.3.4.3 Message contents ..... 2149
17.5.3.5 Test requirement ..... 2152
17.5.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA ..... 2154
17.5.4.1 Test purpose ..... 2154
17.5.4.2 Test applicability ..... 2154
17.5.4.3 Minimum conformance requirements ..... 2154
17.5.4.4 Test description ..... 2154
17.5.4.4.1 Initial conditions ..... 2154
17.5.4.4.2 Test procedure ..... 2154
17.5.4.4.3 Message contents ..... 2155
17.5.4.5 Test requirement ..... 2158

**Annex A (informative): Connection Diagrams ..... 2161**

**Annex B (normative): Converting A-GNSS UE-assisted measurement reports into position estimates ..... 2172**

B.1 Introduction ..... 2172
B.2 UTRAN UE measurement reports for A-GPS L1 C/A only ..... 2172
B.3 UTRAN UE measurement reports for A-GNSS ..... 2172
B.4 E-UTRAN and NR UE measurement reports ..... 2174
B.5 WLS position solution ..... 2174

**Annex C (normative): General test conditions and declarations ..... 2177**
C.1 Acceptable uncertainty of Test System ..... 2177
C.1.1 Measurement of test environments ..... 2177
C.1.2 A-GNSS Minimum Performance requirements ..... 2177
C.1.3 ECID and OTDOA Measurement requirements ..... 2178
C.1.4 MBS Minimum Performance requirements ..... 2189
C.1.5 WLAN and BLE measurement requirements ..... 2189
C.1.6 NR PRS-based measurement requirements ..... 2189
C.2 Test Parameter Relaxations (This clause is informative) ..... 2198
C.2.1 A-GNSS Minimum Performance requirements ..... 2198
C.2.2 ECID and OTDOA Measurement requirements ..... 2199
C.2.3 MBS Minimum Performance requirements ..... 2205
C.2.4 WLAN and BLE measurement requirements ..... 2206
C.2.5 NR PRS-based Measurement requirements ..... 2206
C.3 Interpretation of measurement results ..... 2214
C.4 Derivation of Test Requirements (This clause is informative) ..... 2214

**Annex D (normative): Rules for statistical testing ..... 2241**
D.1 Test Method ..... 2241
D.2 Error Ratio (ER) ..... 2241
D.3 Test Design ..... 2241
D.3.1 Confidence level ..... 2241
D.3.2 Introduction: Supplier Risk versus Customer Risk ..... 2241
D.3.3 Supplier Risk versus Customer Risk ..... 2242
D.3.4 Introduction: Standard test versus early decision concept ..... 2242
D.3.5 Standard test versus early decision concept ..... 2243
D.3.6 Selectivity ..... 2243
D.3.7 Design of the test ..... 2244
D.4 Pass fail decisions ..... 2245
D.4.1 Numerical definition of the pass fail limits for A-GNSS test cases ..... 2245
D.4.2 Pass fail decision rules for A-GNSS test cases ..... 2246
D.4.3 Numerical definition of the pass fail limits for ECID, OTDOA, MBS, WLAN and BLE test cases ..... 2246
D.4.4 Pass fail decision rules for ECID, OTDOA, MBS, WLAN and BLE test cases ..... 2247
D.4.5 Background information to the pass fail limits ..... 2247

**Annex E (normative): Conditions for ECID and OTDOA requirements ..... 2249**
E.1 Conditions for E-CID UE Rx – Tx time difference Measurements ..... 2249
E.1.1 Conditions for E-CID UE Rx – Tx time difference by UE Category M1/M2 ..... 2249
E.2 Conditions for OTDOA intra-frequency RSTD Measurements ..... 2250
E.2.1 Conditions for OTDOA intra-frequency RSTD Measurements by UE Category M1 and M2 ..... 2250
E.3 Conditions for OTDOA inter-frequency RSTD Measurements ..... 2251
E.3.1 Conditions for OTDOA inter-frequency RSTD Measurements by UE Category M1 and M2 ..... 2251
E.4 Conditions for UE Rx-Tx Time Difference Measurement under Time Domain Measurement Resource Restriction with CRS Assistance Information ..... 2252
E.5 Conditions for NB-IOT OTDOA intra-frequency RSTD Measurements ..... 2252
E.6 Conditions for NB-IOT OTDOA inter-frequency RSTD Measurements ..... 2252

**Annex F (normative): UTRAN Generic procedures ..... 2253**
F.1 General ..... 2253
F.2 UTRAN connection set up ..... 2253
F.2.1 Initial conditions ..... 2253
F.2.2 Procedures ..... 2253
F.2.3 Specific message contents ..... 2254
F.3 UTRAN connection release ..... 2254
F.3.1 Procedure ..... 2254
F.3.2 Specific message contents ..... 2254

**Annex G (normative): Environmental conditions ..... 2255**
G.1 General ..... 2255
G.2 Environmental requirements ..... 2255
G.2.1 Temperature ..... 2255
G.2.2 Voltage ..... 2255

**Annex H (informative): MBS Beacon parameters ..... 2256**
H.1 General ..... 2256
H.2 Beacon parameters ..... 2256

**Annex I (normative): Conditions for NR PRS-based measurements ..... 2257**
I.1 General ..... 2257
I.2 General ..... 2257

**Annex J (informative): Change history ..... 2259**

History ..... 2277

# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

# Introduction

The present document is part 1 of a multi-parts TS:

- 3GPP TS 37.571-1: User Equipment (UE) conformance specification for UE positioning; Part 1: Conformance test specification.
- 3GPP TS 37.571-2: User Equipment (UE) conformance specification for UE positioning; Part 2: Protocol conformance.
- 3GPP TS 37.571-3: User Equipment (UE) conformance specification for UE positioning; Part 3: Implementation Conformance Statement (ICS).
- 3GPP TS 37.571-4: User Equipment (UE) conformance specification for UE positioning; Part 4: Test suites.
- 3GPP TS 37.571-5: User Equipment (UE) conformance specification for UE positioning; Part 5: Test scenarios and assistance data.

# 1 Scope

The present document specifies the procedures for the conformance test of the measurement requirements for FDD or TDD mode of UTRA and FDD or TDD mode of E-UTRA, NB-IOT and NR for the User Equipment (UE) that supports one or more of the defined positioning methods. These positioning methods are:

- for UTRA: Assisted Global Positioning System (A-GPS), Assisted Global Navigation Satellite Systems (A-GNSS),
- for E-UTRA: Assisted Global Navigation Satellite System (A-GNSS), Observed Time Difference of Arrival (OTDOA), Enhanced Cell ID (ECID), Metropolitan Beacon System (MBS), Wireless Local Area Network (WLAN), Bluetooth Low Energy (BLE),
- for NB-IOT: Observed Time Difference of Arrival (OTDOA), [others FFS], and
- for NR: Assisted Global Navigation Satellite System (A-GNSS), Observed Time Difference of Arrival (OTDOA), Enhanced Cell ID (ECID), Metropolitan Beacon System (MBS), Wireless Local Area Network (WLAN), Bluetooth Low Energy (BLE).

Tests are only applicable to those mobiles that are intended to support the appropriate functionality. To indicate the circumstances in which tests apply, this is noted in the “Test applicability" part of the test.

The Implementation Conformance Statement (ICS) pro-forma could be found in the 3rd part of the present document.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in the same Release as the present document unless the context in which the reference is made suggests a different Release is relevant (information on the applicable release in a particular context can be found in e.g. test case title, description or applicability, message description or content).

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 36.101: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception".

[3] 3GPP TS 36.171: "Evolved Universal Terrestrial Radio Access (E-UTRA); Requirements for Support of Assisted Global Navigation Satellite System (A-GNSS)".

[4] 3GPP TS 36.355: "Evolved Universal Terrestrial Radio Access (E-UTRA); LTE Positioning Protocol (LPP)".

[5] 3GPP TS 36.302: "Evolved Universal Terrestrial Radio Access (E-UTRA); Services provided by the physical layer".

[6] 3GPP TS 36.214: "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical layer; Measurements".

[7] ETSI TR 102 273-1-2: "Electromagnetic compatibility and Radio spectrum Matters (ERM); Improvement on Radiated Methods of Measurement (using test site) and evaluation of the corresponding measurement uncertainties; Part 1: Uncertainties in the measurement of mobile radio equipment characteristics; Sub-part 2: Examples and annexes".

[8] IS-GPS-200, Revision D, Navstar GPS Space Segment/Navigation User Interfaces, March 7th, 2006.

[9] P. Axelrad, R.G. Brown, "GPS Navigation Algorithms", in Chapter 9 of "Global Positioning System: Theory and Applications", Volume 1, B.W. Parkinson, J.J. Spilker (Ed.), Am. Inst. of Aeronautics and Astronautics Inc., 1996.

[10] S.K. Gupta, "Test and Evaluation Procedures for the GPS User Equipment", ION-GPS Red Book, Volume 1, p.119.

[11] 3GPP TS 36.509: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Special conformance testing functions for User Equipment (UE)".

[12] IS-GPS-705, Navstar GPS Space Segment/User Segment L5 Interfaces, September 22, 2005.

[13] IS-GPS-800, Navstar GPS Space Segment/User Segment L1C Interfaces, September 4, 2008.

[14] IS-QZSS, Quasi Zenith Satellite System Navigation Service Interface Specifications for QZSS, Ver.1.1, July 31, 2009.

[15] Galileo OS Signal in Space ICD (OS SIS ICD), Issue 1.2, February 2014, European Union.

[16] Global Navigation Satellite System GLONASS Interface Control Document, Version 5.1, 2008.

[17] Specification for the Wide Area Augmentation System (WAAS), US Department of Transportation, Federal Aviation Administration, DTFA01-96-C-00025, 2001.

[18] 3GPP TS 36.508: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing)".

[19] 3GPP TS 25.172: "Requirements for support of Assisted Galileo and Additional Navigation Satellite Systems (A-GANSS); Frequency Division Duplex (FDD)".

[20] 3GPP TS 37.571-5: "User Equipment (UE) conformance specification for UE positioning; Part 5: Test scenarios and assistance data”.

[21] 3GPP TS 36.104: "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) radio transmission and reception".

[22] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification".

[23] 3GPP TS 36.133: "Evolved Universal Terrestrial Radio Access (E-UTRA); Requirements for support of radio resource management".

[24] 3GPP TS 36.521-1: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification Radio transmission and reception Part 1: Conformance Testing".

[25] 3GPP TS 36.521-3: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification; Radio transmission and reception; Part 3: Radio Resource Management (RRM) conformance testing".

[26] 3GPP TS 36.211: "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical Channels and Modulation".

[27] 3GPP TR 25.990: "Vocabulary for UTRAN".

[28] 3GPP TS 34.108: "Common test environments for User Equipment (UE) conformance testing".

[29] 3GPP TS 34.109: "Terminal logical test interface; Special conformance testing functions".

[30] 3GPP TS 25.331: "Radio Resource Control (RRC) protocol specification".

[31] 3GPP TS 25.171: "Requirements for support of Assisted Global Positioning System (A-GPS); Frequency Division Duplex (FDD)".

[32] 3GPP TS 25.302: "Services provided by the physical layer".

[33] 3GPP TS 25.215: "Physical layer; Measurements (FDD)".

[34] 3GPP TS 36.321: "Medium Access Control (MAC) protocol specification".

[35] 3GPP TS 36.423: "X2 application protocol (X2AP)".

[36] 3GPP TS 25.173: "Requirements for support of Assisted Galileo and Additional Navigation Satellite Systems (A-GANSS); Time Division Duplex (TDD)".

[37] BDS-SIS-ICD-B1I-3.0: "BeiDou Navigation Satellite System Signal In Space Interface Control Document Open Service Signal B1I (Version 3.0)".

[38] ATIS-0500027: "Recommendations for Establishing Wide Scale Indoor Location Performance", May 2015.

[39] 3GPP TS 37.171: "Universal Terrestrial Radio Access (UTRA) and Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) performance requirements for RAT-Independent Positioning Enhancements".

[40] IEEE Standard 802.11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications.

[41] 3GPP TS 36.305: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Stage 2 functional specification of User Equipment (UE) positioning in E-UTRAN".

[42] 3GPP TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".

[43] 3GPP TS 38.171: "NR; Requirements for Support of Assisted Global Navigation Satellite System (A-GNSS)".

[44] 3GPP TS 38.509: "5GS; Special conformance testing functions for User Equipment (UE)".

[45] 3GPP TS 38.508-1: "5GS; User Equipment (UE) conformance specification; Part 1: Common test environment".

[46] 3GPP TS 36.212 "Evolved Universal Terrestrial Radio Access (E-UTRA); Multiplexing and channel coding".

[47] 3GPP TS 38.533 "NR; User Equipment (UE) conformance specification; Radio Resource Management (RRM)".

[48] BDS-SIS-ICD-B1C-1.0: "BeiDou Navigation Satellite System Signal In Space Interface Control Document Open Service Signal B1C (Version 1.0)".

[49] 3GPP TS 37.355: "LTE Positioning Protocol (LPP)".

[50] 3GPP TS 38.133: "NR; Requirements for support of radio resource management".

[51] 3GPP TS 38.305: "NG Radio Access Network (NG-RAN); Stage 2 functional specification of User Equipment (UE) positioning in NG-RAN".

[52] 3GPP TS 38.331: "NR Radio Resource Control (RRC) protocol specification".

[53] 3GPP TS 38.211: "NR; Physical channels and modulation”.

[54] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone".

[55] 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone".

[56] 3GPP TS 38.214: "NR; Physical layer procedures for data".

[57] 3GPP TS 38.215: "NR; Physical layer measurements".

[58] BDS-SIS-ICD-B2a-1.0: "BeiDou Navigation Satellite System Signal In Space Interface Control Document Open Service Signal B2a (Version 1.0)", December, 2017.

[59] BDS-SIS-ICD-B3I-1.0: "BeiDou Navigation Satellite System Signal In Space Interface Control Document Open Service Signal B3I (Version 1.0)", February, 2018.

[60] 3GPP TS 38.521-4: "NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 4: Performance requirements".

[61] 3GPP TS 23.273: "5G System (5GS) Location Services (LCS); Stage 2".

[62] BDS-SIS-ICD-B2b-1.0: "BeiDou Navigation Satellite System Signal In Space Interface Control Document Open Service Signal B2b (Version 1.0)", July, 2020.

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1], 3GPP TR 25.990 [27], TS 36.101 [2], 3GPP TS 36.104 [21] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

**Horizontal Dilution Of Precision (HDOP):** measure of position determination accuracy that is a function of the geometrical layout of the satellites used for the fix, relative to the receiver antenna

## 3.2 Symbols

For the purposes of the present document, the abbreviations given in TR 21.905 [1], 3GPP TR 25.990 [27] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

| | |
|---|---|
| B1I | BeiDou B1I navigation signal with carrier frequency of 1561.098 MHz |
| B1C | BeiDou B1C navigation signal with carrier frequency of 1575.420 MHz. |
| B2a | BeiDou B2a navigation signal with carrier frequency of 1176.450 MHz. |
| B2b | BeiDou B2b navigation signal with carrier frequency of 1207.14 MHz. |
| B3I | BeiDou B3I navigation signal with carrier frequency of 1268.520 MHz. |
| E1 | Galileo E1 navigation signal with carrier frequency of 1575.420 MHz. |
| E5 | Galileo E5 navigation signal with carrier frequency of 1191.795 MHz. |
| E6 | Galileo E6 navigation signal with carrier frequency of 1278.750 MHz. |
| G1 | GLONASS navigation signal in the L1 sub-bands with carrier frequencies 1602 MHz ± k × 562.5 kHz. |
| G2 | GLONASS navigation signal in the L2 sub-bands with carrier frequencies 1246 MHz ± k × 437.5 kHz. |
| k | GLONASS channel number, k = -7…13. |
| L1 C/A | GPS or QZSS L1 navigation signal carrying the Coarse/Acquisition code with carrier frequency of 1575.420 MHz. |
| L1C | GPS or QZSS L1 Civil navigation signal with carrier frequency of 1575.420 MHz. |
| L2C | GPS or QZSS L2 Civil navigation signal with carrier frequency of 1227.600 MHz. |
| L5 | GPS or QZSS L5 navigation signal with carrier frequency of 1176.450 MHz. |
| PRP | Received (linear) average power of the resource elements that carry E-UTRA PRS, measured at the UE antenna connector. |
| **G** | Geometry Matrix. |
| $\rho_{GNSS_m,i}$ | Measured pseudo-range of satellite *i* of $GNSS_m$. |
| **W** | Weighting Matrix. |
| $\mathbf{1}_{GNSS_m,i}$ | Line of sight unit vector from the user to the satellite *i* of $GNSS_m$. |
| **x** | State vector of user position and clock bias. |
| $T_S$ | Basic time unit, defined in TS 36.211 [26], clause 4. |

| | |
|---|---|
| Ês | Received energy per RE (power normalized to the subcarrier spacing) during the useful part of the symbol, i.e. excluding the cyclic prefix, at the UE antenna connector. |
| Io | The total received power density, including signal and interference, as measured at the UE antenna connector. |
| Iot | The received power spectral density of the total noise and interference for a certain RE (power integrated over the RE and normalized to the subcarrier spacing) as measured at the UE antenna connector. |
| $N_{oc}$ | The power spectral density of a white noise source (average power per RE normalised to the subcarrier spacing), simulating interference from cells that are not defined in a test procedure, as measured at the UE antenna connector. |
| $\mathrm{PRS}\,\hat{E}_s / \mathrm{Iot}$ | The ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry PRS. |

## 3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

| | |
|---|---|
| A-GANSS | Assisted Galileo and Additional Navigation Satellite Systems |
| A-Galileo | Assisted-Galileo |
| A-GLONASS | Assisted-- GLObal'naya NAvigatsionnaya Sputnikovaya Sistema (English: Global Navigation Satellite System) |
| A-GNSS | Assisted Global Navigation Satellite System |
| A-GPS | Assisted - Global Positioning System |
| AP | Access Point |
| AWGN | Additive White Gaussian Noise |
| BDS | BeiDou Navigation Satellite System |
| BLE | Bluetooth Low Energy |
| BSS | Bluetooth System Simulator |
| BSSID | Basic Service Set IDentification |
| C/A | Coarse/Acquisition |
| DL-AoD | Downlink Angle-of-Departure |
| DL-PRS | Downlink Positioning Reference Signal |
| DL-TDOA | Downlink Time Difference Of Arrival |
| DRX | Discontinuous Reception |
| DUT | Device Under Test |
| ECEF | Earth Centred, Earth Fixed |
| ECID | Enhanced Cell Identification |
| EPRE | Energy Per Resource Element |
| EN-DC | E-UTRA-NR Dual Connectivity |
| E-UTRA | Evolved UMTS Terrestrial Radio Access |
| E-UTRAN | Evolved UMTS Terrestrial Radio Access Network |
| FDD | Frequency Division Duplex |
| GANSS | Galileo and Additional Navigation Satellite Systems |
| GEO | Geostationary Earth Orbit |
| GLONASS | GLObal'naya NAvigatsionnaya Sputnikovaya Sistema (English: Global Navigation Satellite System) |
| GNSS | Global Navigation Satellite System |
| GPS | Global Positioning System |
| GSS | GNSS System Simulator |
| HD-FDD | Half Duplex - Frequency Division Duplex |
| HDOP | Horizontal Dilution Of Precision |
| ICD | Interface Control Document |
| ICS | Implementation Conformance Statement |
| IGSO | Inclined Geosynchronous Satellite OrbitIS |
| IS | Interface Specification |
| LOS | Line Of Sight |
| LPP | LTE Positioning Protocol |
| MBS | Metropolitan Beacon System |
| MSS | MBS System Simulator |

| | |
|---|---|
| MEO | Medium Earth Orbit |
| Multi-RTT | Multi-Round Trip Time |
| NB-IOT | Narrow Band - Internet Of Things |
| NE-DC | NR-E-UTRA Dual Connectivity |
| NGEN-DC | NG-RAN E-UTRA-NR Dual Connectivity |
| NG-RAN E-UTRA | NG-RAN E-UTRA Radio Access |
| NG-RAN NR | NG-RAN NR Radio Access |
| NPRS | NB-IOT Positioning Reference Signal |
| NR | New Radio |
| NR-DC | NR-NR Dual Connectivity |
| NR E-CID | NR Enhanced Cell ID (positioning method) |
| OCNG | OFDMA Channel Noise Generator |
| OCNS | Orthogonal Channel Noise Simulator |
| OTA | Over The Air |
| OTDOA | Observed Time Difference Of Arrival |
| PBCH | Physical Broadcast Channel |
| PCC | Primary Component Carrier |
| PCell | Primary Cell |
| PCFICH | Physical Control Format Indicator Channel |
| PDCCH | Physical Downlink Control Channel |
| PDSCH | Physical Downlink Shared Channel |
| PHICH | Physical Hybrid ARQ Indictor Channel |
| PPM | Parts per million |
| PRS | Positioning Reference Signal |
| PRS-RSRP | Positioning Reference Signal based Reference Signal Received Power |
| PSS | Primary Synchronization Signal |
| QZSS | Quasi-Zenith Satellite System |
| RB | Resource Block |
| RE | Resource Element |
| RRC | Radio Resource Control |
| RSSI | Received Signal Strength Indicator |
| RSTD | Reference Signal Time Difference |
| SBAS | Space Based Augmentation System |
| SCC | Secondary Component Carrier |
| SCell | Secondary Cell |
| SFN | System Frame Number |
| SNR | Signal to Noise Ratio |
| SS | System simulator |
| SSS | Secondary Synchronization Signal |
| SV | Space Vehicle |
| SV ID | Space Vehicle Identity |
| TBS | Terrestrial Beacon System |
| TDD | Time Division Duplex |
| TOD | Time Of Day |
| TOW | Time Of Week |
| TTFF | Time To First Fix |
| UE | User Equipment |
| UL-SRS | Uplink Sounding Reference Signal |
| UUID | Universal Unique Identifier |
| UTRA | Universal Terrestrial Radio Access |
| UTRAN | Universal Terrestrial Radio Access Network |
| WGS-84 | World Geodetic System 1984 |
| WLAN | Wireless Local Area Network |
| WLS | Weighted Least Square |
| WSS | WLAN System Simulator |

# 4 General test conditions

## 4.1 Introduction

This clause defines the various common test conditions required for the various measurement requirements in the remainder of the document.

## 4.2 GNSS test conditions

### 4.2.0 General

In this clause the terms GNSS and A-GNSS also include the cases where the only satellite system used is GPS unless otherwise stated.

### 4.2.1 GNSS signals

The GNSS signal is defined at the A-GNSS antenna connector of the UE. For UE with integral antenna only, a reference antenna with a gain of 0 dBi is assumed.

### 4.2.2 GNSS frequency

The GNSS signals shall be transmitted with a frequency accuracy of ± 0.025 PPM.

### 4.2.3 GNSS static propagation conditions

The propagation for the static performance measurement is an Additive White Gaussian Noise (AWGN) environment. No fading and multi-paths exist for this propagation model.

### 4.2.4 GNSS multi-path conditions

Doppler frequency difference between direct and reflected signal paths is applied to the carrier and code frequencies. The Carrier and Code Doppler frequencies of LOS and multi-path for GNSS signals are defined in table 4.2.1.

**Table 4.2.1: Multi-path Conditions for GNSS Signals**

| Initial relative Delay [GNSS chip] | Carrier Doppler frequency of tap [Hz] | Code Doppler frequency of tap [Hz] | Relative mean Power [dB] |
|---|---|---|---|
| 0 | Fd | Fd / N | 0 |
| X | Fd - 0.1 | (Fd-0.1) /N | Y |
| NOTE: Discrete Doppler frequency is used for each tap. | | | |

Where the X and Y depends on the GNSS signal type and is shown in Table 4.2.2, and N is the ratio between the transmitted carrier frequency of the signals and the transmitted chip rate as shown in Table 4.2.3 (where k in Table 4.2.3 is the GLONASS frequency channel number).

**Table 4.2.2: Values of X and Y for GNSS Signals**

| System | Signals | X [m] | Y [dB] |
|---|---|---|---|
| Galileo | E1 | 125 | -4.5 |
| | E5a | 15 | -6 |
| | E5b | 15 | -6 |
| GPS/Modernized GPS | L1 C/A | 0.5 chip / 150m | -6 |
| | L1C | 125 | -4.5 |
| | L2C | 150 | -6 |
| | L5 | 15 | -6 |

| | | | |
|---|---|---|---|
| GLONASS | G1 | 275 | -12.5 |
| | G2 | 275 | -12.5 |
| BDS | B1I | 75 | -4.5 |
| | B1C | 125 | -4.5 |
| | B2a | 15 | -6 |
| | B2b | 15 | -6 |
| | B3I | 15 | -6 |

**Table 4.2.3: Values of N for GNSS Signals**

| System | Signals | N |
|---|---|---|
| Galileo | E1 | 1540 |
| | E5a | 115 |
| | E5b | 118 |
| GPS/Modernized GPS | L1 C/A | 1540 |
| | L1C | 1540 |
| | L2C | 1200 |
| | L5 | 115 |
| GLONASS | G1 | 3135.03 + k · 1.10 |
| | G2 | 2438.36 + k · 0.86 |
| BDS | B1I | 763 |
| | B1C | 1540 |
| | B2a | 115 |
| | B2b | 118 |
| | B3I | 124 |

The initial carrier phase difference between taps shall be randomly selected between 0 and 2 $\pi$ radians. The initial value shall have uniform random distribution.

## 4.2.5 UEs supporting multiple satellite signals

For UEs supporting multiple satellite signals, different minimum performance requirements may be associated with different signals. The satellite simulator shall generate all signals supported by the UE. Signals not supported by the UE do not need to be simulated. The relative power levels of each signal type for each GNSS are defined in Table 4.2.4. The individual test scenarios in clauses 6, 7 and 13 define the reference signal power level for each satellite. The power level of each simulated satellite signal type shall be set to the reference signal power level defined in each test scenario in clauses 6, 7 and 13 plus the relative power level defined in Table 4.2.4.

**Table 4.2.4: Relative signal power levels for each signal type for each GNSS**

| | Galileo | | GPS/Modernized GPS | | GLONASS | | QZSS | | SBAS | | BDS | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Signal power levels relative to reference power levels | E1 | 0 dB | L1 C/A | 0 dB | G1 | 0 dB | L1 C/A | 0 dB | L1 | 0 dB | B1I | D1 | 0 dB |
| | | | | | | | | | | | | D2 | +5 dB |
| | E6 | +2 dB | L1C | +1.5 dB | G2 | -6 dB | L1C | +1.5 dB | | | B1C | $D_{MEO}$ | +4 dB |
| | | | | | | | | | | | | $D_{IGSO}$ | +2 dB |
| | E5 | +2 dB | L2C | -1.5 dB | | | L2C | -1.5 dB | | | B2a | $D_{MEO}$ | +7 dB |
| | | | | | | | | | | | | $D_{IGSO}$ | +5 dB |
| | | | | | | | | | | | B2b | $D_{MEO}$ | +7 dB |
| | | | | | | | | | | | | $D_{IGSO}$ | +5 dB |
| | | | L5 | +3.6 dB | | | L5 | +3.6 dB | | | B3I | D1 | 0 dB |
| | | | | | | | | | | | | D2 | +5 dB |

NOTE 1: For test cases which involve “Modernized GPS”, the satellite simulator shall also generate the GPS L1 C/A signal if the UE supports “GPS” in addition to “Modernized GPS”.

NOTE 2: The signal power levels in the Test Parameter Tables represent the total signal power of the satellite per channel not e.g. pilot and data channels separately.

NOTE 3: For test cases which involve "BDS", D1 represents MEO/IGSO satellites for B1I, and B3I signal types and D2 represents GEO satellites for B1I and B3I signal types.

NOTE 4: For test cases which involve "BDS", $D_{MEO}$ represents MEO satellites for B1C, B2a and B2b signal types and $D_{IGSO}$ represents IGSO satellites for B1C, B2a and B2b signal type.

### 4.2.6 GNSS multi System Time Offsets

If more than one GNSS is used in a test, the accuracy of the GNSS-GNSS Time Offsets used at the system simulator shall be better than 3 ns.

## 4.3 UTRA test conditions

### 4.3.1 UTRA frequency band and frequency range

The UTRA tests in clauses 5 and 6 in the present document are performed at mid range of the UTRA operating frequency band of the UE. The UARFCNs to be used for mid range are defined in 3GPP TS 34.108 [28], clause 5.1.1.

If the UE supports multiple UTRA frequency bands then the Sensitivity tests in clauses 5.2 and 6.2 shall be repeated in each supported UTRA frequency band.

### 4.3.2 UTRA frequency

For the UTRA tests in clause 5 the UTRA frequency shall be offset with respect to the nominal frequency by an amount equal to the sum of +0.025 PPM and the offset in PPM of the actual transmitted GPS carrier frequency with respect to the nominal GPS frequency.

### 4.3.3 Sensors

The UTRA tests in clause 6 shall be met without the use of any data coming from sensors that can aid the positioning. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 34.109 [29] for the purpose of disabling any such sensors.

## 4.4 E-UTRA test conditions

### 4.4.1 E-UTRA frequency band and frequency range

The E-UTRA A-GNSS tests in clause 7, MBS tests in clause 11, WLAN and BLE tests in clause 12 are performed on the mid range EARFCN of the E-UTRA operating frequency band of the UE and the channel bandwidth as defined in TS 36.508 [18] clause 4.3.1.

If the UE supports multiple E-UTRA frequency bands then the A-GNSS Sensitivity tests in clause 7.1 shall be repeated in each supported E-UTRA frequency band.

The E-UTRA ECID tests in clause 8 and the OTDOA tests in clauses 9 and 10 are performed on the EARFCN(s) of the E-UTRA operating frequency band of the UE and the channel bandwidth(s) specified in the test cases and as defined in TS 36.508 [18] clause 4.3.1 and 4.4.2.

### 4.4.2 Groups of bands

The E-UTRA tests use the band groupings defined in TS 36.521-3 [25] clause 3.5.1 in order to increase the readability of the specification.

**Table 4.4.2-1: Void**

**Table 4.4.2-2: Void**

**Table 4.4.2-3: Void**

### 4.4.3 Sensors

All the minimum performance requirements in clause 7 shall be met without the use of any data coming from sensors that can aid the positioning. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] for the purpose of disabling any such sensors.

4.4A LPP transport mechanism for E-UTRA

The E-UTRA A-GNSS minimum performance requirements tested in this specification are agnostic as to whether the LPP session is performed over the Control Plane or the User Plane. Thus, the E-UTRA A-GNSS test cases in clause 7 support both configurations. The user can select either of the two options to run the test.

## 4.5 A-GNSS test conditions

### 4.5.1 General

Clauses 5, 6, 7 and 13 define the minimum performance requirements for both UE based and UE assisted A-GNSS UEs. If a UE supports both modes then it shall be tested in both modes.

### 4.5.2 UTRAN measurement parameters

#### 4.5.2.1 UE based A-GNSS measurement parameters

In case of UE-based A-GNSS, the measurement parameters are contained in the RRC UE POSITIONING POSITION ESTIMATE INFO IE. The measurement parameter is the horizontal position estimate reported by the UE and expressed in latitude/longitude.

#### 4.5.2.2 UE assisted A-GNSS measurement parameters

In case of UE-assisted A-GNSS, the measurement parameters are contained in the RRC UE POSITIONING GANSS MEASURED RESULTS IE and/or the RRC UE POSITIONING GPS MEASURED RESULTS IE. The measurement parameters are the UE GANSS Code Phase measurements and/or the UE GPS Code Phase measurements, as specified in 3GPP TS 25.302 [32] and 3GPP TS 25.215 [33]. The UE GANSS Code Phase measurements and/or the UE GPS Code Phase measurements are converted into a horizontal position estimate using the procedure detailed in Annex B.

#### 4.5.2.3 2D position error

The 2D position error is defined by the horizontal difference in meters between the ellipsoid point reported or calculated from the UE Measurement Report and the actual simulated position of the UE in the test case considered.

#### 4.5.2.4 Response time

Max Response Time is defined as the time starting from the moment that the UE has received the final RRC measurement control message containing reporting criteria different from "No Reporting" sent before the UE sends the measurement report containing the position estimate or the GANSS and/or GPS measured result, and ending when the UE starts sending the measurement report containing the position estimate or the GANSS and/or GPS measured result on the Uu interface. The response times specified for all test cases are Time-to-First-Fix (TTFF) unless otherwise stated, i.e. the UE shall not re-use any information on GNSS time, location or other aiding data that was previously acquired or calculated and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' specified in 3GPP TS 34.109 [29], clause 5.4, has been defined for the purpose of deleting this information.

### 4.5.3 E-UTRAN and NR measurement parameters

#### 4.5.3.1 UE based A-GNSS measurement parameters

In case of UE-based A-GNSS, the measurement parameters are contained in the LPP *GNSS-LocationInformation* IE which is included in the *A-GNSS-ProvideLocationInformation* IE provided in the LPP message of type PROVIDE LOCATION INFORMATION. The measurement parameter in case of UE-based A-GNSS is the horizontal position estimate reported by the UE and expressed in latitude/longitude.

#### 4.5.3.2 UE assisted A-GNSS measurement parameters

In case of UE-assisted A-GNSS, the measurement parameters are contained in the LPP *GNSS-SignalMeasurementInformation* IE which is included in the *A-GNSS-ProvideLocationInformation* IE provided in the LPP message of type PROVIDE LOCATION INFORMATION. The measurement parameters in case of UE-assisted A-GNSS are the UE GNSS code phase measurements, as specified in TS 36.302 [5] and TS 36.214 [6]. The UE GNSS code phase measurements are converted into a horizontal position estimate using the procedure detailed in Annex B.

#### 4.5.3.3 2D Error definition

The 2D position error is defined by the horizontal difference in meters between the ellipsoid point reported or calculated from the LPP message of type PROVIDE LOCATION INFORMATION and the actual position of the UE in the test case considered.

#### 4.5.3.4 Response time

Max Response Time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response times specified for all test cases are Time-to-First-Fix (TTFF) unless otherwise stated, i.e. the UE shall not re-use any information on GNSS time, location or other aiding data that was previously acquired or calculated and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 and in TS 38.509 [44] clause 6.3.5 for the purpose of deleting this information.

### 4.5.4 Converting A-GNSS UE-assisted measurement reports into position estimates

To convert the A-GNSS UE measurement reports in case of UE-assisted mode of A-GNSS into position errors, a transformation between the "measurement domain" (code-phases, etc.) into the "state" domain (position estimate) is necessary. Such a transformation procedure is outlined in Annex B.

## 4.6 ECID test conditions

### 4.6.1 Simulated cells

For the ECID performance test cases in clause 8.1, a cell environment as defined in 3GPP TS 36.508 [18] with Cell 1 is used. The default parameters for simulated cells are the same as specified in 3GPP TS 36.508 [18].

### 4.6.2 Propagation conditions

#### 4.6.2.1 Static

See TS 36.521-1 [24] clause B.1.

### 4.6.2.2 Multi-path fading

See TS 36.521-1[24] clauses B.2, B.2.1 and B.2.2.

## 4.6.3 UE Rx - Tx time difference reporting range

The reporting range of FDD UE Rx - Tx time difference is defined from 0 to $20472T_s$ with $2T_s$ resolution for UE Rx - Tx time difference less than $4096T_s$ and 8Ts for UE Rx - Tx time difference equal to or greater than $4096T_s$.

The mapping of measured quantity for FDD is defined in Table 4.6.3-1.

**Table 4.6.3-1: FDD UE Rx - Tx time difference measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_FDD_0000 | $T_{UE\ Rx-Tx} < 2$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_FDD_0001 | $2 \le T_{UE\ Rx-Tx} < 4$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_FDD_0002 | $4 \le T_{UE\ Rx-Tx} < 6$ | $T_s$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_FDD_2046 | $4092 \le T_{UE\ Rx-Tx} < 4094$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_FDD_2047 | $4094 \le T_{UE\ Rx-Tx} < 4096$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_FDD_2048 | $4096 \le T_{UE\ Rx-Tx} < 4104$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_FDD_2049 | $4104 \le T_{UE\ Rx-Tx} < 4112$ | $T_s$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_FDD_4093 | $20456 \le T_{UE\ Rx-Tx} < 20464$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_FDD_4094 | $20464 \le T_{UE\ Rx-Tx} < 20472$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_FDD_4095 | $20472 \le T_{UE\ Rx-Tx}$ | $T_s$ |

The reporting range of TDD UE Rx - Tx time difference is defined from 624 to $21096T_s$ with $2T_s$ resolution for UE Rx - Tx time difference less than $4720T_s$ and 8Ts for UE Rx - Tx time difference equal to or greater than $4720T_s$.

The mapping of measured quantity for TDD is defined in Table 4.6.3-2.

**Table 4.6.3-2: TDD UE Rx - Tx time difference measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_TDD_0000 | $T_{UE\ Rx-Tx} < 626$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_TDD_0001 | $626 \le T_{UE\ Rx-Tx} < 628$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_TDD_0002 | $628 \le T_{UE\ Rx-Tx} < 630$ | $T_s$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_TDD_2046 | $4716 \le T_{UE\ Rx-Tx} < 4718$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_TDD_2047 | $4718 \le T_{UE\ Rx-Tx} < 4720$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_TDD_2048 | $4720 \le T_{UE\ Rx-Tx} < 4728$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_TDD_2049 | $4728 \le T_{UE\ Rx-Tx} < 4736$ | $T_s$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_TDD_4093 | $21080 \le T_{UE\ Rx-Tx} < 21088$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_TDD_4094 | $21088 \le T_{UE\ Rx-Tx} < 21096$ | $T_s$ |
| RX-TX_TIME_DIFFERENCE_TDD_4095 | $21096 \le T_{UE\ Rx-Tx}$ | $T_s$ |

# 4.7 OTDOA test conditions

## 4.7.1 Simulated cells

For the intra-frequency OTDOA measurement test cases in clause 9.1, a multi cell environment as defined in 3GPP TS 36.508 [18] with Cell 1, Cell 2, and Cell 4 (if needed in the test) is used.

For the inter-frequency OTDOA measurement test cases in clause 9.2, a multi cell environment as defined in 3GPP TS 36.508 [18] with Cell 1 (called Cell 1 in the tests), Cell 3 (called Cell 2 in the tests), and Cell 6 (called Cell 3 in the tests) (if needed in the test) is used.

For the intra-frequency OTDOA measurement test cases for UE Category M1/M2 in clause 9.3, a multi cell environment as defined in 3GPP TS 36.508 [18] with Cell 1, Cell 2, and Cell 4 (if needed in the test) is used.

For the inter-frequency OTDOA measurement test cases for UE Category M1/M2 in clause 9.4, a multi cell environment as defined in 3GPP TS 36.508 [18] with Cell 1 (called Cell 1 in the tests), Cell 3 (called Cell 2 in the tests), and Cell 6 (called Cell 3 in the tests) (if needed in the test) is used.

For the intra-frequency NB-IOT OTDOA measurement accuracy test cases in clause 9.5, a multi cell environment with LTE Cell 1 and Cell 1a (see 3GPP TS 36.508 [18] Clause 4.4.2) and NB-IOT Ncell 1 and Ncell 1a (see 3GPP TS 36.508 [18] Clause 8.1.4.2) is used.

For the intra-frequency NB-IOT OTDOA measurement reporting delay test cases in clause 9.5, a multi cell environment with NB-IOT Ncell 1, Ncell 1a and Ncell 2 (see 3GPP TS 36.508 [18] Clause 8.1.4.2) is used.

For the inter-frequency NB-IOT OTDOA measurement accuracy test cases in clause 9.6, a multi cell environment with LTE Cell 1 and Cell 1a (see 3GPP TS 36.508 [18] Clause 4.4.2) and NB-IOT Ncell 1 and Ncell 1a (see 3GPP TS 36.508 [18] Clause 8.1.4.2) is used.

For the inter-frequency NB-IOT OTDOA measurement reporting delay test cases in clause 9.9, a multi cell environment with NB-IOT Ncell 1, Ncell 1a and Ncell 2 (see 3GPP TS 36.508 [18] Clause 8.1.4.2) is used.

For the OTDOA measurement test cases for Carrier Aggregation in clause 10, a multi cell environment is used with Cell 1 as the PCell on the PCC, Cell 2 is an active SCell on the SCC, and Cell 3 is a neighbour cell on the SCC. For the OTDOA measurement test cases for 3 DL Carrier Aggregation in clause 10, a multi cell environment is used with Cell 1 as the PCell on the PCC, Cell 2 is an active SCell on SCC1, Cell 3 is an active SCell on SCC2and Cell 4 is a neighbour cell on SCC2.

The default parameters for simulated cells are the same as specified in 3GPP TS 36.508 [18], with the following exceptions:

- All cells transmit PRS according to the PRS configuration provided in the OTDOA assistance data defined for each test. The positioning subframes are low-interference subframes, i.e. contain no PDSCH transmissions.
- The physical layer cell identities are selected such that the relative shifts of PRS patterns among cells used in the tests are as given by the test parameters of the individual test cases.
- The cells shall be synchronized and the timing offset (the RSTD) between the cells referenced to the UE’s antenna input is given in the individual test cases.

## 4.7.2 Propagation conditions

### 4.7.2.1 Static

See TS 36.521-1 [24] clause B.1.

### 4.7.2.2 Multi-path fading

See TS 36.521-1[24] clauses B.2, B.2.1 and B.2.2.

## 4.7.3 Response time

The response time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response time specified for the Measurement Reporting Delay test cases assumes that the UE shall not reuse any RSTD information or other aiding data that was previously acquired and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 and in TS 38.509 [44] clause 6.3.5 for the purpose of deleting this information.

## 4.7.4 RSTD reporting range

The reporting range of RSTD is defined from -15391$T_s$ to 15391$T_s$ with 1$T_s$ resolution for absolute value of RSTD less or equal to 4096$T_s$ and 5Ts for absolute value of RSTD greater than 4096$T_s$.

The mapping of measured quantity is defined in Table 4.7.4-1.

**Table 4.7.4-1: RSTD report mapping**

| Reported Value | Measured Quantity Value | Unit |
|---|---|---|
| RSTD_0000 | -15391 > RSTD | $T_s$ |
| RSTD_0001 | -15391 ≤ RSTD < -15386 | $T_s$ |
| … | … | … |
| RSTD_2258 | -4106 ≤ RSTD < -4101 | $T_s$ |
| RSTD_2259 | -4101 ≤ RSTD < -4096 | $T_s$ |
| RSTD_2260 | -4096 ≤ RSTD < -4095 | $T_s$ |
| RSTD_2261 | -4095 ≤ RSTD < -4094 | $T_s$ |
| … | … | … |
| RSTD_6353 | -3 ≤ RSTD < -2 | $T_s$ |
| RSTD_6354 | -2 ≤ RSTD < -1 | $T_s$ |
| RSTD_6355 | -1 ≤ RSTD ≤ 0 | $T_s$ |
| RSTD_6356 | 0 < RSTD ≤ 1 | $T_s$ |
| RSTD_6357 | 1 < RSTD ≤ 2 | $T_s$ |
| RSTD_6358 | 2 < RSTD ≤ 3 | $T_s$ |
| … | … | … |
| RSTD_10450 | 4094 < RSTD ≤ 4095 | $T_s$ |
| RSTD_10451 | 4095 < RSTD ≤ 4096 | $T_s$ |
| RSTD_10452 | 4096 < RSTD ≤ 4101 | $T_s$ |
| RSTD_10453 | 4101 < RSTD ≤ 4106 | $T_s$ |
| … | … | … |
| RSTD_12709 | 15381 < RSTD ≤ 15386 | $T_s$ |
| RSTD_12710 | 15386 < RSTD ≤ 15391 | $T_s$ |
| RSTD_12711 | 15391 < RSTD | $T_s$ |

## 4.7.5 RSTD Carrier Aggregation Test Cases with Different Channel Bandwidth Combinations

RSTD carrier aggregation test cases may be defined with different channel bandwidth combinations to verify the same requirement.

If multiple carrier aggregation test cases with different channel bandwidth combinations are defined to verify the same requirement that is channel bandwidth independent, then the UE needs to be tested only with one bandwidth combination out of the bandwidth combination sets supported by that UE.

# 4.8 MBS test conditions

## 4.8.1 MBS signals

A single or multi MBS beacon environment, depending on the test, is used.

The MBS signal is defined at the antenna connector of the UE. For UE with integral antenna only, a reference antenna with a gain of 0 dBi is assumed.

The beacons shall be synchronized, and the beacon code phase delays are defined in each test. The MBS signals shall be transmitted with a frequency accuracy of ± 2.5 PPM from the specified MBS carrier centre frequency.

## 4.8.2 Propagation conditions

### 4.8.2.1 Static

See TS 36.521-1 [24] clause B.1.

### 4.8.2.2 Multi-path fading

According to the Extended Pedestrian A model with a Maximum Doppler frequency of 5Hz (EPA 5Hz) in TS 36.521-1 [24] clauses B.2, B.2.1 and B.2.2.

## 4.8.3 Response time

The response time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response time specified for the Measurement Reporting Delay test case assumes that the UE shall not reuse any information that was previously acquired and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 and in TS 38.509 [44] clause 6.3.5 for the purpose of deleting this information.

# 4.9 WLAN test conditions

## 4.9.1 Simulated WLAN Access Points

A multi-WLAN AP environment is used.

The WLAN signal is defined at the antenna connector of the UE. For UE with integral antenna only, a reference antenna with a gain of 0 dBi is assumed.

The beacon signals from multiple WLAN APs shall be available at the UE with a periodicity of at least 102.4 ms (Beacon Interval). In order to ensure that the UE is in passive scan mode, this interval can be reduced. Beacon signals from different APs shall be received at different time slots or in non-overlapping frequency channels. Non-overlapping frequency channels shall be at least 25 MHz apart in the WLAN 2.4 GHz band and at least 20 MHz apart in the WLAN 5 GHz band.

The WLAN Test Frequency IDs to be used during the tests are specified in the test cases and are as defined in TS 36.508 [18] clause 4.3.1.6.

## 4.9.2 Propagation conditions

### 4.9.2.1 Static

See TS 36.521-1 [24] clause B.1.

### 4.9.3 Response time

The response time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response time specified for the Measurement Reporting Delay test case assumes that the UE shall not reuse any information that was previously acquired and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 and in TS 38.509 [44] clause 6.3.5 for the purpose of deleting this information.

### 4.9.4 Void

## 4.10 BLE test conditions

### 4.10.1 Simulated BLE

A multi-BLE device environment is used.

The BLE signal is defined at the antenna connector of the UE. For UE with integral antenna only, a reference antenna with a gain of 0 dBi is assumed.

The beacon signals from multiple BLE devices shall be available at the UE with a broadcast interval of 100 ms. Signals from different BLE devices shall be received at different time slots or in non-overlapping BLE advertising frequency channels. The BLE advertising channels are Channel 37 (2402 MHz), Channel 38 (2426 MHz) and Channel 39 (2480 MHz). The beacons shall be of type Non-Connectable Advertising beacons.

### 4.10.2 Propagation conditions

#### 4.10.2.1 Static

See TS 36.521-1 [24] clause B.1.

### 4.10.3 Response time

The response time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response time specified for the Measurement Reporting Delay test case assumes that the UE shall not reuse any information that was previously acquired and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 and in TS 38.509 [44] clause 6.3.5 for the purpose of deleting this information.

## 4.11 NB-IOT test conditions

### 4.11.1 Groups of bands

The NB-IOT tests use the band groupings defined in TS 36.521-3 [25] clause 3.5.1 in order to increase the readability of the specification

**Table 4.11.1-1: Void**

### 4.11.2 NB-IOT inband mode

The E-UTRA donor cell shall use the settings defined in Clause 4.4.1 unless otherwise stated.

# 4.12 NR test conditions

## 4.12.1 NR terminology

The terminology used in this specification for NR architecture options is described below.

**Table 4.12.1-1: NR terminology**

| Terminology | Abbreviation |
|---|---|
| NSA | |
| E-UTRA-NR Dual Connectivity | EN-DC |
| NR-E-UTRA Dual Connectivity | NE-DC |
| NG-RAN E-UTRA-NR Dual Connectivity | NGEN-DC |
| SA | |
| NG-RAN NR Radio Access | NG-RAN NR |
| NG-RAN E-UTRA Radio Access | NG-RAN E-UTRA |

## 4.12.2 NR frequency band and frequency range

The A-GNSS tests in clause 13, MBS tests in clause 11, WLAN tests in clause 15 and BLE tests in clause 16 are, where relevant, performed on the NR test frequency and default channel bandwidth of the NR operating frequency band of the UE as defined in TS 38.508-1 [45] clause 4.3.1. .

The A-GNSS requirements and tests in clause 13 apply for NR UE in FR1 and FR2.

If connectivity is *NR* (see TS 38.508-1 [45] clause 4.5) and if the UE supports multiple NR frequency bands then the A-GNSS Sensitivity tests in clause 13.2 shall be repeated in each supported NR frequency band.

If connectivity is *EN-DC* (see TS 38.508-1 [45] clause 4.5) and if the UE supports multiple EN-DC configurations, then the A-GNSS Sensitivity tests in clause 13.2 shall be performed in one EN-DC band combination in each of the applicable frequency group combination as specified in clause 4.12.6.

The NR OTDOA tests in clause 14 are performed on the ARFCN(s) of the operating frequency band of the UE and the channel bandwidth(s) specified in the test cases and as defined in FFS.

## 4.12.3 Groups of bands

The NR tests use the band groupings defined in TS 38.533 [47] clause 3A.4 in order to increase the readability of the specification.

**Table 4.12.3-1: Void**

**Table 4.12.3-2: Void**

## 4.12.4 Sensors

All the minimum performance requirements in clause 13 shall be met without the use of any data coming from sensors that can aid the positioning. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 38.509 [44] clause 6.3.5 for the purpose of disabling any such sensors.

## 4.12.5 Default signal conditions for FR2

For NR FR2, the connection between the SS and the DUT shall be OTA.

For the RAT-Independent test cases defined in clause 13, the SS shall ensure that a stable OTA link between the SS and the DUT can be established and maintained throughout the test. This link shall be sufficient to provide stable LPP message transmissions between the SS and the DUT. The connection for the other technologies (i.e. non-NR) used for the tests in clause 13 (e.g. LTE, GNSS, WLAN …) shall be conducted.

For the RAT-Dependent test cases defined in clauses 14, 15 and 16, a calibrated NR FR2 signal is required. The requirements OTA test method are defined in clause 7.1.3 of TS 38.508-1 [45].

### 4.12.6 Frequency Bands for Testing

#### 4.12.6.1 EN-DC band combination groups

For the A-GNSS sensitivity requirements in EN-DC operation mode with uplink assigned to E-UTRA and NR frequency bands, the A-GNSS Sensitivity tests in clause 13.2 shall be performed in one EN-DC band combination in each of the supported frequency group combination specified in TS 38.171 [43] Table B.1.13.1-1, where the frequency groups are defined in TS 38.171 [43] Table B.1.13.1-2.

#### 4.12.6.2 Applicable EN-DC band combinations for performing A-GNSS Sensitivity Requirements

The A-GNSS Sensitivity tests in clause 13.2 when in EN-DC operation mode shall be performed in EN-DC band combinations that can generate second or third order intermodulation products falling into the following GNSS receiver bands for the particular GNSS (where supported by the UE):

- GPS L1 C/A: 1574.3970 – 1576.4430 MHz
- Galileo E1 / GPS L1C: 1573.3740 – 1577.4660 MHz
- GLONASS G1: 1597.5515 – 1605.8860 MHz
- BDS B1I: 1559.0520 – 1563.1440 MHz

For each frequency group combination in TS 38.171 [43] Table B.1.13.2-1, in the case that the UE supports only one GNSS, only one EN-DC band combination shall be used for testing for the supported GNSS. In the case the UE supports more than one GNSS then the one EN-DC band combination used for testing shall be common across the supported GNSSs unless there is no common EN-DC band combination in which case the tests shall be repeated as necessary.

#### 4.12.6.3 Test frequencies for EN-DC band combinations

For performing the A-GNSS Sensitivity tests in clause 13.2 in EN-DC operation mode, the E-UTRA and NR frequencies and channel configurations shall be selected to ensure the intermodulation products fall into the GNSS receiver bands as defined in TS 38.171 [43] clause B.1.13.2 for the particular GNSS.

4.13 LPP transport mechanism for NR

The NR A-GNSS minimum performance requirements tested in this specification are agnostic as to whether the LPP session is performed over the Control Plane or the User Plane. Thus, the NR A-GNSS test cases in clause 13 support both configurations. The user can select either of the two options to run the test.

## 4.14 Multi-RTT test conditions

### 4.14.1 Simulated cells

For the Multi-RTT measurement test cases in clause 15 a cell environment as defined in 3GPP TS 38.508-1 [45] with NR Cell 1 and NR cell 2 are used. The default parameters for simulated cells are the same as specified in 3GPP TS 38.508-1 [45].

### 4.14.2 Propagation conditions

See TS 38.533 [47] clause C 2.

## 4.14.3 Measurement Reporting Requirements

### 4.14.3.1 Measurement Reporting Requirements in RRC_CONNECTED state

This requirement assumes that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x TTIDCCH where TTIDCCH is the duration of subframe or slot or subslot when the measurement report is transmitted on the PUSCH with subframe or slot or subslot duration. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

### 4.14.3.2 Measurement Reporting Requirements in RRC_INACTIVE state

See TS 38.133 [50] clause 5.6.4.4.

## 4.14.4 Measurement Period Requirements

### 4.14.4.1 Measurement Period Requirements in RRC_CONNECTED state within Measurement Gaps

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [49], UE shall be able to measure multiple (up to the UE capability specified in TS 38.133 clause 9.9.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [57] in configured positioning frequency layers within the measurement period $T_{UERxTx,Total}$ ms.

$T_{UERxTx,\ Total} = \sum_{i=1}^{L} T_{UERxTx,i} + (L-1) * \max(T_{effect,i})$.

where $i$ is the index of positioning frequency layer,

$T_{UERxTx,i}$ is the measurement period for UE Rx-Tx time difference measurements in positioning frequency layer $i$ as further defined in this clause,

L is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer $i$ as defined further in this clause.

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where

$CSSF_i$ is the carrier-specific scaling factor for NR PRS-based measurement in the positioning frequency layer $i$ as defined in TS 38.133 clause 9.1.5.2,

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [49] in *NR-Multi-RTT-RequestLocationInformation*;

otherwise,

$k_{multiTEG}$=$N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}$=$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [49] in *NR-Multi-RTT-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-Multi-RTT-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in clause 4.2.7.2 of TS 37.355 [49],

N' is UE capability for number of DL PRS resources that it can process in a slot corresponding to *maxNumOfDL-PRS-ResProcessedPerSlot* as specified in clause 6.4.3 of TS 37.355 [49],

$N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:

- $N_{sample}$ = 4 if the UE is not capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [49].
- $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [49] and LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34] and the following conditions are met:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.

- $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [49] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34] but the following conditions are not met:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time, $T_{last,i} = T_i + T_{available_PRS,i}$,

$T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

where

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$

$MGRP_i$ is the measurement gap repetition periodicity in positioning frequency layer *i*.

$T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

### 4.14.4.2 Measurement Period Requirements in RRC_CONNECTED state without Measurement Gaps

See TS 38.133 [50] clause 9.9.4.6.

### 4.14.4.2A Measurement Period Requirements in RRC_CONNECTED state with Bandwidth Aggregation

See TS 38.133 [50] clause 9.9.4.9.

### 4.14.4.3 Measurement Period Requirements in RRC_INACTIVE state

See TS 38.133 [50] clause 5.6.4.5

### 4.14.4.4 Measurement Period Requirements in RRC_INACTIVE state with Bandwidth Aggregation

See TS 38.133 [50] clause 5.6.4.6

## 4.14.5 Measurement Accuracy Requirements

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall not apply, if:

$N_{TA_offset}$ defined in TS 38.133 Table 7.1.2-2 changes during the UE Rx-Tx measurement period or

if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the network-configured Timing Advance.

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that:

- The UE transmits SRS within [-160, 160] msec of at least one DL PRS resource of each of the TRPs in the assistance data.

If the uplink transmission timing changes during the UE Rx-Tx measurement period due to the autonomous timing adjustment defined in TS 38.133 clause 7.1.2 then:

- UE Rx-Tx measurement accuracy requirements shall apply for a cell, which is also the downlink reference cell (defined in TS 38.133 section 7.1.1) for SRS transmission even if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

- UE Rx-Tx measurement accuracy requirements shall not apply for a cell, which is not the downlink reference cell (defined in TS 38.133 section 7.1.1) for SRS transmission, if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

When a serving cell change occurs during the UE Rx-Tx measurement period, the UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that the serving cell change does not impact SRS configuration for the UE Rx-Tx measurement.

Note: The requirements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

The accuracy requirements in Table 4.14.5-1 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band.

AWGN propagation condition.

**Table 4.14.5-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ± 78+δ | -3 | ≥24 | 15 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | |
| | | | | | NR_TDD_FR1_C | -120 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | NR_FDD_FR1_G | -118 | |
| | | | | | NR_FDD_FR1_H | -117.5 | |
| ± 59+δ | | ≥52 | | ≥1 | Note 6 | Note 6 | Note 6 |
| ± 30+δ | | >104 | | ≥1 | Note 6 | Note 6 | Note 6 |
| ± 57+δ | | ≥24 | 30 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -118 | -50 |
| | | | | | NR_FDD_FR1_B | -117.5 | |
| | | | | | NR_TDD_FR1_C | -117 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -116.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -116 | |
| | | | | | NR_FDD_FR1_F | -115.5 | |
| | | | | | NR_FDD_FR1_G | -115 | |
| | | | | | NR_FDD_FR1_H | -114.5 | |
| ± 30+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 15+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 29+δ | | ≥24 | 60 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -115 | -50 |
| | | | | | NR_FDD_FR1_B | -114.5 | |
| | | | | | NR_TDD_FR1_C | -114 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -113.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -113 | |
| | | | | | NR_FDD_FR1_F | -113.5 | |
| | | | | | NR_FDD_FR1_G | -113 | |
| | | | | | NR_FDD_FR1_H | -111.5 | |
| ± 15+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 101+δ | -13 | ≥24 | 15 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 75+δ | | ≥52 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 37+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 58+δ | | ≥24 | 30 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 39+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 36+δ | | ≥24 | 60 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 8+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in TS 38.133 Section 3.5.
NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 4.14.5-5.

The accuracy requirements in Table 4.14.5-1a for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.

PRP$|_{\text{dBm}}$ according to TS 38.133 Annex B.2.14 for a corresponding Band.

Number of measurement samples is less than 4

AWGN propagation condition.

**Table 4.14.5-1a: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN with reduced measurement samples**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{\text{rep}}^{\text{PRS}} * L_{\text{PRS}}/K_{\text{comb}}^{\text{PRS}}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ± 59+δ | 0 | ≥52 | 15 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| ± 30+δ6 | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 30+δ | | ≥48 | 30 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| ± 15+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 15+δ | | ≥64 | 60 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 75+δ | -6 | ≥52 | 15 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 37+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 39+δ | | ≥48 | 30 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥64 | 60 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 8+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{\mathrm{rep}}^{\mathrm{PRS}}$, $L_{\mathrm{PRS}}$ , $K_{\mathrm{comb}}^{\mathrm{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [49].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 4.14.5-5.

The relative accuracy requirements in Table 4.14.5-1b for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.

$\mathrm{PRP}|_{\mathrm{dBm}}$ according to TS 38.133 Annex B.2.14 for a corresponding Band.

AWGN propagation condition.

the two UE Rx-Tx time difference measurements are associated with the same RxTx TEG

**Table 4.14.5-1b: UE Rx-Tx time difference relative measurement accuracy in FR1 in AWGN with TEG reporting**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 132 + Δ Note 7 | (PRS Ês/Iot)$_i$ ≥ -6dB<br>(PRS Ês/Iot)$_i$ ≥ -13dB | 15 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| 98 + Δ | | | ≥ 52 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 42 + Δ | | | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 75 + Δ | | 30 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| 48 + Δ | | | ≥ 48 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 50 + Δ | | 60 | ≥24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 24 + Δ | | | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 10 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of resource j and resource i.

NOTE 2: Minimum number of PRS resource repetitions among resource j and resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.

NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in TS 38.133 [50] clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Δis the value of the timing error margin for the RxTx TEG, reported via *nr-UE-RxTxTEG-TimingErrorMargin*. Δ cannot be larger than the sum of the margins in table 4.14.5-5 (dependent on PRS/SRS BW) for any pair of individual UE Rx-Tx time difference measurements associated with the RxTx TEG. .

The accuracy requirements in Table 4.14.5-2 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band.

Fading propagation condition.

**Table 4.14.5-2: UE Rx-Tx time difference measurement accuracy in FR1 in fading**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition $(T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ± 137+δ | -3 | ≥24 | 15 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | |
| | | | | | NR_TDD_FR1_C | -120 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | NR_FDD_FR1_G | -118 | |
| | | | | | NR_FDD_FR1_H | -117.5 | |
| ± 96+δ | | ≥52 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 62+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 87+δ | | ≥24 | 30 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -118 | -50 |
| | | | | | NR_FDD_FR1_B | -117.5 | |
| | | | | | NR_TDD_FR1_C | -117 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -116.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -116 | |
| | | | | | NR_FDD_FR1_F | -115.5 | |
| | | | | | NR_FDD_FR1_G | -115 | |
| | | | | | NR_FDD_FR1_H | -114.5 | |
| ± 68+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 44+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 59+δ | | ≥24 | 60 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -115 | -50 |
| | | | | | NR_FDD_FR1_B | -114.5 | |
| | | | | | NR_TDD_FR1_C | -114 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -113.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -113 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_FDD_FR1_F | -113.5 | |
| | | | | | NR_FDD_FR1_G | -113 | |
| | | | | | NR_FDD_FR1_H | -111.5 | |
| ± 42+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 36+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 180+δ | -13 | ≥24 | 15 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 98+δ | | ≥52 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 68+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 87+δ | | ≥24 | 30 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 85+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 44+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 139+δ | | ≥24 | 60 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 46+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 30+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in TS 38.133 Section 3.5.
NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 4.14.5-5.

The accuracy requirements in Table 4.14.5-3 for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band.

AWGN propagation condition.

**Table 4.14.5-3: UE Rx-Tx time difference measurement accuracy in FR2 in AWGN**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{\text{rep}}^{\text{PRS}} * L_{\text{PRS}} / K_{\text{comb}}^{\text{PRS}}$ Note 3 | Io Note 4 range | |
| | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| ± 22+δ | -3 | ≥24 | 60 | ≥4 | Same value as PRP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ± 15+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 12+δ | | ≥32 | 120 | ≥1 | Same value as PRP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ± 7+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 35+δ | -13 | ≥24 | 60 | ≥1 | NOTE 6 | NOTE 6 |
| ± 15+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 14+δ | | ≥32 | 120 | ≥1 | NOTE 6 | NOTE 6 |
| ± 9+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in TS 38.133 Section 3.5.

NOTE 3: $T_{\mathrm{rep}}^{\mathrm{PRS}}$, $L_{\mathrm{PRS}}$, $K_{\mathrm{comb}}^{\mathrm{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 4.14.5-6.

The accuracy requirements in Table 4.14.5-3a for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

PRP|$_{\mathrm{dBm}}$ according to TS 38.133 Annex B.2.14 for a corresponding Band

Number of measurement samples is less than 4

AWGN propagation condition.

**Table 4.14.5-3a: UE Rx-Tx time difference measurement accuracy in FR2 in AWGN with reduced measurement samples**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{\mathrm{rep}}^{\mathrm{PRS}} * L_{\mathrm{PRS}} / K_{\mathrm{comb}}^{\mathrm{PRS}}$ Note 3 | Io Note 4 range | |
| | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| ± 15+δ | 0 | ≥64 | 60 | ≥1 | Same value as PRP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥64 | 120 | ≥1 | Same value as PRP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 15+δ | -6 | ≥64 | 60 | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 9+δ | | ≥64 | 120 | ≥1 | NOTE 6 | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in TS 38.133 Section 3.5.
NOTE 3: $T_{\mathrm{rep}}^{\mathrm{PRS}}$, $L_{\mathrm{PRS}}$, $K_{\mathrm{comb}}^{\mathrm{PRS}}$ are configured by higher layer parameter dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN defined in TS 37.355 [49].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 4.14.5-6.

The relative accuracy requirements in Table 4.14.5-3b for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

PRP|$_{\mathrm{dBm}}$ according to TS 38.133 Annex B.2.14 for a corresponding Band

AWGN propagation condition.

the two UE Rx-Tx time difference measurements are associated with the same RxTx TEG

**Table 4.14.5-3b: UE Rx-Tx time difference relative measurement accuracy in FR2 in AWGN with TEG reporting**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |
| 35 + Δ Note 6 | (PRS Ês/Iot)$_j$ ≥ -6dB | 60 | ≥ 24 | ≥ 4 | Same value as PRS_RP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 24 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 11 + Δ | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 24+ Δ | (PRS Ês/Iot)$_i$ ≥ -13dB | 120 | ≥ 32 | ≥ 4 | Same value as PRS_RP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 13 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 6 + Δ | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of resource j and resource i.
NOTE 2: Minimum number of PRS resource repetitions among resource j and resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: Δis the value of the timing error margin for the RxTx TEG, reported via *nr-UE-RxTxTEG-TimingErrorMargin*. Δ cannot be larger than the sum of the margins in table 4.14.5-6 (dependent on PRS/SRS BW) for any pair of individual UE Rx-Tx time difference measurements associated with the RxTx TEG.

The accuracy requirements in Table 4.14.5-4 for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

PRP$|_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band.

Fading propagation condition.

**Table 4.14.5-4: UE Rx-Tx time difference measurement accuracy in FR2 in fading**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$ Note 3 | Io Note 4 range | |
| | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | dBm / SCS$_{PRS}$ | dBm/BW$_{Channel}$ |
| ± 75+δ | -3 | ≥24 | 60 | ≥4 | Same value as PRP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ± 72+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 57+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 61+δ | | ≥32 | 120 | ≥1 | Same value as PRP in TS 38.133 Table B.2.14-2, according to UE | -50 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | Power class, operating band and angle of arrival | |
| ± 64+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 55+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 92+δ | -13 | ≥24 | 60 | ≥4 | NOTE 6 | NOTE 6 |
| ± 70+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 57+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 60+δ | | ≥32 | 120 | ≥1 | NOTE 6 | NOTE 6 |
| ± 66+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 62+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.<br>NOTE 2: NR operating band groups are as defined in TS 38.133 Section 3.5.<br>NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [49].<br>NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.<br>NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].<br>NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.<br>NOTE 7: δ is the margin determined from Table 4.14.5-6. | | | | | | |

**Table 4.14.5-5: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (PRB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 24 | N/A | N/A | 160 |
| ≥ 52 | ≥ 24 | N/A | 80 |
| ≥ 104 | ≥ 48 | ≥ 24 | 56 |
| N/A | ≥ 132 | ≥ 64 | 24 |
| N/A | N/A | ≥ 132 | 24 |
| NOTE 1: Tc is the basic timing unit defined in TS 38.211 [53].<br>NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies. | | | |

**Table 4.14.5-6: Margin for UE Rx-Tx time difference measurement accuracy in FR2**

| Min(PRS BW, SRS BW) (MHz) | | Margin (Tc [Note 1]) |
|---|---|---|
| SCS = 60 kHz | SCS = 120 kHz | |
| ≥ 24 | N/A | 76 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 24 |
| N/A | ≥ 128 | 20 |
| NOTE 1: Tc is the basic timing unit defined in TS 38.211 [53].<br>NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies. | | |

## 4.14.5B Measurement Accuracy Requirements with Bandwidth Aggregation

See TS 38.133 [50] clause 10.1.25A.2

## 4.14.6 Reporting mapping

### 4.14.6.1 Absolute UE Rx-Tx Measurement Report Mapping

The reporting range for the absolute UE Rx-Tx time difference measurement ($T_{\text{UE Rx-Tx}}$) is defined from $-985024 \times T_c$ to $985024 \times T_c$ with the resolution step of $2^k \times T_c$, where:

$T_c$ is defined in TS 38.211 [53],

$k_{min} \leq k \leq k_{max}$,

$k_{min}$=2 and $k_{max}$=5, when at least one of the PRS and the SRS resources configured for $T_{UE\ Rx-Tx}$ is in FR1,

$k_{min}$=0 and $k_{max}$=5, when both PRS and SRS resources configured for $T_{UE\ Rx-Tx}$ are in FR2,

$k \geq$ *timingReportingGranularityFactor* [49] configured by LMF via LPP for the UE Rx-Tx time difference measurement.

The $T_{UE\ Rx-Tx}$ report mapping for $k$ = 0, 1, 2, 3, 4, and 5 are specified in Tables 4.14.6.1-1, 14.14.6.1-2, 4.14.6.1-3, 4.14.6.1-4, 4.14.6.1-5, and 4.14.6.1-6, respectively.

**Table 4.14.6.1-1: Absolute UE Rx-Tx time difference measurement report mapping for *k*=0**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_0000 | $T_{UE\ Rx-Tx} < -985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0001 | $-985024 \leq T_{UE\ Rx-Tx} < -985023$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0002 | $-985023 \leq T_{UE\ Rx-Tx} < -985022$ | $T_c$ |
|  |  | … |
| RX-TX_TIME_DIFFERENCE_985024 | $-1 \leq T_{UE\ Rx-Tx} < 0$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_985025 | $0 \leq T_{UE\ Rx-Tx} < 1$ | $T_c$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_1970047 | $985022 \leq T_{UE\ Rx-Tx} < 985023$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_1970048 | $985023 \leq T_{UE\ Rx-Tx} < 985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_1970049 | $985024 \leq T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.1-2: Absolute UE Rx-Tx time difference measurement report mapping for *k*=1**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_0000 | $T_{UE\ Rx-Tx} < -985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0001 | $-985024 \leq T_{UE\ Rx-Tx} < -985022$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0002 | $-985022 \leq T_{UE\ Rx-Tx} < -985020$ | $T_c$ |
|  |  | … |
| RX-TX_TIME_DIFFERENCE_492512 | $-2 \leq T_{UE\ Rx-Tx} < 0$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_492513 | $0 \leq T_{UE\ Rx-Tx} < 2$ | $T_c$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_985023 | $985020 \leq T_{UE\ Rx-Tx} < 985022$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_985024 | $985022 \leq T_{UE\ Rx-Tx} < 985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_985025 | $985024 \leq T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.1-3: Absolute UE Rx-Tx time difference measurement report mapping for *k*=2**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_0000 | $T_{UE\ Rx-Tx} < -985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0001 | $-985024 \leq T_{UE\ Rx-Tx} < -985020$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0002 | $-985020 \leq T_{UE\ Rx-Tx} < -985016$ | $T_c$ |
|  |  | … |
| RX-TX_TIME_DIFFERENCE_246256 | $-4 \leq T_{UE\ Rx-Tx} < 0$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_246257 | $0 \leq T_{UE\ Rx-Tx} < 4$ | $T_c$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_492511 | $985016 \leq T_{UE\ Rx-Tx} < 985020$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_492512 | $985020 \leq T_{UE\ Rx-Tx} < 985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_492513 | $985024 \leq T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.1-4: Absolute UE Rx-Tx time difference measurement report mapping for *k*=3**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_0000 | $T_{UE\ Rx-Tx} < -985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0001 | $-985024 \leq T_{UE\ Rx-Tx} < -985016$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0002 | $-985016 \leq T_{UE\ Rx-Tx} < -985008$ | $T_c$ |
|  |  | … |

| | | |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_123128 | $-8 \le T_{UE\ Rx-Tx} < 0$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_123129 | $0 \le T_{UE\ Rx-Tx} < 8$ | $T_c$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_246255 | $985008 \le T_{UE\ Rx-Tx} < 985016$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_246256 | $985016 \le T_{UE\ Rx-Tx} < 985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_246257 | $985024 \le T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.1-5: Absolute UE Rx-Tx time difference measurement report mapping for *k*=4**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_0000 | $T_{UE\ Rx-Tx} < -985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0001 | $-985024 \le T_{UE\ Rx-Tx} < -985008$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0002 | $-985008 \le T_{UE\ Rx-Tx} < -984992$ | $T_c$ |
| □ | □ | … |
| RX-TX_TIME_DIFFERENCE_61564 | $-16 \le T_{UE\ Rx-Tx} < 0$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_61565 | $0 \le T_{UE\ Rx-Tx} < 16$ | $T_c$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_123127 | $984992 \le T_{UE\ Rx-Tx} < 985008$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_123128 | $985008 \le T_{UE\ Rx-Tx} < 985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_123129 | $985024 \le T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.1-6: Absolute UE Rx-Tx time difference measurement report mapping for *k*=5**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| RX-TX_TIME_DIFFERENCE_0000 | $T_{UE\ Rx-Tx} < -985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0001 | $-985024 \le T_{UE\ Rx-Tx} < -984992$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_0002 | $-984992 \le T_{UE\ Rx-Tx} < -984960$ | $T_c$ |
| □ | □ | … |
| RX-TX_TIME_DIFFERENCE_30782 | $-32 \le T_{UE\ Rx-Tx} < 0$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_30783 | $0 \le T_{UE\ Rx-Tx} < 32$ | $T_c$ |
| … | … | … |
| RX-TX_TIME_DIFFERENCE_61563 | $984960 \le T_{UE\ Rx-Tx} < 984992$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_61564 | $984992 \le T_{UE\ Rx-Tx} < 985024$ | $T_c$ |
| RX-TX_TIME_DIFFERENCE_61565 | $985024 \le T_{UE\ Rx-Tx}$ | $T_c$ |

### 4.14.6.2 Differential UE Rx-Tx Measurement Report Mapping

The reporting range for differential UE Rx-Tx time difference measurement ($\Delta T_{UE\ Rx-Tx}$) is defined from 0 up to $8191 \times T_c$ where:

$\Delta T_{UE\ Rx-Tx} = T_{UE\ Rx-Tx1} - T_{UE\ Rx-Tx2;}$ where:

$T_{UE\ Rx-Tx1} > T_{UE\ Rx-Tx2}$,

$T_{UE\ Rx-Tx1}$ is the first absolute UE Rx-Tx time difference measurement,

$T_{UE\ Rx-Tx1}$ is the second absolute UE Rx-Tx time difference measurement,

$T_c$ is defined in TS 38.211 [53],

$k_{min} \le k \le k_{max}$,

$k_{min}$=2 and $k_{max}$=5, when at least one of the PRS and the SRS resources configured for $\Delta T_{UE\ Rx-Tx}$ is in FR1,

$k_{min}$=0 and $k_{max}$=5, when all the PRS and SRS resources configured for $\Delta T_{UE\ Rx-Tx}$ are in FR2,

$k \ge$ *timingReportingGranularityFactor* [49] configured by LMF via LPP for the UE Rx-Tx time difference measurement.

The $\Delta T_{UE\ Rx-Tx}$ report mapping for $k$ = 0, 1, 2, 3, 4, and 5 are specified in Tables 4.14.6.2-1, 4.14.6.2-2, 4.14.6.2-3, 4.14.6.2-4, 4.14.6.2-5, and 4.14.6.2-6, respectively.

**Table 4.14.6.2-1: Differential UE Rx-Tx time difference measurement report mapping for *k*=0**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| DIFF_RX-TX_TIME_DIFFERENCE_0000 | $0 \le \Delta T_{UE\ Rx-Tx} < 1$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0001 | $1 \le \Delta T_{UE\ Rx-Tx} < 2$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0002 | $2 \le \Delta T_{UE\ Rx-Tx} < 3$ | $T_c$ |
| □ | □ | … |
| DIFF_RX-TX_TIME_DIFFERENCE_8189 | $8189 \le \Delta T_{UE\ Rx-Tx} < 8190$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_8190 | $8190 \le \Delta T_{UE\ Rx-Tx} < 8191$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_8191 | $8191 \le \Delta T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.2-2: Differential UE Rx-Tx time difference measurement report mapping for *k*=1**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| DIFF_RX-TX_TIME_DIFFERENCE_0000 | $0 \le \Delta T_{UE\ Rx-Tx} < 2$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0001 | $2 \le \Delta T_{UE\ Rx-Tx} < 4$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0002 | $4 \le \Delta T_{UE\ Rx-Tx} < 6$ | $T_c$ |
| □ | □ | … |
| DIFF_RX-TX_TIME_DIFFERENCE_4093 | $8186 \le \Delta T_{UE\ Rx-Tx} < 8188$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_4094 | $8188 \le \Delta T_{UE\ Rx-Tx} < 8190$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_4095 | $8190 \le \Delta T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.2-3: Differential UE Rx-Tx time difference measurement report mapping for *k*=2**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| DIFF_RX-TX_TIME_DIFFERENCE_0000 | $0 \le \Delta T_{UE\ Rx-Tx} < 4$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0001 | $4 \le \Delta T_{UE\ Rx-Tx} < 8$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0002 | $8 \le \Delta T_{UE\ Rx-Tx} < 12$ | $T_c$ |
| □ | □ | … |
| DIFF_RX-TX_TIME_DIFFERENCE_2045 | $8180 \le \Delta T_{UE\ Rx-Tx} < 8184$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_2046 | $8184 \le \Delta T_{UE\ Rx-Tx} < 8188$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_2047 | $8188 \le \Delta T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.2-4: Differential UE Rx-Tx time difference measurement report mapping for *k*=3**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| DIFF_RX-TX_TIME_DIFFERENCE_0000 | $0 \le \Delta T_{UE\ Rx-Tx} < 8$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0001 | $8 \le \Delta T_{UE\ Rx-Tx} < 16$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0002 | $16 \le \Delta T_{UE\ Rx-Tx} < 24$ | $T_c$ |
| □ | □ | … |
| DIFF_RX-TX_TIME_DIFFERENCE_1021 | $8168 \le \Delta T_{UE\ Rx-Tx} < 8176$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_1022 | $8176 \le \Delta T_{UE\ Rx-Tx} < 8184$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_1023 | $8184 \le \Delta T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.2-5: Differential UE Rx-Tx time difference measurement report mapping for *k*=4**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|
| DIFF_RX-TX_TIME_DIFFERENCE_0000 | $0 \le \Delta T_{UE\ Rx-Tx} < 16$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0001 | $16 \le \Delta T_{UE\ Rx-Tx} < 32$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0002 | $32 \le \Delta T_{UE\ Rx-Tx} < 48$ | $T_c$ |
| □ | □ | … |
| DIFF_RX-TX_TIME_DIFFERENCE_509 | $8144 \le \Delta T_{UE\ Rx-Tx} < 8160$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_510 | $8160 \le \Delta T_{UE\ Rx-Tx} < 8176$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_511 | $8176 \le \Delta T_{UE\ Rx-Tx}$ | $T_c$ |

**Table 4.14.6.2-6: Differential UE Rx-Tx time difference measurement report mapping for *k*=5**

| Reported Quantity Value | Measured Quantity Value | Unit |
|---|---|---|

| | | |
|---|---|---|
| DIFF_RX-TX_TIME_DIFFERENCE_0000 | $0 \le \Delta T_{UE\ Rx\text{-}Tx} < 32$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0001 | $32 \le \Delta T_{UE\ Rx\text{-}Tx} < 64$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_0002 | $64 \le \Delta T_{UE\ Rx\text{-}Tx} < 96$ | $T_c$ |
| □ | □ | … |
| DIFF_RX-TX_TIME_DIFFERENCE_253 | $8096 \le \Delta T_{UE\ Rx\text{-}Tx} < 8128$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_254 | $8128 \le \Delta T_{UE\ Rx\text{-}Tx} < 8160$ | $T_c$ |
| DIFF_RX-TX_TIME_DIFFERENCE_255 | $8160 \le \Delta T_{UE\ Rx\text{-}Tx}$ | $T_c$ |

### 4.14.6.3 Additional Path Report Mapping for UE Rx-Tx Time Difference

The reporting range for the additional path reporting for an UE Rx-Tx time difference measurement is defined up to the range from $-8175 \times T_c$ to $8175 \times T_c$ with the resolution step of $2^k \times T_c$, where

$T_c$ is defined in TS 38.211 [53],

$k_{min} \le k \le k_{max}$,

$k_{min}$=2 and $k_{max}$=5, when at least one of the PRS resource and SRS resource configured for the UE Rx-Tx time difference measurement is in FR1,

$k_{min}$=0 and $k_{max}$=5, when both of the PRS resource and SRS resource configured for the UE Rx-Tx time difference measurement is in FR2,

$k \ge$ *timingReportingGranularityFactor* [49] configured by LMF via LPP for the UE Rx-Tx time difference measurement.

The UE can report the timing of up to two additional paths with respect to the path timing determining the UE Rx-Tx time difference measurement.

The report mappings for different *k* values are specified in Tables 4.14.6.3-1 – 4.14.6.3-6.

**Table 4.14.6.3-1: Report mapping for *k*=0**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_00000 | Δpath < -8175 | $T_c$ |
| path_00001 | -8175 ≤ Δpath < -8174 | $T_c$ |
| path_00002 | -8174 ≤ Δpath < -8173 | $T_c$ |
| … | … | … |
| path_08175 | -1 ≤ Δpath < 0 | $T_c$ |
| path_08176 | 0 ≤ Δpath < 1 | $T_c$ |
| … | … | … |
| path_16349 | 8173 ≤ Δpath < 8174 | $T_c$ |
| path_16350 | 8174 ≤ Δpath < 8175 | $T_c$ |
| path_16351 | 8175 ≤ Δpath | $T_c$ |

**Table 4.14.6.3-2: Report mapping for *k*=1**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8175 | $T_c$ |
| path_0001 | -8175 ≤ Δpath < -8173 | $T_c$ |
| path_0002 | -8173 ≤ Δpath < -8171 | $T_c$ |
| … | … | … |
| path_4088 | -1 ≤ Δpath < 1 | $T_c$ |
| … | … | … |
| path_8174 | 8171 ≤ Δpath < 8173 | $T_c$ |
| path_8175 | 8173 ≤ Δpath < 8175 | $T_c$ |
| path_8176 | 8175 ≤ Δpath | $T_c$ |

**Table 4.14.6.3-3: Report mapping for *k*=2**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8174 | $T_c$ |
| path_0001 | -8174 ≤ Δpath < -8170 | $T_c$ |
| path_0002 | -8170 ≤ Δpath < -8166 | $T_c$ |
| … | … | … |
| path_2044 | -2 ≤ Δpath < 2 | $T_c$ |
| … | … | … |
| path_4086 | 8166 ≤ Δpath < 8170 | $T_c$ |
| path_4087 | 8170 ≤ Δpath < 8174 | $T_c$ |
| path_4088 | 8174 ≤ Δpath | $T_c$ |

**Table 4.14.6.3-4: Report mapping for *k*=3**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8172 | $T_c$ |
| path_0001 | -8172 ≤ Δpath < -8164 | $T_c$ |
| path_0002 | -8164 ≤ Δpath < -8156 | $T_c$ |
| … | … | … |
| path_1022 | -4 ≤ Δpath < 4 | $T_c$ |
| … | … | … |
| path_2042 | 8156 ≤ Δpath < 8164 | $T_c$ |
| path_2043 | 8164 ≤ Δpath < 8172 | $T_c$ |
| path_2044 | 8172 ≤ Δpath | $T_c$ |

**Table 4.14.6.3-5: Report mapping for *k*=4**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8168 | $T_c$ |
| path_0001 | -8168 ≤ Δpath < -8152 | $T_c$ |
| path_0002 | -8152 ≤ Δpath < -8136 | $T_c$ |
| … | … | … |
| path_511 | -8 ≤ Δpath < 8 | $T_c$ |
| … | … | … |
| path_1020 | 8136 ≤ Δpath < 8152 | $T_c$ |
| path_1021 | 8152 ≤ Δpath < 8168 | $T_c$ |
| path_1022 | 8168 ≤ Δpath | $T_c$ |

**Table 4.14.6.3-6: Report mapping for *k*=5**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_000 | Δpath < -8160 | $T_c$ |
| path_001 | -8160 ≤ Δpath < -8128 | $T_c$ |
| path_002 | -8128 ≤ Δpath < -8096 | $T_c$ |
| … | … | … |
| path_256 | 0 ≤ Δpath < 32 | $T_c$ |
| … | … | … |
| path_509 | 8096 ≤ Δpath < 8128 | $T_c$ |
| path_510 | 8128 ≤ Δpath < 8160 | $T_c$ |
| path_511 | 8160 ≤ Δpath | $T_c$ |

# 4.14A Multi-RTT test conditions for Redcap UE

## 4.14A.1 Simulated cells

For the Multi-RTT measurement test cases in clause 15 a cell environment as defined in 3GPP TS 38.508-1 [45] with NR Cell 1 and NR cell 2 are used. The default parameters for simulated cells are the same as specified in 3GPP TS 38.508-1 [45].

## 4.14A.2 Propagation conditions

See TS 38.533 [47] clause C 2.

## 4.14A.3 Measurement Reporting Requirements

### 4.14A.3.1 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH with Measurement Gaps

See TS 38.133 [50] clause 9.9A.4.4.

### 4.14A.3.2 Measurement Reporting Requirements in RRC_INACTIVE state

See TS 38.133 [50] clause 5.6A.6.4.

## 4.14A.4 Measurement Period Requirements

### 4.14A.4.1 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH within Measurement Gaps

See TS 38.133 [50] clause 9.9A.4.5.

### 4.14A.4.2 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH without Measurement Gaps

See TS 38.133 [50] clause 9.9A.4.6.

### 4.14A.4.3 Measurement Reporting Requirements in RRC_CONNECTED state without RX FH within both Measurement Gaps and PPW

See TS 38.133 [50] clause 9.9A.4.7.

### 4.14A.4.4 Measurement Reporting Requirements in RRC_CONNECTED state with RX FH

See TS 38.133 [50] clause 9.9A.4.8.

### 4.14A.4.5 Measurement Period Requirements in RRC_INACTIVE state without RX FH

See TS 38.133 [50] clause 5.6A.6.5.

### 4.14A.4.6 Measurement Period Requirements in RRC_INACTIVE state with RX FH

See TS 38.133 [50] clause 5.6A.6.6.

## 4.14A.5 Measurement Accuracy Requirements

### 4.14A.5.1 Measurement Accuracy Requirements for 2RX RedCap UE without RX FH

See TS 38.133 [50] clause 10.1A.18.2.1.

### 4.14A.5.2 Measurement Accuracy Requirements for 1RX RedCap UE without RX FH

See TS 38.133 [50] clause 10.1A.18.2.2.

### 4.14A.5.3 Measurement Accuracy Requirements for 2RX RedCap UE with RX FH

See TS 38.133 [50] clause 10.1A.18.2.3.

### 4.14A.5.4 Measurement Accuracy Requirements for 1RX RedCap UE with RX FH

See TS 38.133 [50] clause 10.1A.18.2.4.

## 4.14A.6 Reporting mapping

### 4.14A.6.1 Absolute UE Rx-Tx Measurement Report Mapping

See TS 38.133 [50] clause 10.1A.18.3.1.

### 4.14A.6.2 Differential UE Rx-Tx Measurement Report Mapping

See TS 38.133 [50] clause 10.1A.18.3.2.

### 4.14A.6.3 Additional Path Report Mapping for UE Rx-Tx Time Difference

See TS 38.133 [50] clause 10.1A.18.3.3.

# 4.15 DL-TDOA test conditions

## 4.15.1 Simulated cells

For the DL-TDOA measurement test cases in clause 14, a multi cell environment as defined in 3GPP TS 38.508-1 [45] with NR Cell 1, NR Cell 2 and NR Cell 3 (if needed in the test) are used. The default parameters for simulated cells are the same as specified in 3GPP TS 38.508-1 [45].

## 4.15.2 Propagation conditions

See TS 38.533 [47] clause C 2.

## 4.15.3 Measurement Reporting Requirements

### 4.15.3.1 Measurement Reporting Requirements in RRC_CONNECTED state

The measurement reporting delay is defined as the time between the moment when the periodic measurement report is triggered and the moment when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$ where $TTI_{DCCH}$ is the duration of subframe or slot or subslot when the measurement report is transmitted on the PUSCH with subframe or slot or subslot duration. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

### 4.15.3.2 Measurement Reporting Requirements in RRC_INACTIVE state

See TS 38.133 [50] clause 5.6.2.4.

## 4.15.4 Measurement Period Requirements

### 4.15.4.1 Measurement Period Requirements within Measurement Gaps

When physical layer receives last of *NR-TDOA-ProvideAssistanceData* message and *NR-TDOA-RequestLocationInformation* message from LMF via LPP [49], the UE shall be able to measure multiple (up to the UE capability specified in TS 38.133 Clause 9.9.2.3) DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined as:

$$\mathrm{T_{RSTD,Total}} = \sum_{\mathrm{i=1}}^{\mathrm{L}} \mathrm{T_{RSTD,i}} + (\mathrm{L}-1) * \max\left(\mathrm{T_{effect,i}}\right)$$

Where ,

$i$ is the index of positioning frequency layer,

$L$ is total number of positioning frequency layers, and

$\mathrm{T_{effect},i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$\mathrm{T_{RSTD,i}}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below:

$$\mathrm{T_{RSTD,i}} = \left(k_{multiTEG,i} * \mathrm{CSSF_{PRS,i}} * \mathrm{ceil}(\mathrm{K_{p,PRS,i}}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * \mathrm{T_{effect,i}} + \mathrm{T_{last,i}}$$

,

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1, $N_{RxBeam,i}$ = 1;

and in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$CSSF_{PRS,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in positioning frequency layer $i$ as defined in TS 38.133 clause 9.1.5.2.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*;

otherwise,

$k_{multiTEG}$=$N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}$=$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*, and in case ‘n0’ is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

- For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioningfrequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:
  - $N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and
  - $N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule
  - Requirements do not apply if $N_{available}$ =0.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time. If all of the PRS resources to be measured are available in the same MG occasion during $T_{availabe}$, $T_{last,i} = T_i$ +MGL. Otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$ ,

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$.

$MGRP_{\mathrm{i}}$ is the repetition periodicity of the measurement gap applicable for measurement in the PRS frequency layer i.

$T_{PRS,\mathrm{i}}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,\mathrm{i}}$, where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where

$T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- Note: For the purpose of calculating $T_{PRS,\mathrm{i}}$, only the PRS resources fully or partially covered by the MG are considered.

$\{N, T\}$ is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49].

$N'$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [49].

### 4.15.4.2 Measurement Period Requirements without Measurement Gaps

See TS 38.133 [50] clause 9.9.2.7.

### 4.15.4.3 Measurement Period Requirements in RRC_INACTIVE state

See TS 38.133 [50] clause 5.6.2.5

### 4.15.4.4 Measurement Period Requirements in RRC_INACTIVE state with Bandwidth Aggregation

When UE physical layer receives last of *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from LMF via LPP [49] with a request to perform measurement by aggregating PRS resources from multiple PFLs via *nr-DL-PRS-JointMeasurementRequestedPFL-List*, the UE shall be able to perform RSTD measurement by aggregating PRS resources from multiple PFLs (up to the UE capability specified in TS 38.133 [50] clause 9.9.2.3) defined in TS 38.215 [57], during the measurement period $T_{RSTD_aggregate,\ Total}$ defined as:

$$T_{RSTD_aggregate,\ Total} = T_{non_aggregate_RSTD} + T_{aggregate_RSTD} + T_{margin},$$

where,

$T_{non_aggregate_RSTD}$ is the total measurement period for RSTD measurement on PFLs that do not contain PRS resources for aggregation.

Calculation of $T_{non_aggregate_RSTD}$ is based on TS 38.133 [50]clause 9.9.2.5 with the following exceptions,

- only PFLs containing resource set(s) not linked to any other resource set(s) are considered in *L*
- on each PFL $i$, only resource set(s) not linked to any other resource set(s) are considered in $T_{available_PRS,\mathrm{i}}$, $N_{PRS,\mathrm{i}}^{slot}$ and $L_{available_PRS,\mathrm{i}}$
- $N_{sample}$ = 2 if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* specified in TS 37.355 [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples,

- $(L-1)*\max(T_{effect,i})$ is replace by $(L-1)*T'_{margin}$.

$T_{non_aggregate_RSTD}$ is equal to zero if UE is not configured to perform non-aggregated measurements by the LMF.

If UE is configured to perform positioning measurements on PFLs that contain PRS resources for aggregation and on PFLs that do not contain PRS resources for aggregation, $T_{margin}= T'_{margin}$ is delay margin to account for delay between RSTD measurement performed by UE on PFLs that contain PRS resources for aggregation and RSTD measurement performed by UE on PFLs that do not contain PRS resources for aggregation.

If UE is only configured to perform measurements on PFLs that contain PRS resources for aggregation or if UE is only configured to perform measurements on PFLs that do not contain PRS resources for aggregation, then $T_{margin} = 0$.

$T'_{margin} = \max(Teffect, i)$, when both aggregated PFLs and non-aggregated PFLs are configured for positioning measurement.

If UE is capable of performing latency reduced positioning measurements and is configured to perform latency reduced positioning measurement by LMF via *reducedDL-PRS-ProcessingSamples-r17* [49], then $N_{sample} = 2$ shall be considered in calculations of $T_{aggregate_RSTD}$ and $T_{non\text{-}aggregate_RSTD}$.

If UE is not capable to perform latency reduced positioning measurements or is not configured to perform latency reduced positioning measurement by LMF via *reducedDL-PRS-ProcessingSamples-r17* [49], then $N_{sample} = 4$ shall be considered in calculations of $T_{aggregate_RSTD}$ and $T_{non\text{-}aggregate_RSTD}$.

$T_{aggregate_RSTD}$, total measurement period for RSTD measurements performed by UE by aggregating PRS resources from multiple PFLs as defined in TS 38.214 [56], is defined as:

$$T_{aggregate_RSTD} = \sum_{i=1}^{G} T_{RSTD_aggregate,i} + (G-1)*T'_{margin},$$

where

i is the index of effective PFL, corresponding to the group of PFLs containing linked PRS resource sets to be aggregated for RSTD measurement,

G denotes the number of effective PFLs that UE is configured to perform aggregated measurements on. G is configured by LMF and is provisioned to UE via *nr-DL-PRS-AggregationInfo*.

$T_{RSTD_aggregate,i}$ is the measurement period for PRS RSTD measurement in the effective PFL *i* as specified below:

$$T_{RSTD_aggregate,i} = \left(k_{multiTEG_agg,i} \times CSSF_{PRS_agg,i} \times ceil(K_{p,PRS_agg,i}) \times N_{RxBeam,i} \times \left\lceil \frac{N^{slot}_{PRS_agg,i}}{N'_{agg}} \right\rceil \left\lceil \frac{L_{available_PRS_agg,i}}{N_{agg}} \right\rceil \times N_{sample_agg} - 1\right) \times T_{effect_agg,i} + T_{last_agg,i},$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor.

In FR1, $N_{RxBeam,i} = 1$.

In FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing effective PFL i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$CSSF_{PRS_agg,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in effective PFL i as defined in TS 38.133 [50] clause 9.1.5.2.

$k_{multiTEG_agg,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG_agg,i} = 1$ if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*;

Otherwise,

$k_{multiTEG_agg,i} = N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG_{agg},i}=\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure aggregated PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS_agg,i}$ is a scaling factor for effective PFL i to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS_agg,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS_agg,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS_agg,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the effective PFL, and starting at the beginning of any associated gap occasions covering the PRS occasion:

- $N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and
- $N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule
- Requirements do not apply if $N_{available} = 0$.

$N^{slot}_{PRS_agg,i}$ is the maximum number of DL PRS resources in effective PFL i configured in a slot.

$L_{available_PRS_agg,i}$ is the time duration of available PRS in the effective PFL i to be measured during $T_{available_PRS_agg,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS_agg,i}$, only the PRS resources unmuted and fully or partially overlapped with MG and satisfying the conditions for PRS BW aggregation are considered.

$N_{sample_agg}$ is the number of PRS RSTD measurement samples.

$T_{last_agg,i}$ is the measurement duration for the last PRS RSTD sample in the effective PFL i, including the sampling time and processing time. If all of the PRS resources to be measured are available in the same MG occasion during $T_{availabe_agg}$, $T_{last_agg,i} = T_{agg,i}$ + MGL. Otherwise, $T_{last_agg,i} = T_{agg,i} + T_{available_PRS_agg,i}$,

$T_{effect_agg,i}$ is the periodicity of the PRS RSTD measurement in the effective PFL i defined as:

$$T_{effect_agg,i} = \left\lceil \frac{T_{agg,i}}{T_{available_PRS_agg,i}} \right\rceil \times T_{available_PRS_agg,i}$$

where,

$T_{available_PRS_agg,i} = LCM(T_{PRS_agg,i}, MGRP_i)$, the least common multiple between $T_{PRS_agg,i}$ and $MGRP_i$. $MGRP_i$ is the repetition periodicity of the measurement gap applicable for measurement in the effective PFL i.

$T_{PRS_agg,i}$ is the periodicity of DL PRS resources meeting the bandwidth aggregation conditions with muting on effective PFL i. If more than one PRS periodicities are configured in effective PFL i, the least common multiple of PRS periodicities $T^{PRS\ with\ muting}_{per_agg}$ among all DL PRS resource sets in the effective PFL is used to derive $T_{PRS_agg,i}$, where,

$T_{per_agg}^{PRS\ with\ muting} = N_{muting} \times T_{per_agg}^{PRS}$, is the PRS periodicity with muting per PRS resource configured for aggregation,

$T_{per_agg}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS_agg} \times L_{muting}$, where $T_{muting}^{PRS_agg}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- Note: For the purpose of calculating $T_{PRS_agg,i}$, only the PRS resources fully or partially covered by the MG and PRS resources that satisfy the conditions for PRS BW aggregation are considered.

$\{N_{agg}, T_{agg,i}\}$ is UE capability combination per band to aggregate PRS resources from multiple PFLs within an effective PFL where $N_{agg}$ is a duration of DL PRS symbols in ms corresponding to *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsN-r18* in TS 37.355 [49] processed every $T_{agg,i}$ ms corresponding to *prs-durationOfTwoPRS-BWA-ProcessingSymbolsT-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsT* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *maximumOfTwoAggregatedDL-PRS-Bandwidth* or *maximumOfThreeAggregatedDL-PRS-Bandwidth* in TS 37.355 [49]

$N'_{agg}$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR1-r18* for FR1 and *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR2-r18* for FR2 specified in TS 37.355 [49].

The requirements in this clause for aggregated measurements apply provided that the linked PRS resource sets on multiple PFLs for aggregated measurements satisfy all the conditions specified in clause 5.1.6.5.3 in TS 38.214 [56].

Except for deferred MT-LR as defined in clause 4.1a.5 TS 23.273, the time $T_{RSTD_aggregate,Total}$ starts from the first MG instance aligned with DL PRS resources in the assistance data for aggregation after both the *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the physical layer of UE via LPP [49] including a request to perform measurement by aggregating PRS resources from multiple PFLs via *nr-DL-PRS-JointMeasurementRequestedPFL-List*.

For deferred MT-LR with other event than “Periodic Location” as defined in clause 4.1a.5.1 TS 23.273, the time $T_{RSTD_aggregate,Total}$ starts from the first MG instance aligned with DL PRS resources in the assistance data after the associated event(s) occurs.

For deferred MT-LR with event “Periodic Location” as defined in clause 4.1a.5.1 TS 23.273, the UE shall perform the aggregated RSTD measurement in each reporting period and activate the location report at the time when the periodic timer expires.

When PRS-RSRP is also reported by UE together with RSTD measurement based on aggregated DL-PRS resources from multiple-PFLs, PRS-RSRP are performed over the measurement period defined in TS 38.133 [50] clause 9.9.2.10.

When PRS-RSRPP is also reported by UE together with RSTD measurement based on aggregated DL-PRS resources from multiple-PFLs, PRS-RSRPP are performed over the measurement period defined in TS 38.133 [50] clause 9.9.2.10.

If during the measurement period of one or more effective PFLs, the MG pattern is reconfigured, the measurement period can be longer.

The measurement requirements in this clause apply, provided no PRS symbols are dropped during the measurement period $T_{RSTD_aggregate,Total}$ within measurement gaps due to collisions with other signals; otherwise, the measurement period can be longer.

If CSSF changes during the measurement period, the measurement period could be longer.

The measurement requirements do not apply for aggregated PRS resources, if the PRS resources to be aggregated are across two sampling duration of $N_{agg}$ within duration $L_{available_PRS_agg,i}$.

The measurement requirements do not apply, if time span of the instance of the PRS resources to be aggregated (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability $N_{agg}$.

The requirements in TS 38.133 [50] clause 9.9.2.10 do not apply if the PRS configuration given by higher layer parameters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-TDOA-ProvideCapabilities*, and it is up to UE implementation which aggregated PRS resources are measured, subject to UE measurement capabilities.

If handover occurs while aggregated RSTD measurements are being performed, then the UE shall continue and complete the on-going aggregated RSTD measurements. The aggregated RSTD measurement period can be longer. The UE shall meet the RSTD measurement accuracy requirements in TS 38.133 [50] clause 10.1.23A.2.

## 4.15.5 Measurement Accuracy Requirements

The accuracy requirements for RSTD measurement shall be within $\pm(X+Y+Z)\ T_c$.

X is defined in Table 4.15.5-1 for AWGN channel and Table 4.15.5-3 for fading channel for FR1, provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

X is defined in Table 4.15.5-2 for AWGN channel and Table 4.15.5-4 for fading channel for FR2, provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

X is defined in Table 4.15.5-7 for AWGN channel in FR1 provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

X is defined in Table 4.15.5-8 for AWGN channel in FR2 provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

Note: The requirements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

When UE measures RSTD on PRS resources belonging to different PFLs, then the RSTD accuracy is defined as the accuracy corresponding to the largest accuracy value among different PFLs.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 160 ms.

- When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in Table 4.15.5-5 for FR1 and Table 4.15.5-6 for FR2, respectively.
- Δ is zero for single PFL, and is defined in Table 4.15.5-5a for FR1 and Table 4.15.5-6a for FR2, respectively, for dual PFL.

If the measurements of reference cell and neighbour cell belong to the same Rx TEG, i.e. associated and reported with a common Rx TEG ID, then the sum of Y+Z+Δ is equal to the timing error margin of the Rx TEG reported in *nr-UE-RxTEG-TimingErrorMargin*. The timing error margin reported via *nr-UE-RxTEG-TimingErrorMargin* cannot be larger than the value of (Y+Z+Δ) defined when the UE does not associate the measurements with the same Rx TEG.

**Table 4.15.5-1: RSTD absolute accuracy in FR1 for AWGN channel**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 132 + Δ Note 7 | (PRS Ês/Iot)$_{ref}$ ≥ -6dB<br>(PRS Ês/Iot)$_i$ ≥ -13dB | 15 | ≥24 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 98 + Δ | | | ≥52 | ≥1 | Note 6 | Note 6 | Note 6 |
| 42 + Δ | | | ≥104 | ≥1 | Note 6 | Note 6 | Note 6 |
| 75 + Δ | | 30 | ≥24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -118 | -50 |
| | | | | | NR_FDD_FR1_B | -117.5 | -50 |
| | | | | | NR_TDD_FR1_C | -117 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -116.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -116 | -50 |
| | | | | | NR_FDD_FR1_F | -115.5 | -50 |
| | | | | | NR_FDD_FR1_G | -115 | -50 |
| | | | | | NR_FDD_FR1_H | -114.5 | -50 |
| 48 + Δ | | | ≥ 48 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 50 + Δ | | 60 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -115 | -50 |
| | | | | | NR_FDD_FR1_B | -114.5 | -50 |
| | | | | | NR_TDD_FR1_C | -114 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -113.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -113 | -50 |
| | | | | | NR_FDD_FR1_F | -113.5 | -50 |
| | | | | | NR_FDD_FR1_G | -113 | -50 |
| | | | | | NR_FDD_FR1_H | -111.5 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 24 + Δ | | | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 10 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3 Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in TS 38.133[50] clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Δ= 0 for single PFL, Δ= TBD for dual PFL.

**Table 4.15.5-2: RSTD absolute accuracy in FR2 for AWGN channel**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{\text{rep}}^{\text{PRS}} * L_{\text{PRS}}/K_{\text{comb}}^{\text{PRS}}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | PRB | | dBm/SCS | $dBm/BW_{Channel}$ |
| 35 + Δ Note 6 | (PRS Ês/Iot)$_{ref}$ ≥ -6dB<br>(PRS Ês/Iot)$_i$ ≥ -13dB | 60 | ≥ 24 | ≥ 4 | Same value as PRS_RP in TS 38.133[50] Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 24 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 11 + Δ | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 24 + Δ | | 120 | ≥ 32 | ≥ 4 | Same value as PRS_RP in TS 38.133[50] Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 13 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 6 + Δ | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: Δ= 0 for single PFL, Δ= TBD for dual PFL.

**Table 4.15.5-3: RSTD absolute accuracy in FR1 for fading channel**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{\text{rep}}^{\text{PRS}} * L_{\text{PRS}}/K_{\text{comb}}^{\text{PRS}}$) Note 2 | Io Note 3 range | | |
| | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | $dBm/BW_{Channel}$ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 247 + Δ Note 7 | (PRS Ês/Iot)$_{ref}$ ≥ -6dB<br><br>(PRS Ês/Iot)$_i$ ≥ -13dB | 15 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 140 + Δ | | | ≥ 52 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 86 + Δ | | | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 118 + Δ | | 30 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -118 | -50 |
| | | | | | NR_FDD_FR1_B | -117.5 | -50 |
| | | | | | NR_TDD_FR1_C | -117 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -116.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -116 | -50 |
| | | | | | NR_FDD_FR1_F | -115.5 | -50 |
| | | | | | NR_FDD_FR1_G | -115 | -50 |
| | | | | | NR_FDD_FR1_H | -114.5 | -50 |
| 109 + Δ | | | ≥ 48 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 28 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 147 + Δ | | 60 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -115 | -50 |
| | | | | | NR_FDD_FR1_B | -114.5 | -50 |
| | | | | | NR_TDD_FR1_C | -114 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -113.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -113 | -50 |
| | | | | | NR_FDD_FR1_F | -113.5 | -50 |
| | | | | | NR_FDD_FR1_G | -113 | -50 |
| | | | | | NR_FDD_FR1_H | -111.5 | -50 |
| 27 + Δ | | | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 21 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3 Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in TS 38.133[50] clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Δ= 0 for single PFL, Δ= TBD for dual PFL.

**Table 4.15.5-4: RSTD absolute accuracy in FR2 for fading channel**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |

| | | | | | | |
|---|---|---|---|---|---|---|
| 83 + Δ Note 6 | (PRS Ês/Iot)$_{ref}$ ≥ -6dB | 60 | ≥ 24 | ≥ 4 | Same value as PRS_RP in TS 38.133[50] Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 64 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 46 + Δ | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 48 + Δ | (PRS Ês/Iot)$_i$ ≥ -13dB | 120 | ≥ 32 | ≥ 4 | Same value as PRS_RP in TS 38.133[50] Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 54 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 36 + Δ | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$ , $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: Δ= 0 for single PFL, Δ= TBD for dual PFL.

**Table 4.15.5-5: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 24 | N/A | N/A | 120 |
| ≥ 52 | ≥ 24 | N/A | 72 |
| ≥ 104 | ≥ 48 | ≥ 24 | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

**Table 4.15.5-5a: Margin Δ for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 24 | N/A | N/A | 128 |
| ≥ 52 | ≥ 24 | N/A | 64 |
| ≥ 104 | ≥ 48 | ≥ 24 | 32 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 8 |

**Table 4.15.6-6: Margin for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 24 | N/A | 72 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 16 |
| N/A | ≥ 128 | 12 |

**Table 4.15.5-6a: Margin Δ for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 24 | N/A | 32 |
| ≥ 64 | ≥ 32 | 16 |

| | | |
|---|---|---|
| ≥ 132 | ≥ 64 | 8 |
| N/A | ≥ 128 | 4 |

**Table 4.15.5-7: RSTD absolute accuracy in FR1 for AWGN channel with reduced number of samples**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 98 | (PRS Ês/Iot)$_{ref}$ ≥ -3dB<br><br>(PRS Ês/Iot)$_i$ ≥ -6dB | 15 | ≥ 52 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| 42 | | | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 48 | | 30 | ≥ 48 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| 24 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 | | 60 | ≥ 64 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 10 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in TS 38.133 clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].

NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Void

**Table 4.15.5-8: RSTD absolute accuracy in FR2 for AWGN channel with reduced number of samples**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |
| 24 | (PRS Ês/Iot)$_{ref}$ ≥ -3dB | 60 | ≥ 64 | ≥ 1 | Same value as PRS_RP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | Note 5 |
| 11 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 13 | (PRS Ês/Iot)$_i$ ≥ -6dB | 120 | ≥ 64 | ≥ 1 | Same value as PRS_RP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | Note 5 |
| 6 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: Void

## 4.15.5A Measurement Accuracy Requirements with Bandwidth Aggregation

When UE measures RSTD on PRS resources belonging to different PFLs or different PFL combinations, then the RSTD accuracy is defined as the accuracy corresponding to the largest accuracy value among different PFLs or different PFL combinations.

The requirements in this clause for 3-PFL and 2-PFL apply provided that:

- PRS resources linked for aggregation saftisfy all the conditions specified in TS 38.214 [56] clause 5.1.6.5.3.
- The spacing between the center frequencies of adjacent PFLs containing PRS resources linked for aggregation does not exceed the nominal channel spacing for intra-band contiguous CA defined in TS 38.101-1 [54], clause 5.4A.1 for FR1, and in TS 38.101-2, [55] clause 5.4A.1 for FR2-1.

The accuracy requirements for RSTD measurement based on PRS Aggregation shall be within ±(X+Y+Z+Δ) $T_c$.

X is defined in table 4.15.5A-1 for AWGN channel and table 4.15.5A-3 for fading channel for FR1, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 [50] annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

X is defined in table 4.15.5A-2 for AWGN channel and table 4.15.5A-4 for fading channel for FR2, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 [50] annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

X is defined in table 4.15.5A-5 for AWGN channel in FR1, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 [50] annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

X is defined in table 4.15.5A-6 for AWGN channel in FR2, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [55] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 [50] annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

NOTE: The requriements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL or same PFL combination, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 160 ms.
- When UE measures RSTD on PRS resources belonging different PFLs or different PFL combinations, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in table 4.15.5A-7 for FR1 and table 4.15.5A-9 for FR2, respectively, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.
- Δ is zero when UE measures RSTD on PRS resources belonging to same PFL or same PFL combination and is defined in table 4.15.5A-8 for FR1 and table 4.15.5A-10 for FR2, respectively, when UE measures RSTD on PRS resources belonging different PFLs or different PFL combinations, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.

If the measurements of reference cell and neighbour cell belong to the same Rx TEG, i.e. associated and reported with a common Rx TEG ID, then the sum of Y+Z+Δ is equal to the timing error margin of the Rx TEG reported in *nr-UE-RxTEG-TimingErrorMargin*. The timing error margin reported via *nr-UE-RxTEG-TimingErrorMargin* cannot be larger than the value of (Y+Z+Δ) defined when the UE does not associate the measurements with the same Rx TEG.

**Table 4.15.5A-1: RSTD absolute accuracy in FR1 for AWGN channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |

| | | | | | Note 2 | | | |
|---|---|---|---|---|---|---|---|---|
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 20 | 16 | (PRS Ês/Iot)$_{ref}$ ≥-6dB<br><br>(PRS Ês/Iot)$_i$ ≥-13dB | 15 | ≥ 104 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -126.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -126 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -124 | -50 |
| | | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -120.5 | -50 |
| 11 | 6 | | 30 | ≥ 132 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -123.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -123 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -121 | -50 |
| | | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -117.5 | -50 |
| 6 | 4 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 11 | 6 | | 60 | ≥ 64 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -120.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -120 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -118 | -50 |
| | | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -114.5 | -50 |
| 5 | 4 | | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.

NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.

NOTE 3: Io is assumed to have constant EPRE across the bandwidth.

NOTE 4: NR operating band groups in FR1 are as defined in TS 38.133 [50] clause 3.5.2.

NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].

NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.

**Table 4.15.5A-2: RSTD absolute accuracy in FR2 for AWGN channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | $dBm/BW_{Channel}$ |
| 11 | 6 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 60 | ≥ 64 | ≥ 1 | Same value as PRS_RP in TS 38.133 [50] table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 5 | 4 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 6 | 4 | (PRS Ês/Iot)$_i$ ≥-13dB | 120 | ≥ 64 | ≥ 1 | Same value as PRS_RP in TS 38.133 [50] table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 3 | 2 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.

**Table 4.15.5A-3: RSTD absolute accuracy in FR1 for fading channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | $dBm/BW_{Channel}$ |
| 38 | 30 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 15 | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 29 | 29 | | 30 | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 27 | 26 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 29 | 35 | | 60 | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 27 | 31 | (PRS Ês/Iot)$_i$ ≥-13dB | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in TS 38.133 [50] clause 3.5.2.

NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 4.15.5A-1.

**Table 4.15.5A-4: RSTD absolute accuracy in FR2 for fading channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |
| 71 | 51 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 60 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 55 | 38 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 72 | 49 | | 120 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 56 | 53 | (PRS Ês/Iot)$_i$ ≥-13dB | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 4.15.5A-2.

**Table 4.15.5A-5: RSTD absolute accuracy in FR1 for AWGN channel with reduced number of samples**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 24 | 20 | (PRS Ês/Iot)$_{ref}$ ≥-3dB | 15 | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 13 | 7 | | 30 | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 6 | 5 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 13 | 7 | | 60 | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 6 | 5 | (PRS Ês/Iot)$_i$ ≥-6dB | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in TS 38.133 [50] clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 4.15.5A-1.

**Table 4.15.5A-6: RSTD absolute accuracy in FR2 for AWGN channel with reduced number of samples**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/$BW_{Channel}$ |
| 13 | 7 | (PRS Ês/Iot)$_{ref}$ ≥ -3dB<br>(PRS Ês/Iot)$_i$ ≥ -6dB | 60 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 6 | 5 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 6 | 4 | | 120 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 3 | 2 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 4.15.5A-2.

**Table 4.15.5A-7: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 104 | N/A | N/A | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

**Table 4.15.5A-8: Margin Δ for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 104 | N/A | N/A | 32 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 8 |

**Table 4.15.5A-9: Margin for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 64 | N/A | 32 |
| ≥ 132 | ≥ 64 | 16 |
| N/A | ≥ 128 | 12 |

**Table 4.15.5A-10: Margin Δ for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 64 | N/A | 16 |
| ≥ 132 | ≥ 64 | 8 |
| N/A | ≥ 128 | 4 |

## 4.15.6 Reporting mapping

### 4.15.6.1 Absolute DL RSTD Measurement Reporting

The reporting range for the DL RSTD measurement is defined from $-985024\times T_c$ to $985024\times T_c$ with the resolution step of $2^k\times T_c$, where

$T_c$ is defined in TS 38.211 [53],

$k_{min}\leq k\leq k_{max}$,

$k_{min}$=2 and $k_{max}$=5, when configured PRS resource of at least one of the reference cell and neighbour cell measured for the RSTD measurement is in FR1,

$k_{min}$=0 and $k_{max}$=5, when configured PRS resource of both the reference cell and neighbour cell measured for the RSTD measurement are in FR2,

$k\geq$ *timingReportingGranularityFactor* [49] configured by LMF via LPP for the RSTD measurement.

The measurement report mapping for different *k* values are specified in Tables 4.15.6.1-1 – 4.15.6.1-6.

**Table 4.15.6.1-1: Report mapping for *k*=0**

| Reported Quantity Value, RSTD_i | Measured Quantity Value, RSTD | Unit |
|---|---|---|
| RSTD_0000000 | RSTD < -985024 | $T_c$ |
| RSTD_0000001 | -985024 ≤ RSTD < -985023 | $T_c$ |
| RSTD_0000002 | -985023 ≤ RSTD < -985022 | $T_c$ |
| … | … | … |
| RSTD_0985024 | -1 ≤ RSTD < 0 | $T_c$ |
| RSTD_0985025 | 0 ≤ RSTD < 1 | $T_c$ |
| … | … | … |
| RSTD_1970047 | 985022 ≤ RSTD < 985023 | $T_c$ |
| RSTD_1970048 | 985023 ≤ RSTD < 985024 | $T_c$ |
| RSTD_1970049 | 985024 ≤ RSTD | $T_c$ |

**Table 4.15.6.1-2: Report mapping for *k*=1**

| Reported Quantity Value, RSTD_i | Measured Quantity Value, RSTD | Unit |
|---|---|---|
| RSTD_000000 | RSTD < -985024 | $T_c$ |
| RSTD_000001 | -985024 ≤ RSTD < -985022 | $T_c$ |
| RSTD_000002 | -985022 ≤ RSTD < -985020 | $T_c$ |
| … | … | … |
| RSTD_492512 | -2 ≤ RSTD < 0 | $T_c$ |
| RSTD_492513 | 0 ≤ RSTD < 2 | $T_c$ |
| … | … | … |
| RSTD_985023 | 985020 ≤ RSTD < 985022 | $T_c$ |
| RSTD_985024 | 985022 ≤ RSTD < 985024 | $T_c$ |
| RSTD_985025 | 985024 ≤ RSTD | $T_c$ |

**Table 4.15.6.1-3: Report mapping for *k*=2**

| Reported Quantity Value, RSTD_i | Measured Quantity Value, RSTD | Unit |
|---|---|---|
| RSTD_000000 | RSTD < -985024 | $T_c$ |
| RSTD_000001 | -985024 ≤ RSTD < -985020 | $T_c$ |
| RSTD_000002 | -985020 ≤ RSTD < -985016 | $T_c$ |
| … | … | … |
| RSTD_246256 | -4 ≤ RSTD < 0 | $T_c$ |

| | | |
|---|---|---|
| RSTD_246257 | 0 ≤ RSTD < 4 | $T_c$ |
| … | … | … |
| RSTD_492511 | 985016 ≤ RSTD < 985020 | $T_c$ |
| RSTD_492512 | 985020 ≤ RSTD < 985024 | $T_c$ |
| RSTD_492513 | 985024 ≤ RSTD | $T_c$ |

**Table 4.15.6.1-4: Report mapping for *k*=3**

| Reported Quantity Value RSTD_i | Measured Quantity Value, RSTD | Unit |
|---|---|---|
| RSTD_000000 | RSTD < -985024 | $T_c$ |
| RSTD_000001 | -985024 ≤ RSTD < -985016 | $T_c$ |
| RSTD_000002 | -985016 ≤ RSTD < -985008 | $T_c$ |
| … | … | … |
| RSTD_123128 | -8 ≤ RSTD < 0 | $T_c$ |
| RSTD_123129 | 0 ≤ RSTD < 8 | $T_c$ |
| … | … | … |
| RSTD_246255 | 985008 ≤ RSTD < 985016 | $T_c$ |
| RSTD_246256 | 985016 ≤ RSTD < 985024 | $T_c$ |
| RSTD_246257 | 985024 ≤ RSTD | $T_c$ |

**Table 4.15.6.1-5: Report mapping for *k*=4**

| Reported Quantity Value, RSTD_i | Measured Quantity Value, RSTD | Unit |
|---|---|---|
| RSTD_000000 | RSTD < -985024 | $T_c$ |
| RSTD_000001 | -985024 ≤ RSTD < -985008 | $T_c$ |
| RSTD_000002 | -985008 ≤ RSTD < -984992 | $T_c$ |
| … | … | … |
| RSTD_061564 | -16 ≤ RSTD < 0 | $T_c$ |
| RSTD_061565 | 0 ≤ RSTD < 16 | $T_c$ |
| … | … | … |
| RSTD_123127 | 984992 ≤ RSTD < 985008 | $T_c$ |
| RSTD_123128 | 985008 ≤ RSTD < 985024 | $T_c$ |
| RSTD_123129 | 985024 ≤ RSTD | $T_c$ |

**Table 4.15.6.1-6: Report mapping for *k*=5**

| Reported Quantity Value, RSTD_i | Measured Quantity Value, RSTD | Unit |
|---|---|---|
| RSTD_00000 | RSTD < -985024 | $T_c$ |
| RSTD_00001 | -985024 ≤ RSTD < -984992 | $T_c$ |
| RSTD_00002 | -984992 ≤ RSTD < -984960 | $T_c$ |
| … | … | … |
| RSTD_30782 | -32 ≤ RSTD < 0 | $T_c$ |
| RSTD_30783 | 0 ≤ RSTD < 32 | $T_c$ |
| … | … | … |
| RSTD_61563 | 984960 ≤ RSTD < 984992 | $T_c$ |
| RSTD_61564 | 984992 ≤ RSTD < 985024 | $T_c$ |
| RSTD_61565 | 985024 ≤ RSTD | $T_c$ |

### 4.15.6.2 Differential Reporting for DL RSTD Measurement

A first DL RSTD measurement is reported by means of differential reporting, i.e. as ΔRSTD, relative to a second DL RSTD measurement (RSTD2), provided that:

- the absolute measured quantity value of the second DL RSTD measurement (RSTD2) is not larger than the absolute measured quantity value of the first DL RSTD measurement (RSTD1), i.e., ΔRSTD=RSTD1-RSTD2≥0, and

- the absolute value of the second DL RSTD measurement (RSTD2) is reported together with ΔRSTD for the first DL RSTD measurement.

The reporting range for differential reporting ΔRSTD of the first DL RSTD measurement is defined from 0 up to $8191 \times T_c$ with the resolution step of $2^k \times T_c$, where

$T_c$ is defined in TS 38.211 [53],

$k_{min} \leq k \leq k_{max}$,

$k_{min}$=2 and $k_{max}$=5, when configured PRS resource of at least one of the reference cell and neighbour cell measured for the first RSTD measurement or second RSTD measurement is in FR1,

$k_{min}$=0 and $k_{max}$=5, when configured PRS resource of both the reference cell and neighbour cell measured for both of the first RSTD measurement and the second RSTD measurement are in FR2,

$k \geq$ *timingReportingGranularityFactor* [49] configured by LMF via LPP for the RSTD measurement.

The measurement report mapping for different *k* values are specified in Tables 4.15.6.2-1 – 4.15.6.2-6.

**Table 4.15.6.2-1: Report mapping for *k*=0**

| Reported Quantity Value, DIFFRSTD_i | ΔRSTD = RSTD1 – RSTD2 | Unit |
|---|---|---|
| DIFFRSTD_0000 | 0 ≤ ΔRSTD < 1 | $T_c$ |
| DIFFRSTD_0001 | 1 ≤ ΔRSTD < 2 | $T_c$ |
| DIFFRSTD_0002 | 2 ≤ ΔRSTD < 3 | $T_c$ |
| … | … | … |
| DIFFRSTD_8189 | 8189 ≤ ΔRSTD < 8190 | $T_c$ |
| DIFFRSTD_8190 | 8190 ≤ ΔRSTD < 8191 | $T_c$ |
| DIFFRSTD_8191 | 8191 ≤ ΔRSTD | $T_c$ |

**Table 4.15.6.2-2: Report mapping for *k*=1**

| Reported Quantity Value, DIFFRSTD_i | ΔRSTD = RSTD1 – RSTD2 | Unit |
|---|---|---|
| DIFFRSTD_0000 | 0 ≤ ΔRSTD < 2 | $T_c$ |
| DIFFRSTD_0001 | 2 ≤ ΔRSTD < 4 | $T_c$ |
| DIFFRSTD_0002 | 4 ≤ ΔRSTD < 6 | $T_c$ |
| … | … | … |
| DIFFRSTD_4093 | 8186 ≤ ΔRSTD < 8188 | $T_c$ |
| DIFFRSTD_4094 | 8188 ≤ ΔRSTD < 8190 | $T_c$ |
| DIFFRSTD_4095 | 8190 ≤ ΔRSTD | $T_c$ |

**Table 4.15.6.2-3: Report mapping for *k*=2**

| Reported Quantity Value, DIFFRSTD_i | ΔRSTD = RSTD1 – RSTD2 | Unit |
|---|---|---|
| DIFFRSTD_0000 | 0 ≤ ΔRSTD < 4 | $T_c$ |
| DIFFRSTD_0001 | 4 ≤ ΔRSTD < 8 | $T_c$ |
| DIFFRSTD_0002 | 8 ≤ ΔRSTD < 12 | $T_c$ |
| … | … | … |
| DIFFRSTD_2045 | 8180 ≤ ΔRSTD < 8184 | $T_c$ |
| DIFFRSTD_2046 | 8184 ≤ ΔRSTD < 8188 | $T_c$ |
| DIFFRSTD_2047 | 8188 ≤ ΔRSTD | $T_c$ |

**Table 4.15.6.2-4: Report mapping for *k*=3**

| Reported Quantity Value, DIFFRSTD_i | ΔRSTD = RSTD1 – RSTD2 | Unit |
|---|---|---|
| DIFFRSTD_0000 | 0 ≤ ΔRSTD < 8 | $T_c$ |
| DIFFRSTD_0001 | 8 ≤ ΔRSTD < 16 | $T_c$ |
| DIFFRSTD_0002 | 16 ≤ ΔRSTD < 24 | $T_c$ |
| … | … | … |
| DIFFRSTD_1021 | 8168 ≤ ΔRSTD < 8176 | $T_c$ |
| DIFFRSTD_1022 | 8176 ≤ ΔRSTD < 8184 | $T_c$ |
| DIFFRSTD_1023 | 8184 ≤ ΔRSTD | $T_c$ |

**Table 4.15.6.2-5: Report mapping for *k*=4**

| Reported Quantity Value, DIFFRSTD_i | ΔRSTD = RSTD1 – RSTD2 | Unit |
|---|---|---|
| DIFFRSTD_000 | 0 ≤ ΔRSTD < 16 | $T_c$ |
| DIFFRSTD_001 | 16 ≤ ΔRSTD < 32 | $T_c$ |
| DIFFRSTD_002 | 32 ≤ ΔRSTD < 48 | $T_c$ |
| … | … | … |
| DIFFRSTD_509 | 8144 ≤ ΔRSTD < 8160 | $T_c$ |
| DIFFRSTD_510 | 8160 ≤ ΔRSTD < 8176 | $T_c$ |
| DIFFRSTD_511 | 8176 ≤ ΔRSTD | $T_c$ |

**Table 4.15.6.2-6: Report mapping for *k*=5**

| Reported Quantity Value, DIFFRSTD_i | ΔRSTD = RSTD1 – RSTD2 | Unit |
|---|---|---|
| DIFFRSTD_000 | 0 ≤ ΔRSTD < 32 | $T_c$ |
| DIFFRSTD_001 | 32 ≤ ΔRSTD < 64 | $T_c$ |
| DIFFRSTD_002 | 64 ≤ ΔRSTD < 96 | $T_c$ |
| … | … | … |
| DIFFRSTD_253 | 8096 ≤ ΔRSTD < 8128 | $T_c$ |
| DIFFRSTD_254 | 8128 ≤ ΔRSTD < 8160 | $T_c$ |
| DIFFRSTD_255 | 8160 ≤ ΔRSTD | $T_c$ |

### 4.15.6.3 Additional Path Report Mapping for DL RSTD

The reporting range for the additional path reporting for an RSTD measurement is defined up to the range from -8175×$T_c$ to 8175×$T_c$ with the resolution step of $2^k$×$T_c$, where

$T_c$ is defined in TS 38.211 [53],

$k_{min} \leq k \leq k_{max}$,

$k_{min}$=2 and $k_{max}$=5, when configured PRS resource of at least one of the reference cell and neighbour cell measured for the RSTD measurement is in FR1,

$k_{min}$=0 and $k_{max}$=5, when configured PRS resource of both the reference cell and neighbour cell measured for the RSTD measurement are in FR2,

$k \geq$ *timingReportingGranularityFactor* [49] configured by LMF via LPP for the RSTD measurement.

The UE can report the timing of up to two additional paths with respect to the path timing determining the RSTD measurement.

The report mappings for different *k* values are specified in Tables 10.1.23.3.3-1 – 10.1.23.3.3-6.

**Table 10.1.23.3.3-1: Report mapping for *k*=0**

| Reported Quantity Value, | Measured Quantity Value, | Unit |
|---|---|---|

| path_i | Δpath | |
|---|---|---|
| path_00000 | Δpath < -8175 | $T_c$ |
| path_00001 | -8175 ≤ Δpath < -8174 | $T_c$ |
| path_00002 | -8174 ≤ Δpath < -8173 | $T_c$ |
| … | … | … |
| path_08175 | -1 ≤ Δpath < 0 | $T_c$ |
| path_08176 | 0 ≤ Δpath < 1 | $T_c$ |
| … | … | … |
| path_16349 | 8173 ≤ Δpath < 8174 | $T_c$ |
| path_16350 | 8174 ≤ Δpath < 8175 | $T_c$ |
| path_16351 | 8175 ≤ Δpath | $T_c$ |

**Table 10.1.23.3.3-2: Report mapping for *k*=1**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8175 | $T_c$ |
| path_0001 | -8175 ≤ Δpath < -8173 | $T_c$ |
| path_0002 | -8173 ≤ Δpath < -8171 | $T_c$ |
| … | … | … |
| path_4088 | -1 ≤ Δpath < 1 | $T_c$ |
| … | … | … |
| path_8174 | 8171 ≤ Δpath < 8173 | $T_c$ |
| path_8175 | 8173 ≤ Δpath < 8175 | $T_c$ |
| path_8176 | 8175 ≤ Δpath | $T_c$ |

**Table 10.1.23.3.3-3: Report mapping for *k*=2**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8174 | $T_c$ |
| path_0001 | -8174 ≤ Δpath < -8170 | $T_c$ |
| path_0002 | -8170 ≤ Δpath < -8166 | $T_c$ |
| … | … | … |
| path_2044 | -2 ≤ Δpath < 2 | $T_c$ |
| … | … | … |
| path_4086 | 8166 ≤ Δpath < 8170 | $T_c$ |
| path_4087 | 8170 ≤ Δpath < 8174 | $T_c$ |
| path_4088 | 8174 ≤ Δpath | $T_c$ |

**Table 10.1.23.3.3-4: Report mapping for *k*=3**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8172 | $T_c$ |
| path_0001 | -8172 ≤ Δpath < -8164 | $T_c$ |
| path_0002 | -8164 ≤ Δpath < -8156 | $T_c$ |
| … | … | … |
| path_1022 | -4 ≤ Δpath < 4 | $T_c$ |
| … | … | … |
| path_2042 | 8156 ≤ Δpath < 8164 | $T_c$ |
| path_2043 | 8164 ≤ Δpath < 8172 | $T_c$ |
| path_2044 | 8172 ≤ Δpath | $T_c$ |

**Table 10.1.23.3.3-5: Report mapping for *k*=4**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_0000 | Δpath < -8168 | $T_c$ |

| | | |
|---|---|---|
| path_0001 | -8168 ≤ Δpath < -8152 | $T_c$ |
| path_0002 | -8152 ≤ Δpath < -8136 | $T_c$ |
| … | … | … |
| path_511 | -8 ≤ Δpath < 8 | $T_c$ |
| … | … | … |
| path_1020 | 8136 ≤ Δpath < 8152 | $T_c$ |
| path_1021 | 8152 ≤ Δpath < 8168 | $T_c$ |
| path_1022 | 8168 ≤ Δpath | $T_c$ |

**Table 10.1.23.3.3-6: Report mapping for *k*=5**

| Reported Quantity Value, path_i | Measured Quantity Value, Δpath | Unit |
|---|---|---|
| path_000 | Δpath < -8160 | $T_c$ |
| path_001 | -8160 ≤ Δpath < -8128 | $T_c$ |
| path_002 | -8128 ≤ Δpath < -8096 | $T_c$ |
| … | … | … |
| path_256 | 0 ≤ Δpath < 32 | $T_c$ |
| … | … | … |
| path_509 | 8096 ≤ Δpath < 8128 | $T_c$ |
| path_510 | 8128 ≤ Δpath < 8160 | $T_c$ |
| path_511 | 8160 ≤ Δpath | $T_c$ |

# 4.16 DL-AoD test conditions

## 4.16.1 Simulated cells

For the DL-AoD measurement test cases in clause 16, a multi cell environment as defined in 3GPP TS 38.508-1 [45] with NR Cell 1 and NR Cell 2 are used. The default parameters for simulated cells are the same as specified in 3GPP TS 38.508-1 [45].

## 4.16.2 Propagation conditions

For PRS-RSRP test cases see TS 38.533 [47] clause C 2.

For PRS-RSRPP test cases see TS 38.521-4 [60] clause B.2.4.

## 4.16.3 Measurement Reporting Requirements

### 4.16.3.1 Measurement Reporting Requirements in RRC_CONNECTED state

The measurement reporting delay is defined as the time between the moment when the periodic measurement report is triggered and the moment when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$ where $TTI_{DCCH}$ is the duration of subframe or slot or subslot when the measurement report is transmitted on the PUSCH with subframe or slot or subslot duration. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

### 4.16.3.2 Measurement Reporting Requirements in RRC_INACTIVE state

See TS 38.133 [50] clause 5.6.3.4.

## 4.16.4 PRS-RSRP Measurement Period Requirements

### 4.16.4.1 Measurement Period Requirements within Measurement Gaps

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [49], the UE shall be able to measure multiple (up to the UE capability specified in TS 38.133 Clause 9.9.3.3) PRS-RSRP measurements, defined in TS 38.215 [57], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,\ total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max\left(T_{effect,i}\right)$$

where

*i* is the index of positioning frequency layer,

L is total number of positioning frequency layers,

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left(CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last}$$

where

$CSSF_i$ is the carrier specific scaling factor for PRS-RSRP measurements specified in TS 38.133 clause 9.1.5.2,

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioining frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in TS 38.133 clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49],

N' is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [49],

$N_{sample}$ is the number of PRS-RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time, if not all PRS resources to be measured are available in the same measurement gap occasion during $T_{available_PRS,i}$, otherwise $T_{last,i} = T_i + MGL$,

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,

$T_i$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ is the least common multiple between $T_{PRS,i}$ and $MGRP_i$ ,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$MGRP_i$ is the measurement gap repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

### 4.16.4.2 Measurement Period Requirements without Measurement Gaps

See TS 38.133 [50] clause 9.9.3.6.

### 4.16.4.3 Measurement Period Requirements in RRC_INACTIVE state

See TS 38.133 [50] clause 5.6.3.5

## 4.16.5 PRS-RSRP Measurement Accuracy Requirements

The absolute accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 4.16.5-1 are valid under the following conditions:

Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band

UE does not support positioning measurements with reduced number of samples, or LMF does not indicate UE to perform positioning measurements with reduced number of samples

**Table 4.16.5-1: PRS-RSRP absolute accuracy for FR1**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | | | | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | NR operating band groups Note 8 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 6 | dBm/30kHz Note 6 | dBm/60kHz Note 6 | |
| ±3.5 | ±8 | ≥-3dB | ≥24 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±8.5 | ±13 | ≥-13dB | 24 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±6 | ±10.5 | | 52< BW≤ 104 | All | Note 4 | | | | |
| ±4.5 | ±9 | | BW >104 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in TS 37.355 [49].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 PRB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in TS 38.133[50] Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in TS 38.133[50] Section 3.5.2.

The absolute accuracy requirements for PRS-RSRP measurement for FR2 defined in Table 4.16.5-2 are valid under the following conditions:

Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.

PRP 1,2|$_{dBm}$ according to TS 38.133[50] Annex B.2.14 for a corresponding Band

UE does not support positioning measurements with reduced number of samples, or LMF does not indicate UE to perform positioning measurements with reduced number of samples

**Table 4.16.5-2: PRS-RSRP absolute accuracy for FR2**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Minimum Io Note 1 dBm / $SCS_{PRS}$ | Maximum Io |
| dB | dB | dB | PRB | - | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| | | | | | dBm/120kHz Note 6 / dBm/60kHz Note 6 | |
| ±5 | ±8 | ≥-3dB | ≥24 | All | Same value as PRP in TS 38.133[50]Table B.2.14 -2, according to UE Power class, operating band and angle of arrival | -50 |
| | | | | | Note 4 | |
| | | | | | Note 4 | |
| ±8.5 | ±11.5 | ≥-13dB | 24 ≤ BW ≤ 64 | All | Note 4 | |
| ±6 | ±9 | | BW >64 | All | Note 4 | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in TS 37.355[49].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 PRB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by Δ>0, when applicable, as described in TS 38.133[50] Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in TS 38.133[50] Section 3.5.2.

The absolute accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 4.16.5-2a are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP 1,2|$_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

The absolute accuracy requirements for PRS-RSRP measurement for FR2 defined in Table 4.16.5-2b are valid under the following conditions:

- Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.
- PRP 1,2|$_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

**Table 4.16.5-2a: PRS-RSRP absolute accuracy for FR1 with reduced sample number**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Io Note 6 range | | | | |
| | | | | | NR operating band groups Note 7 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
| dB | dB | dB | PRB | - | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| | | | | | | dBm/15kHz Note 5 | dBm/30kHz Note 5 | dBm/60kHz Note 5 | |
| ±3.5 | ±8 | ≥0 | ≥48 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| ±8.5 | ±13 | ≥-6 | 48 ≤ BW ≤ 52 | All | Note 3 | | | | |
| ±6 | ±10.5 | | 52< BW≤ 104 | All | Note 3 | | | | |
| ±4.5 | ±9 | | BW >104 | All | Note 3 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [49].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 PRB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in TS 38.133 Sections B.3.2 and B.3.3.
NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 7: NR operating band groups are as defined in TS 38.133 Section 3.5.2.

**Table 4.16.5-2b: PRS-RSRP absolute accuracy for FR2 with reduced sample number**

| Accuracy | | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Io Note 6 range | | |
| | | | | | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | Maximum Io |
| dB | dB | dB | PRB | - | dBm / $SCS_{PRS}$ | | dBm/$BW_{Channel}$ |
| | | | | | dBm/120kHz Note 5 | dBm/60kHz Note 5 | |
| ±5 | ±8 | ≥0 | ≥48 | All | Same value as PRP in TS 38.133 Table B.2.14-2, according to UE Power class, operating band and angle of arrival | | -50 |
| ±8.5 | ±11.5 | ≥-6 | 48 ≤ BW ≤ 64 | All | Note 3 | | |
| ±6 | ±9 | | BW >64 | All | Note 3 | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [49].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 PRB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in TS 38.133 Sections B.3.2 and B.3.3.
NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.

NOTE 7: NR operating band groups are as defined in TS 38.133 Section 3.5.2.

The relative PRS-RSRP accuracy is defined as accuracy of the difference between two PRS-RSRP measurements.

The relative PRS-RSRP accuracy requirements apply for the cases when PRS-RSRP is measured from PRS resources in the same PRS resource set in FR1 or FR2, and measured with same Rx beam in case of FR2.

The accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 4.16.5-3 are valid under the following conditions:

Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to TS 38.133[50] Annex B.2.14 for a corresponding Band

UE does not support positioning measurements with reduced number of samples, or LMF does not indicate UE to perform positioning measurements with reduced number of samples

**Table 4.16.5-3: PRS-RSRP relative accuracy for FR1**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | | | | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | NR operating band groups Note 8 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 6 | dBm/30kHz Note 6 | dBm/60kHz Note 6 | |
| ±3.5 | ±5.0 | ≥-3dB | ≥24 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±9.5 | ±11.0 | ≥-13dB | 24 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±6.5 | ±8.0 | | 52< BW≤ 104 | All | Note 4 | | | | |
| ±5.0 | ±6.5 | | BW >104 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [49].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 PRB.

NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in TS 38.133[50] Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in TS 38.133[50] Section 3.5.2.

The accuracy requirements for PRS-RSRP measurement for FR2 defined in Table 4.16.5-3 are valid under the following conditions:

Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to TS 38.133[50] Annex B.2.14 for a corresponding Band

UE does not support positioning measurements with reduced number of samples, or LMF does not indicate UE to perform positioning measurements with reduced number of samples

**Table 4.16.5-4: PRS-RSRP relative accuracy for FR2**

<table>
<tr><th colspan="2">Accuracy</th><th colspan="5">Conditions</th></tr>
<tr><th rowspan="2">Normal condition</th><th rowspan="2">Extreme condition</th><th rowspan="2">PRS Ês/Iot</th><th rowspan="2">PRS BW</th><th rowspan="2">Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$)</th><th colspan="2">Io Note 7 range</th></tr>
<tr><th>Minimum Io Note 1 dBm / SCS<sub>PRS</sub></th><th>Maximum Io</th></tr>
<tr><th>dB</th><th>dB</th><th>dB</th><th>PRB</th><th>-</th><th colspan="2">dBm / SCS<sub>PRS</sub></th><th>dBm/BW<sub>Channel</sub></th></tr>
<tr><th></th><th></th><th></th><th></th><th></th><th>dBm/120kHz Note 6</th><th>dBm/60kHz Note 6</th><th></th></tr>
<tr><td>±5.0</td><td>±8.0</td><td>≥-3dB</td><td>≥24</td><td>All</td><td colspan="2">Same value as PRP in TS 38.133[50] Table B.2.14-2, according to UE Power class, operating band and angle of arrival</td><td>-50</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td colspan="3">Note 4</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td colspan="3">Note 4</td></tr>
<tr><td>±10</td><td>±13</td><td rowspan="2">≥-13dB</td><td>24 ≤ BW ≤ 64</td><td>All</td><td colspan="3">Note 4</td></tr>
<tr><td>±7.5</td><td>±10.5</td><td>BW >64</td><td>All</td><td colspan="3">Note 4</td></tr>
</table>

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [49].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 PRB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in TS 38.133[50] Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in TS 38.133[50] Section 3.5.2.

The absolute accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 4.16.5-5 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

The absolute accuracy requirements for PRS-RSRP measurement for FR2 defined in Table 4.16.5-6 are valid under the following conditions:

- Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to TS 38.133 Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

**Table 4.16.5-5: PRS-RSRP relative accuracy for FR1 with reduced sample number**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) | Io Note 6 range | | | | |
| | | | | | NR operating band groups Note 7 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 5 | dBm/30kHz Note 5 | dBm/60kHz Note 5 | |
| ±3.5 | ±5.0 | ≥0 | ≥48 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| ±9.5 | ±11.0 | ≥-6 | 48 ≤ BW ≤ 52 | All | Note 3 | | | | |
| ±6.5 | ±8.0 | | 52< BW≤ 104 | All | Note 3 | | | | |
| ±5.0 | ±6.5 | | BW >104 | All | Note 3 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [49].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 PRB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 7: NR operating band groups are as defined in Section 3.5.2.

**Table 4.16.5-6: PRS-RSRP relative accuracy for FR2 with reduced sample number**

| Accuracy | | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) | Io Note 6 range | | |
| | | | | | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | Maximum Io |
| | | | | | dBm / $SCS_{PRS}$ | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | dBm/120kHz Note 5 | dBm/60kHz Note 5 | |
| ±5.0 | ±8.0 | ≥0 | ≥48 | All | Same value as PRP in Table B.2.14-2, according to UE | | -50 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | Power class, operating band and angle of arrival | |
| ±10 | ±13 | ≥-6 | 48 ≤ BW ≤ 64 | All | Note 3 | |
| ±7.5 | ±10.5 | | BW >64 | All | Note 3 | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 PRB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

## 4.16.6 PRS-RSRP Reporting mapping

### 4.16.6.1 Absolute PRS-RSRP Measurement Report Mapping

The reporting range of absolute PRS-RSRP measurement is defined from -156 dBm to -31 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 4.16.6.1-1. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 4.16.6.1-1: Measurement report mapping for PRS-RSRP**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| PRS_RSRP_0 | PRS-RSRP<-156 | dBm |
| PRS_RSRP_1 | -156≤PRS-RSRP<-155 | dBm |
| PRS_RSRP_2 | -155≤PRS-RSRP<-154 | dBm |
| PRS_RSRP_3 | -154≤PRS-RSRP<-153 | dBm |
| PRS_RSRP_4 | -153≤PRS-RSRP<-152 | dBm |
| PRS_RSRP_5 | -152≤PRS-RSRP<-151 | dBm |
| PRS_RSRP_6 | -151≤PRS-RSRP<-150 | dBm |
| PRS_RSRP_7 | -150≤PRS-RSRP<-149 | dBm |
| PRS_RSRP_8 | -149≤PRS-RSRP<-148 | dBm |
| PRS_RSRP_9 | -148≤PRS-RSRP<-147 | dBm |
| PRS_RSRP_10 | -147≤PRS-RSRP<-146 | dBm |
| PRS_RSRP_11 | -146≤PRS-RSRP<-145 | dBm |
| PRS_RSRP_12 | -145≤PRS-RSRP<-144 | dBm |
| PRS_RSRP_13 | -144≤PRS-RSRP<-143 | dBm |
| PRS_RSRP_14 | -143≤PRS-RSRP<-142 | dBm |
| PRS_RSRP_15 | -142≤PRS-RSRP<-141 | dBm |
| PRS_RSRP_16 | -141≤PRS-RSRP<-140 | dBm |
| PRS_RSRP_17 | -140≤PRS-RSRP<-139 | dBm |
| PRS_RSRP_18 | -139≤PRS-RSRP<-138 | dBm |
| … | … | … |
| PRS_RSRP_111 | -46≤PRS-RSRP<-45 | dBm |
| PRS_RSRP_112 | -45≤PRS-RSRP<-44 | dBm |
| PRS_RSRP_113 | -44≤PRS-RSRP<-43 | dBm |
| PRS_RSRP_114 | -43≤PRS-RSRP<-42 | dBm |
| PRS_RSRP_115 | -42≤PRS-RSRP<-41 | dBm |
| PRS_RSRP_116 | -41≤PRS-RSRP<-40 | dBm |
| PRS_RSRP_117 | -40≤PRS-RSRP<-39 | dBm |
| PRS_RSRP_118 | -39≤PRS-RSRP<-38 | dBm |
| PRS_RSRP_119 | -38≤PRS-RSRP<-37 | dBm |
| PRS_RSRP_120 | -37≤PRS-RSRP<-36 | dBm |
| PRS_RSRP_121 | -36≤PRS-RSRP<-35 | dBm |
| PRS_RSRP_122 | -35≤PRS-RSRP<-34 | dBm |
| PRS_RSRP_123 | -34≤PRS-RSRP<-33 | dBm |
| PRS_RSRP_124 | -33≤PRS-RSRP<-32 | dBm |
| PRS_RSRP_125 | -32≤PRS-RSRP<-31 | dBm |

| | | |
|---|---|---|
| PRS_RSRP_126 | -31≤PRS-RSRP | dBm |

### 4.16.6.2 Differential Report Mapping for PRS-RSRP Measurement

The reporting range of differential PRS-RSRP is defined from -30 dB to 0 dB with 1 dB resolution when *nr-DL-AoD-RequestLocationInformation* message is received.

The mapping of measured quantity is defined in Table 4.16.6.2-1. The range in the signalling may be larger than the guaranteed accuracy range.

The reporting range of differential PRS-RSRP is defined from -30 dB to 30 dB with 1 dB resolution when *nr-DL-TDOA-RequestLocationInformation* or *nr-Multi-RTT-RequestLocationInformation* is received.

The mapping of measured quantity is defined in Table 4.16.6.2-2. The range in the signalling may be larger than the guaranteed accuracy range or the range supported by the UE receiver for differential RSRP measured on different PRS resources in frequency domain at the same time.

**Table 4.16.6.2-1: Measurement report mapping for differential PRS-RSRP**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| DIFFRSRP_0 | -30≥ΔRSRP | dB |
| DIFFRSRP_1 | -29≥ΔRSRP>-30 | dB |
| DIFFRSRP_2 | -28≥ΔRSRP>-29 | dB |
| DIFFRSRP_3 | -27≥ΔRSRP>-28 | dB |
| DIFFRSRP_4 | -26≥ΔRSRP>-27 | dB |
| DIFFRSRP_5 | -25≥ΔRSRP>-26 | dB |
| DIFFRSRP_6 | -24≥ΔRSRP>-25 | dB |
| DIFFRSRP_7 | -23≥ΔRSRP>-24 | dB |
| DIFFRSRP_8 | -22≥ΔRSRP>-23 | dB |
| DIFFRSRP_9 | -21≥ΔRSRP>-22 | dB |
| DIFFRSRP_10 | -20≥ΔRSRP>-21 | dB |
| DIFFRSRP_11 | -19≥ΔRSRP>-20 | dB |
| DIFFRSRP_12 | -18≥ΔRSRP>-19 | dB |
| DIFFRSRP_13 | -17≥ΔRSRP>-18 | dB |
| DIFFRSRP_14 | -16≥ΔRSRP>-17 | dB |
| DIFFRSRP_15 | -15≥ΔRSRP>-16 | dB |
| DIFFRSRP_16 | -14≥ΔRSRP>-15 | dB |
| DIFFRSRP_17 | -13≥ΔRSRP>-14 | dB |
| DIFFRSRP_18 | -12≥ΔRSRP>-13 | dB |
| DIFFRSRP_19 | -11≥ΔRSRP>-12 | dB |
| DIFFRSRP_20 | -10≥ΔRSRP>-11 | dB |
| DIFFRSRP_21 | -9≥ΔRSRP>-10 | dB |
| DIFFRSRP_22 | -8≥ΔRSRP>-9 | dB |
| DIFFRSRP_23 | -7≥ΔRSRP>-8 | dB |
| DIFFRSRP_24 | -6≥ΔRSRP>-7 | dB |
| DIFFRSRP_25 | -5≥ΔRSRP>-6 | dB |
| DIFFRSRP_26 | -4≥ΔRSRP>-5 | dB |
| DIFFRSRP_27 | -3≥ΔRSRP>-4 | dB |
| DIFFRSRP_28 | -2≥ΔRSRP>-3 | dB |
| DIFFRSRP_29 | -1≥ΔRSRP>-2 | dB |
| DIFFRSRP_30 | 0≥ΔRSRP>-1 | dB |

**Table 4.16.6.2-2: Measurement report mapping for differential PRS-RSRP**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| DIFFRSRP_0 | -30≥ΔRSRP | dB |
| DIFFRSRP_1 | -29≥ΔRSRP>-30 | dB |
| DIFFRSRP_2 | -28≥ΔRSRP>-29 | dB |
| DIFFRSRP_3 | -27≥ΔRSRP>-28 | dB |
| DIFFRSRP_4 | -26≥ΔRSRP>-27 | dB |
| DIFFRSRP_5 | -25≥ΔRSRP>-26 | dB |
| DIFFRSRP_6 | -24≥ΔRSRP>-25 | dB |
| DIFFRSRP_7 | -23≥ΔRSRP>-24 | dB |

| | | |
|---|---|---|
| DIFFRSRP_8 | -22≥ΔRSRP>-23 | dB |
| DIFFRSRP_9 | -21≥ΔRSRP>-22 | dB |
| DIFFRSRP_10 | -20≥ΔRSRP>-21 | dB |
| DIFFRSRP_11 | -19≥ΔRSRP>-20 | dB |
| DIFFRSRP_12 | -18≥ΔRSRP>-19 | dB |
| DIFFRSRP_13 | -17≥ΔRSRP>-18 | dB |
| DIFFRSRP_14 | -16≥ΔRSRP>-17 | dB |
| … | … | … |
| DIFFRSRP_25 | -5≥ΔRSRP>-6 | dB |
| DIFFRSRP_26 | -4≥ΔRSRP>-5 | dB |
| DIFFRSRP_27 | -3≥ΔRSRP>-4 | dB |
| DIFFRSRP_28 | -2≥ΔRSRP>-3 | dB |
| DIFFRSRP_29 | -1≥ΔRSRP>-2 | dB |
| DIFFRSRP_30 | 0≥ΔRSRP>-1 | dB |
| DIFFRSRP_31 | 1≥ΔRSRP>0 | dB |
| DIFFRSRP_32 | 2≥ΔRSRP>1 | dB |
| DIFFRSRP_33 | 3≥ΔRSRP>2 | dB |
| DIFFRSRP_34 | 4≥ΔRSRP>3 | dB |
| DIFFRSRP_35 | 5≥ΔRSRP>4 | dB |
| DIFFRSRP_36 | 6≥ΔRSRP>5 | dB |
| … | … | … |
| DIFFRSRP_47 | 17≥ΔRSRP>16 | dB |
| DIFFRSRP_48 | 18≥ΔRSRP>17 | dB |
| DIFFRSRP_49 | 19≥ΔRSRP>18 | dB |
| DIFFRSRP_50 | 20≥ΔRSRP>19 | dB |
| DIFFRSRP_51 | 21≥ΔRSRP>20 | dB |
| DIFFRSRP_52 | 22≥ΔRSRP>21 | dB |
| DIFFRSRP_53 | 23≥ΔRSRP>-22 | dB |
| DIFFRSRP_54 | 24≥ΔRSRP>23 | dB |
| DIFFRSRP_55 | 25≥ΔRSRP>24 | dB |
| DIFFRSRP_56 | 26≥ΔRSRP>25 | dB |
| DIFFRSRP_57 | 27≥ΔRSRP>26 | dB |
| DIFFRSRP_58 | 28≥ΔRSRP>27 | dB |
| DIFFRSRP_59 | 29≥ΔRSRP>28 | dB |
| DIFFRSRP_60 | 30≥ΔRSRP>29 | dB |
| DIFFRSRP_61 | ΔRSRP>30 | dB |

## 4.16.7 PRS-RSRPP Measurement Period Requirements

For PRS measurement within MG configured to UE, measurement period requirements for PRS-RSRP defined in clause 4.16.4.1 is re-used for PRS-RSRPP.

For PRS measurement without MG configured to UE, measurement period requirements for PRS-RSRP defined in clause 4.16.4.2 is re-used for PRS-RSRPP.

For PRS measurement in RRC_INACTIVE state, measurement period requirements for PRS-RSRP defined in 4.16.4.2 is re-used for PRS-RSRPP.

## 4.16.8 PRS-RSRPP Measurement Accuracy Requirements

See TS 38.133 [50] clause 10.1.38.2.

## 4.16.9 PRS-RSRPP Reporting mapping

### 4.16.9.1 Absolute PRS-RSRPP Measurement Report Mapping

The reporting range of absolute PRS-RSRPP measurement is defined from -156 dBm to -31 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 4.16.9.1-1. The range in the signalling may be larger than the guaranteed accuracy range.

The UE capable of *additionalPathsExtSupport-r17* can report the PRS-RSRPP measurement of up to its supported number of additional paths.

**Table 4.16.9.1-1: Measurement report mapping for PRS-RSRPP**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| PRS_RSRPP_0 | PRS-RSRPP<-156 | dBm |
| PRS_RSRPP_1 | -156≤PRS-RSRPP<-155 | dBm |
| PRS_RSRPP_2 | -155≤PRS-RSRPP<-154 | dBm |
| PRS_RSRPP_3 | -154≤PRS-RSRPP<-153 | dBm |
| PRS_RSRPP_4 | -153≤PRS-RSRPP<-152 | dBm |
| PRS_RSRPP_5 | -152≤PRS-RSRPP<-151 | dBm |
| PRS_RSRPP_6 | -151≤PRS-RSRPP<-150 | dBm |
| PRS_RSRPP_7 | -150≤PRS-RSRPP<-149 | dBm |
| PRS_RSRPP_8 | -149≤PRS-RSRPP<-148 | dBm |
| PRS_RSRPP_9 | -148≤PRS-RSRPP<-147 | dBm |
| PRS_RSRPP_10 | -147≤PRS-RSRPP<-146 | dBm |
| PRS_RSRPP_11 | -146≤PRS-RSRPP<-145 | dBm |
| PRS_RSRPP_12 | -145≤PRS-RSRPP<-144 | dBm |
| PRS_RSRPP_13 | -144≤PRS-RSRPP<-143 | dBm |
| PRS_RSRPP_14 | -143≤PRS-RSRPP<-142 | dBm |
| PRS_RSRPP_15 | -142≤PRS-RSRPP<-141 | dBm |
| PRS_RSRPP_16 | -141≤PRS-RSRPP<-140 | dBm |
| PRS_RSRPP_17 | -140≤PRS-RSRPP<-139 | dBm |
| PRS_RSRPP_18 | -139≤PRS-RSRPP<-138 | dBm |
| … | … | … |
| PRS_RSRPP_111 | -46≤PRS-RSRPP<-45 | dBm |
| PRS_RSRPP_112 | -45≤PRS-RSRPP<-44 | dBm |
| PRS_RSRPP_113 | -44≤PRS-RSRPP<-43 | dBm |
| PRS_RSRPP_114 | -43≤PRS-RSRPP<-42 | dBm |
| PRS_RSRPP_115 | -42≤PRS-RSRPP<-41 | dBm |
| PRS_RSRPP_116 | -41≤PRS-RSRPP<-40 | dBm |
| PRS_RSRPP_117 | -40≤PRS-RSRPP<-39 | dBm |
| PRS_RSRPP_118 | -39≤PRS-RSRPP<-38 | dBm |
| PRS_RSRPP_119 | -38≤PRS-RSRPP<-37 | dBm |
| PRS_RSRPP_120 | -37≤PRS-RSRPP<-36 | dBm |
| PRS_RSRPP_121 | -36≤PRS-RSRPP<-35 | dBm |
| PRS_RSRPP_122 | -35≤PRS-RSRPP<-34 | dBm |
| PRS_RSRPP_123 | -34≤PRS-RSRPP<-33 | dBm |
| PRS_RSRPP_124 | -33≤PRS-RSRPP<-32 | dBm |
| PRS_RSRPP_125 | -32≤PRS-RSRPP<-31 | dBm |
| PRS_RSRPP_126 | -31≤PRS-RSRPP | dBm |

### 4.16.9.2 Differential Report Mapping for PRS-RSRPP Measurement

The reporting range of differential PRS-RSRPP is defined from -30 dB to 30 dB with 1 dB resolution.

The mapping of measured quantity is defined in Table 4.16.9.2-1. The range in the signalling may be larger than the guaranteed accuracy range.

For differential reporting, PRS-RSRPP is reported as the difference in dB with respect to the first reported PRS-RSRPP.

**Table 4.16.9.2-1: Measurement report mapping for differential PRS-RSRPP**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| DIFFRSRPP_0 | -30≥ΔRSRPP | dB |
| DIFFRSRPP_1 | -29≥ΔRSRPP>-30 | dB |
| DIFFRSRPP_2 | -28≥ΔRSRPP>-29 | dB |
| DIFFRSRPP_3 | -27≥ΔRSRPP>-28 | dB |
| DIFFRSRPP_4 | -26≥ΔRSRPP>-27 | dB |
| DIFFRSRPP_5 | -25≥ΔRSRPP>-26 | dB |
| DIFFRSRPP_6 | -24≥ΔRSRPP>-25 | dB |
| DIFFRSRPP_7 | -23≥ΔRSRPP>-24 | dB |
| DIFFRSRPP_8 | -22≥ΔRSRPP>-23 | dB |
| DIFFRSRPP_9 | -21≥ΔRSRPP>-22 | dB |
| DIFFRSRPP_10 | -20≥ΔRSRPP>-21 | dB |
| DIFFRSRPP_11 | -19≥ΔRSRPP>-20 | dB |
| DIFFRSRPP_12 | -18≥ΔRSRPP>-19 | dB |
| DIFFRSRPP_13 | -17≥ΔRSRPP>-18 | dB |
| DIFFRSRPP_14 | -16≥ΔRSRPP>-17 | dB |
| … | … | … |
| DIFFRSRPP_25 | -5≥ΔRSRPP>-6 | dB |
| DIFFRSRPP_26 | -4≥ΔRSRPP>-5 | dB |
| DIFFRSRPP_27 | -3≥ΔRSRPP>-4 | dB |
| DIFFRSRPP_28 | -2≥ΔRSRPP>-3 | dB |
| DIFFRSRPP_29 | -1≥ΔRSRPP>-2 | dB |
| DIFFRSRPP_30 | 0≥ΔRSRPP>-1 | dB |
| DIFFRSRPP_31 | 1≥ΔRSRPP>0 | dB |
| DIFFRSRPP_32 | 2≥ΔRSRPP>1 | dB |
| DIFFRSRPP_33 | 3≥ΔRSRPP>2 | dB |
| DIFFRSRPP_34 | 4≥ΔRSRPP>3 | dB |
| DIFFRSRPP_35 | 5≥ΔRSRPP>4 | dB |
| DIFFRSRPP_36 | 6≥ΔRSRPP>5 | dB |
| … | … | … |
| DIFFRSRPP_47 | 17≥ΔRSRPP>16 | dB |
| DIFFRSRPP_48 | 18≥ΔRSRPP>17 | dB |
| DIFFRSRPP_49 | 19≥ΔRSRPP>18 | dB |
| DIFFRSRPP_50 | 20≥ΔRSRPP>19 | dB |
| DIFFRSRPP_51 | 21≥ΔRSRPP>20 | dB |
| DIFFRSRPP_52 | 22≥ΔRSRPP>21 | dB |
| DIFFRSRPP_53 | 23≥ΔRSRPP>-22 | dB |
| DIFFRSRPP_54 | 24≥ΔRSRPP>23 | dB |
| DIFFRSRPP_55 | 25≥ΔRSRPP>24 | dB |
| DIFFRSRPP_56 | 26≥ΔRSRPP>25 | dB |
| DIFFRSRPP_57 | 27≥ΔRSRPP>26 | dB |
| DIFFRSRPP_58 | 28≥ΔRSRPP>27 | dB |
| DIFFRSRPP_59 | 29≥ΔRSRPP>28 | dB |
| DIFFRSRPP_60 | 30≥ΔRSRPP>29 | dB |
| DIFFRSRPP_61 | ΔRSRPP>30 | dB |

# 4A Support of 4 Rx capable UEs

## 4A.0 Introduction

In this section, the method for applying 2RX tests to UEs supporting 4RX antenna ports is specified.

# 4A.1 RAT Independent Tests

All tests in Sections 5 to 7, 11 and 12 are applicable for all types of UEs independently of the number of RX antennas. Only one of the E-UTRAN/UTRAN RX antennas shall be connected to the SS.

# 4A.2 RAT Dependent Tests

All tests in Sections 8 to 10 are specified for UEs supporting either category 0 (1RX) or 2RX. No tests cases are currently specified in Sections 8 to 10 that are applicable only to 4RX antenna ports, so 4RX capable UEs are always tested by reusing tests that were originally specified for 2RX UEs.

## 4A.2.1 Principle of testing

### 4A.2.1.1 Single carrier tests

ECID (Section 8) and OTDOA Measurement Accuracy test cases shall be tested on all bands supported by the UE. For bands where 2RX is supported, the test shall be performed with the antenna connection specified in 4A.2.1.3. For bands where only 4RX is supported, the test shall be performed with the antenna connection specified in 4A.2.1.4.

OTDOA Measurement Reporting Delay test cases shall be tested on any band where 2RX is supported, using the antenna connection specified in 4A.2.1.3. If 2RX is not supported on any band, any 4RX band shall be tested, using the antenna connection specified in 4A.2.1.4.

### 4A.2.1.2 Carrier Aggregation tests

For all carrier aggregation tests, the antenna connection for each cell needs to be considered separately. If a PCell or SCell is in a band where 2RX is supported, the test shall be performed using the antenna connection specified in 4A.2.1.3 for that cell. Otherwise, the test shall be performed using the antenna connection specified in 4A.2.1.4 for that cell.

### 4A.2.1.3 Antenna connection for bands where 2RX is supported

For bands where 2RX is supported, it is left to the UE declaration and AP configuration to decide which 2 of the 4 Rx ports are connected with data source from system simulator. The remaining 2 Rx ports shall be connected with zero input**.** No test parameters or requirements are modified.

### 4A.2.1.4 Antenna connection for bands where 4RX is supported

For bands where 4RX is supported, all 4 Rx are connected with data source from system simulator**.** The system simulator shall provide independent noise and fading (low correlation) for each antenna port.

# 4B Applicability of tests for types and Categories of UE

# 4B.1 Introduction

In this clause, the applicability of the tests defined in clauses 7 to 16 of this specification are detailed for various types and Categories of UE for information.

# 4B.2 Applicability of requirements and tests

The applicability of the requirements for various types and Categories of UE for the tests in this specification are defined as follows:

- for the tests in clause 7 (E-UTRA A-GNSS): applicabilities are defined in TS 36.171 [3] clauses 4.1 and 4.1.1

- for the tests in clauses 8 (E-UTRA ECID), 9 (E-UTRA OTDOA), 10 (E-UTRA OTDOA for Carrier Aggregation): applicabilities are defined in TS 36.133 [23] clause 3.6.1
- for the tests in clauses 11 (E-UTRA and NR MBS), 12 (E-UTRA WLAN and BLE): applicabilities are defined in TS 37.171 [39]
- for the tests in clause 13 (NR A-GNSS): applicabilities are defined in TS 38.171 [43] clause 4.1
- for the tests in clause 14 (NR OTDOA): applicabilities are defined in TS 38.133 [23] clause 3.6
- for the tests in clauses 15 (NR WLAN), 16 (NR BLE): applicabilities are defined in TS 37.171 [39].

These are summarised below for the relevant tests in this specification.

**Table 4B.2-1: Applicability of tests for various types and Categories of UE (informative)**

| Tests | Types and Categories of UE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | LTE UE other than types and Categories listed here | UE Category 0 (Note 1) | UE Category 1 | UE Category 1bis (Note 2) | UE Category M1 | UE Category M2 | UE Category NB1 and NB2 (Note 3) | UE configured with NR EN-DC | UE supporting NR SA mode or NE-DC or NGEN-DC |
| Clause 7 (E-UTRA A-GNSS) | All | All | All | All | All (UE must also support VoLTE) | All (UE must also support VoLTE) | None | N/A | N/A |
| Clause 8 (E-UTRA ECID), | All except those defined for types and Categories listed here | All | All | Only tests defined for UE Category 1bis | Only tests defined for UE Category M1 | Only tests defined for UE Category M2 | None | Requirements and tests defined in RAN 4. Tests here are FFS | N/A |
| Clause 9 (E-UTRA OTDOA) | All except those defined for types and Categories listed here | All | All | Only tests defined for UE Category 1bis | Only tests defined for UE Category M1 | Only tests defined for UE Category M2 | Only tests defined for NB-IOT | Requirements and tests defined in RAN 4. Tests here are FFS | N/A |
| Clause 10 (E-UTRA OTDOA for Carrier Aggregation) | All | All | All | None | None | None | None | Requirements and tests defined in RAN 4. Tests here are FFS | N/A |
| Clause 11 (E-UTRA and NR MBS) | All except those defined for NR | All except those defined for NR | All except those defined for NR | All except those defined for NR | All except those defined for NR | All except those defined for NR | FFS | All those defined for NR | All those defined for NR |
| Clause 12 (E-UTRA WLAN and BLE) | All | All | All | All | All | All | FFS | Requirements (only) defined in RAN 4. Tests are FFS | Requirement (only) define in RAN 4. Tests are FFS |
| Clause 13 (NR A-GNSS) | N/A | N/A | N/A | N/A | N/A | N/A | N/A | All | All |
| Clause 14 (NR OTDOA) | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A. | FFS |
| Clause 15 (NR WLAN) | N/A | N/A | N/A | N/A | N/A | N/A | N/A | FFS | FFS |
| Clause 16 (NR BLE) | N/A | N/A | N/A | N/A | N/A | N/A | N/A | FFS | FFS |
| Note 1: The requirements for a UE Category 0 are derived assuming UE Category 0 and a single antenna receiver.<br>Note 2: The requirements for UE Category 1bis are derived assuming UE Category 1bis and a single antenna receiver.<br>Note 3: The requirements for UE Category NB1/NB2 are derived assuming UE Category NB1/NB2 and a single antenna receiver. | | | | | | | | | |

Editor's note: the current versions of this specification and TS 37.571-3 do not accurately reflect the above table. The above table should therefore be used where it differs from the applicabilities given in this specification and TS 37.571-3.

# 5 UTRA A-GPS Minimum Performance requirements

## 5.1 General

This clause defines the minimum performance requirements for FDD UTRA UEs where the only Assisted Global Navigation Satellite System (A-GNSS) supported is Assisted Global Positioning System (A-GPS) L1 C/A. The procedures for UEs that support other or additional A-GNSSs are specified in clause 6. Note that for TDD UTRA UEs where the only Assisted Global Navigation Satellite System (A-GNSS) supported is Assisted Global Positioning System (A-GPS) L1 C/A there are no requirements.

This clause defines requirements for both UE based and UE assisted modes; if a UE supports both modes then it shall be tested in both modes

The requirements in this clause are defined for CELL_DCH and CELL_FACH states. All tests shall be performed in CELL_DCH state and the Nominal Accuracy Performance test case shall be also performed in CELL_FACH state.

## 5.2 Sensitivity

### 5.2.1 Sensitivity Coarse Time Assistance

#### 5.2.1.1 Definition and applicability

Sensitivity with coarse time assistance is the minimum level of GPS satellite signals required for the UE to make an A-GPS position estimate to a specific accuracy and within a specific response time when the network only provides coarse time assistance.

The requirements and this test apply to all types of UTRA for the FDD UE that supports A-GPS.

#### 5.2.1.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 5.2.1.2 for the parameters specified in table 5.2.1.1.

**Table 5.2.1.1: Test parameters for Sensitivity Coarse Time Assistance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±2 |
| GPS Signal for one satellite | dBm | -142 |
| GPS Signal for remaining satellites | dBm | -147 |

**Table 5.2.1.2: Minimum requirements for Sensitivity Coarse Time Assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.171 [31], clause 5.1.1.1.

#### 5.2.1.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GPS satellite signal conditions that represent weak signal conditions and with only Coarse Time Assistance provided by the SS.

### 5.2.1.4 Method of test

#### 5.2.1.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GPS test parameters as specified in table 5.2.1.3 for GPS scenario #1. Select the first satellite PRN defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5 for the one satellite with the higher level.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

#### 5.2.1.4.2 Procedure

1. Start GPS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 5.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 5.2.1.2.4
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 5.2.2 and 5.2.6 for UE based testing; or clauses 5.2.4 and 5.2.6 for UE assisted testing with the value of GPS TOW msec offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 5.2.6.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.1 or 7.5.4.
3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 5.2.1.4 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 5.2.1.4 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.2.1.4 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GPS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE, used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.2.1.4 and record one Good Result or Bad Result as appropriate.
5. Repeat steps 1 to 4 using GPS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Select the first satellite PRN defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5 for the one satellite with the higher level. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec offset in step 2.
6. Repeat steps 1 to 5 until the statistical requirements of clause 5.2.1.5 are met. Each time scenario #1 or #2 is used, the start time of the GPS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used select the next satellite PRN from the one used previously, defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5, for the one satellite with the higher level.
7. Release the connection using the procedure in clause F.3.

### 5.2.1.5 Test Requirements

For the parameters specified in table 5.2.1.3 the UE shall meet the requirements and the success rate specified in table 5.2.1.4 with a confidence level of 95% according to annex D.

**Table 5.2.1.3: Test parameters for Sensitivity Coarse Time Assistance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| GPS Signal for one satellite | dBm | -141 |
| GPS Signal for remaining satellites | dBm | -146 |

**Table 5.2.1.4: Test requirements for Sensitivity Coarse Time Assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

## 5.2.2 Sensitivity Fine Time Assistance

### 5.2.2.1 Definition and applicability

Sensitivity with fine time assistance is the minimum level of GPS satellite signals required for the UE to make an A-GPS position estimate to a specific accuracy and within a specific response time when the network provides fine time assistance in addition to coarse time assistance.

The requirements and this test apply to all types of UTRA for the FDD UE that supports only A-GPS and that is capable of providing an enhanced performance when the network provides Fine Time Assistance.

### 5.2.2.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 5.2.2.2 for the parameters specified in table 5.2.2.1.

**Table 5.2.2.1: Test parameters for Sensitivity Fine Time Assistance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse time assistance error range | seconds | ±2 |
| GPS Fine Time assistance error range | μs | ±10 |
| GPS Signal for all satellites | dBm | -147 |

**Table 5.2.2.2: Minimum requirements for Sensitivity Fine Time Assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.171 [31], clause 5.1.2.1.

### 5.2.2.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GPS satellite signal conditions that represent weak signal conditions and with Fine Time Assistance provided by the SS.

### 5.2.2.4 Method of test

#### 5.2.2.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GPS test parameters as specified in table 5.2.2.3 for GPS scenario #1.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

#### 5.2.2.4.2 Procedure

1. Start GPS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 5.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 5.2.1.2.4
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 5.2.2 and 5.2.6 for UE based testing; or clauses 5.2.4 and 5.2.6 for UE assisted testing with the values of GPS TOW msec and UTRAN GPS timing of cell frames offset by random values as specified in 3GPP TS 37.571-5 [20] clause 5.2.6.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.1 or 7.5.4.
3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 5.2.2.4 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 5.2.2.4 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.2.2.4 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GPS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.2.2.4 and record one Good Result or Bad Result as appropriate.
5. Repeat steps 1 to 4 using GPS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec and UTRAN GPS timing of cell frames offsets in step 2.
6. Repeat steps 1 to 5 until the statistical requirements of clause 5.2.2.5 are met. Each time scenario #1 or #2 is used, the start time of the GPS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.
7. Release the connection using the procedure in clause F.3.

### 5.2.2.5 Test Requirements

For the parameters specified in table 5.2.2.3 the UE shall meet the requirements and the success rate specified in table 5.2.2.4 with a confidence level of 95% according to annex D.

**Table 5.2.2.3: Test parameters for Sensitivity Fine Time Assistance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse time assistance error range | seconds | ±1.8 |
| GPS Fine Time assistance error range | μs | ±9 |
| GPS Signal for all satellites | dBm | -146 |

**Table 5.2.2.4: Test requirements for Sensitivity Fine Time Assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

# 5.3 Nominal Accuracy

## 5.3.1 Definition and applicability

Nominal accuracy is the accuracy of the UE's A-GPS position estimate under ideal GPS signal conditions.

The requirements and this test apply to all types of UTRA for the FDD UE that supports A-GPS.

## 5.3.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 5.3.2 for the parameters specified in table 5.3.1.

**Table 5.3.1: Test parameters for Nominal Accuracy**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±2 |
| GPS Signal for all satellites | dBm | -130 |

**Table 5.3.2: Minimum requirements for Nominal Accuracy**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 30 m | 20 s |

The reference for this requirement is 3GPP TS 25.171 [31], clause 5.2.1.

## 5.3.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GPS satellite signal conditions that represent ideal conditions.

## 5.3.4 Method of test

### 5.3.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GPS test parameters as specified in table 5.3.3 for GPS scenario #1.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 5.3.4.2 Procedure

1. Start GPS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 5.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 5.2.1.2.4

2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 5.2.2 and 5.2.6 for UE based testing; or clauses 5.2.4 and 5.2.6 for UE assisted testing with the value of GPS TOW msec offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 5.2.6.2; using the exception to the RRC MEASUREMENT CONTROL message listed in table 5.3.2A; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.1 or 7.5.4.

**Table 5.3.2A: Contents of RRC MEASUREMENT CONTROL message**

| Information Element | Value/Remark |
|---|---|
| - UE positioning reporting quantity<br>- Horizontal accuracy | <br>10 (15.9 m) |

3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 5.3.4 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 5.3.4 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.

4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.3.4 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GPS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.3.4 and record one Good Result or Bad Result as appropriate.

5. Repeat steps 1 to 4 using GPS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec offset in step 2.

6. Repeat steps 1 to 5 until the statistical requirements of clause 5.3.5 are met. Each time scenario #1 or #2 is used, the start time of the GPS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.

7. Release the connection using the procedure in clause F.3.

### 5.3.5 Test Requirements

For the parameters specified in table 5.3.3 the UE shall meet the requirements and the success rate specified in table 5.3.4 with a confidence level of 95% according to annex D.

**Table 5.3.3: Test parameters for Nominal Accuracy**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| GPS Signal for all satellites | dBm | -130 |

**Table 5.3.4: Test requirements for Nominal Accuracy**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 31.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

## 5.4 Dynamic Range

### 5.4.1 Definition and applicability

Dynamic Range is the maximum difference in level of the GPS signals from a number of satellites that allows the UE to make an A-GPS position estimate with a specific accuracy and a specific response time.

The requirements and this test apply to all types of UTRA for the FDD UE that supports A-GPS.

### 5.4.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 5.4.2 for the parameters specified in table 5.4.1.

**Table 5.4.1: Test parameters for Dynamic Range**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 6 |
| HDOP Range | - | 1.4 to 2.1 |
| GPS Coarse Time assistance error range | seconds | ±2 |
| Propagation conditions | - | AWGN |
| GPS Signal for 1st satellite | dBm | -129 |
| GPS Signal for 2nd satellite | dBm | -135 |
| GPS Signal for 3rd satellite | dBm | -141 |
| GPS Signal for 4th satellite | dBm | -147 |
| GPS Signal for 5th satellite | dBm | -147 |
| GPS Signal for 6th satellite | dBm | -147 |

**Table 5.4.2: Minimum requirements for Dynamic Range**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.171 [31], clause 5.3.1.

## 5.4.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GPS satellite signal conditions that have a wide dynamic range. Strong satellites are likely to degrade the acquisition of weaker satellites due to their cross-correlation products.

## 5.4.4 Method of test

### 5.4.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GPS test parameters as specified in table 5.4.3 for GPS scenario #1. Select the first three satellite PRNs defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5 for the three satellites with the higher levels.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 5.4.4.2 Procedure

1. Start GPS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 5.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 5.2.1.2.4
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 5.2.2 and 5.2.6 for UE based testing; or clauses 5.2.4 and 5.2.6 for UE assisted testing with the value of GPS TOW msec offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 5.2.6.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.1 or 7.5.4.
3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 5.4.4 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 5.4.4 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.4.4 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GPS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.4.4 and record one Good Result or Bad Result as appropriate.
5. Repeat steps 1 to 4 using GPS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Select the first three satellite PRNs defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5 for the three satellites with the higher levels. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec offset in step 2.
6. Repeat steps 1 to 5 until the statistical requirements of clause 5.4.5 are met. Each time scenario #1 or #2 is used, the start time of the GPS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, increment the set of three satellite PRNs by one from the ones used previously, defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5, for the three satellites with the higher

levels (i.e. if the set of satellites is a, b, c, d, e, f and the first set used was a, b, c, the second set shall be b, c, d and so on).

7. Release the connection using the procedure in clause F.3.

### 5.4.5 Test Requirements

For the parameters specified in table 5.4.3 the UE shall meet the requirements and the success rate specified in table 5.4.4 with a confidence level of 95% according to annex D.

**Table 5.4.3: Test parameters for Dynamic Range**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 6 |
| HDOP Range | - | 1.4 to 2.1 |
| GPS Coarse Time assistance error range | seconds | ±2+TT |
| Propagation conditions | - | AWGN |
| GPS Signal for 1st satellite | dBm | -128.2 |
| GPS Signal for 2nd satellite | dBm | -134 |
| GPS Signal for 3rd satellite | dBm | -140 |
| GPS Signal for 4th satellite | dBm | -146 |
| GPS Signal for 5th satellite | dBm | -146 |
| GPS Signal for 6th satellite | dBm | -146 |

**Table 5.4.4: Test requirements for Dynamic Range**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

## 5.5 Multi-path Performance

### 5.5.1 Definition and applicability

Multi-path performance measures the accuracy and response time of the UE's A-GPS position estimate in a specific GPS signal multi-path environment.

The requirements and this test apply to all types of UTRA for the FDD UE that supports A-GPS.

### 5.5.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 5.5.2 for the parameters specified in table 5.5.1.

**Table 5.5.1: Test parameters for Multi-path Performance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites (see note) | - | 5 |
| GPS Coarse Time assistance error range | seconds | ±2 |
| HDOP Range | - | 1.8 to 2.5 |
| GPS signal for Satellite 1, 2 (see note) | dBm | -130 |
| GPS signal for Satellite 3, 4, 5 (see note) | dBm | LOS signal of -130 dBm, multi-path signal of -136 dBm |
| NOTE: Satellites 1, 2 no multi-path. Satellites 3, 4, 5 multi-path defined in clause 4.2.4. | | |

**Table 5.5.2: Minimum requirements for Multi-path Performance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.171 [31], clause 5.4.1.

## 5.5.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GPS satellite signal conditions that represent simple multi-path conditions.

## 5.5.4 Method of test

### 5.5.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GPS test parameters as specified in table 5.5.3 for GPS scenario #1. Select the first two satellite PRNs defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5 for the two satellites with the higher levels.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 5.5.4.2 Procedure

1. Start GPS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 5.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 5.2.1.2.4. The initial carrier phase difference between taps of the multi-path model shall be randomly selected between 0 and $2\pi$ radians by selecting the next random number from a standard uniform random number generator, in the range 0 to $2\pi$, representing radians with a resolution of 0.1, representing 0.1 radians.
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 5.2.2 and 5.2.6 for UE based testing; or clauses 5.2.4 and 5.2.6 for UE assisted testing with the value of GPS TOW msec offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 5.2.6.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.1 or 7.5.4.
3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 5.5.4 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 5.5.4 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.5.4 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GPS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.5.4 and record one Good Result or Bad Result as appropriate.
5. Repeat steps 1 to 4 using GPS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Select the first two satellite PRNs defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5 for the two satellites with the higher levels. Use new random values for

the UE location and altitude, and the initial carrier phase difference between taps of the multi-path model in step 1 and for the GPS TOW msec offset in step 2.

6. Repeat steps 1 to 5 until the statistical requirements of clause 5.5.5 are met. Each time scenario #1 or #2 is used, the start time of the GPS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, increment the set of two satellite PRNs by one from the ones used previously, defined in the table in 3GPP TS 37.571-5 [20] clause 5.2.1.2.5, for the two satellites with the higher level (i.e. if the set of satellites is a, b, c, d, e and the first set used was a, b the second set shall be b, c and so on).

7. Release the connection using the procedure in clause F.3.

### 5.5.5 Test Requirements

For the parameters specified in table 5.5.3 the UE shall meet the requirements and the success rate specified in table 5.5.4 with a confidence level of 95% according to annex D.

**Table 5.5.3: Test parameters for Multi-path Performance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites (see note) | - | 5 |
| GPS Coarse Time assistance error range | seconds | ±2+TT |
| HDOP Range | - | 1.8 to 2.5 |
| GPS signal for Satellite 1, 2 (see note) | dBm | -130 |
| GPS signal for Satellite 3, 4, 5 (see note) | dBm | LOS signal of -130 dBm, multi-path signal of -136.2 dBm |
| NOTE: Satellites 1, 2 no multi-path. Satellites 3, 4, 5 multi-path defined in clause 4.2.4. | | |

**Table 5.5.4: Test requirements for Multi-path Performance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

## 5.6 Moving Scenario and Periodic Update Performance

### 5.6.1 Definition and applicability

Moving scenario and periodic update performance measures the accuracy of the UE's A-GPS position estimates and the periodic update capability of the UE in a moving scenario.

The requirements and this test apply to all types of UTRA for the FDD UE that supports only A-GPS.

### 5.6.2 Minimum requirements

The position estimates, after the first reported position estimate, shall meet the accuracy requirement in table 5.6.2 with the periodical reporting interval of 2 seconds for the parameters specified in table 5.6.1.

NOTE: In the actual testing the UE may report error messages until it has been able to acquire GPS measured results or a position estimate. The SS shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 5.6.2.

**Table 5.6.1: Test parameters for Moving Scenario and Periodic Update Performance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 5 |

| Parameters | Unit | Value |
|---|---|---|
| HDOP Range | - | 1.8 to 2.5 |
| Propagation condition | - | AWGN |
| GPS signal for all satellites | dBm | -130 |

**Table 5.6.2: Minimum requirements for Moving Scenario and Periodic Update Performance**

| Success Rate | 2-D position error |
|---|---|
| 95 % | 100 m |

The reference for this requirement is 3GPP TS 25.171 [31], clause 5.5.1.

## 5.6.3 Test purpose

To verify the UE's position estimates, after the first reported position estimate, meet the minimum requirements under GPS satellite signal conditions that simulate a moving scenario. A good tracking performance, with regular position estimate reporting is essential for certain location services.

## 5.6.4 Method of test

### 5.6.4.1 Initial conditions

Test environment: normal; see Annex G.

The UE is requested to use periodical reporting with a reporting interval of 2 seconds.

The GPS signals simulate the UE moving on a rectangular trajectory of 940 m by 1 440 m with rounded corners defined in figure 5.6.1 and table 5.6.3. The initial reference is first defined followed by acceleration to final speed of 100 km/h in 250 m. The UE then maintains the speed for 400 m. This is followed by deceleration to final speed of 25 km/h in 250 m. The UE then turn 90 degrees with turning radius of 20 m at 25 km/h. This is followed by acceleration to final speed of 100 km/h in 250 m. The sequence is repeated to complete the rectangle.

**Table 5.6.3: Trajectory Parameters for Moving Scenario and Periodic Update Performance test case**

| Parameter | Distance (m) | Speed (km/h) |
|---|---|---|
| $l_{11}$, $l_{15}$, $l_{21}$, $l_{25}$ | 20 | 25 |
| $l_{12}$, $l_{14}$, $l_{22}$, $l_{24}$ | 250 | 25 to 100 and 100 to 25 |
| $l_{13}$ | 400 | 100 |
| $l_{23}$ | 900 | 100 |

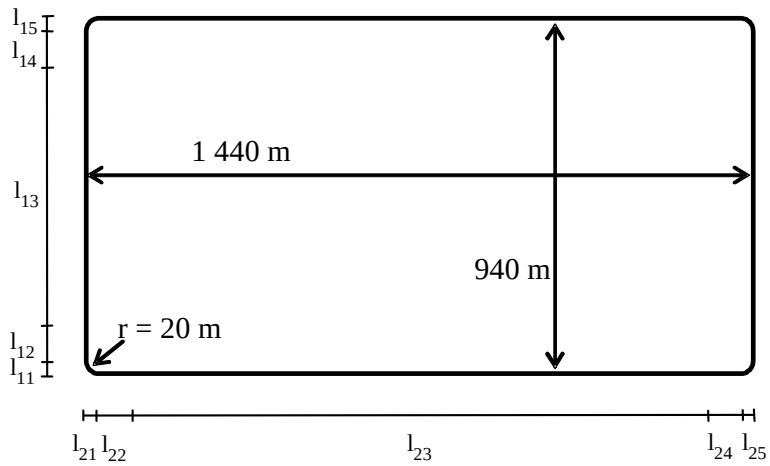



**Figure 5.6.1: Rectangular Trajectory for Moving Scenario and Periodic Update Performance test case**

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GPS test parameters as specified in table 5.6.4 for GPS scenario #3.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 5.6.4.2 Procedure

1. Start GPS scenario #3 as specified in 3GPP TS 37.571-5 [20], clause 5.2.1.2
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 5.2.2 and 5.2.6 for UE based testing; or clauses 5.2.4 and 5.2.6 for UE assisted testing; as required to obtain fixes using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.2 or 7.5.5.
3. Ignore any error messages that the UE may report in MEASUREMENT REPORT messages until it has been able to acquire the GPS signals and reports the first GPS measured result or position estimate.
4. Discard the first GPS measured result or position estimate.
5. Record the time of reception of the next MEASUREMENT REPORT message after reception of the first GPS measured result or position estimate.
6. After the reception of the first GPS measured result or position estimate reported in a MEASUREMENT REPORT message, every time the UE returns a GPS measured result or position estimate in the MEASUREMENT REPORT message record the time of reception and the result. If the difference between the time of reception and the time of reception of the previous result is less than 1.5 seconds or greater than 2.5 seconds, or if the UE reports a UE positioning error in any MEASUREMENT REPORT messages, then record one Bad Result. Otherwise process the result as specified in step 7.
7. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE at the time of applicability reported in the position estimate and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.6.5 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GPS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE at the time of applicability reported in the GPS measured results and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 5.6.5 and record one Good Result or Bad Result as appropriate.
8. If the UE sends the first MEASUREMENT REPORT that contains a measured result or position estimate later than 240s after the start of the GPS scenario, fail the UE and stop the test early. Otherwise collect MEASUREMENT REPORTs during 900s, starting from the time recorded in step 5. If at any time the difference between the times of reception of two consecutive results is greater than 240s, fail the UE and stop the test early. Use the collected Good Results and Bad Results to determine the PASS/FAIL according to clause 5.6.5.
9. Release the connection using the procedure in clause F.3.

## 5.6.5 Test Requirements

For the parameters specified in table 5.6.4, after the first reported position estimate, the UE shall meet the accuracy requirement and the success rate specified in table 5.6.5 with a periodical reporting interval of 2 seconds +/- 20% plus measurement system uncertainty of 100ms.

NOTE: Due to the statistical nature of the results it is not possible to design a test with predefined confidence level for the success rate in Table 5.6.5, therefore a simple PASS/FAIL of the results gathered against this success rate is used.

**Table 5.6.4: Test parameters for Moving Scenario and Periodic Update Performance**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 5 |
| HDOP Range | - | 1.8 to 2.5 |
| Propagation condition | - | AWGN |
| GPS signal for all satellites | dBm | -130 |

**Table 5.6.5: Test requirements for Moving Scenario and Periodic Update Performance**

| Success Rate | 2-D position error |
|---|---|
| 95 % | 101.3 m |

NOTE 1: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

NOTE 2: In the actual testing the UE may report error messages until it has been able to acquire GPS measured results or a position estimate. The test equipment shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 5.6.5.

# 6 UTRA A-GNSS Minimum Performance requirements

## 6.1 General

This clause defines the minimum performance requirements for both UE based and UE assisted A-GNSS UTRA UEs. If a UE supports both modes then it shall be tested in both modes. It excludes performance requirements for FDD UEs where the only A-GNSS supported is A-GPS L1 C/A which are specified in clause 5. It excludes performance requirements for TDD UEs where the only A-GNSS supported is A-GPS L1 C/A for which there is no requirement.

The requirements are defined for CELL_DCH and CELL_FACH states. All tests shall be performed in CELL_DCH state and the Nominal Accuracy Performance test case shall be also performed in CELL_FACH state.

## 6.2 Sensitivity

### 6.2.1 Sensitivity Coarse Time Assistance

#### 6.2.1.1 Definition and applicability

Sensitivity with coarse time assistance is the minimum level of GNSS satellite signals required for the UE to make an A-GNSS position estimate to a specific accuracy and within a specific response time when the network only provides coarse time assistance.

The requirements and this test apply to all types of UTRA for the UE that supports A-GNSS.

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 6.2.1.1.

**Table 6.2.1.1: Sub-Test Case Number Definition**

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS |
| 2 | UE supporting A-Galileo |
| 3 | UE supporting A-GPS and Modernized GPS |
| 4 | UE supporting A-GPS and A-GLONASS |
| 8 | UE supporting A-GPS and A-Galileo |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS |

#### 6.2.1.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 6.2.1.2-3 for the parameters specified in table 6.2.1.2-1.

**Table 6.2.1.2-1: Test parameters for Sensitivity Coarse Time Assistance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.2.1.2-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| GPS(1) | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| GLONASS | Reference high signal power level | dBm | -142 |

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Reference low signal power level | dBm | -147 |
| BDS | Reference high signal power level | dBm | -136 |
| | Reference low signal power level | dBm | -145 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.2.1.2-2: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1(1) | GNSS-2 | GNSS-3 |
| Single constellation | High signal level | 1 | - | - |
| | Low signal level | 5 | - | - |
| Dual constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 3 | - |
| Triple constellation | High signal level | 1 | - | - |
| | Low signal level | 1 | 2 | 2 |
| Note: For GPS capable receivers, GNSS-1, i.e. the system having the satellite with high signal level, shall be GPS. | | | | |

**Table 6.2.1.2-3: Minimum requirements for Sensitivity Coarse Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.172 [19], clause 5.1.1.1, and 3GPP TS 25.173 [36], clause 5.1.1.1.

### 6.2.1.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GNSS satellite signal conditions that represent weak signal conditions and with only Coarse Time Assistance provided by the SS.

### 6.2.1.4 Method of test

#### 6.2.1.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GNSS test parameters as specified in table 6.2.1.5-1 for GNSS scenario #1. For GNSS-1, select the first satellite SV ID defined in the relevant table in 3GPP TS 37.571-5 [20] clause 6.2.1.2 for the one satellite with the higher level.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

#### 6.2.1.4.2 Procedure

1. Start GNSS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 6.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 6.2.1.2.6.
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 6.2.2 and 6.2.7 for UE based testing; or clauses 6.2.4 and 6.2.7 for UE assisted testing with the value of GPS TOW msec or GANSS TOD offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 6.2.7.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.6 or 7.5.8.

3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 6.2.1.5-3 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 6.2.1.5-3 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.

4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 6.2.1.5-3 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE, used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 6.2.1.5-3 and record one Good Result or Bad Result as appropriate.

5. Repeat steps 1 to 4 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. For GNSS-1, select the first satellite SV ID defined in the relevant table in 3GPP TS 37.571-5 [20] clause 6.2.1.2 for the one satellite with the higher level. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec or GANSS TOD offset in step 2.

6. Repeat steps 1 to 5 until the statistical requirements of clause 6.2.1.5 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used for GNSS-1, select the next satellite SV ID from the one used previously, defined in the relevant table in 3GPP TS 37.571-5 [20] clause 6.2.1.2, for the one satellite with the higher level.

7. Release the connection using the procedure in clause F.3.

### 6.2.1.5 Test Requirements

For the parameters specified in table 6.2.1.5-1 the UE shall meet the requirements and the success rate specified in table 6.2.1.5-3 with a confidence level of 95% according to Annex D.

**Table 6.2.1.5-1: Test parameters for Sensitivity Coarse Time Assistance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.2.1.5-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| Galileo | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| GPS(1) | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| GLONASS | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| BDS | Reference high signal power level | dBm | -135 |
| | Reference low signal power level | dBm | -144 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.2.1.5-2: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1(1) | GNSS-2 | GNSS-3 |
| Single constellation | High signal level | 1 | - | - |
| | Low signal level | 5 | - | - |
| Dual constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 3 | - |

| | | | | |
|---|---|---|---|---|
| Triple constellation | High signal level | 1 | - | - |
| | Low signal level | 1 | 2 | 2 |
| Note: For GPS capable receivers, GNSS-1, i.e. the system having the satellite with high signal level, shall be GPS. | | | | |

**Table 6.2.1.5-3: Test requirements for Sensitivity Coarse Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

## 6.2.2 Sensitivity Fine Time Assistance

### 6.2.2.1 Definition and applicability

Sensitivity with fine time assistance is the minimum level of GNSS satellite signals required for the UE to make an A-GNSS position estimate to a specific accuracy and within a specific response time when the network provides fine time assistance in addition to coarse time assistance.

The requirements and this test apply to all types of UTRA for the UE that supports A-GNSS and that is capable of providing an enhanced performance when the network provides Fine Time Assistance.

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 6.2.2.1.

**Table 6.2.2.1: Sub-Test Case Number Definition**

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS |
| 2 | UE supporting A-Galileo |
| 3 | UE supporting A-GPS and Modernized GPS |
| 4 | UE supporting A-GPS and A-GLONASS |
| 8 | UE supporting A-GPS and A-Galileo |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS |

### 6.2.2.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 6.2.2.2-3 for the parameters specified in table 6.2.2.2-1.

**Table 6.2.2.2-1: Test parameters for Sensitivity Fine Time Assistance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.2.2.2-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| | GNSS fine time assistance error range | µs | ±10 |
| Galileo | Reference signal power level | dBm | -147 |
| GPS(1) | Reference signal power level | dBm | -147 |
| GLONASS | Reference signal power level | dBm | -147 |
| BDS | Reference signal power level | dBm | -147 |

| System | Parameters | Unit | Value |
|---|---|---|---|
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.2.2.2-2: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | 6 | - | - |
| Dual constellation | 3 | 3 | - |
| Triple constellation | 2 | 2 | 2 |

**Table 6.2.2.2-3: Minimum requirements for Sensitivity Fine Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.172 [19], clause 5.1.2.1, and 3GPP TS 25.173 [36], clause 5.1.2.1.

## 6.2.2.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GNSS satellite signal conditions that represent weak signal conditions and with Fine Time Assistance provided by the SS.

## 6.2.2.4 Method of test

### 6.2.2.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GNSS test parameters as specified in table 6.2.2.5-1 for GNSS scenario #1.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 6.2.2.4.2 Procedure

1. Start GNSS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 6.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 6.2.1.2.6.
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 6.2.2 and 6.2.7 for UE based testing; or clauses 6.2.4 and 6.2.7 for UE assisted testing with the values of GPS TOW msec or GANSS TOD, and UTRAN GPS timing of cell frames or UTRAN GANSS timing of cell frames offset by random values as specified in 3GPP TS 37.571-5 [20] clause 6.2.7.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.6 or 7.5.8.
3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 6.2.2.5-3 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 6.2.2.5-3 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause

6.1.1.3. Compare the 2D position error against the value in table 6.2.2.5-3 and record one Good Result or Bad Result as appropriate; or

For UE assisted testing convert the GNSS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.2.2.5-3 and record one Good Result or Bad Result as appropriate.

5. Repeat steps 1 to 4 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec or GANSS TOD, and UTRAN GPS timing of cell frames or UTRAN GANSS timing of cell frames offsets in step 2.

6. Repeat steps 1 to 5 until the statistical requirements of clause 6.2.2.5 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.

7. Release the connection using the procedure in clause F.3.

## 6.2.2.5 Test Requirements

For the parameters specified in table 6.2.2.5-1 the UE shall meet the requirements and the success rate specified in table 6.2.2.5-3 with a confidence level of 95% according to Annex D.

**Table 6.2.2.5-1: Test parameters for Sensitivity Fine Time Assistance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.2.2.5-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| | GNSS fine time assistance error range | μs | ±9 |
| Galileo | Reference signal power level | dBm | -146 |
| GPS(1) | Reference signal power level | dBm | -146 |
| GLONASS | Reference signal power level | dBm | -146 |
| BDS | Reference signal power level | dBm | -146 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.2.2.5-2: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | 6 | - | - |
| Dual constellation | 3 | 3 | - |
| Triple constellation | 2 | 2 | 2 |

**Table 6.2.2.5-3: Test requirements for Sensitivity Fine Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

# 6.3 Nominal Accuracy

## 6.3.1 Definition and applicability

Nominal accuracy is the accuracy of the UE's A-GNSS position estimate under ideal GNSS signal conditions.

The requirements and this test apply to all types of UTRA for the UE that supports A-GNSS.

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 6.3.1.

**Table 6.3.1: Sub-Test Case Number Definition**

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS |
| 2 | UE supporting A-Galileo |
| 3 | UE supporting A-GPS and Modernized GPS |
| 4 | UE supporting A-GPS and A-GLONASS |
| 8 | UE supporting A-GPS and A-Galileo |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS |

## 6.3.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 6.3.2-3 for the parameters specified in table 6.3.2-1.

**Table 6.3.2-1: Test parameters for Nominal Accuracy**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.3.2-2 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| SBAS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| Note 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>Note 2: 7 satellites apply only for SBAS case. | | | |

If SBAS is supported, the SBAS satellite with the highest elevation will be added to the scenario.

**Table 6.3.2-2: Satellite allocation**

| | Satellite allocation for each constellation | | | |
|---|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) | SBAS |
| Single constellation | 6 | -- | -- | 1 |
| Dual constellation | 3 | 3 | -- | 1 |
| Triple constellation | 2 | 2 | 2 | 1 |
| Note: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 6.3.2-3: Minimum requirements for Nominal Accuracy**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 15 m | 20 s |

The reference for this requirement is 3GPP TS 25.172 [19], clause 5.2.1, and 3GPP TS 25.173 [36], clause 5.2.1.

## 6.3.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GNSS satellite signal conditions that represent ideal conditions.

## 6.3.4 Method of test

### 6.3.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GNSS test parameters as specified in table 6.3.4.2 for GNSS scenario #3.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 6.3.4.2 Procedure

1. Start GNSS scenario #3 as specified in 3GPP TS 37.571-5 [20] clause 6.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 6.2.1.2.6.
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 6.2.2 and 6.2.7 for UE based testing; or clauses 6.2.4 and 6.2.7 for UE assisted testing with the value of GPS TOW msec or GANSS TOD offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 6.2.7.2; using the exception to the RRC MEASUREMENT CONTROL message listed in table 6.3.5-1; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.6 or 7.5.8.

**Table 6.3.4.2: Contents of RRC MEASUREMENT CONTROL message**

| Information Element | Value/Remark |
|---|---|
| - UE positioning reporting quantity<br>- Horizontal accuracy | <br>'6' (7.7m) |

3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 6.3.5-3 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 6.3.5-3 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.3.5-3 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 6.1.1.3.

Compare the 2D position error against the value in table 6.3.5-3 and record one Good Result or Bad Result as appropriate.

5. Repeat steps 1 to 4 using GNSS scenario #4 instead of #3 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec or GANSS TOD offset in step 2.

6. Repeat steps 1 to 5 until the statistical requirements of clause 6.3.5 are met. Each time scenario #3 or #4 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.

7. Release the connection using the procedure in clause F.3.

## 6.3.5 Test Requirements

For the parameters specified in table 6.3.5-1 the UE shall meet the requirements and the success rate specified in table 6.3.5-3 with a confidence level of 95% according to Annex D.

**Table 6.3.5-1: Test parameters for Nominal Accuracy**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.3.5-3 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| SBAS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| Note 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>Note 2: 7 satellites apply only for SBAS case. | | | |

If SBAS is supported, the SBAS satellite with the highest elevation will be added to the scenario.

**Table 6.3.5-2: Satellite allocation**

| | Satellite allocation for each constellation | | | |
|---|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) | SBAS |
| Single constellation | 6 | -- | -- | 1 |
| Dual constellation | 3 | 3 | -- | 1 |
| Triple constellation | 2 | 2 | 2 | 1 |
| Note: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 6.3.5-3: Test requirements for Nominal Accuracy**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 16.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

# 6.4 Dynamic Range

## 6.4.1 Definition and applicability

Dynamic Range is the maximum difference in level of the GNSS signals from a number of satellites that allows the UE to make an A-GNSS position estimate with a specific accuracy and a specific response time.

The requirements and this test apply to all types of UTRA for the UE that supports A-GNSS.

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 6.4.1.

**Table 6.4.1: Sub-Test Case Number Definition**

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS |
| 2 | UE supporting A-Galileo |
| 3 | UE supporting A-GPS and Modernized GPS |
| 4 | UE supporting A-GPS and A-GLONASS |
| 8 | UE supporting A-GPS and A-Galileo |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS |

## 6.4.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 6.4.2-3 for the parameters specified in table 6.4.2-1.

**Table 6.4.2-1: Test parameters for Dynamic Range**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.4.2-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference high signal power level | dBm | -127.5 |
| | Reference low signal power level | dBm | -147 |
| GPS(1) | Reference high signal power level | dBm | -129 |
| | Reference low signal power level | dBm | -147 |
| GLONASS | Reference high signal power level | dBm | -131.5 |
| | Reference low signal power level | dBm | -147 |
| BDS | Reference high signal power level | dBm | -133.5 |
| | Reference low signal power level | dBm | -145 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.4.2-2: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | High signal level | 2 | -- | -- |
| | Low signal level | 4 | -- | -- |
| Dual constellation | High signal level | 1 | 1 | -- |
| | Low signal level | 2 | 2 | -- |
| Triple constellation | High signal level | 1 | 1 | 1 |
| | Low signal level | 1 | 1 | 1 |
| Note: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 6.4.2-3: Minimum requirements for Dynamic Range**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.172 [19], clause 5.3.1, and 3GPP TS 25.173 [36], clause 5.3.1.

## 6.4.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GNSS satellite signal conditions that have a wide dynamic range. Strong satellites are likely to degrade the acquisition of weaker satellites due to their cross-correlation products.

## 6.4.4 Method of test

### 6.4.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GNSS test parameters as specified in table 6.4.5-1 for GNSS scenario #1. Randomly select from the satellite SV IDs defined in the table in 3GPP TS 37.571-5 [20] clause 6.2.1.2 for the satellites with the higher levels.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 6.4.4.2 Procedure

1. Start GNSS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 6.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 6.2.1.2.6.
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 6.2.2 and 6.2.7 for UE based testing; or clauses 6.2.4 and 6.2.7 for UE assisted testing with the value of GPS TOW msec or GANSS TOD offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 6.2.7.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.6 or 7.5.8.
3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 6.4.5-3 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 6.4.5-3 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.4.5-3 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.4.5-3 and record one Good Result or Bad Result as appropriate.
5. Repeat steps 1 to 4 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Randomly select from the satellite SV IDs defined in the

table in 3GPP TS 37.571-5 [20] clause 6.2.1.2 for the satellites with the higher levels. Use new random values for the UE location and altitude in step 1 and for the GPS TOW msec or GANSS TOD offset in step 2.

6. Repeat steps 1 to 5 until the statistical requirements of clause 6.4.5 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, randomly select from the set of satellite SV IDs defined in the table in 3GPP TS 37.571-5 [20] clause 6.2.1.2, for the satellites with the higher levels.

7. Release the connection using the procedure in clause F.3.

## 6.4.5 Test Requirements

For the parameters specified in table 6.4.5-1 the UE shall meet the requirements and the success rate specified in table 6.4.5-3 with a confidence level of 95% according to Annex D.

**Table 6.4.5-1: Test parameters for Dynamic Range**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.4.5-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference high signal power level | dBm | -126.7 |
| | Reference low signal power level | dBm | -146 |
| GPS(1) | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| GLONASS | Reference high signal power level | dBm | -130.7 |
| | Reference low signal power level | dBm | -146 |
| BSD | Reference high signal power level | dBm | -132.7 |
| | Reference low signal power level | dBm | -144 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.4.5-2: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | High signal level | 2 | -- | -- |
| | Low signal level | 4 | -- | -- |
| Dual constellation | High signal level | 1 | 1 | -- |
| | Low signal level | 2 | 2 | -- |
| Triple constellation | High signal level | 1 | 1 | 1 |
| | Low signal level | 1 | 1 | 1 |
| Note: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 6.4.5-3: Test requirements for Dynamic Range**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

# 6.5 Multi-path Performance

## 6.5.1 Definition and applicability

Multi-path performance measures the accuracy and response time of the UE's A-GNSS position estimate in a specific GNSS signal multi-path environment.

The requirements and this test apply to all types of UTRA for the UE that supports A-GNSS.

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 6.5.1.

**Table 6.5.1: Sub-Test Case Number Definition**

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS |
| 2 | UE supporting A-Galileo |
| 3 | UE supporting A-GPS and Modernized GPS |
| 4 | UE supporting A-GPS and A-GLONASS |
| 8 | UE supporting A-GPS and A-Galileo |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS |

## 6.5.2 Minimum requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 6.5.2-3 for the parameters specified in table 6.5.2-1.

**Table 6.5.2-1: Test parameters for Multi-path Performance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.5.2-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference signal power level | dBm | -127 |
| GPS(1) | Reference signal power level | dBm | -128.5 |
| GLONASS | Reference signal power level | dBm | -131 |
| BDS | Reference signal power level | dBm | -133 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.5.2-2: Channel model allocation**

| | | Channel model allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | One-tap channel | 2 | -- | -- |
| | Two-tap channel | 4 | -- | -- |
| Dual constellation | One-tap channel | 1 | 1 | -- |
| | Two-tap channel | 2 | 2 | -- |
| Triple constellation | One-tap channel | 1 | 1 | 1 |
| | Two-tap channel | 1 | 1 | 1 |
| Note: One-tap channel: no multi-path. Two-tap channel: multi-path defined in clause 4.2.4 | | | | |

**Table 6.5.2-3: Minimum requirements for Multi-path Performance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 100 m | 20 s |

The reference for this requirement is 3GPP TS 25.172 [19], clause 5.4.1, and 3GPP TS 25.173 [36], clause 5.4.1.

## 6.5.3 Test purpose

To verify the UE's first position estimate meets the minimum requirements under GNSS satellite signal conditions that represent simple multi-path conditions.

## 6.5.4 Method of test

### 6.5.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GNSS test parameters as specified in table 6.5.5-1 for GNSS scenario #1. Randomly select from the satellite SV IDs defined in the table in 3GPP TS 37.571-5 [20] clause 6.2.1.2 for the satellites with one-tap channel.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 6.5.4.2 Procedure

1. Start GNSS scenario #1 as specified in 3GPP TS 37.571-5 [20] clause 6.2.1.2 with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in 3GPP TS 37.571-5 [20] clause 6.2.1.2.6. The initial carrier phase difference between taps of the multi-path model shall be randomly selected between 0 and $2\pi$ radians by selecting the next random number from a standard uniform random number generator, in the range 0 to $2\pi$, representing radians with a resolution of 0.1, representing 0.1 radians.
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 6.2.2 and 6.2.7 for UE based testing; or clauses 6.2.4 and 6.2.7 for UE assisted testing with the value of GPS TOW msec or GANSS TOD offset by a random value as specified in 3GPP TS 37.571-5 [20] clause 6.2.7.2; as required to obtain a fix using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.6 or 7.5.8.
3. If the UE returns a valid result in the MEASUREMENT REPORT message within the Max response time specified in table 6.5.5-4 then record the result and process it as specified in step 4. If the UE does not return a valid result within the Max response time specified in table 6.5.5-4 or reports a UE positioning error in the MEASUREMENT REPORT message then record one Bad Result.
4. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.5.5-4 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.5.5-4 and record one Good Result or Bad Result as appropriate.
5. Repeat steps 1 to 4 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Randomly select from the satellite SV IDs defined in the

table in 3GPP TS 37.571-5 [20] clause 6.2.1.2 for the satellites with one-tap channel. Use new random values for the UE location and altitude, and the initial carrier phase difference between taps of the multi-path model in step 1 and for the GPS TOW msec or GANSS TOD offset in step 2.

6. Repeat steps 1 to 5 until the statistical requirements of clause 6.5.5 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, randomly select from the satellite SV IDs defined in the table in 3GPP TS 37.571-5 [20] clause 6.2.1.2, for the satellites with one-tap channel.

7. Release the connection using the procedure in clause F.3.

## 6.5.5 Test Requirements

For the parameters specified in table 6.5.5-1 the UE shall meet the requirements and the success rate specified in table 6.5.5-4 with a confidence level of 95% according to Annex D.

**Table 6.5.5-1: Test parameters for Multi-path Performance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.5.5-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| Galileo | Reference signal power level | dBm | -127 |
| GPS(1) | Reference signal power level | dBm | -128.5 |
| GLONASS | Reference signal power level | dBm | -131 |
| BDS | Reference signal power level | dBm | -133 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.5.5-2: Channel model allocation**

| | | Channel model allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | One-tap channel | 2 | -- | -- |
| | Two-tap channel | 4 | -- | -- |
| Dual constellation | One-tap channel | 1 | 1 | -- |
| | Two-tap channel | 2 | 2 | -- |
| Triple constellation | One-tap channel | 1 | 1 | 1 |
| | Two-tap channel | 1 | 1 | 1 |
| Note: One-tap channel: no multi-path. Two-tap channel: multi-path defined in clause 4.2.4 with Relative mean Power (Y) defined in Table 6.5.5-3. | | | | |

**Table 6.5.5-3: Relative mean Power (Y) for use in Table 6.5.5-2**

| System | Signals | Y [dB] |
|---|---|---|
| Galileo | E1 | -4.7 |
| | E5a | -6.2 |
| | E5b | -6.2 |
| GPS/Modernized GPS | L1 C/A | -6.2 |
| | L1C | -4.7 |
| | L2C | -6.2 |
| | L5 | -6.2 |
| GLONASS | G1 | -12.7 |
| | G2 | -12.7 |
| BDS | B1I | -4.7 |

**Table 6.5.5-4: Test requirements for Multi-path Performance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

# 6.6 Moving Scenario and Periodic Update Performance

## 6.6.1 Definition and applicability

Moving scenario and periodic update performance measures the accuracy of the UE's A-GNSS position estimates and the periodic update capability of the UE in a moving scenario.

The requirements and this test apply to all types of UTRA for the UE that supports A-GNSS.

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 6.6.1.

**Table 6.6.1: Sub-Test Case Number Definition**

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS |
| 2 | UE supporting A-Galileo |
| 3 | UE supporting A-GPS and Modernized GPS |
| 4 | UE supporting A-GPS and A-GLONASS |
| 8 | UE supporting A-GPS and A-Galileo |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS |

## 6.6.2 Minimum requirements

The position estimates, after the first reported position estimate, shall meet the accuracy requirement in table 6.6.2-3 with the periodical reporting interval of 2 seconds for the parameters specified in table 6.6.2-1.

NOTE: In the actual testing the UE may report error messages until it has been able to acquire GNSS measured results or a position estimate. The SS shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 6.6.2-1.

**Table 6.6.2-1: Test parameters for Moving Scenario and Periodic Update Performance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.6.2-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP Range per system | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.6.2-2: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS 1[(1)] | GNSS 2[(1)] | GNSS 3[(1)] |
| Single constellation | 6 | -- | -- |
| Dual constellation | 3 | 3 | -- |
| Triple constellation | 2 | 2 | 2 |
| Note: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | |

**Table 6.6.2-3: Minimum requirements for Moving Scenario and Periodic Update Performance**

| System | Success rate | 2-D position error | Periodical reporting interval |
|---|---|---|---|
| All | 95 % | 50 m | 2 s |

The reference for this requirement is 3GPP TS 25.172 [19], clause 5.5.1, and 3GPP TS 25.173 [36], clause 5.5.1.

## 6.6.3 Test purpose

To verify the UE's position estimates, after the first reported position estimate, meet the minimum requirements under GNSS satellite signal conditions that simulate a moving scenario. A good tracking performance, with regular position estimate reporting is essential for certain location services.

## 6.6.4 Method of test

### 6.6.4.1 Initial conditions

Test environment: normal; see Annex G.

The UE is requested to use periodical reporting with a reporting interval of 2 seconds.

The GNSS signals simulate the UE moving on a rectangular trajectory of 940 m by 1 440 m with rounded corners defined in figure 6.6.1 and table 6.6.4.1. The initial reference is first defined followed by acceleration to final speed of 100 km/h in 250 m. The UE then maintains the speed for 400 m. This is followed by deceleration to final speed of 25 km/h in 250 m. The UE then turn 90 degrees with turning radius of 20 m at 25 km/h. This is followed by acceleration to final speed of 100 km/h in 250 m. The sequence is repeated to complete the rectangle.

**Table 6.6.4.1: Trajectory Parameters for Moving Scenario and Periodic Update Performance test case**

| Parameter | Distance (m) | Speed (km/h) |
|---|---|---|
| $l_{11}$, $l_{15}$, $l_{21}$, $l_{25}$ | 20 | 25 |
| $l_{12}$, $l_{14}$, $l_{22}$, $l_{24}$ | 250 | 25 to 100 and 100 to 25 |
| $l_{13}$ | 400 | 100 |
| $l_{23}$ | 900 | 100 |

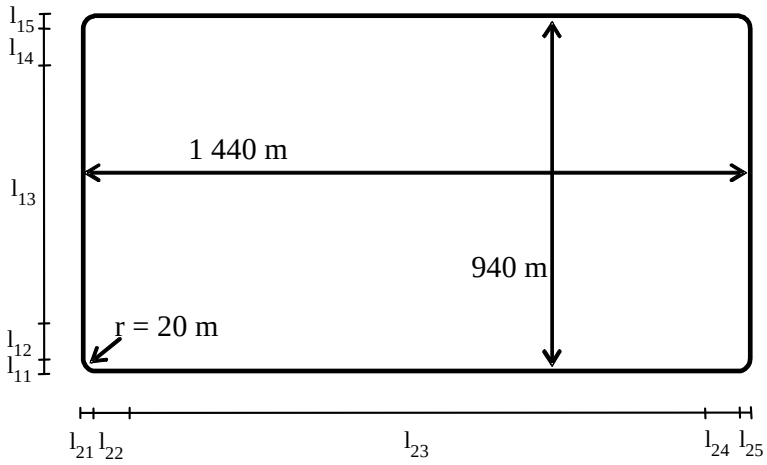



**Figure 6.6.1: Rectangular Trajectory for Moving Scenario and Periodic Update Performance test case**

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in figures A.1 or A.2.
2. Set the GNSS test parameters as specified in table 6.6.5-1 for GNSS scenario #5.
3. Switch on the UE.
4. Set up a connection using the procedure in clause F.2.

### 6.6.4.2 Procedure

1. Start GNSS scenario #5 as specified in 3GPP TS 37.571-5 [20], clause 6.2.1.2.
2. Send a RESET UE POSITIONING STORED INFORMATION message followed by RRC MEASUREMENT CONTROL messages containing appropriate assistance data; as specified in 3GPP TS 37.571-5 [20], clauses 6.2.2 and 6.2.7 for UE based testing; or clauses 6.2.4 and 6.2.7 for UE assisted testing; using the exception to the RRC MEASUREMENT CONTROL message listed in table 6.6.4.2; as required to obtain fixes using the procedure specified in 3GPP TS 34.108 [28], clauses 7.5.7 or 7.5.9.

**Table 6.6.4.2: Contents of RRC MEASUREMENT CONTROL message**

| Information Element | Value/Remark |
|---|---|
| - UE positioning reporting quantity<br>- Horizontal accuracy | <br>'13' (24.5m) |

3. Ignore any error messages that the UE may report in MEASUREMENT REPORT messages until it has been able to acquire the GNSS signals and reports the first GNSS measured result or position estimate.
4. Discard the first GNSS measured result or position estimate.
5. Record the time of reception of the next MEASUREMENT REPORT message after reception of the first GNSS measured result or position estimate.
6. After the reception of the first GNSS measured result or position estimate reported in a MEASUREMENT REPORT message, every time the UE returns a GNSS measured result or position estimate in the MEASUREMENT REPORT message record the time of reception and the result. If the difference between the time of reception and the time of reception of the previous result is less than 1.5 seconds or greater than 2.5 seconds, or if the UE reports a UE positioning error in any MEASUREMENT REPORT messages, then record one Bad Result. Otherwise process the result as specified in step 7.
7. For UE based testing compare the reported position estimate in the MEASUREMENT REPORT message against the simulated position of the UE at the time of applicability reported in the position estimate and

calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.6.9 and record one Good Result or Bad Result as appropriate; or

For UE assisted testing convert the GNSS measured results reported in the MEASUREMENT REPORT message to a 2D position using the method described in Annex B and then compare the result against the simulated position of the UE at the time of applicability reported in the GNSS measured results and calculate the 2D position error as specified in clause 6.1.1.3. Compare the 2D position error against the value in table 6.6.9 and record one Good Result or Bad Result as appropriate.

8. If the UE sends the first MEASUREMENT REPORT that contains a measured result or position estimate later than 240s after the start of the GNSS scenario, fail the UE and stop the test early. Otherwise collect MEASUREMENT REPORTs during 900s, starting from the time recorded in step 5. If at any time the difference between the times of reception of two consecutive results is greater than 240s, fail the UE and stop the test early. Use the collected Good Results and Bad Results to determine the PASS/FAIL according to clause 6.6.5.

9. Release the connection using the procedure in clause F.3.

## 6.6.5 Test Requirements

For the parameters specified in table 6.6.5-1, after the first reported position estimate, the UE shall meet the accuracy requirement and the success rate specified in table 6.6.5-3 with a periodical reporting interval of 2 seconds +/- 20% plus measurement system uncertainty of 100ms.

NOTE: Due to the statistical nature of the results it is not possible to design a test with predefined confidence level for the success rate in Table 6.6.5-3; therefore a simple PASS/FAIL of the results gathered against this success rate is used.

**Table 6.6.5-1: Test parameters for Moving Scenario and Periodic Update Performance**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 6.6.5-2 |
| | Total number of generated satellites | - | 6 |
| | HDOP Range per system | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

**Table 6.6.5-2: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | 6 | -- | -- |
| Dual constellation | 3 | 3 | -- |
| Triple constellation | 2 | 2 | 2 |
| Note: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | |

**Table 6.6.5-3: Test requirements for Moving Scenario and Periodic Update Performance**

| System | Success rate | 2-D position error |
|---|---|---|
| All | 95 % | 51.3 m |

NOTE 1: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause C.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause C.4.

NOTE 2: In the actual testing the UE may report error messages until it has been able to acquire GNSS measured results or a position estimate. The test equipment shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 6.6.5-3.

# 7 E-UTRA A-GNSS minimum performance requirements

## 7.0 General

This clause defines the minimum performance requirements for both UE based and UE assisted A-GNSS FDD and TDD E-UTRA UEs. If a UE supports both UE based and UE assisted modes then it shall be tested in both modes.

## 7.1 Sensitivity

### 7.1.1 Sensitivity Coarse time assistance

#### 7.1.1.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 7.1.1.1

**Table 7.1.1.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS |
| 15 | UE supporting A-GPS and A-NAVIC |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

#### 7.1.1.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with only coarse time assistance.

#### 7.1.1.3 Test applicability

This test applies to all types of E-UTRA UE that supports A-GNSS except Category M1 and Category M2 devices that do not support VoLTE.

#### 7.1.1.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 7.1.1.2 for the parameters specified in table 7.1.1.3 or 7.1.1.4.

**Table 7.1.1.2: Requirements for Sensitivity Coarse time assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 7.1.1.3: Parameters for Sensitivity Coarse time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse time assistance error range | seconds | ±2 |
| GPS L1 C/A Signal for one satellite | dBm | -142 |
| GPS L1 C/A Signal for remaining satellites | dBm | -147 |

**Table 7.1.1.4: Parameters for Sensitivity Coarse time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.1.1.5 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| GPS(1) | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| GLONASS | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| BDS | Reference high signal power level | dBm | -136 |
| | Reference low signal power level | dBm | -145 |
| QZSS | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| NAVIC | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.1.1.5: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1(1) | GNSS-2 | GNSS-3 |
| Single constellation | High signal level | 1 | - | - |
| | Low signal level | 5 | - | - |
| Dual constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 3 | - |
| Triple constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 2 | 2 |
| Note 1: Up to Rel-14: for GPS capable receivers, GNSS-1, i.e. the system having the satellite with high signal level, shall be GPS.<br>Rel-15 onwards: GNSS-1, i.e. the system having the satellite with high signal level, shall be selected by the device manufacturer. | | | | |

The normative reference for this requirement is TS 36.171 [3] clause 5.1.1 and 6.1.1.

## 7.1.1.5 Test description

### 7.1.1.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 36.508 [18] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 7.1.1.6 or 7.1.1.7 for GNSS scenario #1 in TS 37.571-5 [20]. For GNSS-1, select the first satellite SV ID defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the one satellite with the higher level.
3. The parameter settings for the cell are set up according to TS 36.508 [18] clause 4.4.3, single cell scenario.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 7.1.1.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]
2. Send a RESET UE POSITIONING STORED INFORMATION message.
3. Depending on how the LPP session is performed:

- In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

- In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.
5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.
7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 7.1.1.9 then record the result and process it as specified in step 8. If the UE does not return a valid result within the Max response time specified in table 7.1.1.9 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

7a. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

8. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.1.1.9 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Signal Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE, used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.1.1.9 and record one Good Result or Bad Result as appropriate.

9. Repeat steps 1 to 8 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. For GNSS-1, select the first satellite SV ID defined in the

relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the one satellite with the higher level. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time offset in step 5.

10. Repeat steps 1 to 9 until the statistical requirements of clause 7.1.1.6 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, for GNSS-1 select the next satellite SV ID from the one used previously, defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20], for the one satellite with the higher level.

11. Release the signalling connection.

#### 7.1.1.5.3 Message contents

Message contents are according to TS 36.508 [18] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| a-gnss-RequestCapabilities | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy | '19' (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |

| | | |
|---|---|---|
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

### 7.1.1.6 Test requirement

For the parameters specified in table 7.1.1.6 or 7.1.1.7 the UE shall meet the requirements and the success rate specified in table 7.1.1.9 with a confidence level of 95% according to Annex D.

**Table 7.1.1.6: Test parameters for Sensitivity Coarse time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| GPS L1 C/A Signal for one satellite | dBm | -141 |
| GPS L1 C/A Signal for remaining satellites | dBm | -146 |

**Table 7.1.1.7: Test parameters for Sensitivity Coarse time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.1.1.8 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| Galileo | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| GPS(1) | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| GLONASS | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| BDS | Reference high signal power level | dBm | -135 |
| | Reference low signal power level | dBm | -144 |
| QZSS | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| Navic | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.1.1.8: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1(1) | GNSS-2 | GNSS-3 |
| Single constellation | High signal level | 1 | - | - |
| | Low signal level | 5 | - | - |
| Dual constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 3 | - |
| Triple constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 2 | 2 |

| | |
|---|---|
| Note 1: | Up to Rel-14: for GPS capable receivers, GNSS-1, i.e. the system having the satellite with high signal level, shall be GPS.<br>Rel-15 onwards: GNSS-1, i.e. the system having the satellite with high signal level, shall be selected by the device manufacturer. |

**Table 7.1.1.9: Test requirements for Sensitivity Coarse Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

## 7.1.2 Sensitivity Fine time assistance

### 7.1.2.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 7.1.2.1

**Table 7.1.2.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

### 7.1.2.2 Test purpose

To verify the performance of the first position estimate, when the UE is additionally provided with fine time assistance.

### 7.1.2.3 Test applicability

This test applies to all types of E-UTRA UE that supports A-GNSS and that is capable of providing an enhanced performance when the network provides Fine Time Assistance, except Category M1 and Category M2 devices that do not support VoLTE.

### 7.1.2.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 7.1.2.2 for the parameters specified in table 7.1.2.3 or 7.1.2.4.

**Table 7.1.2.2: Requirements for Sensitivity Fine time assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 7.1.2.3: Parameters for Sensitivity Fine time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse time assistance error range | seconds | ±2 |
| GPS L1 C/A Fine time assistance error range | µs | ±10 |
| GPS L1 C/A Signal for all satellites | dBm | -147 |

**Table 7.1.2.4: Parameters for Sensitivity Fine time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.1.2.5 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| | GNSS fine time assistance error range | µs | ±10 |
| Galileo | Reference signal power level | dBm | -147 |
| GPS(1) | Reference signal power level | dBm | -147 |
| GLONASS | Reference signal power level | dBm | -147 |
| BDS | Reference signal power level | dBm | -147 |
| QZSS | Reference signal power level | dBm | -147 |
| Navic | Reference signal power level | dBm | -147 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.1.2.5: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | 6 | - | - |
| Dual constellation | 3 | 3 | - |
| Triple constellation | 3 | 2 | 2 |

The normative reference for this requirement is TS 36.171 [3] clause 5.1.2 and 6.1.2.

## 7.1.2.5 Test description

### 7.1.2.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 36.508 [18] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 7.1.2.6 or 7.1.2.7 for GNSS scenario #1 in TS 37.571-5 [20].
3. The parameter settings for the cell are set up according to TS 36.508 [18] clause 4.4.3, single cell scenario.
4. Switch on the UE.

5. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

#### 7.1.2.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]

2. Send a RESET UE POSITIONING STORED INFORMATION message.

3. Depending on how the LPP session is performed:

- In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

- In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.

5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time and GNSS Reference Time for one cell offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 7.1.2.9 then record the result and process it as specified in step 8. If the UE does not return a valid result within the Max response time specified in table 7.1.2.9 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

7a. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

8. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.1.2.9 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.1.2.9 and record one Good Result or Bad Result as appropriate.

9. Repeat steps 1 to 8 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time GNSS Reference Time for one cell offsets in step 5.

10. Repeat steps 1 to 9 until the statistical requirements of clause 7.1.2.6 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.

11. Release the signalling connection.

#### 7.1.2.5.3 Message contents

Message contents are according to TS 36.508 [18] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| a-gnss-RequestCapabilities | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy | '19' (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 7.1.2.6 Test requirement

For the parameters specified in table 7.1.2.6 or 7.1.2.7 the UE shall meet the requirements and the success rate specified in table 7.1.2.9 with a confidence level of 95% according to Annex D.

**Table 7.1.2.6: Test parameters for Sensitivity Fine time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |

| Parameters | Unit | Value |
|---|---|---|
| GPS Coarse time assistance error range | seconds | ±1.8 |
| GPS Fine Time assistance error range | µs | ±9 |
| GPS L1 C/A Signal for all satellites | dBm | -146 |

**Table 7.1.2.7: Test parameters for Sensitivity Fine time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.1.2.8 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| | GNSS fine time assistance error range | µs | ±9 |
| Galileo | Reference signal power level | dBm | -146 |
| GPS(1) | Reference signal power level | dBm | -146 |
| GLONASS | Reference signal power level | dBm | -146 |
| BDS | Reference signal power level | dBm | -146 |
| QZSS | Reference signal power level | dBm | -146 |
| Navic | Reference signal power level | dBm | -146 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.1.2.8: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | 6 | - | - |
| Dual constellation | 3 | 3 | - |
| Triple constellation | 3 | 2 | 2 |

**Table 7.1.2.9: Test requirements for Sensitivity Fine Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

# 7.2 Nominal Accuracy

## 7.2.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 7.2.1

**Table 7.2.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |

| | |
|---|---|
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

## 7.2.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with ideal GNSS signal conditions.

## 7.2.3 Test applicability

This test applies to all types of E-UTRA UE that supports A-GNSS, except Category M1 and Category M2 devices that do not support VoLTE.

## 7.2.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 7.2.2 or 7.2.3 for the parameters specified in table 7.2.4 or 7.2.5.

**Table 7.2.2: Requirements for Nominal Accuracy - Sub-Test 1**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 30 m | 20 s |

**Table 7.2.3: Requirements for Nominal Accuracy - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 15 m | 20 s |

**Table 7.2.4: Parameters for Nominal Accuracy - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±2 |
| GPS L1 C/A Signal for all satellites | dBm | -130 |

**Table 7.2.5: Parameters for Nominal Accuracy - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.2.6 |
| | Total number of generated satellites | - | 6, 7(2) or 8(3) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| SBAS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |

| System | Parameters | Unit | Value |
|---|---|---|---|
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites apply only for SBAS case or sub-tests with 3 different GNSSs.<br>NOTE 3: 8 satellites apply only for sub-tests with 3 different GNSSs and SBAS. | | | |

If SBAS is supported, the SBAS satellite with the highest elevation will be added to the scenario.

**Table 7.2.6: Satellite allocation**

| | Satellite allocation for each constellation | | | |
|---|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) | SBAS |
| Single constellation | 6 | -- | -- | 1 |
| Dual constellation | 3 | 3 | -- | 1 |
| Triple constellation | 3 | 2 | 2 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

The normative reference for this requirement is TS 36.171 [3] clause 5.2 and 6.2.

## 7.2.5 Test description

### 7.2.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 36.508 [18] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 7.2.7 or 7.2.8 for GNSS scenario #3 in TS 37.571-5 [20].
3. The parameter settings for the cell are set up according to TS 36.508 [18] clause 4.4.3, single cell scenario.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 7.2.5.2 Test procedure

1. Start GNSS scenario #3 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]
2. Send a RESET UE POSITIONING STORED INFORMATION message.
3. Depending on how the LPP session is performed:

- In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

- In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.
5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as

specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the (first) LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 7.2.10 then record the result and process it as specified in step 8. If the UE does not return a valid result within the Max response time specified in table 7.2.10 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

7a. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

8. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.2.10 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table7.2.10 and record one Good Result or Bad Result as appropriate.

9. Repeat steps 1 to 8 using GNSS scenario #4 instead of #3 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time offset in step 5.

10. Repeat steps 1 to 9 until the statistical requirements of clause 7.2.6 are met. Each time scenario #3 or #4 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.

11. Release the signalling connection.

### 7.2.5.3 Message contents

Message contents are according to TS 36.508 [18] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| a-gnss-RequestCapabilities | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | ‘locationEstimateRequired’ or ‘locationMeasurementsRequired’ | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | ‘onlyReturnInformationRequested’ | |
| > qos | | |
| >> horizontalAccuracy (Sub-Test 1) | ’10’ (15.9m) | |
| >> horizontalAccuracy (Sub-Tests 2 to 5 and 8 to 13) | ’6’ (7.7m) | |

| | | |
|---|---|---|
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps' and possibly 'sbas'<br>Sub-test 2: 'glonass' and possibly 'sbas'<br>Sub-test 3: 'galileo' and possibly 'sbas'<br>Sub-test 4: 'gps' and possibly 'sbas'<br>Sub-test 5: 'gps' and 'glonass' and possibly 'sbas'<br>Sub-test 8: 'gps' and 'galileo' and possibly 'sbas'<br>Sub-test 9: 'bds' and possibly 'sbas'<br>Sub-test 10: 'gps'and'bds'and possibly 'sbas'<br>Sub-test 11: 'gps' and 'glonass' and 'bds' and possibly 'sbas'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | Depending on UE capabilities |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 7.2.6 Test requirement

For the parameters specified in table 7.2.7 or 7.2.8 the UE shall meet the requirements and the success rate specified in table 7.2.10 or 7.2.11 with a confidence level of 95% according to Annex D.

**Table 7.2.7: Test parameters Nominal Accuracy - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| GPS L1 C/A Signal for all satellites | dBm | -130 |

**Table 7.2.8: Test parameters Nominal Accuracy - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.2.9 |
| | Total number of generated satellites | - | 6, 7(2) or 8(3) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |

| System | Parameters | Unit | Value |
|---|---|---|---|
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| SBAS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| Navic | Reference signal power level for all satellites | dBm | -128.5 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites apply only for SBAS case or sub-tests with 3 different GNSSs.<br>NOTE 3: 8 satellites apply only for sub-tests with 3 different GNSSs and SBAS. | | | |

If SBAS is supported, the SBAS satellite with the highest elevation will be added to the scenario.

**Table 7.2.9: Satellite allocation**

| | Satellite allocation for each constellation | | | |
|---|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) | SBAS |
| Single constellation | 6 | - | - | 1 |
| Dual constellation | 3 | 3 | - | 1 |
| Triple constellation | 3 | 2 | 2 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 7.2.10: Test requirements for Nominal Accuracy – Sub-Test 1**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 31.3 m | 20.3 s |

**Table 7.2.11: Test requirements for Nominal Accuracy – Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 16.3 m | 20.3 s |

# 7.3 Dynamic Range

## 7.3.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 7.3.1

**Table 7.3.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

## 7.3.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with GNSS signals with large dynamic ranges.

## 7.3.3 Test applicability

This test applies to all types of E-UTRA UE that supports A-GNSS, except Category M1 and Category M2 devices that do not support VoLTE.

## 7.3.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 7.3.2 for the parameters specified in table 7.3.3 or 7.3.4.

**Table 7.3.2: Requirements for Dynamic Range**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 7.3.3: Parameters for Dynamic Range - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 6 |
| HDOP Range | - | 1.4 to 2.1 |
| GPS Coarse Time assistance error range | seconds | ±2 |
| Propagation conditions | - | AWGN |
| GPS L1 C/A Signal for 1st satellite | dBm | -129 |
| GPS L1 C/A Signal for 2nd satellite | dBm | -135 |
| GPS L1 C/A Signal for 3rd satellite | dBm | -141 |
| GPS L1 C/A Signal for 4th satellite | dBm | -147 |
| GPS L1 C/A Signal for 5th satellite | dBm | -147 |
| GPS L1 C/A Signal for 6th satellite | dBm | -147 |

**Table 7.3.4: Parameters for Dynamic Range - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.3.5 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference high signal power level | dBm | -127.5 |
| | Reference low signal power level | dBm | -147 |
| GPS(1) | Reference high signal power level | dBm | -129 |
| | Reference low signal power level | dBm | -147 |
| GLONASS | Reference high signal power level | dBm | -131.5 |
| | Reference low signal power level | dBm | -147 |
| BDS | Reference high signal power level | dBm | -133.5 |
| | Reference low signal power level | dBm | -145 |
| QZSS | Reference high signal power level | dBm | -129 |
| | Reference low signal power level | dBm | -147 |
| Navic | Reference high signal power level | dBm | -129 |
| | Reference low signal power level | dBm | -147 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.3.5: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS 1[(1)] | GNSS 2[(1)] | GNSS 3[(1)] |
| Single constellation | High signal level | 2 | -- | -- |
| | Low signal level | 4 | -- | -- |
| Dual constellation | High signal level | 1 | 1 | -- |
| | Low signal level | 2 | 2 | -- |
| Triple constellation | High signal level | 1 | 1 | 1 |
| | Low signal level | 2 | 1 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

The normative reference for this requirement is TS 36.171 [3] clause 5.3 and 6.3.

## 7.3.5 Test description

### 7.3.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 36.508 [18] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 7.3.6 or 7.3.7 for GNSS scenario #1 in TS 37.571-5 [20]. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with the higher levels.
3. The parameter settings for the cell are set up according to TS 36.508 [18] clause 4.4.3, single cell scenario.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 7.3.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]
2. Send a RESET UE POSITIONING STORED INFORMATION message.
3. Depending on how the LPP session is performed:

- In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

- In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.
5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the (first) LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 7.3.9 then record the result and process it as specified in step 8. If the UE does not return a valid result within the Max response time specified in table 7.3.9 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

7a. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

8. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.3.9 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.3.9 and record one Good Result or Bad Result as appropriate.

9. Repeat steps 1 to 8 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with the higher levels. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time offset in step 5.

10. Repeat steps 1 to 9 until the statistical requirements of clause 7.3.6 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20], for the satellites with the higher levels.

11. Release the signalling connection.

### 7.3.5.3 Message contents

Message contents are according to TS 36.508 [18] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| a-gnss-RequestCapabilities | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy | '19' (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |

| | | |
|---|---|---|
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 7.3.6 Test requirement

For the parameters specified in table 7.3.6 or 7.3.7 the UE shall meet the requirements and the success rate specified in table 7.3.9 with a confidence level of 95% according to Annex D.

**Table 7.3.6: Test parameters for Dynamic Range - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 6 |
| HDOP Range | - | 1.4 to 2.1 |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| Propagation conditions | - | AWGN |
| GPS L1 C/A Signal for 1st satellite | dBm | -128.2 |
| GPS L1 C/A Signal for 2nd satellite | dBm | -134 |
| GPS L1 C/A Signal for 3rd satellite | dBm | -140 |
| GPS L1 C/A Signal for 4th satellite | dBm | -146 |
| GPS L1 C/A Signal for 5th satellite | dBm | -146 |
| GPS L1 C/A Signal for 6th satellite | dBm | -146 |

**Table 7.3.7: Test parameters for Dynamic Range - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.3.8 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| Galileo | Reference high signal power level | dBm | -126.7 |
| | Reference low signal power level | dBm | -146 |
| GPS(1) | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| GLONASS | Reference high signal power level | dBm | -130.7 |
| | Reference low signal power level | dBm | -146 |
| **BDS** | Reference high signal power level | dBm | -132.7 |

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Reference low signal power level | dBm | -144 |
| QZSS | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| Navic | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.3.8: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | High signal level | 2 | -- | -- |
| | Low signal level | 4 | -- | -- |
| Dual constellation | High signal level | 1 | 1 | -- |
| | Low signal level | 2 | 2 | -- |
| Triple constellation | High signal level | 1 | 1 | 1 |
| | Low signal level | 2 | 1 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 7.3.9: Test requirements for Dynamic Range**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

# 7.4 Multi-Path scenario

## 7.4.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 7.4.1

**Table 7.4.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS | | |
|---|---|---|---|
| 1 | UE supporting A-GPS L1 C/A | | |
| 2 | UE supporting A-GLONASS | | |
| 3 | UE supporting A-Galileo | | |
| 4 | UE supporting A-GPS and Modernized GPS | | |
| 5 | UE supporting A-GPS and A-GLONASS (Note) | | |
| 8 | UE supporting A-GPS and A-Galileo (Note) | | |
| 9 | UE supporting A-BDS | | |
| 10 | UE supporting A-GPS and A-BDS (Note) | | |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) | | |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) | | |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) | | |
| QZSS | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| Navic | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | | | |

## 7.4.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with GNSS signals with multi-path components.

## 7.4.3 Test applicability

This test applies to all types of E-UTRA UE that supports A-GNSS, except Category M1 and Category M2 devices that do not support VoLTE.

## 7.4.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 7.4.2 for the parameters specified in table 7.4.3 or 7.4.4.

**Table 7.4.2: Requirements for Multi-Path scenario**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 7.4.3: Parameters for Multi-Path scenario - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites (Satellites 1, 2 unaffected by multi-path)<br>(Satellites 3, 4, 5 affected by multi-path) | - | 5 |
| GPS Coarse time assistance error range | seconds | ±2 |
| HDOP Range | - | 1.8 to 2.5 |
| GPS L1 C/A Signal for satellite 1, 2 | dBm | -130 |
| GPS L1 C/A Signal for satellite 3, 4, 5 | dBm | LOS signal of -130 dBm, multi-path signal of -136 dBm |

**Table 7.4.4: Parameters for Multi-Path scenario - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.4.5 |
| | Total number of generated satellites | - | 6 or 7[(2)] |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference signal power level | dBm | -127 |
| GPS[(1)] | Reference signal power level | dBm | -128.5 |
| GLONASS | Reference signal power level | dBm | -131 |
| BDS | Reference signal power level | dBm | -133 |
| QZSS | Reference signal power level | dBm | -128.5 |
| Navic | Reference signal power level | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.4.5: Channel model allocation**

| | | Channel model allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | One-tap channel | 2 | -- | -- |
| | Two-tap channel | 4 | -- | -- |
| Dual constellation | One-tap channel | 1 | 1 | -- |
| | Two-tap channel | 2 | 2 | -- |
| Triple constellation | One-tap channel | 1 | 1 | 1 |

| | | | | |
|---|---|---|---|---|
| | Two-tap channel | 2 | 1 | 1 |
| Note: One-tap channel: no multi-path. Two-tap channel: multi-path defined in clause 4.2.4 | | | | |

The normative reference for this requirement is TS 36.171 [3] clause 5.4 and 6.4.

## 7.4.5 Test description

### 7.4.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 36.508 [18] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 7.4.6 or 7.4.7 for GNSS scenario #1 in TS 37.571-5 [20]. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with one-tap channels.
3. The parameter settings for the cell are set up according to TS 36.508 [18] clause 4.4.3, single cell scenario.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 7.4.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]. The initial carrier phase difference between taps of the multi-path model shall be randomly selected between 0 and $2\pi$ radians by selecting the next random number from a standard uniform random number generator, in the range 0 to $2\pi$, representing radians with a resolution of 0.1, representing 0.1 radians.
2. Send a RESET UE POSITIONING STORED INFORMATION message.
3. Depending on how the LPP session is performed:

- In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

- In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.
5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.
7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 7.4.10 then record the result and process it as specified in step 8. If the UE does

not return a valid result within the Max response time specified in table 7.4.10 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

7a. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

8. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.4.10 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.4.10 and record one Good Result or Bad Result as appropriate.

9. Repeat steps 1 to 8 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with the one-tap channels. Use new random values for the UE location and altitude, and the initial carrier phase difference between taps of the multi-path model in step 1 and for the GNSS Reference Time offset in step 5.

10. Repeat steps 1 to 9 until the statistical requirements of clause 7.4.6 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20], for the satellites with the one-tap channels.

11. Release the signalling connection

### 7.4.5.3 Message contents

Message contents are according to TS 36.508 [18] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| a-gnss-RequestCapabilities | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | ‘locationEstimateRequired’ or ‘locationMeasurementsRequired’ | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | ‘onlyReturnInformationRequested’ | |
| > qos | | |
| >> horizontalAccuracy | ‘19’ (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | ‘20’ | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |

| | | |
|---|---|---|
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: ‘gps’<br>Sub-test 2: ‘glonass’<br>Sub-test 3: ‘galileo’<br>Sub-test 4: ‘gps’<br>Sub-test 5: ‘gps’ and ‘glonass’<br>Sub-test 8: ‘gps’ and ‘galileo’<br>Sub-test 9: ‘bds’<br>Sub-test 10: ‘gps’and’bds’<br>Sub-test 11: ‘gps’ and ‘glonass’ and ‘bds’<br>Sub-test 12: ‘gps’ and 'galileo' and ‘glonass’<br>Sub-test 13: ‘gps’ and ‘galileo’ and ‘bds’<br>Sub-test 14: ‘gps’ and ‘qzss’<br>Sub-test 15: ‘gps’ and ‘navic’ | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 7.4.6 Test requirement

For the parameters specified in table 7.4.6 or 7.4.7 the UE shall meet the requirements and the success rate specified in table 7.4.10 with a confidence level of 95% according to Annex D.

**Table 7.4.6: Test parameters for Multi-Path scenario - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites (see note) | - | 5 |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| HDOP Range | - | 1.8 to 2.5 |
| GPS L1 C/A Signal for Satellite 1, 2 (see note) | dBm | -130 |
| GPS L1 C/A Signal for Satellite 3, 4, 5 (see note) | dBm | LOS signal of -130 dBm, multi-path signal of -136.2 dBm |
| NOTE: Satellites 1, 2 no multi-path. Satellites 3, 4, 5 multi-path defined in clause 4.2.4. | | |

**Table 7.4.7: Test parameters for Multi-Path scenario - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.4.8 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range per system | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| Navic | Reference signal power level for all satellites | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

Table 7.4.8: Channel model allocation

| | Channel model allocation for each constellation | | |
|---|---|---|---|
| | GNSS-1 | GNSS-2 | GNSS-3 |

| | | | | |
|---|---|---|---|---|
| Single constellation | One-tap channel | 2 | -- | -- |
| | Two-tap channel | 4 | -- | -- |
| Dual constellation | One-tap channel | 1 | 1 | -- |
| | Two-tap channel | 2 | 2 | -- |
| Triple constellation | One-tap channel | 1 | 1 | 1 |
| | Two-tap channel | 2 | 1 | 1 |
| Note: One-tap channel: no multi-path. Two-tap channel: multi-path defined in clause 4.2.4 with Relative mean Power (Y) defined in Table 7.4.9. | | | | |

**Table 7.4.9: Relative mean Power (Y) for use in Table 7.4.8**

| | | |
|---|---|---|
| Galileo | E1 | -4.7 |
| | E5a | -6.2 |
| | E5b | -6.2 |
| GPS/Modernized GPS/QZSS | L1 C/A | -6.2 |
| | L1C | -4.7 |
| | L2C | -6.2 |
| | L5 | -6.2 |
| GLONASS | G1 | -12.7 |
| | G2 | -12.7 |
| BDS | B1I | -4.7 |
| | B1C | -4.7 |
| | B2a | -6.2 |
| | B2b | -6.2 |
| | B3I | -6.2 |
| Navic | L5 | -6,2 |

**Table 7.4.10: Test requirements for Multi-Path scenario**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

# 7.5 Moving scenario and periodic update (Rel-9 to Rel-13)

## 7.5.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 7.5.1.

**Table 7.5.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

## 7.5.2 Test purpose

To verify the performance when the UE is requested to use periodical reporting with a reporting interval of 2 seconds.

## 7.5.3 Test applicability

This test applies to all types of E-UTRA UE that supports A-GNSS with LPP Release 9 to 13, except Category M1 and Category M2 devices that do not support VoLTE.

## 7.5.4 Minimum conformance requirements

The position estimates, after the first reported position estimate, shall meet the accuracy requirement in table 7.5.2 or 7.5.3 with the periodical reporting interval of 2 seconds for the parameters specified in table 7.5.4 or 7.5.5.

NOTE: In the actual testing the UE may report error messages until it has been able to acquire GNSS measured results or a position estimate. The SS shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 7.5.2 or 7.5.3.

**Table 7.5.2: Requirements for Moving scenario and periodic update - Sub-Test 1**

| Success Rate | 2-D position error | Periodical reporting interval |
|---|---|---|
| 95 % | 100 m | 2 s |

**Table 7.5.3: Requirements for Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| Success Rate | 2-D position error | Periodical reporting interval |
|---|---|---|
| 95 % | 50 m | 2 s |

**Table 7.5.4: Parameters for Moving scenario and periodic update - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 5 |
| HDOP Range | - | 1.8 to 2.5 |
| Propagation condition | - | AWGN |
| GPS L1 C/A signal for all satellites | dBm | -130 |

**Table 7.5.5: Parameters for Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.5.6 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| Galileo | Reference signal power level | dBm | -127 |
| GPS(1) | Reference signal power level | dBm | -128.5 |
| GLONASS | Reference signal power level | dBm | -131 |
| BDS | Reference signal power level | dBm | -133 |
| QZSS | Reference signal power level | dBm | -128.5 |
| Navic | Reference signal power level | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.5.6: Satellite allocation**

| | Satellite allocation for each constellation |
|---|---|

| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
|---|---|---|---|
| Single constellation | 6 | -- | -- |
| Dual constellation | 3 | 3 | -- |
| Triple constellation | 3 | 2 | 2 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | |

The normative reference for this requirement is TS 36.171 [3] clause 5.5 and 6.5.

## 7.5.5 Test description

### 7.5.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 36.508 [18] clause 4.3.1

The UE moves on a rectangular trajectory of 940 m by 1 440 m with rounded corner defined in Figure 7.1. The initial reference is first defined followed by acceleration to final speed of 100 km/h in 250 m. The UE then maintains the speed for 400 m. This is followed by deceleration to final speed of 25 km/h in 250 m. The UE then turn 90 degrees with turning radius of 20 m at 25 km/h. This is followed by acceleration to final speed of 100 km/h in 250 m. The sequence is repeated to complete the rectangle.

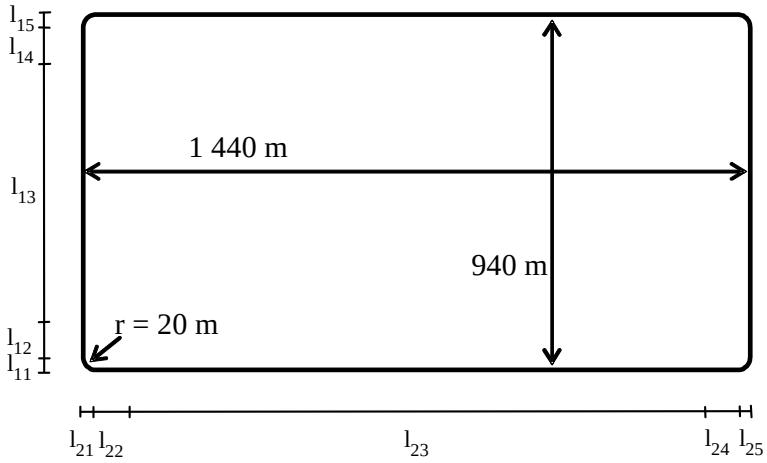



**Figure 7.1: Rectangular trajectory of the moving scenario and periodic update test case**

**Trajectory Parameters**

| Parameter | Distance (m) | Speed (km/h) |
|---|---|---|
| $l_{11}$, $l_{15}$, $l_{21}$, $l_{25}$ | 20 | 25 |
| $l_{12}$, $l_{14}$, $l_{22}$, $l_{24}$ | 250 | 25 to 100 and 100 to 25 |
| $l_{13}$ | 400 | 100 |
| $l_{23}$ | 900 | 100 |

1. Connect SS and GSS to the UE antenna cofor nnector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 7.5.7 or 7.5.8 for GNSS scenario #5 in TS 37.571-5 [20].
3. The parameter settings for the cell are set up according to TS 36.508 [18] clause 4.4.3, single cell scenario.

4. Switch on the UE.

5. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 7.5.5.2 Test procedure

1. Start GNSS scenario #5 as specified in clause 6.2.1.2 of TS 37.571-5 [20]

2. Send a RESET UE POSITIONING STORED INFORMATION message.

3. Depending on how the LPP session is performed:

- In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

- In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.

5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. Ignore any Error messages that the UE may report in LPP PROVIDE LOCATION INFORMATION messages until it has been able to acquire the GNSS signals and reports the first GNSS Measurement Information or Location Information.

8. Discard the first GNSS Measurement Infofor rmation or Location Information.

9. Record the time of reception of the next LPP PROVIDE LOCATION INFORMATION message after reception of the first GNSS Measurement Information or Location Information.

10. After the reception of the first GNSS Measurement Information or Location Information reported in a LPP PROVIDE LOCATION INFORMATION message, every time the UE returns a GNSS Measurement Information or Location Information in the LPP PROVIDE LOCATION INFORMATION message record the time of reception and the result. If the difference between the time of reception and the time of reception of the previous result is less than 1.5 seconds or greater than 2.5 seconds, or if the UE reports an Error in any LPP PROVIDE LOCATION INFORMATION messages, then record one Bad Result. Otherwise process the result as specified in step 11.

10a. If the UE messages at steps 7 to 10 include the ackRequested IE set to TRUE, then the SS shall send LPP acknowledgment messages as required.

11. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE at the time of applicability reported in the Location Information, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.5.10 or 7.5.11 and record one Good Result or Bad Result as appropriate; or

    For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE at the time of applicability reported in the GNSS Measurement Information, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.5.10 or 7.5.11 and record one Good Result or Bad Result as appropriate.

12. If the UE sends the first LPP PROVIDE LOCATION INFORMATION that contains GNSS Measurement Information or Location Information later than 240s after the start of the GNSS scenario, fail the UE and stop the test early. Otherwise collect LPP PROVIDE LOCATION INFORMATION results during 900s, starting from the time recorded in step 9. If at any time the difference between the times of reception of two consecutive results is

greater than 240s, fail the UE and stop the test early. Use the collected Good Results and Bad Results to determine the PASS/FAIL according to clause 7.5.6.

13. Release the signalling connection.

### 7.5.5.3 Message contents

Message contents are according to TS 36.508 [18] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| a-gnss-RequestCapabilities | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > periodicalReporting | | |
| >> reportingAmount | 'ra-Infinity' | Infinite means during the complete test time |
| >> reportingInterval | 'ri0-5' | 2 seconds |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy (Sub-Test 1) | '19' (51.2m) | |
| >> horizontalAccuracy (Sub-Tests 2 to 5 and 8 to 13) | '13' (24.5m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | Not present | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 7.5.6 Test requirement

For the parameters specified in table 7.5.7 or 7.5.8 the UE shall meet the requirements and the success rate specified in table 7.5.10 or 7.5.11 after the first reported position estimates.

NOTES: 1. In the testing the UE may report error messages until it has been able to acquire GNSS measured results or a position estimate. The test equipment shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 7.5.10 or 7.5.11.

2. Due to the statistical nature of the results it is not possible to design a test with predefined confidence level for the success rate in table 7.5.10 or 7.5.11, therefore a simple PASS/FAIL of the results gathered against this success rate is used.

**Table 7.5.7: Test parameters Moving scenario and periodic update - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 5 |
| HDOP Range | - | 1.8 to 2.5 |
| Propagation condition | - | AWGN |
| GPS L1 C/A Signal for all satellites | dBm | -130 |

**Table 7.5.8: Test parameters Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 7.5.9 |
| | Total number of generated satellites | - | 6 or 7[(2)] |
| | HDOP Range per system | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GPS[(1)] | Reference signal power level for all satellites | dBm | -128.5 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| Navic | Reference signal power level for all satellites | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 7.5.9: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS 1[(1)] | GNSS 2[(1)] | GNSS 3[(1)] |
| Single constellation | 6 | -- | -- |
| Dual constellation | 3 | 3 | -- |
| Triple constellation | 3 | 2 | 2 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | |

**Table 7.5.10: Test requirements for Moving scenario and periodic update - Sub-Test 1**

| System | Success rate | 2-D position error | Periodical reporting interval |
|---|---|---|---|
| All | 95 % | 101.3 m | Between 1.5 s and 2.5s |

**Table 7.5.11: Test requirements for Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Success rate | 2-D position error | Periodical reporting interval |
|---|---|---|---|
| All | 95 % | 51.3 m | Between 1.5 s and 2.5s |

## 7.5A Moving scenario and periodic update (Rel-14 onwards)

### 7.5A.1 Sub-tests

Same as defined in clause 7.5.1.

### 7.5A.2 Test purpose

Same as defined in clause 7.5.2.

### 7.5A.3 Test applicability

This test applies to all types of E-UTRA UE with LPP Release 14 onwards that supports A-GNSS with periodical reporting, except Category M1 and Category M2 devices that do not support VoLTE.

NOTE: The capability to support periodical reporting is indicated in LPP [4] by either omitting the field *periodicalReportingNotSupported-r14* in the LPP PROVIDE CAPABILITIES message, or by including the field *periodicalReportingNotSupported-r14* in the LPP PROVIDE CAPABILITIES message but with bits for UE-assisted or UE-based mode set to zero.

### 7.5A.4 Minimum conformance requirements

Same as defined in clause 7.5.4.

### 7.5A.5 Test description

Same as defined in clause 7.5.5.

### 7.5A.6 Test requirement

Same as defined in clause 7.5.6.

# 8 E-UTRA ECID measurement requirements

## 8.0 General

This clause defines the minimum performance requirements for ECID FDD and TDD E-UTRA UEs and UEs supporting NR EN-DC.

## 8.1 UE Rx – Tx Time Difference

### 8.1.1 E-UTRAN FDD UE Rx – Tx time difference case (Rel-9 to Rel-11)

#### 8.1.1.1 Test purpose

The purpose of this test is to verify that the E-UTRAN FDD UE Rx – Tx time difference measurement accuracy is within the specified limits in TS 36.133 [23] clause 9.1.9.

#### 8.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 to release 11 that supports ECID positioning.

### 8.1.1.3 Minimum conformance requirements

NOTE: This measurement is used for UE positioning purposes.

The UE RX-TX time difference is measured from the PCell.

The accuracy requirements in Table 8.1.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

No changes to the uplink transmission timing are applied during the measurement period.

$RSRP|_{dBm}$ according to clause E.1 for a corresponding Band.

**Table 8.1.1.3-1: UE Rx – Tx time difference measurement accuracy**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | Ês/Iot | Downlink transmission bandwidth of PCell | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| Ts Note 2 | dB | MHz | | dBm/15kHz Note 5 | $dBm/BW_{Channel}$ |
| ±20 | ≥-3 dB | ≤ 3 MHz | FDD_A Note 7, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G Note 4 | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±10 | ≥-3 dB | ≥ 5 MHz | Note 3 | Note 3 | Note 3 |

NOTE 1: When in dBm/15kHz, the minimum Io condition is expressed as the average Io per RE over all REs in that symbol. Io may be different in different symbols within a subframe.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with downlink bandwidth ≤ 3 MHz.
NOTE 4: Except Band 29.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 7: Except Band 32.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.9 and A.9.7.1.

### 8.1.1.4 Test description

The test consists of two sub-tests; the difference between the sub-tests is the bandwidth, 1.4 MHz and 10 MHz. Each sub-test has two test points with time delays starting at 32 $T_s$ and 5008 $T_s$ respectively. There is only one active cell in the tests. The tested UE is connected with the serving cell, configured to transmit SRS signals periodically, and signalled to report UE Rx – Tx time difference measurement. The test equipment measures the transmit timing of the UE using the transmitted SRS. The test equipment then compares this timing to the UE Rx-Tx measurement reported by the UE.

#### 8.1.1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel Bandwidth to be tested: 1.4 and 10 MHz. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test, as defined in TS 36.101 [2] clause 5.6.1, then the corresponding sub-test shall be omitted.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in Annex A figure A.5a for 4RX capable UE without any 2RX bands. Otherwise use Annex A figure A.5.

2. Propagation conditions are set according to clause 4.6.2.1.

3. Message contents are defined in clause 8.1.1.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test.

### 8.1.1.4.2 Test procedure

1. Bring the UE to State 3A or 3A-RF according to TS 36.508 [18] clause 4.5.3A or 5.2A.2, using a value of initial timing advance command $T_A = 2$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 32\ T_s$. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_s$.

2. Set the parameters according to Sub-test 1 in Tables 8.1.1.5-1 and 8.1.1.5-2 as appropriate. Propagation conditions are set according to clause 4.6.2.1.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+8\ T_S$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

4a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

4b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the ECID capabilities supported by the UE in the *ECID-ProvideCapabilities* IE.

5. The SS shall transmit an LPP REQUEST LOCATION INFORMATION message, including the *ECID-RequestLocationInformation* IE. If the UE message at step 4b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.

6. The UE shall transmit an LPP PROVIDE LOCATION INFORMATION message including the *ecid-SignalMeasurementInformation* IE.

7. As soon as possible after step 6 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.

8. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

9. The SS shall check the reported value of *ue-RxTxTimeDiff* in the *ecid-SignalMeasurementInformation* IE provided by the UE in step 6 and compare it with the value measured in step 7. The SS shall check that the reported value is within the limits specified in table 8.1.1.5-3 for Sub-test 1 compared to the measured value. If the reported value is within the limits the number of successful results for “Sub-test 1 – Test point 1” is increased by one. If the reported value is not within the limits, or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 6, or does not respond at step 6 within the time given by the *time* IE in the *ECID-RequestLocationInformation* IE in step 5, then the number of unsuccessful results for “Sub-test 1 – Test point 1” is increased by one.

10. Repeat steps 3-9 until the confidence level according to Annex D.4.3 is achieved.
NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

11. Repeat steps 1-10 for “Sub-test 1 – Test point 2”. Set a value of initial timing advance command $T_A = 313$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 5008\ T_s$ in step 1.

12. Repeat steps 1-11 for Sub-test 2 (consisting of Test point 1 and Test point 2) in Tables 8.1.1.5-1 and 8.1.1.5-2 as appropriate. In step 3 the SS adjusts the downlink timing for Cell 1 to a delay of $+4\ T_S$ compared to the current value.

If both test points of a sub-test pass, the sub-test passes. If one test point of a sub-test fails, the sub-test fails.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

#### 8.1.1.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 4.6 with the following exceptions:

**Table 8.1.1.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw7 for sub-test 1<br>bw5 for sub-test 2 | Set according to specific sub-test | |
| srs-SubframeConfig | Sc1 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 8.1.1.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 0 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 8.1.1.4.3-2a: LPP REQUEST CAPABILITIES: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| **Information Element** | **Value/remark** |
|---|---|
| ecid-RequestCapabilities | TRUE |

**Table 8.1.1.4.3-3: *ECID-RequestLocationInformation*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |

| | | | |
|---|---|---|---|
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 2 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | 0 0 1 | ueRxTxReq | |
| } | | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.1.4.3-4: *ECID-ProvideLocationInformation*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation ::= SEQUENCE { | | | |
| ecid-SignalMeasurementInformation ::= SEQUENCE { | | | |
| primaryCellMeasuredResults | Not Present | | |
| MeasuredResultsList ::= SEQUENCE (SIZE(1..32)) OF MeasuredResultsElement | | | |
| MeasuredResultsElement ::= SEQUENCE { | | | |
| physCellId | | | |
| cellGlobalId | | | |
| arfcnEUTRA | | | |
| systemFrameNumber | | | |
| rsrp-Result | Not Present | | |
| rsrq-Result | Not Present | | |
| ue-RxTxTimeDiff | | Set according to specific sub-test and test point. | |
| } | | | |
| } | | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.1.4.3-5: *CQI-ReportConfig-DEFAULT*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for Sub-test 1 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

## 8.1.1.5 Test requirement

Table 8.1.1.5-1 defines the primary level settings including test tolerances for all sub-tests.

**Table 8.1.1.5-1: FDD UE Rx – Tx time difference test parameters**

| Parameter | Unit | Sub-test 1 | Sub-test 2 |
|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | 1 |
| $BW_{channel}$ | MHz | 1.4 | 10 |
| DRX | | OFF | |
| PDSCH Reference measurement channel defined in TS 36.521-3 [25] clause A.1.1 | | R.2 FDD | R.0 FDD |
| PDSCH allocation | $n_{PRB}$ | 2—3 | 13—36 |

| | | | |
|---|---|---|---|
| PDCCH/PCFICH/PHICH Reference measurement channel defined in TS 36.521-3 [25] clause A.2.1 | | R.8 FDD | R.6 FDD |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.3 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | -98 |
| RSRP Note 3 | dBm/15 kHz | -101 | -101 |
| $\hat{E}_s / N_{oc}$ | dB | 2.7 | 2.7 |
| Io Note 3 | dBm/1.08 MHz | -76.55 | N/A |
| | dBm/9 MHz | N/A | -67.35 |
| $\hat{E}_s / I_{ot}$ | dB | -2.7 | -2.7 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the resources in the active cell are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.1.5-2: Sounding Reference Symbol Configuration to be used in FDD UE Rx – Tx time difference test**

| Field | Sub-test 1 | Sub-test 2 | Comment |
|---|---|---|---|
| | Value | | |
| srsBandwidthConfiguration | bw7 | bw5 | |
| srsSubframeConfiguration | sc1 | | |
| ackNackSrsSimultaneousTransmission | FALSE | | |
| srsMaxUpPTS | N/A | | Not applicable for FDD |
| srsBandwidth | 0 | | No hopping |
| srsHoppingBandwidth | hbw0 | | |
| frequencyDomainPosition | 0 | | |
| Duration | TRUE | | Indefinite duration |
| Srs-ConfigurationIndex | 0 | | SRS periodicity of 2ms. |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | | No cyclic shift |
| SRS-AntennaPort | an1 | | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in 3GPP TS 36.331 [22]. | | | |

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.1.5-3.

**Table 8.1.1.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Sub-test 1 | Sub-test 2 |
|---|---|---|
| Lowest reported value | (Measured value from step 7 - 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | (Measured value from step 7 - 13) $T_s$ convertedto RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

| | | |
|---|---|---|
| Highest reported value | (Measured value from step 7 + 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

NOTE: Each sub-test in table 8.1.1.5-3 has two test points starting at 32 $T_s$ and 5008 $T_s$.

The test tolerances are defined in Annex C.

For the overall test to pass, the ratio of successful reported values in each test point of each sub-test shall be more than 90% with a confidence level of 95%. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test then Sub-test 1 shall be omitted.

# 8.1.1A E-UTRAN FDD UE Rx – Tx time difference case (Rel-12 onwards)

## 8.1.1A.1 Test purpose

Same as defined in clause 8.1.1.1.

## 8.1.1A.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 onwards that supports ECID positioning.

## 8.1.1A.3 Minimum conformance requirements

Same as defined in clause 8.1.1.3 except that Table 8.1.1A.3-1 replaces Table 8.1.1.3-1.

**Table 8.1.1A.3-1: UE Rx – Tx time difference measurement accuracy from Release 12 onwards**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | Ês/Iot | Downlink bandwidth | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| Ts Note 2 | dB | MHz | | dBm/15kHz Note 5 | $dBm/BW_{Channel}$ |
| ±20 | ≥-3 dB | ≥1.4 MHz | FDD_A Note 7, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G Note 4 | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±14 | ≥-3 dB | ≥ 3 MHz | Note 3 | Note 3 | Note 3 |
| ±10 | ≥-3 dB | ≥ 5 MHz | Note 3 | Note 3 | Note 3 |
| ±7 | ≥-3 dB | ≥10 MHz | Note 3 | Note 3 | Note 3 |

NOTE 1: When in dBm/15kHz, the minimum Io condition is expressed as the average Io per RE over all REs in that symbol. Io may be different in different symbols within a subframe.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with downlink bandwidth ≥1.4 MHz.
NOTE 4: Except Band 29.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 7: Except Band 32.

## 8.1.1A.4 Test description

Same as defined in clause 8.1.1.4.

#### 8.1.1A.4.1 Initial conditions

Same as defined in clause 8.1.1.4.1.

#### 8.1.1A.4.2 Test procedure

Same as defined in clause 8.1.1.4.2.

#### 8.1.1A.4.3 Message contents

Same as defined in clause 8.1.1.4.3.

### 8.1.1A.5 Test requirement

Same as defined in clause 8.1.1.5 except that Table 8.1.1A.5-3 replaces Table 8.1.1.5-3.

**Table 8.1.1A.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Sub-test 1 | Sub-test 2 |
|---|---|---|
| Lowest reported value | (Measured value from step 7 - 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | (Measured value from step 7 - 10) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| Highest reported value | (Measured value from step 7 + 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | (Measured value from step 7 + 10) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

## 8.1.1B E-UTRAN FDD UE Rx – Tx time difference case for UE Category 1bis

### 8.1.1B.1 Test purpose

Same as defined in clause 8.1.1.1.

### 8.1.1B.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 onwards of category 1bis that supports ECID positioning.

### 8.1.1B.3 Minimum conformance requirements

Same as defined in clause 8.1.1A.3.

### 8.1.1B.4 Test description

Same as defined in clause 8.1.1.4.

#### 8.1.1B.4.1 Initial conditions

Same as defined in clause 8.1.1.4.1.

#### 8.1.1B.4.2 Test procedure

Same as defined in clause 8.1.1.4.2.

#### 8.1.1B.4.3 Message contents

Same as defined in clause 8.1.1.4.3.

### 8.1.1B.5 Test requirement

Same as defined in clause 8.1.1A.5 except that the Cell Antenna Configuration is 1x1 instead of the default 1x2.

## 8.1.2 E-UTRAN TDD UE Rx – Tx time difference case (Rel-9 to Rel-11)

### 8.1.2.1 Test purpose

The purpose of this test is to verify that the E-UTRAN TDD UE Rx – Tx time difference measurement accuracy is within the specified limits in TS 36.133 [23] clause 9.1.9.

### 8.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 to release 11 with LPP release 13 onwards that supports ECID positioning. Note that for LPP releases before release 13 the UE TDD Rx - Tx time difference measurement report mapping is ambiguous and therefore this test shall not be used.

### 8.1.2.3 Minimum conformance requirements

NOTE: This measurement is used for UE positioning purposes.

The UE RX-TX time difference is measured from the PCell.

The accuracy requirements in Table 8.1.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

No changes to the uplink transmission timing are applied during the measurement period.

$RSRP|_{dBm}$ according to clause E.1 for a corresponding Band.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.9 and A.9.7.1.

### 8.1.2.4 Test description

The test consists of two sub-tests; the difference between the sub-tests is the bandwidth, 1.4 MHz and 10 MHz. Each sub-test has two test points with time delays starting at 32 $T_s$ and 5008 $T_s$ respectively. There is only one active cell in the tests. The tested UE is connected with the serving cell, configured to transmit SRS signals periodically, and signalled to report UE Rx – Tx time difference measurement. The test equipment measures the transmit timing of the UE using the transmitted SRS. The test equipment then compares this timing to the UE Rx-Tx measurement reported by the UE.

#### 8.1.2.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.2.

Channel Bandwidth to be tested: 1.4 and 10 MHz. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test, as defined in TS 36.101 [2] clause 5.6.1, then the corresponding sub-test shall be omitted.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in Annex A figure A.5a for 4RX capable UE without any 2RX bands. Otherwise use Annex A figure A.5.

2. Propagation conditions are set according to clause 4.6.2.1.

3. Message contents are defined in clause 8.1.2.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test.

### 8.1.2.4.2 Test procedure

1. Bring the UE to State 3A or 3A-RF according to TS 36.508 [18] clause 4.5.3A or 5.2A.2, using a value of initial timing advance command $T_A = 2$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 32\ T_s$. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_s$.

2. Set the parameters according to Sub-test 1 in Tables 8.1.2.5-1 and 8.1.5.2-2 as appropriate. Propagation conditions are set according to clause 4.6.2.1.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+8\ T_S$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

4a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

4b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the ECID capabilities supported by the UE in the *ECID-ProvideCapabilities* IE. The IE *ueRxTxSupTDD-r13* shall be present (TRUE).

5. The SS shall transmit a LPP REQUEST LOCATION INFORMATION message, including the *ECID-RequestLocationInformation* IE. If the UE message at step 4b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.

6. The UE shall transmit an LPP PROVIDE LOCATION INFORMATION message including the *ecid-SignalMeasurementInformation* IE.

7. As soon as possible after step 6 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.

8. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

9. The SS shall check the reported value of *ue-RxTxTimeDiff* in the *ecid-SignalMeasurementInformation* IE provided by the UE in step 6 and compare it with the value measured in step 7. The SS shall check that the reported values are within the limits specified in table 8.1.2.5-3 for Sub-test 1 compared to the measured value. If the reported value is within the limits the number of successful results for "Sub-test 1 – Test point 1" is increased by one. If the reported value is not within the limits, or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 6, or does not respond at step 6 within the time given by the *time* IE in the *ECID-RequestLocationInformation* IE in step 5, then the number of unsuccessful results for "Sub-test 1 – Test point 1" is increased by one.

10. Repeat steps 3-9 until the confidence level according to Annex D.4.3 is achieved.
    NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

11. Repeat steps 1-10 for "Sub-test 1 – Test point 2". Set a value of initial timing advance command $T_A = 313$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 5008\ T_s$ in step 1.

12. Repeat steps 1-11 for Sub-test 2 (consisting of Test point 1and Test point 2) in Tables 8.1.2.5-1 and 8.1.2.5-2 as appropriate. In step 3 the SS adjusts the downlink timing for Cell 1 to a delay of $+4\ T_S$ compared to the current value.

If both test points of a sub-test pass, the sub-test passes. If one test point of a sub-test fails, the sub-test fails.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 8.1.2.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 4.6 with the following exceptions:

**Table 8.1.2.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw7 for sub-test 1<br>bw5 for sub-test 2 | Set according to specific sub-test | |
| srs-SubframeConfig | Sc1 | | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | TRUE | | TDD |
| } | | | |

**Table 8.1.2.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 10 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 8.1.2.4.3-2a: LPP REQUEST CAPABILITIES: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Information Element | Value/remark |
|---|---|
| ecid-RequestCapabilities | TRUE |

**Table 8.1.2.4.3-2b: LPP PROVIDE CAPABILITIES: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Information Element | Value/remark |
|---|---|
| ueRxTxSupTDD-r13 | TRUE |

**Table 8.1.2.4.3-3: *ECID-RequestLocationInformation*: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 2 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | 0 0 1 | ueRxTxReq | |
| } | | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.2.4.3-4: *ECID-ProvideLocationInformation*: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| ecid-SignalMeasurementInformation ::= SEQUENCE { | | | |
| primaryCellMeasuredResults | Not Present | | |
| MeasuredResultsList ::= SEQUENCE (SIZE(1..32)) OF MeasuredResultsElement | | | |
| MeasuredResultsElement ::= SEQUENCE { | | | |
| physCellId | | | |
| cellGlobalId | | | |
| arfcnEUTRA | | | |
| systemFrameNumber | | | |
| rsrp-Result | Not Present | | |
| rsrq-Result | Not Present | | |
| ue-RxTxTimeDiff | | Set according to specific sub-test and test point. | |
| } | | | |
| } | | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.2.4.3-5: *CQI-ReportConfig-DEFAULT*: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for Sub-test 1 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

## 8.1.2.5 Test requirement

Table 8.1.2.5-1 defines the primary level settings including test tolerances for all sub-tests.

**Table 8.1.2.5-1: Cell specific test parameters for UE Rx-Tx time difference measurement**

| Parameter | Unit | Sub-test 1 | Sub-test 2 |
|---|---|---|---|
| E-UTRAN RF Channel Number | - | 1 | 1 |
| $BW_{channel}$ | MHz | 1.4 | 10 |
| Uplink-downlink configuration of cell Note 1 | | 1 | 1 |
| Special subframe configuration of cell Note 1 | | 6 | 6 |
| PDSCH Reference measurement channel defined in TS 36.521-3 [25] clause A.1.2 | - | R.2 TDD | R.0 TDD |
| PDSCH allocation | $n_{PRB}$ | 2-3 | 13-36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in TS 36.521-3 [25] clause A.2.2 | - | R.8 TDD | R.6 TDD |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | - | OP.3 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |

| | | | |
|---|---|---|---|
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 2 | dB | | |
| OCNG_RB Note 2 | dB | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | -98 |
| RSRP Note 4 | dBm/15 kHz | -100.7 | -100.7 |
| $\hat{E}_s / N_{oc}$ | dB | -2.7 | -2.7 |
| Io Note 4 | dBm/1.08 MHz | -77.55 | N/A |
| | dBm/9 MHz | N/A | -67.35 |
| $\hat{E}_s / I_{ot}$ | dB | -2.7 | -2.7 |
| Propagation Condition | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211 [26].

Note 2: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.2.5-2: Sounding Reference Symbol Configuration to be used in TDD UE Rx – Tx time difference test**

| Field | Sub-test 1 | Sub-test 2 | Comment |
|---|---|---|---|
| | Value | | |
| srsBandwidthConfiguration | bw7 | bw5 | |
| srsSubframeConfiguration | sc1 | | |
| ackNackSrsSimultaneousTransmission | FALSE | | |
| srsMaxUpPTS | TRUE | | |
| srsBandwidth | 0 | | No hopping |
| srsHoppingBandwidth | hbw0 | | |
| frequencyDomainPosition | 0 | | |
| Duration | TRUE | | Indefinite duration |
| Srs-ConfigurationIndex | 10 | | SRS periodicity of 10ms. |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | | No cyclic shift |
| SRS-AntennaPort | an1 | | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in 3GPP TS 36.331 [22]. | | | |

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.2.5-3.

**Table 8.1.2.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Sub-test 1 | Sub-test 2 |
|---|---|---|
| Lowest reported value | (Measured value from step 7 - 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 | (Measured value from step 7 - 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |
| Highest reported value | (Measured value from step 7 + 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 | (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |

NOTE: Each sub-test in table 8.1.2.5-3 has two test points starting at 32 $T_s$ and 5008 $T_s$.

The test tolerances are defined in Annex C.

For the overall test to pass, the ratio of successful reported values in each test point of each sub-test shall be more than 90% with a confidence level of 95%. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test then Sub-test 1 shall be omitted.

## 8.1.2A E-UTRAN TDD UE Rx – Tx time difference case (Rel-12 onwards)

### 8.1.2A.1 Test purpose

Same as defined in clause 8.1.2.1.

### 8.1.2A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 onwards with LPP release 13 onwards that supports ECID positioning. Note that for LPP releases before release 13 the UE TDD Rx - Tx time difference measurement report mapping is ambiguous and therefore this test shall not be used.

### 8.1.2A.3 Minimum conformance requirements

Same as defined in clause 8.1.2.3 except that Table 8.1.1A.3-1 replaces Table 8.1.1.3-1.

### 8.1.2A.4 Test description

Same as defined in clause 8.1.2.4.

#### 8.1.2A.4.1 Initial conditions

Same as defined in clause 8.1.2.4.1.

#### 8.1.2A.4.2 Test procedure

Same as defined in clause 8.1.2.4.2.

#### 8.1.2A.4.3 Message contents

Same as defined in clause 8.1.2.4.3.

### 8.1.2A.5 Test requirement

Same as defined in clause 8.1.2.5 except that Table 8.1.2A.5-3 replaces Table 8.1.2.5-3.

**Table 8.1.2A.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Sub-test 1 | Sub-test 2 |
|---|---|---|
| Lowest reported value | (Measured value from step 7 - 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 | (Measured value from step 7 - 10) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |
| Highest reported value | (Measured value from step 7 + 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 | (Measured value from step 7 + 10) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |

## 8.1.2B E-UTRAN TDD UE Rx – Tx time difference case for UE Category 1bis

### 8.1.2B.1 Test purpose

Same as defined in clause 8.1.2.1.

### 8.1.2B.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 onwards of category 1bis that supports ECID positioning.

### 8.1.2B.3 Minimum conformance requirements

Same as defined in clause 8.1.2A.3.

### 8.1.2B.4 Test description

Same as defined in clause 8.1.2.4.

#### 8.1.2B.4.1 Initial conditions

Same as defined in clause 8.1.2.4.1.

#### 8.1.2B.4.2 Test procedure

Same as defined in clause 8.1.2.4.2.

#### 8.1.2B.4.3 Message contents

Same as defined in clause 8.1.2.4.3.

### 8.1.2B.5 Test requirement

Same as defined in clause 8.1.2A.5 except that the Cell Antenna Configuration is 1x1 instead of the default 1x2.

## 8.1.3 E-UTRAN FDD UE Rx–Tx Time Difference under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 8.1.3.1 Test purpose

To verify that the E-UTRAN FDD UE Rx – Tx time difference measurement accuracy is within the specified limits under a time-domain measurement resource restriction pattern, and when non-MBSFN ABS is configured in the interfering cells.

### 8.1.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and ECID positioning. Applicability requires support for FGI bit 115.

### 8.1.3.3 Minimum conformance requirements

NOTE: This measurement is used for UE positioning purposes.

The requirements in this section apply for UE configured with a time-domain measurement resource restriction pattern for PCell measurements. The UE Rx-Tx time difference is measured from the Pcell.

The accuracy requirements in Table 8.1.3.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports,
- Conditions defined in TS 36.101[2] Clause 7.3 for reference sensitivity are fulfilled,
- No changes to the uplink transmission timing are applied during the measurement period,

$RSRP|_{dBm}$ according to Annex E.4 for a corresponding Band,

- The time domain measurement resource restriction pattern configured for the PCell indicates at least one subframe per radio frame for performing the PCell measurements,
- Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 8.1.3.3-1: UE Rx–Tx time difference measurement accuracy under time domain measurement resource restriction**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | **Ês/Iot** Note 6 | **Downlink transmission bandwidth of PCell** | **Io** Note 1, 5 **range** | | |
| | | | **E-UTRA operating band groups** Note 8 | **Minimum Io** | **Maximum Io** |
| **Ts** Note 2 | **dB** | **MHz** | | **dBm/15kHz** Note 7 | **dBm/BW$_{Channel}$** |
| ±20 | ≥-3 dB | ≤ 3 MHz | FDD_A Note 9, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G Note 4 | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±10 | ≥-3 dB | ≥ 5 MHz | Note 3 | Note 3 | Note 3 |

NOTE 1: When in dBm/15kHz, the minimum Io condition is expressed as the average Io per RE over all REs in that symbol. Io may be different in different symbols within a subframe.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with downlink bandwidth ≤ 3 MHz.
NOTE 4: Except Band 29.
NOTE 5: Io is defined for the subframes indicated by the time-domain measurement resource restriction pattern for serving cell measurements. The specified Io range applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 6: CRS Ês/Iot is in subframes indicated for PCell measurements by the time-domain measurement resource restriction pattern.
NOTE 7: The condition level is increased by $\Delta>0$, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 8: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 9: Except Band 32.

NOTE: It is up to the UE implementation whether the UE Rx-Tx time difference measurement is performed in any subframe or in subframes indicated by the time-domain measurement resource restriction pattern.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.9.3 and A.9.7.3.

## 8.1.3.4 Test description

The test has two test points with time delays starting at 32 $T_S$ and 5008 $T_S$, respectively.

In this test case, there are two synchronous cells, Cell 1 and Cell 2, on the same RF channel. Cell 1 is the PCell on which UE Rx-Tx is measured, and Cell 2 is the interfering cell. Non-MBSFN ABS pattern is configured in Cell 2 during the entire test.

The tested UE is connected to the PCell and configured to transmit SRS signals periodically. The SRS configuration is provided to the UE before the measurement starts. The UE is configured to report UE Rx–Tx time difference measurement. The test equipment measures the transmit timing of the UE using the transmitted SRS, and measures the

receive timing using the downlink CRS. The test equipment then compares the difference of these two timings to the UE Rx-Tx measurement reported by the UE. The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on PCell. The information for both patterns shall be provided to the UE before the measurement starts.

### 8.1.3.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: According to TS 36.521-1 [24] Annex E table E-1 and TS 36.508 [18] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [18] Annex A figure A.54 (without faders).

2. Propagation conditions are set according to clause 4.6.2.1.

3. Message contents are defined in clause 8.1.3.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test. Cell 2 is the neighbour cells. Both cells are on the same RF channel.

**Table 8.1.3.4.1-1: General test parameters for FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The measured cell |
| Neighbour cell | | Cell 2 | The cell interfering to Cell 1 |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in TS 36.521-3 [25] Table C.3.1.1.1-1 |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used |
| Downlink Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | µs | 3 | Synchronous cells |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 !=0 | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '10000000100000001000 00001000000010000000' | Non-MBSFN ABS. FDD ABS Pattern Info IE, as defined in TS 36.423 [35], clause 9.2.54. Configured in Cell 2. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod 40 = 0. No MBSFN subframes are configured in Cell 1 or Cell 2 during the ABS subframes of Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000100000001000 00001000000010000000' | Configured for measurements on Cell 1. |

### 8.1.3.4.2 Test procedure

1. Bring the UE to State 3A or 3A-RF according to TS 36.508 [18] clause 4.5.3A or 5.2A.2 with exceptions listed in 7.2A.6, using a value of initial timing advance command $T_A = 2$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 32\ T_s$. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_s$.

2. Set the parameters according to Tables 8.1.3.5-1 and 8.1.3.5-2. Propagation conditions are set according to clause 4.6.2.1.

3. The SS adjusts the downlink timing for Cell 1 to a delay of +4 $T_S$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the ECID capabilities supported by the UE in the *ECID-ProvideCapabilities* IE.
7. The SS shall transmit an LPP REQUEST LOCATION INFORMATION message, including the *ECID-RequestLocationInformation* IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.
8. The UE shall transmit an LPP PROVIDE LOCATION INFORMATION message including the *ecid-SignalMeasurementInformation* IE.
9. As soon as possible after step 8 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.
10. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.
11. The SS shall check the reported value of *ue-RxTxTimeDiff* in the *ecid-SignalMeasurementInformation* IE provided by the UE in step 8 and compare it with the value measured in step 9. The SS shall check that the reported value is within the limits specified in table 8.1.3.5-3 compared to the measured value. If the reported value is within the limits the number of successful results for the test point is increased by one. If the reported value is not within the limits, or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 6, or does not respond at step 8 within the time given by the *responseTime* IE in the *ECID-RequestLocationInformation* IE in step 7, then the number of unsuccessful results for the test point test is increased by one.
12. Repeat steps 3-11 until the confidence level according to Annex D.4.3 is achieved.
    NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.
13. Repeat steps 1-12 for test point 2.

### 8.1.3.4.3 Message contents

**Table 8.1.3.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc1 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |
| } | | | |

**Table 8.1.3.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |

| | | | |
|---|---|---|---|
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 0 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 8.1.3.4.3-3: LPP REQUEST CAPABILITIES: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Information Element | Value/remark |
|---|---|
| ecid-RequestCapabilities | TRUE |

**Table 8.1.3.4.3-4: *ECID-RequestLocationInformation*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 2 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | 0 0 1 | ueRxTxReq | |
| } | | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.3.4.3-5: *ECID-ProvideLocationInformation*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation ::= SEQUENCE { | | | |
| ecid-SignalMeasurementInformation ::= SEQUENCE { | | | |
| primaryCellMeasuredResults | Not Present | | |
| MeasuredResultsList ::= SEQUENCE (SIZE(1..32)) OF MeasuredResultsElement | | | |
| MeasuredResultsElement ::= SEQUENCE { | | | |
| physCellId | | | |
| cellGlobalId | | | |
| arfcnEUTRA | | | |
| systemFrameNumber | | | |
| rsrp-Result | Not Present | | |
| rsrq-Result | Not Present | | |
| ue-RxTxTimeDiff | | Set according to specific sub-test and test point. | |
| } | | | |
| } | | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.3.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB(n, m):* Additional FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | Cell1 |
| } | | | |
| } | | | |
| } | | | |

## 8.1.3.5 Test requirement

Table 8.1.3.5-1 defines the primary level settings including test tolerances for the test.

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.3.5-3.

**Table 8.1.3.5-1: Cell-specific test parameters for FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | 1 |
| Channel bandwidth ($BW_{channel}$) | MHz | 10 | 10 |
| PDSCH Reference measurement channel defined in TS 36.521-3 [25] A.1.1 | | R.0 FDD | N/A |
| PDSCH allocation | $n_{PRB}$ | 13—36 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in TS 36.521-3 [25] A.2.1 | | R.6 FDD | N/A |
| OCNG Patterns defined in TS 36.521-3 [25] D.1.1 (OP.1 FDD) and in D.1.6 (OP.6 FDD) | | OP.5 FDD | OP.6 FDD |
| PBCH_RA | dB | 0 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1 in TS 36.521-3 [25]. |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | -98 |
| CRS $\hat{E}_s / N_{oc}$ | dB | -2.7 | 1 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 3 | dB | -2.7 | -0.87 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ Note 3 | dB | -6.24 | -0.87 |
| RSRP Note 4 | dBm/15 kHz | -100.7 | -97 |
| $(\mathrm{Io})_{meas}$ Note 4 | dBm/9 MHz | -67.8 | -67.8 |
| $(\mathrm{Io})_{nonABS}$ Note 4 | dBm/9 MHz | -65.75 | -65.75 |
| Propagation condition | | AWGN | |

NOTE 1: OCNG shall be used such that the resources in the active cell are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: $\left(\hat{E}_s / I_{ot}\right)_{meas}$ is calculated in CRS REs in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ is calculated in CRS REs in the subframes not indicated for PCell measurements by measurement resource restriction pattern.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. $(\mathrm{Io})_{meas}$ is calculated in CRS symbols in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $(\mathrm{Io})_{nonABS}$ is calculated in CRS symbols in the subframes not indicated for PCell measurements by measurement resource restriction pattern.

**Table 8.1.3.5-2: Sounding Reference Symbol Configuration to be used in FDD UE Rx–Tx time difference test**

| Field | Value | Comment |
|---|---|---|
| UL bandwidth | 50 RBs | Same as the DL bandwidth |
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc1 | |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | Not applicable for FDD |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| srs-ConfigIndex | 0 | SRS periodicity of 2ms |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| srsAntennaPort | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [22]. | | |

**Table 8.1.3.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Test requirement |
|---|---|
| Lowest reported value | (Measured value from step 7 - 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| Highest reported value | (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

NOTE: The test in table 8.1.3.5-3 has two test points starting at 32 $T_s$ and 5008 $T_s$.

The test tolerances are defined in Annex C.

For the overall test to pass, the ratio of successful reported values in each test point shall be more than 90% with a confidence level of 95%.

## 8.1.4 E-UTRAN TDD UE Rx–Tx Time Difference under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 8.1.4.1 Test purpose

To verify that the E-UTRAN TDD UE Rx – Tx time difference measurement accuracy is within the specified limits under a time-domain measurement resource restriction pattern, and when non-MBSFN ABS is configured in the interfering cells.

### 8.1.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 with LPP release 13 onwards and ECID positioning. Applicability requires support for FGI bit 115. Note that for LPP releases before release 13 the UE Rx - Tx time difference measurement report mapping is ambiguous and therefore this test shall not be used.

### 8.1.4.3 Minimum conformance requirements

NOTE: This measurement is used for UE positioning purposes.

The requirements in this section apply for UE configured with a time-domain measurement resource restriction pattern for PCell measurements. The UE Rx-Tx time difference is measured from the Pcell.

The accuracy requirements in Table 8.1.4.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports,
- Conditions defined in TS 36.101[2] Clause 7.3 for reference sensitivity are fulfilled,
- No changes to the uplink transmission timing are applied during the measurement period,

$RSRP|_{dBm}$ according to Annex E.4 for a corresponding Band,

- The time domain measurement resource restriction pattern configured for the PCell indicates at least one subframe per radio frame for performing the PCell measurements,
- Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 8.1.4.3-1: UE Rx–Tx time difference measurement accuracy under time domain measurement resource restriction**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | $\hat{E}s/Iot$ Note 6 | Downlink transmission bandwidth of PCell | Io Note 1, 5 range | | |
| | | | E-UTRA operating band groups Note 8 | Minimum Io | Maximum Io |
| Ts Note 2 | dB | MHz | | dBm/15kHz Note 7 | $dBm/BW_{Channel}$ |
| ±20 | ≥-3 dB | ≤ 3 MHz | FDD_A Note 9, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G Note 4 | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±10 | ≥-3 dB | ≥ 5 MHz | Note 3 | Note 3 | Note 3 |

NOTE 1: When in dBm/15kHz, the minimum Io condition is expressed as the average Io per RE over all REs in that symbol. Io may be different in different symbols within a subframe.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with downlink bandwidth ≤ 3 MHz.
NOTE 4: Except Band 29.

| | |
|---|---|
| NOTE 5: | Io is defined for the subframes indicated by the time-domain measurement resource restriction pattern for serving cell measurements. The specified Io range applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe. |
| NOTE 6: | CRS Ês/Iot is in subframes indicated for PCell measurements by the time-domain measurement resource restriction pattern. |
| NOTE 7: | The condition level is increased by Δ>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3. |
| NOTE 8: | E-UTRA operating band groups are as defined in Section 4.4.2. |
| NOTE 9: | Except Band 32. |

NOTE: It is up to the UE implementation whether the UE Rx-Tx time difference measurement is performed in any subframe or in subframes indicated by the time-domain measurement resource restriction pattern.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.9.3 and A.9.7.4.

## 8.1.4.4 Test description

The test has two test points with time delays starting at 32 $T_S$ and 5008 $T_S$, respectively.

In the test, there are two synchronous cells, Cell 1 and Cell 2, on the same RF channel. Cell 1 is the PCell on which UE Rx-Tx is measured, and Cell 2 is the interfering cell. Non-MBSFN ABS pattern is configured in Cell 2 during the entire test.

The tested UE is connected to the PCell and configured to transmit SRS signals periodically. The SRS configuration is provided to the UE before the measurement starts. The UE is configured to report UE Rx–Tx time difference measurement. The test equipment measures the transmit timing of the UE using the transmitted SRS, and measures the receive timing using the downlink CRS. The test equipment then compares the difference of these two timings to the UE Rx-Tx measurement reported by the UE. The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN TDD UE Rx-Tx time difference measurements on PCell. The information for both patterns shall be provided to the UE before the measurement starts.

### 8.1.4.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: According to TS 36.521-1 [24] Annex E table E-1 and TS 36.508 [18] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [18] Annex A figure A.54 (without faders).

2. Propagation conditions are set according to clause 4.6.2.1.

3. Message contents are defined in clause 8.1.4.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test. Cell 2 is the neighbour cells. Both cells are on the same RF channel.

**Table 8.1.4.4.1-1: General test parameters for E-UTRAN TDD UE Rx-Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | Cell to be measured |
| Neighbour cell | | Cell 2 | The cell interfering to Cell 1 |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in TS 36.521-3 [25] Table C.3.1.1.1-1 |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used |
| Downlink Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| CP length | | Normal | For both cells in the test |

| | | | |
|---|---|---|---|
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in TS 36.211 [26]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2- in TS 36.211 [26]. |
| DRX | | | OFF |
| Time offset between cells | µs | 3 | Synchronous cells |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 !=0 | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met. |
| ABS pattern | | '00000000010000000001' | Non-MBSFN ABS. TDD ABS Pattern Info IE, as defined in TS 36.423 [35], clause 9.2.54. Configured in Cell 2. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod 20 = 0. No MBSFN subframes are configured in the ABS subframes in Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '00000000010000000001' | Configured for measurements on Cell 1. |

### 8.1.4.4.2 Test procedure

1. Bring the UE to State 3A or 3A-RF according to TS 36.508 [18] clause 4.5.3A or 5.2A.2 with exceptions listed in 7.2A.6, using a value of initial timing advance command $T_A = 2$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 32\ T_s$. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_s$.

2. Set the parameters according to Tables 8.1.4.5-1 and 8.1.4.5-2. Propagation conditions are set according to clause 4.6.2.1.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+4\ T_S$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

5. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the ECID capabilities supported by the UE in the *ECID-ProvideCapabilities* IE. The IE *ueRxTxSupTDD-r13* shall be present (TRUE).

7. The SS shall transmit an LPP REQUEST LOCATION INFORMATION message, including the *ECID-RequestLocationInformation* IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.

8. The UE shall transmit an LPP PROVIDE LOCATION INFORMATION message including the *ecid-SignalMeasurementInformation* IE.

9. As soon as possible after step 8 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.

10. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

11. The SS shall check the reported value of *ue-RxTxTimeDiff* in the *ecid-SignalMeasurementInformation* IE provided by the UE in step 8 and compare it with the value measured in step 9. The SS shall check that the reported value is within the limits specified in table 8.1.4.5-3 compared to the measured value. If the reported value is within the limits the number of successful results for the test point is increased by one. If the reported value is not within the limits, or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 6, or does not respond at step 8 within the time given by the *responseTime* IE in the *ECID-RequestLocationInformation* IE in step 7, then the number of unsuccessful results for the test point test is increased by one.

12. Repeat steps 3-11 until the confidence level according to Annex D.4.3 is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

13. Repeat steps 1-12 for test point 2.

### 8.1.4.4.3 Message contents

**Table 8.1.4.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc1 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |
| } | | | |

**Table 8.1.4.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 0 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 8.1.4.4.3-3: LPP REQUEST CAPABILITIES: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Information Element | Value/remark |
|---|---|
| ecid-RequestCapabilities | TRUE |

**Table 8.1.4.4.3-3a: LPP PROVIDE CAPABILITIES: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Information Element | Value/remark |
|---|---|
| ueRxTxSupTDD-r13 | TRUE |

**Table 8.1.4.4.3-4: *ECID-RequestLocationInformation*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 2 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | 0 0 1 | ueRxTxReq | |
| } | | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.4.4.3-5: *ECID-ProvideLocationInformation*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation ::= SEQUENCE { | | | |
| ecid-SignalMeasurementInformation ::= SEQUENCE { | | | |
| primaryCellMeasuredResults | Not Present | | |
| MeasuredResultsList ::= SEQUENCE (SIZE(1..32)) OF MeasuredResultsElement | | | |
| MeasuredResultsElement ::= SEQUENCE { | | | |
| physCellId | | | |
| cellGlobalId | | | |
| arfcnEUTRA | | | |
| systemFrameNumber | | | |
| rsrp-Result | Not Present | | |
| rsrq-Result | Not Present | | |
| ue-RxTxTimeDiff | | Set according to specific sub-test and test point. | |
| } | | | |
| } | | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.4.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB(n, m):* Additional TDD UE Rx−Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | Cell 1 |
| | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.1.4.5 Test requirement

Table 8.1.4.5-1 defines the primary level settings including test tolerances for the test.

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.4.5-3.

**Table 8.1.4.5-1: Cell-specific test parameters for TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| PDSCH Reference measurement channel defined in TS 36.521-3 [25] A.1.2 | | R.0 TDD | N/A |
| PDSCH allocation | $n_{PRB}$ | 13—36 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in TS 36.521-3 [25] A.2.2 | | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in TS 36.521-3 [25] D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1 in TS 36.521-3 [25]. |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 |
| CRS $\hat{E}_s / N_{oc}$ | dB | -2.7 | 1 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 3 | dB | -2.7 | -0.87 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ Note 3 | dB | -6.24 | -0.87 |
| RSRP Note 4 | dBm/15 kHz | -100.7 | -97 |
| $(\mathrm{Io})_{meas}$ Note 4 | dBm/9 MHz | -67.8 | -67.8 |
| $(\mathrm{Io})_{nonABS}$ Note 4 | dBm/9 MHz | -65.75 | -65.75 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the resources in the active cell are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\left(\hat{E}_s / I_{ot}\right)_{meas}$ is calculated in CRS REs in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ is calculated in CRS REs in the subframes not indicated for PCell measurements by measurement resource restriction pattern.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. $(\mathrm{Io})_{meas}$ is calculated in CRS symbols in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $(\mathrm{Io})_{nonABS}$ is calculated in CRS symbols

**Table 8.1.4.5-2: Sounding Reference Symbol Configuration to be used in TDD UE Rx–Tx time difference test**

| Field | Value | Comment |
|---|---|---|
| UL bandwidth | 50 RBs | Same as the DL bandwidth |
| srsBandwidthConfiguration | bw5 | |

| | | |
|---|---|---|
| srsSubframeConfiguration | sc1 | |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | TRUE | |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 10 | SRS periodicity of 10ms for all Tests. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [22]. | | |

**Table 8.1.4.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Test requirement |
|---|---|
| Lowest reported value | (Measured value from step 7 - 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |
| Highest reported value | (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |

NOTE: The test in table 8.1.4.5-3 has two test points starting at 32 $T_s$ and 5008 $T_s$.

The test tolerances are defined in Annex C.

For the overall test to pass, the ratio of successful reported values in each test point shall be more than 90% with a confidence level of 95%.

## 8.1.5 E-UTRAN FDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

### 8.1.5.1 Test purpose

The purpose of this test is to verify that the E-UTRAN FDD UE Rx – Tx time difference measurement accuracy is within the specified limits in TS 36.133 [23] clause 9.1.9.4 when the UE is provided with a time-domain measurement resource restriction pattern and CRS assistance information, and when non-MBSFN ABS is configured in the interfering cells.

### 8.1.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 and forward that supports ECID positioning and CRS interference handling. Applicability requires support of FGI bit 115.

### 8.1.5.3 Minimum conformance requirements

NOTE: This measurement is used for UE positioning purposes.

The UE RX-TX time difference is measured from the PCell.

For UE configured with a time-domain measurement resource restriction pattern for PCell measurements, the accuracy requirements in Table 8.1.5.3-1 apply provided that the following conditions are met for the PCell:

PCell cell specific reference signals are transmitted from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled,

No changes to the uplink transmission timing are applied during the measurement period,

$RSRP|_{dBm}$ according to clause E.4 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern, and

The UE is provided via PCell with the CRS assistance information (TS 36.331 [22]) and the CRS assistance information is valid throughout the entire evaluation period.

The requirements in this section shall also be met when the number of transmit antenna ports TS 36.211 [26] of one or more cells whose CRS assistance information is provided TS 36.331 [22] is different from the number of transmit antenna ports of the measured cell.

When the CRS assistance information is provided, the transmission bandwidth in all intra-frequency cells in the CRS assistance information is the same or larger than the transmission bandwidth of the PCell for which measurement is performed.

**Table 8.1.5.3-1: UE Rx – Tx time difference measurement accuracy**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | CRS Ês/Iot Note 9 | Downlink transmission bandwidth of PCell | Io range Note 8 | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | MHz | | dBm/15kHz Note 5 | dBm/$BW_{Channel}$ |
| ±20 | ≥-7.76 dB | ≤ 3 MHz | FDD_A Note 7, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G Note 4 | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±10 | ≥-7.76 dB | ≥ 5 MHz | Note 3 | Note 3 | Note 3 |

NOTE 1: This Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with downlink bandwidth ≤ 3 MHz.
NOTE 4: Except Band 29.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 7: Except Band 32.
NOTE 8: Io is defined in subframes indicated for PCell measurements by the time domain measurement resource restriction pattern. The specified Io range applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 9: CRS Ês/Iot is in subframes indicated for PCell measurements by the time-domain measurement resource restriction pattern.

NOTE: It is up to the UE implementation whether the UE Rx-Tx time difference measurement is performed in any subframe or in subframes indicated by the time-domain measurement resource restriction pattern.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.9.4 and A.9.7.5.

## 8.1.5.4 Test description

The test has two test points with time delays starting at 32 $T_S$ and 5008 $T_S$, respectively. In this test case, there are three cells, Cell 1, Cell 2 and Cell 3, on the same RF channel. Cell 1 is the PCell on which UE Rx-Tx is measured. Cell 2 and

Cell 3 are the interfering cells. A non-MBSFN ABS pattern is configured in each of the Cell 2 and Cell 3 during the entire test. The tested UE is connected to the PCell and configured to transmit SRS signals periodically. The SRS configuration is provided to the UE before the measurement starts. The UE is configured to report UE Rx–Tx time difference measurement. The test equipment measures the transmit timing of the UE using the transmitted SRS. The test equipment then compares this timing to the UE Rx-Tx measurement reported by the UE. The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on PCell. The UE is also provided via higher layers with the CRS assistance information for Cell 2. The information for both measurement patterns and the CRS assistance information shall be provided via RRC to the UE before the measurement starts.

### 8.1.5.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: According to TS 36.521-1 [24] Annex E table E-1 and TS 36.508 [18] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [18] Annex A figure A.54 (without faders).

2. Propagation conditions are set according to clause 4.6.2.1.

3. Message contents are defined in clause 8.1.5.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel. Cell 3 in the test is the Cell 4 defined in clause 4.7.1

**Table 8.1.5.4.1-1: General test parameters for FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The measured cell |
| Neighbour cell | | Cell 2 and Cell 3 | Cell 2 is the first interfering cell to Cell 1, whilst Cell 3 is the second interfering cell to Cell 1. |
| ABS transmission configuration | | Non-MBSFN ABS | As defined in TS 36.521-3 [25] Table C.3.1.2-1. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used |
| Downlink Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For all cells in the test |
| CP length | | Normal | For all cells in the test |
| DRX | | | OFF |
| Time offset between cells | µs | Cell 2 offset with respect to Cell 1: 3<br>Cell 3 offset with respect to Cell 1: 2 | Three synchronous cells |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 =0<br>($PCI_{cell1}$ - $PCI_{cell3}$ )mod6 !=0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that all conditions are met |
| ABS pattern | | '10000001000000010000000010000000' | Non-MBSFN ABS. FDD ABS Pattern Info IE, as defined in TS 36.423 [35], clause 9.2.54. The first/leftmost bit corresponds to the PCell subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes. Configured in Cell 2 and Cell 3 during the testing. |
| Time-domain measurement resource restriction pattern for PCell measurements | | '10000001000000010000000010000000' | Configured for measurements on Cell 1. |
| physCellId | | see PCI conditions above | |

| | | | | |
|---|---|---|---|---|
| CRS assistance information | antennaPortsCount | | 1 | The CRS assistance information is provided for Cell 2 and Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000'. |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

### 8.1.5.4.2 Test procedure

1. Bring the UE to State 3A or 3A-RF according to TS 36.508 [18] clause 4.5.3A or 5.2A.2 with exceptions listed in 7.2A.6, using a value of initial timing advance command $T_A = 2$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 32\ T_s$. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_s$.

2. Set the parameters according to Tables 8.1.5.5-1 and 8.1.5.5-2. Propagation conditions are set according to clause 4.6.2.1.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+4\ T_S$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

5. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the ECID capabilities supported by the UE in the *ECID-ProvideCapabilities* IE.

7. The SS shall transmit an LPP REQUEST LOCATION INFORMATION message, including the *ECID-RequestLocationInformation* IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.

8. The UE shall transmit an LPP PROVIDE LOCATION INFORMATION message including the *ecid-SignalMeasurementInformation* IE.

9. As soon as possible after step 8 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.

10. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

11. The SS shall check the reported value of *ue-RxTxTimeDiff* in the *ecid-SignalMeasurementInformation* IE provided by the UE in step 8 and compare it with the value measured in step 9. The SS shall check that the reported value is within the limits specified in table 8.1.5.5-3 compared to the measured value. If the reported value is within the limits the number of successful results for the test point is increased by one. If the reported value is not within the limits, or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 6, or does not respond at step 8 within the time given by the *time* IE in the *ECID-RequestLocationInformation* IE in step 7, then the number of unsuccessful results for the test point test is increased by one.

12. Repeat steps 3-11 until the confidence level according to Annex D.4.3 is achieved.
    NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

13. Repeat steps 1-12 for test point 2.

### 8.1.5.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 5.2A.5.1 with the following exceptions:

**Table 8.1.5.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc1 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 0 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-3: LPP REQUEST CAPABILITIES: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Information Element | Value/remark |
|---|---|
| ecid-RequestCapabilities | TRUE |

**Table 8.1.5.4.3-4: *ECID-RequestLocationInformation*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |

| | | | |
|---|---|---|---|
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 2 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | 0 0 1 | ueRxTxReq | |
| } | | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-5: *ECID-ProvideLocationInformation*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (felCIC)**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation ::= SEQUENCE { | | | |
| ecid-SignalMeasurementInformation ::= SEQUENCE { | | | |
| primaryCellMeasuredResults | Not Present | | |
| MeasuredResultsList ::= SEQUENCE (SIZE(1..32)) OF MeasuredResultsElement | | | |

| | | | |
|---|---|---|---|
| MeasuredResultsElement ::= SEQUENCE { | | | |
| physCellId | | | |
| cellGlobalId | | | |
| arfcnEUTRA | | | |
| systemFrameNumber | | | |
| rsrp-Result | Not Present | | |
| rsrq-Result | Not Present | | |
| ue-RxTxTimeDiff | | Set according to specific sub-test and test point. | |
| } | | | |
| } | | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellsCRS-Info-r11 ::= CHOICE { | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| Release | NULL | | |
| Setup | CRS-AssistanceInfoList-r11 | | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-7: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 | | | |
| CRS-AssistancedInfo-r11 ::= SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$ - $PCI_{cell3}$ )mod6 = 0<br>($PCI_{cell2}$ - $PCI_{cell3}$ )mod6 != 0 | Cell PCIs are selected so that both conditions are met | |
| antennaPortsCount-r11 | an1 | | |
| mbsfn-SubframeConfigList-r11 | MBSFN-SubframeConfigList | | |
| } | | | |

**Table 8.1.5.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| MBSFN-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | Only the CRS information of Cell 2 is provided in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *oneFrame*='000000' BIT STRING (SIZE(6)) | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-9: *RadioResourceConfigDedicated-SRB2-DRB(n, m):* Additional FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | Cell1 |
| } | | | |
| } | | | |
| } | | | |

## 8.1.5.5 Test requirement

Table 8.1.5.5-1 defines the primary level settings including test tolerances for the test.

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.5.5-3.

**Table 8.1.5.5-1: Test parameters test parameters for FDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | 1 | 1 |
| PDSCH Reference measurement channel defined in TS 36.521-3 [25] A.1.1 | | R.0 FDD | N/A | N/A |
| PDSCH allocation | $n_{PRB}$ | 13—36 | N/A | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in TS 36.521-3 [25] A.2.1 | | R.6 FDD | N/A | N/A |
| OCNG Patterns defined in TS 36.521-3 [25] D.1.5 (OP.5 FDD) and in D.1.6 (OP.6 FDD) | | OP.5 FDD | OP.6 FDD | OP.6 FDD |
| PBCH_RA | dB | 0 | Non-ABS and ABS subframe channel powers defined in Table | |
| PBCH_RB | dB | | | |

| | | | | |
|---|---|---|---|---|
| PSS_RA | dB | | C.3.1.1.1-1-1 in TS 36.521-3 [25]. | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | -98 | -98 |
| CRS $\hat{E}_s / N_{oc}$ | dB | -2.60 | 3 | 1 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 3 | dB | -7.36 | 1.10 | -0.90 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ Note 3 | dB | -8.89 | -1.48 | -4.50 |
| RSRP Note 4 | dBm/15 kHz | -100.6 | -95 | -97 |
| $(\mathrm{Io})_{meas}$ Note 4 | dBm/9 MHz | - | - | - |
| $(\mathrm{Io})_{nonABS}$ Note 4 | dBm/9 MHz | -63.40 | -63.40 | -63.40 |
| Propagation condition | | AWGN | | |

Note 1: OCNG shall be used such that the resources in the active cell are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 3: $\left(\hat{E}_s / I_{ot}\right)_{meas}$ is calculated in CRS REs in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ is calculated in CRS REs in the subframes not indicated for PCell measurements by measurement resource restriction pattern.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. $(\mathrm{Io})_{meas}$ is calculated in CRS symbols in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $(\mathrm{Io})_{nonABS}$ is calculated in CRS symbols in the subframes not indicated for PCell measurements by measurement resource restriction pattern.

**Table 8.1.5.5-2: Sounding Reference Symbol Configuration to be used in FDD UE Rx–Tx time difference test**

| Field | Value | Comment |
|---|---|---|
| UL bandwidth | 50 RBs | Same as the DL bandwidth |
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc1 | |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | Not applicable for FDD |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| srs-ConfigIndex | 0 | SRS periodicity of 2ms |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| srsAntennaPort | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [22]. | | |

**Table 8.1.5.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Test requirement |
|---|---|
| Lowest reported value | (Measured value from step 7 - 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| Highest reported value | (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

NOTE: The test in table 8.1.5.5-3 has two test points starting at 32 $T_s$ and 5008 $T_s$.

The test tolerances are defined in Annex C.

For the overall test to pass, the ratio of successful reported values in each test point shall be more than 90% with a confidence level of 95%.

## 8.1.6 E-UTRAN TDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

### 8.1.6.1 Test purpose

The purpose of this test is to verify that the E-UTRAN TDD UE Rx – Tx time difference measurement accuracy is within the specified limits in TS 36.133 [23] clause 9.1.9.4 when the UE is provided with a time-domain measurement resource restriction pattern and CRS assistance information, and when non-MBSFN ABS is configured in the interfering cells.

### 8.1.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward with LPP release 13 onwards that supports ECID positioning and CRS interference handling. Applicability requires support of FGI bit 115. Note that for LPP releases before release 13 the UE Rx - Tx time difference measurement report mapping is ambiguous and therefore this test shall not be used.

### 8.1.6.3 Minimum conformance requirements

NOTE: This measurement is used for UE positioning purposes.

The UE RX-TX time difference is measured from the PCell.

For UE configured with a time-domain measurement resource restriction pattern for PCell measurements, the accuracy requirements in Table 8.1.5.3-1 apply provided that the following conditions are met for the PCell:

PCell cell specific reference signals are transmitted from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled,

No changes to the uplink transmission timing are applied during the measurement period,

$RSRP|_{dBm}$ according to clause E.4 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern, and

The UE is provided via PCell with the CRS assistance information (TS 36.331 [22]) and the CRS assistance information is valid throughout the entire evaluation period.

The requirements in this section shall also be met when the number of transmit antenna ports TS 36.211 [26] of one or more cells whose CRS assistance information is provided TS 36.331 [22] is different from the number of transmit antenna ports of the measured cell.

When the CRS assistance information is provided, the transmission bandwidth in all intra-frequency cells in the CRS assistance information is the same or larger than the transmission bandwidth of the PCell for which measurement is performed.

**Table 8.1.6.3-1: TDD UE Rx – Tx time difference measurement accuracy**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | CRS Ês/Iot Note 9 | Downlink transmission bandwidth of PCell | Io range Note 8 | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| Ts Note 2 | dB | MHz | | dBm/15kHz Note 5 | dBm/BW$_{Channel}$ |
| ±20 | ≥-7.76 dB | ≤ 3 MHz | FDD_A Note 7, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G Note 4 | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±10 | ≥-7.76 dB | ≥ 5 MHz | Note 3 | Note 3 | Note 3 |

NOTE 1: This Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with downlink bandwidth ≤ 3 MHz.
NOTE 4: Except Band 29.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 7: Except Band 32.
NOTE 8: Io is defined in subframes indicated for PCell measurements by the time domain measurement resource restriction pattern. The specified Io range applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 9: CRS Ês/Iot is in subframes indicated for PCell measurements by the time-domain measurement resource restriction pattern.

NOTE: It is up to the UE implementation whether the UE Rx-Tx time difference measurement is performed in any subframe or in subframes indicated by the time-domain measurement resource restriction pattern.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.9.4 and A.9.7.6.

## 8.1.6.4 Test description

The test has two test points with time delays starting at 32 $T_S$ and 5008 $T_S$, respectively. In this test case, there are three cells, Cell 1, Cell 2 and Cell 3, on the same RF channel. Cell 1 is the PCell on which UE Rx-Tx is measured. Cell 2 and Cell 3 are the interfering cells. A non-MBSFN ABS pattern is configured in each of the Cell 2 and Cell 3 during the entire test. The tested UE is connected to the PCell and configured to transmit SRS signals periodically. The SRS configuration is provided to the UE before the measurement starts. The UE is configured to report UE Rx–Tx time difference measurement. The test equipment measures the transmit timing of the UE using the transmitted SRS. The test equipment then compares this timing to the UE Rx-Tx measurement reported by the UE. The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on PCell. The UE is also provided via higher layers with the CRS assistance information for Cell 2. The information for both measurement patterns and the CRS assistance information shall be provided via RRC to the UE before the measurement starts.

### 8.1.6.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: According to TS 36.521-1 [24] Annex E table E-1 and TS 36.508 [18] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [18] Annex A Figure A.54 (without faders).

2. Propagation conditions are set according to clause 4.6.2.1.

3. Message contents are defined in clause 8.1.6.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel. Cell 3 in the test is the Cell 4 defined in clause 4.7.1

**Table 8.1.6.4.1-1: General test parameters for TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Serving cell (PCell) | | | Cell 1 | Cell to be measured |
| Neighbour cell | | | Cell 2 and Cell 3 | Cell 2 is the first interfering cell to Cell 1, whilst Cell 3 is the second interfering cell to Cell 1. |
| ABS transmission configuration | | | Non-MBSFN ABS | As defined in TS 36.521-3 [25] Table C.3.1.2.1-1 |
| E-UTRA RF Channel Number | | | 1 | One TDD carrier frequency is used |
| Downlink Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | For all cells in the test |
| CP length | | | Normal | For all cells in the test |
| Special subframe configuration | | | 6 | For all cells in the test. For special subframe configurations see Table 4.2-1 in TS 36.211 [26]. |
| Uplink/downlink subframe configuration | | | 1 | For all cells in the test. For uplink-downlink subframe configurations see Table 4.2-2 in TS 36.211 [26]. |
| DRX | | | | OFF |
| Time offset between cells | | µs | Cell 2 offset with respect to Cell 1: 3<br>Cell 3 offset with respect to Cell 1: 2 | Three synchronous cells |
| Physical cell ID PCI | | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 =0<br>($PCI_{cell1}$ - $PCI_{cell3}$ )mod6 !=0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that both conditions are met |
| ABS pattern | | | '00000000010000000001' | Non-MBSFN ABS. TDD ABS Pattern Info IE, as defined in TS 36.423 [35], clause 9.2.54.<br>The first/leftmost bit corresponds to the PCell subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes. Configured in Cell 2 and Cell 3 during the testing. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | | '00000000010000000001' | Configured for measurements on Cell 1. |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 and Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig |
| | antennaPortsCount | | 1 | |

| | | | | |
|---|---|---|---|---|
| | mbsfn-SubframeConfigList | | oneFrame = '000000' | element with subframe allocation one Frame='000000'. |

### 8.1.6.4.2 Test procedure

1. Bring the UE to State 3A or 3A-RF according to TS 36.508 [18] clause 4.5.3A or 5.2A.2, with exceptions listed in 7.2A.6 using a value of initial timing advance command $T_A = 2$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 32\ T_s$. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_s$.

2. Set the parameters according to Tables 8.1.6.5-1 and 8.1.6.5-2 as appropriate. Propagation conditions are set according to clause 4.6.2.1.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+4\ T_S$, compared to the current value.

4. Wait for 1.6s to allow for the possibility that the UE makes autonomous timing adjustments.

5. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the ECID capabilities supported by the UE in the *ECID-ProvideCapabilities* IE. The IE *ueRxTxSupTDD-r13* shall be present (TRUE).

7. The SS shall transmit an LPP REQUEST LOCATION INFORMATION message, including the *ECID-RequestLocationInformation* IE. If the UE message at step 4b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.

8. The UE shall transmit an LPP PROVIDE LOCATION INFORMATION message including the *ecid-SignalMeasurementInformation* IE.

9. As soon as possible after step 8 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.

10. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

11. The SS shall check the reported value of *ue-RxTxTimeDiff* in the *ecid-SignalMeasurementInformation* IE provided by the UE in step 8 and compare it with the value measured in step 9. The SS shall check that the reported value is within the limits specified in table 8.1.6.5-3 for test compared to the measured value. If the reported value is within the limits the number of successful results for test is increased by one. If the reported value is not within the limits, or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 8, or does not respond at step 8 within the time given by the *time* IE in the *ECID-RequestLocationInformation* IE in step 7, then the number of unsuccessful results for testis increased by one.

12. Repeat steps 3-11 until the confidence level according to Annex D.4.3 is achieved.
    NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

13. Repeat steps 1-12 for test point 2.

### 8.1.6.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 5.2A.5.1 with the following exceptions:

**Table 8.1.6.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| srs-BandwidthConfig | bw0 | | |
| srs-SubframeConfig | sc1 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw5 | | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 0 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-3: LPP REQUEST CAPABILITIES: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Information Element | Value/remark |
|---|---|
| ecid-RequestCapabilities | TRUE |

**Table 8.1.6.4.3-3a: LPP PROVIDE CAPABILITIES: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Information Element | Value/remark |
|---|---|
| ueRxTxSupTDD-r13 | TRUE |

**Table 8.1.6.4.3-4: *ECID-RequestLocationInformation*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 2 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | 0 0 1 | ueRxTxReq | |
| } | | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-5: *ECID-ProvideLocationInformation*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation ::= SEQUENCE { | | | |
| ecid-SignalMeasurementInformation ::= | | | |

| | | | |
|---|---|---|---|
| SEQUENCE { | | | |
| primaryCellMeasuredResults | Not Present | | |
| MeasuredResultsList ::= SEQUENCE (SIZE(1..32)) OF MeasuredResultsElement | | | |
| MeasuredResultsElement ::= SEQUENCE { | | | |
| physCellId | | | |
| cellGlobalId | | | |
| arfcnEUTRA | | | |
| systemFrameNumber | | | |
| rsrp-Result | Not Present | | |
| rsrq-Result | Not Present | | |
| ue-RxTxTimeDiff | | Set according to specific sub-test and test point. | |
| } | | | |
| } | | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellsCRS-Info-r11 ::= CHOICE { | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| Release | NULL | | |
| Setup | CRS-AssistanceInfoList-r11 | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-7: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 | | | |
| CRS-AssistancedInfo-r11 ::= SEQUENCE { | | | |
| physCellId-r11 | $(PCI_{cell1} - PCI_{cell3})mod6 = 0$<br>$(PCI_{cell2} - PCI_{cell3})mod6 != 0$ | Cell PCIs are selected so that both conditions are met | |
| antennaPortsCount-r11 | an1 | | |
| mbsfn-SubframeConfigList-r11 | MBSFN-SubframeConfigList | | |
| } | | | |

**Table 8.1.6.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MBSFN-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | Only the CRS information of Cell 2 is provided in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000' BIT STRING (SIZE(6)) | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-9: *RadioResourceConfigDedicated-SRB2-DRB(n, m):* Additional TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | Cell 1 |
| | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.1.6.5 Test requirement

Table 8.1.6.5-1 defines the primary level settings including test tolerances for the test.

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.6.5-3.

**Table 8.1.6.5-1: Test parameters test parameters for TDD UE Rx–Tx time difference measurement under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | 1 | 1 |
| PDSCH Reference measurement channel defined in TS 36.521-3 [25] A.1.2 | | R.0 TDD | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| PDSCH allocation | $n_{PRB}$ | 13—36 | N/A | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in TS 36.521-3 [25] A.2.2 | | R.6 TDD | N/A | N/A |
| OCNG Patterns defined in TS 36.521-3 [25] D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | Non-ABS and ABS subframe channel powers defined in TS 36.521-3 [25] Table C.3.1.2.1-1 | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA[Note1] | dB | | | |
| OCNG_RB[Note1] | dB | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 | -98 |
| CRS $\hat{E}_s / N_{oc}$ | dB | -2.60 | 3 | 1 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 3 | dB | -7.36 | 1.10 | -0.90 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ Note 3 | dB | -8.89 | -1.48 | -4.50 |
| RSRP Note 4 | dBm/15 kHz | -100.6 | -95 | -97 |
| $(\mathrm{Io})_{meas}$ Note 4 | dBm/9 MHz | - | - | - |
| $(\mathrm{Io})_{nonABS}$ Note 4 | dBm/9 MHz | -63.40 | -63.40 | -63.40 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that the resources in the active cell are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\left(\hat{E}_s / I_{ot}\right)_{meas}$ is calculated in CRS REs in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $\left(\hat{E}_s / I_{ot}\right)_{nonABS}$ is calculated in CRS REs in the subframes not indicated for PCell measurements by measurement resource restriction pattern.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. $(\mathrm{Io})_{meas}$ is calculated in CRS symbols in the subframes indicated for PCell measurements by measurement resource restriction pattern, whilst $(\mathrm{Io})_{nonABS}$ is calculated in CRS symbols

**Table 8.1.6.5-2: Sounding Reference Symbol Configuration to be used in FDD UE Rx–Tx time difference test**

| Field | Value | Comment |
|---|---|---|
| UL bandwidth | 50 RBs | Same as the DL bandwidth |
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc1 | |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | TRUE | |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |

| | | |
|---|---|---|
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 10 | SRS periodicity of 10ms for all Tests. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [22]. | | |

**Table 8.1.6.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | Test requirement |
|---|---|
| Lowest reported value | (Measured value from step 7 - 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |
| Highest reported value | (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |

NOTE: The test in table 8.1.6.5-3 has two test points starting at 32 $T_s$ and 5008 $T_s$.

The test tolerances are defined in Annex C.

For the overall test to pass, the ratio of successful reported values in each test point shall be more than 90% with a confidence level of 95%.

## 8.1.7 E-UTRAN FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA

### 8.1.7.1 Test purpose

The purpose of this test is to verify that the E-UTRAN FDD UE Rx – Tx time difference measurement accuracy for Category M1/M2 UEs is within the specified limits in TS 36.133 [23] clause 9.1.21.19 and 9.1.25.3.

### 8.1.7.2 Test applicability

This test applies to E-UTRA FDD UE Category M1/M2 release 14 and forward that supports ECID positioning.

### 8.1.7.3 Minimum conformance requirements

NOTE: This measurement is used for UE positioning purposes.

The UE RX-TX time difference is measured from the PCell.

The accuracy requirements in Table 8.1.7.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

No changes to the uplink transmission timing are applied during the measurement period.

$RSRP|_{dBm}$ according to clause E.1-1 for a corresponding Band.

**Table 8.1.7.3-1: UE Rx – Tx time difference measurement accuracy for CEModeA**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | Ês/Iot | Downlink transmission bandwidth of PCell | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |

| Ts Note 2 | dB | MHz | | dBm/15kHz Note 5 | dBm/BW$_{Channel}$ |
|---|---|---|---|---|---|
| ±20 | ≥-3 dB | ≥ 6 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | FDD-M1_B | -120.5 | -50 |
| | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | FDD-M1_D | -119.5 | -50 |
| | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | FDD-M1_F | -118.5 | -50 |
| | | | FDD-M1_G | -118 | -50 |
| | | | FDD-M1_H | -117.5 | -50 |
| | | | FDD-M1_N | -114.5 | -50 |
| ±10 Note 8 | ≥-3 dB | ≥ 24 | Note 3 | Note 3 | Note 3 |

NOTE 1: When in dBm/15kHz, the minimum Io condition is expressed as the average Io per RE over all REs in that symbol. Io may be different in different symbols within a subframe.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with downlink bandwidth ≤ 3 MHz.
NOTE 4: Except Band 29.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 7: Except Band 32.
NOTE 8: Only for Category M2.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.19, 9.1.25.3 and A.9.7.7.

## 8.1.7.4 Test description

There is only one active cell in the tests. The tested UE is connected with the serving cell, configured to transmit SRS signals periodically, and signalled to report UE Rx – Tx time difference measurement. The test equipment measures the transmit timing of the UE using the transmitted SRS. The test equipment then compares this timing to the UE Rx-Tx measurement reported by the UE.

### 8.1.7.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in Annex A figure A.5.

2. Propagation conditions are set according to clause 4.6.2.1.

3. Message contents are defined in clause 8.1.7.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test.

### 8.1.7.4.2 Test procedure

1. Bring the UE to Generic RB Established State 3A-RF-CE according to 3GPP TS 36.508 [18] clause 7.2A.3AA, using a value of initial timing advance command $T_A = 2$ in the Random Access Response which indicates an initial timing advance value $N_{TA} = 32\ T_s$. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_s$.

2. Set the parameters according to Table 8.1.7.5-1 and 8.1.7.5-2 as appropriate. Propagation conditions are set according to clause 4.6.2.1.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+8\ T_S$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

4a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

4b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the ECID capabilities supported by the UE in the *ECID-ProvideCapabilities* IE.

5. The SS shall transmit an LPP REQUEST LOCATION INFORMATION message, including the *ECID-RequestLocationInformation* IE. If the UE message at step 4b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.

6. The UE shall transmit an LPP PROVIDE LOCATION INFORMATION message including the *ecid-SignalMeasurementInformation* IE.

7. As soon as possible after step 6 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.

8. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

9. The SS shall check the reported value of *ue-RxTxTimeDiff* in the *ecid-SignalMeasurementInformation* IE provided by the UE in step 6 and compare it with the value measured in step 7. The SS shall check that the reported value is within the limits specified in table 8.1.7.5-3 compared to the measured value. If the reported value is within the limits the number of successful results is increased by one. If the reported value is not within the limits, or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 6, or does not respond at step 6 within the time given by the *time* IE in the *ECID-RequestLocationInformation* IE in step 5, then the number of unsuccessful results is increased by one.

10. Repeat steps 3-9 until the confidence level according to Annex D.4.3 is achieved.
NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 8.1.7.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 4.6 with the following exceptions:

**Table 8.1.7.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc1 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 8.1.7.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |

| | | | |
|---|---|---|---|
| srs-ConfigIndex | 0 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 8.1.7.4.3-2a: LPP REQUEST CAPABILITIES: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Information Element | Value/remark |
|---|---|
| ecid-RequestCapabilities | TRUE |

**Table 8.1.7.4.3-3: *ECID-RequestLocationInformation*: UE Rx – Tx time difference for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.355 clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 2 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | 0 0 1 | ueRxTxReq | |
| } | | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.1.7.5 Test requirement

Table 8.1.7.5-1 defines the primary level settings including test tolerances.

**Table 8.1.7.5-1: FDD UE Rx – Tx time difference test parameters**

| Parameter | Unit | Test 1 |
|---|---|---|
| E-UTRAN RF Channel Number | | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 |
| DRX | | OFF |
| PRACH Configuration | | PRACH_4CE As specified in TS 36.133 [23] A.3.16 |
| MPDCCH Reference measurement channel Note1 | | R.16 FDD |
| OCNG Pattern Note2 | | OP.21 FDD |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PHICH_RA | dB | |
| PHICH_RB | dB | |
| MPDCCH_RA | dB | |
| MPDCCH_RB | dB | |
| OCNG_RA Note3 | dB | |
| OCNG_RB Note3 | dB | |
| $N_{oc}$ | dBm/15 kHz | -98 |
| $\hat{E}_s / N_{oc}$ | dB | -2.7 |
| $\hat{E}_s / I_{ot}$ | dB | -2.7 |
| Io Note4 | dBm/9 MHz | -65.5 |
| Propagation Condition | | AWGN |

Note 1: For the reference measurement channels, see TS 36.521-3 [25] A.7.1
Note 2: For the OCNG pattern, see clause A.3.2.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 8.1.7.5-2: Sounding Reference Symbol Configuration to be used in FDD UE Rx – Tx time difference test**

| Field | Test 1 | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc1 | |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | Not applicable for FDD |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 0 | SRS periodicity of 2ms for all Tests. |
| transmissionComb | 0 | |

| | | |
|---|---|---|
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [22]. | | |

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.7.5-3.

**Table 8.1.7.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | |
|---|---|
| Lowest reported value | Category M1: (Measured value from step 7 – 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>Category M2: (Measured value from step 7 – 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| Highest reported value | Category M1: (Measured value from step 7 + 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>Category M2: (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

The test tolerances are defined in Annex C.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 8.1.8 E-UTRAN HD-FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA

### 8.1.8.1 Test purpose

The purpose of this test is to verify that the E-UTRAN HD-FDD UE Rx – Tx time difference measurement accuracy for Category M1/M2 UEs is within the specified limits in TS 36.133 [23] clause 9.1.21.19 and 9.1.25.3.

### 8.1.8.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1/M2 release 14 and forward that supports ECID positioning.

### 8.1.8.3 Minimum conformance requirements

Same in section 8.1.7.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.19, 9.1.25.3 and A.9.7.8.

### 8.1.8.4 Test description

Same as in clause 8.1.7.4.

#### 8.1.8.4.1 Initial conditions

Same as in clause 8.1.7.4.1.

#### 8.1.8.4.2 Test procedure

Same as in clause 8.1.7.4.2.

#### 8.1.8.4.3 Message contents

Same as in clause 8.1.7.4.3.

### 8.1.8.5 Test requirement

Table 8.1.8.5-1 defines the primary level settings including test tolerances.

**Table 8.1.8.5-1: HD-FDD UE Rx – Tx time difference test parameters**

| Parameter | Unit | Test 1 |
|---|---|---|
| E-UTRAN RF Channel Number | | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 |
| DRX | | OFF |
| PRACH Configuration | | PRACH_4CE As specified in TS 36.133 [23] A.3.16 |
| MPDCCH Reference measurement channel[Note1] | | R.6 HD-FDD |
| OCNG Pattern[Note2] | | OP.21 FDD |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PHICH_RA | dB | |
| PHICH_RB | dB | |
| MPDCCH_RA | dB | |
| MPDCCH_RB | dB | |
| OCNG_RA[Note3] | dB | |
| OCNG_RB[Note3] | dB | |
| $N_{oc}$ | dBm/15 kHz | -98 |
| $\hat{E}_s / N_{oc}$ | dB | -2.7 |
| $\hat{E}_s / I_{ot}$ | dB | -2.7 |
| Io[Note4] | dBm/9 MHz | -65.5 |
| Propagation Condition | | AWGN |

Note 1: For the reference measurement channels, see TS 36.521-3 [25] A.7.2
Note 2: For the OCNG pattern, see clause A.3.2.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 8.1.8.5-2: Sounding Reference Symbol Configuration to be used in HD-FDD UE Rx – Tx time difference test**

| Field | Test 1 | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc1 | |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | Not applicable for FDD |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 0 | SRS periodicity of 2ms for all Tests. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |

| Note: | For further information see clause 6.3.2 in TS 36.331 [22]. |
|---|---|

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.8.5-3.

**Table 8.1.8.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | |
|---|---|
| Lowest reported value | Category M1: (Measured value from step 7 – 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>Category M2: (Measured value from step 7 - 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| Highest reported value | Category M1: (Measured value from step 7 + 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>Category M2: (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

The test tolerances are defined in Annex C.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 8.1.9 E-UTRAN TDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA

### 8.1.9.1 Test purpose

The purpose of this test is to verify that the E-UTRAN TDD UE Rx – Tx time difference measurement accuracy for Category M1/M2 UEs is within the specified limits in TS 36.133 [23] clause 9.1.21.19 and 9.1.25.3.

### 8.1.9.2 Test applicability

This test applies to E-UTRA TDD UE Category M1/M2 release 14 and forward that supports ECID positioning.

### 8.1.9.3 Minimum conformance requirements

Same in section 8.1.7.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.19, 9.1.25.3 and A.9.7.9.

### 8.1.9.4 Test description

Same as in clause 8.1.7.4.

#### 8.1.9.4.1 Initial conditions

Same as in clause 8.1.7.4.1.

#### 8.1.9.4.2 Test procedure

Same as in clause 8.1.7.4.2.

### 8.1.9.4.3 Message contents

Same as in clause 8.1.7.4.3 with the following exceptions:

**Table 8.1.9.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | Sc3 | | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | FALSE | | TDD |
| } | | | |

**Table 8.1.9.4.3-2: *SoundingRS-RL-ConfigDedicated-DEFAULT*: UE Rx – Tx time difference for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 15 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

## 8.1.9.5 Test requirement

Table 8.1.9.5-1 defines the primary level settings including test tolerances.

**Table 8.1.9.5-1: TDD UE Rx – Tx time difference test parameters**

| Parameter | Unit | Test 1 |
|---|---|---|
| E-UTRAN RF Channel Number | | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 |
| DRX | | OFF |
| PRACH Configuration | | PRACH_4CE As specified in TS 36.133 [23] A.3.16 |
| MPDCCH Reference measurement channel[Note1] | | R.14 TDD |
| OCNG Pattern[Note2] | | OP.11 TDD |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PHICH_RA | dB | |
| PHICH_RB | dB | |
| MPDCCH_RA | dB | |

| | | |
|---|---|---|
| MPDCCH_RB | dB | |
| OCNG_RA[Note3] | dB | |
| OCNG_RB[Note3] | dB | |
| $N_{oc}$ | dBm/15 kHz | -98 |
| $\hat{E}_s / N_{oc}$ | dB | -2.7 |
| $\hat{E}_s / I_{ot}$ | dB | -2.7 |
| Io[Note4] | dBm/9 MHz | -65.5 |
| Propagation Condition | | AWGN |
| Note 1: For the reference measurement channels, see TS 36.521-3 [25] A.7.3<br>Note 2: For the OCNG pattern, see clause A.3.2.<br>Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter. | | |

**Table 8.1.9.5-2: Sounding Reference Symbol Configuration to be used in TDD UE Rx – Tx time difference test**

| Field | Test 1 | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc3 | |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | FALSE | |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | SRS periodicity of 10ms |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [22]. | | |

The UE Rx – Tx time difference measurement accuracy shall fulfil the requirements in Table 8.1.9.5-3.

**Table 8.1.9.5-3: Test requirements UE Rx – Tx time difference measurement accuracy requirements**

| | |
|---|---|
| Lowest reported value | Category M1: (Measured value from step 7 – 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2<br>Category M2: (Measured value from step 7 – 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |
| Highest reported value | Category M1: (Measured value from step 7 + 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2<br>Category M2: (Measured value from step 7 + 13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-2 |

The test tolerances are defined in Annex C.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 9 E-UTRA OTDOA measurement requirements

## 9.0 General

This clause defines the minimum performance requirements for OTDOA FDD and TDD E-UTRA UEs and UEs supporting NR EN-DC.

## 9.1 RSTD Intra-Frequency Measurements

### 9.1.1 FDD RSTD Measurement Reporting Delay

#### 9.1.1.1 Test purpose

To verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

#### 9.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that supports UE-assisted OTDOA except UE Category 1bis.

#### 9.1.1.3 Minimum conformance requirements

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data are provided, the UE shall be able to detect and measure intra-frequency RSTD, specified in 3GPP TS 36.214 [6], for at least *n*=16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{RSTDIntraFreqFDD,E-UTRAN}$ ms as given below (see also Figure 9.1.1.3-1):

$$T_{RSTDIntraFreqFDD,E-UTRAN} = T_{PRS} \cdot (M - 1) + \Delta \quad ms,$$

where

$T_{RSTDIntraFreqFDD,E-UTRAN}$ is the total time for detecting and measuring at least *n* cells,

$T_{PRS}$ is the cell-specific positioning subframe configuration period as defined in 3GPP TS 36.211 [26],

$M$ is the number of PRS positioning occasions as defined in Table 9.1.1.3-1, where each PRS positioning occasion comprises of $N_{PRS}$ ($1 \le N_{PRS} \le 6$) consecutive downlink positioning subframes defined in 3GPP TS 36.211 [26], and

$\Delta = 160 \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single PRS positioning occasion which includes the sampling time and the processing time.

**Table 9.1.1.3-1: Number of PRS positioning occasions within $T_{RSTDIntraFreqFDD,E-UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| | f1 Note 1 |
| 160 ms | 16 |
| >160 ms | 8 |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving FDD carrier frequency f1. | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTDIntraFreqFDD,E-UTRAN}$ provided:

$\left(PRS\ \hat{E}_s / Iot\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(PRS\hat{E}_s / Iot\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(PRS\ \hat{E}_s / Iot\right)_{ref}$ and $\left(PRS\hat{E}_s / Iot\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ PRS positioning occasions,

$PRP\ 1,2|_{dBm}$ according to clause E.2 for a corresponding Band.

The time $T_{RSTDIntraFreqFDD,E-UTRAN}$ starts from the first subframe of the PRS positioning occasion closest in time after both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in 3GPP TS 36.355 [4], are delivered to the physical layer of the UE as illustrated in Figure 9.1.1.3-1.

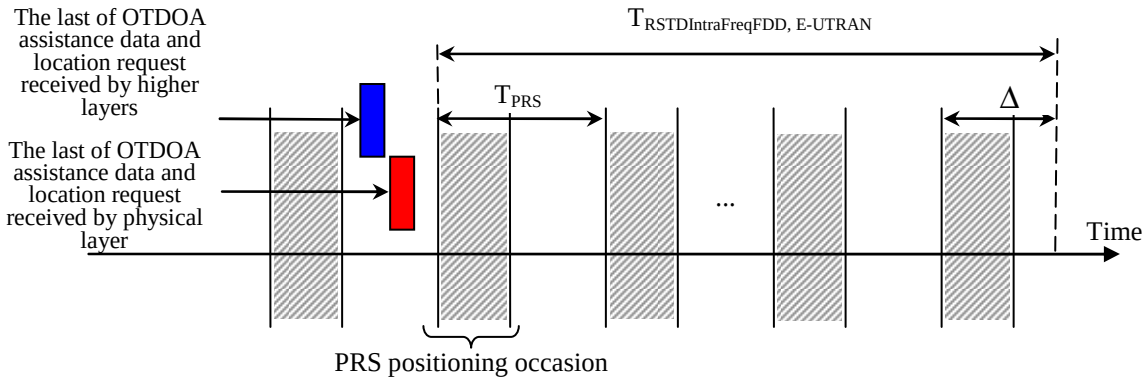



**Figure 9.1.1.3-1: Illustration of the RSTD reporting time requirement in an FDD system**

This requirement assumes that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.5.1 and A.8.12.1.

## 9.1.1.4 Test description

### 9.1.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4.

2. The general test parameter settings are set up according to Table 9.1.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.1.1.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel. Cell 3 in the test is the Cell 4 defined in clause 4.7.1. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.

**Table 9.1.1.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 171 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | μs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | μs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |

| | | | |
|---|---|---|---|
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.28 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.1.4.1.

**Table 9.1.1.4.1-2: DRX parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.1.1.4.2 Test procedure

The test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 9.1.1.4.1-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. Cell 2 is active until the end of T3, and Cell 3 is active until the end of T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, while Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.1.1.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 9.1.1.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).

4. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

6a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the first 7 elements of the sequence, and the position of neighbour Cell 3 is randomly selected to be in the last 8 elements of the sequence, as described in 3GPP TS 37.571-5 [20], clause 7.2.3. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.

9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 9.1.1.5-2.

10. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.1.1.5-2.

11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 9.1.1.5. The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. Repeat steps 2-12 until the confidence level according to Annex D is achieved. For each iteration, at step 7 change the random position of the Cells 2 and 3 in the *OTDOA-NeighbourCellInfoList.*

### 9.1.1.4.3 Message contents

**Table 9.1.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.1.1.4.3-2: *MAC-MainConfig-RBC*: FDD RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |

| | | | |
|---|---|---|---|
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.1.4.3-2a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.1.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 3 | See clause 9.1.1.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.1.4.3-4: Void**

**Table 9.1.1.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.1.4.3-6: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.1.5 Test requirement

Table 9.1.1.5-1 and 9.1.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.1.1.5-1: Cell-specific test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.5 FDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |

| | | | | |
|---|---|---|---|---|
| PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.1.1.5-2: Cell-specific test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.5 FDD | | OP.6 FDD | | OP.6 FDD | N/A |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -1 | -Infinity | -Infinity | -7 | -7 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1.79 | -Infinity | -Infinity | -7 | -9.54 | -Infinity |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Io Note 4 | dBm/ 9 MHz | -69.55 | -67.08 | -69.55 | -67.08 | -69.55 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -105 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -93 | -105 | -105 | -108 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -10 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 3.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 2710 ms. This is rounded up to the next allowed LPP value of 3 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=8 and $n$=16 are the parameters specified in clause 9.1.1.3 and Table 9.1.1.3-1. This gives the total RSTD reporting delay of 2560 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.1.1A FDD RSTD Measurement Reporting Delay for UE Category 1bis

### 9.1.1A.1 Test purpose

To verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

### 9.1.1A.2 Test applicability

This test applies to E-UTRA FDD UE Category 1bis release 13 and forward that supports UE-assisted OTDOA.

### 9.1.1A.3 Minimum conformance requirements

Same as 9.1.1.3 but using Table 9.1.1A.3-1 instead of Table 9.1.1.3-1.

**Table 9.1.1A.3-1: Number of PRS positioning occasions within $T_{RSTD,IntraFreq,FDD,E-UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| | f1 Note 1 |
| 160 ms | 32 |

| | |
|---|---|
| >160 ms | 16 |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving FDD carrier frequency f1. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.5.3 and A.8.12.1.

## 9.1.1A.4 Test description

### 9.1.1A.4.1 Initial conditions

Same as 9.1.1.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4, using only the main Tx/Rx path.

2. The general test parameter settings are set up according to Table 9.1.1A.4.1-1.

**Table 9.1.1A.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 171 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.1.4.1. | | | |

### 9.1.1A.4.2 Test procedure

Same as 9.1.1.4.2.

### 9.1.1A.4.3 Message contents

Same as 9.1.1.4.3 with the following exceptions:

**Table 9.1.1A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.1.1.4.3-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See clause 9.1.1A.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.1.1A.5 Test requirement

Table 9.1.1.5-1 and 9.1.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 5110 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=8 and $n$=16 are the parameters specified in clause 9.1.1A.3 and Table 9.1.1A.3-1. This gives the total RSTD reporting delay of 4960 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.1.2 TDD RSTD Measurement Reporting Delay

### 9.1.2.1 Test purpose

To verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

### 9.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that supports UE-assisted OTDOA except UE Category 1bis.

### 9.1.2.3 Minimum conformance requirements

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data are provided, the UE shall be able to detect and measure intra-frequency RSTD, specified in 3GPP TS 36.214 [6], for at least *n*=16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{RSTDIntraFreqTDD,E\text{-}UTRAN}$ ms as given below:

$$T_{RSTDIntraFreqTDD,E\text{-}UTRAN} = T_{PRS} \cdot (M\text{-}1) + \Delta \quad ms,$$

where

$T_{RSTDIntraFreqTDD,E\text{-}UTRAN}$ is the total time for detecting and measuring at least *n* cells,

$T_{PRS}$ is the cell-specific positioning subframe configuration period as defined in 3GPP TS 36.211 [26],

$M$ is the number of PRS positioning occasions as defined in Table 9.1.2.3-1, where each PRS positioning occasion comprises of $N_{PRS}$ ($1 \le N_{PRS} \le 6$) consecutive downlink positioning subframes defined in 3GPP TS 36.211 [26], and

$\Delta = 160 \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single PRS positioning occasion which includes the sampling time and the processing time.

**Table 9.1.2.3-1: Number of PRS positioning occasions within $T_{RSTD_IntraFreq_FDD_E\text{-}UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| | **f1** Note 1 |
| 160 ms | 16 |
| >160 ms | 8 |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving FDD carrier frequency f1. | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD_IntraFreq_FDD_E\text{-}UTRAN}$ provided:

$\left(\text{PRS } \hat{E}_s / \text{Iot}\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\text{PRS}\hat{E}_s / \text{Iot}\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{PRS } \hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{PRS}\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ PRS positioning occasions,

PRP $1,2|_{dBm}$ according to clause E.2 for a corresponding Band.

The time $T_{RSTD_IntraFreq_FDD_E\text{-}UTRAN}$ starts from the first subframe of the PRS positioning occasion closest in time after both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in 3GPP TS 36.355 [4], are delivered to the physical layer of the UE.

The requirements shall apply for all TDD special subframe configurations specified in 3GPP TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.1.2.3-2.

**Table 9.1.2.3-2: TDD uplink-downlink subframe configurations applicable for TDD intra-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6, 15 | 1, 2, 3, 4 and 5 |
| 25, 50, 75, 100 | 0, 1, 2, 3, 4, 5 and 6 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in 3GPP TS 36.211 [26]. | |

This requirement assumes that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.5.2 and A.8.12.2.

## 9.1.2.4 Test description

### 9.1.2.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.2.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4.

2. The general test parameter settings are set up according to Table 9.1.2.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.1.2.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel. Cell 3 in the test is the Cell 4 defined in clause 4.7.1. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.

**Table 9.1.2.4-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 174 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Section 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Section 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | The same CP length applies for DL and UL |
| DRX | | ON | DRX parameters are further specified in Table 9.1.2.4-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | μs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.28 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.2.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.1.2.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.2.4.1.

**Table 9.1.2.4-2: DRX parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2. |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | disable | |

### 9.1.2.4.2 Test procedure

The test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 9.1.2.4-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. Cell 2 is active until the end of T3, and Cell 3 is active until the end of T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, while Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.1.2.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.1.2.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).

4. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

6a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the first 7 elements of the sequence, and the position of neighbour Cell 3 is randomly selected to be in the last 8 elements of the sequence, as described in 3GPP TS 37.571-5 [20], clause 7.2.3. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.

9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 9.1.2.5-3.

10. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.1.2.5-3.

11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 9.1.1.5. The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. Repeat steps 2-12 until the confidence level according to Annex D is achieved. For each iteration, at step 9 change the random position of the Cells 2 and 3 in the *OTDOA-NeighbourCellInfoList.*

#### 9.1.2.4.3 Message contents

**Table 9.1.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.1.2.4.3-2: *MAC-MainConfig-RBC*: TDD RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |

| | | | |
|---|---|---|---|
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.2.4.3-2a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.1.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 3 | See clause 9.1.2.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 9.1.2.4.3-4: Void**

**Table 9.1.2.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.2.4.3-6: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.2.5 Test requirement

Table 9.1.2.5-1 and 9.1.2.5-2 define the primary level settings including test tolerances for the test.

**Table 9.1.2.5-1: Cell-specific test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.1 TDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |

| | | | | |
|---|---|---|---|---|
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| $\hat{E}_s/N_{oc}$ PRS | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.1.2.5-2: Cell-specific test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.1 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| $\hat{E}_s/N_{oc}$ PRS | dB | -1 | -Infinity | -Infinity | -7 | -7 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1.79 | -Infinity | -Infinity | -7 | -9.54 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.55 | -67.08 | -69.55 | -67.08 | -69.55 | N/A |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -105 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -93 | -105 | -105 | -108 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -10 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 3.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 2710 ms. This is rounded up to the next allowed LPP value of 3 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=8 and $n$=16 are the parameters specified in clause 9.1.2.3 and Table 9.1.2.3-1. This gives the total RSTD reporting delay of 2560 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.1.2A TDD RSTD Measurement Reporting Delay for UE Category 1bis

### 9.1.2A.1 Test purpose

To verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

### 9.1.2A.2 Test applicability

This test applies to E-UTRA TDD UE Category 1bis release 13 and forward that supports UE-assisted OTDOA.

### 9.1.2A.3 Minimum conformance requirements

Same as 9.1.2.3 but using Table 9.1.2A.3-1 instead of Table 9.1.2.3-1.

**Table 9.1.2A.3-1: Number of PRS positioning occasions within $T_{RSTD,IntraFreq,TDD,E-UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| | f1 Note 1 |
| 160 ms | 32 |
| >160 ms | 16 |

Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving TDD carrier frequency f1

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.5.4 and A.8.12.2.

## 9.1.2A.4 Test description

### 9.1.2A.4.1 Initial conditions

Same as 9.1.2.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4, using only the main Tx/Rx path.

2. The general test parameter settings are set up according to Table 9.1.2A.4.1-1.

**Table 9.1.2.4A-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 174 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Section 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Section 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | The same CP length applies for DL and UL |
| DRX | | ON | DRX parameters are further specified in Table 9.1.2.4-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.2.4.3-5 and TS 37.571-5 [20], clause 7.2.2.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.1.2.4.3-5 and TS 37.571-5 [20], clause 7.2.2.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.2.4.1. | | | |

### 9.1.2A.4.2 Test procedure

Same as 9.1.2.4.2

### 9.1.2A.4.3 Message contents

Same as 9.1.2.4.3 with the following exceptions:

**Table 9.1.2A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.1.2.4.3-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See clause 9.1.2A.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

### 9.1.2A.5 Test requirement

Table 9.1.2.5-1 and 9.1.2.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 5110 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=8 and $n$=16 are the parameters specified in clause 9.1.2A.3 and Table 9.1.2A.3-1. This gives the total RSTD reporting delay of 4960 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.1.3 FDD RSTD Measurement Accuracy

### 9.1.3.1 Test purpose

To verify that the RSTD FDD intra-frequency measurement accuracy is within the specified limits.

### 9.1.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that supports UE-assisted OTDOA except UE Category 1bis.

### 9.1.3.3 Minimum conformance requirements

The accuracy requirements in Table 9.1.3.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to clause E.2 for a corresponding Band.

There are no measurement gaps overlapping with the PRS subframes of the measured cell.

The parameter *expectedRSTDUncertainty* signalled over LPP as defined in 3GPP TS 36.355 [4] is less than 5 µs.

**Table 9.1.3.3-1: RSTD measurement accuracy**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | Io [Note 9] range | | |
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | E-UTRA operating band groups [Note 10] | Minimum Io [Note 1] | Maximum Io |

| | | cell *i* Notes 3, 6 | | | | |
|---|---|---|---|---|---|---|
| **Ts** Note 2 | **dB** | **RB** | | | **dBm/15kHz** Note 8 | **dBm/BW$_{Channel}$** |
| ±15 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 6 | 6 | FDD_A, TDD_A | -121 | -50 |
| | | | | FDD_B | -120.5 | -50 |
| | | | | FDD_C, TDD_C | -120 | -50 |
| | | | | FDD_D | -119.5 | -50 |
| | | | | FDD_E, TDD_E | -119 | -50 |
| | | | | FDD_F | -118.5 | -50 |
| | | | | FDD_G | -118 | -50 |
| | | | | FDD_H | -117.5 | -50 |
| | | | | FDD_N | -114.5 | -50 |
| ±10 Note 11 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 15 | 6 | Note 5 | Note 5 | Note 5 |
| ±6 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 25 | ≥ 2 | Note 5 | Note 5 | Note 5 |
| ±5 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 50 | ≥ 1 | Note 5 | Note 5 | Note 5 |
| ±4 Note 11 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 75 | ≥ 1 | Note 5 | Note 5 | Note 5 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: Void.
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 6 RB.
NOTE 6: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 7: Void.
NOTE 8: The condition level is increased by Δ>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 9: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 10: E-UTRA operating band groups are as defined in clause 4.4.2.
NOTE 11: Only applicable from Rel-12 onwards

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.1 and A.9.8.1.

### 9.1.3.4 Test description

#### 9.1.3.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 1.4 MHz (Test 1 and 2) and 10 MHz (Test 3 and 4). In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test, as defined in TS 36.101 [2] clause 5.6.1, then this part of the test shall be omitted.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3a for 4RX capable UE without any 2RX bands. Otherwise use Annex A figure A.3.

2. The general test parameter settings are set up according to Table 9.1.3.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 9.1.3.4.3.

5. All cells are on the same carrier frequency. Cell 1 is the serving cell and OTDOA assistance data reference cell; Cell 2 is the neighbour cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:
   Test 1: -92 Ts (about -3 μs)
   Test 2: 92 Ts (about 3 μs)
   Test 3: 92 Ts (about 3 μs)
   Test 4: -92 Ts (about -3 μs)
   Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.1.3.4-1 for each test.

**Table 9.1.3.4.1-1: General Test Parameters for intra frequency RSTD Tests for E-UTRAN FDD**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 | Test 4 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 FDD | | R.6 FDD | | As specified in TS 36.521-3 [25] clause A.2.1. |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.7 FDD | | OP.6 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | | 50 | | |
| PRS configuration Index $I_{PRS}$ Note 2 | | 12 | | 2 | | As defined in 3GPP TS 36.211 [26] |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 1 | | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | us | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | us | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | us | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTD_IntraFreq_PDE_E\text{-}UTRAN}$ Note 4 | ms | 2560 | | | | Derived according to the RSTD measurement requirements specified in Section 9.1.1.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.3.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.1.3.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.

NOTE 4: The parameter “$T_{RSTD_IntraFreq_PDE_E\text{-}UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.1.3.4.3-2. The value of the LPP time IE is set to $T_{RSTD_IntraFreq_PDE_E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 2710 ms. This is rounded up to the next allowed LPP value of 3 seconds.

### 9.1.3.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.1.3.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 9.1.3.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

3a. The SS shall send an LPP REQUEST CAPABILITIES message.

3b. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

4. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 3b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.

6. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

7. If the UE message at step 6 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

8. The SS shall check the *rstd* value for Cell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.1.3.5-2.

9. Repeat step 2-8 until the confidence level according to Annex D is achieved.

10. Repeat step 1-9 for each sub-test in Table 9.1.3.5-1 as appropriate.

#### 9.1.3.4.3 Message contents

**Table 9.1.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.1.3.4.3-1a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.1.3.4.3-2: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 3 | See Note 4 of Table 9.1.3.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |

| | | | |
|---|---|---|---|
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.3.4.3-3: Void**

**Table 9.1.3.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.3.4.3-5: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(1)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 9.1.3.5-2 for each specific test | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.3.4.3-6: *CQI-ReportConfig-DEFAULT*: FDD RSTD Measurement Accuracy**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for Test 1 and Test 2 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

### 9.1.3.5 Test requirement

Table 9.1.3.5-1 defines the primary level settings including test tolerances for all tests.

The RSTD FDD intra-frequency accuracy test shall meet the reported values in Table 9.1.3.5-2.

**Table 9.1.3.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRAN FDD**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | | Test 4 | |
|---|---|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | 1 | | | | | | | |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| PRS_RA | dB | 0 | 0 | -2.7 | 0.3 | 0 | 0 | -2.7 | 0.3 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | -98 | -98 | -98 | -98 | -98 | -98 | -98 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -2.37 | -8.02 | -5.7 | -12.7 | -2.37 | -8.02 | -5.7 | -12.7 |
| PRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -3 | -10 | -5.7 | -12.7 | -3 | -10 | -5.7 | -12.7 |
| Io Note 3 | dBm/1.08 MHz | -78.92 | -78.92 | -79.2 | -79.2 | N/A | N/A | N/A | N/A |
| | dBm/9 MHz | N/A | N/A | N/A | N/A | -69.72 | -69.72 | -69.99 | -69.99 |
| PRP Note 3 | dBm/15kHz | -100.37 | -106.02 | -103.7 | -110.7 | -100.37 | -106.02 | -103.7 | -110.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -2.37 | -8.02 | -3 | -13 | -2.37 | -8.02 | -3 | -13 |
| RSRP Note 3 | dBm/15kHz | -100.37 | -106.02 | -101 | -111 | -100.37 | -106.02 | -101 | -111 |
| Propagation condition | | AWGN | | | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS.

**Table 9.1.3.5-2: RSTD FDD intra-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Lowest reported value | RSTD_6248 | RSTD_6431 | RSTD_6441 | RSTD_6258 |
| Highest reported value | RSTD_6280 | RSTD_6463 | RSTD_6453 | RSTD_6270 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test then Test 1 and Test 2 shall be omitted.

## 9.1.3A FDD RSTD Measurement Accuracy for UE Category 1bis

### 9.1.3A.1 Test purpose

To verify that the RSTD FDD intra-frequency measurement accuracy is within the specified limits.

### 9.1.3A.2 Test applicability

This test applies to E-UTRA FDD UE Category 1bis release 13 and forward that supports UE-assisted OTDOA.

### 9.1.3A.3 Minimum conformance requirements

Same as 9.1.3.3

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.5 and A.9.8.1.2A.

### 9.1.3A.4 Test description

#### 9.1.3A.4.1 Initial conditions

Same as 9.1.3.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3, using only the main Tx/Rx path.

2. The general test parameter settings are set up according to Table 9.1.3A.4.1-1

**Table 9.1.3A.4.1-1: General Test Parameters for intra frequency RSTD Tests for E-UTRAN FDD for UE Category 1bis**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 | Test 4 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 FDD | | R.6 FDD | | As specified in TS 36.521-3 [25] clause A.2.1. |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.7 FDD | | OP.6 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | | 50 | | |
| PRS configuration Index $I_{PRS}$ Note 2 | | 12 | | 2 | | As defined in 3GPP TS 36.211 [26] |

| | | | | | |
|---|---|---|---|---|---|
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 1 | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '1111111100000000' | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | us | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | us | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | us | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ Note 4 | ms | 5120 | | | | Derived according to the RSTD measurement requirements specified in Section 9.1.1A.3 |

NOTE 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.3.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.1.3.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.3.4.1.

NOTE 4: The parameter "$T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.1.3A.4.3-2. The value of the LPP time IE is set to $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ + $\Delta$T ms, where $\Delta$T = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

### 9.1.3A.4.2 Test procedure

Same as 9.1.3.4.2.

### 9.1.3A.4.3 Message contents

Same as 9.1.3.4.3 with the following exceptions

**Table 9.1.3A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.1.3.4.3-2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See Note 4 of Table 9.1.3A.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.1.3A.5 Test requirement

Same as 9.1.3.5.

## 9.1.4 TDD RSTD Measurement Accuracy

### 9.1.4.1 Test purpose

To verify that the RSTD TDD intra-frequency measurement accuracy is within the specified limits.

### 9.1.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that supports UE-assisted OTDOA except UE Category 1bis.

### 9.1.4.3 Minimum conformance requirements

The accuracy requirements in Table 9.1.3.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to clause E.2 for a corresponding Band.

There are no measurement gaps overlapping with the PRS subframes of the measured cell.

The parameter *expectedRSTDUncertainty* signalled over LPP as defined in 3GPP TS 36.355 [4] is less than 5 µs.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.1 and A.9.8.2.

### 9.1.4.4 Test description

#### 9.1.4.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.2.

Channel bandwidth to be tested: 1.4 MHz (Test 1 and 2) and 10 MHz (Test 3 and 4). In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test, as defined in TS 36.101 [2] clause 5.6.1, then this part of the test shall be omitted.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3a for 4RX capable UE without any 2RX bands. Otherwise use Annex A figure A.3.

2. The general test parameter settings are set up according to Table 9.1.4.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 9.1.4.4.3.

5. All cells are on the same carrier frequency. Cell 1 is the serving cell and OTDOA assistance data reference cell; Cell 2 is the neighbour cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:
   Test 1: -92 Ts (about -3 μs)
   Test 2: 92 Ts (about 3 μs)
   Test 3: 92 Ts (about 3 μs)
   Test 4: -92 Ts (about -3 μs)
   Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.1.4.4-1 for each test.

**Table 9.1.4.4.1-1: General Test Parameters for intra frequency RSTD Tests for E-UTRAN TDD**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 | Test 4 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 TDD | | R.6 TDD | | As specified in TS 36.521-3 [25] clause A.2.2. |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | OP.4 TDD | | OP.2 TDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | | 10 | | |
| Special subframe configuration | | 6 | | 6 | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 3 | | 1 | | As specified in table 4.2-2 in TS 36.211 [26] and table 9.1.2.3-2. The same configuration in both cells. |
| PRS Transmission Bandwidth Note 2 | RB | 6 | | 50 | | |
| PRS configuration Index $I_{PRS}$ Note 2 | | 9 | | 14 | | As defined in 3GPP TS 36.211 [26]. |

| | | | | | | |
|---|---|---|---|---|---|---|
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 1 | | As defined in 3GPP TS 36.211 [26]. |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | us | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | us | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | us | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTDIntraFreqFDD,E-UTRAN}$ Note 4 | ms | 2560 | | | | Derived according to the RSTD measurement requirements specified in Section 9.1.2.3 |

NOTE 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.1.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.4.4.1.

NOTE 4: The parameter "$T_{RSTDIntraFreqFDD,E-UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.1.4.4.3-2. The value of the LPP time IE is set to $T_{RSTDIntraFreqFDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 2710 ms. This is rounded up to the next allowed LPP value of 3 seconds.

### 9.1.4.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.1.4.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.1.4.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

3a. The SS shall send an LPP REQUEST CAPABILITIES message.

3b. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

4. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 3b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of measurement period, where ΔT = 150 ms.
6. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.
7. If the UE message at step 6 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
8. The SS shall check the *rstd* value for Cell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.1.4.5-2.
9. Repeat step 2-8 until the confidence level according to Annex D is achieved.
10. Repeat step 1-9 for each sub-test in Table 9.1.4.5-1 as appropriate.

### 9.1.4.4.3 Message contents

**Table 9.1.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.1.4.4.3-1a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.1.4.4.3-2: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 3 | See Note 4 of Table 9.1.4.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.4.4.3-3: Void**

**Table 9.1.4.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |

| | | | |
|---|---|---|---|
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.4.4.3-5: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(1)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 9.1.4.5-2 for each specific test | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason ‘undefined’ or ‘attemptedButUnableToMeasureSomeNeighbourCells’ | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.4.4.3-6: *CQI-ReportConfig-DEFAULT*: TDD RSTD Measurement Accuracy**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for Test 1 and Test 2 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

## 9.1.4.5 Test requirement

Table 9.1.4.5-1 defines the primary level settings including test tolerances for all tests.

Each RSTD TDD intra-frequency accuracy test shall meet the reported values in Table 9.1.4.5-2.

**Table 9.1.4.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRAN TDD**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | | Test 4 | |
|---|---|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | 1 | | | | | | | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | | |
| PSS_RA | | | | | | | | | |
| SSS_RA | | | | | | | | | |
| PCFICH_RB | | | | | | | | | |
| PHICH_RA | | | | | | | | | |
| PHICH_RB | | | | | | | | | |
| PDCCH_RA | | | | | | | | | |
| PDCCH_RB | | | | | | | | | |
| OCNG_RA Note 1 | | | | | | | | | |
| OCNG_RB Note 1 | | | | | | | | | |
| PRS_RA | dB | 0 | 0 | -2.7 | 0.3 | 0 | 0 | -2.7 | 0.3 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | -98 | -98 | -98 | -98 | -98 | -98 | -98 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -2.37 | -8.02 | -5.7 | -12.7 | -2.37 | -8.02 | -5.7 | -12.7 |
| PRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -3 | -10 | -5.7 | -12.7 | -3 | -10 | -5.7 | -12.7 |
| Io Note 3 | dBm/1.08 MHz | -78.92 | -78.92 | -79.2 | -79.2 | N/A | N/A | N/A | N/A |
| | dBm/9 MHz | N/A | N/A | N/A | N/A | -69.72 | -69.72 | -69.99 | -69.99 |
| PRP Note 3 | dBm/15kHz | -100.37 | -106.02 | -103.7 | -110.7 | -100.37 | -106.02 | -103.7 | -110.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -2.37 | -8.02 | -3 | -13 | -2.37 | -8.02 | -3 | -13 |
| RSRP Note 3 | dBm/15kHz | -100.37 | -106.02 | -101 | -111 | -100.37 | -106.02 | -101 | -111 |
| Propagation condition | | AWGN | | | | | | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). | | | | | | | | | |

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS.

**Table 9.1.4.5-2: RSTD TDD intra-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|
| Lowest reported value | RSTD_6248 | RSTD_6431 | RSTD_6441 | RSTD_6258 |
| Highest reported value | RSTD_6280 | RSTD_6463 | RSTD_6453 | RSTD_6270 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test then Test 1 and Test 2 shall be omitted.

## 9.1.4A TDD RSTD Measurement Accuracy for UE Category 1bis

### 9.1.4A.1 Test purpose

To verify that the RSTD TDD intra-frequency measurement accuracy is within the specified limits.

### 9.1.4A.2 Test applicability

This test applies to E-UTRA TDD UE Category 1bis release 13 and forward that supports UE-assisted OTDOA.

### 9.1.4A.3 Minimum conformance requirements

Same as 9.1.4.3

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.6 and A.9.8.2.2A.

### 9.1.4A.4 Test description

#### 9.1.4A.4.1 Initial conditions

Same as 9.1.4.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3, using only the main Tx/Rx path.
2. The general test parameter settings are set up according to Table 9.1.4A.4.1-1

**Table 9.1.4A.4.1-1: General Test Parameters for intra frequency RSTD Tests for E-UTRAN TDD for UE Category 1bis**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 | Test 4 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 TDD | | R.6 TDD | | As specified in TS 36.521-3 [25] clause A.2.2. |

| | | | | | | |
|---|---|---|---|---|---|---|
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | OP.4 TDD | | OP.2 TDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | | 10 | | |
| Special subframe configuration | | 6 | | 6 | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 3 | | 1 | | As specified in table 4.2-2 in TS 36.211 [26] and table 9.1.2.3-2. The same configuration in both cells. |
| PRS Transmission Bandwidth Note 2 | RB | 6 | | 50 | | |
| PRS configuration Index $I_{PRS}$ Note 2 | | 9 | | 14 | | As defined in 3GPP TS 36.211 [26]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 1 | | As defined in 3GPP TS 36.211 [26]. |
| prs-MutingInfo Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '1111111100000000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | us | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | us | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | us | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTDIntraFreqTDD,E-UTRAN}$ Note 4 | ms | 5120 | | | | Derived according to the RSTD measurement requirements specified in Section 9.1.2A.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.1.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.
NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.1.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.
NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.4.4.1.
NOTE 4: The parameter “$T_{RSTDIntraFreqTDD,E\text{-}UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.1.4A.4.3-1. The value of the LPP time IE is set to $T_{RSTDIntraFreqTDD,E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

### 9.1.4A.4.2 Test procedure

Same as 9.1.3.4.2.

### 9.1.4A.4.3 Message contents

Same as 9.1.4.4.3 with the following exceptions

**Table 9.1.4A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.1.4.4.3-2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See Note 4 of Table 9.1.3A.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.1.4A.5 Test requirement

Same as 9.1.4.5.

# 9.2 RSTD Inter-Frequency Measurements

## 9.2.1 FDD-FDD inter-frequency RSTD measurement reporting delay

### 9.2.1.1 Test purpose

To verify that the FDD-FDD inter-frequency RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

### 9.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that support inter-frequency RSTD measurements except UE Category 1bis.

### 9.2.1.3 Minimum conformance requirements

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data are provided, the UE shall be able to detect and measure inter-frequency RSTD, specified in 3GPP TS 36.214 [6], for at least *n*=16 cells, including the reference cell, within $T_{RSTD_InterFreq_FDD,E\text{-}UTRAN}$ ms as given below:

$$T_{RSTD_InterFreq_FDD,E\text{-}UTRAN} = T_{PRS} \cdot (M - 1) + \Delta \quad ms,$$

where

$T_{RSTD_InterFreq_FDD,E\text{-}UTRAN}$ is the total time for detecting and measuring at least *n* cells,

$T_{PRS}$ is the largest value of the cell-specific positioning subframe configuration period, defined in 3GPP TS 36.211 [26], among the measured *n* cells including the reference cell,

$M$ is the number of PRS positioning occasions as defined in Table 9.2.1.3-1, where each PRS positioning occasion comprises of $N_{PRS}$ ($1 \le N_{PRS} \le 6$) consecutive downlink positioning subframes defined in 3GPP TS 36.211 [26], and $\Delta = 160 \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single PRS positioning occasion which includes the sampling time and the processing time.

**Table 9.2.1.3-1: Number of PRS positioning occasions within $T_{RSTD_InterFreq_FDD,E\text{-}UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ | |
|---|---|---|
| | f2 Note 1 | f1 and f2 Note 2 |
| 160 ms | 16 | 32 |
| >160 ms | 8 | 16 |
| Note 1: When inter-frequency RSTD measurements are performed over the reference cell and neighbour cells, which belong to the FDD inter-frequency carrier frequency f2. | | |
| Note 2: When inter-frequency RSTD measurements are performed over the reference cell and the neighbour cells, which belong to the serving FDD carrier frequency f1 and the FDD inter-frequency carrier frequency f2 respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD_InterFreq_FDD,E\text{-}UTRAN}$ provided:

$\left(PRS\ \hat{E}_s / Iot\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\widehat{PRSE}_s / Iot\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(PRS\ \hat{E}_s / Iot\right)_{ref}$ and $\left(\widehat{PRSE}_s / Iot\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ PRS positioning occasions,

PRP $1,2|_{dBm}$ according to E.3 for a corresponding Band.

$\widehat{PRSE}_s / Iot$ is as defined in Section 9.1.1.3.

The time $T_{RSTDInterFreqFDD,E\text{-}UTRAN}$ starts from the first subframe of the PRS positioning occasion closest in time after both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in 3GPP TS 36.355 [4], are delivered to the physical layer of the UE.

This requirement assumes that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.6.1 and A.8.13.1.

### 9.2.1.4 Test description

#### 9.2.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4.
2. The general test parameter settings are set up according to Table 9.2.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.2.1.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. Cell 1 is on FDD RF channel 1. Cell 2 and Cell 3 are on a FDD RF channel 2. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.
7. The gap pattern configuration # 0 as defined in Table 8.1.2.1-1 in 3GPP TS 36.133 [23] is configured and does not overlap with PRS subframes of Cell 1.

**Table 9.2.1.4.1-1: General test parameters for E-UTRAN FDD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell on RF channel 1 in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cells on RF channel 2. The cells appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| Gap pattern Id | | 0 | As specified in Table 8.1.2.1-1 in TS 36.133 [23]. |
| Gap offset | | 9 | As specified in TS 36.331 [22], Section 6.3.5 |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 181,<br>Cell 2, Cell 3: 171 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.2.1.4.1-2 |
| prs-SubframeOffset Note 2 | | 310 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |

| | | | |
|---|---|---|---|
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.48 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 2: Parameters "PRS Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", "prs-SubframeOffset", "slotNumberOffset" and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.2.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.2.1.4.1.

**Table 9.2.1.4.1-2: DRX parameters for E-UTRAN FDD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.2.1.4.2 Test procedure

The test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 9.2.1.4.1-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. Cell 2 is active until the end of T3, and Cell 3 is active until the end of T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS only in T2, Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.2.1.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T = 150$ ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.2.1.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).
4. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration and the measurement gap configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

6a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the first 7 elements of the sequence, and the position of neighbour Cell 3 is randomly selected to be in the last 8 elements of the sequence, as described in 3GPP TS 37.571-5 [20], clause 7.2.3. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.

9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 9.2.1.5-2.

10. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.2.1.5-2.

11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 9.2.1.5. The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. Repeat steps 2-12 until the confidence level according to Annex D is achieved. For each iteration, at step 7 change the random position of the Cells 2 and 3 in the *OTDOA-NeighbourCellInfoList.*

#### 9.2.1.4.3 Message contents

**Table 9.2.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.2.1.4.3-2: *MAC-MainConfig-RBC*: FDD-FDD Inter-frequency RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |

| | | | |
|---|---|---|---|
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 9.2.1.4.3-3: *MeasGapConfig-GP1*: FDD-FDD inter-frequency RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.6.6, Table 4.6.6-1A: MeasGapConfig-GP1 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| MeasGapConfig-GP1 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 9 | TGRP = 40 ms | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.1.4.3-3a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.2.1.4.3-4: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See clause 9.2.1.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |

| | | | |
|---|---|---|---|
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.1.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.1.4.3-6: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |

| | | | |
|---|---|---|---|
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | RF channel 2 | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | RF channel 2 | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason ‘undefined’ or ‘attemptedButUnableToMeasureSomeNeighbourCells’ | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.2.1.5 Test requirement

Table 9.2.1.5-1 and 9.2.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.2.1.5-1: Cell-specific test parameters for E-UTRAN FDD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.5 FDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | N/A | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that the active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.2.1.5-2: Cell-specific test parameters for E-UTRAN FDD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | N/A |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.5 FDD | | OP.6 FDD | | OP.6 FDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -1 | -Infinity | -Infinity | -7 | -8 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1 | -Infinity | -Infinity | -7 | -8 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.68 | -70.22 | -70.11 | -67.08 | -70.11 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -106 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -96 | -105 | -105 | -109 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -11 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test and assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 5110 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=16 and $n$=16 are the parameters specified in clause 9.2.1.3 and Table 9.2.1.3-1. This gives the total RSTD reporting delay of 4960 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.2.1A FDD-FDD inter-frequency RSTD Measurement Reporting Delay for UE Category 1bis

### 9.2.1A.1 Test purpose

To verify that the FDD-FDD inter-frequency RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

### 9.2.1A.2 Test applicability

This test applies to E-UTRA FDD UE Category 1bis release 13 and forward that support inter-frequency RSTD measurements.

### 9.2.1A.3 Minimum conformance requirements

Same as 9.2.1.3 but using Table 9.2.1A.3-1 instead of Table 9.2.1.3-1.

**Table 9.2.1A.3-1: Number of PRS positioning occasions within $T_{RSTDInterFreqFDD,E-UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ | |
|---|---|---|
| | f2 Note 1 | f1 and f2 Note 2 |
| 160 ms | 32 | 64 |
| >160 ms | 16 | 32 |

Note 1: When inter-frequency RSTD measurements are performed over the reference cell and neighbour cells, which belong to the FDD inter-frequency carrier frequency f2.
Note 2: When inter-frequency RSTD measurements are performed over the reference cell and the neighbour cells, which belong to the serving FDD carrier frequency f1 and the FDD inter-frequency carrier frequency f2 respectively.

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.6.5 and A.8.13.1.

### 9.2.1A.4 Test description

#### 9.2.1A.4.1 Initial conditions

Same as 9.2.1.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4, using only the main Tx/Rx path.

2. The general test parameter settings are set up according to Table 9.2.1A.4.1-1.

**Table 9.2.1A.4.1-1: General test parameters for E-UTRAN FDD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions for UE Category 1bis**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell on RF channel 1 in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cells on RF channel 2. The cells appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| Gap pattern Id | | 0 | As specified in Table 8.1.2.1-1 in TS 36.133 [23]. |
| Gap offset | | 9 | As specified in TS 36.331 [22], Section 6.3.5 |

| | | | |
|---|---|---|---|
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 181,<br>Cell 2, Cell 3: 171 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.2.1.4.1-2 |
| prs-SubframeOffset Note 2 | | 310 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| PRS muting info Note 2 | | Cell 1:<br>‘11111111111111110000000000000000’<br>Cell 2:<br>‘00000000000000001111111111111111’<br>Cell 3:<br>‘11111111111111110000000000000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 4.96 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 4.96 | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2. |
| Note 2: | Parameters “PRS Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, “prs-SubframeOffset”, “slotNumberOffset” and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.2.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.2.1.4.1. |

### 9.2.1A.4.2 Test procedure

Same as 9.2.1.4.2.

### 9.2.1A.4.3 Message contents

Same as 9.2.1.4.3 with the following exceptions:

**Table 9.2.1A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.2.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 11 | See clause 9.2.1A.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.1A.5 Test requirement

Table 9.2.1.5-1 and 9.2.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 11.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 10230 ms. This is rounded up to the next allowed LPP value of 11 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=32 and $n$=16 are the parameters specified in clause 9.2.1A.3 and Table 9.2.1A.3-1. This gives the total RSTD reporting delay of 10080 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.2.2 TDD-TDD inter-frequency RSTD measurement reporting delay

### 9.2.2.1 Test purpose

To verify that the TDD-TDD inter-frequency RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

### 9.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support inter-frequency RSTD measurements except UE Category 1bis.

### 9.2.2.3 Minimum conformance requirements

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data are provided, the UE shall be able to detect and measure inter-frequency RSTD, specified in 3GPP TS 36.214 [6], for at least *n*=16 cells, including the reference cell, within $T_{RSTD_InterFreq_TDD_E\text{-}UTRAN}$ ms as given below:

$$T_{RSTD_InterFreq_TDD_E\text{-}UTRAN} = T_{PRS} \cdot (M - 1) + \Delta \quad ms,$$

where

$T_{RSTD_InterFreq_TDD_E\text{-}UTRAN}$ is the total time for detecting and measuring at least *n* cells,

$T_{PRS}$ is the largest value of the cell-specific positioning subframe configuration period, defined in 3GPP TS 36.211 [26], among the measured *n* cells including the reference cell,

$M$ is the number of PRS positioning occasions as defined in Table 9.2.2.3-1, where each PRS positioning occasion comprises of $N_{PRS}$ ($1 \le N_{PRS} \le 6$) consecutive downlink positioning subframes defined in 3GPP TS 36.211 [26], and

$\Delta = 160 \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single PRS positioning occasion which includes the sampling time and the processing time.

**Table 9.2.2.3-1: Number of PRS positioning occasions within $T_{RSTD_InterFreq_TDD_E\text{-}UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ | |
|---|---|---|
| | f2 Note 1 | f1 and f2 Note 2 |
| 160 ms | 16 | 32 |
| >160 ms | 8 | 16 |

Note 1: When inter-frequency RSTD measurements are performed over the reference cell and neighbour cells, which belong to the TDD inter-frequency carrier frequency f2.

Note 2: When inter-frequency RSTD measurements are performed over the reference cell and the neighbour cells, which belong to the serving TDD carrier frequency f1 and the TDD inter-frequency carrier frequency f2 respectively.

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD_InterFreq_TDD_E\text{-}UTRAN}$ provided:

$\left(\text{PRS } \hat{E}_s / \text{Iot}\right)_{ref} \ge$-6 dB for all Frequency Bands for the reference cell,

$\left(\widehat{PRSE}_s / Iot\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(PRS\ \hat{E}_s / Iot\right)_{ref}$ and $\left(\widehat{PRSE}_s / Iot\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ PRS positioning occasions,

PRP $1,2|_{dBm}$ according to E.3 for a corresponding Band.

$\widehat{PRSE}_s / Iot$ is as defined in Section 9.1.1.3.

The time $T_{RSTDInterFreqTDD,E\text{-}UTRAN}$ starts from the first subframe of the PRS positioning occasion closest in time after both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in 3GPP TS 36.355 [4], are delivered to the physical layer of the UE.

This requirement assumes that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.6.3 and A.8.13.2.

### 9.2.2.4 Test description

#### 9.2.2.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4.
2. The general test parameter settings are set up according to Table 9.2.2.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.2.2.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. Cell 1 is on TDD RF channel 1. Cell 2 and Cell 3 are on a TDD RF channel 2. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.
7. The gap pattern configuration # 0 as defined in Table 8.1.2.1-1 in 3GPP TS 36.133 [23] is configured and does not overlap with PRS subframes of Cell 1.

**Table 9.2.2.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell on RF channel 1 in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cells on RF channel 2. The cells appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| Gap pattern Id | | 0 | As specified in Table 8.1.2.1-1 in TS 36.133 [23]. |
| Gap offset | | 12 | As specified in TS 36.331 [22], Section 6.3.5 |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 184,<br>Cell 2, Cell 3: 174 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Section 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Section 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.2.2.4.1-2 |
| prs-SubframeOffset Note 2 | | 310 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.48 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.2.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 2: Parameters "PRS Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", "prs-SubframeOffset", "slotNumberOffset" and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.2.2.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.2.2.4.1.

**Table 9.2.2.4.1-2: DRX parameters for E-UTRAN TDD-TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.2.2.4.2 Test procedure

The test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 9.2.2.4.1-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. Cell 2 is active until the end of T3, and Cell 3 is active until the end of T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS only in T2, Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.2.2.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.2.2.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).
4. T1 starts.
5. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration and the measurement gap configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.
6. The UE shall transmit RRCConnectionReconfigurationComplete message.

6a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the first 7 elements of the sequence, and the position of neighbour Cell 3 is randomly selected to be in the last 8 elements of the sequence, as described in 3GPP TS 37.571-5 [20], clause 7.2.3. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 9.2.2.5-2.
10. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.2.2.5-2.
11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 9.2.2.5. The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.
12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
13. Repeat steps 2-12 until the confidence level according to Annex D is achieved. For each iteration, at step 7 change the random position of the Cells 2 and 3 in the *OTDOA-NeighbourCellInfoList.*

### 9.2.2.4.3 Message contents

**Table 9.2.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.2.2.4.3-2: *MAC-MainConfig-RBC*: TDD-TDD Inter-frequency RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |

| | | | |
|---|---|---|---|
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 9.2.2.4.3-3: *MeasGapConfig-GP1*: TDD-TDD inter-frequency RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.6.6, Table 4.6.6-1A: MeasGapConfig-GP1 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| MeasGapConfig-GP1 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 12 | TGRP = 40 ms | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.2.4.3-3a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.2.2.4.3-4: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| time | 6 | See clause 9.2.2.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.2.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.2.4.3-6: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | RF channel 2 | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbor | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | RF channel 2 | | |
| rstd | Present | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason ‘undefined’ or ‘attemptedButUnableToMeasureSomeNeighbourCells’ | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.2.5 Test requirement

Table 9.2.2.5-1 and 9.2.2.5-2 define the primary level settings including test tolerances for the test.

**Table 9.2.2.5-1: Cell-specific test parameters for E-UTRAN TDD-TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | N/A | N/A |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.1 TDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | N/A | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that the active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.2.2.5-2: Cell-specific test parameters for E-UTRAN TDD-TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | N/A |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.1 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -1 | -Infinity | -Infinity | -7 | -8 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1 | -Infinity | -Infinity | -7 | -8 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.68 | -70.22 | -70.11 | -67.08 | -70.11 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -106 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -96 | -105 | -105 | -109 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -11 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes

The response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 5110 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=16 and $n$=16 are the parameters specified in clause 9.2.2.3 and Table 9.2.2.3-1. This gives the total RSTD reporting delay of 4960 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.2.2A TDD-TDD inter-frequency RSTD Measurement Reporting Delay for UE Category 1bis

### 9.2.2A.1 Test purpose

To verify that the TDD-TDD inter-frequency RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions.

### 9.2.2A.2 Test applicability

This test applies to E-UTRA TDD UE Category 1bis release 13 and forward that support inter-frequency RSTD measurements.

### 9.2.2A.3 Minimum conformance requirements

Same as 9.2.2.3 but using Table 9.2.2A.3-1 instead of Table 9.2.2.3-1.

**Table 9.2.2A.3-1: Number of PRS positioning occasions within $T_{RSTDInterFreqTDD,E-UTRAN}$**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ | |
|---|---|---|
| | f2 Note 1 | f1 and f2 Note 2 |
| 160 ms | 32 | 64 |
| >160 ms | 16 | 32 |
| Note 1: When inter-frequency RSTD measurements are performed over the reference cell and neighbour cells, which belong to the TDD inter-frequency carrier frequency f2.<br>Note 2: When inter-frequency RSTD measurements are performed over the reference cell and the neighbour cells, which belong to the serving TDD carrier frequency f1 and the FDD inter-frequency carrier frequency f2 respectively. | | |

The normative reference for this requirement is TS 36.133 [23] clause 8.1.2.6.7 and A.8.13.2.

### 9.2.2A.4 Test description

#### 9.2.2A.4.1 Initial conditions

Same as 9.2.2.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4, using only the main Tx/Rx path.
2. The general test parameter settings are set up according to Table 9.2.2A.4.1-1.

**Table 9.2.2A.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions for UE Category 1bis**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell on RF channel 1 in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cells on RF channel 2. The cells appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Bandwidth Note 2 | RB | 50 | PRS are transmitted over the system bandwidth |
| Gap pattern Id | | 0 | As specified in Table 8.1.2.1-1 in TS 36.133 [23]. |

| | | | |
|---|---|---|---|
| Gap offset | | 12 | As specified in TS 36.331 [22], Section 6.3.5 |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 184,<br>Cell 2, Cell 3: 174 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | As defined in TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Section 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Section 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.2.1.4.1-2 |
| prs-SubframeOffset Note 2 | | 310 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| PRS muting info Note 2 | | Cell 1:<br>‘11111111111111110000000000000000’<br>Cell 2:<br>‘00000000000000001111111111111111’<br>Cell 3:<br>‘11111111111111110000000000000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |

| | | | |
|---|---|---|---|
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 4.96 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 4.96 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 2: Parameters “PRS Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, “prs-SubframeOffset”, “slotNumberOffset” and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.2.1.4.3-5 and TS 37.571-5 [20], clause 7.2.2.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.2.1.4.1.

### 9.2.2A.4.2 Test procedure

Same as 9.2.2.4.2.

### 9.2.2A.4.3 Message contents

Same as 9.2.2.4.3 with the following exceptions:

**Table 9.2.2A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.2.2.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 11 | See clause 9.2.2A.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.2A.5 Test requirement

Table 9.2.2.5-1 and 9.2.2.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 11.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 10230 ms. This is rounded up to the next allowed LPP value of 11 seconds. The RSTD measurement reporting

delay in the test is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=32 and $n$=16 are the parameters specified in clause 9.2.2A.3 and Table 9.2.2A.3-1. This gives the total RSTD reporting delay of 10080 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.2.3 Void

## 9.2.4 FDD-FDD inter-frequency RSTD Accuracy

### 9.2.4.1 Test purpose

To verify that the Reference Signal Time Difference (RSTD) FDD-FDD inter-frequency measurement accuracy is within the specified limit for all bands in AWGN channels.

### 9.2.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that support inter-frequency RSTD measurements except UE Category 1bis.

### 9.2.4.3 Minimum conformance requirements

The accuracy of FDD-FDD inter-frequency RSTD measurement shall meet the requirement defined in the Table 9.2.4.3-1 without DRX as well as for all the DRX cycles specified in TS 36.331 [22].

The accuracy requirements in Table 9.2.4.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to clause E.3 for a corresponding Band.

There are no measurement gaps overlapping with the PRS subframes in cells belonging to the serving carrier frequency.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 µs.

**Table 9.2.4.3-1: RSTD measurement accuracy**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | Io Note 8 range | | |
| | PRS Ês/Iot | Minimum PRS bandwidth which is minimum of serving cell channel bandwidth Note 9 and the PRS bandwidths of the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | E-UTRA operating band groups Note 10 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz Note 7 | $dBm/BW_{Channel}$ |
| ±21 | $(PRS Ês/Iot)_{ref}$ ≥-6dB and | ≥ 6 | 4 | FDD_A, TDD_A | -121 | -50 |
| | | | | FDD_B | -120.5 | -50 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | (PRS Ês/Iot)$_i$ ≥-13dB | | | FDD_C, TDD_C | -120 | -50 |
| | | | | FDD_D | -119.5 | -50 |
| | | | | FDD_E, TDD_E | -119 | -50 |
| | | | | FDD_F | -118.5 | -50 |
| | | | | FDD_G | -118 | -50 |
| | | | | FDD_H | -117.5 | -50 |
| | | | | FDD_N | -114.5 | -50 |
| ±16 Note 11 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 15 | 4 | Note 5 | Note 5 | Note 5 |
| ±10 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 25 | ≥ 2 | Note 5 | Note 5 | Note 5 |
| ±9 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 50 | ≥ 1 | Note 5 | Note 5 | Note 5 |
| ±8 Note 11 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 75 | ≥ 1 | Note 5 | Note 5 | Note 5 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in [24].
NOTE 4: Void.
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 6 RB.
NOTE 6: Void.
NOTE 7: The condition level is increased by Δ>0, when applicable, as described in TS 36.521-3 [25] Sections I.4.2 and I.4.3.
NOTE 8: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 9: If a CA capable UE is configured with SCell, the serving cell channel bandwidth is the minimum of the serving cell channel bandwidths in the component carriers involved in the RSTD measurement. If one of the serving cells is not involved in this RSTD measurement for CA, the channel bandwidth of that serving cell is not included in the determination of the minimum PRS bandwidth.
NOTE 10: E-UTRA operating band groups are as defined in clause 4.4.2.
NOTE 11: Only applicable from Rel-12 onwards

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.2 and A.9.8.3.

## 9.2.4.4 Test description

### 9.2.4.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 1.4 MHz (Test 1) and 10 MHz (Test 2). In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test, as defined in TS 36.101 [2] clause 5.6.1, then this part of the test shall be omitted.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3a for 4RX capable UE without any 2RX bands. Otherwise use Annex A figure A.3.

2. The general test parameter settings are set up according to Table 9.2.4.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 9.2.4.4.3.

5. Two cells are on the different carrier frequencies. Cell 1 is the serving cell and OTDOA assistance data reference cell; Cell 2 is the neighbour cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between the two cells as seen at the UE antenna connector) is set to 92 Ts (about 3 μs) between neighbour cell 2 and serving cell 1 for Test 1 and -92 Ts (about -

3 μs) for Test 2.

Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.2.4.4-1.

7. The gap pattern configuration # 0 as defined in Table 8.1.2.1-1 in 3GPP TS 36.133 [23] is configured and does not overlap with PRS subframes of Cell 1.

**Table 9.2.4.4.1-1: General Test Parameters for inter-frequency RSTD Tests for E-UTRAN FDD**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 FDD | R.6 FDD | As specified in TS 36.521-3 [25] clause A.2.1. |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.7 FDD | OP.6 FDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Cell 1 on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 on RF channel number 2 |
| E-UTRA RF Channel Number | | 1,2 | | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | 10 | |
| GapOffset | | 18 | 11 | For Cell 1 |
| Gap Pattern ID | | 0 | 0 | For Cell 1 |
| PRS Bandwidth | RB | 6 | 50 | |
| PRS configuration Index $I_{PRS}$ Note 2 | | Cell 1: 12<br>Cell 2: 19 | Cell 1: 2<br>Cell 2: 12 | As defined in 3GPP TS 36.211 [26] |
| PRS subframe offset | | 7 | 10 | For Cell 2 |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | 1 | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | Cell 1: 0<br>Cell 2: 1 | | |
| Expected RSTD Note 1 | μs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | μs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | μs | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| $T_{RSTDInterFreqFDD,E-UTRAN}$ Note 4 | ms | 5120 | | Derived according to the RSTD measurement requirements specified in Section 8.1.2.6.1 in TS 36.133 [23]. |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” in Table 9.2.4.4.1-1 are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 2: Parameters “PRS Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” in Table 9.2.4.4.1-1 are settable parameters and also parameters signalled in LPP. For all the values to be used in LPP see Table 9.2.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.2.4.4.1.

NOTE 4: The parameter “$T_{RSTDInterFreqTDD,E\text{-}UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.2.4.4.3-3. The value of the LPP time IE is set to $T_{RSTDInterFreqTDD,E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

### 9.2.4.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.2.4.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 9.2.4.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

4. The SS shall transmit an RRCConnectionReconfiguration message with the measurement gap configuration.

5. The UE shall transmit RRCConnectionReconfigurationComplete message.

5a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

5b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 5b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.

8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

10. The SS shall check the *rstd* value for Cell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.2.4.5-2.

11. Repeat step 2-10 until the confidence level according to Annex D is achieved.

12. Repeat step 1-11 for each sub-test in Table 9.2.4.5-1 as appropriate.

9.2.4.4.3 Message contents

**Table 9.2.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.2.4.4.3-2: *MeasGapConfig-GP1*: FDD-FDD inter-frequency RSTD Accuracy**

| Derivation Path: TS 36.508 [18] clause 4.6.6, Table 4.6.6-1A: MeasGapConfig-GP1 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| MeasGapConfig-GP1 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 18 (Test 1)<br>11 (Test 2) | TGRP = 40 ms | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.4.4.3-2a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.2.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See Note 4 of Table 9.2.4.4.1-1 | |

| | | | |
|---|---|---|---|
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.4.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.4.4.3-5: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(1)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | RF channel 2 | | |
| rstd | Set according to Table 9.2.4.5-2 for each specific test | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason ‘undefined’ or ‘attemptedButUnableToMeasureSomeNeighbourCells’ | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.4.4.3-6: *CQI-ReportConfig-DEFAULT*: FDD-FDD inter-frequency RSTD Accuracy**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for Test 1 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |

| | | | |
|---|---|---|---|
| } | | | |

### 9.2.4.5 Test requirement

Table 9.2.4.5-1 defines the primary level settings including test tolerances for all tests.

RSTD FDD-FDD inter-frequency accuracy test shall meet the reported values in Table 9.2.4.5-2.

**Table 9.2.4.5-1: Cell Specific Test Parameters for inter-frequency RSTD Tests for E-UTRAN FDD**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | 1 | 2 | 1 | 2 |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA Note 1 | | | | | |
| OCNG_RB Note 1 | | | | | |
| PRS_RA | dB | -2.7 | 0.3 | -2.7 | 0.3 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| PRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| Io Note 3 | dBm/1.08 MHz | -79.24 | -79.39 | N/A | N/A |
| | dBm/9 MHz | N/A | N/A | -70.03 | -70.18 |
| PRP Note 3 | dBm/15kHz | -103.7 | -110.7 | -103.7 | -110.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -3 | -13 | -3 | -13 |
| RSRP Note 3 | dBm/15kHz | -101 | -111 | -101 | -111 |
| Propagation condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, RSRP, Io and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS

**Table 9.2.4.5-2: RSTD FDD inter-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6424 | RSTD_6253 |
| Highest reported value | RSTD_6470 | RSTD_6275 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test then Test 1 shall be omitted.

# 9.2.4A FDD-FDD inter-frequency RSTD Accuracy for UE Category 1bis

## 9.2.4A.1 Test purpose

To verify that the Reference Signal Time Difference (RSTD) FDD-FDD inter-frequency measurement accuracy is within the specified limit for all bands in AWGN channels.

## 9.2.4A.2 Test applicability

This test applies to E-UTRA FDD UE Category 1bis release 13 and forward that support inter-frequency RSTD measurements.

## 9.2.4A.3 Minimum conformance requirements

Same as 9.2.4.3

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.6 and A.9.8.3.2A.

## 9.2.4A.4 Test description

### 9.2.4A.4.1 Initial conditions

Same as 9.2.4.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3, using only the main Tx/Rx path.

2. The general test parameter settings are set up according to Table 9.2.4A.4.1-1

**Table 9.2.4A.4.1-1: General Test Parameters for inter-frequency RSTD Tests for E-UTRAN FDD for UE Category 1bis**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 FDD | R.6 FDD | As specified in TS 36.521-3 [25] clause A.2.1. |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.7 FDD | OP.6 FDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Cell 1 on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 on RF channel number 2 |
| E-UTRA RF Channel Number | | 1,2 | | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | 10 | |
| GapOffset | | 18 | 11 | For Cell 1 |
| Gap Pattern ID | | 0 | 0 | For Cell 1 |
| PRS Bandwidth | RB | 6 | 50 | |
| PRS configuration Index $I_{PRS}$ Note 2 | | Cell 1: 12<br>Cell 2: 19 | Cell 1: 2<br>Cell 2: 12 | As defined in 3GPP TS 36.211 [26] |
| PRS subframe offset | | 7 | 10 | For Cell 2 |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | 1 | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '1111111100000000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | Cell 1: 0<br>Cell 2: 1 | | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |

| | | | | |
|---|---|---|---|---|
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| $T_{RSTDInterFreqFDD,E-UTRAN}$ Note 4 | ms | 10240 | | Derived according to the RSTD measurement requirements specified in Section 8.1.2.6.5 in TS 36.133 [23]. |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” in Table 9.2.4.4.1-1 are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 2: Parameters “PRS Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” in Table 9.2.4.4.1-1 are settable parameters and also parameters signalled in LPP. For all the values to be used in LPP see Table 9.2.4.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.2.4.4.1.

NOTE 4: The parameter “$T_{RSTDInterFreqFDD,E-UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.2.4.4.3-3. The value of the LPP time IE is set to $T_{RSTDInterFreqFDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 10390 ms. This is rounded up to the next allowed LPP value of 11 seconds.

### 9.2.4A.4.2 Test procedure

Same as 9.2.4.4.2.

### 9.2.4A.4.3 Message contents

Same as 9.2.4.4.3 with the following exceptions

**Table 9.2.4A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.2.4.4.3-2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 11 | See Note 4 of Table 9.2.4A.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

### 9.2.4A.5 Test requirement

Same as 9.2.4.5.

## 9.2.5 TDD-TDD inter-frequency RSTD Accuracy

### 9.2.5.1 Test purpose

To verify that the Reference Signal Time Difference (RSTD) of TDD-TDD inter-frequency measurement accuracy is within the specified limit for all bands in AWGN channels

### 9.2.5.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support inter-frequency RSTD measurements except UE Category 1bis.

### 9.2.5.3 Minimum conformance requirements

This RSTD measurement is used for UE positioning purposes.

The accuracy of TDD-TDD inter-frequency RSTD measurement shall meet the requirement defined in the Table 9.2.4.3-1 without DRX as well as for all the DRX cycles specified in TS 36.331 [22].

The accuracy requirements in Table 9.2.4.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to clause E.3 for a corresponding Band

There are no measurement gaps overlapping with the PRS subframes in cells belonging to the serving carrier frequency.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 µs.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.2 and A.9.8.4.

### 9.2.5.4 Test description

#### 9.2.5.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.2.

Channel bandwidth to be tested: 1.4 MHz (Test 1) and 10 MHz (Test 2). In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test, as defined in TS 36.101 [2] clause 5.6.1, then this part of the test shall be omitted.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3a for 4RX capable UE without any 2RX bands. Otherwise use Annex A figure A.3.

2. The general test parameter settings are set up according to Table 9.2.5.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 9.2.5.4.3.

5. Two cells are on the different carrier frequencies. Cell 1 is the serving cell and OTDOA assistance data reference cell; Cell 2 is the neighbour cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between the two cells as seen at the UE antenna connector) is set to 92 Ts (about 3 μs) between neighbour cell 2 and serving cell 1 for Test 1 and -92 Ts (about -3 μs) for Test 2.

   Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.2.5.4-1.

7. The gap pattern configuration # 0 as defined in Table 8.1.2.1-1 in 3GPP TS 36.133 [23] is configured and does not overlap with PRS subframes of Cell 1.

**Table 9.2.5.4.1-1: General Test Parameters for inter-frequency RSTD Tests for E-UTRAN TDD-TDD**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 TDD | R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2. |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | OP.4 TDD | OP.2 TDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Cell 1 on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 on RF channel number 2 |
| E-UTRA RF Channel Number | | 1,2 | | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | 10 | |
| PRS Bandwidth Note 2 | RB | 6 | 50 | |
| GapOffset | | 34 | 13 | For Cell 1 |
| Gap Pattern ID | | 0 | | For Cell 1 |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 3 | 1 | As specified in table 4.2-2 in TS 36.211 [26] and table 9.1.2.3-2. The same configuration in both cells. |
| PRS configuration Index $I_{PRS}$ Note 2 | | Cell 1: 15<br>Cell 2: 35 | Cell 1: 4<br>Cell 2: 14 | As defined in 3GPP TS 36.211 [26] |
| PRS subframe offset | | 20 | 10 | For Cell 2 |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | 1 | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1:'11110000'<br>Cell 2:'11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | Cell 1: 0<br>Cell 2: 1 | | |
| Expected RSTD Note 1 | μs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | μs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | μs | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| $T_{RSTDInterFreqTDD,E-UTRAN}$ Note 4 | ms | 5120 | | Derived according to the RSTD measurement requirements specified in Section 8.1.2.6.3 in TS 36.133 [23]. |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.5.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 2: Parameters “PRS Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. For all the values to be used in LPP see Table 9.2.5.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.2.5.4.1.

NOTE 4: The parameter “$T_{RSTDInterFreqTDD,E-UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.2.5.4.3-3. The value of the LPP time IE is set to $T_{RSTDInterFreqTDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

### 9.2.5.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.2.5.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 9.2.5.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

4. The SS shall transmit an RRCConnectionReconfiguration message with the measurement gap configuration.

5. The UE shall transmit RRCConnectionReconfigurationComplete message.

5a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

5b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 5b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.

8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

10. The SS shall check the *rstd* value for Cell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.2.5.5-2.

11. Repeat step 2-10 until the confidence level according to Annex D is achieved.

12. Repeat step 1-11 for each sub-test in Table 9.2.5.5-1 as appropriate.

9.2.5.4.3 Message contents

**Table 9.2.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.2.5.4.3-2: *MeasGapConfig-GP1*: TDD-TDD inter-frequency RSTD Accuracy**

| Derivation Path: TS 36.508 [18] clause 4.6.6, Table 4.6.6-1A: MeasGapConfig-GP1 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| MeasGapConfig-GP1 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 34 (Test 1)<br>13 (Test 2) | TGRP = 40 ms | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.5.4.3-2a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| otdoa-RequestCapabilities | TRUE |

**Table 9.2.5.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See Note 4 of Table 9.2.5.4.1-1 | |

| | | | |
|---|---|---|---|
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.5.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause7.2.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.5.4.3-5: LPP ProvideLocation Information**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 1 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(1)) { | | | |
| physCellIdNeighbor | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | RF channel 2 | | |
| rstd | Set according to Table 9.2.5.5-2 for each specific test | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason ‘undefined’ or ‘attemptedButUnableToMeasureSomeNeighbourCells’ | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.5.4.3-6: *CQI-ReportConfig-DEFAULT*: TDD-TDD inter-frequency RSTD Accuracy**

| Derivation Path: TS 36.508 [18] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for Test 1 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |

| | | | |
|---|---|---|---|
| } | | | |

## 9.2.5.5 Test requirement

Table 9.2.5.5-1 defines the primary level settings including test tolerances for all tests.

The RSTD TDD-TDD inter frequency measurement accuracy test shall meet the reported values in Table 9.2.5.5-2.

**Table 9.2.5.5-1: Cell Specific Test Parameters for inter frequency RSTD Tests for E-UTRAN TDD-TDD**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | 1 | 2 | 1 | 2 |
| PBCH_RA | dB | 0 | | | |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA Note 1 | | | | | |
| OCNG_RB Note 1 | | | | | |
| PRS_RA | dB | -2.7 | 0.3 | -2.7 | 0.3 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| PRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| Io Note 3 | dBm/1.08 MHz | -79.24 | -79.39 | N/A | N/A |
| | dBm/9 MHz | N/A | N/A | -70.03 | -70.18 |
| PRP Note 3 | dBm/15kHz | -103.7 | -110.7 | -103.7 | -110.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -3 | -13 | -3 | -13 |
| RSRP Note 3 | dBm/15kHz | -101 | -111 | -101 | -111 |
| Propagation condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS.

**Table 9.2.5.5-2: RSTD TDD inter-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6424 | RSTD_6253 |
| Highest reported value | RSTD_6470 | RSTD_6275 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%. In the case that 1.4 MHz channel bandwidth is not defined for the operating band under test then Test 1 shall be omitted.

# 9.2.5A TDD-TDD inter-frequency RSTD Accuracy for UE Category 1bis

## 9.2.5A.1 Test purpose

To verify that the Reference Signal Time Difference (RSTD) TDD-TDD inter-frequency measurement accuracy is within the specified limit for all bands in AWGN channels.

## 9.2.5A.2 Test applicability

This test applies to E-UTRA TDD UE Category 1bis release 13 and forward that support inter-frequency RSTD measurements.

## 9.2.5A.3 Minimum conformance requirements

Same as 9.2.5.3

The normative reference for this requirement is TS 36.133 [23] clause 9.1.10.6 and A.9.8.4.2A.

## 9.2.5A.4 Test description

### 9.2.5A.4.1 Initial conditions

Same as 9.2.5.4.1 with the following exceptions:

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3, using only the main Tx/Rx path.

2. The general test parameter settings are set up according to Table 9.2.5A.4.1-1

**Table 9.2.5A.4.1-1: General Test Parameters for inter-frequency RSTD Tests for E-UTRAN TDD-TDD for UE Category 1bis**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | R.14 TDD | R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2. |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | OP.4 TDD | OP.2 TDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Cell 1 on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 on RF channel number 2 |
| E-UTRA RF Channel Number | | 1,2 | | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 1.4 | 10 | |
| PRS Bandwidth Note 2 | RB | 6 | 50 | |
| GapOffset | | 34 | 13 | For Cell 1 |
| Gap Pattern ID | | 0 | | For Cell 1 |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 3 | 1 | As specified in table 4.2-2 in TS 36.211 [26] and table 9.1.2.3-2. The same configuration in both cells. |
| PRS configuration Index $I_{PRS}$ Note 2 | | Cell 1: 15<br>Cell 2: 35 | Cell 1: 4<br>Cell 2: 14 | As defined in 3GPP TS 36.211 [26] |
| PRS subframe offset | | 20 | 10 | For Cell 2 |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | 1 | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1:'1111111100000000'<br>Cell 2:'1111111100000000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |

| | | | | |
|---|---|---|---|---|
| Cell ID Note 2 | | Cell 1: 0<br>Cell 2: 1 | | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The list includes the reference cell (received in *OTDOA-ReferenceCellInfo* [4]) on RF channel 1 and 15 other cells on RF channel 2, all received in *OTDOA-ProvideAssistanceData* [4]. |
| $T_{RSTDInterFreqTDD,E\text{-}UTRAN}$ Note 4 | ms | 10240 | | Derived according to the RSTD measurement requirements specified in Section 8.1.2.6.7 in TS 36.133 [23]. |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.2.5.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 2: Parameters “PRS Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. For all the values to be used in LPP see Table 9.2.5.4.3-4 and TS 37.571-5 [20], clause 7.2.2.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.2.5.4.1.

NOTE 4: The parameter “$T_{RSTDInterFreqTDD,E\text{-}UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.2.5A.4.3-3. The value of the LPP time IE is set to $T_{RSTDInterFreqTDD,E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 10390 ms. This is rounded up to the next allowed LPP value of 11 seconds.

### 9.2.5A.4.2 Test procedure

Same as 9.2.5.4.2.

### 9.2.5A.4.3 Message contents

Same as 9.2.5.4.3 with the following exceptions

**Table 9.2.5A.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.2.5.4.3-2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 11 | See Note 4 of Table 9.2.5A.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.2.5A.5 Test requirement

Same as 9.2.5.5.

# 9.3 RSTD Intra-Frequency Measurements for UE Category M1/M2

## 9.3.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A

### 9.3.1.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1

#### 9.3.1.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

#### 9.3.1.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA.

#### 9.3.1.1.3 Minimum conformance requirements

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data are provided, the UE shall be able to detect and measure intra-frequency RSTD, specified in 3GPP TS 36.214 [6], for at least *n*=16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{\text{RSTD IntraFreqFDD, Cat_M}}$ ms as given below (see also Figure 9.3.1.1.3-1):

$$T_{\text{RSTD IntraFreqFDD, Cat_M}} = T_{PRS} \cdot (M-1) + \Delta + T_{\text{MIB}} \text{ ms,}$$

where

$T_{\text{RSTD IntraFreqFDD, Cat_M}}$ is the total time for detecting and measuring at least *n* cells,

$T_{\text{PRS}}$ is the cell-specific positioning subframe configuration period as defined in 3GPP TS 36.211 [26] for UE not configured with measurement gaps for intra-frequency RSTD. For UE configured with measurement gaps for intra-frequency RSTD measurements, $T_{\text{PRS}}$ = max($T_{\text{PRS}}$, MGRP), where MGRP is the Measurement Gap Repetition Period as defined in section 8.1.2.1 of TS 36.133 [23].

$M$ is the number of PRS positioning occasions as defined in Table 9.3.1.1.3-1, , where downlink positioning subframes defined in TS 36.211 [26],

$\Delta = T_{PRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single PRS positioning occasion which includes the sampling time and the processing time,

$N_{\text{PRS}}$ is the cell-specific number of PRS subframes within a PRS occasion as defined in TS 36.355 [4],

$N_{\text{actual_PRS}}$ is the cell-specific number of PRS subframes within a PRS occasion; $N_{\text{actual_PRS}} = N_{\text{PRS}}$ for UE not configured with measurement gaps for intra-frequency RSTD measurements; for UE configured with measurement gaps for intra-frequency RSTD measurements $N_{\text{actual_PRS}}$ is the number of PRS subframes which can be measured by UE within MGL, where $N_{\text{actual_PRS}}$ = (MGL-2) if MGRP≥ $N_{\text{PRS}}$ >(MGL-2), $N_{\text{actual_PRS}}$ = $(MGL-2) \cdot \left\lfloor N_{\text{PRS}} / MGRP \right\rfloor$ if $N_{\text{PRS}}$ >MGRP, and $N_{\text{actual_PRS}} = N_{\text{PRS}}$ if $N_{\text{PRS}}$ ≤(MGL-2).

$N_{PRS_total}$ is the minimum number of PRS subframes per cell measurement as specified in TS 36.133 [23] Section 9.1.21.20.

$T_{\text{PRS}}$ $N_{\text{PRS}}$, and $N_{PRS_total}$ are the parameters of the same cell, for which $T_{PRS} \cdot \left\lceil \frac{N_{PRS_total}}{N_{actual_PRS}} \right\rceil$ is the largest among all the measured cells.

$T_{MIB}$ is the time required for acquiring the MIB information of the target cell. $T_{MIB}$ = 0 if the SFN of at least one cell in OTDOA assistance data is known to the UE.

**Table 9.3.1.1.3-1: Number of PRS positioning occasions within $T_{\text{RSTD IntraFreqFDD, Cat_M}}$**

| Positioning subframe configuration period $T_{\text{PRS}}$ | Number of PRS positioning occasions $M$ |
|---|---|
| | f1 Note 1 |
| 160 ms | $16 \cdot \left\lceil N_{PRS_Total} / N_{actual_PRS} \right\rceil$ |
| >160 ms | $8 \cdot \left\lceil N_{PRS_Total} / N_{actual_PRS} \right\rceil$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving FDD carrier frequency f1. | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD IntraFreqFDD, Cat_M}}$ provided:

$\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ PRS positioning occasions,

PRP $1,2|_{\text{dBm}}$ according to clause E.2 for a corresponding Band.

$\text{PRS}\,\hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry PRS.

The time $T_{\text{RSTD IntraFreqFDD, Cat_M}}$ starts from the first subframe of the PRS positioning occasion closest in time after both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in 3GPP TS 36.355 [4], are delivered to the physical layer of the UE as illustrated in Figure 9.3.1.1.3-1.

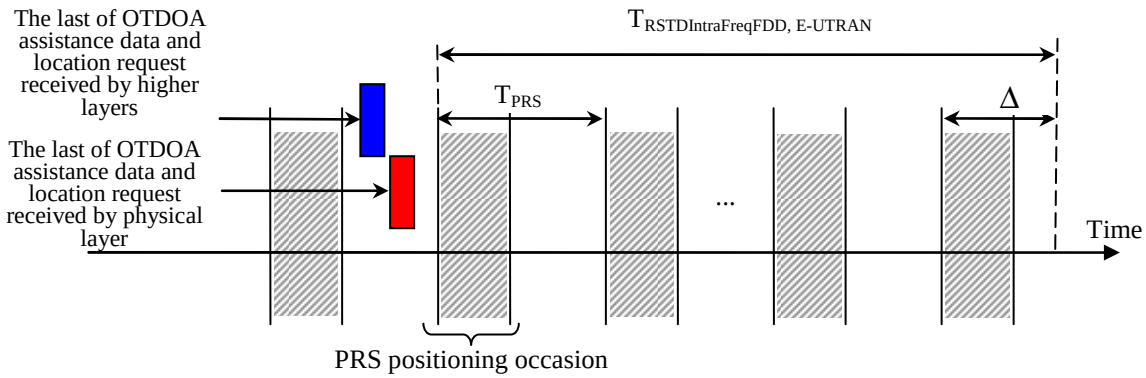



**Figure 9.3.1.1.3-1: Illustration of the RSTD reporting time requirement in an FDD system**

This requirement assumes that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is $N_{rep}$x $TTI_{DCCH}$, where $N_{rep}$ is the maximum number of PUSCH repetitions configured for the UE, otherwise uncertainty is defined as 2 x $TTI_{DCCH}$. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.3.1 and A.8.12.3.

### 9.3.1.1.4 Test description

#### 9.3.1.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4 using only the main Tx/Rx antenna of the UE.
2. The general test parameter settings are set up according to Table 9.3.1.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.3.1.1.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel. Cell 3 in the test is the Cell 4 defined in clause 4.7.1. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.5).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.

**Table 9.3.1.1.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.1.1.4.1. |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. |

**Table 9.3.1.1.4.1-2: DRX parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.3.1.1.4.2 Test procedure

The test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 9.3.1.1.4.1-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. Cell 2 is active until the end of T3, and Cell 3 is active until the end of T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, while Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.3.1.1.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF-CE according to 3GPP TS 36.508 [18] clause 7.2A.3AA.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 9.3.1.1.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).

4. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

6a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the first 7 elements of the sequence, and the position of neighbour Cell 3 is randomly

selected to be in the last 8 elements of the sequence, as described in 3GPP TS 37.571-5 [20], clause 7.2.3. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.

9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 9.3.1.1.5-2.

10. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.3.1.1.5-2.

11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 9.3.1.1.5. The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. Repeat steps 2-12 until the confidence level according to Annex D is achieved. For each iteration, at step 7 change the random position of the Cells 2 and 3 in the *OTDOA-NeighbourCellInfoList.*

#### 9.3.1.1.4.3 Message contents

**Table 9.3.1.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: TS 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.3.1.1.4.3-2: *MAC-MainConfig-RBC*: FDD RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.4.3-3: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 9.3.1.1.4.3-4: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See clause 9.3.1.1.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |

| | | | |
|---|---|---|---|
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.5. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.2.5. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

9.3.1.1.5 Test requirement

Table 9.3.1.1.5-1 and 9.3.1.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.3.1.1.5-1: Cell-specific test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |

| | | |
|---|---|---|
| Propagation Condition | | ETU30 |
| Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | |
| Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | |
| Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | |
| Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. | | |

**Table 9.3.1.1.5-2: Cell-specific test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.6 FDD | | OP.6 FDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -1 | -Infinity | -Infinity | -7 | -7 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1.79 | -Infinity | -Infinity | -7 | -9.54 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.55 | -67.08 | -69.55 | -67.08 | -69.55 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -105 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -93 | -105 | -105 | -108 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -10 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS. |
| Note 2: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes. |

The response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =16 and $n$ =16 are the parameters specified in clause 9.3.1.1.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.1.1.4.1-1. This gives the total RSTD reporting delay of 5210 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.3.1.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2

### 9.3.1.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.3.1.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA.

### 9.3.1.2.3 Minimum conformance requirements

Same as in clause 9.3.1.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.2.3.1 and A.8.12.3.

### 9.3.1.2.4 Test description

#### 9.3.1.2.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.1.2.4.1-1.

**Table 9.3.1.2.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |

| | | | | |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | 2 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.1.1.4.1. |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. |

#### 9.3.1.2.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.1.2.4.1-1 as appropriate

#### 9.3.1.2.4.3 Message contents

Same as in clause 9.3.1.1.4.3.

### 9.3.1.2.5 Test requirement

Same as in clause 9.3.1.1.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A

### 9.3.2.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1

#### 9.3.2.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements normal coverage mode in an environment with fading propagation conditions.

#### 9.3.2.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA.

#### 9.3.2.1.3 Minimum conformance requirements

Same as in clause 9.3.1.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.3.3 and A.8.12.4.

#### 9.3.2.1.4 Test description

##### 9.3.2.1.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.2.1.4.1-1

**Table 9.3.2.1.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.6 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |

| | | | |
|---|---|---|---|
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.3.2.1.4.2 Test procedure

Same as in clause 9.3.1.1.4.2.

#### 9.3.2.1.4.3 Message contents

Same as in clause 9.3.1.1.4.3

#### 9.3.2.1.5 Test requirement

Same as in clause 9.3.1.1.5.

### 9.3.2.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2

#### 9.3.2.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements normal coverage mode in an environment with fading propagation conditions.

#### 9.3.2.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA.

#### 9.3.2.2.3 Minimum conformance requirements

Same as in clause 9.3.1.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.2.3.3 and A.8.12.4.

#### 9.3.2.2.4 Test description

#### 9.3.2.2.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.2.2.4.1-1.

**Table 9.3.2.2.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |

| | | | | |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.6 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | 2 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.1.1.4.1. |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. |

#### 9.3.2.2.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.2.2.4.1-1 as appropriate.

#### 9.3.2.2.4.3 Message contents

Same as in clause 9.3.1.1.4.3.

### 9.3.2.2.5 Test requirement

Same as in clause 9.3.2.1.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.3 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A

### 9.3.3.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1

#### 9.3.3.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

#### 9.3.3.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA.

#### 9.3.3.1.3 Minimum conformance requirements

Same as in clause 9.3.1.1.3.

The intra-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.3.3.1.3-1.

**Table 9.3.3.1.3-1: TDD uplink-downlink subframe configurations applicable for TDD intra-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.3.2 and A.8.12.5.

9.3.3.1.4 Test description

9.3.3.1.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.3.1.4.1-1

**Table 9.3.3.1.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.14 TDD | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 304 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.1.1.4.1.<br>Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. | | | |

### 9.3.3.1.4.2 Test procedure

Same as in clause 9.3.1.1.4.2.

### 9.3.3.1.4.3 Message contents

Same as in clause 9.3.1.1.4.3

### 9.3.3.1.5 Test requirement

Table 9.3.3.1.5-1 and 9.3.3.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.3.3.1.5-1: Cell-specific test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |

| | | | | |
|---|---|---|---|---|
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.3.3.1.5-2: Cell-specific test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -1 | -Infinity | -Infinity | -7 | -7 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1.79 | -Infinity | -Infinity | -7 | -9.54 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.55 | -67.08 | -69.55 | -67.08 | -69.55 | N/A |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -105 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -93 | -105 | -105 | -108 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -10 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where $\Delta T = N_{rep} \times TTI_{DCCH} = N_{rep} \times 75$ ms, giving a value of [5270] ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil \frac{n}{M} \right\rceil$, where $M$ =16 and $n$ =16 are the parameters specified in clause 9.3.1.1.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.3.1.4.1-1. This gives the total RSTD reporting delay of [5210] ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.3.3.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2

### 9.3.3.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.3.3.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA.

### 9.3.3.2.3 Minimum conformance requirements

Same as in clause 9.3.3.1.3.

The intra-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.3.3.2.3-1.

**Table 9.3.3.2.3-1: TDD uplink-downlink subframe configurations applicable for TDD intra-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |

| | |
|---|---|
| 24 | 0, 1, 2, 3, 4, 5 and 6 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.16.2.3.2 and A.8.12.5.

9.3.3.2.4 Test description

9.3.3.2.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.3.2.4.1-1.

**Table 9.3.3.2.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.14 TDD | | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 304 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | 2 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |

| | | | |
|---|---|---|---|
| DRX | | ON | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.1.1.4.1.<br>Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. | | | |

#### 9.3.3.2.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.2.2.4.1-1 as appropriate.

#### 9.3.3.2.4.3 Message contents

Same as in clause 9.3.1.1.4.3.

### 9.3.3.2.5 Test requirement

Same as in clause 9.3.3.1.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.4 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B

### 9.3.4.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1

#### 9.3.4.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in enhanced coverage mode in an environment with fading propagation conditions.

#### 9.3.4.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

#### 9.3.4.1.3 Minimum conformance requirements

Same as 9.3.1.1.3 with the following exceptions:

The conditions under which the UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD IntraFreqFDD, Cat_M}}$ are changed:

$\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{PRS}\,\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ PRS positioning occasions,

PRP $1,2|_{\text{dBm}}$ according to clause E.2 for a corresponding Band.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.3.1 and A.8.12.6.

#### 9.3.4.1.4 Test description

##### 9.3.4.1.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.4.1.4.1-1.

**Table 9.3.4.1.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.18 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |

| | | | |
|---|---|---|---|
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 6.4 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 6.4 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.3.4.1.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but using condition CEModeB.

#### 9.3.4.1.4.3 Message contents

Same as in clause 9.3.1.1.4.3 with the following exceptions:

**Table 9.3.4.1.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 13 | See clause 9.3.4.1.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 9.3.4.1.5 Test requirement

Table 9.3.4.1.5-1 and 9.3.4.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.3.4.1.5-1: Cell-specific test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |

| | | | | |
|---|---|---|---|---|
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.3.4.1.5-2: Cell-specific test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.6 FDD | | OP.6 FDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -12 | -Infinity | -Infinity | -13 | -13 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -12.21 | -Infinity | -Infinity | -13 | -13.27 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.92 | -67.18 | -69.92 | -67.18 | -69.92 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -110 | -Infinity | -Infinity | -108 | -111 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -107 | -104 | -111 | -111 | -114 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | -9 | -9 | -13 | -16 | -16 | -Infinity |

| Propagation Condition | | ETU30 |
|---|---|---|

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 13.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 12950 ms. This is rounded up to the next allowed LPP value of 13 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =16 and $n$ =16 are the parameters specified in clause 9.3.1.1.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.4.1.4.1-1. This gives the total RSTD reporting delay of 12800 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.3.4.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2

### 9.3.4.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in enhanced coverage mode in an environment with fading propagation conditions.

### 9.3.4.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

### 9.3.4.2.3 Minimum conformance requirements

Same as in clause 9.3.4.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.3.1.1 and A.8.12.6.

### 9.3.4.2.4 Test description

#### 9.3.4.2.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.4.2.4.1-1.

**Table 9.3.4.2.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |

| | | | | |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.18 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 6.4 | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 6.4 | 2.56 | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.1.1.4.1. |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. |

### 9.3.4.2.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but using condition CEModeB and with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.4.2.4.1-1 as appropriate

### 9.3.4.2.4.3 Message contents

Same as in clause 9.3.1.1.4.3 with the following exceptions:

**Table 9.3.4.2.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 13 | See clause 9.3.4.1.5 | Test 1 |
| time | 6 | See clause 9.3.4.2.5 | Test 2 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.3.4.2.5 Test requirement

The primary level settings including test tolerances for the test are defined in clause 9.3.4.1.5.

For Test 1, the response time is defined in clause 9.3.4.1.5.

For Test 2, the response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where $\Delta T = N_{rep} \times TTI_{DCCH} = N_{rep} \times 75$ ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD

measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =40 and $n$ =16 are the parameters specified in clause 9.3.1.1.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.4.2.4.1-1. This gives the total RSTD reporting delay of 5210 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.5 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B

### 9.3.5.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1

#### 9.3.5.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in an environment with fading propagation conditions.

#### 9.3.5.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

#### 9.3.5.1.3 Minimum conformance requirements

Same as in clause 9.3.4.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.3.3 and A.8.12.7.

#### 9.3.5.1.4 Test description

##### 9.3.5.1.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.5.1.4.1-1

**Table 9.3.5.1.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.8 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2 |

| | | | |
|---|---|---|---|
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 6.4 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 6.4 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.3.5.1.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but using condition CEModeB.

#### 9.3.5.1.4.3 Message contents

Same as in clause 9.3.4.1.4.3

### 9.3.5.1.5 Test requirement

Same as in clause 9.3.4.1.5.

## 9.3.5.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2

### 9.3.5.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in an environment with fading propagation conditions.

### 9.3.5.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

### 9.3.5.2.3 Minimum conformance requirements

Same as in clause 9.3.4.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.3.1.3 and A.8.12.7.

### 9.3.5.2.4 Test description

#### 9.3.5.2.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.5.2.4.1-1.

**Table 9.3.5.2.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.8 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |

| | | | | |
|---|---|---|---|---|
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 6.4 | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 6.4 | 2.56 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.3.5.2.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but using condition CEModeB and with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.2.2.4.1-1 as appropriate.

#### 9.3.5.2.4.3 Message contents

Same as in clause 9.3.4.2.4.3.

### 9.3.5.2.5 Test requirement

Same as in clause 9.3.4.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.6 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B

### 9.3.6.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1

#### 9.3.6.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in an environment with fading propagation conditions.

#### 9.3.6.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

#### 9.3.6.1.3 Minimum conformance requirements

Same as in clause 9.3.4.1.3.

The intra-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.3.6.1.3-1.

**Table 9.3.6.1.3-1: TDD uplink-downlink subframe configurations applicable for TDD intra-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.3.2 and A.8.12.8.

#### 9.3.6.1.4 Test description

##### 9.3.6.1.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.6.1.4.1-1

**Table 9.3.6.1.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 TDD | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 304 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |

| | | | |
|---|---|---|---|
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 6.4 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 6.4 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

### 9.3.6.1.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but using condition CEModeB.

### 9.3.6.1.4.3 Message contents

Same as in clause 9.3.4.1.4.3

### 9.3.6.1.5 Test requirement

Table 9.3.6.1.5-1 and 9.3.6.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.3.6.1.5-1: Cell-specific test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |

| | | | | |
|---|---|---|---|---|
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.3.6.1.5-2: Cell-specific test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -12 | -Infinity | -Infinity | -13 | -13 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -12.21 | -Infinity | -Infinity | -13 | -13.27 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.92 | -67.18 | -69.92 | -67.18 | -69.92 | -67.18 |
| PRP Note 4 | dBm/ 15 kHz | -110 | -Infinity | -Infinity | -108 | -111 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -107 | -104 | -111 | -111 | -114 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | -9 | -9 | -13 | -16 | -16 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS. |
| Note 2: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes. |

The response time is defined in clause 9.3.4.1.5.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.3.6.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2

### 9.3.6.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in an environment with fading propagation conditions.

### 9.3.6.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

### 9.3.6.2.3 Minimum conformance requirements

Same as in clause 9.3.6.1.3.

The intra-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.3.6.2.3-1.

**Table 9.3.6.2.3-1: TDD uplink-downlink subframe configurations applicable for TDD intra-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| 24 | 0, 1, 2, 3, 4, 5 and 6 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.16.3.1.2 and A.8.12.8.

### 9.3.6.2.4 Test description

#### 9.3.6.2.4.1 Initial conditions

Same as in clause 9.3.1.1.4.1 but replacing Table 9.3.1.1.4.1-1 with Table 9.3.6.2.4.1-1.

**Table 9.3.6.2.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 TDD | | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 304 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |

| | | | | |
|---|---|---|---|---|
| Expected RSTD uncertainty for all neighbour cells Note 1 | μs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 6.4 | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 6.4 | 2.56 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.3.6.2.4.2 Test procedure

Same as in clause 9.3.1.1.4.2 but using condition CEModeB and with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.2.2.4.1-1 as appropriate.

#### 9.3.6.2.4.3 Message contents

Same as in clause 9.3.4.2.4.3.

### 9.3.6.2.5 Test requirement

The primary level settings including test tolerances for the test are defined in clause 9.3.6.1.5.

For Test 1, the response time is defined in clause 9.3.6.1.5.

For Test 2, the response time including test tolerance is 6.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where $\Delta T = N_{rep} \times TTI_{DCCH} = N_{rep} \times 75$ ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =40 and $n$ =16 are the parameters specified in clause 9.3.1.1.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.6.2.4.1-1. This gives the total RSTD reporting delay of 5210 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.9.3.7 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A.

# 9.3.7 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A

## 9.3.7.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1

### 9.3.7.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

### 9.3.7.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA.

### 9.3.7.1.3 Minimum conformance requirements

The accuracy requirements in Table 9.3.7.1.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex E.2 for a corresponding Band

For a UE that does not need a measurement gap for intra-frequency RSTD measurement, there are no measurement gaps overlapping with the PRS subframes of the measured serving cell and PRS are available within the UE measurement bandwidth in all PRS subframes

For a UE that needs a measurement gap for intra-frequency RSTD measurement, the measurement gaps are configured to contain PRS subframes

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.3.7.1.3-1: RSTD intra-frequency measurement accuracy for CEModeA**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io Note 4 range | | |
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* Notes 3, 6 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell and the measured neighbour cell *i* as defined in [24] | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | | dBm/15kHz | $dBm/BW_{Channel}$ |
| ±15 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 6 | ≥ 12 | ≥ 6 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD-M1_B | -120.5 | -50 |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 6: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.20 and A.9.8.20.

#### 9.3.7.1.4 Test description

##### 9.3.7.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 using only the main Tx/Rx antenna of the UE.
2. The general test parameter settings are set up according to Table 9.3.7.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.3.7.1.4.3.
5. All cells are on the same carrier frequency. Cell 1 is the serving cell and OTDOA assistance data reference cell; Cell 2 is the neighbour cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.2.5).
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:
   - Test 1: -92 Ts (about -3 μs)
   - Test 2: 92 Ts (about 3 μs)

   Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.3.7.1.4-1 for each test.

**Table 9.3.7.1.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions.**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| MPDCCH | | R.16 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |

| | | | | |
|---|---|---|---|---|
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 151 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ | ms | 5120 | | Derived according to the RSTD measurement requirements specified in section 9.3.7.1.3 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.7.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.7.1.4.1. |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. |
| Note 5: | The parameter “$T_{\text{RSTD IntraFreqFDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.3.7.1.4.3-3. The value of the LPP time IE is set to $T_{\text{RSTD IntraFreqFDD, E-UTRAN}}$ + $\Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds. |

### 9.3.7.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.3.7.1.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF-CE according to 3GPP TS 36.508 [18] clause 7.2A.3AA.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.3.7.1.5-1. Propagation conditions are set according to clause 4.7.2.1.
4. The SS shall transmit an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms.
8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
11. The SS shall check the *rstd* value for Cell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.3.7.1.5-2.
12. Repeat steps 2-10 until the confidence level according to Annex D is achieved.

13. Repeat step 1-12 for each sub-test in Table 9.3.7.1.5-1 as appropriate.

9.3.7.1.4.3 Message contents

**Table 9.3.7.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path:TS 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.3.7.1.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 9.3.7.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See Note 4 of Table 9.3.7.1.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |

| | | | |
|---|---|---|---|
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.7.1.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.2.5. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.2.5. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

9.3.7.1.5 Test requirement

Table 9.3.7.1.5-1 defines the primary level settings including test tolerances for the test.

The RSTD FDD intra-frequency accuracy test shall meet the reported values in Table 9.3.7.1.5-2

**Table 9.3.7.1.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRAN FDD**

| Parameter | Unit | Test1 | | Test2 | | Test3 Note 4 | | Test4 Note 4 | |
|---|---|---|---|---|---|---|---|---|---|
| | | Cell1 | Cell2 | Cell1 | Cell2 | Cell1 | Cell2 | Cell1 | Cell2 |
| E-UTRA RF Channel Number | | 1 | | | | | | | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | | |
| PSS_RA | | | | | | | | | |
| SSS_RA | | | | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| MPDCCH_RA<br>MPDCCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | | | | | | | | |
| PRS_RA | dB | 0 | 0 | -2.7 | 0.3 | 0 | 0 | -2.7 | 0.3 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 | -98 | -98 | -98 | -98 | -98 | -98 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -2.37 | -8.02 | -5.7 | -12.7 | -2.37 | -8.02 | -5.7 | -12.7 |
| PRS $\hat{E}_s/I_{ot}$ Note3 | dB | -3.01 | -10.01 | -5.7 | -12.7 | -3.01 | -10.01 | -5.7 | -12.7 |
| Io Note3 | dBm/9 MHz | -69.23 | -69.23 | -70 | -70 | -69.23 | -69.23 | -70 | -70 |
| PRP Note3 | dBm/15kHz | -100.37 | -106.02 | -103.7 | -110.7 | -100.37 | -106.02 | -103.7 | -110.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -2.37 | -8.02 | -3 | -13 | -2.37 | -8.02 | -3 | -13 |
| RSRP Note 3 | dBm/15kHz | -100.37 | -106.02 | -101 | -111 | -100.37 | -106.02 | -101 | -111 |
| Propagation condition | | AWGN | | | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS

Note 4: Test3 and Test4 are not used for test 9.3.7.1

**Table 9.3.7.1.5-2: RSTD FDD intra-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6248 | RSTD_6431 |
| Highest reported value | RSTD_6280 | RSTD_6463 |

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.7.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2

### 9.3.7.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

### 9.3.7.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA.

### 9.3.7.2.3 Minimum conformance requirements

The accuracy requirements in Table 9.3.7.2.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex E.2 for a corresponding Band

For a UE that does not need a measurement gap for intra-frequency RSTD measurement, there are no measurement gaps overlapping with the PRS subframes of the measured serving cell and PRS are available within the UE measurement bandwidth in all PRS subframes

For a UE that needs a measurement gap for intra-frequency RSTD measurement, the measurement gaps are configured to contain PRS subframes

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.3.7.2.3-1: RSTD intra-frequency measurement accuracy for CEModeA**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io [Note 4] range | | |
| | PRS $\hat{E}s/Iot$ | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* Notes 3, 6 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell and the measured neighbour cell *i* as defined in [24] | E-UTRA operating band groups [Note 5] | Minimum Io [Note 1] | Maximum Io |
| Ts [Note 2] | dB | RB | | | | dBm/15kHz | $dBm/BW_{Channel}$ |
| ±15 | $(PRS \hat{E}s/Iot)_{ref}$ ≥-6dB and $(PRS \hat{E}s/Iot)_i$ ≥-13dB | ≥ 6 | ≥ 12 | ≥ 6 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD-M1_B | -120.5 | -50 |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |
| ±6 | $(PRS \hat{E}s/Iot)_{ref}$ ≥-6dB and $(PRS \hat{E}s/Iot)_i$ ≥-13dB | ≥ 24 | ≥ 4 | ≥ 2 | Note 7 | Note 7 | Note 7 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 6: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 7: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 6 RB.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.4 and A.9.8.20.

9.3.7.2.4 Test description

9.3.7.2.4.1 Initial conditions

Same as in clause 9.3.7.1.4.1 adding Test 3 and Test 4 and replacing Table 9.3.7.1.4.1-1 with Table 9.3.7.2.4.1-1.

The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:

Test 1: -92 Ts (about -3 μs)
Test 2: 92 Ts (about 3 μs)
Test 3: 92 Ts (about 3 μs)
Test 4: -92 Ts (about -3 μs)

Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.3.7.2.4-1 for each test.

**Table 9.3.7.2.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test1 | Test2 | Test3 | Test4 | |
| M-PDCCH parameters | | R.16 FDD | | R.16 FDD | | As specified in TS 36.521-3 [25] clause A.7.1. |
| *mPDCCH-startSF-UESS* | | 10 | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | 50 Note 4 | | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| PRS configuration Index $I_{PRS}$ Note 2 | | 151 | | 151 | | As defined in TS 36.211 [26] |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 2 | | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 1 Other neighbour cells: randomly between -3 and 3 | Cell 2: -1 Other neighbour cells: randomly between -3 and 3 | Cell 2: -1 Other neighbour cells: randomly between -3 and 3 | Cell 2: 1 Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ Note 5 | ms | 5120 | | | | Derived according to the RSTD measurement requirements specified in section 9.3.7.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter “$T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.3.7.1.4.3-3. The value of the LPP time IE is set to $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

#### 9.3.7.2.4.2 Test procedure

Same as in clause 9.3.7.1.4.2.

#### 9.3.7.2.4.3 Message contents

Same as in clause 9.3.7.1.4.3.

### 9.3.7.2.5 Test requirement

Same as in clause 9.3.7.1.5 adding Test 3 and Test 4 and replacing Table 9.3.7.1.5-2 with Table 9.3.7.2.5-1:

**Table 9.3.7.2.5-1: RSTD FDD intra-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|
| Lowest reported value | RSTD_6257 | RSTD_6440 | RSTD_6440 | RSTD_6257 |
| Highest reported value | RSTD_6271 | RSTD_6454 | RSTD_6454 | RSTD_6271 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.8 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A

### 9.3.8.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1

#### 9.3.8.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits normal coverage mode in an environment with fading propagation conditions.

#### 9.3.8.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA.

#### 9.3.8.1.3 Minimum conformance requirements

Same as in clause 9.3.7.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.20 and A.9.8.21.

#### 9.3.8.1.4 Test description

##### 9.3.8.1.4.1 Initial conditions

Same as in clause 9.3.7.1.4.1 but replacing Table 9.3.7.1.4.1-1 with Table 9.3.8.1.4.1-1

**Table 9.3.8.1.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| MPDCCH | | R.6 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |

| | | | | |
|---|---|---|---|---|
| PRS configuration index $I_{PRS}$ Note 2 | | 151 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| $T_{RSTD\ IntraFreqHD\text{-}FDD,\ E\text{-}UTRAN}$ | ms | 5120 | | Derived according to the RSTD measurement requirements specified in section 9.3.7.1.3 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.7.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.7.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter "$T_{RSTD\ IntraFreqHD\text{-}FDD,\ E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.3.7.1.4.3-3. The value of the LPP time IE is set to $T_{RSTD\ IntraFreqHD\text{-}FDD,\ E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

### 9.3.8.1.4.2 Test procedure

Same as in clause 9.3.7.1.4.2.

### 9.3.8.1.4.3 Message contents

Same as in clause 9.3.7.1.4.3

### 9.3.8.1.5 Test requirement

Same as in clause 9.3.7.1.5.

## 9.3.8.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2

### 9.3.8.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits normal coverage mode in an environment with fading propagation conditions.

### 9.3.8.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA.

### 9.3.8.2.3 Minimum conformance requirements

Same as in clause 9.3.7.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.4 and A.9.8.21.

### 9.3.8.2.4 Test description

#### 9.3.8.2.4.1 Initial conditions

Same as in clause 9.3.7.2.4.1 but replacing Table 9.3.7.2.4.1-1 with Table 9.3.8.2.4.1-1.

**Table 9.3.8.2.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test1 | Test2 | Test3 | Test4 | |
| M-PDCCH parameters | | R.6 HD-FDD | | R.6 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.2. |
| *mPDCCH-startSF-UESS* | | 10 | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | 50 Note 4 | | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| PRS configuration Index $I_{PRS}$ Note 2 | | 151 | | 151 | | As defined in TS 36.211 [26] |

| | | | | | | |
|---|---|---|---|---|---|---|
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 2 | | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTD\ IntraFreqHD\text{-}FDD,\ E\text{-}UTRAN}$ Note 5 | ms | 5120 | | | | Derived according to the RSTD measurement requirements specified in section 9.3.7.2.3 |

NOTE 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter "$T_{RSTD\ IntraFreqHD\text{-}FDD,\ E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.3.7.1.4.3-3. The value of the LPP time IE is set to $T_{RSTD\ IntraFreqHD\text{-}FDD,\ E\text{-}UTRAN} + \Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

#### 9.3.8.2.4.2 Test procedure

Same as in clause 9.3.7.1.4.2.

#### 9.3.8.2.4.3 Message contents

Same as in clause 9.3.7.1.4.3.

### 9.3.8.2.5 Test requirement

Same as in clause 9.3.7.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.9 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A

### 9.3.9.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1

#### 9.3.9.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

#### 9.3.9.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA.

#### 9.3.9.1.3 Minimum conformance requirements

Same as in clause 9.3.7.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.20 and A.9.8.22.

#### 9.3.9.1.4 Test description

##### 9.3.9.1.4.1 Initial conditions

Same as in clause 9.3.7.1.4.1 but replacing Table 9.3.7.1.4.1-1 with Table 9.3.9.1.4.1-1

**Table 9.3.9.1.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| MPDCCH | | R.14 TDD | | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |

| | | | | |
|---|---|---|---|---|
| Uplink-downlink configuration | | 1 | | As specified in table 4.2-2 in TS 36.211 [26] and table 8.1.2.5.2-2. The same configuration in both cells. |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 154 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| $T_{RSTD\ IntraFreqTDD,E\text{-}UTRAN}$ | ms | 5120 | | Derived according to the RSTD measurement requirements specified in section 9.3.7.1.3 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.7.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.7.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter "$T_{RSTD\ IntraFreqTDD,E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.3.7.1.4.3-3. The value of the LPP time IE is set to $T_{RSTD\ IntraFreqTDD,E\text{-}UTRAN}$ + $\Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

### 9.3.9.1.4.2 Test procedure

Same as in clause 9.3.7.1.4.2.

#### 9.3.9.1.4.3 Message contents

Same as in clause 9.3.7.1.4.3

#### 9.3.9.1.5 Test requirement

Same as in clause 9.3.7.1.5.

### 9.3.9.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2

#### 9.3.9.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

#### 9.3.9.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA.

#### 9.3.9.2.3 Minimum conformance requirements

Same as in clause 9.3.7.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.4 and A.9.8.22.

#### 9.3.9.2.4 Test description

#### 9.3.9.2.4.1 Initial conditions

Same as in clause 9.3.7.2.4.1 but replacing Table 9.3.7.2.4.1-1 with Table 9.3.9.2.4.1-1.

**Table 9.3.9.2.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test1 | Test2 | Test3 | Test4 | |
| M-PDCCH parameters | | R.14 TDD | | R.14 TDD | | As specified in TS 36.521-3 [25] clause A.7.3. |
| *mPDCCH-startSF-UESS* | | 10 | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OP.11 FTDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | | 10 | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Special subframe configuration | | 6 | | | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 1 | | | | As specified in table 4.2-2 in TS 36.211 [26] and table 8.1.2.5.2-2. The same configuration in both cells. |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | 50 Note 4 | | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| PRS configuration Index $I_{PRS}$ Note 2 | | 154 | | 154 | | As defined in TS 36.211 [26] |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 2 | | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTD\ IntraFreqTDD,\ E\text{-}UTRAN}$ Note 5 | ms | 5120 | | | | Derived according to the RSTD measurement requirements specified in section 9.3.7.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.
NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.
NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.
NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.
NOTE 5: The parameter “$T_{\text{RSTD IntraFreqTDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.3.7.1.4.3-3. The value of the LPP time IE is set to $T_{\text{RSTD IntraFreqTDD, E-UTRAN}} + \Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 5270 ms. This is rounded up to the next allowed LPP value of 6 seconds.

#### 9.3.9.2.4.2 Test procedure

Same as in clause 9.3.7.1.4.2.

#### 9.3.9.2.4.3 Message contents

Same as in clause 9.3.7.1.4.3.

### 9.3.9.2.5 Test requirement

Same as in clause 9.3.7.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.10 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B

### 9.3.10.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1

#### 9.3.10.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

#### 9.3.10.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

#### 9.3.10.1.3 Minimum conformance requirements

The accuracy requirements in Table 9.3.10.1.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{\text{dBm}}$ according to Annex E.2 for a corresponding Band

For a UE that does not need a measurement gap for intra-frequency RSTD measurement, there are no measurement gaps overlapping with the PRS subframes of the measured serving cell and PRS are available within the UE measurement bandwidth in all PRS subframes

For a UE that needs a measurement gap for intra-frequency RSTD measurement, the measurement gaps are configured to contain PRS subframes

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.3.10.1.3-1: RSTD intra-frequency measurement accuracy for CEModeB**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* Notes 3, 6 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell and the measured neighbour cell *i* as defined in [24] | Io Note 4 range | | |
| | | | | | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | | dBm/15kHz | $dBm/BW_{Channel}$ |
| ±15 | $(PRS Ês/Iot)_{ref}$ ≥-15dB and $(PRS Ês/Iot)_i$ ≥-15dB | ≥ 6 | ≥ 12 | ≥ 6 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD-M1_B | -120.5 | -50 |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 6: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.21 and A.9.8.23.

### 9.3.10.1.4 Test description

#### 9.3.10.1.4.1 Initial conditions

Same as in clause 9.3.7.1.4.1 but replacing Table 9.3.7.1.4.1-1 with Table 9.3.10.1.4.1-1.

**Table 9.3.10.1.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions.**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |

| | | | | |
|---|---|---|---|---|
| MPDCCH | | R.18 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 151 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| $T_{RSTD\ IntraFreqFDD,\ E-UTRAN}$ | ms | 12800 | | Derived according to the RSTD measurement requirements specified in section 9.3.10.1.3 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.10.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.10.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter “$T_{\text{RSTD IntraFreqFDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.3.10.1.4.3-3. The value of the LPP time IE is set to $T_{\text{RSTD IntraFreqFDD, E-UTRAN}}$ + $\Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 12950 ms. This is rounded up to the next allowed LPP value of 13 seconds.

### 9.3.10.1.4.2 Test procedure

Same as in clause 9.3.7.1.4.2 but using condition CEModeB.

### 9.3.10.1.4.3 Message contents

Same as in clause 9.3.7.1.4.3 with the following exceptions

**Table 9.3.10.1.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.7.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 13 | See Note 4 in Table 9.3.10.1.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.3.10.1.5 Test requirement

Table 9.3.10.1.5-1 defines the primary level settings including test tolerances for the test.

The RSTD FDD intra-frequency accuracy test shall meet the reported values in Table 9.3.10.1.5-2

**Table 9.3.10.1.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRAN FDD**

| Parameter | Unit | Test1 | | Test2 | | Test3 Note4 | | Test4 Note4 | |
|---|---|---|---|---|---|---|---|---|---|
| | | Cell1 | Cell2 | Cell1 | Cell2 | Cell1 | Cell2 | Cell1 | Cell2 |
| E-UTRA RF Channel Number | | 1 | | | | | | | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | | |
| PSS_RA | | | | | | | | | |
| SSS_RA | | | | | | | | | |
| MPDCCH_RA | | | | | | | | | |
| MPDCCH_RB | | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | | |
| PRS_RA | dB | 0 | 0 | -2.7 | 0.3 | 0 | 0 | -2.7 | 0.3 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 | -98 | -98 | -98 | -98 | -98 | -98 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -5 | -13 | -14.7 | -14.7 | -5 | -13 | -14.7 | -14.7 |
| PRS $\hat{E}_s/I_{ot}$ Note3 | dB | -5.21 | -14.19 | -14.7 | -14.7 | -5.21 | -14.19 | -14.7 | -14.7 |
| Io Note3 | dBm/9 MHz | -69.79 | -69.79 | -70.06 | -70.06 | -69.79 | -69.79 | -70.06 | -70.06 |
| PRP Note3 | dBm/15kHz | -103 | -111 | -112.7 | -112.7 | -103 | -111 | -112.7 | -112.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -5 | -13 | -12 | -15 | -5 | -13 | -12 | -15 |
| RSRP Note 3 | dBm/15kHz | -103 | -111 | -110 | -113 | -103 | -111 | -110 | -113 |
| Propagation condition | | AWGN | | | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS

Note 4: Test3 and Test4 are not used for test 9.3.10.1

**Table 9.3.10.1.5-2: RSTD FDD intra-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6248 | RSTD_6431 |
| Highest reported value | RSTD_6280 | RSTD_6463 |

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.10.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2

### 9.3.10.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

### 9.3.10.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

9.3.10.2.3 Minimum conformance requirements

The accuracy requirements in Table 9.3.10.2.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex E.2 for a corresponding Band

For a UE that does not need a measurement gap for intra-frequency RSTD measurement, there are no measurement gaps overlapping with the PRS subframes of the measured serving cell and PRS are available within the UE measurement bandwidth in all PRS subframes

For a UE that needs a measurement gap for intra-frequency RSTD measurement, the measurement gaps are configured to contain PRS subframes

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.3.10.2.3-1: RSTD intra-frequency measurement accuracy for CEModeB**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io [Note 4] range | | |
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* Notes 3, 6 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell and the measured neighbour cell *i* as defined in [24] | E-UTRA operating band groups [Note 5] | Minimum Io [Note 1] | Maximum Io |
| Ts [Note 2] | dB | RB | | | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±15 | (PRS Ês/Iot)$_{ref}$ ≥-15dB and (PRS Ês/Iot)$_i$ ≥-15dB | ≥ 6 | ≥ 12 | ≥ 6 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD-M1_B | -120.5 | -50 |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |
| ±6 | (PRS Ês/Iot)$_{ref}$ ≥-15dB and (PRS Ês/Iot)$_i$ ≥-15dB | ≥ 24 | ≥ 4 | ≥ 4 | Note 7 | Note 7 | Note 7 |
| NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol. | | | | | | | |

NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 6: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 7: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 6 RB.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.5 and A.9.8.23.

### 9.3.10.2.4 Test description

#### 9.3.10.2.4.1 Initial conditions

Same as in clause 9.3.10.1.4.1 adding Test 3 and Test 4 and replacing Table 9.3.10.1.4.1-1 with Table 9.3.10.2.4.1-1.

The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:
- Test 1: -92 Ts (about -3 μs)
- Test 2: 92 Ts (about 3 μs)
- Test 3: 92 Ts (about 3 μs)
- Test 4: -92 Ts (about -3 μs)

Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.3.10.2.4-1 for each test.

**Table 9.3.10.2.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test1 | Test2 | Test3 | Test4 | |
| M-PDCCH parameters | | R.18 FDD | | R.18 FDD | | As specified in TS 36.521-3 [25] clause A.7.1. |
| *mPDCCH-startSF-UESS* | | 10 | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | 50 Note 4 | | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| PRS configuration Index $I_{PRS}$ Note 2 | | 151 | | 151 | | As defined in TS 36.211 [26] |

| | | | | | | |
|---|---|---|---|---|---|---|
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 4 | | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTDIntraFreqFDD,E-UTRAN}$ Note 5 | ms | 12800 | | 5120 | | Derived according to the RSTD measurement requirements specified in section 9.3.10.2.3 |

NOTE 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter "$T_{RSTDIntraFreqFDD,E-UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.3.10.2.4.3-1. The value of the LPP time IE is set to $T_{RSTDIntraFreqFDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 12950 ms for Test 1 and 2 and 5270 ms for Test 3 and 4. This is rounded up to the next allowed LPP value of 13 or 6 seconds, respectively.

### 9.3.10.2.4.2 Test procedure

Same as in clause 9.3.7.1.4.2 but using condition CEModeB.

### 9.3.10.2.4.3 Message contents

Same as in clause 9.3.7.1.4.3 with the following exceptions:

**Table 9.3.10.2.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.7.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 13 | See Note 4 in Table 9.3.10.2.4.1-1 | Test 1 or Test 2 |
| time | 6 | See Note 4 in Table 9.3.10.2.4.1-1 | Test 3 or Test 4 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 9.3.10.2.5 Test requirement

Same as in clause 9.3.10.1.5 but replacing Table 9.3.10.1.5-2 with Table 9.3.10.2.5-1:

**Table 9.3.10.2.5-1: RSTD FDD intra-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|
| Lowest reported value | RSTD_6257 | RSTD_6440 | RSTD_6440 | RSTD_6257 |
| Highest reported value | RSTD_6271 | RSTD_6454 | RSTD_6454 | RSTD_6271 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.11 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B

### 9.3.11.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1

#### 9.3.11.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits enhanced coverage mode in an environment with fading propagation conditions.

#### 9.3.11.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

#### 9.3.11.1.3 Minimum conformance requirements

Same as in clause 9.3.10.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.21 and A.9.8.24.

9.3.11.1.4 Test description

9.3.11.1.4.1 Initial conditions

Same as in clause 9.3.10.1.4.1 but replacing Table 9.3.10.1.4.1-1 with Table 9.3.11.1.4.1-1

**Table 9.3.11.1.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| MPDCCH | | R.8 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 151 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 6 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |

| DRX | | OFF | | |
|---|---|---|---|---|
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| $T_{RSTD\ IntraFreqHD-FDD,\ E-UTRAN}$ | ms | 12800 | | Derived according to the RSTD measurement requirements specified in section 9.3.10.1.3 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.10.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.10.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter “$T_{RSTDIntraFreqHD-FDD,E-UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.3.10.1.4.3-3. The value of the LPP time IE is set to $T_{RSTDIntraFreqHD-FDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 12950 ms. This is rounded up to the next allowed LPP value of 13 seconds.

#### 9.3.11.1.4.2 Test procedure

Same as in clause 9.3.7.1.4.2 but using condition CEModeB.

#### 9.3.11.1.4.3 Message contents

Same as in clause 9.3.10.1.4.3.

### 9.3.11.1.5 Test requirement

Same as in clause 9.3.10.1.5.

## 9.3.11.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2

### 9.3.11.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits enhanced coverage mode in an environment with fading propagation conditions.

### 9.3.11.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

### 9.3.11.2.3 Minimum conformance requirements

Same as in clause 9.3.10.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.5 and A.9.8.24.

9.3.11.2.4 Test description

9.3.11.2.4.1 Initial conditions

Same as in clause 9.3.10.2.4.1 but replacing Table 9.3.10.2.4.1-1 with Table 9.3.11.2.4.1-1.

**Table 9.3.11.2.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test1 | Test2 | Test3 | Test4 | |
| M-PDCCH parameters | | R.8 HD-FDD | | R.8 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.2. |
| *mPDCCH-startSF-UESS* | | 10 | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.21 FDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | 50 Note 4 | | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| PRS configuration Index $I_{PRS}$ Note 2 | | 151 | | 151 | | As defined in TS 36.211 [26] |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 4 | | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | μs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTDIntraFreqHD\text{-}FDD,E\text{-}UTRAN}$ Note 5 | ms | 12800 | | 5120 | | Derived according to the RSTD measurement requirements specified in section 9.3.10.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter “$T_{RSTDIntraFreqHD\text{-}FDD,E\text{-}UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.3.10.2.4.3-1. The value of the LPP time IE is set to $T_{RSTDIntraFreqHD\text{-}FDD,E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 12950 ms for Test 1 and 2 and 5270 ms for Test 3 and 4. This is rounded up to the next allowed LPP value of 13 or 6 seconds, respectively.

#### 9.3.11.2.4.2 Test procedure

Same as in clause 9.3.7.1.4.2 but using condition CEModeB.

#### 9.3.11.2.4.3 Message contents

Same as in clause 9.3.10.2.4.3.

### 9.3.11.2.5 Test requirement

Same as in clause 9.3.10.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.12 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B

### 9.3.12.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1

#### 9.3.12.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

#### 9.3.12.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

9.3.12.1.3 Minimum conformance requirements

Same as in clause 9.3.10.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.21 and A.9.8.25.

9.3.12.1.4 Test description

9.3.12.1.4.1 Initial conditions

Same as in clause 9.3.10.1.4.1 but replacing Table 9.3.10.1.4.1-1 with Table 9.3.12.1.4.1-1

**Table 9.3.12.1.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| MPDCCH | | R.16 TDD | | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 1 | | As specified in table 4.2-2 in TS 36.211 [26] and table 8.1.2.5.2-2. The same configuration in both cells. |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{\text{PRS}}$ Note 2 | | 154 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{\text{PRS}} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{\text{PRS}}$ Note 2 | | 6 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | |

| | | | | |
|---|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| $T_{RSTDIntraFreqTDD,E-UTRAN}$ | ms | 12800 | | Derived according to the RSTD measurement requirements specified in section 9.3.10.1.3 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.10.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.10.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter “$T_{RSTDIntraFreqTDD,E-UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.3.10.1.4.3-3. The value of the LPP time IE is set to $T_{RSTDIntraFreqTDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 12950 ms. This is rounded up to the next allowed LPP value of 13 seconds.

#### 9.3.12.1.4.2 Test procedure

Same as in clause 9.3.7.1.4.2 but using condition CEModeB.

#### 9.3.12.1.4.3 Message contents

Same as in clause 9.3.10.1.4.3.

#### 9.3.12.1.5 Test requirement

Same as in clause 9.3.10.1.5.

### 9.3.12.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2

#### 9.3.12.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

#### 9.3.12.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and CE Mode B.

#### 9.3.12.2.3 Minimum conformance requirements

Same as in clause 9.3.20.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.5 and A.9.8.25.

#### 9.3.12.2.4 Test description

##### 9.3.12.2.4.1 Initial conditions

Same as in clause 9.3.10.2.4.1 but replacing Table 9.3.10.2.4.1-1 with Table 9.3.12.2.4.1-1.

**Table 9.3.12.2.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|
| | | Test1 | Test2 | Test3 | Test4 | |
| M-PDCCH parameters | | R.16 TDD | | R.16 TDD | | As specified in TS 36.521-3 [25] clause A.7.3. |
| *mPDCCH-startSF-UESS* | | 10 | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OP.11 TDD | | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Reference cell | | Cell 1 | | | | |
| Neighbour cell | | Cell 2 | | | | |
| E-UTRA RF Channel Number | | 1 | | | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | | 10 | | |
| Special subframe configuration | | 6 | | | | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 1 | | | | As specified in table 4.2-2 in TS 36.211 [26] and table 8.1.2.5.2-2. The same configuration in both cells. |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | 50 Note 4 | | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| PRS configuration Index $I_{PRS}$ Note 2 | | 154 | | 154 | | As defined in TS 36.211 [26] |

| | | | | | | |
|---|---|---|---|---|---|---|
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 6 | | 4 | | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 1 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 3 | |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | 5 | 5 | |
| CP length Note 2 | | Normal | | | | |
| DRX | | OFF | | | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: 3 | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | | | The number of cells includes the reference cell |
| $T_{RSTDIntraFreqTDD,E-UTRAN}$ Note 5 | ms | 12800 | | 5120 | | Derived according to the RSTD measurement requirements specified in section 9.3.10.2.3 |

NOTE 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.3.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter "$T_{RSTDIntraFreqTDD,E-UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.3.10.2.4.3-3. The value of the LPP time IE is set to $T_{RSTDIntraFreqTDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 12950 ms for Test 1 and 2 and 5270 ms for Test 3 and 4. This is rounded up to the next allowed LPP value of 13 or 6 seconds, respectively.

#### 9.3.12.2.4.2 Test procedure

Same as in clause 9.3.7.1.4.2 but using condition CEModeB.

#### 9.3.12.2.4.3 Message contents

Same as in clause 9.3.10.1.4.3.

### 9.3.12.2.5 Test requirement

Same as in clause 9.3.10.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.3.13 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions

### 9.3.13.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.3.13.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA and dense PRS configuration or additional PRS configuration.

### 9.3.13.3 Minimum conformance requirements

Same as in clause 9.3.1.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.3.1, 8.16.2.3.1 and A.8.12.9.

### 9.3.13.4 Test description

#### 9.3.13.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4 using only the main Tx/Rx antenna of the UE.
2. The general test parameter settings are set up according to Table 9.3.13.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.3.13.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel. Cell 3 in the test is the Cell 4 defined in clause 4.7.1. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.5).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.

**Table 9.3.13.4.1-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |

| | | | |
|---|---|---|---|
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | PRS are transmitted in the centre RBs |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd6 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 150 | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 12 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 20 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.13.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |

| | | | |
|---|---|---|---|
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.60 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.13.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.13.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.13.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

**Table 9.3.13.4.1-2: DRX parameters**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.3.13.4.2 Test procedure

Same as in clause 9.3.1.1.4.2.

### 9.3.13.4.3 Message contents

Same as in clause 9.3.1.1.4.3 with the following exceptions:

**Table 9.3.13.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 3 | See clause 9.3.13.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.3.13.5 Test requirement

Table 9.3.1.1.5-1 and 9.3.1.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 3300ms. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 2955 ms. This is rounded up to the next allowed LPP value of 3 seconds. The RSTD measurement reporting delay in the test is derived from the following expression

$T_{RSTDIntraFreqFDD,Cat_M} = T_{PRS} \cdot (M-1) + \Delta$ +$T_{MIB}$, where $T_{PRS}$=320ms, M=8, $\Delta = T_{PRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ and $n$ =16 are the parameters specified in clause 9.3.13.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.13.4.1-1. This gives the total RSTD reporting delay of 2880 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.3.14 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions

### 9.3.14.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.3.14.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA and dense PRS configuration or additional PRS configuration.

### 9.3.14.3 Minimum conformance requirements

Same as in clause 9.3.2.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.3.3, 8.16.2.3.3 and A.8.12.10.

### 9.3.14.4 Test description

#### 9.3.14.4.1 Initial conditions

Same as in clause 9.3.13.4.1 but replacing Table 9.3.13.4.1-1 with Table 9.3.14.4.1-1

**Table 9.3.14.4.1-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.6 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | PRS are transmitted in the centre RBs |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd6 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 150 | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 12 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 20 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.3.14.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |

| | | | |
|---|---|---|---|
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.60 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.14.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.14.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.14.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

**Table 9.3.14.4.1-2: DRX parameters**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.3.14.4.2 Test procedure

Same as in clause 9.3.1.1.4.2.

### 9.3.14.4.3 Message contents

Same as in clause 9.3.13.4.3.

### 9.3.14.5 Test requirement

Same as in clause 9.3.13.5.

## 9.3.15 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions

### 9.3.15.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.3.15.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA and dense PRS configuration or additional PRS configuration.

### 9.3.15.3 Minimum conformance requirements

Same as in clause 9.3.2.1.3

The intra-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.3.3.1.3-1.

**Table 9.3.3.1.3-1: TDD uplink-downlink subframe configurations applicable for TDD intra-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.3.2, 8.16.2.3.2 and A.8.12.11.

### 9.3.15.4 Test description

#### 9.3.15.4.1 Initial conditions

Same as in clause 9.3.13.4.1 but replacing Table 9.3.13.4.1-1 with Table 9.3.15.4.1-1

**Table 9.3.15.4.1-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.14 TDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | PRS are transmitted in the center RBs |
| PRS configuration index $I_{PRS}$ Note 2 | | 304 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |

| | | | |
|---|---|---|---|
| Gap pattern Id | | rstd6 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 143 | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{\mathrm{PRS}}$ Note 2 | | 12 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 20 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [16], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [16], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| DRX | | ON | DRX parameters are further specified in Table 9.3.15.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.60 | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.15.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.15.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.15.4.1. |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. |

**Table 9.3.15.4.1-2: DRX parameters**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.3.15.4.2 Test procedure

Same as in clause 9.3.1.1.4.2.

### 9.3.15.4.3 Message contents

Same as in clause 9.3.13.4.3.

## 9.3.15.5 Test requirement

Table 9.3.3.1.5-1 and 9.3.3.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 3300ms. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where $\Delta T = N_{rep} \times TTI_{DCCH} = N_{rep} \times 75$ ms, giving a value of 2955 ms. This is rounded up to the next allowed LPP value of 3 seconds. The RSTD measurement reporting delay in the test is derived from the following expression

$T_{RSTDIntraFreqFDD,Cat_M} = T_{PRS} \cdot (M-1) + \Delta + T_{MIB}$, where $T_{PRS}$=320ms, M=8, $\Delta = T_{PRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ and $n$ =16 are the parameters specified in clause 9.3.13.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.13.4.1-1. This gives the total RSTD reporting delay of 2880 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 9.3.16 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions

## 9.3.16.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

## 9.3.16.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA and dense PRS configuration or additional PRS configuration and CE Mode B. Test 2 is applicable only to UE Category M2.

## 9.3.16.3 Minimum conformance requirements

Same as in clause 9.3.4.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.3.1, 8.16.3.3.1 and A.8.12.12.

## 9.3.16.4 Test description

### 9.3.16.4.1 Initial conditions

Same as in clause 9.3.13.4.1 but replacing Table 9.3.13.4.1-1 with Table 9.3.16.4.1-1

**Table 9.3.16.4.1-1: General test parameters**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.18 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | 24 | PRS are transmitted in the centre RBs |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd12 | rstd2 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 150 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 30 | 8 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |

| | | | | |
|---|---|---|---|---|
| | | 40 | 10 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.16.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.60 | | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.16.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

**Table 9.3.16.4.1-2: DRX parameters**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

#### 9.3.16.4.2 Test procedure

Same as in clause 9.3.13.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.16.4.1-1 as appropriate

#### 9.3.16.4.3 Message contents

Same as in clause 9.3.13.4.3

### 9.3.16.5 Test requirement

Table 9.3.1.1.5-1 and 9.3.1.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 3300ms. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 2955 ms. This is rounded up to the next allowed LPP value of 3 seconds. The RSTD measurement reporting delay in the test is derived from the following expression

$T_{RSTDIntraFreqFDD,Cat_M} = T_{PRS} \cdot (M-1) + \Delta$ +$T_{MIB}$, where $T_{PRS}$=320ms, M=8, $\Delta = T_{PRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ and $n$ =16 are the parameters specified in clause 9.3.16.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.16.4.1-1. This gives the total RSTD reporting delay of 2880 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.3.17 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions

### 9.3.17.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.3.17.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA and dense PRS configuration or additional PRS configuration and CE Mode B. Test 2 is applicable only to UE Category M2.

### 9.3.17.3 Minimum conformance requirements

Same as in clause 9.3.4.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.3.3, 8.16.3.3.3 and A.8.12.13.

### 9.3.17.4 Test description

#### 9.3.17.4.1 Initial conditions

Same as in clause 9.3.13.4.1 but replacing Table 9.3.13.4.1-1 with Table 9.3.17.4.1-1

**Table 9.3.17.4.1-1: General test parameters**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |

| | | | | |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.8 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | 24 | PRS are transmitted in the centre RBs |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd12 | rstd2 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 150 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 30 | 8 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 40 | 10 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.17.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |

| | | | |
|---|---|---|---|
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.60 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.17.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.17.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.17.4.1.<br>Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. | | | |

**Table 9.3.17.4.1-2: DRX parameters**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.3.17.4.2 Test procedure

Same as in clause 9.3.13.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.17.4.1-1 as appropriate

### 9.3.17.4.3 Message contents

Same as in clause 9.3.13.4.3

## 9.3.17.5 Test requirement

Same as in clause 9.3.16.5.

# 9.3.18 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions

## 9.3.18.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.3.18.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA and dense PRS configuration or additional PRS configuration and CE Mode B. Test 2 is applicable only to UE Category M2.

### 9.3.18.3 Minimum conformance requirements

Same as in clause 9.3.4.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.3.2, 8.16.3.3.2 and A.8.12.14.

### 9.3.18.4 Test description

#### 9.3.18.4.1 Initial conditions

Same as in clause 9.3.13.4.1 but replacing Table 9.3.13.4.1-1 with Table 9.3.18.4.1-1

**Table 9.3.18.4.1-1: General test parameters**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 TDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 6 | 24 | PRS are transmitted in the centre RBs |
| PRS configuration index $I_{PRS}$ Note 2 | | 624 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd15 | rstd10 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 143 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 30 | 8 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |

| | | | | |
|---|---|---|---|---|
| | | 40 | 10 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 2 | 1 | As specified in TS 36.211 [16], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [16], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.18.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.18.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.18.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

**Table 9.3.18.4.1-2: DRX parameters**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |

| | | |
|---|---|---|
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.3.18.4.2 Test procedure

Same as in clause 9.3.13.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.3.18.4.1-1 as appropriate

### 9.3.18.4.3 Message contents

Same as in clause 9.3.1.1.4.3 with the following exceptions:

**Table 9.3.18.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See clause 9.3.18.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.18.5 Test requirement

Table 9.3.3.1.5-1 and 9.3.3.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 6300ms. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 5285 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression

$T_{RSTD\ IntraFreqFDD,Cat_M} = T_{PRS} \cdot (M-1) + \Delta$ +$T_{MIB}$, where $T_{PRS}$=640ms, M=8, $\Delta = T_{PRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ and $n$ =16 are the parameters specified in clause 9.3.13.3, Table 9.3.1.1.3-1 and Note 4 of Table 9.3.13.4.1-1. This gives the total RSTD reporting delay of 5210 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 9.4 RSTD Inter-Frequency Measurements for UE Category M1/M2

## 9.4.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A

### 9.4.1.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1

#### 9.4.1.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

#### 9.4.1.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

#### 9.4.1.1.3 Minimum conformance requirements

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data are provided, the UE shall be able to detect and measure inter-frequency RSTD, specified in 3GPP TS 36.214 [6], for at least *n*=16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{\text{RSTD InterFreqFDD, Cat_M}}$ ms as given below (see also Figure 9.4.1.1.3-1):

$$T_{\text{RSTD InterFreqFDD, Cat_M}} = T_{PRS} \cdot (M-1) + \Delta + T_{\text{MIB}} \text{ ms},$$

where

$T_{\text{RSTD InterFreqFDD, Cat_M}}$ is the total time for detecting and measuring at least *n* cells,

$T_{\text{PRS}}$ is the cell-specific positioning subframe configuration period as defined in 3GPP TS 36.211 [26]; if $T_{\text{PRS}}$<MGRP, $T_{\text{PRS}}$ equals to MGRP; MGRP is the Measurement Gap Repetition Period as defined in 3GPP TS 36.133 [23] section 8.1.2.1,

$M$ is the number of PRS positioning occasions as defined in Table 9.4.1.1.3-1, , where downlink positioning subframes defined in TS 36.211 [16],

$\Delta = T_{PRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single PRS positioning occasion which includes the sampling time and the processing time,

$N_{\text{PRS}}$ is the cell-specific number of PRS subframes within a PRS occasion as defined in TS 36.355 [4],

$N_{\text{actual_PRS}}$ is the cell-specific number of PRS subframes within a PRS occasion and can be measured by UE within MGL; if MGRP>= $N_{\text{PRS}}$>(MGL-2ms), $N_{\text{actual_PRS}}$ equals to (MGL-2ms); if $N_{\text{PRS}}$>MGRP, $N_{\text{actual_PRS}}$ equals to $(MGL-2) \cdot \left\lfloor N_{\text{PRS}} / MGRP \right\rfloor$;

$N_{PRS_total}$ is the minimum number of PRS subframes per cell measurement as specified in TS 36.133 [23] Section 9.1.21.17.

$T_{\text{PRS}}$ $N_{\text{PRS}}$, $N_{\text{actual_PRS}}$ and $N_{PRS_total}$ are the parameters of the same cell, for which $T_{PRS} \cdot \left\lceil \frac{N_{PRS_total}}{N_{actual_PRS}} \right\rceil$ is the largest among all the measured cells.

$T_{MIB}$ is the time required for acquiring the MIB information of the target cell. $T_{MIB}$ = 0 if the SFN of at least one cell in OTDOA assistance data is known to the UE.

**Table 9.4.1.1.3-1: Number of PRS positioning occasions within $T_{\text{RSTD InterFreqFDD, Cat_M}}$**

| Positioning subframe configuration period $T_{\text{PRS}}$ | Number of PRS positioning occasions $M$ |
|---|---|
| | f1 and f2 Note 1 |
| 160 ms | $32 \cdot \left\lceil N_{PRS_Total} / N_{actual_PRS} \right\rceil$ |
| >160 ms | $16 \cdot \left\lceil N_{PRS_Total} / N_{actual_PRS} \right\rceil$ |
| Note 1: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving FDD carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD InterFreqFDD, Cat_M}}$ provided:

$\left(\text{PRS}\,\hat{\text{E}}_{\text{s}} / \text{Iot}\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\text{PRS}\,\hat{\text{E}}_{\text{s}} / \text{Iot}\right)_{i}$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{PRS}\,\hat{\text{E}}_{\text{s}} / \text{Iot}\right)_{ref}$ and $\left(\text{PRS}\,\hat{\text{E}}_{\text{s}} / \text{Iot}\right)_{i}$ conditions apply for all subframes of at least $L = \frac{M}{2}$ PRS positioning occasions,

PRP $1,2|_{\text{dBm}}$ according to clause E.3.1 for a corresponding Band.

$\text{PRS}\,\hat{\text{E}}_{\text{s}} / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry PRS.

The time $T_{\text{RSTD InterFreqFDD, Cat_M}}$ starts from the first subframe of the PRS positioning occasion closest in time after both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in 3GPP TS 36.355 [4], are delivered to the physical layer of the UE.

This requirement assumes that the measurement report is not delayed by other LPP signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is $N_{rep}$x $\text{TTI}_{\text{DCCH}}$, where $N_{rep}$ is the maximum number of PUSCH repetitions configured for the UE, otherwise uncertainty is defined as 2 x $\text{TTI}_{\text{DCCH}}$. This measurement reporting delay excludes any delay caused by no UL resources for UE to send the measurement report.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.4.1 and A.8.13.3.

### 9.4.1.1.4 Test description

#### 9.4.1.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4 using only the main Tx/Rx antenna of the UE.
2. The general test parameter settings are set up according to Table 9.4.1.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.4.1.1.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. Cell 1 is on FDD RF channel 1. Cell 2 and Cell 3 are on a FDD RF channel 2. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.5).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.

**Table 9.4.1.1.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap pattern Id | | 0 | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |

| | | | |
|---|---|---|---|
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 7.68 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 7.68 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

**Table 9.4.1.1.4.1-2: DRX parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |

| | | |
|---|---|---|
| shortDRX | Disable | |

#### 9.4.1.1.4.2 Test procedure

The test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 9.4.1.1.4.1-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. Cell 2 is active until the end of T3, and Cell 3 is active until the end of T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, while Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.4.1.1.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF-CE according to 3GPP TS 36.508 [18] clause 7.2A.3AA.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.4.1.1.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).
4. T1 starts.
5. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration and the measurement gap configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.
6. The UE shall transmit RRCConnectionReconfigurationComplete message.

6a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the first 7 elements of the sequence, and the position of neighbour Cell 3 is randomly selected to be in the last 8 elements of the sequence, as described in 3GPP TS 37.571-5 [20], clause 7.2.5. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 9.4.1.1.5-2.
10. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.4.1.1.5-2.
11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 9.4.1.1.5. The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. Repeat steps 2-12 until the confidence level according to Annex D is achieved. For each iteration, at step 7 change the random position of the Cells 2 and 3 in the *OTDOA-NeighbourCellInfoList.*

#### 9.4.1.1.4.3 Message contents

Same as in clause 9.3.1.1.4.3 with the following exceptions.

**Table 9.4.1.1.4.3-1: *MeasGapConfig-GP1*: FDD-FDD inter-frequency RSTD Measurement Reporting Delay**

| Derivation Path: TS 36.508 [18] clause 4.6.6, Table 4.6.6-1A: MeasGapConfig-GP1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig-GP1 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 9 | TGRP = 40 ms | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.1.1.4.3-2: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 16 | See clause 9.4.1.1.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 9.4.1.1.5 Test requirement

Table 9.4.1.1.5-1 and 9.4.1.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.4.1.1.5-1: Cell-specific test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 2 |

| | | | | |
|---|---|---|---|---|
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.4.1.1.5-2: Cell-specific test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.6 FDD | | OP.6 FDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -1 | -Infinity | -Infinity | -7 | -8 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1 | -Infinity | -Infinity | -7 | -8 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.68 | -70.22 | -70.11 | -67.08 | -70.11 | -67.08 |
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -106 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -96 | -105 | -105 | -109 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -11 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 16.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 15510 ms. This is rounded up to the next allowed LPP value of 16 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =48 and $n$ =16 are the parameters specified in clause 9.4.1.1.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.1.1.4.1-1. This gives the total RSTD reporting delay of 15360 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.4.1.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2

### 9.4.1.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.4.1.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

#### 9.4.1.2.3 Minimum conformance requirements

Same as in clause 9.4.1.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.2.4.1 and A.8.13.3.

#### 9.4.1.2.4 Test description

##### 9.4.1.2.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.1.2.4.1-1.

**Table 9.4.1.2.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Gap pattern Id | | 0 | | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |

| | | | | |
|---|---|---|---|---|
| prs-SubframeOffset | | 10 | | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 7.68 | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 7.68 | 5.12 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

### 9.4.1.2.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.4.1.2.4.1-1 as appropriate

### 9.4.1.2.4.3 Message contents

Same as in clause 9.4.1.1.4.3 with the following exceptions:

**Table 9.4.2.1.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 16 | See clause 9.4.1.1.5 | Test 1 |
| time | 11 | See clause 9.4.1.1.5 | Test 2 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 9.4.1.2.5 Test requirement

The primary level settings including test tolerances for the test are defined in clause 9.4.1.1.5.

For Test 1, the response time is defined in clause 9.4.1.1.5.

For Test 2, the response time including test tolerance is 11.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where $\Delta T = N_{rep} \times TTI_{DCCH} = N_{rep} \times 75$ ms, giving a value of 10390 ms. This is rounded up to the next allowed LPP value of 11 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =32 and $n$ =16 are the parameters specified in clause 9.4.1.1.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.1.2.4.1-1. This gives the total RSTD reporting delay of 10240 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A

### 9.4.2.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1

#### 9.4.2.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements normal coverage mode in an environment with fading propagation conditions.

#### 9.4.2.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

#### 9.4.2.1.3 Minimum conformance requirements

Same as in clause 9.4.1.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.4.3 and A.8.13.4.

#### 9.4.2.1.4 Test description

##### 9.4.2.1.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.2.1.4.1-1

**Table 9.4.2.1.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.6 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap pattern Id | | 0 | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |

| | | | |
|---|---|---|---|
| DRX | | ON | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 7.68 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 7.68 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.4.2.1.4.2 Test procedure

Same as in clause 9.4.1.1.4.2.

#### 9.4.2.1.4.3 Message contents

Same as in clause 9.4.1.1.4.3

### 9.4.2.1.5 Test requirement

Same as in clause 9.4.1.1.5.

## 9.4.2.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2

### 9.4.2.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements normal coverage mode in an environment with fading propagation conditions.

### 9.4.2.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.4.2.2.3 Minimum conformance requirements

Same as in clause 9.4.1.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.2.4.3 and A.8.13.4.

### 9.4.2.2.4 Test description

#### 9.4.2.2.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.2.2.4.1-1.

**Table 9.4.2.2.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.6 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap pattern Id | | 0 | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |

| | | | | |
|---|---|---|---|---|
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| prs-SubframeOffset | | 10 | | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 7.68 | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 7.68 | 5.12 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.4.2.2.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.4.2.2.4.1-1 as appropriate.

#### 9.4.2.2.4.3 Message contents

Same as in clause 9.4.1.2.4.3.

### 9.4.2.2.5 Test requirement

Same as in clause 9.4.1.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.3 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A

### 9.4.3.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1

#### 9.4.3.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

#### 9.4.3.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

#### 9.4.3.1.3 Minimum conformance requirements

Same as in clause 9.4.1.1.3.

The inter-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.4.3.1.3-1.

**Table 9.4.3.1.3-1: TDD uplink-downlink subframe configurations applicable for TDD inter-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.4.2 and A.8.13.5.

#### 9.4.3.1.4 Test description

##### 9.4.3.1.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.3.1.4.1-1

**Table 9.4.2.1.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.14 TDD | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap pattern Id | | 0 | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{\mathrm{PRS}}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{\mathrm{PRS}} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{\mathrm{PRS}}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |

| | | | |
|---|---|---|---|
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 7.68 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 7.68 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.4.3.1.4.2 Test procedure

Same as in clause 9.4.1.1.4.2.

#### 9.4.3.1.4.3 Message contents

Same as in clause 9.4.1.1.4.3

### 9.4.3.1.5 Test requirement

Table 9.4.3.1.5-1 and 9.4.3.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.4.3.1.5-1: Cell-specific test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 2 |
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | N/A | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.4.3.1.5-2: Cell-specific test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -1 | -Infinity | -Infinity | -7 | -7 | -Infinity |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -1.79 | -Infinity | -Infinity | -7 | -9.54 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.55 | -67.08 | -69.55 | -67.08 | -69.55 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -99 | -Infinity | -Infinity | -102 | -105 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -93 | -105 | -105 | -108 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -10 | -10 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 16.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 15510 ms. This is rounded up to the next allowed LPP value of 16 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =48 and $n$ =16 are the parameters specified in clause 9.4.1.1.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.1.1.4.1-1. This gives the total RSTD reporting delay of 15360 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.4.3.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2

### 9.4.3.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.4.3.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.4.3.2.3 Minimum conformance requirements

Same as in clause 9.4.3.1.3.

The intra-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.4.3.2.3-1.

**Table 9.4.3.2.3-1: TDD uplink-downlink subframe configurations applicable for TDD intra-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| 24 | 0, 1, 2, 3, 4, 5 and 6 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.16.2.4.2 and A.8.13.5.

### 9.4.3.2.4 Test description

#### 9.4.3.2.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.3.2.4.1-1.

**Table 9.4.3.2.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.14 TDD | | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Gap pattern Id | | 0 | | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |

| | | | | |
|---|---|---|---|---|
| TDD uplink-downlink configuration | | 1 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| prs-SubframeOffset | | 10 | | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 7.68 | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 7.68 | 5.12 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.4.3.2.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.4.2.2.4.1-1 as appropriate.

#### 9.4.3.2.4.3 Message contents

Same as in clause 9.4.1.2.4.3.

### 9.4.3.2.5 Test requirement

The primary level settings including test tolerances for the test are defined in clause 9.4.3.1.5.

For Test 1, the response time is defined in clause 9.4.3.1.5.

For Test 2, the response time including test tolerance is 11.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 10390 ms. This is rounded up to the next allowed LPP value of 11 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =32 and $n$ =16 are the parameters specified in clause 9.4.1.1.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.3.2.4.1-1. This gives the total RSTD reporting delay of 10240 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.4 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B

### 9.4.4.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1

#### 9.4.4.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in enhanced coverage mode in an environment with fading propagation conditions.

#### 9.4.4.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

#### 9.4.4.1.3 Minimum conformance requirements

Same as 9.4.1.1.3 with the following exceptions:

The conditions under which the UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD IntraFreqFDD, Cat_M}}$ are changed:

$\left(\text{PRS}\,\hat{E}_s/\text{Iot}\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(\text{PRS}\,\hat{E}_s/\text{Iot}\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{PRS}\,\hat{E}_s/\text{Iot}\right)_{ref}$ and $\left(\text{PRS}\,\hat{E}_s/\text{Iot}\right)_i$ conditions apply for all subframes of at least $L=\frac{M}{2}$ PRS positioning occasions,

$PRP\ 1,2|_{dBm}$ according to clause E.3.1 for a corresponding Band.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.7.1 and A.8.13.6.

9.4.4.1.4 Test description

9.4.4.1.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.4.1.4.1-1.

**Table 9.4.4.1.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap pattern Id | | 0 | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 20.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 20.48 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.1.1.4.1.<br>Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. | | | |

### 9.4.4.1.4.2 Test procedure

Same as in clause 9.4.1.4.2 but using condition CEModeB.

### 9.4.4.1.4.3 Message contents

Same as in clause 9.4.1.1.4.3 with the following exceptions:

**Table 9.4.4.1.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.4.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 42 | See clause 9.4.4.1.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

9.4.4.1.5 Test requirement

Table 9.4.4.1.5-1 and 9.4.4.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.4.4.1.5-1: Cell-specific test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 2 |
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.4.4.1.5-2: Cell-specific test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | | OP.6 FDD | | OP.6 FDD | N/A |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| PRS_RA | dB | 0 | N/A | N/A | 0 | 0 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -12 | -Infinity | -Infinity | -13 | -14 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -12 | -Infinity | -Infinity | -13 | -14 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -70.17 | -67.13 | -70.19 | -70.18 | -70.19 | -70.18 |
| PRP Note 4 | dBm/ 15 kHz | -110 | -Infinity | -Infinity | -108 | -112 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -110 | -110 | -114 | -108 | -112 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | -9 | -9 | -14 | -13 | -14 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 42.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 41110 ms. This is rounded up to the next allowed LPP value of 42 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =128 and $n$ =16 are the parameters specified in clause 9.4.1.1.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.4.1.4.1-1. This gives the total RSTD reporting delay of 40960 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.4.4.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2

### 9.4.4.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in enhanced coverage mode in an environment with fading propagation conditions.

### 9.4.4.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

### 9.4.4.2.3 Minimum conformance requirements

Same as in clause 9.4.4.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.3.2.1 and A.8.13.6.

### 9.4.4.2.4 Test description

#### 9.4.4.2.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.4.2.4.1-1.

**Table 9.4.4.2.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Gap pattern Id | | 0 | | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | | As specified in TS 36.331 [22], Clause 6.3.5 |

| | | | | |
|---|---|---|---|---|
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 20.48 | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 20.48 | 5.12 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.4.4.2.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but using condition CEModeB and with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.4.4.2.4.1-1 as appropriate

#### 9.4.4.2.4.3 Message contents

Same as in clause 9.4.1.1.4.3 with the following exceptions:

**Table 9.4.4.2.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.4.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 42 | See clause 9.4.4.1.5 | Test 1 |
| time | 11 | See clause 9.4.4.1.5 | Test 2 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 9.4.4.2.5 Test requirement

The primary level settings including test tolerances for the test are defined in clause 9.4.4.1.5.

For Test 1, the response time is defined in clause 9.4.4.1.5.

For Test 2, the response time including test tolerance is 11.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 10390 ms. This is rounded up to the next allowed LPP value of 11 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =32 and $n$ =16 are the parameters specified in clause 9.4.1.1.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.4.2.4.1-1. This gives the total RSTD reporting delay of 10240 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.5 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B

### 9.4.5.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1

#### 9.4.5.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in an environment with fading propagation conditions.

#### 9.4.5.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

#### 9.4.5.1.3 Minimum conformance requirements

Same as in clause 9.4.4.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.7.3 and A.8.13.7.

#### 9.4.5.1.4 Test description

##### 9.4.5.1.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.5.1.4.1-1

**Table 9.4.5.1.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.8 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap pattern Id | | 0 | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |

| | | | |
|---|---|---|---|
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 20.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 20.48 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.4.5.1.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but using condition CEModeB.

#### 9.4.5.1.4.3 Message contents

Same as in clause 9.4.4.1.4.3

#### 9.4.5.1.5 Test requirement

Same as in clause 9.4.4.1.5.

### 9.4.5.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2

#### 9.4.5.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in an environment with fading propagation conditions.

#### 9.4.5.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

#### 9.4.5.2.3 Minimum conformance requirements

Same as in clause 9.4.4.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.16.3.2.3 and A.8.13.7.

#### 9.4.5.2.4 Test description

##### 9.4.5.2.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.5.2.4.1-1.

**Table 9.4.5.2.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.8 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Gap pattern Id | | 0 | | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | | As specified in TS 36.331 [22], Clause 6.3.5 |

| | | | | |
|---|---|---|---|---|
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 311 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 20.48 | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 20.48 | 5.12 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

#### 9.4.5.2.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but using condition CEModeB and with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.4.2.2.4.1-1 as appropriate.

#### 9.4.5.2.4.3 Message contents

Same as in clause 9.4.4.2.4.3.

### 9.4.5.2.5 Test requirement

Same as in clause 9.4.4.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.6 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B

### 9.4.6.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1

#### 9.4.6.1.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M1 meets the performance requirements in an environment with fading propagation conditions.

#### 9.4.6.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

#### 9.4.6.1.3 Minimum conformance requirements

Same as in clause 9.4.4.1.3.

The inter-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.4.6.1.3-1.

**Table 9.4.6.1.3-1: TDD uplink-downlink subframe configurations applicable for TDD inter-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.7.2 and A.8.13.8.

#### 9.4.6.1.4 Test description

##### 9.4.6.1.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.6.1.4.1-1

**Table 9.4.6.1.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 TDD | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap pattern Id | | 0 | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{\mathrm{PRS}}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{\mathrm{PRS}} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{\mathrm{PRS}}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |

| | | | |
|---|---|---|---|
| T2 | s | 20.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 20.48 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

### 9.4.6.1.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but using condition CEModeB.

### 9.4.6.1.4.3 Message contents

Same as in clause 9.4.4.1.4.3

### 9.4.6.1.5 Test requirement

Table 9.4.6.1.5-1 and 9.4.6.1.5-2 define the primary level settings including test tolerances for the test.

**Table 9.4.6.1.5-1: Cell-specific test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 2 |
| Correlation Matrix and Antenna Configuration | | 1x1 | 1x1 | 1x1 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |

| Propagation Condition | | ETU30 |
|---|---|---|

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 9.4.6.1.5-2: Cell-specific test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions during T2 and T3**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
| Correlation Matrix and Antenna Configuration | | 1x1 | | 1x1 | | 1x1 | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -3 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -12 | -Infinity | -Infinity | -13 | -14 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -12 | -Infinity | -Infinity | -13 | -14 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -70.17 | -67.13 | -70.19 | -70.18 | -70.19 | -70.18 |
| PRP Note 4 | dBm/ 15 kHz | -110 | -Infinity | -Infinity | -108 | -112 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -110 | -110 | -114 | -108 | -112 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | -9 | -9 | -14 | -13 | -14 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS. |
| Note 2: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes. |

The response time is defined in clause 9.4.4.1.5.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.4.6.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2

### 9.4.6.2.1 Test purpose

To verify that the RSTD measurement reporting delay for UE Category M2 meets the performance requirements in an environment with fading propagation conditions.

### 9.4.6.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

### 9.4.6.2.3 Minimum conformance requirements

Same as in clause 9.4.6.1.3.

The inter-frequency requirements in this clause shall apply for all TDD special subframe configurations specified in TS 36.211 [26] and for the TDD uplink-downlink configurations as specified in Table 9.4.6.2.3-1.

**Table 9.4.6.2.3-1: TDD uplink-downlink subframe configurations applicable for TDD inter-frequency requirements**

| PRS Transmission Bandwidth [RB] | Applicable TDD uplink-downlink configurations |
|---|---|
| 6 | 1, 2, 3, 4 and 5 |
| 24 | 0, 1, 2, 3, 4, 5 and 6 |
| Note: Uplink-downlink configurations are specified in Table 4.2-2 in TS 36.211 [26]. | |

The normative reference for this requirement is TS 36.133 [23] clause 8.16.3.2.2 and A.8.13.8.

### 9.4.6.2.4 Test description

#### 9.4.6.2.4.1 Initial conditions

Same as in clause 9.4.1.1.4.1 but replacing Table 9.4.1.1.4.1-1 with Table 9.4.6.2.4.1-1.

**Table 9.4.6.2.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement reporting delay under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 TDD | | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Gap pattern Id | | 0 | | As specified in TS 36.133 [23] Table 8.1.2.1-1. Applies for measurements on Cell 2 and Cell 3 |
| Gap offset | | 9 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 142,<br>Cell 2, Cell 3: 152 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.4.1.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |

| | | | | |
|---|---|---|---|---|
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 20.48 | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 20.48 | 5.12 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.1.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.1.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

### 9.4.6.2.4.2 Test procedure

Same as in clause 9.4.1.1.4.2 but using condition CEModeB and with the following additional step:

14. Repeat step 1-13 for each sub-test in Table 9.4.2.2.4.1-1 as appropriate.

### 9.4.6.2.4.3 Message contents

Same as in clause 9.4.4.2.4.3.

### 9.4.6.2.5 Test requirement

The primary level settings including test tolerances for the test are defined in clause 9.4.6.1.5.

For Test 1, the response time is defined in clause 9.4.6.1.5.

For Test 2, the response time including test tolerance is 11.3 s. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 10390 ms. This is rounded up to the next allowed LPP value of 11 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where $M$ =40 and $n$ =16 are the parameters specified in clause 9.4.1.1.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.6.2.4.1-1. This gives the total RSTD reporting delay of 10240 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

# 9.4.7 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A

## 9.4.7.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1

### 9.4.7.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

### 9.4.7.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.4.7.1.3 Minimum conformance requirements

The accuracy requirements in Table 9.4.7.1.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex E.3.1 for a corresponding Band

There are no measurement gaps overlapping with the PRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.4.7.1.3-1: RSTD inter-frequency measurement accuracy for CEModeA**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io [Note 4] range | | |
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* [Note 3] | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell and the measured neighbour cell *i* as defined in [24] | E-UTRA operating band groups [Note 5] | Minimum Io [Note 1] | Maximum Io |
| Ts [Note 2] | dB | RB | | | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±21 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 6 | ≥ 12 | ≥4 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD-M1_B | -120.5 | -50 |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.17 and A.9.8.26.

### 9.4.7.1.4 Test description

#### 9.4.7.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 using only the main Tx/Rx antenna of the UE.
2. The general test parameter settings are set up according to Table 9.4.7.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.4.7.1.4.3.
5. Cell 1 is the serving cell and OTDOA assistance data reference cell; Cell 2 is the neighbour cell. Cell 1 is on RF channel 1 and Cell 2 is on RF channel 2. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.2.5).
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:
   Test 1: -92 Ts (about -3 μs)
   Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.4.7.1.4-1 for each test.

**Table 9.4.7.1.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions.**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| MPDCCH | | R.16 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | One carrier frequency is used. |
| E-UTRA RF Channel Number | | Cell 1: 1<br>Cell 2: 2 | The two cells are on different frequencies. |

| | | | |
|---|---|---|---|
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| $T_{RSTD\ InterFreqFDD\text{-}FDD,\ E\text{-}UTRAN}$ | ms | 15360 | Derived according to the RSTD measurement requirements specified in section 9.4.7.1.3 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.7.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: 6. For all the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.7.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter "$T_{RSTD\ InterFreqFDD\text{-}FDD,\ E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.4.7.1.4.3-3. The value of the LPP time IE is set to $T_{RSTD\ InterFreqFDD\text{-}FDD,\ E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 15510 ms. This is rounded up to the next allowed LPP value of 16 seconds.

#### 9.4.7.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.4.7.1.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF-CE according to 3GPP TS 36.508 [18] clause 7.2A.3AA.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.4.7.1.5-1. Propagation conditions are set according to clause 4.7.2.1.
4. The SS shall transmit an RRCConnectionReconfiguration message with the measurement gap configuration.
5. The UE shall transmit RRCConnectionReconfigurationComplete message.
6. The SS shall transmit an LPP REQUEST CAPABILITIES message.
7. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
8. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
9. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. The SS shall check the *rstd* value for Cell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.4.7.1.5-2.
13. Repeat steps 2-12 until the confidence level according to Annex D is achieved.
14. Repeat step 1-13 for each sub-test in Table 9.4.7.1.5-1 as appropriate.

#### 9.4.7.1.4.3 Message contents

Same as in clause 9.3.7.1.4.3 with the following exceptions.

**Table 9.4.7.1.4.3-1: *MeasGapConfig-GP1*: FDD-FDD inter-frequency RSTD Measurement Accuracy**

| Derivation Path: TS 36.508 [18] clause 4.6.6, Table 4.6.6-1A: MeasGapConfig-GP1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig-GP1 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 9 | TGRP = 40 ms | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 9.4.7.1.4.3-2: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.7.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 16 | See Note 5 in Table 9.4.7.1.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

9.4.7.1.5 Test requirement

Table 9.4.7.1.5-1 defines the primary level settings including test tolerances for the test.

The RSTD FDD inter-frequency accuracy test shall meet the reported values in Table 9.4.7.1.5-2

**Table 9.4.7.1.5-1: Cell Specific Test Parameters for inter frequency RSTD Tests for E-UTRAN FDD**

| **Parameter** | **Unit** | **Test1** | | **Test2** Note4 | |
|---|---|---|---|---|---|
| | | **Cell1** | **Cell2** | **Cell1** | **Cell2** |
| E-UTRA RF Channel Number | | 1 | 2 | 1 | 2 |
| Gap offset | | 9 | N/A | 9 | N/A |
| Gap pattern | | #0 | N/A | #0 | N/A |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | OP.6 FDD | OP.21 FDD | OP.6 FDD |
| PRS configuration Index $I_{\mathrm{PRS}}$, as defined in TS 36.211 | | 142 | 152 | 142 | 152 |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| MPDCCH_RA | | | | | |
| MPDCCH_RB | | | | | |
| OCNG_RA Note1 | | | | | |
| OCNG_RB Note1 | | | | | |
| PRS_RA | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 | -98 | -98 |
| PRS $\hat{\mathrm{E}}_s/N_{oc}$ | dB | -1 | -11 | -1 | -11 |
| PRS $\hat{\mathrm{E}}_s/\mathrm{I}_{ot}$ Note3 | dB | -1 | -11 | -1 | -11 |

| | | | | | |
|---|---|---|---|---|---|
| Io Note3 | dBm/9 MHz | -69.68 | -70.16 | -69.68 | -70.16 |
| PRP Note3 | dBm/15kHz | -99 | -109 | -99 | -109 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -1 | -11 | -1 | -11 |
| RSRP Note 3 | dBm/15kHz | -99 | -109 | -99 | -109 |
| Propagation condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS

Note 4: Test2 is not used for test 9.4.7.1

**Table 9.4.7.1.5-2: RSTD FDD inter-frequency accuracy requirements for the reported values**

| | Test 1 |
|---|---|
| Lowest reported value | RSTD_6241 |
| Highest reported value | RSTD_6287 |

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.7.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2

### 9.4.7.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

### 9.4.7.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.4.7.2.3 Minimum conformance requirements

The accuracy requirements in Table 9.4.7.2.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex E.2 for a corresponding Band

There are no measurement gaps overlapping with the PRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.4.7.2.3-1: RSTD inter-frequency measurement accuracy for CEModeA**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel | Minimum number of available measurement subframes among the reference cell | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell | Io Note 4 range | | |
| | | | | | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |

| | | bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* Note 3 | and the measured neighbour cell *i* | and the measured neighbour cell *i* as defined in [24] | | | |
|---|---|---|---|---|---|---|---|
| **Ts Note 2** | **dB** | **RB** | | | | **dBm/15kHz** | **dBm/BW$_{Channel}$** |
| ±21 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 6 | ≥ 12 | ≥ 4 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD-M1_B | -120.5 | -50 |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |
| ±10 | (PRS Ês/Iot)$_{ref}$ ≥-6dB and (PRS Ês/Iot)$_i$ ≥-13dB | ≥ 24 | ≥ 4 | ≥ 2 | Note 6 | Note 6 | Note 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 6 RB.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.1 and A.9.8.26.

### 9.4.7.2.4 Test description

#### 9.4.7.2.4.1 Initial conditions

Same as in clause 9.4.7.1.4.1 adding Test 2 and replacing Table 9.4.7.1.4.1-1 with Table 9.4.7.2.4.1-1.

The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:

Test 1: -92 Ts (about -3 μs)
Test 2: 92 Ts (about 3 μs)

Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.4.7.2.4-1 for each test.

**Table 9.4.7.2.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test1 | Test2 | |
| M-PDCCH parameters | | R.16 FDD | R.16 FDD | As specified in TS 36.521-3 [25] clause A.7.1. |
| *mPDCCH-startSF-UESS* | | 10 | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | |
| E-UTRA RF Channel Number | | Cell 1: 1<br>Cell 2: 2 | | The two cells are on different frequencies. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | 50 Note 4 | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The number of cells includes the reference cell |
| $T_{RSTD\ InterFreqFDD\text{-}FDD,\ E\text{-}UTRAN}$ Note 5 | ms | 15360 | 10240 | Derived according to the RSTD measurement requirements specified in section 9.4.7.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.
NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: 6. For all the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.
NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.
NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.
NOTE 5: The parameter “$T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.7.2.4.3-1. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}} + \Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 15510 ms for Test 1 and 10390 ms for Test 2. This is rounded up to the next allowed LPP value of 16 and 11 seconds, respectively.

### 9.4.7.2.4.2 Test procedure

Same as in clause 9.4.7.1.4.2.

### 9.4.7.2.4.3 Message contents

Same as in clause 9.4.7.1.4.3 with the following exceptions:

**Table 9.4.7.2.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.4.7.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 16 | See Note 5 in Table 9.4.7.2.4.1-1 | Test 1 |
| time | 11 | See Note 5 in Table 9.4.7.2.4.1-1 | Test 2 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.4.7.2.5 Test requirement

Same as in clause 9.4.7.1.5 adding Test 2 and replacing Table 9.4.7.1.5-2 with Table 9.4.7.2.5-1:

**Table 9.4.7.2.5-1: RSTD FDD inter-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6252 | RSTD_6435 |
| Highest reported value | RSTD_6276 | RSTD_6459 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.8 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A

### 9.4.8.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1

#### 9.4.8.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits normal coverage mode in an environment with fading propagation conditions.

#### 9.4.8.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

#### 9.4.8.1.3 Minimum conformance requirements

Same as in clause 9.4.7.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.17 and A.9.8.27.

#### 9.4.8.1.4 Test description

##### 9.4.8.1.4.1 Initial conditions

Same as in clause 9.4.7.1.4.1 but replacing Table 9.4.7.1.4.1-1 with Table 9.4.8.1.4.1-1

**Table 9.4.8.1.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| MPDCCH | | R.6 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | One carrier frequency is used. |
| E-UTRA RF Channel Number | | Cell 1: 1<br>Cell 2: 2 | The two cells are on different frequencies. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |

| | | | |
|---|---|---|---|
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| $T_{RSTD\ InterFreqHD\text{-}FDD,\ E\text{-}UTRAN}$ | ms | 15360 | Derived according to the RSTD measurement requirements specified in section 9.4.7.1.3 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.7.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: 6. For all the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.7.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter "$T_{RSTD\ InterFreqHD\text{-}FDD,\ E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.4.7.1.4.3-3. The value of the LPP time IE is set to $T_{RSTD\ InterFreqHD\text{-}FDD,\ E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 15510 ms. This is rounded up to the next allowed LPP value of 16 seconds.

### 9.4.8.1.4.2 Test procedure

Same as in clause 9.4.7.1.4.2.

#### 9.4.8.1.4.3 Message contents

Same as in clause 9.4.7.1.4.3

#### 9.4.8.1.5 Test requirement

Same as in clause 9.4.7.1.5.

### 9.4.8.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2

#### 9.4.8.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits normal coverage mode in an environment with fading propagation conditions.

#### 9.4.8.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

#### 9.4.8.2.3 Minimum conformance requirements

Same as in clause 9.4.7.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.1 and A.9.8.27.

#### 9.4.8.2.4 Test description

#### 9.4.8.2.4.1 Initial conditions

Same as in clause 9.4.7.2.4.1 but replacing Table 9.4.7.2.4.1-1 with Table 9.4.8.2.4.1-1.

**Table 9.4.8.2.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test1 | Test2 | |
| M-PDCCH parameters | | R.6 HD-FDD | R.6 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2. |
| *mPDCCH-startSF-UESS* | | 10 | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | |
| E-UTRA RF Channel Number | | Cell 1: 1<br>Cell 2: 2 | | The two cells are on different frequencies. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | 50 Note 4 | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |

| | | | | |
|---|---|---|---|---|
| prs-SubframeOffset | | 10 | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The number of cells includes the reference cell |
| $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$ Note 5 | ms | 15360 | 10240 | Derived according to the RSTD measurement requirements specified in section 9.4.7.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: 6. For all the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter “$T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.7.2.4.3-1. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}} + \Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 15510 ms for Test 1 and 10390 ms for Test 2. This is rounded up to the next allowed LPP value of 16 and 11 seconds, respectively.

### 9.4.8.2.4.2 Test procedure

Same as in clause 9.4.7.1.4.2.

### 9.4.8.2.4.3 Message contents

Same as in clause 9.4.7.1.4.3.

### 9.4.8.2.5 Test requirement

Same as in clause 9.4.7.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

# 9.4.9 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A

## 9.4.9.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1

### 9.4.9.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

### 9.4.9.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.4.9.1.3 Minimum conformance requirements

Same as in clause 9.4.7.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.17 and A.9.8.28.

### 9.4.9.1.4 Test description

#### 9.4.9.1.4.1 Initial conditions

Same as in clause 9.4.7.1.4.1 but replacing Table 9.4.7.1.4.1-1 with Table 9.4.9.1.4.1-1

**Table 9.4.9.1.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| MPDCCH | | R.14 TDD | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | One carrier frequency is used. |
| E-UTRA RF Channel Number | | Cell 1: 1<br>Cell 2: 2 | The two cells are on different frequencies. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |

| | | | |
|---|---|---|---|
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| $T_{\text{RSTD InterFreqTDD, E-UTRAN}}$ | ms | 15360 | Derived according to the RSTD measurement requirements specified in section 9.4.7.1.3 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.7.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: 6. For all the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.7.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter “$T_{\text{RSTD InterFreqTDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.7.1.4.3-3. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqTDD, E-UTRAN}}$ + $\Delta$T ms, where $\Delta$T = 150 ms, giving a value of 15510 ms. This is rounded up to the next allowed LPP value of 16 seconds.

### 9.4.9.1.4.2 Test procedure

Same as in clause 9.4.7.1.4.2.

#### 9.4.9.1.4.3 Message contents

Same as in clause 9.4.7.1.4.3

### 9.4.9.1.5 Test requirement

Table 9.4.9.1.5-1 defines the primary level settings including test tolerances for the test.

The RSTD TDD inter-frequency accuracy test shall meet the reported values in Table 9.4.9.1.5-2

**Table 9.4.9.1.5-1: Cell Specific Test Parameters for inter frequency RSTD Tests for E-UTRAN TDD**

| Parameter | Unit | Test1 | | Test2 Note4 | |
|---|---|---|---|---|---|
| | | Cell1 | Cell2 | Cell1 | Cell2 |
| E-UTRA RF Channel Number | | 1 | 2 | 1 | 2 |
| Gap offset | | 9 | N/A | 9 | N/A |
| Gap pattern | | #0 | N/A | #0 | N/A |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | OP.2 TDD | OP.11 TDD | OP.2 TDD |
| PRS configuration Index $I_{\mathrm{PRS}}$, as defined in TS 36.211 | | 140 | 150 | 140 | 150 |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| MPDCCH_RA | | | | | |
| MPDCCH_RB | | | | | |
| OCNG_RA Note1 | | | | | |
| OCNG_RB Note1 | | | | | |
| PRS_RA | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 | -98 | -98 |
| PRS $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$ | dB | -1 | -11 | -1 | -11 |
| PRS $\hat{\mathrm{E}}_{\mathrm{s}}/\mathrm{I}_{\mathrm{ot}}$ Note3 | dB | -1 | -11 | -1 | -11 |
| Io Note3 | dBm/9 MHz | -69.68 | -70.16 | -69.68 | -70.16 |
| PRP Note3 | dBm/15kHz | -99 | -109 | -99 | -109 |
| $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$ Note 3 | dB | -1 | -11 | -1 | -11 |
| RSRP Note 3 | dBm/15kHz | -99 | -109 | -99 | -109 |
| Propagation condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$, PRS $\hat{\mathrm{E}}_{\mathrm{s}}/\mathrm{I}_{\mathrm{ot}}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS

Note 4: Test2 is not used for test 9.4.9.1

**Table 9.4.9.1.5-2: RSTD TDD inter-frequency accuracy requirements for the reported values**

| | Test 1 |
|---|---|
| Lowest reported value | RSTD_6241 |
| Highest reported value | RSTD_6287 |

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.9.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2

### 9.4.9.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in normal coverage mode in an environment with fading propagation conditions.

### 9.4.9.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.4.9.2.3 Minimum conformance requirements

Same as in clause 9.4.7.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.1 and A.9.8.28.

### 9.4.9.2.4 Test description

#### 9.4.9.2.4.1 Initial conditions

Same as in clause 9.4.7.2.4.1 but replacing Table 9.4.7.2.4.1-1 with Table 9.4.9.2.4.1-1.

**Table 9.4.9.2.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test1 | Test2 | |
| M-PDCCH parameters | | R.14 TDD | R.14 TDD | As specified in TS 36.521-3 [25] clause A.7.3. |
| *mPDCCH-startSF-UESS* | | 10 | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | |
| E-UTRA RF Channel Number | | Cell 1: 1<br>Cell 2: 2 | | The two cells are on different frequencies. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | 50 Note 4 | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |

| | | | | |
|---|---|---|---|---|
| slotNumberOffset | | 0 | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | |
| TDD uplink-downlink configuration | | 1 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The number of cells includes the reference cell |
| $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$ Note 5 | ms | 15360 | 10240 | Derived according to the RSTD measurement requirements specified in section 9.4.7.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: 6. For all the values to be used in LPP see Table 9.4.7.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter “$T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.7.2.4.3-1. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$ + ΔT ms, where ΔT = 150 ms, giving a value of 15510 ms for Test 1 and 10390 ms for Test 2. This is rounded up to the next allowed LPP value of 16 and 11 seconds, respectively.

### 9.4.9.2.4.2 Test procedure

Same as in clause 9.4.7.1.4.2.

### 9.4.9.2.4.3 Message contents

Same as in clause 9.4.7.1.4.3.

### 9.4.9.2.5 Test requirement

Same as in clause 9.4.9.1.5 adding Test 2 and replacing Table 9.4.9.1.5-2 with Table 9.4.9.2.5-1:

**Table 9.4.9.2.5-1: RSTD TDD inter-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6252 | RSTD_6435 |
| Highest reported value | RSTD_6276 | RSTD_6459 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.10 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B

### 9.4.10.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1

Editor's note: This test is incomplete. The following aspects are missing:

The core requirements in TS 36.133 [23] contain square brackets

#### 9.4.10.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

#### 9.4.10.1.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

#### 9.4.10.1.3 Minimum conformance requirements

The accuracy requirements in Table 9.4.10.1.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex E.3.1 for a corresponding Band

There are no measurement gaps overlapping with the PRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 µs.

**Table 9.4.10.1.3-1: RSTD inter-frequency measurement accuracy for CEModeB**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io Note 4 range | | |
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell and the measured neighbour cell *i* as defined in [24] | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | | dBm/15kHz | $dBm/BW_{Channel}$ |
| [±21] | (PRS Ês/Iot)$_{ref}$ ≥-15dB and (PRS | ≥ 6 | ≥ 30 | ≥ 4 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD- | -120.5 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Ês/Iot)$_i$ ≥-15dB | | | | M1_B | | |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.18 and A.9.8.29.

### 9.4.10.1.4 Test description

#### 9.4.10.1.4.1 Initial conditions

Same as in clause 9.4.7.1.4.1 but replacing Table 9.4.7.1.4.1-1 with Table 9.4.10.1.4.1-1.

**Table 9.4.10.1.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions.**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| MPDCCH | | R.18 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |

| | | | |
|---|---|---|---|
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$ | ms | 40960 | Derived according to the RSTD measurement requirements specified in section 9.4.10.1.3 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.10.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.10.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter "$T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.4.10.1.4.3-3. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}} + \Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 41210 ms. This is rounded up to the next allowed LPP value of 42 seconds.

### 9.4.10.1.4.2 Test procedure

Same as in clause 9.4.7.1.4.2 but using condition CEModeB.

### 9.4.10.1.4.3 Message contents

Same as in clause 9.4.7.1.4.3 with the following exceptions

**Table 9.4.10.1.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.4.7.1.4.3-4 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 42 | See Note 5 in Table 9.4.10.1.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

9.4.10.1.5 Test requirement

Table 9.4.10.1.5-1 defines the primary level settings including test tolerances for the test.

The RSTD FDD inter-frequency accuracy test shall meet the reported values in Table 9.4.10.1.5-2

**Table 9.4.10.1.5-1: Cell Specific Test Parameters for inter frequency RSTD Tests for E-UTRAN FDD**

| Parameter | Unit | Test1 | | Test2 Note4 | |
|---|---|---|---|---|---|
| | | Cell1 | Cell2 | Cell1 | Cell2 |
| E-UTRA RF Channel Number | | 1 | 2 | 1 | 2 |
| Gap offset | | 9 | N/A | 9 | N/A |
| Gap pattern | | #0 | N/A | #0 | N/A |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.21 FDD | OP.6 FDD | OP.21 FDD | OP.6 FDD |
| PRS configuration Index $I_{\mathrm{PRS}}$, as defined in TS 36.211 [26] | | 142 | 152 | 142 | 152 |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| MPDCCH_RA | | | | | |
| MPDCCH_RB | | | | | |
| OCNG_RA Note1 | | | | | |
| OCNG_RB Note1 | | | | | |
| PRS_RA | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 | -98 | -98 |
| PRS $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$ | dB | -11 | -14 | -11 | -14 |
| PRS $\hat{\mathrm{E}}_{\mathrm{s}}/\mathrm{I}_{\mathrm{ot}}$ Note3 | dB | -11 | -14 | -11 | -14 |
| Io Note3 | dBm/9 MHz | -70.16 | -70.19 | -70.16 | -70.19 |
| PRP Note3 | dBm/15kHz | -109 | -112 | -109 | -112 |
| $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$ Note 3 | dB | -11 | -14 | -11 | -14 |
| RSRP Note 3 | dBm/15kHz | -109 | -112 | -109 | -112 |
| Propagation condition | | AWGN | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS |
| Note 4: | Test2 is not used for test 9.4.10.1 |

**Table 9.4.10.1.5-2: RSTD FDD inter-frequency accuracy requirements for the reported values**

| | Test 1 |
|---|---|
| Lowest reported value | [RSTD_6241] |
| Highest reported value | [RSTD_6287] |

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.10.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2

### 9.4.10.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

### 9.4.10.2.2 Test applicability

This test applies to E-UTRA FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

### 9.4.10.2.3 Minimum conformance requirements

The accuracy requirements in Table 9.4.10.2.3-1 are valid under the following conditions:

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex E.2 for a corresponding Band

There are no measurement gaps overlapping with the PRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.4.10.2.3-1: RSTD inter-frequency measurement accuracy for CEModeB**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io Note 4 range | | |
| | PRS Ês/Iot | Minimum PRS bandwidth, which is minimum of serving cell channel bandwidth and the PRS bandwidths of the reference cell and the measured neighbour cell *i* | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i* | The number of consecutive downlink subframes $N_{PRS}$ among the reference cell and the measured neighbour cell *i* as defined in [24] | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |

| | | Note 3 | | | | | |
|---|---|---|---|---|---|---|---|
| **Ts** Note 2 | **dB** | **RB** | | | | **dBm/15kHz** | **dBm/BW$_{Channel}$** |
| ±21 | (PRS Ês/Iot)$_{ref}$ ≥-15dB and (PRS Ês/Iot)$_i$ ≥-[13]dB | ≥ 6 | ≥ 30 | ≥ 4 | FDD-M1_A, TDD-M1_A | -121 | -50 |
| | | | | | FDD-M1_B | -120.5 | -50 |
| | | | | | FDD-M1_C, TDD-M1_C | -120 | -50 |
| | | | | | FDD-M1_D | -119.5 | -50 |
| | | | | | FDD-M1_E, TDD-M1_E | -119 | -50 |
| | | | | | FDD-M1_F | -118.5 | -50 |
| | | | | | FDD-M1_G | -118 | -50 |
| | | | | | FDD-M1_H | -117.5 | -50 |
| | | | | | FDD-M1_N | -114.5 | -50 |
| ±10 | (PRS Ês/Iot)$_{ref}$ ≥-15dB and (PRS Ês/Iot)$_i$ ≥-[13]dB | ≥ 24 | ≥ 8 | ≥ 2 | Note 6 | Note 6 | Note 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4].
NOTE 4: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.4.2.
NOTE 6: FFS?
NOTE 7: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 6 RB.

Editor's note: The values of (PRS Ês/Iot)$_i$: in the above table are inconsistent with other similar values and are therefore placed in square brackets pending further review.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.2 and A.9.8.29.

### 9.4.10.2.4 Test description

#### 9.4.10.2.4.1 Initial conditions

Same as in clause 9.4.10.1.4.1 adding Test 2 and replacing Table 9.4.10.1.4.1-1 with Table 9.4.10.2.4.1-1.

The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to the following values:

Test 1: -92 Ts (about -3 μs)
Test 2: 92 Ts (about 3 μs))

Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.4.10.2.4-1 for each test.

**Table 9.4.10.2.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | Test1 | Test2 | |
|---|---|---|---|---|
| M-PDCCH parameters | | R.18 FDD | R.18 FDD | As specified in TS 36.521-3 [25] clause A.7.1. |
| *mPDCCH-startSF-UESS* | | 10 | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | |
| E-UTRA RF Channel Number | | 1 | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | 50 Note 4 | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The number of cells includes the reference cell |
| $T_{RSTD\ InterFreqFDD\text{-}FDD,\ E\text{-}UTRAN}$ Note 5 | ms | 40960 | 10240 | Derived according to the RSTD measurement requirements specified in section 9.4.10.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.
NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.
NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.
NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.
NOTE 5: The parameter “$T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.10.2.4.3-1. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqFDD-FDD, E-UTRAN}} + \Delta T$ ms, where $\Delta T$ = 150 ms, giving a value of 41210 ms for Test and 10390 ms for Test 2. This is rounded up to the next allowed LPP value of 42 and 11 seconds, respectively.

### 9.4.10.2.4.2 Test procedure

Same as in clause 9.4.7.1.4.2 but using condition CEModeB.

### 9.4.10.2.4.3 Message contents

Same as in clause 9.4.7.1.4.3 with the following exceptions:

**Table 9.4.10.2.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.4.7.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 42 | See Note 5 in Table 9.4.10.2.4.1-1 | Test 1 |
| time | 11 | See Note 5 in Table 9.4.10.2.4.1-1 | Test 2 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.4.10.2.5 Test requirement

Same as in clause 9.4.10.1.5 adding Test 2 and replacing Table 9.4.10.1.5-2 with Table 9.4.10.2.5-1:

**Table 9.4.10.2.5-1: RSTD FDD inter-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6252 | RSTD_6435 |
| Highest reported value | RSTD_6276 | RSTD_6459 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.11 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B

### 9.4.11.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1

Editor's note: This test is incomplete. The following aspects are missing:

The core requirements in TS 36.133 [23] contain square brackets

#### 9.4.11.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits enhanced coverage mode in an environment with fading propagation conditions.

#### 9.4.11.1.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

#### 9.4.11.1.3 Minimum conformance requirements

Same as in clause 9.4.10.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.18 and A.9.8.30.

#### 9.4.11.1.4 Test description

##### 9.4.11.1.4.1 Initial conditions

Same as in clause 9.4.10.1.4.1 but replacing Table 9.4.10.1.4.1-1 with Table 9.4.11.1.4.1-1

**Table 9.4.11.1.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| MPDCCH | | R.8 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.2 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | One carrier frequency is used. |

| | | | |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| $T_{RSTD\ InterFreqHD\text{-}FDD,\ E\text{-}UTRAN}$ | ms | 40960 | Derived according to the RSTD measurement requirements specified in section 9.4.10.1.3 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.10.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.10.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter "$T_{RSTD\ InterFreqHD\text{-}FDD,\ E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 9.4.10.2.4.3-1. The value of the LPP time IE is set to $T_{RSTD\ InterFreqHD\text{-}FDD,\ E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 41210 ms. This is rounded up to the next allowed LPP value of 42 seconds.

#### 9.4.11.1.4.2 Test procedure

Same as in clause 9.4.7.1.4.2 but using condition CEModeB.

#### 9.4.11.1.4.3 Message contents

Same as in clause 9.4.10.1.4.3.

### 9.4.11.1.5 Test requirement

Same as in clause 9.4.10.1.5.

## 9.4.11.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2

### 9.4.11.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits enhanced coverage mode in an environment with fading propagation conditions.

### 9.4.11.2.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

### 9.4.11.2.3 Minimum conformance requirements

Same as in clause 9.4.10.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.2 and A.9.8.30.

### 9.4.11.2.4 Test description

#### 9.4.11.2.4.1 Initial conditions

Same as in clause 9.4.10.2.4.1 but replacing Table 9.4.10.2.4.1-1 with Table 9.4.11.2.4.1-1.

**Table 9.4.11.2.4.1-1: General test parameters for E-UTRAN HD-FDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test1 | Test2 | |
| M-PDCCH parameters | | R.8 HD-FDD | R.8 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.1. |
| *mPDCCH-startSF-UESS* | | 10 | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | |
| E-UTRA RF Channel Number | | 1 | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | 50 Note 4 | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |

| | | | | |
|---|---|---|---|---|
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | Cell 2: -1<br>Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The number of cells includes the reference cell |
| $T_{\text{RSTD InterFreqHD-FDD, E-UTRAN}}$ Note 5 | ms | 40960 | 10240 | Derived according to the RSTD measurement requirements specified in section 9.4.10.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter “$T_{\text{RSTD InterFreqHD-FDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.10.1.4.3-3. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqHD-FDD, E-UTRAN}}$ + ΔT ms, where ΔT = 150 ms, giving a value of 41210 ms for Test and 10390 ms for Test 2. This is rounded up to the next allowed LPP value of 42 and 11 seconds, respectively.

#### 9.4.11.2.4.2 Test procedure

Same as in clause 9.4.7.1.4.2 but using condition CEModeB.

#### 9.4.11.2.4.3 Message contents

Same as in clause 9.4.10.2.4.3.

### 9.4.11.2.5 Test requirement

Same as in clause 9.4.10.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.12 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B

### 9.4.12.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1

Editor's note: This test is incomplete. The following aspects are missing:

The core requirements in TS 36.133 [23] contain square brackets

#### 9.4.12.1.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M1 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

#### 9.4.12.1.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

#### 9.4.12.1.3 Minimum conformance requirements

Same as in clause 9.4.10.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.21.18 and A.9.8.31.

#### 9.4.12.1.4 Test description

##### 9.4.12.1.4.1 Initial conditions

Same as in clause 9.4.10.1.4.1 but replacing Table 9.4.10.1.4.1-1 with Table 9.4.12.1.4.1-1

**Table 9.4.12.1.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| MPDCCH | | R.16 TDD | As specified in TS 36.521-3 [25] clause A.7.3 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cell | | Cell 2 | One carrier frequency is used. |
| E-UTRA RF Channel Number | | 1 | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 4 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |

| | | | |
|---|---|---|---|
| Physical cell ID PCI Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Expected RSTD Note 1 | µs | Cell 2: 1<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| $T_{\text{RSTD InterFreqTDD, E-UTRAN}}$ | ms | 40960 | Derived according to the RSTD measurement requirements specified in section 9.4.10.1.3 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.10.1.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.10.1.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

Note 5: The parameter “$T_{\text{RSTD InterFreqTDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.10.1.4.3-3. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqTDD, E-UTRAN}}$ + ΔT ms, where ΔT = 150 ms, giving a value of 41210 ms. This is rounded up to the next allowed LPP value of 42 seconds.

### 9.4.12.1.4.2 Test procedure

Same as in clause 9.4.7.1.4.2 but using condition CEModeB.

#### 9.4.12.1.4.3 Message contents

Same as in clause 9.4.10.1.4.3.

### 9.4.12.1.5 Test requirement

Table 9.4.12.1.5-1 defines the primary level settings including test tolerances for the test.

The RSTD TDD inter-frequency accuracy test shall meet the reported values in Table 9.4.12.1.5-2

**Table 9.4.12.1.5-1: Cell Specific Test Parameters for inter frequency RSTD Tests for E-UTRAN TDD**

| Parameter | Unit | Test1 | | Test2 Note4 | |
|---|---|---|---|---|---|
| | | Cell1 | Cell2 | Cell1 | Cell2 |
| E-UTRA RF Channel Number | | 1 | 2 | 1 | 2 |
| Gap offset | | 9 | N/A | 9 | N/A |
| Gap pattern | | #0 | N/A | #0 | N/A |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.11 TDD | OP.2 TDD | OP.11 TDD | OP.2 TDD |
| PRS configuration Index $I_{\mathrm{PRS}}$, as defined in TS 36.211 [26] | | 140 | 150 | 140 | 150 |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| MPDCCH_RA | | | | | |
| MPDCCH_RB | | | | | |
| OCNG_RA Note1 | | | | | |
| OCNG_RB Note1 | | | | | |
| PRS_RA | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | -98 | -98 | -98 |
| PRS $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$ | dB | -11 | -14 | -11 | -14 |
| PRS $\hat{\mathrm{E}}_{\mathrm{s}}/\mathrm{I}_{\mathrm{ot}}$ Note3 | dB | -11 | -14 | -11 | -14 |
| Io Note3 | dBm/9 MHz | -70.16 | -70.19 | -70.16 | -70.19 |
| PRP Note3 | dBm/15kHz | -109 | -112 | -109 | -112 |
| $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$ Note 3 | dB | -11 | -14 | -11 | -14 |
| RSRP Note 3 | dBm/15kHz | -109 | -112 | -109 | -112 |
| Propagation condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{\mathrm{E}}_{\mathrm{s}}/N_{oc}$, PRS $\hat{\mathrm{E}}_{\mathrm{s}}/\mathrm{I}_{\mathrm{ot}}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS

Note 4: Test2 is not used for test 9.4.12.1

**Table 9.4.12.1.5-2: RSTD TDD inter-frequency accuracy requirements for the reported values**

| | Test 1 |
|---|---|
| Lowest reported value | [RSTD_6241] |
| Highest reported value | [RSTD_6287] |

The test tolerances are defined in clauses C.1.3 and C4.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.12.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2

### 9.4.12.2.1 Test purpose

To verify that the RSTD measurement accuracy for UE Category M2 is within the specified limits in enhanced coverage mode in an environment with fading propagation conditions.

### 9.4.12.2.2 Test applicability

This test applies to E-UTRA TDD UE Category M2 release 14 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and CE Mode B.

### 9.4.12.2.3 Minimum conformance requirements

Same as in clause 9.4.20.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.25.2 and A.9.8.31.

### 9.4.12.2.4 Test description

#### 9.4.12.2.4.1 Initial conditions

Same as in clause 9.4.10.2.4.1 but replacing Table 9.4.10.2.4.1-1 with Table 9.4.12.2.4.1-1.

**Table 9.4.12.2.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RSTD measurement Accuracy under fading propagation conditions**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test1 | Test2 | |
| M-PDCCH parameters | | R.16 TDD | R.16 TDD | As specified in TS 36.521-3 [25] clause A.7.3. |
| *mPDCCH-startSF-UESS* | | 10 | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Reference cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | |
| E-UTRA RF Channel Number | | 1 | | One carrier frequency is used. |
| System channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 Note 4 | 50 Note 4 | PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 4 | 2 | As defined in TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1: '11110000'<br>Cell 2: '11110000' | | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | (Cell ID of cell 1 – Cell ID of cell 2) mod 6 = 0 | |
| prs-SubframeOffset | | 10 | 10 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |

| | | | | |
|---|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: 1 Other neighbour cells: randomly between -3 and 3 | Cell 2: -1 Other neighbour cells: randomly between -3 and 3 | |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | 5 | |
| TDD uplink-downlink configuration | | 1 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |
| DRX | | OFF | | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: -3 | Cell 2 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | | The number of cells includes the reference cell |
| $T_{\text{RSTD InterFreqTDD, E-UTRAN}}$ Note 5 | ms | 40960 | 10240 | Derived according to the RSTD measurement requirements specified in section 9.4.10.2.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: Test 1: 6, Test 2: 7, Test 3: 6, Test 4: 9. For all the values to be used in LPP see Table 9.4.10.1.4.3-4 and TS 37.571-5 [20], clause 7.2.5.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.1.3.4.1.

NOTE 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

NOTE 5: The parameter “$T_{\text{RSTD InterFreqTDD, E-UTRAN}}$” is not a settable parameter but is used to set the LPP “time” value in Table 9.4.10.2.4.3-1. The value of the LPP time IE is set to $T_{\text{RSTD InterFreqTDD, E-UTRAN}}$ + ΔT ms, where ΔT = 150 ms, giving a value of 41210 ms for Test and 10390 ms for Test 2. This is rounded up to the next allowed LPP value of 42 and 11 seconds, respectively.

### 9.4.12.2.4.2 Test procedure

Same as in clause 9.4.7.1.4.2 but using condition CEModeB.

### 9.4.12.2.4.3 Message contents

Same as in clause 9.4.10.1.4.3.

### 9.4.12.2.5 Test requirement

Same as in clause 9.4.12.1.5 adding Test 2 and replacing Table 9.4.12.1.5-2 with Table 9.4.12.2.5-1:

**Table 9.4.12.2.5-1: RSTD TDD inter-frequency accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Lowest reported value | RSTD_6252 | RSTD_6435 |
| Highest reported value | RSTD_6276 | RSTD_6459 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.13 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions

### 9.4.13.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.4.13.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and dense PRS configuration or additional PRS configuration.

### 9.4.13.3 Minimum conformance requirements

Same as in clause 9.4.1.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.4.1, 8.16.2.1.1 and A.8.13.9.

### 9.4.13.4 Test description

#### 9.4.13.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.4 using only the main Tx/Rx antenna of the UE.
2. The general test parameter settings are set up according to Table 9.4.13.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.4.13.4.3.
5. In the test there are three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the OTDOA assistance data reference as well as the serving cell. Cell 2 and Cell 3 are the neighbour cells. Cell 1 is on RF channel 1. Cell 2 and Cell 3 are on RF channel 2. Cell 3 in the test is the Cell 4 defined in clause 4.7.1. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20] clause 7.2.5).
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour Cell 2 and serving Cell 1; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and serving Cell 1.

**Table 9.4.13.4.1-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |

| | | | |
|---|---|---|---|
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth | RB | 50 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 232,<br>Cell 2, Cell 3: 252 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd6 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 91 | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 12 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 20 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.4.13.4.1-2 |
| prs-SubframeOffset | | 20 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.13.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.13.4.3-5 and TS 37.571-5 [20], clause 7.2.5.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.4.13.4.1.<br>Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. | | | |

**Table 9.4.13.4.1-2: DRX parameters**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 9.4.13.4.2 Test procedure

Same as in clause 9.4.1.1.4.2.

### 9.4.13.4.3 Message contents

Same as in clause 9.4.1.1.4.3 with the following exceptions:

**Table 9.4.13.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.3.1.1.4.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |

| | | | |
|---|---|---|---|
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See clause 9.4.13.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.4.13.5 Test requirement

Table 9.4.1.1.5-1 and 9.4.1.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 6300ms. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 5195 ms. This is rounded up to the next allowed LPP value of 6 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+320\cdot\left\lceil\frac{n}{M}\right\rceil$, where M=16 and $n$ =16 are the parameters specified in clause 9.4.13.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.13.4.1-1. This gives the total RSTD reporting delay of 5120 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 9.4.14 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions

### 9.4.14.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.4.14.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and dense PRS configuration or additional PRS configuration.

### 9.4.14.3 Minimum conformance requirements

Same as in clause 9.4.2.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.4.3, 8.16.2.1.3 and A.8.13.10.

## 9.4.14.4 Test description

### 9.4.14.4.1 Initial conditions

Same as in clause 9.4.13.4.1 but replacing Table 9.4.13.4.1-1 with Table 9.4.14.4.1-1.

**Table 9.4.14.4.1-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.6 HD-FDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth | RB | 50 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 232,<br>Cell 2, Cell 3: 252 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd6 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 91 | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 12 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 20 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| DRX | | ON | DRX parameters are further specified in Table 9.4.14.4.1-2 |

| | | | |
|---|---|---|---|
| prs-SubframeOffset | | 20 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.14.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.14.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.14.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

### 9.4.14.4.2 Test procedure

Same as in clause 9.4.1.1.4.2.

### 9.4.14.4.3 Message contents

Same as in clause 9.4.13.4.3.

## 9.4.14.5 Test requirement

Same as in clause 9.4.13.5.

## 9.4.15 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions

### 9.4.15.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.4.15.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA, inter-frequency RSTD measurements and dense PRS configuration or additional PRS configuration.

### 9.4.15.3 Minimum conformance requirements

Same as in clause 9.4.3.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.2.4.2, 8.16.2.1.2 and A.8.13.11.

### 9.4.15.4 Test description

#### 9.4.15.4.1 Initial conditions

Same as in clause 9.4.13.4.1 but replacing Table 9.4.13.4.1-1 with Table 9.4.15.4.1-1.

**Table 9.4.15.4.1-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.14 TDD | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth | RB | 50 | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 212,<br>Cell 2, Cell 3: 252 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Gap pattern Id | | rstd12 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |

| | | | |
|---|---|---|---|
| Gap offset | | 91 | As specified in TS 36.331 [22], Clause 6.3.5 |
| Number of consecutive downlink positioning subframes $N_{\mathrm{PRS}}$ Note 2 | | 12 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 20 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | |
| TDD uplink-downlink configuration | | 1 | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| DRX | | ON | DRX parameters are further specified in Table 9.4.15.4.1-2 |
| prs-SubframeOffset | | 40 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |

| | | | |
|---|---|---|---|
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.4.15.4.3-5 and TS 37.571-5 [20], clause 7.2.5. | | |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.4.15.4.3-5 and TS 37.571-5 [20], clause 7.2.5. | | |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.4.15.4.1. | | |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. | | |

### 9.4.15.4.2 Test procedure

Same as in clause 9.4.1.1.4.2.

### 9.4.15.4.3 Message contents

Same as in clause 9.4.13.4.3.

## 9.4.15.5 Test requirement

Same as in clause 9.4.13.5.

# 9.4.16 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions

## 9.4.16.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

## 9.4.16.2 Test applicability

This test applies to E-UTRA FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA, CE Mode B, inter-frequency RSTD measurements and dense PRS configuration or additional PRS configuration. Test 2 is applicable only to UE Category M2.

## 9.4.16.3 Minimum conformance requirements

Same as in clause 9.4.4.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.7.1, 8.16.3.2.1 and A.8.13.12.

## 9.4.16.4 Test description

### 9.4.16.4.1 Initial conditions

Same as in clause 9.4.13.4.1 but replacing Table 9.4.13.4.1-1 with Table 9.4.16.4.1-1.

**Table 9.4.16.4.1-1: General test parameters**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |

| | | | | |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.18 FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| | | | | |
| Gap pattern Id | | rstd12 | rstd2 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 151 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 232,<br>Cell 2, Cell 3: 312 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 30 | 8 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 40 | 10 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.16.4.1-2 |
| prs-SubframeOffset | | 80 | | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |

| | | | |
|---|---|---|---|
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is not a settable parameter but is used to set the "true RSTD" values in step 6 of clause 9.3.16.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

### 9.4.16.4.2 Test procedure

Same as in clause 9.4.14.4.2 but with the following additional step:

14. For UE Category M2, repeat step 1-13 for sub-test 2 in Table 9.4.16.4.1-1.

### 9.4.16.4.3 Message contents

Same as in clause 9.4.13.4.3.

## 9.4.16.5 Test requirement

Same as in clause 9.4.13.5 except that for a UE Category M2 for the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.17 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions

### 9.4.17.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.4.17.2 Test applicability

This test applies to E-UTRA HD-FDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA, CE Mode B, inter-frequency RSTD measurements and dense PRS configuration or additional PRS configuration. Test 2 is applicable only to UE Category M2.

### 9.4.17.3 Minimum conformance requirements

Same as in clause 9.4.4.1.3

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.7.3, 8.16.3.2.3 and A.8.13.13.

### 9.4.17.4 Test description

#### 9.4.17.4.1 Initial conditions

Same as in clause 9.4.13.4.1 but replacing Table 9.4.13.4.1-1 with Table 9.4.17.4.1-1.

**Table 9.4.17.4.1-1: General test parameters**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.8 HD-FDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| | | | | |
| Gap pattern Id | | rstd12 | rstd2 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 151 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 | | PRS are transmitted over the system bandwidth |

| | | | | |
|---|---|---|---|---|
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 232,<br>Cell 2, Cell 3: 312 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 30 | 8 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 40 | 10 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 9.3.16.4.1-2 |
| prs-SubframeOffset | | 80 | | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 2.56 | | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 2.56 | | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.16.4.1. |
| Note 4: | If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth. |

#### 9.4.17.4.2 Test procedure

Same as in clause 9.4.14.4.2 but with the following additional step:

14. For UE Category M2, repeat step 1-13 for sub-test 2 in Table 9.4.17.4.1-1.

#### 9.4.17.4.3 Message contents

Same as in clause 9.4.13.4.3.

### 9.4.17.5 Test requirement

Same as in clause 9.4.13.5 except that for a UE Category M2 for the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 9.4.18 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions

### 9.4.18.1 Test purpose

To verify that the RSTD measurement reporting delay with longer PRS occasions for UE Category M1 and M2 meets the performance requirements in normal coverage mode in an environment with fading propagation conditions.

### 9.4.18.2 Test applicability

This test applies to E-UTRA TDD UE Category M1 and M2 release 15 and forward that supports UE-assisted OTDOA, CE Mode B, inter-frequency RSTD measurements and dense PRS configuration or additional PRS configuration. Test 2 is applicable only to UE Category M2.

### 9.4.18.3 Minimum conformance requirements

Same as in clause 9.4.4.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.13.3.7.2, 8.16.3.2.2 and A.8.13.14.

### 9.4.18.4 Test description

#### 9.4.18.4.1 Initial conditions

Same as in clause 9.4.13.4.1 but replacing Table 9.4.13.4.1-1 with Table 9.4.18.4.1-1.

**Table 9.4.18.4.1-1: General test parameters**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |

| | | | | |
|---|---|---|---|---|
| Reference cell | | Cell 1 | | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in 3GPP TS 36.214 [6] and 3GPP TS 36.355 [4]. The reference cell is the serving cell in this test case. |
| Neighbour cells | | Cell 2 and Cell 3 | | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| MPDCCH | | DL Reference Measurement Channel R.16 TDD | | As specified in TS 36.521-3 [25] clause A.7.1 |
| *mPDCCH-startSF-UESS* | | 10 | | Parameter *G* in $T = r_{\max} \cdot G$ which determines subframe *k0* in which MPDCCH starts |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| | | | | |
| Gap pattern Id | | rstd15 | rstd10 | As specified in TS 36.133 [23] Table 8.1.2.1-3. Applies for measurements on Cell 1, Cell 2, and Cell 3 |
| Gap offset | | 131 | | As specified in TS 36.331 [22], Clause 6.3.5 |
| PRS Transmission Bandwidth Note 2 | RB | 50 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | Cell 1: 532,<br>Cell 2, Cell 3: 612 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 30 | 8 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *additional-prs-config* capability |
| | | 40 | 10 | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion. Corresponds to parameter *add-numDL-Frames* in TS 36.355 [4], for UE with *densePrsConfig* capability |
| Physical cell ID PCI Note 2 | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| TDD uplink-downlink configuration | | 2 | 1 | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |

| | | | |
|---|---|---|---|
| DRX | | ON | DRX parameters are further specified in Table 9.3.16.4.1-2 |
| prs-SubframeOffset | | 80 | Number of subframes rounded to the closest integer. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset | | 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell specified in TS 36.355 [4] |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 2 to Cell 1: 1<br>Cell 3 to Cell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| PRS muting info Note 2 | | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 5.12 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For all the values to be used in LPP see Table 9.3.16.4.3-5 and TS 37.571-5 [20], clause 7.2.5.

Note 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is not a settable parameter but is used to set the “true RSTD” values in step 6 of clause 9.3.16.4.1.

Note 4: If the PRS transmission bandwidth is larger than the UE RF bandwidth, the UE is measuring RSTD within its RF bandwidth.

### 9.4.18.4.2 Test procedure

Same as in clause 9.4.14.4.2 but with the following additional step:

14. For UE Category M2, repeat step 1-13 for sub-test 2 in Table 9.4.18.4.1-1.

### 9.4.18.4.3 Message contents

Same as in clause 9.4.13.4.3 with the following exceptions:

**Table 9.4.18.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.4.13.4.3-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 11 | See test requirement in Section 9.4.18.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.4.18.5 Test requirement

Table 9.4.1.1.5-1 and 9.4.1.1.5-2 define the primary level settings including test tolerances for the test.

The response time including test tolerance is 11300ms. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = $N_{rep}$x $TTI_{DCCH}$ = $N_{rep}$x 75 ms, giving a value of 10315 ms. This is rounded up to the next allowed LPP value of 11 seconds. The RSTD measurement reporting delay in the test is derived from the following expression $T_{PRS}(M-1)+640\cdot\left\lceil\frac{n}{M}\right\rceil$, where M=16 and $n$ =16 are the parameters specified in clause 9.4.13.3, Table 9.4.1.1.3-1 and Note 4 of Table 9.4.13.4.1-1. This gives the total RSTD reporting delay of 10240 ms for the 15 neighbour cells including Cell 2 and Cell 3 with respect to the reference cell, Cell 1.

The test tolerances are defined in clauses C.1.3 and C4.

For a UE Category M1 the rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%. For a UE Category M2 for the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

# 9.5 HD-FDD RSTD Intra-Frequency Measurements for NB-IOT

## 9.5.1 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage

### 9.5.1.1 Test purpose

To verify that the RSTD HD-FDD intra-frequency measurement accuracy is within the specified limits for NB-IOT Inband Mode in normal coverage.

### 9.5.1.2 Test applicability

This test applies to NB-IOT E-UTRA HD-FDD UE release 14 and forward that supports UE-assisted OTDOA.

### 9.5.1.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

When the physical layer cell identities of the neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure intra-frequency RSTD, specified in TS 36.214 [6], for at least $n$ = 16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{\text{RSTD IntraFreq, NB}}$ ms as given below:

$$T_{\text{RSTD IntraFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \qquad ms\,,$$

where

$T_{\text{RSTD IntraFreq, NB}}$ is the total time for detecting and measuring at least $n$ cells;

$T_{\text{NPRS}}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{\text{NPRS}}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.5.1.3-1,

$\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{\text{NPRS}}$ is the cell-specific number of NPRS subframes within a NPRS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS_total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.5.1.3-2.

$T_{\text{NPRS}}$ $N_{\text{NPRS}}$, and $N_{NPRS_total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS_total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.5.1.3-1: Number of NPRS positioning occasions within $T_{\text{RSTD IntraFreq, NB}}$**

| | Number of NPRS positioning occasions $M$ |
|---|---|

| Positioning subframe configuration period $T_{\text{NPRS}}$ | f1 Note1 | f1 and f2 Note2 |
|---|---|---|
| 160 ms | $16*\lfloor N_{NPRS_total} / N_{NPRS} \rfloor$ | $32*\lfloor N_{NPRS_total} / N_{NPRS} \rfloor$ |
| >160 ms | $8*\lfloor N_{NPRS_total} / N_{NPRS} \rfloor$ | $16*\lfloor N_{NPRS_total} / N_{NPRS} \rfloor$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1.<br>Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD IntraFreq, NB}}$ provided:

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{\text{dBm}}$ according to Annex E.5 for a corresponding Band

$\text{NPRS}\,\hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{\text{RSTD IntraFreq, NB}}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE and the UE has entered the RRC_IDLE state.

The accuracy requirements in Table 9.5.1.3-2 are valid under the following conditions:

Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

NPRP $1,2|_{\text{dBm}}$ according to Annex E.5 for a corresponding Band

There are no measurement gaps overlapping with the NPRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.5.1.3-2: Intra RSTD measurement accuracy for normal coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | Io Note 4 range | | |
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | dBm/$BW_{Channel}$ |

| | | | | | | |
|---|---|---|---|---|---|---|
| ±20 | (NPRS Ês/Iot)$_{ref}$ ≥-6dB and (NPRS Ês/Iot)$_i$ ≥-13dB | 1 | 320 | NFDD_G | -118 | -70 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1.
NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.1, 9.1.22.10, A.3.23.2 and A.9.8.16.

### 9.5.1.4 Test description

#### 9.5.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz. If the band under test does not support 10 MHz bandwidth, 5 MHz shall be used.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.5.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.5.1.4.3.
5. All cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.5.1.4-1.

**Table 9.5.1.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 2 and eCell 1 | |
| NPDCCH parameters | | R.26 HD-FDD | As defined in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS36.355 [4] |

| | | | |
|---|---|---|---|
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS36.355 [4] |
| NPRS muting info | | nCell 1: '11110000'<br>nCell 2: '11110000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration | | as in the following 2 rows | |
| subframePattern10 | | '0111001110' | Corresponds to subframePattern10-r14 defined in TS 36.355 [4] |
| nprsSequenceInfo | | $BW_{channel}$ 5MHz: 54<br>$BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-1 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Expected RSTD | µs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Measurement period | s | 11.52 | Derived according to the RSTD measurement period in clause 9.5.1.3 |

#### 9.5.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. nCell 1 and nCell 2 (and their LTE donor cells eCell 1 and eCell 2) are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.5.1.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T = 150$ ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-NB with CP CIoT optimisation according to TS 36.508 [18] clause 8.1.5 in nCell 1.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.5.1.5-1 and Table 9.5.1.5-2 as appropriate. Propagation conditions are set according to clause 4.7.2.1.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.

8. When the Inactivity Timer expires, the SS shall send a RRC Connection Release to send the UE to RRC_IDLE state.

9. The UE shall perform location measurements in RRC_IDLE state start a Mobile Originated Data Transport according to TS 23.401 [42] clause 5.3.4B.2.

10. When the signalling connection between the UE and SS is re-established, the UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

12. The SS shall check the *rstd* value for nCell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.5.1.5-3.

13. Repeat step 2-12 until the confidence level according to Annex D is achieved.

#### 9.5.1.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 8.1.4.3 and clause 8.1.6 using condition "Inband_Same" with the following exceptions:

**Table 9.5.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 9.5.1.4.3-1a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 9.5.1.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |

| | | | |
|---|---|---|---|
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 12 | See Measurement Period of Table 9.5.1.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| responseTimeNB-r14 | Not present | Only required if response time shall be >128s | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| messageSizeLimitNB-r14 | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not Present | | |
| tbs-RequestLocationInformation-r13 | Not Present | | |
| wlan-RequestLocationInformation-r13 | Not Present | | |
| bt-RequestLocationInformation-r13 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.1.4.3-1: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | Not present | | |
| otdoa-NeighbourCellInfo | Not present | | |
| otdoa-Error | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-ReferenceCellInfoNB-r14 | As defined in TS 37.571-5 [20], clause 7.4.2. | | |
| otdoa-NeighbourCellInfoNB-r14 | As defined in TS 37.571-5 [20], clause 7.4.2. | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.5.1.5 Test requirement

Table 9.5.1.5-1 and 9.5.1.5-2 define the primary level settings including test tolerances for all tests.

The RSTD FDD intra-frequency accuracy test shall meet the reported values in Table 9.5.1.5-3.

**Table 9.5.1.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 5MHz: 17<br>eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 5MHz: 17<br>eCell 2 $BW_{channel}$ 10MHz: 30 |
| NPBCH_RA<br>NPBCH_RB<br>NPSS_RA<br>NSSS_RA<br>NPDCCH_RA<br>NPDCCH_RB<br>NPDSCH_RA<br>NPDSCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | 0 |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -98 | -98 |
| NPRS_RA | dB | -7.2 | -8.7 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -0.2 | -4.7 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -5.66 | -12.49 |
| Io Note 3 | dBm/ 180kHz | -78.40 | -78.40 |
| NPRP Note 3 | dBm/ 15 kHz | -98.2 | -102.7 |
| NRSRP Note 3 | dBm/ 15 kHz | -91 | -94 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | 7 | 4 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.5.1.5-2: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | | | eCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 or 10 | | | 5 or 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3 | - | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | -98 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | 7 | 7 | 7 | 4 | 4 | 4 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 9.5.1.5-3: RSTD HD-FDD intra-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6426 |
| Highest reported value | RSTD_6468 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 12 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 11670 ms. This is rounded up to the next allowed LPP value of 12 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{\text{RSTD IntraFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \quad ms$, where $T_{\text{NPRS}}$=1280 ms, $M$ =8 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$=320 ms, $N_{\text{NPRS}}$=640 ms and $n$=16. All the parameters are specified in clause 9.5.1.3 and Table 9.5.1.3-1. This gives the total RSTD reporting delay of 11520 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.5.2 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage

### 9.5.2.1 Test purpose

To verify that the RSTD HD-FDD intra-frequency measurement accuracy is within the specified limits for NB-IOT Inband Mode in enhanced coverage.

### 9.5.2.2 Test applicability

This test applies to NB-IOT E-UTRA HD-FDD UE release 14 and forward that supports UE-assisted OTDOA.

### 9.5.2.3 Minimum conformance requirements

Same as clause 9.5.1.3, replacing Table 9.5.1.3-1 with Table 9.5.2.3-1 and Table 9.5.1.3-2 with Table 9.5.2.3-2.

**Table 9.5.2.3-1: Number of NPRS positioning occasions within $T_{\text{RSTD IntraFreq, NB}}$**

| Positioning subframe configuration period $T_{\text{NPRS}}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $32 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| >160 ms | $8 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $16 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

**Table 9.5.2.3-2: Intra RSTD measurement accuracy for enhanced coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | Io Note 4 range | | |
| | | | | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | $dBm/BW_{Channel}$ |
| ±32 | $(NPRS Ês/Iot)_{ref} \geq -15dB$ and $(NPRS Ês/Iot)_i \geq -15dB$ | 1 | 320 | NFDD_G | -118 | -70 |
| NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol. | | | | | | |
| NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26]. | | | | | | |
| NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency. | | | | | | |
| NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe. | | | | | | |
| NOTE 5: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1. | | | | | | |
| NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions. | | | | | | |

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.2, 9.1.22.12, A.3.23.2 and A.9.8.18.

### 9.5.2.4 Test description

#### 9.5.2.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz. If the band under test does not support 10 MHz bandwidth, 5 MHz shall be used.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.5.2.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.5.2.4.3.
5. All cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.5.2.4-1.

**Table 9.5.2.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 2 and eCell 1 | |
| NPDCCH parameters | | R.26 HD-FDD | As defined in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS36.355 [4] |
| NPRS muting info | | nCell 1: '11110000'<br>nCell 2: '11110000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration | | as in the following two rows | |
| subframePattern10 | | '0111001110' | Corresponds to subframePattern10-r14 defined in TS 36.355 [4] |
| nprsSequenceInfo | | $BW_{channel}$ 5MHz: 54<br>$BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-1 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Expected RSTD | μs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | μs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |

| | | | |
|---|---|---|---|
| Measurement period | s | 11.52 | Derived according to the RSTD measurement period in clause 9.5.2.3 |

#### 9.5.2.4.2 Test procedure

The test consists of a set-up period and a measurement period. nCell 1 and nCell 2 (and their LTE donor cells eCell 1 and eCell 2) are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.5.2.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message of the last NPDCCH repetition shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T = 150$ ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-NB with CP CIoT optimisation according to TS 36.508 [18] clause 8.1.5 in nCell 1.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.5.2.5-1 and Table 9.5.2.5-2 as appropriate. Propagation conditions are set according to clause 4.7.2.1.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T = 150$ ms.
8. When the Inactivity Timer expires, the SS shall send a RRC Connection Release to send the UE to RRC_IDLE state.
9. The UE shall perform location measurements in RRC_IDLE state start a Mobile Originated Data Transport according to TS 23.401 [42] clause 5.3.4B.2.
10. When the signalling connection between the UE and SS is re-established, the UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. The SS shall check the *rstd* value for nCell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.5.1.5-3.
13. Repeat step 2-12 until the confidence level according to Annex D is achieved.

#### 9.5.2.4.3 Message contents

Same as clause 9.5.1.4.3.

### 9.5.2.5 Test requirement

Table 9.5.2.5-1 and 9.5.2.5-2 define the primary level settings including test tolerances for all tests.

The RSTD FDD intra-frequency accuracy test shall meet the reported values in Table 9.5.2.5-3.

**Table 9.5.2.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 5MHz: 17<br>eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 5MHz: 17<br>eCell 2 $BW_{channel}$ 10MHz: 30 |
| NPBCH_RA | dB | 0 | 0 |
| NPBCH_RB | | | |
| NPSS_RA | | | |
| NSSS_RA | | | |
| NPDCCH_RA | | | |
| NPDCCH_RB | | | |
| NPDSCH_RA | | | |
| NPDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -110 | -110 |
| NPRS_RA | dB | -12.4 | -0.4 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -14.4 | -12.4 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -14.67 | -14.52 |
| Io Note 3 | dBm/ 180kHz | -100.8 | -110.8 |
| NPRP Note 3 | dBm/ 15 kHz | -124.4 | -122.4 |
| NRSRP Note 3 | dBm/ 15 kHz | -112 | -122 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -2 | -12 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.5.2.5-2: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | | | eCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 or 10 | | | 5 or 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3 | - | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -110 | | | -110 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | -2 | -2 | -2 | -2 | -2 | -2 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 9.5.2.5-3: RSTD HD-FDD intra-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6414 |
| Highest reported value | RSTD_6480 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 12 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 11670 ms. This is rounded up to the next allowed LPP value of 12 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{\text{RSTD IntraFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \quad ms$, where $T_{\text{NPRS}}$=1280 ms, $M$ =8 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$=320 ms, $N_{\text{NPRS}}$=640 ms and $n$=16. All the parameters are specified in clause 9.5.2.3 and Table 9.5.2.3-1. This gives the total RSTD reporting delay of 11520 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.5.3 HD-FDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Standalone Mode in enhanced coverage

### 9.5.3.1 Test purpose

To verify that the RSTD HD-FDD intra-frequency measurement reporting delay is within the specified limits for NB-IOT Standalone Mode in enhanced coverage.

### 9.5.3.2 Test applicability

This test applies to NB-IOT E-UTRA HD-FDD UE release 14 and forward that supports UE-assisted OTDOA.

### 9.5.3.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure intra-frequency RSTD, specified in TS 36.214 [6], for at least $n$ = 16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{\text{RSTD IntraFreq, NB}}$ ms as given below:

$$T_{\text{RSTD IntraFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \qquad ms\,,$$

where

$T_{\text{RSTD IntraFreq, NB}}$ is the total time for detecting and measuring at least $n$ cells;

$T_{\text{NPRS}}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{\text{NPRS}}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.5.3.3-1,

$\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{\text{NPRS}}$ is the cell-specific number of NPRS subframes within a NPRS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS_total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.5.2.3-2.

$T_{\text{NPRS}}$ $N_{\text{NPRS}}$, and $N_{NPRS_total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS_total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.5.3.3-1: Number of NPRS positioning occasions within $T_{\text{RSTD IntraFreq, NB}}$**

| Positioning subframe configuration period $T_{\text{NPRS}}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ | $32 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ |
| >160 ms | $8 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ | $16 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD\ IntraFreq,\ NB}$ provided:

$\left(NPRS\,\hat{E}_s / Iot\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(NPRS\,\hat{E}_s / Iot\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(NPRS\,\hat{E}_s / Iot\right)_{ref}$ and $\left(NPRS\,\hat{E}_s / Iot\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{dBm}$ according to Annex E.5 for a corresponding Band

$NPRS\,\hat{E}_s / Iot$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{RSTD\ IntraFreq,\ NB}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE, and the UE has entered the RRC_IDLE state.

After receiving both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], the UE shall be sent to RRC IDLE state. The maximum allowed RSTD reporting delay is defined as the time between the point when the UE has entered the RRC_IDLE state, and the point when the UE is ready to transmit the measurement report over the air interface and starts to establish a signalling connection with the MME.

The maximum allowed RSTD reporting delay shall be less than $T_{RSTD\,IntraFreq,NB} + T_{RandomAccess_NB\text{-}IoT\text{-}EC}$.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.2, 4.8.2.1, A.3.23.2 and A.4.7.1.

### 9.5.3.4 Test description

#### 9.5.3.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 200 kHz as defined in TS 36.508 [18] clause 8.1.3.1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.5.3.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.5.3.4.3.
5. There are three synchronous cells: nCell 1, nCell 2 and nCell 3. nCell 1 is the reference as well as the serving cell. nCell 2 and nCell 3 are the neighbour cells. All cells are on the same RF channel. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the received time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour nCell 2 and serving nCell 1; and set to -31 Ts (about -1 μs) between neighbour nCell 3 and serving nCell 1.

**Table 9.5.3.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Standalone | |
| Reference cell | | nCell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 36.214 [6] and TS 36.355 [4]. The reference cell is the PCell in this test case. |
| Neighbor cells | | nCell 2 and nCell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| nprsID | | (nprsID of Cell 1 – nprsID of Cell 2)mod6=1<br>and<br>(nprsID of Cell 1 – nprsID of Cell 3)mod6=2 | As defined in TS 36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS 36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 320 | As defined in TS 36.355 [4] |
| NPRS muting info | | nCell 1: ‘11110000’<br>nCell 2: ‘00001111’<br>nCell 3: ‘11110000’ | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| Part A Configuration | | N/A | NPRS is configured based on Part B but not Part A. |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-1 | Refer to A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Radio frame receive time offset between the cells at the UE antenna connector | µs | nCell 2 to nCell 1: 1<br>nCell 3 to nCell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD | µs | nCell 2: 3<br>nCell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| T1 | s | - | The length of the time interval from the beginning of the test until T2 |
| T2 | s | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 5.12 | The length of the time interval that follows immediately after time interval T2 |
| T4 | s | 5.12 | The length of the time interval that follows immediately after time interval T3 |

| | | | |
|---|---|---|---|
| T5 | s | 5.12 | The length of the time interval that follows immediately after time interval T4 |
| T6 | s | ≥ 57 | The length of the time interval that follows immediately after time interval T5 |

#### 9.5.3.4.2 Test procedure

The test consists of six consecutive time intervals, with durations of T1, T2, T3, T4, T5 and T6. nCell 1 is active throughout T1, T2, T3, T4, T5 and T6, whilst nCell 2 and nCell 3 are activated only in the beginning of T2. nCell 2 is active until the end of T5 and nCell 3 is active until the end of T4. nCell 1 transmits NPRS in T2 and T4, while nCell 2 transmits NPRS in T3 and T5 and nCell 3 transmits NPRS only in T2 and T4. Note: The information on when NPRS is muted is conveyed to the UE using PRS muting information.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.5.3.4.3 shall be provided to the UE during the set-up period, T1. After the receipt of the OTDOA assistance data and *OTDOA-RequestLocationInformation*, the UE is provided with a RRC connection release command. The last TTI containing the RRC connection release command shall be provided to the UE ΔT ms before the start of T2 where ΔT = 150 ms. The UE is then expected to enter RRC_IDLE state before T4.

1. Ensure that the UE is in state Generic RB Established State 3A-NB with CP CIoT optimisation according to TS 36.508 [18] clause 8.1.5 in nCell 1.
2. T1 starts.
3. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
4. Set the parameters according to Table 9.5.3.5-1 and Table 9.5.3.5-2 as appropriate. Propagation conditions are set according to clause 4.7.2.1.
5. The SS shall send an LPP REQUEST CAPABILITIES message.
6. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE.
9. The SS shall send a RRC Connection Release to send the UE to RRC_IDLE state and ΔT ms after the last TTI containing the RRC connection release command, T2 starts, where ΔT = 150 ms.
10. At the start of T2, the SS shall switch the power setting from T1 to T2 as specified in Table 9.5.3.5-2.
11. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.5.3.5-2.
12. When T3 expires, the SS shall switch the power setting from T3 to T4 as specified in Table 9.5.3.5-2.
13. When T4 expires, the SS shall switch the power setting from T4 to T5 as specified in Table 9.5.3.5-2.
14. When T5 expires, the SS shall switch the power setting from T5 to T6 as specified in Table 9.5.3.5-1.
15. The UE shall perform location measurements in RRC_IDLE state and then start a Mobile Originated Data Transport according to TS 23.401 [42] clause 5.3.4B.2.
16. When the signalling connection between the UE and SS is re-established, the UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE. The LPP PROVIDE LOCATION INFORMATION shall be transmitted within the response time (see clause 4.7.3)

specified in clause 9.5.3.5. The UE shall perform and report the RSTD measurements for both nCell 2 and nCell 3 with respect to the reference cell in the OTDOA assistance data, nCell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both nCell 2 and nCell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

17. If the UE message at step 16 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

18. Repeat step 2-17 until the confidence level according to Annex D is achieved.

#### 9.5.3.4.3 Message contents

Same as in Clause 9.5.1.3.4.3 with the following exceptions:

**Table 9.5.3.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.5.1.4.3-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 78 | See test requirement in Section 9.5.3.5 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.5.3.5 Test requirement

Table 9.5.3.5-1 and 9.5.3.5-2 define the primary level settings including test tolerances for all tests.

**Table 9.5.3.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for T1 and T6**

| Parameter | Unit | nCell 1 | nCell 2 | nCell 3 |
|---|---|---|---|---|
| NB-IoT RF Channel Number | | 1 | 1 | 1 |
| NB-IoT Channel Bandwidth ($BW_{channel}$) | kHz | 200 | 200 | 200 |
| OCNG Pattern Note 1 | | NOP.3 FDD | N/A | N/A |
| NPDSCH parameters Note 2 | | R.18 HD-FDD | N/A | N/A |
| NPDCCH parameters Note 2 | | R.30 HD-FDD | N/A | N/A |
| NPBCH_RA | dB | 0 | N/A | N/A |
| NPBCH_RB | | | | |

| | | | | |
|---|---|---|---|---|
| NPSS_RA | | | | |
| NSSS_RA | | | | |
| NPDCCH_RA | | | | |
| NPDCCH_RB | | | | |
| NPDSCH_RA | | | | |
| NPDSCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | | |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | dB | -2 | -Infinity | -Infinity |
| Propagation Condition | | AWGN | | |
| Antenna Configuration | | 1x1 | | |
| Timing offset to nCell 1 | μs | N/A | 1 | -1 |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The NPDSCH and NPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

**Table 9.5.3.5-2: Cell Specific Test Parameters for intra frequency RSTD Tests for T2 to T5**

| Parameter | Unit | nCell 1 | | nCell 2 | | nCell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 |
| $BW_{channel}$ | kHz | 200 | | 200 | | 200 | |
| NB-IoT RF Channel Number | | 1 | | 1 | | 1 | |
| OCNG patterns | | NOP.3 FDD | | N/A | NOP.3 FDD | NOP.3 FDD | N/A |
| NPBCH_RA<br>NPBCH_RB<br>NPSS_RA<br>NSSS_RA<br>NPDCCH_RA<br>NPDCCH_RB<br>NPDSCH_RA<br>NPDSCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| NPRS_RA | dB | -3 | N/A | N/A | 0 | 0 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| NPRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| Io Note 4 | dBm/ 180kHz | -87.14 | -87.12 | -87.14 | -87.12 | -87.14 | -87.12 |
| NPRP Note 4 | dBm/ 15 kHz | -113 | -Infinity | -Infinity | -110 | -113 | -Infinity |
| NRSRP Note 4 | dBm/ 15 kHz | -110 | -107 | -113 | -110 | -113 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | -12 | -12 | -15 | -15 | -15 | -Infinity |
| Propagation Condition | | AWGN | | | | | |
| Antenna Configuration | | 1x1 | | | | | |
| Timing offset to nCell 1 | µs | N/A | | 1 | | -1 | |

Note 1: OCNG shall be used such that active cells (all, except nCell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 78.3 s. The response time is equal to the LPP time IE value plus the test tolerance.

The LPP time IE value is derived from the maximum allowed RSTD reporting delay plus T2 and T3 plus ΔT, where ΔT = 150 ms, giving a value of 77550 ms. This is rounded up to the next allowed LPP value of 78 seconds.

The maximum allowed RSTD reporting delay is 67.16 s. This time is measured starting from the beginning of time interval T4, to the moment when the UE starts to send preambles on the PRACH for sending the positioning

measurement report message to nCell1. The maximum allowed RSTD reporting delay is equal to $T_{RSTD_intra_NB\text{-}IoT\text{-}EC}$ $+T_{RandomAccess_NB\text{-}IoT\text{-}EC}$.

The RSTD measurement time $T_{RSTD_intra_NB\text{-}IoT\text{-}EC}$ in the test is derived according to section 9.5.3.3 where it is equal to $T_{RSTD\ IntraFreq,\ NB}$ . This gives the total RSTD measurement time of 11.52s for nCell 2 and nCell 3 with respect to the reference nCell 1

The random access to an already detected cell $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ can be expressed as: $T_{evaluate,\ NB_intra_NB\text{-}IoT\text{-}EC} + T_{SI} + T_{PRACH_NB\text{-}IoT}$,

Where:

$T_{evaluate,\ NB_intra_NB\text{-}IoT\text{-}EC}$ = 12800 ms: see Table 4.6.2.4-1 in clause 4.6.2.4 in TS 36.133 [23].

$T_{SI}$ = 41560 ms and is the time required for receiving all the relevant system information as defined in TS 36.331 [22] for the target NB-IoT FDD cell.

$T_{PRACH_NB\text{-}IoT}$ = 1280 ms and is the additional delay caused by the random access procedure.

This gives $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ = 55.64 s for the random access delay to an already detected cell in the test case.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90% with a confidence level of 95%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 9.1.10.3, i.e., between RSTD_0000 and RSTD_12711.

## 9.5.4 Void

# 9.6 HD-FDD RSTD Inter-Frequency Measurements for NB-IOT

## 9.6.1 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage

### 9.6.1.1 Test purpose

To verify that the RSTD HD-FDD inter-frequency measurement accuracy is within the specified limits for NB-IOT Mode in normal coverage.

### 9.6.1.2 Test applicability

This test applies to NB-IOT E-UTRA HD-FDD UE release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.6.1.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure inter-frequency RSTD, specified in TS 36.214 [6], for at least $n$ = 16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{RSTD\ InterFreq,\ NB}$ ms as given below:

$$T_{RSTD\ InterFreq,NB} = T_{NPRS} \cdot (M\text{-}1) + \Delta \qquad ms\ ,$$

where

$T_{RSTD\ InterFreq,\ NB}$ is the total time for detecting and measuring at least $n$ cells;

$T_{\text{NPRS}}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{\text{NPRS}}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.6.1.3-1,

$\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{\text{NPRS}}$ is the cell-specific number of NPRS subframes within a NPRS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS_total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.6.1.3-2.

$T_{\text{NPRS}}$ $N_{\text{NPRS}}$, and $N_{NPRS_total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS_total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.6.1.3-1: Number of NPRS positioning occasions within $T_{\text{RSTD InterFreq, NB}}$**

| Positioning subframe configuration period $T_{\text{NPRS}}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ | $32 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ |
| >160 ms | $8 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ | $16 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD InterFreq, NB}}$ provided:

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{\text{dBm}}$ according to Annex E.5 for a corresponding Band

$\text{NPRS}\,\hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{\text{RSTD InterFreq, NB}}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message

as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE and the UE has entered the RRC_IDLE state.

The accuracy requirements in Table 9.6.1.3-2 are valid under the following conditions:

Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

NPRP 1,2|$_{dBm}$ according to Annex E.5 for a corresponding Band

There are no measurement gaps overlapping with the NPRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.6.1.3-2: Inter RSTD measurement accuracy for normal coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | Io Note 4 range | | |
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±28 | (NPRS Ês/Iot)$_{ref}$ ≥ -6dB and (NPRS Ês/Iot)$_i$ ≥ -13dB | 1 | 320 | NFDD_G | -118 | -70 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1.
NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.3, 9.1.22.11, A.3.23.2 and A.9.8.17.

## 9.6.1.4 Test description

### 9.6.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz. If the band under test does not support 10 MHz bandwidth, 5 MHz shall be used.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.6.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 9.6.1.4.3.

5. The two NB-IOT cells are on different PRBs of the same LTE carrier frequency. The two LTE Cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.6.1.4-1.

**Table 9.6.1.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 2 and eCell 1 | |
| NPDCCH parameters | | R.26 HD-FDD | As defined in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS36.355 [4] |
| nprs-slotNumberOffset | | 0 | As defined in TS36.355 [4] |
| nprs-SubframeOffset | | 640 | As defined in TS36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS36.355 [4] |
| NPRS muting info | | nCell 1: '1111111100000000'<br>nCell 2: '1111111100000000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration | | as in the following 3 rows | |
| subframePattern10 | | '0111001110' | Corresponds to subframePattern10-r14 defined in TS 36.355 [4] |
| nprsSequenceInfo nCell1 | | $BW_{channel}$ 5MHz: 54<br>$BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| nprsSequenceInfo nCell2 | | $BW_{channel}$ 5MHz: 59<br>$BW_{channel}$ 10MHz: 135 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-1 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Expected RSTD | μs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | μs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Measurement period | s | 20.48 | Derived according to the RSTD measurement period in clause 9.6.1.3 |

#### 9.6.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. nCell 1 and nCell 2 (and their LTE donor cells eCell 1 and eCell 2) are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.6.1.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-NB with CP CIoT optimisation according to TS 36.508 [18] clause 8.1.5 in nCell 1.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.6.1.5-1 and Table 9.6.1.5-2 as appropriate. Propagation conditions are set according to clause 4.7.2.1.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.
8. When the Inactivity Timer expires, the SS shall send a RRC Connection Release to send the UE to RRC_IDLE state.
9. The UE shall perform location measurements in RRC_IDLE state start a Mobile Originated Data Transport according to TS 23.401 [42] clause 5.3.4B.2.
10. When the signalling connection between the UE and SS is re-established, the UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. The SS shall check the *rstd* value for nCell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.6.1.5-3.
13. Repeat step 2-12 until the confidence level according to Annex D is achieved.

#### 9.6.1.4.3 Message contents

Same as clause 9.5.1.4.3 with the following exceptions:

**Table 9.6.1.4.3-1: LPP RequestLocationInformation**

| Derivation Path: Table 9.5.1.4.3-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| qos SEQUENCE { | | | |
| responseTime SEQUENCE { | | | |
| time | 21 | See Measurement Period of Table 9.6.1.4.1-1 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.6.1.5 Test requirement

Table 9.6.1.5-1 and 9.6.1.5-2 define the primary level settings including test tolerances for all tests.

The RSTD FDD inter-frequency accuracy test shall meet the reported values in Table 9.6.1.5-3.

**Table 9.6.1.5-1: Cell Specific Test Parameters for inter frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 5MHz: 17<br>eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 5MHz: 22<br>eCell 2 $BW_{channel}$ 10MHz: 35 |
| NPBCH_RA | dB | 0 | 0 |
| NPBCH_RB | | | |
| NPSS_RA | | | |
| NSSS_RA | | | |
| NPDCCH_RA | | | |
| NPDCCH_RB | | | |
| NPDSCH_RA | | | |
| NPDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -98 | -98 |
| NPRS_RA | dB | -12.7 | -16.7 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -5.7 | -12.7 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -5.7 | -12.7 |
| Io Note 3 | dBm/ 180kHz | -79.90 | -82.90 |
| NPRP Note 3 | dBm/ 15 kHz | -103.7 | -110.7 |
| NRSRP Note 3 | dBm/ 15 kHz | -91 | -94 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | 7 | 4 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.6.1.5-2: Cell Specific Test Parameters for inter frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | | | eCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 or 10 | | | 5 or 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3 | - | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | -98 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | 7 | 7 | 7 | 4 | 4 | 4 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 9.6.1.5-3: RSTD HD-FDD inter-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6417 |
| Highest reported value | RSTD_6477 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 21 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 20630 ms. This is rounded up to the next allowed LPP value of 21 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{\text{RSTD IntraFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \quad ms$, where $T_{\text{NPRS}}$=1280 ms, $M$ =16 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$=320 ms, $N_{\text{NPRS}}$=640 ms and $n$=16. All the parameters are specified in clause 9.6.1.3 and Table 9.6.1.3-1. This gives the total RSTD reporting delay of 20480 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.6.2 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage

### 9.6.2.1 Test purpose

To verify that the RSTD HD-FDD inter-frequency measurement accuracy is within the specified limits for NB-IOT Inband Mode in enhanced coverage.

### 9.6.2.2 Test applicability

This test applies to NB-IOT E-UTRA HD-FDD UE release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.6.2.3 Minimum conformance requirements

Same as clause 9.6.1.3, replacing Table 9.6.1.3-1 with Table 9.6.2.3-1 and Table 9.6.1.3-2 with Table 9.6.2.3-2.

**Table 9.6.2.3-1: Number of NPRS positioning occasions within $T_{\text{RSTD InterFreq, NB}}$**

| Positioning subframe configuration period $T_{\text{NPRS}}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $32*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| >160 ms | $8*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $16*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

**Table 9.6.2.3-2: Inter RSTD measurement accuracy for normal coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | Io Note 4 range | | |
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±40 | $(NPRS\ \hat{E}s/Iot)_{ref} \geq -15dB$ and $(NPRS\ \hat{E}s/Iot)_i \geq -15dB$ | 1 | 320 | NFDD_G | -118 | -70 |
| NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol. | | | | | | |
| NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26]. | | | | | | |
| NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency. | | | | | | |
| NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe. | | | | | | |
| NOTE 5: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1. | | | | | | |
| NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions. | | | | | | |

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.4, 9.1.22.13, A.3.23.2 and A.9.8.19.

### 9.6.2.4 Test description

#### 9.6.2.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz. If the band under test does not support 10 MHz bandwidth, 5 MHz shall be used.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.6.2.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.6.2.4.3.
5. The two NB-IOT cells are on different PRBs of the same LTE carrier frequency. The two LTE Cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.6.2.4-1.

**Table 9.6.2.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 2 and eCell 1 | |
| NPDCCH parameters | | R.26 HD-FDD | As defined in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS36.355 [4] |
| nprs-slotNumberOffset | | 0 | As defined in TS36.355 [4] |
| nprs-SubframeOffset | | 640 | As defined in TS36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS36.355 [4] |
| NPRS muting info | | nCell 1: ‘1111111100000000’<br>nCell 2: ‘1111111100000000’ | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration | | as in the following 3 rows | |
| subframePattern10 | | ‘0111001110’ | Corresponds to subframePattern10-r14 defined in TS 36.355 [4] |
| nprsSequenceInfo nCell1 | | $BW_{channel}$ 5MHz: 54<br>$BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| nprsSequenceInfo nCell2 | | $BW_{channel}$ 5MHz: 59<br>$BW_{channel}$ 10MHz: 135 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-1 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |

| | | | |
|---|---|---|---|
| Expected RSTD | µs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Measurement period | s | 20.48 | Derived according to the RSTD measurement period in clause 9.6.2.3 |

#### 9.6.2.4.2 Test procedure

The test consists of a set-up period and a measurement period. nCell 1 and nCell 2 (and their LTE donor cells eCell 1 and eCell 2) are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.6.2.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message of the last NPDCCH repetition shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-NB with CP CIoT optimisation according to TS 36.508 [18] clause 8.1.5 in nCell 1.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 9.6.2.5-1 and Table 9.6.2.5-2 as appropriate. Propagation conditions are set according to clause 4.7.2.1.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.
8. When the Inactivity Timer expires, the SS shall send a RRC Connection Release to send the UE to RRC_IDLE state.
9. The UE shall perform location measurements in RRC_IDLE state start a Mobile Originated Data Transport according to TS 23.401 [42] clause 5.3.4B.2.
10. When the signalling connection between the UE and SS is re-established, the UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

12. The SS shall check the *rstd* value for nCell 2 in the *OTDOA-SignalMeasurementInformation* IE according to Table 9.6.2.5-3.

13. Repeat step 2-12 until the confidence level according to Annex D is achieved.

#### 9.6.2.4.3 Message contents

Same as clause 9.6.1.4.3.

### 9.6.2.5 Test requirement

Table 9.6.2.5-1 and 9.6.2.5-2 define the primary level settings including test tolerances for all tests.

The RSTD FDD inter-frequency accuracy test shall meet the reported values in Table 9.6.2.5-3.

**Table 9.6.2.5-1: Cell Specific Test Parameters for inter frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 5MHz: 17<br>eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 5MHz: 22<br>eCell 2 $BW_{channel}$ 10MHz: 35 |
| NPBCH_RA | dB | 0 | 0 |
| NPBCH_RB | | | |
| NPSS_RA | | | |
| NSSS_RA | | | |
| NPDCCH_RA | | | |
| NPDCCH_RB | | | |
| NPDSCH_RA | | | |
| NPDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -110 | -110 |
| NPRS_RA | dB | -12.7 | -2.7 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -14.7 | -14.7 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -14.7 | -14.7 |
| Io Note 3 | dBm/ 180kHz | -101.20 | -111.20 |
| NPRP Note 3 | dBm/ 15 kHz | -124.7 | -124.7 |
| NRSRP Note 3 | dBm/ 15 kHz | -112 | -122 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -2 | -12 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.6.2.5-2: Cell Specific Test Parameters for inter frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | | | eCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 or 10 | | | 5 or 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3 | - | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | | $BW_{channel}$ 5MHz: NOP.4 FDD<br>$BW_{channel}$ 10MHz: NOP.1 FDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -110 | | | -110 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | -2 | -2 | -2 | -12 | -12 | -12 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 9.6.2.5-3: RSTD HD-FDD inter-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6405 |
| Highest reported value | RSTD_6489 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 21 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 20630 ms. This is rounded up to the next allowed LPP value of 21 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{\text{RSTD IntraFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \quad ms$, where $T_{\text{NPRS}}$=1280 ms, $M$ =16 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$=320 ms, $N_{\text{NPRS}}$=640 ms and $n$=16. All the parameters are specified in clause 9.6.2.3 and Table 9.6.2.3-1. This gives the total RSTD reporting delay of 20480 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.6.3 HD-FDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Standalone Mode in enhanced coverage

### 9.6.3.1 Test purpose

To verify that the RSTD HD-FDD inter-frequency measurement reporting delay is within the specified limits for NB-IOT Standalone Mode in enhanced coverage.

### 9.6.3.2 Test applicability

This test applies to NB-IOT E-UTRA HD-FDD UE release 14 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.6.3.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure inter-frequency RSTD, specified in TS 36.214 [6], for at least $n$ = 16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{\text{RSTD IntraFreq, NB}}$ ms as given below:

$$T_{\text{RSTD InterFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \qquad ms\,,$$

where

$T_{\text{RSTD InterFreq, NB}}$ is the total time for detecting and measuring at least $n$ cells;

$T_{\text{NPRS}}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{\text{NPRS}}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.6.3.3-1,

$\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{\text{NPRS}}$ is the cell-specific number of NPRS subframes within a NPRS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS_total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.6.2.3-2.

$T_{\text{NPRS}}$ $N_{\text{NPRS}}$, and $N_{NPRS_total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS_total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.6.3.3-1: Number of NPRS positioning occasions within $T_{\text{RSTD InterFreq, NB}}$**

| Positioning subframe configuration period $T_{\text{NPRS}}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $32 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| >160 ms | $8 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $16 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD InterFreq, NB}}$ provided:

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{\text{dBm}}$ according to Annex E.5 for a corresponding Band

$\text{NPRS}\,\hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{\text{RSTD InterFreq, NB}}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE and the UE has entered the RRC_IDLE state.

After receiving both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], the UE shall be sent to RRC IDLE state. The maximum allowed RSTD reporting delay is defined as the time between the point when the UE has entered the RRC_IDLE state, and the point when the UE is ready to transmit the measurement report over the air interface and starts to establish a signalling connection with the MME.

The maximum allowed RSTD reporting delay shall be less than $T_{\text{RSTD InterFreq, NB}} + T_{\text{RandomAccess_NB-IoT-EC}}$.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.4, 4.8.4.1, A.3.23.2 and A.4.7.2.

### 9.6.3.4 Test description

#### 9.6.3.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 200 kHz as defined in TS 36.508 [18] clause 8.1.3.1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.6.3.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.6.3.4.3.
5. There are three synchronous cells: nCell 1, nCell 2 and nCell 3. nCell 1 is the reference as well as the serving cell. nCell 2 and nCell 3 are the neighbour cells. nCell 1 is on one RF channel, nCell 2 and 3 are on a different RF channel. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the received time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between neighbour nCell 2 and serving nCell 1; and set to -31 Ts (about -1 μs) between neighbour nCell 3 and serving nCell 1.

**Table 9.6.3.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Standalone | |
| Reference cell | | nCell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 36.214 [6] and TS 36.355 [4]. The reference cell is the PCell in this test case. |
| Neighbor cells | | nCell 2 and nCell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| nprsID | | (nprsID of Cell 1 – nprsID of Cell 2)mod6=1<br>and<br>(nprsID of Cell 1 – nprsID of Cell 3)mod6=2 | As defined in TS 36.355 [4] |
| nprs-period | ms | 640 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 320 | As defined in TS36.355 [4] |
| nprs-SubframeOffset | | 0 | As defined in TS36.355 [4] |
| NPRS muting info | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| Part A Configuration | | N/A | NPRS is configured based on Part B but not Part A. |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-1 | Refer to A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Radio frame receive time offset between the cells at the UE antenna connector | µs | nCell 2 to nCell 1: 1<br>nCell 3 to nCell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| T1 | s | - | The length of the time interval from the beginning of the test until T2 |
| T2 | s | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 5.12 | The length of the time interval that follows immediately after time interval T2 |

| | | | |
|---|---|---|---|
| T4 | s | 5.12 | The length of the time interval that follows immediately after time interval T3 |
| T5 | s | 5.12 | The length of the time interval that follows immediately after time interval T4 |
| T6 | s | ≥ 57 | The length of the time interval that follows immediately after time interval T5 |

#### 9.6.3.4.2 Test procedure

The test consists of six consecutive time intervals, with durations of T1, T2, T3, T4, T5 and T6. nCell 1 is active throughout T1, T2, T3, T4, T5 and T6, whilst nCell 2 and nCell 3 are activated only in the beginning of T2. nCell 2 is active until the end of T5 and nCell 3 is active until the end of T4. nCell 1 transmits NPRS in T2 and T4, while nCell 2 transmits NPRS in T3 and T5 and nCell 3 transmits NPRS only in T2 and T4. Note: The information on when NPRS is muted is conveyed to the UE using PRS muting information.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 9.6.3.4.3 shall be provided to the UE during the set-up period, T1. After the receipt of the OTDOA assistance data and *OTDOA-RequestLocationInformation*, the UE is provided with a RRC connection release command. The last TTI containing the RRC connection release command shall be provided to the UE ΔT ms before the start of T2 where ΔT = 150 ms. The UE is then expected to enter RRC_IDLE state before T4.

1. Ensure that the UE is in state Generic RB Established State 3A-NB with CP CIoT optimisation according to TS 36.508 [18] clause 8.1.5 in nCell 1.
2. T1 starts.
3. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
4. Set the parameters according to Table 9.6.3.5-1 and Table 9.6.3.5-2 as appropriate. Propagation conditions are set according to clause 4.7.2.1.
5. The SS shall send an LPP REQUEST CAPABILITIES message.
6. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE.
9. The SS shall send a RRC Connection Release to send the UE to RRC_IDLE state and ΔT ms after the last TTI containing the RRC connection release command, T2 starts, where ΔT = 150 ms.
10. At the start of T2, the SS shall switch the power setting from T1 to T2 as specified in Table 9.6.3.5-2.
11. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 9.6.3.5-2.
12. When T3 expires, the SS shall switch the power setting from T3 to T4 as specified in Table 9.6.3.5-2.
13. When T4 expires, the SS shall switch the power setting from T4 to T5 as specified in Table 9.6.3.5-2.
14. When T5 expires, the SS shall switch the power setting from T5 to T6 as specified in Table 9.6.3.5-1.
15. The UE shall perform location measurements in RRC_IDLE state and then start a Mobile Originated Data Transport according to TS 23.401 [42] clause 5.3.4B.2.

16. When the signalling connection between the UE and SS is re-established, the UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE. The LPP PROVIDE LOCATION INFORMATION shall be transmitted within the response time (see clause 4.7.3) specified in clause 9.6.3.5. The UE shall perform and report the RSTD measurements for both nCell 2 and nCell 3 with respect to the reference cell in the OTDOA assistance data, nCell 1. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for both nCell 2 and nCell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

17. If the UE message at step 16 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

18. Repeat step 2-17 until the confidence level according to Annex D is achieved.

#### 9.6.3.4.3 Message contents

Same as in Clause 9.5.3.4.3.

### 9.6.3.5 Test requirement

Table 9.6.3.5-1 and 9.6.3.5-2 define the primary level settings including test tolerances for all tests.

**Table 9.6.3.5-1: Cell Specific Test Parameters for Inter frequency RSTD Tests for T1 and T6**

| Parameter | Unit | nCell 1 | nCell 2 | nCell 3 |
|---|---|---|---|---|
| NB-IoT RF Channel Number | | 1 | 2 | 2 |
| NB-IoT Channel Bandwidth ($BW_{channel}$) | kHz | 200 | 200 | 200 |
| OCNG Pattern Note 1 | | NOP.3 FDD | N/A | N/A |
| NPDSCH parameters Note 2 | | R.18 HD-FDD | N/A | N/A |
| NPDCCH parameters Note 2 | | R.30 HD-FDD | N/A | N/A |
| NPBCH_RA | dB | 0 | N/A | N/A |
| NPBCH_RB | | | | |
| NPSS_RA | | | | |
| NSSS_RA | | | | |
| NPDCCH_RA | | | | |
| NPDCCH_RB | | | | |
| NPDSCH_RA | | | | |
| NPDSCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | | |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | dB | -2 | -Infinity | -Infinity |
| Propagation Condition | | AWGN | | |
| Antenna Configuration | | 1x1 | | |
| Timing offset to nCell 1 | μs | N/A | 1 | -1 |

| | |
|---|---|
| Note 1: | OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel. |
| Note 2: | The NPDSCH and NPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |

**Table 9.6.3.5-2: Cell Specific Test Parameters for Inter frequency RSTD Tests for T2 to T5**

| Parameter | Unit | nCell 1 | | nCell 2 | | nCell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 |
| $BW_{channel}$ | kHz | 200 | | 200 | | 200 | |
| NB-IoT RF Channel Number | | 1 | | 1 | | 1 | |
| OCNG patterns | | NOP.3 FDD | | N/A | NOP.3 FDD | NOP.3 FDD | N/A |
| NPBCH_RA<br>NPBCH_RB<br>NPSS_RA<br>NSSS_RA<br>NPDCCH_RA<br>NPDCCH_RB<br>NPDSCH_RA<br>NPDSCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| NPRS_RA | dB | -3 | N/A | N/A | 0 | 0 | N/A |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| Io Note 3 | dBm/ 180kHz | -87.17 | -87.20 | -87.17 | -87.15 | -87.17 | -87.15 |
| NPRP Note 3 | dBm/ 15 kHz | -113 | -Infinity | -Infinity | -110 | -113 | -Infinity |
| NRSRP Note 3 | dBm/ 15 kHz | -110 | -107 | -113 | -110 | -113 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -12 | -12 | -15 | -15 | -15 | -Infinity |
| Propagation Condition | | AWGN | | | | | |
| Antenna Configuration | | 1x1 | | | | | |
| Timing offset to nCell 1 | µs | N/A | | 1 | | -1 | |

Note 1: OCNG shall be used such that active cells (all, except nCell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 78.3 s. The response time is equal to the LPP time IE value plus the test tolerance.

The LPP time IE value is derived from the maximum allowed RSTD reporting delay plus T2 and T3 plus ΔT, where ΔT = 150 ms, giving a value of 77550 ms. This is rounded up to the next allowed LPP value of 78 seconds.

The maximum allowed RSTD reporting delay is 67.16 s. This time is measured starting from the beginning of time interval T4, to the moment when the UE starts to send preambles on the PRACH for sending the positioning

measurement report message to nCell1. The maximum allowed RSTD reporting delay is equal to $T_{RSTD_inter_NB\text{-}IoT\text{-}EC}$ $+T_{RandomAccess_NB\text{-}IoT\text{-}EC}$.

The RSTD measurement time $T_{RSTD_inter_NB\text{-}IoT\text{-}EC}$ in the test is derived according to section 9.6.3.3 where it is equal to $T_{RSTD\ InterFreq,\ NB}$ . This gives the total RSTD measurement time of 11.52s for nCell 2 and nCell 3 with respect to the reference nCell 1

The random access to an already detected cell $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ can be expressed as: $T_{evaluate,\ NB_inter_NB\text{-}IoT\text{-}EC} + T_{SI} + T_{PRACH_NB\text{-}IoT}$,

Where:

$T_{evaluate,\ NB_inter_NB\text{-}IoT\text{-}EC}$ = 12800 ms: see Table 4.6.2.4-1 in clause 4.6.2.4 in TS 36.133 [23].

$T_{SI}$ = 41560 ms and is the time required for receiving all the relevant system information as defined in TS 36.331 [22] for the target NB-IoT FDD cell.

$T_{PRACH_NB\text{-}IoT}$ = 1280 ms and is the additional delay caused by the random access procedure.

This gives $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ = 55.64 s for the random access delay to an already detected cell in the test case.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90% with a confidence level of 95%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 9.1.10.3, i.e., between RSTD_0000 and RSTD_12711.

## 9.6.4 Void

# 9.7 TDD RSTD Intra-Frequency Measurements for NB-IOT

## 9.7.1 TDD Intra Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage

### 9.7.1.1 Test purpose

To verify that the RSTD TDD intra-frequency measurement accuracy is within the specified limits for NB-IOT Mode in normal coverage.

### 9.7.1.2 Test applicability

This test applies to all types of NB-IoT TDD UE release 15 and forward that supports UE-assisted OTDOA.

### 9.7.1.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

When the physical layer cell identities of the neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure intra-frequency RSTD, specified in TS 36.214 [6], for at least *n* = 16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{RSTD\,IntraFreq,NB}$ ms as given below:

$$T_{RSTD\ IntraFreq,NB} = T_{NPRS} \cdot (M\text{-}1) + \Delta \qquad ms,$$

where

$T_{RSTD\,IntraFreq,NB}$ is the total time for detecting and measuring at least *n* cells;

$T_{\text{NPRS}}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{\text{NPRS}}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.7.1.3-1,

$\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{\text{NPRS}}$ is the cell-specific number of NPRS subframes within a NRPS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS_total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.7.1.3-2.

$T_{\text{NPRS}}$ $N_{\text{NPRS}}$, and $N_{NPRS_total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS_total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.7.1.3-1: Number of NPRS positioning occasions within $T_{\text{RSTDIntraFreq,NB}}$**

| Positioning subframe configuration period $T_{\text{NPRS}}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $32 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| >160 ms | $8 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $16 * \lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTDIntraFreq,NB}}$ provided:

$\left(\text{NPRS } \hat{E}_s / \text{Iot}\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS } \hat{E}_s / \text{Iot}\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS } \hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS } \hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{\text{dBm}}$ according to Annex E.5 for a corresponding Band

$\text{NPRS } \hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{\text{RSTDIntraFreq,NB}}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message

as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE and the UE has entered the RRC_IDLE state.

The accuracy requirements in Table 9.7.1.3-2 are valid under the following conditions:

Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

NPRP $1,2|_{dBm}$ according to Annex E.5 for a corresponding Band

There are no measurement gaps overlapping with the NPRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 μs.

**Table 9.7.1.3-2: Intra RSTD measurement accuracy for normal coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | Io Note 4 range | | |
| | | | | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±20 | $(NPRS\ Ês/Iot)_{ref}$ ≥ -6dB and $(NPRS\ Ês/Iot)_i$ ≥ -13dB | 1 | 320 | NTDD_G | -118 | -70 |
| NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.<br>NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].<br>NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.<br>NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe.<br>NOTE 5: E-UTRA operating band groups are as defined in Section 4.11.1.<br>NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions. | | | | | | |

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.1, 9.1.22.10, A.3.23.2 and A.9.8.32.

### 9.7.1.4 Test description

#### 9.7.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.7.1.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 9.7.1.4.3.

5. All cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.7.1.4-1.

**Table 9.7.1.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 2 and eCell 1 | |
| NPDCCH parameters | | R.26 NB-TDD | Specified in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS 36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS 36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS 36.355 [4] |
| NPRS muting info | | nCell 1: '11110000'<br>nCell 2: '11110000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration: subframePattern10-TDD | | '01100011' | Corresponds to subframePattern10-TDD-r15 defined in TS 36.355 [4] |
| nprsSequenceInfo | | $BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-2 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Expected RSTD | μs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | μs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26] |
| Uplink-downlink configuration | | 2 | As specified in table 4.2-2 in TS 36.211 [26] |
| Measurement period | s | 11.52 | Derived according to the RSTD measurement period in clause 9.7.1.3 |

9.7.1.4.2 Test procedure

Same as in clause 9.5.1.4.2.

9.7.1.4.3 Message contents

Same as in clause 9.5.1.4.3.

9.7.1.5 Test requirement

Table 9.7.1.5-1 and 9.7.1.5-2 define the primary level settings including test tolerances for the test.

The RSTD TDD intra-frequency accuracy test shall meet the reported values in Table 9.7.1.5-3.

**Table 9.7.1.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 10MHz: 30 |
| NPBCH_RA | dB | 0 | 0 |
| NPBCH_RB | | | |
| NPSS_RA | | | |
| NSSS_RA | | | |
| NPDCCH_RA | | | |
| NPDCCH_RB | | | |
| NPDSCH_RA | | | |
| NPDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| NPRS_RA | dB | 0.3 | 0.3 |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -98 | -98 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -5.7 | -12.7 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -5.7 | -12.7 |
| Io Note 3 | dBm/ 180kHz | -86.89 | -86.89 |
| NPRP Note 3 | dBm/ 15 kHz | -103.7 | -110.7 |
| NRSRP Note 3 | dBm/ 15 kHz | -104 | -111 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -6 | -13 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.7.1.5-2: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | eCell 2 |
|---|---|---|---|

| | | T1 | T2 | T3 | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3.6 | - | NOP.1 TDD | | | NOP.1 TDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | -98 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | 0 | 0 | 0 | -7 | -7 | -7 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | us | - | | | 3 | | |

Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 9.7.1.5-3: RSTD TDD intra-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6426 |
| Highest reported value | RSTD_6468 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 12 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 11670 ms. This is rounded up to the next allowed LPP value of 12 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{\text{RSTD IntraFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \quad ms$, where $T_{\text{NPRS}}$=1280 ms, $M$ =8 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$=320 ms, $N_{\text{NPRS}}$=640 ms and $n$=16. All the parameters are specified in clause 9.7.1.3 and Table 9.7.1.3-1. This gives the total RSTD reporting delay of 11520 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.7.2 TDD Intra Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage

### 9.7.2.1 Test purpose

To verify that the RSTD TDD intra-frequency measurement accuracy is within the specified limits for NB-IOT Mode in enhanced coverage.

### 9.7.2.2 Test applicability

This test applies to all types of NB-IoT TDD UE release 15 and forward that supports UE-assisted OTDOA.

9.7.2.3 Minimum conformance requirements

Same as clause 9.7.1.3, replacing Table 9.7.1.3-1 with Table 9.7.2.3-1 and Table 9.7.1.3-2 with Table 9.7.2.3-2.

**Table 9.7.2.3-1: Number of NPRS positioning occasions within $T_{RSTD\ IntraFreq,\ NB}$**

| Positioning subframe configuration period $T_{NPRS}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $32*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| >160 ms | $8*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $16*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD\ IntraFreq,\ NB}$ provided:

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS}\hat{E}_s / \text{Iot}\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS}\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{dBm}$ according to Annex E.5 for a corresponding Band

$\text{NPRS}\hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

**Table 9.7.2.3-2: Intra RSTD measurement accuracy for normal coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | Io Note 4 range | | |
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±32 | $(\text{NPRS}\,\hat{E}s/\text{Iot})_{ref}$ ≥-15dB and $(\text{NPRS}\,\hat{E}s/\text{Iot})_i$ ≥-15dB | 1 | 320 | NTDD_G | -118 | -70 |
| NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol. | | | | | | |
| NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26]. | | | | | | |

NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups are as defined in Section 4.11.1.
NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.2, 9.1.22.12, A.3.23.2 and A.9.8.34.

### 9.7.2.4 Test description

#### 9.7.2.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.7.2.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.7.2.4.3.
5. All cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.7.2.4-1.

**Table 9.7.2.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 2 and eCell 1 | |
| NPDCCH parameters | | R.26 NB-TDD | Specified in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS 36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS 36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS 36.355 [4] |
| NPRS muting info | | nCell 1: '11110000'<br>nCell 2: '11110000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration: subframePattern10-TDD | | '01100011' | Corresponds to subframePattern10-TDD-r15 defined in TS 36.355 [4] |
| nprsSequenceInfo | | $BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| CP length | | Normal | |

| | | | |
|---|---|---|---|
| NPRACH Configuration | | NPRACH.R-2 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Expected RSTD | μs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | μs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26] |
| Uplink-downlink configuration | | 2 | As specified in table 4.2-2 in TS 36.211 [26] |
| Measurement period | s | 11.52 | Derived according to the RSTD measurement period in clause 9.7.2.3 |

### 9.7.2.4.2 Test procedure

Same as in clause 9.5.2.4.2.

### 9.7.2.4.3 Message contents

Same as in clause 9.5.2.4.3.

### 9.7.2.5 Test requirement

Table 9.7.2.5-1 and 9.7.2.5-2 define the primary level settings including test tolerances for the test.

The RSTD TDD intra-frequency accuracy test shall meet the reported values in Table 9.7.2.5-3.

**Table 9.7.2.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 10MHz: 30 |
| NPBCH_RA | dB | 0 | 0 |
| NPBCH_RB | | | |
| NPSS_RA | | | |
| NSSS_RA | | | |
| NPDCCH_RA | | | |
| NPDCCH_RB | | | |
| NPDSCH_RA | | | |
| NPDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| NPRS_RA | dB | 0.3 | 0.3 |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -98 | -98 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -14.3 | -14.3 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -14.3 | -14.3 |
| Io Note 3 | dBm/ 180kHz | -87.14 | -87.14 |

| | | | |
|---|---|---|---|
| NPRP Note 3 | dBm/ 15 kHz | -112.7 | -112.7 |
| NRSRP Note 3 | dBm/ 15 kHz | -113 | -113 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -15 | -15 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.7.2.5-2: Cell Specific Test Parameters for intra frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | | | eCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3.6 | - | NOP.1 TDD | | | NOP.1 TDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | -98 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | -9 | -9 | -9 | -9 | -9 | -9 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | us | - | | | 3 | | |
| Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$. | | | | | | | |

**Table 9.7.2.5-3: RSTD TDD intra-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6414 |
| Highest reported value | RSTD_6480 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 12 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 11670 ms. This is rounded up to the next allowed LPP value of 12 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{RSTD\ IntraFreq,NB} = T_{NPRS} \cdot (M-1) + \Delta \quad ms$, where $T_{NPRS}$=1280 ms, $M$=8 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{NPRS} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$ =320 ms, $N_{NPRS}$=640 ms and $n$=16. All the parameters are specified in clause 9.7.2.3 and Table 9.7.2.3-1. This gives the total RSTD reporting delay of 11520 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.7.3 TDD Intra Frequency RSTD Measurement Reporting Delay for NB-IOT Standalone Mode in enhanced coverage

### 9.7.3.1 Test purpose

To verify that the RSTD TDD intra-frequency measurement reporting delay is within the specified limits for NB-IOT Standalone Mode in enhanced coverage.

### 9.7.3.2 Test applicability

This test applies to all types of NB-IoT TDD UE release 15 and forward that supports UE-assisted OTDOA.

### 9.7.3.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure intra-frequency RSTD, specified in TS 36.214 [6], for at least *n* = 16 cells, including the reference cell, on the same carrier frequency f1 as that of the reference cell within $T_{RSTD\ IntraFreq,\ NB}$ ms as given below:

$$T_{RSTD\ IntraFreq,NB} = T_{NPRS} \cdot (M - 1) + \Delta \quad ms,$$

where

$T_{RSTD\ IntraFreq,\ NB}$ is the total time for detecting and measuring at least *n* cells;

$T_{NPRS}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{NPRS}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.7.3.3-1,

$\Delta = T_{NPRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{NPRS}$ is the cell-specific number of NPRS subframes within a NPRS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS\ _total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.7.3.3-2.

$T_{NPRS}$ $N_{NPRS}$, and $N_{NPRS\ _total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS\ _total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.7.3.3-1: Number of NPRS positioning occasions within $T_{RSTD\ IntraFreq,\ NB}$**

| Positioning subframe configuration period $T_{NPRS}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lvert N_{NPRS\ _total} / N_{NPRS} \rvert$ | $32 * \lvert N_{NPRS\ _total} / N_{NPRS} \rvert$ |
| >160 ms | $8 * \lvert N_{NPRS\ _total} / N_{NPRS} \rvert$ | $16 * \lvert N_{NPRS\ _total} / N_{NPRS} \rvert$ |

| | |
|---|---|
| Note 1: | When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. |
| Note 2: | When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD_IntraFreq,_NB}$ provided:

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS}\hat{E}_s / \text{Iot}\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS}\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{dBm}$ according to Annex E.5 for a corresponding Band

$\text{NPRS}\hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{RSTD_IntraFreq,_NB}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE, and the UE has entered the RRC_IDLE state.

After receiving both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], the UE shall be sent to RRC IDLE state. The maximum allowed RSTD reporting delay is defined as the time between the point when the UE has entered the RRC_IDLE state, and the point when the UE is ready to transmit the measurement report over the air interface and starts to establish a signalling connection with the MME.

The maximum allowed RSTD reporting delay shall be less than $T_{RSTD\,IntraFreq,NB} + T_{RandomAccess_NB\text{-}IoT\text{-}EC}$.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.2, 4.8.2.1, A.3.23.2 and A.4.7.3.

### 9.7.3.4 Test description

#### 9.7.3.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 200 kHz as defined in TS 36.508 [18] clause 8.1.3.1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.7.3.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.7.3.4.3.
5. There are three synchronous cells: nCell 1, nCell 2 and nCell 3. nCell 1 is the reference as well as the serving cell. nCell 2 and nCell 3 are the neighbour cells. All cells are on the same RF channel. The assistance data

neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 31 Ts (about 1 μs between neighbour nCell 2 and serving nCell 1; and set to -31 Ts (about -1 μs) between neighbour nCell 3 and serving nCell 1

**Table 9.7.3.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Standalone | |
| Reference cell | | nCell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 36.214 [6] and TS 36.355 [4]. The reference cell is the PCell in this test case. |
| Neighbor cells | | nCell 2 and nCell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| nprsID | | (nprsID of Cell 1 – nprsID of Cell 2)mod6=1<br>and<br>(nprsID of Cell 1 – nprsID of Cell 3)mod6=2 | As defined in TS 36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS 36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS 36.355 [4] |
| NPRS muting info | | nCell 1: ‘11110000’<br>nCell 2: ‘00001111’<br>nCell 3: ‘11110000’ | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| Part A Configuration | | N/A | NPRS is configured based on Part B but not Part A. |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-2 | Refer to A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Radio frame receive time offset between the cells at the UE antenna connector | μs | nCell 2 to nCell 1: 1<br>nCell 3 to nCell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD | μs | nCell 2: 3<br>nCell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | μs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26] |
| Uplink-downlink configuration | | 2 | As specified in table 4.2-2 in TS 36.211 [26] |

| | | | |
|---|---|---|---|
| T2 | s | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 5.12 | The length of the time interval that follows immediately after time interval T2 |
| T4 | s | 5.12 | The length of the time interval that follows immediately after time interval T3 |
| T5 | s | 5.12 | The length of the time interval that follows immediately after time interval T4 |
| T6 | s | ≥ 58.2 | The length of the time interval that follows immediately after time interval T5 |

#### 9.7.3.4.2 Test procedure

Same as in clause 9.5.3.4.2.

#### 9.7.3.4.3 Message contents

Same as in clause 9.5.3.4.3.

### 9.7.3.5 Test requirement

Table 9.7.3.5-1 and 9.7.3.5-2 define the primary level settings including test tolerances for the test.

**Table 9.7.3.5-1: Cell Specific Test Parameters for intra frequency RSTD Tests for T1 and T6**

| Parameter | Unit | nCell 1 | nCell 2 | nCell 3 |
|---|---|---|---|---|
| NB-IoT RF Channel Number | | 1 | 1 | 1 |
| NB-IoT Channel Bandwidth ($BW_{channel}$) | kHz | 200 | 200 | 200 |
| OCNG Pattern Note 1 | | NOP.3 TDD | N/A | N/A |
| NPDSCH parameters Note 2 | | R.18 NB-TDD | N/A | N/A |
| NPDCCH parameters Note 2 | | R.30 NB-TDD | N/A | N/A |
| NPBCH_RA | dB | 0 | N/A | N/A |
| NPBCH_RB | | | | |
| NPSS_RA | | | | |
| NSSS_RA | | | | |
| NPDCCH_RA | | | | |
| NPDCCH_RB | | | | |
| NPDSCH_RA | | | | |
| NPDSCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | | |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | dB | -2 | -Infinity | -Infinity |
| Propagation Condition | | AWGN | | |
| Antenna Configuration | | 1x1 | | |

| | | | | |
|---|---|---|---|---|
| Timing offset to nCell 1 | µs | N/A | 1 | -1 |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The NPDSCH and NPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

**Table 9.7.3.5-2: Cell Specific Test Parameters for intra frequency RSTD Tests for T2 to T5**

| Parameter | Unit | nCell 1 | | nCell 2 | | nCell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 |
| $BW_{channel}$ | kHz | 200 | | 200 | | 200 | |
| NB-IoT RF Channel Number | | 1 | | 1 | | 1 | |
| OCNG patterns | | NOP.3 TDD | | N/A | NOP.3 TDD | NOP.3 TDD | N/A |
| NPBCH_RA<br>NPBCH_RB<br>NPSS_RA<br>NSSS_RA<br>NPDCCH_RA<br>NPDCCH_RB<br>NPDSCH_RA<br>NPDSCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| NPRS_RA | dB | -3 | N/A | N/A | 0 | 0 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -95 | -98 | -95 | -98 | -95 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| NPRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| Io Note 4 | dBm/ 180kHz | -87.14 | -87.12 | -87.14 | -87.12 | -87.14 | -87.12 |
| NPRP Note 4 | dBm/ 15 kHz | -113 | -Infinity | -Infinity | -110 | -113 | -Infinity |
| NRSRP Note 4 | dBm/ 15 kHz | -110 | -107 | -113 | -110 | -113 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | -12 | -12 | -15 | -15 | -15 | -Infinity |
| Propagation Condition | | AWGN | | | | | |
| Antenna Configuration | | 1x1 | | | | | |
| Timing offset to nCell 1 | µs | N/A | | 1 | | -1 | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 79.3 s. The response time is equal to the LPP time IE value plus the test tolerance.

The LPP time IE value is derived from the maximum allowed RSTD reporting delay plus T2 and T3 plus ΔT, where ΔT = 150 ms, giving a value of 78830 ms. This is rounded up to the next allowed LPP value of 79 seconds.

The maximum allowed RSTD reporting delay is 68.44 s. This time is measured starting from the beginning of time interval T4, to the moment when the UE starts to send preambles on the PRACH for sending the positioning

measurement report message to nCell1. The maximum allowed RSTD reporting delay is equal to $T_{RSTD_intra_NB\text{-}IoT\text{-}EC}$ $+T_{RandomAccess_NB\text{-}IoT\text{-}EC}$.

The RSTD measurement time $T_{RSTD_intra_NB\text{-}IoT\text{-}EC}$ in the test is derived according to section 9.7.3.3 where it is equal to $T_{RSTD\ IntraFreq,\ NB}$. This gives the total RSTD measurement time of 11.52s for nCell 2 and nCell 3 with respect to the reference nCell 1

The random access to an already detected cell $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ can be expressed as: $T_{evaluate,\ NB_intra_NB\text{-}IoT\text{-}EC} + T_{SI} + T_{PRACH_NB\text{-}IoT}$,

Where:

$T_{evaluate,\ NB_intra_NB\text{-}IoT\text{-}EC}$ = 12800 ms: see Table 4.6.2.4-1 in clause 4.6.2.4 in TS 36.133 [23].

$T_{SI}$ = 41560 ms and is the time required for receiving all the relevant system information as defined in TS 36.331 [22] for the target NB-IoT TDD cell.

$T_{PRACH_NB\text{-}IoT}$ = 2560 ms and is the additional delay caused by the random access procedure.

This gives $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ = 56.92 s for the random access delay to an already detected cell in the test case.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90% with a confidence level of 95%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 9.1.10.3, i.e., between RSTD_0000 and RSTD_12711.

# 9.8 TDD RSTD Inter-Frequency Measurements for NB-IOT

## 9.8.1 TDD Inter Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage

### 9.8.1.1 Test purpose

To verify that the RSTD TDD inter-frequency measurement accuracy is within the specified limits for NB-IOT Mode in normal coverage.

### 9.8.1.2 Test applicability

This test applies to all types of NB-IoT TDD UE release 15 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.8.1.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

All positioning subframes indicated in the OTDOA assistance data and specified in sub-clause 9.1.22.11 are available for RSTD measurements in the measured and reference cell.

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure inter-frequency RSTD, specified in TS 36.214 [6], for at least $n$ = 16 cells, including the reference cell within $T_{RSTD\ InterFreq,\ NB}$ ms as given below:

$$T_{RSTD\ InterFreq,NB} = T_{NPRS} \cdot (M\text{-}1) + \Delta \quad ms,$$

where

$T_{RSTD\,InterFreq,\,NB}$ is the total time for detecting and measuring at least *n* cells;

$T_{NPRS}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{NPRS}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.8.1.3-1,

$\Delta = T_{NPRS} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{NPRS}$ is the cell-specific number of NPRS subframes within a NPRS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS_total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.8.1.3-2.

$T_{NPRS}$ $N_{NPRS}$, and $N_{NPRS_total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS_total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.8.1.3-1: Number of NPRS positioning occasions within $T_{RSTD\,InterFreq,\,NB}$**

| Positioning subframe configuration period $T_{NPRS}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ | $32 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ |
| >160 ms | $8 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ | $16 * \lceil N_{NPRS_total} / N_{NPRS} \rceil$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD\,InterFreq,\,NB}$ provided:

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-6 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS}\hat{E}_s / \text{Iot}\right)_i$ ≥-13 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS}\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{dBm}$ according to Annex E.5 for a corresponding Band

$\hat{E}_s / Iot$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{RSTD\,InterFreq,\,NB}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE and the UE has entered the RRC_IDLE state.

The accuracy requirements in Table 9.8.1.3-2 are valid under the following conditions:

Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

NPRP $1,2|_{dBm}$ according to Annex E.5 for a corresponding Band

There are no measurement gaps overlapping with the NPRS subframes of the measured serving cell.

The parameter expectedRSTDUncertainty signalled over LPP by E-SMLC as defined in TS 36.355 [4] is less than 5 µs.

**Table 9.8.1.3-2: Inter RSTD measurement accuracy for normal coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | Io Note 4 range | | |
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | $dBm/BW_{Channel}$ |
| ±28 | $(NPRS\ \hat{E}s/Iot)_{ref} \geq -6dB$ and $(NPRS\ \hat{E}s/Iot)_i \geq -13dB$ | 1 | 320 | NTDD_G | -118 | -70 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1.
NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.3, 9.1.22.11, A.3.23.2 and A.9.8.33.

### 9.8.1.4 Test description

#### 9.8.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.

2. The general test parameter settings are set up according to Table 9.8.1.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 9.8.1.4.3.

5. The two NB-IOT cells are on different PRBs of the same LTE carrier frequency. The two LTE Cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).

6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.8.1.4-1.

**Table 9.8.1.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 1 and eCell 2 | |
| NPDCCH parameters | | R.26 NB-TDD | Specified in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS 36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS 36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS 36.355 [4] |
| nprs-slotNumberOffset | | 0 | As defined in TS 36.355 [4] |
| nprs-SubframeOffset | | 640 | As defined in TS 36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS 36.355 [4] |
| NPRS muting info | | nCell 1: '1111111100000000'<br>nCell 2: '1111111100000000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration: subframePattern10-TDD | | '01100011' | Corresponds to subframePattern10-TDD-r15 defined in TS 36.355 [4] |
| nprsSequenceInfo nCell1 | | $BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| nprsSequenceInfo nCell2 | | $BW_{channel}$ 10MHz: 135 | Corresponds to nprsSequenceInfo defined in TS36.355 [4] |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-2 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Expected RSTD | μs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |

| | | | |
|---|---|---|---|
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26] |
| Uplink-downlink configuration | | 2 | As specified in table 4.2-2 in TS 36.211 [26] |
| Measurement period | s | 20.48 | Derived according to the RSTD measurement period in TS 36.133 [23] 4.8.3 |

#### 9.8.1.4.2 Test procedure

Same as in clause 9.6.1.4.2.

#### 9.8.1.4.3 Message contents

Same as in clause 9.6.1.4.3.

### 9.8.1.5 Test requirement

Table 9.8.1.5-1 and 9.8.1.5-2 define the primary level settings including test tolerances for the test.

The RSTD TDD inter-frequency accuracy test shall meet the reported values in Table 9.8.1.5-3.

**Table 9.8.1.5-1: Cell Specific Test Parameters for inter-frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 10MHz: 35 |
| NPBCH_RA | dB | 0 | 0 |
| NPBCH_RB | | | |
| NPSS_RA | | | |
| NSSS_RA | | | |
| NPDCCH_RA | | | |
| NPDCCH_RB | | | |
| NPDSCH_RA | | | |
| NPDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| NPRS_RA | dB | 0.3 | 0.3 |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -98 | -98 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -5.7 | -12.7 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -5.7 | -12.7 |
| Io Note 3 | dBm/ 15kHz | -86.93 | -86.93 |
| NPRP Note 3 | dBm/ 15 kHz | -103.7 | -110.7 |
| NRSRP Note 3 | dBm/ 15 kHz | -104 | -111 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -6 | -13 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.8.1.5-2: Cell Specific Test Parameters for inter-frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | | | eCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3.6 | - | NOP.1 TDD | | | NOP.1 TDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | -98 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | 3 | 3 | 3 | 3 | 3 | 3 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | us | - | | | 3 | | |

Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 9.8.1.5-3: RSTD TDD inter-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6417 |
| Highest reported value | RSTD_6477 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 21 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 20630 ms. This is rounded up to the next allowed LPP value of 21 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{RSTD\,InterFreq,NB} = T_{NPRS} \cdot (M\text{-}1) + \Delta \quad ms$, where $T_{NPRS}$=1280 ms, $M$=16 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{NPRS} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$ =320 ms, $N_{NPRS}$=640 ms and $n$=16. All the parameters are specified in clause 9.8.1.3 and Table 9.8.1.3-1. This gives the total RSTD reporting delay of 20480 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.8.2 TDD Inter Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage

### 9.8.2.1 Test purpose

To verify that the RSTD TDD inter-frequency measurement accuracy is within the specified limits for NB-IOT Mode in enhanced coverage.

### 9.8.2.2 Test applicability

This test applies to all types of NB-IoT TDD UE release 15 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.8.2.3 Minimum conformance requirements

Same as clause 9.8.1.3, replacing Table 9.8.1.3-1 with Table 9.8.2.3-1 and Table 9.8.1.3-2 with Table 9.8.2.3-2.

**Table 9.8.2.3-1: Number of NPRS positioning occasions within $T_{RSTD\ InterFreq,\ NB}$**

| Positioning subframe configuration period $T_{NPRS}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $32*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| >160 ms | $8*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ | $16*\lvert N_{NPRS_total} / N_{NPRS} \rvert$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{RSTD\ InterFreq,\ NB}$ provided:

$\left(\mathrm{NPRS}\ \hat{E}_s / \mathrm{Iot}\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(\mathrm{NPRS}\hat{E}_s / \mathrm{Iot}\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(\mathrm{NPRS}\ \hat{E}_s / \mathrm{Iot}\right)_{ref}$ and $\left(\mathrm{NPRS}\hat{E}_s / \mathrm{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{\mathrm{dBm}}$ according to Annex E.5 for a corresponding Band

$\mathrm{NPRS}\hat{E}_s / \mathrm{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

**Table 9.8.2.3-2: Inter RSTD measurement accuracy for enhanced coverage**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | NPRS Ês/Iot | UE NPRS measurement bandwidth on the reference cell and the measured neighbour cell *i* Note 3 | Minimum number of available measurement subframes among the reference cell and the measured neighbour cell *i*, $N_{NPRS_total}$ Note 6 | Io Note 4 range | | |
| | | | | E-UTRA operating band groups Note 5 | Minimum Io Note 1 | Maximum Io |
| Ts Note 2 | dB | RB | | | dBm/15kHz | dBm/$BW_{Channel}$ |

| | | | | | | |
|---|---|---|---|---|---|---|
| ±40 | (NPRS Ês/Iot)$_{ref}$ ≥-15dB and (NPRS Ês/Iot)$_i$ ≥-15dB | 1 | 320 | NTDD_G | -118 | -70 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Ts is the basic timing unit defined in TS 36.211 [26].
NOTE 3: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 4: The Io is defined in NPRS positioning subframes. The same Io range applies to NPRS and non-NPRS symbols. Io levels are different in NPRS and non-NPRS symbols within the same subframe.
NOTE 5: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1.
NOTE 6: $N_{NPRS_total}$ can be in one or more NPRS positioning occasions.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.4, 9.1.22.13, A.3.23.2 and A.9.8.35.

### 9.8.2.4 Test description

#### 9.8.2.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 10 MHz.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.8.2.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.8.2.4.3.
5. The two NB-IOT cells are on different PRBs of the same LTE carrier frequency. The two LTE Cells are on the same carrier frequency. nCell 1 is the serving cell and OTDOA assistance data reference cell; nCell 2 is the neighbour cell. eCell 1 and eCell 2 are the LTE donor cells to nCell 1 and nCell 2, respectively. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 92 Ts (about 3 μs). Note that the related expectedRSTD values to be signalled over LPP are defined in Table 9.8.2.4-1.

**Table 9.8.2.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Inband | |
| Reference cell | | nCell 1 | |
| Neighbor cells | | nCell 2, eCell 1 and eCell 2 | |
| NPDCCH parameters | | R.26 NB-TDD | Specified in TS 36.133 [23] section A.3.1.6.1 |
| nprsID | | Test1: (nprsID of Cell 1 – nprsID of Cell 2)mod6=1 | As defined in TS 36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS 36.355 [4] |

| | | | |
|---|---|---|---|
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS 36.355 [4] |
| NPRS muting info | | nCell 1: '1111111100000000'<br>nCell 2: '1111111100000000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [4] |
| PartA Configuration: subframePattern10-TDD | | '01100011' | Corresponds to subframePattern10-TDD-r15 defined in TS 36.355 [4] |
| nprsSequenceInfo nCell1 | | $BW_{channel}$ 10MHz: 130 | Corresponds to nprsSequenceInfo defined in TS 36.355 [4] |
| nprsSequenceInfo nCell2 | | $BW_{channel}$ 10MHz: 135 | Corresponds to nprsSequenceInfo defined in TS36.355 [4] |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-2 | Refer to TS 36.133 [23] section A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Expected RSTD | µs | nCell 2: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26] |
| Uplink-downlink configuration | | 2 | As specified in table 4.2-2 in TS 36.211 [26] |
| Measurement period | s | 20.48 | Derived according to the RSTD measurement period in TS 36.133 [23] 4.8.1 |

#### 9.8.2.4.2 Test procedure

Same as in clause 9.6.2.4.2.

#### 9.8.2.4.3 Message contents

Same as in clause 9.6.2.4.3.

### 9.8.2.5 Test requirement

Table 9.8.2.5-1 and 9.8.2.5-2 define the primary level settings including test tolerances for the test.

The RSTD TDD inter-frequency accuracy test shall meet the reported values in Table 9.8.2.5-3.

**Table 9.8.2.5-1: Cell Specific Test Parameters for inter-frequency RSTD Tests for NB-IOT Cells**

| Parameter | Unit | Test1 | |
|---|---|---|---|
| | | nCell 1 | nCell 2 |
| $BW_{channel}$ | kHz | 180 | 180 |
| PRB location within eCell | | eCell 1 $BW_{channel}$ 10MHz: 30 | eCell 2 $BW_{channel}$ 10MHz: 35 |
| NPBCH_RA | dB | 0 | 0 |
| NPBCH_RB | | | |
| NPSS_RA | | | |
| NSSS_RA | | | |

| | | | |
|---|---|---|---|
| NPDCCH_RA<br>NPDCCH_RB<br>NPDSCH_RA<br>NPDSCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | | | |
| NPRS_RA | dB | 0.3 | 0.3 |
| $N_{oc}$ Note 2 | dBm/<br>15 kHz | -98 | -98 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -14.7 | -14.7 |
| NPRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -14.7 | -14.7 |
| Io Note 3 | dBm/<br>15kHz | -87.14 | -87.14 |
| NPRP Note 3 | dBm/<br>15 kHz | -112.7 | -112.7 |
| NRSRP Note 3 | dBm/ 15 kHz | -113 | -113 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -15 | -15 |
| Propagation Condition | | AWGN | AWGN |
| Antenna Configuration | | 1x1 | 1x1 |
| Timing offset to nCell 1 | us | N/A | 3 |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 2: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

**Table 9.8.2.5-2: Cell Specific Test Parameters for inter-frequency RSTD Tests for E-UTRA Cells**

| Parameter | Unit | eCell 1 | | | eCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| NOCNG Pattern defined in TS 36.133 [23] section A.3.2.3.6 | - | NOP.1 TDD | | | NOP.1 TDD | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | -98 | | |
| $\hat{E}_s / N_{oc}$ Note2 | dBm | -9 | -9 | -9 | -9 | -9 | -9 |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to eCell 1 | us | - | | | 3 | | |

Note 1: OCNG shall be used such that the Cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 9.8.2.5-3: RSTD TDD inter-frequency accuracy requirements for the reported values**

| | |
|---|---|
| Lowest reported value | RSTD_6405 |
| Highest reported value | RSTD_6489 |

For the overall test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

The expected response time is 21 s. The response time is equal to the LPP time IE value. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 20630 ms. This is rounded up to the next allowed LPP value of 21 seconds. The RSTD measurement reporting delay in the test is derived from the following expression, $T_{RSTD\ InterFreq,NB} = T_{NPRS} \cdot (M - 1) + \Delta \quad ms$, where $T_{NPRS}$=1280 ms, $M$=16 $\left| N_{NPRS_total} / N_{NPRS} \right|$, $\Delta = T_{NPRS} \cdot \left\lceil \frac{n}{M} \right\rceil$, $N_{NPRS_total}$ =320 ms, $N_{NPRS}$=640 ms and $n$=16. All the parameters are specified in clause 9.8.2.3 and Table 9.8.2.3-1. This gives the total RSTD reporting delay of 20480 ms for the 15 neighbour cells including nCell 2 with respect to the reference cell, nCell 1. This expected response time excludes any delay caused by RRC connection release before the RRC_IDLE mode measurement. This expected response time excludes any delay caused by establishing a signalling connection with the SS (including random access procedure) as defined in TS 36.305 [41] for LPP measurement reporting. The response time is not explicitly evaluated for this test; hence, no test tolerance values need to be applied.

## 9.8.3 TDD Inter-Frequency RSTD Measurement Reporting Delay for NB-IOT Standalone Mode in enhanced coverage

### 9.8.3.1 Test purpose

To verify that the RSTD TDD inter-frequency measurement reporting delay is within the specified limits for NB-IOT Standalone Mode in enhanced coverage.

### 9.8.3.2 Test applicability

This test applies to all types of NB-IoT TDD UE release 15 and forward that supports UE-assisted OTDOA and inter-frequency RSTD measurements.

### 9.8.3.3 Minimum conformance requirements

The UE shall follow the procedure for RRC_IDLE state positioning measurement as defined in TS 36.305 [41] section 7.1.3.

When the physical layer cell identities of neighbour cells together with the OTDOA assistance data have been provided and the UE has entered the RRC_IDLE state, the UE shall be able to detect and measure inter-frequency RSTD, specified in TS 36.214 [6], for at least $n$ = 16 cells, including the reference cell within $T_{\text{RSTD InterFreq, NB}}$ ms as given below:

$$T_{\text{RSTD InterFreq,NB}} = T_{\text{NPRS}} \cdot (M\text{-}1) + \Delta \qquad ms,$$

where

$T_{\text{RSTD InterFreq, NB}}$ is the total time for detecting and measuring at least $n$ cells;

$T_{\text{NPRS}}$ is the cell-specific positioning subframe configuration period as defined in TS 36.355 [4] if Part B subframe configuration is provided; otherwise if only Part A subframe configuration is provided, the $T_{\text{NPRS}}$ equals to the length of the subframe pattern,

$M$ is the number of NPRS positioning occasions as defined in Table 9.8.3.3-1,

$\Delta = T_{\text{NPRS}} \cdot \left\lceil \frac{n}{M} \right\rceil$ ms is the measurement time for a single NPRS positioning occasion which includes the sampling time and the processing time;

$N_{\text{NPRS}}$ is the cell-specific number of NPRS subframes within a NPRS occasion as defined in TS 36.355 [4] if Part B subframe configuration is provided; if only Part A subframe configuration is provided, the NPRS occasion length is 10 ms,

$N_{NPRS_total}$ is the minimum number of NPRS subframes per cell measurement as defined in Table 9.8.3.3-2.

$T_{\text{NPRS}}$ $N_{\text{NPRS}}$, and $N_{NPRS_total}$ are the parameters of the same cell, for which $T_{NPRS} \cdot \left\lceil \frac{N_{NPRS_total}}{N_{NPRS}} \right\rceil$ is the largest among all the measured cells.

**Table 9.8.3.3-1: Number of NPRS positioning occasions within $T_{\text{RSTD InterFreq, NB}}$**

| Positioning subframe configuration period $T_{\text{NPRS}}$ | Number of NPRS positioning occasions $M$ | |
|---|---|---|
| | f1 Note1 | f1 and f2 Note2 |
| 160 ms | $16 * \lceil N_{total_NPRS} / N_{NPRS} \rceil$ | $32 * \lceil N_{total_NPRS} / N_{NPRS} \rceil$ |
| >160 ms | $8 * \lceil N_{total_NPRS} / N_{NPRS} \rceil$ | $16 * \lceil N_{total_NPRS} / N_{NPRS} \rceil$ |
| Note 1: When only intra-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1. | | |
| Note 2: When intra-frequency RSTD and inter-frequency RSTD measurements are performed over cells belonging to the serving carrier frequency f1 and one inter-frequency carrier frequency f2, respectively. | | |

The UE physical layer shall be capable of reporting RSTD for the reference cell and all the neighbour cells *i* out of at least (*n*-1) neighbour cells within $T_{\text{RSTD InterFreq, NB}}$ provided:

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ ≥-15 dB for all Frequency Bands for the reference cell,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ ≥-15 dB for all Frequency Bands for neighbour cell *i*,

$\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_{ref}$ and $\left(\text{NPRS}\,\hat{E}_s / \text{Iot}\right)_i$ conditions apply for all subframes of at least $L = \frac{M}{2}$ NPRS positioning occasions,

NPRP $1,2|_{\text{dBm}}$ according to Annex E.5 for a corresponding Band

$\text{NPRS}\,\hat{E}_s / \text{Iot}$ is defined as the ratio of the average received energy per PRS RE during the useful part of the symbol to the average received power spectral density of the total noise and interference for this RE, where the ratio is measured over all REs which carry NPRS.

The time $T_{\text{RSTD InterFreq, NB}}$ starts from the point when the UE has received both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], and the message and data have been delivered to the physical layer of the UE and the UE has entered the RRC_IDLE state.

After receiving both the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the OTDOA-ProvideAssistanceData message as specified in TS 36.355 [4], the UE shall be sent to RRC IDLE state. The maximum allowed RSTD reporting delay is defined as the time between the point when the UE has entered the RRC_IDLE state, and the point when the UE is ready to transmit the measurement report over the air interface and starts to establish a signalling connection with the MME.

The maximum allowed RSTD reporting delay shall be less than $T_{\text{RSTD InterFreq, NB}} + T_{\text{RandomAccess_NB-IoT-EC}}$.

The normative reference for this requirement is TS 36.133 [23] clauses 4.8.4, 4.8.4.1, A.3.23.2 and A.4.7.4.

### 9.8.3.4 Test description

#### 9.8.3.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 8.1.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 8.1.3.1.1.

Channel bandwidth to be tested: 200 kHz as defined in TS 36.508 [18] clause 8.1.3.1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.3 but using only the main TX/RX antenna.
2. The general test parameter settings are set up according to Table 9.8.3.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 9.8.3.4.3.
5. There are three synchronous cells: nCell 1, nCell 2 and nCell 3. nCell 1 is the reference as well as the serving cell. nCell 2 and nCell 3 are the neighbour cells. All cells are on the same RF channel. The assistance data neighbour cell list includes in total 15 cells, where 13 of the cells are not simulated (dummy cells; as defined in TS 37.571-5 [20], clause 7.2.2).
6. The true RSTD (which is the receive time difference for frame 0 between nCell 2 and nCell 1 as seen at the UE antenna connector) is set to 31 Ts (about 1 μs between neighbour nCell 2 and serving nCell 1; and set to -31 Ts (about -1 μs) between neighbour nCell 3 and serving nCell 1

**Table 9.8.3.4-1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| NB-IoT operational mode | | Standalone | |
| Reference cell | | nCell 1 | Reference cell is the cell in the OTDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 36.214 [6] and TS 36.355 [4]. The reference cell is the PCell in this test case. |
| Neighbor cells | | nCell 2 and nCell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the OTDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| nprsID | | (nprsID of Cell 1 – nprsID of Cell 2)mod6=1<br>and<br>(nprsID of Cell 1 – nprsID of Cell 3)mod6=2 | As defined in TS 36.355 [4] |
| nprs-period | ms | 1280 | As defined in TS 36.355 [4] |
| nprs-startSF | | sf0 | Subframe offset of the NPRS positioning occasion as defined in TS 36.355 [4] |
| Number of consecutive downlink positioning subframes nprs-NumSF | | 640 | As defined in TS 36.355 [4] |
| nprs-SubframeOffset | | 0 | As defined in TS 36.355 [24] |
| NPRS muting info | | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to nprs-MutingInfoB defined in TS 36.355 [24] |
| Part A Configuration | | N/A | NPRS is configured based on Part B but not Part A. |
| CP length | | Normal | |
| NPRACH Configuration | | NPRACH.R-2 | Refer to A.3.18 |
| DRX cycle length | | 1.28 | The value shall be used for all cells in the test. |
| Radio frame receive time offset between the cells at the UE antenna connector | µs | nCell 2 to nCell 1: 1<br>nCell 3 to nCell 1: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD | µs | Cell 2: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 | Including the reference cell |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26] |
| Uplink-downlink configuration | | 2 | As specified in table 4.2-2 in TS 36.211 [26] |
| T2 | s | 5.12 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 5.12 | The length of the time interval that follows immediately after time interval T2 |

| | | | |
|---|---|---|---|
| T4 | s | 10.24 | The length of the time interval that follows immediately after time interval T3 |
| T5 | s | 10.24 | The length of the time interval that follows immediately after time interval T4 |
| T6 | s | ≥ 58.2 | The length of the time interval that follows immediately after time interval T5 |

#### 9.8.3.4.2 Test procedure

Same as in clause 9.6.3.4.2.

#### 9.8.3.4.3 Message contents

Same as in clause 9.6.3.4.3.

### 9.8.3.5 Test requirement

Table 9.8.3.5-1 and 9.8.3.5-2 define the primary level settings including test tolerances for the test.

**Table 9.8.3.5-1: Cell Specific Test Parameters during T1 and T6**

| Parameter | Unit | nCell 1 | nCell 2 | nCell 3 |
|---|---|---|---|---|
| NB-IoT RF Channel Number | | 1 | 2 | 2 |
| NB-IoT Channel Bandwidth ($BW_{channel}$) | kHz | 200 | 200 | 200 |
| OCNG Pattern Note 1 | | NOP.3 TDD | N/A | N/A |
| NPDSCH parameters Note 2 | | R.18 NB-TDD | N/A | N/A |
| NPDCCH parameters Note 2 | | R.30 NB-TDD | N/A | N/A |
| NPBCH_RA | dB | 0 | N/A | N/A |
| NPBCH_RB | | | | |
| NPSS_RA | | | | |
| NSSS_RA | | | | |
| NPDCCH_RA | | | | |
| NPDCCH_RB | | | | |
| NPDSCH_RA | | | | |
| NPDSCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | | |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | dB | -2 | -Infinity | -Infinity |
| Propagation Condition | | AWGN | | |
| Antenna Configuration | | 1x1 | | |
| Timing offset to nCell 1 | µs | N/A | 1 | -1 |

| | |
|---|---|
| Note 1: | OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel. |
| Note 2: | The NPDSCH and NPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |

**Table 9.8.3.5-2: Cell Specific Test Parameters from T2 to T5**

| Parameter | Unit | nCell 1 | | nCell 2 | | nCell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 | T2 and T4 | T3 and T5 |
| $BW_{channel}$ | kHz | 200 | | 200 | | 200 | |
| NB-IoT RF Channel Number | | 1 | | 2 | | 2 | |
| NPDSCH parameters Note 2 | | R.18 NB-TDD | | N/A | | N/A | |
| NPDCCH parameters Note 2 | | R.30 NB-TDD | | N/A | | N/A | |
| OCNG patterns | | NOP.3 TDD | | N/A | NOP.3 TDD | NOP.3 TDD | N/A |
| NPBCH_RA<br>NPBCH_RB<br>NPSS_RA<br>NSSS_RA<br>NPDCCH_RA<br>NPDCCH_RB<br>NPDSCH_RA<br>NPDSCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| NPRS_RA | dB | -3 | N/A | N/A | 0 | 0 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | -95 |
| NPRS $\hat{E}_s/N_{oc}$ | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| NPRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -15 | -Infinity | -Infinity | -15 | -15 | -Infinity |
| Io Note 4 | dBm/ 180kHz | -87.17 | -87.20 | -87.17 | -87.15 | -87.17 | -87.15 |
| NPRP Note 4 | dBm/ 15 kHz | -113 | -Infinity | -Infinity | -110 | -113 | -Infinity |
| NRSRP Note 4 | dBm/ 15 kHz | -110 | -110 | -113 | -110 | -113 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | -12 | -12 | -15 | -15 | -15 | -Infinity |
| Propagation Condition | | AWGN | | | | | |
| Antenna Configuration | | 1x1 | | | | | |
| Timing offset to nCell 1 | µs | N/A | | 1 | | -1 | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted NPRS.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If NPRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, NPRS $\hat{E}_s/I_{ot}$, Io, NRSRP and NPRP levels have been derived from other parameters and are given for information purpose. If NPRS_RA is "N/A", Io and NRSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 90.3 s. The response time is equal to the LPP time IE value plus the test tolerance.

The LPP time IE value is derived from the maximum allowed RSTD reporting delay plus T2 and T3 plus ΔT, where ΔT = 150 ms, giving a value of 89070 ms. This is rounded up to the next allowed LPP value of 90 seconds.

The maximum allowed RSTD reporting delay is 78.68 s. This time is measured starting from the beginning of time interval T4, to the moment when the UE starts to send preambles on the PRACH for sending the positioning measurement report message to nCell1. The maximum allowed RSTD reporting delay is equal to $T_{RSTD\ _inter_NB\text{-}IoT\text{-}EC}$ $+T_{RandomAccess_NB\text{-}IoT\text{-}EC}$.

The RSTD measurement time $T_{RSTD\ _inter_NB\text{-}IoT\text{-}EC}$ in the test is derived according to section 9.8.3.3 where it is equal to $T_{RSTD\ InterFreq,\ NB}$. This gives the total RSTD measurement time of 21.76s for nCell 2 and nCell 3 with respect to the reference nCell 1

The random access to an already detected cell $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ can be expressed as: $T_{evaluate,\ NB_inter_NB\text{-}IoT\text{-}EC} + T_{SI} + T_{PRACH_NB\text{-}IoT}$,

Where:

$T_{evaluate,\ NB_inter_NB\text{-}IoT\text{-}EC}$ = 12800 ms: see Table 4.6.2.4-1 in clause 4.6.2.4 in TS 36.133 [23].

$T_{SI}$ = 41560 ms and is the time required for receiving all the relevant system information as defined in TS 36.331 [22] for the target NB-IoT TDD cell.

$T_{PRACH_NB\text{-}IoT}$ = 2560 ms and is the additional delay caused by the random access procedure.

This gives $T_{RandomAccess_NB\text{-}IoT\text{-}EC}$ = 56.92 s for the random access delay to an already detected cell in the test case.

The test tolerances are defined in clauses C.1.3 and C4.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90% with a confidence level of 95%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 9.1.10.3, i.e., between RSTD_0000 and RSTD_12711.

# 10 E-UTRA OTDOA measurement requirements for Carrier Aggregation

## 10.0 General

This clause defines the minimum performance requirements for OTDOA FDD and TDD E-UTRA UEs and UEs supporting NR EN-DC, with Carrier Aggregation.

## 10.1 FDD RSTD Measurement Reporting Delay for Carrier Aggregation

### 10.1.1 Test purpose

To verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions. This test case verifies the measurement period requirements for RSTD measurements performed on the secondary component carrier and also the measurement period requirements for RSTD measurements performed on both the primary and secondary component carriers.

### 10.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.1.3 Minimum conformance requirements

### 10.1.3.1 Measurements on the secondary component carrier

The RSTD measurements when all cells are on the configured secondary component carrier shall meet all applicable requirements (FDD) specified in TS 36.133 [23] section 8.1.2.5, i.e., E-UTRAN intra-frequency RSTD measurement period applies, regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34].

### 10.1.3.2 Measurements on both primary component carrier and secondary component carrier

The RSTD measurements of cells on both primary component carrier and configured secondary component carrier shall meet all applicable requirements (FDD) specified in TS 36.133 [23] section 8.1.2.6, i.e., E-UTRAN inter-frequency RSTD measurement period applies regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34], with the following exception

- the number of PRS positioning occasions is as specified in Table 10.1.3.2-1 shall apply.

**Table 10.1.3.2-1: Number of PRS positioning occasions within measurement period**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| 160 ms | 32 |
| >160 ms | 16 |

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.1.

## 10.1.4 Test description

### 10.1.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in TS 36.508 [18] Annex A, Figure group A.42 as appropriate.
2. The general test parameter settings are set up according to Table 10.1.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 10.1.4.3.
5. In the tests, there are two configured component carriers: PCC and SCC, and three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is PCell on the PCC, Cell 2 is an active SCell on the SCC, and Cell 3 is a neighbour cell on the SCC. In both tests, Cell 2 is the OTDOA assistance data reference cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 clause 7.3.2) for Test 1 and where 13 of the cells are not simulated for Test 2. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2 and Cell 3 are powered OFF.
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between Cell 1 and OTDOA assistance data reference cell, Cell 2; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and OTDOA assistance data reference cell, Cell 2.

**Table 10.1.4.1-1: General test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions for Carrier Aggregation**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCell | | Cell 1 | | PCell is on RF channel 1 (PCC). |
| SCell | | Cell 2 | | SCell on RF channel 2 (SCC). Cell 2 is the assistance data reference cell. |
| Other neighbour cell | | Cell 3 | | Neighbour cell on RF channel 2 (SCC). |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 171 for all cells on PCC<br>181 for all cells on SCC | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 2 – PCI of Cell 3)mod6=0 | | The PCI of Cell 1 is selected randomly. PCIs of Cell 2 and Cell 3 are selected randomly such that the relative subcarrier shifts of PRS patterns among these cells are as given by the condition. |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 10.1.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 1 to Cell 2: 1<br>Cell 3 to Cell 2: -1 | Cell 1 to Cell 2: 1<br>Cell 3 to Cell 2: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | Cell 1: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 cells in total | | The list includes the reference cell and 15 other cells. Cell 1 (when included) appears at random places in the first half of the neighbour cell list in the OTDOA assistance data. Cell 3 always appears at random places in the second half of the list |
| | | OTDOA neighbour cells include Cell 3 and other 14 cells on SCC | OTDOA neighbour cells include Cell 1 and other 7 cells on PCC, and Cell 3 and other 6 cells on SCC | |

| | | | | |
|---|---|---|---|---|
| prs-SubframeOffset Note 2 | | Cells on PCC: 310<br>Cells on SCC, except reference cell: 0 | | Subframe offset, counted in full subframes. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | Cells on PCC: 0<br>Cells on SCC, except reference cell: 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell. The corresponding parameter in the OTDOA assistance data is slotNumberOffset specified in TS 36.355 [4]. |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000' | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | S | 3 | | The length of the time interval from the beginning of each test |
| T2 | S | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | S | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.1.4.3-4 and TS 37.571-5 [20], clause 7.3.2.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", "prs-SubframeOffset", "slotNumberOffset" and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For the values to be used in LPP see Table 10.1.4.3-4 and TS 37.571-5 [20], clause 7.3.2.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is used to set the "true RSTD" values in step 6 of clause 10.1.4.1. | | | | |

**Table 10.1.4.1-2: DRX parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions for Carrier Aggregation**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

## 10.1.4.2 Test procedure

The test case includes two tests. Test 1 is designed for the scenario where the UE receives OTDOA assistance data with cells on SCC, and the UE is expected to report RSTD measurements performed on SCC only. Test 2 is designed for the scenario where the UE receives OTDOA assistance data with cells on PCC and SCC, and the UE is expected to report RSTD measurements performed on PCC and on SCC.

Each test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 10.1.4.1-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 is active only in T2 and T3, and Cell 3 is active only during T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the OTDOA assistance data reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, while Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 10.1.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of T2, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.

2. Configure Cell 2 and Cell 3 on the SCC according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [18] clause 5.2A.4.

4. The SS activates the SCell (Cell 2) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.

5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

6. Set the parameters according to Table 10.1.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).

7. T1 starts.

8. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

9a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

9b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 3 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the last 8 elements of the sequence for Test 1 and in the 7 elements of the relevant sequence for Test 2, and the position of neighbour Cell 1 is randomly selected to be in the first 7 elements of the relevant sequence for Test 2, as described in 3GPP TS 37.571-5 [20], clause 7.3.2. If the UE message at step 9b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 10.1.5-2.

13. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 10.1.5-2.

14. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 10.1.5.

For Test 1 the UE shall perform and report the RSTD measurement for Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 2. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with the *rstd* field included within the response time then the number of failure tests is increased by one.

For Test 2 the UE shall perform and report the RSTD measurements for Cell 1 with respect to the reference cell in the OTDOA assistance data, Cell 2 and also Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 2. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 1 (with respect to Cell 2) and Cell 3 (with respect to Cell 2) within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

15. If the UE message at step 14 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

16. Repeat steps 5-15 until the confidence level according to Annex D is achieved. For each iteration, at step 10 change the random positions of the Cell 3 and Cell 1(for Test 2 only) in the relevant sequence in the *OTDOA-NeighbourCellInfoList.*

17. Repeat from clause 10.1.4.1 for Test 2.

### 10.1.4.3 Message contents

**Table 10.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.1.4.3-2: *MAC-MainConfig-RBC*: FDD RSTD Measurement Reporting Delay for Carrier Aggregation**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 10.1.4.3-2a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | Test 1: 3<br>Test 2: 6 | See clause 10.1.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.1.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.1.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 2 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbour | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | With respect to Cell 2 | |
| rstd-Quality | | | |
| } | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbour | Cell 1 | Test 2 only | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | Test 2 only<br>With respect to Cell 2 | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason ‘undefined’ or ‘attemptedButUnableToMeasureSomeNeighbourCells’ | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.1.5 Test requirement

Table 10.1.5-1 and 10.1.5-2 define the primary level settings including test tolerances for the tests.

**Table 10.1.5-1: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T1 for Carrier Aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | N/A | N/A |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.5 FDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | N/A | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 10.1.5-2: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for Carrier Aggregation**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |

| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
|---|---|---|---|---|---|---|---|
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.5 FDD | | OP.6 FDD | | OP.6 FDD | N/A |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | N/A |
| PRS_RA | dB | -6 | N/A | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -4 | -Infinity | -Infinity | -1 | -8 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -4 | -Infinity | -Infinity | -1 | -8 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.94 | N/A | N/A | -66.68 | -70.11 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -102 | -Infinity | -Infinity | -96 | -106 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -96 | -105 | -99 | -109 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -4 | -11 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 3.3s for Test 1 and 6.3s for Test 2. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 2710 ms for Test 1 and 5110ms for Test 2. This is rounded up to the next allowed LPP value of 3 seconds for Test 1 and 6 seconds for Test 2.

The RSTD measurement reporting delay in the tests is derived from the following expression,

$T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$ =8 and $n$ =16 for Test 1, and $M$ =16 and $n$ =16 for Test 2 are the parameters specified in clause 10.1.3.1 for Test 1 and clause 10.1.3.2 for Test 2.

This gives the total RSTD reporting delay of 2560 ms for Test 1 for the 15 neighbour cells including Cell 3 with respect to the reference cell, Cell 2.

This gives the total RSTD reporting delay of 4960 ms for Test 2 for the 15 neighbour cells including Cell 1 and Cell 3 with respect to the reference cell, Cell 2.

The test tolerances are defined in clauses C.1.3 and C.4.

For the overall test to pass, the rate of successful tests during repeated tests in both Test 1 and Test 2 shall be more than 90% with a confidence level of 95%.

# 10.1A FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz Bandwidth

## 10.1A.1 Test purpose

Same as defined in clause 10.1.1.

NOTE: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5.

## 10.1A.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.1A.3 Minimum conformance requirements

Same as defined in clause 10.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.3.

## 10.1A.4 Test description

### 10.1A.4.1 Initial conditions

Same as defined in clause 10.1.4.1 except that the values of the parameters in Table 10.1A.4.1-1 will replace the values of the corresponding parameters in Table 10.1.4.1-1.

Channel bandwidth to be tested: 20 MHz.

**Table 10.1A.4.1-1: General test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation for 20 MHz**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 FDD | | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | | |
| PRS Transmission Bandwidth | RB | 100 | | PRS are transmitted over the system bandwidth |
| Note 1: See Table 10.1.4.1-1 for the other parameters.<br>Note 2: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5. | | | | |

### 10.1A.4.2 Test procedure

Same as defined in clause 10.1.4.2.

### 10.1A.4.3 Message contents

Same as defined in clause 10.1.4.3.

## 10.1A.5 Test requirement

Same as defined in clause 10.1.5 except that the values of the parameters in Table 10.1A.5-1 and Table 10.1A.5-2 will replace the values of the corresponding parameters in Table 10.1.5-1 and Table 10.1.5-2, respectively.

**Table 10.1A.5-1: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation for 20 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.13 FDD | N/A | N/A |
| Io Note 1 | dBm/ 18 MHz | -64.21 | N/A | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.1.5-1 for the other parameters. | | | | |

**Table 10.1A.5-2: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation for 20 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.13 FDD | | OP.14 FDD | | OP.14 FDD | N/A |
| Io Note 1 | dBm/ 18 MHz | -66.93 | N/A | N/A | -63.67 | -67.09 | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.1.5-2 for the other parameters. | | | | | | | |

# 10.1B FDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz + 5 MHz Bandwidth

## 10.1B.1 Test purpose

Same as defined in clause 10.1.1.

NOTE: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5.

## 10.1B.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.1B.3 Minimum conformance requirements

Same as defined in clause 10.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.7.

## 10.1B.4 Test description

### 10.1B.4.1 Initial conditions

Same as defined in clause 10.1.4.1 except that the values of the parameters in Table 10.1B.4.1-1 will replace the values of the corresponding parameters in Table 10.1.4.1-1.

Channel bandwidth to be tested: 5 MHz.

**Table 10.1B.4.1-1: General test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | | |
| PRS Transmission Bandwidth | RB | 25 | | PRS are transmitted over the system bandwidth |
| PRS occasion length $N_{PRS}$ | | 2 | | |
| Note 1: See Table 10.1.4.1-1 for the other parameters.<br>Note 2: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in section 4.7.5. | | | | |

### 10.1B.4.2 Test procedure

Same as defined in clause 10.1.4.2.

### 10.1B.4.3 Message contents

Same as defined in clause 10.1.4.3.

## 10.1B.5 Test requirement

Same as defined in clause 10.1.5 except that the values of the parameters in Table 10.1B.5-1 and Table 10.1B.5-2 will replace the values of the corresponding parameters in Table 10.1.5-1 and Table 10.1.5-2, respectively.

**Table 10.1B.5-1: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| OCNG patterns defined in TS 36.521-3 [25] clause D.1.18 | | OP.18 FDD | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| Io Note 1 | dBm/ 4.5 MHz | -70.23 | N/A | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.1.5-1 for the other parameters. | | | | |

**Table 10.1B.5-2: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.18 FDD | | OP.19 FDD | | OP.19 FDD | N/A |
| Io Note 1 | dBm/ 4.5 MHz | -72.95 | N/A | N/A | -69.69 | -73.12 | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.1.5-2 for the other parameters. | | | | | | | |

# 10.1C FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth

## 10.1C.1 Test purpose

Same as defined in clause 10.1.1.

NOTE: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5.

## 10.1C.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.1C.3 Minimum conformance requirements

Same as defined in clause 10.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.5.

## 10.1C.4 Test description

### 10.1C.4.1 Initial conditions

Same as defined in clause 10.1.4.1 except that the values of the parameters in Table 10.1C.4.1-1 will replace the values of the corresponding parameters in Table 10.1.4.1-1.

Channel bandwidth to be tested: Cell 1: 10 MHz, Cell 2 and Cell 3: 5 MHz.

**Table 10.1C.4.1-1: General test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | Test 1 | Test 2 | |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | Cell 1: R.6 FDD<br>Cell 2: R.11 FDD<br>Cell 3: R.11 FDD | Cell 1: R.6 FDD<br>Cell 2: R.11 FDD<br>Cell 3: R.11 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | Cell 1: 10<br>Cell 2: 5<br>Cell 3: 5 | Cell 1: 10<br>Cell 2: 5<br>Cell 3: 5 | |
| PRS Transmission Bandwidth | RB | Cell 1: 50<br>Cell 2: 25<br>Cell 3: 25 | Cell 1: 50<br>Cell 2: 25<br>Cell 3: 25 | PRS are transmitted over the system bandwidth |
| PRS occasion length $N_{\mathrm{PRS}}$ | | Cell 1: 1<br>Cell 2: 2<br>Cell 3: 2 | Cell 1: 1<br>Cell 2: 2<br>Cell 3: 2 | |
| Note 1: See Table 10.1.4.1-1 for the other parameters.<br>Note 2: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in section 4.7.5. | | | | |

### 10.1C.4.2 Test procedure

Same as defined in clause 10.1.4.2.

### 10.1C.4.3 Message contents

Same as defined in clause 10.1.4.3.

## 10.1C.5 Test requirement

Same as defined in clause 10.1.5 except that the values of the parameters in Table 10.1C.5-1 and Table 10.1C.5-2 will replace the values of the corresponding parameters in Table 10.1.5-1 and Table 10.1.5-2, respectively.

**Table 10.1C.5-1: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| Io Note 1 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| | dBm/ 4.5 MHz | N/A | N/A | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.1.5-1 for the other parameters. | | | | |

**Table 10.1C.5-2: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 | | OP.5 FDD | | OP.19 FDD | | OP.19 FDD | N/A |
| Io Note 1 | dBm/ 9 MHz | -69.94 | N/A | N/A | N/A | N/A | N/A |
| | dBm/ 4.5 MHz | N/A | N/A | N/A | -69.69 | -73.12 | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.1.5-2 for the other parameters. | | | | | | | |

# 10.2 TDD RSTD Measurement Reporting Delay for Carrier Aggregation

## 10.2.1 Test purpose

To verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions. This test case verifies the measurement period requirements for RSTD measurements performed on the secondary component carrier and also the measurement period requirements for RSTD measurements performed on both the primary and secondary component carriers.

## 10.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.2.3 Minimum conformance requirements

### 10.2.3.1 Measurements on the secondary component carrier

The RSTD measurements when all cells are on the configured secondary component carrier shall meet all applicable requirements (TDD) specified in TS 36.133 [23] section 8.1.2.5, i.e., E-UTRAN intra-frequency RSTD measurement period applies, regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34].

### 10.2.3.2 Measurements on both primary component carrier and secondary component carrier

The RSTD measurements of cells on both primary component carrier and configured secondary component carrier shall meet all applicable requirements (TDD) specified in TS 36.133 [23] section 8.1.2.6, i.e., E-UTRAN inter-frequency RSTD measurement period applies regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34], with the following exceptions

- the number of PRS positioning occasions is as specified in Table 10.2.3.2-1 shall apply, and
- TDD uplink-downlink subframes configurations as specified in TS 36.133 [23] section 8.1.2.5.2, Table 8.1.2.5.2-2 shall apply.

**Table 10.2.3.2-1: Number of PRS positioning occasions within measurement period**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| 160 ms | 32 |
| >160 ms | 16 |

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.2.

## 10.2.4 Test description

### 10.2.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.2.

Channel bandwidth to be tested: 10 MHz as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in TS 36.508 [18] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 10.2.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 10.2.4.3.

5. In the tests, there are two configured component carriers: PCC and SCC, and three synchronized cells: Cell 1, Cell 2 and Cell 3. Cell 1 is PCell on the PCC, Cell 2 is an active SCell on the SCC, and Cell 3 is a neighbour cell on the SCC. In both tests, Cell 2 is the OTDOA assistance data reference cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 clause 7.3.2) for Test 1 and where 13 of the cells are not simulated for Test 2. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2 and Cell 3 are powered OFF.

6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between Cell 1 and OTDOA assistance data reference cell, Cell 2; and set to -31 Ts (about -1 μs) between neighbour Cell 3 and OTDOA assistance data reference cell, Cell 2.

**Table 10.2.4.1-1: General test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions for Carrier Aggregation**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCell | | Cell 1 | | PCell is on RF channel 1 (PCC). |
| SCell | | Cell 2 | | SCell on RF channel 2 (SCC). Cell 2 is the assistance data reference cell. |
| Other neighbour cell | | Cell 3 | | Neighbour cell on RF channel 2 (SCC). |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| PRS Transmission Bandwidth Note 2 | RB | 50 | | PRS are transmitted over the system bandwidth |
| PRS configuration index $I_{PRS}$ Note 2 | | 174 for all cells on PCC<br>184 for all cells on SCC | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in 3GPP TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive downlink positioning subframes $N_{PRS}$ Note 2 | | 1 | | As defined in 3GPP TS 36.211 [26]. The number of subframes in a positioning occasion |
| Physical cell ID PCI Note 2 | | (PCI of Cell 2 – PCI of Cell 3)mod6=0 | | The PCI of Cell 1 is selected randomly. PCIs of Cell 2 and Cell 3 are selected randomly such that the relative subcarrier shifts of PRS patterns among these cells are as given by the condition |
| TDD uplink-downlink configuration | | 1 | | As specified in TS 36.211 [26], Section 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |

| | | | | |
|---|---|---|---|---|
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Section 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 10.2.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 1 to Cell 2: 1<br>Cell 3 to Cell 2: -1 | Cell 1 to Cell 2: 1<br>Cell 3 to Cell 2: -1 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | Cell 1: -2<br>Cell 3: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 cells in total | | The list includes the reference cell and 15 other cells.<br>Cell 1 (when included) appears at random places in the first half of the neighbour cell list in the OTDOA assistance data.<br>Cell 3 always appears at random places in the second half of the list |
| | | OTDOA neighbour cells include Cell 3 and other 14 cells on SCC | OTDOA neighbour cells include Cell 1 and other 7 cells on PCC, and Cell 3 and other 6 cells on SCC | |
| prs-SubframeOffset Note2 | | Cells on PCC: 310<br>Cells on SCC, except reference cell: 0 | | Subframe offset, counted in full subframes. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | Cells on PCC: 0<br>Cells on SCC, except reference cell: 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell. The corresponding parameter in the OTDOA assistance data is slotNumberOffset specified in TS 36.355 [4]. |
| PRS muting info Note 2 | | Cell 1: ‘11110000’<br>Cell 2: ‘00001111’<br>Cell 3: ‘11110000’ | Cell 1: ‘1111111100000000’<br>Cell 2: ‘0000000011111111’<br>Cell 3: ‘1111111100000000’ | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T2 |

| | |
|---|---|
| Note 1: | Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.2.4.3-4 and TS 37.571-5 [20], clause 7.3.2. |
| Note 2: | Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive downlink positioning subframes”, “Physical cell ID PCI”, “CP length”, “prs-SubframeOffset”, “slotNumberOffset” and “PRS muting info” are settable parameters and also parameters signalled in LPP. The values to be used for “Physical cell ID PCI” are as follows: Cell 1: 0, Cell 2: 6, Cell 3: 12. For the values to be used in LPP see Table 10.2.4.3-4 and TS 37.571-5 [20], clause 7.3.2. |
| Note 3: | The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is used to set the “true RSTD” values in step 6 of clause 10.2.4.1. |

**Table 10.2.4.1-2: DRX parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions for Carrier Aggregation**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in 3GPP TS 36.331 [22], clause 6.3.2. |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | disable | |

## 10.2.4.2 Test procedure

The test case includes two tests. Test 1 is designed for the scenario where the UE receives OTDOA assistance data with cells on SCC, and the UE is expected to report RSTD measurements performed on SCC only. Test 2 is designed for the scenario where the UE receives OTDOA assistance data with cells on PCC and SCC, and the UE is expected to report RSTD measurements performed on PCC and on SCC.

Each test consists of three consecutive time intervals, with duration of T1, T2 and T3 defined in Table 10.2.4-1. Cell 1 is active in T1, T2 and T3, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. Cell 2 is active until the end of T3, and Cell 3 is active until the end of T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, while Cell 2 transmits PRS only in T3, and Cell 3 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 10.2.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. Configure Cell 2 and Cell 3 on the SCC according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [18] clause 5.2A.4.
4. The SS activates the SCell (Cell 2) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.
5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
6. Set the parameters according to Table 10.2.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).
7. T1 starts.
8. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

9a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

9b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 3 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the last 8 elements of the sequence for Test 1 and in the 7 elements of the relevant sequence for Test 2, and the position of neighbour Cell 1 is randomly selected to be in the first 7 elements of the relevant sequence for Test 2, as described in 3GPP TS 37.571-5 [20], clause 7.3.2. If the UE message at step 9b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 10.2.5-3.

13. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 10.2.5-3.

14. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 10.2.5.

    For Test 1 the UE shall perform and report the RSTD measurement for Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 2. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with the *rstd* field included within the response time then the number of failure tests is increased by one.

    For Test 2 the UE shall perform and report the RSTD measurements for Cell 1 with respect to the reference cell in the OTDOA assistance data, Cell 2 and also Cell 3 with respect to the reference cell in the OTDOA assistance data, Cell 2. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 1 (with respect to Cell 2) and Cell 3 (with respect to Cell 2) within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with both the *rstd* fields included within the response time then the number of failure tests is increased by one.

15. If the UE message at step 14 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

16. Repeat steps 5-15 until the confidence level according to Annex D is achieved. For each iteration, at step 10 change the random positions of the Cell 3 and Cell 1(for Test 2 only) in the relevant sequence in the *OTDOA-NeighbourCellInfoList.*

17. Repeat from clause 10.2.4.1 for Test 2.

### 10.2.4.3 Message contents

**Table 10.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.2.4.3-2: *MAC-MainConfig-RBC*: TDD RSTD Measurement Reporting Delay for Carrier Aggregation**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |

| | | | |
|---|---|---|---|
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 10.2.4.3-2a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | Test 1: 3<br>Test 2: 6 | See clause 10.2.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |

| | | | |
|---|---|---|---|
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.2.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.2.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 2 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbour | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | With respect to Cell 2 | |
| rstd-Quality | | | |
| } | | | |
| neighbourMeasurementList SEQUENCE (SIZE(n)) { | | | |
| physCellIdNeighbour | Cell 1 | Test 2 only | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | Test 2 only<br>With respect to Cell 2 | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.2.5 Test requirement

Table 10.2.5-1 and 10.2.5-2 define the primary level settings including test tolerances for the test.

**Table 10.2.5-1: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T1 for Carrier Aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | N/A | N/A |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |

| | | | | |
|---|---|---|---|---|
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.1 TDD | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | N/A | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 10.2.5-2: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for Carrier Aggregation**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.1 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| PBCH_RA | dB | 0 | | 0 | | 0 | N/A |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| PRS_RA | dB | -6 | N/A | N/A | 3 | 3 | N/A |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -4 | -Infinity | -Infinity | -1 | -8 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -4 | -Infinity | -Infinity | -1 | -8 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.94 | N/A | N/A | -66.68 | -70.11 | N/A |
| PRP Note 4 | dBm/ 15 kHz | -102 | -Infinity | -Infinity | -96 | -106 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -96 | -105 | -99 | -109 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -4 | -11 | -Infinity |
| Propagation Condition | | ETU30 | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 3.3s for Test 1 and 6.3s for Test 2. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 2710 ms for Test 1 and 5110ms for Test 2. This is rounded up to the next allowed LPP value of 3 seconds for Test 1 and 6 seconds for Test 2.

The RSTD measurement reporting delay in the tests is derived from the following expression, $T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$ =8 and $n$ =16 for Test 1, and $M$ =16 and $n$ =16 for Test 2 are the parameters specified in clause 10.2.3.1 for Test 1 and clause 10.2.3.2 for Test 2.

This gives the total RSTD reporting delay of 2560 ms for Test 1 for the 15 neighbour cells including Cell 3 with respect to the reference cell, Cell 2.

This gives the total RSTD reporting delay of 4960 ms for Test 2 for the 15 neighbour cells including Cell 1 and Cell 3 with respect to the reference cell, Cell 2.

The test tolerances are defined in clauses C.1.3 and C.4.

For the overall test to pass, the rate of successful tests during repeated tests in both Test 1 and Test 2 shall be more than 90% with a confidence level of 95%.

# 10.2A TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz Bandwidth

## 10.2A.1 Test purpose

Same as defined in clause 10.2.1.

NOTE: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5.

## 10.2A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.2A.3 Minimum conformance requirements

Same as defined in clause 10.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.4.

## 10.2A.4 Test description

### 10.2A.4.1 Initial conditions

Same as defined in clause 10.2.4.1 except that the values of the parameters in Table 10.2A.4.1-1 will replace the values of the corresponding parameters in Table 10.2.4.1-1.

Channel bandwidth to be tested: 20 MHz.

**Table 10.2A.4.1-1: General test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation for 20 MHz**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 TDD | | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | | |
| PRS Transmission Bandwidth | RB | 100 | | PRS are transmitted over the system bandwidth |
| Note 1: See Table 10.2.4.1-1 for the other parameters.<br>Note 2: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5. | | | | |

### 10.2A.4.2 Test procedure

Same as defined in clause 10.2.4.2.

### 10.2A.4.3 Message contents

Same as defined in clause 10.2.4.3.

## 10.2A.5 Test requirement

Same as defined in clause 10.2.5 except that the values of the parameters in Table 10.2A.5-1 and Table 10.2A.5-2 will replace the values of the corresponding parameters in Table 10.2.5-1 and Table 10.2.5-2, respectively.

**Table 10.2A.5-1: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation for 20 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.7 TDD | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| Io Note 1 | dBm/ 18 MHz | -64.21 | N/A | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-1 for the other parameters. | | | | |

**Table 10.2A.5-2: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation for 20 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.7 TDD | | OP.8 TDD | | OP.8 TDD | N/A |
| Io Note 1 | dBm/ 18 MHz | -66.93 | N/A | N/A | -63.67 | -67.09 | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-2 for the other parameters. | | | | | | | |

# 10.2B TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 5 MHz + 5 MHz Bandwidth

## 10.2B.1 Test purpose

Same as defined in clause 10.2.1.

NOTE: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5.

## 10.2B.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.2B.3 Minimum conformance requirements

Same as defined in clause 10.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.8.

## 10.2B.4 Test description

### 10.2B.4.1 Initial conditions

Same as defined in clause 10.2.4.1 except that the values of the parameters in Table 10.2B.4.1-1 will replace the values of the corresponding parameters in Table 10.2.4.1-1.

Channel bandwidth to be tested: 5 MHz.

**Table 10.2B.4.1-1: General test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | Test 1 | Test 2 | |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 TDD | | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | | |
| PRS Transmission Bandwidth | RB | 25 | | PRS are transmitted over the system bandwidth |
| PRS occasion length $N_{PRS}$ | | 2 | | |
| Note 1: See Table 10.2.4.1-1 for the other parameters.<br>Note 2: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in section 4.7.5. | | | | |

### 10.2B.4.2 Test procedure

Same as defined in clause 10.2.4.2.

### 10.2B.4.3 Message contents

Same as defined in clause 10.2.4.3.

## 10.2B.5 Test requirement

Same as defined in clause 10.2.5 except that the values of the parameters in Table 10.2B.5-1 and Table 10.2B.5-2 will replace the values of the corresponding parameters in Table 10.2.5-1 and Table 10.2.5-2, respectively.

**Table 10.2B.5-1: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.9 TDD | N/A | N/A |
| Io Note 1 | dBm/ 4.5 MHz | -70.23 | N/A | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-1 for the other parameters. | | | | |

**Table 10.2B.5-2: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.9 TDD | | OP.10 TDD | | OP.10 TDD | N/A |
| Io Note 1 | dBm/ 4.5 MHz | -72.95 | N/A | N/A | -69.69 | -73.12 | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-2 for the other parameters. | | | | | | | |

# 10.2C TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz + 5 MHz Bandwidth

## 10.2C.1 Test purpose

Same as defined in clause 10.2.1.

NOTE: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5.

## 10.2C.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.2C.3 Minimum conformance requirements

Same as defined in clause 10.2.3.

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.6.

## 10.2C.4 Test description

### 10.2C.4.1 Initial conditions

Same as defined in clause 10.2.4.1 except that the values of the parameters in Table 10.2C.4.1-1 will replace the values of the corresponding parameters in Table 10.2.4.1-1.

Channel bandwidth to be tested: Cell 1: 10 MHz, Cell 2 and Cell 3: 5 MHz.

**Table 10.2C.4.1-1: General test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | Cell 1: R.6 TDD<br>Cell 2: R.11 TDD<br>Cell 3: R.11 TDD | Cell 1: R.6 TDD<br>Cell 2: R.11 TDD<br>Cell 3: R.11 TDD | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | Cell 1: 10<br>Cell 2: 5<br>Cell 3: 5 | Cell 1: 10<br>Cell 2: 5<br>Cell 3: 5 | |
| PRS Transmission Bandwidth | RB | Cell 1: 50<br>Cell 2: 25<br>Cell 3: 25 | Cell 1: 50<br>Cell 2: 25<br>Cell 3: 25 | PRS are transmitted over the system bandwidth |
| PRS occasion length $N_{PRS}$ | | Cell 1: 1<br>Cell 2: 2<br>Cell 3: 2 | Cell 1: 1<br>Cell 2: 2<br>Cell 3: 2 | |
| Note 1: See Table 10.2.4.1-1 for the other parameters.<br>Note 2: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in section 4.7.5. | | | | |

### 10.2C.4.2 Test procedure

Same as defined in clause 10.2.4.2.

### 10.2C.4.3 Message contents

Same as defined in clause 10.2.4.3.

## 10.2C.5 Test requirement

Same as defined in clause 10.2.5 except that the values of the parameters in Table 10.2C.5-1 and Table 10.2C.5-2 will replace the values of the corresponding parameters in Table 10.2.5-1 and Table 10.2.5-2, respectively.

**Table 10.2C.5-1: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| Io Note 1 | dBm/ 9 MHz | -67.22 | N/A | N/A |
| | dBm/ 4.5 MHz | N/A | N/A | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-1 for the other parameters. | | | | |

**Table 10.2C.5-2: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |
| OCNG patterns defined in TS 36.521-3 [25] clause D.2 | | OP.1 TDD | | OP.10 TDD | | OP.10 TDD | N/A |
| Io Note 1 | dBm/ 9 MHz | -69.94 | N/A | N/A | N/A | N/A | N/A |
| | dBm/ 4.5 MHz | N/A | N/A | N/A | -69.69 | -73.12 | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-2 for the other parameters. | | | | | | | |

# 10.2D TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz +10 MHz Bandwidth

## 10.2D.1 Test purpose

Same as defined in clause 10.2.1

NOTE: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in clause 4.7.5.

## 10.2D.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.2D.3 Minimum conformance requirements

Same as defined in clause 10.2.3

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4 and A.8.17.9.

# 10.2D.4 Test description

## 10.2D.4.1 Initial conditions

Same as defined in clause 10.2.4.1 except that the values of the parameters in Table 10.2D.4.1-1 will replace the values of the corresponding parameters in Table 10.2.4.1-1.

Channel bandwidth to be tested: Cell 1: 20 MHz, Cell 2 and Cell 3: 10 MHz.

**Table 10.2D.4.1-1: General test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation for 20 MHz+10 MHz**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | Cell 1: R.10 TDD<br>Cell 2: R.6 TDD<br>Cell 3: R.6 TDD | Cell 1: R.10 TDD<br>Cell 2: R.6 TDD<br>Cell 3: R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | Cell 1: 20<br>Cell 2: 10<br>Cell 3: 10 | Cell 1: 20<br>Cell 2: 10<br>Cell 3: 10 | |
| PRS Transmission Bandwidth | RB | Cell 1: 100<br>Cell 2: 50<br>Cell 3: 50 | Cell 1: 100<br>Cell 2: 50<br>Cell 3: 50 | PRS are transmitted over the system bandwidth |
| Note 1: See Table 10.2.4.1-1 for the other parameters.<br>Note 2: This test verifies the requirement which is independent of channel bandwidth and is performed according to the principle defined in section 4.7.5. | | | | |

## 10.2D.4.2 Test procedure

Same as defined in clause 10.2.4.2

## 10.2D.4.3 Message contents

Same as defined in clause 10.2.4.3

# 10.2D.5 Test requirement

Same as defined in clause 10.2.5 except that the values of the parameters in Table 10.2D.5-1 and Table 10.2D.5-2 will replace the values of the corresponding parameters in Table 10.2.5-1 and Table 10.2.5-2, respectively.

**Table 10.2D.5-1: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation for 20 MHz+10 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| Io Note 1 | dBm/ 18 MHz | -64.21 | N/A | N/A |
| | dBm/ 9 MHz | N/A | N/A | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-1 for the other parameters. | | | | |

**Table 10.2D.5-2: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation for 20 MHz+10 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| OCNG patterns defined in TS36.521-3 clause D.2 | | OP.7 TDD | | OP.2 TDD | | OP.2 TDD | N/A |
| Io Note 1 | dBm/ 18 MHz | -66.93 | N/A | N/A | N/A | N/A | N/A |
| | dBm/ 9 MHz | N/A | N/A | N/A | -66.68 | -70.11 | N/A |
| Note 1: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 2: See Table 10.2.5-2 for the other parameters. | | | | | | | |

# 10.3 FDD RSTD Measurement Accuracy for Carrier Aggregation

## 10.3.1 Test purpose

To verify that the FDD RSTD measurement accuracy is within the specified limits when both the reference cell and neighbouring cell belong to the secondary component carrier.

## 10.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.3.3 Minimum conformance requirements

The UE may operate in either E-UTRA inter-band or intra-band carrier aggregation mode. The requirements in this section shall apply regardless whether the configured downlink secondary cell is activated or deactivated by the MAC-CE command (3GPP TS 36.321 [34]). The requirements apply for bandwidths defined in the bandwidth combination set for the CA configurations supported by the UE as defined in TS 36.101 [2].

The RSTD measurements, which are obtained when both the reference cell and neighbouring cell belong to the secondary component carrier, shall meet the intra-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.5.

## 10.3.4 Test description

### 10.3.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: 10 MHz.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as follows:

   - For UEs supporting only 2Rx in all the bands under test, use TS 36.508 [18] Annex A, Figure group A.41 as appropriate.

   - For UEs supporting 4Rx in any of the bands under test use TS 36.508 [18] Annex A, Figure A.90. Use the 2Rx module for cells on bands supporting 2Rx and the 4Rx module for cells on bands supporting 4Rx.

2. The general test parameter settings are set up according to Table 10.3.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 10.3.4.3.

5. There are three synchronized cells on two different carrier frequencies. Cell 1 is the PCell on primary component carrier F1 (RF channel number 1), Cell 2 is the SCell and OTDOA assistance data reference cell on secondary component carrier F2 (RF channel number 2), and Cell 3 is the neighbour cell on F2. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2 and Cell 3 are powered OFF.

Cell 3 is included in the OTDOA assistance data neighbour cell list, whilst Cell 1 is not included in the OTDOA assistance data. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.3.2).

Note that the measurement gap is not configured in the test because of UE carrier aggregation capability.

6. The true RSTD (which is the receive time difference for frame 0 between the two cells as seen at the UE antenna connector) is set to 31 Ts ( about 1 μs) between neighbour Cell 3 and OTDOA assistance data reference cell, Cell 2.

Note that the related expectedRSTD value to be signalled over LPP is defined in Table 10.3.4.1-1.

**Table 10.3.4.1-1: General Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | R.6 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.6 FDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Assistance data reference cell | | Cell 2 | Cell 2 is the SCell on RF channel number 2 |
| PCell | | Cell 1 | Cell 1 on RF channel number 1 |
| Neighbour cell | | Cell 3 | Cell 3 on RF channel number 2 |
| E-UTRA RF Channel Number | | 1,2 | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| PRS Transmission Bandwidth Note 2 | RB | 50 | PRS Bandwidth: bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| PRS configuration Index $I_{PRS}$ Note 2 | | 2 | As defined in 3GPP TS 36.211 [26] |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 1 | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1:'11110000'<br>Cell 2:'11110000'<br>Cell 3:'11110000' | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 2 – Cell ID of cell 3) mod 6 = 3 | PCI of cell 1 is selected randomly. |
| Expected RSTD Note 1 | μs | Cell 3: -2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | μs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | μs | Cell 1 to Cell 2: -1<br>Cell 3 to Cell 2: 1 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the assistance-data-reference cell and 15 other cells. All cells provided in OTDOA assistance data are on RF channel 2. |
| $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ Note 4 | ms | 2560 | Derived according to the RSTD measurement requirements specified in Section 10.1.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.3.4.3-3 and TS 37.571-5 [20], clause 7.3.2.
NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: 7, Cell 3: 10. For the values to be used in LPP see Table 10.3.4.3-3 and TS 37.571-5 [20], clause 7.3.2.
NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is used to set the “true RSTD” value in step 6 of clause 10.3.4.1.
NOTE 4: The parameter “$T_{\text{RSTD IntraFreqFDD, E-UTRAN}}$ ” is not a settable parameter but is used to set the LPP “time” value in Table 10.3.4.3-2. The value of the LPP time IE is set to $T_{\text{RSTD IntraFreqFDD, E-UTRAN}}$ + ΔT ms, where ΔT = 150 ms, giving a value of 2710 ms. This is rounded up to the next allowed LPP value of 3 seconds.

## 10.3.4.2 Test procedure

The RSTD measurements are performed between Cell 2 and Cell 3 to verify that when both the reference cell and neighbouring cell belong to the secondary component carrier the RSTD measurement accuracy can meet the intra-frequency RSTD accuracy requirements defined in section 10.3.3.

The test consists of a set-up period and a measurement period. All cells are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 10.3.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. Configure Cell 2 and Cell 3 on the SCC according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [18] clause 5.2A.4.
4. The SS activates the SCell (Cell 2) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.
5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
6. Set the parameters according to Table 10.3.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

6a. The SS shall send an LPP REQUEST CAPABILITIES message.

6b. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.
9. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

10. If the UE message at step 9 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

11. The SS shall check the *rstd* value for Cell 3 in the *OTDOA-SignalMeasurementInformation* IE according to Table 10.3.5-2.

12. Repeat step 5-11 until the confidence level according to Annex D is achieved.

### 10.3.4.3 Message contents

**Table 10.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.3.4.3-1a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.3.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 3 | See Note 4 of Table 10.3.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.3.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.3.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |

| | | | |
|---|---|---|---|
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 2 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(1)) { | | | |
| physCellIdNeighbour | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 10.3.5-2 | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.3.5 Test requirement

Table 10.3.5-1 defines the primary level settings including test tolerances for the test.

The FDD RSTD accuracy test shall meet the reported values in Table 10.3.5-2.

**Table 10.3.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 2 |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA[Note 1] | | | | |

| | | | | |
|---|---|---|---|---|
| OCNG_RB Note 1 | | | | |
| PRS_RA | dB | -3 | 0.3 | 0.3 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -6 | -5.7 | -12.7 |
| PRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -6 | -5.7 | -12.7 |
| Io Note 3 | dBm/9 MHz | -70.04 | -69.99 | -69.99 |
| PRP Note 3 | dBm/15 kHz | -104 | -103.7 | -110.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -3 | -6 | -13 |
| RSRP Note 3 | dBm/15 kHz | -101 | -104 | -111 |
| Propagation condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS.

**Table 10.3.5-2: RSTD FDD accuracy requirements for the reported values for Carrier Aggregation**

| | Value |
|---|---|
| Lowest reported value | RSTD_6380 |
| Highest reported value | RSTD_6392 |

For the test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

# 10.3A FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-10 and Rel-11)

## 10.3A.1 Test purpose

Same as defined in clause 10.3.1.

## 10.3A.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and 11 that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.3A.3 Minimum conformance requirements

Same as defined in clause 10.3.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.7.

## 10.3A.4 Test description

### 10.3A.4.1 Initial conditions

Same as defined in clause 10.3.4.1 except that the values of the parameters in Table 10.3A.4.1-1 will replace the values of the corresponding parameters in Table 10.3.4.1-1.

Channel bandwidth to be tested: 20 MHz.

**Table 10.3A.4.1-1: General Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation for 20 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | R.10 FDD | As specified in clause TS 36.521-3 [25] clause A.2.1 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.14 FDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | |
| PRS Bandwidth | RB | 100 | PRS Bandwidth bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| Note 1: See Table 10.3.4.1-1 for other general test parameters. | | | |

### 10.3A.4.2 Test procedure

Same as defined in clause 10.3.4.2.

### 10.3A.4.3 Message contents

Same as defined in clause 10.3.4.3.

## 10.3A.5 Test requirement

Same as defined in clause 10.3.5 except that the value of the parameter in Table 10.3A.5-1 will replace the value of the corresponding parameter in Table 10.3.5-1.

**Table 10.3A.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation for 20 MHz**

| Parameter | Unit | Cell1 | Cell2 | Cell3 |
|---|---|---|---|---|
| Io Note1 | dBm/18 MHz | -67.03 | -66.98 | -66.98 |
| Note 1: Io level has been derived from other parameters for information purposes. It is not settable parameter itself. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS<br>Note 2: See Table 10.3.5-1 for other cell specific test parameters. | | | | |

# 10.3A_1 FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-12 onwards)

## 10.3A_1.1 Test purpose

Same as defined in clause 10.3A.1.

## 10.3A_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.3A_1.3 Minimum conformance requirements

Same as defined in clause 10.3A.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.7.

## 10.3A_1.4 Test description

### 10.3A_1.4.1 Initial conditions

Same as defined in clause 10.3A.4.1.

### 10.3A_1.4.2 Test procedure

Same as defined in clause 10.3A.4.2.

### 10.3A_1.4.3 Message contents

Same as defined in clause 10.3A.4.3.

## 10.3A_1.5 Test requirement

Same as defined in clause 10.3A.5 except that in addition Table 10.3A_1.5-1 will replace Table 10.3.5-2.

**Table 10.3A_1.5-1: RSTD FDD accuracy requirements for the reported values for Carrier Aggregation**

| | Value |
|---|---|
| Lowest reported value | RSTD_6381 |
| Highest reported value | RSTD_6391 |

# 10.3B FDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz + 5 MHz Bandwidth

## 10.3B.1 Test purpose

Same as defined in clause 10.3.1.

## 10.3B.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.3B.3 Minimum conformance requirements

Same as defined in clause 10.3.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.11.

## 10.3B.4 Test description

### 10.3B.4.1 Initial conditions

Same as defined in clause 10.3.4.1 except that the values of the parameters in Table 10.3B.4.1-1 will replace the values of the corresponding parameters in Table 10.3.4.1-1.

Channel bandwidth to be tested: 5 MHz.

**Table 10.3B.4.1-1: General Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | R.11 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | OP.19 FDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). There is no PDSCH allocated in the subframe transmitting PRS. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| PRS Bandwidth | RB | 25 | PRS Bandwidth bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ | | 2 | As defined in 3GPP TS 36.211 [26] |
| Note 1: See Table 10.3.4.1-1 for other general test parameters. | | | |

## 10.3B.4.2 Test procedure

Same as defined in clause 10.3.4.2.

## 10.3B.4.3 Message contents

Same as defined in clause 10.3.4.3.

# 10.3B.5 Test requirement

Same as defined in clause 10.3.5 except that the value of the parameter in Table 10.3B.5-1 will replace the value of the corresponding parameter in Table 10.3.5-1 and the FDD RSTD accuracy shall meet the reported values in Table 10.3B.5-2.

**Table 10.3B.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Cell1 | Cell2 | Cell3 |
|---|---|---|---|---|
| Io Note1 | dBm/4.5 MHz | -73.05 | -73.00 | -73.00 |
| Note 1: Io level has been derived from other parameters for information purposes. It is not settable parameter itself. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS<br>Note 2: See Table 10.3.5-1 for other cell specific test parameters. | | | | |

**Table 10.3B.5-2: RSTD FDD accuracy requirements for the reported values for Carrier Aggregation for 5 MHz+5 MHz bandwidth**

| | Value |
|---|---|
| Lowest reported value | RSTD_6379 |
| Highest reported value | RSTD_6393 |

# 10.3C FDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz + 5 MHz Bandwidth

## 10.3C.1 Test purpose

Same as defined in clause 10.3.1.

## 10.3C.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.3C.3 Minimum conformance requirements

Same as defined in clause 10.3.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.9.

## 10.3C.4 Test description

### 10.3C.4.1 Initial conditions

Same as defined in clause 10.3.4.1 except that the values of the parameters in Table 10.3C.4.1-1 will replace the values of the corresponding parameters in Table 10.3.4.1-1.

Channel bandwidth to be tested: Cell 1: 10 MHz, Cell 2 and Cell 3: 5 MHz.

**Table 10.3C.4.1-1: General Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | Cell1: R.6 FDD<br>Cell2: R.11 FDD<br>Cell3: R.11 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 | | Cell1: OP.6 FDD<br>Cell2: OP.19 FDD<br>Cell3: OP.19 FDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). There is no PDSCH allocated in the subframe transmitting PRS. |
| Channel Bandwidth ($BW_{channel}$) | MHz | Cell1: 10<br>Cell2: 5<br>Cell3: 5 | |
| PRS Bandwidth | RB | Cell1: 50<br>Cell2: 25<br>Cell3: 25 | PRS Bandwidth bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ | | 2 | As defined in 3GPP TS 36.211 [26] |
| Note 1: See Table 10.3.4.1-1 for other general test parameters. | | | |

### 10.3C.4.2 Test procedure

Same as defined in clause 10.3.4.2.

### 10.3C.4.3 Message contents

Same as defined in clause 10.3.4.3.

## 10.3C.5 Test requirement

Same as defined in clause 10.3.5 except that the value of the parameter in Table 10.3C.5-1 will replace the value of the corresponding parameter in Table 10.3.5-1 and the FDD RSTD accuracy shall meet the reported values in Table 10.3C.5-2.

**Table 10.3C.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation for 10 MHz +5 MHz**

| Parameter | Unit | Cell1 | Cell2 | Cell3 |
|---|---|---|---|---|
| Io Note1 | dBm/9 MHz | -70.04 | N/A | N/A |
| | dBm/4.5 MHz | N/A | -73.00 | -73.00 |
| Note 1: Io level has been derived from other parameters for information purposes. It is not settable parameter itself. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS<br>Note 2: See Table 10.3.5-1 for other cell specific test parameters. | | | | |

**Table 10.3C.5-2: RSTD FDD accuracy requirements for the reported values for Carrier Aggregation for 10 MHz+5 MHz bandwidth**

| | Value |
|---|---|
| Lowest reported value | RSTD_6379 |
| Highest reported value | RSTD_6393 |

# 10.4 TDD RSTD Measurement Accuracy for Carrier Aggregation

## 10.4.1 Test purpose

To verify that the TDD RSTD measurement accuracy is within the specified limits when both the reference cell and neighbouring cell belong to the secondary component carrier.

## 10.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.4.3 Minimum conformance requirements

The UE may operate in either E-UTRA inter-band or intra-band carrier aggregation mode. The requirements in this section shall apply regardless whether the configured downlink secondary cell is activated or deactivated by the MAC-CE command (3GPP TS 36.321 [34]). The requirements apply for bandwidths defined in the bandwidth combination set for the CA configurations supported by the UE as defined in TS 36.101 [2].

The RSTD measurements, which are obtained when both the reference cell and neighbouring cell belong to the secondary component carrier, shall meet the intra-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.6.

## 10.4.4 Test description

### 10.4.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.2.

Channel bandwidth to be tested: 10 MHz.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as follows:

   - For UEs supporting only 2Rx in all the bands under test, use TS 36.508 [18] Annex A, Figure group A.41 as appropriate.

   - For UEs supporting 4Rx in any of the bands under test use TS 36.508 [18] Annex A, Figure A.90. Use the 2Rx module for cells on bands supporting 2Rx and the 4Rx module for cells on bands supporting 4Rx.

2. The general test parameter settings are set up according to Table 10.4.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 10.4.4.3.

5. There are three synchronized cells on two different carrier frequencies. Cell 1 is the PCell on primary component carrier F1 (RF channel number 1), Cell 2 is the SCell and OTDOA assistance data reference cell on secondary component carrier F2 (RF channel number 2), and Cell 3 is the neighbour cell on F2. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2 and Cell 3 are powered OFF.

Cell 3 is included in the OTDOA assistance data neighbour cell list, whilst Cell 1 is not included in the OTDOA assistance data. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.3.2).

Note that the measurement gap is not configured in the test because of UE carrier aggregation capability.

6. The true RSTD (which is the receive time difference for frame 0 between the two cells as seen at the UE antenna connector) is set to 31 Ts ( about 1 μs) between neighbour Cell 3 and OTDOA assistance data reference cell, Cell 2.

Note that the related expectedRSTD value to be signalled over LPP is defined in Table 10.4.4.1-1.

**Table 10.4.4.1-1: General Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | R.6 TDD | As specified in TS 36.521-3 [25] clause A.2.2 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | OP.2 TDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Assistance data reference cell | | Cell 2 | Cell 2 is the SCell on RF channel number 2 |
| PCell | | Cell 1 | Cell 1 on RF channel number 1 |
| Neighbour cell | | Cell 3 | Cell 3 on RF channel number 2 |
| E-UTRA RF Channel Number | | 1,2 | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [26] and table 8.1.2.5.2-2 in TS 36.133 [23]. The same configuration in both cells. |
| PRS Transmission Bandwidth Note 2 | RB | 50 | PRS Bandwidth: bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |

| | | | |
|---|---|---|---|
| PRS configuration Index $I_{PRS}$ Note 2 | | Cell 1: 14<br>Cell 2: 14<br>Cell 3: 14 | As defined in 3GPP TS 36.211 [26] |
| Number of consecutive positioning downlink subframes $N_{PRS}$ Note 2 | | 1 | As defined in 3GPP TS 36.211 [26] |
| prs-MutingInfo Note 2 | | Cell 1:'11110000'<br>Cell 2:'11110000'<br>Cell 3:'11110000' | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 2 – Cell ID of cell 3) mod 6 = 3 | PCI of cell 1 is selected randomly. |
| Expected RSTD Note 1 | µs | Cell 3: -2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 1 to Cell 2: -1<br>Cell 3 to Cell 2: 1 | PRS are transmitted from synchronous cells |
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the assistance-data-reference cell and 15 other cells. All cells provided in OTDOA assistance data are on RF channel 2. |
| $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ Note 4 | ms | 2560 | Derived according to the RSTD measurement requirements specified in Section 10.2.3 |

NOTE 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.4.4.3-3 and TS 37.571-5 [20], clause 7.3.2.

NOTE 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive positioning downlink subframes", "prs-MutingInfo", "Cell ID" and "CP length" are settable parameters and also parameters signalled in LPP. The values to be used for "Cell ID" are as follows: Cell 1: 0, Cell 2: 7, Cell 3: 10. For the values to be used in LPP see Table 10.4.4.3-3 and TS 37.571-5 [20], clause 7.3.2.

NOTE 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is used to set the "true RSTD" value in step 6 of clause 10.4.4.1.

NOTE 4: The parameter "$T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$" is not a settable parameter but is used to set the LPP "time" value in Table 10.4.4.3-2. The value of the LPP time IE is set to $T_{RSTD\ IntraFreqFDD,\ E\text{-}UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 2710 ms. This is rounded up to the next allowed LPP value of 3 seconds.

## 10.4.4.2 Test procedure

The RSTD measurements are performed between Cell 2 and Cell 3 to verify that when both the reference cell and neighbouring cell belong to the secondary component carrier the RSTD measurement accuracy can meet the intra-frequency RSTD accuracy requirements defined in section 10.4.3.

The test consists of a set-up period and a measurement period. All Cells are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 10.4.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.

2. Configure Cell 2 and Cell 3 on the SCC according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [18] clause 5.2A.

4. The SS activates the SCell (Cell 2) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.

5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

6. Set the parameters according to Table 10.4.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

6a. The SS shall send an LPP REQUEST CAPABILITIES message.

6b. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.

9. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

10. If the UE message at step 9 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

11. The SS shall check the *rstd* value for Cell 3 in the *OTDOA-SignalMeasurementInformation* IE according to Table 10.4.5-2.

12. Repeat step 5-11 until the confidence level according to Annex D is achieved.

### 10.4.4.3 Message contents

**Table 10.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.4.4.3-1a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.4.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 3 | See Note 4 of Table 10.4.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.4.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.4.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 2 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE (SIZE(1)) { | | | |
| physCellIdNeighbour | Cell 3 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 10.4.5-2 | | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason ‘undefined’ or ‘attemptedButUnableToMeasureSomeNeighbourCells’ | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.4.5 Test requirement

Table 10.4.5-1 defines the primary level settings including test tolerances for the test.

The TDD RSTD accuracy test shall meet the reported values in Table 10.4.5-2.

**Table 10.4.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 2 |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| PRS_RA | dB | -3 | 0.3 | 0.3 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -6 | -5.7 | -12.7 |
| PRS $\hat{E}_s/I_{ot}$ Note 3 | dB | -6 | -5.7 | -12.7 |
| Io Note 3 | dBm/9 MHz | -70.04 | -69.99 | -69.99 |
| PRP Note 3 | dBm/15 kHz | -104 | -103.7 | -110.7 |
| $\hat{E}_s/N_{oc}$ Note 3 | dB | -3 | -6 | -13 |
| RSRP Note 3 | dBm/15 kHz | -101 | -104 | -111 |
| Propagation condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS.

**Table 10.4.5-2: RSTD TDD accuracy requirements for the reported values for Carrier Aggregation**

| | Value |
|---|---|
| Lowest reported value | RSTD_6380 |
| Highest reported value | RSTD_6392 |

For the test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95%.

# 10.4A TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-10 and Rel-11)

## 10.4A.1 Test purpose

Same as defined in clause 10.4.1.

## 10.4A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and 11 that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.4A.3 Minimum conformance requirements

Same as defined in clause 10.4.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.8.

## 10.4A.4 Test description

### 10.4A.4.1 Initial conditions

Same as defined in clause 10.4.4.1 except that the values of the parameters in Table 10.4A.4.1-1 will replace the values of the corresponding parameters in Table 10.4.4.1-1.

Channel bandwidth to be tested: 20 MHz.

**Table 10.4A.4.1-1: General Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 20 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | R.10 TDD | As specified in clause TS 36.521-3 [25] clause A.2.2 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | OP.8 TDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | |
| PRS Bandwidth | RB | 100 | PRS Bandwidth: bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| Note 1: See Table 10.4.4.1-1 for other general test parameters. | | | |

### 10.4A.4.2 Test procedure

Same as defined in clause 10.4.4.2.

### 10.4A.4.3 Message contents

Same as defined in clause 10.4.4.3.

## 10.4A.5 Test requirement

Same as defined in clause 10.4.5 except that the value of the parameter in Table 10.4A.5-1 will replace the value of the corresponding parameter in Table 10.4.5-1.

**Table 10.4A.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 20 MHz**

| Parameter | Unit | Cell1 | Cell2 | Cell3 |
|---|---|---|---|---|
| Io Note1 | dBm/18 MHz | -67.03 | -66.98 | -66.98 |
| Note 1: Io level has been derived from other parameters for information purposes. It is not settable parameter itself. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS<br>Note 2: See Table 10.4.5-1 for other cell specific test parameters. | | | | |

# 10.4A_1 TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz Bandwidth (Rel-12 onwards)

## 10.4A_1.1 Test purpose

Same as defined in clause 10.4A.1.

## 10.4A_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.4A_1.3 Minimum conformance requirements

Same as defined in clause 10.4A.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.8.

## 10.4A_1.4 Test description

### 10.4A_1.4.1 Initial conditions

Same as defined in clause 10.4A.4.1.

### 10.4A_1.4.2 Test procedure

Same as defined in clause 10.4A.4.2.

### 10.4A_1.4.3 Message contents

Same as defined in clause 10.4A.4.3.

## 10.4A_1.5 Test requirement

Same as defined in clause 10.4A.5 except that in addition Table 10.4A_1.5-1 will replace Table 10.4.5-2.

**Table 10.4A_1.5-1: RSTD FDD accuracy requirements for the reported values for Carrier Aggregation**

| | Value |
|---|---|
| Lowest reported value | RSTD_6381 |
| Highest reported value | RSTD_6391 |

# 10.4B TDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz + 5 MHz bandwidth

## 10.4B.1 Test purpose

Same as defined in clause 10.4.1.

## 10.4B.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.4B.3 Minimum conformance requirements

Same as defined in clause 10.4.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.12.

## 10.4B.4 Test description

### 10.4B.4.1 Initial conditions

Same as defined in clause 10.4.4.1 except that the values of the parameters in Table 10.4B.4.1-1 will replace the values of the corresponding parameters in Table 10.4.4.1-1.

Channel bandwidth to be tested: 5 MHz.

**Table 10.4B.4.1-1: General Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 5 MHz + 5 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | R.11 TDD | As specified in clause TS 36.521-3 [25] clause A.2.2 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | OP.10 TDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). There is no PDSCH allocated in the subframe transmitting PRS. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| PRS Bandwidth | RB | 25 | PRS Bandwidth bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{PRS}$ | | 2 | As defined in 3GPP TS 36.211 [26] |
| Note 1: See Table 10.4.4.1-1 for other general test parameters. | | | |

### 10.4B.4.2 Test procedure

Same as defined in clause 10.4.4.2.

### 10.4B.4.3 Message contents

Same as defined in clause 10.4.4.3.

## 10.4B.5 Test requirement

Same as defined in clause 10.4.5 except that the value of the parameter in Table 10.4B.5-1 will replace the value of the corresponding parameter in Table 10.4.5-1 and the TDD RSTD accuracy shall meet the reported values in Table 10.4B.5-2.

**Table 10.4B.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 5 MHz + 5 MHz bandwidth**

| Parameter | Unit | Cell1 | Cell2 | Cell3 |
|---|---|---|---|---|
| Io Note1 | dBm/4.5 MHz | -73.05 | -73.00 | -73.00 |
| Note 1: Io level has been derived from other parameters for information purposes. It is not settable parameter itself. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS<br>Note 2: See Table 10.4.5-1 for other cell specific test parameters. | | | | |

**Table 10.4B.5-2: RSTD TDD accuracy requirements for the reported values for Carrier Aggregation for 5 MHz+5 MHz bandwidth**

| | Value |
|---|---|
| Lowest reported value | RSTD_6379 |
| Highest reported value | RSTD_6393 |

# 10.4C TDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz + 5 MHz Bandwidth

## 10.4C.1 Test purpose

Same as defined in clause 10.4.1.

## 10.4C.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.4C.3 Minimum conformance requirements

Same as defined in clause 10.4.3.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.10.

## 10.4C.4 Test description

### 10.4C.4.1 Initial conditions

Same as defined in clause 10.4.4.1 except that the values of the parameters in Table 10.4C.4.1-1 will replace the values of the corresponding parameters in Table 10.4.4.1-1.

Channel bandwidth to be tested: Cell 1: 10 MHz, Cell 2 and Cell 3: 5 MHz.

**Table 10.4C.4.1-1: General Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | Cell1: R.6 TDD<br>Cell2: R.11 TDD<br>Cell3: R.11 TDD | As specified in clause TS 36.521-3 [25] clause A.2.2 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | Cell1: OP.2 TDD<br>Cell2: OP.10 TDD<br>Cell3: OP.10 TDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). There is no PDSCH allocated in the subframe transmitting PRS. |
| Channel Bandwidth ($BW_{channel}$) | MHz | Cell1: 10<br>Cell2: 5<br>Cell3: 5 | |
| PRS Bandwidth | RB | Cell1: 50<br>Cell2: 25<br>Cell3: 25 | PRS Bandwidth bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| Number of consecutive positioning downlink subframes $N_{\mathrm{PRS}}$ | | 2 | As defined in 3GPP TS 36.211 [26] |
| Note 1: See Table 10.4.4.1-1 for other general test parameters. | | | |

## 10.4C.4.2 Test procedure

Same as defined in clause 10.4.4.2.

## 10.4C.4.3 Message contents

Same as defined in clause 10.4.4.3.

# 10.4C.5 Test requirement

Same as defined in clause 10.4.5 except that the value of the parameter in Table 10.4C.5-1 will replace the value of the corresponding parameter in Table 10.4.5-1 and the TDD RSTD accuracy shall meet the reported values in Table 10.4C.5-2.

**Table 10.4C.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 10 MHz + 5 MHz**

| Parameter | Unit | Cell1 | Cell2 | Cell3 |
|---|---|---|---|---|
| Io Note1 | dBm/9 MHz | -70.04 | N/A | N/A |
| | dBm/4.5 MHz | N/A | -73.00 | -73.00 |
| Note 1: Io level has been derived from other parameters for information purposes. It is not settable parameter itself. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS<br>Note 2: See Table 10.4.5-1 for other cell specific test parameters. | | | | |

**Table 10.4C.5-2: RSTD TDD accuracy requirements for the reported values for Carrier Aggregation for 10 MHz+5 MHz bandwidth**

| | Value |
|---|---|
| Lowest reported value | RSTD_6379 |
| Highest reported value | RSTD_6393 |

# 10.4D TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz+10 MHz Bandwidth

## 10.4D.1 Test purpose

Same as defined in clause 10.4.1

## 10.4D.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that supports UE-assisted OTDOA for Carrier Aggregation.

## 10.4D.3 Minimum conformance requirements

Same as defined in clause 10.4.3

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.13.

## 10.4D.4 Test description

### 10.4D.4.1 Initial conditions

Same as defined in clause 10.4.4.1 except that the values of the parameters in Table 10.4D.4.1-1 will replace the values of the corresponding parameters in Table 10.4.4.1-1.

Channel bandwidth to be tested: Cell 1: 20 MHz, Cell 2 and Cell 3: 10 MHz.

**Table 10.4D.4.1-1: General Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 20 MHz+10 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | Cell 1: R.10 TDD<br>Cell 2: R.6 TDD<br>Cell 3: R.6 TDD | As specified in clause TS 36.521-3 [25] clause A.2.2 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.2 | | Cell 1: OP.8 TDD<br>Cell 2: OP.2 TDD<br>Cell 3: OP.2 TDD | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). There is no PDSCH allocated in the subframe transmitting PRS. |
| Channel Bandwidth ($BW_{channel}$) | MHz | Cell 1: 20<br>Cell 2: 10<br>Cell 3: 10 | |
| PRS Bandwidth | RB | Cell 1: 100<br>Cell 2: 50<br>Cell 3: 50 | PRS Bandwidth bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS36.355 [4]. |
| Note 1: See Table 10.4.4.1-1 for other general test parameters. | | | |

### 10.4D.4.2 Test procedure

Same as defined in clause 10.4.4.2

### 10.4D.4.3 Message contents

Same as defined in clause 10.4.4.3

## 10.4D.5 Test requirement

Same as defined in clause 10.4.5 except that the value of the parameter in Table 10.4D.5-1 will replace the value of the corresponding parameter in Table 10.4.5-1.

**Table 10.4D.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation for 20 MHz+10 MHz**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| Io Note1 | dBm/ 18 MHz | -67.03 | N/A | N/A |
| | dBm/ 9 MHz | N/A | -69.99 | -69.99 |
| Note 1: Io level has been derived from other parameters for information purposes. It is not settable parameter itself. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS<br>Note 2: See Table 10.4.5-1 for other cell specific test parameters. | | | | |

# 10.5 FDD 3 DL CA RSTD Measurement Reporting Delay

## 10.5.1 Test Purpose

The purpose of the test case is to verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions. This test case verifies the measurement period requirements for RSTD measurements performed on the same secondary component carrier, the measurement period requirements for RSTD measurements performed on both the primary and secondary component carriers, and also the measurement period requirements for RSTD measurements performed on different secondary component carriers.

## 10.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports UE-assisted OTDOA for 3DL Carrier Aggregation.

## 10.5.3 Minimum conformance requirements

### 10.5.3.1 Measurements on the secondary component carrier

The RSTD measurements when all cells are on the configured secondary component carrier shall meet all applicable requirements (FDD) specified in TS 36.133 [23] section 8.1.2.5, i.e., E-UTRAN intra-frequency RSTD measurement period applies, regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34].

### 10.5.3.2 Measurements on both primary component carrier and secondary component carrier

The RSTD measurements of cells on both primary component carrier and configured secondary component carrier shall meet all applicable requirements (FDD) specified in TS 36.133 [23] section 8.1.2.6, i.e., E-UTRAN inter-frequency RSTD measurement period applies regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34], with the following exception

- the number of PRS positioning occasions is as specified in Table 10.5.3.2-1 shall apply.

**Table 10.5.3.2-1: Number of PRS positioning occasions within measurement period**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| 160 ms | 32 |

| | |
|---|---|
| >160 ms | 16 |

### 10.5.3.3 Measurements on different secondary component carriers

The RSTD measurements of cells on a configured secondary component carrier and another configured secondary component carrier shall meet all applicable requirements (FDD-FDD inter-Frequency OTDOA) specified in TS 36.133 [23] section 8.1.2.6, i.e., E-UTRAN inter-frequency RSTD measurement period applies regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34], with the following exceptions

- the number of PRS positioning occasions is as specified in Table 10.5.3.3-1 shall apply.

**Table 10.5.3.3-1: Number of PRS positioning occasions within measurement period**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| 160 ms | 32 |
| >160 ms | 16 |

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4, 8.4.5 and A.8.17.10.

## 10.5.4 Test description

### 10.5.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: the largest aggregated bandwidth combination supported by the UE of the Channel bandwidths defined in Table 10.5.4.1-1. The Channel bandwidths for CA Intra-Band combinations are, as defined in TS 36.508 [18] clause 4.3.1 and for CA Inter-Band combinations are defined in TS 36.521-1[24] clause 5.4.2A. In case of multiple possible Channel bandwidth combinations, the first combination listed in the above mentioned clauses shall be selected.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in TS 36.508 [18] Annex A, Figure A.68 as appropriate.
2. The general test parameter settings are set up according to Table 10.5.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 10.5.4.3.
5. In the tests, there are three configured component carriers: PCC, SCC1 and SCC2, and four synchronized cells: Cell 1, Cell 2, Cell 3 and Cell 4. Cell 1 is PCell on the PCC, Cell 2 is SCell on the SCC1, Cell 3 is SCell on the SCC2 and Cell 4 is a neighbour cell on the SCC2. In all tests, Cell 3 is the OTDOA assistance data reference cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 clause 7.3.2) for Test 1 and where 12 of the cells are not simulated for Test 2. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2, Cell 3 and Cell 4 are powered OFF.
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between Cell 1 and OTDOA assistance data reference cell, Cell 3, and set to -31 Ts (about -1 μs) between Cell 2 and OTDOA assistance data reference cell, Cell 3, and set to 92 Ts (about 3 μs) between neighbour Cell 4 and OTDOA assistance data reference cell, Cell 3.

**Table 10.5.4.1-1: General test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCell | | Cell 1 | | PCell is on RF channel 1 (PCC). |
| SCell 1 | | Cell 2 | | SCell 1 on RF channel 2 (SCC1). |
| SCell 2 | | Cell 3 | | SCell 2 on RF channel 3 (SCC2). Cell 3 is the assistance data reference cell. |
| Other neighbour cell | | Cell 4 | | Neighbour cell on RF channel 3 (SCC2). |
| PCFICH/PDCCH/PHICH parameters (PCFICH/PDCCH/PHICH parameters depend on selected channel bandwidth) | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5,10,20 | | |
| PRS Transmission Bandwidth (PRS transmission bandwidth depends on selected channel bandwidth) Note 2 | RB | 5MHz: 25<br>10MHz: 50<br>20MHz:100 | | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{PRS}$. ( $N_{PRS}$ depends on selected channel bandwidth) Note 2 | | 5MHz: 2<br>10MHz: 1<br>20MHz:1 | | As defined in TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS configuration index $I_{PRS}$ Note 2 | | 171 for all cells on PCC<br>181 for all cells on SCC1<br>191 for all cells on SCC2 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in TS 36.211 [26], Table 6.10.4.3-1 |
| Physical cell ID PCI Note 2 | | (PCI of Cell 3 – PCI of Cell 4)mod6=0 | | The PCIs of Cell 1 and Cell 2 are selected randomly. PCIs of Cell 3 and Cell 4 are selected randomly such that the relative subcarrier shifts of PRS patterns among these cells are as given by the condition |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 10.5.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 1 to Cell 3: 1<br>Cell 2 to Cell 3: -1<br>Cell 4 to Cell 3: 3 | Cell 1 to Cell 3: 1<br>Cell 2 to Cell 3: -1<br>Cell 4 to Cell 3: | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 4: 2<br>Other neighbour cells: randomly between -3 and 3 | Cell 1: -2<br>Cell 2: 0<br>Cell 4: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| | | 16 cells in total | | |

| | | | | |
|---|---|---|---|---|
| Number of cells provided in OTDOA assistance data | | OTDOA neighbour cells include Cell 4 and other 14 cells on SCC2 | OTDOA neighbour cells include Cell 1 and other 3 cells on PCC, Cell 2 and other 3 cells on SCC1 and Cell 4 and other 6 cells on SCC2 | The list includes the reference cell and 15 other cells. Cell 1 and Cell 2 (when included) appears at random places in the first half of the neighbour cell list in the OTDOA assistance data. Cell 4 always appears at random places in the second half of the list. |
| prs-SubframeOffset Note 2 | | Cells on PCC: 300<br>Cells on SCC1: 310<br>Cells on SCC2, except reference cell: 0 | | Subframe offset, counted in full subframes. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | Cells on PCC: 0<br>Cells on SCC1: 0<br>Cells on SCC2, except reference cell: 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell. The corresponding parameter in the OTDOA assistance data is slotNumberOffset specified in TS 36.355 [4]. |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000'<br>Cell 4: '00001111' | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000'<br>Cell 4: '0000000011111111' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T2 |

Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.5.4.3-4 and TS 37.571-5 [20], clause 7.3.2.

Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", "prs-SubframeOffset", "slotNumberOffset" and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 3, Cell 3: 6, Cell 4: 12. For the values to be used in LPP see Table 10.5.4.3-4 and TS 37.571-5 [20], clause 7.3.2.

Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is used to set the "true RSTD" values in step 6 of clause 10.5.4.1.

**Table 10.5.4.1-2: DRX parameters for the test of E-UTRAN FDD intra-frequency RSTD measurement reporting delay under fading propagation conditions for carrier aggregation**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in TS 36.331 [22], Clause 6.3.2 |
| Drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

### 10.5.4.2 Test procedure

The test case includes two tests. Test 1 is designed for the scenario where the UE receives OTDOA assistance data with cells only on SCC2, and the UE is expected to report RSTD measurements performed on SCC2 only. Test 2 is designed for the scenario where the UE receives OTDOA assistance data with cells on PCC, SCC1 and SCC2, and the UE is expected to report RSTD measurements performed on PCC, SCC1 and SCC2.

Each test consists of three consecutive time intervals, with duration of T1, T2 and T3. Cell 1 is active in T1, T2 and T3, Cell 2 is active only in T2 and T3, Cell 3 is active only during T2 and T3, and Cell 4 is active only during T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the OTDOA assistance data reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, Cell 2 transmits PRS only in T3, Cell 3 transmits PRS only in T3, and Cell 4 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in 10.5.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.

2. Configure Cell 2 on SCC1 and Cell 3 and Cell 4 on SCC2 according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCells (Cell 2 and Cell 3) on the SCCs as per TS 36.508 [18] clause 5.2A.4.

4. The SS activates the SCells (Cell 2 and Cell 3) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.

5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

6. Set the parameters according to Table 10.5.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).

7. T1 starts.

8. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

9a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

9b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 4 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the last 8 elements of the sequence for Test 1 and in the 7 elements of the relevant sequence for Test 2, and the position of Cell 1 and the position of Cell 2 are randomly selected to be in the 4 elements of the relevant sequence for Test 2, as described in 3GPP TS 37.571-5 [20], clause 7.3.2. If the UE message at step 9b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 150 ms.

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 10.5.5-2.

13. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 10.5.5-2.

14. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 10.5.5.

For Test 1 the UE shall perform and report the RSTD measurement for Cell 4 with respect to the reference cell in the OTDOA assistance data, Cell 3. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 4 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with the *rstd* field included within the response time then the number of failure tests is increased by one.

For Test 2 the UE shall perform and report the RSTD measurements for Cell 1 with respect to the reference cell in the OTDOA assistance data, Cell 3 and also Cell 2 with respect to the reference cell in the OTDOA assistance data, Cell 3 and also Cell 4 with respect to the reference cell in the OTDOA assistance data, Cell 3. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 1 (with respect to Cell 3) and Cell 2 (with respect to Cell 3) and Cell 4 (with respect to Cell 3) within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with the three *rstd* fields included within the response time then the number of failure tests is increased by one.

15. If the UE message at step 14 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

16. Repeat steps 5-15 until the confidence level according to Annex D is achieved. For each iteration, at step 10 change the random positions of Cell 4 and Cell 1(for Test 2 only) and Cell 2(for Test 2 only) in the relevant sequence in the *OTDOA-NeighbourCellInfoList.*

17. Repeat from clause 10.5.4.1 for Test 2.

### 10.5.4.3 Message contents

**Table 10.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.5.4.3-2: *MAC-MainConfig-RBC*: FDD RSTD Measurement Reporting Delay for Carrier Aggregation**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 10.5.4.3-2a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.5.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | Test 1: 3<br>Test 2: 6 | See clause 10.5.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.5.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |

| | | | |
|---|---|---|---|
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.5.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 3 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE{ | | | |
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 4 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |

| | | | |
|---|---|---|---|
| rstd | Present | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| NeighbourMeasurementElement SEQUENCE { | | Test 2 only | |
| physCellIdNeighbour | Cell 1 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | Test 2 only<br>With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| NeighbourMeasurementElement SEQUENCE { | | Test 2 only | |
| physCellIdNeighbour | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | Test 2 only<br>With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.5.5 Test Requirements

Table 10.5.5-1 and 10.5.5-2 define the primary level settings including test tolerances for the tests.

**Table 10.5.5-1: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | N/A | N/A | N/A |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1. (OCNG patterns depend on selected channel bandwidth) | | 5MHz: OP.18 FDD<br>10MHz: OP.5 FDD<br>20MHz: OP.13 FDD | N/A | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A | N/A |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | N/A | N/A | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 +10log ($N_{RB,c}$ /50) | N/A | N/A | N/A |
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 10.5.5-2: Cell-specific test parameters for E-UTRAN FDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | | Cell 4 | |
|---|---|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | | 3 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 (There is no PDSCH allocated in the subframe transmitting PRS) (OCNG patterns depend on selected channel bandwidth) | | 5MHz: OP.18 FDD 10MHz: OP.5 FDD 20MHz: OP.13 FDD | | 5MHz: OP.19 FDD 10MHz: OP.6 FDD 20MHz: OP.14 FDD | | 5MHz: OP.19 FDD 10MHz: OP.6 FDD 20MHz: OP.14 FDD | | 5MHz: OP.19 FDD 10MHz: OP.6 FDD 20MHz: OP.14 FDD | N/A |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | | 0 | N/A |
| PRS_RA | dB | -6 | N/A | N/A | 3 | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | -95 | -98 | -95 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| PRS $\hat{E}_s/N_{oc}$ | dB | -4 | -Infinity | -Infinity | -1 | -Infinity | -1 | -8 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -4 | -Infinity | -Infinity | -1 | -Infinity | -1 | -8 | -Infinity |
| Io Note 4 | dBm/9 MHz | -69.94 +10log ($N_{RB,c}$ /50) | N/A | N/A | -66.68 +10log ($N_{RB,c}$ /50) | N/A | -66.68 +10log ($N_{RB,c}$ /50) | -70.11 +10log ($N_{RB,c}$ /50) | N/A |
| PRP Note 4 | dBm/15 kHz | -102 | -Infinity | -Infinity | -96 | -Infinity | -96 | -106 | -Infinity |
| RSRP Note 4 | dBm/15 kHz | -96 | -96 | -105 | -99 | -105 | -99 | -109 | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -4 | -7 | -4 | -11 | -Infinity |
| Propagation Condition | | ETU30 | | | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 3.3 s for Test 1 and 6.3 s for Test 2. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 2710 ms for Test 1 and 5110ms for Test 2. This is rounded up to the next allowed LPP value of 3 seconds for Test 1 and 6 seconds for Test 2.

The RSTD measurement reporting delay in the tests are derived from the following expression,

$T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=8 and $n$ =16 for Test 1, and $M$=16 and $n$ =16 for Test 2 are the parameters specified in clause 10.5.3.1 for Test 1 and clause 10.5.3.2 for Test 2.

This gives the total RSTD reporting delay of 2560 ms for Test 1 for the 15 neighbour cells including Cell 4 with respect to the reference cell, Cell 3.

This gives the total RSTD reporting delay of 4960 ms for Test 2 for the 15 neighbour cells including Cell 1, Cell 2 and Cell 4 with respect to the reference cell, Cell 3.

The test tolerances are defined in clauses C.1.3 and C.4.

For the overall test to pass, the rate of successful tests during repeated tests in both Test 1 and Test 2 shall be more than 90% with a confidence level of 95%.

# 10.6 TDD 3 DL CA RSTD Measurement Reporting Delay

## 10.6.1 Test Purpose

The purpose of the test case is to verify that the RSTD measurement reporting delay meets the requirements in an environment with fading propagation conditions. This test case verifies the measurement period requirements for RSTD measurements performed on the same secondary component carrier, the measurement period requirements for RSTD

measurements performed on both the primary and secondary component carriers, and also the measurement period requirements for RSTD measurements performed on different secondary component carriers.

## 10.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports UE-assisted OTDOA for 3DL Carrier Aggregation.

## 10.6.3 Minimum conformance requirements

### 10.6.3.1 Measurements on the secondary component carrier

The RSTD measurements when all cells are on the configured secondary component carrier shall meet all applicable requirements (TDD) specified in TS 36.133 [23] section 8.1.2.5, i.e., E-UTRAN intra-frequency RSTD measurement period applies, regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34].

### 10.6.3.2 Measurements on both primary component carrier and secondary component carrier

The RSTD measurements of cells on both primary component carrier and configured secondary component carrier shall meet all applicable requirements (TDD) specified in TS 36.133 [23] section 8.1.2.6, i.e., E-UTRAN inter-frequency RSTD measurement period applies regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34], with the following exception

- the number of PRS positioning occasions is as specified in Table 10.6.3.2-1 shall apply.
- TDD uplink-downlink subframes configurations as specified in TS 36.133 [23] section 8.1.2.5.2, Table 8.1.2.5.2-2 shall apply.

**Table 10.6.3.2-1: Number of PRS positioning occasions within measurement period**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| 160 ms | 32 |
| >160 ms | 16 |

### 10.6.3.3 Measurements on different secondary component carriers

The RSTD measurements of cells on a configured secondary component carrier and another configured secondary component carrier shall meet all applicable requirements (TDD-TDD inter-Frequency OTDOA) specified in TS 36.133 [23] section 8.1.2.6, i.e., E-UTRAN inter-frequency RSTD measurement period applies regardless of whether the SCell on the corresponding frequency is activated or deactivated by the MAC-CE commands as specified in 3GPP TS 36.321 [34], with the following exceptions

- the number of PRS positioning occasions is as specified in Table 10.6.3.3-1 shall apply.
- TDD uplink-downlink subframes configurations as specified in TS 36.133 [23] section 8.1.2.5.2, Table 8.1.2.5.2-2 shall apply.

**Table 10.6.3.3-1: Number of PRS positioning occasions within measurement period**

| Positioning subframe configuration period $T_{PRS}$ | Number of PRS positioning occasions $M$ |
|---|---|
| 160 ms | 32 |
| >160 ms | 16 |

The normative reference for this requirement is TS 36.133 [23] clause 8.4.3, 8.4.4, 8.4.5 and A.8.17.11.

## 10.6.4 Test description

### 10.6.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidth to be tested: the largest aggregated bandwidth combination (where all channels have the same bandwidth) supported by the UE of the Channel bandwidths defined in Table 10.6.4.1-1. The Channel bandwidths for CA Intra-Band combinations are defined in TS 36.508 [18] clause 4.3.1 and for CA Inter-Band combinations are defined in TS 36.521-1 [24] clause 5.4.2A. In case of multiple possible Channel bandwidth combinations, the first combination listed in the above mentioned clauses shall be selected.

1. Connect the SS, faders and AWGN noise sources to the UE antenna connector or antenna connectors as shown in TS 36.508 [18] Annex A, Figure A.68 as appropriate.
2. The general test parameter settings are set up according to Table 10.6.4.1-1.
3. Propagation conditions are set according to clause 4.7.2.1.
4. Message contents are defined in clause 10.6.4.3.
5. In the tests, there are three configured component carriers: PCC, SCC1 and SCC2, and four synchronized cells: Cell 1, Cell 2, Cell 3 and Cell 4. Cell 1 is PCell on the PCC, Cell 2 is SCell on the SCC1, Cell 3 is SCell on the SCC2 and Cell 4 is a neighbour cell on the SCC2. In all tests, Cell 3 is the OTDOA assistance data reference cell. The assistance data neighbour cell list includes in total 15 cells, where 14 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 clause 7.3.2) for Test 1 and where 12 of the cells are not simulated for Test 2. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2, Cell 3 and Cell 4 are powered OFF.
6. The true RSTD (which is the receive time difference for frame 0 between two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between Cell 1 and OTDOA assistance data reference cell, Cell 3, and set to -31 Ts (about -1 μs) between Cell 2 and OTDOA assistance data reference cell, Cell 3, and set to 92 Ts (about 3 μs) between neighbour Cell 4 and OTDOA assistance data reference cell, Cell 3.

**Table 10.6.4.1-1: General test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions for carrier aggregation**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PCell | | Cell 1 | | PCell is on RF channel 1 (PCC). |
| SCell 1 | | Cell 2 | | SCell 1 on RF channel 2 (SCC1). |
| SCell 2 | | Cell 3 | | SCell 2 on RF channel 3 (SCC2). Cell 3 is the assistance data reference cell. |
| Other neighbour cell | | Cell 4 | | Neighbour cell on RF channel 3 (SCC2). |
| PCFICH/PDCCH/PHICH parameters (PCFICH/PDCCH/PHICH parameters depend on selected channel bandwidth) | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | As specified in TS 36.521-3 [25] clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5MHz or 10MHz or 20MHz | | All channels in a test have the same bandwidth. |

| | | | | |
|---|---|---|---|---|
| PRS Transmission Bandwidth (PRS transmission bandwidth depends on selected channel bandwidth) Note 2 | RB | 5MHz: 25<br>10MHz: 50<br>20MHz:100 | | PRS are transmitted over the system bandwidth |
| Number of consecutive downlink positioning subframes $N_{\text{PRS}}$. ( $N_{\text{PRS}}$ depends on selected channel bandwidth) Note 2 | | 5MHz: 2<br>10MHz: 1<br>20MHz:1 | | As defined in TS 36.211 [26]. The number of subframes in a positioning occasion |
| PRS configuration index $I_{\text{PRS}}$ Note 2 | | 174 for all cells on PCC<br>184 for all cells on SCC1<br>194 for all cells on SCC2 | | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{\text{PRS}} - 160$ DL subframes, as defined in TS 36.211 [26], Table 6.10.4.3-1 |
| Physical cell ID PCI Note 2 | | (PCI of Cell 3 – PCI of Cell 4)mod6=0 | | The PCIs of Cell 1 and Cell 2 are selected randomly. PCIs of Cell 3 and Cell 4 are selected randomly such that the relative subcarrier shifts of PRS patterns among these cells are as given by the condition |
| TDD uplink-downlink configuration | | 1 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to a configuration with 5 ms switch-point periodicity and two downlink consecutive subframes |
| TDD special subframe configuration | | 6 | | As specified in TS 36.211 [26], Clause 4.2; corresponds to DwPTS of $19760 \cdot T_s$ and UpPTS of $4384 \cdot T_s$ |
| CP length Note 2 | | Normal | | |
| DRX | | ON | | DRX parameters are further specified in Table 10.6.4.1-2 |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 1 to Cell 3: 1<br>Cell 2 to Cell 3: -1<br>Cell 4 to Cell 3: 3 | Cell 1 to Cell 3: 1<br>Cell 2 to Cell 3: -1<br>Cell 4 to Cell 3: | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 4: 2<br>Other neighbour cells: randomly between -3 and 3 | Cell 1: -2<br>Cell 2: 0<br>Cell 4: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | 16 cells in total | | The list includes the reference cell and 15 other cells.<br>Cell 1 and Cell 2 (when included) appears at random places in the first half of the neighbour cell list in the OTDOA assistance data. Cell 4 always appears at random places in the second half of the list. |
| | | OTDOA neighbour cells include Cell 4 and other 14 cells on SCC2 | OTDOA neighbour cells include Cell 1 and other 3 cells on PCC, Cell 2 and other 3 cells on SCC1 and Cell 4 and other 6 cells on SCC2 | |

| | | | | |
|---|---|---|---|---|
| prs-SubframeOffset Note 2 | | Cells on PCC: 300<br>Cells on SCC1: 310<br>Cells on SCC2, except reference cell: 0 | | Subframe offset, counted in full subframes. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | Cells on PCC: 0<br>Cells on SCC1: 0<br>Cells on SCC2, except reference cell: 0 | | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell. The corresponding parameter in the OTDOA assistance data is slotNumberOffset specified in TS 36.355 [4]. |
| PRS muting info Note 2 | | Cell 1: '11110000'<br>Cell 2: '00001111'<br>Cell 3: '11110000'<br>Cell 4: '00001111' | Cell 1: '1111111100000000'<br>Cell 2: '0000000011111111'<br>Cell 3: '1111111100000000'<br>Cell 4: '0000000011111111' | Corresponds to prs-MutingInfo defined in TS 36.355 [4] |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T1 |
| T3 | s | 1.28 | 2.48 | The length of the time interval that follows immediately after time interval T2 |
| Note 1: Parameters "Expected RSTD" and "Expected RSTD uncertainty for all neighbour cells" are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.6.4.3-4 and TS 37.571-5 [20], clause 7.3.2.<br>Note 2: Parameters "PRS Transmission Bandwidth", "PRS configuration index", "Number of consecutive downlink positioning subframes", "Physical cell ID PCI", "CP length", "prs-SubframeOffset", "slotNumberOffset" and "PRS muting info" are settable parameters and also parameters signalled in LPP. The values to be used for "Physical cell ID PCI" are as follows: Cell 1: 0, Cell 2: 3, Cell 3: 6, Cell 4: 12. For the values to be used in LPP see Table 10.6.4.3-4 and TS 37.571-5 [20], clause 7.3.2.<br>Note 3: The parameter "Radio frame receive time offset between the cells at the UE antenna connector" is used to set the "true RSTD" values in step 6 of clause 10.6.4.1. | | | | |

**Table 10.6.4.1-2: DRX parameters for the test of E-UTRAN TDD intra-frequency RSTD measurement reporting delay under fading propagation conditions for carrier aggregation**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in TS 36.331 [22], Clause 6.3.2 |
| Drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | sf1 | |
| longDRX-CycleStartOffset | sf320 | |
| shortDRX | Disable | |

## 10.6.4.2 Test procedure

The test case includes two tests. Test 1 is designed for the scenario where the UE receives OTDOA assistance data with cells only on SCC2, and the UE is expected to report RSTD measurements performed on SCC2 only. Test 2 is designed for the scenario where the UE receives OTDOA assistance data with cells on PCC, SCC1 and SCC2, and the UE is expected to report RSTD measurements performed on PCC, SCC1 and SCC2.

Each test consists of three consecutive time intervals, with duration of T1, T2 and T3. Cell 1 is active in T1, T2 and T3, Cell 2 is active only in T2 and T3, Cell 3 is active only during T2 and T3, and Cell 4 is active only during T2. The beginning of the time interval T2 shall be aligned 5 ms before the first PRS positioning subframe of a positioning occasion in the OTDOA assistance data reference cell, where 5 ms is the necessary test tolerance. Cell 1 transmits PRS in T2, Cell 2 transmits PRS only in T3, Cell 3 transmits PRS only in T3, and Cell 4 transmits PRS only in T2.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in 10.6.4.3 shall be provided to the UE during T1. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of T2, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. Configure Cell 2 on SCC1 and Cell 3 and Cell 4 on SCC2 according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCells (Cell 2 and Cell 3) on the SCCs as per TS 36.508 [18] clause 5.2A.4.
4. The SS activates the SCells (Cell 2 and Cell 3) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.
5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
6. Set the parameters according to Table 10.6.5-1. Propagation conditions are set according to clause 4.7.2.2 (ETU30).
7. T1 starts.
8. The SS shall transmit an RRCConnectionReconfiguration message with the DRX configuration. PDCCHs indicating new transmissions shall be sent continuously until the start of T2 to ensure that the UE would not enter the DRX state before T2.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.

9a. The SS shall transmit an LPP REQUEST CAPABILITIES message.

9b. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE.

10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of neighbour Cell 4 in the *OTDOA-NeighbourCellInfoList* is randomly selected to be in the last 8 elements of the sequence for Test 1 and in the 7 elements of the relevant sequence for Test 2, and the position of Cell 1 and the position of Cell 2 are randomly selected to be in the 4 elements of the relevant sequence for Test 2, as described in 3GPP TS 37.571-5 [20], clause 7.3.2. If the UE message at step 9b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms.
12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 10.6.5-2.
13. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 10.6.5-2.
14. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *OTDOA-ProvideLocationInformation* IE within the response time (see clause 4.7.3) specified in clause 10.6.5.

For Test 1 the UE shall perform and report the RSTD measurement for Cell 4 with respect to the reference cell in the OTDOA assistance data, Cell 3. If the UE transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 4 within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with the *rstd* field included within the response time then the number of failure tests is increased by one.

For Test 2 the UE shall perform and report the RSTD measurements for Cell 1 with respect to the reference cell in the OTDOA assistance data, Cell 3 and also Cell 2 with respect to the reference cell in the OTDOA assistance data, Cell 3 and also Cell 4 with respect to the reference cell in the OTDOA assistance data, Cell 3. If the UE

transmits an *OTDOA-ProvideLocationInformation* IE including the *rstd* field for Cell 1 (with respect to Cell 3) and Cell 2 (with respect to Cell 3) and Cell 4 (with respect to Cell 3) within the response time then the number of successful tests is increased by one. If the UE fails to report the *OTDOA-ProvideLocationInformation* IE with the three *rstd* fields included within the response time then the number of failure tests is increased by one.

15. If the UE message at step 14 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

16. Repeat steps 5-15 until the confidence level according to Annex D is achieved. For each iteration, at step 10 change the random positions of Cell 4 and Cell 1(for Test 2 only) and Cell 2(for Test 2 only) in the relevant sequence in the *OTDOA-NeighbourCellInfoList.*

17. Repeat from clause 10.6.4.1 for Test 2.

### 10.6.4.3 Message contents

**Table 10.6.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.6.4.3-2: *MAC-MainConfig-RBC*: TDD RSTD Measurement Reporting Delay for Carrier Aggregation**

| Derivation Path: TS 36.508 [18] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | sf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf320 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |

**Table 10.6.4.3-2a: LPP Request Capabilities**

| **Information Element** | **Value/remark** |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.6.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | Test 1: 3<br>Test 2: 6 | See clause 10.6.5 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.6.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |

| | | | |
|---|---|---|---|
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.6.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 3 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE{ | | | |
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 4 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| NeighbourMeasurementElement SEQUENCE { | | Test 2 only | |
| physCellIdNeighbour | Cell 1 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | Test 2 only<br>With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| NeighbourMeasurementElement SEQUENCE { | | Test 2 only | |

| | | | |
|---|---|---|---|
| physCellIdNeighbour | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Present | Test 2 only<br>With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.6.5 Test Requirements

Table 10.6.5-1 and 10.6.5-2 define the primary level settings including test tolerances for the tests.

**Table 10.6.5-1: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T1 for carrier aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | N/A | N/A | N/A |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1. (OCNG patterns depend on selected channel bandwidth) | | 5MHz: OP.10 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | N/A | N/A | N/A |
| PBCH_RA | dB | 0 | N/A | N/A | N/A |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA Note 1 | | | | | |
| OCNG_RB Note 1 | | | | | |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -95 | N/A | N/A | N/A |
| PRS $\hat{E}_s/N_{oc}$ | dB | -Infinity | -Infinity | -Infinity | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -67.22 +10log ($N_{RB,c}$ /50) | N/A | N/A | N/A |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s/N_{oc}$ | dB | 0 | -Infinity | -Infinity | -Infinity |
| Propagation Condition | | ETU30 | | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 10.6.5-2: Cell-specific test parameters for E-UTRAN TDD RSTD measurement reporting delay under fading propagation conditions during T2 and T3 for carrier aggregation**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | | Cell 4 | |
|---|---|---|---|---|---|---|---|---|---|
| | | T2 | T3 | T2 | T3 | T2 | T3 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | | 3 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in TS 36.521-3 [25] clause D.1 (There is no PDSCH allocated in the subframe transmitting PRS) (OCNG patterns depend on selected channel bandwidth) | | 5MHz: OP.10 TDD 10MHz: OP.1 TDD 20MHz: OP.7 TDD | | 5MHz: OP.10 TDD 10MHz: OP.1 TDD 20MHz: OP.7 TDD | | 5MHz: OP.10 TDD 10MHz: OP.1 TDD 20MHz: OP.7 TDD | | 5MHz: OP.10 TDD 10MHz: OP.1 TDD 20MHz: OP.7 TDD | N/A |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>OCNG_RA Note 1<br>OCNG_RB Note 1 | dB | 0 | | 0 | | 0 | | 0 | N/A |
| PRS_RA | dB | -6 | N/A | N/A | 3 | N/A | 3 | 3 | N/A |
| $N_{oc}$ Note 3 | dBm/ 15 kHz | -98 | -98 | -98 | -95 | -98 | -95 | -98 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | dB | -4 | -Infinity | -Infinity | -1 | -Infinity | -1 | -8 | -Infinity |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -4 | -Infinity | -Infinity | -1 | -Infinity | -1 | -8 | -Infinity |
| Io Note 4 | dBm/ 9 MHz | -69.94 +10log ($N_{RB,c}$ /50) | N/A | N/A | -66.68 +10log ($N_{RB,c}$ /50) | N/A | -66.68 +10log ($N_{RB,c}$ /50) | -70.11 +10log ($N_{RB,c}$ /50) | N/A |
| PRP Note 4 | dBm/ 15 kHz | -102 | -Infinity | -Infinity | -96 | -Infinity | -96 | -106 | -Infinity |
| RSRP Note 4 | dBm/ 15 kHz | -96 | -96 | -105 | -99 | -105 | -99 | -109 | -Infinity |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $\hat{E}_s/N_{oc}$ Note 4 | dB | 2 | 2 | -7 | -4 | -7 | -4 | -11 | -Infinity |
| Propagation Condition | | ETU30 | | | | | | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: If PRS_RA is not "N/A", $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, Io, RSRP and PRP levels have been derived from other parameters and are given for information purpose. If PRS_RA is "N/A", Io and RSRP levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. Interference conditions shall be applied to all PRS symbols of DL positioning subframes.

The response time including test tolerance is 3.3 s for Test 1 and 6.3 s for Test 2. The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD reporting delay plus ΔT, where ΔT = 150 ms, giving a value of 2710 ms for Test 1 and 5110ms for Test 2. This is rounded up to the next allowed LPP value of 3 seconds for Test 1 and 6 seconds for Test 2.

The RSTD measurement reporting delay in the tests are derived from the following expression,

$T_{PRS}(M-1)+160\left\lceil \frac{n}{M} \right\rceil$, where $M$=8 and $n$ =16 for Test 1, and $M$=16 and $n$ =16 for Test 2 are the parameters specified in clause 10.6.3.1 for Test 1 and clause 10.6.3.2 for Test 2.

This gives the total RSTD reporting delay of 2560 ms for Test 1 for the 15 neighbour cells including Cell 4 with respect to the reference cell, Cell 3.

This gives the total RSTD reporting delay of 4960 ms for Test 2 for the 15 neighbour cells including Cell 1, Cell 2 and Cell 4 with respect to the reference cell, Cell 3.

The test tolerances are defined in clauses C.1.3 and C.4.

For the overall test to pass, the rate of successful tests during repeated tests in both Test 1 and Test 2 shall be more than 90% with a confidence level of 95%.

# 10.7 FDD RSTD Measurement Accuracy for 3DL Carrier Aggregation

## 10.7.1 Test purpose

To verify that the FDD RSTD measurement accuracy is within the specified limits.

## 10.7.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports UE-assisted OTDOA for 3DL Carrier Aggregation.

## 10.7.3 Minimum conformance requirements

This section contains RSTD measurement accuracy requirements for a UE configured with one or two downlink SCell(s). The UE may operate in one of the E-UTRA carrier aggregations listed in TS 36.133 [23] section 8.3.1. The requirements in this section shall apply regardless of whether the configured downlink secondary cell is activated or deactivated by the MAC-CE command [34]. The requirements apply for bandwidths defined in the bandwidth combination set for the CA configurations supported by the UE defined in TS 36.101 [2].

The RSTD measurements, which are obtained when both the reference cell and neighbouring cell belong to the primary component carrier, shall meet the intra-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.10.1.

The RSTD measurements, which are obtained when both the reference cell and neighbouring cell belong to the same secondary component carrier, shall meet the intra-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.10.1.

The RSTD measurements, which are obtained when the reference cell and neighbouring cell do not belong to the same carrier, shall meet the inter-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.10.2.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.14.

## 10.7.4 Test description

### 10.7.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidths to be tested: the largest and the smallest aggregated bandwidth combinations supported by the UE of the Channel bandwidths defined in Table 10.7.4.1-1. The Channel bandwidths for CA Intra-Band combinations are as defined in TS 36.508 [18] clause 4.3.1 and for CA Inter-Band combinations are defined in TS 36.521-1 [24] clause 5.4.2A. In case of multiple possible Channel bandwidth combinations, the first combination listed in the above mentioned clauses shall be selected.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as follows:

   - For UEs supporting only 2Rx in all the bands under test, use TS 36.508 [18] Annex A, Figure group A.68 as appropriate.

   - For UEs supporting 4Rx in any of the bands under test use TS 36.508 [18] Annex A, Figure A.90. Use the 2Rx module for cells on bands supporting 2Rx and the 4Rx module for cells on bands supporting 4Rx.

2. The general test parameter settings are set up according to Table 10.7.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 10.7.4.3.

5. There are four synchronized cells on three different carrier frequencies. Cell 1 is the PCell on primary component carrier F1 (RF channel number 1), Cell 2 is an SCell on secondary component carrier F2 (RF channel number 2), Cell 3 is an SCell and reference cell on secondary component carrier F3 (RF channel number 3), and Cell 4 is the neighbour cell on F3. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2, Cell 3 and Cell 4 are powered OFF.

   Cell 1, Cell 2, Cell 3, and Cell 4 are included in the OTDOA assistance data neighbour cell list.

   The assistance data neighbour cell list includes in total 15 cells, where 12 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.3.2).

   Note that the measurement gap is not configured in the test because of UE carrier aggregation capability.

6. The true RSTD (which is the receive time difference for frame 0 between the two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between Cell 1 and OTDOA assistance data reference cell, Cell 3, and set to -31 Ts (about -1 μs) between Cell 2 and OTDOA assistance data reference cell, Cell 3, and set to 92 Ts (about 3 μs) between neighbour Cell 4 and OTDOA assistance data reference cell, Cell 3.

Note that the related expectedRSTD values to be signalled over LPP are defined in Table 10.7.4.1-1.

**Table 10.7.4.1-1: General Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCell | | Cell 1 | Cell 1 on RF channel number 1 |
| SCell 1 | | Cell 2 | Cell 2 is an SCell on RF channel number 2 |
| SCell 2 (Assistance data reference cell) | | Cell 3 | Cell 3 is an SCell on RF channel number 3 |
| Neighbour cell | | Cell 4 | Cell 4 on RF channel number 3 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5,10,20 | |
| PRS Transmission Bandwidth (PRS transmission bandwidth depends on selected channel bandwidth) Note 2 | RB | 5MHz: 25<br>10MHz: 50<br>20MHz:100 | PRS are transmitted over the system bandwidth. PRS Bandwidth: bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |
| PCFICH/PDCCH/PHICH parameters (PCFICH/PDCCH/PHICH parameters depend on selected channel bandwidth) | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 (There is no PDSCH allocated in the subframe transmitting PRS) (OCNG Patterns depend on selected channel bandwidth) | | 5MHz: OP.18 FDD<br>10MHz: OP.5 FDD<br>20MHz: OP.13 FDD | OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| PRS configuration Index $I_{PRS}$ Note 2 | | 171 for all cells on PCC<br>181 for all cells on SCC1<br>191 for all cells on SCC2 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{PRS} - 160$ DL subframes, as defined in TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive positioning downlink subframes $N_{PRS}$( $N_{PRS}$ depends on selected channel bandwidth) Note 2 | | 5MHz: 2<br>10MHz: 1<br>20MHz:1 | As defined in 3GPP TS 36.211 [26] |
| prs-SubframeOffset Note 2 | | Cells on PCC: 300<br>Cells on SCC1: 310<br>Cells on SCC2, except reference cell: 0 | Subframe offset, counted in full subframes. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | Cells on PCC: 0<br>Cells on SCC1: 0<br>Cells on SCC2, except reference cell: 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell. The corresponding parameter in the OTDOA assistance data is slotNumberOffset specified in TS 36.355 [4]. |
| prs-MutingInfo Note 2 | | Cell 1:'11110000'<br>Cell 2:'11110000'<br>Cell 3:'11110000'<br>Cell 4:'11110000' | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Cell ID Note 2 | | (Cell ID of cell 4 – Cell ID of cell 3) mod 6 = 3 | PCIs of cell 1 and cell 2 are selected randomly. |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 1 to Cell 3: 1<br>Cell 2 to Cell 3: -1<br>Cell 4 to Cell 3: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 1: -2<br>Cell 2: 0<br>Cell 4: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |

| | | | |
|---|---|---|---|
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the assistance-data-reference cell and 15 other cells. Cell 1 and Cell 2 appear at random places in the first half of the neighbour cell list in the OTDOA assistance data. Cell 4 always appears at random places in the second half of the list. |
| $T_{RSTDInterFreqFDD,E-UTRAN}$ Note 4 | ms | 4960 | Derived according to the RSTD measurement requirements specified in Section 10.5.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.7.4.3-3 and TS 37.571-5 [20], clause 7.3.2.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-SubframeOffset”, “slotNumberOffset”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: 3, Cell 3: 7, Cell 4: 10. For the values to be used in LPP see Table 10.7.4.3-3 and TS 37.571-5 [20], clause 7.3.2.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is used to set the “true RSTD” value in step 6 of clause 10.7.4.1.

NOTE 4: The parameter “$T_{RSTDInterFreqFDD,E-UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 10.7.4.3-2. The value of the LPP time IE is set to $T_{RSTDInterFreqFDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 5110 ms. This is rounded up to the next allowed LPP value of 6 seconds.

## 10.7.4.2 Test procedure

The RSTD measurements are performed:

- between Cell 4 and Cell 3 to verify the accuracy of RSTD measurement when the reference cell and neighbouring cell belong to the same secondary component carrier can meet the intra-frequency RSTD accuracy requirements defined in section 10.7.3.
- between Cell 1 and Cell 3 to verify the accuracy of RSTD measurement between the PCell and an SCell can meet the inter-frequency RSTD accuracy requirements defined in section 10.7.3.
- between Cell 2 and Cell 3 to verify the accuracy of RSTD measurement between two SCells can meet the inter-frequency RSTD accuracy requirements defined in section 10.7.3.

The test consists of a set-up period and a measurement period. All cells are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 10.7.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. Configure Cell 2 on SCC1 and Cell 3 and Cell 4 on SCC2 according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCells (Cell 2 and Cell 3) on the SCCs as per TS 36.508 [18] clause 5.2A.4.

4. The SS activates the SCells (Cell 2 and Cell 3) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.

5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

6. Set the parameters according to Table 10.7.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

6a. The SS shall send an LPP REQUEST CAPABILITIES message.

6b. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of Cell 1 in the *OTDOA-NeighbourCellInfoList* and the position of Cell 2 are randomly selected in the relevant sequence and the position of Cell 4 is randomly selected in the relevant sequence as described in 3GPP TS 37.571-5 [20], clause 7.3.2. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T$ = 150 ms.

9. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

10. If the UE message at step 9 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

11. The SS shall check the *rstd* values for Cell 1, Cell 2 and Cell 4 in the *OTDOA-SignalMeasurementInformation* IE according to Table 10.7.5-2.

12. Repeat step 5-11 until the confidence level according to Annex D is achieved. For each iteration, at step 7 change the random positions of Cell 1 and Cell 2 and Cell 4 in the relevant sequence in the *OTDOA-NeighbourCellInfoList.*

13. Repeat complete test for the other channel bandwidth(s) supported by the UE (if any).

### 10.7.4.3 Message contents

**Table 10.7.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.7.4.3-1a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.7.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See Note 4 of Table 10.7.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.7.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |

| | | | |
|---|---|---|---|
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.7.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 3 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE{ | | | |
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 1 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 10.7.5-2 | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |

| | | | |
|---|---|---|---|
| rstd | Set according to Table 10.7.5-2 | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 4 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 10.7.5-2 | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.7.5 Test requirement

Table 10.7.5-1 defines the primary level settings including test tolerances for the test.

The FDD RSTD accuracy test shall meet the reported values in Table 10.7.5-2.

**Table 10.7.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN FDD for Carrier Aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 3 | 3 |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA Note1 | | | | | |
| OCNG_RB Note1 | | | | | |
| PRS_RA | dB | -3 | 0 | 0.3 | 0.3 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -6 | -6 | -5.7 | -12.7 |
| PRS $\hat{E}_s / I_{ot}$ Note3 | dB | -6 | -6 | -5.7 | -12.7 |
| Io Note3 | dBm/9 MHz | -70.04 +10log ($N_{RB,c}$ /50) | -70.04 +10log ($N_{RB,c}$ /50) | -69.99 +10log ($N_{RB,c}$ /50) | -69.99 +10log ($N_{RB,c}$ /50) |
| PRP Note3 | dBm/15kHz | -104 | -104 | -103.7 | -110.7 |
| RSRP Note3 | dBm/15kHz | -101 | -104 | -104 | -111 |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s/N_{oc}$ Note3 | dB | -3 | -6 | -6 | -13 |
| Propagation condition | | AWGN | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes).

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, RSRP, Io and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS.

**Table 10.7.5-2: RSTD FDD accuracy requirements for the reported values for Carrier Aggregation**

| | Value Cell 1 | Value Cell 2 | Value Cell 4 |
|---|---|---|---|
| Lowest reported value (depends on selected channel bandwidth) | 5MHz: RSTD_6374<br>10MHz: RSTD_6375<br>20MHz: RSTD_6376 | 5MHz: RSTD_6313<br>10MHz: RSTD_6314<br>20MHz: RSTD_6315 | 5MHz: RSTD_6440<br>10MHz: RSTD_6441<br>20MHz: RSTD_6442 |
| Highest reported value(depends on selected channel bandwidth) | 5MHz: RSTD_6398<br>10MHz: RSTD_6397<br>20MHz: RSTD_6396 | 5MHz: RSTD_6337<br>10MHz: RSTD_6336<br>20MHz: RSTD_6335 | 5MHz: RSTD_6454<br>10MHz: RSTD_6453<br>20MHz: RSTD_6452 |

For the test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95% for each of Cell 1, Cell 2 and Cell 4 for each supported channel bandwidth.

# 10.8 TDD RSTD Measurement Accuracy for 3DL Carrier Aggregation

## 10.8.1 Test purpose

To verify that the TDD RSTD measurement accuracy is within the specified limits.

## 10.8.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports UE-assisted OTDOA for 3DL Carrier Aggregation.

## 10.8.3 Minimum conformance requirements

This section contains RSTD measurement accuracy requirements for a UE configured with one or two downlink SCell(s). The UE may operate in one of the E-UTRA carrier aggregations listed in TS 36.133 [23] section 8.3.1. The requirements in this section shall apply regardless of whether the configured downlink secondary cell is activated or deactivated by the MAC-CE command [34]. The requirements apply for bandwidths defined in the bandwidth combination set for the CA configurations supported by the UE defined in TS 36.101 [2].

The RSTD measurements, which are obtained when both the reference cell and neighbouring cell belong to the primary component carrier, shall meet the intra-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.10.1.

The RSTD measurements, which are obtained when both the reference cell and neighbouring cell belong to the same secondary component carrier, shall meet the intra-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.10.1.

The RSTD measurements, which are obtained when the reference cell and neighbouring cell do not belong to the same carrier, shall meet the inter-frequency RSTD accuracy requirements defined in TS 36.133 [23] section 9.1.10.2.

The normative reference for this requirement is TS 36.133 [23] clause 9.1.12 and A.9.8.15.

## 10.8.4 Test description

### 10.8.4.1 Initial conditions

Test Environment: Normal; as defined in TS 36.508 [18] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.1.

Channel bandwidths to be tested: the largest and the smallest aggregated bandwidth combinations supported by the UE of the Channel bandwidths defined in Table 10.8.4.1-1. The Channel bandwidths for CA Intra-Band combinations are as defined in TS 36.508 [18] clause 4.3.1 and for CA Inter-Band combinations are defined in TS 36.521-1[24] clause 5.4.2A. In case of multiple possible Channel bandwidth combinations, the first combination listed in the above mentioned clauses shall be selected.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as follows:

   - For UEs supporting only 2Rx in all the bands under test, use TS 36.508 [18] Annex A, Figure group A.68 as appropriate.

   - For UEs supporting 4Rx in any of the bands under test use TS 36.508 [18] Annex A, Figure A.90. Use the 2Rx module for cells on bands supporting 2Rx and the 4Rx module for cells on bands supporting 4Rx.

2. The general test parameter settings are set up according to Table 10.8.4.1-1.

3. Propagation conditions are set according to clause 4.7.2.1.

4. Message contents are defined in clause 10.8.4.3.

5. There are four synchronized cells on three different carrier frequencies. Cell 1 is the PCell on primary component carrier F1 (RF channel number 1), Cell 2 is an SCell on secondary component carrier F2 (RF channel number 2), Cell 3 is an SCell and reference cell on secondary component carrier F3 (RF channel number 3), and Cell 4 is the neighbour cell on F3. PCell (Cell 1) is the cell used for connection setup with the power level and mapping set according to TS 36.521-1 [24] Annex C.0 and C.1 as appropriate for this test. Cell 2, Cell 3 and Cell 4 are powered OFF.

Cell 1, Cell 2, Cell 3, and Cell 4 are included in the OTDOA assistance data neighbour cell list.

The assistance data neighbour cell list includes in total 15 cells, where 12 of the cells are not simulated (dummy cells; as defined in 3GPP TS 37.571-5 [20], clause 7.3.2).

Note that the measurement gap is not configured in the test because of UE carrier aggregation capability.

6. The true RSTD (which is the receive time difference for frame 0 between the two cells as seen at the UE antenna connector) is set to 31 Ts (about 1 μs) between Cell 1 and OTDOA assistance data reference cell, Cell 3, and set to -31 Ts (about -1 μs) between Cell 2 and OTDOA assistance data reference cell, Cell 3, and set to 92 Ts (about 3 μs) between neighbour Cell 4 and OTDOA assistance data reference cell, Cell 3.

Note that the related expectedRSTD values to be signalled over LPP are defined in Table 10.8.4.1-1.

**Table 10.8.4.1-1: General Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PCell | | Cell 1 | Cell 1 on RF channel number 1 |
| SCell 1 | | Cell 2 | Cell 2 is an SCell on RF channel number 2 |
| SCell 2 (Assistance data reference cell) | | Cell 3 | Cell 3 is an SCell on RF channel number 3 |
| Neighbour cell | | Cell 4 | Cell 4 on RF channel number 3 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5,10,20 | |
| PRS Transmission Bandwidth (PRS transmission bandwidth depends on selected channel bandwidth) Note 2 | RB | 5MHz: 25<br>10MHz: 50<br>20MHz:100 | PRS are transmitted over the system bandwidth. PRS Bandwidth: bandwidth is as indicated in *prs-Bandwidth* in the OTDOA assistance data defined in 3GPP TS 36.355 [4]. |

| | | | |
|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters (PCFICH/PDCCH/PHICH parameters depend on selected channel bandwidth) | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | As specified in TS 36.521-3 [25] clause A.2.1 |
| OCNG Patterns defined in TS 36.521-3 [25] clause D.1 (There is no PDSCH allocated in the subframe transmitting PRS) (OCNG Patterns depend on selected channel bandwidth) | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| PRS configuration Index $I_{\mathrm{PRS}}$ Note 2 | | 171 for all cells on PCC<br>181 for all cells on SCC1<br>191 for all cells on SCC2 | This corresponds to periodicity of 320 ms and PRS subframe offset of $I_{\mathrm{PRS}} - 160$ DL subframes, as defined in TS 36.211 [26], Table 6.10.4.3-1 |
| Number of consecutive positioning downlink subframes $N_{\mathrm{PRS}}$( $N_{\mathrm{PRS}}$ depends on selected channel bandwidth) Note 2 | | 5MHz: 2<br>10MHz: 1<br>20MHz:1 | As defined in 3GPP TS 36.211 [26] |
| prs-SubframeOffset Note 2 | | Cells on PCC: 300<br>Cells on SCC1: 310<br>Cells on SCC2, except reference cell: 0 | Subframe offset, counted in full subframes. The corresponding parameter in the OTDOA assistance data is prs-SubframeOffset specified in TS 36.355 [4] |
| slotNumberOffset Note 2 | | Cells on PCC: 0<br>Cells on SCC1: 0<br>Cells on SCC2, except reference cell: 0 | The slot number offset at the transmitter between a neighbour cell and the assistance data reference cell. The corresponding parameter in the OTDOA assistance data is slotNumberOffset specified in TS 36.355 [4]. |
| prs-MutingInfo Note 2 | | Cell 1:‘11110000’<br>Cell 2:‘11110000’<br>Cell 3:’11110000’<br>Cell 4:’11110000’ | See section 6.5.1.2 in 3GPP TS 36.355 [4] for more information |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [26]. The same configuration in both cells. |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [26] and table 8.1.2.5.2-2 in TS 36.133 [23]. The same configuration in both cells. |
| Cell ID Note 2 | | (Cell ID of cell 4 – Cell ID of cell 3) mod 6 = 3 | PCIs of cell 1 and cell 2 are selected randomly. |
| Radio frame receive time offset between the cells at the UE antenna connector Note 3 | µs | Cell 1 to Cell 3: 1<br>Cell 2 to Cell 3: -1<br>Cell 4 to Cell 3: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD Note 1 | µs | Cell 1: -2<br>Cell 2: 0<br>Cell 4: 2<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells Note 1 | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 36.355 [4] is the expectedRSTD-Uncertainty index |
| CP length Note 2 | | Normal | |
| DRX | | OFF | |

| | | | |
|---|---|---|---|
| Number of cells provided in OTDOA assistance data | | 16 | The list includes the assistance-data-reference cell and 15 other cells. Cell 1 and Cell 2 appear at random places in the first half of the neighbour cell list in the OTDOA assistance data. Cell 4 always appears at random places in the second half of the list. |
| $T_{RSTDInterFreqTDD,E-UTRAN}$ Note 4 | ms | 4960 | Derived according to the RSTD measurement requirements specified in Section 10.6.3 |

NOTE 1: Parameters “Expected RSTD” and “Expected RSTD uncertainty for all neighbour cells” are not settable parameters. These are parameters signalled in LPP only. For the values to be used in LPP see Table 10.8.4.3-3 and TS 37.571-5 [20], clause 7.3.2.

NOTE 2: Parameters “PRS Transmission Bandwidth”, “PRS configuration index”, “Number of consecutive positioning downlink subframes”, “prs-SubframeOffset”, “slotNumberOffset”, “prs-MutingInfo”, “Cell ID” and “CP length” are settable parameters and also parameters signalled in LPP. The values to be used for “Cell ID” are as follows: Cell 1: 0, Cell 2: 3, Cell 3: 7, Cell 4: 10. For the values to be used in LPP see Table 10.8.4.3-3 and TS 37.571-5 [20], clause 7.3.2.

NOTE 3: The parameter “Radio frame receive time offset between the cells at the UE antenna connector” is used to set the “true RSTD” value in step 6 of clause 10.8.4.1.

NOTE 4: The parameter “$T_{RSTD\ InterFreqT\ DD,\ E-UTRAN}$” is not a settable parameter but is used to set the LPP “time” value in Table 10.8.4.3-2. The value of the LPP time IE is set to $T_{RSTDInterFreqTDD,E-UTRAN}$ + ΔT ms, where ΔT = 150 ms, giving a value of 5110 ms. This is rounded up to the next allowed LPP value of 6 seconds.

## 10.8.4.2 Test procedure

The RSTD measurements are performed:

- between Cell 4 and Cell 3 to verify the accuracy of RSTD measurement when the reference cell and neighbouring cell belong to the same secondary component carrier can meet the intra-frequency RSTD accuracy requirements defined in section 10.8.3.
- between Cell 1 and Cell 3 to verify the accuracy of RSTD measurement between the PCell and an SCell can meet the inter-frequency RSTD accuracy requirements defined in section 10.8.3.
- between Cell 2 and Cell 3 to verify the accuracy of RSTD measurement between two SCells can meet the inter-frequency RSTD accuracy requirements defined in section 10.8.3.

The test consists of a set-up period and a measurement period. All cells are active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

NOTE: The information on when PRS is muted is conveyed to the UE using PRS muting information in the OTDOA assistance data.

The OTDOA-RequestLocationInformation message and the OTDOA assistance data as defined in clause 10.8.4.3 shall be provided to the UE during the set-up period. The last TTI containing the OTDOA-RequestLocationInformation message shall be provided to the UE ΔT ms before the start of the measurement period, where ΔT = 150 ms is the maximum processing time of the OTDOA-RequestLocationInformation message and the OTDOA assistance data in the UE.

1. Ensure that the UE is in state Generic RB Established State 3A-RF according to 3GPP TS 36.508 [18] clause 7.2A.3.
2. Configure Cell 2 on SCC1 and Cell 3 and Cell 4 on SCC2 according to TS 36.521-3 [25] Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCells (Cell 2 and Cell 3) on the SCCs as per TS 36.508 [18] clause 5.2A.4.
4. The SS activates the SCells (Cell 2 and Cell 3) by sending the Activation/Deactivation MAC control element according to TS 36.321 [34] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [23] clause 8.3.3.2.
5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

6. Set the parameters according to Table 10.8.5-1 as appropriate. Propagation conditions are set according to clause 4.7.2.1.

6a. The SS shall send an LPP REQUEST CAPABILITIES message.

6b. The UE shall send an LPP PROVIDE CAPABILITIES message indicating the OTDOA capabilities supported by the UE in the *OTDOA-ProvideCapabilities* IE

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *OTDOA-ProvideAssistanceData* IE. The position of Cell 1 in the *OTDOA-NeighbourCellInfoList* and the position of Cell 2 are randomly selected in the relevant sequence and the position of Cell 4 is randomly selected in the relevant sequence as described in 3GPP TS 37.571-5 [20], clause 7.3.2. If the UE message at step 6b includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *OTDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 150 ms.

9. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *OTDOA-ProvideLocationInformation* IE.

10. If the UE message at step 9 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

11. The SS shall check the *rstd* values for Cell 1, Cell 2 and Cell 4 in the *OTDOA-SignalMeasurementInformation* IE according to Table 10.8.5-2.

12. Repeat step 5-11 until the confidence level according to Annex D is achieved. For each iteration, at step 7 change the random positions of Cell 1 and Cell 2 and Cell 4 in the relevant sequence in the *OTDOA-NeighbourCellInfoList.*

13. Repeat complete test for the other channel bandwidth(s) supported by the UE (if any).

### 10.8.4.3 Message contents

**Table 10.8.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 10.8.4.3-1a: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *otdoa-RequestCapabilities* | TRUE |

**Table 10.8.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 6 | See Note 4 of Table 10.8.4.1-1 | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.8.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| otdoa-ReferenceCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-NeighbourCellInfo | As defined in TS 37.571-5 [20], clause 7.3.2. | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-ProvideAssistanceData | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 10.8.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation SEQUENCE { | | | |
| otdoaSignalMeasurementInformation SEQUENCE { | | | |
| systemFrameNumber | | | |
| physCellIdRef | Cell 3 | | |
| cellGlobalIdRef | | | |
| earfcnRef | | | |
| referenceQuality | | | |
| neighbourMeasurementList SEQUENCE{ | | | |
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 1 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 10.8.5-2 | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 2 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 10.8.5-2 | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |

| | | | |
|---|---|---|---|
| NeighbourMeasurementElement SEQUENCE { | | | |
| physCellIdNeighbour | Cell 4 | | |
| cellGlobalIdNeighbour | | | |
| earfcnNeighbour | | | |
| rstd | Set according to Table 10.8.5-2 | With respect to Cell 3 | |
| rstd-Quality | | | |
| } | | | |
| } | | | |
| } | | | |
| otdoa-Error | May be present with error reason 'undefined' or 'attemptedButUnableToMeasureSomeNeighbourCells' | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 10.8.5 Test requirement

Table 10.8.5-1 defines the primary level settings including test tolerances for the test.

The TDD RSTD accuracy test shall meet the reported values in Table 10.8.5-2.

**Table 10.8.5-1: Cell Specific Test Parameters for RSTD Test for E-UTRAN TDD for Carrier Aggregation**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | 3 | 3 |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA Note1 | | | | | |
| OCNG_RB Note1 | | | | | |
| PRS_RA | dB | -3 | 0 | 0.3 | 0.3 |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -6 | -6 | -5.7 | -12.7 |
| PRS $\hat{E}_s / I_{ot}$ Note3 | dB | -6 | -6 | -5.7 | -12.7 |
| Io Note3 | dBm/9 MHz | -70.04 +10log ($N_{RB,c}$ /50) | -70.04 +10log ($N_{RB,c}$ /50) | -69.99 +10log ($N_{RB,c}$ /50) | -69.99 +10log ($N_{RB,c}$ /50) |
| PRP Note3 | dBm/15kHz | -104 | -104 | -103.7 | -110.7 |
| RSRP Note3 | dBm/15kHz | -101 | -104 | -104 | -111 |
| $\hat{E}_s / N_{oc}$ Note3 | dB | -3 | -6 | -6 | -13 |
| Propagation condition | | AWGN | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols (other than those in the PRS subframes). |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | $\hat{E}_s/N_{oc}$, PRS $\hat{E}_s/I_{ot}$, RSRP, Io and PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io values are derived in the case that there is no PBCH, PSS or SSS in the OFDM symbols carrying PRS. |

**Table 10.8.5-2: RSTD TDD accuracy requirements for the reported values for Carrier Aggregation**

| | Value Cell 1 | Value Cell 2 | Value Cell 4 |
|---|---|---|---|
| Lowest reported value (depends on selected channel bandwidth) | 5MHz: RSTD_6374<br>10MHz: RSTD_6375<br>20MHz: RSTD_6376 | 5MHz: RSTD_6313<br>10MHz: RSTD_6314<br>20MHz: RSTD_6315 | 5MHz: RSTD_6440<br>10MHz: RSTD_6441<br>20MHz: RSTD_6442 |
| Highest reported value(depends on selected channel bandwidth) | 5MHz: RSTD_6398<br>10MHz: RSTD_6397<br>20MHz: RSTD_6396 | 5MHz: RSTD_6337<br>10MHz: RSTD_6336<br>20MHz: RSTD_6335 | 5MHz: RSTD_6454<br>10MHz: RSTD_6453<br>20MHz: RSTD_6452 |

For the test to pass, the ratio of successful reported values shall be more than 90% with a confidence level of 95% for each of Cell 1, Cell 2 and Cell 4 for each supported channel bandwidth.

# 11 E-UTRA and NR MBS measurement requirements

## 11.0 General

This clause defines the minimum performance requirements for MBS FDD and TDD E-UTRA UEs, and NR UEs.

The requirements in this clause that apply for NR UE include NG-RAN NR, EN-DC, NE-DC, NG-RAN E-UTRA and NGEN-DC.

Editor's Note: For NR tests, the initial test conditions apply for NR UE in FR1. For FR2, initial test conditions are FFS.

## 11.1 MBS Measurement Reporting Delay (Release 13 only)

### 11.1.1 Test purpose

The purpose of the test is to verify that the MBS measurements meet the measurement time requirements specified in clause 4.2.3 of TS 37.171 [39] in an environment with fading propagation conditions specified in clause 4.8.2.2 (EPA 5 Hz).

### 11.1.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 13 only.

### 11.1.3 Minimum conformance requirements

The MBS measurement reporting delay (response time) shall be $\leq$12000 msec.

The normative reference for this requirement is TS 37.171 [39] clauses 4.2.3 and A.3.1.

## 11.1.4 Test description

### 11.1.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and MSS to the UE antenna connector or antenna connectors as shown in figures A.6 or A.7.
2. Switch on the UE.
3. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 11.1.4.2 Test procedure

1. Set the MSS test parameters as specified in clause 11.1.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS shall send an LPP REQUEST CAPABILITIES message.
4. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the TBS capabilities supported by the UE in the *TBS-ProvideCapabilities* IE.
5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.
6. The UE shall perform and report the code phase measurement for the simulated beacon. If the UE transmits a *TBS-ProvideLocationInformation* IE including the *transmitterID* and *codePhase* fields for the simulated beacon within the required response time in 11.1.5, then the number of successful tests is increased by one. Otherwise the number of failure tests is increased by one.
7. If the UE message at step 6 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
8. Repeat steps 1-7 until the confidence level according to Annex D is achieved. For each iteration, at step 1 reselect the PN code assigned to the MBS beacon.
9. Release the signalling connection.

### 11.1.4.3 Message contents

**Table 11.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: TS 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 1 0 | MBS | |

**Table 11.1.4.3-2: LPP RequestCapabilities**

| Information Element | Value/remark |
|---|---|
| *tbs-RequestCapabilities-r13* | TRUE |

**Table 11.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |

| | | | |
|---|---|---|---|
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| Time | 12 | | |
| responseTimeEarlyFix-r12 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| Environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 SEQUENCE { | | | |
| mbsSgnMeasListReq-r13 | TRUE | | |
| } | | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 11.1.4.3-4: LPP ProvideLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| Acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | | | |
| tbs-MeasurementInformation-r13 SEQUENCE { | | | |
| measurementReferenceTime-r13 | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| } | | | |
| tbs-Error-r13 | May be present with error reason 'undefined' or 'thereWereNotEnoughMBSBeaconsReceived' | | |
| } | | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.1.5 Test requirement

The details of the beacon parameters are in Table 11.1.5-1 and Table 11.1.5-2.

**Table 11.1.5-1: General test parameters for the beacon to be simulated for the measurement reporting delay test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Number of beacons | Integer | 1 | Beacon transmitted in any beacon slot, but static for the test, in the MBS beacon transmission period. Other slots contain no simulated beacons[Note 1] |
| Centre Frequency | MHz | 925.977 | |
| RF Channel | N/A | EPA 5Hz | |
| MBS Beacon Configuration | N/A | TB1 (2 MHz) | For details see Annex H |
| MBS Packet Type | N/A | Type 2 | For details see Annex H |
| Beacon PN Code | Integer | Chosen for the beacon from the PN code list for TB1 | For details see Annex H[Note 1] |
| Transmit power | dBm | -30 | |
| Response time | Seconds | 12 | Value of Time used in LPP RequestLocationInformation message in Table 11.1.4.3-3 |
| Note 1: The slots and PN codes are chosen at random, such that the UE does not and cannot have advanced knowledge of what that slot/PN code might be. | | | |

**Table 11.1.5-2: MBS Beacon Payload fields for the beacon to be simulated for the measurement reporting delay test**

| MBS Tx ID (see Annex H) | Slot Index (see Annex H) | All Other fields (see Annex H) |
|---|---|---|
| Equal to Slot number | Equal to Slot number | min value (bit_value = 0)Note 1 |

Note 1: bit_value is the conversion of the binary number represented by the corresponding bits in the payload to decimal.

The MBS measurement reporting delay (response time) shall be $\leq$ 12300 msec.

The test tolerances are defined in clauses C.1.4 and C.4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 11.1A MBS Measurement Reporting Delay (Release 14 Onwards)

## 11.1A.1 Test purpose

Same as defined in clause 11.1.1

## 11.1A.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 14 onwards.

## 11.1A.3 Minimum conformance requirements

Same as defined in clause 11.1.3

## 11.1A.4 Test description

### 11.1A.4.1 Initial conditions

Same as defined in clause 11.1.4.1

### 11.1A.4.2 Test procedure

Same as defined in clause 11.1.4.2, except step 4a is introduced and step 5 is modified as follows:

4a. The SS shall send an LPP PROVIDE ASSISTANCE DATA message to provide the MBS assistance data in accordance with TS 37.571-5 [20], and with the values defined therein. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE.

### 11.1A.4.3 Message contents

Same as defined in clause 11.1.4.3, with the addition of the LPP Provide Assistance Data.

**Table 11.1A.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-ProvideAssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | Rel-14 onwards |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | | | Rel-14 onwards |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| } | | | |
| } | | | |
| tbs-Error-r14 | Not Present | | |
| } | | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 | Not Present | | Rel-14 onwards |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.1A.5 Test requirement

Same as defined in clause 11.1.5, with the beacon parameters set according to the assistance data in TS 37.571-5 [20] clause 8.

# 11.1B MBS Measurement Reporting Delay (NR)

## 11.1B.1 Test purpose

Same as defined in clause 11.1.1

## 11.1B.2 Test applicability

This test applies to all types of NR UE that supports UE-assisted MBS with LPP Release 13 onwards.

## 11.1B.3 Minimum conformance requirements

Same as defined in clause 11.1.3

## 11.1B.4 Test description

### 11.1B.4.1 Initial conditions

Same as defined in clause 11.1.4.1, except in the case of NR UE, step 3 is modified as follows:

3. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4.

### 11.1B.4.2 Test procedure

For NR UE that supports LPP Release 13 only, same as defined in clause 11.1.4.2. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.1A.4.2.

### 11.1B.4.3 Message contents

For NR UE that supports LPP Release 13 only, same as defined in clause 11.1.4.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.1A.4.3.

## 11.1B.5 Test requirement

For NR UE that supports LPP Release 13 only, same as defined in clause 11.1.5. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.1A.5.

# 11.2 MBS Sensitivity Measurement Accuracy (Release 13 only)

## 11.2.1 Test purpose

The purpose of this test is to verify that the MBS Code Phase measurement accuracy is within the specified limits under target sensitivity conditions. This test will verify the requirements in clause 5.2 of TS 37.171 [39] for MBS measurements. The channel type for this test is AWGN, as specified in clause 4.8.2.1.

## 11.2.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 13 only.

## 11.2.3 Minimum conformance requirements

The MBS code phase measurement accuracy shall fulfil the requirement given in Table 11.2.3-1.

**Table 11.2.3-1: Accuracy requirements for sensitivity scenario**

| Signal Strength (dBm) | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|
| -130 | $1.66 \times 10^{-4}$ | $2.35 \times 10^{-4}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | |

The accuracy requirement for the difference of code phase measurements is derived from the requirement for the code phase measurement accuracy, assuming a scaling factor of $\sqrt{2}$ due to the compounding of two error distributions.

The normative reference for this requirement is TS 37.171 [39] clause 5.2 (Sensitivity) and clause A.4.2.

## 11.2.4 Test description

### 11.2.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and MSS to the UE antenna connector or antenna connectors as shown in figures A.6 or A.7.
2. Switch on the UE.
3. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 11.2.4.2 Test procedure

1. Set the MSS test parameters as specified in clause 11.2.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS shall send an LPP REQUEST CAPABILITIES message.
4. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the TBS capabilities supported by the UE in the *TBS-ProvideCapabilities* IE.
5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.
6. The UE shall perform and report the code phase measurement for each simulated beacon. If the UE transmits a *TBS-ProvideLocationInformation* IE including the *transmitterID* and *codePhase* field for the two simulated beacons and the difference between *codePhase* field values for the two beacons meet the corresponding requirements in Table 11.2.5-3, then the number of successful tests is increased by one. Otherwise the number of failure tests is increased by one.
7. If the UE message at step 6 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
8. Repeat steps 1-7 until the confidence level according to Annex D is achieved. For each iteration, at step 1 reselect the PN code assigned to each MBS beacon.
9. Release the signalling connection.

### 11.2.4.3 Message contents

**Table 11.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: TS 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 1 0 | MBS | |

**Table 11.2.4.3-2: LPP RequestCapabilities**

| Information Element | Value/remark |
|---|---|
| *tbs-RequestCapabilities-r13* | TRUE |

**Table 11.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| Time | 12 | | |
| responseTimeEarlyFix-r12 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| Environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 SEQUENCE { | | | |
| mbsSgnMeasListReq-r13 | TRUE | | |
| } | | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 11.2.4.3-4: LPP ProvideLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| Acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | | | |
| tbs-MeasurementInformation-r13 SEQUENCE { | | | |
| measurementReferenceTime-r13 | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| } | | | |
| tbs-Error-r13 | May be present with error reason 'undefined' or 'thereWereNotEnoughMBSBeaconsReceived' | | |
| } | | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.2.5 Test requirement

The details of the beacon parameters are in Table 11.2.5-1 and Table 11.2.5-2.

**Table 11.2.5-1: General test parameters for the beacons to be simulated for measurement accuracy in Sensitivity test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Number of Beacons | | 2 | Beacons transmitted in any two beacon slots in the beacon transmission period, but static for the test. Other slots contain no simulated beacons[Note 1] |

| | | | |
|---|---|---|---|
| Centre Frequency | MHz | 925.977 | |
| RF Channel | N/A | AWGN | |
| MBS Beacon Configuration | N/A | TB1 (2 MHz) | For details see Annex H |
| MBS Packet Type | N/A | Type 2 | For details see Annex H\ |
| Beacon PN Code | Integer | Chosen for each beacon from the PN code list for TB1 | Each of the 2 beacons uses a different PN code. For details see Annex H[Note 1] |
| Response time | Seconds | 12 | Value of Time used in LPP RequestLocationInformation message in Table 11.2.4.3-3 |
| Note 1: The slots and PN codes are chosen at random, such that the UE does not and cannot have advanced knowledge of what that slot/PN code might be. | | | |

**Table 11.2.5-2: MBS Beacon Payload fields, code phase delay difference and transmit powers for the beacons to be simulated for measurement accuracy in sensitivity test**

| MBS Tx ID (See Annex H) | **Slot Index (See Annex H)** | **All Other fields (See Annex H)** | **Code phase delay difference between beacons (ms)** | **Transmit Power (dBm)** |
|---|---|---|---|---|
| Equal to Slot number | Equal to Slot number | min value (bit_value = 0)[Note 1] | 0 | -128 |
| Note 1: bit_value is the conversion of the binary number represented by the corresponding bits in the payload to decimal. | | | | |

The MBS code phase measurement accuracy shall fulfil the requirements given in Table 11.2.5-3.

**Table 11.2.5-3: Accuracy requirements for Sensitivity scenario**

| **Code phase measurement accuracy (ms)[Note 1]** | **Accuracy requirement for the difference of code phase measurements (ms)[Note 2]** |
|---|---|
| $1.66 \times 10^{-4}$ | $2.40 \times 10^{-4}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | |

The test tolerances are defined in clauses C.1.4 and C.4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 11.2A MBS Sensitivity Measurement Accuracy (Release 14 Onwards)

## 11.2A.1 Test purpose

Same as defined in clause 11.2.1

## 11.2A.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 14 onwards.

## 11.2A.3 Minimum conformance requirements

Same as defined in clause 11.2.3 except that the accuracy requirements are:

**Table 11.2A.3-1: Accuracy requirements for sensitivity scenario**

| MBS Configuration | Signal Strength (dBm) | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|---|
| TB1 (2 MHz) | -130 | $1.66 \times 10^{-4}$ | $2.35 \times 10^{-4}$ |
| TB2 (5 MHz) | -130 | $6.6 \times 10^{-5}$ | $9.3 \times 10^{-5}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | | |

# 11.2A.4 Test description

## 11.2A.4.1 Initial conditions

Same as defined in clause 11.2.4.1

## 11.2A.4.2 Test procedure

Same as defined in clause 11.2.4.2, except step 4a is introduced and step 5 is modified as follows:

4a. The SS shall send an LPP PROVIDE ASSISTANCE DATA message to provide the MBS assistance data in accordance with TS 37.571-5 [20], and with the values defined therein. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE.

## 11.2A.4.3 Message contents

Same as defined in clause 11.2.4.3, with the addition of the LPP Provide Assistance Data

**Table 11.2A.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-ProvideAssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | Rel-14 onwards |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | | | Rel-14 onwards |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb1 | |

| | | | |
|---|---|---|---|
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| } | | | |
| } | | | |
| tbs-Error-r14 | Not Present | | |
| } | | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 | Not Present | | Rel-14 onwards |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.2A.5 Test requirement

Same as defined in clause 11.2.5, with the beacon parameters set according to the assistance data in TS 37.571-5 [20] clause 8 and with the measurement accuracy requirement in Table 11.2A.5-1.

**Table 11.2A.5-1: Accuracy requirements for Sensitivity scenario**

| MBS Configuration | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|
| TB1 (2 MHz) | $1.66 \times 10^{-4}$ | $2.40 \times 10^{-4}$ |
| TB2 (5 MHz) | $6.6 \times 10^{-5}$ | $9.8 \times 10^{-5}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | |

# 11.2B MBS Sensitivity Measurement Accuracy (NR)

## 11.2B.1 Test purpose

Same as defined in clause 11.2.1

## 11.2B.2 Test applicability

This test applies to all types of NR UE that supports UE-assisted MBS with LPP Release 13 onwards.

## 11.2B.3 Minimum conformance requirements

For NR UE that supports LPP Release 13 only, same as defined in clause 11.2.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.2A.3

## 11.2B.4 Test description

### 11.2B.4.1 Initial conditions

Same as defined in clause 11.2.4.1, except in the case of NR UE, step 3 is modified as follows:

3. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4.

### 11.2B.4.2 Test procedure

For NR UE that supports LPP Release 13 only, same as defined in clause 11.2.4.2. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.2A.4.2.

### 11.2B.4.3 Message contents

For NR UE that supports LPP Release 13 only, same as defined in clause 11.2.4.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.2A.4.3

## 11.2B.5 Test requirement

For NR UE that supports LPP Release 13 only, same as defined in clause 11.2.5. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.2A.5

# 11.3 MBS Nominal Measurement Accuracy (Release 13 only)

## 11.3.1 Test purpose

The purpose of this test is to verify that the MBS Code Phase measurement accuracy is within the specified limits under ideal conditions. This test will verify the requirements in clauses 5.3 of 37.171 [39] for MBS measurements. The channel type for this test is AWGN, as specified in clause 4.8.2.1.

## 11.3.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 13 only.

## 11.3.3 Minimum conformance requirements

The MBS code phase measurement accuracy shall fulfil the requirement given in Table 11.3.3-1.

**Table 11.3.3-1: Accuracy requirements for Nominal scenario**

| Signal Strength (dBm) | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|
| -30 | $5.0 \times 10^{-5}$ | $7.1 \times 10^{-5}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | |

The accuracy requirement for the difference of code phase measurements is derived from the requirement for the code phase measurement accuracy, assuming a scaling factor of $\sqrt{2}$ due to the compounding of two error distributions.

The normative reference for this requirement is TS 37.171 [39] clause 5.3 (Nominal) and clause A.4.2.

## 11.3.4 Test description

### 11.3.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and MSS to the UE antenna connector or antenna connectors as shown in figures A.6 or A.7.
2. Switch on the UE.
3. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 11.3.4.2 Test procedure

1. Set the MSS test parameters as specified in clause 11.3.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS shall send an LPP REQUEST CAPABILITIES message.
4. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the TBS capabilities supported by the UE in the *TBS-ProvideCapabilities* IE.
5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.
6. The UE shall perform and report the code phase measurement for each simulated beacon. If the UE transmits a *TBS-ProvideLocationInformation* IE including the *transmitterID* and *codePhase* field for the two simulated beacons and the difference between *codePhase* field values for the two beacons meet the corresponding requirements in Table 11.3.5-3, then the number of successful tests is increased by one. Otherwise the number of failure tests is increased by one.
7. If the UE message at step 6 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
8. Repeat steps 1-7 until the confidence level according to Annex D is achieved. For each iteration, at step 1 reselect the PN code assigned to each MBS beacon.
9. Release the signalling connection.

### 11.3.4.3 Message contents

**Table 11.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: TS 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 1 0 | MBS | |

**Table 11.3.4.3-2: LPP RequestCapabilities**

| Information Element | Value/remark |
|---|---|
| *tbs-RequestCapabilities-r13* | TRUE |

**Table 11.3.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |

| | | | |
|---|---|---|---|
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| Time | 12 | | |
| responseTimeEarlyFix-r12 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| Environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 SEQUENCE { | | | |
| mbsSgnMeasListReq-r13 | TRUE | | |
| } | | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 11.3.4.3-4: LPP ProvideLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| Acknowledgement | | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | | | |
| tbs-MeasurementInformation-r13 SEQUENCE { | | | |
| measurementReferenceTime-r13 | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| } | | | |
| tbs-Error-r13 | May be present with error reason ‘undefined’ or ‘thereWereNotEnoughMBSBeaconsReceived’ | | |
| } | | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.3.5 Test requirement

The details of the beacon parameters are in Table 11.3.5-1 and Table 11.3.5-2.

**Table 11.3.5-1: General test parameters for the beacons to be simulated for measurement accuracy in Nominal test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Number of Beacons | | 2 | Beacons transmitted in any two beacon slots in the beacon transmission period, but static for the test. Other slots contain no simulated beacons[Note 1] |
| Centre Frequency | MHz | 925.977 | |
| RF Channel | N/A | AWGN | |
| MBS Beacon Configuration | N/A | TB1 (2 MHz) | For details see Annex H |
| MBS Packet Type | N/A | Type 2 | For details see Annex H |

| | | | |
|---|---|---|---|
| Beacon PN Code | Integer | Chosen for each beacon from the PN code list for TB1 | Each of the 2 beacons uses a different PN code. For details see Annex H[Note 1] |
| Response time | Seconds | 12 | Value of Time used in LPP RequestLocationInformation message in Table 11.3.4.3-3 |
| Note 1: The slots and PN codes are chosen at random, such that the UE does not and cannot have advanced knowledge of what that slot/PN code might be. | | | |

**Table 11.3.5-2: MBS Beacon Payload fields, code phase delay difference and transmit powers for the beacons to be simulated for measurement accuracy in Nominal test**

| MBS Tx ID (See Annex H) | Slot Index (See Annex H) | All Other fields (See Annex H) | Code phase delay difference between beacons (ms) | Transmit Power (dBm) |
|---|---|---|---|---|
| Equal to Slot number | Equal to Slot number | min value (bit_value = 0)[Note 1] | 0 | -30 |
| Note: bit_value is the conversion of the binary number represented by the corresponding bits in the payload to decimal. | | | | |

The MBS code phase measurement accuracy shall fulfil the requirements given in Table 11.3.5-3.

**Table 11.3.5-3: Accuracy requirements for Nominal scenario**

| Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|
| $5.0 \times 10^{-5}$ | $7.6 \times 10^{-5}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | |

The test tolerances are defined in clauses C.1.4 and C.4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 11.3A MBS Nominal Measurement Accuracy (Release 14 Onwards)

## 11.3A.1 Test purpose

Same as defined in clause 11.3.1

## 11.3A.2 Test applicability

This test applies to all types of E-UTRA UE supports UE-assisted MBS with LPP Release 14 onwards.

## 11.3A.3 Minimum conformance requirements

Same as defined in clause 11.3.3 except that the accuracy requirements are:

**Table 11.3A.3-1: Accuracy requirements for Nominal scenario**

| MBS Configuration | Signal Strength (dBm) | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|---|
| TB1 (2 MHz) | -30 | $5.0 \times 10^{-5}$ | $7.1 \times 10^{-5}$ |
| TB2 (5 MHz) | -30 | $2.0 \times 10^{-5}$ | $2.8 \times 10^{-5}$ |
| Note 1: Provided for reference only. | | | |

| Note 2: To be used for testing. |
|---|

## 11.3A.4 Test description

### 11.3A.4.1 Initial conditions

Same as defined in clause 11.3.4.1

### 11.3A.4.2 Test procedure

Same as defined in clause 11.3.4.2, except step 4a is introduced and step 5 is modified as follows:

4a. The SS shall send an LPP PROVIDE ASSISTANCE DATA message to provide the MBS assistance data in accordance with TS 37.571-5 [20], and with the values defined therein. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE.

### 11.3A.4.3 Message contents

Same as defined in clause 11.3.4.3, with the addition of the LPP Provide Assistance Data.

**Table 11.3A.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-ProvideAssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | Rel-14 onwards |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | | | Rel-14 onwards |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |

| | | | |
|---|---|---|---|
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| } | | | |
| } | | | |
| tbs-Error-r14 | Not Present | | |
| } | | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 | Not Present | | Rel-14 onwards |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.3A.5 Test requirement

Same as defined in clause 11.3.5, with the beacon parameters set according to the assistance data in TS 37.571-5 [20] clause 8 and with the measurement accuracy requirement in Table 11.3A.5-1.

**Table 11.3A.5-1: Accuracy requirements for Nominal scenario**

| MBS Configuration | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|
| TB1 (2 MHz) | $5.0 \times 10^{-5}$ | $7.6 \times 10^{-5}$ |
| TB2 (5 MHz) | $2.0 \times 10^{-5}$ | $3.3 \times 10^{-5}$ |
| Note 1: Provided for reference only.<br>Note 2: To be used for testing. | | |

# 11.3B MBS Nominal Measurement Accuracy (NR)

## 11.3B.1 Test purpose

Same as defined in clause 11.3.1

## 11.3B.2 Test applicability

This test applies to all types of NR UE that supports UE-assisted MBS with LPP Release 13 onwards.

## 11.3B.3 Minimum conformance requirements

For NR UE that supports LPP Release 13 only, same as defined in clause 11.3.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.3A.3

## 11.3B.4 Test description

### 11.3B.4.1 Initial conditions

Same as defined in clause 11.3.4.1, except in the case of NR UE, step 3 is modified as follows:

3. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4.

### 11.3B.4.2 Test procedure

For NR UE that supports LPP Release 13 only, same as defined in clause 11.3.4.2. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.3A.4.2

### 11.3B.4.3 Message contents

For NR UE that supports LPP Release 13 only, same as defined in clause 11.3.4.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.3A.4.3

## 11.3B.5 Test requirement

For NR UE that supports LPP Release 13 only, same as defined in clause 11.3.5. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.3A.5

# 11.4 MBS Dynamic Range Measurement Accuracy (Release 13 only)

## 11.4.1 Test purpose

The purpose of this test is to verify that the MBS Code Phase measurement accuracy is within the specified limits under maximum dynamic range conditions. This test will verify the requirements in clauses 5.4 of TS 37.171 [39] for MBS measurements. The channel type for this test is AWGN, as specified in clause 4.8.2.1.

## 11.4.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 13 only.

## 11.4.3 Minimum conformance requirements

The MBS code phase measurement accuracy shall fulfil the requirements given in Table 11.4.3-1.

**Table 11.4.3-1: Accuracy requirements for Dynamic Range scenario**

| Signal Strength (dBm) | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|
| -30 | $5.0 \times 10^{-5}$ | $7.\ 1 \times 10^{-5}$ |
| -130 | $1.66 \times 10^{-4}$ | $2.35 \times 10^{-4}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | |

The accuracy requirement for the difference of code phase measurements is derived from the requirement for the code phase measurement accuracy, assuming a scaling factor of $\sqrt{2}$ due to the compounding of two error distributions.

The normative reference for this requirement is TS 37.171 [39] clause 5.4 (Dynamic Range) and clause A.4.2.

## 11.4.4 Test description

### 11.4.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and MSS to the UE antenna connector or antenna connectors as shown in figures A.6 or A.7.
2. Switch on the UE.
3. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 11.4.4.2 Test procedure

1. Set the MSS test parameters as specified in clause 11.4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS shall send an LPP REQUEST CAPABILITIES message.
4. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the TBS capabilities supported by the UE in the *TBS-ProvideCapabilities* IE.
5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.
6. The UE shall perform and report the code phase measurement for each simulated beacon. If the UE transmits a *TBS-ProvideLocationInformation* IE including the *transmitterID* and *codePhase* field for the four simulated beacons and the difference between *codePhase* field values for the two high power beacons and the difference in the *codePhase* field values for the two low power beacons meet the corresponding requirements in Table 11.4.5-3, then the number of successful tests is increased by one. Otherwise the number of failure tests is increased by one.
7. If the UE message at step 6 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
8. Repeat steps 1-7 until the confidence level according to Annex D is achieved. For each iteration, at step 1 reselect the PN code assigned to each MBS beacon.
9. Release the signalling connection.

### 11.4.4.3 Message contents

**Table 11.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: TS 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 1 0 | MBS | |

**Table 11.4.4.3-2: LPP RequestCapabilities**

| Information Element | Value/remark |
|---|---|
| *tbs-RequestCapabilities-r13* | TRUE |

**Table 11.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| Time | 12 | | |
| responseTimeEarlyFix-r12 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| Environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 SEQUENCE { | | | |
| mbsSgnMeasListReq-r13 | TRUE | | |
| } | | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 11.4.4.3-4: LPP ProvideLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | (0..255) | | |
| Acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | | | |
| tbs-MeasurementInformation-r13 SEQUENCE { | | | |
| measurementReferenceTime-r13 | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| } | | | |
| tbs-Error-r13 | May be present with error reason ‘undefined’ or ‘thereWereNotEnoughMBSBeaconsReceived’ | | |
| } | | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.4.5 Test requirement

The details of the beacon parameters are in Table 11.4.5-1 and Table 11.4.5-2.

**Table 11.4.5-1: General test parameters for the beacons to be simulated for measurement accuracy in Dynamic Range test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Number of Beacons | | 4 | Beacons 1 to 4. Transmitted in any four consecutive beacon slots in the beacon transmission period, but static for the test. Other slots contain no simulated beacons[Note 1] |
| Centre Frequency | MHz | 925.977 | |
| RF Channel | N/A | AWGN | |
| MBS Beacon Configuration | N/A | TB1 (2 MHz) | For details see Annex H |
| MBS Packet Type | N/A | Type 2 | For details see Annex H |
| Beacon PN Code | Integer | Chosen for each beacon from the PN code list for TB1 | Each of the 4 beacons uses a different PN code. For details see Annex H[Note 1] |
| Response time | Seconds | 12 | Value of Time used in LPP RequestLocationInformation message in Table 11.4.4.3-3 |
| Note 1: The slots and PN codes are chosen at random, such that the UE does not and cannot have advanced knowledge of what that slot/PN code might be. | | | |

**Table 11.4.5-2: MBS Beacon Payload fields, code phase delay difference and transmit powers for the beacons to be simulated for measurement accuracy in Dynamic Range test**

| Beacon | MBS Tx ID (See Annex H) | Slot Index (See Annex H) | All Other fields (See Annex H) | Code phase delay difference between beacons (ms) | Transmit Power (dBm) |
|---|---|---|---|---|---|
| 1 | Equal to Slot number | Equal to Slot number | min value (bit_value = 0)[Note 1] | Beacon 1 to beacon 3: 0 [Note 2] | -30 (high power) |
| 2 | Equal to Slot number | Equal to Slot number | min value (bit_value = 0)[Note 1] | Beacon 2 to beacon 4: 0 [Note 2] | -128 (low power) |
| 3 | Equal to Slot number | Equal to Slot number | min value (bit_value = 0)[Note 1] | Beacon 1 to beacon 3: 0 [Note 2] | -30 (high power) |
| 4 | Equal to Slot number | Equal to Slot number | min value (bit_value = 0)[Note 1] | Beacon 2 to beacon 4: 0 [Note 2] | -128 (low power) |
| Note 1: bit_value is the conversion of the binary number represented by the corresponding bits in the payload to decimal.<br>Note 2: The code phase delay difference between beacon 1 and 3 and beacon 2 and 4 shall be set to some non-zero value. | | | | | |

The MBS code phase measurement accuracy shall fulfil the requirements given in Table 11.4.5-3.

**Table 11.4.5-3: Accuracy requirements for Dynamic Range scenario**

| Beacon Signal Strength | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|
| High Power (-30 dBm) | $5.0 \times 10^{-5}$ | $7.6 \times 10^{-5}$ |
| Low Power (-130 dBm) | $1.66 \times 10^{-4}$ | $2.40 \times 10^{-4}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | |

The test tolerances are defined in clauses C.1.4 and C.4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 11.4A MBS Dynamic Range Measurement Accuracy (Release 14 Onwards)

## 11.4A.1 Test purpose

Same as defined in clause 11.4.1

## 11.4A.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 14 onwards.

## 11.4A.3 Minimum conformance requirements

Same as defined in clause 11.4.3 except that the accuracy requirements are:

**Table 11.4A.3-1: Accuracy requirements for Dynamic Range scenario**

| MBS Configuration | Signal Strength (dBm) | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|---|
| TB1 (2 MHz) | -30 | $5.0 \times 10^{-5}$ | $7.1 \times 10^{-5}$ |
| | -130 | $1.66 \times 10^{-4}$ | $2.35 \times 10^{-4}$ |
| TB2 (5 MHz) | -30 | $2.0 \times 10^{-5}$ | $2.8 \times 10^{-5}$ |
| | -130 | $6.6 \times 10^{-5}$ | $9.3 \times 10^{-5}$ |
| Note 1: Provided for reference only.<br>Note 2: To be used for testing. | | | |

## 11.4A.4 Test description

### 11.4A.4.1 Initial conditions

Same as defined in clause 11.4.4.1

### 11.4A.4.2 Test procedure

Same as defined in clause 11.4.4.2, except step 4a is introduced and step 5 is modified as follows:

4a. The SS shall send an LPP PROVIDE ASSISTANCE DATA message to provide the MBS assistance data in accordance with TS 37.571-5 [20], and with the values defined therein. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation* IE.

### 11.4A.4.3 Message contents

Same as defined in clause 11.4.4.3, with the addition of the LPP Provide Assistance Data.

**Table 11.4A.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

| | | | |
|---|---|---|---|
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-ProvideAssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | Rel-14 onwards |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | | | Rel-14 onwards |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 3 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 4 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 3 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 4 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| tbs-Error-r14 | Not Present | | |
| } | | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 | Not Present | | Rel-14 onwards |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.4A.5 Test requirement

Same as defined in clause 11.4.5, with the beacon parameters set according to the assistance data in TS 37.571-5 [20] clause 8 and with the measurement accuracy requirementsin Table 114A.5-1.

**Table 11.4A.5-1: Accuracy requirements for Dynamic Range scenario**

| MBS Configuration | Beacon Signal Strength | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|---|
| TB1 (2 MHz) | High Power (-30 dBm) | $5.0 \times 10^{-5}$ | $7.6 \times 10^{-5}$ |
| | Low Power (-130 dBm) | $1.66 \times 10^{-4}$ | $2.40 \times 10^{-4}$ |
| TB2 (5 MHz) | High Power (-30 dBm) | $2.0 \times 10^{-5}$ | $3.3 \times 10^{-5}$ |
| | Low Power (-130 dBm) | $6.6 \times 10^{-5}$ | $9.8 \times 10^{-5}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | | |

# 11.4B MBS Dynamic Range Measurement Accuracy (NR)

## 11.4B.1 Test purpose

Same as defined in clause 11.4.1

## 11.4B.2 Test applicability

This test applies to all types of NR UE that supports UE-assisted MBS with LPP Release 13 onwards.

## 11.4B.3 Minimum conformance requirements

For NR UE that supports LPP Release 13 only, same as defined in clause 11.4.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.4A.3

## 11.4B.4 Test description

### 11.4B.4.1 Initial conditions

Same as defined in clause 11.4.4.1, except in the case of NR UE, step 3 is modified as follows:

3. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4.

### 11.4B.4.2 Test procedure

For NR UE that supports LPP Release 13 only, same as defined in clause 11.4.4.2. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.4A.4.2

### 11.4B.4.3 Message contents

For NR UE that supports LPP Release 13 only, same as defined in clause 11.4.4.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.4A.4.3

## 11.4B.5 Test requirement

For NR UE that supports LPP Release 13 only, same as defined in clause 11.4.5. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.4A.5

# 11.5 MBS Measurement Accuracy in Multipath (Release 13 only)

## 11.5.1 Test purpose

The purpose of this test is to verify that the MBS Code Phase measurement accuracy is within the specified limits in a multipath environment. This test will verify the requirements in clause 5.5 of TS 37.171 [39] for MBS measurements. The channel type for the test is EPA 5 Hz, as specified in clause 4.8.2.2.

## 11.5.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 13 only.

## 11.5.3 Minimum conformance requirements

The MBS code phase measurement accuracy shall fulfil the requirements in Table 11.5.3-1.

**Table 11.5.3-1: Accuracy requirements for Multipath scenario**

| Direct Path Signal Strength (dBm) | Code phase measurement accuracy (ms)[Note 1] | Accuracy requirement for the difference of code phase measurements (ms)[Note 2] |
|---|---|---|
| -30 | $1.66 \times 10^{-4}$ | $2.35 \times 10^{-4}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | | |

The accuracy requirement for the difference of code phase measurements is derived from the requirement for the code phase measurement accuracy, assuming a scaling factor of $\sqrt{2}$ due to the compounding of two error distributions.

The normative reference for this requirement is TS 37.171 [39] clause 5.5 (Multipath) and clause A.4.3.

## 11.5.4 Test description

### 11.5.4.1 Initial conditions

Test environment: normal; see Annex G.

1. Connect SS and MSS to the UE antenna connector or antenna connectors as shown in figures A.6 or A.7.
2. Switch on the UE.
3. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

### 11.5.4.2 Test procedure

1. Set the MSS test parameters as specified in clause 11.5.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS shall send an LPP REQUEST CAPABILITIES message.
4. The UE shall transmit an LPP PROVID CAPABILITIES message indicating the TBS capabilities supported by the UE in the *TBS-ProvideCapabilities* IE.
5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *TBS-RequestLocationInformation*. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.
6. The UE shall perform and report the code phase measurement for each simulated beacon. If the UE transmits a *TBS-ProvideLocationInformation* IE including the *transmitterID* and *codePhase* field for the two simulated and the difference between *codePhase* field values for the two beacons meets the requirement in Table 11.5.5-3, then the number of successful tests is increased by one. Otherwise the number of failure tests is increased by one.
7. If the UE message at step 6 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
8. Repeat steps 1-7 until the confidence level according to Annex D is achieved. For each iteration, at step 1 reselect the PN code assigned to each MBS beacon.
9. Release the signalling connection.

### 11.5.4.3 Message contents

**Table 11.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: TS 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 1 0 | MBS | |

**Table 11.5.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| *tbs-RequestCapabilities-r13* | TRUE |

**Table 11.5.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |

| | | | |
|---|---|---|---|
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| Time | 12 | | |
| responseTimeEarlyFix-r12 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| Environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 SEQUENCE { | | | |
| mbsSgnMeasListReq-r13 | TRUE | | |
| } | | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 11.5.4.3-4: LPP ProvideLocationInformation**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| Acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| tbs-MeasurementInformation-r13 SEQUENCE { | | | |
| measurementReferenceTime-r13 | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| mbs-SgnMeasList-r13 SEQUENCE (SIZE(n)) { | | | |
| transmitterID-r13 | Present | | |
| codePhase-r13 | Present | | |
| codePhaseRMSError-r13 | | | |
| } | | | |
| } | | | |
| tbs-Error-r13 | May be present with error reason 'undefined' or 'thereWereNotEnoughMBSBeaconsReceived' | | |
| } | | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 11.5.5 Test requirement

The details of the beacon parameters are in Table 11.5.5-1 and Table 11.5.5-2.

**Table 11.5.5-1: General test parameters for the beacons to be simulated for measurement accuracy in Multipath test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Number of beacons | Integer | 2 | Beacons transmitted in the any two beacon slots in the beacon transmission period, but static for the test. Other slots contain no simulated beacons.Note 1 |
| Centre Frequency | MHz | 925.977 | |
| RF Channel | N/A | EPA 5 Hz | |
| MBS Beacon Configuration | N/A | TB1 (2 MHz) | For details see Annex H |
| MBS Packet Type | N/A | Type 2 | For details see Annex H |
| Beacon PN Code | Integer | Chosen for each beacon from the PN code list for TB1 | Each of the 2 beacons uses a different PN code For details see Annex HNote 1 |
| Response time | Seconds | 12 | Value of Time used in LPP RequestLocationInformation message in Table 11.5.4.3-3 |
| Note 1: The slots and PN codes are chosen at random, such that the UE does not and cannot have advanced knowledge of what that slot/PN code might be. | | | |

**Table 11.5.5-2: MBS Beacon Payload fields, and code phase delay difference and transmit powers for the beacons to be simulated for measurement accuracy in Multipath test**

| MBS Tx ID (See Annex H) | Slot Index (See Annex H) | All Other fields (See Annex H) | Code phase delay difference between beacons (ms) | Transmit Power (dBm) |
|---|---|---|---|---|
| Equal to Slot number | Equal to Slot number | min value (bit_value = 0)Note 1 | 0 | -30 |
| Note 1: bit_value is the conversion of the binary number represented by the corresponding bits in the payload to decimal. | | | | |

The MBS code phase measurement accuracy shall fulfil the requirements in Table 11.5.5-3.

**Table 11.5.5-3: Accuracy requirements for Multipath scenario**

| Code phase measurement accuracy (ms)Note 1 | Accuracy requirement for the difference of code phase measurements (ms)Note 2 |
|---|---|
| $1.66 \times 10^{-4}$ | $2.40 \times 10^{-4}$ |
| Note 1: Provided for reference only<br>Note 2: To be used for testing | |

The test tolerances are defined in clauses C.1.4 and C.4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 11.5A MBS Measurement Accuracy in Multipath (Release 14 Onwards)

## 11.5A.1 Test purpose

Same as defined in clause 11.5.1

## 11.5A.2 Test applicability

This test applies to all types of E-UTRA UE that supports UE-assisted MBS with LPP Release 14 onwards.

## 11.5A.3 Minimum conformance requirements

Same as defined in clause 11.5.3

## 11.5A.4 Test description

### 11.5A.4.1 Initial conditions

Same as defined in clause 11.5.4.1

### 11.5A.4.2 Test procedure

Same as defined in clause 11.5.4.2, except step 4a is introduced and step 5 is modified as follows:

4a. The SS shall send an LPP PROVIDE ASSISTANCE DATA message to provide the MBS assistance data in accordance with TS 37.571-5 [20], and with the values defined therein. If the UE message at step 4 includes the *ackRequested* IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

5. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the TBS RequestLocationInformation.

### 11.5A.4.3 Message contents

Same as defined in clause 11.5.4.3, with the addition of the LPP Provide Assistance Data.

**Table 11.5A.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 36.355 [4] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-ProvideAssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | Rel-14 onwards |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | | | Rel-14 onwards |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb1 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb2 | |
| mbs-AlmanacAssistance-r14 | Not Present | | |
| mbs-AcquisitionAssistance-r14 | As defined in TS 37.571-5 [20], clause 8 | | |
| } | | | |
| } | | | |
| } | | | |
| tbs-Error-r14 | Not Present | | |
| } | | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 | Not Present | | Rel-14 onwards |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

### 11.5A.5 Test requirement

Same as defined in clause 11.5.5, with the beacon parameters set according to the assistance data in TS 37.571-5 [20] clause 8.

## 11.5B MBS Measurement Accuracy in Multipath (NR)

### 11.5B.1 Test purpose

Same as defined in clause 11.5.1.

### 11.5B.2 Test applicability

This test applies to all types of NR UE that supports UE-assisted MBS with LPP Release 13 onwards.

### 11.5B.3 Minimum conformance requirements

Same as defined in clause 11.5.3.

### 11.5B.4 Test description

#### 11.5B.4.1 Initial conditions

For NR UE that supports LPP Release 13 only, same as defined in clause 11.5.4.1. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.5A.4.1.

#### 11.5B.4.2 Test procedure

For NR UE that supports LPP Release 13 only, same as defined in clause 11.5.4.2. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.5A.4.2.

#### 11.5B.4.3 Message contents

For NR UE that supports LPP Release 13 only, same as defined in clause 11.5.4.3. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.5A.4.3.

### 11.5B.5 Test requirement

For NR UE that supports LPP Release 13 only, same as defined in clause 11.5.5. For NR UE that supports LPP Release 14 onwards, same as defined in clause 11.5A.5.

# 12 E-UTRA WLAN and BLE measurement requirements

## 12.0 General

This clause defines the minimum performance requirements for WLAN and/or BLE FDD and TDD E-UTRA UEs and NR UEs. Details for NR UEs are FFS.

# 12.1 WLAN Access Point Identification and Reporting Delay

## 12.1.1 WLAN AP Identification and reporting delay under nominal conditions

### 12.1.1.1 Test purpose

The purpose of this test is to verify that the E-UTRAN UE WLAN AP measurements fulfil the performance requirements for WLAN AP identification under nominal conditions in TS 37.171 [39] clause 7.3 and reporting delay in TS 37.171 [39] clause 4.3.

### 12.1.1.2 Test applicability

This test applies to all types of E-UTRA UE that support LPP release 14 and forward and WLAN positioning. Optionally, this test can be run by LPP release 13 UEs.

### 12.1.1.3 Minimum conformance requirements

Under nominal conditions of the WLAN signal, the UE shall be able to identify 6 WLAN APs. The minimum requirements for Nominal conditions are shown in Table 12.1.1.3-1. In these requirements, AWGN channel model is used and the signal level is above the noise floor.

**Table 12.1.1.3-1: Requirements for WLAN Access Point Identification under Nominal conditions**

| Number of WLAN APs | Signal Strength (dBm) | % of reported Access Points |
|---|---|---|
| 6 | -60 | 90 |

For LTE, the WLAN measurement time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response times specified for all test cases are based on new measurements unless otherwise stated, i.e. the UE shall not re use any information on measurements or other aiding data that was previously acquired or calculated and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 for the purpose of deleting this information. No WLAN assistance data is provided to the UE.

The signals from the WLAN APs shall be available at the UE for the duration of the measurement time. Each WLAN AP transmits a beacon signal with a beacon interval smaller or equal to 102.4 ms. The beacon frames from different access points shall be transmitted in different time slots or non-overlapping frequency channels. The beacon frames have variable time duration of ~1ms.

The WLAN Measurement Reporting Delay is given as:

$$T_{WLAN_meas} = \tau + 20 \text{ sec}$$

where:

$T_{WLAN_meas}$ is the total time for detecting and measuring the WLAN Access Points

$\tau$ is the elapsed time from the trigger of the measurement to the start of the first WLAN transmission period and is shown in Figure 12.1.1.3-1.

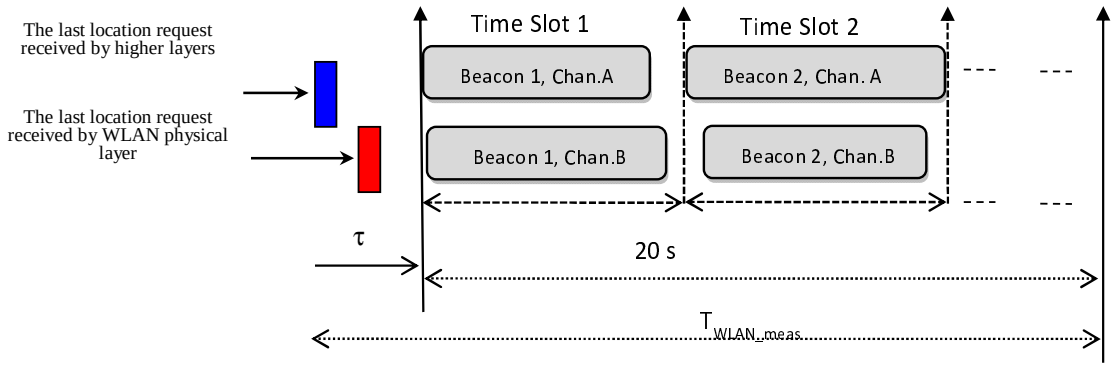



**Figure 12.1.1.3-1: Illustration of the WLAN Measurement Time**

The normative reference for the WLAN reporting delay requirement is 3GPP TS 37.171 [39] clause 4.3 and the normative reference for the WLAN AP Identification under nominal conditions requirement is 3GPP TS 37.171 [39] clause 7.3.

### 12.1.1.4 Test description

There is one active LTE cell and 6 WLAN APs transmitting beacon signals at least every 102.4 ms. The APs are transmitting in 3 non-overlapping frequency channels in the same WLAN Frequency Band. Non-overlapping frequency channels are those whose centre frequencies are separated by at least 25 MHz in the WLAN 2.4 GHz band and by at least 20 MHz in the WLAN 5 GHz band. There are 2 APs in every channel. The tested UE is connected to the serving cell and signalled to report WLAN AP measurements. The test consists of two successive time periods, with duration of T1 and T2, respectively. WLAN-RequestLocationInformation message shall be provided to the UE during T1. WLAN Access Points only transmit signal during T2. The test equipment compares the BSSID reported by the UE in the WLAN AP measurements with the BSSID of the APs simulated in the test.

#### 12.1.1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

E-UTRA frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

WLAN Channel Numbers to be tested: as specified in Table 12.1.1.5-2 and as defined in TS 36.508 [18] clause 4.3.1.6.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in Annex A figure A.8.
2. Propagation conditions are set according to clause 4.9.2.1.
3. Message contents are defined in clause 12.1.1.4.3.
4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test. After the connection is established, the parameter settings for the cell are set according to Table 12.1.1.5-2.
5. Switch on the UE.
6. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

#### 12.1.1.4.2 Test procedure

1. Set the SS test parameters as specified in clause 12.1.1.5. The BSSID of the simulated APs shall be generated in a random manner.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. T1 starts.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the WLAN capabilities supported by the UE in the *WLAN-ProvideCapabilities* IE.
6. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *WLAN-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms. If the UE message at step 5 includes the *ackRequested IE* set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.

6a. If the UE sends a LPP REQUEST ASSISTANCE DATA message requesting WLAN assistance data, the SS shall send a LPP PROVIDE ASSISTANCE DATA message, including *wlan-ProvideAssistanceData-r14* IE with no WLAN assistance data and the *wlan-Error-r14* IE with *WLAN-LocationServerErrorCauses-r13* with *cause-r13* set to *requestedADNotAvailable-v1420.*

If the UE message includes the *ackRequested IE* set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. When T1 expires, the SS shall switch the WLAN power setting from T1 to T2 as specified in Table 12.1.1.5-2.
8. The UE shall perform and report the WLAN AP measurements for the simulated WLAN APs. The UE shall transmit a *WLAN-ProvideLocationInformation* IE including the *wlan-MeasurementList-r13* field. If the report is sent within the maximum response time specified in Clause 12.1.1.5 and it includes WLAN Measurements for at least the percentage of the simulated APs indicated in Table 12.1.1.3-1, the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one. The verification shall be done by comparing the reported list of bssid-r13 against the simulated BSSIDs.
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
10. Repeat steps 1-9 until the confidence level according to Annex D, clauses D.4.3 and D.4.4 is achieved. For each iteration, at step 1 reselect a new list of WLAN APs. The BSSID of the new APs shall be different from the previous set of simulated BSSIDs.

10a. The test is repeated for both the 2.4GHz and 5GHz WLAN bands if supported by the UE.

11. Release the signalling connection.

### 12.1.1.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 4.6 with the following exceptions:

**Table 12.1.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 1 1 | WLAN | |

**Table 12.1.1.4.3-2: *LPP-RequestCapabilities***

| Derivation Path: 36.355 [4] clause 6.3 | |
|---|---|
| **Information Element** | **Value/remark** |
| *wlan-RequestCapabilities-r13* | TRUE |

**Table 12.1.1.4.3-3: *LPP-RequestLocationInformation***

| Derivation Path: TS 36.355 clause 6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| verticalCoordinateRequest | FALSE | | |
| responseTime SEQUENCE { | | | |
| time | 21 | See clause 12.1.1.5 | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| } | | | |
| wlan-RequestLocationInformation-r13 ::= SEQUENCE { | | | |
| requestedMeasurements-r13 | If reporting of RSSI is supported by the UE: 1 0 If reporting of RSSI is not supported by the UE: 0 0 | RSSI Requested if reporting of RSSI is supported by the UE as indicated by IE rssi-r13 in the LPP PROVIDE CAPABILITIES message | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 12.1.1.5 Test requirement

The UE shall send *WLAN-ProvideLocationInformation* within a maximum response time of 21.15 seconds (including test tolerance of 300ms) from the beginning of T2. The maximum response time is equal to the LPP time IE value of 21 seconds, minus ΔT, where ΔT = 150 ms, plus the test tolerance of 300ms. The LPP time IE value is derived from $T_{WLAN_meas}$, where $T_{WLAN_meas} = \tau + 20$ seconds as described in clause 12.1.1.3, and where the value of $\tau$ is equal to ΔT, where ΔT = 150 ms, plus one beacon interval, which is taken as 100 ms, giving a value for $\tau$ of 250 ms. This value for $T_{WLAN_meas}$ of 20.25 seconds is then rounded up to the next allowed LPP value of 21 seconds.

The *wlan-MeasurementInformation* IE shall include WLAN measurements for each AP indicating at least wlan-AP-Identifier (BSSID) and RSSI (if reporting of RSSI is supported by the UE as indicated by the UE in the LPP PROVIDE CAPABILITIES message). The list of reported BSSIDs shall contain at least the BSSID of 90% of the WLAN APs simulated in the test, as defined in Table 12.1.1.3-1.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

**Table 12.1.1.5-1: General WLAN AP test parameters for WLAN AP Identification and reporting delay under nominal conditions test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Number of Access Points | N/A | 6 | AP 1-AP 6 |
| Time Slot 1 | ms | 1 | AP 1, AP 2 |

| | | | |
|---|---|---|---|
| Time Slot 2 | ms | 1 | AP 3, AP 4 |
| Time Slot 3 | ms | 1 | AP 5 |
| Time Slot 4 | ms | 1 | AP 6 |
| T1 | s | 5 | During this time the WLAN signals are not transmitted |
| T2 | s | 25 | UE should report WLAN measurement information within 21.15s (including test tolerance) |

**Table 12.1.1.5-2: Cell specific and WLAN AP specific test parameters for WLAN AP Identification and reporting delay under nominal conditions test**

| Parameter | Unit | Cell 1 | | AP.1, 4 | | AP.2, 5 | | AP.3, 6 | |
|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| WLAN Test Frequency ID | | N/A | | 1 | | 2 | | 3 | |
| PDSCH parameters: DL Reference Measurement Channel Note 6 | | R.0 FDD R.0 TDD | | N/A | | N/A | | N/A | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel Note 6 | | R.6 FDD R.6 TDD | | N/A | | N/A | | N/A | |
| OCNG Patterns Note 6 | | OP.1 FDD OP.1 TDD | | N/A | | N/A | | N/A | |
| PBCH_RA | dB | 0 | | N/A | | N/A | | N/A | |
| PBCH_RB | dB | | | | | | | | |
| PSS_RA | dB | | | | | | | | |
| SSS_RA | dB | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | |
| PHICH_RA | dB | | | | | | | | |
| PHICH_RB | dB | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | |
| $N_{oc1}$ Note 2 | dBm/15 KHz | -98 | | N/A | | N/A | | N/A | |
| $N_{oc2}$ Note 3 | dBm/20 MHz | N/A | | -75 | | -75 | | -75 | |
| $\hat{E}_s/N_{oc1}$ | dB | 3 | 3 | N/A | | N/A | | N/A | |
| $\hat{E}_s/I_{ot}$ Note 4 | dB | 3 | 3 | | | | | | |
| RSRP Note 4 | dBm/15 kHz | -95 | -95 | | | | | | |
| SCH_RP Note 4 | dBm/15 kHz | -95 | -95 | | | | | | |
| Io Note 3 | dBm/Ch BW | -65.5 | -65.5 | | | | | | |
| WLAN Received Power Level | dBm | N/A | N/A | - inf | -60 | - inf | -60 | - inf | -60 |
| WLAN SNR Note 4 | dB | N/A | | 15 | | 15 | | 15 | |
| Propagation Condition | | AWGN | | | | | | | |
| Antenna Configuration | | 1x2 | | - | | - | | - | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc1}$ to be fulfilled.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over the bandwidth and time and shall be modelled as AWGN of appropriate power for $N_{oc2}$ to be fulfilled.
Note 4: Es/Iot, RSRP, SCH_RP, Io and WLAN SNR have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 6: If Cell 1 is LTE FDD, the FDD OCNG and RMCs shall be used. If Cell 1 is LTE TDD, the TDD OCNG and RMCs shall be used.

## 12.1.2 WLAN AP Identification and reporting delay under dynamic range conditions

### 12.1.2.1 Test purpose

The purpose of this test is to verify that the E-UTRAN UE WLAN AP measurements fulfil the performance requirements for WLAN AP identification under dynamic range conditions in TS 37.171 [39] clause 7.4 and reporting delay in TS 37.171 [39] clause 4.3.

### 12.1.2.2 Test applicability

This test applies to all types of E-UTRA UE that support LPP release 14 and forward and WLAN positioning. Optionally, this test can be run by LPP release 13 UEs.

### 12.1.2.3 Minimum conformance requirements

The WLAN Access Point identification under dynamic range conditions verifies the UE capability to identify and report WLAN APs when the received power difference between WLAN APs is large. The power difference between APs follows the adjacent channel rejection criteria defined by IEEE in [40].

The UE shall be able to identify at least 3 WLAN AP located in 3 adjacent channels where the separation between channels is ≥ 20 MHz and the middle channel is received with high power and the side channels are received with low power.

**Table 12.1.2.3-1: Requirements for WLAN Access Point Identification under Dynamic Range conditions**

| Number of WLAN APs | Signal Strength (dBm) | % of reported Access Points |
|---|---|---|
| 3 | See [40] | 100 |

For LTE, the WLAN measurement time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response times specified for all test cases are based on new measurements unless otherwise stated, i.e. the UE shall not re use any information on measurements or other aiding data that was previously acquired or calculated and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 for the purpose of deleting this information. No WLAN assistance data is provided to the UE.

The signals from the WLAN APs shall be available at the UE for the duration of the measurement time. Each WLAN AP transmits a beacon signal with a beacon interval smaller or equal to $T_{WLAN_TP}$ (102.4 ms). The beacon frames from different access points shall be transmitted in different time slots or non-overlapping frequency channels. The beacon frames have variable time duration of ~1ms.

The WLAN Measurement Reporting Delay is given as:

$$T_{WLAN_meas} = \tau + 20 \text{ .sec}$$

where:

$T_{WLAN_meas}$ is the total time for detecting and measuring the WLAN Access Points

$\tau$ is the elapsed time from the trigger of the measurement to the start of the first WLAN transmission period and is shown in Figure 12.1.1.3-1.

The normative reference for the WLAN reporting delay requirement is 3GPP TS 37.171 [39] clause 4.3 and the normative reference for the WLAN AP Identification under dynamic range requirement is 3GPP TS 37.171 [39] clause 7.4.

### 12.1.2.4 Test description

In this test, there are LTE cell1 and 3 WLAN APs transmitting beacon signals at least every 102.4 ms. The APs are transmitting in 3 non-overlapping frequency channels in the same WLAN Frequency Band. Non-overlapping frequency channels are those whose centre frequencies are separated by at least 25 MHz in the WLAN 2.4 GHz band and by at least 20 MHz in the WLAN 5 GHz band. There is 1 AP in every channel. The tested UE is connected to the serving cell and signalled to report WLAN AP measurements. The test consists of two successive time periods, with duration of T1 and T2, respectively. WLAN-RequestLocationInformation message shall be provided to the UE during T1. WLAN Access Points only transmit signal during T2. The test equipment compares the BSSID reported by the UE in the WLAN AP measurements with the BSSID of the APs simulated in the test.

#### 12.1.2.4.1 Initial conditions

Same as in Clause 12.1.1.4.1

#### 12.1.2.4.2 Test procedure

Same as in clause 12.1.1.4.2 with the exception that SS test parameters are specified in clause 12.1.2.5 and the percentage of reported WLAN APs to count an iteration as successful is defined in Table 12.1.2.3-1

#### 12.1.2.4.3 Message contents

Same as in clause 12.1.1.4.3.

### 12.1.2.5 Test requirement

The UE shall send *WLAN-ProvideLocationInformation* within a maximum response time of 21.15 seconds (including test tolerance of 300ms) from the beginning of T2. See clause 12.1.1.5 for more details.

The *wlan-MeasurementInformation* IE shall include WLAN measurements for each AP indicating at least wlan-AP-Identifier (BSSID) and RSSI (if reporting of RSSI is supported by the UE as indicated by the UE in the LPP PROVIDE CAPABILITIES message). The list of reported BSSIDs shall contain the BSSID of 100% of the WLAN APs simulated in the test, as defined in Table 12.1.2.3-1.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

**Table 12.1.2.5-1: General test parameters for WLAN AP Identification and reporting delay under dynamic range conditions test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Number of Access Points | N/A | 3 | AP 1-AP 3 |
| Time Slot 1 | ms | 1 | AP 1, AP 2, AP 3 |
| T1 | s | 5 | During this time the WLAN signals are not transmitted |
| T2 | s | 25 | UE should report WLAN measurement information within 21.15s (including test tolerance) |

**Table 12.1.2.5-2: Cell specific test parameters for WLAN AP Identification and reporting delay under dynamic range conditions test**

| Parameter | Unit | Cell 1 | | AP 1 | | AP 2 | | AP 3 | |
|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| WLAN Channel Number | | N/A | | 1 | | 2 | | 3 | |
| PDSCH parameters: DL Reference Measurement Channel Note 6 | | R.0 FDD R.0 TDD | | N/A | | N/A | | N/A | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel Note 6 | | R.6 FDD R.6 TDD | | N/A | | N/A | | N/A | |
| OCNG Patterns Note 6 | | OP.1 FDD OP.1 TDD | | N/A | | N/A | | N/A | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| PBCH_RA | dB | 0 | | N/A | | | N/A | | | N/A | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | |
| $N_{oc1}$ Note 2 | dBm/15 KHz | -98 | | N/A | | | N/A | | | N/A | |
| $N_{oc2}$ Note 3 | dBm/20 MHz | N/A | | -85 | | | -85 | | | -85 | |
| $\hat{E}_s/N_{oc1}$ | dB | 3 | 3 | N/A | | | N/A | | | N/A | |
| $\hat{E}_s/I_{ot}$ Note 4 | dB | 3 | 3 | | | | | | | | |
| RSRP Note 4 | dBm/15 kHz | -95 | -95 | | | | | | | | |
| SCH_RP Note 4 | dBm/15 kHz | -95 | -95 | | | | | | | | |
| Io Note 3 | dBm/Ch BW | -65.5 | -65.5 | | | | | | | | |
| WLAN Received Power Level | dBm | N/A | N/A | -inf | WLAN 2.4 GHz band: -73 WLAN 5 GHz band: -78 | | -inf | WLAN 2.4 GHz band: -39 WLAN 5 GHz band: -63 | | -inf | WLAN 2.4 GHz band: -73 WLAN 5 GHz band: -78 |
| WLAN SNR Note 4 | db | N/A | | WLAN 2.4 GHz band: 12 WLAN 5 GHz band: 7 | | | WLAN 2.4 GHz band: 46 WLAN 5 GHz band: 22 | | | WLAN 2.4 GHz band: 12 WLAN 5 GHz band: 7 | |
| Propagation Condition | | AWGN | | | | | | | | | |
| Antenna Configuration | | 1x2 | | - | | | - | | | - | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc1}$ to be fulfilled.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over the bandwidth and time and shall be modelled as AWGN of appropriate power for $N_{oc2}$ to be fulfilled.
Note 4: Es/Iot, RSRP, SCH_RP, Io and WLAN SNR have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 6: If Cell 1 is LTE FDD, the FDD OCNG and RMCs shall be used. If Cell 1 is LTE TDD, the TDD OCNG and RMCs shall be used.

# 12.2 BLE Identification and Reporting Delay

## 12.2.1 Bluetooth identification

### 12.2.1.1 Test purpose

The purpose of this test is to verify that the E-UTRAN UE BLE measurements fulfil the performance requirements for Bluetooth Access Point identification and reporting delay in TS 37.171 [39] clause 4.4.

### 12.2.1.2 Test applicability

This test applies to all types of E-UTRA UE that support LPP release 14 and forward and BLE positioning. Optionally, this test can be run by LPP release 13 UEs.

### 12.2.1.3 Minimum conformance requirements

In the RRC_CONNECTED state the measurement period for Bluetooth Access Point identification shall be $T_{BT_meas}$. The value of $T_{BT_meas}$ is 10.24s, and can be extended to 40.96s if extended inquiry is allowed, provided that the following conditions are met:

- At least one Bluetooth beacon signal is transmitted on one of the Bluetooth advertising channels with a broadcast interval of 100 ms.

The UE physical layer shall be capable of reporting Bluetooth Access Point(s) measurements to higher layers within the measurement period of $T_{BT_meas}$. For LTE, the BLE measurement time is defined as the time starting from the moment that the UE has received the LPP message of type REQUEST LOCATION INFORMATION, and ending when the UE starts sending the LPP message of type PROVIDE LOCATION INFORMATION on the Uu interface. The response times specified for all test cases are based on new measurements unless otherwise stated, i.e. the UE shall not re use any information on measurements or other aiding data that was previously acquired or calculated and stored internally in the UE. A dedicated test message 'RESET UE POSITIONING STORED INFORMATION' has been defined in TS 36.509 [11] clause 6.9 for deleting this information.

The signals from the BLE devices shall be available at the UE for the duration of the measurement time. Each BLE device transmits a beacon signal with a broadcast interval of $T_{BLE_TP}$ of 100 ms. Beacon frames from different BLE devices shall be transmitted in different time slots or non-overlapping frequency channels.

The normative reference for this requirement is 3GPP TS 37.171 [39] clause 4.4.

### 12.2.1.4 Test description

There is one active LTE cell and 6 BLE devices transmitting advertising non-connectable beacon signals at least every 100 ms. The BLE devices are transmitting in 3 non-overlapping BLE advertising frequency channels. The BLE advertising channels are Channel 37 (2402 MHz), Channel 38 (2426 MHz) and Channel 39 (2480 MHz). There are 2 BLE devices transmitting in each channel. The tested UE is connected to the serving cell and signalled to report BLE measurements. The test consists of two successive time periods, with duration of T1 and T2, respectively. BT-RequestLocationInformation message shall be provided to the UE during T1. BLE devices only transmit signal during T2. The test equipment compares the UUID reported by the UE in the BLE measurements with the UUID of the BLE devices simulated in the test.

#### 12.2.1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [18] clause 4.1.

E-UTRA frequencies to be tested: Mid Range, as defined in TS 36.508 [18] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in Annex A figure A.9.

2. Propagation conditions are set according to clause 4.10.2.1.

3. Message contents are defined in clause 12.2.1.4.3.

4. Cell 1 is the serving cell. Cell 1 is the cell used for connection setup with the power levels set according to TS 36.521-3 [25] clauses C.0 and C.1 for this test. After the connection is established, the parameter settings for the cell are set according to Table 12.2.1.5-2.

5. Switch on the UE.

6. Establish a signalling connection according to the generic procedure in TS 36.508 [18] clause 4.5.3 (State 3, Generic RB established) on a channel in the Mid EARFCN range.

#### 12.2.1.4.2 Test procedure

1. Set the SS test parameters as specified in clause 12.2.1.5. The UUID of the simulated BLE devices shall be generated in a random manner.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. T1 starts.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the BLE capabilities supported by the UE in the *BT-ProvideCapabilities* IE.
6. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *BT-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 150 ms. If the UE message at step 5 includes the *ackRequested IE* set to TRUE, then the SS shall send an acknowledgment in the LPP REQUEST LOCATION INFORMATION message.
7. When T1 expires, the SS shall switch the BLE power setting from T1 to T2 as specified in Table 12.2.1.5-2.
8. The UE shall perform and report the BLE measurements for the simulated BLE devices. The UE shall transmit a *BT-ProvideLocationInformation* IE including the *BT-MeasurementList-r13* field. If the report is sent within the maximum response time specified in Clause 12.2.1.5 and it includes BT Measurements for all of the simulated BLE devices, the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one. The verification shall be done by comparing the reported list of *btAddr-r13* against the simulated UUIDs.
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
10. Repeat steps 1-9 until the confidence level according to Annex D, clauses D.4.3 and D.4.4 is achieved. For each iteration, at step 1 reselect a new list of BLE devices. The UUID of the new BLE devices shall be different from the previous set of simulated UUIDs.
11. Release the signalling connection.

#### 12.2.1.4.3 Message contents

Message contents are according to TS 36.508 [18] clause 4.6 with the following exceptions:

**Table 12.2.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 36.509 [11] clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 0 0 | BLE | |

**Table 12.2.1.4.3-2: *LPP-RequestCapabilities***

| Derivation Path: 36.355 [4] clause 6.3 | |
|---|---|
| **Information Element** | **Value/remark** |
| *bt-RequestCapabilities-r13* | TRUE |

**Table 12.2.1.4.3-3: *LPP-RequestLocationInformation***

| Derivation Path: TS 36.355 clause 6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| additionalInformation | onlyReturnInformationRequested | | |

| | | | |
|---|---|---|---|
| qos SEQUENCE { | | | |
| verticalCoordinateRequest | FALSE | | |
| responseTime SEQUENCE { | | | |
| time | 11 | See clause 12.2.1.5 | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| } | | | |
| BT-RequestLocationInformation-r13 ::= SEQUENCE { | | | |
| requestedMeasurements-r13 | If reporting of RSSI is supported by the UE: 1<br>If reporting of RSSI is not supported by the UE: 0 | RSSI Requested if reporting of RSSI is supported by the UE as indicated by IE rssi-r13 in the LPP PROVIDE CAPABILITIES message | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 12.2.1.5 Test requirement

The UE shall send *BT-ProvideLocationInformation*, within a maximum response time of 11.15 seconds (including test tolerance of 300ms) from the beginning of time period T2. The maximum response time is equal to the LPP time IE value of 11 seconds, minus ΔT, where ΔT = 150 ms, plus the test tolerance of 300ms. The LPP time IE value is derived from $T_{BT_meas}$, where $T_{BT_meas}$ is 10.24 seconds as described in clause 12.2.1.3, plus one broadcast interval, which is taken as 100 ms, giving a total value of 10.34 seconds which is then rounded up to the next allowed LPP value of 11 seconds.

The *BT-ProvideLocationInformation* IE shall include BT Measurements for all of the simulated BLE devices identified by the corresponding UUID.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

**Table 12.2.1.5-1: General test parameters for Bluetooth Identification**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Bluetooth Low Energy (BLE) Devices | | BLE 1, BLE 2, BLE 3, BLE 4, BLE 5 and BLE 6 | BLE 1 and BLE 2 are on Bluetooth Advertising Channel 1 (2402 MHz).<br>BLE 3 and BLE 4 are on Bluetooth Advertising Channel 2 (2426 MHz).<br>BLE 5 and BLE 6 are on Bluetooth Advertising Channel 3 (2480 MHz). |
| Bluetooth Advertising Channel Numbers and frequencies | | Channel 37:2402 MHz,<br>Channel 38:2426 MHz,<br>Channel 39:2480 MHz | |
| Bluetooth beacon signal broadcast interval | ms | 100 ms | |
| T1 | s | 5 | During this time the BLE signals are not transmitted |
| T2 | s | 15 | UE should report Bluetooth measurement information within 10.54s. |

**Table 12.2.1.5-2: Cell specific test parameters for Bluetooth Identification**

| Parameter | Unit | Cell 1 | | BLE 1, BLE 2 | | BLE 3, BLE 4 | | BLE 5, BLE 6 | |
|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| Bluetooth Advertising Channel Number | | N/A | | 37 | | 38 | | 39 | |
| PDSCH parameters: DL Reference Measurement Channel Note 6 | | R.0 FDD R.0 TDD | | N/A | | N/A | | N/A | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel Note 6 | | R.6 FDD R.6 TDD | | N/A | | N/A | | N/A | |
| OCNG Patterns Note 6 | | OP.1 FDD OP.1 TDD | | N/A | | N/A | | N/A | |
| PBCH_RA | dB | 0 | | N/A | | N/A | | N/A | |
| PBCH_RB | dB | | | | | | | | |
| PSS_RA | dB | | | | | | | | |
| SSS_RA | dB | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | |
| PHICH_RA | dB | | | | | | | | |
| PHICH_RB | dB | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | |
| $N_{oc1}$ Note 2 | dBm/15 KHz | -98 | | N/A | | N/A | | N/A | |
| $N_{oc2}$ Note 3 | dBm/2MHz | N/A | | -84 | | -84 | | -84 | |
| $\hat{E}_s/N_{oc1}$ | dB | 3 | 3 | N/A | | N/A | | N/A | |
| $\hat{E}_s/I_{ot}$ Note 4 | dB | 3 | 3 | | | | | | |
| RSRP Note 4 | dBm/15 kHz | -95 | -95 | | | | | | |
| SCH_RP Note 4 | dBm/15 kHz | -95 | -95 | | | | | | |
| Io Note 3 | dBm/Ch BW | -65.5 | -65.5 | | | | | | |
| Bluetooth RSSI Note 4 | dBm/2 MHz | N/A | N/A | -infinity | -60 | -infinity | -60 | -infinity | -60 |
| SINR Note 4 | dB | N/A | N/A | -infinity | -63.2 | -infinity | -63.2 | -infinity | -63.2 |
| Propagation Condition | | AWGN | | | | | | | |
| Antenna Configuration | | 1x2 | | - | | - | | - | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc1}$ to be fulfilled.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over the bandwidth and time and shall be modelled as AWGN of appropriate power for $N_{oc2}$ to be fulfilled.
Note 4: Es/Iot, RSRP, SCH_RP, Io and Bluetooth RSSI have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5 The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 6: If Cell 1 is LTE FDD, the FDD OCNG and RMCs shall be used. If Cell 1 is LTE TDD, the TDD OCNG and RMCs shall be used

# 13 NR A-GNSS minimum performance requirements

## 13.1 General

This clause defines the minimum performance requirements for both UE based and UE assisted A-GNSS FDD and TDD NR UEs. If a UE supports both UE based and UE assisted modes then it shall be tested in both modes.

The requirements in this section apply for NR UE in FR1.

For ease of use of this document a number of Test Configurations corresponding to Network Deployment Types are defined in Table 13.1.1.

**Table 13.1.1: Test Configuration**

| Test Configuration | Network Deployment Type |
|---|---|
| A | EN-DC |
| B | NG-RAN NR |
| FFS | NE-DC |

## 13.2 Sensitivity

### 13.2.1 Sensitivity Coarse time assistance

#### 13.2.1.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 13.2.1.1

**Table 13.2.1.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS |
| 15 | UE supporting A-GPS and A-NAVIC |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

#### 13.2.1.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with only coarse time assistance.

#### 13.2.1.3 Test applicability

This test applies to all types of NR UE that support EN-DC or NG-RAN NR, and A-GNSS.

### 13.2.1.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 13.2.1.2 for the parameters specified in table 13.2.1.3 or 13.2.1.4.

**Table 13.2.1.2: Requirements for Sensitivity Coarse time assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 13.2.1.3: Parameters for Sensitivity Coarse time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse time assistance error range | seconds | ±2 |
| GPS L1 C/A Signal for one satellite | dBm | -142 |
| GPS L1 C/A Signal for remaining satellites | dBm | -147 |

**Table 13.2.1.4: Parameters for Sensitivity Coarse time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.2.1.5 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| GPS(1) | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| GLONASS | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| BDS | Reference high signal power level | dBm | -136 |
| | Reference low signal power level | dBm | -145 |
| QZSS | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| NAVIC | Reference high signal power level | dBm | -142 |
| | Reference low signal power level | dBm | -147 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs. | | | |

**Table 13.2.1.5: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1(1) | GNSS-2 | GNSS-3 |
| Single constellation | High signal level | 1 | - | - |
| | Low signal level | 5 | - | - |
| Dual constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 3 | - |
| Triple constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 2 | 2 |

| | |
|---|---|
| Note 1: | Up to Rel-14: for GPS capable receivers, GNSS-1, i.e. the system having the satellite with high signal level, shall be GPS.<br>Rel-15 onwards: GNSS-1, i.e. the system having the satellite with high signal level, shall be selected by the device manufacturer. |

The normative reference for this requirement is TS 38.171 [43] clause 5.1.1 and 6.1.1.

### 13.2.1.5 Test description

#### 13.2.1.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 13.2.1.7 or 13.2.1.8 for GNSS scenario #1 in TS 37.571-5 [20]. For GNSS-1, select the first satellite SV ID defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the one satellite with the higher level.
3. For Test Configuration A (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.2, non-standalone E-UTRA single cell and NR single cell network scenarios.

   For Test Configuration B (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.1, standalone NR single cell network scenarios.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4 (RRC_CONNECTED).

#### 13.2.1.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]
2. Send a RESET UE POSITIONING STORED INFORMATION message.
3. Depending on how the LPP session is performed:

- In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

- In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.
5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 13.2.1.10 then record the result and process it as specified in step 9. If the UE does not return a valid result within the Max response time specified in table 13.2.1.10 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

8. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

9. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.2.1.10 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Signal Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE, used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.2.1.10 and record one Good Result or Bad Result as appropriate.

10. Repeat steps 1 to 9 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. For GNSS-1, select the first satellite SV ID defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the one satellite with the higher level. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time offset in step 5.

11. Repeat steps 1 to 10 until the statistical requirements of clause 13.2.1.7 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, for GNSS-1 select the next satellite SV ID from the one used previously, defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20], for the one satellite with the higher level.

12. Release the signalling connection.

#### 13.2.1.5.3 Message contents

Message contents are according to TS 38.508-1 [45] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| *a-gnss-RequestCapabilities* | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy | '19' (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |

| | | |
|---|---|---|
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

### 13.2.1.6 Test requirement

For the parameters specified in table 13.2.1.7 or 13.2.1.8 the UE shall meet the requirements and the success rate specified in table 13.2.1.10 with a confidence level of 95% according to Annex D.

**Table 13.2.1.7: Test parameters for Sensitivity Coarse time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| GPS L1 C/A Signal for one satellite | dBm | -141 |
| GPS L1 C/A Signal for remaining satellites | dBm | -146 |

**Table 13.2.1.8: Test parameters for Sensitivity Coarse time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.2.1.9 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| Galileo | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| GPS(1) | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| GLONASS | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| BDS | Reference high signal power level | dBm | -135 |
| | Reference low signal power level | dBm | -144 |
| QZSS | Reference high signal power level | dBm | -141 |

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Reference low signal power level | dBm | -146 |
| Navic | Reference high signal power level | dBm | -141 |
| | Reference low signal power level | dBm | -146 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 13.2.1.9: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1(1) | GNSS-2 | GNSS-3 |
| Single constellation | High signal level | 1 | - | - |
| | Low signal level | 5 | - | - |
| Dual constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 3 | - |
| Triple constellation | High signal level | 1 | - | - |
| | Low signal level | 2 | 2 | 2 |
| Note 1: Up to Rel-14: for GPS capable receivers, GNSS-1, i.e. the system having the satellite with high signal level, shall be GPS.<br>Rel-15 onwards: GNSS-1, i.e. the system having the satellite with high signal level, shall be selected by the device manufacturer. | | | | |

**Table 13.2.1.10: Test requirements for Sensitivity Coarse Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

## 13.2.2 Sensitivity Fine time assistance

### 13.2.2.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 13.2.2.1

**Table 13.2.2.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

### 13.2.2.2 Test purpose

To verify the performance of the first position estimate, when the UE is additionally provided with fine time assistance.

### 13.2.2.3 Test applicability

This test applies to all types of NR UE that support EN-DC and A-GNSS and that are capable of providing an enhanced performance when the network provides Fine Time Assistance.

This test applies to all types of NR UE with LPP Release 15 onwards that support NG-RAN NR and A-GNSS and that are capable of providing an enhanced performance when the network provides Fine Time Assistance.

### 13.2.2.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 13.2.2.2 for the parameters specified in table 13.2.2.3 or 13.2.2.4.

**Table 13.2.2.2: Requirements for Sensitivity Fine time assistance**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 13.2.2.3: Parameters for Sensitivity Fine time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse time assistance error range | seconds | ±2 |
| GPS L1 C/A Fine time assistance error range | µs | ±10 |
| GPS L1 C/A Signal for all satellites | dBm | -147 |

**Table 13.2.2.4: Parameters for Sensitivity Fine time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.2.2.5 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| | GNSS fine time assistance error range | µs | ±10 |
| Galileo | Reference signal power level | dBm | -147 |
| GPS(1) | Reference signal power level | dBm | -147 |
| GLONASS | Reference signal power level | dBm | -147 |
| BDS | Reference signal power level | dBm | -147 |
| QZSS | Reference signal power level | dBm | -147 |
| Navic | Reference signal power level | dBm | -147 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 13.2.2.5: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | 6 | - | - |
| Dual constellation | 3 | 3 | - |
| Triple constellation | 3 | 2 | 2 |

The normative reference for this requirement is TS 38.171 [43] clause 5.1.2 and 6.1.2.

### 13.2.2.5 Test description

#### 13.2.2.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 13.2.2.7 or 13.2.2.8 for GNSS scenario #1 in TS 37.571-5 [20].
3. For Test Configuration A (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.2, non-standalone E-UTRA single cell and NR single cell network scenarios.

   For Test Configuration B (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.1, standalone NR single cell network scenarios.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4 (RRC_CONNECTED).

#### 13.2.2.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]
2. Send a RESET UE POSITIONING STORED INFORMATION message.
3. Depending on how the LPP session is performed:

   - In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

   - In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.
4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.
5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time and GNSS Reference Time for one cell offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.
7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 13.2.2.10 then record the result and process it as specified in step 9. If the UE does not return a valid result within the Max response time specified in table 13.2.2.10 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.
8. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.
9. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position

error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.2.2.10 and record one Good Result or Bad Result as appropriate; or

For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.2.2.10 and record one Good Result or Bad Result as appropriate.

10. Repeat steps 1 to 9 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time GNSS Reference Time for one cell offsets in step 5.

11. Repeat steps 1 to 10 until the statistical requirements of clause 13.2.2.7 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.

12. Release the signalling connection.

### 13.2.2.5.3 Message contents

Message contents are according to TS 38.508-1 [45] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| *a-gnss-RequestCapabilities* | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy | '19' (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds' | |

| | | |
|---|---|---|
| | Sub-test 12: ‘gps’ and 'galileo' and ‘glonass’<br>Sub-test 13: ‘gps’ and ‘galileo’ and ‘bds’<br>Sub-test 14: ‘gps’ and ‘qzss’<br>Sub-test 15: ‘gps’ and ‘navic’ | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

### 13.2.2.6 Test requirement

For the parameters specified in table 13.2.2.7 or 13.2.2.8 the UE shall meet the requirements and the success rate specified in table 13.2.2.10 with a confidence level of 95% according to Annex D.

**Table 13.2.2.7: Test parameters for Sensitivity Fine time assistance - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse time assistance error range | seconds | ±1.8 |
| GPS Fine Time assistance error range | μs | ±9 |
| GPS L1 C/A Signal for all satellites | dBm | -146 |

**Table 13.2.2.8: Test parameters for Sensitivity Fine time assistance - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.2.2.9 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| | GNSS fine time assistance error range | μs | ±9 |
| Galileo | Reference signal power level | dBm | -146 |
| GPS(1) | Reference signal power level | dBm | -146 |
| GLONASS | Reference signal power level | dBm | -146 |
| BDS | Reference signal power level | dBm | -146 |
| QZSS | Reference signal power level | dBm | -146 |
| Navic | Reference signal power level | dBm | -146 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 13.2.2.9: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | 6 | - | - |
| Dual constellation | 3 | 3 | - |
| Triple constellation | 3 | 2 | 2 |

**Table 13.2.2.10: Test requirements for Sensitivity Fine Time Assistance**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

# 13.3 Nominal Accuracy

## 13.3.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 13.3.1

**Table 13.3.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

## 13.3.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with ideal GNSS signal conditions.

## 13.3.3 Test applicability

This test applies to all types of NR UE that support EN-DC or NG-RAN NR, and A-GNSS.

## 13.3.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 13.3.2 or 13.3.3 for the parameters specified in table 13.3.4 or 13.3.5.

**Table 13.3.2: Requirements for Nominal Accuracy - Sub-Test 1**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 30 m | 20 s |

**Table 13.3.3: Requirements for Nominal Accuracy - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 15 m | 20 s |

**Table 13.3.4: Parameters for Nominal Accuracy - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±2 |
| GPS L1 C/A Signal for all satellites | dBm | -130 |

**Table 13.3.5: Parameters for Nominal Accuracy - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.3.6 |
| | Total number of generated satellites | - | 6, 7(2) or 8(3) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| SBAS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites apply only for SBAS case or sub-tests with 3 different GNSSs.<br>NOTE 3: 8 satellites apply only for sub-tests with 3 different GNSSs and SBAS. | | | |

If SBAS is supported, the SBAS satellite with the highest elevation will be added to the scenario.

**Table 13.3.6: Satellite allocation**

| | Satellite allocation for each constellation | | | |
|---|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) | SBAS |
| Single constellation | 6 | -- | -- | 1 |
| Dual constellation | 3 | 3 | -- | 1 |
| Triple constellation | 3 | 2 | 2 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

The normative reference for this requirement is TS 38.171 [43] clause 5.2 and 6.2.

## 13.3.5 Test description

### 13.3.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 13.3.8 or 13.3.9 for GNSS scenario #3 in TS 37.571-5 [20].
3. For Test Configuration A (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.2, non-standalone E-UTRA single cell and NR single cell network scenarios.

For Test Configuration B (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.1, standalone NR single cell network scenarios.

4. Switch on the UE.

5. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4 (RRC_CONNECTED).

#### 13.3.5.2 Test procedure

1. Start GNSS scenario #3 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]

2. Send a RESET UE POSITIONING STORED INFORMATION message.

3. Depending on how the LPP session is performed:

   - In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

   - In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.

5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the (first) LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 13.3.11 then record the result and process it as specified in step 9. If the UE does not return a valid result within the Max response time specified in table 13.3.11 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

8. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

9. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.3.11 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.3.11 and record one Good Result or Bad Result as appropriate.

10. Repeat steps 1 to 9 using GNSS scenario #4 instead of #3 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time offset in step 5.

11. Repeat steps 1 to 10 until the statistical requirements of clause 13.3.6 are met. Each time scenario #3 or #4 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again.

12. Release the signalling connection.

### 13.3.5.3 Message contents

Message contents are according to TS 38.508-1 [45] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| *a-gnss-RequestCapabilities* | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy (Sub-Test 1) | '10' (15.9m) | |
| >> horizontalAccuracy (Sub-Tests 2 to 5 and 8 to 13) | '6' (7.7m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps' and possibly 'sbas'<br>Sub-test 2: 'glonass' and possibly 'sbas'<br>Sub-test 3: 'galileo' and possibly 'sbas'<br>Sub-test 4: 'gps' and possibly 'sbas'<br>Sub-test 5: 'gps' and 'glonass' and possibly 'sbas'<br>Sub-test 8: 'gps' and 'galileo' and possibly 'sbas'<br>Sub-test 9: 'bds' and possibly 'sbas''<br>Sub-test 10: 'gps'and'bds'and possibly 'sbas<br>Sub-test 11: 'gps' and 'glonass' and 'bds' and possibly 'sbas'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | Depending on UE capabilities |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 13.3.6 Test requirement

For the parameters specified in table 13.3.8 or 13.3.9 the UE shall meet the requirements and the success rate specified in table 13.3.11 or 13.3.12 with a confidence level of 95% according to Annex D.

**Table 13.3.8: Test parameters for Nominal Accuracy - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 8 |
| HDOP Range | - | 1.1 to 1.6 |
| Propagation conditions | - | AWGN |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| GPS L1 C/A Signal for all satellites | dBm | -130 |

**Table 13.3.9: Test parameters for Nominal Accuracy - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.3.10 |
| | Total number of generated satellites | - | 6, 7[(2)] or 8[(3)] |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| GPS[(1)] | Reference signal power level for all satellites | dBm | -128.5 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| SBAS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| Navic | Reference signal power level for all satellites | dBm | -128.5 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites apply only for SBAS case or sub-tests with 3 different GNSSs.<br>NOTE 3: 8 satellites apply only for sub-tests with 3 different GNSSs and SBAS. | | | |

If SBAS is supported, the SBAS satellite with the highest elevation will be added to the scenario.

**Table 13.3.10: Satellite allocation**

| | Satellite allocation for each constellation | | | |
|---|---|---|---|---|
| | GNSS 1[(1)] | GNSS 2[(1)] | GNSS 3[(1)] | SBAS |
| Single constellation | 6 | - | - | 1 |
| Dual constellation | 3 | 3 | - | 1 |
| Triple constellation | 3 | 2 | 2 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 13.3.11: Test requirements for Nominal Accuracy – Sub-Test 1**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 31.3 m | 20.3 s |

**Table 13.3.12: Test requirements for Nominal Accuracy – Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 16.3 m | 20.3 s |

# 13.4 Dynamic Range

## 13.4.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 13.4.1

**Table 13.4.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

## 13.4.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with GNSS signals with large dynamic ranges.

## 13.4.3 Test applicability

This test applies to all types of NR UE that support EN-DC or NG-RAN NR, and A-GNSS.

## 13.4.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 13.4.2 for the parameters specified in table 13.4.3 or 13.4.4.

**Table 13.4.2: Requirements for Dynamic Range**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 13.4.3: Parameters for Dynamic Range - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 6 |
| HDOP Range | - | 1.4 to 2.1 |
| GPS Coarse Time assistance error range | seconds | ±2 |
| Propagation conditions | - | AWGN |
| GPS L1 C/A Signal for 1st satellite | dBm | -129 |
| GPS L1 C/A Signal for 2nd satellite | dBm | -135 |
| GPS L1 C/A Signal for 3rd satellite | dBm | -141 |
| GPS L1 C/A Signal for 4th satellite | dBm | -147 |
| GPS L1 C/A Signal for 5th satellite | dBm | -147 |
| GPS L1 C/A Signal for 6th satellite | dBm | -147 |

**Table 13.4.4: Parameters for Dynamic Range - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.4.6 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference high signal power level | dBm | -127.5 |
| | Reference low signal power level | dBm | -147 |
| GPS(1) | Reference high signal power level | dBm | -129 |
| | Reference low signal power level | dBm | -147 |
| GLONASS | Reference high signal power level | dBm | -131.5 |
| | Reference low signal power level | dBm | -147 |
| BDS | Reference high signal power level | dBm | -133.5 |
| | Reference low signal power level | dBm | -145 |
| QZSS | Reference high signal power level | dBm | -129 |
| | Reference low signal power level | dBm | -147 |
| Navic | Reference high signal power level | dBm | -129 |
| | Reference low signal power level | dBm | -147 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 13.4.5: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | High signal level | 2 | -- | -- |
| | Low signal level | 4 | -- | -- |
| Dual constellation | High signal level | 1 | 1 | -- |
| | Low signal level | 2 | 2 | -- |
| Triple constellation | High signal level | 1 | 1 | 1 |
| | Low signal level | 2 | 1 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

The normative reference for this requirement is TS 38.171 [43] clause 5.3 and 6.3.

## 13.4.5 Test description

### 13.4.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.

2. Set the GNSS test parameters as specified in table 13.4.7 or 13.4.8 for GNSS scenario #1 in TS 37.571-5 [20]. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with the higher levels.

3. For Test Configuration A (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.2, non-standalone E-UTRA single cell and NR single cell network scenarios.

For Test Configuration B (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.1, standalone NR single cell network scenarios.

4. Switch on the UE.

5. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5.4 (RRC_CONNECTED).

### 13.4.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]

2. Send a RESET UE POSITIONING STORED INFORMATION message.

3. Depending on how the LPP session is performed:

   - In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

   - In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.

5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the (first) LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 13.4.10 then record the result and process it as specified in step 9. If the UE does not return a valid result within the Max response time specified in table 13.4.10 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

8. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

9. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 7.3.9 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.4.10 and record one Good Result or Bad Result as appropriate.

10. Repeat steps 1 to 9 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with the higher levels. Use new random values for the UE location and altitude in step 1 and for the GNSS Reference Time offset in step 5.

11. Repeat steps 1 to 10 until the statistical requirements of clause 13.4.6 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20], for the satellites with the higher levels.

12. Release the signalling connection.

### 13.4.5.3 Message contents

Message contents are according to TS 38.508-1 [45] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| *a-gnss-RequestCapabilities* | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy | '19' (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 13.4.6 Test requirement

For the parameters specified in table 13.4.7 or 13.4.8 the UE shall meet the requirements and the success rate specified in table 13.4.10 with a confidence level of 95% according to Annex D.

**Table 13.4.7: Test parameters for Dynamic Range - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 6 |
| HDOP Range | - | 1.4 to 2.1 |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| Propagation conditions | - | AWGN |
| GPS L1 C/A Signal for 1st satellite | dBm | -128.2 |
| GPS L1 C/A Signal for 2nd satellite | dBm | -134 |
| GPS L1 C/A Signal for 3rd satellite | dBm | -140 |
| GPS L1 C/A Signal for 4th satellite | dBm | -146 |
| GPS L1 C/A Signal for 5th satellite | dBm | -146 |
| GPS L1 C/A Signal for 6th satellite | dBm | -146 |

**Table 13.4.8: Test parameters for Dynamic Range - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.4.9 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±1.8 |
| Galileo | Reference high signal power level | dBm | -126.7 |
| | Reference low signal power level | dBm | -146 |
| GPS(1) | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| GLONASS | Reference high signal power level | dBm | -130.7 |
| | Reference low signal power level | dBm | -146 |
| BDS | Reference high signal power level | dBm | -132.7 |
| | Reference low signal power level | dBm | -144 |
| QZSS | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| Navic | Reference high signal power level | dBm | -128.2 |
| | Reference low signal power level | dBm | -146 |
| NOTE 1: ”GPS” here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 13.4.9: Power level and satellite allocation**

| | | Satellite allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | High signal level | 2 | -- | -- |
| | Low signal level | 4 | -- | -- |
| Dual constellation | High signal level | 1 | 1 | -- |
| | Low signal level | 2 | 2 | -- |
| Triple constellation | High signal level | 1 | 1 | 1 |
| | Low signal level | 2 | 1 | 1 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | | |

**Table 13.4.10: Test requirements for Dynamic Range**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

# 13.5 Multi-Path scenario

## 13.5.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 13.5.1

**Table 13.5.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

## 13.5.2 Test purpose

To verify the performance of the first position estimate, when the UE is provided with GNSS signals with multi-path components.

## 13.5.3 Test applicability

This test applies to all types of NR UE that support EN-DC or NG-RAN NR, and A-GNSS.

## 13.5.4 Minimum conformance requirements

The first fix position estimates shall meet the accuracy and response time requirements in table 13.5.2 for the parameters specified in table 13.5.3 or 13.5.4.

**Table 13.5.2: Requirements for Multi-Path scenario**

| Success rate | 2-D position error | Max response time |
|---|---|---|
| 95 % | 100 m | 20 s |

**Table 13.5.3: Parameters for Multi-Path scenario - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites (Satellites 1, 2 unaffected by multi-path)<br>(Satellites 3, 4, 5 affected by multi-path) | - | 5 |
| GPS Coarse time assistance error range | seconds | ±2 |
| HDOP Range | - | 1.8 to 2.5 |
| GPS L1 C/A Signal for satellite 1, 2 | dBm | -130 |
| GPS L1 C/A Signal for satellite 3, 4, 5 | dBm | LOS signal of -130 dBm, multi-path signal of -136 dBm |

**Table 13.5.4: Parameters for Multi-Path scenario - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.5.5 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference signal power level | dBm | -127 |
| GPS(1) | Reference signal power level | dBm | -128.5 |
| GLONASS | Reference signal power level | dBm | -131 |
| BDS | Reference signal power level | dBm | -133 |
| QZSS | Reference signal power level | dBm | -128.5 |
| Navic | Reference signal power level | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs | | | |

**Table 13.5.5: Channel model allocation**

| | | Channel model allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | One-tap channel | 2 | -- | -- |
| | Two-tap channel | 4 | -- | -- |
| Dual constellation | One-tap channel | 1 | 1 | -- |
| | Two-tap channel | 2 | 2 | -- |
| Triple constellation | One-tap channel | 1 | 1 | 1 |
| | Two-tap channel | 2 | 1 | 1 |
| Note: One-tap channel: no multi-path. Two-tap channel: multi-path defined in clause 4.2.4 | | | | |

The normative reference for this requirement is TS 38.171 [43] clause 5.4 and 6.4.

## 13.5.5 Test description

### 13.5.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 13.5.7 or 13.5.8 for GNSS scenario #1 in TS 37.571-5 [20]. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with one-tap channels.
3. For Test Configuration A (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.2, non-standalone E-UTRA single cell and NR single cell network scenarios.

   For Test Configuration B (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.1, standalone NR single cell network scenarios.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 38.508-1 [18] clause 4.5. 4 (RRC_CONNECTED).

### 13.5.5.2 Test procedure

1. Start GNSS scenario #1 as specified in clause 6.2.1.2 of TS 37.571-5 [20] with the UE location randomly selected to be within 3 km of the Reference Location and the altitude of the UE randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid using the method described in clause 6.2.1.2.6 of TS 37.571-5 [20]. The initial carrier phase difference between taps of the multi-path model shall be randomly selected between 0 and $2\pi$ radians by selecting the next random number from a standard uniform random number generator, in the range 0 to $2\pi$, representing radians with a resolution of 0.1, representing 0.1 radians.

2. Send a RESET UE POSITIONING STORED INFORMATION message.

3. Depending on how the LPP session is performed:

   - In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

   - In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.

4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.

5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20] with the value of GNSS Reference Time offset by a random value as specified in clause 6.2.7.2 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. If the UE returns a valid result in the LPP PROVIDE LOCATION INFORMATION message within the Max response time specified in table 13.5.10 then record the result and process it as specified in step 9. If the UE does not return a valid result within the Max response time specified in table 13.5.10 or reports an Error in the LPP PROVIDE LOCATION INFORMATION message then record one Bad Result.

8. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send a LPP acknowledgment message.

9. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.5.10 and record one Good Result or Bad Result as appropriate; or

   For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE used in step 1, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.5.10 and record one Good Result or Bad Result as appropriate.

10. Repeat steps 1 to 9 using GNSS scenario #2 instead of #1 so that the reference location changes sufficiently such that the UE shall have to use the new assistance data. Randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20] for the satellites with the one-tap channels. Use new random values for the UE location and altitude, and the initial carrier phase difference between taps of the multi-path model in step 1 and for the GNSS Reference Time offset in step 5.

11. Repeat steps 1 to 10 until the statistical requirements of clause 13.5.6 are met. Each time scenario #1 or #2 is used, the start time of the GNSS scenario shall be advanced by 2 minutes from the time used previously for that scenario. Once a scenario reaches the end of its viable running time, restart it from its nominal start time again. Each time scenario #1 or #2 is used, randomly select from the satellite SV IDs defined in the relevant table of Satellites to be simulated in clause 6.2.1.2 in TS 37.571-5 [20], for the satellites with the one-tap channels.

12. Release the signalling connection

#### 13.5.5.3 Message contents

Message contents are according to TS 38.508-1 [45] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| *a-gnss-RequestCapabilities* | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy | '19' (51.2m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | | |
| >>>time | '20' | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

### 13.5.6 Test requirement

For the parameters specified in table 13.5.7 or 13.5.8 the UE shall meet the requirements and the success rate specified in table 13.5.10 with a confidence level of 95% according to Annex D.

**Table 13.5.7: Test parameters for Multi-Path scenario - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites (see note) | - | 5 |
| GPS Coarse Time assistance error range | seconds | ±1.8 |
| HDOP Range | - | 1.8 to 2.5 |
| GPS L1 C/A Signal for Satellite 1, 2 (see note) | dBm | -130 |
| GPS L1 C/A Signal for Satellite 3, 4, 5 (see note) | dBm | LOS signal of -130 dBm, multi-path signal of -136.2 dBm |
| NOTE: Satellites 1, 2 no multi-path. Satellites 3, 4, 5 multi-path defined in clause 4.2.4. | | |

**Table 13.5.8: Test parameters for Multi-Path scenario - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.5.9 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range per system | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| | GNSS coarse time assistance error range | seconds | ±2 |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| Navic | Reference signal power level for all satellites | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs. | | | |

**Table 13.5.9: Channel model allocation**

| | | Channel model allocation for each constellation | | |
|---|---|---|---|---|
| | | GNSS-1 | GNSS-2 | GNSS-3 |
| Single constellation | One-tap channel | 2 | -- | -- |
| | Two-tap channel | 4 | -- | -- |
| Dual constellation | One-tap channel | 1 | 1 | -- |
| | Two-tap channel | 2 | 2 | -- |
| Triple constellation | One-tap channel | 1 | 1 | 1 |
| | Two-tap channel | 2 | 1 | 1 |
| Note: One-tap channel: no multi-path. Two-tap channel: multi-path defined in clause 4.2.4 with Relative mean Power (Y) defined in Table 13.5.9A. | | | | |

**Table 13.5.9A: Relative mean Power (Y) for use in Table 135.9**

| System | Signals | Y [dB] |
|---|---|---|
| Galileo | E1 | -4.7 |
| | E5a | -6.2 |
| | E5b | -6.2 |
| GPS/Modernized GPS/QZSS | L1 C/A | -6.2 |
| | L1C | -4.7 |
| | L2C | -6.2 |
| | L5 | -6.2 |
| GLONASS | G1 | -12.7 |
| | G2 | -12.7 |
| BDS | B1I | -4.7 |
| | B1C | -4.7 |
| | B2a | -6.2 |
| | B2b | -6.2 |
| | B3I | -6.2 |
| Navic | L5 | -6,2 |

**Table 13.5.10: Test requirements for Multi-Path scenario**

| System | Success rate | 2-D position error | Max response time |
|---|---|---|---|
| All | 95 % | 101.3 m | 20.3 s |

# 13.6 Moving scenario and periodic update (Rel-9 to Rel-13)

## 13.6.1 Sub-tests

This test includes sub-tests dependent on the GNSS supported by the UE. Each sub-test is identified by a Sub-Test Number as defined in Table 13.6.1

**Table 13.6.1: Sub-Test Number Definition**

| Sub-Test Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GPS L1 C/A |
| 2 | UE supporting A-GLONASS |
| 3 | UE supporting A-Galileo |
| 4 | UE supporting A-GPS and Modernized GPS |
| 5 | UE supporting A-GPS and A-GLONASS (Note) |
| 8 | UE supporting A-GPS and A-Galileo (Note) |
| 9 | UE supporting A-BDS |
| 10 | UE supporting A-GPS and A-BDS (Note) |
| 11 | UE supporting A-GPS and A-GLONASS and A-BDS (Note) |
| 12 | UE supporting A-GPS and A-Galileo and A-GLONASS (Note) |
| 13 | UE supporting A-GPS and A-Galileo and A-BDS (Note) |
| 14 | UE supporting A-GPS and A-QZSS (Note) |
| 15 | UE supporting A-GPS and A-Navic (Note) |
| Note: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities. | |

## 13.6.2 Test purpose

To verify the performance when the UE is requested to use periodical reporting with a reporting interval of 2 seconds.

## 13.6.3 Test applicability

This test applies to all types of NR UE that support EN-DC or NG-RAN NR, and A-GNSS, with LPP Release 9 to 13.

## 13.6.4 Minimum conformance requirements

The position estimates, after the first reported position estimate, shall meet the accuracy requirement in table 13.6.2 or 13.6.3 with the periodical reporting interval of 2 seconds for the parameters specified in table 13.6.4 or 13.6.5.

NOTE: In the actual testing the UE may report error messages until it has been able to acquire GNSS measured results or a position estimate. The SS shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 13.6.2 or 13.6.3.

**Table 13.6.2: Requirements for Moving scenario and periodic update - Sub-Test 1**

| Success Rate | 2-D position error | Periodical reporting interval |
|---|---|---|
| 95 % | 100 m | 2 s |

**Table 13.6.3: Requirements for Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| Success Rate | 2-D position error | Periodical reporting interval |
|---|---|---|
| 95 % | 50 m | 2 s |

**Table 13.6.4: Parameters for Moving scenario and periodic update - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 5 |
| HDOP Range | - | 1.8 to 2.5 |
| Propagation condition | - | AWGN |
| GPS L1 C/A signal for all satellites | dBm | -130 |

**Table 13.6.5: Parameters for Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.6.6 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP range | | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| Galileo | Reference signal power level | dBm | -127 |
| GPS(1) | Reference signal power level | dBm | -128.5 |
| GLONASS | Reference signal power level | dBm | -131 |
| BDS | Reference signal power level | dBm | -133 |
| QZSS | Reference signal power level | dBm | -128.5 |
| Navic | Reference signal power level | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs. | | | |

**Table 13.6.6: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | 6 | -- | -- |
| Dual constellation | 3 | 3 | -- |
| Triple constellation | 3 | 2 | 2 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | |

The normative reference for this requirement is TS 38.171 [43] clause 5.5 and 6.5.

## 13.6.5 Test description

### 13.6.5.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

Channel Bandwidth to be tested: as defined in TS 38.508-1 [45] clause 4.3.1.

The UE moves on a rectangular trajectory of 940 m by 1 440 m with rounded corner defined in Figure 13.1. The initial reference is first defined followed by acceleration to final speed of 100 km/h in 250 m. The UE then maintains the speed for 400 m. This is followed by deceleration to final speed of 25 km/h in 250 m. The UE then turn 90 degrees with

turning radius of 20 m at 25 km/h. This is followed by acceleration to final speed of 100 km/h in 250 m. The sequence is repeated to complete the rectangle.

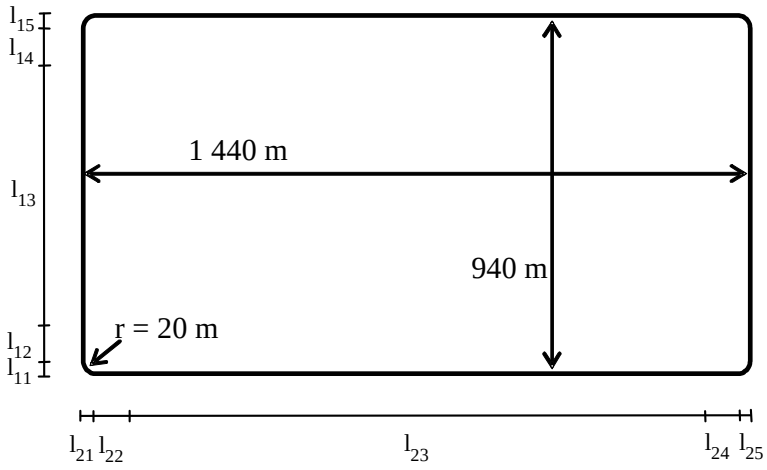



**Figure 13.1: Rectangular trajectory of the moving scenario and periodic update test case**

**Trajectory Parameters**

| Parameter | Distance (m) | Speed (km/h) |
|---|---|---|
| $l_{11}$, $l_{15}$, $l_{21}$, $l_{25}$ | 20 | 25 |
| $l_{12}$, $l_{14}$, $l_{22}$, $l_{24}$ | 250 | 25 to 100 and 100 to 25 |
| $l_{13}$ | 400 | 100 |
| $l_{23}$ | 900 | 100 |

1. Connect SS and GSS to the UE antenna connector or antenna connectors as shown in Annex A.
2. Set the GNSS test parameters as specified in table 13.6.8 or 13.6.9 for GNSS scenario #5 in TS 37.571-5 [20].
3. For Test Configuration A (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.2, non-standalone E-UTRA single cell and NR single cell network scenarios.

   For Test Configuration B (Table 13.1.1), the parameter settings for the cell are set up according to TS 38.508-1 [45] clause 4.4.1.1, standalone NR single cell network scenarios.
4. Switch on the UE.
5. Establish a signalling connection according to the generic procedure in TS 38.508-1 [45] clause 4.5. 4 (RRC_CONNECTED).

### 13.6.5.2 Test procedure

1. Start GNSS scenario #5 as specified in clause 6.2.1.2 of TS 37.571-5 [20]
2. Send a RESET UE POSITIONING STORED INFORMATION message.
3. Depending on how the LPP session is performed:

   - In the case of C-Plane, the SS shall send an LPP REQUEST CAPABILITIES message.

   - In the case of U-Plane, the SS shall establish a SUPL session with the UE for positioning.
4. The UE returns an LPP PROVIDE CAPABILITIES message indicating the assistance data supported by the UE in the Assistance Data Support List in the A GNSS Provide Capabilities IE.

5. Send an LPP PROVIDE ASSISTANCE DATA message to provide the assistance data that is supported by the UE as indicated in the step 4 and in accordance with clause 6.2.6 of TS 37.571-5 [20], and with the values defined in clause 6.2.7 of TS 37.571-5 [20]. If the LPP session is performed over C-Plane, then if the UE message at step 4 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

6. Send an LPP REQUEST LOCATION INFORMATION message to obtain a fix.

7. Ignore any Error messages that the UE may report in LPP PROVIDE LOCATION INFORMATION messages until it has been able to acquire the GNSS signals and reports the first GNSS Measurement Information or Location Information.

8. Discard the first GNSS Measurement Information or Location Information.

9. Record the time of reception of the next LPP PROVIDE LOCATION INFORMATION message after reception of the first GNSS Measurement Information or Location Information.

10. After the reception of the first GNSS Measurement Information or Location Information reported in a LPP PROVIDE LOCATION INFORMATION message, every time the UE returns a GNSS Measurement Information or Location Information in the LPP PROVIDE LOCATION INFORMATION message record the time of reception and the result. If the difference between the time of reception and the time of reception of the previous result is less than 1.5 seconds or greater than 2.5 seconds, or if the UE reports an Error in any LPP PROVIDE LOCATION INFORMATION messages, then record one Bad Result. Otherwise process the result as specified in step 12.

11. If the UE messages at steps 7 to 10 include the ackRequested IE set to TRUE, then the SS shall send LPP acknowledgment messages as required.

12. For UE based testing compare the reported Location Information in the LPP PROVIDE LOCATION INFORMATION message against the simulated position of the UE at the time of applicability reported in the Location Information, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.6.11 or 13.6.12 and record one Good Result or Bad Result as appropriate; or

    For UE assisted testing convert the GNSS Measurement Information reported in the LPP PROVIDE LOCATION INFORMATION message to a 2D position using the method described in clause 4.4.3 and then compare the result against the simulated position of the UE at the time of applicability reported in the GNSS Measurement Information, and calculate the 2D position error as specified in clause 4.5.2.3. Compare the 2D position error against the value in table 13.6.11 or 13.6.12 and record one Good Result or Bad Result as appropriate.

13. If the UE sends the first LPP PROVIDE LOCATION INFORMATION that contains GNSS Measurement Information or Location Information later than 240s after the start of the GNSS scenario, fail the UE and stop the test early. Otherwise collect LPP PROVIDE LOCATION INFORMATION results during 900s, starting from the time recorded in step 9. If at any time the difference between the times of reception of two consecutive results is greater than 240s, fail the UE and stop the test early. Use the collected Good Results and Bad Results to determine the PASS/FAIL according to clause 13.6.6.

14. Release the signalling connection.

### 13.6.5.3 Message contents

Message contents are according to TS 38.508-1 [45] clauses 4.6 and 4.7 and as follows:

RESET UE POSITIONING STORED INFORMATION

| Information Element | Value/remark |
|---|---|
| UE POSITIONING TECHNOLOGY | AGNSS |

LPP REQUEST CAPABILITIES

| Information Element | Value/remark |
|---|---|
| *a-gnss-RequestCapabilities* | TRUE |

LPP REQUEST LOCATION INFORMATION

| Information Element | Value/remark | Comment |
|---|---|---|
| commonIEsRequestLocationInformation | | |
| > locationInformationType | 'locationEstimateRequired' or 'locationMeasurementsRequired' | Depending on test case and UE capabilities, i.e. support for UE-based or UE-assisted |
| > periodicalReporting | | |
| >> reportingAmount | 'ra-Infinity' | Infinite means during the complete test time |
| >> reportingInterval | 'ri0-5' | 2 seconds |
| > additionalInformation | 'onlyReturnInformationRequested' | |
| > qos | | |
| >> horizontalAccuracy (Sub-Test 1) | '19' (51.2m) | |
| >> horizontalAccuracy (Sub-Tests 2 to 5 and 8 to 13) | '13' (24.5m) | |
| >> verticalCoordinateRequest | FALSE | |
| >> responseTime | Not present | |
| a-gnss-RequestLocationInformation | | |
| > gnss-PositioningInstructions | | |
| >> gnssMethods | | |
| >>> gnss-ids | Sub-test 1: 'gps'<br>Sub-test 2: 'glonass'<br>Sub-test 3: 'galileo'<br>Sub-test 4: 'gps'<br>Sub-test 5: 'gps' and 'glonass'<br>Sub-test 8: 'gps' and 'galileo'<br>Sub-test 9: 'bds'<br>Sub-test 10: 'gps'and'bds'<br>Sub-test 11: 'gps' and 'glonass' and 'bds'<br>Sub-test 12: 'gps' and 'galileo' and 'glonass'<br>Sub-test 13: 'gps' and 'galileo' and 'bds'<br>Sub-test 14: 'gps' and 'qzss'<br>Sub-test 15: 'gps' and 'navic' | |
| >> fineTimeAssistanceMeasReq | FALSE | |
| >> adrMeasReq | FALSE | |
| >> multiFreqMeasReq | TRUE or FALSE | Depending on UE capabilities |
| >> assistanceAvailability | FALSE | |

## 13.6.6 Test requirement

For the parameters specified in table 13.6.8 or 13.6.9 the UE shall meet the requirements and the success rate specified in table 13.6.11 or 13.6.12 after the first reported position estimates.

NOTES: 1. In the testing the UE may report error messages until it has been able to acquire GNSS measured results or a position estimate. The test equipment shall only consider the first measurement report different from an error message as the first position estimate in the requirement in table 13.6.11 or 13.6.12.

2. Due to the statistical nature of the results it is not possible to design a test with predefined confidence level for the success rate in table 13.6.11 or 13.6.12, therefore a simple PASS/FAIL of the results gathered against this success rate is used.

**Table 13.6.8: Test parameters for Moving scenario and periodic update - Sub-Test 1**

| Parameters | Unit | Value |
|---|---|---|
| Number of generated satellites | - | 5 |
| HDOP Range | - | 1.8 to 2.5 |
| Propagation condition | - | AWGN |

| Parameters | Unit | Value |
|---|---|---|
| GPS L1 C/A Signal for all satellites | dBm | -130 |

**Table 13.6.9: Test parameters for Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Parameters | Unit | Value |
|---|---|---|---|
| | Number of generated satellites per system | - | See Table 13.6.10 |
| | Total number of generated satellites | - | 6 or 7(2) |
| | HDOP Range per system | - | 1.4 to 2.1 |
| | Propagation conditions | - | AWGN |
| Galileo | Reference signal power level for all satellites | dBm | -127 |
| GPS(1) | Reference signal power level for all satellites | dBm | -128.5 |
| GLONASS | Reference signal power level for all satellites | dBm | -131 |
| BDS | Reference signal power level for all satellites | dBm | -133 |
| QZSS | Reference signal power level for all satellites | dBm | -128.5 |
| Navic | Reference signal power level for all satellites | dBm | -128.5 |
| NOTE 1: "GPS" here means GPS L1 C/A, Modernized GPS, or both, dependent on UE capabilities.<br>NOTE 2: 7 satellites are used for sub-tests with 3 different GNSSs. | | | |

**Table 13.6.10: Satellite allocation**

| | Satellite allocation for each constellation | | |
|---|---|---|---|
| | GNSS 1(1) | GNSS 2(1) | GNSS 3(1) |
| Single constellation | 6 | -- | -- |
| Dual constellation | 3 | 3 | -- |
| Triple constellation | 3 | 2 | 2 |
| NOTE 1: GNSS refers to global systems i.e., GPS, Galileo, GLONASS and BDS. | | | |

**Table 13.6.11: Test requirements for Moving scenario and periodic update - Sub-Test 1**

| System | Success rate | 2-D position error | Periodical reporting interval |
|---|---|---|---|
| All | 95 % | 101.3 m | Between 1.5 s and 2.5s |

**Table 13.6.12: Test requirements for Moving scenario and periodic update - Sub-Tests 2 to 5 and Sub-Tests 8 to 13**

| System | Success rate | 2-D position error | Periodical reporting interval |
|---|---|---|---|
| All | 95 % | 51.3 m | Between 1.5 s and 2.5s |

# 13.7 Moving scenario and periodic update (Rel-14 onwards)

## 13.7.1 Sub-tests

Same as defined in clause 13.6.1.

## 13.7.2 Test purpose

Same as defined in clause 13.6.2.

### 13.7.3 Test applicability

This test applies to all types of NR UE that support EN-DC or NG-RAN NR, and A-GNSS with periodical reporting with LPP Release 14 onwards.

NOTE: The capability to support periodical reporting is indicated in LPP [4] by either omitting the field *periodicalReportingNotSupported-r14* in the LPP PROVIDE CAPABILITIES message, or by including the field *periodicalReportingNotSupported-r14* in the LPP PROVIDE CAPABILITIES message but with bits for UE-assisted or UE-based mode set to zero.

### 13.7.4 Minimum conformance requirements

Same as defined in clause 13.6.4.

### 13.7.5 Test description

Same as defined in clause 13.6.5.

### 13.7.6 Test requirement

Same as defined in clause 13.6.6.

# 14 NR RSTD measurement requirements

## 14.1 General

This clause defines the minimum performance requirements for NR UEs capable of performing NR RSTD measurements for DL-TDOA.

## 14.2 NR RSTD measurement period test cases in RRC_CONNECTED

### 14.2.1 NR RSTD measurement period test case for single positioning frequency layer in FR1 SA

#### 14.2.1.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] Clause 9.9.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

#### 14.2.1.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

#### 14.2.1.3 Minimum conformance requirements

When physical layer receives last of *NR-TDOA-ProvideAssistanceData* message and *NR-TDOA-RequestLocationInformation* message from LMF via LPP [49], the UE shall be able to measure and report multiple (up to the UE capability specified in TS 38.133 [50] Clause 9.9.2.3) DL RSTD measurements, during the measurement period $T_{RSTD,Total}$ defined as:

$$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD,i} + (L-1) * \max(T_{effect,i})$$

Where ,

$i$ is the index of positioning frequency layer,

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$T_{RSTD,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below:

$$T_{RSTD,i} = \left(CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last},$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1, $N_{RxBeam,i}$ = 1; and in FR2, $N_{RxBeam,i}$ = 8.

$CSSF_{PRS,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in positioning frequency layer $i$ as defined in TS 38.133[50] clause 9.1.5.2.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56].

$N_{sample}$ is the number of PRS RSTD samples and $N_{sample}$ = 4.

$T_{last}$ is the measurement duration for the last PRS RSTD sample, including the sampling time and processing time, $T_{last} = T_i + T_{available_PRS,i}$ ,

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$.

$MGRP_i$ is the repetition periodicity of the measurement gap applicable for measurement in the PRS frequency layer i.

$T_{PRS,i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive the measurement period of that positioning frequency layer *i*. Where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. If bitmap $\{b^1\}$ for higher-layer parameter *DL-PRS-MutingPattern* is provided, and $T_{per}^{PRS} * T_{muting}^{PRS} \leq 10240ms$, then $N_{muting} = T_{muting}^{PRS} * min(L, \frac{10240}{T_{per}^{PRS} * T_{muting}^{PRS}})$; otherwise, if bitmap $\{b^1\}$ is not provided or $T_{per}^{PRS} * T_{muting}^{PRS} > 10240ms$, then $N_{muting} = 1$.

$T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and L is the size of the bitmap $\{b^1\}$.

- Note: For the purpose of calculating $T_{PRS,i}$, only the PRS resources fully or partially covered by the MG are considered.

$\{N, T\}$ is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-*

*ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49].

$N'$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [49].

The time $T_{RSTD,i}$ starts from the first MG instance aligned with a DL PRS resource(s) of positioning frequency layer *i* closest in time after both the *NR-TDOA-ProvideAssistanceData* message and *NR-TDOA-RequestLocationInformation* message are delivered from LMF to the physical layer of UE via LPP.

### 14.2.1.4 Test description

#### 14.2.1.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 14.2.1.4.1-1.

**Table 14.2.1.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.2.1.5-1, Table 14.2.1.5-2 and Table 14.2.1.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.4.1.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the RSTD reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1

#### 14.2.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 is active in T1 and T2, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. The beginning of the time interval T2 shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

The *NR-DL-TDOA-ProvideAssistanceData* and *NR-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.10.1, shall be provided to the UE during T1. The last TTI containing the *NR-DL-TDOA-RequestLocationInformation* shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

The UE is configured with measurement gap pattern ID # 24 or #0 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On,* according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 14.2.1.5-1, Table 14.2.1.5-2 and Table 14.2.1.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 14.2.1.5-2 and Table 14.2.1.5-3.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. Repeat steps 2-11 in Tables 14.2.1.4-1 until the confidence level according to Annex D is achieved.

### 14.2.1.4.3 Message contents

**Table 14.2.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.1.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 | Not present | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.1.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.1.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.1.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 16 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | 0 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 38 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 SEQUENCE { | | | |
| mcc-r15 | Cell 3 | | |
| mnc-r15 | Cell 3 | | |
| nr-cellidentity-r15 | Cell 3 | | |
| } | | | |
| nr-ARFCN-r16 | Cell 3 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 38 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[4-16] | As specificied in TS 37.571-5 [20] | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.1.4.3-6: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| | | | |
| | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 3 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| | | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.1.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.1.5 Test requirement

Table 14.2.1.5-1, Table 14.2.1.5-2 and Table 14.2.1.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.1.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 36.214 [6] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the DL-TDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |

| | | | | |
|---|---|---|---|---|
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |
| | Config 2 | | CCR.1.1 TDD | |
| | Config 3 | | CCR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Active DL BWP configuration | Config 1,2,3 | | DLBWP.1.1 | |
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.1 FR1 | |
| | Config 2 | | PRS.1.1 FR1 | |
| | Config 3 | | PRS.2.1 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#0 | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | μs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | μs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | μs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Corresponds to prs-MutingInfo defined in TS 37.355 [49]<br>Cell 1 and Cell 3 will be configured with different Comb patterns or resource offsets |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |

**Table 14.2.1.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| EPRE ratio of PSS to SSS | | dB | 0 | N/A | N/A |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -52.56 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols where PRS is transmitted.

**Table 14.2.1.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| | | T2 | T2 | T2 |
| NR RF Channel Number | | 1 | 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5 | -11.20 | -11.20 |
| | Config 2 | dB | -5 | -11.20 | -11.20 |
| | Config 3 | dB | -5 | -11.20 | -11.20 |
| SSB $\hat{E}_s/N_{oc}$ | Config 1~3 | dB | 10 | 3 | 3 |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -65.29 | -65.29 | -65.29 |
| | Config 2 | dBm/ 19.08MHz | -65.29 | -65.29 | -65.29 |
| | Config 3 | dBm/ 47.88MHz | -61.29 | -61.29 | -61.29 |
| SSB_RP Note4 | Config 1 | dBm/SCS | -88 | -95 | -95 |
| | Config 2 | dBm/SCS | -88 | -95 | -95 |
| | Config 3 | dBm/SCS | -85 | -92 | -92 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -5.61 | -12.64 | -12.64 |
| Propagation Condition | | | AWGN | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS. |
| Note 2: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. |

The RSTD measurement time fulfils the requirements specified in clause 14.2.1.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.1.3 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50 ms. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left( CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} = 4$, $N_{sample} = 4$, $L_{available_PRS,i} = 1.143ms \ for \ SCS \ 15kHz \ and \ 0.571 \ for \ SCS \ 30kHz$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left( \left\lceil \frac{4}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * 4 - 1 \right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities.

Depending on the UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]:

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 0.648s and 204.9s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.1.4.3-3. The LPP time IE ranges between 1s and 210s.

Similarly for Type 2 Buffering, Best Case result of 0.648s is 1s and for the worst-case result of 327840ms, rounded to the next multiple of ten-seconds, the final value is 33s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 210.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 0.648s and 102.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.1.4.3-3. The LPP time IE ranges between 1s and 103s.

Similarly for Type 2 Buffering, Best Case result of 0.648s is 1s and for the worst-case result of 164000ms, rounded to the next multiple of ten-seconds, the final value is 17s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 103.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.1.5-4, Table 14.2.1.5-5 and Table 14.2.1.5-6 for reference.

**Table 14.2.1.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| *maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n4 | 4 |
| n8 | 2 |
| >=n16 | 1 |

**Table 14.2.1.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\left\lceil \frac{1}{N} \right\rceil$ |
|---|---|
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |

**Table 14.2.1.5-5a**

| *durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 1 Buffering* Note 1 | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 5 |
| nDot5 | 3 |
| >=n1 | 2 |
| *durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 1 Buffering* Note 1 | $\left\lceil \frac{0.571}{N} \right\rceil$ |
| nDot125 | 5 |
| nDot25 | 3 |
| nDot5 | 2 |
| >=n1 | 1 |
| *durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 2 Buffering* Note 1 | $\left\lceil \frac{2}{N} \right\rceil$ |
| nDot125 | 16 |
| nDot25 | 8 |
| nDot5 | 4 |
| >=n1 | 2 |
| *durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 2 Buffering* Note 1 | $\left\lceil \frac{0.5}{N} \right\rceil$ |
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26], | |

**Table 14.2.1.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |

| | | |
|---|---|---|
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.2 NR RSTD measurement period test case for dual positioning frequency layer in FR1 SA

### 14.2.2.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] Clause 9.9.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when dual positioning frequency layer is configured.

### 14.2.2.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

### 14.2.2.3 Minimum conformance requirements

Same as in clause 14.2.1.3.

### 14.2.2.4 Test description

#### 14.2.2.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 14.2.2.4.1-1.

**Table 14.2.2.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Low for RF channel #1 and High for RF channel #2, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.2.2.5-1, Table 14.2.2.5-2 and Table 14.2.2.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.4.1.3.

5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the RSTD reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. Cell 1 and Cell 2 are on RF channel #1 in FR1, Cell 3 on RF channel #2 on FR1.

#### 14.2.2.4.2 Test procedure

Same as clause 14.2.1.4.2.

#### 14.2.2.4.3 Message contents

Same as 14.2.4.1.3 except that Table 14.2.2.4.3-1 replaces 14.2.4.1.3-5 and Table 14.2.2.4.3-2 replaces 14.2.4.1.3-6.

**Table 14.2.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.2.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | | 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.2.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| | | | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 3 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 3 |
| dl-PRS-ID-r16 | 3 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 SEQUENCE { | | | |
| mcc-r15 | Cell 3 | | |
| mnc-r15 | Cell 3 | | |
| nr-cellidentity-r15 | Cell 3 | | |
| } | | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.2.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 3 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Config 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.2.5 Test requirement

Table 14.2.2.5-1, Table 14.2.2.5-2 and Table 14.2.2.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.2.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 36.214 [6] and TS 37.355 [49]. The reference cell is the PCell in this test case. |

| | | | | |
|---|---|---|---|---|
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at random places in the neighbour cell list in the DL-TDOA assistance data, but Cell 2 always appears in the first half of the list, whilst Cell 3 appears in the second half of the list. |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |
| | Config 2 | | CCR.1.1 TDD | |
| | Config 3 | | CCR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Active DL BWP configuration | Config 1,2,3 | | DLBWP.1.1 | |
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.1 FR1 | |
| | Config 2 | | PRS.1.1 FR1 | |
| | Config 3 | | PRS.2.1 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#0 | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the OTDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in OTDOA assistance data | | | 4 | Including the reference cell |

| | | | |
|---|---|---|---|
| PRS muting info | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Corresponds to prs-MutingInfo defined in TS 37.355 [49]<br>Cell 1 and Cell 3 will be configured with different Comb patterns or resource offsets |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |

**Table 14.2.2.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 2 |
| Positioning frequency layer | | | 1 | 1 | 2 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| EPRE ratio of PSS to SSS | | dB | 0 | N/A | N/A |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -66.96 |
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | -66.96 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -62.97 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.2.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 2 |
| Positioning frequency layer | | | 1 | 1 | 2 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.0 | -11.20 | -12.60 |
| | Config 2 | dB | -5.0 | -11.20 | -12.60 |
| | Config 3 | dB | -5.0 | -11.20 | -12.60 |
| Io Note 4 | Config 1 | dBm/ 19.08MHz z | -65.52 | -65.52 | -66.72 |
| | Config 2 | dBm/ 19.08MHz z | -65.52 | -65.52 | -66.72 |
| | Config 3 | dBm/ 47.88MHz | -61.52 | -61.52 | -62.73 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -5.53 | -12.57 | -12.60 |
| Propagation Condition | | | AWGN | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS. |
| Note 2: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols where PRS is transmitted. |

The RSTD measurement time fulfils the requirements specified in clause 14.2.2.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.2.3 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50 ms. The RSTD measurement period follows the equation:

$$T_{RSTD,Total} = T_{RSTD,1} + T_{RSTD,2} + Max(T_{effect,i})$$

Given that both PFLs have the same settings, it can be simplified:

$$T_{RSTD,Total} = 2 * T_{RSTD,i} + T_{effect,i}$$

$$T_{RSTD,i} = \left(CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} 2$, $N_{sample} = 4$, $L_{available_PRS,i} = 0.571ms\ for\ SCS\ 15kHz\ and\ 0.286\ for\ SCS\ 30kHz$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$T_{RSTD,i} = \left(\left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities.

Depending on the UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]:

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 1.456s and 104s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.2.4.3-3. The LPP time IE ranges between 2s and 104s.

Similarly for Type 2 Buffering, Best Case result of 1.456s is 2s and for the worst-case result of 328000ms, rounded to the next multiple of ten-seconds, the final value is 64s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 104.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 1.456s and 63.04s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.2.4.3-3. The LPP time IE ranges between 2s and 64s.

Similarly for Type 2 Buffering, Best Case result of 1.456s is 2s and for the worst-case result of 164160ms, rounded to the next multiple of ten-seconds, the final value is 17s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 64.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.2.5-4, Table 14.2.2.5-5 and Table 14.2.2.5-6 for reference.

**Table 14.2.2.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n4 | **2** |
| >=n8 | 1 |

**Table 14.2.2.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{1}{N} \right\rceil$ |
|---|---|
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |

**Table 14.2.2.5-5a**

| ***durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 1 Buffering*** Note 1 | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 3 |
| nDot5 | 2 |
| >=n1 | 1 |
| ***durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 1 Buffering*** Note 1 | $\left\lceil \frac{0.286}{N} \right\rceil$ |
| nDot125 | 3 |
| nDot25 | 2 |
| nDot5 | 1 |
| ***durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 2 Buffering*** Note 1 | $\left\lceil \frac{2}{N} \right\rceil$ |
| nDot125 | 16 |
| nDot25 | 8 |
| nDot5 | 4 |
| >=n1 | 2 |
| ***durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 2 Buffering*** Note 1 | $\left\lceil \frac{1}{N} \right\rceil$ |
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26], | |

**Table 14.2.2.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|

| | | |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.3 NR RSTD measurement period test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.2.3.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 9.9.2 in an environment with AWGN propagation conditions. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.2.3.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

### 14.2.3.3 Minimum conformance requirements

Same as in clause 14.2.1.3.

### 14.2.3.4 Test description

**The supported test configurations in listed in Table 14.2.3.4-1Table 14.2.3.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 14.2.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.

2. The general test parameter settings are set up according to Table 14.2.3.5-1, Table 14.2.3.5-2 and Table 14.2.3.5-3.

3. Propagation conditions are set according to clause 4.15.2.

4. Message contents are defined in clause 14.2.3.4.3.

5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the RSTD reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR2.

### 14.2.3.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 is active in T1 and T2, whilst Cell 2 and Cell 3 are activated only in the beginning of T2. The beginning of the time interval T2 shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

The *NR-DL-TDOA-ProvideAssistanceData* and *NR-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.10.1 shall be provided to the UE during T1. The last TTI containing the *NR-DL-TDOA-RequestLocationInformation* shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 14.2.3.5-1, Table 14.2.3.5-2 and Table 14.2.3.5-3. Propagation conditions are set according to clause 4.15.2.

4. T1 starts.

5. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.

7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.

9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 14.2.3.5-2 and Table 14.2.3.5-3.

10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

12. Repeat steps 2-11 in Tables 14.2.3.4-1 until the confidence level according to Annex D is achieved.

#### 14.2.3.4.3 Message contents

**Table 14.2.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.3.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.3.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.3.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |

| | | | |
|---|---|---|---|
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | | | |
| SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 | Not present | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.3.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.3.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.3.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.3.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | | 0 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.3.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.3.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.3.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| Scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.3.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.2.3.5 Test requirement

Table 14.2.3.5-1, Table 14.2.3.5-2 and Table 14.2.3.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.3.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355[49]. The reference cell is the PCell in this test case. |
| Neighbour cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.2 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.1 FR2 | As specified in TS 38.133 [50] clause A.3.31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#13 | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Corresponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |

| | | | |
|---|---|---|---|
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | Rough | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.2.3.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| $BW_{channel}$ | | MHz | 200: $N_{RB,c}$ = 132 | 200: $N_{RB,c}$ = 132 | 200: $N_{RB,c}$ = 132 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -57 | -57 | -57 |
| SSB RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. |

**Table 14.2.3.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| $BW_{channel}$ | | MHz | 200: NRB,c = 132 | 200: NRB,c = 132 | 200: NRB,c = 132 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89.41 | -89.41 | -89.41 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -4.844 | -11.02 | -11.02 |
| Io | Config 1 | dBm/ 190.08MHz | -55.66 | -55.66 | -55.66 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -5.7 | -12.7 | -12.7 |
| PRP Note 4 | | dBm/SCS | -94.2 | -100.4 | -100.4 |

| Propagation Condition | | AWGN |
|---|---|---|

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. The Io is calculated based only on the symbols in which PRS is transmitted.

Note 4: PRP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.

Note 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] Table 6.2.1.3-4.

The RSTD measurement time fulfils the requirements specified in clause 14.2.3.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.3.3 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The RSTD measurement period follows the equation:

$$\mathrm{T_{RSTD,i}} = \left(\mathrm{CSSF_{PRS,i}} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * \mathrm{T_{effect,i}} + \mathrm{T_{last,i}}$$

Where $\mathrm{CSSF_{PRS,i}} = 1, N_{RxBeam,i} = 8, N_{PRS,i}^{slot} = 3, L_{available_PRS,\mathrm{i}} = 1, N_{sample} = 4$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $\mathrm{T_{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM\left(T_{PRS,i}, MGRP_i\right)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $\mathrm{T_{available_PRS,i}} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{0.214}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.178s and 245.97s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.3.4.3-3. The LPP time IE ranges between 6s and 250s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 250.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.3.5-4, Table 14.2.3.5-5 and Table 14.2.3.5-6 for reference.

**Table 14.2.3.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |

| | |
|---|---|
| n2 | 2 |
| >=n4 | 1 |

**Table 14.2.3.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.214}{N} \right\rceil$ |
|---|---|
| nDot125 | 2 |
| >= nDot25 | 1 |

**Table 14.2.3.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.4 NR RSTD measurement period test case for dual positioning frequency layers in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.2.4.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 9.9.2 in an environment with AWGN propagation conditions. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when dual positioning frequency layers are configured.

### 14.2.4.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

### 14.2.4.3 Minimum conformance requirements

Same as in clause 14.2.1.3.

### 14.2.4.4 Test description

The supported test configurations in listed in Table 14.2.4.4-1.

**Table 14.2.4.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 14.2.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.4.5-1, Table 14.2.4.5-2 and Table 14.2.4.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.4.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the 2 RF channels distributed in dual positioning frequency layers.

#### 14.2.4.4.2 Test procedure

Same as in clause 14.2.3.4.2.

#### 14.2.4.4.3 Message contents

**Table 14.2.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.4.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | | | |
| SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 | Not present | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.4.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | Not present | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.4.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.4.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.4.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | 0 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.4.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 µs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.4.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.4.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 3 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.4.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.2.4.5 Test requirement

Table 14.2.4.5-1, Table 14.2.4.5-2 and Table 14.2.4.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.4.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 49]. The reference cell is the PCell in this test case. |
| Neighbour cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.2 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |

| | | | | |
|---|---|---|---|---|
| PRS Configuration | Config 1 | | PRS.1.1 FR2 | As specified in TS 38.133 [50] clause A.3.31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#13 | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | μs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | μs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355[49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | μs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355[49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Corresponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | | Rough | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.2.4.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| NR RF Channel Number | | 1 | 1 | 2 |
| Positioning frequency layer | | 1 | 1 | 2 |
| BWchannel | MHz | 200: NRB,c = 132 | 200: NRB,c = 132 | 200: NRB,c = 132 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | OP.1 | N/A | N/A |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |

| | | | | | |
|---|---|---|---|---|---|
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/190.08MHz | -57.00 | -57.00 | -57.00 |
| SSB_RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB_RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.4.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| | | T2 | T2 | T2 |
| RF Channel Number | | 1 | 1 | 2 |
| Positioning frequency layer | | 1 | 1 | 2 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| EPRE ratio of PBCH DMRS to SSS | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |

| | | | | | |
|---|---|---|---|---|---|
| EPRE ratio of PDSCH to PDSCH | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89.1 | -89.1 | -89 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.17 | -11.36 | -12.47 |
| Io | Config 1 | dBm/ 190.08MHz | -55.68 | -55.68 | -56.73 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -5.7 | -12.7 | -12.7 |
| PRP Note 4 | | dBm/SCS kHz | -94.2 | -100.4 | -101.4 |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.

Note 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] Table 6.2.1.3-4.

The RSTD measurement time fulfils the requirements specified in clause 14.2.4.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.4.3 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left( CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$ for $T_{RSTD,1}$, $N_{PRS,i}^{slot} = 1$ for $T_{RSTD,2}$, $L_{available_PRS,i} = 0.142$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 10.456s and 206.45s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.3.4.3-3. The LPP time IE ranges between 11s and 210s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 11.3s and 210.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.4.5-4, Table 14.2.4.5-5 and Table 14.2.3.5-6 for reference.

**Table 14.2.4.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 14.2.4.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.142}{N} \right\rceil$ |
|---|---|
| nDot125 | 2 |
| >= nDot25 | 1 |

**Table 14.2.3.4-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.5 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 14.2.5.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] Clause 9.9.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.2.5.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 14.2.5.3 Minimum conformance requirements

When physical layer receives last of *NR-TDOA-ProvideAssistanceData* message and *NR-TDOA-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.2.3) DL RSTD measurements, defined in TS 38.215 [4], during the measurement period $T_{RSTD,Total}$ defined as:

$$\mathrm{T_{RSTD,Total}} = \sum_{i=1}^{L} \mathrm{T_{RSTD,i}} + (\mathrm{L}-1) * \max\left(\mathrm{T_{effect,i}}\right)$$

Where ,

$i$ is the index of positioning frequency layer,

$L$ is total number of positioning frequency layers, and

$\mathrm{T_{effect},i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$\mathrm{T_{RSTD,i}}$ is the measurement period for PRS RSTD measurement in positioning frequency layer $i$ as specified below:

$$\mathrm{T_{RSTD,i}} = \left(k_{multiTEG,i} * \mathrm{CSSF_{PRS,i}} * \mathrm{ceil}(\mathrm{K_{p,PRS,i}}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * \mathrm{T_{effect,i}} + \mathrm{T_{last,i}}$$

,

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1, $N_{RxBeam,i}$ = 1;

and in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$CSSF_{PRS,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in positioning frequency layer $i$ as defined in TS 38.133[50] clause 9.1.5.2.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] in *NR-DL-TDOA-RequestLocationInformation*;

otherwise,

$k_{multiTEG,i}$ =$N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG,i}$ =$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gaps.

- For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioningfrequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:
  - $N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and
  - $N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule
  - Requirements do not apply if $N_{available}$ =0.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the

sampling time and processing time. If all of the PRS resources to be measured are available in the same MG occasion during $T_{availabe}$, $T_{last,i} = T_i$ +MGL. Otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$ ,

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$.

$MGRP_{\mathrm{i}}$ is the repetition periodicity of the measurement gap applicable for measurement in the PRS frequency layer i.

$T_{PRS,\mathrm{i}}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,\mathrm{i}}$, where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where

$T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- Note: For the purpose of calculating $T_{PRS,i}$, only the PRS resources fully or partially covered by the MG are considered.

$\{N, T\}$ is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49].

$N'$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [49].

## 14.2.5.4 Test description

### 14.2.5.4.1 Initial conditions

The supported test configuration is listed in Table 14.2.5.4-1.

**Table 14.2.5.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.2.5.5-1, Table 14.2.5.5-2 and Table 14.2.5.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.5.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

#### 14.2.5.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2 and Cell 3. All three cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources.

The NR-DL-TDOA-ProvideAssistanceData and nr-DL-TDOA-RequestLocationInformation as defined in TS 37.355 [49] clause 6.5.12.1, shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

The UE is configured with measurement gap pattern ID # 24 or #0 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.2.5.5-1, Table 14.2.5.5-2 and Table 14.2.5.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 14.2.5.5-3.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. Repeat steps 2-11 in Table 14.2.5.4-1 until the confidence level according to Annex D is achieved.

#### 14.2.5.4.3 Message contents

**Table 14.2.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.5.4.3-2: LPP Request Capabilities**

| **Information Element** | **Value/remark** |
|---|---|

| | |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.5.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.5.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.5.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.5.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.5.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.5.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 SEQUENCE { | | | |
| mcc-r15 | Cell 3 | | |
| mnc-r15 | Cell 3 | | |
| nr-cellidentity-r15 | Cell 3 | | |
| } | | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[4-16] | As specificied in TS 37.571-5 [20] | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.5.4.3-6: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n160-r16 | 10 | | |

| | | | |
|---|---|---|---|
| }, | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 3 |
| | 01 | | Cell 2 |
| }, | | | |
| }, | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| }, | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| }, | | | |
| }, | | | |
| }, | | | |
| }, | | | |
| } | | | |

**Table 14.2.5.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| }, | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.5.5 Test requirement

Table 14.2.5.5-1, Table 14.2.5.5-2 and Table 14.2.5.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.5.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in clause A.3.1.2.1 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |

| | | | | |
|---|---|---|---|---|
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Active DL BWP configuration | Config 1,2,3 | | DLBWP.1.1 | |
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.2 FR1 | As specified in clause A.3.31 |
| | Config 2 | | PRS.1.2 FR1 | |
| | Config 3 | | PRS.2.2 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#0 | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Corresponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |

**Table 14.2.5.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |

| | | | | | |
|---|---|---|---|---|---|
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -52.56 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -88 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.5.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | 2.2 | 0.05 | 0.05 |
| | Config 2 | dB | 2.2 | 0.05 | 0.05 |
| | Config 3 | dB | 2.2 | 0.05 | 0.05 |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -60.25 | -60.25 | -60.25 |
| | Config 2 | dBm/ 19.08MHz | -60.25 | -60.25 | -60.25 |
| | Config 3 | dBm/ 47.88MHz | -56.25 | -56.25 | -56.25 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -3 | -6 | -6 |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

The RSTD measurement time fulfils the requirements specified in clause 14.2.5.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration X1 specified in section 9.9.2.5 starting from the beginning of time interval T2, where X1 is 320ms.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50 ms. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left(k_{multiTEG,i} * CSSF_{PRS,i} * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

,

Since the test case is for single positioning frequency layer, i = 1 and L = 1, so $T_{RSTD,Total} = T_{RSTD,1}$.

Analysing the equation for $T_{RSTD,i}$:

$k_{multiTEG,i}$=1 ,$CSSF_{PRS,i} = 1$ $N_{RxBeam,i} = 1$ $N_{PRS,i}^{slot} = 3$ $N_{sample}$= 1(for reduced sample number).

For SCS 15KHz:

$$L_{available_{PRS},i} = \frac{10}{10 * 1 * 14} * 4 * 1 * 3 = 0.857ms$$

For SCS=30KHz

$$L_{available_{PRS},i} = \frac{10}{20 * 1 * 14} * 4 * 1 * 3 = 0.428ms$$

$T_{last,i} = T_i + T_{available_PRS,i}$

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$

Therefore, the first part of the equation:

$$\left(1 * 1 * 1 * 1 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS},i}}{N} \right\rceil * 1 - 1\right)$$

Where the remaining parameters depend on the UE capabilities. For SCS 15KHz The LPP time IE ranges between 0.168s and 27.04s. For SCS 30KHz The LPP time IE ranges between 0.168s and 15.5s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.5.4.3-3. The LPP time IE ranges between 1s and 28s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 28.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.5.5-4, Table 14.2.5.5-5 , Table 14.2.5.5-6 and Table 14.2.5.5-7 for reference.

**Table 14.2.5.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| N' | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |
| n2 | 2 |
| >=n4 | 1 |

**Table 14.2.5.5-5: value of N based on *durationOfPRS-ProcessingSysmbols* for SCS 15KHz**

| N | $\left\lceil \frac{0.857}{N} \right\rceil$ |
|---|---|
| nDot125 | 7 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |

**Table 14.2.5.5-6: value of N based on *durationOfPRS-ProcessingSysmbols* for SCS 30KHz**

| N | $\left\lceil \frac{0.428}{N} \right\rceil$ |
|---|---|
| nDot125 | 4 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table: 14.2.5.5-7: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.6 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA

### 14.2.6.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the gapless RSTD measurement period requirement specified in TS 38.133[50] Clause 9.9.2.7 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.2.6.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and prs-Processing Window.

### 14.2.6.3 Minimum conformance requirements

When physical layer receives last of *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in TS 38.133[50] Clause 9.9.2.3) DL RSTD measurements, defined in TS 38.215 [4], during the measurement period $T_{RSTD,Total}$ defined as:

$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{RSTD,Total} = \max(T_{RSTD_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

Where,

$i$ is the index of positioning frequency layer, and

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i, and

$T_{uncertainty,i}$ is the time from the start of the first PPW occasion for positioning frequency layer i to the start of measurement period $T_{RSTD,Total}$.

A positioning frequency layer is in Case 1 if UE reports *ppw-durationOfPRS-Processing1-r17* for the band containing the positioning frequency layer, and a positioning frequency layer is in Case 2 if UE reports *ppw-durationOfPRS-Processing2-r17* for the band containing the positioning frequency layer.

$T_{RSTD_wo_gap,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below.

$$T_{RSTD_wo_gap,i} = \left(k_{multiTEG,i} * N_{RxBeam,i} * \left\lfloor \frac{N_{PRS,i}^{slot}}{N'} \right\rfloor \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i},$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1, $N_{RxBeam,i}$ = 1; and in FR2,$N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not supported *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] in *NR-DL-TDOA-RequestLocationInformation*;

otherwise,

$k_{multiTEG,i}$ =$N_{TEG,i}$, if the UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG,i}$ =$\left\lfloor \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rfloor$ if the UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$,

only the PRS resources unmuted and fully or partially overlapped with PPW are considered, if positioning frequency layer i is in Case 1, or

only the PRS resources unmuted and fully or partially overlapped with the first (PPWL-T2) ms of PPW are considered, if positioning frequency layer i is in Case 2, where PPWL is the PPW length and T2 corresponds to *ppw-durationOfPRS-ProcessingSymbolsT2*.

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ =2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
- Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

$T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time.

If positioning frequency layer *i* is in Case 1 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i} = T_i$ +PPWL, else

if positioning frequency layer *i* is in Case 2 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i}$ = PPWL;

otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$.

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *ppw-durationOfPRS-ProcessingSymbolsT* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponds to the sum of *ppw-durationOfPRS-ProcessingSymbolsT2* and *ppw*-durationOfPRS-*ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$, the least common multiple between $T_{PRS,i}$ and $PPWRP_i$.

$PPWRP_i$ is the repetition periodicity of the PRS processing window applicable for measurements in the positioning frequency layer *i*.

$T_{PRS,i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,i}$, where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*,

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

Note: For the purpose of calculating $T_{PRS,\mathrm{i}}$, only the PRS resources that meet the conditions for PRS measurement outside measurement gaps as defined in clause TS 38.133[50] 9.9.1.2 are considered.

$N$ is a duration of DL PRS symbols in ms corresponding to *ppw-durationOfPRS-ProcessingSymbolsN* in TS 37.355 [34] if positioning frequency layer i is in Case 1, or corresponding to *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer i is in Case 2.

$N'$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [34].

## 14.2.6.4 Test description

### 14.2.6.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.6.4-1.

**Table 14.2.6.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.6.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.2.6.5-1, Table 14.2.6.5-2 and Table 14.2.6.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.6.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel distributed in single positioning frequency layer.

### 14.2.6.4.2 Test procedure

The test consists of two sub-tests; Sub-test 1 is for Nsample = 4 and sub-test 2 is for Nsample = 1. Sub-test 2 is only conducted when UE also supporting supported DL-PRS-ProcessingSamples-RRC-CONNECTED. The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2 and Cell 3. All three cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first PRS processing window instance containing the PRS resources.

The *NR-DL-TDOA-ProvideAssistanceData* and *nr-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.12.1, shall be provided to the UE during T1. The last TTI containing the two messages shall be provided

to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 14.2.6.5-1, Table 14.2.6.5-2 and Table 14.2.6.5-3. Propagation conditions are set according to clause 4.15.2.

4. T1 starts.

5. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.

7. The SS shall transmit an RRCReconfiguration message with the PPW configuration.

8. The UE shall transmit RRCReconfigurationComplete message.

9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.

11. The SS sends a Positioning Measurement Gap Activation command containing an ID to activate the associated measurement gap

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 14.2.6.5-2 and Table 14.2.6.5-3.

13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 2-14 in Table 14.2.6.4-1 until the confidence level according to Annex D is achieved.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.2.6.4.3 Message contents

**Table 14.2.6.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.6.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.6.9.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | OCTET STRING (CONTAINING CellGroupConfig) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.6.4.3-4: CellGroupConfig (RRCReconfiguration, Table 14.2.9.4.3-3)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 CHOICE { | The periodicity is 160ms and the starting slot offset is 11 ms for any SCS configuration | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| } | | | |
| length-r17 | 10 | | |
| type-r17 | type1A | | |
| priority-r17 | st1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.6.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6. | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |

| | | | |
|---|---|---|---|
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.6.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | Sub-test 1 |

| | | | |
|---|---|---|---|
| | requested | | Sub-test 2 |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.6.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.6.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.6.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.6.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | As specified in Table 14.2.6.5-1 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.6.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | As specified in Table 14.2.6.5-1 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.6.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.6.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and cell 3 |
| | 01 | | Cell 2 |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.6.4.3-9: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |

| | | | |
|---|---|---|---|
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.6.5 Test requirement

Table 14.2.6.5-1, Table 14.2.6.5-2 and Table 14.2.6.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.6.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |

| | | | | |
|---|---|---|---|---|
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in clause A.3.1.2.1 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Active DL BWP configuration | Config 1,2,3 | | DLBWP.1.1 | Cell 2 and cell 3 also transmit PRS on active DL BWP configured to the UE |
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.4 FR1 | As specified in clause A.3.31 |
| | Config 2 | | PRS.1.4 FR1 | |
| | Config 3 | | PRS.2.4 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| PPW configuration | | | PPW.1 | As defined in A.3.33 |
| Time offset between serving and neighbour cells | µs | – set to the UE reported capability for receive time difference threshold if the UE reported value is < 3µs<br>– 3µs otherwise | | PRS are transmitted from synchronous cells |
| Expected RSTD | µs | 0 | | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | Same as time offset | | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | 4 | | Including the reference cell |
| PRS muting info | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | | Corresponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | | The length of the time interval from the beginning of each test |
| T2 | ms | 10 | | The length of the time interval that follows after time interval T1 and processing time of assistance data. In this test UE is configured to measure single PFL within the configured PPW. |

**Table 14.2.6.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -52.56 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -88 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.6.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 |
| NR RF Channel Number | | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | | OP.1 | | OP.1 | |
| PRACH configuration | | | FR1 PRACH configuration 1 | | FR1 PRACH configuration 1 | | FR1 PRACH configuration 1 | |
| $N_{oc}$ Note 3 | Config 1,2 | dBm/SCS | -98 | | -98 | | -98 | |
| | Config 3 | dBm/SCS | -95 | | -95 | | -95 | |
| PRS $\hat{E}_s/N_{oc}$ | Config 1,2,3 | dB | -5.05 | 1 | -11.22 | -1.2 | -11.22 | -1.2 |
| Io Note 4 | Config 1,2 | dBm/ 19.08MHz | -65.30 | -61.18 | -65.30 | -61.18 | -65.30 | -61.18 |
| | Config 3 | dBm/ 47.88MHz | -61.31 | -57.19 | -61.31 | -57.19 | -61.31 | -57.19 |
| PRS $\hat{E}_s/I_{ot}$ | Config 1,2,3 | dB | -6 | -3 | -13 | -6 | -13 | -6 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| PRS-RSRP[Note 4] | Config 1,2 | dBm/SCS | -103.45 | -97 | -109.67 | -99.2 | -109.67 | -99.2 |
| | Config 3 | | -100.45 | -94 | -106.67 | -96.2 | -106.67 | -96.2 |
| SS-RSRP[Note 4] | Config 1,2 | dBm/SCS | -88 | -97 | -88 | -99.2 | -88 | -99.2 |
| | Config 3 | | -85 | -94 | -85 | -96.2 | -85 | -96.2 |
| Propagation Condition | | | AWGN | | | | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RSTD measurement time fulfils the requirements specified in clause 14.2.6.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in section 9.9.2.7 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The LPP response time calculation procedure could be divided into two situations based on whether the UE support ppw-durationOfPRS-Processing1-r17 or ppw-durationOfPRSProcessing2-r17.The RSTD measurement period follows the equation:

$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{RSTD,Total} = \max(T_{RSTD_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

$$T_{RSTD_wo_gap,i} = \left(k_{multiTEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i},$$

Since the test case is for single positioning frequency layer, i = 1 and L = 1, so $T_{RSTD,\ Total} = T_{RSTD_wo_gap,\ 1}$.

Analysing the equation for $T_{RSTD_wo_gap,\ 1}$: $N_{RxBeam,i} = 1$ $k_{multiTEG,i}=1$ $N_{PRS,i}^{slot} = 3$ $N_{sample} = 4$.

For SCS 15KHz:

$$L_{available_{PRS,i}} = \frac{10}{10 * 1 * 14} * 4 * 1 * 3 = 0.857ms$$

For SCS=30KHz

$$L_{available_{PRS,i}} = \frac{10}{20 * 1 * 14} * 4 * 1 * 3 = 0.428ms$$

Therefore, the first part of the equation:

$$\left(1 * 1 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS,i}}}{N} \right\rceil * 4 - 1\right)$$

$T_{last,i} = T_i$ +PPWL

$$T_{effect,i} = \left\lceil \frac{T_j}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

$T_{available_PRS,\mathrm{i}} = LCM\left(T_{PRS,\mathrm{i}}, PPWRP_{\mathrm{i}}\right)$; where Tprs = 160 ms, and the repetition periodicity $\mathrm{PPWRP_i}$ =160ms. Therefore, $T_{available_PRS,i} = 160$

For case 1:

$$\left(1 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS,i}}}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. For SCS 15KHz The LPP time IE ranges between 0.49s and 108s.

For SCS 30KHz The LPP time IE ranges between 0.49s and 61.4s.

The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.5.4.3-3. The LPP time IE ranges between 1s and 108s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 108.3s.

Case 2:

$\mathrm{T_{last,i}} = \mathrm{PPWL}$

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,\mathrm{i}} = LCM\left(T_{PRS,\mathrm{i}}, PPWRP_{\mathrm{i}}\right)$; where Tprs = 160 ms, and the repetition periodicity $\mathrm{PPWRP_i}$=160ms. Therefore, $T_{available_PRS,i} = 160$

Therefore, $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ and $\mathrm{T_{last,i}}$ = PPWL take following values depending on Ti

$$\left(1 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,\mathrm{i}}}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. For SCS 15KHz The LPP time IE ranges between 0.49s and 13.29s.

For SCS 30KHz The LPP time IE ranges between 0.49s and 7.53s.

The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.5.4.3-3. The LPP time IE ranges between 1s and 14s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 14.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.6.5-4, Table 14.2.9.6-5, Table 14.2.6.5-6, Table 14.2.6.5-7 and Table 14.2.6.5-8 for reference.

**Table 14.2.6.5-4: value of N' based on *ppw-maxNumOfDL-PRS-ResProcessedPerSlot***

| N' | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |
| n2 | 3 |
| >=n4 | 1 |

**Table 14.2.6.5-5: value of N based on ppw-*durationOfPRS-ProcessingSymbolsN* for SCS 15KHz**

| N | $\left\lceil \frac{0.857}{N} \right\rceil$ |
|---|---|

| | |
|---|---|
| msDot125 | 7 |
| msDot25 | 4 |
| msDot5 | 2 |
| >=ms1 | 1 |

**Table 14.2.6.5-6: value of N based on *ppw-durationOfPRS-ProcessingSymbolsN* for SCS 30KHz**

| N | $\lceil \frac{0.428}{N} \rceil$ |
|---|---|
| msDot125 | 4 |
| msDot25 | 2 |
| >=msDot5 | 1 |

**Table 14.2.6.5-7: Teffect and Tlast based on *ppw-durationOfPRS-ProcessingSymbolsT* For Case 1**

| ppw-durationOfPRS-ProcessingSymbolsT-r17 | Teffect | Tlast |
|---|---|---|
| ms1 | 160 | 11 |
| ms2 | 160 | 12 |
| ms4 | 160 | 14 |
| ms8 | 160 | 18 |
| ms16 | 160 | 26 |
| ms20 | 160 | 30 |
| ms30 | 160 | 40 |
| ms40 | 160 | 50 |
| ms80 | 160 | 90 |
| ms160 | 160 | 170 |
| ms320 | 320 | 330 |
| ms640 | 640 | 650 |
| ms1280 | 1280 | 1290 |

**Table 14.2.6.5-8: Teffect and Tlast based on the sum of ppw-durationOfPRS-ProcessingSymbolsT2 and ppw-durationOfPRS-ProcessingSymbolsN2 for Case 2**

| Ti | Teffect | Tlast |
|---|---|---|
| 4.125ms | 160 | 10 |
| 20ms | 160 | 10 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.7 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR1 SA

### 14.2.7.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the Rx TEG based measurement period requirements specified in TS 38.133 [50] Clause 9.9.2.5 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.2.7.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports UE assisted DL-TDOA positioning and nr-UE-RxTEG-ID-MaxSupport.

### 14.2.7.3 Minimum conformance requirements

Same as defined in clause 14.2.5.3.

## 14.2.7.4 Test description

### 14.2.7.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.7.4-1.

**Table 14.2.7.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.7.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.2.7.5-1, Table 14.2.7.5-2 and Table 14.2.7.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.7.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel distributed in single positioning frequency layer.

### 14.2.7.4.2 Test procedure

Same as defined in clause 14.2.5.4.2.

### 14.2.7.4.3 Message contents

**Table 14.2.7.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.7.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.7.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.7.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | n2 | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTEG-Request-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.7.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.7.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.7.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.7.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | As specified in Table 14.2.7.5-1 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | As specified in Table 14.2.7.5-1 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.7.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | As specified in Table 14.2.7.5-1 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | As specified in Table 14.2.7.5-1 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.7.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.7.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.7.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.7.5 Test requirement

Table 14.2.7.5-1, Table 14.2.7.5-2 and Table 14.2.7.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.7.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in clause A.3.1.2.1 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1 ULBWP.0.1 | |
| Active DL BWP configuration | Config 1,2,3 | | DLBWP.1.1 | |
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.4 FR1 | As specified in clause A.3.31 |
| | Config 2 | | PRS.1.4 FR1 | |
| | Config 3 | | PRS.2.4 FR1 | |

| | | | |
|---|---|---|---|
| Physical cell ID PCI | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | Normal | |
| DRX | | OFF | |
| Measurement gap | | GP#24 or GP#0 | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | 4 | Including the reference cell |
| PRS muting info | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Corresponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | [1.28*$k_{multiTEG}$]Note 1 | The length of the time interval that follows immediately after time interval T1 |

Note 1:

o $k_{multiTEG}=\left\lceil \frac{N_{TEG}}{k_{TEG,simul}} \right\rceil$ if UE is capable of receiving the same DL PRS resource from the same TRP simultaneously from multiple Rx TEGs, where $N_{TEG}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource, and $k_{TEG,simul}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

**Table 14.2.7.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |

| | | | | | |
|---|---|---|---|---|---|
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -52.56 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -88 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.7.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.08 | -11.27 | -11.27 |
| | Config 2 | dB | -5.08 | -11.27 | -11.27 |
| | Config 3 | dB | -5.08 | -11.27 | -11.27 |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -65.31 | -65.31 | -65.31 |
| | Config 2 | dBm/ 19.08MHz | -65.31 | -65.31 | -65.31 |
| | Config 3 | dBm/ 47.88MHz | -61.32 | -61.32 | -61.32 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -6 | -13 | -13 |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

The RSTD measurement time fulfils the requirements specified in clause 14.2.7.3.

The UE shall perform and report the Rx TEG based RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in section 9.9.2.5 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50 ms. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left(k_{multiTEG,i} * CSSF_{PRS,i} * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Since the test case is for single positioning frequency layer, i = 1 and L = 1, so $T_{RSTD,Total} = T_{RSTD,1}$.

Analysing the equation for $T_{RSTD,i}$:

$CSSF_{PRS,i} = 1$ $N_{RxBeam,i} = 1$ $N_{PRS,i}^{slot} = 4$ $N_{sample} = 4$

For SCS 15KHz:

$$L_{available_{PRS,i}} = \frac{10}{10 * 1 * 14} * 4 * 1 * 4 = 1.14ms$$

For SCS=30KHz

$$L_{available_{PRS,i}} = \frac{10}{20 * 1 * 14} * 4 * 1 * 4 = 0.571ms$$

Case 1: $k_{multiTEG,i} = N_{TEG,i}$

Case 2: $k_{multiTEG,i} = \left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$

For case 1: $k_{multiTEG,i}$ can take a maximum value of n8 and minimum value of n2

For case2: $k_{multiTEG,i}$ can take a maximum value of 4 and minimum value would be 1.

Therefore, the first part of the equation:

$$\left(k_{multiTEG} * 1 * \left\lceil \frac{4}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS,i}}}{N} \right\rceil * 4 - 1\right)$$

$T_{last,i} = T_i + T_{available_PRS,i}$

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$

Case 1:

Where the remaining parameters depend on the UE capabilities. For SCS 15KHz The LPP time IE ranges between 1.28s and 1638.5s. For SCS 30KHz The LPP time IE ranges between 1.28s and 819.6s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.7.4.3-3. The LPP time IE ranges between 2s and 1640s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and (1638.3sec)164ten-s.

Case 2:

Where the remaining parameters depend on the UE capabilities. For SCS 15KHz The LPP time IE ranges between 0.648s and 1638.5s. For SCS 30KHz The LPP time IE ranges between 0.64s and 819.6s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.7.4.3-3. The LPP time IE ranges between 1s and 1640s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and (1638.3sec)164ten-s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.7.5-4, Table 14.2.7.5-5 , Table 14.2.7.5-6 and Table 14.2.7.5-7 for reference.

**Table 14.2.7.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| N' | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 1 |
| >=n4 | 1 |

**Table 14.2.7.5-5: value of N based on *durationOfPRS-ProcessingSysmbols* for SCS 15KHz**

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |

**Table 14.2.7.5-6: value of N based on *durationOfPRS-ProcessingSysmbols* for SCS 30KHz**

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table: 14.2.7.5-7: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.8 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.2.8.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] Clause 9.9.2 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured. In this test PRS is transmitted within the active BWP of the UE.

### 14.2.8.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 14.2.8.3 Minimum conformance requirements

When physical layer receives last of *NR-TDOA-ProvideAssistanceData* message and *NR-TDOA-RequestLocationInformation* message from LMF via LPP [49], the UE shall be able to measure multiple DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined as:

$$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD,i} + (L-1) * \max(T_{effect,i})$$

Where ,

$i$ is the index of positioning frequency layer,

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$T_{RSTD,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer $i$ as specified below:

$$T_{RSTD,i} = \left(k_{multiTEG,i} * CSSF_{PRS,i} * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

,

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1, $N_{RxBeam,i}$ = 1;

and in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$CSSF_{PRS,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in positioning frequency layer $i$ as defined in TS 38.133[50] clause 9.1.5.2.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*;

otherwise,

$k_{multiTEG}$=$N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}$=$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

- For a window W of duration max($T_{PRS,i,}$ MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:
  - $N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including those overlapped with other MG occasions within the window, and
  - $N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule
  - Requirements do not apply if $N_{available}$ =0.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

- $N_{sample}$ = 4 otherwise.

$T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time. If all of the PRS resources to be measured are available in the same MG occasion during $T_{availabe}$, $T_{last,i} = T_i$ +MGL. Otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$ ,

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$.

$MGRP_i$ is the repetition periodicity of the measurement gap applicable for measurement in the PRS frequency layer i.

$T_{PRS,i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,i}$, where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where

$T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- Note: For the purpose of calculating $T_{PRS,i}$, only the PRS resources fully or partially covered by the MG are considered.

$\{N, T\}$ is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49].

$N'$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [49].

## 14.2.8.4 Test description

### 14.2.8.4.1 Initial conditions

The supported test configuration is listed in Table 14.2.8.4-1.

**Table 14.2.8.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.8.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.8.5-1, Table 14.2.8.5-2 and Table 14.2.8.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.8.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel distributed in single positioning frequency layer.

### 14.2.8.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2 and Cell 3. All three cells transmit PRS during T2.The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources.

The *NR-DL-TDOA-ProvideAssistanceData* and *nr-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.12.1, shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.2.8.5-1, Table 14.2.8.5-2 and Table 14.2.8.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 14.2.8.5-3.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

12. Repeat steps 2-11 in Table 14.2.8.4-1 until the confidence level according to Annex D is achieved.

### 14.2.8.4.3 Message contents

**Table 14.2.8.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.8.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.8.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.8.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.8.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |

| | | | |
|---|---|---|---|
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.8.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.8.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.8.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.8.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.8.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.8.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.8.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |

| | | | |
|---|---|---|---|
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.8.5 Test requirement

Table 14.2.8.5-1, Table 14.2.8.5-2 and Table 14.2.8.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.8.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.3 FR2 | |
| $BW_{channel}$ | Config 1 | MHz | 200: $N_{RB,c}$ = 132 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |

| | | | | |
|---|---|---|---|---|
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.2 FR2 | As specified in TS 38.133 [50] clause A.3. 31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#13 | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Corresponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | | Rough | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.2.8.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| NR RF Channel Number | | 1 | 1 | 1 |
| Positioning frequency layer | | 1 | 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.5 FDD | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -57.00 | -57.00 | -57.00 |
| SSB RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.8.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | 2.9 | 0.75 | 0.75 |
| Io | Config 1 | dBm/ 190.08MHz | -55.71 | -55.71 | -55.71 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -2.7 | -5.6 | -5.6 |
| PRP Note 4 | | dBm/SCS | -92.1 | -94.2 | -94.2 |
| Propagation Condition | | | AWGN | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS. |
| Note 2: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 3: | Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | PRP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted. |
| Note 5: | Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. |

The RSTD measurement time fulfils the requirements specified in clause 14.2.8.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.8.3 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left( CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 3$, $L_{available_PRS,i} = 1$, $N_{sample} = 1$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_j}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{0.107}{N} \right\rceil * 1 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 1.288s and 30.88s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.8.4.3-3. The LPP time IE ranges between 2s and 31s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 31.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.8.5-4, Table 14.2.8.5-5 and Table 14.2.8.5-6 for reference.

**Table 14.2.8.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |
| n2 | 3 |
| >=n4 | 1 |

**Table 14.2.8.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\lceil \frac{0.107}{N} \rceil$ |
|---|---|
| >=nDot125 | 1 |

**Table 14.2.8.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.9 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.2.9.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the gapless RSTD measurement requirements specified in TS 38.133 [50] Clause 9.9.2.7 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.2.9.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and prs-Processing Window.

### 14.2.9.3 Minimum conformance requirements

When physical layer receives last of *NR-TDOA-ProvideAssistanceData* message and *NR-TDOA-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.2.3) DL RSTD measurements, defined in TS 38.215 [4], during the measurement period $T_{RSTD,Total}$ defined as:

$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{RSTD,Total} = \max(T_{RSTD_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

Where,

$i$ is the index of positioning frequency layer, and

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i, and

$T_{uncertainty,i}$ is the time from the start of the first PPW occasion for positioning frequency layer i to the start of measurement period $T_{RSTD,Total}$.

A positioning frequency layer is in Case 1 if UE reports *ppw-durationOfPRS-Processing1-r17* for the band containing the positioning frequency layer, and a positioning frequency layer is in Case 2 if UE reports *ppw-durationOfPRS-Processing2-r17* for the band containing the positioning frequency layer.

$T_{RSTD_wo_gap,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below.

$$T_{RSTD_wo_gap,i} = \left(k_{multiTEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i} ,$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1, $N_{RxBeam,i}$ = 1; and in FR2,$N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not supported *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] in *NR-DL-TDOA-RequestLocationInformation*;

otherwise,

$k_{multiTEG}$=$N_{TEG,i}$, if the UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}$=$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if the UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$,

only the PRS resources unmuted and fully or partially overlapped with PPW are considered, if positioning frequency layer i is in Case 1, or

only the PRS resources unmuted and fully or partially overlapped with the first (PPWL-T2) ms of PPW are considered, if positioning frequency layer i is in Case 2, where PPWL is the PPW length and T2 corresponds to *ppw-durationOfPRS-ProcessingSymbolsT2*.

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ =2 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
- Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

$T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time.

If positioning frequency layer *i* is in Case 1 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i} = T_i$ +PPWL, else

if positioning frequency layer *i* is in Case 2 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i}$ = PPWL;

otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$.

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *ppw-durationOfPRS-ProcessingSymbolsT* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponds to the sum of *ppw-durationOfPRS-ProcessingSymbolsT2* and *ppw*-durationOfPRS-*ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$, the least common multiple between $T_{PRS,i}$ and $PPWRP_i$.

$PPWRP_i$ is the repetition periodicity of the PRS processing window applicable for measurements in the positioning frequency layer *i*.

$T_{PRS,i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,i}$, where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*,

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

Note: For the purpose of calculating $T_{PRS,i}$, only the PRS resources that meet the conditions for PRS measurement outside measurement gaps as defined in clause 9.9.1.2 are considered.

$N$ is a duration of DL PRS symbols in ms corresponding to *ppw-durationOfPRS-ProcessingSymbolsN* in TS 37.355 [34] if positioning frequency layer i is in Case 1, or corresponding to *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer i is in Case 2.

*N'* is UE capability for number of DL PRS resources that it can process in a slot as indicated by *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [34].

## 14.2.9.4 Test description

### 14.2.9.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.9.4-1.

**Table 14.2.9.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.9.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.9.5-1, Table 14.2.9.5-2 and Table 14.2.9.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.9.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel distributed in single positioning frequency layer.

### 14.2.9.4.2 Test procedure

The test consists of two sub-tests; Sub-test 1 is for Nsample = 4 and sub-test 2 is for Nsample = 1. Sub-test 2 is only conducted when UE also supporting supportedDL-PRS-ProcessingSamples-RRC-CONNECTED. The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2 and Cell 3. All three cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first PRS processing window instance containing the PRS resources.

The *NR-DL-TDOA-ProvideAssistanceData* and *nr-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.12.1, shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

3. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.2.9.5-1, Table 14.2.9.5-2 and Table 14.2.9.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
7. The SS shall transmit an RRCReconfiguration message with the PPW configuration.

8. The UE shall transmit RRCReconfigurationComplete message.

9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.

11. The SS sends a Positioning Measurement Gap Activation command containing an ID to activate the associated measurement gap

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 14.2.9.5-2 and Table 14.2.9.5-3.

13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 2-14 in Table 14.2.9.4-1 until the confidence level according to Annex D is achieved.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.2.9.4.3 Message contents

**Table 14.2.9.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.9.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.9.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | OCTET STRING (CONTAINING CellGroupConfig) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.9.4.3-4: CellGroupConfig (RRCReconfiguration, Table 14.2.9.4.3-3)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 CHOICE { | The periodicity is 160ms and the starting slot offset is 11 ms for any SCS configuration | | |
| scs120 CHOICE { | | | |
| n1280 | 4 | | |
| } | | | |
| } | | | |
| length-r17 | 80 | | |
| type-r17 | type1A | | |
| priority-r17 | st1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.9.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.9.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.9.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.9.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | About 0 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 17 | About 0.58 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.9.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | About 0 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 17 | About 0.58 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.9.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.9.4.3-7: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| Scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.9.4.3-8: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |

| | | | |
|---|---|---|---|
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.9.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | Sub-test 1 |
| | requested | | Sub-test 2 |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.9.4.3-9: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.2.9.5 Test requirement

Table 14.2.9.5-1, Table 14.2.9.5-2 and Table 14.2.9.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.9.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ | Config 1 | MHz | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | Config 1 | | SSB.3 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.4. FR2 | As specified in TS 38.133 [50] clause A.3. 31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| PPW configuration | | | Table A.3.X33-1: Reference PPW configuration | As defined in TS 38.133 [50] A.3.33 |
| Time offset between serving and neighbour cells | | µs | set to the UE reported capability for receive time difference threshold if the UE reported value is < 3µs<br>3µs otherwise | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | 0 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | Same as time offset between serving and neighbour cells | The corresponding parameter in the DL-TDOA assistance ta specified in TS |

| | | | |
|---|---|---|---|
| | | | 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | 4 | Including the reference cell |
| PRS muting info | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Corresponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | ms | 10 | The length of the time interval that follows after time interval T1 and processing time of assistance data. In this test UE is configured to measure single PFL within the configured PPW. |
| AoA setup | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | Rough | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.2.9.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.5 FDD | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -57.00 | -57.00 | -57.00 |
| SSB RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.9.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 |
| RF Channel Number | | | 1 | | 1 | | 1 | |
| Positioning frequency layer | | | 1 | | 1 | | 1 | |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | | OP.1 | | OP.1 | |
| PRACH configuration | | | FR2 PRACH configuration 1 | | FR2 PRACH configuration 1 | | FR2 PRACH configuration 1 | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89.41 | -95 | -89.41 | -95 | -84.41 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -4.84 | 42.9 | -11.02 | 0.75 | -11.02 | 0.75 |
| Io | Config 1 | dBm/ 190.08MHz | -55.66 | -55.71 | -55.66 | -55.71 | -55.66 | -55.71 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -5.7 | -276 | -12.7 | -5.6 | -12.7 | -5.6 |
| PRS-RSRP Note 4 | | dBm/SCS | -94.2 | -92.1 | -100.4 | -94.2 | -100.4 | -94.2 |
| Propagation Condition | | | AWGN | | | | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: PRP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.
Note 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The RSTD measurement time fulfils the requirements specified in clause 14.2.9.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.9.3 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The LPP response time calculation procedure could be divided into two situations based on whether the UE support ppw-durationOfPRS-Processing1-r17 or ppw-durationOfPRS-Processing2-r17.The RSTD measurement period follows the equation:

$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{RSTD,Total} = \max(T_{RSTD_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

Where $N_{RxBeam,i}$ = 8, $k_{multiTEG,i}$=1, $N_{PRS,i}^{slot}$ = 3, $L_{available_PRS,i}$ = 0.214, $N_{sample}$= 4, $T_{uncertainty,i}$ = 0. N is the parameter *ppw-durationOfPRS-ProcessingSymbolsN* from TS 37.355 [49], N' is the parameter *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i}$ = $T_i$ +PPWL and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$; where Tprs = 160 ms, and the repetition periodicity $PPWRP_i$=160ms. Therefore, $T_{available_PRS,i} = 160$

$T_i$ depends on the UE parameter *ppw-durationOfPRS-ProcessingSymbolsT* from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(8 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{0.214}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 4.971s and 245.77s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.9.4.3-3. The LPP time IE ranges between 5s and 250s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 5.3s and 25.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.9.5-4, Table 14.2.9.5-5 and Table 14.2.9.5-6 for reference.

**Table 14.2.9.5-4: value of N' based on *ppw-maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |
| n2 | 3 |
| >=n4 | 1 |

**Table 14.2.9.5-5: value of N based on *ppw-durationOfPRS-ProcessingSymbolsN***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.214}{N} \right\rceil$ |
|---|---|
| nDot125 | 2 |
| >= nDot25 | 1 |

**Table 14.2.9.5-6: value of $T_{effect}$ and $T_{last}$ based on ppw-durationOfPRS-ProcessingSymbolsT-r17**

| **ppw-durationOfPRS-ProcessingSymbolsT-r17** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| ms1 | 160 | 11 |
| ms2 | 160 | 12 |
| ms4 | 160 | 14 |
| ms8 | 160 | 18 |
| ms16 | 160 | 26 |
| ms20 | 160 | 30 |
| ms30 | 160 | 40 |
| ms40 | 160 | 50 |
| ms80 | 160 | 90 |
| ms160 | 160 | 170 |
| ms320 | 320 | 330 |
| ms640 | 640 | 650 |
| ms1280 | 1280 | 1290 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.10 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.2.10.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the Rx TEG based measurement period requirements specified in TS 38.133 [50] Clause 9.9.2.5 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.2.10.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports UE assisted DL-TDOA positioning and nr-UE-RxTEG-ID-MaxSupport.

### 14.2.10.3 Minimum conformance requirements

Same as defined in clause 14.2.8.3.

### 14.2.10.4 Test description

#### 14.2.10.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.10.4-1.

**Table 14.2.10.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.10.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.10.5-1, Table 14.2.10.5-2 and Table 14.2.10.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.10.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All cells are on the same RF channel distributed in single positioning frequency layer.

### 14.2.10.4.2 Test procedure

Same as defined in clause 14.2.8.4.2.

### 14.2.10.4.3 Message contents

**Table 14.2.10.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.10.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.10.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.10.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |

| | | | |
|---|---|---|---|
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | n2 | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.10.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.10.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.10.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.10.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.10.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.10.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.10.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| Scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.10.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.10.5 Test requirement

Table 14.2.10.5-1, Table 14.2.10.5-2 and Table 14.2.10.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.10.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ | Config 1 | MHz | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | Config 1 | | SSB.3 FR2 | |

| | | | | |
|---|---|---|---|---|
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.4. FR2 | As specified in TS 38.133 [50] clause A.3. 31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#13 | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Corresponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 1.28*$k_{multiTEG}$ Note 1 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | | Rough | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

Note 1: $k_{multiTEG}=\left\lceil \frac{N_{TEG}}{k_{TEG,simul}} \right\rceil$ if UE is capable of receiving the same DL PRS resource from the same TRP simultaneously from multiple Rx TEGs, where $N_{TEG}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource, and $k_{TEG,simul}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

**Table 14.2.10.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.5 FDD | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -57.00 | -57.00 | -57.00 |
| SSB RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.10.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89.41 | -89.41 | -89.41 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -4.84 | -11.02 | -11.02 |
| Io | Config 1 | dBm/ 190.08MHz | -55.66 | -55.66 | -55.66 |

| | | | | |
|---|---|---|---|---|
| PRS $\hat{E}_s/I_{ot}$ | dB | -5.7 | -12.7 | -12.7 |
| PRP Note 4 | dBm/SCS | -93.8 | -100 | -100 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: PRP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.
Note 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The RSTD measurement time fulfils the requirements specified in clause 14.2.10.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.10.3 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left(k_{multiTEG,i} * CSSF_{PRS,i} * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 3$, $L_{available_PRS,i} = 0.214$, $N_{sample} = 4$ $k_{multiTEG} = 2$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(2 * 1 * 8 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{0.214}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 10.248s and 491.68s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.10.4.3-3. The LPP time IE ranges between 11s and 500s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 11.3s and 500.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.10.5-4, Table 14.2.10.5-5 and Table 14.2.10.5-6 for reference.

**Table 14.2.10.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| *maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|

| | |
|---|---|
| n1 | 3 |
| n2 | 3 |
| >=n4 | 1 |

**Table 14.2.10.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.214}{N} \right\rceil$ |
|---|---|
| nDot125 | 2 |
| >= nDot25 | 1 |

**Table 14.2.10.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.11 NR RSTD measurement reporting delay test case for PRS aggregation in FR1

### 14.2.11.1 Test purpose

The purpose of the test is to verify that the RSTD measurement by aggregating PRS resources from multiple positioning frequency layers (PFLs) meets the measurement period requirements specified in TS 38.133 [50] clause 9.9.2.10 in an environment with AWGN propagation conditions in FR1.

### 14.2.11.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Connected-r18.*

### 14.2.11.3 Minimum conformance requirements

When UE physical layer receives last of *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from LMF via LPP [49] with a request to perform measurement by aggregating PRS resources from multiple PFLs via *nr-DL-PRS-JointMeasurementRequestedPFL-List*, the UE shall be able to perform RSTD measurement by aggregating PRS resources from multiple PFLs (up to the UE capability specified in TS 38.133 clause 9.9.2.3 ) defined in TS 38.215 [57], during the measurement period $T_{RSTD_aggregate,\ Total}$ defined as:

$$T_{RSTD_aggregate,\ Total} = T_{non_aggregate_RSTD} + T_{aggregate_RSTD} + T_{margin},$$

where,

$T_{non_aggregate_RSTD}$ is the total measurement period for RSTD measurement on PFLs that do not contain PRS resources for aggregation.

Calculation of $T_{non_aggregate_RSTD}$ is based on TS 38.133 clause 9.9.2.5 with the following exceptions,

- only PFLs containing resource set(s) not linked to any other resource set(s) are considered in $L$
- on each PFL $i$, only resource set(s) not linked to any other resource set(s) are considered in $T_{available_PRS,i}$, $N_{PRS,i}^{slot}$ and $L_{available_PRS,i}$
- $N_{sample} = 2$ if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* specified in TS 37.355 [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples,
- $(L-1) * \max(T_{effect,i})$ is replace by $(L-1) * T'_{margin}$.

$T_{non_aggregate_RSTD}$ is equal to zero if UE is not configured to perform non-aggregated measurements by the LMF.

If UE is configured to perform positioning measurements on PFLs that contain PRS resources for aggregation and on PFLs that do not contain PRS resources for aggregation, $T_{margin} = T'_{margin}$ is delay margin to account for delay between RSTD measurement performed by UE on PFLs that contain PRS resources for aggregation and RSTD measurement performed by UE on PFLs that do not contain PRS resources for aggregation.

If UE is only configured to perform measurements on PFLs that contain PRS resources for aggregation or if UE is only configured to perform measurements on PFLs that do not contain PRS resources for aggregation, then $T_{margin} = 0$.

$T'_{margin} = \max(Teffect, i)$, when both aggregated PFLs and non-aggregated PFLs are configured for positioning measurement.

If UE is capable of performing latency reduced positioning measurements and is configured to perform latency reduced positioning measurement by LMF via *reducedDL-PRS-ProcessingSamples-r17* [49], then $N_{sample} = 2$ shall be considered in calculations of $T_{aggregate_RSTD}$ and $T_{non\text{-}aggregate_RSTD}$.

If UE is not capable to perform latency reduced positioning measurements or is not configured to perform latency reduced positioning measurement by LMF via *reducedDL-PRS-ProcessingSamples-r17* [49], then $N_{sample} = 4$ shall be considered in calculations of $T_{aggregate_RSTD}$ and $T_{non\text{-}aggregate_RSTD}$.

$T_{aggregate_RSTD}$, total measurement period for RSTD measurements performed by UE by aggregating PRS resources from multiple PFLs as defined in TS 38.214 [56], is defined as:

$$T_{aggregate_RSTD} = \sum_{i=1}^{G} T_{RSTD_aggregate,i} + (G-1) * T'_{margin},$$

where

i is the index of effective PFL, corresponding to the group of PFLs containing linked PRS resource sets to be aggregated for RSTD measurement,

G denotes the number of effective PFLs that UE is configured to perform aggregated measurements on. G is configured by LMF and is provisioned to UE via *nr-DL-PRS-AggregationInfo.*

$T_{RSTD_aggregate,i}$ is the measurement period for PRS RSTD measurement in the effective PFL *i* as specified below:

$$T_{RSTD_aggregate,i} = \left(k_{multiTEG_agg,i} \times CSSF_{PRS_agg,i} \times ceil(K_{p,PRS_agg,i}) \times N_{RxBeam,i} \times \left\lceil \frac{N_{PRS_agg,i}^{slot}}{N'_{agg}} \right\rceil \left\lceil \frac{L_{available_PRS_agg,i}}{N_{agg}} \right\rceil \times N_{sample_agg} - 1\right) \times T_{effect_agg,i} + T_{last_agg,i},$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor.

In FR1, $N_{RxBeam,i} = 1$.

In FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing effective PFL i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$CSSF_{PRS_agg,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in effective PFL i as defined in TS 38.133 clause 9.1.5.2.

$k_{multiTEG_agg,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG_agg,i}$ = 1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*;

Otherwise,

$k_{multiTEG_agg,i} = N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG_{agg},i}=\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure aggregated PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS_agg,i}$ is a scaling factor for effective PFL i to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS_agg,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS_agg,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS_agg,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the effective PFL, and starting at the beginning of any associated gap occasions covering the PRS occasion:

- $N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and
- $N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule
- Requirements do not apply if $N_{available} = 0$.

$N^{slot}_{PRS_agg,i}$ is the maximum number of DL PRS resources in effective PFL i configured in a slot.

$L_{available_PRS_agg,i}$ is the time duration of available PRS in the effective PFL i to be measured during $T_{available_PRS_agg,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS_agg,i}$, only the PRS resources unmuted and fully or partially overlapped with MG and satisfying the conditions for PRS BW aggregation are considered.

$N_{sample_agg}$ is the number of PRS RSTD measurement samples.

$T_{last_agg,i}$ is the measurement duration for the last PRS RSTD sample in the effective PFL i, including the sampling time and processing time. If all of the PRS resources to be measured are available in the same MG occasion during $T_{availabe_agg}$, $T_{last_agg,i} = T_{agg,i}$ + MGL. Otherwise, $T_{last_agg,i} = T_{agg,i} + T_{available_PRS_agg,i}$,

$T_{effect_agg,i}$ is the periodicity of the PRS RSTD measurement in the effective PFL i defined as:

$$T_{effect_agg,i} = \left\lceil \frac{T_{agg,i}}{T_{available_PRS_agg,i}} \right\rceil \times T_{available_PRS_agg,i}$$

where,

$T_{available_PRS_agg,i} = LCM(T_{PRS_agg,i}, MGRP_i)$, the least common multiple between $T_{PRS_agg,i}$ and $MGRP_i$. $MGRP_i$ is the repetition periodicity of the measurement gap applicable for measurement in the effective PFL i.

$T_{PRS_agg,i}$ is the periodicity of DL PRS resources meeting the bandwidth aggregation conditions with muting on effective PFL i. If more than one PRS periodicities are configured in effective PFL i, the least common multiple of PRS periodicities $T_{per_agg}^{PRS\ with\ muting}$ among all DL PRS resource sets in the effective PFL is used to derive $T_{PRS_agg,i}$, where,

$T_{per_agg}^{PRS\ with\ muting} = N_{muting} \times T_{per_agg}^{PRS}$, is the PRS periodicity with muting per PRS resource configured for aggregation,

$T_{per_agg}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS_agg} \times L_{muting}$, where $T_{muting}^{PRS_agg}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- Note: For the purpose of calculating $T_{PRS_agg,i}$, only the PRS resources fully or partially covered by the MG and PRS resources that satisfy the conditions for PRS BW aggregation are considered.

$\{N_{agg}, T_{agg,i}\}$ is UE capability combination per band to aggregate PRS resources from multiple PFLs within an effective PFL where $N_{agg}$ is a duration of DL PRS symbols in ms corresponding to *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsN-r18* in TS 37.355 [49] processed every $T_{agg,i}$ ms corresponding to *prs-durationOfTwoPRS-BWA-ProcessingSymbolsT-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsT* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *maximumOfTwoAggregatedDL-PRS-Bandwidth* or *maximumOfThreeAggregatedDL-PRS-Bandwidth* in TS 37.355 [49]

$N'_{agg}$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR1-r18* for FR1 and *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR2-r18* for FR2 specified in TS 37.355 [49].

The requirements in this clause for aggregated measurements apply provided that the linked PRS resource sets on multiple PFLs for aggregated measurements satisfy all the conditions specified in clause 5.1.6.5.3 in TS 38.214 [56].

Except for deferred MT-LR as defined in clause 4.1a.5 TS 23.273, the time $T_{RSTD_aggregate,Total}$ starts from the first MG instance aligned with DL PRS resources in the assistance data for aggregation after both the *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the physical layer of UE via LPP [49] including a request to perform measurement by aggregating PRS resources from multiple PFLs via *nr-DL-PRS-JointMeasurementRequestedPFL-List*.

For deferred MT-LR with other event than “Periodic Location” as defined in clause 4.1a.5.1 TS 23.273 , the time $T_{RSTD_aggregate,Total}$ starts from the first MG instance aligned with DL PRS resources in the assistance data after the associated event(s) occurs.

For deferred MT-LR with event “Periodic Location” as defined in clause 4.1a.5.1 TS 23.27] , the UE shall perform the aggregated RSTD measurement in each reporting period and activate the location report at the time when the periodic timer expires.

When PRS-RSRP is also reported by UE together with RSTD measurement based on aggregated DL-PRS resources from multiple-PFLs, PRS-RSRP are performed over the measurement period defined in TS 38.133 clause 9.9.2.10.

When PRS-RSRPP is also reported by UE together with RSTD measurement based on aggregated DL-PRS resources from multiple-PFLs, PRS-RSRPP are performed over the measurement period defined in TS 38.133 clause 9.9.2.10.

If during the measurement period of one or more effective PFLs, the MG pattern is reconfigured, the measurement period can be longer.

The measurement requirements in this clause apply, provided no PRS symbols are dropped during the measurement period $T_{RSTD_aggregate,Total}$ within measurement gaps due to collisions with other signals; otherwise, the measurement period can be longer.

If CSSF changes during the measurement period, the measurement period could be longer.

The measurement requirements do not apply for aggregated PRS resources, if the PRS resources to be aggregated are across two sampling duration of $N_{agg}$ within duration $L_{available_PRS_agg,i}$.

The measurement requirements do not apply, if time span of the instance of the PRS resources to be aggregated (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability $N_{agg}$.

The requirements in TS 38.133 clause 9.9.2.10 do not apply if the PRS configuration given by higher layer parameters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-TDOA-ProvideCapabilities*, and it is up to UE implementation which aggregated PRS resources are measured, subject to UE measurement capabilities.

If handover occurs while aggregated RSTD measurements are being performed, then the UE shall continue and complete the on-going aggregated RSTD measure, ments. The aggregated RSTD measurement period can be longer. The UE shall meet the RSTD measurement accuracy requirements in TS 38.133 clause 10.1.23A.2.

## 14.2.11.4 Test description

### 14.2.11.4.1 Initial conditions

**Table 14.2.11.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth per PFL, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth per PFL, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth per PFL, TDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.11.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.2.11.5-1, Table 14.2.11.5-2 and Table 14.2.11.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.11.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

### 14.2.11.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2 and Cell 3. All three cells transmit PRS resources on multiple positioning frequency layers during T2.

The *NR-DL-TDOA-ProvideAssistanceData* and *nr-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.10, shall be provided to the UE during T1. The UE is capable of performing RSTD measurements by aggregating PRS resources from multiple PFLs and is configured by the LMF to perform measurements by aggregating the PRS resources from multiple positioning frequency layers via *jointMeasurementsReq*. The resources to be aggregated for RSTD measurement are indicated by the LMF to the UE via *nr-DL-PRS-JointMeasurementRequestedPFL-List*.

The last TTI containing the two messages shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the DL-TDOA assistance data and location information request.

The beginning of the time interval T2 shall be aligned with the beginning of the first measurement gap instance containing the PRS resources.

The UE is configured with measurement gap pattern ID # 24 or measurement gap pattern ID # 0 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.2.11.5-1, Table 14.2.11.5-2 and Table 14.2.11.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *nr-DL-PRS-JointMeasurementRequestedPFL-List* in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 14.2.11.5-3.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.4.4). The UE shall perform and report the RSTD measurements by aggregating PRS resources from multiple PFLs for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1.

    If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.
11. If the UE message at step 10 includes the ackRequested IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. Repeat steps 2-11 in Tables 14.2.11.4-1 until the confidence level according to Annex D is achieved.

### 14.2.11.4.3 Message contents

**Table 14.2.11.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.2.11.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.2.11.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.2.11.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRP-Pair-r16 | Not present | | |

| | | | |
|---|---|---|---|
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.11.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.2.11.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | | | |
| assistanceDataValidityArea-r17 | | | |

| | | | |
|---|---|---|---|
| nr-PeriodicAssistData-r18 | | | |
| } | | | |
| bt-ProvideAssistanceData-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.11.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.11.4.3-6 | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.11.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.11.4.3-6 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.11.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.11.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.11.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 3 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[3] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 3 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.11.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-Assistance DataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-Assistance DataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-Assistance DataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-Assistance DataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-Assistance DataPerTRP-r16[3] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | NR Cell 1 |
| | 0 | | NR Cell 2 |
| | 1 | | NR Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.11.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not Present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not Present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not Present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.11.5 Test requirement

Table 14.2.11.5-1 to Table 14.2.11.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.11.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first place in the respective neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ per carrier | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |
| | Config 2 | | CCR.1.1 TDD | |
| | Config 3 | | CCR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Active DL BWP configuration | Config 1,2,3 | | DLBWP.1.1 | |
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration per PFL | Config 1 | | PRS.1.2 FR1 | As specified in TS 38.133 [50] clause A.3.31. |
| | Config 2 | | PRS.1.2 FR1 | |
| | Config 3 | | PRS.2.2 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2) mod 6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3) mod 6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#0 | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | μs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |

| | | | |
|---|---|---|---|
| Expected RSTD | μs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expected RSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | μs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the *expectedRSTD-Uncertainty* index |
| Number of cells provided in DL-TDOA assistance data | | 3 | Including the reference cell |
| PRS muting info | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Correponds to *NR-MutingPattern* defined in TS 37.355 [49] |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |

**Table 14.2.11.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1×2 Low | 1×2 Low | 1×2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| EPRE ratio of PSS to SSS | | dB | 0 | N/A | N/A |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -66.96 |
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | --66.96 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -62.97 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |

| | | | | | |
|---|---|---|---|---|---|
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |
| NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.<br>NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.<br>NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>NOTE 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. | | | | | |

**Table 14.2.11.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number Note 5 | | | 1,2 | 1,2 | 1,2 |
| Positiong frequency layer (PFL) | | | 1,2 | 1,2 | 1,2 |
| Correlation Matrix and Antenna Configuration | | | 1×2 Low | 1×2 Low | 1×2 Low |
| OCNG patterns defined in clause A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5 | -11.17 | -11.17 |
| | Config 2 | dB | -5 | -11.17 | -11.17 |
| | Config 3 | dB | -5 | -11.17 | -11.17 |
| Io Note 4 | Config 1 | dBm/ 19.08MHz per PFL | -65.28 | -65.28 | -65.28 |
| | Config 2 | dBm/ 19.08MHz per PFL | -65.28 | -65.28 | -65.28 |
| | Config 3 | dBm/ 47.88MHz per PFL | -58.29 | -58.29 | -58.29 |

| | | | | |
|---|---|---|---|---|
| PRS $\hat{E}_s/I_{ot}$ | dB | -5.62 | -12.61 | -12.61 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated, and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.
Note 5: Carriers are intra-band and contiguous.

The RSTD measurement time fulfils the requirements specified in clause 14.2.11.3.

The UE shall perform and report the RSTD measurements by aggregating PRS resources from multiple positioning frequency layers for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in TS 38.133 [50] section 9.9.2.10 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133 [50] clause 10.1.23A.3, i.e., between RSTD_000000000 and RSTD_126083073.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE is derived from the UE RSTD measurement period plus ΔT, where ΔT = 50 ms. The RSTD measurement period follows the equation:

$$T_{RSTD_aggregate,i} = \left( k_{multiTEG_agg,i} \times CSSF_{PRS_agg,i} \times ceil(K_{p,PRS_agg,i}) \times N_{RxBeam,i} \times \left\lceil \frac{N^{slot}_{PRS_agg,i}}{N'_{agg}} \right\rceil \left\lceil \frac{L_{available_PRS_agg,i}}{N_{agg}} \right\rceil \times N_{sample_agg} - 1 \right) \times T_{effect_agg,i} + T_{last_agg,i}$$

Where $k_{multiTEG_agg,i} = 1$, $CSSF_{PRS_agg,i} = 1$, $K_{p,PRS_agg,i} = 1$, $N_{RxBeam,i} = 1$, $N^{slot}_{PRS_agg,i} = 1$, $L_{avavailable_PRS_agg,,i} = 0.857$ for 15kHz and 0.429 for 30kHz, $N_{sample_agg} = 4$. $N'_{agg}$ is the parameter maxNumOfAggregatedDL-PRS-ResourcePerSlot from TS 37.355 [49], Nagg is the parameter prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN from TS 37.355 [49], $T_{last_agg,i} = Tagg,i + T_{available_PRS_agg,i}$ and $T_{effect_agg,i} = \left\lceil \frac{T_{agg,i}}{T_{available_PRS_agg,i}} \right\rceil \times T_{available_PRS_agg,i}$.

$T_{available_PRS_agg,i} = LCM(T_{PRS_agg,i}, MGRP_i)$; where $T_{PRS_agg,i}$ = 320 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS_agg,i} = 320$ ms.

Finally, it results in the following equation:

$$T_{RSTD_aggregate,i} = \left( 1 \times 1 \times 1 \times 1 \times \left\lceil \frac{1}{N'_{agg}} \right\rceil \left\lceil \frac{L_{available_PRS_agg,i}}{N_{agg}} \right\rceil \times 4 - 1 \right) \times T_{effect_agg,i} + T_{last_agg,i}$$

where the remaining parameters depend on the UE capabilities.

Depending on the UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]:

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 2.896s and 73.6s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.11.4.3-3. The LPP time IE ranges between 3s and 74s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 3.3s and 74.3s.

Similarly for Type 2 Buffering, Best Case result of 2.896s is 3s and for the worst-case result of 83.84s is 84s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 3.3s and 84.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 2.896s and 42.88s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.11.4.3-3. The LPP time IE ranges between 3s and 43s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 3.3s and 43.3s.

Similarly for Type 2 Buffering, Best Case result of 2.896s is 3s and for the worst-case result of 42.88s is 43s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 3.3s and 43.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.2.11.5-4, Table 14.2.11.5-5 and Table 14.2.11.5-6 for reference.

**Table 14.2.11.5-4: value of N' based on *maxNumOfAggregatedDL-PRS-ResourcePerSlot***

| $N'_{agg}$ | $\left\lceil \frac{1}{N'_{agg}} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 14.2.11.5-5: value of N based on *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN***

| **$N_{agg}$ for SCS 15kHz *for Type 1 Buffering*** Note 1 | $\left\lceil \frac{0.857}{N} \right\rceil$ |
|---|---|
| msDot125 | 7 |
| msDot25 | 4 |
| msDot5 | 2 |
| >=ms1 | 1 |
| **$N_{agg}$ for SCS 30kHz *for Type 1 Buffering*** Note 1 | $\left\lceil \frac{0.429}{N} \right\rceil$ |
| msDot125 | 4 |
| msDot25 | 2 |
| >=msDot5 | 1 |
| **$N_{agg}$ for SCS 15kHz *for Type 2 Buffering*** Note 1 | $\left\lceil \frac{1}{N} \right\rceil$ |
| msDot125 | 8 |
| msDot25 | 4 |
| msDot5 | 2 |
| >=ms1 | 1 |
| **$N_{agg}$ for SCS 30kHz *for Type 1 Buffering*** Note 1 | $\left\lceil \frac{0.5}{N} \right\rceil$ |
| msDot125 | 4 |
| msDot25 | 2 |
| >=msDot5 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26], | |

**Table 14.2.11.5-6: value of $T_{effect_agg,i}$ and $T_{last_agg,i}$ based on $T_{agg,i}$**

| $T_{agg,i}$ | $T_{effect_agg,i}$ | $T_{last_agg,i}$ |
|---|---|---|
| ms8 | 320 | 328 |
| ms16 | 320 | 336 |
| ms20 | 320 | 340 |
| ms30 | 320 | 350 |
| ms40 | 320 | 360 |
| ms80 | 320 | 400 |
| ms160 | 320 | 480 |
| ms320 | 320 | 640 |

| | | |
|---|---|---|
| ms640 | 640 | 960 |
| ms1280 | 1280 | 1600 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.2.12 NR RSTD measurement reporting delay test case for PRS aggregation in FR2

Editor's Note: This test case is incomplete. Following aspects are either missing or FFS: Test requirement is incomplete and MU/TT analysis is missing

### 14.2.12.1 Test purpose

The purpose of the test is to verify that the RSTD measurement performed by UE by aggregating PRS resources from multiple positioning frequency layers (PFLs) meets the requirements specified in TS 38.133 [50] clause 9.9.2.10 in an environment with AWGN propagation conditions in FR2 in standalone scenario when multiple positioning frequency layer is configured.

### 14.2.12.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Connected-r18.*

### 14.2.12.3 Minimum conformance requirements

Same as in clause 14.2.11.3.

### 14.2.12.4 Test description

#### 14.2.12.4.1 Initial conditions

**Table 14.2.12.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth per PFL, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.12.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.12.5-1, Table 14.2.12.5-2 and Table 14.2.12.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.12.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

### 14.2.12.4.2 Test procedure

Same test procedure as defined in clause 14.2.11.4.2 with following exceptions:

Table 14.2.12.4-1 replaces Table 14.2.11.4-1; Table 14.2.12.5-1 replaces 14.2.11.5-1; Table 14.2.12.5-2 repalces Table 14.2.11.5-2; Table 14.2.12.5-3 replaces Table 14.2.11.5-3;

### 14.2.12.4.3 Message contents

Same message contents as defined in clause 14.2.11.4.3, with the following exceptions:

Table 14.2.12.4.3-1 replaces Table 14.2.11.4.3-5; Table 14.2.12.4.3-2 replacesTable 14.2.11.4.3-6.

**Table 14.2.12.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.12.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.12.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.12.4.3-2 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.12.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.12.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.2.12.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 3 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[3] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 3 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.2.12.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-Assistance |

| | | | |
|---|---|---|---|
| | | | DataPerTRP-r16[3] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | NR Cell 1 |
| | 0 | | NR Cell 2 |
| | 1 | | NR Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.2.12.5 Test requirement

Table 14.2.12.5-1 to Table 14.2.12.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.12.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.3 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |
| PRS Configuration per PFL | Config 1 | | PRS.1.1 FR2 | As specified in TS 38.133 [50] clause A.3.31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |

| | | | |
|---|---|---|---|
| Measurement gap | | GP#24 or GP#13 | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | 3 | Including the reference cell |
| PRS muting info | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Correponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | Rough | Information about types of UE beam is given in clause TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.2.12.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| NR RF Channel Number | | 1 | 1 | 1 |
| Positiong frequency layer | | 1 | 1 | 1 |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | 200: $N_{RB,c}$ = 132 | 200: $N_{RB,c}$ = 132 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | OP.1 | N/A | N/A |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |

| | | | | | |
|---|---|---|---|---|---|
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -53.99 | -53.99 | -53.99 |
| SSB_RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | Config 1 | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: SSB_RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.12.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| | | T2 | T2 | T2 |
| RF Channel Number NOTE 6 | | 1,2 | 1,2 | 1,2 |
| Positioning frequency layer (PFL) | | 1,2 | 1,2 | 1,2 |
| $BW_{channel}$ per PFL | MHz | 200: $N_{RB,c}$ = 132 | 200: $N_{RB,c}$ = 132 | 200: $N_{RB,c}$ = 132 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | OP.1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |

| | | | | | |
|---|---|---|---|---|---|
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 + TT | -89 + TT | -89 + TT |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| Io Note4 | Config 1 | dBm/ 190.08MHz per PFL | -55.48 + TT | -55.48 + TT | -55.48 + TT |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -6 + TT | -13 + TT | -13 + TT |
| PRP Note 4 | | dBm/SCS | -94.45 + TT | -100.67 + TT | -100.67 + TT |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: PRP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.

Note 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] table 6.2.1.3-4.

Note 6: Carriers are intra-band and contiguous to each other.

The RSTD measurement time fulfils the requirements specified in 14.2.12.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 9.9.2.10 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2\times TTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133 [50] clause 10.1.23A.2, i.e., between RSTD_000000000 and RSTD_126083073.

## 14.2.13 NR RSTD measurement reporting delay test case in FR1 for Redcap UE without Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.2.13.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC CONNECTED state meets the requirements specified in TS 38.133 clause 9.9A.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.2.13.2 Test applicability

This test applies to RedCap UE release 18 onwards that support DL-TDOA positioning.

### 14.2.13.3 Minimum conformance requirements

The requirements in this clause shall apply provided the RedCap UE has received *NR-DL-TDOA-RequestLocationInformation* message from LMF via LPP requesting the RedCap UE to measure and report DL RSTD measurements defined in TS 38.215 [57].

The requirements in this clause shall apply provided the RedCap UE does not support, or support but not configured, to measure PRS resources with Frequency Hopping (FH), subject to UE capability indicated via *NR-DL-AoD-ProvideCapabilities*.

The requirements in this clause apply for periodic and triggered RSTD measurements without FH, provided:

- PRS-RSTD related side conditions for 2 Rx RedCap UE given in TS 38.133 [50] clause 10.1.23.2 for FR1 and FR2 are fulfilled, for a corresponding band, or
- PRS-RSTD related side conditions for 1 Rx RedCap UE given in TS 38.133 [50] clause 10.1A.16.2.1 for FR1 are fulfilled, for a corresponding band.

RedCap UE PRS RSTD measurement capability is as indicated by the RedCap UE in *NR-DL-TDOA-ProvideCapabilities*, according to TS 37.355 [49].

The requirements in TS 38.133 [50] clause 9.9.2.4 shall apply except for RSTD measurement accuracy. The RSTD measurements performed and reported according to this clause shall meet the RSTD measurement accuracy requirements in TS 38.133 [50] clause 10.1A.16.2 for each measured DL PRS resource.

When physical layer receives last of *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from LMF via LPP [34], the RedCap UE shall be able to measure multiple (up to the RedCap UE capability specified in clause TS 38.133 [50] 9.9A.2.3) DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined as:

$$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD,i} + (L-1) * \max\left(T_{effect,i}\right),$$

where

$i$ is the index of positioning frequency layer,

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$T_{RSTD,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer $i$ as specified below:

$T_{RSTD,i} = \left(k_{multiTEG,i} * CSSF_{PRS_RedCap,i} * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$ ,

where:

$N_{RxBeam,i}$ is the RedCap UE Rx beam sweeping factor. In FR1 (for 2 Rx or 1 Rx RedCap UE), $N_{RxBeam,i}$ = 1;

and in FR2 (for 2 Rx RedCap UE only), $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$CSSF_{PRS_RedCap,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in positioning frequency layer $i$ as defined in TS 38.133 [50] clause 9.1A.5.2.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in *NR-DL-TDOA-RequestLocationInformation*;

otherwise,

$k_{multiTEG}$=$N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}$=$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

- For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:
  - $N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and
  - $N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule
  - Requirements do not apply if $N_{available}$ =0.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED*, and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED*, and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the

sampling time and processing time. If all of the PRS resources to be measured are available in the same MG occasion during $T_{availabe}$, $T_{last,i} = T_i$ +MGL. Otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$ ,

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$.

$MGRP_i$ is the repetition periodicity of the measurement gap applicable for measurement in the PRS frequency layer i.$T_{PRS,i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,i}$, where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where

$T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- Note: For the purpose of calculating $T_{PRS,i}$, only the PRS resources fully or partially covered by the MG are considered.

$\{N, T\}$ is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49].

$N'$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [49].

Except for deferred MT-LR as defined in clause 4.1a.5 TS 23.273 [44], the time $T_{RSTD,Total}$ starts from the first MG instance aligned with a DL PRS resource(s) in the assistance data after both the *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the physical layer of UE via LPP [34].

For deferred MT-LR with other event than "Periodic Location" as defined in clause 4.1a.5.1 TS 23.273 [61], the time $T_{RSTD,Total}$ starts from the first MG instance aligned with a DL PRS resource(s) in the assistance data after the associated event(s) occurs.

For deferred MT-LR with event "Periodic Location" as defined in clause 4.1a.5.1 TS 23.273 [61], the UE shall perform the RSTD measurement in each reporting period and activate the location report at the time when the periodic timer expires.

Note: No per-positioning frequency layer requirement is applied in scenarios when multiple positioning frequency layers are configured.

When PRS-RSRP is configured for DL-TDOA, RSTD and RSRP are performed over the same measurement period.

When PRS-RSRPP is configured for DL-TDOA, RSTD and RSRPP are performed over the same measurement period.

If during the measurement period of one or more positioning frequency layers, the MG pattern is reconfigured, the measurement period can be longer.

The measurement requirements in this clause apply, provided no PRS symbols are dropped during the measurement period $T_{RSTD,Total}$ within measurement gaps due to collisions with other signals; otherwise, the measurement period can be longer.

If CSSF changes during the measurement period, the measurement period could be longer.

The measurement requirements do not apply for a PRS resource, if the PRS resource is across two sampling duration of N within duration $L_{available_PRS,i}$.

The measurement requirements do not apply for a PRS resource, if time span of the PRS resource instance (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability N.

The requirements in clause 9.9A.2 do not apply if the PRS configuration given by higher layer paramters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-TDOA-ProvideCapabilities*, and it is up to UE implementation which PRS resources are measured, subject to UE measurement capabilities.

If handover occurs while RSTD measurements are being performed, then the UE shall continue and complete the on-going RSTD measurements. The RSTD measurement period can be longer. The UE shall meet the RSTD measurement accuracy requirements in clause 10.1A.16.2.

When physical layer receives last of *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from LMF via LPP [34], the RedCap UE shall be able to measure multiple (up to the RedCap UE capability specified in clause 9.9A.2.3) DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined as:

$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{RSTD,Total} = \max(T_{RSTD_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

Where,

$i$ is the index of positioning frequency layer, and

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i, and

$T_{uncertainty,i}$ is the time from the start of the first PPW occasion for positioning frequency layer i to the start of measurement period $T_{RSTD,Total}$.

A positioning frequency layer is in Case 1 if UE reports *ppw-durationOfPRS-Processing1-r17* for the band containing the positioning frequency layer, and a positioning frequency layer is in Case 2 if UE reports *ppw-durationOfPRS-Processing2-r17* for the band containing the positioning frequency layer.

$T_{RSTD_wo_gap,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below.

$$T_{RSTD_wo_gap,i} = \left(k_{multiTEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i},$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1 (for 2 Rx or 1 Rx RedCap UE), $N_{RxBeam,i}$ = 1; and in FR2 (for 2 Rx RedCap UE only), $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not supported *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] in *NR-DL-TDOA-RequestLocationInformation*;

otherwise,

$k_{multiTEG}$=$N_{TEG,i}$, if the UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}$=$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if the UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

Where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in *NR-DL-TDOA-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$,

only the PRS resources unmuted and fully or partially overlapped with PPW are considered, if positioning frequency layer i is in Case 1, or

only the PRS resources unmuted and fully or partially overlapped with the first (PPWL-T2) ms of PPW are considered, if positioning frequency layer i is in Case 2, where PPWL is the PPW length and T2 corresponds to *ppw-durationOfPRS-ProcessingSymbolsT2*.

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$= 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* , and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and

- Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.

- $N_{sample}$=2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED*, and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:

  - PRS bandwidth is within the active BWP and

- Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.

- $N_{sample}$= 4 otherwise.

$T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time.

If positioning frequency layer *i* is in Case 1 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i} = T_i$ +PPWL, else

if positioning frequency layer *i* is in Case 2 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i}$ = PPWL;

otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$.

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where,

$T_i$ corresponds to *ppw-durationOfPRS-ProcessingSymbolsT* in TS 37.355 [] if positioning frequency layer *i* is in Case 1, or corresponds to the sum of *ppw-durationOfPRS-ProcessingSymbolsT2* and *ppw*-durationOfPRS-*ProcessingSymbolsN2* in TS 37.355 [49] if positioning frequency layer *i* is in Case 2,

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$, the least common multiple between $T_{PRS,i}$ and $PPWRP_i$.

$PPWRP_i$ is the repetition periodicity of the PRS processing window applicable for measurements in the positioning frequency layer *i*.

$T_{PRS,i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,i}$, where,

$T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*,

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

Note: For the purpose of calculating $T_{PRS,i}$, only the PRS resources that meet the conditions for PRS measurement outside measurement gaps as defined in clause 9.9.1.2 are considered.

$N$ is a duration of DL PRS symbols in ms corresponding to *ppw-durationOfPRS-ProcessingSymbolsN* in TS 37.355 [34] if positioning frequency layer i is in Case 1, or corresponding to *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer i is in Case 2.

$N'$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* specified in TS 37.355 [49].

The time $T_{RSTD_wo_gap,i}$ starts from the first instance of the activated PPW for measurement of positioning frequency layer *i* aligned with a DL PRS resource(s) in the assistance data after both the *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the physical layer of UE via LPP [34].

When PRS-RSRP is configured for DL-TDOA, RSTD and RSRP are performed over the same measurement period.

When PRS-RSRPP is configured for DL-TDOA, RSTD and RSRPP are performed over the same measurement period.

If during the measurement period of one or more positioning frequency layers, the PPW is re-configured or reactivated, the measurement period can be longer.

The measurement requirements in this clause apply, provided no PRS symbols are dropped during the measurement period $T_{RSTD_wo_gap,i}$ within PPW due to collisions with other signals; otherwise, the measurement period can be longer.

The measurement requirements do not apply for a PRS resource, if the PRS resource is across two sampling duration of N within duration $L_{available_PRS,i}$.

The measurement requirements do not apply for a PRS resource, if time span of the PRS resource instance (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability N.

The requirements in clause 9.9A.2 do not apply if the PRS configuration given by higher layer paramters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-TDOA-ProvideCapabilities*, and it is up to UE implementation which PRS resources are measured, subject to UE measurement capabilities.

If handover occurs while RSTD measurements are being performed, then the UE shall continue and complete the on-going RSTD measurements. The RSTD measurement period can be longer. The UE shall meet the RSTD measurement accuracy requirements in clause 10.1A.16.2.1.

If any positioning frequency layer is in Case 2, the requirements in this clause apply provided that the PPWL corresponding to the positioning frequency layer is larger than (T2+X) ms.

The requirements in this clause apply provided that a single positioning frequency layer is configured for measurement in each PPW.

If the RedCap UE is configured with both MG applicable to positioning measurement and PPW, the UE shall measure positioning frequency layer *i* within

- MG, if PRS resources on positioning frequency layer *i* are overlapped with MG, or
- PPW, if PRS resources on positioning frequency layer *i* are overlapped with PPW.

The measurement period $T_{RSTD,Total}$ is defined as:

$$T_{RSTD,Total} = T_{RSTD,Total,MG} + T_{RSTD,Total,PPW} + T_{guard},$$

Where

- $T_{RSTD,Total,MG}$ is defined in TS 38.133 [50] clause 9.9A.2.5.1 and includes all positioning frequency layers to be measured within MG,
- $T_{RSTD,Total,PPW}$ is defined in TS 38.133 [50] clause 9.9A.2.5.2 and includes all positioning frequency layers to be measured within PPW,
- $T_{guard}$ is the maximum $T_{effect}$ defined in TS 38.133 [50] clauses 9.9A.2.5.1 and 9.9A.2.5.2 among all PFLs.

The requirements apply provided the following conditions are satisfied:

- MGs and PPWs do not overlap in time;
- Each PFL in the assistance data can be measured completely (all PRS resources) either within MG or within activated PPW.

## 14.2.13.4 Test description

### 14.2.13.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.13.4-1.

**Table 14.2.13.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.13.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.13.5-1, Table 14.2.13.5-2 and Table 14.2.13.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.13.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

### 14.2.13.4.2 Test procedure

TBD

### 14.2.13.4.3 Message contents

TBD

## 14.2.13.5 Test requirement

Table 14.2.13.5-1, Table 14.2.13.5-2 and Table 14.2.13.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.13.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |

| | | | | |
|---|---|---|---|---|
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.1 RedCap FR1 | |
| | Config 4 | | SSB.1 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| | Config 4 | | SMTC.2 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| | Config 4 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in clause A.3.1.2 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| | Config 4 | | CR.1.1 FDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |
| | Config 2 | | CCR.1.1 TDD | |
| | Config 3 | | CCR.2.1 TDD | |
| | Config 4 | | CCR.1.1 FDD | |
| Initial BWP configuration | Config 1,2,3,4 | | DLBWP.0.1 ULBWP.0.1 | |
| Active DL BWP configuration | Config 1,2,3,4 | | DLBWP.1.1 | |
| Active UL BWP configuration | Config 1,2,3,4 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.1 FR1 | As specified in clause A.3.31 |
| | Config 2 | | PRS.1.1 FR1 | |
| | Config 3 | | PRS.2.1 FR1 | |
| | Config 4 | | PRS.1.1 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#0 | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 3 | Including the reference cell |

| | | | |
|---|---|---|---|
| PRS muting info | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Correponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |

**Table 14.2.13.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| EPRE ratio of PSS to SSS | | dB | 0 | N/A | N/A |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| | Config 4 | dBm/SCS | -98 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 9.36MHz | -59.63 | -59.63 | -59.63 |
| | Config 2 | dBm/ 9.36MHz | -59.63 | -59.63 | -59.63 |
| | Config 3 | dBm/ 18.36MHz | -56.71 | -56.71 | -56.71 |
| | Config 4 | dBm/ 9.36MHz | -59.63 | -59.63 | -59.63 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |

| | | | | | |
|---|---|---|---|---|---|
| | Config 4 | dBm/SCS | -88 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.13.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| | Config 4 | dBm/SCS | -98 | -98 | -98 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.45 | -11.67 | -11.67 |
| | Config 2 | dB | -5.45 | -11.67 | -11.67 |
| | Config 3 | dB | -5.45 | -11.67 | -11.67 |

| | | | | | |
|---|---|---|---|---|---|
| | Config 4 | dB | -5.45 | -11.67 | -11.67 |
| SSB $\hat{E}_s/N_{oc}$ | Config 1,2,3,4 | dB | 10 | 3 | 3 |
| Io Note 4 | Config 1 | dBm/ 9.36MHz | -68.52 | -68.52 | -68.52 |
| | Config 2 | dBm/ 9.36MHz | -68.52 | -68.52 | -68.52 |
| | Config 3 | dBm/ 18.36MHz | -65.61 | -65.61 | -65.61 |
| | Config 4 | dBm/ 9.36MHz | -68.52 | -68.52 | -68.52 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -95 | -95 |
| | Config 2 | dBm/SCS | -88 | -95 | -95 |
| | Config 3 | dBm/SCS | -85 | -92 | -92 |
| | Config 4 | dBm/SCS | -88 | -95 | -95 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -6.00 | -12.98 | -12.98 |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.

The RSTD measurement time without FH for RedCap fulfils the requirements specified in clause 14.2.13.2.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.2.13.2 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90% with a confidence level of 95%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133 [50] clause 10.1A.16.3, i.e., between RSTD_0000000 and RSTD1970049.

## 14.2.14 RSTD measurement reporting delay test case in FR2 for Redcap UE without Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.2.14.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC CONNECTED state meets the requirements specified in TS 38.133 [50] clause 9.9A.2 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.2.14.2 Test applicability

This test applies to RedCap UE release 18 onwards that support DL-TDOA positioning.

### 14.2.14.3 Minimum conformance requirements

Same as in clause 14.2.13.3.

### 14.2.14.4 Test description

#### 14.2.14.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.14.4-1.

**Table 14.2.14.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 100 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.14.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.14.5-1, Table 14.2.14.5-2 and Table 14.2.14.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.14.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR2.

#### 14.2.14.4.2 Test procedure

TBD

#### 14.2.14.4.3 Message contents

TBD

### 14.2.14.5 Test requirement

Table 14.2.14.5-1, Table 14.2.14.5-2 and Table 14.2.14.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.14.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.3 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.1 FR2 | As specified in clause A.3.31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#13 | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 3 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |

| | | | |
|---|---|---|---|
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | Setup 1 | As defined in clause A.3.15.1 |
| Beam assumption | | Rough | Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.2.14.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| $BW_{channel}$ | | MHz | 100: $N_{RB,c}$ = 66 | 100: $N_{RB,c}$ = 66 | 100: $N_{RB,c}$ = 66 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 95.04MHz | -57 | -57 | -57 |
| SSB_RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | Config 1 | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: SSB_RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.14.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| $BW_{channel}$ | | MHz | 100: $N_{RB,c}$ = 66 | 100: $N_{RB,c}$ = 66 | 100: $N_{RB,c}$ = 66 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | -89 | -89 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.45 | -11.67 | -11.67 |
| Io Note4 | Config 1 | dBm/ 95.04MHz | -58.49 | -58.49 | -58.49 |

| | | | | |
|---|---|---|---|---|
| PRS $\hat{E}_s/I_{ot}$ | dB | -6 | -13 | -13 |
| PRP Note 4 | dBm/SCS | -94.45 | -100.67 | -100.67 |
| Propagation Condition | | AWGN | | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: PRP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.
NOTE 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

The RSTD measurement time without FH for RedCap fulfils the requirements specified in clause 9.9A.2.5.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 9.9.2A.5 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90% with the confidence level of 95%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in clause 10.1A.16.3, i.e., between RSTD_0000000 and RSTD_1970049.

## 14.2.15 NR RSTD measurement reporting delay test case in FR1 for Redcap UE with Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.2.15.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] clause 9.9A.2.6 in an environment with AWGN propagation conditions in FR1 in standalone scenario when frequency hopping is configured.

### 14.2.15.2 Test applicability

This test applies to RedCap UE release 18 onwards that support DL-TDOA positioning.

### 14.2.15.3 Minimum conformance requirements

The requirements in this clause shall apply provided the RedCap UE has received *NR-DL-TDOA-RequestLocationInformation* message from LMF via LPP requesting the RedCap UE to measure and report DL RSTD measurements defined in TS 38.215 [57].

The requirements in TS 38.133 [50] clause 9.9A.2.5 shall apply provided the RedCap UE does not support, or support but not configured, to measure PRS resources with Frequency Hopping (FH), subject to UE capability indicated via *NR-DL-AoD-ProvideCapabilities*.

The requirements in TS 38.133 [50] clause 9.9A.2.6 shall apply provided the RedCap UE supports and is configured to measure PRS resources with Frequency Hopping (FH), subject to UE capability indicated via *NR-DL-AoD-ProvideCapabilities*.

The requirements in this clause apply for periodic and triggered RSTD measurements with FH, provided:

- PRS-RSTD related side conditions for 2 Rx RedCap UE given in TS 38.133 [50] clause 10.1A.16.2.2 for FR1 and FR2 are fulfilled, for a corresponding band. or
- PRS-RSTD related side conditions for 1 Rx RedCap UE given in TS 38.133 [50] clause 10.1A.16.2.2 for FR1 are fulfilled, for a corresponding band.

RedCap UE PRS RSTD measurement capability is as indicated by the RedCap UE in *NR-DL-TDOA-ProvideCapabilities*, according to TS 37.355[49].

The requirements in TS 38.133 [50] clause 9.9.2.4 shall apply except for RSTD measurement accuracy. The RSTD measurements performed and reported according to this clause shall meet the RSTD measurement accuracy requirements in clause 10.1A.16.2 for each measured DL PRS resource.

When UE physical layer receives last of *NR-TDOA-ProvideAssistanceData* message and *NR-TDOA-RequestLocationInformation* message from LMF via LPP, requesting RedCap UE to measure DL RSTD measurement with FH, the RedCap UE shall be able to measure multiple (up to the RedCap UE capability specified in clause 9.9A.2.3) DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined in TS 38.133 [50] clause 9.9A.2.5.1 with using the following definition for $N_{sample}$ and $L_{available_PRS,\mathrm{i}}$:

- $N_{sample}$ is the number of PRS RSTD measurement samples, where
  - $N_{sample}$ = 2 if the RedCap UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED*, and the LMF requests the UE to perform positioning measurements with reduced number of samples.
  - $N_{sample}$ = 4 otherwise.
- A measurement sample with FH is defined as a PRS measurement over multiple hops within a single measurement gap occasion.
- $L_{available_PRS,\mathrm{i}}$ is the time duration of available PRS in the positioning frequency layer i to be measured during a MG, and is calculated by:

$$L_{available_PRS,i} = \sum_{j=1}^{J} N_{hop,i,j} * L_{per-hop,i,j}$$

  where,

  - $N_{hop,i,j}$ is the number of hops that UE can do in an MG occasion *j* as defined in the following, and
  - $L_{per-hop,i,j}$ is the time duration of available PRS resources in the positioning frequency layer *i* in each hop to be measured in MG instance *j* during $T_{available_PRS,\mathrm{i}}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{per-hop,i,j}$, only the PRS resources unmuted and fully or partially overlapped with the sampling duration in each hop are considered;
  - *J* is the number of MG instance during $T_{available_PRS,\mathrm{i}}$.

The sampling window per hop is the first $T_{sample,hop}$ symbols in each hop, where $T_{sample,hop} = T_{hop} - RRT_{FH}$, $T_{hop}$ is the applicable length per hop as defined in table 14.2.15.3-1, and $RRT_{FH}$ is the retuning time between Rx hops as reported by UE in *rf-RxRetuneTimeFR1* or *rf-RxRetuneTimeFR2* via *NR-DL-PRS-ProcessingCapability*. The first hop

within a MG instance starts at no earlier than the earliest arrival time of the first unmuted PRS resource fully or partially overlapped with the MG instance taking into account the *expectedRSTD* and *expectedRSTD-uncertainty* in the PRS assistance data.

**Table 14.2.15.3-1: Applicable number of hops per slot and applicable length of each hop**

| $RRT_{FH}$ | (comb size, Number of PRS symbols) | Applicable number of hops per slot $(N_{hops}^{slot})$ | Applicable length per hop ($T_{hop}$) in number of symbols |
|---|---|---|---|
| $RRT_{FH} \leq 2$ symbols | (2, 12) with SCS 15 kHz, 30 kHz, 60 kHz in FR2, 120 kHz | 2 | 7 |
| | All others | 1 | 14 |
| 2 symbols $< RRT_{FH}$ $\leq 6$ symbols | (≤ 6, any) | 1 | 14 |
| | (12, 12) | ½ | 28 |
| $RRT_{FH} > 6$ symbols | Any combination | ½ | 28 |

The number of hops within a single MG occasion $N_{hop,i,j}$ is defined as

$$N_{hop,i,j} = min\left(N_{hops,effect,i,j}, \left\lfloor \frac{N3_i}{L_{per-hop,i,j}} \right\rfloor, N_{hop,max,i}\right)$$

where

- $N_{hop,max,i}$ is the maximum number of Rx hops signaled by UE in *maximumFH-Hops* via *NR-DL-PRS-ProcessingCapability* as specified in for positioning frequency layer *i*
- $N3_i$ is indicated via UE capability *processingPRS-SymbolsDurationN3-r18* for positioning frequency layer *i*
- $N_{hops,effect,i,j}$ is the effective number of Rx hops within a MG instance *j*,
- $N_{hops,effect,i,j} = 2 * N_{rep}^{PRS}$, if $N_{hops}^{slot} = 2$,
- $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot} = 1$,
- $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot} = 1/2$ and $M_{rep}^{PRS} > 1$,
- $N_{hops,effect,i,j} = \left\lfloor \frac{(N_{rep}^{PRS}-1)}{2} \right\rfloor + 1$, if $N_{hops}^{slot} = 1/2$ and $M_{rep}^{PRS} = 1$,
  - where $N_{rep}^{PRS}$ is the number of inter-slot PRS repetitions within the MG occasion, excluding the gap retuning times, which are unmuted and fully or partially overlapped with the sampling duration within each hop. $M_{rep}^{PRS}$ is the PRS repetition interval configured by *dl-PRS-ResourceTimeGap*, $N_{hops}^{slot}$ is the applicable number of hops per slot as defined in table 14.2.15.3-1.

The measurement period requirement with the above modifications applies provided all PRS resources in each PFL have the same number of PRS repetitions $(N_{rep}^{PRS})$ within an MG instance and the same PRS repetition interval $(M_{rep}^{PRS})$, otherwise the measurement period can be longer.

UE shall be able to measure PRS resources with multiple hops at least over the BW of $BW_{total}$ defined as

$$BW_{total} = min\left(BW_{PRS}, N_{hop} * BW_{per-hop} - \left(N_{hop} - 1\right) * BW_{overlap}\right)$$

where

$BW_{PRS}$ is the minimum among

- configured PRS BW, and
- UE capability of maximum PRS BW across all hops indicated via *maximumPRS-BandwidthAcrossAllHopsFR1-r18* or *maximumPRS-BandwidthAcrossAllHopsFR2-r18*, and
- total BW of all hops requested by LMF via *nr-DL-PRS-RxHoppingTotalBandwidth-r18.*

$N_{hop,i,j}$ is number of hops within a single MG occasion as define above,

$BW_{per-hop}$ is the UE capability on PRS BW per hop indicated via *supportedBandwidthPRS-r16*,

$BW_{overlap}$ $BW_{overlap}$ is the UE capability on BW of the overlapping RB indicated via *numOfOverlappingPRB-r18*.

## 14.2.15.4 Test description

### 14.2.15.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.15.4-1.

**Table 14.2.15.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, cell bandwidth 50 MHz, FDD duplex mode |
| 2 | 15 kHz SSB SCS, cell bandwidth 50 MHz, TDD duplex mode |
| 3 | 30 kHz SSB SCS, cell bandwidth 100 MHz, TDD duplex mode |
| 4 | 15 kHz SSB SCS, cell bandwidth 50 MHz, HD-FDD duplex mode |
| NOTE 1: The UE is only required to be tested in one of the supported test configurations.<br>NOTE 2: UE with 1 Rx or 2 Rx is required to meet the same requirements specified in this clause.<br>NOTE 3: This test case is applicable to the UE supporting per hop bandwidth ≥ 10 MHz for SCS 15 kHz.<br>NOTE 4: This test case is applicable to the UE supporting per hop bandwidth = 20 MHz for SCS 30 kHz. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.15.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.2.15.5-1, Table 14.2.15.5-2 and Table 14.2.15.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.2.15.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

### 14.2.15.4.2 Test procedure

TBD

### 14.2.15.4.3 Message contents

TBD

## 14.2.15.5 Test requirement

Table 14.2.15.5-1, Table 14.2.15.5-2 and Table 14.2.15.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.15.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| CD-SSB configuration | Config 1, 2, 4 | | SSB.1 FR1 | |
| | Config 3 | | SSB.1 RedCap FR1 | |
| NCD-SSB configuration | Config 1, 2, 4 | | SSB.6 RedCap FR1 | |
| | Config 3 | | SSB.7 RedCap FR1 | |
| SMTC configuration | Config 1, 4 | | SMTC.4 RedCap | |
| | Config 2 | | SMTC.2 RedCap | |
| | Config 3 | | SMTC.2 RedCap | |
| PRS Configuration | Config 1, 4 | | PRS.1.5 FR1 | PRS configured with frequency hopping as specified in clause A.3.31 |
| | Config 2 | | PRS.1.5 FR1 | |
| | Config 3 | | PRS.2.6 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#0 | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 3 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |

| | | | |
|---|---|---|---|
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |

**Table 14.2.15.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| PDSCH RMC configuration | Config 1, 4 | | SR.1.1 FDD | N/A | N/A |
| | Config 2 | | SR.1.1 TDD | | |
| | Config 3 | | SR.2.1 TDD | | |
| RMSI CORESET RMC configuration | Config 1, 4 | | CR.1.1 FDD | N/A | N/A |
| | Config 2 | | CR.1.1 TDD | | |
| | Config 3 | | CR.2.1 TDD | | |
| Dedicated CORESET RMC configuration | Config 1, 4 | | CCR.1.1 FDD | N/A | N/A |
| | Config 2 | | CCR.1.1 TDD | | |
| | Config 3 | | CCR.2.1 TDD | | |
| OCNG Patterns | Config 1,2,3,4 | | OP.1 | OP.1 | OP.1 |
| RMSI CORESET RMC configuration | Config 1, 4 | | CR.1.1 FDD | N/A | N/A |
| | Config 2 | | CR.1.1 TDD | | |
| | Config 3 | | CR.2.1 TDD | | |
| Initial BWP configuration | Config 1,2,3,4 | | DLBWP.0.1 ULBWP.0.1 | N/A | N/A |
| Active DL BWP configuration | Config 1,2,3,4 | | DLBWP.1.3 RedCap | N/A | N/A |
| Active UL BWP configuration | Config 1,2,3,4 | | ULBWP.1.3 RedCap | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1, 4 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1, 4 | dBm/ 48.6MHz | -52.48 | -52.48 | -52.48 |
| | Config 2 | dBm/ 48.6MHz | -52.48 | -52.48 | -52.48 |
| | Config 3 | dBm/ 98.28MHz | -49.43 | -49.43 | -49.43 |
| SSB RP Note4 | Config 1, 4 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.15.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1, 4 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1, 4 | dB | -5.45 | -11.67 | -11.67 |
| | Config 2 | dB | -5.45 | -11.67 | -11.67 |
| | Config 3 | dB | -5.45 | -11.67 | -11.67 |
| SSB $\hat{E}_s/N_{oc}$ | Config 1,2,3,4 | dB | 10 | 3 | 3 |
| Io Note 4 | Config 1, 4 | dBm/ 48.6MHz | -61.37 | -61.37 | -61.37 |
| | Config 2 | dBm/ 48.6MHz | -61.37 | -61.37 | -61.37 |
| | Config 3 | dBm/ 98.28MHz | -61.32 | -61.32 | -61.32 |
| SSB RP Note4 | Config 1, 4 | dBm/SCS | -88 | -95 | -95 |
| | Config 2 | dBm/SCS | -88 | -95 | -95 |
| | Config 3 | dBm/SCS | -85 | -92 | -92 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -6.00 | -12.98 | -12.98 |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.

## 14.2.16 NR RSTD measurement reporting delay test case in FR2 for Redcap UE with Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.2.16.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the TBD.

### 14.2.16.2 Test applicability

This test applies to all types of NR UE release 18 onwards that support TBD.

### 14.2.16.3 Minimum conformance requirements

Same as in 14.2.15.3.

### 14.2.16.4 Test description

#### 14.2.16.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.10.4-1.

**Table 14.2.16.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, cell bandwidth 400 MHz, TDD duplex mode |
| NOTE 1: This test case is applicable to the UE supporting per hop bandwidth = 100 MHz for SCS 120 kHz. | |

#### 14.2.16.4.2 Test procedure

TBD

#### 14.2.16.4.3 Message contents

TBD

### 14.2.16.5 Test requirement

Table 14.2.16.5-1, Table 14.2.16.5-2 and Table 14.2.16.5-3 define the primary level settings including test tolerances for the test.

**Table 14.2.16.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.1 RedCap FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 RedCap | |
| PDSCH RMC configuration | Config 1 | | SR.3.2 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.3.1 TDD | |
| PRS Configuration | Config 1 | | PRS.1.6 FR2 | PRS configured with frequency hopping as specified in clause A.3.31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | | OFF | |
| Measurement gap | | | GP#24 or GP#13 | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 3 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |

| | | | |
|---|---|---|---|
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | Setup 1 | As defined in clause A.3.15.1 |
| Beam assumption | | Rough | Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.2.16.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| $BW_{channel}$ | | MHz | 400: $N_{RB,c}$ = 264 | 400: $N_{RB,c}$ = 264 | 400: $N_{RB,c}$ = 264 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 380.16MHz | -53.99 | -53.99 | -53.99 |
| SSB_RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | Config 1 | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: SSB_RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.2.16.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| | | T2 | T2 | T2 |
| RF Channel Number | | 1 | 1 | 1 |
| Positiong frequency layer | | 1 | 1 | 1 |
| $BW_{channel}$ | MHz | 400: $N_{RB,c}$ = 264 | 400: $N_{RB,c}$ = 264 | 400: $N_{RB,c}$ = 264 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |

| | | | | | |
|---|---|---|---|---|---|
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | -89 | -89 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.44 | -11.67 | -11.67 |
| Io Note4 | Config 1 | dBm/ 380.16MHz | -52.68 | -52.68 | -52.68 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | dB | -6 | -13 |
| PRP Note 4 | | dBm/SCS | dBm/SCS | -94.44 | -100.67 |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.

NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 4: PRP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters. The Io is calculated based only on the symbols in which PRS is transmitted.

NOTE 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

The RSTD measurement time fulfils the requirements specified in clause 9.9A.2.6.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 9.9A.2.6 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90% with the confidence level of 95%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in clause 10.1A.16.3, i.e., between RSTD_0000000 and RSTD_1970049.

# 14.3 NR RSTD measurement accuracy test case in RRC_CONNECTED

## 14.3.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA

### 14.3.1.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions.

### 14.3.1.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

### 14.3.1.3 Minimum conformance requirements

The RSTD measurement reported by the UE shall fulfil the accuracy requirements defined by ±(X+Y+Z) $T_c$.

X is defined in Table 14.3.1.3-1 for AWGN channel, provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [54] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex I.2 for a corresponding Band for each relevant PRS resource configured for measurement.

The RSTD measurement reported by the UE shall fulfil the accuracy requirements defined in Table 14.3.1.3-2 for AWGN channel, provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex I.2 for a corresponding Band for each relevant PRS resource configured for measurement.

When UE measures RSTD on PRS resources belonging to different PFLs, then the RSTD accuracy is defined as the accuracy corresponding to the largest accuracy value among different PFLs.

When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 160 ms.

When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 1280 ms

Z is defined in Table 14.3.1.3-2 a for FR1.

**Table 14.3.1.3-1: RSTD absolute accuracy in FR1 for AWGN channel**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | $dBm/BW_{Channel}$ |
| 132 + Δ Note 7 | (PRS Ês/Iot)$_{ref}$ ≥ -6dB | 15 | ≥24 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 98 + Δ | (PRS Ês/Iot)$_i$ ≥ -13dB | | ≥52 | ≥1 | Note 6 | Note 6 | Note 6 |
| 42 + Δ | | | ≥104 | ≥1 | Note 6 | Note 6 | Note 6 |
| 75 + Δ | | 30 | ≥24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -118 | -50 |
| | | | | | NR_FDD_FR1_B | -117.5 | -50 |
| | | | | | NR_TDD_FR1_C | -117 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -116.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -116 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_FDD_FR1_F | -115.5 | -50 |
| | | | | | NR_FDD_FR1_G | -115 | -50 |
| | | | | | NR_FDD_FR1_H | -114.5 | -50 |
| 48 + Δ | | | ≥ 48 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 50 + Δ | | 60 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -115 | -50 |
| | | | | | NR_FDD_FR1_B | -114.5 | -50 |
| | | | | | NR_TDD_FR1_C | -114 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -113.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -113 | -50 |
| | | | | | NR_FDD_FR1_F | -113.5 | -50 |
| | | | | | NR_FDD_FR1_G | -113 | -50 |
| | | | | | NR_FDD_FR1_H | -111.5 | -50 |
| 24 + Δ | | | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 10 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Δ= 0 for single PFL, Δ is defined in Table 14.3.1.3-3 for dual PFL.

**Table 14.3.1.3-2: Void**

**Table 14.3.1.3-2a: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 24 | N/A | N/A | 120 |
| ≥ 52 | ≥ 24 | N/A | 72 |
| ≥ 104 | ≥ 48 | ≥ 24 | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

**Table 14.3.1.3-3: Margin Δ for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 24 | N/A | N/A | 128 |
| ≥ 52 | ≥ 24 | N/A | 64 |
| ≥ 104 | ≥ 48 | ≥ 24 | 32 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 8 |

### 14.3.1.4 Test description

#### 14.3.1.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 14.3.1.4.1-1.

**Table 14.3.1.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 14.3.1.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.1.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the RSTD reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR1. GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.3.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.3.1.4.3 shall be provided to the UE during the set-up period. The last TTI containing the NR-DL-TDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms is the maximum processing time of the NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data in the UE.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On* according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.3.1.5-1 as appropriate. Propagation conditions are set according to clause 4.15.2.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.

6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-TDOA-ProvideAssistanceData-r16 IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-TDOA-RequestLocationInformation-r16 IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the nr-DL-TDOA-ProvideLocationInformation-r16 IE.
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.3.1.5-2.
11. Repeat step 2-10 until the confidence level according to Annex D is achieved.
12. Repeat step 1-11 for the other sub-test defined in Table 14.3.1.4-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.3.1.4.3 Message contents

**Table 14.3.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.1.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.3.1.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 | Not present | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.1.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.1.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.1.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Sub-test 1-1, Sub-test 1-2, Sub test 2-1 and Sub-test 2-2 |
| | kHz30 | | Sub-test 3-1 and Sub-test 3-2 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |

| | | | |
|---|---|---|---|
| | 21 | 104 PRBs | Sub-test 1-2 and Sub-test 2-2 |
| | 28 | 132 PRBs | Sub-test 3-2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | n4 | | Sub-test 1-2, Sub-test 2-2 and Sub-test 3-2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.1.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.1.4.3-6: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Sub-tests 1-1, Sub-test 1-2, Sub-test 2-1 and Sub-test 2-2 |
| N320-r16 | 20 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Sub-tests 3-1 and Sub–test 3-2 |
| n160-r16 | 10 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| | n2 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| n4-r16 | 0 | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1-1 Cell 1, Sub-test 1-2 Cell 1, Sub-test 2-1 Cell 1, Sub-test 2-2 Cell 1, Sub-test 3-1 Cell 1,and Sub-test 3-2 Cell 1 |

| | | | |
|---|---|---|---|
| | 2 | | Sub-test 1-1 Cell 2, Sub-test 1-2 Cell 2, Sub-test 2-1 Cell 2, Sub-test 2-2 Cell 2, Sub-test 3-1 Cell 2, and Sub-test 3-2 Cell 2. |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.1.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.1.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.1.3.

**Table 14.3.1.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | 1~3 | | freq1 | Freq1 | freq1 | Freq1 |
| $BW_{channel}$ | 1 | MHz | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 2 | | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 3 | | 50: $N_{RB,c}$ = 133 | | 50: $N_{RB,c}$ = 133 | |
| Duplex mode | 1 | | FDD | | FDD | |
| | 2 | | TDD | | TDD | |
| | 3 | | TDD | | TDD | |
| TDD configuration | 1 | | N/A | | N/A | |
| | 2 | | TDDConf.1.1 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | | TDDConf.2.1 | |
| Measurement gap | 1, 2, 3 | | GP#24 or GP#0 | | GP#24 or GP#0 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | 3 | | SR.2.1 FDD | | SR.2.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - | CR.1.1 TDD | - |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 3 | | CR.2.1 FDD | - | CR.2.1 FDD | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - | CCR.2.1 TDD | - |
| SSB configuration | 1 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 3 | | SSB.2 FR1 | | SSB.2 FR1 | |
| OCNG Patterns | 1~3 | | OP.1 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - | TRS.1.1 FDD | |
| | 2 | | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP Configuration | 1~3 | | DLBWP.0.1 ULBWP.0.1 | | DLBWP.0.1 ULBWP.0.1 | |
| Dedicated BWP configuration | 1~3 | | DLBWP.1.1 ULBWP.1.1 | | DLBWP.1.1 ULBWP.1.1 | |
| Time offset with Cell 1 | 1 | µs | - | 3 | - | 3 |
| | 2,3 | | - | 3 | - | 3 |
| SMTC configuration | 1 | | SMTC.2 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.1 FR1 | | PRS.1.2 FR1 | |
| | 2 | | PRS.1.1 FR1 | | PRS.1.2 FR1 | |
| | 3 | | PRS.2.1 FR1 | | PRS.2.2 FR1 | |
| PRS Resource slot offset | slot | 1, 2, 3 | 0 | 2 | 0 | 2 |
| Expected RSTD | 1, 2, 3 | µs | N/A | | 3 | |
| Expected RSTD uncertainty | 1, 2, 3 | µs | N/A | | 5 | |
| EPRE ratio of PSS to SSS | 1~3 | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | 1~3 | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | 1,2 | dBm/ SCS | -98 | | -98 | |
| | 3 | | -95 | | -95 | |
| PRS $\hat{E}_s / I_{ot}$ | 1~3 | dB | -5.7 | -12.6 | -5.7 | -12.6 |
| PRP Note3 | 1,2 | dBm/SCS | -103.4 | -109.5 | -103.4 | -109.5 |
| | 3 | | -100.4 | -106.5 | -100.4 | -106.5 |
| Io Note3 | 1,2 | dBm/ 19.08MHz | -65.62 | -65.62 | -65.62 | -65.62 |
| | 3 | dBm/ 47.88MHz | -61.63 | -61.63 | -61.63 | -61.63 |
| PRS $\hat{E}_s / N_{oc}$ | 1~3 | dB | -5.4 | -11.5 | -5.4 | -11.5 |
| SSB $\hat{E}_s / N_{oc}$ | 1~3 | dB | -5.4 | -11.5 | -5.4 | -11.5 |
| Propagation condition | 1~3 | - | AWGN | | AWGN | |
| Antenna configuration | 1~3 | | 1x2 | | 1x2 | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols where PRS is transmitted. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification. |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.2.1.5, with the following difference: $N_{PRS,i}^{slot} = 2$. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(\left\lceil\frac{2}{N'}\right\rceil\left\lceil\frac{L_{available_{PRS,i}}}{N}\right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.1.4.3-3).

**Table 14.3.1.5-2: RSTD accuracy requirements for the reported values**

| Test Configuration | Subtest | Lowest reported value [$T_c$] | Highest reported value [$T_c$] |
|---|---|---|---|
| 1 & 2 | Test 1 | 5582 | 6214 |
| | Test 2 | 5756 | 6040 |
| 3 | Test 1 | 5687 | 6109 |
| | Test 2 | 5794 | 6002 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 14.3.2 NR RSTD measurement accuracy test case for dual positioning frequency layer in FR1 SA

### 14.3.2.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions.

### 14.3.2.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

### 14.3.2.3 Minimum conformance requirements

Same as in clause 14.3.1.3.

### 14.3.2.4 Test description

#### 14.3.2.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 14.3.2.4.1-1.

**Table 14.3.2.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Low for RF channel #1 and High for RF channel #2 as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.2.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 14.3.2.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.2.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the RSTD reference as well as the PCell on NR RF channel #1 in FR1. Cell 2 is a neighbour cell on a different NR RF channel #2 in FR1. GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.2.4.2 Test procedure

Same as clause 14.3.1.4.2.

### 14.3.2.4.3 Message contents

Same as clause 14.3.1.4.3 except that Table 14.3.2.4.3-1 replaces Table 14.3.1.4.3-5.

**Table 14.3.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Sub-test 1-1, Sub-test 1-2, Sub test 2-1 and Sub-test 2-2 |
| | kHz30 | | Sub-test 3-1 and Sub-test 3-2 |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1-1, Sub-test 1-2 and Sub-test 1-3 |
| | 21 | 104 PRBs | Sub-test 1-2 and Sub-test 2-2 |
| | 28 | 132 PRBs | Sub-test 3-2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | n4 | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.1.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Sub-test 1-1, Sub-test 1-2, Sub test 2-1 and Sub-test 2-2 |
| | kHz30 | | Sub-test 3-1 and Sub-test 3-2 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |

| | | | |
|---|---|---|---|
| | 21 | 104 PRBs | Sub-test 1-2 and Sub-test 2-2 |
| | 28 | 132 PRBs | Sub-test 3-2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | n4 | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.1.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.2.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.2.3.

**Table 14.3.2.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | 1~3 | | freq1 | freq2 | freq1 | freq2 |
| $BW_{channel}$ | 1 | MHz | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 2 | | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 3 | | 50: $N_{RB,c}$ = 133 | | 50: $N_{RB,c}$ = 133 | |
| Duplex mode | 1 | | FDD | | FDD | |
| | 2 | | TDD | | TDD | |
| | 3 | | TDD | | TDD | |
| TDD configuration | 1 | | N/A | | N/A | |
| | 2 | | TDDConf.1.1 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | | TDDConf.2.1 | |
| | 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |

| | | | | | | |
|---|---|---|---|---|---|---|
| PDSCH Reference measurement channel | 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | 3 | | SR.2.1 FDD | | SR.2.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - | CR.1.1 TDD | - |
| | 3 | | CR.2.1 FDD | - | CR.2.1 FDD | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - | CCR.2.1 TDD | - |
| SSB configuration | 1 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 3 | | SSB.2 FR1 | | SSB.2 FR1 | |
| OCNG Patterns | 1~3 | | OP.1 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - | TRS.1.1 FDD | |
| | 2 | | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP Configuration | 1~3 | | DLBWP.0.1 ULBWP.0.1 | | DLBWP.0.1 ULBWP.0.1 | |
| Dedicated BWP configuration | 1~3 | | DLBWP.1.1 ULBWP.1.1 | | DLBWP.1.1 ULBWP.1.1 | |
| Time offset with Cell 1 | 1 | µs | - | 3 | - | 3 |
| | 2,3 | | - | 3 | - | 3 |
| SMTC configuration | 1 | | SMTC.2 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.1 FR1 | | PRS.1.2 FR1 | |
| | 2 | | PRS.1.1 FR1 | | PRS.1.2 FR1 | |
| | 3 | | PRS.2.1 FR1 | | PRS.2.2 FR1 | |
| PRS Resource slot offset | 1, 2, 3 | slot | 0 | 2 | 0 | 2 |
| Expected RSTD | 1, 2, 3 | µs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3 | µs | N/A | 5 | N/A | 5 |
| EPRE ratio of PSS to SSS | 1~3 | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS[Note 1] | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS[Note 1] | | | | | | |
| EPRE ratio of PRS to SSS | 1, 2, 3 | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$[Note2] | 1,2 | dBm/ SCS | -98 | | -98 | |
| | 3 | | -95 | | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | 1~3 | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| PRS-RSRP[Note3] | 1,2 | dBm/SCS | -103.7 | -110.7 | -103.7 | -110.7 |
| | 3 | | -100.7 | -107.7 | -100.7 | -107.7 |
| Io[Note3] | 1,2 | dBm/ 19.08MHz | -65.92 | -66.73 | -65.92 | -66.73 |
| | 3 | dBm/ 47.88MHz | -61.92 | -62.73 | -61.92 | -62.73 |
| PRS $\hat{E}_s/N_{oc}$ | 1~3 | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| Propagation condition | 1~3 | - | AWGN | | AWGN | |
| Antenna configuration | 1~3 | | 1x2 | | 1x2 | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols where PRS is transmitted. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification. |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.2.2.1.5, with the following difference: $N_{PRS,i}^{slot} = 1$. Therefore, the value of the LPP Time IE is given by the following equation:

$$T_{RSTD} = \left(\left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS,i}}}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.1.4.3-3).

**Table 14.3.2.5-2: RSTD accuracy requirements for the reported values**

| Test Configuration | Subtest | Lowest reported value [$T_c$] | Highest reported value [$T_c$] |
|---|---|---|---|
| 1 & 2 | Test 1 | 5230 | 6566 |
| | Test 2 | 5500 | 6296 |
| 3 | Test 1 | 5399 | 6397 |
| | Test 2 | 5554 | 6242 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 14.3.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.3.3.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when dsingle positioning frequency layers are configured.

### 14.3.3.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

### 14.3.3.3 Minimum conformance requirements

Same as in clause 14.3.1.3.

### 14.3.3.4 Test description

The supported test configurations in listed in Table 14.3.3.4-1.

**Table 14.3.3.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

#### 14.3.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.3.5-1 and Table 14.3.3.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.3.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. GP#13 is configured for the test.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.3.3.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.1 FR2 and PRS.1.2 FR2. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.3.3.4.3 shall be provided to the UE during the set-up period. The last TTI containing the NR-DL-TDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms is the maximum processing time of the NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data in the UE.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.3.3.5-1 and Table 14.3.3.5-2 as appropriate. Propagation conditions are set according to clause 4.15.2.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-TDOA-ProvideAssistanceData-r16 IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-TDOA-RequestLocationInformation-r16 IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.

8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the nr-DL-TDOA-ProvideLocationInformation-r16 IE.

9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.3.3.5-3.

11. Repeat step 2-10 until the confidence level according to Annex D is achieved.

12. Repeat step 1-11 for the other sub-test defined in Table 14.3.3.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.3.3.4.3 Message contents

**Table 14.3.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.3.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.3.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| time | See 14.3.3.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 | Not present | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.3.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |

| | | | |
|---|---|---|---|
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.3.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.3.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.3.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.3.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.3.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and |

| | | | |
|---|---|---|---|
| | | | Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.3.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.3.3.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.3.3.

**Table 14.3.3.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Measurement gap | | GP#24 or GP#13 | | GP#24 or GP#13 | |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - | Not applicable | - |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.1 | OP.1 | OP.1 | OP.1 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.1 FR2 | PRS.1.1 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset | slot | 0 | 2 | 0 | 2 |
| Expected RSTD | µs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |

| | | | | | |
|---|---|---|---|---|---|
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| EPRE ratio of PRS to SSS | dB | 0 | 0 | 0 | 0 |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.. | | | | | |

**Table 14.3.3.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 5 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89.1 | | -89.1 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -5.17 | -11.36 | -5.17 | -11.36 |
| PRP Note2 | dBm/SCS | -94.2 | -100.4 | -94.2 | -100.4 |
| PRS $\hat{E}_s / I_{ot}$ | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| Io Note2 | dBm/190.08 MHz Note3 | -55.68 | -55.68 | -55.68 | -55.68 |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | |
| Note 2: PRP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted. | | | | | |
| Note 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone | | | | | |
| Note 4: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [2], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. | | | | | |
| Note 5: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation | | | | | |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.2.3.5 with the following difference: $N_{PRS,i}^{slot} = 2$. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.3.4.3-3).

**Table 14.3.3.5-3: RSTD accuracy requirements for the reported values**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Sub-test 1 | (5898 – 120) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 120) $T_c$ converted to RSTD measurement according to clause 4.15.6 |
| Sub-test 2 | (5898 – 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.3.4 NR RSTD measurement accuracy test case for dual positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.3.4.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when dual positioning frequency layers are configured.

### 14.3.4.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning.

### 14.3.4.3 Minimum conformance requirements

Same as in clause 14.3.1.3.

### 14.3.4.4 Test description

The supported test configurations in listed in Table 14.3.4.4-1.

**Table 14.3.4.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

#### 14.3.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.

2. The general test parameter settings are set up according to Table 14.3.4.5-1 and Table 14.3.4.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.4.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell on NR RF channel #1 in FR2. Cell 2 is a neighbour cell on a different NR RF channel #2 in FR2. GP#13 is configured for the test.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.4.4.2 Test procedure

Same as clause 14.3.3.4.2.

### 14.3.4.4.3 Message contents

**Table 14.3.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.4.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| time | see 14.3.4.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 | Not present | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.4.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |

| | | | |
|---|---|---|---|
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.4.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.4.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.4.4.3-6 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.4.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.4.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |

| | | | |
|---|---|---|---|
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.4.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.4.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.4.3.

**Table 14.3.4.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | | freq1 | freq2 | freq1 | freq2 |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Measurement gap | | GP#24 or GP#13 | | | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |

| | | | | | |
|---|---|---|---|---|---|
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - | Not applicable | - |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.1 | OP.1 | OP.1 | OP.1 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.1 FR2 | PRS.1.1 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset | slot | 0 | 2 | 0 | 2 |
| Expected RSTD | µs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| EPRE ratio of PRS to SSS | dB | 0 | 0 | 0 | 0 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS. | | | | | |

**Table 14.3.4.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to clause A.3.15.1 | | | |
| Assumption for UE beams Note 5 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89 | | -89 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -5.47 | -12.47 | -5.47 | -12.47 |
| PRP Note2 | dBm/SCS | -94.4 | -101.4 | -94.4 | -101.4 |
| $\hat{E}_s / I_{ot}$ BB Note4 | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| Io Note2 | dBm/190.08 MHz Note3 | -55.89 | -55.89 | -55.89 | -55.89 |

| | |
|---|---|
| Note 1: | Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 2: | PRP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted. |
| Note 3: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 4: | Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] Table 6.2.1.3-4. |
| Note 5: | Information about types of UE beam is given in B.2.1.3, and does not limit UE implementation or test system implementation |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.2.4.5 with the following difference: $N_{PRS,i}^{slot} = 1$. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 8 * \left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.4.4.3-3).

**Table 14.3.4.5-3: RSTD accuracy requirements for the reported values**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Sub-test 1 | [5898 – 360] $T_c$ converted to RSTD measurement according to clause 4.15.6 | [5898 + 360] $T_c$ converted to RSTD measurement according to clause 4.15.6 |
| Sub-test 2 | [5898 – 310] $T_c$ converted to RSTD measurement according to clause 4.15.6 | [5898 + 310] $T_c$ converted to RSTD measurement according to clause 4.15.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.3.5 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 14.3.5.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions.

### 14.3.5.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and supportedDL-PRS-ProcessingSamples.

### 14.3.5.3 Minimum conformance requirements

The accuracy requirements for RSTD measurement shall be within ±(X+Y+Z+Δ) $T_c$.

X is defined in TS 38.133 [50] Table 10.1.23.2-7 for AWGN channel in FR1 provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

Note: The requirements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

When UE measures RSTD on PRS resources belonging to different PFLs, then the RSTD accuracy is defined as the accuracy corresponding to the largest accuracy value among different PFLs.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 160 ms.
- When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in Table 14.3.5.3-1
- Δ is zero for single PFL, and is defined in Table 14.3.5.3-1, for dual PFL.

**Table 14.3.5.3-1: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 24 | N/A | N/A | 120 |
| ≥ 52 | ≥ 24 | N/A | 72 |
| ≥ 104 | ≥ 48 | ≥ 24 | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

**Table 14.3.5.3-2: Margin Δ for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 24 | N/A | N/A | 128 |
| ≥ 52 | ≥ 24 | N/A | 64 |
| ≥ 104 | ≥ 48 | ≥ 24 | 32 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 8 |

**Table 14.3.5.3-3: RSTD absolute accuracy in FR1 for AWGN channel with reduced number of samples**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | dBm/$BW_{Channel}$ |
| 98 | $(PRS Ês/Iot)_{ref}$ ≥ -3dB | 15 | ≥ 52 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | (PRS Ês/Iot)$_i$ ≥ -6dB | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| 42 | | | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 48 | | 30 | ≥ 48 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| 24 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 | | 60 | ≥ 64 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 10 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}, L_{\text{PRS}}, K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Void

## 14.3.5.4 Test description

### 14.3.5.4.1 Initial conditions

The supported test configuration is listed in Table 14.3.5.4-1.

**Table 14.3.5.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 14.3.5.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.5.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR1. GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured for the test.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.5.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.3.5.4.3 shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 if UE supports MG#24, otherwise #0 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.3.5.5-1. Propagation conditions are set according to clause 4.15.2.
4. The SS shall transmit an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3).
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.3.5.5-2.
11. Repeat steps 2-10 in Table 14.3.5.4-1 until the confidence level according to Annex D is achieved.
12. Repeat step 1-11 for the other sub-test defined in Table 14.3.5.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

#### 14.3.5.4.3 Message contents

**Table 14.2.3.7.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.5.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.5.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.3.5.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.5.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.5.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.5.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Sub-test 1-1, Sub-test 1-2, Sub test 2-1 and Sub-test 2-2 |
| | kHz30 | | Sub-test 1-3 and Sub-test 2-3 |
| dl-PRS-ResourceBandwidth-r16 | 13 | 52 PRBs | Subtest 1 : Config 1 and Config 2 |

| | | | |
|---|---|---|---|
| | 7 | 48 PRBs | Subtest 1 : Config 3 |
| | 21 | 104 PRBs | Subtest 2: Config 1 and Config 2 |
| | 28 | 132 PRBs | Subtest 2: Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Subtest1: config 1,2 &3 |
| | n4 | | Sub-test 2: Config 1,2&3 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.5.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.5.4.3-6: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Sub-tests 1-1, Sub-test 1-2, Sub-test 2-1 and Sub-test 2-2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Sub-tests 3-1 and Sub–test 3-2 |
| n160-r16 | 10 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | Test1 : Config 1,2&3 |
| | n2 | | Test2 : Config 1,2&3 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | 10 | Config 1,2,&3 | Cell 1 Subtest 1&2 |
| | 01 | Config 1,2,&3 | Cell 2 Subtest 1&2 |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 01 | | Test 1 – Config 1,2&3 |
| n4-r16 | 01 | | Test 2 – Config 1,2&3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Subtest 1 – Cell 1& 2 – Config 1,2&3 |
| | 2 | | Subtest 2 – Cell 1& 2 – Config 1,2&3 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.5.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.5.5 Test requirement

Table 14.3.5.5-1 defines the primary level settings including test tolerances for the test.

**Table 14.3.5.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | 1~3 | | freq1 | Freq1 | freq1 | Freq1 |
| $BW_{channel}$ | 1 | MHz | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 2 | | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 3 | | 50: $N_{RB,c}$ = 133 | | 50: $N_{RB,c}$ = 133 | |
| Duplex mode | 1 | | FDD | | FDD | |
| | 2 | | TDD | | TDD | |
| | 3 | | TDD | | TDD | |
| TDD configuration | 1 | | N/A | | N/A | |
| | 2 | | TDDConf.1.1 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | 3 | | SR.2.1 FDD | | SR.2.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - | CR.1.1 TDD | - |
| | 3 | | CR.2.1 FDD | - | CR.2.1 FDD | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - | CCR.2.1 TDD | - |

| | | | | | | |
|---|---|---|---|---|---|---|
| SSB configuration | 1 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 3 | | SSB.2 FR1 | | SSB.2 FR1 | |
| OCNG Patterns | 1~3 | | OP.1 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - | TRS.1.1 FDD | |
| | 2 | | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP Configuration | 1~3 | | DLBWP.0.1 ULBWP.0.1 | | DLBWP.0.1 ULBWP.0.1 | |
| Dedicated BWP configuration | 1~3 | | DLBWP.1.1 ULBWP.1.1 | | DLBWP.1.1 ULBWP.1.1 | |
| Time offset with Cell 1 | 1 | μs | - | 3 | - | 3 |
| | 2,3 | | - | 3 | - | 3 |
| SMTC configuration | 1 | | SMTC.2 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.3 FR1 | | PRS.1.4 FR1 | |
| | 2 | | PRS.1.3 FR1 | | PRS.1.4 FR1 | |
| | 3 | | PRS.2.3 FR1 | | PRS.2.4 FR1 | |
| PRS BW | 1,2 | | 52 PRBs | | - | |
| | 3 | | 48 PRBs | | - | |
| PRS muting info | 1~3 | | '10' | '01' | '10' | '01' |
| Expected RSTD | 1, 2, 3 | μs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3 | μs | N/A | 5 | N/A | 5 |
| EPRE ratio of PSS to SSS | 1~3 | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| $N_{oc}$ Note2 | 1,2 | dBm/ SCS | -98 | | -98 | |
| | 3 | | -95 | | -95 | |
| $\hat{E}_s / I_{ot}$ | 1~3 | dB | -3 | -6 | -3 | -6 |
| PRS-RSRP Note3 | 1,2 | dBm/SCS | -99.44 | -101.65 | -99.44 | -101.65 |
| | 3 | | -96.44 | -98.65 | -96.44 | -98.65 |
| SS-RSRP Note3 | 1,2 | dBm/SCS | -98.95 | -101.10 | -98.95 | -101.10 |
| | 3 | | -95.9 | -98.10 | -95.9 | -98.10 |
| Io Note3 | 1,2 | dBm/ 19.08MHz | -63.35 | -63.35 | -63.35 | -63.35 |
| | 3 | dBm/ 47.88MHz | -59.35 | -59.35 | -59.35 | -59.35 |
| $\hat{E}_s / N_{oc}$ | 1~3 | dB | -0.95 | -3.10 | -0.95 | -3.10 |
| Propagation condition | 1~3 | - | AWGN | | AWGN | |
| Antenna configuration | 1~3 | | 1x2 | | 1x2 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | |

The

| | |
|---|---|
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification. |

RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.5.3.

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.3.1.5.

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.3.4.3-3).

**Table 14.3.5.5-2: RSTD accuracy requirements for the reported values**

| Test Configuration | Subtest | Lowest reported value [$T_c$] | Highest reported value [$T_c$] |
|---|---|---|---|
| 1 & 2 | Test 1 | 5664 | 6132 |
| | Test 2 | 5756 | 6040 |
| 3 | Test 1 | 5750 | 6046 |
| | Test 2 | 5794 | 6002 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.3.6 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR1 SA

### 14.3.6.1 Test purpose

The purpose of the test is to verify that the RSTD measurements when the measurements are within the same Rx TEG meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions.

### 14.3.6.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports UE assisted DL-TDOA positioning and nr-UE-RxTEG-ID-MaxSupport.

### 14.3.6.3 Minimum conformance requirements

Same as defined in clause 14.3.5.3.

### 14.3.6.4 Test description

#### 14.3.6.4.1 Initial conditions

The supported test configuration is listed in Table 14.3.6-1.

**Table 14.3.6.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |

| | |
|---|---|
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.6.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 14.3.6.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.6.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. GP#24 is configured if UE supports GP#24. otherwise, GP#0 is configured for the test.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.6.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.3.6.4.3 shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 if UE supports MG#24, otherwise #0 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.3.6.5-1 as appropriate. Propagation conditions are set according to clause 4.15.2.
4. The SS shall transmit an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports nr-UE-RxTEG-ID-MaxSupport in the *NR-UE-TEG-Capability* IE and DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-UE-RxTEG-Request-r17 in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3).
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.3.6.5-3.

11. Repeat steps 2-10 in Table 14.3.6.4-1 until the confidence level according to Annex D is achieved.

12. Repeat step 1-11 for the other sub-test defined in Table 14.3.6.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.3.6.4.3 Message contents

**Table 14.3.6.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.6.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.6.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.3.6.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if | |

| | | | |
|---|---|---|---|
| | | response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.6.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.6.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.6.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Test 1 – Config 1&2<br>Test 2 – Config 1&2 |
| | kHz30 | | Test 1 – Config 3<br>Test 2 – Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Subtest 1-Config 1&2 |
| | 1 | 24 PRBs | Subtest 1-Config 3 |

| | | | |
|---|---|---|---|
| | 28 | 132 PRBs | Sub-test 2-Config 3 |
| | 21 | 104 PRBs | Subtest 2-Config 1&2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Test 1 – Config 1,2&3 |
| | n4 | | Test 2 – Config 1,2&3 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.6.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.6.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.6.4.3-6: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Test 1-Config 1&2 Tests 2-Config 1&2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Test 1-Config 3 Tests 2-Config 3 |
| n160-r16 | 10 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Test 2 : Config 1,2 &3 |
| | n2 | | Test 1 : Config 1,2 &3 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Test 1 |
| n4-r16 | 0 | | Test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Subtest 1-Cell 1 Config 1,2 &3 Subtest 2-cell 1 Config 1,2&3 |
| | 2 | | Subtest 1-Cell 2 Config 1,2 &3 Subtest 2-cell 2 Config 1,2&3 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.6.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |

| | | | |
|---|---|---|---|
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.6.5 Test requirement

Table 14.3.6.5-1 define the primary level settings including test tolerances for the test.

**Table 14.3.6.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | 1~3 | | freq1 | Freq1 | freq1 | Freq1 |
| $BW_{channel}$ | 1 | MHz | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 2 | | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 3 | | 50: $N_{RB,c}$ = 133 | | 50: $N_{RB,c}$ = 133 | |
| Duplex mode | 1 | | FDD | | FDD | |
| | 2 | | TDD | | TDD | |
| | 3 | | TDD | | TDD | |
| TDD configuration | 1 | | N/A | | N/A | |
| | 2 | | TDDConf.1.1 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | 3 | | SR.2.1 FDD | | SR.2.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - | CR.1.1 TDD | - |
| | 3 | | CR.2.1 FDD | - | CR.2.1 FDD | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - | CCR.1.1 TDD | - |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 3 | | CCR.2.1 TDD | - | CCR.2.1 TDD | - |
| SSB configuration | 1 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 3 | | SSB.2 FR1 | | SSB.2 FR1 | |
| OCNG Patterns | 1~3 | | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | 1~3 | dB | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS configuration | 1 | | TRS.1.1 FDD | - | TRS.1.1 FDD | |
| | 2 | | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP Configuration | 1~3 | | DLBWP.0.1 ULBWP.0.1 | | DLBWP.0.1 ULBWP.0.1 | |
| Dedicated BWP configuration | 1~3 | | DLBWP.1.1 ULBWP.1.1 | | DLBWP.1.1 ULBWP.1.1 | |
| Time offset with Cell 1 | 1 | μs | - | 3 | - | 3 |
| | 2,3 | | - | 3 | - | 3 |
| SMTC configuration | 1 | | SMTC.2 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.1 FR1 | | PRS.1.2 FR1 | |
| | 2 | | PRS.1.1 FR1 | | PRS.1.2 FR1 | |
| | 3 | | PRS.2.1 FR1 | | PRS.2.2 FR1 | |
| PRS Resource slot offset | 1~3 | slot | 0 | 2 | 0 | 2 |
| Expected RSTD | 1, 2, 3 | μs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3 | μs | N/A | 5 | N/A | 5 |
| $N_{oc}$ Note2 | 1,2 | dBm/ SCS | -98 | | -98 | |
| | 3 | | -95 | | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | 1~3 | dB | -6 | -13 | -6 | -13 |
| PRP Note3 | 1,2 | dBm/SCS | -103.4 | -109.5 | -103.4 | -109.5 |
| | 3 | | -100.4 | -106.5 | -100.4 | -106.5 |
| Io Note3 | 1,2 | dBm/ 19.08MHz | -65.63 | -65.63 | -65.63 | -65.63 |
| | 3 | dBm/ 47.88MHz | -61.63 | -61.63 | -61.63 | -61.63 |
| PRS $\hat{E}_s/N_{oc}$ | 1~3 | dB | -5.4 | -11.5 | -5.4 | -11.5 |
| Propagation condition | 1~3 | - | AWGN | | AWGN | |
| Antenna configuration | 1~3 | | 1x2 | | 1x2 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | |

| | |
|---|---|
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | RRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification. |

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.6.3.

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.3.1.5.

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.6.4.3-3).

**Table 14.3.6.5-2: RSTD accuracy requirements for the reported values**

| Test Configuration | Subtest | Lowest reported value [$T_c$] | Highest reported value [$T_c$] |
|---|---|---|---|
| 1 & 2 | Test 1 | 5582 | 6214 |
| | Test 2 | 5756 | 6040 |
| 3 | Test 1 | 5687 | 6109 |
| | Test 2 | 5794 | 6002 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.3.7 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.3.7.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions.

### 14.3.7.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 14.3.7.3 Minimum conformance requirements

The accuracy requirements for RSTD measurement shall be within ±(X+Y+Z+Δ) $T_c$.

…

X is defined in Table 10.1.23.2-8 for AWGN channel in FR2 provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

Note: The requirements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

When UE measures RSTD on PRS resources belonging to different PFLs, then the RSTD accuracy is defined as the accuracy corresponding to the largest accuracy value among different PFLs.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 160 ms.
- When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in Table 10.1.23.2-5 for FR1 and Table 10.1.23.2-6 for FR2, respectively.
- Δ is zero for single PFL, and is defined in Table 10.1.23.2-5a for FR1 and Table 10.1.23.2-6a for FR2, respectively, for dual PFL.

…

**Table 10.1.23.2-6: Margin for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 24 | N/A | 72 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 16 |
| N/A | ≥ 128 | 12 |

**Table 10.1.23.2-6a: Margin Δ for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 24 | N/A | 32 |
| ≥ 64 | ≥ 32 | 16 |
| ≥ 132 | ≥ 64 | 8 |
| N/A | ≥ 128 | 4 |

…

**Table 10.1.23.2-8: RSTD absolute accuracy in FR2 for AWGN channel with reduced number of samples**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |
| 24 | | 60 | ≥ 64 | ≥ 1 | Same value as PRS_RP in Table B.2.14-2, according to | Note 5 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | (PRS Ês/Iot)$_{ref}$ ≥ -3dB<br><br>(PRS Ês/Iot)$_i$ ≥ -6dB | | | | UE Power class, operating band and angle of arrival | |
| 11 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 13 | | 120 | ≥ 64 | ≥ 1 | Same value as PRS_RP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | Note 5 |
| 6 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |
| NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i. | | | | | | |
| NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$ , $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively. | | | | | | |
| NOTE 3: Io is assumed to have constant EPRE across the bandwidth. | | | | | | |
| NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6]. | | | | | | |
| NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS. | | | | | | |
| NOTE 6: Void | | | | | | |

## 14.3.7.4 Test description

### 14.3.7.4.1 Initial conditions

The supported test configuration is listed in Table 14.3.7.4-1.

**Table 14.3.7.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.7.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.7.5-1 and Table 14.3.7.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.7.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. GP#24 is configured if UE supports GP#24, otherwise, GP#13 is configured for the test.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.7.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.1 FR2 and PRS.1.2 FR2. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.3.7.4.3 shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 14.3.7.5-1 and Table 14.3.7.5-2 as appropriate. Propagation conditions are set according to clause 4.15.2.

4. The SS shall transmit an LPP REQUEST CAPABILITIES message.

5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.

6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.

8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3).

9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.3.7.5-3.

11. Repeat steps 2-11 in Table 14.3.7.4-1 until the confidence level according to Annex D is achieved.

12. Repeat step 1-11 for the other sub-test defined in Table 14.3.7.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.3.7.4.3 Message contents

**Table 14.2.3.7.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.7.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.7.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.3.7.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.7.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.7.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 14.3.7.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.7.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.7.4.3-6 | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.7.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.7.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |

| | | | |
|---|---|---|---|
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.7.5 Test requirement

Table 14.3.7.5-1 and Table 14.3.7.5-2 define the primary level settings including test tolerances for the test.

**Table 14.3.7.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - | Not applicable | - |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.2 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS BW | | 64 PRBs | 64 PRBs | 128 PRBs | 128 PRBs |
| PRS Resource slot offset | slot | 0 | 2 | 0 | 2 |
| Expected RSTD | µs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | |

**Table 14.3.7.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to clause A.3.15.1 | | | |
| Assumption for UE beams Note 5 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/15kHz Note3 | -91.6 | | -91.6 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -0.56 | -2.72 | -0.56 | -2.72 |
| PRS-RSRP Note2 | dBm/SCS | -92.1 | -94.3 | -92.1 | -94.3 |
| SS-RSRP Note2 | dBm/SCS | -89.7 | N/A | -89.7 | N/A |
| PRS $\hat{E}_s / I_{ot}$ BB Note4 | dB | -2.96 | -5.68 | -2.69 | -5.68 |
| Io Note2 | dBm/190.08 MHz Note3 | -55.74 | -55.74 | -55.74 | -55.74 |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | |
| Note 2: SSB_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |
| Note 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone | | | | | |
| Note 4: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [2], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] Table 6.2.1.3-4. | | | | | |
| Note 5: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation | | | | | |

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.7.3.

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.2.8.5 with the following difference: $N_{PRS,i}^{slot} = 2$. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.107}{N} \right\rceil * 1 - 1\right) * T_{effect} + \mathrm{T}_{\mathrm{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.3.4.3-3).

**Table 14.3.7.5-3: RSTD accuracy requirements for the reported values**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Sub-test 1 | (5898 – 93) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 93) $T_c$ converted to RSTD measurement according to clause 4.15.6 |
| Sub-test 2 | (5898 – 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.3.8 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.3.8.1 Test purpose

The purpose of the test is to verify that the RSTD measurements when the measurements are within the same Rx TEG meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions.

### 14.3.8.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports UE assisted DL-TDOA positioning and nr-UE-RxTEG-ID-MaxSupport.

### 14.3.8.3 Minimum conformance requirements

Same as defined in clause 14.3.7.3.

### 14.3.8.4 Test description

#### 14.3.8.4.1 Initial conditions

The supported test configuration is listed in Table 14.3.8-1.

**Table 14.3.8.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.8.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.8.5-1 and Table 14.3.8.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.8.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. GP#24 is configured if UE supports GP#24. otherwise, GP#13 is configured for the test.

6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.8.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.1 FR2 and PRS.1.2 FR2. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.3.8.4.3 shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 14.3.8.5-1 and Table 14.3.8.5-2 as appropriate. Propagation conditions are set according to clause 4.15.2.

4. The SS shall transmit an LPP REQUEST CAPABILITIES message.

5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports nr-UE-RxTEG-ID-MaxSupport in the *NR-UE-TEG-Capability* IE and DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.

6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-UE-RxTEG-Request-r17 in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.

8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3).

9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.3.8.5-3.

11. Repeat steps 2-11 in Table 14.3.8.4-1 until the confidence level according to Annex D is achieved.

12. Repeat step 1-11 for the other sub-test defined in Table 14.3.8.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.3.8.4.3 Message contents

**Table 14.3.8.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.8.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.8.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.3.8.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.8.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.8.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-PositionCalculationAssistance-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.8.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.8.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.8.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.8.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.8.4.3-7: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | Present | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.8.5 Test requirement

Table 14.3.8.5-1 and Table 14.3.8.5-2 define the primary level settings including test tolerances for the test.

**Table 14.3.8.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - | Not applicable | - |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.1 | OP.1 | OP.1 | OP.1 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.1 FR2 | PRS.1.1 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset | slot | 0 | 2 | 0 | 2 |
| Expected RSTD | μs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | μs | N/A | 5 | N/A | 5 |

| | | | | | |
|---|---|---|---|---|---|
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | |

**Table 14.3.8.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to clause A.3.15.1 | | | |
| Assumption for UE beams Note 5 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89.1 | | -89.1 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -5.17 | -11.36 | -5.17 | -11.36 |
| PRP Note2 | dBm/SCS | -94.2 | -100.4 | -94.28 | -100.4 |
| PRS $\hat{E}_s / I_{ot}$ Note4 | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| Io Note2 | dBm/190.08 MHz Note3 | -55.68 | -55.68 | -55.68 | -55.68 |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | |
| Note 2: PRP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |
| Note 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone | | | | | |
| Note 4: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. | | | | | |
| Note 5: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation | | | | | |

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.8.3.

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.2.10.5 with the following difference: $N_{PRS,i}^{slot} = 2$. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(k_{multiTEG} * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right)$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 10.248s and 164s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.3.8.4.3-3. The LPP time IE ranges between 11s and 170s.

**Table 14.3.8.5-3: RSTD accuracy requirements for the reported values**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Sub-test 1 | (5898 – 120) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 120) $T_c$ converted to RSTD measurement according to clause 4.15.6 |
| Sub-test 2 | (5898 – 86) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 86) $T_c$ converted to RSTD measurement according to clause 4.15.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.3.9 NR RSTD measurement accuracy test case for PRS aggregation in FR1 SA

### 14.3.9.1 Test purpose

The purpose of the test is to verify that the RSTD measurement by aggregating PRS resoureces from multiple positioning frequency layers (PFLs) meets the measurement accuracy requirements specified in TS 38.133 [50] clause 10.1.23A.2 in an environment with AWGN propagation conditions.

### 14.3.9.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Connected-r18*.

### 14.3.9.3 Minimum conformance requirements

When UE measures RSTD on PRS resources belonging to different PFLs or different PFL combinations, then the RSTD accuracy is defined as the accuracy corresponding to the largest accuracy value among different PFLs or different PFL combinations.

The requirements in this clause for 3-PFL and 2-PFL apply provided that:

- PRS resources linked for aggregation saftisfy all the conditions specified in TS 38.214 [56] clause 5.1.6.5.3.
- The spacing between the center frequencies of adjacent PFLs containing PRS resources linked for aggregation does not exceed the nominal channel spacing for intra-band contiguous CA defined in TS 38.101-1 [18], clause 5.4A.1 for FR1, and in TS 38.101-2, [19] clause 5.4A.1 for FR2-1.

The accuracy requirements for RSTD measurement based on PRS Aggregation shall be within $\pm(X+Y+Z+\Delta)$ $T_c$.

X is defined in table 14.3.9.3-1 for AWGN channel and table 14.3.9.3-3 for fading channel for FR1, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133[50] annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

X is defined in table 14.3.9.3-2 for AWGN channel and table 14.3.9.3-4 for fading channel for FR2, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.

- Conditions for RSTD measurements are fulfilled according to annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.

- UE does not perform positioning measurement with reduced number of samples.

X is defined in table 14.3.9.3-5 for AWGN channel in FR1, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.

- Conditions for RSTD measurements are fulfilled according to annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.

- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

X is defined in table 14.3.9.3-6 for AWGN channel in FR2, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.

- Conditions for RSTD measurements are fulfilled according to annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.

- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

NOTE: The requriements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL or same PFL combination, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 160 ms.

- When UE measures RSTD on PRS resources belonging different PFLs or different PFL combinations, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 1280 ms.

- Z is defined in table 14.3.9.3-7 for FR1 and table 14.3.9.3-9 for FR2, respectively, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.

- Δ is zero when UE measures RSTD on PRS resources belonging to same PFL or same PFL combination and is defined in table 14.3.9.3-8 for FR1 and table 14.3.9.3-10 for FR2, respectively, when UE measures RSTD on PRS resources belonging different PFLs or different PFL combinations, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.

If the measurements of reference cell and neighbour cell belong to the same Rx TEG, i.e. associated and reported with a common Rx TEG ID, then the sum of Y+Z+Δ is equal to the timing error margin of the Rx TEG reported in *nr-UE-RxTEG-TimingErrorMargin*. The timing error margin reported via *nr-UE-RxTEG-TimingErrorMargin* cannot be larger than the value of (Y+Z+Δ) defined when the UE does not associate the measurements with the same Rx TEG.

**Table 14.3.9.3-1: RSTD absolute accuracy in FR1 for AWGN channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | $dBm/BW_{Channel}$ |
| 20 | 16 | | 15 | ≥ 104 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, | -127 | -50 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | NR_SDL_FR1_A | | |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -126.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -126 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -124 | -50 |
| | | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -120.5 | -50 |
| 11 | 6 | (PRS $\hat{E}s/Iot)_{ref}$ ≥-6dB<br><br>(PRS $\hat{E}s/Iot)_i$ ≥-13dB | 30 | ≥ 132 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -123.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -123 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -121 | -50 |
| | | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -117.5 | -50 |
| 6 | 4 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 11 | 6 | | 60 | ≥ 64 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -120.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -120 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -118 | -50 |
| | | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -114.5 | -50 |
| 5 | 4 | | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.

**Table 14.3.9.3-2: RSTD absolute accuracy in FR2 for AWGN channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS}$ * $L_{PRS}$/ $K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/$BW_{Channel}$ |
| 11 | 6 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 60 | ≥ 64 | ≥ 1 | Same value as PRS_RP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 5 | 4 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 6 | 4 | (PRS Ês/Iot)$_i$ ≥-13dB | 120 | ≥ 64 | ≥ 1 | Same value as PRS_RP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 3 | 2 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.

**Table 14.3.9.3-3: RSTD absolute accuracy in FR1 for fading channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS}$ * $L_{PRS}$/ $K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | dBm/$BW_{Channel}$ |
| 38 | 30 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 15 | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 29 | 29 | | 30 | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 27 | 26 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 29 | 35 | (PRS Ês/Iot)$_i$ ≥-13dB | 60 | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 27 | 31 | | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 14.3.9.3-1.

**Table 14.3.9.3-4: RSTD absolute accuracy in FR2 for fading channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/$BW_{Channel}$ |
| 71 | 51 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 60 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 55 | 38 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 72 | 49 | | 120 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 56 | 53 | (PRS Ês/Iot)$_i$ ≥-13dB | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor*, *dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 14.3.9.3-2.

**Table 14.3.9.3-5: RSTD absolute accuracy in FR1 for AWGN channel with reduced number of samples**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | dBm/$BW_{Channel}$ |
| 24 | 20 | (PRS Ês/Iot)$_{ref}$ ≥-3dB | 15 | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 13 | 7 | | 30 | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 6 | 5 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 13 | 7 | | 60 | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 6 | 5 | (PRS Ês/Iot)$_i$ ≥-6dB | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 14.3.9.3-1.

**Table 14.3.9.3-6: RSTD absolute accuracy in FR2 for AWGN channel with reduced number of samples**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | | | | Io Note 3 range |

| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Minimum Io | Maximum Io |
|---|---|---|---|---|---|---|---|
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |
| 13 | 7 | (PRS Ês/Iot)$_{ref}$ ≥ -3dB | 60 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 6 | 5 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 6 | 4 | | 120 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 3 | 2 | (PRS Ês/Iot)$_i$ ≥ -6dB | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 14.3.9.3-2.

**Table 14.3.9.3-7: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 104 | N/A | N/A | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

**Table 14.3.9.3-8: Margin Δ for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 104 | N/A | N/A | 32 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 8 |

**Table 14.3.9.3-9: Margin for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 64 | N/A | 32 |
| ≥ 132 | ≥ 64 | 16 |
| N/A | ≥ 128 | 12 |

**Table 114.3.9.3-10: Margin Δ for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 64 | N/A | 16 |
| ≥ 132 | ≥ 64 | 8 |
| N/A | ≥ 128 | 4 |

The normative reference for this requirement is TS 38.133 [50] clause 10.1.23A.2.

## 14.3.9.4 Test description

### 14.3.9.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 14.3.9.4.1-1.

**Table 14.3.9.4.1-1: Supported test configurations for PRS aggregation**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth per PFL, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth per PFL, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth per PFL, TDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.9.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 14.3.9.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 14.3.9.4.3.1
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.

### 14.3.9.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The *NR-TDOA-ProvideAssistanceData* and *NR-TDOA-RequestLocationInformation* message as defined in TS 37.355 [49], shall be provided to the UE before the start of the test. The test duration should be larger than the UE measurement period as defined in TS 38.133[50] clause 9.9.2.

The last TTI containing the NR-DL-TDOA-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms is the maximum processing time of the NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data in the UE.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On* according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.3.9.5-1 as appropriate. Propagation conditions are set according to clause 4.15.2.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *nr-DL-PRS-JointMeasurementRequestedPFL-List* in the *NR-DL-TDOA-RequestLocationInformation* such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.

8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the *NR-DL-TDOA-ProvideLocationInformation* IE.

9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.3.9.5-2.

11. Repeat step 2-10 until the confidence level according to Annex D is achieved.

#### 14.3.9.4.3 Message contents

**Table 14.3.9.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.3.9.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.3.9.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.3.9.5 | Result of the response time calculation rounded up to the next | |

| | | | |
|---|---|---|---|
| | | second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRP-Pair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.9.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.3.9.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | | | |
| assistanceDataValidityArea-r17 | | | |
| nr-PeriodicAssistData-r18 | | | |
| } | | | |
| bt-ProvideAssistanceData-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.9.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.9.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.9.4.3-6 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.9.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.9.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.9.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS- |

| | | | |
|---|---|---|---|
| | | | Assistance DataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.9.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not Present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not Present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not Present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.9.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.9.3.

**Table 14.3.9.5-1: RSTD accuracy test parameters for PRS aggregation**

| Parameter | Config | Unit | Cell 1 | Cell 2 |
|---|---|---|---|---|
| PRS ARFCN | 1,2,3 | | freq1, freq2 | freq1, freq2 |
| $BW_{channel}$ per carrier | 1 | MHz | 20: $N_{RB,c}$ = 106 | |
| | 2 | | 20: $N_{RB,c}$ = 106 | |
| | 3 | | 50: $N_{RB,c}$ = 133 | |
| Duplex mode | 1 | | FDD | |
| | 2 | | TDD | |
| | 3 | | TDD | |
| TDD configuration | 1 | | N/A | |
| | 2 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | |
| Measurement gap | 1, 2, 3 | | GP#24 or GP#0 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | |
| | 3 | | SR.2.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - |
| | 3 | | CR.2.1 FDD | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - |
| SSB configuration | 1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | |
| | 3 | | SSB.2 FR1 | |
| OCNG Patterns | 1,2,3 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - |
| | 2 | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | |
| Initial BWP Configuration | 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |

| | | | | |
|---|---|---|---|---|
| Dedicated BWP configuration | 1,2,3 | | DLBWP.1.1<br>ULBWP.1.1 | |
| Time offset with Cell 1 | 1 | µs | - | 3 |
| | 2,3 | | - | 3 |
| SMTC configuration | 1 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.2 FR1 | |
| | 2 | | PRS.1.2 FR1 | |
| | 3 | | PRS.2.2 FR1 | |
| PRS Resource slot offset | 1, 2, 3 | slot | 0 | 4 |
| Expected RSTD | 1, 2, 3 | µs | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3 | µs | N/A | 5 |
| EPRE ratio of PSS to SSS | 1,2,3 | dB | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS | | | | |
| EPRE ratio of OCNG to OCNG DMRS | | | | |
| EPRE ratio of PRS to SSS | 1,2,3 | dB | 0 | 0 |
| $N_{oc}$ Note2 | 1,2 | dBm/ SCS | -98 | |
| | 3 | | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | 1,2,3 | dB | -5.64 | -12.65 |
| PRP | 1,2 | dBm/SCS | -103.4 | -109.5 |
| | 3 | | -100.4 | -106.5 |
| Io | 1,2 | dBm/ 19.08MHz per PFL | -62.61 | -62.61 |
| | 3 | dBm/ 47.88MHz per PFL | -58.62 | -58.62 |
| PRS $\hat{E}_s/N_{oc}$ | 1,2,3 | dB | -5.35 | -11.54 |
| SSB $\hat{E}_s/N_{oc}$ | 1,2,3 | dB | -5.7 | |
| Propagation condition | 1,2,3 | - | AWGN | |

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.9.3.

**Table 14.3.9.5-2: RSTD accuracy requirements for the reported values**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Config1, Config2 | (5898 – 376) $T_c$ | (5898 + 376) $T_c$ |
| Config 3 | (5898 – 331) $T_c$ | (5898 + 331) $T_c$ |

The test tolerances are defined in clauses C.1.6 and C.2.5.

# The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.14.3.10NR RSTD measurement accuracy test case for PRS aggregation in FR2 SA

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test requirements and test tolerance analysis is incomplete

## 14.3.10.1 Test purpose

The purpose of the test is to verify that the RSTD measurement by aggregating PRS resoureces from multiple positioning frequency layers (PFLs) meets the measurement accuracy requirements specified in TS 38.133 [50] clause 10.1.23A.2 in an environment with AWGN propagation conditions.

## 14.3.10.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Connected-r18*.

## 14.3.10.3 Minimum conformance requirements

Same minimum requirements as in clause 14.3.9.3

## 14.3.10.4 Test description

### 14.3.10.4.1 Initial conditions

**Table 14.3.10.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth per PFL, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.10.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.10.5-1 and Table 14.3.10.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.10.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cells. GP#24 is configured if UE supports GP#24, otherwise, GP#13 is configured for the test.

### 14.3.10.4.2 Test procedure

Same test procedure as defined in clause 14.3.9.4.2 with following exceptions:

3. Set the parameters according to Table 14.3.10.5-1 and Table 14.3.10.5-2 as appropriate. Propagation conditions are set according to clause 4.15.2.

10. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 4.3.10.5-3.

### 14.3.10.4.3 Message contents

Same message contents as defined in clause 14.3.9.4.3 with following exceptions:

Table 14.3.10.4.3-1 replaces Table 14.3.9.4.3-5; Table 14.3.10.4.3-2 replaces Table 14.3.9.4.3-6.

**Table 14.3.10.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.10.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.10.4.3-2 | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.10.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.3.10.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.3.10.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-Assistance DataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-Assistance DataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-Assistance DataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS- |

| | | | |
|---|---|---|---|
| | | | Assistance DataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.3.10.5 Test requirements

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.3.10.3.

**Table 14.3.10.5-1: RSTD accuracy test parameters for PRS aggregation**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| PRS ARFCN | | freq1, freq2 | |
| Duplex mode | | TDD | |
| TDD configuration | | TDDConf.3.1 | |
| $BW_{channel}$ per carrier | MHz | 200: $N_{RB,c}$ = 132 | |
| Measurement gap | | GP#24 or GP#13 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - |
| TRS configuration | | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.1 | OP.1 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.1 FR2 | PRS.1.1 FR2 |
| PRS Resource slot offset | slot | 0 | 4 |
| Expected RSTD | µs | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | |
| EPRE ratio of PRS to SSS | dB | 0 | 0 |
| Propagation conditions | | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.<br>Note 2: Carriers with PRS resources for aggregation are intra-band and contiguous to each other. | | | |

**Table 14.3.10.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | |
| Assumption for UE beams Note 5 | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89 | |
| PRS $\hat{E}_s/N_{oc}$ | dB | -5.7 + TT | -11.9 + TT |
| PRP Note2 | dBm/SCS | -94.7 + TT | -100.9 + TT |
| PRS $\hat{E}_s/I_{ot}$ | dB | -6 + TT | -13 + TT |
| Io Note2 per PFL | dBm/190.08 MHz Note3 | -55.75 + TT | -55.75 + TT |

Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over sub-carriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 2: PRP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted.
Note 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
Note 4: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.
Note 5: Information about types of UE beam is given in clause B.2.1.3, and does not limit UE implementation or test system implementation.

The value of the LPP Time IE depends on the UE capabilities. The calculation is FFS

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 14.3.9.4.3-3).

The RSTD accuracy requirements for the reported values is FFS.

## 14.3.11 NR RSTD measurement accuracy test case in FR1 for Redcap UE without Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.3.11.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC CONNECTED state meets the accuracy requirements specified in TS 38.133 [50] clause 10.1A.16.2 in an environment with AWGN propagation conditions.

### 14.3.11.2 Test applicability

This test applies to RedCap UE release 18 onwards that support DL-TDOA positioning.

### 14.3.11.3 Minimum conformance requirements

The accuracy requirements for RSTD measurement shall be within $\pm(X+Y+Z+\Delta)$ $T_c$. The values of Y, Z and Δ and Rx TEG based requirement are as defined in TS 38.133 [50] clause 10.1.23.2. The requirements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2, respectively.

### 14.3.11.4 Test description

#### 14.3.11.4.1 Initial conditions

The supported test configurations in listed in Table 14.3.11.4-1.

**Table 14.3.11.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.11.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.11.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.11.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR1. GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.3.11.4.2 Test procedure

TBD

#### 14.3.11.4.3 Message contents

TBD

### 14.3.11.5 Test requirement

Table 14.3.11.5-1 define the primary level settings including test tolerances for the test.

**Table 14.3.11.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| PRS ARFCN | 1,2,3,4 | | freq1 | Freq1 |
| $BW_{channel}$ | 1 | MHz | 10: $N_{RB,c}$ = 52 | |
| | 2 | | 10: $N_{RB,c}$ = 52 | |
| | 3 | | 20: $N_{RB,c}$ = 51 | |
| | 4 | | 10: $N_{RB,c}$ = 52 | |
| Duplex mode | 1 | | FDD | |
| | 2 | | TDD | |
| | 3 | | TDD | |
| | 4 | | HD-FDD | |
| TDD configuration | 1 | | N/A | |
| | 2 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | |
| | 4 | | N/A | |
| Measurement gap | 1, 2, 3, 4 | | GP#24 or GP#0 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | |
| | 3 | | SR.2.1 TDD | |
| | 4 | | SR.1.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - |
| | 3 | | CR.2.1 TDD | - |
| | 4 | | CR.1.1 FDD | |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - |
| | 4 | | CCR.1.1 FDD | |
| SSB configuration | 1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | |
| | 3 | | SSB.1 RedCap FR1 | |
| | 4 | | SSB.1 FR1 | |
| OCNG Patterns | 1,2,3,4 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - |
| | 2 | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | |
| | 4 | | TRS.1.1 FDD | |
| Initial BWP Configuration | 1,2,3,4 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Dedicated BWP configuration | 1,2,3,4 | | DLBWP.1.1<br>ULBWP.1.1 | |
| Time offset with Cell 1 | 1,2,3,4 | µs | - | 3 |
| SMTC configuration | 1,4 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.1 FR1 | |
| | 2 | | PRS.1.1 FR1 | |
| | 3 | | PRS.2.1 FR1 | |
| | 4 | | PRS.1.1 FR1 | |
| PRS Resource slot offset | 1, 2, 3, 4 | slot | 0 | 4 |
| Expected RSTD | 1, 2, 3, 4 | µs | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3, 4 | µs | N/A | 5 |
| EPRE ratio of PSS to SSS | 1,2,3,4 | dB | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |

| | | | | |
|---|---|---|---|---|
| EPRE ratio of PDCCH to PDCCH DMRS<br>EPRE ratio of PDSCH DMRS to SSS<br>EPRE ratio of PDSCH to PDSCH DMRS<br>EPRE ratio of OCNG DMRS to SSS Note 1<br>EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | 1,2,3,4 | dB | 0 | 0 |
| $N_{oc}$ Note2 | 1,2,4 | dBm/ SCS | -98 | |
| | 3 | | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | 1,2,3,4 | dB | -6 | -13 |
| PRP Note3 | 1,2,4 | dBm/SCS | -103.7 | -109.9 |
| | 3 | | -100.7 | -106.9 |
| Io Note3 | 1,2,4 | dBm/ 9.36MHz | -68.8 | -68.8 |
| | 3 | dBm/ 18.36MHz | -65.88 | -65.88 |
| PRS $\hat{E}_s/N_{oc}$ | 1,2,3,4 | dB | -5.7 | -11.9 |
| Propagation condition | 1,2,3,4 | - | AWGN | |
| Antenna configuration | 1,2,3,4 | | 1x2 | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification.

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.4.11.3.

TBD

## 14.3.12 NR RSTD measurement accuracy test case in FR2 for Redcap UE without Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.3.12.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC CONNECTED state meets the accuracy requirements specified in TS 38.133 [50] clause 10.1A.16.2 in an environment with AWGN propagation conditions.

### 14.3.12.2 Test applicability

This test applies to RedCap UE release 18 onwards that support DL-TDOA positioning.

### 14.3.12.3 Minimum conformance requirements

Same as in 14.3.11.3.

### 14.3.12.4 Test description

#### 14.3.12.4.1 Initial conditions

The supported test configurations in listed in Table 14.3.12.4-1.

**Table 14.3.12.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 100 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.12.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.12.5-1 and Table 14.3.12.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.12.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. GP#24 is configured if UE supports GP#24, otherwise, GP#13 is configured for the test.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.3.12.4.2 Test procedure

TBD

#### 14.3.12.4.3 Message contents

TBD

### 14.3.12.5 Test requirement

Table 14.3.12.5-1 and Table 14.3.12.5-2: define the primary level settings including test tolerances for the test.

**Table 14.3.12.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |
| PRS ARFCN | | freq1 | |
| Duplex mode | | TDD | |
| TDD configuration | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 100: $N_{RB,c}$ = 66 | |
| Measurement gap | | GP#24 or GP#13 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - |
| TRS configuration | | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.1 | OP.1 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.1 FR2 | PRS.1.1 FR2 |
| PRS Resource slot offset | slot | 0 | 4 |
| Expected RSTD | µs | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | |
| EPRE ratio of PRS to SSS | dB | 0 | 0 |
| Propagation conditions | | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 |
| NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS. | | | |

**Table 14.3.12.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |

| | | | |
|---|---|---|---|
| Angle of arrival configuration | | Setup 1 according to clause A.3.15.1 | |
| Assumption for UE beams Note 5 | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -5.7 | -11.9 |
| PRP Note2 | dBm/SCS | -94.7 | -100.9 |
| PRS $\hat{E}_s / I_{ot}$ | dB | -6 | -13 |
| Io Note2 | dBm/95.04 MHz Note3 | -58.76 | -58.76 |

NOTE 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 2: PRP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted.
NOTE 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
NOTE 4: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.
NOTE 5: Information about types of UE beam is given in annex B.2.1.3 and does not limit UE implementation or test system implementation.

The RSTD measurement accuracy shall fulfil the absolute requirement in clause 14.3.12.3.

TBD

## 14.3.13 NR RSTD measurement accuracy test case in FR1 for Redcap UE with Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.3.13.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC CONNECTED state meets the accuracy requirements specified in TS 38.133 [50] clause 10.1A.16.2 in an environment with AWGN propagation conditions.

### 14.3.13.2 Test applicability

This test applies to RedCap UE release 18 onwards that support DL-TDOA positioning.

### 14.3.13.3 Minimum conformance requirements

The accuracy requirements for RSTD measurement shall be within ±(X+Y+Z+Δ) $T_c$. The values of Y, Z and Δ and Rx TEG based requirement are as defined in TS 38.133 [50] clause 10.1.23.2. For Rx FH, PRS BW in TS 38.133 [50] in

tables 10.1.23.2-5, 10.1.23.2-5a, 10.1.23.2-6, and 10.1.23.2-6a refer to per hop BW. The requirements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2, respectively.

## 14.3.13.4 Test description

### 14.3.13.4.1 Initial conditions

The supported test configurations in listed in Table 14.3.13.4-1.

**Table 14.3.14.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, cell bandwidth 50 Hz, FDD duplex mode |
| 2 | 15 kHz SSB SCS, cell bandwidth 50 Hz, TDD duplex mode |
| 3 | 30 kHz SSB SCS, cell bandwidth 100 Hz, TDD duplex mode |
| 4 | 15 kHz SSB SCS, cell bandwidth 50 Hz, HD-FDD duplex mode |
| NOTE 1: The UE is only required to be tested in one of the supported test configurations.<br>NOTE 2: UE with 1 Rx or 2 Rx is required to meet the same requirements specified in this clause.<br>NOTE 3: This test case is applicable to the UE supporting per hop bandwidth ≥ 10 MHz for SCS 15 kHz.<br>NOTE 4: This test case is applicable to the UE supporting per hop bandwidth = 20 MHz for SCS 30 kHz. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.13.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.13.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.13.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR1. GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.13.4.2 Test procedure

TBD

### 14.3.13.4.3 Message contents

TBD

## 14.3.13.5 Test requirement

Table 14.3.13.5-1 define the primary level settings including test tolerances for the test.

**Table 14.3.13.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| PRS ARFCN | 1,2,3,4 | | freq1 | Freq1 |
| $BW_{channel}$ | 1 | MHz | 50: $N_{RB,c}$ = 270 | |

| | | | | |
|---|---|---|---|---|
| | 2 | | 50: $N_{RB,c}$ = 270 | |
| | 3 | | 100: $N_{RB,c}$ = 273 | |
| | 4 | | 50: $N_{RB,c}$ = 270 | |
| Duplex mode | 1 | | FDD | |
| | 2 | | TDD | |
| | 3 | | TDD | |
| | 4 | | HD-FDD | |
| TDD configuration | 1 | | N/A | |
| | 2 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | |
| | 4 | | N/A | |
| Measurement gap | 1, 2, 3, 4 | | GP#24 or GP#0 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | |
| | 3 | | SR.2.1 TDD | |
| | 4 | | SR.1.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - |
| | 3 | | CR.2.1 TDD | - |
| | 4 | | CR.1.1 FDD | |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - |
| | 4 | | CCR.1.1 FDD | |
| SSB configuration | 1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | |
| | 3 | | SSB.1 RedCap FR1 | |
| | 4 | | SSB.1 FR1 | |
| OCNG Patterns | 1,2,3,4 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - |
| | 2 | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | |
| | 4 | | TRS.1.1 FDD | |
| Initial BWP Configuration | 1,2,3,4 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Dedicated BWP configuration | 1,2,3,4 | | DLBWP.1.1<br>ULBWP.1.1 | |
| Time offset with Cell 1 | 1,2,3,4 | µs | - | 3 |
| SMTC configuration | 1,4 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.5 FR1 | |
| | 2 | | PRS.1.5 FR1 | |
| | 3 | | PRS.2.6 FR1 | |
| | 4 | | PRS.1.5 FR1 | |
| PRS Resource slot offset | 1, 2, 3, 4 | slot | 0 | 4 |
| Expected RSTD | 1, 2, 3, 4 | µs | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3, 4 | µs | N/A | 5 |
| EPRE ratio of PSS to SSS | 1,2,3,4 | dB | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | 1,2,3,4 | dB | 0 | 0 |

| | | | | |
|---|---|---|---|---|
| $N_{oc}$ Note2 | 1,2,4 | dBm/ SCS | -98 | |
| | 3 | | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | 1,2,3,4 | dB | -6 | -13 |
| PRP Note3 | 1,2,4 | dBm/SCS | -103.7 | -109.9 |
| | 3 | | -100.7 | -106.9 |
| Io Note3 | 1,2,4 | dBm/ 48.6MHz | -61.64 | -61.64 |
| | 3 | dBm/ 98.28MHz | -58.51 | -58.51 |
| PRS $\hat{E}_s/N_{oc}$ | 1,2,3,4 | dB | -5.7 | -11.9 |
| Propagation condition | 1,2,3,4 | - | AWGN | |
| Antenna configuration | 1,2,3,4 | | 1x2 | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification.

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.4.13.3.

TBD

## 14.3.14 NR RSTD measurement accuracy test case in FR2 for Redcap UE with Rx FH

Editor's Note: This test case is incomplete in following aspects:

- Message content is missing;
- Apllicability is missing;
- Cell configuration is missing;
- Test procedure is missing;
- TT analysis is missing.

### 14.3.14.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC CONNECTED state meets the accuracy requirements specified in TS 38.133 [50] clause 10.1A.16.2 in an environment with AWGN propagation conditions.

### 14.3.14.2 Test applicability

This test applies to RedCap UE release 18 onwards that support DL-TDOA positioning.

### 14.3.14.3 Minimum conformance requirements

Same as in 14.3.13.3.

## 14.3.14.4 Test description

### 14.3.14.4.1 Initial conditions

The supported test configurations in listed in Table 14.2.10.4-1.

**Table 14.3.14.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, cell bandwidth 400 MHz, TDD duplex mode |
| NOTE 1: This test case is applicable to the UE supporting per hop bandwidth = 100 MHz for SCS 120 kHz. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.3.14.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.3.14.5-1 and Table 14.3.14.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.3.14.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. GP#24 is configured if UE supports GP#24, otherwise, GP#13 is configured for the test.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

### 14.3.14.4.2 Test procedure

TBD

### 14.3.14.4.3 Message contents

TBD

## 14.3.14.5 Test requirement

Table 14.3.14.5-1 and Table 14.3.14.5-2: define the primary level settings including test tolerances for the test.

**Table 14.3.14.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| PRS ARFCN | | freq1 | |
| Duplex mode | | TDD | |
| TDD configuration | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 400: $N_{RB,c}$ = 264 | |
| Measurement gap | | GP#24 or GP#13 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - |
| TRS configuration | | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - |

| | | | |
|---|---|---|---|
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.1 | OP.1 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.6 FR2 | PRS.1.6 FR2 |
| PRS Resource slot offset | slot | 0 | 4 |
| Expected RSTD | µs | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | |
| EPRE ratio of PRS to SSS | dB | 0 | 0 |
| Propagation conditions | | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 |
| NOTE 1: OCNG shall be used such that both cells are fully allocated, and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS. | | | |

**Table 14.3.14.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| Angle of arrival configuration | | Setup 1 according to clause A.3.15.1 | |
| Assumption for UE beams Note 5 | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -5.7 | -11.9 |
| PRP Note2 | dBm/SCS | -94.7 | -100.9 |
| PRS $\hat{E}_s / I_{ot}$ | dB | -6 | -13 |
| Io Note2 | dBm/380.16MHz Note3 | -52.74 | -52.74 |
| NOTE 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | |
| NOTE 2: PRP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted. | | | |
| NOTE 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone. | | | |
| NOTE 4: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4. | | | |
| NOTE 5: Information about types of UE beam is given in annex B.2.1.3 and does not limit UE implementation or test system implementation. | | | |

The RSTD measurement accuracy shall fulfil the absolute requirement in clause 14.3.14.3.

TBD

# 14.4 NR RSTD measurement period test cases in RRC_INACTIVE

## 14.4.1 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR1 SA

### 14.4.1.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] clause 5.6.2.5 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.4.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and DL-PRS measurement in RRC_INACTIVE.

### 14.4.1.3 Minimum conformance requirements

After receiving both *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from the LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 5.6.2.3) DL RSTD measurements, defined in TS 38.215 [4], during the measurement period $T_{RSTD,Total}$ defined as:

$$\mathrm{T_{RSTD,Total}} = \sum_{i=1}^{L} \mathrm{T_{RSTD,i}} + (\mathrm{L}-1) * \max\left(\mathrm{T_{effect,i}}\right)$$

Where:

- $i$ is the index of positioning frequency layer,
- $L$ is total number of positioning frequency layers, and
- $\mathrm{T_{effect},i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$\mathrm{T_{RSTD,i}}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below:

$$\mathrm{T_{RSTD,i}} = \left(K_{\mathrm{carrier_PRS}} * N_{Rx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil * \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * \mathrm{T_{effect,i}} + \mathrm{T_{last,i}} ,$$

Where:

- $N_{RxBeam,i}$ is the UE Rx beam sweeping factor:
  - $N_{RxBeam,i}$ = 1 if positioning frequency layer *i* is in FR1, and if positioning frequency layer *i* is in FR2
    - $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*.
  - $N_{RxBeam,i}$ equals to 8, otherwise.
- $K_{\mathrm{carrier_PRS}}$ is a scaling factor for PRS-based NR positioning measurements in RRC_INACTIVE. If the UE supports *parallelPRS-MeasRRC-Inactive-r17*, $\mathrm{K_{carrier_PRS}}$ = 1; otherwise,
- If Srxlev ≤ $\mathrm{S_{nonIntraSearchP}}$ or Squal ≤ $\mathrm{S_{nonIntraSearchQ}}$, $K_{\mathrm{carrier_PRS}} = K_{\mathrm{carrier}} + 1$, where $K_{\mathrm{carrier}}$ is defined in clause 4.2.2.4
- If Srxlev > $\mathrm{S_{nonIntraSearchP}}$ and Squal > $\mathrm{S_{nonIntraSearchQ}}$, $K_{\mathrm{carrier_PRS}} = N_{\mathrm{layers}} + 1$, where $N_{\mathrm{layers}}$ is defined in clause 4.2.2.7.
- $N_{Rx,TEG,i}$ is the Rx TEG specific scaling factor:

- $N_{Rx,TEG,i}$ =1 if the UE is not configured by the LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34].

- $N_{Rx,TEG,i}$ is defined as follows if the UE is configured by the LMF with *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] to perform measurement on same DL PRS resource of a TRP using different Rx TEGs in *NR-DL-TDOA-RequestLocationInformation* [34]:

  - $N_{Rx,TEG,i} = P$, if the UE is not capable of measuring same DL PRS resource simultaneously from multiple Rx TEGs, where P is the number of UE Rx TEGs that the UE is requested by LMF to measure the same DL-PRS Resource of a TRP indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in [34], and in case 'n0' is indicated, P is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.

  - $N_{Rx,TEG,i} = \left\lfloor \frac{P}{Q} \right\rfloor$, if the UE is capable of measuring the same DL PRS resource simultaneously from multiple Rx TEGs, where $Q$ is the number of UE Rx TEGs for measuring the same DL-PRS Resource simultaneously indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul-r17* in [34].

- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

- $L_{available_PRS,\mathrm{i}}$ is the time duration of available PRS in positioning frequency layer *i* to be measured within $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,\mathrm{i}}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.

- $N_{sample}$ is the number of PRS RSTD samples, where

  - $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 4 otherwise.

- $T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time, $\mathrm{T_{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ ,

- $\mathrm{T_{effect,}}_i$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{\mathrm{effect,i}} = \left\lceil \frac{T_{\mathrm{i}}}{T_{available_PRS,\mathrm{i}}} \right\rceil * T_{available_PRS,\mathrm{i}}$$

Where:

- $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [34],

- $T_{available_PRS,\mathrm{i}} = LCM\left(T_{PRS,\mathrm{i}}, T_{DRX}\right)$, the least common multiple between $T_{PRS,\mathrm{i}}$ and the DRX cycle length $T_{DRX}$

- $T_{PRS,\mathrm{i}}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,\mathrm{i}}$, where,

- $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,
- $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.
- $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where
- $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.
- $\{N, T\}$ is the UE capability combination per band for RRC_INACTIVE state where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [34], T (ms) corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [34], [ and T-N (>0) is the time required to process duration N of DL PRS symbols already buffered in memory], for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],
- $N'$ is UE capability for number of DL PRS resources that it can process in a slot [in RRC_INACTIVE state as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* specified in TS 37.355 [34].

The time $T_{RSTD,Total}$ starts from the first DRX cycle containing the DL PRS resource(s) in the assistance data after both the *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the UE via LPP [34].

### 14.4.1.4 Test description

The supported test configurations in listed in Table 14.4.1.4-1

**Table 14.4.1.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

#### 14.4.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.4.1.5-1, Table 14.4.1.5-2 and Table 14.4.1.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.4.1.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the RSTD reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

#### 14.4.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2 and Cell3. During T2 UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-DL-TDOA-ProvideAssistanceData* as defined in TS 37.355 [49] clause 6.5.10.1 shall be provided to the UE during T1. The last TTI containing the *NR-DL-TDOA-ProvideAssistanceData* and *NR-DL-TDOA-RequestLocationInformation* shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

The beginning of the time interval T2 shall be aligned with the first DRX cycle containing a DL PRS resource(s).

0. Configure the Cell for RA-SDT with SIB1 configured according to TS 38.508-1 [45], clause 4.6.1 with condition SDT.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.4.1.5-1, Table 14.4.1.5-2 and Table 14.4.1.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
12. When T1 expires, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and switch the power setting from T1 to T2 as specified in Table 14.4.1.5-2 and Table 14.4.1.5-3.
13. Wait for the UE monitors for occurrence of the periodic event requested.
14. The UE shall transmit the UE transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 18.
15. The SS transmits an *RRCResume* message.
16. The UE transmits an *RRCResumeComplete* message.
17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3).

18. The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

19. Repeat steps 2-17 in Tables 14.4.1.4-1 until the confidence level according to Annex D is achieved.

#### 14.4.1.4.3 Message contents

**Table 14.4.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.4.1.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.1.4.3-3: CellGroupConfig (Table 14.4.1.4.3-2)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms1280 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.1.4.3-4: LPP Request Capabilities**

| **Information Element** | **Value/remark** |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.4.1.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.4.1.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.1.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.1.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.1.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.1.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.1.4.3-7: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| N2-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| N2-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.1.4.3-8: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 14.4.1.4.3-9 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 14.4.1.4.3-9: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.4.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.1.4.3-10: RRCRelease**

| |
|---|
| Derivation Path: TS 38.508-1 [45], Table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT |

**Table 14.4.1.4.3-11: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.4.1.5 Test requirement

Table 14.4.1.5-1, Table 14.4.1.5-2 and Table 14.4.1.5-3 define the primary level settings including test tolerances for the test.

**Table.14.4.1.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| **Reference cell** | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| **Neighbor cells** | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in clause A.3.1.2.1 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1 ULBWP.0.1 | |

| | | | | |
|---|---|---|---|---|
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.1 FR1 | As specified in clause A.3.31 |
| | Config 2 | | PRS.1.1 FR1 | |
| | Config 3 | | PRS.2.1 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | s | 1.28 | |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Corresponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 5 | The length of the time interval that follows immediately after time interval T1. |

**Table Table.14.4.1.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -52.56 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |

| | | | | | |
|---|---|---|---|---|---|
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.4.1.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.08 | -11.27 | -11.27 |
| | Config 2 | dB | -5.08 | -11.27 | -11.27 |
| | Config 3 | dB | -5.08 | -11.27 | -11.27 |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -65.31 | -65.31 | -65.31 |
| | Config 2 | dBm/ 19.08MHz | -65.31 | -65.31 | -65.31 |
| | Config 3 | dBm/ 47.88MHz | -61.32 | -61.32 | -61.32 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -6 | -13 | -13 |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T2) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

The RSTD measurement time fulfils the requirements specified in Clause 5.6.2.5.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in section 5.6.2.5 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be higher than the time duration above because of the uncertainty in acquiring the first available PRACH occasion to transition to RRC_CONNECTED state to report the measurements.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD1970049.

$T_{RSTD,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below:

$$T_{RSTD,i} = \left(K_{carrier_PRS} * N_{Rx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil * \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i},$$

Since the test case is for single positioning frequency layer, i = 1 and L = 1, so $T_{RSTD,Total} = T_{RSTD,1}$.

Analysing the equation for $T_{RSTD,i}$:

, $K_{carrier_PRS} = 1$;

$N_{RxBeam,i} = 1$

$N_{Rx,TEG,i} = 1$

$N_{PRS,i}^{slot} = 4$

For SCS 15KHz:

$$L_{available_{PRS},i} = \frac{10}{10 * 1 * 14} * 4 * 1 * 4 = 1.14ms$$

For SCS=30KHz

$$L_{available_{PRS},i} = \frac{10}{20 * 1 * 14} * 4 * 1 * 4 = 0.571ms$$

$N_{sample} = 4$.

$T_{last,i} = T_i + T_{available_PRS,i}$

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$ ; where Tprs = 160 ms, and , $T_{DRX} = 1.28sec = 1280msec$ Therefore, $T_{available_PRS,i} = 1280$

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 5.12s and 206.08s.

For SCS 30KHz The LPP time IE ranges between 5.12s and 103.6s.

The LPP time IE ranges between 6s and 207s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 207.3

The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.4.1.5-4, Table 14.4.1.5-5 , Table 14.4.1.5-6 and Table 14.4.1.5-7 for reference.

**Table 14.4.1.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17***

| N' | $\lceil 4/N' \rceil$ |
|---|---|
| n1 | 4 |

| | |
|---|---|
| n2 | 2 |
| >=n4 | 1 |

**Table 14.4.1.5-5: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 15KHz***

| N | $\lceil \frac{1.143}{N} \rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 14.4.1.5-6: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 30KHz***

| N | $\lceil \frac{0.571}{N} \rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 14.4.1.5-7: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.4.2 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 14.4.2.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] Clause 5.6.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.4.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning, DL-PRS measurement in RRC_INACTIVE and reduced number of samples for PRS measurement in RRC_INACTIVE state.

### 14.4.2.3 Minimum conformance requirements

Same as in clause 14.4.1.3.

### 14.4.2.4 Test description

The supported test configurations in listed in Table 14.4.2.4-1

**Table 14.4.2.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS,50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

#### 14.4.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.4.2.5-1, Table 14.4.2.5-2 and Table 14.4.2.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.4.2.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the RSTD reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

#### 14.4.2.4.2 Test procedure

Same as in clause 14.4.3.4.2.

#### 14.4.2.4.3 Message contents

As defined in clause 14.4.1.4.3, with the following exceptions:

Table 14.4.2.4.3-1 replaces Table 14.4.1.4.3-6 and Table 14.4.2.4.3-2 replaces Table 14.4.1.4.3-7 and Table 14.4.2.4.3-3 replaces Table 14.4.1.4.3-9..

**Table 14.4.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 24 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.1.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.1.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.1.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 14.4.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |

| | | | |
|---|---|---|---|
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.4.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |

| | | | |
|---|---|---|---|
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.4.2.5 Test requirement

Table 14.4.2.5-1, Table 14.4.2.5-2 and Table 14.4.2.5-3 define the primary level settings including test tolerances for the test.

**Table 14.4.2.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [4] and TS 37.355 [34]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET | Config 1 | | CR.1.1 FDD | As specified in clause A.3.1.2.1 |
| | Config 2 | | CR.1.1 TDD | |

| | | | | |
|---|---|---|---|---|
| RMC configuration | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |
| PRS Configuration | Config 1 | | PRS.1.2 FR1 | As specified in clause A.3.31. |
| | Config 2 | | PRS.1.2 FR1 | |
| | Config 3 | | PRS.2.2 FR1 | |
| PRS Resource slot offset | Config 1, 2, 3 | slots | Cell 1, 2, 3: 0 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX cycle length | | s | 1.28 | |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [34] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [34] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Corresponds to prs-MutingInfo defined in TS 37.355 [34] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 5 | The length of the time interval that follows immediately after time interval T1 |

**Table 14.4.2.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positioning frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |

| | | | | | |
|---|---|---|---|---|---|
| SSB $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -63.94 | -63.94 | -63.94 |
| | Config 2 | dBm/ 19.08MHz | -63.94 | -63.94 | -63.94 |
| | Config 3 | dBm/ 47.88MHz | -59.96 | -59.96 | -59.96 |
| SSB RP Note4 | Config 1 | dBm/SCS | -98 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -98 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -95 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.4.2.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | 2.20 | 0.10 | 0.10 |
| | Config 2 | dB | 2.20 | 0.10 | 0.10 |
| | Config 3 | dB | 2.20 | 0.10 | 0.10 |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -60.23 | -60.23 | -60.23 |
| | Config 2 | dBm/ 19.08MHz | -60.23 | -60.23 | -60.23 |
| | Config 3 | dBm/ 47.88MHz | -56.23 | -56.23 | -56.23 |

The RSTD measurement time fulfils the requirements specified in Clause 5.6.2.5.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in section 5.6.2.5 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be higher than the time duration above because of the uncertainty in acquiring the first available PRACH occasion to transition to RRC_CONNECTED state to report the measurements.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

Since the test case is for single positioning frequency layer, i = 1 and L = 1, so $T_{RSTD,Total} = T_{RSTD,1}$.

Analysing the equation for $T_{RSTD,i}$:

, $K_{carrier_PRS} = 1$;

$N_{RxBeam,i} = 1$

$N_{Rx,TEG,i} = 1$

$N_{PRS,i}^{slot} = 4$

$N_{sample} = 1$.

$T_{last,i} = T_i + T_{available_PRS,i}$

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$ ; where Tprs = 160 ms, and, $T_{DRX} = 1.28sec = 1280msec$ Therefore, $T_{available_PRS,i} = 1280$

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 1.28s and 52.48s.

For SCS 30KHz The LPP time IE ranges between 1.28s and 26.88s.

The LPP time IE ranges between 2s and 53s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 53.3

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.4.2.5-4, Table 14.4.2.5-5 , Table 14.4.2.5-6 and Table 14.4.2.5-7 for reference.

**Table 14.4.2.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17***

| N' | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 14.4.2.5-5: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 15KHz***

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 14.4.2.5-6: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 30KHz***

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|

| | |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 14.4.2.5-7: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.4.3 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.4.3.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] clause 5.6.2.5 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.4.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and DL-PRS measurement in RRC_INACTIVE.

### 14.4.3.3 Minimum conformance requirements

After receiving both *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from the LMF via LPP [49], the UE shall be able to measure multiple (up to the UE capability specified in Clause 5.6.2.3) DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined as:

$$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD,i} + (L-1) * \max(T_{effect,i})$$

Where:

- $i$ is the index of positioning frequency layer,

- $L$ is total number of positioning frequency layers, and

- $T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$T_{RSTD,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer *i* as specified below:

$$T_{RSTD,i} = \left(K_{carrier_PRS} * N_{Rx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil * \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i},$$

Where:

- $N_{RxBeam,i}$ is the UE Rx beam sweeping factor:
  - $N_{RxBeam,i}$ = 1 if positioning frequency layer *i* is in FR1, and if positioning frequency layer *i* is in FR2
    - $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*.
  - $N_{RxBeam,i}$ equals to 8, otherwise.
- $K_{carrier_PRS}$ is a scaling factor for PRS-based NR positioning measurements in RRC_INACTIVE. If the UE supports *parallelPRS-MeasRRC-Inactive-r17*, $K_{carrier_PRS}$ = 1; otherwise,
- If Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$, $K_{carrier_PRS} = K_{carrier} + 1$, where $K_{carrier}$ is defined in clause 4.2.2.4
- If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$, $K_{carrier_PRS} = N_{layers} + 1$, where $N_{layers}$ is defined in clause 4.2.2.7.
- $N_{Rx,TEG,i}$ is the Rx TEG specific scaling factor:
  - $N_{Rx,TEG,i}$ =1 if the UE is not configured by the LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49].
  - $N_{Rx,TEG,i}$ is defined as follows if the UE is configured by the LMF with *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49] to perform measurement on same DL PRS resource of a TRP using different Rx TEGs in *NR-DL-TDOA-RequestLocationInformation* [49]:
    - $N_{Rx,TEG,i} = P$, if the UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, where P is the number of UE Rx TEGs that the UE is requested by LMF to measure the same DL-PRS Resource of a TRP indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in [49], and in case 'n0' is indicated, P is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.
    - $N_{Rx,TEG,i} = \left\lceil \frac{P}{Q} \right\rceil$, if the UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs, where $Q$ is the number of UE Rx TEGs for measuring the same DL-PRS Resource simultaneously indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul-r17* in [49].
- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.
- $L_{available_PRS,i}$ is the time duration of available PRS in positioning frequency layer *i* to be measured $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.
- $N_{sample}$ is the number of PRS RSTD samples, where
  - $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
    - PRS bandwidth is within the initial BWP and
    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the initial BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

- $T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time, $T_{\text{last,i}} = T_{\text{i}} + T_{available_PRS,\text{i}}$ ,
- $T_{\text{effect},i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{\text{effect,i}} = \left\lceil \frac{T_{\text{i}}}{T_{available_PRS,\text{i}}} \right\rceil * T_{available_PRS,\text{i}}$$

Where:

- $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [49],
- $T_{available_PRS,\text{i}} = LCM\left(T_{PRS,\text{i}}, T_{DRX}\right)$, the least common multiple between $T_{PRS,\text{i}}$ and the DRX cycle length $T_{DRX}$
- $T_{PRS,\text{i}}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,\text{i}}$, where,

- $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,
- $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.
- $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where
- $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.
- $\{N, T\}$ is the UE capability combination per band for RRC_INACTIVE state where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [49], T (ms) corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [49], [ and T-N (>0) is the time required to process duration N of DL PRS symbols already buffered in memory], for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49],
- $N'$ is UE capability for number of DL PRS resources that it can process in a slot [in RRC_INACTIVE state as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* specified in TS 37.355 [49].

## 14.4.3.4 Test description

The supported test configurations in listed in Table 14.4.3.4-1

**Table 14.4.3.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

### 14.4.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.4.3.5-1, Table 14.4.3.5-2 and Table 14.4.3.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.4.3.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the RSTD reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR2.

### 14.4.3.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2 and Cell3. During T2 UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-DL-TDOA-ProvideAssistanceData* as defined in TS 37.355 [49] clause 6.5.10.1 shall be provided to the UE during T1. The last TTI containing the *NR-DL-TDOA-ProvideAssistanceData* and *NR-DL-TDOA-RequestLocationInformation* shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

The beginning of the time interval T2 shall be aligned with the first DRX cycle containing a DL PRS resource(s).

0 Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 14.4.3.4.3-0.

1 Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.4.3.5-1, Table 14.4.3.5-2 and Table 14.4.3.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 and the position of neighbour Cell 3 are described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.

12. When T1 expires, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and switch the power setting from T1 to T2 as specified in Table 14.4.3.5-2 and Table 14.4.3.5-3.

13. Wait for the UE monitors for occurrence of the periodic event requested.

14. The UE shall transmit the UE transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 17.

15. The SS transmits an *RRCResume* message.

16. The UE transmits an *RRCResumeComplete* message.

17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for both Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for both Cell 2 and Cell 3 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

18. Repeat steps 2-17 in Tables 14.4.3.4-1 until the confidence level according to Annex D is achieved.

### 14.4.3.4.3 Message contents

**Table 14.4.3.4.3-0: SIB1**

| Derivation Path: TS 38.508-1 [45], clause 4.6.1 with condition SDT |
|---|

**Table 14.4.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.4.3.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.3.4.3-3: CellGroupConfig (Table 14.4.3.4.3-2)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |

| | | | |
|---|---|---|---|
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.3.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.4.3.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.4.3.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-TDOA |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.3.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.3.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.3.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.3.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.3.4.3-7: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| Scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.3.4.3-8: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 14.4.3.4.3-9 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 14.4.3.4.3-9: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.4.3.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.3.4.3-10: RRCRelease**

| |
|---|
| Derivation Path: TS 38.508-1 [45], Table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT |

**Table 14.4.3.4.3-11: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |

| | | | |
|---|---|---|---|
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.4.3.5 Test requirement

Table 14.4.3.5-1, Table 14.4.3.5-2 and Table 14.4.3.5-3 define the primary level settings including test tolerances for the test.

**Table 14.4.3.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ | | MHz | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | Config 1 | | SSB.2 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |

| | | | | |
|---|---|---|---|---|
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.4. FR2 | As specified in TS 38.133 [50] clause A.3. 31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | s | 0.64 | |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 125 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | | Rough | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.4.3.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| NR RF Channel Number | | 1 | 1 | 1 |
| Positiong frequency layer | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | OP.5 FDD | N/A | N/A |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -54.00 | -54.00 | -54.00 |
| SSB RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.4.3.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89.41 | -89.41 | -89.41 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -4.84 | -11.02 | -11.02 |
| Io | Config 1 | dBm/ 19.08MHz | -55.66 | -55.66 | -55.66 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -5.7 | -12.7 | -12.7 |
| PRS-RSRP | | dBm/SCS | -94.2 | -100.4 | -100.4 |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells (all, except Cell 3 in T3) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

The RSTD measurement time fulfils the requirements specified in clause 14.4.3.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.4.3.3 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left(K_{carrier_{PRS}} * N_{Rx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil * \left\lceil \frac{L_{available_{PRS,i}}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{Rx,TEG,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 3$, $L_{available_PRS,i} = 0.035$, $N_{sample} = 4$. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 640ms depending on UE capabilities. Therefore, $T_{available_PRS,i} = 640$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 20.488s and 123.52s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.4.3.4.3-9. The LPP time IE ranges between 21s and 124s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 21.3s and 124.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.4.3.5-4, Table 14.4.3.5-5 and Table 14.4.3.5-6 for reference.

**Table 14.4.3.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |
| n2 | 2 |
| >=n4 | 1 |

**Table 14.4.3.5-5: value of N based on *durationOfPRS-ProcessingSymbols***

| ***durationOfPRS-ProcessingSymbols*** | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >= nDot025 | 1 |

**Table 14.4.3.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |

| | | |
|---|---|---|
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.e

## 14.4.4 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.4.4.1 Test purpose

The purpose of the test is to verify that the RSTD measurement meets the requirements specified in TS 38.133 [50] Clause 5.6.2 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.4.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning, DL-PRS measurement in RRC_INACTIVE and reduced number of samples for PRS measurement in RRC_INACTIVE state.

### 14.4.4.3 Minimum conformance requirements

Same as in clause 14.4.3.3.

### 14.4.4.4 Test description

The supported test configurations in listed in Table 14.4.4.4-1

**Table 14.4.4.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 14.4.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.

2. The general test parameter settings are set up according to Table 14.4.4.5-1, Table 14.4.4.5-2 and Table 14.4.4.5-3.

3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.4.4.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the RSTD reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR2.

### 14.4.4.4.2 Test procedure

Same as in clause 14.4.3.4.2.

### 14.4.4.4.3 Message contents

As defined in clause 14.4.3.4.3, with the following exceptions:

Table 14.4.4.4.3-1 replaces Table 14.4.3.4.3-7 and Table 14.4.4.4.3-2 replaces Table 14.4.3.4.3-9.

**Table 14.4.4.4.3-1: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| Scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| | 10 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.4.4.3-2: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.4.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.4.4.5 Test requirement

**Table 14.4.4.5-1, Table 14.4.4.5-2 and Table 14.4.4.5-3 define the primary level settings including test tolerances for the test.Table 14.4.4.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.2 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| Initial BWP configuration | Config 1 | | DLBWP.0.1<br>ULBWP.0.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |

| | | | | |
|---|---|---|---|---|
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.2 FR2 | As specified in TS 38.133 [50] clause A.3. 31. |
| PRS Resource slot offset | Config 1 | slots | Cell 1, 2, 3: 0 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX cycle length | | s | 0.64 | |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 4 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 32 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | | Setup 1 | As defined in TS 38.133 [50] A.3.15.1 |
| Beam assumption | | | Rough | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.4.4.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| NR RF Channel Number | | 1 | 1 | 1 |
| Positiong frequency layer | | 1 | 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -54.00 | -54.00 | -54.00 |
| SSB RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.4.4.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -95 | -95 | -95 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | 2.9 | 0.75 | 0.75 |
| Io Note 4 | Config 1 | dBm/ 190.08MHz | -55.71 | -55.71 | -55.71 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -2.7 | -5.6 | -5.6 |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that all active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols, except for slots where PRS is transmitted by any of the cells.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Io levels apply only for symbols where PRS is transmitted.

The RSTD measurement time fulfils the requirements specified in clause 14.4.4.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.4.4.3 starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in TS 38.133[50] Clause 10.1.23.3, i.e., between RSTD_0000000 and RSTD_1970049.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period. The RSTD measurement period follows the equation:

$$T_{RSTD,i} = \left(K_{carrier_{PRS}} * N_{Rx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil * \left\lceil \frac{L_{available_{PRS,i}}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{Rx,TEG,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 3$, $L_{available_PRS,i} = 0.035$, $N_{sample} = 1$. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 640ms depending on UE capabilities. Therefore, $T_{available_PRS,i} = 640$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{3}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 1 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 31.36s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 32s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 32.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 14.4.4.5-4, Table 14.4.4.5-5 and Table 14.4.4.5-6 for reference.

**Table 14.4.4.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* | $\left\lceil \frac{3}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |
| n2 | 2 |
| >=n4 | 1 |

**Table 14.4.4.5-5: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >= nDot125 | 1 |

**Table 14.4.4.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |

| | | |
|---|---|---|
| n40 | 640 | 680 |
| n80 | 640 | 720 |
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.4.5 NR RSTD measurement reporting delay test case for PRS aggregation in FR1 SA in RRC_INACTIVE state

Editor's Note: This test case has been completed for the following aspects:

- Test requirement and MU/TT are not complete

### 14.4.5.1 Test purpose

The purpose of the test is to verify that the RSTD measurement with PRS aggregation in RRC_INACTIVE state meets the requirements specified in TS 38.133[50] clause 5.6.2.6 in AWGN propagation condition in FR1 in standalone scenario when two intra-band contiguous positioning frequency layers (PFL) are configured.

### 14.4.5.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Inactive-r18*.

### 14.4.5.3 Minimum conformance requirements

The requirements in this clause apply provided that UE receives requests from LMF to perform PRS measurement on aggregated positioning frequency layers (PFLs) via *nr-DL-PRS-JointMeasurementRequestedPFL-List [49]*.

After receiving both *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from the LMF via LPP [49], the UE shall be able to measure multiple (up to the UE capability specified in TS 38.133[50] clause 5.6.2.3) DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined as:

$$T_{RSTD,total} = T_{RSTD,aggr} + T_{RSTD,non-aggr} + T_{margin}$$

Where:

- $T_{RSTD,aggr}$ is the total measurement period for aggregated measurements as defined in this clause, and
- $T_{RSTD,non-aggr}$ is the total measurement period for non-aggregated measurements as defined in this clause, and
- When both $T_{RSTD,aggr}$ and $T_{RSTD,non-aggr}$ are non-zero, then $T_{margin} = T'_{margin}$; otherwise $T_{margin} = 0$, where $T'_{margin} = \max(T_{effect,j})$, andthe maximum $T_{effect,j}$ is across all the PFL combination(s) and non-aggregated PFL(s) configured for positioning measurements.

$T_{RSTD,non-aggr}$ is zero if every resource set on every PFL is linked for aggregation to at least one other resource set on another PFL. Otherwise, $T_{RSTD,non-aggr}$ is as defined in clause 5.6.2.5 with the following exceptions,

- only PFLs containing resource set(s) not linked to any other resource set(s) are considered in $L$,
- on each PFL $i$, only resource set(s) not linked to any other resource set(s) are considered in $T_{available_PRS,i}$, $N^{slot}_{PRS,i}$ and $L_{available_PRS,i}$,

- $N_{sample}$ = 2 if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *supportedDL-PRS-ProcessingSamples-RRC-Inactive* specified in TS 37.355 [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples,

- $(\mathrm{L}-1)*\max(\mathrm{T}_{\mathrm{effect,i}})$ is replace by $(\mathrm{L}-1)*\mathrm{T}'_{\mathrm{margin}}$.

$\mathrm{T}_{\mathrm{RSTD,aggr}}$ is zero if no resource sets on any PFL are linked for aggregation with other resource sets on other PFLs. Otherwise, $\mathrm{T}_{\mathrm{RSTD,aggr}}$ is defined as

$$\mathrm{T}_{\mathrm{RSTD,aggr}} = \sum_{m=1}^{M} \mathrm{T}_{\mathrm{RSTD,aggr,m}} + (\mathrm{M}-1)*\mathrm{T}'_{\mathrm{margin}} \quad ,$$

where:

- $m$ is the index of PFL combination,

- $M$ is total number of PFL combinations,

- $\mathrm{T}_{\mathrm{effect,aggr,m}}$ is the periodicity of the PRS measurement in PFL combination $m$ as defined in this clause,

- $\mathrm{T}_{\mathrm{RSTD,aggr,m}}$ is the measurement period for PRS RSTD measurement in PFL combination $m$ as specified below.

$$\mathrm{T}_{\mathrm{RSTD,aggr,m}} = \left(\mathrm{K}_{\mathrm{carrier,aggr}} * \mathrm{N}_{\mathrm{Rx,TEG,aggr,m}} * \mathrm{N}_{\mathrm{RxBeam,aggr,m}} * \left\lceil \frac{\mathrm{N}^{\mathrm{slot}}_{\mathrm{PRS,aggr,m}}}{\mathrm{N}'_{\mathrm{aggr,m}}} \right\rceil * \left\lceil \frac{\mathrm{L}_{\mathrm{available}_{\mathrm{PRS}},\mathrm{aggr,m}}}{\mathrm{N}_{\mathrm{aggr,m}}} \right\rceil * \mathrm{N}_{\mathrm{sample}} - 1\right) * \mathrm{T}_{\mathrm{effect,aggr,m}} + \mathrm{T}_{\mathrm{last,aggr,m}} \ ,$$

where:

- $\mathrm{K}_{\mathrm{carrier,aggr}}$ is a scaling factor for PRS measurements in RRC_INACTIVE,

  - $\mathrm{K}_{\mathrm{carrier,aggr}} = 1$ if the UE supports *parallelPRS-MeasRRC-Inactive-r17*,

  - Otherwise,

    - If Srxlev $\leq \mathrm{S}_{\mathrm{nonIntraSearchP}}$ or Squal $\leq \mathrm{S}_{\mathrm{nonIntraSearchQ}}$, $\mathrm{K}_{\mathrm{carrier,aggr}} = K_{\mathrm{carrier}} + 1$, where $K_{\mathrm{carrier}}$ is defined in clause 4.2.2.4

    - If Srxlev > $\mathrm{S}_{\mathrm{nonIntraSearchP}}$ and Squal > $\mathrm{S}_{\mathrm{nonIntraSearchQ}}$, $\mathrm{K}_{\mathrm{carrier,aggr}} = N_{\mathrm{layers}} + 1$, where $N_{\mathrm{layers}}$ is defined in clause 4.2.2.7.

  - $\mathrm{N}_{\mathrm{Rx,TEG,aggr,m}}$ is a scaling factor for PRS measurements with multiple Rx TEGs

    - $\mathrm{N}_{\mathrm{Rx,TEG,aggr,m}}$ =1 if the UE is not configured by the LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49].

    - $\mathrm{N}_{\mathrm{Rx,TEG,aggr,m}}$ is defined as follows if the UE is configured by the LMF with *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] to perform measurement on same DL PRS resource of a TRP using different Rx TEGs in *NR-DL-TDOA-RequestLocationInformation* [49]:

      - $N_{Rx,TEG,i} = P$, if the UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, where P is the number of UE Rx TEGs that the UE is requested by LMF to measure the same DL-PRS Resource of a TRP indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in [34], and in case 'n0' is indicated, P is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.

      - $N_{Rx,TEG,i} = \left\lceil \frac{P}{Q} \right\rceil$, if the UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs, where $Q$ is the number of UE Rx TEGs for measuring the same DL-PRS Resource simultaneously indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul-r17* in [34].

  - $\mathrm{N}_{\mathrm{RxBeam,aggr,m}}$ is a scaling factor for PRS measurements with multiple Rx beams, and is defined as

    - $\mathrm{N}_{\mathrm{RxBeam,aggr,m}}$ = 1 if PFL combination $m$ is in FR1,

- $N_{RxBeam,aggr,m}$ is defined as follows if PFL combination *m* is in FR2
  - $N_{RxBeam,aggr,m}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing PFL combination *m*, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*,
  - $N_{RxBeam,aggr,m}$ equals to 8 otherwise

- $N^{slot}_{PRS,aggr,m}$ is the maximum number of DL PRS resources in PFL combination *m* configured in a slot, and only the PRS resources in resource set(s) linked to other resource set in PFL combination *m* are counted

- $N'_{aggr,m}$ is the UE capability on maximum number of DL PRS resources that can be processed in a slot for PFL combination *m* as indicated by *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR1-r18* for FR1 and *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR2-r18* for FR2 specified in TS 37.355 [34].

- $L_{available_{PRS},aggr,m}$ is the time duration of available PRS resources in PFL combination *m* to be measured during $T_{PRS,aggr,m}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_{PRS},aggr,m}$, only unmuted PRS resources in resource set(s) linked to other resource set in PFL combination *m* and that are not fully overlapped with other higher-priority DL signals/channels are considered.

- $N_{aggr,m}$ is the UE capability on duration of DL PRS resources in ms for PFL combination *m* as indicated by *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsN-r18* specified in TS 37.355 [49].

- $N_{sample}$ is number of PRS measurement samples,
  - $N_{sample}$= 2 if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *reducedNumOfSampleInMeasurementWithPRS-BWA-RRC-IdleAndInactive* specified in TS 37.355 [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples,
  - $N_{sample}$= 4 otherwise.

- $T_{effect,aggr,m}$ is the periodicity of the PRS measurement in PFL combination $m$,

$$T_{effect,aggr,m} = \left\lceil \frac{T_{aggr,m}}{T_{available,PRS,aggr,m}} \right\rceil * T_{available,PRS,aggr,m}$$

  - $T_{aggr,m}$ is the UE capability on time for processing of DL PRS resources in ms for PFL combination *m* as indicated by *prs-durationOfTwoPRS-BWA-ProcessingSymbolsT-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsT* specified in TS 37.355 [49].
  - $T_{available,PRS,aggr,m} = LCM(T_{PRS,aggr,m}, T_{DRX})$, the least common multiple between $T_{PRS,aggr,m}$ and the DRX cycle length $T_{DRX}$, where $T_{PRS,aggr,m}$ is the periodicity of DL PRS resource with muting on PFL combination $m$.
    - If more than one PRS periodicities are configured in PFL combination $m$, the least common multiple of PRS periodicities $T_{per}^{\text{PRS with muting}}$ among all DL PRS resource sets that are linked to other resource set in PFL combination $m$, is used to derive $T_{PRS,aggr,m}$, and for each applicable PRS resource set,
      - $T_{per}^{\text{PRS with muting}} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource, and
      - $T_{per}^{PRS}$ is the periodicity of PRS resource set given by the higher-layer parameter *DL-PRS-Periodicity*, and
      - $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- $T_{last,aggr,m}$ is the measurement duration for the last PRS sample in PFL combination $m$, including the sampling time and processing time, $T_{last,aggr,m} = T_{aggr,m} + T_{available,PRS,aggr,m}$.

The requirements in this clause for aggregated measurements apply provided that the linked PRS resource sets on multiple PFLs for aggregated measurements are transmitted by the TRP using single Tx chain as defined in clause 5.1.6.5.3 in TS 38.214 [26].

If PRS resources in one or more of the aggregated PFLs in PFL combination $m$ are dropped because of collision with other signals, $T_{RSTD,aggr,m}$ can be longer than specified above.

The time $T_{RSTD,Total}$ starts from the first DRX cycle containing the DL PRS resource(s) in the assistance data after both the *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the UE via LPP [34].

NOTE: No separate requirement on non-aggregated measurement based on $T_{RSTD,non-aggr}$ or on aggregated measurement based on $T_{RSTD,aggr}$ is applied when both $T_{RSTD,aggr}$ and $T_{RSTD,non-aggr}$ are non-zero.

If the DRX cycle is reconfigured during the RSTD measurement period, then the measurement period can be longer.

When PRS-RSRP is configured for DL-TDOA, RSTD and PRS-RSRP are performed over the same measurement period. When PRS-RSRPP is configured for DL-TDOA, RSTD and PRS-RSRPP are performed over the same measurement period.

The measurement requirements do not apply to any PRS resource that always collides with other higher-priority DL signals/channels, as specified in TS 38.133[50] clause 5.6.1.

Longer RSTD measurement period is expected when there are collisions between PRS resources and other higher-priority DL signals/channels.

If $K_{carrier_PRS}$ changes for any PFL or any PFL combination during the measurement period, the measurement period could be longer.

The measurement requirements do not apply for a PRS resource, if the PRS resource is across two sampling duration of N within duration $L_{available_PRS,i}$.

The measurement requirements do not apply for a PRS resource, if time span of the PRS resource instance (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability N.

The requirements in TS 38.133[50] clause 5.6.2 do not apply if the PRS configuration given by higher layer parameters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-TDOA-ProvideCapabilities*, and it is up to UE implementation which PRS resources are measured, subject to UE measurement capabilities.

If cell re-selection occurs while RSTD measurements are being performed, then the UE shall continue and complete the on-going RSTD measurements after the cell selection is completed. The RSTD measurement period can be longer.

If the RRC state transition occurs from RRC_INACTIVE to RRC_CONNECTED state during the RSTD measurement period, then the UE shall continue the RSTD measurement in the RRC_CONNECTED state. The RSTD measurement period can be longer.

The UE shall meet the RSTD measurement accuracy requirements in TS 38.133[50] clause 10.1.23A.2.

The normative reference for this requirement is TS 38.133 [50] clause 5.6.2.6.

## 14.4.5.4 Test description

### 14.4.5.4.1 Initial conditions

The supported test configuration is listed in Table 14.4.5.4.1-1.

**Table 14.4.5.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|

| | |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.5.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.16.
2. The general test parameter settings are set up according to Table 14.4.5.5-1, Table 14.4.5.5-2 and Table 14.4.5.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.4.5.4.3.
5. In the test there are 6 synchronous cells in the test: Cell 1, Cell 2, Cell 3 Cell 4, Cell 5 and Cell 6. Cell 1 is the PCell on NR RF channel 1 in FR1. Cell 2 and Cell 3 are neighbour cells on the same RF channel as Cell 1. Cell 4, Cell 5 and Cell 6 are the neighbour cells on a different NR RF channel, i.e., RF channel 2, in FR1.

### 14.4.5.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2, Cell 3, Cell 4, Cell 5 and Cell 6. During T2 UE shall be in RRC_INACTIVE state and all 6 cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-DL-TDOA-ProvideAssistanceData* and *nr-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49], clause 6.5.12.1, shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T = 50$ ms is the maximum processing time of the *DL-TDOA assistance* data and location information request.

The beginning of the time interval T2 shall be aligned with the first DRX cycle containing a DL PRS resource(s).

The UE is configured with DRX cycle of 1.28s.

0. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 14.4.5.4.3-0.
1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.4.5.5-1, Table 14.4.5.5-2 and Table 14.4.5.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T = 50$

ms. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *nr-DL-PRS-JointMeasurementRequestedPFL-List* in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.

11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message

12. When T1 expires, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and switch the power setting from T1 to T2 as specified in Table 14.4.5.5-2 and Table 14.4.5.5-3.

13. Wait for the UE monitors for occurrence of the periodic event requested

14. The UE shall transmit an RRCResumeRequest message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 17

15. The SS transmits an RRCResume message.

16. The UE transmits an RRCResumeComplete message

17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements by aggregating PRS resources from multiple PFLs for Cell 2 and Cell 5, Cell 3 and Cell 6 respectively with respect to the Cell 1 and Cell 4 from which the transmitted PRS resources are also aggregated in the DL-TDOA assistance data.

If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for Cell 2 and Cell 5, Cell 3 and Cell 6 respectively within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

18. If the UE message at step 17 includes the ackRequested IE set to TRUE, the SS shall send a LPP acknowledgement message.

19. Repeat steps 2-18 in Table 14.4.5.4.1-1 until the confidence level according to Annex D is achieved.

### 14.4.5.4.3 Message contents

**Table 14.4.5.4.3-0: SIB1**

| Derivation Path: TS 38.508-1 [45], clause 4.6.1 with condition SDT |
|---|

**Table 14.4.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.4.5.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.4.5.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.5.4.3-4: CellGroupConfig (Table 14.4.5.4.3-2)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms1280 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.5.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.4.5.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| nr-PeriodicAssistData-r18 | Not present | | |
| } | | | |
| bt-ProvideAssistanceData-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.5.4.3-6: NR-DL-PRS-AssistanceData (Table 14.4.5.4.3-5)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Configuration 1 and Configuration 3 |
| | 21 | 104 PRBs | Configuration 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Configuration 1 and Configuration 3 |
| | n4 | | Configuration 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.5.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.5.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.5.4.3-7 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Configuration 1 and Configuration 3 |
| | 21 | 104 PRBs | Configuration 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 4 | | |

| | | | |
|---|---|---|---|
| dl-PRS-CombSizeN-r16 | n2 | | Configuration 1 and Configuration 3 |
| | n4 | | Configuration 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 4 |
| dl-PRS-ID-r16 | 3 | | |
| nr-PhysCellID-r16 | Cell 4 | | |
| nr-CellGlobalID-r16 | Cell 4 | | |
| nr-ARFCN-r16 | Cell 4 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.5.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 5 |
| dl-PRS-ID-r16 | 4 | | |
| nr-PhysCellID-r16 | Cell 5 | | |
| nr-CellGlobalID-r16 | Cell 5 | | |
| nr-ARFCN-r16 | Cell 5 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.5.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 3 | Cell 6 |
| dl-PRS-ID-r16 | 5 | | |
| nr-PhysCellID-r16 | Cell 6 | | |
| nr-CellGlobalID-r16 | Cell 6 | | |
| nr-ARFCN-r16 | Cell 6 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.5.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 3 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 3 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 4 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[3] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 3 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 5 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.5.4.3-7: NR-DL-PRS-Info (Table 14.4.5.4.3-6)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |

| | | | |
|---|---|---|---|
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Configuration 1 and Configuration 3 |
| | n1 | | Configuration 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 4 |
| | 01 | | Cell 2 and Cell 5 |
| | 10 | | Cell 3 and Cell 6 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-AssistanceDataPerTRP-r16[3] |
| | 3 | | NR-DL-PRS-AssistanceDataPerTRP-r16[4] |
| | 4 | | NR-DL-PRS-AssistanceDataPerTRP-r16[5] |
| | 5 | | NR-DL-PRS-AssistanceDataPerTRP-r16[6] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Cell 1 and Cell 4 |
| | 0 | | Cell 2 and Cell 5 |
| | 1 | | Cell 3 and Cell 6 |

| | | | |
|---|---|---|---|
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Configuration 1 and Configuration 3 |
| | 4 | | Configuration 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.5.4.3-8: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 14.4.5.4.3-9 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 14.4.5.4.3-9: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |

| | | | |
|---|---|---|---|
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.4.5.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRP-Pair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.5.4.3-10: RRCRelease**

| |
|---|
| Derivation Path: TS 38.508-1 [45], Table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT |

**Table 14.4.5.4.3-11: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 6 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |

| | | | |
|---|---|---|---|
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 4 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 4 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 5 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 5 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 6 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 6 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not Present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not Present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not Present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

### 14.4.5.5 Test requirement

Table 14.4.5.5-1 to Table 14.4.5.5-3 defines the primary level settings including test tolerances for the test.

**Table 14.4.5.5-1: General test parameters for RSTD measurement with PRS aggregation reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1, Cell 4 | Reference cells are the cells in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell 1 is the PCell in this test case. |
| Neighbor cells | | | Cell 2, Cell 3, Cell 4, Cell 5 and Cell 6 | Cell 2, Cell 3, Cell 4, Cell 5 and Cell 6 appear at the first, second, third, fourth and fifth places in the neighbour cell list in the DL-TDOA assistance data. |
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.2 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| Initial BWP configuration | Config 1,2,3 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Active UL BWP configuration | Config 1,2,3 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.1 FR1 | As specified in TS 38.133 [50] clause A.3.31 |
| | Config 2 | | PRS.1.2 FR1 | |
| | Config 3 | | PRS.2.1 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 4)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 5)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by then test parameters |

| | | | |
|---|---|---|---|
| | | and<br>(PCI of Cell 1 – PCI of Cell 6)mod6=0 | |
| CP length | | Normal | |
| DRX | s | 1.28 | |
| Radio frame receive time offset between the cells at the UE antenna connector | µs | Cell 2, Cell 4, Cell 5 to Cell 1: 0<br>Cell 3, Cell 6 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | µs | Cell 2 to Cell 6: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | 6 | Including the reference cells |
| PRS muting info | | Cell 1, Cell 4: ‘10’<br>Cell 2, Cell 5: ‘01’<br>Cell 3, Cell 6: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | Cell 1, Cell 4: 0<br>Cell 2, Cell 5: 0<br>Cell 3, Cell 6: 1 | Cell 1, Cell 2, Cell 3, Cell 4, Cell 5 and Cell 6 |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 5 | The length of the time interval that follows immediately after time interval T1. |

**Table 14.4.5.5-2: Cell-specific test parameters for RSTD measurement with PRS aggregation reporting delay during T1**

| Parameter | | Unit | Cell 1, Cell 4 | Cell 2, Cell 5 | Cell 3, Cell 6 |
|---|---|---|---|---|---|
| NR RF Channel Number # | | | 1, 2 | 1, 2 | 1, 2 |
| Positiong frequency layer | | | PFL 1, 2 | PFL 1, 2 | PFL 1, 2 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in clause A.3.2.1 Note 1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3, 5 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ Note 5 | | dB | 10 | -Infinity | -Infinity |
| Io Note 4, 5 | Config 1 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 2 | dBm/ 19.08MHz | -56.54 | -56.54 | -56.54 |
| | Config 3 | dBm/ 47.88MHz | -52.56 | -52.56 | -52.56 |
| SSB RP Note4, 5 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.
Note 5: The values are defined separately for the cells from which the transmitted PRS resources are aggregated.

**Table 14.4.5.5-3: Cell-specific test parameters for RSTD measurement with PRS aggregation reporting delay during T2**

| Parameter | | Unit | Cell 1, Cell 4 | Cell 2, Cell 5 | Cell 3, Cell 6 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number # | | | 1, 2 | 1, 2 | 1, 2 |
| Positiong frequency layer | | | PFL 1, 2 | PFL 1, 2 | PFL 1, 2 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in clause A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3, 4 | Config 1 | dBm/SCS | -98 + TT | -98 + TT | -98 + TT |
| | Config 2 | dBm/SCS | -98 + TT | -98 + TT | -98 + TT |
| | Config 3 | dBm/SCS | -95 + TT | -95 + TT | -95 + TT |
| PRS $\hat{E}_s/N_{oc}$ Note 4 | Config 1 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| | Config 2 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| | Config 3 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| Io Note 3, 4 | Config 1 | dBm/ 19.08MHz | -65.43 + TT | -65.43 + TT | -65.43 + TT |
| | Config 2 | dBm/ 96.48MHz | -65.43 + TT | -65.43 + TT | -65.43 + TT |
| | Config 3 | dBm/ 47.88MHz | -61.44 + TT | -61.44 + TT | -61.44 + TT |
| PRS $\hat{E}_s/I_{ot}$ Note 4 | | dB | -6 + TT | -13 + TT | -13 + TT |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: The values are defined separately for the cells from which the transmitted PRS resources are aggregated.

The RSTD measurement time with PRS aggregation in RRC_INACTIVE state fulfils the requirements specified in clause 14.4.5.3.

The UE shall perform and report the RSTD measurements by aggregating PRS resources from Cell 2 and Cell 5, Cell 3 and Cell 6 respectively with respect to the Cell 1 and Cell 4 from which the transmitted PRS resources are also aggregated, within the time duration specified in section 5.6.2.6 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be higher than the time duration above because of the uncertainty in acquiring the first available PRACH occasion to transition to RRC_CONNECTED state to report the measurements.

The rate of the correct events observed during repeated tests shall be at least 90%, where the reported RSTD measurement with PRS aggregation for each correct event shall be within the RSTD reporting range specified in TS 38.133 [50] clause 10.1.23A.3.

## 14.4.6 NR RSTD measurement reporting delay test case for PRS aggregation in FR2 SA in RRC_INACTIVE state

Editor's Note: This test case has been completed for the following aspects:

- Test requirement and MU/TT are not complete

### 14.4.6.1 Test purpose

The purpose of the test is to verify that the RSTD measurement with PRS aggregation in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.2.6 in AWGN propagation condition in FR2 in standalone scenario when two intra-band contiguous positioning frequency layers (PFL) are configured.

### 14.4.6.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Inactive-r18*.

### 14.4.6.3 Minimum conformance requirements

Same as in clause 14.4.5.3.

### 14.4.6.4 Test description

#### 14.4.6.4.1 Initial conditions

The supported test configuration is listed in Table 14.4.6.4.1-1.

**Table 14.4.6.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.6.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.4.6.5-1, Table 14.4.6.5-2 and Table 14.4.6.5-3.
3. Propagation conditions are set according to clause 4.15.2.

4. Message contents are defined in clause 14.4.6.4.3.

5. In the test there are 6 synchronous cells in the test: Cell 1, Cell 2, Cell 3 Cell 4, Cell 5 and Cell 6. Cell 1 is the PCell on NR RF channel 1 in FR2. Cell 2 and Cell 3 are neighbour cells on the same RF channel as Cell 1. Cell 4, Cell 5 and Cell 6 are the neighbour cells on a different NR RF channel, i.e., RF channel 2, in FR2.

### 14.4.6.4.2 Test procedure

Same test procedure as defined in clause 14.4.5.4.2 with following exceptions:

Table 14.4.6.4.1-1 replaces Table 14.4.5.4.1-1; Table 14.4.6.5-1 replaces 14.4.5.5-1; Table 14.4.6.5-2 repalces Table 14.4.5.5-2; Table 14.4.6.5-3 replaces Table 14.4.5.5-3.

### 14.4.6.4.3 Message contents

Same message contents as defined in clause 14.4.5.4.3, with the following exceptions:

Table 14.4.6.4.3-0 replaces Table 14.4.5.4.3-4; Table 14.4.6.4.3-1 replaces Table 14.4.5.4.3-6; Table 14.4.6.4.3-2 replaces Table 14.4.5.4.3-7; Table 14.4.6.4.3-3 replacesTable 14.4.5.4.3-9.

**Table 14.4.6.4.3-0: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.6.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.6.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.6.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.6.4.3-2 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined | | |

| | | | |
|---|---|---|---|
| | for the DL frequency of the Cell 4 | | |
| | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 4 |
| dl-PRS-ID-r16 | 3 | | |
| nr-PhysCellID-r16 | Cell 4 | | |
| nr-CellGlobalID-r16 | Cell 4 | | |
| nr-ARFCN-r16 | Cell 4 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.6.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 5 |
| dl-PRS-ID-r16 | 4 | | |
| nr-PhysCellID-r16 | Cell 5 | | |
| nr-CellGlobalID-r16 | Cell 5 | | |
| nr-ARFCN-r16 | Cell 5 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.6.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 3 | Cell 6 |
| dl-PRS-ID-r16 | 5 | | |
| nr-PhysCellID-r16 | Cell 6 | | |
| nr-CellGlobalID-r16 | Cell 6 | | |
| nr-ARFCN-r16 | Cell 6 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.6.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 3 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 3 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 4 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[3] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 3 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 5 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.6.4.3-2: NR-DL-PRS-Info (Table 14.4.6.4.3-1)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| po2-r16 | 10 | | Cell 1 and Cell 4 |
| | 01 | | Cell 2 and Cell 5 |
| | 10 | | Cell 3 and Cell 6 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-AssistanceDataPerTRP-r16[3] |
| | 3 | | NR-DL-PRS-AssistanceDataPerTRP-r16[4] |
| | 4 | | NR-DL-PRS-AssistanceDataPerTRP-r16[5] |
| | 5 | | NR-DL-PRS-AssistanceDataPerTRP-r16[6] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 4 |
| | 0 | | Cell 2 and Cell 5 |
| | 1 | | Cell 3 and Cell 6 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.6.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.4.6.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |

| | | | |
|---|---|---|---|
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRP-Pair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.4.6.5 Test requirement

Table 14.4.6.5-1 to Table 14.4.6.5-3 defines the primary level settings including test tolerances for the test.

**Table 14.4.6.5-1: General test parameters for RSTD measurement with PRS aggregation reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1, Cell 4 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2, Cell 3, Cell 4, Cell 5 and Cell 6 | Cell 2, Cell 3, Cell 4, Cell 5 and Cell 6 appear at the first, second, third, fourth and fifth places in the neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ | | MHz | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | Config 1 | | SSB.2 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |

| | | | | |
|---|---|---|---|---|
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.4. FR2 | As specified in TS 38.133 [50]clause A.3. 31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 and (PCI of Cell 1 – PCI of Cell 4)mod6=0 and (PCI of Cell 1 – PCI of Cell 5)mod6=0 and (PCI of Cell 1 – PCI of Cell 6)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | s | 0.64 | |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2, Cell 4, Cell 5 to Cell 1: 0<br>Cell 3, Cell 6 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2 to 6: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 6 | Including the reference cells |
| PRS muting info | | | Cell 1, Cell 4: ‘10’<br>Cell 2, Cell 5: ‘01’<br>Cell 3, Cell 6: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1, Cell 4: 0<br>Cell 2, Cell 5: 0<br>Cell 3, Cell 6: 1 | |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | | Setup 1 | As defined in TS 38.133 [50]A.3.15.1 |
| Beam assumption | | | Rough | Information about types of UE beam is given in TS 38.133 [50]clause B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.4.6.5-2: Cell-specific test parameters for RSTD measurement with PRS aggregation reporting delay during T1**

| Parameter | | Unit | Cell 1, Cell 4 | Cell 2, Cell 5 | Cell 3, Cell 6 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1, 2 | 1, 2 | 1, 2 |
| Positiong frequency layer | | | PFL 1, 2 | PFL 1, 2 | PFL 1, 2 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 Note 1 | | | OP.5 FDD | N/A | N/A |
| $N_{oc}$ Note 3, 5 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ Note 5 | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4, 5 | Config 1 | dBm/ 190.08MHz | -54.00 | -54.00 | -54.00 |
| SSB RP Note4, 5 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ Note 5 | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.
Note 5: The values are defined separately for the cells from which the transmitted PRS resources are aggregated.

**Table 14.4.6.5-3: Cell-specific test parameters for RSTD measurement with PRS aggregation reporting delay during T2**

| Parameter | | Unit | Cell 1, Cell 4 | Cell 2, Cell 5 | Cell 3, Cell 6 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| RF Channel Number | | | 1, 2 | 1, 2 | 1, 2 |
| Positiong frequency layer | | | PFL 1, 2 | PFL 1, 2 | PFL 1, 2 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in TS 38.133 [50] A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3, 4 | Config 1 | dBm/SCS | -89 + TT | -89 + TT | -89 + TT |
| PRS $\hat{E}_s/N_{oc}$ Note 4 | Config 1 | dB | -5.44 + TT | -11.67 + TT | -11.67 + TT |
| Io Note 4 | Config 1 | dBm/ 19.08MHz | -55.48 + TT | -55.48 + TT | -55.48 + TT |

| | | | | |
|---|---|---|---|---|
| PRS $\hat{E}_s/I_{ot}$ Note 4 | dB | -6 + TT | -13 + TT | -13 + TT |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
Note 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
Note 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: The values are defined separately for the cells from which the transmitted PRS resources are aggregated.

## 14.4.7 NR RSTD measurement reporting delay test case in FR1 for Redcap UE without Rx FH

Editor's Note: This test case is incomplete in following aspects:

- TT analysis is missing.

### 14.4.7.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC INACTIVE state meets the requirements specified in TS 38.331 [50] clause 5.6A.4.5 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.4.7.2 Test applicability

This test applies to NR FR1 RedCap UEs release 18 onwards that support DL-TDOA positioning.

### 14.4.7.3 Minimum conformance requirements

After receiving both *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message from the LMF via LPP [34], and UE is not configured to perform measurement with RX FH, the UE shall be able to measure multiple (up to the UE capability specified in clause 5.6A.4.3) DL RSTD measurements, defined in TS 38.215 [57], during the measurement period $T_{RSTD,Total}$ defined as:

$$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD,i} + (L-1) \times \max(T_{effect,i})$$

Where:

- $i$ is the index of positioning frequency layer,
- $L$ is total number of positioning frequency layers, and
- $T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$T_{RSTD,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer $i$ as specified below:

$$T_{RSTD,i} = \left(K_{carrier_PRS_RedCap} \times N_{Rx,TEG,i} \times N_{RxBeam,i} \times \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \times \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil \times N_{sample} - 1\right) \times T_{effect,i} + T_{last,i},$$

Where:

- $N_{RxBeam,i}$ is the UE Rx beam sweeping factor:
  - $N_{RxBeam,i}$ = 1 if positioning frequency layer $i$ is in FR1 or UE has only 1Rx branch,

- $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2*, if positioning frequency layer *i* is in FR2 or the UE has 2Rx branches, and the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*.

- $N_{RxBeam,i}$ equals to 8, otherwise.

- $K_{\text{carrier_PRS_RedCap}}$ is a scaling factor for PRS-based NR positioning measurements in RRC_INACTIVE. If the UE supports *parallelPRS-MeasRRC-Inactive-r17*, $K_{\text{carrier_PRS_RedCap}}$ = 1; otherwise,

  - If Srxlev ≤ $S_{\text{nonIntraSearchP}}$ or Squal ≤ $S_{\text{nonIntraSearchQ}}$, $K_{\text{carrier_PRS_RedCap}} = K_{\text{carrier_RedCap}} + 1$, where $K_{\text{carrier_RedCap}}$ is defined in clause 4.2B.2.4

  - If Srxlev > $S_{\text{nonIntraSearchP}}$ and Squal > $S_{\text{nonIntraSearchQ}}$, $K_{\text{carrier_PRS_RedCap}} = N_{\text{layers}} + 1$, where $N_{\text{layers}}$ is defined in TS 38.133 [50] clause 4.2.2.7.

- $N_{Rx,TEG,i}$ is the Rx TEG specific scaling factor:

  - $N_{Rx,TEG,i}$ =1 if the UE is not configured by the LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49].

  - $N_{Rx,TEG,i}$ is defined as follows if the UE is configured by the LMF with *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] to perform measurement on same DL PRS resource of a TRP using different Rx TEGs in *NR-DL-TDOA-RequestLocationInformation* [49]:

    - $N_{Rx,TEG,i} = P$, if the UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, where P is the number of UE Rx TEGs that the UE is requested by LMF to measure the same DL-PRS Resource of a TRP indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in [34], and in case 'n0' is indicated, P is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.

    - $N_{Rx,TEG,i} = \left\lfloor \frac{P}{Q} \right\rfloor$, if the UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs, where $Q$ is the number of UE Rx TEGs for measuring the same DL-PRS Resource simultaneously indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul-r17* in [49].

- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources in positioning frequency layer *i* configured in a slot.

- $L_{available_PRS,\text{i}}$ is the time duration of available PRS in positioning frequency layer *i* to be measured $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,\text{i}}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.

- $N_{sample}$ is the number of PRS RSTD samples, where

  - $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 4 otherwise.

- $T_{last,i}$ is the measurement duration for the last PRS RSTD sample in positioning frequency layer *i*, including the sampling time and processing time, $T_{\mathrm{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ ,

- $T_{\mathrm{effect},i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{\mathrm{effect,i}} = \left\lceil \frac{T_{\mathrm{i}}}{T_{available_PRS,\mathrm{i}}} \right\rceil \times T_{available_PRS,\mathrm{i}}$$

where:

- $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [49],

- $T_{available_PRS,\mathrm{i}} = LCM(T_{PRS,\mathrm{i}}, T_{DRX})$, the least common multiple between $T_{PRS,\mathrm{i}}$ and the DRX cycle length $T_{DRX}$

  - When UE is configured with RAN eDRX_INACTIVE ≤ 10.24 s:

    $T_{DRX}$ is calculated as T defined in TS 38.304 [1] when CN eDRX_INACTIVE ≤ 10.24 s.

    $T_{DRX}$ is calculated as max($T_{inside}$, $T_{outside}$), where $T_{inside}$ and $T_{outside}$ of the CN PTW as defined in TS 38.304 [1].

  - When UE is configured with RAN eDRX_INACTIVE > 10.24 s:

    $T_{DRX}$ is calculated as max($T_{DRX_RAN}$, $T_{DRX_CN}$), $T_{DRX_RAN}$ and $T_{DRX_CN}$ are DRX cycles with the RAN PTW and CN PTW defined in TS 38.304 .

  Otherwise, $T_{DRX}$ is the DRX cycle of the UE in the serving cell.

- $T_{PRS,\mathrm{i}}$ is the periodicity of DL PRS resource with muting on positioning frequency layer *i*.

If more than one PRS periodicities are configured in positioning frequency layer *i*, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets in the positioning frequency layer is used to derive $T_{PRS,\mathrm{i}}$, where,

- $T_{per}^{PRS\ with\ muting} = N_{muting} \times T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource,

- $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

- $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} \times L_{muting}$, where

- $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- $\{N, T\}$ is the UE capability combination per band for RRC_INACTIVE state where N is the duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [34], T (ms) corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [49], and T-N (>0) is the time required to process duration N of DL PRS symbols already buffered in memory, for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49] .

- $N'$ is UE capability for number of DL PRS resources that it can process in a slot in RRC_INACTIVE state as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* specified in TS 37.355 [49].

If the following conditions are met, the time $T_{\mathrm{RSTD,Total}}$ starts from the first DL PRS resource(s) instances inside a PTW after both the *NR-DL-TDOA-ProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the UE via LPP [49].

- UE is configured with RAN eDRX > 10.24 s, and

- periodic PRS measurement reporting is configured, and

- RAN eDRX cycle is smaller or equal to the PRS measurement reporting periodicity configured via higher layer parameter *reportingInterval* in TS 37.355 [49], and

- there is one or more PRS resources occurring in PTW.

Otherwise, the time $T_{RSTD,Total}$ starts from the first DL PRS resource(s) after both the *NR-DL-TDOAProvideAssistanceData* message and *NR-DL-TDOA-RequestLocationInformation* message are delivered from LMF to the UE via LPP [49].

NOTE: No per-positioning frequency layer requirement is applied in scenarios when multiple positioning frequency layers are configured.

If the DRX cycle is reconfigured during the RSTD measurement period, then the measurement period can be longer.

If eDRX_INACTIVE cycle is reconfigured during the RSTD measurement period, then the measurement period can be longer.

When PRS-RSRP is configured for DL-TDOA, RSTD and PRS-RSRP are performed over the same measurement period.

When PRS-RSRPP is configured for DL-TDOA, RSTD and PRS-RSRPP are performed over the same measurement period.

The measurement requirements do not apply to any PRS resource that always collides with other higher-priority DL signals/channels, as specified in TS 38.133 [50] clause 5.6A.1.

Longer RSTD measurement period is expected when there are collisions between PRS resources and other higher-priority DL signals/channels.

If $K_{\text{carrier_PRS_RedCap}}$ changes for any PFL during the measurement period, the measurement period could be longer.

The measurement requirements do not apply for a PRS resource, if the PRS resource is across two sampling duration of N within duration $L_{available_PRS,i}$.

The measurement requirements do not apply for a PRS resource, if time span of the PRS resource instance (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability N.

The requirements in TS 38.133 [50] clause 5.6A.4.5 do not apply if the PRS configuration given by higher layer parameters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-TDOA-ProvideCapabilities*, and it is up to UE implementation which PRS resources are measured, subject to UE measurement capabilities.

If cell re-selection occurs while RSTD measurements are being performed, then the UE shall continue and complete the on-going RSTD measurements after the cell selection is completed. The RSTD measurement period can be longer.

If the RRC state transition occurs from RRC_INACTIVE to RRC_CONNECTED state during the RSTD measurement period, then the UE shall continue the RSTD measurement in the RRC_CONNECTED state. The RSTD measurement period can be longer.

The normative reference for this requirement is TS 38.133 [50] clause 5.6A.4.5.

## 14.4.7.4 Test description

### 14.4.7.4.1 Initial conditions

The supported test configurations are listed in

**Table 14.4.7.4.1-1**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.7.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 14.4.7.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.4.7.4.3.
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR1.

### 14.4.7.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2 and Cell 3. During T2 UE shall be in RRC_INACTIVE state and all three cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-DL-TDOA-ProvideAssistanceData* and *nr-DL-TDOA-RequestLocationInformation* as defined in TS 37.355 [49], clause 6.5.12.1, shall be provided to the RedCap UE during T1

The beginning of the time interval T2 shall be aligned with the first DRX cycle containing a DL PRS resource(s).

The UE is configured with DRX cycle of 1.28s.

0. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 14.4.5.4.3-0.
1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.4.7.5-1, Table 14.4.7.5-2 and Table 14.4.7.5-3. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message
12. When T1 expires, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and switch the power setting from T1 to T2 as specified in Table 14.4.7.5-3.

13. Wait for the UE monitors for occurrence of the periodic event requested

14. The UE shall transmit an RRCResumeRequest message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 17

15. The SS transmits an RRCResume message.

16. The UE transmits an RRCResumeComplete message

17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3). The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 respectively with respect to the Cell 1 from which the transmitted PRS resources are also aggregated in the DL-TDOA assistance data.

If the UE transmits an *NR-DL-TDOA-ProvideLocationInformation* IE including the nr-RSTD field for Cell 2and Cell 3 respectively within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-TDOA-ProvideLocationInformation* IE with both the nr-RSTD fields included within the response time then the number of failure tests is increased by one.

18. If the UE message at step 17 includes the ackRequested IE set to TRUE, the SS shall send a LPP acknowledgement message.

19. Repeat steps 2-18 until the confidence level according to Annex D is achieved.

#### 14.4.7.4.3 Message contents

**Table 14.4.7.4.3-1: SIB1**

| Derivation Path: TS 38.508-1 [45], clause 4.6.1 with condition SDT |
|---|

**Table 14.4.7.4.3-2: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 1 | DL-TDOA | |

**Table 14.4.7.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.7.4.3-4: CellGroupConfig (Table 14.4.7.4.3-3)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |

| | | | |
|---|---|---|---|
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms1280 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.7.4.3-5: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-TDOA-RequestCapabilities-r16 | TRUE |

**Table 14.4.7.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 | Not present | | |
| tbs-ProvideAssistanceData-r14 | Not present | | |
| wlan-ProvideAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 14.4.7.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| nr-PeriodicAssistData-r18 | Not present | | |
| } | | | |
| bt-ProvideAssistanceData-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 14.4.7.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1, Config 2 and Config 4 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| | | | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.7.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.7.4.3-8 | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.7.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.7.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1, Config 2 and Config 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 2 |
| | 01 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |

| | | | |
|---|---|---|---|
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-AssistanceDataPerTRP-r16[3] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.7.4.3-9: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 14.4.7.4.3-10 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 14.4.7.4.3-10: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 14.4.7.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 0 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRP-Pair-r16 | Not present | | |

| | | | |
|---|---|---|---|
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.7.4.3-11: RRCRelease**

| |
|---|
| Derivation Path: TS 38.508-1 [45], Table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT |

**Table 14.4.7.4.3-12: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 3 entries | | |

| | | | |
|---|---|---|---|
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[3] SEQUENCE { | | entry 3 | |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not checked | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.4.7.5 Test requirement

Table 14.4.7.5-1 to Table 14.4.7.5-3 defines the primary level settings including test tolerances for the test.

**Table 14.4.7.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Reference cell | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |

| | | | | |
|---|---|---|---|---|
| SSB configuration | Config 1 | | SSB.1 FR1 | |
| | Config 2 | | SSB.1 FR1 | |
| | Config 3 | | SSB.1 RedCap FR1 | |
| | Config 4 | | SSB.1 FR1 | |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2 | | SMTC.1 | |
| | Config 3 | | SMTC.1 | |
| | Config 4 | | SMTC.2 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR.2.1 TDD | |
| | Config 4 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.1.1 FDD | As specified in TS 38.133 [50] clause A.3.1.2 |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR.2.1 TDD | |
| | Config 4 | | CR.1.1 FDD | |
| Dedicated CORESET RMC configuration | Config 1 | | CCR.1.1 FDD | |
| | Config 2 | | CCR.1.1 TDD | |
| | Config 3 | | CCR.2.1 TDD | |
| | Config 4 | | CCR.1.1 FDD | |
| Initial BWP configuration | Config 1,2,3,4 | | DLBWP.0.1<br>ULBWP.0.1 | |
| Active UL BWP configuration | Config 1,2,3,4 | | ULBWP.1.1 | |
| PRS Configuration | Config 1 | | PRS.1.1 FR1 | As specified in TS 38.133 [50] clause A.3.31 |
| | Config 2 | | PRS.1.1 FR1 | |
| | Config 3 | | PRS.2.1 FR1 | |
| | Config 4 | | PRS.1.1 FR1 | |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0<br>and<br>(PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | s | 1.28 | |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance ta specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 3 | Including the reference cell |
| PRS muting info | | | Cell 1: ‘10’<br>Cell 2: ‘01’<br>Cell 3: ‘10’ | Correponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | | s | 3 | The length of the time interval from the beginning of each test |
| T2 | | s | 5 | The length of the time interval that follows immediately after time interval T1. |

**Table 14.4.7.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | | |
| | Config 2 | dBm/SCS | -98 | | |
| | Config 3 | dBm/SCS | -95 | | |
| | Config 4 | dBm/SCS | -98 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| SSB $\hat{E}_s/N_{oc}$ | | dB | 10 | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 9.36MHz | -59.63 | -59.63 | -59.63 |
| | Config 2 | dBm/ 9.36MHz | -59.63 | -59.63 | -59.63 |
| | Config 3 | dBm/ 18.36MHz | -56.71 | -56.71 | -56.71 |
| | Config 4 | dBm/ 9.36MHz | -59.63 | -59.63 | -59.63 |
| SSB RP Note4 | Config 1 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 2 | dBm/SCS | -88 | -Infinity | -Infinity |
| | Config 3 | dBm/SCS | -85 | -Infinity | -Infinity |
| | Config 4 | dBm/SCS | -88 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.4.7.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| | | | T2 | T2 | T2 |
| NR RF Channel Number | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 | FR1 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -98 | -98 | -98 |
| | Config 2 | dBm/SCS | -98 | -98 | -98 |

| | | | | | |
|---|---|---|---|---|---|
| | Config 3 | dBm/SCS | -95 | -95 | -95 |
| | Config 4 | dBm/SCS | -98 | -98 | -98 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| | Config 2 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| | Config 3 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| | Config 4 | dB | -5.45 + TT | -11.67 + TT | -11.67 + TT |
| Io Note 4 | Config 1 | dBm/ 9.36MHz | -68.52 + TT | -68.52 + TT | -68.52 + TT |
| | Config 2 | dBm/ 9.36MHz | -68.52 + TT | -68.52 + TT | -68.52 + TT |
| | Config 3 | dBm/ 18.36MHz | -65.61 + TT | -65.61 + TT | -65.61 + TT |
| | Config 4 | dBm/ 9.36MHz | -68.52 + TT | -68.52 + TT | -68.52 + TT |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -6 + TT | -13 + TT | -13 + TT |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cells (all, except Cell 3 in T2) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

The RSTD measurement time without FH for RedCap fulfils the requirements specified in clause 14.4.8.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.4.8.3 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be higher than the time duration above because of the uncertainty in acquiring the first available PRACH occasion to transition to RRC_CONNECTED state to report the measurements.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in clause 10.1A.16.3, i.e., between RSTD_0000000 and RSTD_1970049.

## 14.4.8 NR RSTD measurement reporting delay test case in FR2 for Redcap UE without Rx FH

Editor's Note: This test case is incomplete in following aspects:

- TT analysis is missing.

### 14.4.8.1 Test purpose

The purpose of the test is to verify that the RSTD measurement for RedCap UE without FH in RRC INACTIVE state meets the requirements specified in TS 38.331 [50] clause 5.6A.4.5 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.4.8.2 Test applicability

This test applies to NR FR2 RedCap UEs release 18 onwards that support DL-TDOA positioning.

### 14.4.8.3 Minimum conformance requirements

Same as in clause 14.4.7.3.

### 14.4.8.4 Test description

#### 14.4.8.4.1 Initial conditions

The supported test configurations are listed in Table 14.4.8.4.1-1.

**Table 14.4.8.4.1-1: Supported test configurations for NR RSTD**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 100 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.4.8.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.4.8.5-1, Table 14.4.8.5-2 and Table 14.4.8.5-3.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.4.8.4.3
5. In the test there are three synchronous cells: Cell 1, Cell 2 and Cell 3. Cell 1 is the reference as well as the PCell. Cell 2 and Cell 3 are the neighbour cells. All 3 cells are on the same RF channel in FR2.

#### 14.4.8.4.2 Test procedure

Same test procedure as defined in clause 14.4.7.4.2 with following exceptions:

The UE is configured with DRX cycle of 0.64s.

Table 14.4.8.5-1 replaces Table14.4.7.5-1, Table 14.4.8.5-2 replaces Table14.4.7.5-2, Table 14.4.8.5-3 replaces Table14.4.7.5-3.

#### 14.4.8.4.3 Message contents

Same message contents as defined in clause 14.4.7.4.3, with the following exceptions:

Table 14.4.8.4.3-1 replaces Table 14.4.7.4.3-7 and Table 14.4.8.4.3-2 replaces Table 14.4.7.4.3-8.

**Table 14.4.8.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.8.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.8.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] SEQUENCE { | | entry 3 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.4.8.4.3-2 | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 14.4.8.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 2 |
| | 01 | | Cell 3 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-Assistance DataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-Assistance DataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-Assistance DataPerTRP-r16[3] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 and Cell 2 |
| | 1 | | Cell 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.4.8.5 Test requirement

Table 14.4.8.5-1 to Table 14.4.8.5-3 defines the primary level settings including test tolerances for the test.

**Table 14.4.8.5-1: General test parameters for RSTD measurement reporting delay**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Reference cell | | | Cell 1 | Reference cell is the cell in the DL-TDOA assistance data with respect to which the RSTD measurement is defined, as specified in TS 38.215 [57] and TS 37.355 [49]. The reference cell is the PCell in this test case. |
| Neighbor cells | | | Cell 2 and Cell 3 | Cell 2 and Cell 3 appear at the first and second places in the neighbour cell list in the DL-TDOA assistance data. |
| $BW_{channel}$ | | MHz | 100: $N_{RB,c}$ = 66 | |
| SSB configuration | Config 1 | | SSB.2 FR2 | |
| SMTC configuration | Config 1 | | SMTC.1 | |
| PDSCH RMC configuration | Config 1 | | SR.1.1 FDD | |
| RMSI CORESET RMC configuration | Config 1 | | CR.3.1 TDD | As specified in TS 38.133 [50] clause A.3.1.2.1 |
| Dedicated CORESET RMC configuration | Config 1 | | CR.1.1 FDD | |
| PRS Configuration | Config 1 | | PRS.1.5 FR2 | As specified in TS 38.133 [50] clause A.3. 31 |
| Physical cell ID PCI | | | (PCI of Cell 1 – PCI of Cell 2)mod6=0 and (PCI of Cell 1 – PCI of Cell 3)mod6=0 | The cell PCIs are selected such that the relative shifts of PRS patterns among cells are as given by the test parameters |
| CP length | | | Normal | |
| DRX | | s | 0.64 | |
| Radio frame receive time offset between the cells at the UE antenna connector | | µs | Cell 2 to Cell 1: 0<br>Cell 3 to Cell 1: 3 | PRS are transmitted from synchronous cells |
| Expected RSTD | | µs | Cell 2: 3<br>Cell 3: 3<br>Other neighbour cells: randomly between -3 and 3 | The expected RSTD is what is expected at the receiver. The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD indicator |
| Expected RSTD uncertainty for all neighbour cells | | µs | 5 | The corresponding parameter in the DL-TDOA assistance data specified in TS 37.355 [49] is the expectedRSTD-Uncertainty index |
| Number of cells provided in DL-TDOA assistance data | | | 3 | Including the reference cell |

| | | | |
|---|---|---|---|
| PRS muting info | | Cell 1: '10'<br>Cell 2: '01'<br>Cell 3: '10' | Correponds to prs-MutingInfo defined in TS 37.355 [49] |
| PRS resource RE offset | | Cell 1: 0<br>Cell 2: 0<br>Cell 3: 1 | Cell 1 and Cell 3 are configured with different resource offsets |
| T1 | s | 3 | The length of the time interval from the beginning of each test |
| T2 | s | 1.28 | The length of the time interval that follows immediately after time interval T1 |
| AoA setup | | Setup 1 | As defined in TS 38.133 [50] clause A.3.15.1 |
| Beam assumption | | Rough | Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation |

**Table 14.4.8.5-2: Cell-specific test parameters for RSTD measurement reporting delay during T1**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| NR RF Channel Number | | | 1 | 1 | 1 |
| Positiong frequency layer | | | 1 | 1 | 1 |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.5 FDD | N/A | N/A |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | | |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -Infinity | -Infinity | -Infinity |
| Io Note 4 | Config 1 | dBm/ 95.04MHz | -57 | -57 | -57 |
| SSB RP Note4 | Config 1 | dBm/SCS | -89 | -Infinity | -Infinity |
| $\hat{E}_s/N_{oc}$ | | dB | 0 | -Infinity | -Infinity |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: SSB RP and Io levels have been derived from other parameters and are given for information purpose. These are not settable test parameters.

**Table 14.4.8.5-3: Cell-specific test parameters for RSTD measurement reporting delay during T2**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| | | T2 | T2 | T2 |
| RF Channel Number | | 1 | 1 | 1 |
| Positiong frequency layer | | 1 | 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 1x2 Low | 1x2 Low |
| OCNG patterns defined in A.3.2.1 | | | OP.1 | OP.1 | OP.1 |
| PRACH configuration | | | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 | FR2 PRACH configuration 1 |
| $N_{oc}$ Note 3 | Config 1 | dBm/SCS | -89 | -89 | -89 |
| PRS $\hat{E}_s/N_{oc}$ | Config 1 | dB | -5.44 + TT | -11.67 + TT | -11.67 + TT |
| Io | Config 1 | dBm/ 95.04MHz | -58.49 + TT | -58.49 + TT | -58.49 + TT |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -6 + TT | -13 + TT | -13 + TT |
| Propagation Condition | | | AWGN | | |

NOTE 1: OCNG shall be used such that active cells (all, except Cell 3 in T2) are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.
NOTE 3: Interference from other cells and noise sources not specified in the test are assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

The RSTD measurement time without FH for RedCap fulfils the requirements specified in clause 14.4.8.3.

The UE shall perform and report the RSTD measurements for Cell 2 and Cell 3 with respect to the reference cell in the DL-TDOA assistance data, Cell 1, within the time duration specified in clause 14.4.8.3 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be higher than the time duration above because of the uncertainty in acquiring the first available PRACH occasion to transition to RRC_CONNECTED state to report the measurements.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported RSTD measurement for each correct event shall be within the RSTD reporting range specified in clause 10.1A.16.3, i.e., between RSTD_0000000 and RSTD_1970049.

# 14.5 NR RSTD measurement accuracy test case in RRC_INACTIVE

## 14.5.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA

### 14.5.1.1 Test purpose

The purpose of the test is to verify that the RSTD measurement in RRC_INACTIVE state meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.5.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and DL-PRS measurement in RRC_INACTIVE.

### 14.5.1.3 Minimum conformance requirements

The accuracy requirements for RSTD measurement shall be within ±(X+Y+Z+Δ) $T_c$.

X is defined in TS 38.133[50] Table 10.1.23.2-1 for AWGN channel and TS 38.133[50] Table 10.1.23.2-3 for fading channel for FR1, provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

…

X is defined in Table TS 38.133[50]10.1.23.2-7 for AWGN channel in FR1 provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 160 ms.
- When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in TS 38.133[50] Table 10.1.23.2-5 for FR1.
- Δ is zero for single PFL, and is defined in TS 38.133[50] Table 10.1.23.2-5a for FR1, respectively, for dual PFL.

**Table14.5.1.3-1: RSTD absolute accuracy in FR1 for AWGN channel**

| c | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io Note 3 range | | |
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | dBm/$BW_{Channel}$ |
| 132 | (PRS Ês/Iot)$_{ref}$ ≥ -6dB<br>(PRS Ês/Iot)$_{f}$ ≥ -13dB | 15 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 98 | | | ≥ 52 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 42 | | | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 75 | | 30 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| 48 | | | ≥ 48 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 50 | | 60 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 24 | | | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 10 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Void

**Table14.5.1.3-1: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 24 | N/A | N/A | 120 |
| ≥ 52 | ≥ 24 | N/A | 72 |
| ≥ 104 | ≥ 48 | ≥ 24 | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

## 14.5.1.4 Test description

The supported test configurations in listed in Table 14.5.1.4-1

**Table 14.5.1.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |

| | |
|---|---|
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

#### 14.5.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.5.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 14.5.1.5-1 and Table 14.5.1.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.5.1.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR1. The UE is configured with DRX cycle of 0.64s.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.5.1.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.1 FR1 and PRS.1.2 FR1. The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state and all cells start to transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.5.1.4.3 shall be provided to the UE during the set-up period. The beginning of the measurement period shall be aligned with the first DRX cycle containing a DL PRS resource(s).

0. Configure the Cell for RA-SDT with SIB1 configured according to TS 38.508-1 [45], clause 4.6.1 with condition SDT.
1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set parameters other than PRSconfiguration according to the Table 14.5.1.5-1 and Table 14.5.1.5-2. Propagation conditions are set according to clause 4.15.2.
4. Set-up period starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE

message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of measurement period, where ΔT = 50 ms.

11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.

12. When set-up period ends, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and set the PRS parameters as specified in Table 14.5.1.5-1 and Table 14.5.1.5-2.

13. Wait for the UE monitors for occurrence of the periodic event requested.

14. The UE shall transmit the UE transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 17.

15. The SS transmits an *RRCResume* message.

16. The UE transmits an *RRCResumeComplete* message.

17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.

18. If the UE message at step 17 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

19. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.5.1.5-3.

20. Repeat steps 2-19 in Table 14.5.1.4-1 until the confidence level according to Annex D is achieved.

21. Repeat step 2-20 for the other sub-test defined in Table 14.5.1.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.5.1.4.3 Message contents

As defined in clause 14.4.1.4.3, with the following exceptions:

Table 14.5.1.4.3-1 replaces Table 14.4.1.4.3-6, Table 14.5.1.4.3-2 replaces Table 14.4.1.4.3-7 and Table 14.5.1.4.3-3 replaces Table 14.4.1.4.3-11.

**Table 14.5.1.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |

| | | | |
|---|---|---|---|
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1 Config 1, 2& 3 |
| | 21 | 104 PRBs | Sub-test 2 && config 1, 2 |
| | 28 | 132 PRBs | Sub-test 2 && config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.1.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.1.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.1.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| | 1 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 |
| | 4 | | Sub-test 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.1.4.3-3: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.5.1.5 Test requirement

Table 14.5.1.5-1, define the primary level settings for the test.

**Table 14.5.1.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | 1~3 | | freq1 | Freq1 | freq1 | Freq1 |
| $BW_{channel}$ | 1 | MHz | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 2 | | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 3 | | 50: $N_{RB,c}$ = 133 | | 50: $N_{RB,c}$ = 133 | |
| Duplex mode | 1 | | FDD | | FDD | |
| | 2 | | TDD | | TDD | |
| | 3 | | TDD | | TDD | |
| TDD configuration | 1 | | N/A | | N/A | |
| | 2 | | TDDConf.1.1 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | 3 | | SR.2.1 FDD | | SR.2.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - | CR.1.1 TDD | - |
| | 3 | | CR.2.1 FDD | - | CR.2.1 FDD | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - | CCR.2.1 TDD | - |
| SSB configuration | 1 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 3 | | SSB.2 FR1 | | SSB.2 FR1 | |
| OCNG Patterns | 1~3 | | OP.1 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - | TRS.1.1 FDD | |
| | 2 | | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP Configuration | 1~3 | | DLBWP.0.1 ULBWP.0.1 | | DLBWP.0.1 ULBWP.0.1 | |
| Time offset with Cell 1 | 1 | µs | - | 3 | - | 3 |
| | 2,3 | | - | 3 | - | 3 |
| SMTC configuration | 1 | | SMTC.2 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | | SMTC.1 | |
| PRS configuration | 1 | | PRS.1.3 FR1 | | PRS.1.4 FR1 | |
| | 2 | | PRS.1.3 FR1 | | PRS.1.4 FR1 | |
| | 3 | | PRS.2.3 FR1 | | PRS.2.4 FR1 | |
| PRS muting info | 1~3 | | ‘10’ | ‘01’ | ‘10’ | ‘01’ |
| Expected RSTD | 1, 2, 3 | µs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3 | µs | N/A | 5 | N/A | 5 |

The

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of PSS to SSS | 1~3 | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| $N_{oc}$ Note2 | 1,2 | dBm/ SCS | -98 | | -98 | |
| | 3 | | -95 | | -95 | |
| $\hat{E}_s / I_{ot}$ | 1~3 | dB | -6 | -13 | -6 | -13 |
| PRS-RSRP Note3 | 1,2 | dBm/SCS | -103.7 | -109.9 | -103.7 | -109.9 |
| | 3 | | -100.7 | -106.9 | -100.7 | -106.9 |
| Io Note3 | 1,2 | dBm/ 19.08MHz | -65.62 | -65.62 | -65.62 | -65.62 |
| | 3 | dBm/ 47.88MHz | -61.63 | -61.63 | -61.63 | -61.63 |
| $\hat{E}_s / N_{oc}$ | 1~3 | dB | -5.4 | -11.5 | -5.4 | -11.5 |
| Propagation condition | 1~3 | - | AWGN | | AWGN | |
| Antenna configuration | 1~3 | | 1x2 | | 1x2 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Void.

RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.5.1.3.

The value of the LPP Time IE depends on the UE capabilities

Calculations for RSTD accuracy is as mentioned in below Table 14.5.1.5-2:

**Table 14.5.1.5-2: RSTD accuracy requirements for the reported values**

| Test Configuration | Subtest | Lowest reported value [$T_c$] | Highest reported value [$T_c$] |
|---|---|---|---|
| 1 & 2 | Test 1 | 5582 | 6214 |
| | Test 2 | 5756 | 6040 |
| 3 | Test 1 | 5687 | 6109 |
| | Test 2 | 5794 | 6002 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.5.2 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA

### 14.5.2.1 Test purpose

The purpose of the test is to verify that the RSTD measurement in RRC_INACTIVE state with reduced number of samples meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

### 14.5.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning, DL-PRS measurement in RRC_INACTIVE and reduced number of samples for PRS measurement in RRC_INACTIVE state.

### 14.5.2.3 Minimum conformance requirements

The

The accuracy requirements for RSTD measurement shall be within $\pm(X+Y+Z+\Delta)$ $T_c$.

X is defined in TS 38.133[50] Table 10.1.23.2-1 for AWGN channel and TS 38.133[50] Table 10.1.23.2-3 for fading channel for FR1, provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

…

X is defined in Table TS 38.133[50]10.1.23.2-7 for AWGN channel in FR1 provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 160 ms.
- When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in TS 38.133[50] Table 10.1.23.2-5 for FR1.
- Δ is zero for single PFL, and is defined in TS 38.133[50] Table 10.1.23.2-5a for FR1, respectively, for dual PFL.

…

**Table14.5.2.3-1: RSTD absolute accuracy in FR1 for AWGN channel with reduced number of samples**

| Accuracy | Conditions | | | | |
|---|---|---|---|---|---|
| | PRS $\hat{E}s/Iot$ | | | | Io Note 3 range |

| | | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | NR operating band groups Note 4 | Minimum Io | Maximum Io |
|---|---|---|---|---|---|---|---|
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 98 | (PRS Ês/Iot)$_{ref}$ ≥ -3dB<br>(PRS Ês/Iot)$_i$ ≥ -6dB | 15 | ≥ 52 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| 42 | | | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 48 | | 30 | ≥ 48 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| 24 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 | | 60 | ≥ 64 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 10 | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}, L_{PRS}, K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Void

**Table14.5.2.3-2: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |

| | | | |
|---|---|---|---|
| ≥ 24 | N/A | N/A | 120 |
| ≥ 52 | ≥ 24 | N/A | 72 |
| ≥ 104 | ≥ 48 | ≥ 24 | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

### 14.5.2.4 Test description

The supported test configurations in listed in Table 14.5.2.4-1

**Table 14.5.2.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

#### 14.5.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.5.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 14.5.2.5-1 and Table 14.5.2.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.5.2.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. The UE is configured with DRX cycle of 0.64s.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.5.2.4.2 Test procedure

Same as in clause 14.5.1.4.2.

#### 14.5.2.4.3 Message contents

As defined in clause 14.4.1.4.3, with the following exceptions:

Table 14.5.2.4.3-1 replaces Table 14.4.1.4.3-6, Table 14.5.2.4.3-2 replaces Table 14.4.1.4.3-7 and Table 14.5.2.4.3-3 replaces Table 14.4.1.4.3-11.

**Table 14.5.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1 |
| | 21 | 104 PRBs | Sub-test 2 Config 1& 2 |
| | 28 | 132 PRBs | Sub-test 2 Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.2.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.2.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| | 1 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 |
| | 4 | | Sub-test 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.2.4.3-3: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.5.2.5 Test requirement

Table 14.5.2.5-1, define the primary level settings for the test.

**Table 14.5.2.5-1 RSTD accuracy test parameters**

| Parameter | Config | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | 1~3 | | freq1 | Freq1 | freq1 | Freq1 |
| $BW_{channel}$ | 1 | MHz | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 2 | | 20: $N_{RB,c}$ = 106 | | 20: $N_{RB,c}$ = 106 | |
| | 3 | | 50: $N_{RB,c}$ = 133 | | 50: $N_{RB,c}$ = 133 | |
| Duplex mode | 1 | | FDD | | FDD | |
| | 2 | | TDD | | TDD | |
| | 3 | | TDD | | TDD | |
| TDD configuration | 1 | | N/A | | N/A | |
| | 2 | | TDDConf.1.1 | | TDDConf.1.1 | |
| | 3 | | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | 3 | | SR.2.1 FDD | | SR.2.1 FDD | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | 2 | | CR.1.1 TDD | - | CR.1.1 TDD | - |
| | 3 | | CR.2.1 FDD | - | CR.2.1 FDD | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | 2 | | CCR.1.1 TDD | - | CCR.1.1 TDD | - |
| | 3 | | CCR.2.1 TDD | - | CCR.2.1 TDD | - |
| SSB configuration | 1 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 2 | | SSB.1 FR1 | | SSB.1 FR1 | |
| | 3 | | SSB.2 FR1 | | SSB.2 FR1 | |
| OCNG Patterns | 1~3 | | OP.1 | | OP.1 | |
| TRS configuration | 1 | | TRS.1.1 FDD | - | TRS.1.1 FDD | |
| | 2 | | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | 3 | | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP Configuration | 1~3 | | DLBWP.0.1 ULBWP.0.1 | | - | |
| Time offset with Cell 1 | 1 | μs | - | 3 | - | 3 |
| | 2,3 | | - | 3 | - | 3 |
| SMTC configuration | 1 | | SMTC.2 | | SMTC.2 | |
| | 2,3 | | SMTC.1 | | SMTC.1 | |

The

| | | | | | | |
|---|---|---|---|---|---|---|
| PRS configuration | 1 | | PRS.1.3 FR1 | | PRS.1.4 FR1 | |
| | 2 | | PRS.1.3 FR1 | | PRS.1.4 FR1 | |
| | 3 | | PRS.2.3 FR1 | | PRS.2.4 FR1 | |
| PRS BW | 1,2,3 | | 52 PRBs for test 1 | | | |
| PRS muting info | 1~3 | | '10' | '01' | '10' | '01' |
| Expected RSTD | 1, 2, 3 | μs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | 1, 2, 3 | μs | N/A | 5 | N/A | 5 |
| EPRE ratio of PSS to SSS | 1~3 | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| $N_{oc}$ Note2 | 1,2 | dBm/ SCS | -98 | | -98 | |
| | 3 | | -95 | | -95 | |
| $\hat{E}_s / I_{ot}$ | 1~3 | dB | -3 | -6 | -3 | -6 |
| PRS-RSRP Note3 | 1,2 | dBm/SCS | -99.0 | -101.1 | -99.0 | -101.1 |
| | 3 | | -95.9 | -98.1 | -95.9 | -98.1 |
| SS-RSRP Note3 | 1,2 | dBm/SCS | -99.44 | -101.65 | -99.44 | -101.65 |
| | 3 | | -96.44 | -98.65 | -96.44 | -98.65 |
| Io Note3 | 1,2 | dBm/ 19.08MHz | -63.35 | -63.35 | -63.35 | -63.35 |
| | 3 | dBm/ 47.88MHz | -59.35 | -59.35 | -59.35 | -59.35 |
| $\hat{E}_s / N_{oc}$ | 1~3 | dB | -0.95 | -3.10 | -0.95 | -3.10 |
| Propagation condition | 1~3 | - | AWGN | | AWGN | |
| Antenna configuration | 1~3 | | 1x2 | | 1x2 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Void.

RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.5.2.3.

The value of the LPP Time IE depends on the UE capabilities.

Calculations for RSTD accuracy is as mentioned in below Table 14.5.2.5-2:

| Test Configuration | Subtest | Lowest reported value [$T_c$] | Highest reported value [$T_c$] |
|---|---|---|---|
| 1 & 2 | Test 1 | 5664 | 6132 |
| | Test 2 | 5756 | 6040 |
| 3 | Test 1 | 5750 | 6046 |
| | Test 2 | 5794 | 6002 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.5.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.5.3.1 Test purpose

The purpose of the test is to verify that the RSTD measurement in RRC_INACTIVE state meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.5.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning and DL-PRS measurement in RRC_INACTIVE.

### 14.5.3.3 Minimum conformance requirements

The accuracy requirements for RSTD measurement shall be within ±(X+Y+Z+Δ) $T_c$.

…

X is defined in Table 10.1.23.2-2 for AWGN channel and Table 10.1.23.2-4 for fading channel for FR2, provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.

- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.

- UE does not perform positioning measurement with reduced number of samples.

…

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 160 ms.

- When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in Table 10.1.23.2-5 for FR1 and Table 10.1.23.2-6 for FR2, respectively.
- Δ is zero for single PFL, and is defined in Table 10.1.23.2-5a for FR1 and Table 10.1.23.2-6a for FR2, respectively, for dual PFL.

…

**Table 10.1.23.2-2: RSTD absolute accuracy in FR2 for AWGN channel**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |
| 35 | (PRS Ês/Iot)$_{ref}$ ≥ -6dB | 60 | ≥ 24 | ≥ 4 | Same value as PRS_RP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 24 | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 11 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 24 | (PRS Ês/Iot)$_i$ ≥ -13dB | 120 | ≥ 32 | ≥ 4 | Same value as PRS_RP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 13 | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 6 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: Void

…

**Table 10.1.23.2-6: Margin for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 24 | N/A | 72 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 16 |
| N/A | ≥ 128 | 12 |

## 14.5.3.4 Test description

The supported test configurations in listed in Table 14.5.3.4-1

**Table 14.5.3.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 14.5.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.5.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.5.3.5-1 and Table 14.5.3.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.5.3.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. The UE is configured with DRX cycle of 0.64s.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.5.3.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.1 FR2 and PRS.1.2 FR2. The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state and all cells start to transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The NR-DL-TDOA-RequestLocationInformation message and the DL-TDOA assistance data as defined in clause 14.5.3.4.3 shall be provided to the UE during the set-up period. The beginning of the measurement period shall be aligned with the first DRX cycle containing a DL PRS resource(s).

0 Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 14.4.3.4.3-0.

1 Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On,* according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set parameters other than PRSconfiguration according to the Table 14.5.3.5-1 and Table 14.5.3.5-2. Propagation conditions are set according to clause 4.15.2.
4. Set-up period starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of measurement period, where ΔT = 50 ms.

11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
12. When set-up period ends, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and set the PRS parameters as specified in Table 14.5.3.5-1 and Table 14.5.3.5-2.
13. Wait for the UE monitors for occurrence of the periodic event requested.
14. The UE shall transmit the UE transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 17.
15. The SS transmits an *RRCResume* message.
16. The UE transmits an *RRCResumeComplete* message.
17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.
18. If the UE message at step 17 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
19. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.5.3.5-3.
20. Repeat steps 2-19 in Table 14.5.3.4-1 until the confidence level according to Annex D is achieved.
21. Repeat step 2-20 for the other sub-test defined in Table 14.5.3.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 14.5.3.4.3 Message contents

As defined in clause 14.4.3.4.3, with the following exceptions:

Table 14.5.3.4.3-1 replaces Table 14.4.3.4.3-7, Table 14.5.3.4.3-2 replaces Table 14.4.3.4.3-9 and Table 14.5.3.4.3-3 replaces Table 14.4.3.4.3-11.

**Table 14.5.3.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.3.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.3.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.3.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.3.4.3-3: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |

| | | | |
|---|---|---|---|
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.5.3.5 Test requirement

**The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.5.3.3.Table 14.5.3.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |

| | | | | | |
|---|---|---|---|---|---|
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.1 FR2 | PRS.1.1 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset | slot | 0 | 4 | 0 | 4 |
| Expected RSTD | µs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | |

**Table 14.5.3.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to clause A.3.15.1 | | | |
| Assumption for UE beams Note 5 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89.1 | | -89.1 | |
| $\hat{E}_s / N_{oc}$ | dB | -5.17 | -11.36 | -5.17 | -11.36 |
| PRS-RSRP Note2 | dBm/SCS | -94.2 | -100.4 | -94.2 | -100.4 |
| $\hat{E}_s / I_{ot}$ BB Note4 | dB | -5.7 | -12.7 | -5.7 | -12.7 |
| Io Note2 | dBm/190.08 MHz Note3 | -55.68 | -55.68 | -55.68 | -55.68 |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | |

| | |
|---|---|
| Note 2: | SSB_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 3: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 4: | Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. |
| Note 5: | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.4.3.5 with the following difference: $N_{PRS,i}^{slot} = 2, L_{available_PRS,i} = 0.071$. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation.

**Table 14.5.3.5-3: RSTD accuracy requirements for the reported values**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Sub-test 1 | (5898 – 120) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 120) $T_c$ converted to RSTD measurement according to clause 4.15.6 |
| Sub-test 2 | (5898 – 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.5.4 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 14.5.4.1 Test purpose

The purpose of the test is to verify that the RSTD measurement in RRC_INACTIVE state with reduced number of samples meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23.2 in an environment with AWGN propagation conditions in FR2 in standalone scenario when single positioning frequency layer is configured.

### 14.5.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-TDOA positioning, DL-PRS measurement in RRC_INACTIVE and reduced number of samples for PRS measurement in RRC_INACTIVE state.

### 14.5.4.3 Minimum conformance requirements

The accuracy requirements for RSTD measurement shall be within ±(X+Y+Z+Δ) $T_c$.

…

X is defined in Table 10.1.23.2-8 for AWGN channel in FR2 provided that the following conditions are met.

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to Annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

…

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and Δ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 160 ms.
- When UE measures RSTD on PRS resources belonging different PFLs, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbour cell, which are used for a single RSTD estimate, is no greater than 1280 ms.
- Z is defined in Table 10.1.23.2-5 for FR1 and Table 10.1.23.2-6 for FR2, respectively.
- Δ is zero for single PFL, and is defined in Table 10.1.23.2-5a for FR1 and Table 10.1.23.2-6a for FR2, respectively, for dual PFL.

…

**Table 10.1.23.2-8: RSTD absolute accuracy in FR2 for AWGN channel with reduced number of samples**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS $\hat{E}_s/I_{ot}$ | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | PRB | | dBm/SCS | $dBm/BW_{Channel}$ |
| 24 | $(\hat{E}_s/I_{ot})_{ref}$ ≥ -3dB | 60 | ≥ 64 | ≥ 1 | Same value as PRS_RP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | Note 5 |
| 11 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 13 | $(\hat{E}_s/I_{ot})_i$ ≥ -6dB | 120 | ≥ 64 | ≥ 1 | Same value as PRS_RP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | Note 5 |
| 6 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: Void

…

**Table 10.1.23.2-6: Margin for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |

| | | |
|---|---|---|
| ≥ 24 | N/A | 72 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 16 |
| N/A | ≥ 128 | 12 |

### 14.5.4.4 Test description

The supported test configurations in listed in Table 14.5.4.4-1

**Table 14.5.4.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 14.5.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.5.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.5.4.5-1 and Table 14.5.4.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.5.4.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR2. The UE is configured with DRX cycle of 0.64s.
6. The true RSTD (which is the receive time difference for frame 0 between cell 2 and cell 1 as seen at the UE antenna connector) is set to 5898 Tc (about 3 μs)

#### 14.5.4.4.2 Test procedure

Same as in clause 14.5.3.4.2.

#### 14.5.4.4.3 Message contents

As defined in clause 14.4.3.4.3, with the following exceptions:

Table 14.5.4.4.3-1 replaces Table 14.4.3.4.3-7, Table 14.5.4.4.3-2 replaces Table 14.4.3.4.3-9 and Table 14.5.4.4.3-3 replaces Table 14.4.3.4.3-11.

**Table 14.5.4.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.4.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.4.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.4.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.4.4.3-3: LPP ProvideLocation Information**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | | | |
| nr-UE-Rx-TEG-ID-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | | | |
| nr-DL-TDOA-Error-r16 | | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 14.5.4.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.5.4.3.

**Table 14.5.4.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| PRS ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | DLBWP.0.1 | DLBWP.0.1 | DLBWP.0.1 |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| PRS configuration | | PRS.1.3 FR2 | PRS.1.3 FR2 | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS BW | | 64 PRBs | 64 PRBs | 128 PRBs | 128 PRBs |
| PRS Resource slot offset | slot | 0 | 4 | 0 | 4 |
| Expected RSTD | µs | N/A | 3 | N/A | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 | N/A | 5 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | |

**Table 14.5.4.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 5 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -91.6 | | -91.6 | |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -0.56 | -2.72 | -0.56 | -2.72 |
| PRS-RSRP Note2 | dBm/SCS | -92.1 | -94.3 | -92.1 | -94.3 |
| SS-RSRP Note2 | dBm/SCS | -90.44 | -92.65 | -90.44 | -92.65 |
| $\hat{E}_s / I_{ot}$ BB Note4 | dB | -2.69 | -5.68 | -2.69 | -5.68 |
| Io Note2 | dBm/190.08 MHz Note3 | -55.74 | -55.74 | -55.74 | -55.74 |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | |
| Note 2: SSB_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |
| Note 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone | | | | | |
| Note 4: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in TS 38.133 [50] clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. | | | | | |
| Note 5: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation | | | | | |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 14.4.3.5 with the following difference: $N_{PRS,i}^{slot} = 2, L_{available_PRS,i} = 0.071 \quad N_{sample} = 1$. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 1 - 1\right) * T_{effect} + \mathrm{T}_{\mathrm{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation.

**Table 14.5.4.5-3: RSTD accuracy requirements for the reported values**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Sub-test 1 | (5898 – 93) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 93) $T_c$ converted to RSTD measurement according to clause 4.15.6 |
| Sub-test 2 | (5898 – 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 | (5898 + 82) $T_c$ converted to RSTD measurement according to clause 4.15.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 14.5.5 RSTD measurement accuracy for PRS aggregation in FR1 in RRC_INACTIVE state

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test requirements and test tolerance analysis is incomplete

### 14.5.5.1 Test purpose

The purpose of the test is to verify that the RSTD measurement with PRS aggregation on two PFLs meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23A.2 in an environment with AWGN propagation conditions.

### 14.5.5.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Inactive-r18*.

### 14.5.5.3 Minimum conformance requirements

When UE measures RSTD on PRS resources belonging to different PFLs or different PFL combinations, then the RSTD accuracy is defined as the accuracy corresponding to the largest accuracy value among different PFLs or different PFL combinations.

The requirements in this clause for 3-PFL and 2-PFL apply provided that:

- PRS resources linked for aggregation saftisfy all the conditions specified in TS 38.214 [56] clause 5.1.6.5.3.
- The spacing between the center frequencies of adjacent PFLs containing PRS resources linked for aggregation does not exceed the nominal channel spacing for intra-band contiguous CA defined in TS 38.101-1 [18], clause 5.4A.1 for FR1, and in TS 38.101-2, [19] clause 5.4A.1 for FR2-1.

The accuracy requirements for RSTD measurement based on PRS Aggregation shall be within $\pm(X+Y+Z+\Delta)\ T_c$.

X is defined in table 14.5.5.3-1 for AWGN channel and table 14.5.5.3-3 for fading channel for FR1, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to TS 38.133 [50] annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

X is defined in table 14.5.5.3-2 for AWGN channel and table 14.5.5.3-4 for fading channel for FR2, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE does not perform positioning measurement with reduced number of samples.

X is defined in table 14.5.5.3-5 for AWGN channel in FR1, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

X is defined in table 14.5.5.3-6 for AWGN channel in FR2, provided that the following conditions are met:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- Conditions for RSTD measurements are fulfilled according to annex B.2.14 for a corresponding Band for each relevant PRS resource configured for measurement.
- UE supports positioning measurement with reduced number of sample and is indicated by LMF to perform positioning measurement with reduced number of samples.

NOTE: The requriements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

If the UE doesn't support Rx TEG reporting for RSTD measurement or when the measurements of reference cell and neighbour cell belong to different Rx TEGs, Y, Z and $\Delta$ are defined as follows:

- When UE measures RSTD on PRS resources belonging to same PFL or same PFL combination, Y=32 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 160 ms.

- When UE measures RSTD on PRS resources belonging different PFLs or different PFL combinations, Y=256 Tc, provided that the time offset between the two PRS resource instances from the reference cell and the neighbor cell, which are used for a single RSTD estimate, is no greater than 1280 ms.

- Z is defined in table 14.5.5.3-7 for FR1 and table 14.5.5.3-9 for FR2, respectively, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.

- Δ is zero when UE measures RSTD on PRS resources belonging to same PFL or same PFL combination and is defined in table 14.5.5.3-8 for FR1 and table 14.5.5.3-10 for FR2, respectively, when UE measures RSTD on PRS resources belonging different PFLs or different PFL combinations, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.

If the measurements of reference cell and neighbour cell belong to the same Rx TEG, i.e. associated and reported with a common Rx TEG ID, then the sum of Y+Z+Δ is equal to the timing error margin of the Rx TEG reported in *nr-UE-RxTEG-TimingErrorMargin*. The timing error margin reported via *nr-UE-RxTEG-TimingErrorMargin* cannot be larger than the value of (Y+Z+Δ) defined when the UE does not associate the measurements with the same Rx TEG.

**Table 14.5.5.3-1: RSTD absolute accuracy in FR1 for AWGN channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | dBm/$BW_{Channel}$ |
| 20 | 16 | (PRS Ês/Iot)$_{ref}$ ≥-6dB<br>(PRS Ês/Iot)$_i$ ≥-13dB | 15 | ≥ 104 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -126.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -126 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -124 | -50 |
| | | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -120.5 | -50 |
| 11 | 6 | | 30 | ≥ 132 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -123.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -123 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -121 | -50 |
| | | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -117.5 | -50 |
| 6 | 4 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 11 | 6 | | 60 | ≥ 64 | ≥ 1 | NR_FDD_FR1_A, NR_TDD_FR1_A, | -121 | -50 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | NR_SDL_FR1_A | | |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -120.5 | -50 |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -120 | -50 |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -118 | -50 |
| | | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| | | | | | | NR_FDD_FR1_N | -114.5 | -50 |
| 5 | 4 | | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.

**Table 14.5.5.3-2: RSTD absolute accuracy in FR2 for AWGN channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{\text{rep}}^{\text{PRS}} * L_{\text{PRS}} / K_{\text{comb}}^{\text{PRS}}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{\text{Channel}}$ |
| 11 | 6 | (PRS Ês/Iot)$_{\text{ref}}$ ≥-6dB | 60 | ≥ 64 | ≥ 1 | Same value as PRS_RP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 5 | 4 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 6 | 4 | (PRS Ês/Iot)$_i$ ≥-13dB | 120 | ≥ 64 | ≥ 1 | Same value as PRS_RP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 3 | 2 | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.

**Table 14.5.5.3-3: RSTD absolute accuracy in FR1 for fading channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) Note 2 | Io Note 3 range | | |
| | | | | | | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | dBm/$BW_{Channel}$ |
| 38 | 30 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 15 | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 29 | 29 | | 30 | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 27 | 26 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 29 | 35 | (PRS Ês/Iot)$_i$ ≥-13dB | 60 | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 27 | 31 | | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 10.1.23A.2-1.

**Table 14.5.5.3-4: RSTD absolute accuracy in FR2 for fading channel**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/$BW_{Channel}$ |
| 71 | 51 | (PRS Ês/Iot)$_{ref}$ ≥-6dB | 60 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 55 | 38 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 72 | 49 | | 120 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 56 | 53 | (PRS Ês/Iot)$_i$ ≥-13dB | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 10.1.23A.2-2.

**Table 14.5.5.3-5: RSTD absolute accuracy in FR1 for AWGN channel with reduced number of samples**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL | PRS resource repetition | Io Note 3 range | | |
| | | | | | | NR operating | Minimum Io | Maximum Io |

| | | | | Note 1 | ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | band groups Note 4 | | |
|---|---|---|---|---|---|---|---|---|
| Tc Note 5 | | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 24 | 20 | (PRS Ês/Iot)$_{ref}$ ≥-3dB | 15 | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 13 | 7 | | 30 | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 6 | 5 | | | 272 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 13 | 7 | | 60 | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 6 | 5 | (PRS Ês/Iot)$_i$ ≥-6dB | | 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 10.1.23A.2-1.

**Table 14.5.5.3-6: RSTD absolute accuracy in FR2 for AWGN channel with reduced number of samples**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS SCS | PRS bandwidth per PFL Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | | dB | kHz | PRB | | dBm/SCS | dBm/BW$_{Channel}$ |
| 13 | 7 | (PRS Ês/Iot)$_{ref}$ ≥-3dB | 60 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 6 | 5 | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 6 | 4 | | 120 | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 3 | 2 | (PRS Ês/Iot)$_i$ ≥-6dB | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth per PFL, which is minimum of the PRS bandwidths of the reference resource and the measured neighbour resource i among all the aggregated PFLs.
NOTE 2: Minimum number of PRS resource repetitions among the reference resource and the measured neighbour resource i. $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS as defined in table 10.1.23A.2-2.

**Table 14.5.5.3-7: Margin for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | | | Margin (Tc) |
|---|---|---|---|
| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
| ≥ 104 | N/A | N/A | 36 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 12 |

**Table 14.5.5.3-8: Margin Δ for RSTD measurement accuracy in FR1**

| PRS BW (PRB number) | Margin (Tc) |
|---|---|

| SCS=15kHz | SCS=30kHz | SCS=60kHz | |
|---|---|---|---|
| ≥ 104 | N/A | N/A | 32 |
| N/A | ≥ 132 | ≥ 64 | 16 |
| N/A | N/A | ≥ 132 | 8 |

**Table 14.5.5.3-9: Margin for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 64 | N/A | 32 |
| ≥ 132 | ≥ 64 | 16 |
| N/A | ≥ 128 | 12 |

**Table 14.5.5.3-10: Margin Δ for RSTD measurement accuracy in FR2**

| PRS BW (PRB number) | | Margin (Tc) |
|---|---|---|
| SCS=60kHz | SCS=120kHz | |
| ≥ 64 | N/A | 16 |
| ≥ 132 | ≥ 64 | 8 |
| N/A | ≥ 128 | 4 |

## 14.5.5.4 Test description

### 14.5.5.4.1 Initial conditions

The supported test configuration is listed in Table 14.5.5.4.1-1.

**Table 14.5.5.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.5.5.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.15.
2. The general test parameter settings are set up according to Table 14.5.5.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 14.5.5.4.3.
5. In the test there are four synchronous cells: Cell 1, Cell 2, Cell 3 and Cell 4. Cell 1 is the reference as well as the PCell on NR RF channel #1 in FR1. Cell 2 is a neighbour cell on the same NR RF channel as Cell 1. Cell 3 and Cell 4 are neighbor cells in a different NR RF channel #2 in FR1.

### 14.5.5.4.2 Test procedure

The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state and all cells start to transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-TDOA-ProvideAssistanceData* and *NR-TDOA-RequestLocationInformation* message as defined in TS 37.355 [49], shall be provided to the UE before the start of the test. The test duration should be larger than the UE measurement period as defined in TS 38.133[50] clause 5.6.2.6. The UE is configured with DRX cycle of 1.28 s.

0. Configure the Cell for RA-SDT with SIB1 configured according to the message content in clause 14.5.5.4.3.
1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.5.5.5-1. Propagation conditions are set according to clause 4.15.2.
4. Set-up period starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10, The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *nr-DL-PRS-JointMeasurementRequestedPFL-List* in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of measurement period, where ΔT = 50 ms.
11, The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message
12, When set-up period ends, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and set the PRS parameters as specified in Table 14.5.5.5-1
13, Wait for the UE monitors for occurrence of the periodic event requested
14, The UE shall transmit an RRCResumeRequest message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 17
15, The SS transmits an RRCResume message.
16, The UE transmits an RRCResumeComplete message
17, The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.
18, If the UE message at step 17 includes the ackRequested IE set to TRUE, the SS shall send a LPP acknowledgement message.
19, The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.5.5.5-1.
20. Repeat steps 2-19 in Table 14.5.5.4.1-1 until the confidence level according to Annex D is achieved.

### 14.5.5.4.3 Message contents

As defined in clause 14.4.5.4.3, with the following exceptions:

Table 14.5.5.4.3-1 replaces Table 14.4.5.4.3-6, Table 14.5.5.4.3-2 replaces Table 14.4.5.4.3-7 and Table 14.5.5.4.3-3 replaces Table 14.4.1.4.3-11.

**Table 14.5.5.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | Not present | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | Not present | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.5.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.5.4.3-2 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 3 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.5.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 4 |
| dl-PRS-ID-r16 | 3 | | |
| nr-PhysCellID-r16 | Cell 4 | | |
| nr-CellGlobalID-r16 | Cell 4 | | |
| nr-ARFCN-r16 | Cell 4 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.5.4.3-2 | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 3 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.5.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |

| | | | |
|---|---|---|---|
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-AssistanceDataPerTRP-r16[3] |
| | 3 | | NR-DL-PRS-AssistanceDataPerTRP-r16[4] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 and Cell 3 |
| | 4 | | Cell 2 and Cell 4 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.5.4.3-3: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 | | | |

| | | | |
|---|---|---|---|
| SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 4 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |

| | | | |
|---|---|---|---|
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 4 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 4 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not Present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not Present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not Present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.5.5.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.5.5.3.

**Table 14.5.5.5-1: RSTD accuracy test parameters**

| Parameter | Config | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|---|
| PRS ARFCN | 1,2,3 | | freq1 | freq1 | freq2 | freq2 |
| $BW_{channel}$ | 1 | MHz | 20: $N_{RB,c}$ = 106 | | | |
| | 2 | | 20: $N_{RB,c}$ = 106 | | | |
| | 3 | | 50: $N_{RB,c}$ = 133 | | | |
| Duplex mode | 1 | | FDD | | | |
| | 2 | | TDD | | | |
| | 3 | | TDD | | | |
| TDD configuration | 1 | | N/A | | | |
| | 2 | | TDDConf.1.1 | | | |
| | 3 | | TDDConf.2.1 | | | |
| PDSCH Reference measurement channel | 1 | | SR.1.1 FDD | - | - | - |
| | 2 | | SR.1.1 TDD | - | - | |
| | 3 | | SR.2.1 FDD | - | - | |
| RMSI CORESET Reference Channel | 1 | | CR.1.1 FDD | - | - | - |
| | 2 | | CR.1.1 TDD | - | - | - |
| | 3 | | CR.2.1 FDD | - | - | - |
| Dedicated CORESET Reference Channel | 1 | | CCR.1.1 FDD | - | - | - |
| | 2 | | CCR.1.1 TDD | - | - | - |
| | 3 | | CCR.2.1 TDD | - | - | - |
| SSB configuration | 1 | | SSB.1 FR1 | | | |
| | 2 | | SSB.1 FR1 | | | |
| | 3 | | SSB.2 FR1 | | | |
| OCNG Patterns | 1,2,3 | | OP.1 | | | |
| TRS configuration | 1 | | TRS.1.1 FDD | - | - | - |
| | 2 | | TRS.1.1 TDD | - | - | - |
| | 3 | | TRS.1.2 TDD | - | - | - |
| Initial BWP Configuration | 1,2,3 | | DLBWP.0.1 ULBWP.0.1 | | | |
| Dedicated BWP configuration | 1,2,3 | | DLBWP.1.1 ULBWP.1.1 | | | |
| Time offset with Cell 1 | 1 | μs | - | 3 | 0 | 3 |
| | 2,3 | | - | 3 | 0 | 3 |

| | | | | | | |
|---|---|---|---|---|---|---|
| PRS configuration | 1 | | PRS.1.2 FR1 | | | |
| | 2 | | PRS.1.2 FR1 | | | |
| | 3 | | PRS.2.2 FR1 | | | |
| PRS Resource slot offset | 1, 2, 3 | slot | 0 | 4 | 0 | 4 |
| Expected RSTD | 1, 2, 3 | µs | N/A | 3 | 0 | 3 |
| Expected RSTD uncertainty | 1, 2, 3 | µs | N/A | 5 | 0 | 5 |
| EPRE ratio of PSS to SSS | 1,2,3 | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | 1,2,3 | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | 1,2 | dBm/ SCS | -98 | | -98 | |
| | 3 | | -95 | | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | 1,2,3 | dB | -6 + TT | -13 + TT | -6+ TT | -13 + TT |
| PRP Note3 | 1,2 | dBm/SCS | -104 + TT | -111 + TT | -104 + TT | -111 + TT |
| | 3 | | -101 + TT | -108 + TT | -101 + TT | -108 + TT |
| Io Note3 | 1,2 | dBm/ 19.08MHz | -65.98 + TT | -66.74 + TT | -65.98 + TT | -66.74 + TT |
| | 3 | dBm/ 47.88MHz | -62.00 + TT | -62.76 + TT | -62.00 + TT | -62.76 + TT |
| PRS $\hat{E}_s/N_{oc}$ | 1,2,3 | dB | -6 + TT | -13 + TT | -6 + TT | -13 + TT |
| Propagation condition | 1,2,3 | - | AWGN | | AWGN | |
| Antenna configuration | 1,2,3 | | 1x2 | | 1x2 | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The Io is calculated based only on the symbols in which PRS is transmitted.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification.

## 14.5.6 RSTD measurement accuracy for PRS aggregation in FR2 in RRC_INACTIVE state

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test requirements and test tolerance analysis is incomplete

### 14.5.6.1 Test purpose

The purpose of the test is to verify that the RSTD measurement with PRS aggregation on two PFLs meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.23A.2 in an environment with AWGN propagation conditions.

### 14.5.6.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports DL-TDOA positioning and *supportOfDL-PRS-BWA-RRC-Inactive-r18*.

### 14.5.6.3 Minimum conformance requirements

Same as in clause 14.5.5.3.

### 14.5.6.4 Test description

#### 14.5.6.4.1 Initial conditions

The supported test configuration is listed in Table 14.5.6.4.1-1.

**Table 14.5.6.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 14.5.6.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 14.5.6.5-1, Table 14.5.6.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 14.5.6.4.3.
5. In the test there are four synchronous cells: Cell 1, Cell 2, Cell 3 and Cell 4. Cell 1 is the reference as well as the PCell on NR RF channel #1 in FR2. Cell 2 is a neighbour cell on the same NR RF channel as Cell 1. Cell 3 and Cell 4 are neighbor cells in a different NR RF channel #2 in FR2.

#### 14.5.6.4.2 Test procedure

The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state and all cells start to transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-TDOA-ProvideAssistanceData* and *NR-TDOA-RequestLocationInformation* message as defined in TS 37.355 [49], shall be provided to the UE before the start of the test. The test duration should be larger than the UE measurement period as defined in TS 38.133[50] clause 5.6.2.6.

0. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 14.4.6.4.3-0.
1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 14.5.6.5-1 and Table 14.5.6.5-2. Propagation conditions are set according to clause 4.15.2.
4. Set-up period starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.

6. The UE shall transmit RRCReconfigurationComplete message
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-TDOA capabilities supported by the UE in the *NR-DL-TDOA-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-TDOA-ProvideAssistanceData* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *nr-DL-PRS-JointMeasurementRequestedPFL-List* in the *NR-DL-TDOA-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of measurement period, where ΔT = 50 ms.
11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message
12. When set-up period ends, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and set the PRS parameters as specified in Table 14.5.6.5-1
13. Wait for the UE monitors for occurrence of the periodic event requested
14. The UE shall transmit an RRCResumeRequest message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 17
15. The SS transmits an RRCResume message.
16. The UE transmits an RRCResumeComplete message
17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-TDOA-ProvideLocationInformation* IE within the response time (see clause 4.15.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.
18. If the UE message at step 17 includes the ackRequested IE set to TRUE, the SS shall send a LPP acknowledgement message.
19. The SS shall check the nr-RSTD-r16 value for Cell 2 in the nr-DL-TDOA-SignalMeasurementInformation-r16 according to Table 14.5.6.5-1.
20. Repeat steps 2-19 in Table 14.5.6.4.1-1 until the confidence level according to Annex D is achieved.

### 14.5.6.4.3 Message contents

As defined in clause 14.4.6.4.3, with the following exceptions:

Table 14.5.6.4.3-1 replaces Table 14.4.6.4.3-1, Table 14.5.6.4.3-2 replaces Table 14.4.6.4.3-2; Table 14.5.6.4.3-3 replaces Table 14.4.5.4.3-11.

**Table 14.5.6.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | Not present | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | Not present | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.6.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.6.4.3-2 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 3 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 3 |
| dl-PRS-ID-r16 | 2 | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Cell 3 | | |
| nr-ARFCN-r16 | Cell 3 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.6.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 4 |
| dl-PRS-ID-r16 | 3 | | |
| nr-PhysCellID-r16 | Cell 4 | | |
| nr-CellGlobalID-r16 | Cell 4 | | |
| nr-ARFCN-r16 | Cell 4 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 14.5.6.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 3 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 14.5.6.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| | 2 | | NR-DL-PRS-AssistanceDataPerTRP-r16[3] |
| | 3 | | NR-DL-PRS-AssistanceDataPerTRP-r16[4] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 14.5.6.4.3-3: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 4 entries | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 3 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 3 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 4 | |
| dl-PRS-ID-r16 | | | |
| nr-PhysCellID-r16 | Cell 4 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not Present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not Present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not Present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not Present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 14.5.6.5 Test requirement

The RSTD measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 14.5.6.3.

**Table 14.5.6.5-1: RSTD accuracy test parameters**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|
| PRS ARFCN | | freq1 | freq1 | freq2 | freq2 |
| Duplex mode | | TDD | | | |
| TDD configuration | | TDDConf.3.1 | | | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | - | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | - | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | - | - |
| TRS configuration | | TRS.2.1 TDD | - | - | - |
| TCI state | | TCI.State.0 | - | - | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | - | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | - | - |
| Control channel RMC | | CCR.3.1 TDD | - | - | - |

| | | | | | |
|---|---|---|---|---|---|
| OCNG Patterns | | OP.3 | | | |
| SSB configuration | | SSB.3 FR2 | | | |
| SMTC configuration | | SMTC.1 | | | |
| PRS configuration | | PRS.1.2 FR2 | | | |
| PRS Resource slot offset | slot | 0 | 4 | 0 | 4 |
| Expected RSTD | µs | N/A | 3 | 0 | 3 |
| Expected RSTD uncertainty | µs | N/A | 5 | 0 | 5 |
| Time offset with Cell 1 | µs | - | 3 | 0 | 3 |
| EPRE ratio of PSS to SSS | dB | 0 | | | |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | AWGN | | | |
| Antenna configuration | | 1x2 | | | |
| Note 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS. | | | | | |

**Table 14.5.6.5-2: RSTD accuracy OTA related test parameters**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 5 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/SCS Note3 | -89 | | -89 | |
| $\hat{E}_s / N_{oc}$ | dB | -5.7+TT | -11.9+TT | -5.7+TT | -11.9+TT |
| PRS-RSRP Note2 | dBm/SCS | -94.7+TT | -100.9+TT | -94.7+TT | -100.9+TT |
| $\hat{E}_s / I_{ot}$ BB Note4 | dB | -6+TT | -13+TT | -6+TT | -13+TT |
| Io Note2 | dBm/190.08 MHz Note3 | -55.75+TT | -55.75+TT | -55.75+TT | -55.75+TT |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | |
| Note 2: SSB_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |
| Note 3: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone. | | | | | |
| Note 4: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 36.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4. | | | | | |
| Note 5: Information about types of UE beam is given in clause B.2.1.3, and does not limit UE implementation or test system implementation | | | | | |

# 15 UE Rx-Tx time difference measurement requirements

## 15.1 General

This clause defines the minimum performance requirements for Multi-RTT NR UEs.

## 15.2 UE Rx-Tx time difference measurement period test cases in RRC_CONNECTED

### 15.2.1 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR1 SA

#### 15.2.1.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement period meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

#### 15.2.1.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports Multi-RTT positioning.

#### 15.2.1.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [49], UE shall be able to measure multiple (up to the UE capability specified in TS 38.133 [50] clause 9.9.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [4] in configured positioning frequency layers within the measurement period $T_{UERxTx,Total}$ ms.

$$T_{UERxTx,\ Total} = \sum_{i=1}^{L} T_{UERxTx,i} + (L-1) * \max\left(T_{effect,i}\right).$$

where $i$ is the index of positioning frequency layer,

$T_{UERxTx,i}$ is the measurement period for UE Rx-Tx time difference measurements in positioning frequency layer $i$ as further defined in this clause,

L is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer $i$ as defined further in this clause.

$$T_{UERxTx,i} = \left(CSSF_i * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where

$CSSF_i$ is the carrier-specific scaling factor for NR PRS-based measurement in the positioning frequency layer $i$ as defined in TS 38.133 [50] clause 9.1.5.2,

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer $i$ is in FR1 and $N_{RxBeam,i}$=8 if positioning frequency layer $i$ is in FR2,

$L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer $i$, to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of

TS 38.214 [56]. For calculation of $L_{available_PRS,\mathrm{i}}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$\mathrm{N}_{\mathrm{PRS,i}}^{\mathrm{slot}}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in clause 4.2.7.2 of TS 37.355 [49],

N′ is UE capability for number of DL PRS resources that it can process in a slot corresponding to *maxNumOfDL-PRS-ResProcessedPerSlot* as specified in clause 6.4.3 of TS 37.355 [49],

$N_{sample}$ is the number of UE Rx-Tx time difference measurement samples and $N_{sample}$ = 4,

$T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time, $T_{last,i} = T_i + T_{available_PRS,i}$,

$\mathrm{T}_{\mathrm{effect,i}}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$\mathrm{T}_{\mathrm{effect,i}} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

where

$\mathrm{T}_{\mathrm{i}}$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [49],

$T_{available_PRS,\mathrm{i}} = LCM\left(T_{PRS,\mathrm{i}}, MGRP_{\mathrm{i}}\right)$, the least common multiple between $\mathrm{T}_{\mathrm{PRS,i}}$ and $MGRP_{\mathrm{i}}$

$MGRP_{\mathrm{i}}$ is the measurement gap repetition periodicity in positioning frequency layer i.

$\mathrm{T}_{\mathrm{PRS,i}}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $\mathrm{T}_{\mathrm{PRS,i}}$, where

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

Note: For the purpose of calculating $\mathrm{T}_{\mathrm{PRS,i}}$, only the PRS resources fully or partially covered by the MG are considered.

The time $\mathrm{T}_{\mathrm{UERxTx,Total}}$ starts from the first MG instance aligned with DL PRS resources in the assistance data after both the *NR-Multi-RTT-RequestLocationInformation* message and *NR-Multi-RTT-ProvideAssistanceData* message from LMF via LPP [34] are delivered to the physical layer of UE.

Note: No per-positioning frequency layer requirement is applied in scenarios when multiple positioning frequency layers are configured.

The UE Rx-Tx time difference measurement period is restarted if HO occurs during the measurement period and after SRS reconfiguration on the target cell is complete.

The measurement requirements do not apply for a PRS resource:

- if the PRS resource is across two sampling duration of N within duration $L_{available_PRS,\mathrm{i}}$ or

- if time span of the PRS resource instance (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability N.

The normative reference for this requirement is TS 38.133 [50] clause 9.9.4.5.

### 15.2.1.4 Test description

The test defines three possible test configurations; specified in Table 15.2.1.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.2.1.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

#### 15.2.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.1.5-1 and Table 15.2.1.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.1.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

#### 15.2.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 and Cell 2 mute PRS transmission during T1 and transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources. The UE is configured to transmit SRS during T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. T1 starts.
6. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
7. The UE shall transmit RRCReconfigurationComplete message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.
9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T = 50$ MS. If the UE message at step 9 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-Multi-RTT-RequestLocationInformation* IE.

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 15.2.1.5-2.

13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.2.1.4.3). The UE shall perform and report the UE rx-tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with both the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.2.1.4.3 Message contents

**Table 15.2.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.1.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.1.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.1.4.3-3: SRS-Config (Table 15.2.1.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.1.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Configuration 1 and Configuration 2 |
| sl320 | 46 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 15.2.1.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.1.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.1.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.1.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.1.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.1.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.1.4.3-8 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.1.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| N320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 2 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.1.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.1.5 Test requirement

Table 15.2.1.5-1 and Table 15.2.1.5-2 define the primary level settings not including the test tolerances for the test.

**Table 15.2.1.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | OFF | |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 15.2.1.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1<br>ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -1.98 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.56 | N/A | -64.56 |
| | dBm/19.08 MHz | 2 | | -64.56 | | -64.56 |
| | dBm/47.88 MHz | 3 | | -60.58 | | -60.58 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.1.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( CSSF_i * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 1.142$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_j}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{1.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 0.648s and 102.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.1.4.3-5. The LPP time IE ranges between 1s and 103s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 103.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.1.5-3, Table 15.2.1.5-4 and Table 15.2.1.5-5 for reference.

**Table 15.2.1.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| n2 | 1 |
| >=n4 | 1 |

**Table 15.2.1.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{1.142}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 5 |
| nDot5 | 3 |
| n2 | 2 |
| >=n2 | 1 |

**Table 15.2.1.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.2 UE Rx-Tx time difference measurement period for dual positioning frequency layer in FR1 SA

### 15.2.2.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement period meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 in AWGN propagation condition in FR1 in standalone scenario when dual positioning frequency layers are configured.

### 15.2.2.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports Multi-RTT positioning.

### 15.2.2.3 Minimum conformance requirements

Same as defined in clause 15.2.1.3.

### 15.2.2.4 Test description

The test defines three possible test configurations; specified in Table 15.2.2.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.2.2.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

#### 15.2.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Low Range for Cell 1 and High Range for Cell2, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.2.5-1 and Table 15.2.2.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.2.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). Cell 1 and Cell2 are on different RF channels in FR1.

#### 15.2.2.4.2 Test procedure

Same as defined in clause 15.2.1.4.2.

#### 15.2.2.4.3 Message contents

**Table 15.2.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.2.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |

| | | | |
|---|---|---|---|
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.2.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.2.4.3-3: SRS-Config (Table 15.2.2.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.2.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Configuration 1 and Configuration 2 |
| sl320 | 46 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.2.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.2.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |

| | | | |
|---|---|---|---|
| responseTime SEQUENCE { | | | |
| time | 20 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.2.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |

| | | | |
|---|---|---|---|
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.2.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.2.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.2.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.2.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.2.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 2 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.2.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.2.2.5 Test requirement

Table 15.2.2.5-1 and Table 15.2.2.5-2 define the primary level settings not including the test tolerances for the test.

**Table 15.2.2.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1, 2, 3 | 1 | For Cell 1 |
| RF Channel Number | | 1, 2, 3 | 2 | For Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | OFF | |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 15.2.2.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -3 | -Infinity | -13 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2.4 | -Infinity | -12.4 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100.4 | -Infinity | -110.4 |
| | | 2 | -Infinity | -100.4 | -Infinity | -110.4 |
| | | 3 | -Infinity | -97.4 | -Infinity | -107.4 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.98 | N/A | -66.71 |
| | dBm/19.08 MHz | 2 | | -64.98 | | -66.71 |
| | dBm/47.88 MHz | 3 | | -61.00 | | -62.73 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.2.2.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,\,Total} = T_{UERxTx,\,1} + T_{UERxTx,\,2} + Max(T_{effect,i}).$$

$$T_{UERxTx,i} = \left( CSSF_i * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} = 1$, $L_{available_PRS,i} = 0.571$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * \left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{0.571}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 1.456s and 52.8s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next

multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.2.4.3-5. The LPP time IE ranges between 2s and 53s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 53.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.2.5-3, Table 15.2.2.5-4 and Table 15.2.2.5-5 for reference.

**Table 15.2.2.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| *maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{1}{N'} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.2.2.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 3 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 15.2.2.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.3 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.2.3.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement period meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.2.3.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports Multi-RTT positioning.

### 15.2.3.3 Minimum conformance requirements

Same as defined in clause 15.2.1.3.

### 15.2.3.4 Test description

The supported test configurations in listed in Table 15.2.3.4-1.

**Table 15.2.3.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 15.2.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.2.3.5-1 and Table 15.2.3.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.3.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

#### 15.2.3.4.2 Test procedure

Same as defined in clause 15.2.1.4.2.

#### 15.2.3.4.3 Message contents

**Table 15.2.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.3.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.3.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.3.4.3-3: SRS-Config (Table 15.2.3.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |

| | | | |
|---|---|---|---|
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.2.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.3.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.3.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.3.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded | |

| | | | |
|---|---|---|---|
| | | up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.3.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |

| | | | |
|---|---|---|---|
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.3.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.3.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.3.4.3-8 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.3.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.3.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.3.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.3.5 Test requirement

Table 15.2.3.5-1 and Table 15.2.3.5-2 define the primary level settings including the test tolerances for the test.

**Table 15.2.3.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | | 1 | SSB.2 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | µs | 1 | 3 | Synchronous cells |
| Expected RSTD | µs | 1 | 3 | |
| Expected RSTD uncertainty | µs | 1 | 5 | |
| Number of cells provided in assistance data | | 1 | 4 | Including the reference cell |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 15.2.3.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | | | |
| Beam Assumption Note 7 | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSSNote 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting info | | 1 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -90.3 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -99.3 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.69 | -Infinity | -12.31 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| SSB $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -2 | -Infinity | -10 |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -92.3 | -Infinity | -100.3 |
| Io | dBm/190.08 MHz | 1 | N/A | -55.89 | N/A | -55.89 |
| Propagation Condition | | 1 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.3.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( CSSF_i * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.142$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 164s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.1.4.3-3. The LPP time IE ranges between 6s and 170s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 170.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.3.5-3, Table 15.2.3.5-4 and Table 15.2.3.5-5 for reference.

**Table 15.2.3.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| n2 | 1 |
| >=n4 | 1 |

**Table 15.2.3.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.142}{N} \right\rceil$ |
|---|---|
| nDot125 | 2 |
| >=nDot25 | 1 |

**Table 15.2.3.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.4 UE Rx-Tx time difference measurement period for dual positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.2.4.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement period meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 in AWGN propagation condition in FR2 in standalone scenario when dual positioning frequency layers are configured.

### 15.2.4.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports Multi-RTT positioning.

### 15.2.4.3 Minimum conformance requirements

Same as defined in clause 15.2.1.3.

### 15.2.4.4 Test description

The supported test configurations in listed in Table 15.2.4.4-1.

**Table 15.2.4.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 15.2.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Low Range for Cell 1 and High Range for Cell2, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.2.4.5-1 and Table 15.2.4.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.4.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). Cell 1 and Cell2 are on different RF channels in FR2.

#### 15.2.4.4.2 Test procedure

Same as defined in clause 15.2.1.4.2.

15.2.4.4.3 Message contents

**Table 15.2.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.4.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.4.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.4.4.3-3: SRS-Config (Table 15.2.4.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.2.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.4.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.4.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |

| | | | |
|---|---|---|---|
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.4.5 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.4.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.4.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.4.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.4.4.3-8 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.4.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.4.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| po2-r16 | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.4.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |

| | | | |
|---|---|---|---|
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.4.5 Test requirement

Table 15.2.4.5-1 and Table 15.2.4.5-2 define the primary level settings including the test tolerances for the test.

**Table 15.2.4.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1 | 1 | For Cell 1 |
| RF Channel Number | | 1 | 2 | For Cell 2 |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | |

| | | | | |
|---|---|---|---|---|
| SSB configuration | | 1 | SSB.2 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | µs | 1 | 3 | Synchronous cells |
| Expected RSTD | µs | 1 | 3 | |
| Expected RSTD uncertainty | µs | 1 | 5 | |
| Number of cells provided in assistance data | | 1 | 4 | Including the reference cell |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 15.2.4.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | | | |
| Beam Assumption Note 7 | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSSNote 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting info | | 1 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -89 | -89.65 | -89 | -89 |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | -98.65 | -98 | -98 |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.69 | -Infinity | -12.7 |

| | | | | | | |
|---|---|---|---|---|---|---|
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2.43 | -Infinity | -11.82 |
| SSB $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -2 | -Infinity | -10 |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -92 | -Infinity | -101.4 |
| Io | dBm/190.08 MHz | 1 | N/A | -55.66 | N/A | -56.73 |
| Propagation Condition | | 1 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.4.2.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$\mathrm{T_{UERxTx,i}} = \left(\mathrm{CSSF_i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,\mathrm{i}}}{N} \right\rceil * N_{sample} - 1\right) * \mathrm{T_{effect,i}} + \mathrm{T_{last,i}}$$

Where $\mathrm{CSSF_{PRS,i}} = 1, N_{RxBeam,i} = 8, N_{PRS,i}^{slot} = 1, L_{available_PRS,\mathrm{i}} = 0.071, N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $\mathrm{T_{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,i} = LCM\left(T_{PRS,i}, MGRP_i\right)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(8 * \left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 11s and 84s. The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.4.4.3-5.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 11.3s and 84.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.4.5-3, Table 15.2.4.5-4 and Table 15.2.4.5-5 for reference.

**Table 15.2.4.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{1}{N'} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.2.4.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\lceil \frac{0.071}{N} \rceil$ |
|---|---|
| >=nDot125 | 1 |

**Table 15.2.4.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.5 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 15.2.5.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 with Nsample = 1 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured

### 15.2.5.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 15.2.5.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [34], UE shall be able to measure multiple (up to the UE capability specified in clause 9.9.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [4] in configured positioning frequency layers within the measurement period $T_{UERxTx,Total}$ ms.

$$T_{UERxTx,\ Total} = \sum_{i=1}^{L} T_{UERxTx,i} + (L-1) * \max(T_{effect,i}).$$

where $i$ is the index of positioning frequency layer,

$T_{UERxTx,i}$ is the measurement period for UE Rx-Tx time difference measurements in positioning frequency layer $i$ as further defined in this clause,

L is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer $i$ as defined further in this clause.

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where

$CSSF_i$ is the carrier-specific scaling factor for NR PRS-based measurement in the positioning frequency layer *i* as defined in clause 9.1.5.2,

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*;

otherwise,

$k_{multiTEG,i} = N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG,i} = \left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-Multi-RTT-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in clause 4.2.7.2 of TS 37.355 [34],

N' is UE capability for number of DL PRS resources that it can process in a slot corresponding to *maxNumOfDL-PRS-ResProcessedPerSlot* as specified in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:

- $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [34] and LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34] and the following conditions are met:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [34] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34] but the following conditions are not met:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time, $T_{last,i} = T_i + T_{available_PRS,i}$,

$T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

where

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$

$MGRP_i$ is the measurement gap repetition periodicity in positioning frequency layer *i*.

$T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

## 15.2.5.4 Test description

### 15.2.5.4.1 Initial conditions

The supported test configuration is listed in Table 15.2.5.4-1.

**Table 15.2.5.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.5.5-1 and Table 15.2.5.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.5.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 15.2.5.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 and Cell 2 mute PRS transmission during T1 and transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources. The UE is configured to transmit SRS during T2.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. T1 starts.
6. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
7. The UE shall transmit RRCReconfigurationComplete message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.
9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 9 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T = 50$ ms.

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 15.2.5.5-2.

13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.2.5.4.3). The UE shall perform and report the UE rx-tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.2.5.4.3 Message contents

**Table 15.2.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.5.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.5.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.5.4.3-3: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |

| | | | |
|---|---|---|---|
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.5.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Config 1 and Config 2 |
| sl320 | 46 | | Config 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.5.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.5.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.5.5 | | |
| responseTimeEarlyFix-r12 | Not present | Result of the response time calculation rounded up to the next second | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.5.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.5.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.5.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.5.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.5.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.5.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.5.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.5.5 Test requirement

Table 15.2.5.5-1 and Table 15.2.5.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.5.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [34]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [34]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | OFF | |

| | | | | |
|---|---|---|---|---|
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 15.2.5.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0.43 | -Infinity | -5.53 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 3.10 | -Infinity | -0.70 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -94.90 | -Infinity | -98.70 |
| | | 2 | -Infinity | -94.90 | -Infinity | -98.70 |
| | | 3 | -Infinity | -91.90 | -Infinity | -95.70 |

| | | | | | | |
|---|---|---|---|---|---|---|
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -100 | -100 | -Infinity | -104 |
| | | 2 | -100 | -100 | -Infinity | -104 |
| | | 3 | -97 | -97 | -Infinity | -101 |
| Io | dBm/19.08 MHz | 1 | N/A | -61.05 | N/A | -61.05 |
| | dBm/19.08 MHz | 2 | | -61.05 | | -61.05 |
| | dBm/47.88 MHz | 3 | | -57.07 | | -57.07 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP and PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE Rx-Tx time difference measurement time fulfils the requirements specified in clause 9.9.4.5 with Nsample=1.

The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2 within the specified UE Rx-Tx time difference measurement time starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported UE Rx-Tx measurement for each correct event shall be within the UE Rx-Tx reporting range specified in clause 10.1.25.3.1.

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Since the test case is for single positioning frequency layer, i = 1 and L = 1, so $T_{UERxTx,\,Total} = T_{UERxTx,\,1}$.

Analysing the equation for $T_{UERxTx,i}$:

$CSSF_{PRS,i} = 1$

$N_{RxBeam,i} = 1$

$k_{multiTEG,i}=1$

$(K_{p,PRS,i}=1$

$N_{PRS,i}^{slot} = 4$

$N_{sample}= 1.$

$T_{last,i} = T_i + T_{available_PRS,i}$

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.168s and 51.36s.

For SCS 30KHz The LPP time IE ranges between 0.168s and 25.7s.

The LPP time IE ranges between 1s and 52s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 52.3

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.5.5-3, Table 15.2.5.5-4, Table 15.2.5.5-4 and Table 15.2.5.5-6 for reference.

**Table 15.2.5.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| N' | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 15.2.5.5-4: value of N based on *durationOfPRS-ProcessingSysmbols* for SCS 15KHz**

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 15.2.5.5-5: value of N based on *durationOfPRS-ProcessingSysmbols* for SCS 30KHz**

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 15.2.5.5-6: value of $T_{effect}$ and $T_{last}$ based on durationOfPRS-ProcessingSymbolsInEveryTms**

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

## 15.2.6 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA

### 15.2.6.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement meets the requirements specified in TS 38.133 [50] clause 9.9.4.6 in AWGN propagation condition in FR1 in standalone scenario.

### 15.2.6.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and prs Processing Window.

### 15.2.6.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [34], UE shall be able to measure multiple (up to the UE capability specified in clause 9.9.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [4] in configured positioning frequency layers within the measurement period $T_{UERxTx_wo_gap,Total}$ ms.

$T_{UERxTx_wo_gap,\ Total} = \sum_{i=1}^{L} T_{UERxTx_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{UERxTx_wo_gap,\ Total} = \max(T_{UERxTx_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2.

Where,

$i$ is the index of positioning frequency layer, and

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i, and

$T_{uncertainty,i}$ is the time from the start of the first PPW occasion for positioning frequency layer i to the start of measurement period $T_{UERxTx_wo_gap,\ Total}$.

A positioning frequency layer is in Case 1 if UE reports *ppw-durationOfPRS-Processing1-r17* for the band containing the positioning frequency layer, and a positioning frequency layer is in Case 2 if UE reports *ppw-durationOfPRS-Processing2-r17* for the band containing the positioning frequency layer.

$T_{UERxTx_wo_gap,i}$ is the measurement period for UE Rx-Tx time difference measurement in positioning frequency layer *i* as specified below.

$$T_{UERxTx_wo_gap,i} = \left(k_{multiTEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where

$T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer *i* as defined further in this clause.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not supported *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*;

otherwise,

$k_{multiTEG,i} = N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG,i} = \left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*, and in case 'n0' is

indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1. If positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-Multi-RTT-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$,

only the unmuted PRS resources that meet the applicability conditions and fully or partially overlapped with PRS processing window are considered, if positioning frequency layer i is in Case 1, or

only the PRS resources unmuted and fully or partially overlapped with the first (PPWL-T2) ms of PPW are considered, if positioning frequency layer i is in Case 2, where PPWL is the PPW length and T2 corresponds to *ppw-durationOfPRS-ProcessingSymbolsT2*.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *ppw-durationOfPRS-ProcessingSymbolsN* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponding to *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

- N′ is UE capability for number of DL PRS resources that it can process in a slot corresponding to *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* as specified in clause 6.4.3 of TS 37.355 [34],

- $N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:

  - $N_{sample}$ = 4 if the UE is not capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [34].

  - $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [34] and LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34] and the following conditions are met:

    - PRS bandwidth is within the active BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* defined in [34] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34] but the following conditions are not met:

    - PRS bandwidth is within the active BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 4 otherwise.

- $T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time.

  If positioning frequency layer *i* is in Case 1 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i} = T_i$ +PPWL, else

if positioning frequency layer *i* is in Case 2 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i}$ = PPWL;

otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$.

- $T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_{PRS},i}} \right\rceil * T_{available_{PRS},i}$$

where

$T_i$ corresponds to *ppw-durationOfPRS-ProcessingSymbolsT* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponds to the sum of *ppw-durationOfPRS-ProcessingSymbolsT2* and *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$, the least common multiple between $T_{PRS,i}$ and $PPWRP_i$

$PPWRP_i$ is the PRS processing window repetition periodicity in positioning frequency layer *i*.

$T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

## 15.2.6.4 Test description

### 15.2.6.4.1 Initial conditions

The supported test configuration is listed in Table 15.2.6.4-1.

**Table 15.2.6.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.6.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.6.5-1 and Table 15.2.6.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.6.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 15.2.6.4.2 Test procedure

The test consists of two sub-tests; Sub-test 1 is for Nsample = 4 and sub-test 2 is for Nsample = 1. Sub-test 2 is only conducted when UE also supporting supportedDL-PRS-ProcessingSamples-RRC-CONNECTED. The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 and Cell 2 mute PRS transmission during T1 and transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources. The UE is configured to transmit SRS during T2.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. T1 starts.
6. The SS shall transmit an LPP REQUEST CAPABILITIES message.
7. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
8. The SS shall transmit an RRCReconfiguration message with SRS configuration and PPW configuration.
9. The UE shall transmit RRCReconfigurationComplete message.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 15.2.6.5-2.
13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.2.6.4.3). The UE shall perform and report the UE rx-tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.
14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

15.2.6.4.3 Message contents

**Table 15.2.6.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.6.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 CHOICE { | The periodicity is 160ms and the starting slot offset is 11 ms for any SCS configuration | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| } | | | |
| length-r17 | 80 | | |
| type-r17 | type1A | | |
| priority-r17 | st1 | | |
| } | | | |
| } | | | |
| } | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.6.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.6.4.3-3: SRS-Config (Table 15.2.6.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.6.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Config 1 and Config 2 |
| sl320 | 46 | | Config 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.6.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.6.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.6.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | Sub-test 1 |
| | requested | | Sub-test 2 |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.6.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |

| | | | |
|---|---|---|---|
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.6.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.6.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.6.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | As defined in Table 15.2.6.5-1 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | As defined in Table 15.2.6.5-1 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.6.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.6.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.6.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.6.5 Test requirement

Table 15.2.6.5-1 and Table 15.2.6.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.6.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [34]. |

| | | | | |
|---|---|---|---|---|
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [34]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| PRS processing window | | 1,2,3 | PPW.1 | As defined in A.3.33 |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | OFF | |
| Time offset between serving and neighbour cells | µs | 1 | 3 Note1 | |
| | | 2 | 3 Note1 | |
| | | 3 | 3 or 2.34 Note1 | |
| Expected RSTD | µs | 1,2,3 | 0 | |
| Expected RSTD uncertainty | µs | 1,2,3 | Same as time offset between serving and neighbour cells | |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |
| NOTE 1: The value is up to the UE capability. The possible UE capability value: (1/4 symbol, 1/2 symbol, CP length, half of slot). When the UE reported value is > 3us, the time offset between serving and neighbour cells is set to 3us; when the UE reported value is < 3us, the time offset between serving and neighbour cells is set to the UE reported value. | | | | |

**Table 15.2.6.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | Sub-test 1 | Sub-test 2 | Sub-tets 1 | Sub-test 2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| | | | | | | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | 0.43 | -2.41 | -5.53 | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | 3.10 | -2 | -0.7 | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -94.90 | -100 | -98.70 | -108 |
| | | 2 | -94.90 | -100 | -98.70 | -108 |
| | | 3 | -91.90 | -97 | -95.70 | -105 |
| Io | dBm/19.08 MHz | 1 | -61.05 | -64.57 | -61.05 | -64.57 |
| | dBm/19.08 MHz | 2 | -61.05 | -64.57 | -61.05 | -64.57 |
| | dBm/47.88 MHz | 3 | -57.07 | -60.59 | -57.07 | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE Rx-Tx time difference measurement time fulfils the requirements specified in clause 9.9.4.6.

The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2 within the specified UE Rx-Tx time difference measurement time starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported UE Rx-Tx measurement for each correct event shall be within the UE Rx-Tx reporting range specified in clause 10.1.25.3.1.

$T_{UERxTx_wo_gap,\ Total} = \sum_{i=1}^{L} T_{UERxTx_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{UERxTx_wo_gap,\ Total} = \max(T_{UERxTx_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2.

$N_{RxBeam,i}$ = 1

$k_{multiTEG,i}$=1

$N_{PRS,i}^{slot} = 4$

$N_{sample}$= 4.

$T_{last,i}$ = $T_i$ +PPWL

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$; where Tprs = 160 ms, and the repetition periodicity $PPWRP_i$=160ms. Therefore, $T_{available_PRS,i} = 160$

Solving for Case 1:

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.491s and 204.4s.

For SCS 30KHz The LPP time IE ranges between 0.491s and 102.4s.

The LPP time IE ranges between 1s and 205s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 205.3

Solving for Case 2:

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.491s and 25.4s.

For SCS 30KHz The LPP time IE ranges between 0.491s and 12.6s.

The LPP time IE ranges between 1s and 26s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 26.3

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.6.5-3, Table 15.2.6.5-4, Table 15.2.6.5-4 and Table 15.2.6.5-6 for reference.

**Table 15.2.6.5-3: value of N' based on *ppw-maxNumOfDL-PRS-ResProcessedPerSlot***

| N' | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 15.2.6.5-4: value of N based on ppw-durationOfPRS-ProcessingSymbolsN for SCS 15KHz**

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 15.2.6.5-4: value of N based on ppw-durationOfPRS-ProcessingSymbolsN for SCS 30KHz**

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 15.2.6.5-5: value of $T_{effect}$ and $T_{last}$ based on ppw-durationOfPRS-ProcessingSymbolsT-r17 for Case 1**

| ppw-durationOfPRS-ProcessingSymbolsT-r17 | Teffect | Tlast |
|---|---|---|
| ms1 | 160 | 11 |
| ms2 | 160 | 12 |
| ms4 | 160 | 14 |
| ms8 | 160 | 18 |
| ms16 | 160 | 26 |
| ms20 | 160 | 30 |
| ms30 | 160 | 40 |
| ms40 | 160 | 50 |
| ms80 | 160 | 90 |
| ms160 | 160 | 170 |
| ms320 | 320 | 330 |
| ms640 | 640 | 650 |
| ms1280 | 1280 | 1290 |

**Table 15.2.6.5-6: value of $T_{effect}$ and $T_{last}$ based on ppw-durationOfPRS-ProcessingSymbolsT2 for Case 2**

| Ti | Teffect | Tlast |
|---|---|---|
| 4.125ms | 160 | 10 |
| 20ms | 160 | 10 |

## 15.2.7 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with RxTx TEG in FR1 SA

### 15.2.7.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured, and when UE is requested to measure a PRS resource with RxTx TEG.

### 15.2.7.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and nr-UE-RxTxTEG-ID-MaxSupport-r17.

### 15.2.7.3 Minimum conformance requirements

Same as defined in clause 15.2.5.3.

### 15.2.7.4 Test description

#### 15.2.7.4.1 Initial conditions

The supported test configuration is listed in Table 15.2.7.4-1.

**Table 15.2.7.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.7.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.7.5-1 and Table 15.2.7.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.7.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 15.2.7.4.2 Test procedure

Same as defined in clause 15.2.5.4.2.

### 15.2.7.4.3 Message contents

**Table 15.2.7.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.7.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.7.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.7.4.3-3: SRS-Config (Table 15.2.7.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |

| | | | |
|---|---|---|---|
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.5.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Config 1 and Config 2 |
| sl320 | 46 | | Config 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.7.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.7.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.7.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | case1 | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | n0 | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.7.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |

| | | | |
|---|---|---|---|
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.7.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.7.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.5.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.5.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.7.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.7.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |

| | | | |
|---|---|---|---|
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.7.5 Test requirement

Table 15.2.7.5-1 and Table 15.2.7.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.7.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [34]. |

| | | | | |
|---|---|---|---|---|
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [34]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | OFF | |
| Time offset between serving and neighbour cells | μs | 1, 2, 3 | 3 | Synchronous cells |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 15.2.7.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS muting info | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |
| | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.59 | | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.1.3.

The UE Rx-Tx time difference measurement time fulfils the requirements specified in clause 9.9.4.5, with $k_{multiTEG}$=2 if UE does not support or indicate value 'n1' for *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*, and $k_{multiTEG}$=1 otherwise.

The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2 within the specified UE Rx-Tx time difference measurement time starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported UE Rx-Tx measurement for each correct event shall be within the UE Rx-Tx reporting range specified in clause 10.1.25.3.1.

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Analysing the equation for $T_{UERxTx_wo_gap,i}$:

$N_{RxBeam,i}$ = 1

$N_{PRS,i}^{slot} = 4$

$N_{sample}$= 4.

Case 1 :$k_{multiTEG,i} = N_{TEG,i}$

Case 2: $k_{multiTEG,i} = \left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$

$T_{last,i} = T_i + T_{available_PRS,i}$

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$

Case 1:

Where the remaining parameters depend on the UE capabilities.For SCS 15KHz The LPP time IE ranges between 1.28s and 1638.5s. For SCS 30KHz The LPP time IE ranges between 1.28s and 819.6s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.7.4.3-3. The LPP time IE ranges between 2s and 1640s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and (1638.3sec)164ten-s.

Case 2:

Where the remaining parameters depend on the UE capabilities.For SCS 15KHz The LPP time IE ranges between 0.648s and 1638.5s. For SCS 30KHz The LPP time IE ranges between 0.64s and 819.6s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.7.4.3-3. The LPP time IE ranges between 1s and 431s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and (1638.3sec)164ten-s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.7.5-4, Table 15.2.7.5-5 , Table 15.2.7.5-6 and Table 15.2.7.5-7 for reference.

**Table 15.2.7.5-3: $N'$ based on maxNumOfDL-PRS-ResProcessedPerSlot**

| N' | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 15.2.7.5-4: N based on *durationOfPRS-ProcessingSysmbols for SCS 15KHz***

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 15.2.7.5-5: N based on durationOfPRS-ProcessingSysmbols for SCS 30KHz**

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 15.2.7.5-6: Teffect and Tlast based on durationOfPRS – ProcessingSymbolsInEveryTms**

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.8 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.2.8.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 with Nsample = 1 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured

### 15.2.8.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 15.2.8.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [34], UE shall be able to measure multiple (up to the UE capability specified in clause 9.9.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [4] in configured positioning frequency layers within the measurement period $T_{UERxTx,Total}$ ms.

$$T_{UERxTx,\ Total} = \sum_{i=1}^{L} T_{UERxTx,i} + (L-1) * \max(T_{effect,i}).$$

where $i$ is the index of positioning frequency layer,

$T_{UERxTx,i}$ is the measurement period for UE Rx-Tx time difference measurements in positioning frequency layer $i$ as further defined in this clause,

L is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer $i$ as defined further in this clause.

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where

$CSSF_i$ is the carrier-specific scaling factor for NR PRS-based measurement in the positioning frequency layer $i$ as defined in clause 9.1.5.2,

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*;

otherwise,

$k_{multiTEG}=N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}=\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule.

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in clause 4.2.7.2 of TS 37.355 [34],

N' is UE capability for number of DL PRS resources that it can process in a slot corresponding to *maxNumOfDL-PRS-ResProcessedPerSlot* as specified in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:

- $N_{sample}$ = 4 if the UE is not capable of *supportedDL-PRS-ProcessingSamples* defined in [34].

- $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples* defined in [34] and LMF requests the UE to perform positioning measurements with reduced number of samples by *requestedDL-PRS-ProcessingSamples* [34] and the following conditions are met:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples* defined in [34] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *requestedDL-PRS-ProcessingSamples* [34] but the following conditions are not met:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time, $T_{last,i} = T_i + T_{available_PRS,i}$,

$T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

where

$T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$, the least common multiple between $T_{PRS,i}$ and $MGRP_i$

$MGRP_i$ is the measurement gap repetition periodicity in positioning frequency layer *i*.

$T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

## 15.2.8.4 Test description

### 15.2.8.4.1 Initial conditions

The supported test configuration is listed in Table 15.2.8.4-1.

**Table 15.2.8.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.8.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.2.8.5-1 and Table 15.2.8.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.8.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.2.8.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 and Cell 2 mute PRS transmission during T1 and transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources. The UE is configured to transmit SRS during T2.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. T1 starts.
6. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
7. The UE shall transmit RRCReconfigurationComplete message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.
9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 9 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T = 50$ ms.
12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 15.2.8.5-2.
13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.2.8.4.3). The UE shall perform and report the UE rx-tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.
14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.2.8.4.3 Message contents

**Table 15.2.8.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.8.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.8.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.8.4.3-3: SRS-Config (Table 15.2.8.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.8.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.8.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.8.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.8.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |

| | | | |
|---|---|---|---|
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.8.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.8.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 15.2.8.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.8.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.8.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.8.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.8.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |

| | | | |
|---|---|---|---|
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.8.5 Test requirement

Table 15.2.8.5-1 and Table 15.2.8.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.8.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | | 1 | SSB.3 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | μs | 1 | 3 | Synchronous cells |
| Number of cells provided in assistance data | | 1 | 4 | Including the reference cell |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 15.2.8.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption Note 7 | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS BW | | | 64 PRBs | | 64 PRBs | |
| PRS muting info | | 1 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -94 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -103 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0.32 | -Infinity | -5.64 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 3.56 | -Infinity | -0.28 |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -90.4 | -Infinity | -94.2 |
| | | | | | | |
| Io | dBm/190.08 MHz | 1 | N/A | -55.73 | N/A | -55.73 |
| Propagation Condition | | 1 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP, PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone

Note 7: Information about types of UE beam is given TS 38.133 [50] in B.2.1.3, and does not limit UE implementation or test system implementation

Note 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor ΔMBP from TS 38.101-2 [55] Table 6.2.1.3-4.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.8.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left(k_{multiTEG,i} * CSSF_i * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $K_{p,PRS,i} = 1$, $N_{RxBeam,i} = 8$, $CSSF_{PRS,i} = 1$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.071$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 2.408s and 20.64s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.8.4.3-5. The LPP time IE ranges between 3s and 21s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 3.3s and 21.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.8.5-3, Table 15.2.8.5-4 and Table 15.2.8.5-5 for reference.

**Table 15.2.8.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| *maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 15.2.8.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\left\lceil \frac{0.071}{N} \right\rceil$ |
|---|---|
| >= nDot125 | 1 |

**Table 15.2.8.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.9 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.2.9.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement meets the requirements specified in TS 38.133 [50] clause 9.9.4.6 in AWGN propagation condition in FR2 in standalone scenario.

### 15.2.9.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and prs Processing Window.

### 15.2.9.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [34], UE shall be able to measure multiple (up to the UE capability specified in clause 9.9.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [4] in configured positioning frequency layers within the measurement period $T_{UERxTx_wo_gap,Total}$ ms.

$T_{UERxTx_wo_gap,\ Total} = \sum_{i=1}^{L} T_{UERxTx_wo_gap,i} + (L-1) * \max(T_{effect,i})$, if any of the positioning frequency layers are in Case 1, or

$T_{UERxTx_wo_gap,\ Total} = \max(T_{UERxTx_wo_gap,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2.

Where,

$i$ is the index of positioning frequency layer, and

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i, and

$T_{uncertainty,i}$ is the time from the start of the first PPW occasion for positioning frequency layer i to the start of measurement period $T_{UERxTx_wo_gap,\ Total}$.

A positioning frequency layer is in Case 1 if UE reports *ppw-durationOfPRS-Processing1-r17* for the band containing the positioning frequency layer, and a positioning frequency layer is in Case 2 if UE reports *ppw-durationOfPRS-Processing2-r17* for the band containing the positioning frequency layer.

$T_{UERxTx_wo_gap,i}$ is the measurement period for UE Rx-Tx time difference measurement in positioning frequency layer *i* as specified below.

$$T_{UERxTx_wo_gap,i} = \left(k_{multiTEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where

$T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer *i* as defined further in this clause.

$k_{multiTEG,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$=1 if UE is not supported *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*;

otherwise,

$k_{multiTEG}$=$N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG}$=$\left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*, and in case 'n0' is

indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1. If positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$,

only the unmuted PRS resources that meet the applicability conditions and fully or partially overlapped with PRS processing window are considered, if positioning frequency layer i is in Case 1, or

only the PRS resources unmuted and fully or partially overlapped with the first (PPWL-T2) ms of PPW are considered, if positioning frequency layer i is in Case 2, where PPWL is the PPW length and T2 corresponds to *ppw-durationOfPRS-ProcessingSymbolsT2*.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *ppw-durationOfPRS-ProcessingSymbolsN* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponding to *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

- N′ is UE capability for number of DL PRS resources that it can process in a slot corresponding to *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* as specified in clause 6.4.3 of TS 37.355 [34],

- $N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:

  - $N_{sample}$ = 4 if the UE is not capable of *supportedDL-PRS-ProcessingSamples* defined in [34].

  - $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples* defined in [34] and LMF requests the UE to perform positioning measurements with reduced number of samples by *requestedDL-PRS-ProcessingSamples* [34] and the following conditions are met:

    - PRS bandwidth is within the active BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples* defined in [34] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *requestedDL-PRS-ProcessingSamples* [34] but the following conditions are not met:

    - PRS bandwidth is within the active BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 4 otherwise.

- $T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time.

  If positioning frequency layer *i* is in Case 1 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i} = T_i$ +PPWL, else

if positioning frequency layer *i* is in Case 2 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i}$ = PPWL;

otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$.

- $T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_{PRS,i}}} \right\rceil * T_{available_{PRS,i}}$$

where

$T_i$ corresponds to *ppw-durationOfPRS-ProcessingSymbolsT* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponds to the sum of *ppw-durationOfPRS-ProcessingSymbolsT2* and *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$, the least common multiple between $T_{PRS,i}$ and $PPWRP_i$

$PPWRP_i$ is the PRS processing window repetition periodicity in positioning frequency layer *i*.

$T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

## 15.2.9.4 Test description

### 15.2.9.4.1 Initial conditions

The supported test configuration is listed in Table 15.2.9.4-1.

**Table 15.2.9.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.9.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.2.9.5-1 and Table 15.2.9.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.9.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

#### 15.2.9.4.2 Test procedure

The test consists of two sub-tests; Sub-test 1 is for Nsample = 4 and sub-test 2 is for Nsample = 1. Sub-test 2 is only conducted when UE also supporting supportedDL-PRS-ProcessingSamples-RRC-CONNECTED. The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 and Cell 2 mute PRS transmission during T1 and transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources. The UE is configured to transmit SRS during T2.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

3. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

5. T1 starts.

6. The SS shall transmit an LPP REQUEST CAPABILITIES message.

7. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.

8. The SS shall transmit an RRCReconfiguration message with SRS configuration and PPW configuration.

9. The UE shall transmit RRCReconfigurationComplete message.

10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 7 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 15.2.9.5-2.

13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.2.9.4.3). The UE shall perform and report the UE rx-tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

15.2.9.4.3 Message contents

**Table 15.2.9.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.9.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 CHOICE { | The periodicity is 160ms and the starting slot offset is 11 ms for any SCS configuration | | |
| scs120 CHOICE { | | | |
| n1280 | 4 | | |
| } | | | |
| } | | | |
| length-r17 | 80 | | |
| type-r17 | type1A | | |
| priority-r17 | st1 | | |
| } | | | |
| } | | | |
| } | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.9.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.9.4.3-3: SRS-Config (Table 15.2.9.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |

| | | | |
|---|---|---|---|
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.9.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.9.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.9.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.9.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | Sub-test 1 |
| | requested | | Sub-test 2 |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.9.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.9.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.9.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.9.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | About 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 17 | About 0.58 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.9.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.9.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.9.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |

| | | | |
|---|---|---|---|
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.9.5 Test requirement

**Table 15.2.9.5-1 and Table 15.2.9.5-2 define the primary level settings including test tolerances for the test.Table 15.2.9.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | | 1 | SSB.3 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| PRS processing window | | 1 | TS 38.133 [50] Table A.3.33-1: Reference PPW configuration | As defined in TS 38.133 [50] A.3.33 |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | µs | 1 | 0.58, 2.25 or 3 Note 1 | |
| Expected RSTD | µs | 1 | 0 | |
| Expected RSTD uncertainty | µs | 1 | Same as time offset between serving and neighbour cells | |
| Number of cells provided in assistance data | | 1 | 4 | Including the reference cell |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |

NOTE 1: The value is up to the UE capability. The possible UE capability value: (1/4 symbol, 1/2 symbol, CP length, half of slot). When the UE reported value is > 3us, the time offset between serving and neighbour cells is set to 3us; when the UE reported value is < 3us, the time offset between serving and neighbour cells is set to the UE reported value.

**Table 15.2.9.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | Sub-test 1 | Sub-test 2 | Sub-tets 1 | Sub-test 2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption Note 7 | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS BW | | 1 | 64PRBs | 64 PRBs | 64 PRBs | 64 PRBs |
| PRS muting info | | 1 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -94 | -90.1 | -94 | -90.1 |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -103 | -99.1 | -103 | -99.1 |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | 0.32 | -2.68 | -5.64 | -12.30 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | 3.56 | -2 | -0.28 | -10 |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -90.4 | -92.1 | -94.2 | -100.1 |
| Io | dBm/95.04 MHz | 1 | -55.73 | -55.69 | -55.73 | -55.69 |
| Propagation Condition | | 1 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone

Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

Note 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.9.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $K_{p,PRS,i} = 1$, $N_{RxBeam,i} = 8$, $CSSF_{PRS,i} = 1$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.071$, $N_{sample} = 1$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 2.408s and 20.64s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.9.4.3-5. The LPP time IE ranges between 3s and 21s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 3.3s and 21.3s.

For sub-test 2, N_sample= 4. Finally, it results in the following equation:

$$\left(1 * 1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 82.08s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.9.4.3-5. The LPP time IE ranges between 6.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.9.5-3, Table 15.2.9.5-4 and Table 15.2.9.5-5 for reference.

**Table 15.2.9.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| *maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 15.2.9.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\left\lceil \frac{0.071}{N} \right\rceil$ |
|---|---|
| >= nDot125 | 1 |

**Table 15.2.9.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |

| | | |
|---|---|---|
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.10 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with RxTx TEG in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.2.10.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement meets the requirements specified in TS 38.133 [50] clause 9.9.4.5 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured, and when UE is requested to measure a PRS resource with RxTx TEG.

### 15.2.10.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and nr-UE-RxTxTEG-ID-MaxSupport-r17.

### 15.2.10.3 Minimum conformance requirements

Same as defined in clause 15.2.8.3.

### 15.2.10.4 Test description

#### 15.2.10.4.1 Initial conditions

The supported test configuration is listed in Table 15.2.10.4-1.

**Table 15.2.10.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.10.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.

2. The general test parameter settings are set up according to Table 15.2.10.5-1 and Table 15.2.10.5-2.

3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.10.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.2.10.4.2 Test procedure

Same as defined in clause 15.2.8.4.2.

### 15.2.10.4.3 Message contents

**Table 15.2.10.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.10.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.10.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.10.4.3-3: SRS-Config (Table 15.2.10.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.10.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.10.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.10.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.10.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | case1 | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | n0 | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.10.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.10.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.10.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.10.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.10.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.10.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.10.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |

| | | | |
|---|---|---|---|
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.10.5 Test requirement

Table 15.2.10.5-1 and Table 15.2.10.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.10.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | | 1 | SSB.3 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | µs | 1 | 3 | Synchronous cells |
| Number of cells provided in assistance data | | 1 | 4 | Including the reference cell |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 15.2.10.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption Note 7 | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting info | | 1 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -90.1 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -99.1 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.68 | -Infinity | -12.30 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -92.1 | -Infinity | -100.1 |
| Io | dBm/190.08 MHz | 1 | N/A | -55.69 | N/A | -55.69 |
| Propagation Condition | | 1 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone

Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

Note 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.10.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $K_{p,PRS,i} = 2$, $N_{RxBeam,i} = 8$, $CSSF_{PRS,i} = 1$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.071$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 2 * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 10.248s and 164s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.10.4.3-5. The LPP time IE ranges between 11s and 170s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 11.3s and 170.3s.

**Table 15.2.10.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 15.2.10.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.071}{N} \right\rceil$ |
|---|---|
| >= nDot125 | 1 |

**Table 15.2.10.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.11 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR1 SA

### 15.2.11.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 9.9.4.9 for UE Rx-Tx measurements with PRS bandwidth aggregation. The tests are conducted under AWGN propagation condition with

the UE operating in FR1 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR1.

### 15.2.11.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_CONNECTED.

### 15.2.11.3 Minimum conformance requirements

When UE physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [49] with a request to perform measurement by aggregating PRS resources from multiple PFLs via *nr-DL-PRS-JointMeasurementRequestedPFL-List*, the UE shall be able to perform UE Rx-Tx time difference measurement by aggregating PRS resources from multiple PFLs (up to the UE capability specified in clause 9.9.2.3) defined in TS 38.215 [4], during the measurement period $T_{UE_RxTx_aggregated,\ Total}$ defined as:

$$T_{UE_RxTx_aggregated,\ Total} = T_{non\text{-}aggregate_RxTx} + T_{aggregated_RxTx} + T_{margin},$$

where,

$T_{non_aggregate_RxTx}$ is the total measurement period for UE Rx-Tx time difference measurement on PFLs that do not contain PRS resources for aggregation.

Calculation of $T_{non_aggregate}$ is based on clause 9.9.4.5 with the following exceptions,

- only PFLs containing resource set(s) not linked to any other resource set(s) are considered in $L$
- on each PFL $i$, only resource set(s) not linked to any other resource set(s) are considered in $T_{available_PRS,i}$, $N_{PRS,i}^{slot}$ and $L_{available_PRS,i}$
- $N_{sample} = 2$ if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* specified in TS 37.355 [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples,
- $(L-1) * \max(T_{effect,i})$ is replaced by $(L-1) * T'_{margin}$.

$T_{non_aggregate_RxTx}$ is equal to zero if UE is not configured to perform non-aggregated measurements by the LMF.

$T_{aggregate_RxTx}$ is the total measurement period for UE Rx-Tx time difference measurements performed by UE by aggregating PRS resources from multiple PFLs as defined in TS 38.214 [26].

If UE is configured to perform measurements on PFLs that contain PRS resources for aggregation and on PFLs that do not contain PRS resources for aggregation, $T_{margin} = T'_{margin}$ is delay margin to account for delay between Rx-Tx time difference measurement performed by UE on PFLs that contain PRS resources for aggregation and Rx-Tx time difference measurement performed by UE on PFLs that do not contain PRS resources for aggregation.

If UE is only configured to perform measurements on PFLs that contain PRS resources for aggregation or if UE is only configured to perform measurements on PFLs that do not contain PRS resources for aggregation, then $T_{margin} = 0$.

$T'_{margin} = \max(Teffect, i)$, by considering both aggregated PFLs and non-aggregated PFLs configured for positioning measurement.

If UE is capable of performing latency reduced positioning measurements and is configured to perform latency reduced positioning measurement by LMF via *reducedDL-PRS-ProcessingSamples-r17* [34], then $N_{sample} = 2$ shall be considered in calculations of $T_{aggregate_RxTx}$ and $T_{non\text{-}aggregate_RxTx}$.

If UE is not capable to perform latency reduced positioning measurements or is not configured to perform latency reduced positioning measurement by LMF via *reducedDL-PRS-ProcessingSamples-r17* [34], then $N_{sample} = 4$ shall be considered in calculations of $T_{aggregate_RxTx}$ and $T_{non\text{-}aggregate_RxTx}$.

$T_{aggregate_RxTx}$, total measurement period for UE Rx-Tx time difference measurements performed by UE by aggregating PRS resources from multiple PFLs as defined in TS 38.214 [26], is defined as:

$$T_{aggregate_RxTx} = \sum_{i=1}^{G} T_{RxTx_aggregate,i} + (G-1) * T'_{margin} \quad ,$$

where

i is the index of effective PFL, corresponding to the group of PFLs containing linked PRS resource sets to be aggregated for UE Rx-Tx time difference measurement,

G denotes the number of effective PFLs that UE is configured to perform aggregated measurements on. G is configured by LMF and is provisioned to UE via *nr-DL-PRS-AggregationInfo.*

$T_{RxTx_aggregate,i}$ is the measurement period for UE Rx-Tx time difference measurement in the effective positioning frequency layer *i* as specified below:

$$T_{RxTx_aggregate,i} = \left(k_{multiTEG_agg,i} \times CSSF_{PRS_agg,i} \times ceil(K_{p,PRS_agg,i}) \times N_{RxBeam_agg,i} \times \left\lceil \frac{N^{slot}_{PRS_agg,i}}{N'_{agg}} \right\rceil \left\lceil \frac{L_{available_PRS_agg,i}}{N_{agg}} \right\rceil \times N_{sample_agg} - 1\right) \times T_{effect_agg,i} + T_{last_agg,i} \ ,$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor.

In FR1, $N_{RxBeam,i}$ = 1.

In FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing effective positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-Multi-RTT-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$CSSF_{PRS_agg,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in effective positioning frequency layer i as defined in clause 9.1.5.2.

$k_{multiTEG_agg,i}$ is the scaling factor for measurement of same PRS resource with multiple Rx TEGs.

$k_{multiTEG,i}$ = 1 if UE is not requested by LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*;

Otherwise,

$k_{multiTEG,i} = N_{TEG,i}$, if UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, and

$k_{multiTEG,i} = \left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$ if UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs.

where

$N_{TEG,i}$ is the number of Rx TEGs with which UE is requested to measure aggregated PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* [34] in *NR-Multi-RTT-RequestLocationInformation*, and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*, and

$k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

$K_{p,PRS_agg,i}$ is a scaling factor for effective positioning frequency layer i to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS_agg,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS_agg,i}$ = 1 for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS_agg,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the effective positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

- $N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and
- $N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

- Requirements do not apply if $N_{available} = 0$.

$N^{slot}_{PRS_agg,i}$ is the maximum number of DL PRS resources in effective positioning frequency layer i configured in a slot.

$L_{available_PRS_agg,i}$ is the time duration of available PRS in the effective positioning frequency layer i to be measured during $T_{available_PRS_agg,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS_agg,i}$, only the PRS resources unmuted and fully or partially overlapped with MG and satisfying the conditions for PRS BW aggregation are considered.

$N_{sample_agg}$ is the number of UE Rx-Tx time difference measurement samples.

$T_{last_agg,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the effective positioning frequency layer i, including the sampling time and processing time. If all of the PRS resources to be measured are available in the same MG occasion during $T_{availabe_agg}$, $T_{last_agg,i} = T_{agg,i}$ + MGL. Otherwise, $T_{last_agg,i} = T_{agg,i} + T_{available_PRS_agg,i}$,

$T_{effect_agg,i}$ is the periodicity of the UE Rx-Tx time difference measurement in the effective positioning frequency layer i defined as:

$$T_{effect_agg,i} = \left\lceil \frac{T_{agg,i}}{T_{available_PRS_agg,i}} \right\rceil \times T_{available_PRS_agg,i}$$

where,

$T_{available_PRS_agg,i} = LCM(T_{PRS_agg,i}, MGRP_i)$, the least common multiple between $T_{PRS_agg,i}$ and $MGRP_i$. $MGRP_i$ is the repetition periodicity of the measurement gap applicable for measurement in the effective PRS frequency layer i.

$T_{PRS_agg,i}$ is the periodicity of DL PRS resources meeting the bandwidth aggregation conditions with muting on effective positioning frequency layer i. If more than one PRS periodicities are configured in effective positioning frequency layer i, the least common multiple of PRS periodicities $T^{PRS\ with\ muting}_{per_agg}$ among all DL PRS resource sets in the effective positioning frequency layer is used to derive $T_{PRS_agg,i}$, where,

$T^{PRS\ with\ muting}_{per_agg} = N_{muting} \times T^{PRS}_{per_agg}$, is the PRS periodicity with muting per PRS resource configured for aggregation,

$T^{PRS}_{per_agg}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T^{PRS_agg}_{muting} \times L_{muting}$, where $T^{PRS_agg}_{muting}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

Note: For the purpose of calculating $T_{PRS_agg,i}$, only the PRS resources fully or partially covered by the MG and PRS resources that satisfy the conditions for PRS BW aggregation are considered.

$\{N_{agg}, T_{agg,i}\}$ is UE capability combination per band to aggregate PRS resources from multiple PFLs within an effective PFL where $N_{agg}$ is a duration of DL PRS symbols in ms corresponding to *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsN* in TS 37.355 [34] processed every $T_{agg,i}$ ms corresponding to *prs-durationOfTwoPRS-BWA-ProcessingSymbolsT* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsT* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding

to *maximumOfTwoAggregatedDL-PRS-Bandwidth* or *maximumOfThreeAggregatedDL-PRS-Bandwidth* in TS 37.355 [34]

$N'_{agg}$ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfAggregatedDL-PRS-ResourcePerSlot* specified in TS 37.355 [34].

The normative reference for this requirement is TS 38.133 [50] clause 9.9.4.9.

## 15.2.11.4 Test description

The test defines three possible test configurations; specified in Table 15.2.11.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.2.11.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

### 15.2.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.11.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.11.5-1 and Table 15.2.11.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.11.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.2.11.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Cell 1 and Cell 2 mute PRS transmission during T1 and transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources. The UE is configured to transmit SRS during T2.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. T1 starts.
6. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
7. The UE shall transmit RRCReconfigurationComplete message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.

9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.

10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE and *NR-DL-PRS-AggregationInfo-r18* linking each PRS resource in PFL1 to the corresponding PRS resource in PFL2 such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 MS. If the UE message at step 9 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

11. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the NR-Multi-RTT-RequestLocationInformation IE and nr-DL-PRS-JointMeasurementRequest-r18 IE.

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 15.2.11.5-2.

13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.2.11.4.3). The UE shall perform and report the UE rx-tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with both the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.2.11.4.3 Message contents

**Table 15.2.11.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.2.11.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.2.11.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.11.4.3-3: SRS-Config (Table 15.2.11.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.11.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| sl160 | 20 | | Configuration 1 and Configuration 2 |
| sl320 | 40 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.2.11.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.2.11.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.11.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |

| | | | |
|---|---|---|---|
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.11.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.2.11.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.11.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |

| | | | |
|---|---|---|---|
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.11.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.11.4.3-8 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.11.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.11.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.11.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-Assistance |

| | | | |
|---|---|---|---|
| | | | DataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.11.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.11.5 Test requirement

Table 15.2.11.5-1 and Table 15.2.11.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.11.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2, all signals/channels except PRS in PFL2 are allocated within this channel. |
| | | | 2 | For both Cell 1 and Cell 2, only PRS in PFL2 is allocated within this channel. |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | For both RF channels |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |

| | | | | |
|---|---|---|---|---|
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | OFF | |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells/TRPs |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |
| NOTE 1: GP#24 is configured if the UE supports it, otherwise GP#0 is configured. | | | | |

**Table 15.2.11.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns Note 1 | | 1, 2, 3 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1,4 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1, 4 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration (for both PFL1 and PFL2) | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS muting info (*dl-PRS-MutingOption1* for both PFL1 and PFL2) | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| Io Note 3 (on symbols where PRS is not allocated) | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |
| | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.59 | | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved on all OFDM symbols except those in which PRS is allocated.

NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 3: PRP and Io levels have been derived from other parameters and they are provided for information only. They are not settable parameters themselves.

NOTE 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.11.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{RxTx_aggregate,i} = \left(k_{multiTEG_agg,i} \times CSSF_{PRS_agg,i} \times ceil(K_{p,PRS_agg,i}) \times N_{RxBeam_agg,i} \times \left\lceil \frac{N_{PRS_agg,i}^{slot}}{N'_{agg}} \right\rceil \left\lceil \frac{L_{available_PRS_agg,i}}{N_{agg}} \right\rceil \times N_{sample_agg} - 1\right) \times T_{effect_agg,i} + T_{last_agg,i},$$

Where $k_{multiTEG_agg,i} = 1$, $CSSF_{PRS_agg,i} = 1$, $K_{p,PRS_agg,i} = 1$, $N_{RxBeam_agg,i} = 1$, $N_{PRS_agg,i}^{slot} = 1$, $L_{avavailable_PRS_agg,,i} = 0.285$, $N_{sample_agg} = 4$. $N'_{agg}$ is the parameter maxNumOfAggregatedDL-PRS-ResourcePerSlot from TS 37.355 [49], Nagg is the parameter prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN from TS 37.355 [49], $T_{last_agg,i} = Tagg,i + T_{available_PRS_agg,i}$ and $T_{effect_agg,i} = \left\lceil \frac{T_{agg,i}}{T_{available_PRS_agg,i}} \right\rceil \times T_{available_PRS_agg,i}$.

$T_{available_PRS_agg,i} = LCM(T_{PRS_agg,i}, MGRP_i)$; where $T_{PRS_agg,i}$ = 320 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS_agg,i} = 320$ ms.

Finally, it results in the following equation:

$$(1 \times 1 \times 1 \times 1 \times \left\lceil \frac{1}{N'_{agg}} \right\rceil \left\lceil \frac{0.285}{N_{agg}} \right\rceil \times 4\text{-}1) \times T_{effect_agg,i} + T_{last_agg,i}$$

Depending on the UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]:

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 1.288s and 25.92s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should

be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 2s and 26s.

Similarly for Type 2 Buffering, Best Case result of 1.288s is 2s and for the worst-case result of 41280ms, rounded to the next second, the final value is 42s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 42.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 1.288s and 15.68s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 2s and 16s.

Similarly for Type 2 Buffering, Best Case result of 1.288s is 1s and for the worst-case result of 20800ms, rounded to the next second, the final value is 21s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 21.3s.

**Table 15.2.11.5-3: value of N' based on *maxNumOfAggregatedDL-PRS-ResourcePerSlot***

| $N'_{agg}$ | $\left\lceil \frac{1}{N'_{agg}} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.2.11.5-4: value of $N_{agg}$ based on test configuration and UE buffer type**

| **$N_{agg}$ for SCS 15kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.57}{N} \right\rceil$ |
|---|---|
| msDot125 | 5 |
| msDot25 | 3 |
| msDot5 | 2 |
| >=ms1 | 1 |
| **$N_{agg}$ for SCS 30kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.285}{N} \right\rceil$ |
| msDot125 | 3 |
| msDot25 | 2 |
| >=msDot5 | 1 |
| $N_{agg}$ **for SCS 15kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{1}{N} \right\rceil$ |
| msDot125 | 8 |
| msDot25 | 4 |
| msDot5 | 2 |
| >=ms1 | 1 |
| $N_{agg}$ **for SCS 30kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{0.5}{N} \right\rceil$ |
| msDot125 | 4 |
| msDot25 | 2 |
| >=msDot5 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26] | |

**Table 15.2.11.5-5: value of $T_{effect_agg,i}$ and $T_{last_agg,i}$ based on $T_{agg,i}$**

| $T_{agg,i}$ | $T_{effect_agg,i}$ | $T_{last_agg,i}$ |
|---|---|---|
| ms8 | 320 | 328 |
| ms16 | 320 | 336 |
| ms20 | 320 | 340 |
| ms30 | 320 | 350 |
| ms40 | 320 | 360 |
| ms80 | 320 | 400 |
| ms160 | 320 | 480 |
| ms320 | 320 | 640 |
| ms640 | 640 | 960 |
| ms1280 | 1280 | 1600 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.12 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

### This test case is incomplete for test frequencies > 40.8 GHz15.2.12.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 9.9.4.9 for UE Rx-Tx measurements with PRS bandwidth aggregation. The tests are conducted under AWGN propagation condition with the UE operating in FR2 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR2.

### 15.2.12.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_CONNECTED.

### 15.2.12.3 Minimum conformance requirements

Same as defined in clause 15.2.11.3.

### 15.2.12.4 Test description

The supported test configuration is listed in Table 15.2.12.4-1.

**Table 15.2.12.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 15.2.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.12.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.2.12.5-1 and Table 15.2.12.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.12.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.2.12.4.2 Test procedure

Same as defined in clause 15.2.11.4.2.

### 15.2.12.4.3 Message contents

As defined in clause 15.2.11.4.3, with the following exceptions:

Table 15.2.12.4.3-1 replaces Table 15.2.11.4.3-3, Table 15.2.12.4.3-2 replaces Table 15.2.11.4.3-7 and Table 15.2.12.4.3-3 replaces Table 15.2.11.4.3-8.

**Table 15.2.12.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.12.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.2.12.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.12.4.3-8 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.12.4.3-8 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | N2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.12.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.12.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.12.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | n2 | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| N2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.12.5 Test requirement

Table 15.2.12.5-1 and Table 15.2.12.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.12.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2, all signals/channels except PRS in PFL2 are allocated within this channel. |
| | | | 2 | For both Cell 1 and Cell 2, only PRS in PFL2 is allocated within this channel. |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | For both RF channels |
| SSB configuration | | 1 | SSB.2 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | μs | 1 | 3 | Synchronous cells/TRPs |
| Expected RSTD | μs | 1 | 3 | |

| | | | | |
|---|---|---|---|---|
| Expected RSTD uncertainty | μs | 1 | 5 | |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |
| Note 1: GP#24 is configured if the UE supports it, otherwise GP#13 is configured. | | | | |

**Table 15.2.12.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in clause TS 38.133 clause A.3.15 | | | |
| Beam Assumption[Note 5] | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns[Note 1] | | 1 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS[Note 1, 4] | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS [Note 1, 4] | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration (for both PFL1 and PFL2) | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting info (*dl-PRS-MutingOption1* for both PFL1 and PFL2) | | 1 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ [Note 2] | dBm/SCS | 1 | -89 | -90.1 | -89 | -90.1 |
| $N_{oc}$ [Note 2] | dBm/15 kHz | 1 | -98 | -99.1 | -98 | -99.1 |
| PRS $\hat{E}_s / I_{ot}$ | dB | 1 | -Infinity | -2.69 | -Infinity | -12.1 |
| PRS $\hat{E}_s / N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -9.8 |
| PRP [Note 3] | dBm/SCS kHz | 1 | -Infinity | -92.1 | -Infinity | -99.9 |
| Io [Note 3] (on symbols where PRS is not allocated) | dBm/190.08 MHz | 1 | N/A | -55.68 | N/A | -55.68 |
| Propagation Condition | | 1 | AWGN | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved on all OFDM symbols except those in which PRS is allocated. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | PRP and Io levels have been derived from other parameters and they are provided for information only. They are not settable parameters themselves. |
| Note 4: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 5: | Information about types of UE beam is given in clause TS 38.133 clause B.2.1.3, and does not limit UE implementation or test system implementation |
| Note 6: | Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] table 6.2.1.3-4. |
| Note 7: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone. |
| Note 8: | As observed with 0 dBi gain antenna at the centre of the quiet zone. |

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.12.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{RxTx_aggregate,i} = \left(k_{multiTEG_agg,i} \times CSSF_{PRS_agg,i} \times ceil(K_{p,PRS_agg,i}) \times N_{RxBeam_agg,i} \times \left\lceil \frac{N_{PRS_agg,i}^{slot}}{N'_{agg}} \right\rceil \left\lceil \frac{L_{available_PRS_agg,i}}{N_{agg}} \right\rceil \times N_{sample_agg} - 1\right) \times T_{effect_agg,i} + T_{last_agg,i},$$

Where $k_{multiTEG_agg,i} = 1$, $CSSF_{PRS_agg,i} = 1$, $K_{p,PRS_agg,i} = 1$, $N_{RxBeam_agg,i} = 8$, $N_{PRS_agg,i}^{slot} = 1$, $L_{avavailable_PRS_agg,,i} = 0.142$, $N_{sample_agg} = 4$. $N'_{agg}$ is the parameter maxNumOfAggregatedDL-PRS-ResourcePerSlot from TS 37.355 [49], Nagg is the parameter prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN from TS 37.355 [49], $T_{last_agg,i}$ = Tagg,i + $T_{available_PRS_agg,i}$ and $T_{effect_agg,i} = \left\lceil \frac{T_{agg,i}}{T_{available_PRS_agg,i}} \right\rceil \times T_{available_PRS_agg,i}$.

$T_{available_PRS_agg,i} = LCM(T_{PRS_agg,i}, MGRP_i)$; where $T_{PRS_agg,i}$ = 640 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS_agg,i} = 640$ ms.

Finally, it results in the following equation:

$$(1 \times 1 \times 1 \times 8 \times \left\lceil \frac{1}{N'_{agg}} \right\rceil \left\lceil \frac{0.142}{N_{agg}} \right\rceil \times 4\text{-}1) \times T_{effect_agg,i} + T_{last_agg,i}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 41.646s and 166.4s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.12.4.3-5. The LPP time IE ranges between 42s and 170s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 42.3s and 85.3s.

**Table 15.2.12.5-3: value of N' based on *maxNumOfAggregatedDL-PRS-ResourcePerSlot***

| $N'_{agg}$ | $\left\lceil \frac{1}{N'_{agg}} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.2.12.5-4: value of N based on *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN***

| $N_{agg}$ | $\left\lceil \frac{0.142}{N} \right\rceil$ |
|---|---|

| | |
|---|---|
| msDot125 | 2 |
| >= msDot25 | 1 |

## 15.2.13 UE Rx-Tx time difference measurement reporting delay test case in FR1 for Redcap UE without Rx FH

### 15.2.13.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement without RX FH reported by the RedCap UE meets the requirements specified in TS 38.133 [50] clause 9.9A.4.5 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured. The measurement reporting delay test defined in this clause is valid for both 1 Rx and 2 Rx RedCap UEs.

### 15.2.13.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method.

### 15.2.13.3 Minimum conformance requirements

Same as defined in clause 15.2.11.3.

### 15.2.13.4 Test description

The test defines three possible test configurations; specified in Table 15.2.13.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.2.13.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

#### 15.2.13.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.13.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.13.5-1 and Table 15.2.13.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.13.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.2.13.4.2 Test procedure

Same as defined in clause 15.2.11.4.2.

### 15.2.13.4.3 Message contents

As defined in clause 15.2.11.4.3, with the following exceptions:

Table 15.2.13.4.3-1 replaces Table 15.2.11.4.3-3, Table 15.2.13.4.3-2 replaces Table 15.2.11.4.3-5, Table 15.2.13.4.3-3 replaces Table 15.2.11.4.3-7, Table 15.2.13.4.3-4 replaces Table 15.2.11.4.3-8 and Table 15.2.13.4.3-5 replaces Table 15.2.11.4.3-9.

**Table 15.2.13.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.13.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 20 | | Configuration 1, Configuration 2 and Configuration 4 |
| sl320 | 40 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.2.13.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.13.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |

| | | | |
|---|---|---|---|
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.13.5 Test requirement

Table 15.2.13.5-1 and Table 15.2.13.5-2 define the primary level settings including test tolerances for the test.

**Table A.16.6.7.1.1-2: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1,2,3,4 | Cell 1 | Cell 1 is the reference cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1,2,3,4 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* TS 37.355 [49]. |

| | | | | |
|---|---|---|---|---|
| RF Channel Number | | 1,2,3,4 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1,4 | 10: $N_{RB,c}$ = 52 | |
| | | 2 | 10: $N_{RB,c}$ = 52 | |
| | | 3 | 20: $N_{RB,c}$ = 52 | |
| SSB configuration | | 1,4 | SSB.4 RedCap FR1 | |
| | | 2 | SSB.4 RedCap FR1 | |
| | | 3 | SSB.5 RedCap FR1 | |
| SMTC configuration | | 1,4 | SMTC.1 RedCap | |
| | | 2 | SMTC.1 RedCap | |
| | | 3 | SMTC.1 RedCap | |
| Measurement gap | | 1,2,3,4 | GP#24 or GP#0 Note 1 | |
| CP length | | 1,2,3,4 | Normal | |
| DRX | | 1,2,3,4 | OFF | |
| Time offset between serving and neighbour cells | μs | 1,2,3,4 | 3 | Synchronous cells |
| T1 | s | 1,2,3,4 | 5 | |
| T2 | s | 1,2,3,4 | 10 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table A.16.6.7.1.1-3: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1,4 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1,4 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1,4 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1,4 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1,2,3,4 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1,2,3,4 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1,4 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1,2,3,4 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1,2,3,4 | DLBWP.1.1 RedCap | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 RedCap | | N/A | |
| PRS configuration | | 1,4 | PRS.1.1 FR1 | | PRS.1.1 FR1 | |
| | | 2 | PRS.1.1 FR1 | | PRS.1.1 FR1 | |
| | | 3 | PRS.2.1 FR1 | | PRS.2.1 FR1 | |
| PRS muting info | | 1,2,3,4 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1,4 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/SCS | 1,4 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1,4 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1,4 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1,4 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| Io | dBm/9.36 MHz | 1,4 | N/A | -67.67 | N/A | -67.67 |
| | dBm/9.36 MHz | 2 | | -67.67 | | -67.67 |
| | dBm/18.72 MHz | 3 | | -64.67 | | -64.67 |
| Propagation Condition | | 1,2,3,4 | AWGN | | | |
| NOTE 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | | | | |
| NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | | |
| NOTE 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | | |
| NOTE 4: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS. | | | | | | |

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.13.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$\mathrm{T_{UERxTx,i}} = \left( k_{multiTEG,i} * \mathrm{CSSF_i} * \mathrm{ceil}(\mathrm{K_{p,PRS,i}}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,\mathrm{i}}}{N} \right\rceil * N_{sample} - 1 \right) * \mathrm{T_{effect,i}} + \mathrm{T_{last,i}}$$

Where $\mathrm{CSSF_{PRS,i}} = 1$, $\mathrm{K_{p,PRS,i}} = 1$, $N_{RxBeam,i} = 1$, $\mathrm{CSSF_{PRS,i}} = 1$, $N_{PRS,i}^{slot} = 1$, $L_{available_PRS,\mathrm{i}} = 1.142$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $\mathrm{T_{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where $T_{PRS,i}$ = 320 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 320$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 1 * 1 * \left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{1.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 1.288s and 51.52s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 2s and 52s.

Similarly for Type 2 Buffering, Best Case result of 1.288s is 2s and for the worst-case result of 82240ms, rounded to the next second, the final value is 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 83.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 1.288s and 82.24s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 2s and 83s.

Similarly for Type 2 Buffering, Best Case result of 1.288s is 2s and for the worst-case result of 41280ms, rounded to the next second, the final value is 42s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 42.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.13.5-3, Table 15.2.13.5-4 and Table 15.2.13.5-5 for reference.

**Table 15.2.13.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{1}{N'} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.2.13.5-4: value of N based on test configuration and UE buffer type**

| **N for SCS 15kHz for Type 1 Buffering** [Note 1] | $\left\lceil \frac{1.142}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 5 |
| nDot5 | 3 |
| n1 | 2 |
| >=n2 | 1 |
| **N for SCS 30kHz for Type 1 Buffering** [Note 1] | $\left\lceil \frac{0.571}{N} \right\rceil$ |
| nDot125 | 5 |
| nDot25 | 3 |
| nDot5 | 2 |
| >=n1 | 1 |
| N **for SCS 15kHz for Type 2 Buffering** [Note 1] | $\left\lceil \frac{2}{N} \right\rceil$ |
| nDot125 | 16 |
| nDot25 | 8 |
| nDot5 | 4 |
| n1 | 2 |
| >=n2 | 1 |
| N **for SCS 30kHz for Type 2 Buffering** [Note 1] | $\left\lceil \frac{1}{N} \right\rceil$ |
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26] | |

**Table 15.2.13.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.14 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE without Rx FH

Editor's note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.2.14.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement without RX FH in RRC_CONNECTED mode meets the requirements specified in TS 38.133 [50] clause 9.9A.4.5 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured. The measurement reporting delay test defined in this clause is valid for both 1 Rx and 2 Rx RedCap UEs.

### 15.2.14.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method.

### 15.2.14.3 Minimum conformance requirements

Same as defined in clause 15.2.11.3.

### 15.2.14.4 Test description

The supported test configuration is listed in Table 15.2.14.4-1.

**Table 15.2.14.4-1: Test Configuration**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

#### 15.2.14.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.14.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.2.14.5-1 and Table 15.2.14.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.14.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.2.14.4.2 Test procedure

Same as defined in clause 15.2.11.4.2.

### 15.2.14.4.3 Message contents

As defined in clause 15.2.11.4.3, with the following exceptions:

Table 15.2.14.4.3-1 replaces Table 15.2.11.4.3-3, Table 15.2.14.4.3-2 replaces Table 15.2.11.4.3-5, Table 15.2.14.4.3-3 replaces Table 15.2.11.4.3-7, Table 15.2.14.4.3-4 replaces Table 15.2.11.4.3-8 and Table 15.2.14.4.3-5 replaces Table 15.2.11.4.3-9.

**Table 15.2.14.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.14.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.2.14.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.14.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.14.4.3-3: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined | | |

| | | | |
|---|---|---|---|
| | for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.14.4.3-4 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.14.4.3-4 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.14.4.3-4: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |

| | | | |
|---|---|---|---|
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.14.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.14.5 Test requirement

Table 15.2.14.5-1 and Table 15.2.14.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.14.5-1: General test parameters.**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |

| | | | | |
|---|---|---|---|---|
| $BW_{channel}$ | MHz | 1 | 100: $N_{RB,c}$ = 66 | |
| SSB configuration | | 1 | SSB.2 RedCap FR2 | |
| SMTC configuration | | 1 | SMTC.1 RedCap | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | µs | 1 | 3 | Synchronous cells |
| Expected RSTD | µs | 1 | 3 | |
| Expected RSTD uncertainty | µs | 1 | 5 | |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 15.2.14.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 clause A.3.15 | | | |
| Beam Assumption Note 7 | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 RedCap | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 RedCap | | N/A | |
| PRS configuration | | 1 | PRS.1.5 FR2 | | PRS.1.5 FR2 | |
| PRS muting info | | 1 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -89 | | | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -91 | -Infinity | -99 |
| Io | dBm/92.16 MHz | 1 | N/A | -57.63 | N/A | -57.63 |
| Propagation Condition | | 1 | AWGN | | | |

NOTE 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over sub-carriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
NOTE 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
NOTE 7: Information about types of UE beam is given in TS 38.133 annex B.2.1.3, and does not limit UE implementation or test system implementation
NOTE 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] table 6.2.1.3-4.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.14.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $k_{multiTEG,i} = 1$, $K_{p,PRS,i} = 1$, $N_{RxBeam,i} = 8$, $CSSF_{PRS,i} = 1$, $N_{PRS,i}^{slot} = 1$, $L_{available_PRS,i} = 0.035$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355[49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where $T_{PRS,i}$ = 320 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 320$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355[49]

Finally, it results in the following equation:

$$\left(1 * 1 * 1 * 8 * \left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

For Type 1 duration calculation with UE symbol level buffering capability, the LPP time IE ranges between 10.248s and 41.28s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 11s and 42s.

Similarly for Type 2 Buffering, Best Case result of 10.248s is 11s and for the worst-case result of 41280ms, rounded to the next second, the final value is 42s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 11.3s and 42.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.14.5-3, Table 15.2.14.5-4 and Table 15.2.14.5-5 for reference.

**Table 15.2.14.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| *maxNumOfDL-PRS-ResProcessedPerSlot* | $\lceil \frac{1}{N'} \rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.2.14.5-4: value of N based on test configuration and UE buffer type**

| **N for SCS 120kHz for Type 1 Buffering** Note 1 | $\lceil \frac{0.035}{N} \rceil$ |
|---|---|
| >=nDot125 | 1 |
| N **for SCS 120kHz for Type 2 Buffering** Note 1 | $\lceil \frac{0.125}{N} \rceil$ |
| >=nDot125 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26] | |

**Table 15.2.14.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 320 | 328 |
| n16 | 320 | 336 |
| n20 | 320 | 340 |
| n30 | 320 | 350 |
| n40 | 320 | 360 |
| n80 | 320 | 400 |
| n160 | 320 | 480 |
| n320 | 320 | 640 |
| n640 | 640 | 960 |
| n1280 | 1280 | 1600 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.2.15 UE Rx-Tx time difference measurement reporting delay test case in FR1 for Redcap UE with Rx FH

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.2.15.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement with Rx FH in RRC_CONNECTED state meets the requirements specified in TS 38.133 [50] clause 9.9A.4.8 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.2.15.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and maximumPRS-BandwidthAcrossAllHopsFR1 for at least one band and PRS frequency hopping measurement and report

### 15.2.15.3 Minimum conformance requirements

The requirements in this clause apply when a RedCap UE is requested by the LMF to perform measurements with FH, and UE reports measurement based on multiple hops.

The requirements in clause 9.9A.4.5 shall apply with the following modifications.

- $N_{sample}$ = 2 if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* specified in TS 37.355 [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples. Otherwise, $N_{sample}$ = 4.

- A measurement sample with FH is defined as a PRS measurement over multiple hops within a single measurement gap.

- $L_{available_PRS,i} = \sum_{j=1}^{J} N_{hop,i,j} * L_{per-hop,i,j}$, where

  - $N_{hop,i,j}$ is the number of hops that UE can do in an MG occasion *j* as defined in the following, and

  - $L_{per-hop,i,j}$ is the time duration of available PRS resources in the positioning frequency layer *i* in each hop to be measured in MG instance *j* during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{per-hop,i,j}$, only the PRS resources unmuted and fully or partially overlapped with the sampling duration in each hop are considered;

  - *J* is the number of MG instance during $T_{available_PRS,i}$.

The sampling window per hop is the first $T_{sample,hop}$ symbols in each hop, where $T_{sample,hop} = T_{hop} - RRT_{FH}$, $T_{hop}$ is the applicable length per hop as defined in table 9.9A.4.8-1, and $RRT_{FH}$ is the retuning time between Rx hops as reported by UE in *rf-RxRetunTimeFR1* or *rf-RxRetunTimeFR2* via *NR-DL-PRS-ProcessingCapability* specified in TS 37.355 [49]. The first hop within a MG instance starts at no earlier than the earliest arrival time of the first unmuted PRS resource fully or partially overlapped with the MG instance taking into account the *expectedRSTD* and *expectedRSTD-uncertainty* in the PRS assistance data.

**Table 9.9A.4.8-1: Applicable number of hops per slot and applicable length of each hop**

| $RRT_{FH}$ | (comb size, Number of PRS symbols) | Applicable number of hops per slot ($N_{hops}^{slot}$) | Applicable length per hop ($T_{hop}$) in number of symbols |
|---|---|---|---|
| $RRT_{FH} \leq 2$ symbols | (2, 12) with SCS 15 kHz, 30 kHz, 60 kHz in FR2, 120 kHz | 2 | 7 |
| | All others | 1 | 14 |
| 2 symbols $< RRT_{FH} \leq 6$ symbols | (≤ 6, any) | 1 | 14 |
| | (12, 12) | ½ | 28 |
| $RRT_{FH} > 6$ symbols | Any combination | ½ | 28 |

The number of hops within a single MG occasion $N_{hop,i,j}$ is defined as

$$N_{hop,i,j} = min\left(N_{hops,effect,i,j}, \left\lfloor \frac{N3_i}{L_{per-hop,i,j}} \right\rfloor, N_{hop,max,i}\right)$$

where

- $N_{hop,max}$ is the maximum number of Rx hops signaled by UE in *maximumFH-Hops* via *NR-DL-PRS-ProcessingCapability* as specified in TS 37.355 [49] for positioning frequency layer *i*

- $N3_i$ is indicated via UE capability *processingPRS-SymbolsDurationN3-r18* for positioning frequency layer *i*

- $N_{hops,effect,i,j}$ is the effective number of Rx hops within a MG instance,

- $N_{hops,effect,i,j} = 2 * N_{rep}^{PRS}$, if $N_{hops}^{slot}$ = 2,

- $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot}$ = 1,

- $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot}$ = 1/2 and $M_{rep}^{PRS}$ >1,

- $N_{hops,effect,i,j} = \left\lfloor \frac{(N_{rep}^{PRS}-1)}{2} \right\rfloor + 1$, if $N_{hops}^{slot}$ = 1/2 and $M_{rep}^{PRS}$ =1,

- where $N_{rep}^{PRS}$ is the number of inter-slot PRS repetitions within the MG occasion excluding the gap retuning times, which are unmuted and fully or partially overlapped with the sampling duration within each hop. $M_{rep}^{PRS}$ is

the PRS repetition interval configured by *dl-PRS-ResourceTimeGap* and $N_{hops}^{slot}$ is the applicable number of hops per slot as defined in table 9.9A.4.8-1.

The measurement period requirement with the above modifications applies provided all PRS resources in each PFL have the same number of PRS repetitions $\left(N_{rep}^{PRS}\right)$ within an MG instance and the same PRS repetition interval $\left(M_{rep}^{PRS}\right)$, otherwise the measurement period can be longer.

UE shall be able to measure PRS resources with multiple hops at least over the BW of $BW_{total}$ defined as

$$BW_{total} = min\left(BW_{PRS}, N_{hop,i,j} * BW_{per-hop} - \left(N_{hop} - 1\right) * BW_{overlap}\right)$$

where

- $BW_{PRS}$ is the minimum among
- configured PRS BW, and
- UE capability of maximum PRS BW across all hops indicated via *maximumPRS-BandwidthAcrossAllHopsFR1-r18* or *maximumPRS-BandwidthAcrossAllHopsFR2-r18*, and
- total BW of all hops requested by LMF via *nr-DL-PRS-RxHoppingTotalBandwidth-r18* as specified in TS 37.355 [49],

$N_{hop,i,j}$ is number of hops within a single MG occasion as define above,

$BW_{per-hop}$ is the UE capability on PRS BW per hop indicated via *supportedBandwidthPRS-r16* as specified in TS 37.355 [49],

$BW_{overlap}$ is the UE capability on BW of the overlapping RB indicated via *numOfOverlappingPRB-r18* as specified in TS 37.355 [49].

The normative reference for this requirement is TS 38.133 [50] clause 9.9A.4.8.

## 15.2.15.4 Test description

The test defines three possible test configurations; specified in Table 15.2.15.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.2.15.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, cell bandwidth 50 MHz, FDD duplex mode |
| 2 | 15 kHz SSB SCS, cell bandwidth 50 MHz, TDD duplex mode |
| 3 | 30 kHz SSB SCS, cell bandwidth 100 MHz, TDD duplex mode |
| 4 | 15 kHz SSB SCS, cell bandwidth 50 MHz, HD-FDD duplex mode |
| NOTE 1: The RedCap UE is only required to be tested in one of the supported test configurations.<br>NOTE 2: This test case is applicable to the UE supporting per hop bandwidth ≥ 10 MHz for SCS 15 kHz.<br>NOTE 3: This test case is applicable to the UE supporting per hop bandwidth = 20 MHz for SCS 30 kHz. | |

### 15.2.15.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.15.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.2.15.5-1 and Table 15.2.15.5-2.
3. Propagation conditions are set according to clause 4.14.2.

4. Message contents are defined in clause 15.2.15.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.2.15.4.2 Test procedure

Same as defined in clause 15.2.11.4.2.

### 15.2.15.4.3 Message contents

As defined in clause 15.2.11.4.3, with the following exceptions:

Table 15.2.15.4.3-1 replaces Table 15.2.11.4.3-3, Table 15.2.15.4.3-2 replaces Table 15.2.11.4.3-5, Table 15.2.15.4.3-3 replaces Table 15.2.11.4.3-7, Table 15.2.15.4.3-4 replaces Table 15.2.11.4.3-8 and Table 15.2.15.4.3-5 replaces Table 15.2.11.4.3-9.

**Table 15.2.15.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |

| | | | |
|---|---|---|---|
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.15.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 20 | | Configuration 1, Configuration 2 and Configuration 4 |
| sl320 | 40 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.2.15.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.15.5 | Result of the response time calculation rounded up to the next | |

| | | | |
|---|---|---|---|
| | | second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | mhz50 | | Configuration 1, Configuration 2 and Configuration 4 |
| | mhz100 | | Configuration 3 |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.15.4.3-3: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1, Configuration 2 and Configuration 4 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 62 | 268 PRBs | Configuration 1, Configuration 2 and Configuration 4 |
| | 63 | 272 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.15.4.3-4 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.15.4.3-4 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 15.2.15.4.3-4: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1, Configuration 2 and Configuration 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |

| | | | |
|---|---|---|---|
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | Configuration 1, Configuration 2 and Configuration 4 |
| | 0 | | Configuration 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | Configuration 1, Configuration 2 and Configuration 4 |
| | 0 | | Configuration 3 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.15.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.2.15.5 Test requirement

FFS

## 15.2.16 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE with Rx FH

Editor's note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.2.16.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement with Rx FH in RRC_CONNECTED state meets the requirements specified in TS 38.133 [50] clause 9.9A.4.8 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.2.16.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and maximumPRS-BandwidthAcrossAllHopsFR2 for at least one band and PRS frequency hopping measurement and report

### 15.2.16.3 Minimum conformance requirements

Same as defined in clause 15.2.15.3.

### 15.2.16.4 Test description

The supported test configuration is listed in Table 15.2.16.4-1.

**Table 15.2.16.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, cell bandwidth 400 MHz, TDD duplex mode |
| NOTE 1: This test case is applicable to the UE supporting per hop bandwidth = 100 MHz for SCS 120 kHz. | |

#### 15.2.16.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.2.16.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.2.16.5-1 and Table 15.2.16.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.2.16.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.2.16.4.2 Test procedure

Same as defined in clause 15.2.11.4.2.

#### 15.2.16.4.3 Message contents

As defined in clause 15.2.11.4.3, with the following exceptions:

Table 15.2.16.4.3-1 replaces Table 15.2.11.4.3-3, Table 15.2.16.4.3-2 replaces Table 15.2.11.4.3-5, Table 15.2.16.4.3-3 replaces Table 15.2.11.4.3-7, Table 15.2.16.4.3-4 replaces Table 15.2.11.4.3-8 and Table 15.2.16.4.3-5 replaces Table 15.2.11.4.3-9.

**Table 15.2.16.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.16.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.2.16.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.2.15.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr2 | mhz100 | | |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.16.4.3-3: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 61 | 264 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.16.4.3-4 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.2.16.4.3-4 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 15.2.16.4.3-4: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | NR Cell 1 |
| | 01 | | NR Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.2.16.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.2.16.5 Test requirement

Table 15.2.16.5-1 and Table 15.2.16.5-2 define the primary level settings including test tolerances for the test.

**Table 15.2.16.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 400: $N_{RB,c}$ = 264 | |
| SSB configuration | | 1 | SSB.3 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | µs | 1 | 3 | Synchronous cells |
| PRS RX hopping request | | 1 | requested | |
| T1 | s | 1 | 5 | |

| | | | | |
|---|---|---|---|---|
| T2 | s | 1 | 10 | |
| NOTE 1: GP#24 is configured if RedCap UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 15.2.16.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 clause A.3.15 | | | |
| Beam Assumption[Note 7] | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS[Note 1] | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS [Note 1] | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 RedCap | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 RedCap | | N/A | |
| PRS configuration | | 1 | PRS.1.6 FR2 | | PRS.1.6 FR2 | |
| PRS muting info | | 1 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ [Note 2] | dBm/SCS | 1 | -93.73 | | | |
| $N_{oc}$ [Note 2] | dBm/15 kHz | 1 | -102.73 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.69 | -Infinity | -8.83 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -1.2 | -Infinity | -6 |
| PRP [Note 3] | dBm/SCS | 1 | -Infinity | -94.9 | -Infinity | -99.7 |
| Io | dBm/ 380.16MHz | 1 | N/A | -55.66 | N/A | -55.66 |
| Propagation Condition | | 1 | AWGN | | | |

NOTE 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
NOTE 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
NOTE 7: Information about types of UE beam is given in TS 38.133 annex B.2.1.3, and does not limit UE implementation or test system implementation.
NOTE 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] table 6.2.1.3-4.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.2.16.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( k_{multiTEG,i} * CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $k_{multiTEG,i} = 1$, $K_{p,PRS,i} = 1$, $N_{RxBeam,i} = 8$, $CSSF_{PRS,i} = 1$, $N_{PRS,i}^{slot} = 1$, , $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_j}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where $T_{PRS,i}$ = 320 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 320$ ms. $T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

$$L_{available_{PRS},i} = \sum_{j=1}^{J} N_{hop,i,j} * L_{per-hop,i,j}$$

$$N_{hop,i,j} = min\left( N_{hops,effect,i,j}, \left\lfloor \frac{N3_i}{L_{per-hop,i,j}} \right\rfloor, N_{hop,max,i} \right)$$

$N3_i$ depends on UE parameter processingPRS-SymbolsDurationN3-r18 from TS 37.355 [49]. $N_{hop,max}$ depends on UE parameter maximumFH-Hops from TS 37.355 [49].

For type 1 duration calculation, we can get the minimum $L_{available_{PRS},i} = 0.28$ ms and the maximum $L_{available_{PRS},i} = 0.84$ ms.

For type 2 duration calculation, we can get the minimum $L_{available_{PRS},i} = 0.5$ ms and the maximum $L_{available_{PRS},i} = 7$ ms.

Finally, it results in the following equation:

$$\left(1 * 1 * 1 * 8 * \left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS},i}}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

For Type 1 duration calculation with UE symbol level buffering capability, the LPP time IE ranges between 10.248s and 205.12s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 11s and 210s.

Similarly for Type 2 Buffering, Best Case result of 10.248s is 11s and for the worst-case result of 328000ms, rounded to the next second, the final value is 330s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 11.3s and 330.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.16.5-3, Table 15.2.16.5-4 and Table 15.2.16.5-5 for reference.

**Table 15.2.16.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{1}{N'} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.2.16.5-4: value of N based on test configuration and UE buffer type**

| **N for SCS 120kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.28}{N} \right\rceil$ |
|---|---|
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |
| **N for SCS 120kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.84}{N} \right\rceil$ |
| nDot125 | 5 |
| nDot25 | 3 |
| nDot5 | 2 |
| >=n1 | 1 |
| N **for SCS 120kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{0.5}{N} \right\rceil$ |
| nDot125 | 4 |
| nDot25 | 2 |
| >=nDot5 | 1 |
| N **for SCS 120kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{1}{N} \right\rceil$ |
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26] | |

**Table 15.2.16.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 320 | 328 |
| n16 | 320 | 336 |
| n20 | 320 | 340 |
| n30 | 320 | 350 |
| n40 | 320 | 360 |
| n80 | 320 | 400 |
| n160 | 320 | 480 |
| n320 | 320 | 640 |
| n640 | 640 | 960 |
| n1280 | 1280 | 1600 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 15.3 UE Rx-Tx time difference measurement accuracy test cases in RRC_CONNECTED

## 15.3.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA

### 15.3.1.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.3.1.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports Multi-RTT positioning.

### 15.3.1.3 Minimum conformance requirements

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall not apply, if:

$N_{TA_offset}$ defined in Table 7.1.2-2 changes during the UE Rx-Tx measurement period or

if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the network-configured Timing Advance.

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that:

- The UE transmits SRS within [-160, 160] msec of at least one DL PRS resource of each of the TRPs in the assistance data.

FFS: whether UE Rx-Tx time difference measurement accuracy requirements in this clause shall also apply if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the autonomous timing adjustment defined in clause 7.1.2.

The UE shall continue and complete a UE Rx-Tx measurement while meeting UE Rx-Tx measurement accuracy requirements defined in this clause when a serving cell change occurs during the UE Rx-Tx measurement provided that the serving cell change does not impact the SRS configuration for the UE Rx-Tx measurement.

Note: The requirements for fading channel in this clause are derived based on TDL-A (30 ns delay spread, 5Hz) and TDL-C (60 ns delay spread, 300 Hz) channel models for FR1 and FR2 respectively.

The accuracy requirements in Table 10.1.25.2-1 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 for a corresponding Band.

AWGN propagation condition.

**Table 10.1.25.2-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ± 78+δ | -3 | ≥24 | 15 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, | -127 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_SDL_FR1_A | | |
| | | | | | NR_FDD_FR1_B | -126.5 | |
| | | | | | NR_TDD_FR1_C | -126 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | |
| | | | | | NR_FDD_FR1_F | -124.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | |
| | | | | | NR_FDD_FR1_H | -123.5 | |
| ± 59+δ | | ≥52 | | ≥1 | Note 6 | Note 6 | Note 6 |
| ± 30+δ | | >104 | | ≥1 | Note 6 | Note 6 | Note 6 |
| ± 57+δ | | ≥24 | 30 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | |
| | | | | | NR_TDD_FR1_C | -123 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | |
| | | | | | NR_FDD_FR1_F | -121.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | |
| | | | | | NR_FDD_FR1_H | -120.5 | |
| ± 30+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 15+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 29+δ | | ≥24 | 60 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | |
| | | | | | NR_TDD_FR1_C | -120 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | |
| | | | | | NR_FDD_FR1_H | -117.5 | |
| ± 15+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 101+δ | -13 | ≥24 | 15 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 75+δ | | ≥52 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 37+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 58+δ | | ≥24 | 30 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 39+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 36+δ | | ≥24 | 60 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 8+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [34].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 10.1.25.2-5.

…

…

**Table 10.1.25.2-5: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (PRB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 24 | N/A | N/A | 160 |
| ≥ 52 | ≥ 24 | N/A | 80 |
| ≥ 104 | ≥ 48 | ≥ 24 | 56 |
| N/A | ≥ 132 | ≥ 64 | 24 |
| N/A | N/A | ≥ 132 | 24 |
| NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].<br>NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies. | | | |

## 15.3.1.4 Test description

The test defines three possible test configurations; specified in Table 15.3.1.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.3.1.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

### 15.3.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.3.1.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.1.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 15.3.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-Multi-RTT-ProvideAssistanceData and NR-Multi-RTT-RequestLocationInformation as defined in clause 15.3.1.4.3 shall be provided to the UE during the set-up period. The last TTI containing the NR-Multi-RTT-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T$ = 50 ms is the maximum processing time of the NR-Multi-RTT-RequestLocationInformation message and the Multi-RTT assistance data in the UE.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE.If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE.
12. As soon as possible after step 11 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.
13. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
14. The SS shall check the reported value of nr-UE-RxTxTimeDiff in the *NR-Multi-RTT-SignalMeasurementInformation* IE provided by the UE in step 11 and compare it with the value measured in step 12. The SS shall check that the reported value is within the limits specified in table 15.3.1.5-2 compared to the measured value. If the reported value is within the limits the number of successful results is increased by one. If the reported value is not within the limits or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 11, or does not respond at step 11 within the time given by the *time* IE in the *CommonIEsRequestLocationInformation* IE in step 10, then the number of unsuccessful results for is increased by one.
15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.
16. Repeat step 1-15 for the other sub-test defined in Table 15.3.1.4-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.3.1.4.3 Message contents

**Table 15.3.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.3.1.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.3.1.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.1.4.3-3: SRS-Config (Table 15.3.1.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.1.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Sub-test 1-1, Sub-test 1-2, Sub test 2-1 and Sub-test 2-2 |
| sl320 | 46 | | Sub-test 3-1 and Sub-test 3-2 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.1.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.3.1.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 11 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.1.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.3.1.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.1.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Sub-test 1-1, Sub-test 1-2, Sub test 2-1 and Sub-test 2-2 |
| | kHz30 | | Sub-test 3-1 and Sub-test 3-2 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1-1,, Sub test 2-1 and Sub-test 3-1 |
| | 21 | 104 PRBs | Sub-test 1-2 and Sub-test 2-2 |
| | 28 | 132 PRBs | Sub-test 3-2 |
| dl-PRS-StartPRB-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | n4 | | Sub-test 1-2, Sub-test 2-2 and Sub-test 3-2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.1.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.1.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.1.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Sub-tests 1-1, Sub-test 1-2, Sub-test 2-1 and Sub-test 2-2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Sub-tests 3-1 and Sub–test 3-2 |
| n320-r16-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| | n2 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| n4-r16 | 0 | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1-1 Cell 1, Sub-test 1-2 Cell 1, Sub-test 2-1 Cell 1, Sub-test 2-2 Cell 1, Sub-test 3-1 Cell 1,and Sub-test 3-2 Cell 1 |
| | 2 | | Sub-test 1-1 Cell 2, Sub-test 1-2 Cell 2, Sub-test 2-1 Cell 2, |

| | | | |
|---|---|---|---|
| | | | Sub-test 2-2 Cell 2, Sub-test 3-1 Cell 2, and Sub-test 3-2 Cell 2. |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.1.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Set according to specific test configuration. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Set according to specific test configuration. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.1.5 Test requirement

Table 15.3.1.5-1 defines the primary level settings including the test tolerances for the test.

**Table 15.3.1.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| RF Channel Number | | 1,2,3 | 1 | 1 | 1 | 1 |
| Measurement gap | | 1,2,3 | GP#24 or GP#0 Note 4 | | GP#24 or GP#0 Note 4 | |
| DRX | | 1,2,3 | OFF | | OFF | |
| Time offset with Cell 1 | μs | 1, 2, 3 | N/A | 3 | N/A | 3 |
| TDD configuration | | 1 | N/A | N/A | N/A | N/A |
| | | 2 | TDDConf.1.1 | TDDConf.1.1 | TDDConf.1.1 | TDDConf.1.1 |
| | | 3 | TDDConf.2.1 | TDDConf.2.1 | TDDConf.2.1 | TDDConf.2.1 |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | N/A | SR.1.1 FDD | N/A |
| | | 2 | SR.1.1 TDD | | SR.1.1 TDD | |
| | | 3 | SR.2.1 TDD | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | N/A | CR.1.1 FDD | N/A |
| | | 2 | CR.1.1 TDD | | CR.1.1 TDD | |
| | | 3 | CR.2.1 TDD | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | N/A | CCR.1.1 FDD | N/A |
| | | 2 | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | | 3 | CCR.2.1 TDD | | CCR.2.1 TDD | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | OP.1 | OP.1 | OP.1 |
| TRS Configuration | | 1 | TRS.1.1 FDD | N/A | TRS.1.1 FDD | N/A |
| | | 2 | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | | 3 | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | N/A | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | N/A | DLBWP.1.1 | N/A |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | N/A | ULBWP.1.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR1 | PRS.1.1 FR1 | PRS.1.2 FR1 | PRS.1.2 FR1 |
| | | 2 | PRS.1.1 FR1 | PRS.1.1 FR1 | PRS.1.2 FR1 | PRS.1.2 FR1 |
| | | 3 | PRS.2.1 FR1 | PRS.2.1 FR1 | PRS.2.2 FR1 | PRS.2.2 FR1 |
| PRS Resource slot offset | slot | 1, 2, 3 | 0 | 2 | 0 | 2 |
| SRS configuration | | 1 | POS-SRS.1 | N/A | POS-SRS.1 | N/A |
| | | 2 | POS-SRS.1 | N/A | POS-SRS.1 | N/A |
| | | 3 | POS-SRS.2 | N/A | POS-SRS.2 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | -98 | |
| | | 2 | -98 | | -98 | |
| | | 3 | -95 | | -95 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | -98 | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.31 | -16.12 | -2.31 | -12.6 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -1.9 | -10 | -1.9 | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -100 | -108 | -100 | -108 |
| | | 2 | -100 | -108 | -100 | -108 |
| | | 3 | -97 | -105 | -97 | -105 |
| Io | dBm/9.36 MHz | 1 | -64.54 | -64.54 | -64.54 | -64.54 |
| | dBm/9.36 MHz | 2 | -64.54 | -64.54 | -64.54 | -64.54 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | dBm/38.16 MHz | 3 | -60.55 | -60.55 | -60.55 | -60.55 |
| Propagation Condition | | 1, 2, 3 | AWGN | | AWGN | |
| Note 1: Void.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: PP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 4: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | | | |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 15.2.1.5, with the following difference: $N_{PRS,i}^{slot}$= 2. Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(\left\lceil\frac{2}{N'}\right\rceil\left\lceil\frac{L_{available_{PRS,i}}}{N}\right\rceil * 4 - 1\right) * T_{\text{effect}} + \text{T}_{\text{last}}$$

**Table 15.3.1.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | Sub-test 1 | (Measured value from step 12 - 270) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 270) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 12 - 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 2 | Sub-test 1 | (Measured value from step 12 - 270) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 270) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 12 - 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 | Sub-test 1 | (Measured value from step 12 - 169) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 169) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 12 - 71) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 71) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.1.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.3.2 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA

### 15.3.2.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.3.2.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports Multi-RTT positioning.

### 15.3.2.3 Minimum conformance requirements

Same as in clause 15.3.1.3.

### 15.3.2.4 Test description

The supported test configurations in listed in Table 15.3.2.4-1.

**Table 15.3.2.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

#### 15.3.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.3.2.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.2.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

#### 15.3.2.4.2 Test procedure

Same as in clause 15.3.1.4.2.

#### 15.3.2.4.3 Message contents

**Table 15.3.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.3.2.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.3.2.4.3-3 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.2.4.3-3: SRS-Config (Table 15.3.2.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.2.5-1 | | |
| } | | | |

| | | | |
|---|---|---|---|
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.2.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.3.2.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 42 | | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.2.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.3.2.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.2.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.2.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.2.4.3-8 | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.2.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-test 2 |
| | n2 | | Sub-test 1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.2.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |

| | | | |
|---|---|---|---|
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.3.2.5 Test requirement

Table 15.3.2.5-1 defines the primary level settings including the test tolerances for the test.

**Table 15.3.2.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 clause A.3.15 | | Setup 1 as specified in TS 38.133 clause A.3.15 | |
| Beam Assumption Note 7 | | 1 | Rough | Rough | Rough | Rough |
| Measurement gap | | 1 | GP#24 or GP#13 Note 8 | | GP#24 or GP#13 Note 8 | |
| DRX | | 1 | OFF | | OFF | |
| Time offset with Cell 1 | µs | 1 | N/A | 3 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of OCNG to OCNG DMRS Note 1<br>EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | N/A | TRS.2.1 TDD | N/A |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | N/A | DLBWP.1.1 | N/A |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | N/A | ULBWP.1.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR2 | PRS.1.1 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset | slot | 1 | 0 | 2 | 0 | 2 |
| SRS configuration | | 1 | POS-SRS.3 | N/A | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -89 | | -89 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | -98 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.62 | -12.26 | -2.62 | -12.26 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -10 | -2 | -10 |
| PRP Note 3 | dBm/SCS kHz | 1 | -91 | -99 | -91 | -99 |
| Io | dBm/190.08 MHz | 1 | -54.59 | -54.59 | -54.59 | -54.59 |
| Propagation Condition | | 1 | AWGN | | AWGN | |

Note 1: Void.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133[]B.2.1.3, and does not limit UE implementation or test system implementation
Note 8: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured.

**Table 15.3.2.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | Sub-test 1 | (Measured value from step 12 - 76) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 76) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 12 - 56) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 56) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.2.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.3.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA

### 15.3.3.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy with reduced number of samples is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.3.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 15.3.3.3 Minimum conformance requirements

The accuracy requirements in Table 10.1.25.2-1 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 for a corresponding Band

Number of measurement samples is less than 4

AWGN propagation condition.

**Table 10.1.25.2-3a: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN with reduced measurement samples**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ± 59+δ | 0 | ≥52 | 15 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| ± 30+δ6 | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 30+δ | | ≥48 | 30 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| ± 15+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 15+δ | | ≥64 | 60 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 75+δ | -6 | ≥52 | 15 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 37+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 39+δ | | ≥48 | 30 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥64 | 60 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 8+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 10.1.25.2-5.

…

**Table 10.1.25.2-6: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (RB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 24 | N/A | N/A | 160 |
| ≥ 52 | ≥ 24 | N/A | 80 |
| ≥ 104 | ≥ 48 | ≥ 24 | 56 |
| N/A | ≥ 132 | ≥ 64 | 24 |
| N/A | N/A | ≥ 132 | 24 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.

## 15.3.3.4 Test description

### 15.3.3.4.1 Initial conditions

The supported test configuration is listed in Table 15.3.3.4-1.

**Table 15.3.3.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |

| | |
|---|---|
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.3.3.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.3.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 15.3.3.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 or #13 before the test.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms.
11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.3.3.4.3).
12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the NR-Multi-RTT-SignalMeasurementInformation according to Table 15.3.3.5-2.

14. Repeat steps 1-13 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

#### 15.3.3.4.3 Message contents

**Table 15.3.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.3.3.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.3.3.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.3.4.3-3: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.3.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Config 1 and config 2 |
| sl320 | 46 | | Config 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.3.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.3.3.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.3.5 | Result of the response time calculation rounded up to the next second. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |

| | | | |
|---|---|---|---|
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.3.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.3.3.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.3.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1&2 |
| | kHz30 | | Config3 |
| dl-PRS-ResourceBandwidth-r16 | 13 | 52 PRBs | Config 1&2 |
| | 7 | 48 PRBs | Config3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.3.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.3.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.3.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 10 | | For config 1,2 &3 |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell1 |
| | 2 | | Cell2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.3.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.3.5 Test requirement

Table15.3.3.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.3.3.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| RF Channel Number | | 1,2,3 | 1 | 1 |
| Measurement gap | | 1,2,3 | GP#24 or GP#0 Note 4 | |
| DRX | | 1,2,3 | OFF | |
| Time offset with Cell 1 | μs | 1, 2, 3 | N/A | 3 |
| TDD configuration | | 1 | N/A | N/A |
| | | 2 | TDDConf.1.1 | TDDConf.1.1 |
| | | 3 | TDDConf.2.1 | TDDConf.2.1 |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | N/A |
| | | 2 | SR.1.1 TDD | |
| | | 3 | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | N/A |
| | | 2 | CR.1.1 TDD | |
| | | 3 | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | N/A |
| | | 2 | CCR.1.1 TDD | |
| | | 3 | CCR.2.1 TDD | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | N/A |
| | | 2 | TRS.1.1 TDD | |
| | | 3 | TRS.1.2 TDD | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | N/A |

| | | | | |
|---|---|---|---|---|
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | N/A |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 2 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 3 | PRS.2.1 FR1 | PRS.2.1 FR1 |
| PRS BW | | 1 | 52 PRBs | 52 PRBs |
| | | 2 | 52 PRBs | 52 PRBs |
| | | 3 | 48 PRBs | 48 PRBs |
| PRS Resource slot offset | slot | 1, 2, 3 | 0 | 2 |
| SRS configuration | | 1 | POS-SRS.1 | N/A |
| | | 2 | POS-SRS.1 | N/A |
| | | 3 | POS-SRS.2 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | |
| | | 2 | -98 | |
| | | 3 | -95 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | |
| | | 2 | | |
| | | 3 | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | 0 | -6 |
| | | 2 | | |
| | | 3 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | 2.93 | -0.9 |
| | | 2 | | |
| | | 3 | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -95.07 | -98.90 |
| | | 2 | -95.07 | -98.90 |
| | | 3 | -92.07 | -95.90 |
| Io | dBm/19.08 MHz | 1 | -61.18 | -61.18 |
| | dBm/19.08 MHz | 2 | -61.18 | -61.18 |
| | dBm/47.88 MHz | 3 | -57.20 | -57.20 |
| Propagation Condition | | 1, 2, 3 | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.

**Table 15.3.3.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1&2 | cell 1 | (Measured value from step 13 - 171) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 171) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | cell 2 | (Measured value from step 13 - 187) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 187) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 | cell 1 | (Measured value from step 13 – 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | cell 2 | (Measured value from step 13 - 127) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 127) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.3.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.3.4 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR1 SA

### 15.3.4.1 Test purpose

The purpose of the test is to verify that the relative UE Rx-Tx time difference measurement accuracy when the two measurements are within the same RxTx TEG is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.3.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and nr-UE-RxTxTEG-ID-MaxSupport-r17.

### 15.3.4.3 Minimum conformance requirements

The relative accuracy requirements in Table 10.1.25.2-1 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 for a corresponding Band

AWGN propagation condition.

the two UE Rx-Tx time difference measurements are associated with the same RxTx TEG

**Table 10.1.25.2-3b: UE Rx-Tx time difference relative measurement accuracy in FR1 in AWGN with TEG reporting**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Io Note 3 range | | |
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | NR operating band groups Note 4 | Minimum Io | Maximum Io |
| Tc Note 5 | dB | kHz | PRB | | | dBm/SCS | dBm/BW$_{Channel}$ |
| 132 + Δ Note 7 | (PRS Ês/Iot)$_i$ ≥ -6dB<br><br>(PRS Ês/Iot)$_i$ ≥ -13dB | 15 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| 98 + Δ | | | ≥ 52 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 42 + Δ | | | ≥ 104 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 75 + Δ | | 30 | ≥ 24 | ≥ 4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| 48 + Δ | | | ≥ 48 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 24 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 50 + Δ | | | | | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | 60 | ≥24 | ≥ 4 | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| 24 + Δ | | | ≥ 64 | ≥ 1 | Note 6 | Note 6 | Note 6 |
| 10 + Δ | | | ≥ 132 | ≥ 1 | Note 6 | Note 6 | Note 6 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of resource j and resource i.
NOTE 2: Minimum number of PRS resource repetitions among resource j and resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: NR operating band groups in FR1 are as defined in clause 3.5.2.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: Δis the value of the timing error margin for the RxTx TEG, reported via *nr-UE-RxTxTEG-TimingErrorMargin*. Δ cannot be larger than the sum of the margins in table 10.1.25.2-5 (dependent on PRS/SRS BW) for any pair of individual UE Rx-Tx time difference measurements associated with the RxTx TEG. .

…

**Table 10.1.25.2-6: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (RB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 24 | N/A | N/A | 160 |
| ≥ 52 | ≥ 24 | N/A | 80 |
| ≥ 104 | ≥ 48 | ≥ 24 | 56 |
| N/A | ≥ 132 | ≥ 64 | 24 |
| N/A | N/A | ≥ 132 | 24 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.

## 15.3.4.4 Test description

### 15.3.4.4.1 Initial conditions

The supported test configuration is listed in Table 15.3.4.4-1.

**Table 15.3.4.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|

| | |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.3.4.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.4.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 15.3.4.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.1 FR1 and PRS.1.2 FR1. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-UE-RxTxTEG-Request-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms.
11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.3.4.4.3).
12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the NR-Multi-RTT-SignalMeasurementInformation according to Table 15.3.4.5-3.

14. Repeat steps 1-13 until the confidence level according to Annex D is achieved.

15. Repeat step 1-14for the other sub-test defined in Table 15.3.4.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.3.4.4.3 Message contents

**Table 15.3.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.3.4.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.3.4.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.4.4.3-3: SRS-Config (Table 15.3.4.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.4.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 23 | | Config 1 and config 2 |
| Sl320 | 46 | | Config 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.4.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.3.4.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | case1 | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | n2 | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.4.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.3.4.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.4.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1&2 |
| | kHz30 | | Config3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1 |
| | 22 | 128 PRBs | Sub-test 2 Config 1 & config 2 |
| | 28 | 132 PRBs | Sub-test 2 Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.4.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.4.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.4.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 & config2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | n1 | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and |

| | | | |
|---|---|---|---|
| | | | Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.4.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.4.5 Test requirement

Table 15.3.4.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.3.4.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| RF Channel Number | | 1,2,3 | 1 | 1 | 1 | 1 |
| Measurement gap | | 1,2,3 | GP#24 or GP#0 Note 4 | | GP#24 or GP#0 Note 4 | |
| DRX | | 1,2,3 | OFF | | OFF | |
| Time offset with Cell 1 | µs | 1, 2, 3 | N/A | 3 | N/A | 3 |
| TDD configuration | | 1 | N/A | N/A | N/A | N/A |
| | | 2 | TDDConf.1.1 | TDDConf.1.1 | TDDConf.1.1 | TDDConf.1.1 |
| | | 3 | TDDConf.2.1 | TDDConf.2.1 | TDDConf.2.1 | TDDConf.2.1 |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | N/A | SR.1.1 FDD | N/A |
| | | 2 | SR.1.1 TDD | | SR.1.1 TDD | |
| | | 3 | SR.2.1 TDD | | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | N/A | CR.1.1 FDD | N/A |
| | | 2 | CR.1.1 TDD | | CR.1.1 TDD | |
| | | 3 | CR.2.1 TDD | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | N/A | CCR.1.1 FDD | N/A |
| | | 2 | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | | 3 | CCR.2.1 TDD | | CCR.2.1 TDD | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | OP.1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | N/A | TRS.1.1 FDD | N/A |
| | | 2 | TRS.1.1 TDD | | TRS.1.1 TDD | |
| | | 3 | TRS.1.2 TDD | | TRS.1.2 TDD | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | N/A | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | N/A | DLBWP.1.1 | N/A |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | N/A | ULBWP.1.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR1 | PRS.1.1 FR1 | PRS.1.2 FR1 | PRS.1.2 FR1 |
| | | 2 | PRS.1.1 FR1 | PRS.1.1 FR1 | PRS.1.2 FR1 | PRS.1.2 FR1 |
| | | 3 | PRS.2.1 FR1 | PRS.2.1 FR1 | PRS.2.2 FR1 | PRS.2.2 FR1 |
| PRS Resource slot offset | slot | 1, 2, 3 | 0 | 2 | 0 | 2 |
| SRS configuration | | 1 | POS-SRS.1 | N/A | POS-SRS.1 | N/A |
| | | 2 | POS-SRS.1 | N/A | POS-SRS.1 | N/A |
| | | 3 | POS-SRS.2 | N/A | POS-SRS.2 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | -98 | |
| | | 2 | -98 | | -98 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 3 | -95 | | -95 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | -98 | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.41 | -12.12 | -2.41 | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -10 | -2 | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -100 | -108 | -100 | -108 |
| | | 2 | -100 | -108 | -100 | -108 |
| | | 3 | -97 | -105 | -97 | -105 |
| Io | dBm/19.08 MHz | 1 | -64.57 | -64.57 | -64.57 | -64.57 |
| | dBm/19.08 MHz | 2 | -64.57 | -64.57 | -64.57 | -64.57 |
| | dBm/47.88 MHz | 3 | -60.59 | -60.59 | -60.59 | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | AWGN | |

Note 1: Void.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.

**Table 15.3.4.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | Sub-test 1 | (Measured value from step 13 - 324) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 324) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 13 - 130) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 130) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 2 | Sub-test 1 | (Measured value from step 13 - 324) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 324) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 13 - 130) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 130) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 | Sub-test 1 | (Measured value from step 13 - 187) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 187) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 13 - 80) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 13 + 80) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.4.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.3.5 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

### 15.3.5.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy with reduced number of samples is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.3.5.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 15.3.5.3 Minimum conformance requirements

The accuracy requirements in Table 10.1.25.2-3a for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 for a corresponding Band

Number of measurement samples is less than 4

AWGN propagation condition.

**Table 10.1.25.2-3a: UE Rx-Tx time difference measurement accuracy in FR2 in AWGN with reduced measurement samples**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$ Note 3 | Io Note 4 range | |
| | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | dBm / $SCS_{PRS}$ | $dBm/BW_{Channel}$ |
| ± 15+δ | 0 | ≥64 | 60 | ≥1 | Same value as PRP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥64 | 120 | ≥1 | Same value as PRP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 15+δ | -6 | ≥64 | 60 | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 9+δ | | ≥64 | 120 | ≥1 | NOTE 6 | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN defined in TS 37.355 [34].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.

NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 10.1.25.2-6.

…

**Table 10.1.25.2-6: Margin for UE Rx-Tx time difference measurement accuracy in FR2**

| Min(PRS BW, SRS BW) (MHz) | | Margin (Tc [Note 1]) |
|---|---|---|
| SCS = 60 kHz | SCS = 120 kHz | |
| ≥ 24 | N/A | 76 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 24 |
| N/A | ≥ 128 | 20 |
| NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].<br>NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies. | | |

## 15.3.5.4 Test description

### 15.3.5.4.1 Initial conditions

The supported test configuration is listed in Table 15.3.5.4-1.

**Table 15.3.5.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, e200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.3.5.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.5.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.3.5.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 or #13 before the test.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of +512 $T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T$ = 50 ms.
11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.3.5.4.3).
12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
13. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the NR-Multi-RTT-SignalMeasurementInformation according to Table 15.3.5.5-3.
14. Repeat steps 1-13 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.3.5.4.3 Message contents

**Table 15.3.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.3.5.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |

| | | | |
|---|---|---|---|
| setup | As defined in Table 15.3.5.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.5.4.3-3: SRS-Config (Table 15.3.5.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.5.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.5.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.3.5.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.5.5 | Result of the response time calculation rounded up to the next second. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |

| | | | |
|---|---|---|---|
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.5.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.3.5.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.5.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.5.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.5.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.5.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell1 |
| | 2 | | Cell2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 15.3.5.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.5.5 Test requirement

Table15.3.5.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.3.5.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | |
| Beam Assumption[Note 7] | | 1 | Rough | Rough |
| Measurement gap | | 1 | GP#24 or GP#13 [Note 8] | |
| DRX | | 1 | OFF | |
| Time offset with Cell 1 | μs | 1 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |

| | | | | |
|---|---|---|---|---|
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | N/A |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | N/A |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR2 | PRS.1.1 FR2 |
| PRS BW | | 1 | 64 PRBs | 64 PRBs |
| PRS Resource slot offset | slot | 1 | 0 | 2 |
| SRS configuration | | 1 | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -94 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -103 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | 0.32 | -5.64 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | 3. 56 | -0.28 |
| PRP Note 3 | dBm/SCS kHz | 1 | -90.4 | -94.2 |
| Io | dBm/190.08 MHz | 1 | -55.73 | -55.73 |
| Propagation Condition | | 1 | AWGN | |

Note 1: Void.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation
Note 8: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured.
Note 9: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The UE Rx – Tx time difference measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 15.3.5.3.

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 15.2.8.5 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 1 - 1\right)$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 12.88s and 21.92s.

**Table 15.3.5.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|

| | | |
|---|---|---|
| test 1 | (5898 – 65) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (5898 + 65) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.3.6 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

### 15.3.6.1 Test purpose

The purpose of the test is to verify that the relative UE Rx-Tx time difference measurement accuracy when the two measurements are within the same RxTx TEG is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.3.6.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and nr-UE-RxTxTEG-ID-MaxSupport-r17.

### 15.3.6.3 Minimum conformance requirements

The relative accuracy requirements in Table 10.1.25.2-3b for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 for a corresponding Band

AWGN propagation condition.

the two UE Rx-Tx time difference measurements are associated with the same RxTx TEG

**Table 10.1.25.2-3b: UE Rx-Tx time difference relative measurement accuracy in FR2 in AWGN with TEG reporting**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS SCS | PRS bandwidth Note 1 | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) Note 2 | Io Note 3 range | |
| | | | | | Minimum Io | Maximum Io |
| Tc Note 4 | dB | kHz | RB | | dBm/SCS | $dBm/BW_{Channel}$ |
| 35 + Δ Note 6 | (PRS Ês/Iot)$_j$ ≥ -6dB | 60 | ≥ 24 | ≥ 4 | Same value as PRS_RP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| 24 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 11 + Δ | | | ≥ 132 | ≥ 1 | Note 5 | Note 5 |
| 24+ Δ | (PRS Ês/Iot)$_i$ ≥ -13dB | 120 | ≥ 32 | ≥ 4 | Same value as PRS_RP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 13 + Δ | | | ≥ 64 | ≥ 1 | Note 5 | Note 5 |
| 6 + Δ | | | ≥ 128 | ≥ 1 | Note 5 | Note 5 |

NOTE 1: Minimum PRS bandwidth, which is minimum of the PRS bandwidths of resource j and resource i.
NOTE 2: Minimum number of PRS resource repetitions among resource j and resource i. $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34], respectively.
NOTE 3: Io is assumed to have constant EPRE across the bandwidth.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest RB number for the corresponding SCS.
NOTE 6: Δis the value of the timing error margin for the RxTx TEG, reported via *nr-UE-RxTxTEG-TimingErrorMargin*. Δ cannot be larger than the sum of the margins in table 10.1.25.2-6 (dependent on PRS/SRS BW) for any pair of individual UE Rx-Tx time difference measurements associated with the RxTx TEG.

…

**Table 10.1.25.2-6: Margin for UE Rx-Tx time difference measurement accuracy in FR2**

| Min(PRS BW, SRS BW) (MHz) | | Margin (Tc [Note 1]) |
|---|---|---|
| SCS = 60 kHz | SCS = 120 kHz | |
| ≥ 24 | N/A | 76 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 24 |
| N/A | ≥ 128 | 20 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.

## 15.3.6.4 Test description

### 15.3.6.4.1 Initial conditions

The supported test configuration is listed in Table 15.3.6.4-1.

**Table 15.3.6.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.6.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.3.6.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.6.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.3.6.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.1 FR2 and PRS.1.2 FR2. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period.

The UE is configured with measurement gap pattern ID # 24 or #13 before T2.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.

4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

5. The SS shall transmit an RRCReconfiguration message with the SRS configuration.

6. The UE shall transmit RRCReconfigurationComplete message.

7. The SS shall transmit an LPP REQUEST CAPABILITIES message.

8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.

9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-UE-RxTxTEG-Request-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms.

11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.3.6.4.3).

12. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

13. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the NR-Multi-RTT-SignalMeasurementInformation according to Table 15.3.6.5-3.

14. Repeat steps 1-13 until the confidence level according to Annex D is achieved.

15. Repeat step 1-14for the other sub-test defined in Table 15.3.6.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.3.6.4.3 Message contents

**Table 15.3.6.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.3.6.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.3.6.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.6.4.3-3: SRS-Config (Table 15.3.6.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 0 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.6.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.6.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.3.6.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |

| | | | |
|---|---|---|---|
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.6.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | case1 | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | n2 | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.6.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.3.6.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.6.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.6.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.6.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.6.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.6.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 CHOICE { | | | |
| case1-r17 SEQUENCE { | | | |
| nr-UE-RxTx-TEG-ID-r17 | INTEGER (0..255) | | |
| } | | | |
| } | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.6.5 Test requirement

Table 15.3.6.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.3.6.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | | Setup 1 as specified in clause A.3.15 | |
| Beam Assumption[Note 7] | | 1 | Rough | Rough | Rough | Rough |
| Measurement gap | | 1 | GP#24 or GP#13 [Note 8] | | GP#24 or GP#13 [Note 8] | |
| DRX | | 1 | OFF | | OFF | |
| Time offset with Cell 1 | μs | 1 | N/A | 3 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | N/A | TRS.2.1 TDD | N/A |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | N/A | DLBWP.1.1 | N/A |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | N/A | ULBWP.1.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR2 | PRS.1.1 FR2 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset | slot | 1 | 0 | 2 | 0 | 2 |
| SRS configuration | | 1 | POS-SRS.3 | N/A | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -89.1 | | -89.1 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98.1 | | -98.1 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -5.69 | -12.69 | -5.69 | -12.69 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -5.17 | -11.36 | -5.17 | -11.36 |
| PRP Note 3 | dBm/SCS kHz | 1 | -94.2 | -100.4 | -94.2 | -100.4 |
| Io | dBm/190.08 MHz | 1 | -55.68 | -55.68 | -55.68 | -55.68 |
| Propagation Condition | | 1 | AWGN | | AWGN | |

Note 1: Void.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation
Note 8: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured.
Note 9: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The UE Rx-Tx time difference measurement accuracy for Cell 2 shall fulfil the absolute requirement in clause 15.3.6.3.

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 15.2.10.5 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(k_{multiTEG} * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right)$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 10.248s and 164s.

**Table 15.3.6.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| Sub-test 1 | (5898 – 88) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (5898 + 88) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| Sub-test 2 | (5898 – 58) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (5898 + 58) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.3.7 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR1 SA

### 15.3.7.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25A. The tests are conducted under AWGN propagation condition with the UE operating in FR1 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR1.

### 15.3.7.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_CONNECTED.

### 15.3.7.3 Minimum conformance requirements

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall not apply, if:

$N_{TA_offset}$ defined in table 7.1.2-2 changes during the UE Rx-Tx measurement period or

if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the network-configured Timing Advance.

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that:

- The UE transmits SRS within -160, 160 msec of at least one DL PRS resource of each of the TRPs in the assistance data.
- PRS resources linked for aggregation saftisfy all the conditions specified in TS 38.214 [26] clause 5.1.6.5.3.
- the spacing between the center frequencies of adjacent PFLs containing PRS resources linked for aggregation does not exceed the nominal channel spacing for intra-band contiguous CA defined in TS 38.101-1 [18], clause 5.4A.1 for FR1 and in TS 38.101-2 [19], clause 5.4A.1 for FR2-1.

If the uplink transmission timing changes during the UE Rx-Tx measurement period due to the autonomous timing adjustment defined in clause 7.1.2 then:

- UE Rx-Tx measurement accuracy requirements shall apply for a cell, which is also the downlink reference cell (defined in clause 7.1.1) for SRS transmission even if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

- UE Rx-Tx measurement accuracy requirements shall not apply for a cell, which is not the downlink reference cell (defined in clause 7.1.1) for SRS transmission, if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

When a serving cell change occurs during the UE Rx-Tx measurement period, the UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that the serving cell change does not impact SRS configuration for the UE Rx-Tx measurement.

The accuracy requirements in table 10.1.25A.2-1 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to annex B.2.14 for a corresponding Band.

AWGN propagation condition.

**Table 10.1.25A.2-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS bandwidth per PFL | PRS SCS | PRS resource repetition $(T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ±12 + δ | ±10 + δ | -3 | ≥104 | 15 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -126.5 | |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -126 | |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | |
| | | | | | | NR_FDD_FR1_F | -124.5 | |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -124 | |
| | | | | | | NR_FDD_FR1_H | -123.5 | |
| | | | | | | NR_FDD_FR1_N | -120.5 | |
| ±6 + δ | ±3 + δ | | ≥132 | 30 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -123.5 | |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -123 | |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | |
| | | | | | | NR_FDD_FR1_F | -121.5 | |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -121 | |
| | | | | | | NR_FDD_FR1_H | -120.5 | |
| | | | | | | NR_FDD_FR1_N | -117.5 | |
| ±3 + δ | ±2 + δ | | ≥272 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±6 + δ | ±3 + δ | | ≥64 | 60 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -120.5 | |
| | | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -120 | |
| | | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -118 | |
| | | | | | | NR_FDD_FR1_H | -117.5 | |
| | | | | | | NR_FDD_FR1_N | -114.5 | |
| ±3 + δ | ±2 + δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±14+δ | ±13+δ | -13 | ≥104 | 15 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±8+δ | ±4+δ | | ≥132 | 30 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±4+δ | ±3+δ | | ≥272 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±9+δ | ±4+δ | | ≥ 64 | 60 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±4+δ | ±3+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$ , $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from table 10.1.25A.2-5, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.

…

The accuracy requirements in table 10.1.25A.2-3 for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.

PRP|$_{\text{dBm}}$ according to annex B.2.14 for a corresponding Band.

AWGN propagation condition.

**Table 10.1.25A.2-3: UE Rx-Tx time difference measurement accuracy in FR2 in AWGN**

| Accuracy for 2 PFLs | Accuracy for 3 PFLs | Conditions | | | | | |
|---|---|---|---|---|---|---|---|
| | | PRS Ês/Iot | PRS bandwidth per PFL | PRS SCS | PRS resource repetition ($T_{\text{rep}}^{\text{PRS}} * L_{\text{PRS}} / K_{\text{comb}}^{\text{PRS}}$ Note 3 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | Tc Note 5 | dB | PRB | kHz | | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| ±6+δ | ±3+δ | -3 | ≥64 | 60 | ≥1 | Same value as PRP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ±3+δ | ±2+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ±3+δ | ±2+δ | | ≥64 | 120 | ≥1 | Same value as PRP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ±2+δ | ±1+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| ±9+δ | ±4+δ | -13 | ≥64 | 60 | ≥1 | NOTE 6 | NOTE 6 |
| ±4+δ | ±3+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ±4+δ | ±3+δ | | ≥64 | 120 | ≥1 | NOTE 6 | NOTE 6 |
| ±2+δ | ±1+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$, $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from table 10.1.25A.2-6, where the PRS BW refers to the sum of the PRS PRB numbers across the aggregated PFLs.

…

**Table 10.1.25A.2-5: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (PRB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 104 | N/A | N/A | 56 |
| N/A | ≥ 132 | ≥ 64 | 24 |
| N/A | N/A | ≥ 132 | 24 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.

**Table 10.1.25A.2-6: Margin for UE Rx-Tx time difference measurement accuracy in FR2**

| Min(PRS BW, SRS BW) (PRB) | | Margin (Tc [Note 1]) |
|---|---|---|
| SCS = 60 kHz | SCS = 120 kHz | |
| ≥ 64 | N/A | 32 |
| ≥ 132 | ≥ 64 | 24 |
| N/A | ≥ 128 | 20 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.

The normative reference for this requirement is TS 38.133 [50] clause 10.1.25A.

## 15.3.7.4 Test description

The test defines three possible test configurations; specified in Table 15.3.7.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.3.7.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Note: The UE is only required to be tested in one of the supported test configurations.

### 15.3.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.7.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.3.7.5-1 and Table 15.3.7.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.7.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.3.7.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-Multi-RTT-ProvideAssistanceData and NR-Multi-RTT-RequestLocationInformation as defined in clause 15.3.7.4.3 shall be provided to the UE during the set-up period. The last TTI containing the NR-Multi-RTT-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T$ = 50 ms is the maximum processing time of the NR-Multi-RTT-RequestLocationInformation message and the Multi-RTT assistance data in the UE. The UE is configured with measurement gap pattern ID #0 or ID #24 before the start of the test.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A$ = 31 which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. The SS adjusts the downlink timing for Cell 1 to a delay of +512 $T_c$, compared to the current value.
4. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
5. The SS shall transmit an RRCReconfiguration message with the SRS configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the NR-Multi-RTT-ProvideAssistanceData IE and NR-DL-PRS-AggregationInfo-r18 linking each PRS resource in PFL1 to the corresponding PRS resource in PFL2.If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the NR-Multi-RTT-RequestLocationInformation IE and nr-DL-PRS-JointMeasurementRequest-r18 IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
11. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the NR-Multi-RTT-ProvideLocationInformation IE.
12. As soon as possible after step 11 the SS shall measure the transmit timing of the UE using the transmitted SRS, relative to the current downlink timing.
13. If the UE message at step 11 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
14. The SS shall check the reported value of nr-UE-RxTxTimeDiff in the NR-Multi-RTT-SignalMeasurementInformation IE provided by the UE in step 11 and compare it with the value measured in step 12. The SS shall check that the reported value is within the limits specified in table 15.3.7.5-2 compared to the measured value. If the reported value is within the limits the number of successful results is increased by one. If the reported value is not within the limits or the UE reports an error in the LPP PROVIDE LOCATION INFORMATION message in step 11, or does not respond at step 11 within the time given by the time IE in the

CommonIEsRequestLocationInformation IE in step 10, then the number of unsuccessful results for is increased by one.

15. Repeat steps 1-14 until the confidence level according to Annex D is achieved.

16. Repeat step 1-15 for the other sub-test defined in Table 15.3.7.4-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.3.7.4.3 Message contents

**Table 15.3.7.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.3.7.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45],, table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup SEQUENCE { | CellGroupConfig | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | As defined in Table 15.3.7.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.7.4.3-3: SRS-Config (Table 15.3.7.4.3-2)**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.7.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 20 | | Configuration 1 and Configuration 2 |
| sl320 | 40 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.3.7.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.3.7.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.7.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |

| | | | |
|---|---|---|---|
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.7.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |

| | | | |
|---|---|---|---|
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.3.7.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.7.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.7.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.7.4.3-8 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.7.4.3-8 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.7.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.7.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.7.4.3-9: LPP ProvideLocation Information**

| |
|---|
| Derivation Path: 37.355 clause 6.2 |

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.7.5 Test requirement

Table 15.3.7.5-1 and Table 15.3.7.5-2 define the primary level settings including test tolerances for the test.

**Table 15.3.7.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2, all signals/channels except PRS in PFL2 are allocated within this channel. |
| | | | 2 | For both Cell 1 and Cell 2, only PRS in PFL2 is allocated within this channel. |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | For both RF channels |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | OFF | |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells/TRPs |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |
| NOTE 1: GP#24 is configured if the UE supports it, otherwise GP#0 is configured. | | | | |

**Table 15.3.7.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns Note 1 | | 1, 2, 3 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1,4 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1, 4 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration (for both PFL1 and PFL2) | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS muting info (*dl-PRS-MutingOption1* for both PFL1 and PFL2) | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s / I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s / N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Io Note 3 (on symbols where PRS is not allocated) | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.59 | | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |
| NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved on all OFDM symbols except those in which PRS is allocated. | | | | | | |
| NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | | |
| NOTE 3: PRP and Io levels have been derived from other parameters and they are provided for information only. They are not settable parameters themselves. | | | | | | |
| NOTE 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | | | | |

**Table 15.3.7.5-3: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1&2 | NR cell 1 | (Measured value from step 12 - 70) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 70) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 72) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 72) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 | NR cell 1 | (Measured value from step 12 - 62) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 62) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 64) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 64) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.7.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.3.8 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

### This test case is incomplete for test frequencies > 40.8 GHz15.3.8.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25A. The tests are conducted under AWGN propagation condition with the UE operating in FR2 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR2.

### 15.3.8.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_CONNECTED.

### 15.3.8.3 Minimum conformance requirements

Same as defined in clause 15.3.7.3.

### 15.3.8.4 Test description

The supported test configuration is listed in Table 15.3.8.4-1.

**Table 15.3.8.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 15.3.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.8.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.3.8.5-1 and Table 15.3.8.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.8.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.3.8.4.2 Test procedure

Same as defined in clause 15.3.7.4.2.

#### 15.3.8.4.3 Message contents

As defined in clause 15.3.7.4.3, with the following exceptions:

Table 15.3.8.4.3-1 replaces Table 15.3.7.4.3-3, Table 15.3.8.4.3-2 replaces Table 15.3.8.4.3-7 and Table 15.3.8.4.3-3 replaces Table 15.3.7.4.3-8.

**Table 15.3.8.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.8.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.3.8.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.8.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.8.4.3-8 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |

| | | | |
|---|---|---|---|
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.8.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.8.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.8.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |

| | | | |
|---|---|---|---|
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.8.5 Test requirement

**Table 15.3.8.5-1 and Table 15.3.8.5-2 define the primary level settings including test tolerances for the test.Table 15.3.8.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2, all signals/channels except PRS in PFL2 are allocated within this channel. |
| | | | 2 | For both Cell 1 and Cell 2, only PRS in PFL2 is allocated within this channel. |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | For both RF channels |
| SSB configuration | | 1 | SSB.2 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| Measurement gap | | 1 | GP#24 or GP#13 Note 1 | |
| CP length | | 1 | Normal | |
| DRX | | 1 | OFF | |
| Time offset between serving and neighbour cells | μs | 1 | 3 | Synchronous cells/TRPs |
| Expected RSTD | μs | 1 | 3 | |
| Expected RSTD uncertainty | μs | 1 | 5 | |
| Note 1: GP#24 is configured if the UE supports it, otherwise GP#13 is configured. | | | | |

**Table 15.3.8.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | Cell 2 |
|---|---|---|---|---|
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | |
| Beam Assumption Note 4 | | 1 | Rough | Rough |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A |
| OCNG Patterns Note 1 | | 1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |

| | | | | |
|---|---|---|---|---|
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | N/A |
| Initial BWP configuration | | 1 | DLBWP.0.1<br>ULBWP.0.1 | N/A |
| Active BWP configuration | | 1 | DLBWP.1.1<br>ULBWP.1.1 | N/A |
| PRS configuration (for both PFL1 and PFL2) | | 1 | PRS.1.2 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset (for both PFL1 and PFL2) | slot | 1 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -90.1 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.69 | -12.10 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -9.8 |
| PRP Note 3 | dBm/SCS kHz | 1 | -91 | -99 |
| Io Note 3 (on symbols where PRS is not allocated) | dBm/190.08 MHz | 1 | -54.59 | -54.59 |
| Propagation Condition | | 1 | AWGN | |

Note 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved on all OFDM symbols except those in which PRS is allocated.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRP and Io levels have been derived from other parameters and they are provided for information only. They are not settable parameters themselves.

Note 4: Information about types of UE beam is given in TS 38.133 clause B.2.1.3, and does not limit UE implementation or test system implementation

Note 5: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [33], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [33] table 6.2.1.3-4.

Note 6: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.

Note 7: As observed with 0 dBi gain antenna at the centre of the quiet zone.

**Table 15.3.8.5-3: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | NR cell 1 | (Measured value from step 12 - 54) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 54) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 54) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 54) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.8.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.3.9 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH

### 15.3.9.1 Test purpose

The purpose of the test is to verify that the accuracy of the UE Rx-Tx time difference measurement without RX FH reported by the RedCap UE is within the specified limits in TS 38.133 [50] clause 10.1A.18.2. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.3.9.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method.

### 15.3.9.3 Minimum conformance requirements

For UE Rx-Tx time difference measurement performed by 2RX RedCap UE without RX FH, the accuracy requirements corresponding to the PRS bandwidth supported by the RedCap UE for PRS measurement without RX FH in clause 10.1.25.2 shall apply.

The accuracy requirements in table 10.1A.18.2.2-1 for FR1 are valid under the following conditions:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- $PRP|_{dBm}$ according to annex B.2.14 for a corresponding Band.
- AWGN propagation condition.

**Table 10.1A.18.2.2-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ±89+δ | -3 | ≥24 | 15 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -126.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -126 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | |
| | | | | | NR_FDD_FR1_F | -124.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -124 | |
| | | | | | NR_FDD_FR1_H | -123.5 | |
| | | | | | NR_FDD_FR1_N | -120.5 | |
| ±51+δ | | ≥52 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±25+δ | | 104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±49+δ | | ≥24 | 30 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -123.5 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -123 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | |
| | | | | | NR_FDD_FR1_F | -121.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -121 | |
| | | | | | NR_FDD_FR1_H | -120.5 | |
| | | | | | NR_FDD_FR1_N | -117.5 | |
| ±27+δ | | 48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±27+δ | | 24 | 60 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -120.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -120 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -118 | |
| | | | | | NR_FDD_FR1_H | -117.5 | |
| | | | | | NR_FDD_FR1_N | -114.5 | |
| ±98+δ | -13 | ≥24 | 15 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±61+δ | | ≥52 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±32+δ | | 104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±55+δ | | ≥24 | 30 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±34+δ | | 48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ±31+δ | | 24 | 60 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{\mathrm{rep}}^{\mathrm{PRS}}$, $L_{\mathrm{PRS}}$ , $K_{\mathrm{comb}}^{\mathrm{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from table 10.1A.18.2.2-3.

**Table 10.1A.18.2.2-3: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (PRB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 24 | N/A | N/A | 160 |
| ≥ 52 | ≥ 24 | N/A | 80 |
| 104 | 48 | 24 | 56 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.

### 15.3.9.4 Test description

The test defines three possible test configurations; specified in Table 15.3.9.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.3.9.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

#### 15.3.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.9.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.3.9.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.9.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.3.9.4.2 Test procedure

Same as defined in clause 15.3.7.4.2.

#### 15.3.9.4.3 Message contents

As defined in clause 15.3.7.4.3, with the following exceptions:

Table 15.3.9.4.3-1 replaces Table 15.3.7.4.3-3, Table 15.3.9.4.3-2 replaces Table 15.3.7.4.3-7, Table 15.3.9.4.3-3 replaces Table 15.3.7.4.3-8 and Table 15.3.9.4.3-4 replaces Table 15.3.7.4.3-5.

**Table 15.3.9.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |

| | | | |
|---|---|---|---|
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.9.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 20 | | Configuration 1, configuration 2 and Configuration 4 |
| sl320 | 40 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.3.9.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1, Configuration 2 and Configuration 4 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.9.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.9.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 15.3.9.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1, Configuration 2 and Configuration 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.9.4.3-4: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.9.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |

| | | | |
|---|---|---|---|
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.9.5 Test requirement

Table 15.3.9.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.3.9.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Cell 1 | Cell 2 |
|---|---|---|---|---|
| RF Channel Number | | 1,2,3,4 | 1 | 1 |
| $BW_{channel}$ | MHz | 1 | 10: $N_{RB,c}$ = 52 | |
| | | 2 | 10: $N_{RB,c}$ = 52 | |
| | | 3 | 20: $N_{RB,c}$ = 51 | |
| | | 4 | 10: $N_{RB,c}$ = 52 | |
| Measurement gap | | 1,2,3,4 | GP#24 or GP#0 Note 4 | |
| DRX | | 1,2,3,4 | OFF | |
| Time offset with Cell 1 | µs | 1,2,3,4 | N/A | 3 |
| TDD configuration | | 1,4 | N/A | N/A |
| | | 2 | TDDConf.1.1 | TDDConf.1.1 |
| | | 3 | TDDConf.2.1 | TDDConf.2.1 |
| PDSCH RMC configuration | | 1,4 | SR.1.1 FDD | N/A |
| | | 2 | SR.1.1 TDD | |
| | | 3 | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | | 1,4 | CR.1.1 FDD | N/A |
| | | 2 | CR.1.1 TDD | |
| | | 3 | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | | 1,4 | CCR.1.1 FDD | N/A |
| | | 2 | CCR.1.1 TDD | |
| | | 3 | CCR.2.1 TDD | |
| OCNG Patterns | | 1,2,3,4 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1,2,3,4 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |

| | | | | |
|---|---|---|---|---|
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1,4 | TRS.1.1 FDD | N/A |
| | | 2 | TRS.1.1 TDD | |
| | | 3 | TRS.1.2 TDD | |
| Initial BWP configuration | | 1,2,3,4 | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1,2,3,4 | DLBWP.1.1 RedCap | N/A |
| Active UL BWP configuration | | 1,2,3,4 | ULBWP.1.1 RedCap | N/A |
| PRS configuration | | 1,4 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 2 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 3 | PRS.2.1 FR1 | PRS.2.1 FR1 |
| PRS Resource slot offset | slot | 1,2,3,4 | 0 | 4 |
| SRS configuration | | 1,4 | POS-SRS.1 | N/A |
| | | 2 | POS-SRS.1 | N/A |
| | | 3 | POS-SRS.2 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1,4 | -98 | |
| | | 2 | -98 | |
| | | 3 | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1,2,3,4 | -2.41 | -12.12 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1,2,3,4 | -2 | -10 |
| PRP Note 3 | dBm/SCS kHz | 1,4 | -100 | -108 |
| | | 2 | -100 | -108 |
| | | 3 | -97 | -105 |
| Io | dBm/9.36 MHz | 1,4 | -67.67 | -67.67 |
| | dBm/9.36 MHz | 2 | -67.67 | -67.67 |
| | dBm/18.36 MHz | 3 | -64.75 | -64.75 |
| Propagation Condition | | 1,2,3,4 | AWGN | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

NOTE 4: PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

NOTE 5: The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification.

NOTE 6: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.

**Table 15.3.9.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Cell | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1&2&4 (15kHz) | NR cell 1 | (Measured value from step 12 - 270) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 270) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 281) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 281) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 293) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 293) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 290) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 290) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 (30kHz) | NR cell 1 | (Measured value from step 12 - 169) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 169) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 161) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 161) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 170) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 170) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 167) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 167) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.9.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.3.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

### 15.3.10.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy without RX FH in RRC_CONNECTED state is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1A.18.2. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.3.10.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method.

### 15.3.10.3 Minimum conformance requirements

For UE Rx-Tx time difference measurement performed by 2RX RedCap UE without RX FH, the accuracy requirements corresponding to the PRS bandwidth supported by the RedCap UE for PRS measurement without RX FH in clause 10.1.25.2 shall apply.

The normative reference for this requirement is TS 38.133 [50] clause 10.1A.18.2.

### 15.3.10.4 Test description

The supported test configuration is listed in Table 15.3.10.4-1.

**Table 15.3.10.4-1: Test Configuration**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

#### 15.3.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.10.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.3.10.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.10.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.3.10.4.2 Test procedure

Same as defined in clause 15.3.7.4.2.

#### 15.3.10.4.3 Message contents

As defined in clause 15.3.7.4.3, with the following exceptions:

Table 15.3.10.4.3-1 replaces Table 15.3.7.4.3-3, Table 15.3.10.4.3-2 replaces Table 15.3.7.4.3-7, Table 15.3.10.4.3-3 replaces Table 15.3.7.4.3-8 and Table 15.3.10.4.3-4 replaces Table 15.3.7.4.3-5.

**Table 15.3.10.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |

| | | | |
|---|---|---|---|
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.3.10.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.3.10.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.10.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.10.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.10.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | Cell 1 |
| | 1 | | Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.10.4.3-4: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation | | | |

| | | | |
|---|---|---|---|
| SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.10.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.10.5 Test requirement

Table 15.3.10.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.3.10.5-1: UE Rx-Tx time difference measurement accuracy test parameters.**

| Parameter | Unit | Test configuration | Cell 1 | Cell 2 |
|---|---|---|---|---|
| AoA setup | | 1 | Setup 1 as specified in clause TS 38.133 A.3.15 | |
| $BW_{channel}$ | MHz | 1 | 100: $N_{RB,c}$ = 66 | |
| Beam Assumption Note 6 | | 1 | Rough | Rough |
| Measurement gap | | 1 | GP#24 or GP#13 Note 7 | |
| DRX | | 1 | OFF | |
| Time offset with Cell 1 | µs | 1 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |

| | | | | |
|---|---|---|---|---|
| EPRE ratio of OCNG DMRS to SSS | | | | |
| EPRE ratio of OCNG to OCNG DMRS | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | N/A |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1 | DLBWP.1.1 RedCap | N/A |
| Active UL BWP configuration | | 1 | ULBWP.1.1 RedCap | N/A |
| PRS configuration | | 1 | PRS.1.5 FR2 | PRS.1.5 FR2 |
| PRS Resource slot offset | slot | 1 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.3 | N/A |
| $N_{oc}$ Note 1 | dBm/SCS | 1 | -89 | |
| $N_{oc}$ Note 1 | dBm/15 kHz | 1 | -98 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.41 | -12.12 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -10 |
| PRP Note 2 | dBm/SCS kHz | 1 | -91 | -99 |
| Io | dBm/92.16 MHz | 1 | -57.6 | -57.6 |
| Propagation Condition | | 1 | AWGN | |

NOTE 1: Interference from other cells and noise sources not specified in the test is assumed to be constant over sub-carriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 2: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
NOTE 5: As observed with 0 dBi gain antenna at the centre of the quiet zone
NOTE 6: Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation.
NOTE 7: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured.
NOTE 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

**Table 15.3.10.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | NR cell 1 | (Measured value from step 12 - 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

| | | | |
|---|---|---|---|
| | NR cell 2 | (Measured value from step 12 - 65) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 65) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.10.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.3.11 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE with Rx FH

*a) Editor's note: This test case is incomplete. The following aspect is either missing or TBD:*

*b) - Test tolerance are not added in and analysis is missing*

### 15.3.11.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy in RRC_CONNECTED with FH by a RedCap UE is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1A.18.2.3 and 10.1A.18.2.4. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.3.11.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and maximumPRS-BandwidthAcrossAllHopsFR1 for at least one band and PRS frequency hopping measurement and report.

### 15.3.11.3 Minimum conformance requirements

The accuracy requirements in table 10.1A.18.2.3-1 for FR1 are valid under the following conditions:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- $PRP|_{dBm}$ according to annex B.2.14 for a corresponding Band.
- AWGN propagation condition.
- The $BW_{total}$ as defined in clause 9.9A.4.8 for RRC_CONNECTED and in clause 5.6A.6.6 for RRC_INACTIVE is no less than the "Total PRS bandwidth after FH".

**Table 10.1A.18.2.3-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | PRS bandwidth per hop | PRS SCS | Total PRS bandwidth after FH | NR operating band groups[Note 2] | Io[Note 3] range | |
| | | | | | | Minimum Io[Note 1] | Maximum Io |
| Tc[Note 4] | dB | PRB | kHz | PRB | | dBm / $SCS_{PRS}$ | dBm/BW |
| ±23+δ | -3 | ≥52 | 15 | 268 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -126.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -126 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | |
| | | | | | NR_FDD_FR1_F | -124.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -124 | |
| | | | | | NR_FDD_FR1_H | -123.5 | |
| | | | | | NR_FDD_FR1_N | -120.5 | |
| ±17+δ | | 48 | 30 | 272 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -123.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -123 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | |
| | | | | | NR_FDD_FR1_F | -121.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -121 | |
| | | | | | NR_FDD_FR1_H | -120.5 | |
| | | | | | NR_FDD_FR1_N | -117.5 | |
| ±21+δ | | 24 | 60 | 132 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -120.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -120 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -118 | |
| | | | | | NR_FDD_FR1_H | -117.5 | |
| | | | | | NR_FDD_FR1_N | -114.5 | |
| ±54+δ | -13 | ≥52 | 15 | 268 | NOTE 5 | NOTE 5 | NOTE 5 |
| ±38+δ | | 48 | 30 | 272 | NOTE 5 | NOTE 5 | NOTE 5 |
| ±40+δ | | 24 | 60 | 132 | NOTE 5 | NOTE 5 | NOTE 5 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: δ is the margin determined from table 10.1A.18.2.3-5.

The accuracy requirements in table 10.1A.18.2.3-3 for FR2 are valid under the following conditions:

- Conditions defined in clause 7.3 of TS 38.101-2 [19] for reference sensitivity are fulfilled.
- $PRP|_{dBm}$ according to annex B.2.14 for a corresponding Band.
- AWGN propagation condition.
- The $BW_{total}$ as defined in clause 9.9A.4.8 for RRC_CONNECTED and in clause 5.6A.6.6 for RRC_INACTIVE is no less than the “Total PRS bandwidth after FH”.

**Table 10.1A.18.2.3-3: UE Rx-Tx time difference measurement accuracy in FR2 in AWGN**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth per hop | PRS SCS | Total PRS bandwidth after FH | Io[Note 4] range | |
| | | | | | Minimum Io[Note 1] | Maximum Io |
| Tc[Note 5] | dB | PRB | kHz | PRB | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| ±11+δ | -3 | ≥64 | 60 | 264 | Same value as PRP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ±7+δ | | 64 | 120 | 264 | Same value as PRP in table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ±33+δ | -13 | ≥64 | 60 | 264 | NOTE 5 | NOTE 5 |
| ±27+δ | | 64 | 120 | 264 | NOTE 5 | NOTE 5 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: δ is the margin determined from table 10.1A.18.2.3-6.

**Table 10.1A.18.2.3-5: Margin for UE Rx-Tx time difference measurement accuracy in FR1 with FH**

| Min(PRS BW, SRS BW) (PRB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 52 | N/A | N/A | 80 |
| N/A | 48 | 24 | 56 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.
NOTE 3: PRS BW and SRS BW refer to BW per hop.

**Table 10.1A.18.2.3-6: Margin for UE Rx-Tx time difference measurement accuracy in FR2 with FH**

| Min(PRS BW, SRS BW) (PRB) | | Margin (Tc [Note 1]) |
|---|---|---|
| SCS = 60 kHz | SCS = 120 kHz | |
| ≥ 64 | N/A | 32 |
| N/A | 64 | 24 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.
NOTE 3: PRS BW and SRS BW refer to BW per hop.

…

The accuracy requirements in table 10.1A.18.2.4-1 for FR1 are valid under the following conditions:

- Conditions defined in clause 7.3 of TS 38.101-1 [18] for reference sensitivity are fulfilled.
- $PRP|_{dBm}$ according to annex B.2.14 for a corresponding Band.
- AWGN propagation condition.

**Table 10.1A.18.2.4-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | Io[Note 3] range |

| | PRS Ês/Iot | PRS bandwidth per hop | PRS SCS | Total PRS bandwidth after FH | NR operating band groups[Note 2] | Minimum Io[Note 1] | Maximum Io |
|---|---|---|---|---|---|---|---|
| Tc[Note 4] | dB | PRB | kHz | PRB | | dBm / $SCS_{PRS}$ | dBm/BW |
| ±23+δ | | ≥52 | 15 | 268 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -126.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -126 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | |
| | | | | | NR_FDD_FR1_F | -124.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -124 | |
| | | | | | NR_FDD_FR1_H | -123.5 | |
| | | | | | NR_FDD_FR1_N | -120.5 | |
| ±14+δ | -3 | 48 | 30 | 272 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -123.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -123 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | |
| | | | | | NR_FDD_FR1_F | -121.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -121 | |
| | | | | | NR_FDD_FR1_H | -120.5 | |
| | | | | | NR_FDD_FR1_N | -117.5 | |
| ±22+δ | | 24 | 60 | 132 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B, NR_TDD_FR1_B | -120.5 | |
| | | | | | NR_FDD_FR1_C, NR_TDD_FR1_C | -120 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G, NR_SDL_FR1_G | -118 | |
| | | | | | NR_FDD_FR1_H | -117.5 | |
| | | | | | NR_FDD_FR1_N | -114.5 | |
| ±60+δ | -13 | ≥52 | 15 | 268 | NOTE 5 | NOTE 5 | NOTE 5 |
| ±42+δ | | 48 | 30 | 272 | NOTE 5 | NOTE 5 | NOTE 5 |
| ±45+δ | | 24 | 60 | 132 | NOTE 5 | NOTE 5 | NOTE 5 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 4: Tc is the basic timing unit defined in TS 38.211 [6].

NOTE 5: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 6: δ is the margin determined from table 10.1A.18.2.3-5.

## 15.3.11.4 Test description

The test defines three possible test configurations; specified in Table 15.3.11.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.3.11.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | NR 15 kHz SSB SCS, cell bandwidth 50 MHz, FDD duplex mode |
| 2 | NR 15 kHz SSB SCS, cell bandwidth 50 MHz, TDD duplex mode |
| 3 | NR 30 kHz SSB SCS, cell bandwidth 100 MHz, TDD duplex mode |
| 4 | NR 15 kHz SSB SCS, cell bandwidth 50 MHz, HD-FDD duplex mode |
| NOTE 1: The UE is only required to be tested in one of the supported test configurations in each supported band.<br>NOTE 2: This test case is applicable to the UE supporting per hop bandwidth ≥ 10 MHz for SCS 15 kHz.<br>NOTE 3: This test case is applicable to the UE supporting per hop bandwidth = 20 MHz for SCS 30 kHz. | |

### 15.3.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.11.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.3.11.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.11.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.3.11.4.2 Test procedure

Same as defined in clause 15.3.7.4.2.

### 15.3.11.4.3 Message contents

As defined in clause 15.3.7.4.3, with the following exceptions:

Table 15.3.11.4.3-1 replaces Table 15.3.7.4.3-3, Table 15.3.11.4.3-2 replaces Table 15.3.7.4.3-5, Table 15.3.11.4.3-3 replaces Table 15.3.7.4.3-7 and Table 15.3.11.4.3-4 replaces Table 15.3.7.4.3-8.

**Table 15.3.11.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |

| | | | |
|---|---|---|---|
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.2.15.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 20 | | Configuration 1, Configuration 2 and Configuration 4 |
| sl320 | 40 | | Configuration 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 15.3.11.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.11.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |

| | | | |
|---|---|---|---|
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | mhz50 | | Configuration 1, Configuration 2 and Configuration 4 |
| | mhz100 | | Configuration 3 |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.3.11.4.3-3: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1, Configuration 2 and |

| | | | |
|---|---|---|---|
| | | | Configuration 4 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 62 | 268 PRBs | Configuration 1, Configuration 2 and Configuration 4 |
| | 63 | 272 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.11.4.3-4 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.11.4.3-4 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 15.3.11.4.3-4: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1, Configuration 2 and Configuration 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | Configuration 1, Configuration 2 and Configuration 4 |
| | 0 | | Configuration 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.11.5 Test requirement

FFS

## 15.3.12 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

### 15.3.12.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy with FH by a RedCap UE in RRC_CONNECTED is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1A.18.2.3 and 10.1A.18.2.4. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.3.12.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and maximumPRS-BandwidthAcrossAllHopsFR2 for at least one band and PRS frequency hopping measurement and report.

### 15.3.12.3 Minimum conformance requirements

Same as defined in clause 15.2.15.3.

### 15.3.12.4 Test description

The supported test configuration is listed in Table 15.3.12.4-1.

**Table 15.3.12.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, cell bandwidth 400 MHz, TDD duplex mode |
| NOTE 1: This test case is applicable to the UE supporting per hop bandwidth = 100 MHz for SCS 120 kHz. | |

#### 15.3.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.3.12.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.3.12.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.3.12.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.3.12.4.2 Test procedure

Same as defined in clause 15.3.7.4.2.

### 15.3.12.4.3 Message contents

As defined in clause 15.3.7.4.3, with the following exceptions:

Table 15.2.16.4.3-1 replaces Table 15.2.11.4.3-3, Table 15.2.16.4.3-2 replaces Table 15.2.11.4.3-5, Table 15.2.16.4.3-3 replaces Table 15.2.11.4.3-7, Table 15.2.16.4.3-4 replaces Table 15.2.11.4.3-8 and Table 15.2.16.4.3-5 replaces Table 15.2.11.4.3-9.

**Table 15.3.12.4.3-1: SRS-Config**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.312.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.3.12.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.3.12.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr2 | mhz400 | | |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 115.3.12.4.3-3: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 61 | 264 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.12.4.3-4 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.3.12.4.3-4 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 15.3.12.4.3-4: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | NR Cell 1 |
| | 01 | | NR Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.3.12.5 Test requirement

Table 15.3.12.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.3.12.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | |
| $BW_{channel}$ | MHz | 1 | 400: $N_{RB,c}$ = 264 | |
| Beam Assumption Note 7 | | 1 | Rough | Rough |
| Measurement gap | | 1 | GP#24 or GP#13 Note 8 | |
| DRX | | 1 | OFF | |
| Time offset with Cell 1 | µs | 1 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |

| | | | | |
|---|---|---|---|---|
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | N/A |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | N/A |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | N/A |
| PRS configuration | | 1 | PRS.1.6 FR2 | PRS.1.6 FR2 |
| PRS Resource slot offset | slot | 1 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -93.73 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -102.73 | |
| PRS $\hat{E}_s / I_{ot}$ | dB | 1 | -2.69 | -8.83 |
| PRS $\hat{E}_s / N_{oc}$ | dB | 1 | -1.2 | -6 |
| PRP Note 3 | dBm/SCS | 1 | -94.9 | -99.7 |
| Io | dBm/380.16MHz | 1 | -55.66 | -55.66 |
| Propagation Condition | | 1 | AWGN | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
NOTE 6: As observed with 0 dBi gain antenna at the centre of the quiet zone.
NOTE 7: Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation.
NOTE 8: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured.
NOTE 9: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

**Table 15.3.12.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Cell | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | NR cell 1 | (Measured value from step 12 - 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 83) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 83) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.3.12.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

# 15.4 UE Rx-Tx time difference measurement period test cases in RRC_INACTIVE

## 15.4.1 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR1 SA

### 15.4.1.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.4 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.4.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and supports DL-PRS measurement in RRC_INACTIVE state.

### 15.4.1.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [34], UE shall be able to measure multiple (up to the UE capability specified in clause 5.6.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [4] in configured positioning frequency layers within the measurement period $T_{UERxTx,Total}$ ms.

$$T_{UERxTx,\ Total} = \sum_{i=1}^{L} T_{UERxTx,i} + (L-1) * \max(T_{effect,i})$$

Where:

- $i$ is the index of positioning frequency layer,
- $T_{UERxTx,i}$ is the measurement period for UE Rx-Tx time difference measurements in positioning frequency layer *i* as further defined in this clause,
- L is total number of positioning frequency layers,
- $T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer *i* as defined further in this clause.

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where:

- $K_{carrier_PRS}$ =1 if the UE is capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [34].
- $K_{carrier_PRS} = N_{layer} + 1$ if the UE is not capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [34] and if Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$; where $N_{layer}$is defined in clause 4.2.2.7.
- $K_{carrier_PRS} = K_{carrier} + 1$ if the UE is not capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [34] and if Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$; where $K_{carrier}$is defined in clause 4.2.2.4.
- $N_{RxBeam,i}$ is the scaling factor for UE Rx beam sweeping:

- $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1, and if positioning frequency layer *i* is in FR2.
  - $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-Multi-RTT -RequestLocationInformation*.
  - $N_{RxBeam,i}$ equals to 8, otherwise.

$N_{RxTx,TEG,i}$ is the Rx TEG specific scaling factor:

- $N_{RxTx,TEG,i}$ = 1 if UE is not configured by LMF with *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* or measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 [34].

Otherwise,

- $N_{RxTx,TEG,i} = N_{TEG,i}$, if UE is not capable of measuring same DL PRS resource simultaneously from multiple Rx TEGs, where $N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34], and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.
- $N_{RxTx,TEG,i} = \left\lfloor \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rfloor$, if UE is capable of measuring same DL PRS resource simultaneously from multiple Rx TEGs, where $k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

- $L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.
- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,
- {N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in clause 4.2.7.2 of TS 37.355 [34],
- N' is UE capability for number of DL PRS resources that it can process in a slot corresponding to *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* as specified in clause 6.4.3 of TS 37.355 [34],
- $N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:
  - $N_{sample}$ = 4 if the UE is not capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [34] or not configured to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [34].
  - $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [34] and LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [34] and the following conditions are met:
    - PRS bandwidth is within the initial BWP and
    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
  - $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [34] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [34] but the following conditions are not met:
    - PRS bandwidth is within the initial BWP and

- Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

- $T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time, $T_{last,i} = T_i + T_{available_PRS,i}$ ,

- $T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where:

- $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [34],

- $T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, the least common multiple between $T_{PRS,i}$ and $T_{DRX}$.

- $T_{DRX}$ is the DRX cycle of the UE in the serving cell.

- $T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where:

  - $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

  - $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

## 15.4.1.4 Test description

### 15.4.1.4.1 Initial conditions

The test defines three possible test configurations; specified in Table 15.4.1.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.4.1.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.4.1.5-1 and Table 15.4.1.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.1.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

#### 15.4.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During T2 UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

3. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 15.4.1.4.3-1.
4. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
5. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
6. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
7. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
6. Set the parameters for T1 according to Table 15.4.1.5-1 and Table 15.4.1.5-2.
7. T1 starts.
8. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
9. The UE shall transmit RRCReconfigurationComplete message.
10. The SS shall transmit an LPP REQUEST CAPABILITIES message.
11. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilitie* IE.
12. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 11 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
13. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
14. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
15. When T1 expires, the SS shall transmit an RRCRelease message with the UL configuration and SDT configuration to transit the UE to RRC_INACTIVE state.
16. Switch the power setting from T1 to T2 as specified in Table 15.4.1.5-1 and Table 15.4.1.5-2.
17. Wait for the UE monitors for occurrence of the periodic event requested.
18. The UE shall transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 21.
18. The SS transmits an *RRCResume* message.
19. The UE transmits an *RRCResumeComplete* message.
20. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.4.1.4.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.
21. The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time, then the number of successful tests is increased by one. If the UE fails to

report the *NR-Multi-RTT-ProvideLocationInformation* IE with the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.

22. If the UE message at step 20 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

23. UE shall be released and moved to RRC_IDLE.

24. Repeat steps 1-23 in Tables 15.4.1.4-1 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.4.1.4.3 Message contents

**Table 15.4.1.4.3-1: SIB1**

| Derivation Path: TS 38.508-1 [45], clause 4.6.1 with condition SDT |
|---|

**Table 15.4.1.4.3-2: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 1 1 0 | Multi-RTT | |

**Table 15.4.1.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.1.4.3-4: CellGroupConfig (Table 15.4.1.4.3-3)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms1280 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 15.4.1.4.3-5: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-Multi-RTT-RequestCapabilities-r16 | TRUE |

**Table 15.4.1.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 15.4.1.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.1.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.1.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.1.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.1.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.1.4.3-9: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 15.4.1.4.3-12 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 15.4.1.4.3-10: *RRCRelease***

| Derivation Path: TS 38.508-1 [45], table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCRelease ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcRelease SEQUENCE { | | | |
| suspendConfig SEQUENCE { | | | |
| srs-PosRRC-Inactive-r17 CHOICE { | | | |
| setup SEQUENCE { | | | |
| SRS-PosRRC-Inactive-r17 | OCTET STRING containing SRS-PosRRC-InactiveConfig-r17 as defined in Table 15.4.1.4.3-11 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.1.4.3-11: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.1.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 20 | | Config 1 |
| sl320 | 40 | | Config 3 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.4.1.4.3-12: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |

| | | | |
|---|---|---|---|
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.1.4.3-13: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 2 entries | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |

| | | | |
|---|---|---|---|
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-Multi-RTT-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable. | | |
| nr-AdditionalPathList-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-TimingQuality-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | | | |
| | | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | | | |
| nr-UE-RxTx-TEG-Info-r17 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-AdditionalPathListExt-r17 | | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | | | |
| nr-SRS-TxTEG-Set-r17 | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.4.1.5 Test requirement

Table 15.4.1.5-1 and Table 15.4.1.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.1.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |

| | | | | |
|---|---|---|---|---|
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX cycle | | 1, 2, 3 | 1.28s | |
| Time offset between serving and neighbour cells | μs | 1, 2, 3 | 3 | Synchronous cells |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |

**Table15.4.1.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.39 | -Infinity | -12.13 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -1.98 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -99.98 | -Infinity | -108 |
| | | 2 | -Infinity | -99.98 | -Infinity | -108 |
| | | 3 | -Infinity | -96.98 | -Infinity | -105 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.56 | N/A | -64.56 |
| | dBm/19.08 MHz | 2 | | -64.56 | | -64.56 |
| | dBm/47.88 MHz | 3 | | -60.58 | | -60.58 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2 within the specified UE Rx-Tx time difference measurement time starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported UE Rx-Tx measurement for each correct event shall be within the UE Rx-Tx reporting range specified in clause TS 38.133 clause 10.1.25.3.1.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{Rx,TEG,i} = 1, N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} = 0.285, L_{available_PRS,i} = 0.071, N_{sample} = 4$. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 1280ms depending on UE capabilities. Therefore, $T_{available_PRS,i} = 1280$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 1 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.285}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 32s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next

multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 21s and 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 32s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.4.1.5-3, Table 15.4.1.5-4 and Table 15.4.1.5-5 for reference.

**Table 15.4.1.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 3 |
| >=n2 | 2 |

**Table 15.4.1.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\left\lceil \frac{0.285}{N} \right\rceil$ |
|---|---|
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table 15.4.1.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.2 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 15.4.2.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.4 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.4.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning, supports DL-PRS measurement in RRC_INACTIVE state and supportedDL-PRS-ProcessingSamples-RRC-Inactive.

### 15.4.2.3 Minimum conformance requirements

Same as clause 15.4.1.3.

### 15.4.2.4 Test description

#### 15.4.2.4.1 Initial conditions

The test defines three possible test configurations; specified in Table 15.4.2.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.4.2.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.4.2.5-1 and Table 15.4.2.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.2.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

#### 15.4.2.4.2 Test procedure

Same as clause 15.4.1.4.2.

#### 15.4.2.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.2.4.3-1 replaces Table 15.4.1.4.3-11.

**Table 15.4.2.4.3-1: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |

| | | | |
|---|---|---|---|
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.4.2.5 Test requirement

Table 15.4.2.5-1 and Table 15.4.2.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.2.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX cycle length | s | 1, 2, 3 | 1.28 | |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| T1 | s | 1, 2, 3 | 3 | |
| T2 | s | 1, 2, 3 | 5 | |

**Table 15.4.2.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1<br>ULBWP.0.1 | | N/A | |
| PRS configuration Note 5 | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS Resource slot offset | slots | 1, 2, 3 | 0 | | 0 | |
| PRS muting info | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0.32 | -Infinity | -5.64 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 2.88 | -Infinity | -0.96 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -95.12 | -Infinity | -98.96 |
| | | 2 | -Infinity | -95.12 | -Infinity | -98.96 |
| | | 3 | -Infinity | -92.12 | -Infinity | -95.96 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -98 | -98 | -98 | -98 |
| | | 2 | -98 | -98 | -98 | -98 |
| | | 3 | -95 | -95 | -95 | -95 |
| Io Note 3 | dBm/19.08 MHz | 1 | N/A | -61.22 | N/A | -61.22 |
| | dBm/19.08 MHz | 2 | | -61.22 | | -61.22 |
| | dBm/47.88 MHz | 3 | | -57.24 | | -57.24 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

| | |
|---|---|
| Note 1: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels apply only for symbols where PRS is transmitted. |
| Note 4: | OCNG shall be used such that all active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols, except for slots where PRS is transmitted by any of the cells. |
| Note 5: | Void. |

The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2 within the specified UE Rx-Tx time difference measurement time starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported UE Rx-Tx measurement for each correct event shall be within the UE Rx-Tx reporting range specified in clause TS 38.133 clause 10.1.25.3.1.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{Rx,TEG,i} = 1, N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} = 0.285, L_{available_PRS,i} = 0.071, N_{sample} = 1$. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 1280ms depending on UE capabilities. Therefore, $T_{available_PRS,i} = 1280$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 1 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.285}{N} \right\rceil * 1 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 1.288s and 8.96s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 2s and 9s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 9.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.4.2.5-3, Table 15.4.2.5-4 and Table 15.4.2.5-5 for reference.

**Table 15.4.2.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 15.4.2.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\lceil \frac{0.285}{N} \rceil$ |
|---|---|
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table 15.4.2.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.3 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.4.3.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.4 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.4.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and supports DL-PRS measurement in RRC_INACTIVE state.

### 15.4.3.3 Minimum conformance requirements

Same as clause 15.4.1.3.

### 15.4.3.4 Test description

#### 15.4.3.4.1 Initial conditions

The supported test configurations in listed in Table 15.4.3.4.1-1

**Table 15.4.3.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.3.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.4.3.5-1 and Table 15.4.3.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.3.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.4.3.4.2 Test procedure

Same as clause 15.4.1.4.2.

### 15.4.3.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.3.4.3-1 replaces Table 15.4.1.4.3-3, Table 15.4.3.4.3-2 replaces Table 15.4.1.4.3-6, Table 15.4.3.4.3-3 replaces Table 15.4.1.4.3-7 and Table 15.4.3.4.3-4 replaces Table 15.4.1.4.3-10.

**Table 15.4.3.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.3.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.3.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.3.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.3.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.3.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.3.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

## 15.4.3.5 Test requirement

Table 15.4.3.5-1 and Table 15.4.3.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.3.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | | 1 | SSB.2 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| CP length | | 1 | Normal | |
| DRX cycle | | 1 | 0.64s | |
| Time offset between serving and neighbour cells | µs | 1 | 3 | Synchronous cells |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 83 | |

**Table 15.4.3.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | | | |
| Beam Assumption Note 7 | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting info | | 1 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -90.3 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -99.3 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.69 | -Infinity | -12.31 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -92.3 | -Infinity | -100.3 |
| Io | dBm/190.08 MHz | 1 | N/A | -55.89 | N/A | -55.89 |
| Propagation Condition | | 1 | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation
Note 8: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The UE Rx-Tx time difference measurement time fulfils the requirements specified in clause 15.4.3.3.

The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2 within the specified UE Rx-Tx time difference measurement time starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported UE Rx-Tx measurement for each correct event shall be within the UE Rx-Tx reporting range specified in clause TS 38.133 clause 10.1.25.3.1.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{Rx,TEG,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.071$, $N_{sample} = 4$. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 640ms depending on UE capabilities. Therefore, $T_{available_PRS,i} = 640$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 20.488s and 82.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 21s and 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 21.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.4.3.5-3, Table 15.4.3.5-4 and Table 15.4.3.5-5 for reference.

**Table 15.4.3.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 15.4.3.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >= nDot025 | 1 |

**Table 15.4.3.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |

| | | |
|---|---|---|
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.4 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.4.4.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.4 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.4.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning, supports DL-PRS measurement in RRC_INACTIVE state and supportedDL-PRS-ProcessingSamples-RRC-Inactive.

### 15.4.4.3 Minimum conformance requirements

Same as clause 15.4.1.3.

### 15.4.4.4 Test description

#### 15.4.4.4.1 Initial conditions

The supported test configurations in listed in Table 15.4.4.4.1-1

**Table 15.4.4.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.4.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.

2. The general test parameter settings are set up according to Table 15.4.4.5-1 and Table 15.4.4.5-2.

3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.4.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.4.4.4.2 Test procedure

Same as clause 15.4.1.4.2.

### 15.4.4.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.4.4.3-1 replaces Table 15.4.1.4.3-3, Table 15.4.4.4.3-2 replaces Table 15.4.1.4.3-6, Table 15.4.4.4.3-3 replaces Table 15.4.1.4.3-7, Table 15.4.4.4.3-4 replaces Table 15.4.1.4.3-10, Table 15.4.4.4.3-4 replaces Table 15.4.1.4.3-10 and Table 15.4.4.4.3-5 replaces Table 15.4.1.4.3-11.

**Table 15.4.4.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.4.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.4.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.4.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.4.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.4.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |

| | | | |
|---|---|---|---|
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.4.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.4.4.4.3-5: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next | |

| | | | |
|---|---|---|---|
| | | multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.4.4.5 Test requirement

Table 15.4.4.5-1 and Table 15.4.4.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.4.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | |
| SSB configuration | | 1 | SSB.2 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| CP length | | 1 | Normal | |
| DRX cycle length | s | 1 | 0.64 | |
| Time offset between serving and neighbour cells | µs | 1 | 3 | Synchronous cells |
| T1 | s | 1 | 3 | |
| T2 | s | 1 | 22 | |

**Table 15.4.4.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | | | |
| Beam Assumption[Note 7] | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| PRS configuration[Note 9] | | 1 | PRS.1.2 FR2 | | PRS.1.2 FR2 | |
| PRS Resource slot offset | slots | 1 | 0 | | 0 | |
| PRS muting info | | 1 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ [Note 2] | dBm/SCS | 1 | -94 | | | |
| $N_{oc}$ [Note 2] | dBm/15 kHz | 1 | -103 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0.32 | -Infinity | -5.64 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 3.56 | -Infinity | -0.28 |
| PRS-RSRP [Note 3] | dBm/SCS kHz | 1 | -Infinity | -90.4 | -Infinity | -94.2 |
| SS-RSRP [Note 3] | dBm/SCS kHz | 1 | -89 | -86.77 | -89 | -90.73 |
| Io | dBm/190.08 MHz | 1 | N/A | -55.73 | N/A | -55.73 |
| Propagation Condition | | 1 | AWGN | | | |

| | |
|---|---|
| Note 1: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels apply only for symbols where PRS is transmitted. |
| Note 4: | PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 6: | As observed with 0 dBi gain antenna at the centre of the quiet zone |
| Note 7: | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |
| Note 8: | OCNG shall be used such that all active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols, except for slots where PRS is transmitted by any of the cells. |
| Note 9: | Void. |

The UE Rx-Tx time difference measurement time fulfils the requirements specified in clause 15.4.4.3.

The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2 within the specified UE Rx-Tx time difference measurement time starting from the beginning of time interval T2.

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported UE Rx-Tx measurement for each correct event shall be within the UE Rx-Tx reporting range specified in clause TS 38.133 clause 10.1.25.3.1.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{Rx,TEG,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.035$, $N_{sample} = 1$. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 640ms depending on UE capabilities. Therefore, $T_{available_PRS,i} = 640$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 1 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 21.12s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 22s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 22.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.4.4.5-3, Table 15.4.4.5-4 and Table 15.4.4.5-5 for reference.

**Table 15.4.4.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 15.4.4.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >= nDot025 | 1 |

**Table 15.4.4.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.5 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR1 SA

### 15.4.5.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 5.6.4.6 for UE Rx-Tx measurements with PRS bandwidth aggregation. The tests are conducted under AWGN propagation condition with the UE operating in FR1 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR1.

### 15.4.5.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_INACTIVE.

### 15.4.5.3 Minimum conformance requirements

The requirements in this clause apply provided that UE receives requests from LMF to perform PRS measurement on aggregated positioning frequency layers (PFLs) via *nr-DL-PRS-JointMeasurementRequestedPFL-List* [49].

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [49], the UE shall be able to measure multiple (up to the UE capability specified in clause 5.6.4.3) UE Rx-Tx time difference measurements, defined in TS 38.215 [57], during the measurement period $T_{UERxTx,Total}$ defined as:

$$T_{UERxTx,total} = T_{UERxTx,aggr} + T_{UERxTx,non-aggr} + T_{margin}$$

Where:

- $T_{UERxTx,aggr}$ is the total measurement period for aggregated measurements as defined in this clause, and
- $T_{UERxTx,non-aggr}$ is the total measurement period for non-aggregates measurements as defined in this clause, and

- $T_{margin} = T'_{margin}$ is $\max(T_{effect,j})$, if both $T_{UERxTx,aggr}$ and $T_{UERxTx,non-aggr}$ are non-zero, and $T_{margin} = 0$ otherwise, where $T'_{margin} = \max(T_{effect,j})$ and $T_{effect,j}$ is the $T_{effect}$ from both PFL combination(s) and non-aggregated PFL(s) which are configured for positioning measurement.

$T_{UERxTx,non-aggr}$ is zero if every resource set on every PFL is linked for aggregation with at least one other resource set on another PFL. Otherwise, $T_{UERxTx,non-aggr}$ is as defined in TS 38.133 [50] clause 5.6.4.5 with the following exceptions,

- only PFLs containing resource set(s) not linked to any other resource set(s) are considered in $L$
- on each PFL $i$, only resource set(s) not linked to any other resource set(s) are considered in $T_{available_PRS,i}$, $N^{slot}_{PRS,i}$ and $L_{available_PRS,i}$
- $N_{sample} = 2$ if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *supportedDL-PRS-ProcessingSamples-RRC-Inactive* specified in TS 37.355 [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples,
- $(L-1) * \max(T_{effect,i})$ is replace by $(L-1) * T'_{margin}$.

$T_{UERxTx,aggr}$ is zero if no resourse sets on any PFL are linked for aggregation with other resource sets on other PFLs. Otherwise, $T_{UERxTx,aggr}$ is defined as

$$T_{UERxTx,aggr} = \sum_{m=1}^{M} T_{UERxTx,aggr,m} + (M-1) * T'_{margin} \quad ,$$

where:

- $m$ is the index of PFL combination,
- $M$ is total number of PFL combinations,
- $T_{effect,aggr,m}$ is the periodicity of the PRS measurement in PFL combination $m$ as defined in this clause,
- $T_{UERxTx,aggr,m}$ is the measurement period for UE Rx-Tx time difference measurement in PFL combination $m$ as specified below.

$$T_{UERxTx,aggr,m} = \left( K_{carrier,aggr} * N_{Rx,TEG,aggr,m} * N_{RxBeam,aggr,m} * \left\lceil \frac{N^{slot}_{PRS,aggr,m}}{N'_{aggr,m}} \right\rceil * \left\lceil \frac{L_{available_{PRS},aggr,m}}{N_{aggr,m}} \right\rceil * N_{sample} - 1 \right) * T_{effect,aggr,m} + T_{last,aggr,m}$$

where:

- $K_{carrier,aggr}$ is a scaling factor for PRS measurements in RRC_INACTIVE, and is defined as:
  - $K_{carrier,aggr} = 1$ if the UE supports *parallelPRS-MeasRRC-Inactive-r17*,
  - Otherwise,
    - If Srxlev $\leq S_{nonIntraSearchP}$ or Squal $\leq S_{nonIntraSearchQ}$, $K_{carrier,aggr} = K_{carrier} + 1$, where $K_{carrier}$ is defined in TS 38.133 [50] clause 4.2.2.4
    - If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$, $K_{carrier,aggr} = N_{layers} + 1$, where $N_{layers}$ is defined in TS 38.133 [50] clause 4.2.2.7.
- $N_{Rx,TEG,aggr,m}$ is a scaling factor for PRS measurements with multiple Rx TEGs, and is defined as:
  - $N_{Rx,TEG,aggr,m}$ =1 if the UE is not configured by the LMF to measure a PRS resource with multiple Rx TEGs via *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49].
  - $N_{Rx,TEG,aggr,m}$ is defined as follows if the UE is configured by the LMF with *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [34] to perform measurement on same DL PRS resource of a TRP using different Rx TEGs in *NR-DL-TDOA-RequestLocationInformation* [49]:

- $N_{Rx,TEG,i} = P$, if the UE is not capable of receiving same DL PRS resource simultaneously from multiple Rx TEGs, where P is the number of UE Rx TEGs that the UE is requested by LMF to measure the same DL-PRS Resource of a TRP indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* in [49], and in case 'n0' is indicated, P is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.

- $N_{Rx,TEG,i} = \left\lfloor \frac{P}{Q} \right\rfloor$, if the UE is capable of receiving the same DL PRS resource simultaneously from multiple Rx TEGs, where $Q$ is the number of UE Rx TEGs for measuring the same DL-PRS Resource simultaneously indicated by *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul-r17* in [49].

- $N_{RxBeam,aggr,m}$ is a scaling factor for PRS measurements with multiple Rx beams, and is defined as

  - $N_{RxBeam,aggr,m}$ = 1 if PFL combination *m* is in FR1,

  - $N_{RxBeam,aggr,m}$ is defined as follows if PFL combination *m* is in FR2

    - $N_{RxBeam,aggr,m}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing PFL combination *m*, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*,

    - $N_{RxBeam,aggr,m}$ equals to 8 otherwise

- $N_{PRS,aggr,m}^{slot}$ is the maximum number of DL PRS resources in PFL combination *m* configured in a slot, and only the PRS resources in resource set(s) linked to other resource set in PFL combination *m* are counted

- $N'_{aggr,m}$ is the UE capability on maximum number of DL PRS resources that can be processed in a slot for PFL combination *m* as indicated by *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR1-r18* for FR1 and *maxNumOfAggregatedDL-PRS-ResourcePerSlot-FR2-r18* for FR2 specified in TS 37.355 [49].

- $L_{available_{PRS},aggr,m}$ is the time duration of available PRS resources in PFL combination *m* to be measured during $T_{PRS,aggr,m}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_{PRS},aggr,m}$, only unmuted PRS resources in resource set(s) linked to other resource set in PFL combination *m* and that are not fully overlapped with other higher-priority DL signals/channels are considered.

- $N_{aggr,m}$ is the UE capability on duration of DL PRS resources in ms for PFL combination *m* as indicated by *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsN-r18* specified in TS 37.355 [49].

- $N_{sample}$ is number of PRS measurement samples,

  - $N_{sample}$ = 2 if the UE supports the capability of positioning measurements with reduced number of samples as indicated by *reducedNumOfSampleInMeasurementWithPRS-BWA-RRC-IdleAndInactive* specified in TS 37.355 [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples,

  - $N_{sample}$ = 4 otherwise.

- $T_{effect,aggr,m}$ is the periodicity of the PRS measurement in PFL combination $m$,

$$T_{effect,aggr,m} = \left\lceil \frac{T_{aggr,m}}{T_{available,PRS,aggr,m}} \right\rceil * T_{available,PRS,aggr,m}$$

  - $T_{aggr,m}$ is the UE capability on time for processing of DL PRS resources in ms for PFL combination *m* as indicated by *prs-durationOfTwoPRS-BWA-ProcessingSymbolsT-r18* or *prs-durationOfThreePRS-BWA-ProcessingSymbolsT* specified in TS 37.355 [49].

  - $T_{available,PRS,aggr,m} = LCM(T_{PRS,aggr,m}, T_{DRX})$, the least common multiple between $T_{PRS,aggr,m}$ and the DRX cycle length $T_{DRX}$, where $T_{PRS,aggr,m}$ is the periodicity of DL PRS resource with muting on PFL combination $m$.

- If more than one PRS periodicities are configured in PFL combination $m$, the least common multiple of PRS periodicities $T_{per}^{PRS\ with\ muting}$ among all DL PRS resource sets that are linked to other resource set in PFL combination $m$, is used to derive $T_{PRS,aggr,m}$, and for each applicable PRS resource set,
  - $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, is the PRS periodicity with muting per PRS resource, and
  - $T_{per}^{PRS}$ is the periodicity of PRS resource set given by the higher-layer parameter *DL-PRS-Periodicity*, and
  - $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

- $T_{last,aggr,m}$ is the measurement duration for the last PRS sample in PFL combination $m$, including the sampling time and processing time, $T_{last,aggr,m} = T_{aggr,m} + T_{available,PRS,aggr,m}$.

The normative reference for this requirement is TS 38.133 [50] clause 5.6.4.6.

## 15.4.5.4 Test description

The test defines three possible test configurations; specified in Table 15.4.5.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.4.5.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

### 15.4.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.4.5.5-1 and Table 15.4.5.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.5.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.4.5.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During T2 UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

8. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 15.4.5.4.3-1.
9. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.

10. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

11. The SS adjusts the downlink timing for Cell 1 to a delay of +512 $T_c$, compared to the current value.

12. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

6. Set the parameters for T1 according to Table 15.4.5.5-1 and Table 15.4.5.5-2.

7. T1 starts.

8. The SS shall transmit an RRCReconfiguration message with the DRX configuration.

9. The UE shall transmit RRCReconfigurationComplete message.

10. The SS shall transmit an LPP REQUEST CAPABILITIES message.

11. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilitie* IE.

12. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE and *NR-DL-PRS-AggregationInfo-r18* linking each PRS resource in PFL1 to the corresponding PRS resource in PFL2. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 11 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

13. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-Multi-RTT-RequestLocationInformation* IE and and nr-DL-PRS-JointMeasurementRequest-r18 IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.

14. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.

15. When T1 expires, the SS shall transmit an RRCRelease message with the UL configuration and SDT configuration to transit the UE to RRC_INACTIVE state.

16. Switch the power setting from T1 to T2 as specified in Table 15.4.5.5-1 and Table 15.4.5.5-2.

17. Wait for the UE monitors for occurrence of the periodic event requested.

18. The UE shall transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 21.

18. The SS transmits an *RRCResume* message.

19. The UE transmits an *RRCResumeComplete* message.

20. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.4.5.4.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.

21. The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.

22. If the UE message at step 20 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

24. UE shall be released and moved to RRC_IDLE.

24. Repeat steps 1-23 in Tables 15.4.5.4-1 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.4.5.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.5.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.4.5.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.4.5.4.3-3 replaces Table 15.4.1.4.3-12 and Table 15.4.5.4.3-4 replaces Table 15.4.1.4.3-13.

**Table 15.4.5.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.5.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.5.4.3-2 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.5.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.5.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.5.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |

| | | | |
|---|---|---|---|
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.5.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.5.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.5.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.4.5.5 Test requirement

Table 15.4.5.5-1 and Table 15.4.5.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.5.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1, 2, 3 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |

| | | | | |
|---|---|---|---|---|
| CP length | | 1, 2, 3 | Normal | |
| DRX cycle | | 1, 2, 3 | 1.28s | |
| eDRX cycle length (for both RAN and CN) | s | 1, 2, 3 | 40.96 | |
| PTW window length | s | 1, 2, 3 | 1.28 | |
| Reporting periodicity | s | 1, 2, 3 | 20 | *reportingInterval* for periodic reporting defined in TS 37.355 [49] |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 5 | |
| T2 | s | 1, 2, 3 | 10 | |

**Table 15.4.5.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS muting info | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.1 | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |
| | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.59 | | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.4.5.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,aggr,m} = \left( K_{carrier,aggr} * N_{Rx,TEG,aggr,m} * N_{RxBeam,aggr,m} * \left\lceil \frac{N_{PRS,aggr,m}^{slot}}{N'_{aggr,m}} \right\rceil * \left\lceil \frac{L_{available_{PRS},aggr,m}}{N_{aggr,m}} \right\rceil * N_{sample} - 1 \right) * T_{effect,aggr,m} + T_{last,aggr,m}$$

Where $K_{carrier,aggr} = 1, N_{Rx,TEG,aggr,m} = 1{,}1, N_{RxBeam,aggr,m} = 1$, $N_{sample_agg} = 4$. $N_{PRS,aggr,m}^{slot} = 1$, $L_{avavailable_PRS_agg,i} = 0.285\text{ms}$ for type 1 duration calculation and $L_{avavailable_PRS_agg,i} = 1\text{ms}$ for type 2 duration calculation. $N'_{aggr,m}$ is the parameter maxNumOfAggregatedDL-PRS-ResourcePerSlot from TS 37.355 [49], $N_{aggr,m}$ is the parameter prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN from TS 37.355 [49]. $T_{last,aggr,m} = T_{aggr,m} + T_{available,PRS,aggr,m}$ and $T_{effect,aggr,m} = \left\lceil \frac{T_{aggr,m}}{T_{available,PRS,aggr,m}} \right\rceil * T_{available,PRS,aggr,m}$.

$T_{available,PRS,aggr,m} = LCM(T_{PRS,aggr,m}, T_{DRX})$; $T_{DRX} = 1280\text{ms}$. Therefore, $T_{available,PRS,aggr,m} = 1280$ ms.

Finally, it results in the following equation:

$$(1 \times 1 \times 1 \times \left\lceil \frac{1}{N'_{aggr,m}} \right\rceil \left\lceil \frac{0.285}{N_{aggr,m}} \right\rceil \times 4\text{-}1) T_{effect,aggr,m} + T_{last,aggr,m}$$

Depending on the UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]:

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 11.536s and 34.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 35s.

Similarly for Type 2 Buffering, Best Case result of 11.536s is 12s and for the worst-case result of 85760ms, rounded to the next second, the final value is 86s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 86.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 11.536s and 24.32s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 12s and 25s.

Similarly for Type 2 Buffering, Best Case result of 11.536s is 12s and for the worst-case result of 44880ms, rounded to the next second, the final value is 45s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 12.3s and 45.3s.

**Table 15.4.5.5-3: value of N'aggr.m based on *maxNumOfAggregatedDL-PRS-ResourcePerSlot***

| $N'_{aggr,m}$ | $\left\lceil \frac{1}{N'_{aggr,m}} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.4.5.5-4: value of $N_{aggr,m}$ based on test configuration and UE buffer type**

| **$N_{aggr,m}$ for SCS 15kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.285}{N_{aggr,m}} \right\rceil$ |
|---|---|
| msDot125 | 3 |
| msDot25 | 2 |
| >=msDot5 | 1 |
| **$N_{aggr,m}$ for SCS 30kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.142}{N_{aggr,m}} \right\rceil$ |
| nDot125 | 2 |
| >=nDot25 | 1 |
| **$N_{aggr,m}$ for SCS 15kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{1}{N_{aggr,m}} \right\rceil$ |
| msDot125 | 8 |
| msDot25 | 4 |
| msDot5 | 2 |
| >=ms1 | 1 |
| **$N_{aggr,m}$ for SCS 30kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{0.5}{N_{aggr,m}} \right\rceil$ |
| msDot125 | 4 |
| msDot25 | 2 |
| >=msDot5 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26] | |

**Table 15.4.5.5-5: value of $T_{aggr,m}$ and $T_{effect,aggr,m}$ based on $T_{last,aggr,m}$**

| $T_{aggr,m}$ | $T_{effect,aggr,m}$ | $T_{last,aggr,m}$ |
|---|---|---|
| ms8 | 640 | 648 |
| ms16 | 640 | 656 |
| ms20 | 640 | 660 |
| ms30 | 640 | 670 |
| ms40 | 640 | 680 |
| ms80 | 640 | 720 |
| ms160 | 640 | 800 |
| ms320 | 640 | 960 |

| | | |
|---|---|---|
| ms640 | 640 | 1280 |
| ms1280 | 1280 | 1920 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.6 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.4.6.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 5.6.4.6 for UE Rx-Tx measurements with PRS bandwidth aggregation. The tests are conducted under AWGN propagation condition with the UE operating in FR2 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR2.

### 15.4.6.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_INACTIVE.

### 15.4.6.3 Minimum conformance requirements

Same as clause 15.4.5.3.

### 15.4.6.4 Test description

The supported test configurations in listed in Table 15.4.6.4.1-1

**Table 15.4.6.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.6.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.4.6.5-1 and Table 15.4.6.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.6.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.4.6.4.2 Test procedure

Same as clause 15.4.5.4.2.

### 15.4.6.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.6.4.3-1 replaces Table 15.4.1.4.3-4, Table 15.4.6.4.3-2 replaces Table 15.4.1.4.3-7, Table 15.4.6.4.3-3 replaces Table 15.4.1.4.3-8, Table 15.4.6.4.3-4 replaces Table 15.4.1.4.3-11, Table 15.4.6.4.3-5 replaces Table 15.4.1.4.3-12 and Table 15.4.6.4.3-6 replaces Table 15.4.1.4.3-13.

**Table 15.4.6.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.6.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.6.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.6.4.3-3 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | N2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.6.4.3-3 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.6.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.6.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| N2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.6.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.3.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.4.6.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |

| | | | |
|---|---|---|---|
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.6.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |

| | | | |
|---|---|---|---|
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.6.4.3-6: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.4.6.5 Test requirement

Table 15.4.6.5-1 and Table 15.4.6.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.6.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |

| | | | | |
|---|---|---|---|---|
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2, all signals/channels except PRS in PFL2 are allocated within this channel. |
| | | | 2 | For both Cell 1 and Cell 2, only PRS in PFL2 is allocated within this channel. |
| $BW_{channel}$ | MHz | 1 | 200: $N_{RB,c}$ = 132 | For both RF channels |
| SSB configuration | | 1 | SSB.2 FR2 | |
| SMTC configuration | | 1 | SMTC.1 | |
| CP length | | 1 | Normal | |
| DRX | s | 1 | 0.64 | |
| Time offset between serving and neighbour cells | μs | 1 | 3 | Synchronous cells/TRPs |
| Expected RSTD | μs | 1 | 3 | |
| Expected RSTD uncertainty | μs | 1 | 5 | |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |

**Table 15.4.6.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in clause TS 38.133 A.3.15 | | | |
| Beam Assumption[Note 5] | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns[Note 1] | | 1 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS[Note 1, 4] | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS [Note 1, 4] | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1 | ULBWP.1.1 | | N/A | |
| PRS configuration (for both PFL1 and PFL2) | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting info (*dl-PRS-MutingOption1* for both PFL1 and PFL2) | | 1 | ‘10’ | | ‘01’ | |

| | | | | | | |
|---|---|---|---|---|---|---|
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -89 | -90.1 | -89 | -90.1 |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.69 | -Infinity | -12.1 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -9.8 |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -92.1 | -Infinity | -99.9 |
| Io Note 3 (on symbols where PRS is not allocated) | dBm/190.08 MHz | 1 | N/A | -55.68 | N/A | -55.68 |
| Propagation Condition | | 1 | AWGN | | | |

NOTE 1: OCNG shall be used such that active cell (Cell 1) is fully allocated and a constant total transmitted power spectral density is achieved on all OFDM symbols except those in which PRS is allocated.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters and they are provided for information only. They are not settable parameters themselves.
NOTE 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 5: Information about types of UE beam is given in TS 38.133 clause B.2.1.3, and does not limit UE implementation or test system implementation
NOTE 6: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] table 6.2.1.3-4.
NOTE 7: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
NOTE 8: As observed with 0 dBi gain antenna at the centre of the quiet zone.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.4.6.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,aggr,m} = \left( K_{carrier,aggr} * N_{Rx,TEG,aggr,m} * N_{RxBeam,aggr,m} * \left\lceil \frac{N_{PRS,aggr,m}^{slot}}{N'_{aggr,m}} \right\rceil * \left\lceil \frac{L_{available_{PRS},aggr,m}}{N_{aggr,m}} \right\rceil * N_{sample} - 1 \right) * T_{effect,aggr,m} + T_{last,aggr,m}$$

Where $K_{carrier,aggr} = 1$, $N_{Rx,TEG,aggr,m} = 1,1$, $N_{RxBeam,aggr,m} = 8$, $N_{sample_agg} = 4$. $N_{PRS,aggr,m}^{slot} = 1$, $L_{avavailable_PRS_agg,i} = 0.071\text{ms}$ for type 1 duration calculation and $L_{avavailable_PRS_agg,i} = 0.25\text{ms}$ for type 2 duration calculation. $N'_{aggr,m}$ is the parameter maxNumOfAggregatedDL-PRS-ResourcePerSlot from TS 37.355 [49], $N_{aggr,m}$ is the parameter prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN from TS 37.355 [49]. $T_{last,aggr,m} = T_{aggr,m} + T_{available,PRS,aggr,m}$ and $T_{effect,aggr,m} = \left\lceil \frac{T_{aggr,m}}{T_{available,PRS,aggr,m}} \right\rceil * T_{available,PRS,aggr,m}$.

$T_{available,PRS,aggr,m} = LCM(T_{PRS,aggr,m}, T_{DRX})$; $T_{DRX} = 640\text{ms}$, $T_{PRS_agg,i} = 640$ ms. Therefore, $T_{available,PRS,aggr,m} = 640$ ms.

Finally, it results in the following equation:

$$(1 \times 1 \times 8 \times \left\lceil \frac{1}{N'_{aggr,m}} \right\rceil \left\lceil \frac{0.071}{N_{aggr,m}} \right\rceil \times 4\text{-}1) T_{effect,aggr,m} + T_{last,aggr,m}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 20.488s and 82.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 15.2.12.4.3-5. The LPP time IE ranges between 21s and 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 21.3s and 83.3s.

**Table 15.4.6.5-3: value of N'aggr.m based on *maxNumOfAggregatedDL-PRS-ResourcePerSlot***

| $N'_{aggr,m}$ | $\left\lceil \frac{1}{N'_{aggr,m}} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.4.6.5-4: value of Naggr.m based on *prs-durationOfTwoPRS-BWA-ProcessingSymbolsN or prs-durationOfThreePRS-BWA-ProcessingSymbolsN***

| $N_{aggr,m}$ for Type 1 Buffering Note 1 | $\left\lceil \frac{0.071}{N_{aggr,m}} \right\rceil$ |
|---|---|
| >=msDot125 | 1 |
| $N_{aggr,m}$ for Type 1 Buffering Note 1 | $\left\lceil \frac{0.25}{N_{aggr,m}} \right\rceil$ |
| msDot125 | 2 |
| >=msDot25 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26] | |

**Table 15.4.6.5-5: value of $T_{aggr,m}$ and $T_{effect,aggr,m}$ based on $T_{last,aggr,m}$**

| $T_{aggr,m}$ | $T_{effect,aggr,m}$ | $T_{last,aggr,m}$ |
|---|---|---|
| ms8 | 640 | 648 |
| ms16 | 640 | 656 |
| ms20 | 640 | 660 |
| ms30 | 640 | 670 |
| ms40 | 640 | 680 |
| ms80 | 640 | 720 |
| ms160 | 640 | 800 |
| ms320 | 640 | 960 |
| ms640 | 640 | 1280 |
| ms1280 | 1280 | 1920 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.7 UE Rx-Tx time difference measurement reporting delay in FR1 for Redcap UE without Rx FH

### 15.4.7.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement without RX FH reported by the RedCap UE meets the requirements specified in TS 38.133 [50] clause 5.6A.6.5 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured. The measurement reporting delay test defined in this lause is valid for both 1 Rx and 2 Rx RedCap UEs.

### 15.4.7.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE.

### 15.4.7.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [49], UE shall be able to measure multiple (up to the UE capability specified in clause 5.6.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [57] in configured positioning frequency layers within the measurement period $T_{UERxTx,Total}$ ms.

$$T_{UERxTx,\ Total} = \sum_{i=1}^{L} T_{UERxTx,i} + (L-1) * \max(T_{effect,i})$$

Where:

- $i$ is the index of positioning frequency layer,
- $T_{UERxTx,i}$ is the measurement period for UE Rx-Tx time difference measurements in positioning frequency layer *i* as further defined in this clause,
- L is total number of positioning frequency layers,
- $T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer *i* as defined further in this clause.

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where:

- $K_{carrier_PRS}$=1 if the UE is capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [34].
- $K_{carrier_PRS} = N_{layer} + 1$ if the UE is not capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [49] and if Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$; where $N_{layer}$is defined in clause 4.2.2.7.
- $K_{carrier_PRS} = K_{carrier} + 1$ if the UE is not capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [49] and if Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$; where $K_{carrier}$is defined in clause 4.2.2.4.
- $N_{RxBeam,i}$ is the scaling factor for UE Rx beam sweeping:
  - $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1, and if positioning frequency layer *i* is in FR2.
    - $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-Multi-RTT -RequestLocationInformation*.
    - $N_{RxBeam,i}$ equals to 8, otherwise.

  $N_{RxTx,TEG,i}$ is the Rx TEG specific scaling factor:

    - $N_{RxTx,TEG,i}$ = 1 if UE is not configured by LMF with *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49].

    Otherwise,

    - $N_{RxTx,TEG,i} = N_{TEG,i}$, if UE is not capable of measuring same DL PRS resource simultaneously from multiple Rx TEGs, where $N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49], and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.
    - $N_{RxTx,TEG,i} = \left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$, if UE is capable of measuring same DL PRS resource simultaneously from multiple Rx TEGs, where $k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.
- $L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.
- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

- {N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in clause 4.2.7.2 of TS 37.355 [49],

- N' is UE capability for number of DL PRS resources that it can process in a slot corresponding to *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* as specified in clause 6.4.3 of TS 37.355 [49],

- $N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:

  - $N_{sample}$ = 4 if the UE is not capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [49] or not configured to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [49].

  - $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [34] and LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [49] and the following conditions are met:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [49] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [49] but the following conditions are not met:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time, $T_{last,i} = T_i + T_{available_PRS,i}$ ,

- $T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where:

- $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [49],

- $T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, the least common multiple between $T_{PRS,i}$ and $T_{DRX}$.

  - When UE is configured with RAN eDRX ≤ 10.24 s:

    - $T_{DRX}$ is defined as T in TS 38.304 [1] when CN eDRX ≤ 10.24 s.

    - $T_{DRX}$ is the maximum of the T inside and outside of the CN PTW when CN eDRX > 10.24 s, where T inside and outside of the CN PTW are defined in TS 38.304 [1].

  - When UE is configured with RAN eDRX > 10.24 s:

    - $T_{DRX}$ is the maximum of the DRX cycles within the CN PTW and the RAN PTW.

- Otherwise, $T_{DRX}$ is the DRX cycle of the UE in the serving cell.

- $T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where:

- $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

- $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

## 15.4.7.4 Test description

The test defines three possible test configurations; specified in Table 15.4.7.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.4.7.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

### 15.4.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.7.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.4.7.5-1 and Table 15.4.7.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.7.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.4.7.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

### 15.4.7.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.7.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.4.7.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.4.7.4.3-3 replaces Table 15.4.1.4.3-12 and Table 15.4.7.4.3-4 replaces Table 15.4.1.4.3-13.

**Table 15.4.7.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1, Config 2 and Config 4 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.7.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.7.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.7.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1, Config 2 and Config 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.7.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |

| | | | |
|---|---|---|---|
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.7.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 15.4.7.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.4.7.5 Test requirement

Table 15.4.7.5-1 and Table 15.4.7.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.7.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1,2,3,4 | Cell 1 | Cell 1 is the reference cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| Neighbour cell | | 1,2,3,4 | Cell 2 | Cell 2 is a neighbour cell in *NR-Multi-RTT-ProvideAssistanceData* [49]. |
| RF Channel Number | | 1,2,3,4 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1,4 | 10: $N_{RB,c}$ = 52 | |
| | | 2 | 10: $N_{RB,c}$ = 52 | |
| | | 3 | 20: $N_{RB,c}$ = 52 | |
| SSB configuration | | 1,4 | SSB.4 RedCap FR1 | |
| | | 2 | SSB.4 RedCap FR1 | |
| | | 3 | SSB.5 RedCap FR1 | |
| SMTC configuration | | 1,4 | SMTC.1 RedCap | |
| | | 2 | SMTC.1 RedCap | |
| | | 3 | SMTC.1 RedCap | |
| CP length | | 1,2,3,4 | Normal | |
| DRX cycle | | 1,2,3,4 | 1.28s | |
| Time offset between serving and neighbour cells | µs | 1,2,3,4 | 3 | Synchronous cells |
| T1 | s | 1,2,3,4 | 5 | |
| T2 | s | 1,2,3,4 | 10 | |

**Table 15.4.7.5-2: Cell specific test parameters.**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1,4 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1,4 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| RMSI CORESET RMC configuration | | 1,4 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1,4 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1,2,3,4 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1,2,3,4 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1,4 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1,2,3,4 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| PRS configuration | | 1,4 | PRS.1.1 FR1 | | PRS.1.1 FR1 | |
| | | 2 | PRS.1.1 FR1 | | PRS.1.1 FR1 | |
| | | 3 | PRS.2.1 FR1 | | PRS.2.1 FR1 | |
| PRS muting info | | 1,2,3,4 | '10' | | '01' | |
| SRS configuration | | 1,4 | POS-SRS.1 | | N/A | |
| | | 2 | POS-SRS.1 | | N/A | |
| | | 3 | POS-SRS.2 | | N/A | |
| $N_{oc}$ Note 2 | dBm/SCS | 1,4 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1,4 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1,4 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1,4 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| Io | dBm/9.36 MHz | 1,4 | N/A | -67.67 | N/A | -67.67 |
| | dBm/9.36 MHz | 2 | | -67.67 | | -67.67 |
| | dBm/18.72 MHz | 3 | | -64.67 | | -64.67 |
| Propagation Condition | | 1,2,3,4 | AWGN | | | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

| | |
|---|---|
| NOTE 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| NOTE 3: | PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| NOTE 4: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.4.7.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{RxTx,TEG,i} = 1$, $N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} = 1$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$ ; where $T_{PRS,i}$ = 160 ms, and and $T_{DRX}$ = 1280ms. Therefore, $T_{available_PRS,i}$ = 1280 ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 1 * \left\lceil \frac{1}{N'} \right\rceil * \left\lceil \frac{L_{available_{PRS,i}}}{N} \right\rceil * 4 - 1\right)$$

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 5.128s and 26.88s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 27s.

Similarly for Type 2 Buffering, Best Case result of 5.128s is 6s and for the worst-case result of 83200ms, rounded to the next second, the final value is 84s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 84.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 5.128s and 16.64s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 17s.

Similarly for Type 2 Buffering, Best Case result of 5.128s is 6s and for the worst-case result of 42240ms, rounded to the next second, the final value is 43s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 43.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.13.5-3, Table 15.2.13.5-4 and Table 15.2.13.5-5 for reference.

**Table 15.4.7.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{1}{N'} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 15.4.7.5-4: value of N based on test configuration and UE buffer type**

| **N for SCS 15kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 3 |
| nDot5 | 2 |
| >=n1 | 1 |
| **N for SCS 30kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.285}{N} \right\rceil$ |
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot1 | 1 |
| N **for SCS 15kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{2}{N} \right\rceil$ |
| nDot125 | 16 |
| nDot25 | 8 |
| nDot5 | 4 |
| n1 | 2 |
| >=n2 | 1 |
| N **for SCS 30kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{1}{N} \right\rceil$ |
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26] | |

**Table 15.4.7.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.8 UE Rx-Tx time difference measurement reporting delay in FR2 for Redcap UE without Rx FH

Editor's note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 15.4.8.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement without RX FH in RRC_INACTIVE state meets the requirements specified in clause TS 38.133 [50] 5.6A.4.5 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.4.8.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE.

### 15.4.8.3 Minimum conformance requirements

Same as clause 15.4.7.3.

### 15.4.8.4 Test description

The supported test configuration is listed in Table 15.4.8.4-1.

**Table 15.4.8.4-1: Test Configuration**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

#### 15.4.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.8.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.4.8.5-1 and Table 15.4.8.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.8.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.4.8.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

#### 15.4.8.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.8.4.3-1 replaces Table 15.4.1.4.3-4, Table 15.4.8.4.3-2 replaces Table 15.4.1.4.3-7, Table 15.4.8.4.3-3 replaces Table 15.4.1.4.3-8, Table 15.4.8.4.3-4 replaces Table 15.4.1.4.3-11, Table 15.4.8.4.3-5 replaces Table 15.4.1.4.3-12 and Table 15.4.8.4.3-5 replaces Table 15.4.1.4.3-13.

**Table 15.4.8.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |

| | | | |
|---|---|---|---|
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.8.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.8.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.8.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.8.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.8.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.3.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.4.8.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.8.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.8.4.3-6: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.4.8.5 Test requirement

Table 15.4.8.5-1 and Table 15.4.8.5-2 define the primary level settings including test tolerances for the test.

**Table 15.4.8.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | Cell 1 | Cell 1 is the PCell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| Neighbour cell | | 1 | Cell 2 | Cell 2 is a neighbour cell in NR-Multi-RTT-ProvideAssistanceData [49]. |
| RF Channel Number | | 1 | 1 | For both Cell 1 and Cell 2 |
| $BW_{channel}$ | MHz | 1 | 100: $N_{RB,c} = 66$ | |

| | | | | |
|---|---|---|---|---|
| SSB configuration | | 1 | SSB.2 RedCap FR2 | |
| SMTC configuration | | 1 | SMTC.1 RedCap | |
| CP length | | 1 | Normal | |
| DRX cycle | | 1 | 0.64 s | |
| Time offset between serving and neighbour cells | μs | 1 | 3 | Synchronous cells |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 20 | |

**Table 15.4.8.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 clause A.3.15 | | | |
| Beam Assumption[Note 7] | | 1 | Rough | | Rough | |
| TDD configuration | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | | N/A | |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | | N/A | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | N/A | |
| OCNG Patterns | | 1 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS[Note 1] | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS[Note 1] | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | | N/A | |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.5 FR2 | | PRS.1.5 FR2 | |
| PRS muting info | | 1 | ‘10’ | | ‘01’ | |
| SRS configuration | | 1 | POS-SRS.3 | | N/A | |
| $N_{oc}$ [Note 2] | dBm/SCS | 1 | -89 | | | |
| $N_{oc}$ [Note 2] | dBm/15 kHz | 1 | -98 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| PRP [Note 3] | dBm/SCS kHz | 1 | -Infinity | -91 | -Infinity | -99 |
| Io | dBm/92.16 MHz | 1 | N/A | -57.6 | N/A | -57.6 |
| Propagation Condition | | 1 | AWGN | | | |
| NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | |

NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
NOTE 6: As observed with 0 dBi gain antenna at the centre of the quiet zone.
NOTE 7: Information about types of UE beam is given in TS 38.133 annex B.2.1.3, and does not limit UE implementation or test system implementation.
NOTE 8: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.4.8.3.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period plus ΔT, where ΔT = 50ms. The UE Rx-Tx time difference measurement period follows the equation:

$$T_{UERxTx,i} = \left( K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{RxTx,TEG,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.071$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$ ; where $T_{PRS,i}$ = 320 ms, and $T_{DRX}$ = 640 ms. Therefore, $T_{available_PRS,i} = 640$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

For Type 1 duration calculation with UE symbol level buffering capability, the LPP time IE ranges between 20.488s and 80.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 21s and 83s.

Similarly for Type 2 Buffering, Best Case result of 20.488s is 11s and for the worst-case result of 80.56ms.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 21.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.4.8.5-3, Table 15.4.8.5-4 and Table 15.4.8.5-5 for reference.

**Table 15.4.8.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 15.4.8.5-4: value of N based on test configuration and UE buffer type**

| **N for SCS 120kHz for Type 1 Buffering** Note 1 | $\left\lceil \frac{0.071}{N} \right\rceil$ |
|---|---|
| >=nDot125 | 1 |
| N **for SCS 120kHz for Type 2 Buffering** Note 1 | $\left\lceil \frac{0.125}{N} \right\rceil$ |
| >=nDot125 | 1 |

NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]

**Table 15.4.8.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.4.9 UE Rx-Tx time difference measurement reporting delay in FR1 for Redcap UE with Rx FH

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.4.9.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement with Rx FH in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6A.6.6 in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.4.9.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and DL-PRS measurement in RRC_INACTIVE and maximumPRS-BandwidthAcrossAllHopsFR1 for at least one band and PRS frequency hopping measurement and report.

### 15.4.9.3 Minimum conformance requirements

When physical layer receives *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [49], and UE is configured to perform PRS-RSRP measurement with RX FH via *nr-DL-PRS-RxHoppingRequest-r18* as defined in TS 37.355 [49], the RedCap UE shall be able to measure multiple (up to the RedCap UE capability specified in clause 5.6A.6.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [57], during the measurement period $T_{UE\ RxTx,Total}$ defined in clause 5.6.4.5 with using the following definition for $N_{sample}$ and $L_{available_PRS,i}$:

$N_{sample}$ is the number of PRS RSTD measurement samples, where

- $N_{sample}$ = 2 if the RedCap UE supports *reducedNumOfSampleForMeasurementWithFH-RRC-IdleAndInactive* [49], and the LMF requests the UE to perform positioning measurements with reduced number of samples.
- $N_{sample}$ = 4 otherwise.

A measurement sample under RX FH is defined as a PRS measurement over multiple hops.

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured, and is calculated by:

$L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured, and is calculated by:

$$L_{available_PRS,i} = \sum_{j=1}^{J} N_{hop,i,j} * L_{per-hop,i,j} \quad ,$$

where

- $N_{hop,i,j}$ is the number of hops that UE can perform within a time window occasion *j* as defined in the following, and
- $L_{per-hop,i,j}$ is the time duration of available PRS resources in the positioning frequency layer *i* in each hop to be measured in time window occasion *j* during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{per-hop,i}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered;
- *J* is the number of time window occasions during $T_{available_PRS,i}$.

The sampling duration per hop is the first $T_{sample,hop}$ symbols in each hop, where $T_{sample,hop} = T_{hop} - RRT_{FH}$, $T_{hop}$ is the applicable length per hop as defined in table 5.6A.6.6-1, and $RRT_{FH}$ is the retuning time between Rx hops indicated by UE in *rf-RxRetuneTimeFR1* or *rf-RxRetuneTimeFR2* via *NR-DL-PRS-ProcessingCapability* [49]. The first hop starts at no earlier than the earliest arrival time of the first unmuted PRS resource fully or partially overlapped with the measurement time window, defined in clause 5.6A.1, taking into account the *expectedRSTD* and *expectedRSTD-uncertainty* in the PRS assistance data.

**Table 15.4.9.3-1: Applicable number of hops per slot and applicable length of each hop**

| $RRT_{FH}$ | (comb size, Number of PRS symbols) | Applicable number of hops per slot ($N_{hops}^{slot}$) | Applicable length per hop ($T_{per-hop}$) in number of symbols |
|---|---|---|---|
| $RRT_{FH} \leq 2$ symbols | (2, 12) with SCS 15kHz, 30kHz in FR1, and 60kHz, 120kHz in FR2 | 2 | 7 |
| | All others | 1 | 14 |
| 2 symbols $< RRT_{FH} \leq 6$ symbols | (≤ 6, any) | 1 | 14 |
| | (12, 12) | ½ | 28 |
| $RRT_{FH} > 6$ symbols | Any combination | ½ | 28 |

The number of hops within a single time window $N_{hop,i,j}$ is defined as

$$N_{hop,i,j} = min\left(N_{hops,effect,i,j}, \left\lfloor \frac{N3_i}{L_{per-hop,i,j}} \right\rfloor, N_{hop,max,i}\right),$$

where

- $N_{hop,max,i}$ is the maximum number of Rx hops as reported by UE in *maximumFH-Hops* via *NR-DL-PRS-ProcessingCapability* as specified in [49] for positioning frequency layer *i* ,
- $N3_i$ is indicated via UE capability *processingPRS-SymbolsDurationN3-r18* for positioning frequency layer *i*,
- $N_{hops,effect,i,j}$ is the effective number of Rx hops within a single time window occasion *j*,
  - $N_{hops,effect,i,j} = 2 * N_{rep}^{PRS}$, if $N_{hops}^{slot} = 2$,
  - $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot} = 1$,
  - $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot} = 1/2$ and $M_{rep}^{PRS} > 1$,
  - $N_{hops,effect,i,j} = \left\lfloor \frac{(N_{rep}^{PRS}-1)}{2} \right\rfloor + 1$, if $N_{hops}^{slot} = 1/2$ and $M_{rep}^{PRS} = 1$,
  - where $N_{rep}^{PRS}$ is the number of inter-slot PRS repetitions within a single time window, which are unmuted and fully or partially overlapped with the sampling duration within each hop. $M_{rep}^{PRS}$ is the PRS repetition interval configured by *dl-PRS-ResourceTimeGap*, $N_{hops}^{slot}$ is the applicable number of hops per slot as defined in table 5.6A.6.6-1.

The measurement period requirement with the above modifications applies provided all PRS resources in each PFL have the same number of PRS repetitions $\left(N_{rep}^{PRS}\right)$ within a time window instance and the same PRS repetition interval $\left(M_{rep}^{PRS}\right)$, otherwise the measurement period can be longer.

UE shall be able to measure PRS resources with multiple hops at least over the BW of $BW_{total}$ defined as

$$BW_{total} = min\left(BW_{PRS}, N_{hop,i,j} * BW_{per-hop} - \left(N_{hop,i,j} - 1\right) * BW_{overlap}\right),$$

where

$BW_{PRS}$ is the minimum among:

- configured PRS BW, and
- UE capability of maximum PRS BW across all hops indicated via *maximumPRS-BandwidthAcrossAllHopsFR1-r18* or *maximumPRS-BandwidthAcrossAllHopsFR2-r18* as specified in [49], and
- total BW of all hops requested by LMF via *nr-DL-PRS-RxHoppingTotalBandwidth-r18* as specified in [49],

$N_{hop,i,j}$ is number of hops within a single time window as define above,

$BW_{per-hop}$ is the UE capability on PRS BW per hop indicated via *supportedBandwidthPRS-r16* as specified in [49],

$BW_{overlap}$ is the UE capability on BW of the overlapping RB indicated via *numOfOverlappingPRB-r18* as specified in [34].

## 15.4.9.4 Test description

The test defines three possible test configurations; specified in Table 15.4.9.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.4.9.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, cell bandwidth 50 MHz, FDD duplex mode |
| 2 | 15 kHz SSB SCS, cell bandwidth 50 MHz, TDD duplex mode |
| 3 | 30 kHz SSB SCS, cell bandwidth 100 MHz, TDD duplex mode |
| 4 | 15 kHz SSB SCS, cell bandwidth 50 MHz, HD-FDD duplex mode |
| NOTE 1: The RedCap UE is only required to be tested in one of the supported test configurations.<br>NOTE 2: This test case is applicable to the UE supporting per hop bandwidth ≥ 10 MHz for SCS 15 kHz.<br>NOTE 3: This test case is applicable to the UE supporting per hop bandwidth = 20 MHz for SCS 30 kHz. | |

### 15.4.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.9.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.4.9.5-1 and Table 15.4.9.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.9.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.4.9.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

### 15.4.9.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.9.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.4.9.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.4.9.4.3-3 replaces Table 15.4.1.4.3-12 and Table 15.4.9.4.3-4 replaces Table 15.4.1.4.3-13.

**Table 15.4.9.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1, Config 2 and Config 4 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 62 | 268 PRBs | Config 1, Config 2 and Config 4 |
| | 63 | 272 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.9.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.9.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.9.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1, Config 2 and Config 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | Config 1, Config 2 and Config 4 |
| | 0 | | Configuration 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | Config 1, Config 2 |

| | | | |
|---|---|---|---|
| | | | and Config 4 |
| | 0 | | Configuration 3 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.9.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.9.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |

| | | | |
|---|---|---|---|
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | mhz50 | | Configuration 1, Configuration 2 and Configuration 4 |
| | mhz100 | | Configuration 3 |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.9.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |

| | | | |
|---|---|---|---|
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.4.9.5 Test requirement

FFS

## 15.4.10 UE Rx-Tx time difference measurement reporting delay in FR2 for Redcap UE with Rx FH

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.4.10.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx measurement with Rx FH meets the requirements specified in TS 38.133 [50] clause 5.6A.6.6 in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.4.10.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and DL-PRS measurement in RRC_INACTIVE and maximumPRS-BandwidthAcrossAllHopsFR2 for at least one band and PRS frequency hopping measurement and report.

### 15.4.10.3 Minimum conformance requirements

Same as clause 15.4.9.3.

### 15.4.10.4 Test description

The supported test configuration is listed in Table 15.4.10.4-1.

**Table 15.4.10.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, cell bandwidth 400 MHz, TDD duplex mode |
| NOTE 1: This test case is applicable to the UE supporting per hop bandwidth = 100 MHz for SCS 120 kHz. | |

#### 15.4.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.10.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.4.10.5-1 and Table 15.4.10.5-2.
3. Propagation conditions are set according to clause 4.14.2.

4. Message contents are defined in clause 15.4.10.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.4.10.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

### 15.4.10.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.10.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.4.10.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.4.10.4.3-3 replaces Table 15.4.1.4.3-11, Table 15.4.10.4.3-4 replaces Table 15.4.1.4.3-12, and Table 15.4.10.4.3-5 replaces Table 15.4.1.4.3-13.

**Table 15.4.10.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 61 | 264 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.10.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.10.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.10.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.10.4.3-3: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.10.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.4.10.4.3-4: LPP RequestLocationInformation**

| |
|---|
| Derivation Path: 37.355 clause 6.2 |

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.10.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |

| | | | |
|---|---|---|---|
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr2 | mhz400 | | |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.10.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.4.10.5 Test requirement

FFS

## 15.4.11 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for normal UE in FR1 SA

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.4.11.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 5.6.4.5 for UE Rx-Tx measurements in RRC_INACTIVE with eDRX. The tests are conducted under AWGN propagation condition with the UE operating in FR1 stand-alone mode and configured to perform UE Rx-Tx measurements on a single positioning frequency layer (PFL) in FR1.

### 15.4.11.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE and extendedDRX-CycleInactive-r18.

### 15.4.11.3 Minimum conformance requirements

When physical layer receives last of *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message from LMF via LPP [49], UE shall be able to measure multiple (up to the UE capability specified in clause 5.6.4.3) UE Rx-Tx time difference measurements as defined in TS 38.215 [57] in configured positioning frequency layers within the measurement period $T_{UERxTx,Total}$ ms.

$$T_{UERxTx,\ Total} = \sum_{i=1}^{L} T_{UERxTx,i} + (L-1) * \max\left(T_{effect,i}\right)$$

Where:

- $i$ is the index of positioning frequency layer,
- $T_{UERxTx,i}$ is the measurement period for UE Rx-Tx time difference measurements in positioning frequency layer $i$ as further defined in this clause,
- L is total number of positioning frequency layers,
- $T_{effect,i}$ is the periodicity of the UE Rx-Tx time difference measurement in positioning frequency layer $i$ as defined further in this clause.

$$T_{UERxTx,i} = \left(K_{carrier_PRS} * N_{RxTx,TEG,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where:

- $K_{carrier_PRS}$=1 if the UE is capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [49].
- $K_{carrier_PRS} = N_{layer} + 1$ if the UE is not capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [49] and if Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$; where $N_{layer}$is defined in clause 4.2.2.7.
- $K_{carrier_PRS} = K_{carrier} + 1$ if the UE is not capable of *parallelPRS-MeasRRC-Inactive-r17* defined in [49] and if Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$; where $K_{carrier}$is defined in clause 4.2.2.4.
- $N_{RxBeam,i}$ is the scaling factor for UE Rx beam sweeping:
  - $N_{RxBeam,i}$=1 if positioning frequency layer $i$ is in FR1, and if positioning frequency layer $i$ is in FR2.
    - $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-Multi-RTT -RequestLocationInformation*.
    - $N_{RxBeam,i}$ equals to 8, otherwise.

  $N_{RxTx,TEG,i}$ is the Rx TEG specific scaling factor:

  - $N_{RxTx,TEG,i}$ = 1 if UE is not configured by LMF with *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49].

  Otherwise,

- $N_{RxTx,TEG,i} = N_{TEG,i}$, if UE is not capable of measuring same DL PRS resource simultaneously from multiple Rx TEGs, where $N_{TEG,i}$ is the number of Rx TEGs or RxTx TEGs with which UE is requested to measure a PRS resource indicated via *measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17* or *measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17* [49], and in case 'n0' is indicated, $N_{TEG,i}$ is the maximum number of Rx TEGs with which UE can support to measure the same PRS resource as reported in *NR-UE-TEG-Capability*.

- $N_{RxTx,TEG,i} = \left\lceil \frac{N_{TEG,i}}{k_{TEG,simul,i}} \right\rceil$, if UE is capable of measuring same DL PRS resource simultaneously from multiple Rx TEGs, where $k_{TEG,simul,i}$ is the number of Rx TEGs UE can measure simultaneously which is reported via *measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul*.

- $L_{available_PRS,i}$ is the time duration of available PRS resources in the positioning frequency layer *i*, to be measured during $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.

- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

- {N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in clause 4.2.7.2 of TS 37.355 [49],

- N' is UE capability for number of DL PRS resources that it can process in a slot corresponding to *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* as specified in clause 6.4.3 of TS 37.355 [49],

- $N_{sample}$ is the number of UE Rx-Tx time difference measurement samples:

  - $N_{sample}$ = 4 if the UE is not capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [49] or not configured to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [49].

  - $N_{sample}$ = 1 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [49] and LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [49] and the following conditions are met:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 2 if the UE is capable of *supportedDL-PRS-ProcessingSamples-RRC-Inactive* defined in [49] and the LMF requests the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples-r17* [49] but the following conditions are not met:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $T_{last,i}$ is the measurement duration for the last UE Rx-Tx time difference measurement sample in the positioning layer i, including the sampling time and processing time, $T_{last,i} = T_i + T_{available_PRS,i}$ ,

- $T_{effect,i}$ is periodicity of UE Rx-Tx time difference measurement in positioning frequency layer *i*:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$$

Where:

- $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [49],

- $T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$, the least common multiple between $T_{PRS,i}$ and $T_{DRX}$.
  - When UE is configured with RAN eDRX ≤ 10.24 s:
    - $T_{DRX}$ is defined as T in TS 38.304 [1] when CN eDRX ≤ 10.24 s.
    - $T_{DRX}$ is the maximum of the T inside and outside of the CN PTW when CN eDRX > 10.24 s, where T inside and outside of the CN PTW are defined in TS 38.304 [1].
  - When UE is configured with RAN eDRX > 10.24 s:
    - $T_{DRX}$ is the maximum of the DRX cycles within the CN PTW and the RAN PTW.
- Otherwise, $T_{DRX}$ is the DRX cycle of the UE in the serving cell.
- $T_{PRS,i}$ is the PRS resource periodicity in positioning frequency layer *i*. If the positioning frequency layer *i* has more than one DL PRS resource sets with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among DL PRS resource sets is used to derive $T_{PRS,i}$, where:
  - $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.
  - $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$

If the following conditions are met, the time $T_{UE\,Rx\,Tx,\,Total}$ starts from the first DL PRS resource(s) instances inside a PTW after both the *NR-Multi-RTT-ProvideAssistanceData* message and *NR-Multi-RTT-RequestLocationInformation* message are delivered from LMF to the UE via LPP [49].

- UE is configured with RAN eDRX > 10.24 s, and
- periodic PRS measurement reporting is configured, and
- RAN eDRX cycle is smaller or equal to the PRS measurement reporting periodicity configured via higher layer parameter *reportingInterval* in TS 37.355 [49], and
- there is one or more PRS resources occurring in PTW.

## 15.4.11.4 Test description

The test defines three possible test configurations; specified in Table 15.4.11.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.4.11.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB and PRS SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB and PRS SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB and PRS SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

### 15.4.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.11.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.4.11.5-1 and Table 15.4.11.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.11.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.4.11.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

### 15.4.11.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.11.4.3-1 replaces Table 15.4.1.4.3-10, Table 15.4.11.4.3-2 replaces Table 15.4.1.4.3-12 and Table 15.4.11.4.3-3 replaces Table 15.4.1.4.3-13.

**Table 15.4.11.4.3-1: *RRCRelease***

| Derivation Path: TS 38.508-1 [45], table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCRelease ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcRelease SEQUENCE { | | | |
| suspendConfig SEQUENCE { | | | |
| srs-PosRRC-Inactive-r17 CHOICE { | | | |
| setup SEQUENCE { | | | |
| SRS-PosRRC-Inactive-r17 | OCTET STRING containing SRS-PosRRC-InactiveConfig-r17 as defined in Table 15.4.1.4.3-11 | | |
| } | | | |
| } | | | |
| resumeIndication-r18 | Not present | | |
| srs-PosRRC-InactiveEnhanced-r18 | Not present | | |
| ran-ExtendedPagingCycleConfig SEQUENCE { | | | |
| extendedPagingCycle-r18 | hf4 | | |
| pagingPTWLength-r18 | ms1280 | | |
| } | | | |
| multicastConfigInactive-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.11.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |

| | | | |
|---|---|---|---|
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting SEQUENCE { | | | |
| reportingAmount | ra-Infinity | | |
| reportingInterval | ri16 | | |
| } | | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.11.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |

| | | | |
|---|---|---|---|
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.11.4.3-3: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.4.11.5 Test requirement

FFS

## 15.4.12 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for RedCap UE in FR1 SA

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.4.12.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 5.6A.6.5 for UE Rx-Tx measurements in RRC_INACTIVE with eDRX. The tests are conducted under AWGN propagation condition with the UE operating in FR1 stand-alone mode and configured to perform UE Rx-Tx measurements on a single positioning frequency layer (PFL) in FR1.

### 15.4.12.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE and extendedDRX-CycleInactive-r18.

### 15.4.12.3 Minimum conformance requirements

Same as clause 15.4.7.3.

### 15.4.12.4 Test description

The test defines three possible test configurations; specified in Table 15.4.12.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.4.11.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | NR 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | NR 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | NR 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | NR 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations in each supported band | |

#### 15.4.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.12.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.4.121.5-1 and Table 15.4.12.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.12.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.4.12.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

#### 15.4.12.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.12.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.4.12.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.4.12.4.3-3 replaces Table 15.4.1.4.3-10, Table 15.4.12.4.3-4 replaces Table 15.4.1.4.3-12 and Table 15.4.12.4.3-5 replaces Table 15.4.1.4.3-13.

**Table 15.4.12.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1, Config 2 and Config 4 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1, Config 2 and Config 4 |
| | 1 | 24 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | Config 1, Config 2 and Config 4 |
| | n2 | | Config 3 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.12.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.12.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.12.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1, Config 2 and Config 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Config 1, Config 2 and Config 4 |
| | n2 | | Configuration 3 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 4 | | Config 1, Config 2 and Config 4 |
| n2-r16 | 0 | | Configuration 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.12.4.3-3: *RRCRelease***

| Derivation Path: TS 38.508-1 [45], table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCRelease ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcRelease SEQUENCE { | | | |
| suspendConfig SEQUENCE { | | | |
| srs-PosRRC-Inactive-r17 CHOICE { | | | |
| setup SEQUENCE { | | | |
| SRS-PosRRC-Inactive-r17 | OCTET STRING containing SRS-PosRRC-InactiveConfig-r17 as defined in Table 15.4.1.4.3-11 | | |
| } | | | |
| } | | | |
| resumeIndication-r18 | Not present | | |
| srs-PosRRC-InactiveEnhanced-r18 | Not present | | |
| ran-ExtendedPagingCycleConfig SEQUENCE { | | | |
| extendedPagingCycle-r18 | hf4 | | |
| pagingPTWLength-r18 | ms1280 | | |
| } | | | |
| multicastConfigInactive-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.12.4.3-4: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting SEQUENCE { | | | |
| reportingAmount | ra-Infinity | | |
| reportingInterval | ri16 | | |
| } | | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.12.5 | Result of the response time calculation rounded up to the next | |

| | | | |
|---|---|---|---|
| | | second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.12.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |

| | | | |
|---|---|---|---|
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.4.12.5 Test requirement

FFS

## 15.4.13 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for normal UE in FR2 SA

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.4.13.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 5.6.4.5 for UE Rx-Tx measurements in RRC_INACTIVE with eDRX. The tests are conducted under AWGN propagation condition with the UE operating in FR2 stand-alone mode and configured to perform UE Rx-Tx measurements on a single positioning frequency layer (PFL) in FR2.

### 15.4.13.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE and extendedDRX-CycleInactive-r18.

### 15.4.13.3 Minimum conformance requirements

Same as clause 15.4.11.3.

### 15.4.13.4 Test description

The supported test configuration is listed in Table 15.4.13.4-1.

**Table 15.4.13.4-1: Test Configuration**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

#### 15.4.13.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.13.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.

2. The general test parameter settings are set up according to Table 15.4.13.5-1 and Table 15.4.13.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.13.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.4.13.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

### 15.4.13.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.13.4.3-1 replaces Table 15.4.1.4.3-4, Table 15.4.13.4.3-2 replaces Table 15.4.1.4.3-7, Table 15.4.13.4.3-3 replaces Table 15.4.1.4.3-8, Table 15.4.13.4.3-4 replaces Table 15.4.1.4.3-11, Table 15.4.13.4.3-5 replaces Table 15.4.1.4.3-10, Table 15.4.13.4.3-6 replaces Table 15.4.1.4.3-12 and Table 15.4.13.4.3-7 replaces Table 15.4.1.4.3-13.

**Table 15.4.13.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.13.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.13.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.13.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.13.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.13.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |

| | | | |
|---|---|---|---|
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.13.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.4.13.4.3-5: *RRCRelease***

| Derivation Path: TS 38.508-1 [45], table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCRelease ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcRelease SEQUENCE { | | | |
| suspendConfig SEQUENCE { | | | |
| srs-PosRRC-Inactive-r17 CHOICE { | | | |
| setup SEQUENCE { | | | |
| SRS-PosRRC-Inactive-r17 | OCTET STRING containing SRS-PosRRC-InactiveConfig-r17 as defined in Table 15.4.13.4.3-4 | | |
| } | | | |
| } | | | |
| resumeIndication-r18 | Not present | | |
| srs-PosRRC-InactiveEnhanced-r18 | Not present | | |
| ran-ExtendedPagingCycleConfig SEQUENCE { | | | |
| extendedPagingCycle-r18 | hf4 | | |
| pagingPTWLength-r18 | ms1280 | | |
| } | | | |
| multicastConfigInactive-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.13.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |

| | | | |
|---|---|---|---|
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting SEQUENCE { | | | |
| reportingAmount | ra-Infinity | | |
| reportingInterval | ri16 | | |
| } | | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.13.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |

| | | | |
|---|---|---|---|
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.13.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.4.13.5 Test requirement

FFS

## 15.4.14 UE Rx-Tx time difference measurement reporting delay test case for case 2 when eDRX > 10.24s for RedCap UE in FR2 SA

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.4.14.1 Test purpose

The purpose of the test is to verify the measurement requirements specified in TS 38.133 [50] clause 5.6A.6.5 for UE Rx-Tx measurements in RRC_INACTIVE with eDRX. Refer to clause A.7.8.3.3.1 for test configuration and procedure.

### 15.4.14.2 Test applicability

This test applies to all types of NR Redcap UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE and extendedDRX-CycleInactive-r18.

### 15.4.14.3 Minimum conformance requirements

Same as clause 15.4.7.3.

### 15.4.14.4 Test description

The supported test configuration is listed in Table 15.4.14.4-1.

**Table 15.4.14.4-1: Test Configuration**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

#### 15.4.14.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.4.14.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.4.14.5-1 and Table 15.4.14.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.4.14.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.4.14.4.2 Test procedure

Same as defined in clause 15.4.1.4.2.

#### 15.4.14.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.4.14.4.3-1 replaces Table 15.4.1.4.3-4, Table 15.4.14.4.3-2 replaces Table 15.4.1.4.3-7, Table 15.4.14.4.3-3 replaces Table 15.4.1.4.3-8, Table 15.4.14.4.3-4 replaces Table 15.4.1.4.3-11, Table 15.4.14.4.3-5 replaces Table 15.4.1.4.3-10, Table 15.4.14.4.3-6 replaces Table 15.4.1.4.3-12 and Table 15.4.14.4.3-7 replaces Table 15.4.1.4.3-13.

**Table 15.4.14.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.14.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.14.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.4.14.4.3-3 | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.14.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.14.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |

| | | | |
|---|---|---|---|
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.4.14.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.4.14.4.3-5: *RRCRelease***

| Derivation Path: TS 38.508-1 [45], table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCRelease ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcRelease SEQUENCE { | | | |
| suspendConfig SEQUENCE { | | | |
| srs-PosRRC-Inactive-r17 CHOICE { | | | |
| setup SEQUENCE { | | | |
| SRS-PosRRC-Inactive-r17 | OCTET STRING containing SRS- | | |

| | | | |
|---|---|---|---|
| | PosRRC-InactiveConfig-r17 as defined in Table 15.4.14.4.3-5 | | |
| } | | | |
| } | | | |
| resumeIndication-r18 | Not present | | |
| srs-PosRRC-InactiveEnhanced-r18 | Not present | | |
| ran-ExtendedPagingCycleConfig SEQUENCE { | | | |
| extendedPagingCycle-r18 | hf4 | | |
| pagingPTWLength-r18 | ms1280 | | |
| } | | | |
| multicastConfigInactive-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.14.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting SEQUENCE { | | | |
| reportingAmount | ra-Infinity | | |
| reportingInterval | ri16 | | |
| } | | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.14.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.4.14.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |

| | | | |
|---|---|---|---|
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

### 15.4.14.5 Test requirement

FFS

# 15.5 UE Rx-Tx time difference measurement accuracy test cases in RRC_INACTIVE

## 15.5.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA

### 15.5.1.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.5.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and *supportedDL-PRS-ProcessingSamples-RRC-Inactive*. In addition, this test shall be verified with either SDT or non-SDT supporting UEs.

### 15.5.1.3 Minimum conformance requirements

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall not apply, if:

$N_{TA_offset}$ defined in Table 7.1.2-2 changes during the UE Rx-Tx measurement period or

if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the network-configured Timing Advance.

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that:

- The UE transmits SRS within [-160, 160] msec of at least one DL PRS resource of each of the TRPs in the assistance data.

If the uplink transmission timing changes during the UE Rx-Tx measurement period due to the autonomous timing adjustment defined in clause 7.1.2 of TS38.133 [50] then:

- UE Rx-Tx measurement accuracy requirements shall apply for a cell, which is also the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission even if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

- UE Rx-Tx measurement accuracy requirements shall not apply for a cell, which is not the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission, if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

When a serving cell change occurs during the UE Rx-Tx measurement period, the UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that the serving cell change does not impact SRS configuration for the UE Rx-Tx measurement.

The relative UE Rx-Tx measurement accuracy in this clause is defined as accuracy of the difference between two UE Rx-Tx measurements.

The accuracy requirements in Table 15.5.1.3-2 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 of TS38.133 [50] for a corresponding Band.

AWGN propagation condition.

**Table 15.5.1.3-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition $(T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ± 78+δ | -3 | ≥24 | 15 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | |
| | | | | | NR_TDD_FR1_C | -126 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | |
| | | | | | NR_FDD_FR1_F | -124.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | |
| | | | | | NR_FDD_FR1_H | -123.5 | |
| ± 59+δ | | ≥52 | | ≥1 | Note 6 | NOTE 6 | NOTE 6 |
| ± 30+δ | | >104 | | ≥1 | Note 6 | NOTE 6 | NOTE 6 |
| ± 57+δ | | ≥24 | 30 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | |
| | | | | | NR_TDD_FR1_C | -123 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | |
| | | | | | NR_FDD_FR1_F | -121.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | |
| | | | | | NR_FDD_FR1_H | -120.5 | |
| ± 30+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 15+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 29+δ | | ≥24 | 60 | ≥4 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | |
| | | | | | NR_TDD_FR1_C | -120 | |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | |
| | | | | | NR_FDD_FR1_F | -118.5 | |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | |
| | | | | | NR_FDD_FR1_H | -117.5 | |
| ± 15+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 101+δ | -13 | ≥24 | 15 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 75+δ | | ≥52 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 37+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 58+δ | | ≥24 | 30 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ± 39+δ | | ≥48 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 36+δ | | ≥24 | 60 | ≥4 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥ 64 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 8+δ | | ≥ 132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.<br>NOTE 2: NR operating band groups are as defined in Section 3.5.<br>NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$ , $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [34].<br>NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.<br>NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].<br>NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.<br>NOTE 7: δ is the margin determined from Table 10.1.25.2-5. | | | | | | | |

**Table 15.5.1.3-2: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (RB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 24 | N/A | N/A | 160 |
| ≥ 52 | ≥ 24 | N/A | 80 |
| ≥ 104 | ≥ 48 | ≥ 24 | 56 |
| N/A | ≥ 132 | ≥ 64 | 24 |
| N/A | N/A | ≥ 132 | 24 |
| NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].<br>NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies. | | | |

## 15.5.1.4 Test description

### 15.5.1.4.1 Initial conditions

The supported test configuration is listed in Table 15.5.1.4.1.-1.

**Table 15.5.1.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.1.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.5.1.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.1.4.3.

5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). Both cells are on the same RF channel in FR1.

#### 15.5.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period. During the measurement period, UE shall be in RRC_INACTIVE state.

1. Configure the Cell for RA-SDT with SIB1 configured according to the message contents 15.5.1.4.3.
2. State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A$ = 31 which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
3. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
4. The SS adjusts the downlink timing for Cell 1 to a delay of +512 $T_c$, compared to the current value.
5. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
6. The SS shall transmit an *RRCReconfiguration* message with DRX configuration.
7. The UE shall transmit *RRCReconfigurationComplete* message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.
9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall send a LCS Periodic-Triggered Invoke Request message with LPP LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
12. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
13. The SS transmits an *RRCRelease* message including UL configuration and SDT configuration to transit the UE to RRC_INACTIVE state.
14. Wait for the UE as it monitors for occurrence of the periodic event requested.
15. The UE shall transmit an *RRCResumeRequest* message. If the UE supports SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 19.
16. The SS transmits an *RRCResume* message
17. The UE transmits an *RRCResumeComplete* message.
18. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.5.1.4.3) and *RRCResumeComplete* message.
19. If the UE message at step 15 or 18 includes the *ackRequested* IE set to TRUE, the SS shall send LPP acknowledgment message.
20. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the *NR-Multi-RTT-SignalMeasurementInformation* according to Table 15.5.1.5-2.

21. UE shall be released and moved to RRC_IDLE.

22. Repeat steps 1-21 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.5.1.4.3 Message contents

Message contents are the same as in 15.4.1.5.4.3 with the following exceptions:

**Table 15.5.1.4.3-1: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.5.1.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| Sl160 | 20 | | Config 1 &2 |
| Sl320 | 40 | | Config 3 |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |

**Table 15.5.1.4.3-2: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.1.4.3-3: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Subtest 1-Config 1& 2<br>Subtest 2-Config 1&2 |
| | kHz30 | | Subtest 1-Config 3 |

| | | | |
|---|---|---|---|
| | | | Subtest 2-Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1 - Config 1, 2 &3 |
| | 21 | 104 PRBs | Sub-test 2 - Config 1&2 |
| | 28 | 132 PRBs | Subtest 2-Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.1.4.3-4 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.1.4.3-4 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.1.4.3-4: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | |
| n160-r16 | 10 | | Config 1&2 |
| } | | | |
| Scs30-r16 CHOICE { | | | |
| n160-r16 | 10 | | Config 3 |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-test 2 |
| | n2 | | Sub-test 1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.5.1.5 Test requirement

The UE Rx-Tx time difference accuracy test parameters are given in Table 15.5.1.5-1.

**Table 15.5.1.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| RF Channel Number | | 1,2,3 | 1 | 1 | 1 | 1 |
| DRX | s | 1,2,3 | 1.28 | | 1.28 | |
| Time offset with Cell 1 | μs | 1, 2, 3 | N/A | 3 | N/A | 3 |
| TDD configuration | | 1 | N/A | N/A | N/A | N/A |
| | | 2 | TDDConf.1.1 | TDDConf.1.1 | TDDConf.1.1 | TDDConf.1.1 |
| | | 3 | TDDConf.2.1 | TDDConf.2.1 | TDDConf.2.1 | TDDConf.2.1 |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | N/A | SR.1.1 FDD | N/A |
| | | 2 | SR.1.1 TDD | | SR.1.1 TDD | |
| | | 3 | SR.2.1 TDD | | SR.2.1 TDD | |

| | | | | | | |
|---|---|---|---|---|---|---|
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | N/A | CR.1.1 FDD | N/A |
| | | 2 | CR.1.1 TDD | | CR.1.1 TDD | |
| | | 3 | CR.2.1 TDD | | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | N/A | CCR.1.1 FDD | N/A |
| | | 2 | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | | 3 | CCR.2.1 TDD | | CCR.2.1 TDD | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | OP.1 | OP.1 | OP.1 |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | N/A | DLBWP.0.1 ULBWP.0.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR1 | PRS.1.1 FR1 | PRS.1.2 FR1 | PRS.1.2 FR1 |
| | | 2 | PRS.1.1 FR1 | PRS.1.1 FR1 | PRS.1.2 FR1 | PRS.1.2 FR1 |
| | | 3 | PRS.2.1 FR1 | PRS.2.1 FR1 | PRS.2.2 FR1 | PRS.2.2 FR1 |
| PRS Resource slot offset | slot | 1, 2, 3 | 0 | 4 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.1 | N/A | POS-SRS.1 | N/A |
| | | 2 | POS-SRS.1 | N/A | POS-SRS.1 | N/A |
| | | 3 | POS-SRS.2 | N/A | POS-SRS.2 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | -98 | |
| | | 2 | -98 | | -98 | |
| | | 3 | -95 | | -95 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | -98 | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.41 | -12.12 | -2.41 | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -10 | -2 | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -100 | -108 | -100 | -108 |
| | | 2 | -100 | -108 | -100 | -108 |
| | | 3 | -97 | -105 | -97 | -105 |
| Io | dBm/19.08 MHz | 1 | -64.57 | -64.57 | -64.57 | -64.57 |
| | dBm/19.08 MHz | 2 | -64.57 | -64.57 | -64.57 | -64.57 |
| | dBm/47.88 MHz | 3 | -60.59 | -60.59 | -60.59 | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | AWGN | |

Note 1: Void.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 15.5.1.5-2: UE Rx - Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1&2 | Sub-test 1 cell 1 | (Measured value from step 20 - 270) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 270) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 cell 1 | (Measured value from step 20 - 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 1 cell 2 | (Measured value from step 20 - 293) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 293) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 cell 2 | (Measured value from step 20 - 125) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 125) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 | Sub-test 1 cell 1 | (Measured value from step 20 - 169) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 169) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 cell 1 | (Measured value from step 20 - 71) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 71) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 1 cell 2 | (Measured value from step 20 - 170) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 170) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 cell 2 | (Measured value from step 20 - 72) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 20 + 72) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

The UE Rx-Tx time difference measurement period fulfills the requirements specified in clause 15.5.1.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 15.5.2 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR1 SA

### 15.5.2.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy with reduced number of samples in RRC_INACTIVE state is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.5.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that support Multi-RTT positioning and *supportedDL-PRS-ProcessingSamples-RRC-Inactive*. In addition, this test shall be verified with either SDT or non-SDT supporting UEs.

### 15.5.2.3 Minimum conformance requirements

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall not apply, if:

$N_{TA_offset}$ defined in Table 7.1.2-2 in TS38.133 [50] changes during the UE Rx-Tx measurement period or

if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the network-configured Timing Advance.

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that:

- The UE transmits SRS within [-160, 160] msec of at least one DL PRS resource of each of the TRPs in the assistance data.

If the uplink transmission timing changes during the UE Rx-Tx measurement period due to the autonomous timing adjustment defined in clause 7.1.2 of TS38.133 [50] then:

- UE Rx-Tx measurement accuracy requirements shall apply for a cell, which is also the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission even if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

- UE Rx-Tx measurement accuracy requirements shall not apply for a cell, which is not the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission, if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

When a serving cell change occurs during the UE Rx-Tx measurement period, the UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that the serving cell change does not impact SRS configuration for the UE Rx-Tx measurement.

The relative UE Rx-Tx measurement accuracy in this clause is defined as accuracy of the difference between two UE Rx-Tx measurements.

The accuracy requirements in Table 15.5.2.3-1 for FR1 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 of TS38.133 [50] for a corresponding Band

Number of measurement samples is less than 4.

AWGN propagation condition.

**Table 15.5.2.3-1: UE Rx-Tx time difference measurement accuracy in FR1 in AWGN with reduced measurement samples**

| Accuracy | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$ Note 3 | NR operating band groups Note 2 | Io Note 4 range | |
| | | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | | dBm / $SCS_{PRS}$ | dBm/BW |
| ± 59+δ | 0 | ≥52 | 15 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -50 |
| ± 30+δ6 | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 30+δ | | ≥48 | 30 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -124 | -50 |
| | | | | | NR_FDD_FR1_B | -123.5 | -50 |
| | | | | | NR_TDD_FR1_C | -123 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -122.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -122 | -50 |
| | | | | | NR_FDD_FR1_F | -121.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -121 | -50 |
| | | | | | NR_FDD_FR1_H | -120.5 | -50 |
| ± 15+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ± 15+δ | | ≥64 | 60 | ≥1 | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -117.5 | -50 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 75+δ | -6 | ≥52 | 15 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 37+δ | | >104 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 39+δ | | ≥48 | 30 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 16+δ | | ≥64 | 60 | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |
| ± 8+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5.
NOTE 3: $T_{\text{rep}}^{\text{PRS}}$, $L_{\text{PRS}}$ , $K_{\text{comb}}^{\text{PRS}}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN* defined in TS 37.355 [34].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 10.1.25.2-5.

**Table 15.5.2.3-2: Margin for UE Rx-Tx time difference measurement accuracy in FR1**

| Min(PRS BW, SRS BW) (PRB) | | | Margin (Tc [Note 1]) |
|---|---|---|---|
| SCS = 15 kHz | SCS = 30 kHz | SCS = 60 kHz | |
| ≥ 24 | N/A | N/A | 160 |
| ≥ 52 | ≥ 24 | N/A | 80 |
| ≥ 104 | ≥ 48 | ≥ 24 | 56 |
| N/A | ≥ 132 | ≥ 64 | 24 |
| N/A | N/A | ≥ 132 | 24 |

NOTE 1: Tc is the basic timing unit defined in TS 38.211 [6].
NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies.

## 15.5.2.4 Test description

### 15.5.2.4.1 Initial conditions

The supported test configuration is listed in Table 15.5.2.4.1-1.

**Table 15.5.2.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.2.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.5.2.4-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.2.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 15.5.2.4.2 Test procedure

The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state.

The beginning of the measurement period shall be aligned with the first DRX cycle containing a DL PRS resource(s). The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period.

1. Configure the Cell for RA-SDT with SIB1 configured according to the message contents 15.5.2.4.3.
2. State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A$ = 31 which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
3. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
4. The SS adjusts the downlink timing for Cell 1 to a delay of +512 $T_c$, compared to the current value.
5. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
6. The SS shall transmit an *RRCReconfiguration* message with DRX configuration.
7. The UE shall transmit *RRCReconfigurationComplete* message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.
9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall send a LCS Periodic-Triggered Invoke Request message with LPP LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T$ = 50 ms.
12. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
13. The SS transmits an *RRCRelease* message including the UL configuration and SDT configuration to transit the UE to RRC_INACTIVE state.
14. Wait for the UE as it monitors for occurrence of the periodic event requested.

15. The UE shall transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 19.

16. The SS transmits an *RRCResume* message.

17. The UE transmits an *RRCResumeComplete* message.

18. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.5.2.4.3) and *RRCResumeComplete* message.

19. If the UE message at step 15 or 18 includes the *ackRequested* IE set to TRUE, the SS shall send LPP acknowledgment message.

20. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the *NR-Multi-RTT-SignalMeasurementInformation* according to Table 15.5.2.5-2.

21. UE shall be released and moved to RRC_IDLE.

22. Repeat steps 1-21 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.5.2.4.3 Message contents

Message contents are the same as 15.5.1.4.3 with following exceptions:

**Table 15.5.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1&2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 8 | 52 PRBs | Config 1&2 |
| | 7 | 48 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.2.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.2.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | |
| n160-r16 | 10 | | Config 1&2 |
| } | | | |
| Scs30-r16 CHOICE { | | | |
| n160-r16 | 10 | | Config 3 |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |

| | | | |
|---|---|---|---|
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.5.2.5 Test requirement

Table 15.5.2.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.5.2.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |

| | | | | |
|---|---|---|---|---|
| RF Channel Number | | 1,2,3 | 1 | 1 |
| DRX cycle | | 1,2,3 | 1.28s | |
| Time offset with Cell 1 | µs | 1, 2, 3 | N/A | 3 |
| TDD configuration | | 1 | N/A | N/A |
| | | 2 | TDDConf.1.1 | TDDConf.1.1 |
| | | 3 | TDDConf.2.1 | TDDConf.2.1 |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | N/A |
| | | 2 | SR.1.1 TDD | |
| | | 3 | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | N/A |
| | | 2 | CR.1.1 TDD | |
| | | 3 | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | N/A |
| | | 2 | CCR.1.1 TDD | |
| | | 3 | CCR.2.1 TDD | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | N/A |
| | | 2 | TRS.1.1 TDD | |
| | | 3 | TRS.1.2 TDD | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 2 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 3 | PRS.2.1 FR1 | PRS.2.1 FR1 |
| PRS BW | | 1 | 52 PRBs | 52 PRBs |
| | | 2 | 52 PRBs | 52 PRBs |
| | | 3 | 48 PRBs | 48 PRBs |
| PRS Resource slot offset | slot | 1, 2, 3 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.1 | N/A |
| | | 2 | POS-SRS.1 | N/A |
| | | 3 | POS-SRS.2 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | |
| | | 2 | -98 | |
| | | 3 | -95 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | |
| | | 2 | | |
| | | 3 | | |
| PRS $\hat{E}_s / I_{ot}$ | dB | 1 | 0 | -6 |
| | | 2 | | |
| | | 3 | | |
| PRS $\hat{E}_s / N_{oc}$ | dB | 1 | 2.93 | -0.9 |
| | | 2 | | |
| | | 3 | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -95.07 | -98.90 |
| | | 2 | -95.07 | -98.90 |
| | | 3 | -92.07 | -95.90 |
| Io | dBm/19.08 MHz | 1 | -61.18 | -61.18 |

| | | | | |
|---|---|---|---|---|
| | dBm/19.08 MHz | 2 | -61.18 | -61.18 |
| | dBm/47.88 MHz | 3 | -57.20 | -57.20 |
| Propagation Condition | | 1, 2, 3 | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | |
| Note 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | |

**Table 15.5.2.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1&2 | cell 1 | (Measured value from step 12 - 171) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 171) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | cell 2 | (Measured value from step 12 - 187) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 187) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 | cell 1 | (Measured value from step 12 – 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 118) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | cell 2 | (Measured value from step 12 - 127) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 127) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

The UE Rx-Tx time difference measurement period fulfills the requirements specified in clause 15.5.2.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.5.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA

### 15.5.3.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.5.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports Multi-RTT positioning and *supportedDL-PRS-ProcessingSamples-RRC-Inactive*. In addition, this test shall be verified with either SDT or non-SDT supporting UEs.

### 15.5.3.3 Minimum conformance requirements

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall not apply, if:

$N_{TA_offset}$ defined in Table 7.1.2-2 changes during the UE Rx-Tx measurement period or

if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the network-configured Timing Advance.

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that:

- The UE transmits SRS within [-160, 160] msec of at least one DL PRS resource of each of the TRPs in the assistance data.

If the uplink transmission timing changes during the UE Rx-Tx measurement period due to the autonomous timing adjustment defined in clause 7.1.2 of TS38.133 [50] then:

- UE Rx-Tx measurement accuracy requirements shall apply for a cell, which is also the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission even if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

- UE Rx-Tx measurement accuracy requirements shall not apply for a cell, which is not the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission, if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

When a serving cell change occurs during the UE Rx-Tx measurement period, the UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that the serving cell change does not impact SRS configuration for the UE Rx-Tx measurement.

The relative UE Rx-Tx measurement accuracy in this clause is defined as accuracy of the difference between two UE Rx-Tx measurements.

The accuracy requirements in Table 15.5.3.3-2 for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 of TS38.133 [50] for a corresponding Band.

AWGN propagation condition.

**Table 15.5.3.3-1: UE Rx-Tx time difference measurement accuracy in FR2 in AWGN**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$ Note 3 | Io Note 4 range | |
| | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| ± 22+δ | -3 | ≥24 | 60 | ≥4 | Same value as PRP in Table B.2.14-2 of TS38.133[50], according to UE Power class, operating band and angle of arrival | -50 |
| ± 15+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 12+δ | | ≥32 | 120 | ≥14 | Same value as PRP in Table B.2.14-2 of TS38.133[50], according to UE Power class, operating band and angle of arrival | -50 |
| ± 7+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 35+δ | -13 | ≥24 | 60 | ≥1 | NOTE 6 | NOTE 6 |
| ± 15+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 14+δ | | ≥32 | 120 | ≥4 | NOTE 6 | NOTE 6 |
| ± 9+δ | | ≥64 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5 of TS38.133[50].
NOTE 3: $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter *dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols* and *dl-PRS-CombSizeN* defined in TS 37.355 [49].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].

NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 15.5.3.3-2.

**Table 15.5.3.3-2: Margin for UE Rx-Tx time difference measurement accuracy in FR2**

| Min(PRS BW, SRS BW) (MHz) | | Margin (Tc [Note 1]) |
|---|---|---|
| SCS = 60 kHz | SCS = 120 kHz | |
| ≥ 24 | N/A | 76 |
| ≥ 64 | ≥ 32 | 32 |
| ≥ 132 | ≥ 64 | 24 |
| N/A | ≥ 128 | 20 |
| NOTE 1: Tc is the basic timing unit defined in TS 38.211 [53].<br>NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies. | | |

## 15.5.3.4 Test description

### 15.5.3.4.1 Initial conditions

The supported test configuration is listed in Table 15.5.3.4.1.-1.

**Table 15.5.3.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.3.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.5.3.5-1.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.3.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). Both cells are on the same RF channel in FR2.

### 15.5.3.4.2 Test procedure

The test consists of a set-up period and a measurement period. The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period. During the measurement period, UE shall be in RRC_INACTIVE state.

7. Configure the Cell for RA-SDT with SIB1 configured according to the message contents 15.5.3.4.3.
8. State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A$ = 31 which indicates a timing advance adjustment value $N_{TA}$ = 0 $T_c$.
9. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
10. The SS adjusts the downlink timing for Cell 1 to a delay of +512 $T_c$, compared to the current value.

11. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
12. The SS shall transmit an *RRCReconfiguration* message with DRX configuration.
13. The UE shall transmit *RRCReconfigurationComplete* message.
14. The SS shall transmit an LPP REQUEST CAPABILITIES message.
15. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
16. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
17. The SS shall send a LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
18. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
19. The SS transmits an *RRCRelease* message including UL configuration and SDT configuration to transit the UE to RRC_INACTIVE state.
20. Wait for the UE as it monitors for occurrence of the periodic event requested.
21. The UE shall transmit an *RRCResumeRequest* message. If the UE supports SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 19.
22. The SS transmits an *RRCResume* message
23. The UE transmits an *RRCResumeComplete* message.
24. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.5.3.4.3) and *RRCResumeComplete* message.
25. If the UE message at step 15 or 18 includes the *ackRequested* IE set to TRUE, the SS shall send LPP acknowledgment message.
26. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the *NR-Multi-RTT-SignalMeasurementInformation* according to Table 15.5.3.5-2.
27. UE shall be released and moved to RRC_IDLE.
28. Repeat steps 1-21 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.5.3.4.3 Message contents

Message contents are the same as in 15.4.1.5.4.3 with the following exceptions:

**Table 15.5.3.4.3-1: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |

| | | | |
|---|---|---|---|
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.5.3.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| Sl1280 | 160 | | Config 1 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |

**Table 15.5.3.4.3-2: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.3.4.3-3: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.3.4.3-4 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.3.4.3-4 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.3.4.3-4: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-test 2 |
| | n2 | | Sub-test 1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | Cell 1 |
| | 0 | | Cell 2 |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.5.3.5 Test requirement

The UE Rx-Tx time difference accuracy test parameters are given in Table 15.5.3.5-1.

**Table 15.5.3.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 of TS38.133 [50] | | Setup 1 as specified in clause A.3.15 of TS38.133 [50] | |
| Beam Assumption Note 7 | | 1 | Rough | Rough | Rough | Rough |
| DRX | s | 1 | 0.64 | | 0.64 | |
| Time offset with Cell 1 | μs | 1 | N/A | 3 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 | OP.1 | OP.1 |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A | DLBWP.0.1 ULBWP.0.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR2 | PRS.1.2 FR2 | PRS.1.1 FR2 | PRS.1.2 FR2 |
| PRS Resource slot offset | slot | 1 | 0 | 4 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.3 | N/A | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -89 | | -89 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | -98 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.62 | -12.26 | -2.62 | -12.26 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -10 | -2 | -10 |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -91 | -99 | -91 | -99 |
| Io | dBm/190.08 MHz | 1 | -54.59 | -54.59 | -54.59 | -54.59 |
| Propagation Condition | | 1 | AWGN | | AWGN | |

| | |
|---|---|
| Note 1: | Void. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 6: | As observed with 0 dBi gain antenna at the centre of the quiet zone |
| Note 7: | Information about types of UE beam is given in B.2.1.3 of TS38.133[50], and does not limit UE implementation or test system implementation |
| Note 8: | Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 36.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. |

**Table 15.5.3.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | Sub-test 1 | (Measured value from step 12 - 33) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 33) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | Sub-test 2 | (Measured value from step 12 - 33) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 +33) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.5.3.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 15.5.4 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR2 SA

### 15.5.4.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy with reduced number of samples in RRC_INACTIVE state is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25.2. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 15.5.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that support Multi-RTT positioning and *supportedDL-PRS-ProcessingSamples-RRC-Inactive*. In addition, this test shall be verified with either SDT or non-SDT supporting UEs.

### 15.5.4.3 Minimum conformance requirements

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall not apply, if:

$N_{TA_offset}$ defined in Table 7.1.2-2 in TS38.133 [50] changes during the UE Rx-Tx measurement period or

if the uplink transmission timing changes during the UE Rx-Tx measurement period due to the network-configured Timing Advance.

The UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that:

- The UE transmits SRS within [-160, 160] msec of at least one DL PRS resource of each of the TRPs in the assistance data.

If the uplink transmission timing changes during the UE Rx-Tx measurement period due to the autonomous timing adjustment defined in clause 7.1.2 of TS38.133 [50] then:

- UE Rx-Tx measurement accuracy requirements shall apply for a cell, which is also the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission even if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

- UE Rx-Tx measurement accuracy requirements shall not apply for a cell, which is not the downlink reference cell (defined in section 7.1.1 of TS38.133 [50]) for SRS transmission, if the uplink transmission timing changes during the UE Rx-Tx measurement period due to autonomous adjustment.

When a serving cell change occurs during the UE Rx-Tx measurement period, the UE Rx-Tx time difference measurement accuracy requirements in this clause shall apply provided that the serving cell change does not impact SRS configuration for the UE Rx-Tx measurement.

The relative UE Rx-Tx measurement accuracy in this clause is defined as accuracy of the difference between two UE Rx-Tx measurements.

The accuracy requirements in Table 15.5.4.3-1 for FR2 are valid under the following conditions:

Conditions defined in clause 7.3 of TS 38.101-2 [55] for reference sensitivity are fulfilled.

$PRP|_{dBm}$ according to Annex B.2.14 of TS38.133 [50] for a corresponding Band

Number of measurement samples is less than 4.

AWGN propagation condition.

**Table 15.5.4.3-1: UE Rx-Tx time difference measurement accuracy in FR2 in AWGN with reduced measurement samples**

| Accuracy | Conditions | | | | | |
|---|---|---|---|---|---|---|
| | PRS Ês/Iot | Minimum PRS bandwidth | PRS SCS | PRS resource repetition $(T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$ Note 3 | Io Note 4 range | |
| | | | | | Minimum Io Note 1 | Maximum Io |
| Tc Note 5 | dB | PRB | kHz | | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| ± 15+δ | 0 | ≥64 | 60 | ≥1 | Same value as PRP in Table B.2.14-2 of TS38.133[50], according to UE Power class, operating band and angle of arrival | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥64 | 120 | ≥1 | Same value as PRP in Table B.2.14-2 of TS38.133[50], according to UE Power class, operating band and angle of arrival | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 15+δ | -6 | ≥64 | 60 | ≥1 | NOTE 6 | NOTE 6 |
| ± 7+δ | | ≥132 | | ≥1 | NOTE 6 | NOTE 6 |
| ± 9+δ | | ≥64 | 120 | ≥1 | NOTE 6 | NOTE 6 |
| ± 4+δ | | ≥128 | | ≥1 | NOTE 6 | NOTE 6 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: NR operating band groups are as defined in Section 3.5 of TS38.133[50]
NOTE 3: $T_{rep}^{PRS}$, $L_{PRS}$, $K_{comb}^{PRS}$ are configured by higher layer parameter dl-PRS-ResourceRepetitionFactor, dl-PRS-NumSymbols and dl-PRS-CombSizeN defined in TS 37.355 [49].
NOTE 4: The Io is defined in PRS slots. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same slot.
NOTE 5: Tc is the basic timing unit defined in TS 38.211 [53].
NOTE 6: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth of the smallest PRB number for the corresponding SCS.
NOTE 7: δ is the margin determined from Table 15.5.4.3-2

**Table 15.5.4.3-2: Margin for UE Rx-Tx time difference measurement accuracy in FR2**

| Min(PRS BW, SRS BW) (MHz) | | Margin (Tc [Note 1]) |
|---|---|---|
| SCS = 60 kHz | SCS = 120 kHz | |
| ≥ 24 | N/A | 76 |
| ≥ 64 | ≥ 32 | 32 |
| ≥132 | ≥ 64 | 24 |
| N/A | ≥ 128 | 20 |
| NOTE 1: Tc is the basic timing unit defined in TS 38.211 [53].<br>NOTE 2: If SRS and PRS have different SCS, the margin corresponding to the smallest RS BW in MHz applies. | | |

## 15.5.4.4 Test description

### 15.5.4.4.1 Initial conditions

The supported test configuration is listed in Table 15.5.4.4.1-1.

**Table 15.5.4.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.4.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.5.4.4-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.4.4.3.
5. In the test there are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 15.5.4.4.2 Test procedure

The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state.

The beginning of the measurement period shall be aligned with the first DRX cycle containing a DL PRS resource(s). The *NR-Multi-RTT-ProvideAssistanceData* and *NR-Multi-RTT-RequestLocationInformation* shall be provided to the UE during the set-up period.

1. Configure the Cell for RA-SDT with SIB1 configured according to the message contents 15.5.4.4.3.
2. State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A$ = 31 which indicates a timing advance adjustment value $N_{TA}$ = 0 $T_c$.
3. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
4. The SS adjusts the downlink timing for Cell 1 to a delay of +512 $T_c$, compared to the current value.

5. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.
6. The SS shall transmit an *RRCReconfiguration* message with DRX configuration.
7. The UE shall transmit *RRCReconfigurationComplete* message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.
9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* in the *NR-DL-PRS-ProcessingCapability* IE and the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilities* IE.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-Multi-RTT-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall send a LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the reducedDL-PRS-ProcessingSamples-r17 in the *NR-Multi-RTT-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms.
12. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
13. The SS transmits an *RRCRelease* message including the UL configuration and SDT configuration to transit the UE to RRC_INACTIVE state.
14. Wait for the UE as it monitors for occurrence of the periodic event requested.
15. The UE shall transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 19.
16. The SS transmits an *RRCResume* message.
17. The UE transmits an *RRCResumeComplete* message.
18. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.5.4.4.3) and *RRCResumeComplete* message.
19. If the UE message at step 15 or 18 includes the *ackRequested* IE set to TRUE, the SS shall send LPP acknowledgment message.
20. The SS shall check nr-UE-RxTxTimeDiff-r16 value for Cell 2 in the *NR-Multi-RTT-SignalMeasurementInformation* according to Table 15.5.4.5-2.
21. UE shall be released and moved to RRC_IDLE.
22. Repeat steps 1-21 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.5.4.4.3 Message contents

Message contents are the same as 15.5.3.4.3 with following exceptions:

**Table 15.5.4.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.4.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.4.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.4.4.3-2: NR-DL-PRS-Info**

| |
|---|
| Derivation Path: 37.355 clause 6.4.3 |

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | Cell 1 |
| | 0 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.4.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |

| | | | |
|---|---|---|---|
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.5.4.5 Test requirement

Table 15.5.4.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.5.4.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| AoA setup | | 1 | Setup 1 as specified in clause A.3.15 | |
| Beam Assumption[Note 7] | | 1 | Rough | Rough |
| DRX cycle | | 1 | 0.64s | |
| Time offset with Cell 1 | μs | 1 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | |
| EPRE ratio of OCNG DMRS to SSS[Note 1] | | | | |
| EPRE ratio of OCNG to OCNG DMRS [Note 1] | | | | |
| EPRE ratio of PRS to SSS | | | | |
| TRS Configuration | | 1 | TRS.2.1 TDD | N/A |

| | | | | |
|---|---|---|---|---|
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A |
| PRS configuration | | 1 | PRS.1.1 FR2 | PRS.1.1 FR2 |
| PRS BW | | 1 | 64 PRBs | 64 PRBs |
| PRS Resource slot offset | slot | 1 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -94 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -103 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | 0.32 | -5.64 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | 3.56 | -0.28 |
| PRP Note 3 | dBm/SCS kHz | 1 | -90.4 | -94.2 |
| Io | dBm/190.08 MHz | 1 | -55.73 | -55.73 |
| Propagation Condition | | 1 | AWGN | |

Note 1: Void.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in B.2.1.3, and does not limit UE implementation or test system implementation
Note 8: Calculation of Es/Iot$_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

**Table 15.5.4.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Lowest reported value | Highest reported value |
|---|---|---|
| 1 | (5898 - 33) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (5898 + 33) Tc converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.5.4.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 15.5.5 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR1 SA

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.5.5.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25A. The tests are conducted under AWGN propagation condition with the UE operating in FR1 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR1.

### 15.5.5.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_INACTIVE.

### 15.5.5.3 Minimum conformance requirements

Same as clause 15.3.7.3.

### 15.5.5.4 Test description

The test defines three possible test configurations; specified in Table 15.5.5.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.5.5.4-1: Test Configurations**

| PCell configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

#### 15.5.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.5.5.5-1 and Table 15.5.5.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.5.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.5.5.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-Multi-RTT-ProvideAssistanceData and NR-Multi-RTT-RequestLocationInformation as defined in clause 15.5.5.4.3 shall be provided to the UE during the set-up period. The last TTI containing the NR-Multi-RTT-RequestLocationInformation message shall be provided to the UE $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms is the maximum processing time of the NR-Multi-RTT-RequestLocationInformation message and the Multi-RTT assistance data in the UE.

13. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 15.5.5.4.3-1.
14. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5. Note that in the remainder of the test the timing advance command $T_A = 31$ which indicates a timing advance adjustment value $N_{TA} = 0\ T_c$.
15. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
16. The SS adjusts the downlink timing for Cell 1 to a delay of $+512\ T_c$, compared to the current value.
17. Wait for 1.6 s to allow for the possibility that the UE makes autonomous timing adjustments.

6. Set the parameters for T1 according to Table 15.5.5.5-1 and Table 15.5.5.5-2.
7. T1 starts.
8. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
9. The UE shall transmit RRCReconfigurationComplete message.
10. The SS shall transmit an LPP REQUEST CAPABILITIES message.
11. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the Multi-RTT capabilities supported by the UE in the *NR-Multi-RTT-ProvideCapabilitie* IE.
12. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the NR-Multi-RTT-ProvideAssistanceData IE and NR-DL-PRS-AggregationInfo-r18 linking each PRS resource in PFL1 to the corresponding PRS resource in PFL2.If the UE message at step 11 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
13. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the NR-Multi-RTT-RequestLocationInformation IE and nr-DL-PRS-JointMeasurementRequest-r18 IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
14. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
15. When set-up period expires, the SS shall transmit an RRCRelease message with the UL configuration and SDT configuration to transit the UE to RRC_INACTIVE state.
16. Wait for the UE monitors for occurrence of the periodic event requested.
17. The UE shall transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 21.
18. The SS transmits an *RRCResume* message.
19. The UE transmits an *RRCResumeComplete* message.
20. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-Multi-RTT-ProvideLocationInformation* IE within the response time (see clause 15.5.5.4.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.
21. The UE shall perform and report the UE Rx-Tx time difference measurements for Cell 1 and Cell 2. If the UE transmits an *NR-Multi-RTT-ProvideLocationInformation* IE including the nr-UE-RxTxTimeDiff field for Cell 1 and Cell 2 within the response time, then the number of successful tests is increased by one. If the UE fails to report the *NR-Multi-RTT-ProvideLocationInformation* IE with the nr-UE-RxTxTimeDiff field included within the response time then the number of failure tests is increased by one.
22. If the UE message at step 20 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
25. UE shall be released and moved to RRC_IDLE.
24. Repeat steps 1-23 in Tables 15.5.5.4-1 until the confidence level according to Annex D is achieved.

NOTE: To avoid a large divergence between the sent TA and the set downlink timing, the SS may reset the downlink timing for Cell 1 to its initial value after a certain amount of loops. The loop during the reset does not count for the result statistics.

### 15.5.5.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.5.5.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.5.5.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.5.5.4.3-3 replaces Table 15.4.1.4.3-12 and Table 15.5.5.4.3-4 replaces Table 15.4.1.4.3-13.

**Table 15.5.5.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.5.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.5.4.3-2 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1 and Configuration 2 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Configuration 1 and Configuration 2 |
| | 28 | 132 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.5.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.5.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.5.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1 and Configuration 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.5.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.5.5.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |

| | | | |
|---|---|---|---|
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.5.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.5.5.5 Test requirement

FFS

## 15.5.6 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR2 SA

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.5.6.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1.25A. The tests are conducted under AWGN propagation condition with the UE operating in FR2 stand-alone mode and configured to perform UE Rx-Tx measurements by aggregating two intra-band contiguous positioning frequency layers (PFLs) in FR2.

### 15.5.6.2 Test applicability

This test applies to all types of NR UE release 18 onwards that supports Multi-RTT positioning and DL-PRS bandwidth aggregation in RRC_INACTIVE.

### 15.5.6.3 Minimum conformance requirements

Same as defined in clause 15.3.7.3.

### 15.5.6.4 Test description

The supported test configuration is listed in Table 15.5.6.4-1.

**Table 15.5.6.4-1: Test Configuration**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

#### 15.5.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.6.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.5.6.5-1 and Table 15.5.6.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.6.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.5.6.4.2 Test procedure

Same as defined in clause 15.5.5.4.2.

#### 15.5.6.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.5.6.4.3-1 replaces Table 15.4.1.4.3-4, Table 15.5.6.4.3-2 replaces Table 15.4.1.4.3-7, Table 15.5.6.4.3-3 replaces Table 15.4.1.4.3-8, Table 15.5.6.4.3-4 replaces Table 15.4.1.4.3-11, Table 15.5.6.4.3-5 replaces Table 15.4.1.4.3-12 and Table 15.5.6.4.3-6 replaces Table 15.4.1.4.3-13.

**Table 15.5.6.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.6.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.6.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.6.4.3-3 | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.6.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.6.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 2 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.6.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |

| | | | |
|---|---|---|---|
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerTRP-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerTRP-r16[2] |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | NR Cell 1 |
| | 4 | | NR Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.6.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.508-1 [45], Table 4.6.3-182 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-ResourceSetToReleaseList | Not present | | |
| srs-ResourceSetToAddModList | Not present | | |
| srs-ResourceToReleaseList | Not present | | |
| srs-ResourceToAddModList | Not present | | |
| tpc-Accumulation | Not present | | |
| srs-RequestDCI-1-2-r16 | Not present | | |
| srs-RequestDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToAddModListDCI-0-2-r16 | Not present | | |
| srs-ResourceSetToReleaseListDCI-0-2-r16 | Not present | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 0 | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.5.6.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| txHoppingConfig-r18 | Not present | | |
| } | | | |
| } | | | |
| dci-TriggeringPosResourceSetLink-r18 | Not present | | |
| } | | | |

**Table 15.5.6.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.5.6.5 | Result of the response time calculation rounded up to the next | |

| | | | |
|---|---|---|---|
| | | second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.6.4.3-6: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.5.6.5 Test requirement

FFS

## 15.5.7 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH

### 15.5.7.1 Test purpose

The purpose of the test is to verify that the accuracy of the UE Rx-Tx time difference measurement without RX FH reported by the RedCap UE is within the specified limits in TS 38.133 [50] clause 10.1A.18.2. The test is conducted in AWGN propagation condition in FR1 in standalone scenario when single positioning frequency layer is configured.

### 15.5.7.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE.

### 15.5.7.3 Minimum conformance requirements

Same as defined in clause 15.3.9.3.

### 15.5.7.4 Test description

The test defines three possible test configurations; specified in Table 15.5.7.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.5.7.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

### 15.5.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.7.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 15.5.7.5-1 and Table 15.5.7.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.7.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.5.7.4.2 Test procedure

Same as defined in clause 15.5.1.4.2.

### 15.5.7.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.5.7.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.5.7.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.5.7.4.3-3 replaces Table 15.4.1.4.3-12 and Table 15.5.7.4.3-4 replaces Table 15.4.1.4.3-13.

**Table 15.5.7.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1, Config 2 and Config 4 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.7.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.7.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.7.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1, Config 2 and Config 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.7.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.5.7.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.7.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.5.7.5 Test requirement

Table 15.5.7.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.5.7.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Cell 1 | Cell 2 |
|---|---|---|---|---|
| RF Channel Number | | 1,2,3,4 | 1 | 1 |
| $BW_{channel}$ | MHz | 1 | 10: $N_{RB,c}$ = 52 | |
| | | 2 | 10: $N_{RB,c}$ = 52 | |
| | | 3 | 20: $N_{RB,c}$ = 51 | |
| | | 4 | 10: $N_{RB,c}$ = 52 | |
| DRX | s | 1,2,3,4 | 1.28 | |
| Time offset with Cell 1 | µs | 1,2,3,4 | N/A | 3 |
| TDD configuration | | 1,4 | N/A | N/A |
| | | 2 | TDDConf.1.1 | TDDConf.1.1 |
| | | 3 | TDDConf.2.1 | TDDConf.2.1 |
| PDSCH RMC configuration | | 1,4 | SR.1.1 FDD | N/A |
| | | 2 | SR.1.1 TDD | |
| | | 3 | SR.2.1 TDD | |
| RMSI CORESET RMC configuration | | 1,4 | CR.1.1 FDD | N/A |
| | | 2 | CR.1.1 TDD | |
| | | 3 | CR.2.1 TDD | |
| Dedicated CORESET RMC configuration | | 1,4 | CCR.1.1 FDD | N/A |
| | | 2 | CCR.1.1 TDD | |
| | | 3 | CCR.2.1 TDD | |
| OCNG Patterns | | 1,2,3,4 | OP.1 | OP.1 |
| Initial BWP configuration | | 1,2,3,4 | DLBWP.0.1 ULBWP.0.1 | N/A |
| PRS configuration | | 1,4 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 2 | PRS.1.1 FR1 | PRS.1.1 FR1 |
| | | 3 | PRS.2.1 FR1 | PRS.2.1 FR1 |
| PRS Resource slot offset | slot | 1,2,3,4 | 0 | 4 |
| SRS configuration | | 1,4 | POS-SRS.1 | N/A |
| | | 2 | POS-SRS.1 | N/A |
| | | 3 | POS-SRS.2 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1,4 | -98 | |
| | | 2 | -98 | |
| | | 3 | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1,4 | -2.41 | -12.12 |
| | | 2 | | |
| | | 3 | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1,4 | -2 | -10 |
| | | 2 | | |
| | | 3 | | |
| PRP Note 3 | dBm/SCS kHz | 1,4 | -100 | -108 |
| | | 2 | -100 | -108 |
| | | 3 | -97 | -105 |
| Io | dBm/9.36 MHz | 1,4 | -67.67 | -67.67 |
| | dBm/9.36 MHz | 2 | -67.67 | -67.67 |
| | dBm/18.36 MHz | 3 | -64.75 | -64.75 |
| Propagation Condition | | 1,2,3,4 | AWGN | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 15.5.7.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Cell | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1&2&4 (15kHz) | NR cell 1 | (Measured value from step 12 - 270) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 270) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 281) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 281) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 293) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 293) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 290) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 290) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| 3 (30kHz) | NR cell 1 | (Measured value from step 12 - 169) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 169) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 161) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 161) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 170) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 170) $T_c$ for 2RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | | (Measured value from step 12 – 167) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 167) $T_c$ for 1RX redcap UE converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.5.7.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.5.8 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

### 15.5.8.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement accuracy without RX FH in RRC_INACTIVE mode is within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1A.18.2. The test is conducted in AWGN propagation condition in FR2 in standalone scenario.

### 15.5.8.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning and DL-PRS measurement in RRC_INACTIVE.

### 15.5.8.3 Minimum conformance requirements

Same as clause 15.3.9.3.

### 15.5.8.4 Test description

The supported test configuration is listed in Table 15.5.8.4-1.

**Table 15.5.8.4-1: Test Configuration**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 100 MHz bandwidth, TDD duplex mode |

#### 15.5.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.8.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.5.8.5-1 and Table 15.5.8.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.8.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

#### 15.5.8.4.2 Test procedure

Same as defined in clause 15.5.1.4.2.

#### 15.5.8.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.5.8.4.3-1 replaces Table 15.4.1.4.3-4, Table 15.5.8.4.3-2 replaces Table 15.4.1.4.3-7, Table 15.5.8.4.3-3 replaces Table 15.4.1.4.3-8, Table 15.5.8.4.3-4 replaces Table 15.4.1.4.3-11, Table 15.5.8.4.3-5 replaces Table 15.4.1.4.3-12 and Table 15.5.8.4.3-6 replaces Table 15.4.1.4.3-13.

**Table 15.5.8.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.8.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.8.4.3-3 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.8.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.8.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.8.4.3-4: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.5.8.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.5.8.4.3-5: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.5.8.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.8.4.3-6: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 15.5.8.5 Test requirement

Table 15.5.8.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.5.8.5-1: UE Rx-Tx time difference measurement accuracy test parameters.**

| Parameter | Unit | Test configuration | Cell 1 | Cell 2 |
|---|---|---|---|---|
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 clause A.3.15 | |
| $BW_{channel}$ | MHz | 1 | 100: $N_{RB,c}$ = 66 | |
| Beam Assumption[Note 6] | | 1 | Rough | Rough |
| DRX | s | 1 | 0.64 | |

| | | | | |
|---|---|---|---|---|
| Time offset with Cell 1 | µs | 1 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A |
| PRS configuration | | 1 | PRS.1.5 FR2 | PRS.1.5 FR2 |
| PRS Resource slot offset | slot | 1 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.3 | N/A |
| $N_{oc}$ Note 1 | dBm/SCS | 1 | -89 | |
| $N_{oc}$ Note 1 | dBm/15 kHz | 1 | -98 | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.41 | -12.12 |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | -10 |
| PRS-RSRP Note 2 | dBm/SCS kHz | 1 | -91 | -99 |
| Io | dBm/92.16 MHz | 1 | -57.6 | -57.6 |
| Propagation Condition | | 1 | AWGN | |

NOTE 1: Interference from other cells and noise sources not specified in the test is assumed to be constant over sub-carriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 2: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

NOTE 3: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

NOTE 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.

NOTE 5: As observed with 0 dBi gain antenna at the centre of the quiet zone

NOTE 6: Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation.

NOTE 7: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 36.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

**Table 15.5.8.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Subtest | Lowest reported value | Highest reported value |
|---|---|---|---|
| 1 | NR cell 1 | (Measured value from step 12 - 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

| | | | |
|---|---|---|---|
| | NR cell 2 | (Measured value from step 12 - 65) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 65) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.5.8.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

## 15.5.9 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE with Rx FH

Editor's note: This test case is incomplete. The following aspect is either missing or TBD:

- Test tolerance are not added in and analysis is missing

### 15.5.9.1 Test purpose

The purpose of this test is to verify that the UE Rx-Tx measurement accuracy in FR1 with FH by a RedCap UE in RRC_INACTIVE state is within the specified limits. This test will verify the requirements in TS 38.133 [50] clauses 10.1A.18.2.3 and 10.1A.18.2.4.

### 15.5.9.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and DL-PRS measurement in RRC_INACTIVE and maximumPRS-BandwidthAcrossAllHopsFR1 for at least one band and PRS frequency hopping measurement and report.

### 15.5.9.3 Minimum conformance requirements

Same as clause 15.3.11.3.

### 15.5.9.4 Test description

The test defines three possible test configurations; specified in Table 15.5.9.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 15.5.9.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, cell bandwidth 50 MHz, FDD duplex mode |
| 2 | 15 kHz SSB SCS, cell bandwidth 50 MHz, TDD duplex mode |
| 3 | 30 kHz SSB SCS, cell bandwidth 100 MHz, TDD duplex mode |
| 4 | 15 kHz SSB SCS, cell bandwidth 50 MHz, HD-FDD duplex mode |
| NOTE 1: The RedCap UE is only required to be tested in one of the supported test configurations.<br>NOTE 2: This test case is applicable to the UE supporting per hop bandwidth ≥ 10 MHz for SCS 15 kHz.<br>NOTE 3: This test case is applicable to the UE supporting per hop bandwidth = 20 MHz for SCS 30 kHz. | |

#### 15.5.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.9.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.

2. The general test parameter settings are set up according to Table 15.5.9.5-1 and Table 15.5.9.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.9.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.5.9.4.2 Test procedure

Same as defined in clause 15.5.1.4.2.

### 15.5.9.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.5.9.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.5.9.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.5.9.4.3-3 replaces Table 15.4.1.4.3-12 and Table 15.5.9.4.3-4 replaces Table 15.4.1.4.3-13.

**Table 15.5.9.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1, Config 2 and Config 4 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 62 | 268 PRBs | Config 1, Config 2 and Config 4 |
| | 63 | 272 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.9.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.9.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.9.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1, Config 2 and Config 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| n4-r16 | 1 | | Config 1, Config 2 and Config 4 |
| | 0 | | Configuration 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | Config 1, Config 2 and Config 4 |
| | 0 | | Configuration 3 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.9.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.5.9.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |

| | | | |
|---|---|---|---|
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWith DifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | mhz50 | | Configuration 1, Configuration 2 and Configuration 4 |
| | mhz100 | | Configuration 3 |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindows Config-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.9.4.3-4: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.5.9.5 Test requirement

FFS

## 15.5.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

### 15.5.10.1 Test purpose

The purpose of the test is to verify that the UE Rx-Tx time difference measurement requirements with FH by RedCap UE in RRC_INACTIVE state are within the specified limits. This test will verify the requirements in TS 38.133 [50] clause 10.1A.18.2.3 and 10.1A.18.2.4. The test is conducted in AWGN propagation condition in FR2 in standalone scenario.

### 15.5.10.2 Test applicability

This test applies to all types of NR redcap UE release 18 onwards that supports Multi-RTT positioning method and DL-PRS measurement in RRC_INACTIVE and maximumPRS-BandwidthAcrossAllHopsFR2 for at least one band and PRS frequency hopping measurement and report.

### 15.5.10.3 Minimum conformance requirements

Same as clause 15.3.11.3.

### 15.5.10.4 Test description

The supported test configuration is listed in Table 15.5.10.4-1.

**Table 15.5.10.4-1: Test Configuration**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, cell bandwidth 400 MHz, TDD duplex mode |
| NOTE 1: This test case is applicable to the UE supporting per hop bandwidth = 100 MHz for SCS 120 kHz. | |

### 15.5.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 15.5.10.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 15.5.10.5-1 and Table 15.5.10.5-2.
3. Propagation conditions are set according to clause 4.14.2.
4. Message contents are defined in clause 15.5.10.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2).

### 15.5.10.4.2 Test procedure

Same as defined in clause 15.5.1.4.2.

### 15.5.10.4.3 Message contents

As defined in clause 15.4.1.4.3, with the following exceptions:

Table 15.5.10.4.3-1 replaces Table 15.4.1.4.3-7, Table 15.5.10.4.3-2 replaces Table 15.4.1.4.3-8, Table 15.5.10.4.3-3 replaces Table 15.4.1.4.3-11, Table 15.5.10.4.3-4 replaces Table 15.4.1.4.3-12, and Table 15.5.10.4.3-5 replaces Table 15.4.1.4.3-13.

**Table 15.5.10.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 61 | 264 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.10.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 15.5.10.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.10.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n160-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.10.4.3-3: SRS-PosRRC-InactiveConfig-r17**

| Derivation Path: TS 38.331 [52] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveConfig-r17 ::= SEQUENCE { | | | |
| srs-PosConfigNUL-r17 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16 | 0 | 1 entry | |
| } | | | |
| resourceType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | alpha0 | | |
| p0-r16 | 0 | | |
| pathlossReferenceRS-Pos-r16 | Not present | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 5 | | |
| nrofSymbols-r16 | n4 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | Matches $N_{RB,c}$ Table 15.5.10.5-1 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl1280 | 160 | | |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r17 | Not present | | |
| bwp-NUL-r17 | Not present | | |
| bwp-SUL-r17 | Not present | | |
| inactivePosSRS-TimeAlignmentTimer-r17 | infinity | | |
| inactivePosSRS-RSRP-ChangeThreshold-r17 | Not present | | |
| } | | | |

**Table 15.5.10.4.3-4: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 15.5.10.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r1 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| additionalPathsExt-r17 | Not present | | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | Not present | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr2 | mhz400 | | |
| } | | | |
| } | | | |
| timingReportingGranularityFactorExt-r18 | Not present | | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 | Not present | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 15.5.10.4.3-5: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 | | | |

| | | | |
|---|---|---|---|
| SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 15.5.10.5 Test requirement

Table 15.5.10.5-1 defines the primary level settings including test tolerances for the test.

**Table 15.5.10.5-1: UE Rx-Tx time difference measurement accuracy test parameters**

| Parameter | Unit | Test configuration | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| AoA setup | | 1 | Setup 1 as specified in TS 38.133 clause A.3.15 | |
| $BW_{channel}$ | MHz | 1 | 400: $N_{RB,c}$ = 264 | |
| Beam Assumption Note 7 | | 1 | Rough | Rough |
| DRX | s | 1 | 0.64 | |
| Time offset with Cell 1 | μs | 1 | N/A | 3 |
| TDD configuration | | 1 | TDDConf.3.1 | TDDConf.3.1 |
| PDSCH RMC configuration | | 1 | SR.3.1 TDD | N/A |
| RMSI CORESET RMC configuration | | 1 | CR.3.1 TDD | N/A |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | N/A |
| OCNG Patterns | | 1 | OP.1 | OP.1 |
| Initial BWP configuration | | 1 | DLBWP.0.1 ULBWP.0.1 | N/A |
| PRS configuration | | 1 | PRS.1.6 FR2 | PRS.1.6 FR2 |
| PRS Resource slot offset | slot | 1 | 0 | 4 |
| SRS configuration | | 1 | POS-SRS.3 | N/A |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -93.73 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -102.73 | |
| PRS $\hat{E}_s / I_{ot}$ | dB | 1 | -2.69 | -88.83 |
| PRS $\hat{E}_s / N_{oc}$ | dB | 1 | -1.2 | -6 |
| PRS-RSRP Note 3 | dBm/SCS | 1 | -94.9 | -99.7 |
| Io Note 3 | dBm/380.16MHz | 1 | -55.66 | -55.66 |
| Propagation Condition | | 1 | AWGN | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
NOTE 6: As observed with 0 dBi gain antenna at the centre of the quiet zone.
NOTE 7: Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation.
NOTE 8: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 36.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

**Table 15.5.10.5-2: UE Rx – Tx time difference measurement accuracy requirements**

| Test Configuration | Cell | Lowest reported value | Highest reported value |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1 | NR cell 1 | (Measured value from step 12 - 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 63) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |
| | NR cell 2 | (Measured value from step 12 - 83) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 | (Measured value from step 12 + 83) $T_c$ converted to UE Rx-Tx time difference measurement according to clause 4.14.6 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

The UE Rx-Tx time difference measurement period fulfils the requirements specified in clause 15.5.10.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

# 16 NR PRS-RSRP measurement requirements

## 16.1 General

This clause defines the minimum performance requirements for NR UEs capable of PRS-RSRP measurements.

## 16.2 NR PRS-RSRP measurement period test cases in RRC_CONNECTED

### 16.2.1 PRS-RSRP measurement period test case for single positioning frequency layer in FR1 SA

#### 16.2.1.1 Test purpose

The purpose of the test is to verify that the PRS-RSP measurement meets the requirements specified in TS 38.133 [50] Clause 9.9.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when single positioning frequency layer is configured.

#### 16.2.1.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning and PRS-RSRP measurements in FR1.

#### 16.2.1.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.3.3) PRS-RSRP measurements, defined in TS 38.215 [4], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,\,total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max\left(T_{effect,i}\right)$$

where

*i* is the index of positioning frequency layer,

L is total number of positioning frequency layers,

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left( CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

where

$CSSF_i$ is the carrier specific scaling factor for PRS-RSRP measurements specified in clause 9.1.5.2,

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including those overlapped with other MG occasions within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],

N' is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of PRS-RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = [2] if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time, if not all PRS resources to be measured are available in the same measurement gap occasion during $T_{available_PRS,i}$, otherwise $T_{last,i} = T_i + MGL$,

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer $i$,

$T_i$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ is the least common multiple between $T_{PRS,i}$ and $MGRP_i$ ,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$MGRP_i$ is the measurement gap repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

Note: For the purpose of calculating $T_{PRS,i}$, only the PRS resources fully or partially covered by the MG are considered.

When PRS-RSRP measurements are configured for DL-AoD, except for deferred MT-LR as defined in clause 4.1a.5 [TS 23.273], the time $T_{PRS-RSRP,total}$ starts from the first MG instance aligned with DL PRS resources in the assistance data after both the *NR-DL-AoD-RequestLocationInformation* message and *NR-DL-AoD-ProvideAssistanceData* message from LMF via LPP [34] are delivered to the physical layer of UE.

For deferred MT-LR with other event than “Periodic Location” as defined in clause 4.1a.5.1 [TS 23.273], the time $T_{PRS-RSRP,total}$ starts from the first MG instance aligned with a DL PRS resource(s) in the assistance data after the associated event(s) occurs.

For deferred MT-LR with event “Periodic Location” as defined in clause 4.1a.5.1 [TS 23.273], the UE shall perform the PRS-RSRP measurement in each reporting period and activate the location report at the time when the periodic timer expires.

Note: No per-positioning frequency layer requirement is applied in scenarios when multiple positioning frequency layers are configured.

When the PRS-RSRP measurement is configured together with RSTD measurement then the PRS-RSRP measurement shall meet the RSTD measurement requirements defined in clause 9.9.2.

When the PRS-RSRP measurement is configured together with UE Rx-Tx time difference measurement then the PRS-RSRP measurement shall meet the UE Rx-Tx time difference measurement requirements defined in clause 9.9.4.

If CSSF changes during the measurement period, the measurement period could be longer.

The measurement requirements do not apply for a PRS resource:

– if the PRS resource is across two sampling duration of N within duration $L_{available_PRS,i}$ or

– if time span of the PRS resource instance (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability N.

If during the measurement period of one or more positioning frequency layers, the MG pattern is reconfigured either per UE request or not per UE request, the measurement period can be longer.

The requirements in this section apply, provided no PRS symbols are dropped during the measurement period $T_{PRS-RSRP,total}$ within measurement gaps due to collisions with other signals; otherwise, a longer measurement period may be used.

The requirements in clause 9.9.3 do not apply if the PRS configuration given by higher layer parameters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-AoD-ProvideCapabilities*, and it is up to UE implementation which PRS resources are measured, subject to UE measurement capabilities.

If handover occurs while PRS-RSRP measurements are being performed then the UE shall complete the ongoing PRS-RSRP measurements session. The PRS-RSRP measurement period can be longer. The UE shall meet the PRS-RSRP measurement accuracy requirements in clause 10.1.24.

## 16.2.1.4 Test description

### 16.2.1.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 16.2.1.4.1-1.

**Table 16.2.1.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 16.2.1.5-1 and Table 16.2.1.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.2.1.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the PCell. Cell 2 is the neighbour cell. The cells are on the same RF channel in FR1

### 16.2.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Both cells transmit PRS during T2.

The *NR-DL-AoD-RequestLocationInformation* message and *NR-DL-AoD-ProvideAssistanceData* message as defined in TS 37.355 [49] shall be provided to the UE during T1. The last slot containing the two messages for the assistance data and location information request is denoted as #n.

The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources that is $\Delta T$ after slot #n, where $\Delta T$ = 50 ms is the maximum processing time of the assistance data and location information request.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 16.2.1.5-1 and Table 16.2.1.5-2 as appropriate. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.
5. The SS shall send an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the NR-DL-AoD-ProvideCapabilities IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-AoD-ProvideAssistanceData-r16 IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-AoD-RequestLocationInformation-r16 such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 16.2.1.5-2.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the nr-DL-AoD-ProvideLocationInformation-r16 IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for both Cell 1 and Cell 2. If the UE transmits an nr-DL-AoD-ProvideLocationInformation-r16 IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 1 and Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the nr-DL-AoD-ProvideLocationInformation-r16IE with the DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. Repeat step 2-11 until the confidence level according to Annex D is achieved.

### 16.2.1.4.3 Message contents

**Table 16.2.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.1.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |

| | | | |
|---|---|---|---|
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.1.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 16.2.1.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.1.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 38 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.1.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.1.4.3-6: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| | | | |
| | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | Cell 1 |
| | 2 | | Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| | | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | Cell 1 |
| | 3 | | Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.1.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.1.5 Test requirement

Table 16.2.1.5-1 and Table 16.2.1.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.1.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: NRB,c = 106 | |
| | | 2 | 20: NRB,c = 106 | |
| | | 3 | 50: NRB,c = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | NA | OFF |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 5 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 16.2.1.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |
| | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.59 | | -60.59 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit specified in clause 16.2.1.3, starting from the beginning of time interval T2.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRP measurement period plus ΔT, where ΔT = 50 ms. The RSTD measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} = 4$, $N_{sample} = 4$, $L_{available_PRS,i} = 1.143ms\ for\ SCS\ 15kHz\ and\ 0.571\ for\ SCS\ 30kHz$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(\left\lceil\frac{4}{N'}\right\rceil\left\lfloor\frac{L_{available_PRS,i}}{N}\right\rfloor * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities.

Depending on the UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]:

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 0.648s and 204.9s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 16.2.1.4.3-3. The LPP time IE ranges between 1s and 210s.

Similarly for Type 2 Buffering, Best Case result of 0.648s is 1s and for the worst-case result of 164160ms, rounded to the next multiple of ten-seconds, the final value is 17s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 210.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 0.648s and 102.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 16.2.1.4.3-3. The LPP time IE ranges between 1s and 103s.

Similarly for Type 2 Buffering, Best Case result of 0.648s is 1s and for the worst-case result of 82080ms, rounded to the next multiple of ten-seconds, the final value is 83s

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 103.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.2.1.5-4, Table 16.2.1.5-5 and Table 16.2.1.5-6 for reference.

**Table 16.2.1.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil\frac{4}{N'}\right\rceil$ |
|---|---|
| n4 | 4 |
| n8 | 2 |
| >=n16 | 1 |

**Table 16.2.1.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil\frac{1}{N}\right\rceil$ |
|---|---|
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |

**Table 16.2.1.5-5**

| ***durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 1 Buffering*** Note 1 | $\left\lceil\frac{1.143}{N}\right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 5 |
| nDot5 | 3 |
| >=n1 | 2 |
| ***durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 1 Buffering*** Note 1 | $\left\lceil\frac{0.571}{N}\right\rceil$ |
| nDot125 | 5 |
| nDot25 | 3 |
| nDot5 | 2 |
| >=n1 | 1 |
| ***durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 2 Buffering*** Note 1 | $\left\lceil\frac{1}{N}\right\rceil$ |
| nDot125 | 8 |

| | |
|---|---|
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |
| ***durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 2 Buffering*** Note 1 | $\left\lceil \frac{0.5}{N} \right\rceil$ |
| nDot125 | 4 |
| nDot25 | 2 |
| >=nDot5 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26], | |

**Table 16.2.1.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.2.2 PRS-RSRP measurement period test case for dual positioning frequency layer in FR1 SA

### 16.2.2.1 Test purpose

The purpose of the test is to verify that the PRS-RSP measurement meets the requirements specified in TS 38.133 [50] Clause 9.9.2 in an environment with AWGN propagation conditions in FR1 in standalone scenario when dual positioning frequency layer is configured.

### 16.2.2.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning and PRS-RSRP measurements in FR1.

### 16.2.2.3 Minimum conformance requirements

Same as in clause 16.2.1.3.

### 16.2.2.4 Test description

#### 16.2.2.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 16.2.2.4.1-1.

**Table 16.2.2.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |

| | |
|---|---|
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Low for RF channel #1 and High for RF channel #2, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.12.
2. The general test parameter settings are set up according to Table 16.2.2.5-1 and Table 16.2.2.5-2.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.2.2.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the RSTD reference as well as the PCell on NR RF channel #1 in FR1. Cell 2 is a neighbour cell on a different NR RF channel #2 in FR1.

### 16.2.2.4.2 Test procedure

Same as clause 16.2.1.4.2.

### 16.2.2.4.3 Message contents

**Table 16.2.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.2.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |

| | | | |
|---|---|---|---|
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.2.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 16.2.2.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.2.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.2.4.3-8 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.2.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.2.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 and Cell 3 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | Cell 1 |
| | 2 | | Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | Cell 1 |
| | 3 | | Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.2.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.2.5 Test requirement

Table 16.2.2.5-1 and Table 16.2.2.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.2.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1<br>2: Cell 2 | Cell 1 and Cell 2 are on different positioning frequency layers |
| $BW_{channel}$ | MHz | 1 | 20: NRB,c = 106 | |
| | | 2 | 20: NRB,c = 106 | |
| | | 3 | 50: NRB,c = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | NA | OFF |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 10 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 16.2.2.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -3 | -Infinity | -12.70 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -3 | -Infinity | -12.70 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -101 | -Infinity | -110.70 |
| | | 2 | -Infinity | -101 | -Infinity | -110.70 |
| | | 3 | -Infinity | -98 | -Infinity | -107.70 |
| Io | dBm/19.08 MHz | 1 | N/A | -65.19 | N/A | -66.74 |
| | dBm/19.08 MHz | 2 | | -65.19 | | -66.74 |
| | dBm/47.88 MHz | 3 | | -61.21 | | -62.76 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit specified in clause 16.2.2.3, starting from the beginning of time interval T2.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRP measurement period plus ΔT, where ΔT = 50 ms. The PRS-RSRP measurement period follows the equation:

$$T_{PRS-RSRP,Total} = T_{PRS-RSRP,1} + T_{PRS-RSRP,2} + Max(T_{effect,i})$$

Given that both PFLs have the same settings, it can be simplified:

$$T_{PRS-RSRP,Total} = 2 * T_{PRS-RSRP,i} + T_{effect,i}$$

$$T_{PRS-RSRP,i} = \left(CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 1$, $N_{PRS,i}^{slot} 2$, $N_{sample} = 4$, $L_{available_PRS,i} = 0.571ms\ for\ SCS\ 15kHz\ and\ 0.286\ for\ SCS\ 30kHz$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$T_{PRS-RSRP,i} = \left(\left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities.

Depending on the UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26]:

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 15kHz the LPP time IE ranges between 1.456s and 104s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 16.2.2.4.3-3. The LPP time IE ranges between 2s and 104s.

Similarly for Type 2 Buffering, Best Case result of 1,456s is 2s and for the worst-case result of 164160ms, rounded to the next multiple of ten-seconds, the final value is 17s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 104.3s.

For Type 1 duration calculation with UE symbol level buffering capability, for SCS 30kHz the LPP time IE ranges between 1.456s and 63.04s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 14.2.2.4.3-3. The LPP time IE ranges between 2s and 64s.

Similarly for Type 2 Buffering, Best Case result of 1,456s is 2s and for the worst-case result of 82240ms, rounded to the next multiple of ten-seconds, the final value is 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.2.2.5-4, Table 16.2.2.5-5 and Table 16.2.2.5-6 for reference.

**Table 16.2.2.5-4: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n4 | **2** |
| >=n8 | 1 |

**Table 16.2.2.5-5: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 1 Buffering*** Note 1 | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |

| | |
|---|---|
| nDot25 | 3 |
| nDot5 | 2 |
| >=n1 | 1 |
| ***durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 1 Buffering*** Note 1 | $\left\lceil \frac{0.286}{N} \right\rceil$ |
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |
| ***durationOfPRS-ProcessingSysmbols for SCS 15kHz for Type 2 Buffering*** Note 1 | $\left\lceil \frac{1}{N} \right\rceil$ |
| nDot125 | 8 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n1 | 1 |
| ***durationOfPRS-ProcessingSysmbols for SCS 30kHz for Type 2 Buffering*** Note 1 | $\left\lceil \frac{0.5}{N} \right\rceil$ |
| nDot125 | 4 |
| nDot25 | 2 |
| nDot5 | 1 |
| NOTE 1: UE Buffering defined in clause 5.1.6.5 of TS 38.214 [26], | |

**Table 16.2.2.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.2.3 PRS-RSRP measurement period test case for single positioning frequency layer in FR2 SA

### 16.2.3.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] Clause 9.9.3.5. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when single positioning frequency layer is configured.

### 16.2.3.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-AoD positioning and PRS-RSRP measurements in FR2.

### 16.2.3.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [49], the UE shall be able to measure multiple (up to the UE capability specified in TS 38.133 Clause 9.9.3.3) PRS-RSRP measurements, defined in TS 38.215 [57], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max(T_{effect,i})$$

where

*i* is the index of positioning frequency layer,

L is total number of positioning frequency layers,

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left( CSSF_i * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

where

$CSSF_i$ is the carrier specific scaling factor for PRS-RSRP measurements specified in TS38.133 clause 9.1.5.2,

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and $N_{RxBeam,i}$=8 if positioning frequency layer *i* is in FR2,

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [56]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [49] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [49] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [49],

N′ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [49],

$N_{sample}$ is the number of PRS-RSRP measurement samples and $N_{sample}$ = 4,

$T_{last} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time,

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,

$T_i$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [49],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ is the least common multiple between $T_{PRS,i}$ and $MGRP_i$ ,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$MGRP_i$ is the measurement gap repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

### 16.2.3.4 Test description

The supported test configurations in listed in Table 16.2.3.4-1.

**Table 16.2.3.4-1: Supported test configurations for PRS RSRP measurement for FR2**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

#### 16.2.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.2.3.5-1 and Table 16.2.3.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.2.3.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

#### 16.2.3.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2. Both cells transmit PRS during T2.

The NR-DL-AoD-RequestLocationInformation message and NR-DL-AoD-ProvideAssistanceData message as defined in TS 37.355 shall be provided to the UE during T1. The last slot containing the two messages for the assistance data and location information request is denoted as #n. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources that is $\Delta T$ after slot #n, where $\Delta T$ = 50 ms is the maximum processing time of the assistance data and location information request.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On* according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.2.3.5-1 and Table 16.2.3.5-2 as appropriate. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.
5. The SS shall send an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the NR-DL-AoD-ProvideCapabilities IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-AoD-ProvideAssistanceData-r16 IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-AoD-RequestLocationInformation-r16 such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T$ = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 16.2.3.5-2.

10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the nr-DL-AoD-ProvideLocationInformation-r16 IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for both Cell 1 and Cell 2. If the UE transmits an nr-DL-AoD-ProvideLocationInformation-r16 IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 1 and Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the nr-DL-AoD-ProvideLocationInformation-r16IE with the DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.

11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

12. Repeat step 2-11 until the confidence level according to Annex D is achieved.

#### 16.2.3.4.3 Message contents

**Table 16.2.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.3.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.3.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.3.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |

| | | | |
|---|---|---|---|
| | | Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.3.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 16.2.3.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.3.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.3.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.3.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.3.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.3.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.3.5 Test requirement

Table 16.2.3.5-1 and Table 16.2.3.5-2 define the primary level settings including the test tolerances for the test.

**Table 16.2.3.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (Pcell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | Config 1 | GP#13 or GP#24[Note1] | As specified in TS 38.133 [50] clause 9.1.2-1. |
| Measurement gap offset | | Config 1 | 39 | |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | | Config 1 | OFF | DRX is not used |

| | | | | |
|---|---|---|---|---|
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| Number of cells provided in assistance data | | Config 1 | 4 | Including the reference cell |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 16.2.3.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting configuration | | | Config 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -99.1 | | -99.1 | |
| $N_{oc}$ Note2 | dBm/SCS Note4 | Config 1 | -90.1 | | -90.1 | |
| SSB_RP Note 3 | dBm/SCS Note5 | Config 1 | -94.6 | -94.6 | -Infinity | -101.8 |
| PRP Note 3 | dBm/SCS Note5 | Config 1 | -Infinity | -92.1 | -Infinity | -100.1 |
| PRS $\hat{E}_s/I_{ot}$ | dB | Config 1 | -Infinity | -2.7 | -Infinity | -11.2.3 |
| PRS $\hat{E}_s/N_{oc}$ | dB | Config 1 | -Infinity | -2 | -Infinity | -10 |
| SSB $\hat{E}_s/N_{oc}$ | dB | Config 1 | -1.6 | -1.6 | -Infinity | -8.8 |
| Io Note3 | dBm/95.04 MHz Note5 | Config 1 | -58.07 | -55.69 | -58.07 | -55.69 |
| | | | | | | |
| Propagation Condition | | Config 1 | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: SSB_RP/PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone.
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The PRS RSRP measurement time fulfils the requirements specified in TS 38.133 [50] Clause 9.9.3.5.The UE shall perform and report the PRS RSRP measurements for both Cell1 and Cell within the time duration specified in TS 38.133 [50] section 9.9.3.5 starting from the beginning of time interval T2.

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRP measurement for each correct event shall be within the PRS RSRP reporting range specified in TS 38.133 [50] Clause 10.1.24.3, i.e., between PRS RSRP_0 and PRS RSRP_126.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRP measurement period plus ΔT, where ΔT = 50ms. The PRS-RSRP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( CSSF_i * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.142$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.178s and 164.05s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 16.2.3.4.3-3. The LPP time IE ranges between 6s and170s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 170.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.2.3.5-3, Table 16.2.3.5-4 and Table 16.2.3.5-5 for reference.

**Table 16.2.3.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 16.2.3.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.142}{N} \right\rceil$ |
|---|---|
| nDot125 | 2 |
| >= nDot25 | 1 |

**Table 16.2.3.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.2.4 PRS-RSRP measurement period test case for dual positioning frequency layer in FR2 SA

### 16.2.4.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] Clause 9.9.3.5. The test is conducted in AWGN propagation condition in FR2 in standalone scenario when dual positioning frequency layers are configured.

### 16.2.4.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-AoD positioning and PRS-RSRP measurements in FR2.

### 16.2.4.3 Minimum conformance requirements

Same as in clause 14.2.1.3.

### 16.2.4.4 Test description

The supported test configurations in listed in Table 16.2.4.4-1.

**Table 16.2.4.4-1: Supported test configurations for PRS RSRP measurement for FR2**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

#### 16.2.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Low Range for Cell 1 and High Range for Cell 2, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.2.4.5-1 and Table 16.2.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.2.4.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the different RF channel in FR2.

#### 16.2.4.4.2 Test procedure

Same as defined in clause 16.2.3.4.2.

#### 16.2.4.4.3 Message contents

**Table 16.2.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.4.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |

| | | | |
|---|---|---|---|
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.4.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 16.2.4.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.4.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.4.4.3-8 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.4.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.4.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.4.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.4.5 Test requirement

Table 16.2.4.5-1 and Table 16.2.4.5-2 define the primary level settings including the test tolerances for the test.

**Table 16.2.4.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | Config 1 | GP#13 or GP#24[Note1] | As specified in TS 38.133 [50] clause 9.1.2-1. |
| Measurement gap offset | | Config 1 | 39 | |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | | Config 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | | Config 1 | 3µs | Synchronous cells |
| Expected RSTD | µs | Config 1 | 3 | |
| Expected RSTD uncertainty | µs | Config 1 | 5 | |
| Number of cells provided in assistance data | | Config 1 | 4 | Including the reference cell |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 16.2.4.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | Config 1 | Rough | | Rough | |
| NR RF Channel Number | | Config 1 | 1 | | 2 | |
| TDD configuration | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting configuration | | | Config 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz Note5 | | -98.7 | | -98 | |
| $N_{oc}$ Note2 | | dBm/SCS Note4 | Config 1 | -89.7 | | -89 | |
| SSB_RP Note 3 | | dBm/SCS Note5 | Config 1 | -92.7 | -92.7 | -Infinity | -85.7 |
| PRP Note 3 | | dBm/SCS Note5 | Config 1 | -Infinity | -92.1 | -Infinity | -101.4 |
| PRS $\hat{E}_s / I_{ot}$ | | dB | Config 1 | -Infinity | -2.7 | -Infinity | -12.7 |
| PRS $\hat{E}_s / N_{oc}$ | | dB | Config 1 | -Infinity | -2.43 | -Infinity | -12.47 |
| SSB $\hat{E}_s / N_{oc}$ | | dB | Config 1 | -1.7 | -1.7 | -Infinity | -6 |
| Io Note3 | | dBm/19 0.08 MHz Note5 | Config 1 | -57.67 | -55.71 | -56.97 | -56.73 |
| Propagation Condition | | | Config 1 | AWGN | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | SSB_RPPRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 6: | As observed with 0 dBi gain antenna at the centre of the quiet zone |
| Note 7: | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

The PRS RSRP measurement time fulfils the requirements specified in TS 38.133 [50] Clause 9.9.3.5.The UE shall perform and report the PRS RSRP measurements for both Cell1 and Cell 2 within the time duration specified in TS 38.133 [50] section 9.9.3.5 starting from the beginning of time interval T2.

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRP measurement for each correct event shall be within the PRS RSRP reporting range specified in TS 38.133 [50] Clause 10.1.24.3, i.e., between PRS RSRP_0 and PRS RSRP_126.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRP measurement period plus ΔT, where ΔT = 50ms. The PRS-RSRP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( CSSF_i * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

Where $CSSF_{PRS,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 1$, $L_{available_PRS,i} = 0.071$, $N_{sample} = 4$. N is the parameter durationOfPRS-ProcessingSysmbols from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{1}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 10.466s and 83.57s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 16.2.4.4.3-3. The LPP time IE ranges between 11s and 84s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 11.3s and 84.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.2.4.5-3, Table 16.2.4.5-4 and Table 16.2.4.5-5 for reference.

**Table 16.2.4.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{1}{N'} \right\rceil$ |
|---|---|
| >=n1 | 1 |

**Table 16.2.4.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\left\lceil \frac{0.071}{N} \right\rceil$ |
|---|---|
| >= nDot25 | 1 |

**Table 16.2.4.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.2.5 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR1 SA

### 16.2.5.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements for reduced number of samples specified in TS 38.133 [50] Clause 9.9.3.5 for single positioning frequency layer under AWGN propagation conditions in FR1 in standalone scenario.

### 16.2.5.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 16.2.5.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.3.3) PRS-RSRP measurements, defined in TS 38.215 [4], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,\ total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max\left(T_{effect,i}\right)$$

where

*i* is the index of positioning frequency layer,

L is total number of positioning frequency layers,

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left(CSSF_i * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last}$$

where

$CSSF_i$ is the carrier specific scaling factor for PRS-RSRP measurements specified in clause 9.1.5.2,

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],

N' is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of PRS-RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time, if not all PRS resources to be measured are available in the same measurement gap occasion during $T_{available_PRS,i}$, otherwise $T_{last,i} = T_i + MGL$,

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer $i$,

$T_i$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ is the least common multiple between $T_{PRS,i}$ and $MGRP_i$ ,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$MGRP_i$ is the measurement gap repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

## 16.2.5.4 Test description

### 16.2.5.4.1 Initial conditions

The supported test configurations in listed in Table 16.2.5.4-1.

**Table 16.2.5.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.2.5.5-1 and Table 16.2.5.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.2.5.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.2.5.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2. All two cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.2.5.5-1 and Table 16.2.5.5-2. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 16.2.5.5-2.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for Cell 2 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. Repeat steps 2-11 in Table 16.2.5.4-1 until the confidence level according to Annex D is achieved.

### 16.2.5.4.3 Message contents

**Table 16.2.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.5.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.5.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.2.5.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.5.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | N4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.5.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | As specified in Table 16.2.5.5-1 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | As specified in Table 16.2.5.5-1 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.5.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.5.4.3-5: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 &2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| N4-r16 | 0 | | Cell 1 |
| | 1 | | Cell 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.5.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |

| | | | |
|---|---|---|---|
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.5.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |

| | | | |
|---|---|---|---|
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.5.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.5.5 Test requirement

Table 16.2.5.5-1 and Table 16.2.5.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.5.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |

| | | | | |
|---|---|---|---|---|
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | NA | OFF |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 5 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 16.2.5.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| | | | | | | |
| | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0 | -Infinity | -6 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 2.92 | -Infinity | -0.93 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -95.08 | -Infinity | -98.93 |
| | | 2 | -Infinity | -95.08 | -Infinity | -98.93 |
| | | 3 | -Infinity | -92.07 | -Infinity | -95.92 |
| Io | dBm/19.08 MHz | 1 | N/A | -61.20 | N/A | -61.20 |
| | dBm/19.08 MHz | 2 | | -61.20 | | -61.20 |
| | dBm/47.88 MHz | 3 | | -57.20 | | -57.20 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit for reduced number of samples specified in clause 9.9.3.5, starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of correct events observed during repeated tests shall be at least 90%.

$$T_{PRS-RSRP,i} = \left( CSSF_i * ceil( K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

Analysing the equation for $T_{PRS-RSRP,i}$

$CSSF_{PRS,i} = 1$

$N_{RxBeam,i}$ = 1

$N_{PRS,i}^{slot} = 4$

$N_{sample}$ = 1 (For reduced sample number)

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ and $T_{last,i} = T_i + T_{available_PRS,i}$

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.168s and 51.3s.

For SCS 30KHz The LPP time IE ranges between 0.168s and 25.7s

The LPP time IE ranges between 1s and 52s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 26.3

The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation.

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.7.5-4, Table 15.2.7.5-5 , Table 15.2.7.5-6 and Table 15.2.7.5-7 for reference.

**Table 16.2.5.5-3: $N'$ based on maxNumOfDL-PRS-ResProcessedPerSlot**

| N' | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 16.2.5.5-4: N based on *durationOfPRS-ProcessingSysmbols for SCS 15KHz***

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 16.2.5.5-5: N based on *durationOfPRS-ProcessingSysmbols for SCS 30KHz***

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 16.2.5.5-6: Teffect and Tlast based on durationOfPRS − ProcessingSymbolsInEveryTms**

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.2.6 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA

### 16.2.6.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement outside MG requirements specified in TS 38.133 [50] Clause 9.9.3.6 for single positioning frequency layer under AWGN propagation conditions in FR1 in standalone scenario.

### 16.2.6.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and prs-Processing Window.

### 16.2.6.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.3.3) PRS-RSRP measurements as defined in TS 38.215 [4] without measurement gap, on configured positioning frequency layer *i*, within $T_{PRS-RSRP,total}$ ms.

$T_{PRS-RSRP,\ total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max(T_{effect,i})$ if any of the positioning frequency layers are in Case 1, or

$T_{PRS-RSRP,\ total} = \max(T_{PRS-RSRP,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

Where,

$i$ is the index of positioning frequency layer, and

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i, and

$T_{uncertainty,i}$ is the time from the start of the first PPW occasion for positioning frequency layer i to the start of measurement period $T_{PRS-RSRP,\ total}$.

A positioning frequency layer is in Case 1 if UE reports *ppw-durationOfPRS-Processing1-r17* for the band containing the positioning frequency layer, and a positioning frequency layer is in Case 2 if UE reports *ppw-durationOfPRS-Processing2-r17* for the band containing the positioning frequency layer.

$T_{PRS-RSRP,i}$ is the measurement period for PRS-RSRP measurement in positioning frequency layer *i* as specified below.

$$T_{PRS-RSRP,i} = \left(N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last}$$

where

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1. If positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$,

only unmuted PRS resource instances that meet the applicability conditions and fully or partially overlapped with PRS processing window are considered, if positioning frequency layer i is in Case 1, or

only the PRS resources unmuted and fully or partially overlapped with the first (PPWL-T2) ms of PPW are considered, if positioning frequency layer i is in Case 2, where PPWL is the PPW length and T2 corresponds to *ppw-durationOfPRS-ProcessingSymbolsT2*.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *ppw-durationOfPRS-ProcessingSymbolsN* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponding to *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

N′ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of PRS-RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time.

If positioning frequency layer *i* is in Case 1 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i} = T_i$ +PPWL, else

if positioning frequency layer *i* is in Case 2 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i}$ = PPWL;

otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$.

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,

$T_i$ corresponds to *ppw-durationOfPRS-ProcessingSymbolsT* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponds to the sum of *ppw-durationOfPRS-ProcessingSymbolsT2* and *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$ is the least common multiple between $T_{PRS,i}$ and $PPWRP_i$,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$PPWRP_i$ is the PRS processing window repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. If bitmap $\{b^1\}$ for higher-layer parameter *DL-*$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

## 16.2.6.4 Test description

### 16.2.6.4.1 Initial conditions

The supported test configurations in listed in Table 16.2.6.4-1.

**Table 16.2.6.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.6.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.2.6.5-1, Table 16.2.6.5-2 and Table 16.2.6.5-3.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.2.6.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.2.6.4.2 Test procedure

The test consists of two sub-tests; Sub-test 1 is for Nsample = 1 and sub-test 2 is for Nsample = 4. Sub-test 2 is only conducted when UE also supporting supportedDL-PRS-ProcessingSamples-RRC-CONNECTED. The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2. All two cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first PRS processing window instance containing the PRS resources.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.2.6.5-1, Table 16.2.6.5-2 and Table 16.2.6.5-3. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.

5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
7. The SS shall transmit an RRCReconfiguration message with the PPW configuration.
8. The UE shall transmit RRCReconfigurationComplete message.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
11. The SS sends a Positioning Measurement Gap Activation command containing an ID to activate the associated measurement gap
12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 16.2.6.5-3.
13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for Cell 2 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.
14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
15. Repeat steps 2-14 in Table 16.2.6.4-1 until the confidence level according to Annex D is achieved.

### 16.2.6.4.3 Message contents

**Table 16.2.6.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.6.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.6.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | OCTET STRING (CONTAINING CellGroupConfig) | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.6.4.3-4: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 CHOICE { | The periodicity is 160ms and the starting slot offset is 11 ms for any SCS configuration | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| } | | | |
| length-r17 | 80 | | |
| type-r17 | type1A | | |
| priority-r17 | st1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.6.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |

| | | | |
|---|---|---|---|
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.2.6.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.6.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |

| | | | |
|---|---|---|---|
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.6.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | As specified in Table 16.2.6.5-1 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | As specified in Table 16.2.6.5-1 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.6.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.6.4.3-7: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 &2 |
| n160-r16 | 80 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| N4-r16 | 0 | | Cell 1 |
| | 1 | | Cell 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.6.4.3-8: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.6.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the | |

| | | | |
|---|---|---|---|
| | | response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.6.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.6.5 Test requirement

Table 16.2.6.5-1 and Table 16.2.6.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.6.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| PPW configuration | | 1 | PPW.1 | As defined in A.3.33 |
| | | 2 | PPW.1 | As defined in A.3.33 |
| | | 3 | PPW.1 | As defined in A.3.33 |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | NA | OFF |
| Time offset between serving and neighbour cells | µs | 1 | 3 | |
| | | 2 | 3 | |
| | | 3 | 3 or 2.34 Note 1 | |
| Expected RSTD | µs | 1, 2, 3 | 0 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | Same as time offset between serving and neighbour cells | |
| Number of cells provided in assistance data | | 1, 2, 3 | 4 | Including the reference cell |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 5 | |
| NOTE 1: If UE indicates support of CP length for the receive time difference threshold, the time offset is set to 2.34us, otherwise 3us. | | | | |

**Table 16.2.6.5-2: Cell-specific test parameters during T2 for PRS RSRP measurement reporting delay**

| Parameter | Unit | | Cell 1 | Cell 2 |
|---|---|---|---|---|

| | | Test configuration | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 |
|---|---|---|---|---|---|---|
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | 0 | -2.41 | -6 | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | 2.92 | -2 | -0.93 | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -95.08 | -100 | -98.93 | -108 |
| | | 2 | -95.08 | -100 | -98.93 | -108 |
| | | 3 | -92.07 | -97 | -95.92 | -105 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -101 | -101 | -104 | -108 |
| | | 2 | -101 | -101 | -104 | -108 |
| | | 3 | -98 | -98 | -101 | -105 |
| Io | dBm/19.08 MHz | 1 | -61.20 | -64.57 | -61.20 | -64.57 |
| | dBm/19.08 MHz | 2 | -61.20 | -64.57 | -61.20 | -64.57 |
| | dBm/47.88 MHz | 3 | -57.20 | -60.58 | -57.20 | -60.58 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit specified in clause 9.9.3.6, starting from the beginning of time interval T2, with Nsample=1 for sub-test 1 and Nsample=4 for sub-test 2.

NOTE: The actual overall delays measured in the test may be up to 2x$TTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

$T_{PRS-RSRP,\,total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max(T_{effect,i})$ if any of the positioning frequency layers are in Case 1, or

$T_{PRS-RSRP,\,total} = \max(T_{PRS-RSRP,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

$$T_{PRS-RSRP,i} = \left(N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last}$$

$N_{RxBeam,i} = 1$

$k_{multiTEG,i}=1$

$N_{PRS,i}^{slot} = 4$

$N_{sample}= 4.$

$T_{last,i} = T_i + PPWL$

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$; where Tprs = 160 ms, and the repetition periodicity $PPWRP_i$=160ms. Therefore, $T_{available_PRS,i} = 160$

Solving for Case 1 Nsamsple=1:

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.011s and 51.2s.

For SCS 30KHz The LPP time IE ranges between 0.011s and 25.6s.

The LPP time IE ranges between 1s and 52s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 52.3

Solving for Case 1 Nsamsple=4:

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.491s and 204.4s.

For SCS 30KHz The LPP time IE ranges between 0.491s and 102.4s.

The LPP time IE ranges between 1s and 205s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 205.3

Solving for Case 2 Nsamsple=1:

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.010s and 6.25s.

For SCS 30KHz The LPP time IE ranges between 0.010s and 3.05s.

The LPP time IE ranges between 1s and 7s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 7.3.

Solving for Case 2 Nsamsple=4:

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 0.491s and 25.4s.

For SCS 30KHz The LPP time IE ranges between 0.491s and 12.6s.

The LPP time IE ranges between 1s and 26s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 26.3

The values of N', N and Ti and the effect in the response time equation are defined in Table 15.2.6.5-3, Table 15.2.6.5-4, Table 15.2.6.5-4 and Table 15.2.6.5-6 for reference.

**Table 16.2.6.5-3: value of N' based on *ppw-maxNumOfDL-PRS-ResProcessedPerSlot***

| N' | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 16.2.6.5-4: value of N based on ppw-durationOfPRS-ProcessingSymbolsN for SCS 15KHz**

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 16.2.6.5-5: value of N based on ppw-durationOfPRS-ProcessingSymbolsN for SCS 30KHz**

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 16.2.6.5-6: value of $T_{effect}$ and $T_{last}$ based on ppw-durationOfPRS-ProcessingSymbolsT-r17 for Case 1**

| ppw-durationOfPRS-ProcessingSymbolsT-r17 | Teffect | Tlast |
|---|---|---|
| ms1 | 160 | 11 |
| ms2 | 160 | 12 |
| ms4 | 160 | 14 |
| ms8 | 160 | 18 |
| ms16 | 160 | 26 |
| ms20 | 160 | 30 |
| ms30 | 160 | 40 |
| ms40 | 160 | 50 |
| ms80 | 160 | 90 |
| ms160 | 160 | 170 |
| ms320 | 320 | 330 |
| ms640 | 640 | 650 |
| ms1280 | 1280 | 1290 |

**Table 16.2.6.5-7: value of $T_{effect}$ and $T_{last}$ based on ppw-durationOfPRS-ProcessingSymbolsT2 for Case 2**

| Ti | Teffect | Tlast |
|---|---|---|
| 4.125ms | 160 | 10 |
| 20ms | 160 | 10 |

The rate of correct events observed during repeated tests shall be at least 90%.

## 16.2.7 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

### 16.2.7.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements for reduced number of samples specified in TS 38.133 [50] Clause 9.9.3.5 for single positioning frequency layer under AWGN propagation conditions in FR2 in standalone scenario.

### 16.2.7.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 16.2.7.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.3.3) PRS-RSRP measurements, defined in TS 38.215 [4], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,\ total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max(T_{effect,i})$$

where

*i* is the index of positioning frequency layer,

L is total number of positioning frequency layers,

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left( CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

where

$CSSF_i$ is the carrier specific scaling factor for PRS-RSRP measurements specified in clause 9.1.5.2,

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,\mathrm{i}}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,\mathrm{i}}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$\mathrm{N}_{\mathrm{PRS,i}}^{\mathrm{slot}}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],

N′ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of PRS-RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time, if not all PRS resources to be measured are available in the same measurement gap occasion during $T_{available_PRS,\mathrm{i}}$, otherwise $\mathrm{T_{last,i}} = T_{\mathrm{i}} + \mathrm{MGL}$,

$\mathrm{T_{effect,i}} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,

$\mathrm{T_i}$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34],

$T_{available_PRS,\mathrm{i}} = LCM\left(T_{PRS,\mathrm{i}}, MGRP_{\mathrm{i}}\right)$ is the least common multiple between $T_{PRS,\mathrm{i}}$ and $MGRP_{\mathrm{i}}$ ,

$\mathrm{T_{PRS,i}}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$MGRP_{\mathrm{i}}$ is the measurement gap repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $\mathrm{T_{PRS,i}}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

## 16.2.7.4 Test description

### 16.2.7.4.1 Initial conditions

The supported test configurations in listed in Table 16.2.7.4-1.

**Table 16.2.7.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.7.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.2.7.5-1 and Table 16.2.7.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.2.7.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.2.7.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2. All two cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE ΔT ms before the start of T2, where ΔT = 50 ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.2.7.5-1 and Table 16.2.7.5-2. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.

9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 16.2.7.5-2.

10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for Cell 2 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.

11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

12. Repeat steps 2-11 in Table 16.2.7.4-1 until the confidence level according to Annex D is achieved.

### 16.2.7.4.3 Message contents

**Table 16.2.7.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.7.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.7.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.2.7.4.3-4 | | |

| | | | |
|---|---|---|---|
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.7.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 7 | 48 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.7.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.7.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.7.4.3-5: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.7.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.7.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |

| | | | |
|---|---|---|---|
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.7.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 16.2.7.5 Test requirement

Table 16.2.7.5-1 and Table 16.2.7.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.7.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | Config 1 | GP#13 or GP#24[Note1] | As specified in TS 38.133 [50] clause 9.1.2-1. |
| Measurement gap offset | | Config 1 | 39 | |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | | Config 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| Number of cells provided in assistance data | | Config 1 | 4 | Including the reference cell |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 16.2.7.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS BW | | | Config 1 | 48 PRBs | | 48 PRBs | |
| PRS muting configuration | | | Config 1 | ‘10’ | | ‘01’ | |
| EPRE ratio of PSS to SSS | | dB | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz Note5 | | -103 | | -103 | |
| $N_{oc}$ Note2 | | dBm/SCS Note4 | Config 1 | -94 | | -94 | |
| PRS-RSRP Note 3 | | dBm/SCS Note5 | Config 1 | -Infinity | -90.4 | -Infinity | -94.2 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | Config 1 | -Infinity | 0.3 | -Infinity | -5.6 |
| PRS $\hat{E}_s/N_{oc}$ | | dB | Config 1 | -Infinity | 3.56 | -Infinity | -0.28 |
| Io Note3 | | dBm/19 0.08 MHz Note5 | Config 1 | -55.73 | | -55.73 | |
| Propagation Condition | | | Config 1 | AWGN | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 6: | As observed with 0 dBi gain antenna at the centre of the quiet zone |
| Note 7: | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |
| Note 8: | Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. |

The PRS RSRP measurement time fulfils the requirements specified in clause 16.2.7.3.

The UE shall perform and report the PRS RSRP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified starting from the beginning of time interval T2.

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRP measurement for each correct event shall be within the PRS RSRP reporting range specified in Clause 10.1.24.3, i.e., between PRS RSRP_0 and PRS RSRP_126.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The PRS RSRP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

Where $CSSF_{PRS,i} = 1, N_{RxBeam,i} = 8, N_{PRS,i}^{slot} = 2, L_{available_PRS,i} = 0.071\ ms, N_{sample} = 1$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ ; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 1 - 1\right)$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 1.288s and 20.64s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 16.2.7.4.3-3. The LPP time IE ranges between 2s and 21s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 21.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.2.7.5-3, Table 16.2.7.5-4 and Table 16.2.7.5-5 for reference.

**Table 16.2.7.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 16.2.7.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\lceil \frac{0.071}{N} \rceil$ |
|---|---|
| >=nDot125 | 1 |

**Table 16.2.7.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.2.8 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

### 16.2.8.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement outside MG requirements specified in TS 38.133 [50] Clause 9.9.3.6 for single positioning frequency layer under AWGN propagation conditions in FR2 in standalone scenario.

### 16.2.8.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and prs-Processing Window.

### 16.2.8.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.3.3) PRS-RSRP measurements as defined in TS 38.215 [4] without measurement gap, on configured positioning frequency layer *i*, within $T_{PRS-RSRP,total}$ ms.

$T_{PRS-RSRP,\,total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max(T_{effect,i})$ if any of the positioning frequency layers are in Case 1, or

$T_{PRS-RSRP,\,total} = \max(T_{PRS-RSRP,i} + T_{uncertainty,i})$, if all the positioning frequency layers are in Case 2,

Where,

$i$ is the index of positioning frequency layer, and

$L$ is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i, and

$T_{uncertainty,i}$ is the time from the start of the first PPW occasion for positioning frequency layer i to the start of measurement period $T_{PRS-RSRP,\ total}$.

A positioning frequency layer is in Case 1 if UE reports *ppw-durationOfPRS-Processing1-r17* for the band containing the positioning frequency layer, and a positioning frequency layer is in Case 2 if UE reports *ppw-durationOfPRS-Processing2-r17* for the band containing the positioning frequency layer.

$T_{PRS-RSRP,i}$ is the measurement period for PRS-RSRP measurement in positioning frequency layer *i* as specified below.

$$T_{PRS-RSRP,i} = \left(N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last}$$

where

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1. If positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$,

only unmuted PRS resource instances that meet the applicability conditions and fully or partially overlapped with PRS processing window are considered, if positioning frequency layer i is in Case 1, or

only the PRS resources unmuted and fully or partially overlapped with the first (PPWL-T2) ms of PPW are considered, if positioning frequency layer i is in Case 2, where PPWL is the PPW length and T2 corresponds to *ppw-durationOfPRS-ProcessingSymbolsT2*.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *ppw-durationOfPRS-ProcessingSymbolsN* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponding to *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2,

N′ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of PRS-RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and

- Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

- $N_{sample}$ = 4 otherwise.

$T_{last}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time.

If positioning frequency layer *i* is in Case 1 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i} = T_i$ +PPWL, else

if positioning frequency layer *i* is in Case 2 and all of the PRS resources to be measured are available in the same PPW occasion during $T_{available}$, then $T_{last,i}$ = PPWL;

otherwise, $T_{last,i} = T_i + T_{available_PRS,i}$.

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,

$T_i$ corresponds to *ppw-durationOfPRS-ProcessingSymbolsT* in TS 37.355 [34] if positioning frequency layer *i* is in Case 1, or corresponds to the sum of *ppw-durationOfPRS-ProcessingSymbolsT2* and *ppw-durationOfPRS-ProcessingSymbolsN2* in TS 37.355 [34] if positioning frequency layer *i* is in Case 2

$T_{available_PRS,i} = LCM(T_{PRS,i}, PPWRP_i)$ is the least common multiple between $T_{PRS,i}$ and $PPWRP_i$,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$PPWRP_i$ is the PRS processing window repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. If bitmap $\{b^1\}$ for higher-layer parameter *DL-*
$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

## 16.2.8.4 Test description

### 16.2.8.4.1 Initial conditions

The supported test configurations in listed in Table 16.2.8.4-1.

**Table 16.2.8.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.8.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.

2. The general test parameter settings are set up according to Table 16.2.8.5-1, Table 16.2.8.5-2 and Table 16.2.8.5-3.

3. Propagation conditions are set according to clause 4.16.2.

4. Message contents are defined in clause 16.2.8.4.3.

5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.2.8.4.2 Test procedure

The test consists of two sub-tests; Sub-test 1 is for Nsample = 1 and sub-test 2 is for Nsample = 4. Sub-test 2 is only conducted when UE also supporting supportedDL-PRS-ProcessingSamples-RRC-CONNECTED. The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2. All two cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first PRS processing window instance containing the PRS resources.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 16.2.8.5-1, Table 16.2.8.5-2 and Table 16.2.8.5-3. Propagation conditions are set according to clause 4.16.2.

4. T1 starts.

5. The SS shall transmit an LPP REQUEST CAPABILITIES message.

6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.

7. The SS shall transmit an RRCReconfiguration message with the PPW configuration.

8. The UE shall transmit RRCReconfigurationComplete message.

9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.

11. The SS sends a Positioning Measurement Gap Activation command containing an ID to activate the associated measurement gap

12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 16.2.8.5-3.

13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for Cell 2 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 2-14 in Table 16.2.8.4-1 until the confidence level according to Annex D is achieved.

### 16.2.8.4.3 Message contents

**Table 16.2.8.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.8.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.8.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | OCTET STRING (CONTAINING CellGroupConfig) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.8.4.3-4: CellGroupConfig (RRCReconfiguration, Table 16.2.8.4.3-3)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 CHOICE { | The periodicity is 160ms and the starting slot offset is 11 ms for any SCS configuration | | |
| scs120 CHOICE { | | | |
| n1280 | 4 | | |
| } | | | |
| } | | | |
| length-r17 | 80 | | |
| type-r17 | type1A | | |
| priority-r17 | st1 | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.8.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.2.8.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.8.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.8.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | About 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 17 | About 0.58 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.8.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.8.4.3-7: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.8.4.3-8: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.8.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.8.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.8.5 Test requirement

Table 16.2.8.5-1 and Table 16.2.8.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.8.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| PPW configuration | | Config 1 | PPW.1 | As specified in clause A.3.33 |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| CP length | | Config 1 | Normal | |
| DRX | | Config 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | μs | Config 1 | 0.58, 2.0 or 3 Note 1 | |

| | | | | |
|---|---|---|---|---|
| Expected RSTD | µs | Config 1 | 0 | |
| Expected RSTD uncertainty | µs | Config 1 | Same as time offset between serving and neighbour cells | |
| Number of cells provided in assistance data | | Config 1 | 4 | Including the reference cell |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |
| NOTE 1: If UE indicates support of CP length for the receive time difference threshold, the time offset is set to 0.58us; If UE indicates support of 1/4 symbol length for the receive time difference threshold, the time offset is set to 2.0us, otherwise 3us. | | | | |

**Table 16.2.8.5-2: Cell-specific test parameters during T2 for PRS RSRP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.4 FR2 | | PRS.1.4 FR2 | |
| PRS muting configuration | | | Config 1 | ‘10’ | | ‘01’ | |
| EPRE ratio of PSS to SSS | | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -103 | -99.1 | -103 | -99.1 |
| $N_{oc}$ Note2 | dBm/SCS Note4 | Config 1 | -94 | -90.1 | -94 | -90.1 |
| SS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -96 | -96 | -99 | -103 |
| PRS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -90.4 | -92.1 | -94.2 | -100.1 |
| PRS $\hat{E}_s/I_{ot}$ | dB | Config 1 | 0.3 | -2.7 | -5.6 | -12.3 |
| PRS $\hat{E}_s/N_{oc}$ | dB | Config 1 | 3.56 | -2 | -0.28 | -10 |
| Io Note3 | dBm/95.04 MHz Note5 | Config 1 | -55.73 | -55.69 | -55.73 | -55.69 |
| Propagation Condition | | Config 1 | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation
Note 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The PRS RSRP measurement time fulfils the requirements specified in Clause16.2.8.3, with Nsample=1 for sub-test 1 and Nsample=4 for sub-test 2. The UE shall perform and report the PRS RSRP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration starting from the beginning of time interval T2.

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRP measurement for each correct event shall be within the PRS RSRP reporting range specified in Clause 10.1.24.3, i.e., between PRS RSRP_0 and PRS RSRP_126.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The LPP response time calculation procedure could be divided into two situations based on whether the UE support ppw-durationOfPRS-Processing1-r17 or ppw-durationOfPRS-Processing2-r17.The PRS-RSRP measurement period follows the equation:

$T_{\mathrm{PRS-RSRP,\,total}} = \sum_{i=1}^{L} T_{\mathrm{PRS-RSRP,i}} + (L-1) * \max\left(T_{\mathrm{effect},i}\right)$ if any of the positioning frequency layers are in Case 1, or

$T_{\mathrm{PRS-RSRP,\,total}} = \max\left(T_{\mathrm{PRS-RSRP,i}} + T_{\mathrm{uncertainty,i}}\right)$, if all the positioning frequency layers are in Case 2,

Where $N_{RxBeam,i}$ = 8, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,\mathrm{i}} = 0.071$, $N_{sample}$ = 1 for sub-test 1, $N_{sample}$ = 4 for sub-test 2, $T_{uncertainty,i} = 0$. N is the parameter *ppw-durationOfPRS-ProcessingSymbolsN* from TS 37.355 [49], N' is the parameter *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{\mathrm{last,i}} = T_{\mathrm{i}}$ +PPWL and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,\mathrm{i}} = LCM(T_{PRS,\mathrm{i}}, PPWRP_{\mathrm{i}})$; where Tprs = 160 ms, and the repetition periodicity $\mathrm{PPWRP_i}$=160ms. Therefore, $T_{available_PRS,i} = 160$

$T_i$ depends on the UE parameter *ppw-durationOfPRS-ProcessingSymbolsT* from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * N_{sample} - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 2s and 21s for sub-test 1. The LPP time IE ranges between 6s and 80s for sub-test 2. The value of the LPP time IE is rounded up to the next second. The result is transmitted in the response time IE in the LPP-RequestLocationInformation in Table 16.2.8.4.3-8.

The test tolerance for the response time is 300ms. Therefore, the response time ranges for sub-test 1 is between 2.3s and 21.3s. The response time ranges for sub-test 2 is between 6.3s and 80.3s.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.2.9 PRS-RSRP measurement delay in FR1 for Redcap UE without Rx FH

Editor's Note: This test case is incomplete in following aspects:

- TT analysis is missing.

### 16.2.9.1 Test purpose

The purpose of the test is to verify that the PRS-RSRP measurement meets the delay requirements specified in TS 38.133 [50] clause 9.9A.3.5 in an environment with AWGN propagation conditions. The test is applicable to 1 Rx or 2 Rx RedCap UE.

### 16.2.9.2 Test applicability

This test applies to FR1 Redcap UEs release 18 onwards that supports DL-AoD positioning.

### 16.2.9.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the RedCap UE shall be able to measure multiple (up to the RedCapUE capability specified in TS 38.133[50] clause 9.9A.3.3) PRS-RSRP measurements, defined in TS 38.215 [57], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $\mathrm{T_{PRS-RSRP,total}}$ ms.

$$\mathrm{T_{PRS-RSRP,total}} = \sum_{i=1}^{L} \mathrm{T_{PRS-RSRP,i}} + (L-1) * \max(\mathrm{T_{effect,}}_i)$$

where

*i* is the index of positioning frequency layer,

L is total number of positioning frequency layers,

$\mathrm{T_{effect,}}_i$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$\mathrm{T_{PRS-RSRP,i}} = \left(\mathrm{CSSF_{PRS_RedCap,i}} * \mathrm{ceil}(\mathrm{K_{p,PRS,i}}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,\mathrm{i}}}{N} \right\rceil * N_{sample} - 1\right) * \mathrm{T_{effect,i}} + \mathrm{T_{last}}$$

where

$CSSF_{PRS_RedCap,i}$ is the carrier specific scaling factor for PRS-RSRP measurements specified in TS 38.133[50] clause 9.1A.5.2,

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioining frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 (for 2 Rx or 1 Rx RedCap UE) and if positioning frequency layer *i* is in FR2 (for 2 Rx RedCap UE only), $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],

N′ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of PRS-RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbor cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time, if not all PRS resources to be measured are available in the same measurement gap occasion during $T_{available_PRS,i}$, otherwise $T_{last,i} = T_i + MGL$,

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer $i$,

$T_i$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ is the least common multiple between $T_{PRS,i}$ and $MGRP_i$ ,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$MGRP_i$ is the measurement gap repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

NOTE: For the purpose of calculating $T_{PRS,i}$, only the PRS resources fully or partially covered by the MG are considered.

When PRS-RSRP measurements are configured for DL-AoD, except for deferred MT-LR as defined in clause 4.1a.5 TS 23.273 [44], the time $T_{PRS-RSRP,total}$ starts from the first MG instance aligned with DL PRS resources in the assistance data after both the *NR-DL-AoD-RequestLocationInformation* message and *NR-DL-AoD-ProvideAssistanceData* message from LMF via LPP [34] are delivered to the physical layer of UE.

For deferred MT-LR with other event than “Periodic Location” as defined in clause 4.1a.5.1 TS 23.273 [44], the time $T_{PRS-RSRP,total}$ starts from the first MG instance aligned with a DL PRS resource(s) in the assistance data after the associated event(s) occurs.

For deferred MT-LR with event “Periodic Location” as defined in clause 4.1a.5.1 TS 23.273 [44], the UE shall perform the PRS-RSRP measurement in each reporting period and activate the location report at the time when the periodic timer expires.

Note: No per-positioning frequency layer requirement is applied in scenarios when multiple positioning frequency layers are configured.

When the PRS-RSRP measurement is configured together with RSTD measurement then the PRS-RSRP measurement shall meet the RSTD measurement requirements defined in clause 9.9.2.

When the PRS-RSRP measurement is configured together with UE Rx-Tx time difference measurement then the PRS-RSRP measurement shall meet the UE Rx-Tx time difference measurement requirements defined in clause 9.9.4.

If CSSF changes during the measurement period, the measurement period could be longer.

The measurement requirements do not apply for a PRS resource:

if the PRS resource is across two sampling duration of N within duration $L_{available_PRS,i}$ or

if time span of the PRS resource instance (including at least the minimum number of repetitions specified in the accuracy requirements) is greater than UE reported capability N.

If during the measurement period of one or more positioning frequency layers, the MG pattern is reconfigured either per UE request or not per UE request, the measurement period can be longer.

The requirements in this clause apply, provided no PRS symbols are dropped during the measurement period $T_{PRS-RSRP,total}$ within measurement gaps due to collisions with other signals; otherwise, a longer measurement period may be used.

The requirements in clause 9.9A.3 do not apply if the PRS configuration given by higher layer paramters *NR-DL-PRS-AssistanceData* exceeds any of the UE measurement capabilities given by *NR-DL-PRS-ResourcesCapability* in *NR-DL-AoD-ProvideCapabilities*, and it is up to UE implementation which PRS resources are measured, subject to UE measurement capabilities.

If handover occurs while PRS-RSRP measurements are being performed then the UE shall complete the ongoing PRS-RSRP measurements session. The PRS-RSRP measurement period can be longer. The UE shall meet the PRS-RSRP measurement accuracy requirements in TS 38.133[50] clause 10.1A.17.2.

The normative reference for this requirement is TS 38.133[50] clause 9.9A.3.5.

## 16.2.9.4 Test description

### 16.2.9.4.1 Initial conditions

The test is defined with four possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 16.2.9.4.1-1.

**Table 16.2.9.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 10 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 10 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 4 | 15 kHz SSB SCS, 10 MHz bandwidth, HD-FDD duplex mode |
| NOTE: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.9.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.2.9.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.2.9.4.2.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR1.

### 16.2.9.4.2 Test procedures

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2. Both cells transmit PRS during T2.

The *NR-DL-AoD-RequestLocationInformation* message and *NR-DL-AoD-ProvideAssistanceData* message as defined in TS 37.355 [49] shall be provided to the UE during T1. The last slot containing the two messages for the assistance data and location information request is denoted as #n.

The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources that is $\Delta T$ after slot #n, where $\Delta T$ = 50 ms is the maximum processing time of the assistance data and location information request.

The UE is configured with measurement gap pattern ID # 24 or measurement gap pattern ID # 0 before T2.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.2.9.5-1 and Table 16.2.9.5-2. Propagation conditions are set according to clause 4.15.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 16.2.9.5-2.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.4.1). The UE shall perform and report the PRS-RSRP measurements for both Cell1 and Cell 2.

    If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-RSRP-Result-r16 field then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with the DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.
11. If the UE message at step 10 includes the ackRequested IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. Repeat steps 2-11 until the confidence level according to Annex D is achieved.

### 16.2.9.4.3 Message contents

**Table 16.2.9.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.9.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.9.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

| | | | |
|---|---|---|---|
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.9.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.9.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.2.9.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| nr-IntegrityServiceParameters-r18 | Not present | | |
| nr-IntegrityServiceAlert-r18 | Not present | | |
| nr-IntegrityRiskParameters-r18 | Not present | | |
| nr-IntegrityParametersTRP-LocationInfo-r18 | Not present | | |
| nr-IntegrityParametersDL-PRS-BeamInfo-r18 | Not present | | |
| nr-IntegrityParametersRTD-Info-r18 | Not present | | |
| nr-IntegrityParametersTRP-BeamAntennaInfo-r18 | Not present | | |
| nr-PRU-DL-Info-r18 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.9.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1, Configuration 2 and Configuration 4 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.9.4.3-6 | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.9.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 16.2.9.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1, Configuration 2 and Configuration 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |

| | | | |
|---|---|---|---|
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 2 | | NR Cell 1 |
| | 3 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.9.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | Not checked | | |
| nr-DL-AoD-Error-r16 | Not checked | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not checked | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 16.2.9.5 Test requirement

Table 16.2.9.5-1 and Table 16.2.9.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.9.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3, 4 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3, 4 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3, 4 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1, 2, 4 | 10: $N_{RB,c}$ = 52 | |
| | | 3 | 20: $N_{RB,c}$ = 51 | |
| SSB configuration | | 1, 2, 4 | SSB.1 FR1 | |
| | | 3 | SSB.1 RedCap FR1 | |
| SMTC configuration | | 1, 2, 3, 4 | SMTC.1 RedCap | |
| Measurement gap | | 1, 2, 3, 4 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3, 4 | Normal | |
| DRX | | 1, 2, 3, 4 | N/A | OFF |
| Time offset between serving and neighbour cells | µs | 1, 2, 3, 4 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3, 4 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3, 4 | 5 | |
| T1 | s | 1, 2, 3, 4 | 2 | |
| T2 | s | 1, 2, 3, 4 | 5 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 16.2.9.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1, 4 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1, 4 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1.TDD | | | |
| RMSI CORESET RMC configuration | | 1, 4 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1.TDD | | | |
| Dedicated CORESET RMC configuration | | 1, 4 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3, 4 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3, 4 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1, 4 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.2.1 TDD | | | |
| Initial BWP configuration | | 1, 2, 3, 4 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3, 4 | DLBWP.1.1 RedCap | | N/A | |
| Active UL BWP configuration | | 1, 2, 3, 4 | ULBWP.1.1 RedCap | | N/A | |
| PRS configuration | | 1, 4 | PRS.1.3 FR1 | | PRS.1.3 FR1 | |
| | | 2 | PRS.1.3 FR1 | | PRS.1.3 FR1 | |
| | | 3 | PRS.2.3 FR1 | | PRS.2.3 FR1 | |
| PRS muting info | | 1, 2, 3, 4 | ‘10’ | | ‘01’ | |
| $N_{oc}$ Note 2 | dBm/SCS | 1, 4 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1, 4 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1, 4 | -Infinity | -2.41 +TT | -Infinity | -12.12 +TT |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1, 4 | -Infinity | -2 +TT | -Infinity | -10 +TT |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS kHz | 1, 4 | -Infinity | -100 +TT | -Infinity | -108 +TT |
| | | 2 | -Infinity | -100 +TT | -Infinity | -108 +TT |
| | | 3 | -Infinity | -97 +TT | -Infinity | -105 +TT |
| Io | dBm/9.36 MHz | 1, 4 | N/A | -66.22 +TT | N/A | -66.22 +TT |
| | dBm/9.36 MHz | 2 | | -66.22 +TT | | -66.22 +TT |
| | dBm/18.72 MHz | 3 | | -63.22 +TT | | -63.22 +TT |
| Propagation Condition | | 1, 2, 3, 4 | AWGN | | | |

NOTE 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 3: PRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The PRS-RSRP measurement time without FH for RedCap fulfils the requirements specified in clause 16.2.9.3.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit specified in clause 16.2.9.3, starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time limit above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of correct events observed during repeated tests shall be at least 90%, where the reported PRS-RSRP measurement for each correct event shall be within the reporting range specified in TS 38.133[50] clause 10.1A.17.3.

## 16.2.10 PRS-RSRP measurement delay in FR2 for Redcap UE without Rx FH for single positioning frequency layer

Editor's Note: This test case is incomplete in following aspects:

- TT analysis is missing.

### 16.2.10.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] clause 9.9A.3.5 for single positioning frequency layer under AWGN propagation conditions in FR2 standalone scenario.

### 16.2.10.2 Test applicability

This test applies to FR2 Redcap UEs release 18 onwards that supports DL-AoD positioning.

### 16.2.10.3 Minimum conformance requirements

Same minimum requirements as in clause 16.2.9.3

### 16.2.10.4 Test description

#### 16.2.10.4.1 Initial conditions

The defined test configuration is specified in Table 16.2.10.4.1-1.

**Table 16.2.10.4.1-1: supported test configurations for PRS RSRP measurement for FR2**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 100 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.10.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.2.10.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.2.10.4.2.
5. There are two cells in the test, PCell (Cell 1) and a FR2 neighbour cell (Cell 2) on the same frequency as the PCell.

#### 16.2.10.4.2 Test procedures

Same test procedure as in clause 16.2.9.4.2 with following exceptions:

The UE is configured with measurement gap pattern ID # 24 or measurement gap pattern ID # 13 before T2.

Table 16.2.10.5-1 replaces Table 16.2.9.5-1; Table 16.2.10.5-2 replaces Table 16.2.9.5-2.

### 16.2.10.4.3 Message contents

Same message contents as defined in clause 16.2.9.4.3, with the following exceptions:

Table 16.2.10.4.3-1 replaces Table 16.2.9.4.3-5 and Table 16.2.10.4.3-2 replaces Table 16.2.9.4.3-6.

**Table 16.2.10.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.10.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.10.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 16.2.10.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n120-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.10.5 Test requirement

Table 16.2.10.5-1 and Table 16.2.10.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.10.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | 1 | NR cell 1 (Pcell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | 1 | GP#13 or GP#24[Note1] | As specified in TS 38.133[50] clause 9.1.2-1. |
| Measurement gap offset | | 1 | 39 | |
| SMTC parameters | | 1 | SMTC.1 | As specified in TS 38.133[50] clause A.3.11 |
| SSB parameters | | 1 | SSB.3 FR2 | As specified in TS 38.133[50] clause A.3.10.2 |
| A3-Offset | dB | 1 | -6 | |
| Hysteresis | dB | 1 | 0 | |
| CP length | | 1 | Normal | |
| TimeToTrigger | s | 1 | 0 | |
| Filter coefficient | | 1 | 0 | L3 filtering is not used |
| DRX | | 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | | 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | 1 | 3 | |
| Expected RSTD uncertainty | μs | 1 | 5 | |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 7 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 16.2.10.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | 1 | Setup 1 as specified in TS 38.133[50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | 1 | Rough | | Rough | |
| TDD configuration | | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | 1 | 100: $N_{RB,c}$ = 66 | | 100: $N_{RB,c}$ = 66 | |
| BWP configuration | Initial DL BWP | | 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | 1 | CR.3.1 TDD | | - | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | kHz | 1 | 120 | | 120 | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.1 FR2 | |
| PRS muting configuration | | 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -98 | | -98 | |
| $N_{oc}$ Note2 | dBm/SCS Note4 | 1 | -89 | | -89 | |
| SSB_RP Note 3 | dBm/SCS Note5 | 1 | -91 | -91 | -Infinity | -99 |
| PRP Note 3 | dBm/SCS Note5 | 1 | -Infinity | -91+TT | -Infinity | -99+TT |
| PRS $\hat{E}_s / I_{ot}$ | dB | 1 | -Infinity | -2.41+TT | -Infinity | -12.12+TT |
| PRS $\hat{E}_s / N_{oc}$ | dB | 1 | -Infinity | -2+TT | -Infinity | -10+TT |
| SSB $\hat{E}_s / N_{oc}$ | dB | 1 | -2 | -2 | -Infinity | -10 |
| Io Note3 | dBm/95.04 MHz Note5 | 1 | -57.89 | -57.63+TT | -57.89 | -57.63+TT |
| Propagation Condition | | 1 | AWGN | | | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: SSB_RP/PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
NOTE 6: As observed with 0 dBi gain antenna at the centre of the quiet zone.
NOTE 7: Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation.
NOTE 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

The PRS RSRP measurement time fulfils the requirements specified in claus 16.2.10.3. The UE shall perform and report the PRS RSRP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in clause 9.9.3.5 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to 2x$TTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRP measurement for each correct event shall be within the PRS RSRP reporting range specified in TS 38.133[50] clause 10.1A.17.3.

## 16.2.11 PRS-RSRP measurement delay in FR2 for Redcap UE without Rx FH for dual positioning frequency layer

Editor's Note: This test case is incomplete in following aspects:

- TT analysis is missing.

### 16.2.11.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] clause 9.9A.3.5 for dual positioning frequency layers under AWGN propagation conditions in FR2 standalone scenario.

### 16.2.11.2 Test applicability

This test applies to FR2 Redcap UEs release 18 onwards that supports DL-AoD positioning.

### 16.2.11.3 Minimum conformance requirements

Same minimum requirements as in clause 16.2.9.3

#### 16.2.11.4.1 Initial conditions

The defined test configuration is specified in Table 16.2.11.4.1-1.

**Table 16.2.11.4.1-1: supported test configurations for PRS RSRP measurement for FR2**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 100 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.11.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.2.11.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.2.11.4.2.
5. There are two cells in the test, PCell (Cell 1) and a FR2 neighbour cell (Cell 2) on the same frequency as the PCell.

#### 16.2.11.4.2 Test procedures

Same test procedure as in clause 16.2.9.4.2 with following exceptions:

The UE is configured with measurement gap pattern ID # 24 or measurement gap pattern ID # 13 before T2.

Table 16.2.11.5-1 replaces Table 16.2.9.5-1; Table 16.2.11.5-2 replaces Table 16.2.9.5-2.

### 16.2.11.4.3 Message contents

Same message contents as defined in clause 16.2.9.4.3, with the following exceptions:

Table 16.2.11.4.3-1 replaces Table 16.2.9.4.3-5 and Table 16.2.11.4.3-2 replaces Table 16.2.9.4.3-6.

**Table 16.2.11.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.11.4.3-2 | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 2 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.11.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.11.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n120-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Cell 1 |
| | Not present | | Cell 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | NR Cell 1 |
| n4-r16 | 0 | | NR Cell 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.11.5 Test requirement

Table 16.2.11.5-1 and Table 16.2.11.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.11.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Active cell | | 1 | NR cell 1 (Pcell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | 1 | GP#13 or GP#24[Note1] | As specified in clause 9.1.2-1. |
| Measurement gap offset | | 1 | 39 | |
| SMTC parameters | | 1 | SMTC.1 | As specified in clause A.3.11 |
| SSB parameters | | 1 | SSB.3 FR2 | As specified in clause A.3.10.2 |
| A3-Offset | dB | 1 | -6 | |
| Hysteresis | dB | 1 | 0 | |
| CP length | | 1 | Normal | |
| TimeToTrigger | s | 1 | 0 | |
| Filter coefficient | | 1 | 0 | L3 filtering is not used |
| DRX | | 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | | 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | 1 | 3 | |
| Expected RSTD uncertainty | μs | 1 | 5 | |
| T1 | s | 1 | 5 | |
| T2 | s | 1 | 7 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 16.2.11.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | 1 | Setup 1 as specified in clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | 1 | Rough | | Rough | |
| NR RF Channel Number | | | 1 | 1 | | 2 | |
| TDD configuration | | | 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | 1 | 100: $N_{RB,c}$ = 66 | | 100: $N_{RB,c}$ = 66 | |
| BWP configuration | Initial DL BWP | | 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | kHz | 1 | 120 | | 120 | |
| PRS configuration | | 1 | PRS.1.1 FR2 | | PRS.1.2 FR2 | |
| PRS muting configuration | | 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | dB | 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -98 | | -98 | |
| $N_{oc}$ Note2 | dBm/SCS Note4 | 1 | -89 | | -89 | |
| SSB_RP Note 3 | dBm/SCS Note5 | 1 | -92 | -92 | -Infinity | -102 |
| PRP Note 3 | dBm/SCS Note5 | 1 | -Infinity | -92+TT | -Infinity | -102+TT |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -3+TT | -Infinity | -13+TT |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -3+TT | -Infinity | -13+TT |
| SSB $\hat{E}_s/N_{oc}$ | dB | 1 | -3 | -3 | -Infinity | -13 |
| Io Note3 | dBm/95.04 MHz Note5 | 1 | -58.25 | -58.25 +TT | -60.01 | -59.80+TT |
| Propagation Condition | | 1 | AWGN | | | |

| | |
|---|---|
| NOTE 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| NOTE 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| NOTE 3: | SSB_RP/PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| NOTE 4: | PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| NOTE 5: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone. |
| NOTE 6: | As observed with 0 dBi gain antenna at the centre of the quiet zone. |
| NOTE 7: | Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation. |
| NOTE 8: | Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4. |

The PRS RSRP measurement time fulfils the requirements specified in clause 16.2.11.3.The UE shall perform and report the PRS RSRP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in clause 9.9.3.5 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRP measurement for each correct event shall be within the PRS RSRP reporting range specified in TS 38.133 clause 10.1A.17.3.

## 16.2.12 PRS-RSRP measurement delay in FR1 for Redcap UE with Rx FH

Editor's Note: This test case is incomplete in following aspects:

- TT analysis is missing.

### 16.2.12.1 Test purpose

The purpose of the test is to verify that the PRS-RSRP measurement meets the delay requirements specified in TS 38.133 [50] clause 9.9A.3.6 in an environment with AWGN propagation conditions. The test is applicable to 1 Rx or 2 Rx RedCap UE.

### 16.2.12.2 Test applicability

This test applies to FR1 Redcap UEs release 18 onwards that supports DL-AoD positioning and PRS frequency hopping measurement and report.

### 16.2.12.3 Minimum conformance requirements

When the UE the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [49], requesting RedCap UE to measure PRS-RSRP measurement with FH, the RedCap UE shall be able to measure multiple (up to the RedCapUE capability specified in TS 38.133[50] clause 9.9A.3.3) PRS-RSRP measurements, defined in TS 38.215 [57], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms as defined TS 38.133[50] clause 9.9A.3.5.1 with using the following definition for $N_{sample}$ and $L_{available_PRS,i}$:

- $N_{sample}$ is the number of PRS-RSRP measurement samples, where
  - $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples-RRC-CONNECTED* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples.
  - $N_{sample}$ = 4 otherwise.

- A measurement sample with FH is defined as a PRS measurement over multiple hops within a single measurement gap occasion.

- $L_{available_PRS,i}$ is the time duration of available PRS in the positioning frequency layer i to be measured during a MG, and is calculated by:

$$L_{available_PRS,i} = \sum_{j=1}^{J} N_{hop,i,j} * L_{per-hop,i,j}$$

where,

- $N_{hop,i,j}$ is the number of hops that UE can do in an MG occasion *j* as defined in the following, and

- $L_{per-hop,i,j}$ is the time duration of available PRS resources in the positioning frequency layer *i* in each hop to be measured in MG instance *j* during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{per-hop,i,j}$, only the PRS resources unmuted and fully or partially overlapped with the sampling duration in each hop are considered;

- *J* is the number of MG instance during $T_{available_PRS,i}$.

The sampling window per hop is the first $T_{sample,hop}$ symbols in each hop, where $T_{sample,hop} = T_{hop} - RRT_{FH}$, $T_{hop}$ is the applicable length per hop as defined in table 9.9A.3.6.1-1, and $RRT_{FH}$ is the retuning time between Rx hops as reported by UE in *rf-RxRetuneTimeFR1* or *rf-RxRetuneTimeFR2* via *NR-DL-PRS-ProcessingCapability* [34]. The first hop within a MG instance starts at no earlier than the earliest arrival time of the first unmuted PRS resource fully or partially overlapped with the MG instance taking into account the *expectedRSTD* and *expectedRSTD-uncertainty* in the PRS assistance data.

**Table 16.2.12.3-1 Applicable number of hops per slot and applicable length of each hop**

| $RRT_{FH}$ | (comb size, Number of PRS symbols) | Applicable number of hops per slot ($N_{hops}^{slot}$) | Applicable length per hop ($T_{per}$) in number of symbols |
|---|---|---|---|
| $RRT_{FH} \leq 2$ symbols | (2, 12) with SCS 15 kHz, 30 kHz, 60 kHz in FR2, 120 kHz | 2 | 7 |
| | All others | 1 | 14 |
| 2 symbols $< RRT_{FH} \leq 6$ symbols | (≤ 6, any) | 1 | 14 |
| | (12, 12) | ½ | 28 |
| $RRT_{FH} > 6$ symbols | Any combination | ½ | 28 |

The number of hops within a single MG occasion $N_{hop,i,j}$ is defined as

$$N_{hop,i,j} = min\left(N_{hops,effect,i,j}, \left\lfloor \frac{N3_i}{L_{per-hop,i,j}} \right\rfloor, N_{hop,max,i}\right)$$

where

- $N_{hop,max}$ is the maximum number of Rx hops signaled by UE in *maximumFH-Hops* via *NR-DL-PRS-ProcessingCapability* as specified in [34] for positioning frequency layer *i*

- $N3_i$ is indicated via UE capability *processingPRS-SymbolsDurationN3-r18* for positioning frequency layer *i*

- $N_{hops,effect,i,j}$ is the effective number of Rx hops within a MG instance *j*,

- $N_{hops,effect,i,j} = 2 * N_{rep}^{PRS}$, if $N_{hops}^{slot} = 2$,

- $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot} = 1$,

- $N_{hops,effect,i,j} = N_{rep}^{PRS}$, if $N_{hops}^{slot} = 1/2$ and $M_{rep}^{PRS} > 1$,

- $N_{hops,effect,i,j} = \left\lfloor \frac{(N_{rep}^{PRS}-1)}{2} \right\rfloor + 1$, if $N_{hops}^{slot} = 1/2$ and $M_{rep}^{PRS} = 1$,

- where $N_{rep}^{PRS}$ is the number of inter-slot PRS repetitions within the MG occasion, excluding the gap retuning times, which are unmuted and fully or partially overlapped with the sampling duration within each hop. $M_{rep}^{PRS}$ is

the PRS repetition interval configured by *dl-PRS-ResourceTimeGap*, $N_{hops}^{slot}$ is the applicable number of hops per slot as defined in table 16.2.11.3-1.

The measurement period requirement with the above modifications applies provided all PRS resources in each PFL have the same number of PRS repetitions $\left(N_{rep}^{PRS}\right)$ within an MG instance and the same PRS repetition interval $\left(M_{rep}^{PRS}\right)$, otherwise the measurement period can be longer.

UE shall be able to measure PRS resources with multiple hops at least over the BW of $BW_{total}$ defined as

$$BW_{total} = min\left(BW_{PRS}, N_{hop} * BW_{per-hop} - \left(N_{hop} - 1\right) * BW_{overlap}\right)$$

where

- $BW_{PRS}$ is the minimum among
- configured PRS BW, and
- UE capability of maximum PRS BW across all hops indicated via *maximumPRS-BandwidthAcrossAllHopsFR1-r18* or *maximumPRS-BandwidthAcrossAllHopsFR2-r18*, and
- total BW of all hops requested by LMF via *nr-DL-PRS-RxHoppingTotalBandwidth-r18* as specified in [34],

$N_{hop,i,j}$ is number of hops within a single MG occasion as define above,

$BW_{per-hop}$ is the UE capability on PRS BW per hop indicated via *supportedBandwidthPRS-r16* as specified in [49],

$BW_{overlap}$ is the UE capability on BW of the overlapping RB indicated via *numOfOverlappingPRB-r18* as specified in [49].

## 16.2.12.4 Test description

### 16.2.12.4.1 Initial conditions

The test is defined with four possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 16.2.12.4.1-1.

**Table A.16.2.12.4.1-1: Supported test configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, cell bandwidth 50 MHz, FDD duplex mode |
| 2 | 15 kHz SSB SCS, cell bandwidth 50 MHz, TDD duplex mode |
| 3 | 30 kHz SSB SCS, cell bandwidth 100 MHz, TDD duplex mode |
| 4 | 15 kHz SSB SCS, cell bandwidth 50 MHz, HD-FDD duplex mode |
| NOTE 1: The UE is only required to be tested in one of the supported test configurations.<br>NOTE 2: This test case is applicable to the UE supporting per hop bandwidth ≥ 10 MHz for SCS 15 kHz.<br>NOTE 3: This test case is applicable to the UE supporting per hop bandwidth = 20 MHz for SCS 30 kHz. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.12.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.2.12.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.2.12.4.2.

5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is a neighbour cell. Both cells are on the same NR RF channel in FR1.

### 16.2.12.4.2 Test procedures

Same test procedure as in clause 16.2.9.4.2 with following exceptions:

Table 16.2.12.5-1 replaces Table 16.2.9.5-1; Table 16.2.12.5-2 replaces Table 16.2.9.5-2.

### 16.2.12.4.3 Message contents

**Table 16.2.12.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.2.12.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.2.12.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.12.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |

| | | | |
|---|---|---|---|
| }, | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| }, | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | mhz50 | | Configuration 1, Configuration 2 and Configuration 4 |
| | mhz100 | | Configuration 3 |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.12.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |

| | | | |
|---|---|---|---|
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.2.12.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| nr-IntegrityServiceParameters-r18 | Not present | | |
| nr-IntegrityServiceAlert-r18 | Not present | | |
| nr-IntegrityRiskParameters-r18 | Not present | | |
| nr-IntegrityParametersTRP-LocationInfo-r18 | Not present | | |
| nr-IntegrityParametersDL-PRS-BeamInfo-r18 | Not present | | |
| nr-IntegrityParametersRTD-Info-r18 | Not present | | |
| nr-IntegrityParametersTRP-BeamAntennaInfo-r18 | Not present | | |
| nr-PRU-DL-Info-r18 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| assistanceDataValidityArea-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.12.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Configuration 1, Configuration 2 and Configuration 4 |
| | kHz30 | | Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 62 | 268 PRBs | Configuration 1, Configuration 2 and Configuration 4 |
| | 63 | 272 PRBs | Configuration 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.12.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.12.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 16.2.12.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Configuration 1, Configuration 2 and Configuration 4 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Configuration 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | Configuration 1, Configuration 2 and Configuration 4 |
| | 0 | | Configuration 3 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.12.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | Not checked | | |
| nr-DL-AoD-Error-r16 | Not checked | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not checked | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.2.12.5 Test requirement

Table 16.2.12.5-1 and Table 16.2.12.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.12.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3, 4 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3, 4 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3, 4 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1, 2, 4 | 50: $N_{RB,c}$ = 270 | |
| | | 3 | 100: $N_{RB,c}$ = 273 | |
| SSB configuration | | 1, 2, 4 | SSB.1 FR1 | |
| | | 3 | SSB.1 RedCap FR1 | |
| SMTC configuration | | 1, 2, 3, 4 | SMTC.1 RedCap | |
| Measurement gap | | 1, 2, 3, 4 | GP#24 or GP#0 Note 1 | |

| | | | | |
|---|---|---|---|---|
| CP length | | 1, 2, 3, 4 | Normal | |
| DRX | | 1, 2, 3, 4 | N/A | OFF |
| Time offset between serving and neighbour cells | µs | 1, 2, 3, 4 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3, 4 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3, 4 | 5 | |
| PRS RX hopping request | | 1, 2, 3, 4 | Present | |
| T1 | s | 1, 2, 3, 4 | 2 | |
| T2 | s | 1, 2, 3, 4 | 5 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 16.2.12.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1, 4 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1, 4 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1, 4 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1, 4 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3, 4 | OP.1 | | OP.1 | |
| EPRE ratio of PSS to SSS | dB | 1, 2, 3, 4 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| TRS Configuration | | 1, 4 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3, 4 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3, 4 | DLBWP.1.1 RedCap | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Active UL BWP configuration | | 1, 2, 3, 4 | ULBWP.1.1 RedCap | | N/A | |
| PRS configuration | | 1, 4 | PRS.1.5 FR1 | | PRS.1.5 FR1 | |
| | | 2 | PRS.1.5 FR1 | | PRS.1.5 FR1 | |
| | | 3 | PRS.2.6 FR1 | | PRS.2.6 FR1 | |
| PRS muting info | | 1, 2, 3, 4 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1, 4 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1, 4 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1, 4 | -Infinity | -2.41 +TT | -Infinity | -12.12 +TT |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1, 4 | -Infinity | -2 +TT | -Infinity | -10 +TT |
| | | 2 | | | | |
| | | 3 | | | | |
| PRP Note 3 | dBm/SCS | 1, 4 | -Infinity | -100 +TT | -Infinity | -108 +TT |
| | | 2 | -Infinity | -100 +TT | -Infinity | -108 +TT |
| | | 3 | -Infinity | -97 +TT | -Infinity | -105 +TT |
| SS-RSRP Note 3 | dBm/SCS | 1, 2, 4 | -88 | -88 | -Infinity | -Infinity |
| | | 3 | -85 | -85 | -Infinity | -Infinity |
| Io | dBm/48.6 MHz | 1, 4 | -52.48 +TT | -60.51 +TT | -70.05 +TT | -60.51 +TT |
| | dBm/48.6 MHz | 2 | -52.48 +TT | -60.51 +TT | -70.05 +TT | -60.51 +TT |
| | dBm/98.28 MHz | 3 | -49.43 +TT | -57.46 +TT | -67.13 +TT | -57.46 +TT |
| Propagation Condition | | 1, 2, 3, 4 | AWGN | | | |

NOTE 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: PRP/SS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit specified in clause 16.2.12.3, starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time limit above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of correct events observed during repeated tests shall be at least 90%, where the reported PRS-RSRP measurement for each correct event shall be within the reporting range specified in TS 38.133[50] clause 10.1A.17.3.

## 16.2.13 PRS-RSRP measurement delay in FR2 for Redcap UE with Rx FH

Editor's Note: This test case is incomplete in following aspects:

- TT analysis is missing.

### 16.2.13.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement with FH by a RedCap UE meets requirements specified in TS 38.133[50] clause 9.9A.3.6 for single positioning frequency layer under AWGN propagation conditions in FR2 standalone scenario.

### 16.2.13.2 Test applicability

This test applies to FR2 Redcap UEs release 18 onwards that supports DL-AoD positioning and PRS frequency hopping measurement and report.

### 16.2.13.3 Minimum conformance requirements

Same minimum requirements as in clause 16.2.12.3.

### 16.2.13.4 Test description

#### 16.2.13.4.1 Initial conditions

The defined test configuration is specified in Table 16.2.13.4.1-1.

**Table 16.2.13.4.1-1: Supported test configurations for PRS RSRP measurement**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, UE bandwidth 100 MHz, cell bandwidth 400 MHz, TDD duplex mode |
| NOTE 1: This test case is applicable to the UE supporting per hop bandwidth = 100 MHz for SCS 120 kHz. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid-Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.2.13.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.2.13.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.2.13.4.2.
5. There are two cells in the test, PCell (Cell 1) and a FR2 neighbour cell (Cell 2) on the same frequency as the PCell.

#### 16.2.13.4.2 Test procedures

Same test procedure as in clause 16.2.10.4.2 with following exceptions:

Table 16.2.13.5-1 replaces Table 16.2.10.5-1; Table 16.2.13.5-2 replaces Table 16.2.10.5-2.

#### 16.2.13.4.3 Message contents

Same message contents as defined in clause 16.2.12.4.3, with the following exceptions:

Table 16.2.13.4.3-1 replaces Table 16.2.12.4.3-3, Table 16.2.13.4.3-2 replaces Table 16.2.12.4.3-5 and Table 16.2.13.4.3-3 replaces Table 16.2.12.4.3-6.

**Table 16.2.13.4.3-1: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.2.13.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | Not present | | |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr2 | mhz400 | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.2.13.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 61 | 264 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.13.4.3-3 | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.2.13.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 | Not present | | |
| nr-DL-PRS-AggregationInfo-r18 | Not present | | |
| } | | | |
| } | | | |

**Table 16.2.13.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n6 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | NR Cell 1 |
| | 01 | | NR Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

## 16.2.13.5 Test requirement

Table 16.2.13.5-1 and Table 16.2.13.5-2 define the primary level settings including test tolerances for the test.

**Table 16.2.13.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | Config 1 | GP#13 or GP#24[Note1] | As specified in TS 38.133[50] clause 9.1.2-1. |
| Measurement gap offset | | Config 1 | 39 | |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133[50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133[50] clause A.3.10.2 |
| CP length | | Config 1 | Normal | |
| DRX | | Config 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells. |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| PRS RX hopping request | | Config 1 | Present | |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 16.2.13.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 400: $N_{RB,c}$ = 264 | | 400: $N_{RB,c}$ = 264 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.6 FR2 | | PRS.1.6 FR2 | |
| PRS muting configuration | | | Config 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH DMRS | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz Note5 | | -98 | | -98 | |
| $N_{oc}$ Note2 | | dBm/SCS Note4 | Config 1 | -89 | | -89 | |
| SSB_RP Note 3 | | dBm/SCS Note5 | Config 1 | -91 | -91 | -Infinity | -99 |
| PRP Note 3 | | dBm/SCS Note5 | Config 1 | -Infinity | -91+TT | -Infinity | -99+TT |
| PRS $\hat{E}_s / I_{ot}$ | | dB | Config 1 | -Infinity | -2.41+TT | -Infinity | -12.12+TT |
| PRS $\hat{E}_s / N_{oc}$ | | dB | Config 1 | -Infinity | -2+TT | -Infinity | -10+TT |
| SSB $\hat{E}_s / N_{oc}$ | | dB | Config 1 | -2 | -2 | -Infinity | -10 |
| Io Note3 | | dBm/380.16 MHz Note5 | Config 1 | -53.99 | -51.61 +TT | -53.99 +TT | -51.61 +TT |
| Propagation Condition | | | Config 1 | AWGN | | | |

NOTE 1: OCNG shall be used such that active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the subframes with transmitted PRS.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: SSB_RP/PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone.
NOTE 6: As observed with 0 dBi gain antenna at the centre of the quiet zone.
NOTE 7: Information about types of UE beam is given in annex B.2.1.3, and does not limit UE implementation or test system implementation.
NOTE 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [19], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [19] table 6.2.1.3-4.

The PRS RSRP measurement time fulfils the requirements specified in clause 16.2.13.3. The UE shall perform and report the PRS RSRP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in clause 16.2.13.3 starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the time duration above because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRP measurement for each correct event shall be within the PRS RSRP reporting range specified in TS 38.133[50] clause 10.1A.17.3, i.e., between PRS RSRP_0 and PRS RSRP_126.

# 16.3 NR PRS-RSRP measurement accuracy test case in RRC_CONNECTED

## 16.3.1 PRS-RSRP measurement accuracy with PRS in FR1

### 16.3.1.1 Test purpose

The purpose of the test is to verify that the PRS-RSRP measurement meets the accuracy requirements specified in TS 38.133 [50] clauses 10.1.24.2.1 and 10.1.24.2.2.

### 16.3.1.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-TDOA positioning and PRS-RSRP measurements in FR1.

### 16.3.1.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 16.3.1.3-1 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE does not support positioning measurements with reduced number of samples, or LMF does not indicate UE to perform positioning measurements with reduced number of samples

**Table 16.3.1.3-1: PRS-RSRP absolute accuracy for FR1**

| Accuracy | Conditions |
|---|---|

| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Io Note 7 range | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | NR operating band groups Note 8 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 6 | dBm/30kHz Note 6 | dBm/60kHz Note 6 | dBm/$BW_{Channel}$ |
| ±3.5 | ±8 | ≥-3dB | ≥24 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±8.5 | ±13 | ≥-13dB | 24 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±6 | ±10.5 | | 52< BW≤ 104 | All | Note 4 | | | | |
| ±4.5 | ±9 | | BW >104 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 PRB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

The relative PRS-RSRP accuracy is defined as accuracy of the difference between two PRS-RSRP measurements.

The relative PRS-RSRP accuracy requirements apply for the cases when PRS-RSRP is measured from PRS resources in the same PRS resource set in FR1.

The accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 16.3.1.3-2 are valid under the following conditions:

Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band

**Table 16.3.1.3-2: PRS-RSRP relative accuracy for FR1**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io [Note 7] range | | | | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | NR operating band groups [Note 8] | Minimum Io [Note 1] dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz [Note 6] | dBm/30kHz [Note 6] | dBm/60kHz [Note 6] | |
| ±3.5 | ±5.0 | ≥-3dB | ≥24 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±9.5 | ±11.0 | ≥-13dB | 24 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±6.5 | ±8.0 | | 52< BW≤ 104 | All | Note 4 | | | | |
| ±5.0 | ±6.5 | | BW >104 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

## 16.3.1.4 Test description

### 16.3.1.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 16.3.1.4.1-1.

**Table 16.3.1.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.3.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.3.1.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.3.1.4.3.
5. In the test there are three synchronous cells: Cell 1 and Cell 2. Cell 1 is the PCell. Cell 2 is the neighbour cell. The cells are on the same RF channel in FR1

### 16.3.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell.

The NR-DL-AoD-RequestLocationInformation message and NR-DL-AoD-ProvideAssistanceData message as defined in TS 37.355 shall be provided to the UE during T1. The last slot containing the two messages for the assistance data and location information request is denoted as #n. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources that is $\Delta T$ after slot #n, where $\Delta T = 50$ ms is the maximum processing time of the assistance data and location information request

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.3.1.5-1 as appropriate. Propagation conditions are set according to clause 4.15.2.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the NR-DL-AoD-ProvideCapabilities IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-AoD-ProvideAssistanceData-r16 IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-AoD-RequestLocationInformation-r16 such that the UE receives the message $\Delta T$ ms before the start of the measurement period, where $\Delta T = 50$ ms.
8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the nr-DL-AoD-ProvideLocationInformation-r16 IE.
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

10. The SS shall check the nr-DL-PRS-RSRP-Result-r16 and nr-DL-PRS-RSRP-ResultDiff-r16 value for Cell 1 and Cell 2 in the nr-DL-AoD-SignalMeasurementInformation-r16 according to Tables 16.3.1.5-2 and 16.3.1.5-3.11. Repeat step 2-10 until the confidence level according to Annex D is achieved.

12. Repeat step 1-11 for the other sub-test defined in Table 16.3.1.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 16.3.1.4.3 Message contents

**Table 16.3.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.3.1.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.3.1.4.3-3: LPP RequestLocationInformation**

| Derivation Path: TS 37.355 [49] clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.3.1.5 | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |

| | | | |
|---|---|---|---|
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | 2 | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.1.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 16.3.1.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE- |

| | | | |
|---|---|---|---|
| | | | based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.1.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Sub-test 1-1, Sub-test 1-2, Sub test 2-1 and Sub-test 2-2 |
| | kHz30 | | Sub-test 3-1 and Sub-test 3-2 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | 21 | 104 PRBs | Sub-test 1-2 and Sub-test 2-2 |
| | 28 | 132 PRBs | Sub-test 3-2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | n4 | | Sub-test 1-2, Sub-test 2-2 and |

| | | | |
|---|---|---|---|
| | | | Sub-test 3-2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.1.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.1.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.1.4.3-6: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Sub-tests 1-1, Sub-test 1-2, Sub-test 2-1 and Sub-test 2-2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Sub-tests 3-1 and |

| | | | |
|---|---|---|---|
| | | | Sub–test 3-2 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| | n2 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| n4-r16 | 0 | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1-1 Cell 1, Sub-test 1-2 Cell 1, Sub-test 2-1 Cell 1, Sub-test 2-2 Cell 1, Sub-test 3-1 Cell 1,and Sub-test 3-2 Cell 1 |
| | 2 | | Sub-test 1-1 Cell 2, Sub-test 1-2 Cell 2, Sub-test 2-1 Cell 2, Sub-test 2-2 Cell 2, Sub-test 3-1 Cell 2, and Sub-test 3-2 Cell 2. |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |

| | | | |
|---|---|---|---|
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-tests 1-1, Sub-test 2-1 and Sub-test 3-1 |
| n4-r16 | 1 | | Sub-tests 1-2, Sub-test 2-2 and Sub-test 3-2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1-1 Cell 1, Sub-test 1-2 Cell 1, Sub-test 2-1 Cell 1, Sub-test 2-2 Cell 1, Sub-test 3-1 Cell 1,and Sub-test 3-2 Cell 1 |
| | 2 | | Sub-test 1-1 Cell 2, Sub-test 1-2 Cell 2, Sub-test 2-1 Cell 2, Sub-test 2-2 Cell 2, Sub-test 3-1 Cell 2, and Sub-test 3-2 Cell 2. |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.1.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | 2 entries | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 SEQUENCE (SIZE (1..7)) OF NR-DL-AoD-AdditionalMeasurementElement-r16 { | 1 entry | | |
| nr-DL-AoD-AdditionalMeasurementElement-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-ResultDiff-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 SEQUENCE (SIZE (1..7)) OF NR-DL-AoD-AdditionalMeasurementElement-r16 { | 1 entry | | |
| nr-DL-AoD-AdditionalMeasurementElement-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-ResultDiff-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.3.1.5 Test requirement

Table 16.3.1.5-1 defines the primary level settings including test tolerances for the test.

**Table 16.3.1.5-1: General test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | | freq1 | | freq1 | |
| Duplex mode | Config 1 | | FDD | | | |
| | Config 2,3 | | TDD | | | |
| TDD configuration | Config 1 | | Not Applicable | | | |
| | Config 2 | | TDDConf.1.1 | | | |
| | Config 3 | | TDDConf.2.1 | | | |
| $BW_{channel}$ | Config 1 | MHz | 20: NRB,c = 106 | | | |
| | Config 2 | | 20: NRB,c = 106 | | | |
| | Config 3 | | 50: NRB,c = 133 | | | |
| Downlink initial BWP configuration | | | DLBWP.0.1 | | | |
| Downlink dedicated BWP configuration | | | DLBWP.1.1 | | | |
| Uplink initial BWP configuration | | | ULBWP.0.1 | | | |
| Uplink dedicated BWP configuration | | | ULBWP.1.1 | | | |
| TRS configuration | Config 1 | | TRS.1.1 FDD | NA | TRS.1.1 FDD | NA |
| | Config 2 | | TRS.1.1 TDD | NA | TRS.1.1 TDD | NA |
| | Config 3 | | TRS.1.2 TDD | NA | TRS.1.2 TDD | NA |
| DRX Cycle | | ms | Not Applicable | | | |
| Measurement gap | | | GP#24 or GP#0 Note 7 | | | |
| PDSCH Reference measurement channel | Config 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | Config 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | Config 3 | | SR2.1 TDD | | SR2.1 TDD | |
| RMSI CORESET Reference Channel | Config 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | Config 2 | | CR.1.1 TDD | | CR.1.1 TDD | |
| | Config 3 | | CR2.1 TDD | | CR2.1 TDD | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Control channel RMC | Config 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | Config 2 | | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | Config 3 | | CCR2.1 TDD | | CCR2.1 TDD | |
| PRS configuration | Config 1 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | PRS.2.3 FR1 | PRS.2.3 FR1 | PRS.2.4 FR1 | PRS.2.4 FR1 |
| PRS Resource slot offset (slot) | Config 1,2,3 | slot | 0 | 2 | 0 | 2 |
| SSB configuration | Config 1 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 |
| Time offset with Cell 1 | Config 1 | µs | - | 3 | - | 3 |
| | Config 2,3 | | - | 3 | - | 3 |
| Expected RSTD | Config 1,2,3 | µs | 3 | | | |
| Expected RSTD uncertainty | Config 1,2,3 | µs | 5 | | | |
| SMTC configuration | Config 1 | | SMTC.2 | | | |
| | Config 2,3 | | SMTC.1 | | | |
| OCNG Patterns | | | OCNG pattern 1 | | | |
| PDSCH/PDCCH subcarrier spacing | Config 1,2 | kHz | 15 kHz | | | |
| | Config 3 | | 30 kHz | | | |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| EPRE ratio of PRS to SSS | | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | Config 1,2 | dBm/15KhZ | -98 | | -98 | |
| | Config 3 | | -98 | | -98 | |
| $N_{oc}$ Note2 | Config 1,2 | dBm/SCS | -98 | | -98 | |
| | Config 3 | | -95 | | -95 | |
| PRS $\hat{E}_s/I_{ot}$ | | dB | -2.41 | -12.12 | -2.41 | -12.12 |
| PRS $\hat{E}_s/N_{oc}$ | | dB | -2 | -10 | -2 | -10 |
| PRP Note3 | Config 1, 2 | dBm/SCS | -100 | -108 | -100 | -108 |
| | Config 3 | | -97 | -105 | -97 | -105 |
| Io Note3 | Config 1,2 | dBm/19.08MHz | -64.57 | | -64.57 | |
| | Config 3 | dBm/47.88MHz | -60.59 | | -60.59 | |
| Propagation condition | | | AWGN | | | |
| Antenna configuration | | | 1x2 | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | PRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | PRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Void. |
| Note 6: | The test configuration excludes support for band n51 and it is not required to run this test on band n51 in this release of the specification |
| Note 7: | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 16.2.1.5, with the following difference: , N_(PRS,i)^slot= 2 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(\left\lceil\frac{2}{N'}\right\rceil\left\lceil\frac{L_{available_{PRS,i}}}{N}\right\rceil * 4 - 1\right) * T_{\text{effect}} + \text{T}_{\text{last}}$$

**Table 16.3.1.5-2: PRS-RSRP absolute accuracy requirements for the reported values**

| Test Configuration | Subtest | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | Sub-test 1 | 51 | 62 | 38 | 59 |
| | Sub-test 2 | 51 | 62 | 41 | 56 |
| 2 | Sub-test 1 | 51 | 62 | 38 | 59 |
| | Sub-test 2 | 51 | 62 | 41 | 56 |
| 3 | Sub-test 1 | 54 | 65 | 41 | 62 |
| | Sub-test 2 | 54 | 65 | 45 | 58 |

**Table 16.3.1.5-3: PRS-RSRP relative accuracy requirements for the reported values**

| Test Configuration | Subtest | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | Sub-test 1 | 27 | 30 | 21 | 30 |
| | Sub-test 2 | 27 | 30 | 24 | 30 |
| 2 | Sub-test 1 | 27 | 30 | 21 | 30 |
| | Sub-test 2 | 27 | 30 | 24 | 30 |
| 3 | Sub-test 1 | 27 | 30 | 21 | 30 |
| | Sub-test 2 | 27 | 30 | 25 | 30 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 16.3.2 PRS-RSRP measurement accuracy with PRS in FR2

### 16.3.2.1 Test purpose

The purpose of this test is to verify that the PRS-RSRP measurement accuracy is within the specified limits. This test will verify the requirements in TS 38.133 [50] clauses 10.1.24.2.1 and 10.1.24.2.2.

### 16.3.2.2 Test applicability

This test applies to all types of NR UE release 16 onwards that supports DL-AoD positioning and PRS-RSRP measurements in FR2.

### 16.3.2.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRP measurement for FR2 defined in Table 16.3.2.3-1 are valid under the following conditions:

Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to TS 38.133 [50] Annex B.2.14 for a corresponding Band

**Table 16.3.2.3-1: PRS-RSRP absolute accuracy for FR2**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Minimum Io Note 1 dBm / $SCS_{PRS}$ | Maximum Io |
| dB | dB | dB | PRB | - | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| | | | | | dBm/120kHz Note 6 / dBm/60kHz Note 6 | |
| ±5 | ±8 | ≥-3dB | ≥24 | All | Same value as PRP in TS 38.133 [50] Table B.2.14 -2, according to UE Power class, operating band and angle of arrival | -50 |
| | | | | | Note 4 | |
| | | | | | Note 4 | |
| ±8.5 | ±11.5 | ≥-13dB | 24 ≤ BW ≤ 64 | All | Note 4 | |
| ±6 | ±9 | | BW >64 | All | Note 4 | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in TS 37.355 [49].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by Δ>0, when applicable, as described in TS 38.133 [50] Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in TS 38.133 [50] Section 3.5.2.

The accuracy requirements for PRS-RSRP measurement for FR2 defined in Table 116.3.2.3-2 are valid under the following conditions:

Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.

PRP $1,2|_{dBm}$ according to TS 38.133 [50] Annex B.2.14 for a corresponding Band

**Table 16.3.2.3-2: PRS-RSRP relative accuracy for FR2**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Minimum Io Note 1 dBm / $SCS_{PRS}$ | Maximum Io |
| dB | dB | dB | PRB | - | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| | | | | | dBm/120kHz Note 6 / dBm/60kHz Note 6 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| ±5.0 | ±8.0 | ≥-3dB | ≥24 | All | Same value as PRP in TS 38.133 [50] Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| | | | | | Note 4 | |
| | | | | | Note 4 | |
| ±10 | ±13 | ≥-13dB | 24 ≤ BW ≤ 64 | All | Note 4 | |
| ±7.5 | ±10.5 | | BW >64 | All | Note 4 | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in TS 37.355 [49].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in TS 38.133 [50] Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in TS 38.133 [50] Section 3.5.2.

## 16.3.2.4 Test description

The supported test configurations in listed in Table 16.3.2.4-1.

**Table 16.3.2.4-1: Supported test configurations for PRS RSRP measurement for FR2**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

### 16.3.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.3.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.3.2.5-1 and Table 16.3.2.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.3.2.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 16.3.2.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.3 FR2 and PRS.1.4 FR2. The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-AoD-RequestLocationInformation message and the DL-AoD assistance data as defined in clause 16.3.2.4.3 shall be provided to the UE during the set-up period. The last TTI containing the NR-DL-AoD-RequestLocationInformation message shall be provided to the UE $\Delta$T ms before the start

of the measurement period, where ΔT = 50 ms is the maximum processing time of the NR-DL-AoD-RequestLocationInformation message and the DL-AoD assistance data in the UE.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.3.2.5-1 and Table 16.3.2.5-2 as appropriate. Propagation conditions are set according to clause 4.16.2.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the NR-DL-AoD-ProvideCapabilities IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-AoD-ProvideAssistanceData-r16 IE. If the UE message at step 5 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-AoD-RequestLocationInformation-r16 IE such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the nr-DL-AoD-ProvideLocationInformation-r16 IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for both Cell 1 and Cell 2.
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
10. The SS shall check the nr-DL-PRS-RSRP-Result-r16 and nr-DL-PRS-RSRP-ResultDiff-r16 value for Cell 1 and Cell 2 in the nr-DL-AoD-SignalMeasurementInformation-r16 according to Table 16.3.2.5-3.
11. Repeat step 2-10 until the confidence level according to Annex D is achieved.
12. Repeat step 1-11 for the other sub-test defined in Table 16.3.2.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 16.3.2.4.3 Message contents

**Table 16.3.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.3.2.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.3.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.3.2.5 42+TT | Result of the response time calculation rounded up to the next second | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | 2 | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.2.4.3-4: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData | As defined in Table 16.3.2.4.3-5 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.2.4.3-5: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA + offsetToCarrier, as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.2.4.3-6 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.2.4.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.2.4.3-6: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-test 1 |
| n4-r16 | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 2 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.2.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | 2 entries | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 SEQUENCE (SIZE (1..7)) OF NR-DL-AoD-AdditionalMeasurementElement-r16 { | 1 entry | | |
| NR-DL-AoD-AdditionalMeasurementElement-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-ResultDiff-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 SEQUENCE (SIZE (1..7)) OF NR-DL-AoD-AdditionalMeasurementElement-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-AoD-AdditionalMeasurementElement-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-ResultDiff-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.3.2.5 Test requirement

Table 16.3.2.5-1 and Table 16.3.2.5-2 define the primary level settings including the test tolerances for the test.

**Table 16.3.2.5-1: PRS-RSRP general test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - | Not applicable | - |
| Measurement gap | | GP#13 or GP#24 Note2 | | | |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.1 | OP.1 | OP.1 | OP.1 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| Expected RSTD | µs | 3 | | | |
| Expected RSTD uncertainty | µs | 5 | | | |
| PRS configuration | | PRS.1.3 FR2 | PRS.1.3 FR2 | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS Resource slot offset | slot | 0 | 2 | 0 | 2 |

| | | | | | |
|---|---|---|---|---|---|
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| EPRE ratio of PRS to SSS | dB | 0 | 0 | 0 | 0 |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | | |

**Table 16.3.2.5-2: PRS-RSRP OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 7 | | Rough | | Rough | |
| $N_{OC}$ Note1 | dBm/15kHz Note4 | -99.1 | | -99.1 | |
| $N_{OC}$ Note1 | dBm/SCS Note4 | -90.1 | | -90.1 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | -2 | -10 | -2 | -10 |
| PRP Note2 | dBm/SCS | -92.1 | -100.1 | -92.1 | -100.1 |
| PRS $\hat{E}_s / I_{ot}$ BB Note6 | dB | -2.55 | -12.22 | -2.55 | -12.22 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.69 | | -55.69 | |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 2: PPRP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: Void<br>Note 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone<br>Note 5: Void<br>Note 6: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.<br>Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation | | | | | |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 16.2.3.5 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 16.3.2.4.3-3).

**Table 16.3.2.5-3: PRS RSRP accuracy requirements for the reported values**

| Test Configuration | Report Mapping | Lowest reported value | Highest reported value |
|---|---|---|---|
| Sub-test 1 | Absolute PRS RSRP accuracy for Cell 1 | 54 | 85 |
| | Relative PRS RSRP accuracy for Cell 1 | 22 | 30 |
| | Absolute PRS RSRP accuracy for Cell 2 | 46 | 77 |
| | Relative PRS RSRP accuracy for Cell 2 | 20 | 30 |
| Sub-test 2 | Absolute PRS RSRP accuracy for Cell 1 | 54 | 85 |
| | Relative PRS RSRP accuracy for Cell 1 | 24 | 30 |
| | Absolute PRS RSRP accuracy for Cell 2 | 46 | 77 |
| | Relative PRS RSRP accuracy for Cell 2 | 23 | 30 |

The PRS RSRP measurement period fulfils the requirements specified in clause 4.16.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.3.3 PRS-RSRP measurement accuracy with PRS in FR1 with reduced sample number

### 16.3.3.1 Test purpose

The purpose of this test is to verify that the accuracy of PRS-RSRP measurement with reduced sample number is within the specified limits. This test will verify the requirements in TS 38.133 [50] clauses 10.1.24.2.1 and 10.1.24.2.2.

### 16.3.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 16.3.3.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRP measurement for FR1 defined in Table in TS 38.133 [50] clause 10.1.24.2.1-3 are valid under the following conditions:

- Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

**Table 16.3.3.3-1: PRS-RSRP absolute accuracy for FR1 with reduced sample number**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io Note 6 range | | | | |
| Normal condition | Extreme condition | PRS $\hat{E}_s/I_{ot}$ | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) | NR operating band groups Note 7 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 5 | dBm/30kHz Note 5 | dBm/60kHz Note 5 | |
| ±3.5 | ±8 | ≥0 | ≥48 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| ±8.5 | ±13 | ≥-6 | 48 ≤ BW ≤ 52 | All | Note 3 | | | | |
| ±6 | ±10.5 | | 52< BW≤ 104 | All | Note 3 | | | | |
| ±4.5 | ±9 | | BW >104 | All | Note 3 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 PRB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 7: NR operating band groups are as defined in Section 3.5.2.

**Table 16.3.3.3-2: PRS-RSRP relative accuracy for FR1 with reduced sample number**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | PRS BW | | Io Note 6 range |

| Normal condition | Extreme condition | PRS $\hat{E}_s/I_{ot}$ | | Repetition factor ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) | NR operating band groups Note 7 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 5 | dBm/30kHz Note 5 | dBm/60kHz Note 5 | |
| ±3.5 | ±5.0 | ≥0 | ≥48 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| ±9.5 | ±11.0 | ≥-6 | 48 ≤ BW ≤ 52 | All | Note 3 | | | | |
| ±6.5 | ±8.0 | | 52< BW≤ 104 | All | Note 3 | | | | |
| ±5.0 | ±6.5 | | BW >104 | All | Note 3 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 RB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 7: NR operating band groups are as defined in Section 3.5.2.

## 16.3.3.4 Test description

### 16.3.3.4.1 Initial conditions

The supported test configurations in listed in Table 16.3.3.4-1.

**Table 16.3.3.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | NR 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | NR 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | NR 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations in each supported band | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.3.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.3.3.5-1.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.3.3.4.3.

5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.3.3.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-AoD-ProvideAssistanceData and NR-DL-AoD-RequestLocationInformation as defined in TS 37.355 [49] shall be provided to the UE during the set-up period.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.3.3.5-1. Propagation conditions are set according to clause 4.16.2.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the NR-DL-AoD-RequestLocationInformation IE such that the UE receives the message T ms before the start of the measurement period, where T = 50 ms.
9. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for Cell 2 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.
10. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
11. Repeat steps 2-10 in Table 16.3.3.4-1 until the confidence level according to Annex D is achieved.

### 16.3.3.4.3 Message contents

**Table 16.3.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.3.3.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.3.3.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.3.3.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.3.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | N4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.3.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.3.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.3.4.3-5: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 &2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| N4-r16 | 0 | | Cell 1 |
| | 1 | | Cell 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell1 |
| | 2 | | Cell2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.3.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation | | | |

| | | | |
|---|---|---|---|
| SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.3.3.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.3.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.3.3.5 Test requirement

Table 16.3.3.5-1 define the primary level settings including test tolerances for the test.

**Table 16.3.3.5-1: PRS-RSRP general test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| ell ID | | | 489 | 0 |
| SB ARFCN | | | freq1 | |
| uplex mode | Config 1 | | FDD | |
| | Config 2,3 | | TDD | |
| DD configuration | Config 1 | | Not Applicable | |
| | Config 2 | | TDDConf.1.1 | |
| | Config 3 | | TDDConf.2.1 | |
| $W_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| WP BW | Config 1 | | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| ownlink initial BWP configuration | | | DLBWP.0.1 | |
| ownlink dedicated BWP configuration | | | DLBWP.1.1 | |
| plink initial BWP configuration | | | ULBWP.0.1 | |
| plink dedicated BWP configuration | | | ULBWP.1.1 | |
| RS configuration | Config 1 | | TRS.1.1 FDD | NA |
| | Config 2 | | TRS.1.1 TDD | NA |
| | Config 3 | | TRS.1.2 TDD | NA |

| | | | | | |
|---|---|---|---|---|---|
| RX Cycle | | | ms | Not Applicable | |
| leasurement gap | | | | GP#24 or GP#0 Note 7 | |
| DSCH Reference measurement channel | Config 1 | | | SR.1.1 FDD | - |
| | Config 2 | | | SR.1.1 TDD | |
| | Config 3 | | | SR2.1 TDD | |
| MSI CORESET Reference Channel | Config 1 | | | CR.1.1 FDD | - |
| | Config 2 | | | CR.1.1 TDD | |
| | Config 3 | | | CR2.1 TDD | |
| ontrol channel RMC | Config 1 | | | CCR.1.1 FDD | - |
| | Config 2 | | | CCR.1.1 TDD | |
| | Config 3 | | | CCR2.1 TDD | |
| RS configuration | Config 1 | | | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | | PRS.2.4 FR1 | PRS.2.4 FR1 |
| | | | | | |
| | | | | | |
| | | | | | |
| RS Resource slot offset (slot) | Config 1,2,3 | | slot | 0 | 2 |
| SB configuration | Config 1 | | | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | | SSB.2 FR1 | SSB.2 FR1 |
| ime offset with Cell 1 | Config 1 | | ms | - | 3 |
| | Config 2,3 | | µs | - | 3 |
| MTC configuration | Config 1 | | | SMTC.2 | |
| | Config 2,3 | | | SMTC.1 | |
| CNG Patterns | | | | OCNG pattern 1 | |
| DSCH/PDCCH subcarrier spacing | Config 1,2 | | kHz | 15 kHz | |
| | Config 3 | | | 30 kHz | |
| PRE ratio of PSS to SSS | | | dB | 0 | 0 |
| PRE ratio of PBCH DMRS to SSS | | | | | |
| PRE ratio of PBCH to PBCH DMRS | | | | | |
| PRE ratio of PDCCH DMRS to SSS | | | | | |
| PRE ratio of PDCCH to PDCCH DMRS | | | | | |
| PRE ratio of PDSCH DMRS to SSS | | | | | |
| PRE ratio of PDSCH to PDSCH | | | | | |
| PRE ratio of OCNG DMRS to SSS(Note 1) | | | | | |
| PRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | |
| $\hat{I}_{oc}$ Note2 | Config 1,2 | | dBm/15KhZ | -98 | |
| | Config 3 | | | -98 | |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note2 | Config 1,2 | | dBm/SCS | -98 | |
| | Config 3 | | | -95 | |
| $E_s/I_{ot}$ | | | dB | 0 | -6 |
| $E_s/N_{oc}$ | | | dB | 2.92 | -0.93 |
| SS-RSRP | | | dB | Same as PRS-RSRP | |
| PRS-RSRP Note3 | Config 1, 2 | | dBm/SCS | -95.08 | -98.93 |
| | Config 3 | | | -92.07 | -95.92 |
| $I_o$ Note3 | Config 1,2 | | dBm/19.08MHz | -61.20 | |
| | Config 3 | | dBm/47.88MHz | -57.20 | |
| Propagation condition | | | | AWGN | |
| Antenna configuration | | | | 1x2 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP and PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Void.

Note 6: Void.

Note 7: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 16.2.5.5 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(\left\lceil\frac{16}{N'}\right\rceil\left\lceil\frac{1}{N}\right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T}_{\mathrm{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 16.3.3.4.3-3).

**Table 16.3.3.5-2: PRS-RSRP accuracy requirements for the reported values**

| Test Configuration | Configuration | Report Mapping | Lowest reported value | Highest reported value |
|---|---|---|---|---|
| Sub-test 1 Subtest 1 | Config 1&2 | Absolute PRS RSRP accuracy for Cell 1 | 56 | 66 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 50 | 65 |
| | | Relative PRS RSRP accuracy for Cell 2 | 24 | 30 |
| | Config 3 | Absolute PRS RSRP accuracy for Cell 1 | 59 | 69 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 55 | 67 |
| | | Relative PRS RSRP accuracy for Cell 2 | 25 | 30 |

The PRS RSRP measurement period fulfils the requirements specified in clause 4.16.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.3.4 PRS-RSRP measurement accuracy with PRS in FR2 with reduced sample number

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

### 16.3.4.1 Test purpose

The purpose of this test is to verify that the accuracy of PRS-RSRP measurement with reduced sample number is within the specified limits. This test will verify the requirements in TS 38.133 [50] clauses 10.1.24.2.1 and 10.1.24.2.2.

### 16.3.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 16.3.4.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRP measurement for FR2 defined in Table 10.1.24.2.1-4 are valid under the following conditions:

- Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

…

**Table 10.1.24.2.1-4: PRS-RSRP absolute accuracy for FR2 with reduced sample number**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | | | Io Note 6 range | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Minimum Io Note 1 dBm / $SCS_{PRS}$ | Maximum Io |
| dB | dB | dB | PRB | - | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| | | | | | dBm/120kHz Note 5 / dBm/60kHz Note 5 | |
| ±5 | ±8 | ≥0 | ≥48 | All | Same value as PRP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| ±8.5 | ±11.5 | ≥-6 | 48 ≤ BW ≤ 64 | All | Note 3 | |
| ±6 | ±9 | | BW >64 | All | Note 3 | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 PRB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in Sections B.3.2 and B.3.3.

NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 7: NR operating band groups are as defined in Section 3.5.2.

## 16.3.4.4 Test description

### 16.3.4.4.1 Initial conditions

The supported test configurations in listed in Table 16.3.4.4-1.

**Table 16.3.4.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.3.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.3.4.5-1 and Table 16.3.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.3.4.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.3.4.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The NR-DL-AoD-ProvideAssistanceData and NR-DL-AoD-RequestLocationInformation as defined in TS 37.355 [49] shall be provided to the UE during the set-up period.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.3.4.5-1 and Table 16.3.4.5-2. Propagation conditions are set according to clause 4.16.2.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the reducedDL-PRS-ProcessingSamples-r17 in the NR-DL-AoD-RequestLocationInformation IE such that the UE receives the message T ms before the start of the measurement period, where T = 50 ms.

9. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRP measurements for Cell 2 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.

10. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

11. Repeat steps 2-10 in Table 16.3.4.4-1 until the confidence level according to Annex D is achieved.

#### 16.3.4.4.3 Message contents

**Table 16.3.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.3.4.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.3.4.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.3.4.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE- |

| | | | |
|---|---|---|---|
| | | | based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.4.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.4.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.3.4.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.4.4.3-5: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell1 |
| | 2 | | Cell2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.4.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.3.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.3.4.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.3.4.5 Test requirement

Table 16.3.4.5-1 and Table 16.3.4.5-2 define the primary level settings including test tolerances for the test.

**Table 16.3.4.5-1: PRS-RSRP general test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 |
| SSB ARFCN | | freq1 | |
| Duplex mode | | TDD | |
| TDD configuration | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - |
| Measurement gap | | GP#13 or GP#24 Note2 | |
| TRS configuration | | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | µs | - | 3 |
| PRS configuration | | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS BW | RB | 64 | 64 |
| PRS Resource slot offset | slot | 0 | 2 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | |
| Propagation conditions | | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 |

`Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured.

**Table 16.3.4.5-2: PRS-RSRP OTA related test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | |
| Assumption for UE beams Note 7 | | Rough | |
| $N_{OC}$ Note1 | dBm/15kHz Note4 | -103 | |
| $N_{OC}$ Note1 | dBm/SCS Note4 | -94 | |
| PRS $\hat{E}_s / N_{oc}$ | dB | 3.56 | -0.28 |
| $E_S$ | dBm/SCS Note4 | - | - |
| PRS_RP Note2 | dBm/SCS | -090.4 | -94.2 |
| SSB_RP Note2 | dBm/SCS | -86.77 | -90.73 |
| $\hat{E}_s / I_{ot\ BB}$ Note6 | dB | 0.32 | -5.64 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.73 | |

| | |
|---|---|
| Note 1: | Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 2: | SSB_RP, PRS_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 3: | Void |
| Note 4: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 5: | Void |
| Note 6: | Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. |
| Note 7: | Information about types of UE beam is given in B.2.1.3, and does not limit UE implementation or test system implementation |

The value of the LPP Time IE depends on the UE capabilities. The calculation is the same as in clause 16.2.7.5 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.071}{N} \right\rceil * 1 - 1\right)$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation (see Table 16.3.4.4.3-3).

**Table 16.3.4.5-3: PRS RSRP accuracy requirements for the reported values**

| Test Configuration | Report Mapping | Lowest reported value | Highest reported value |
|---|---|---|---|
| Test 1 | Absolute PRS RSRP accuracy for Cell 1 | 66 | 76 |
| | Relative PRS RSRP accuracy for Cell 1 | 25 | 30 |
| | Absolute PRS RSRP accuracy for Cell 2 | 61 | 73 |
| | Relative PRS RSRP accuracy for Cell 2 | 23 | 30 |

The PRS RSRP measurement period fulfils the requirements specified in clause 4.16.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 16.4 NR PRS-RSRP measurement period test cases in RRC_INACTIVE

## 16.4.1 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR1 SA

### 16.4.1.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] Clause 5.6.3.5 for single positioning frequency layer under AWGN propagation conditions in RRC_INACTIVE in FR1 standalone scenario.

### 16.4.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supports DL-PRS measurement in RRC_INACTIVE state.

### 16.4.1.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 5.6.3.3) PRS-RSRP measurements, defined in TS 38.215 [4], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,\ total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max\left(T_{effect,i}\right)$$

Where:

- *i* is the index of positioning frequency layer,
- L is total number of positioning frequency layers,
- $T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left(K_{carrier_PRS} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$ Where:

- $K_{carrier_PRS}$ is a scaling factor for PRS-based NR positioning measurements in RRC_INACTIVE. If the UE supports *parallelPRS-MeasRRC-Inactive-r17*, $K_{carrier_PRS}$ = 1. Otherwise,
- If Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$, $K_{carrier_PRS}$ equals to the sum of $K_{carrier}$ in 4.2.2.4 and one positioning layer.
- If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$, $K_{carrier_PRS}$ equals to the sum of $N_{layer}$ in 4.2.2.7 and one positioning layer.
- $N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping:
  - $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1, and if positioning frequency layer *i* is in FR2
  - $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-AoD-RequestLocationInformation*.
  - $N_{RxBeam,i}$ equals to 8, otherwise.
- $L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.
- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,
- {N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms-r17*in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],
- N′ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* in clause 6.4.3 of TS 37.355 [34],
- $N_{sample}$ is the number of PRS-RSRP measurement samples and
  - $N_{sample}$ = 1, if UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [34], and the LMF indicates the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34], and the following conditions are met:
    - PRS bandwidth is within the initial BWP and
    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

- $N_{sample}$ = 2, if UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [34], and the LMF indicates the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34], and the following conditions are not met:
  - PRS bandwidth is within the initial BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise

- $T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time,
  - $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,
  - $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [34],
  - $T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$ is the least common multiple between $T_{PRS,i}$ and $T_{DRX}$,
  - $T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,
  - $T_{DRX}$ is the DRX cycle length.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

- $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.
- $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

## 16.4.1.4 Test description

### 16.4.1.4.1 Initial conditions

The supported test configurations in listed in Table 16.4.1.4-1.

**Table 16.4.1.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.4.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.4.1.5-1 and Table 16.4.1.5-2.
3. Propagation conditions are set according to clause 4.16.2.

4. Message contents are defined in clause 16.4.1.4.3.

5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.4.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During T2 UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T = 50$ ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

The beginning of the time interval T2 shall be aligned with the first DRX cycle containing a DL PRS resource(s).

1. Configure the Cell for RA-SDT with SIB1 configured according to TS 38.508-1 [45], clause 4.6.1 with condition SDT
2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On,* according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

3. Set the parameters according to Table 16.4.1.5-1 and Table 16.4.1.5-2. Propagation conditions are set according to clause 4.16.2.

4. T1 starts.

5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.

6. The UE shall transmit RRCReconfigurationComplete message.

7. The SS shall transmit an LPP REQUEST CAPABILITIES message.

8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.

9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T = 50$ ms.

11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.

12. When T1 expires, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and switch the power setting from T1 to T2 as specified in Table 16.4.1.5-1 and Table 16.4.1.5-2.

13. Wait for the UE monitors for occurrence of the periodic event requested.

14. The UE shall transmit the UE transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 18.

15. The SS transmits an *RRCResume* message.

16. The UE transmits an *RRCResumeComplete* message. The time between the start of T2 and the UE send the *RRCResumeRequest* message should meet the requirements specified in TS 38.133 [50] Clause 5.6.3.5.

17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the NR-DL-AoD-ProvideLocationInformation IE within the response time (see clause 4.16.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.

18. The UE shall perform and report the RSTD measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1. If the UE transmits an NR-DL-AoD-ProvideLocationInformation IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the NR-DL-AoD-ProvideLocationInformation IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.

19. If the UE message at step 17 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

20. Repeat steps 2-18 in Tables 16.4.1.4-1 until the confidence level according to Annex D is achieved.

#### 16.4.1.4.3 Message contents

**Table 16.4.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.4.1.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.1.4.3-3: CellGroupConfig (Table 16.4.1.4.3-2)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms1280 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 16.4.1.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.4.1.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.4.1.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 16.4.1.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.1.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.1.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.1.4.3-7: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 &2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | Cell 1 |
| | 1 | | Cell 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.1.4.3-8: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |

| | | | |
|---|---|---|---|
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 16.4.1.4.3-9 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 16.4.1.4.3-9: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.4.1.5 | Result of the response time | |

| | | | |
|---|---|---|---|
| | | calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.1.4.3-10: RRCRelease**

| Derivation Path: TS 38.508-1 [45], Table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT |
|---|

**Table 16.4.1.4.3-11: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.4.1.5 Test requirement

Table 16.4.1.5-1 define the primary level settings including test tolerances for the test

**Table 16.4.1.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | 1.28s | |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 6 | |

**Table 16.4.1.5-2: Cell specific test parameters**

| Parameter | Unit | | Cell 1 | Cell 2 |
|---|---|---|---|---|

| | | Test configuration | T1 | T2 | T1 | T2 |
|---|---|---|---|---|---|---|
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -88 | -88 | -Infinity | -88 |
| | | 2 | -88 | -88 | -Infinity | -88 |
| | | 3 | -85 | -85 | -Infinity | -85 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |
| | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.58 | | -60.58 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit specified in clause 5.6.3.5, starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be higher than the time duration above because of the uncertainty in acquiring the first available PRACH occasion to transition to RRC_CONNECTED state to report the measurements.

$$T_{PRS-RSRP,i} = \left( K_{carrier_PRS} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

$N_{RxBeam,i}$ = 1, $K_{carrier_PRS}$=1

$N_{PRS,i}^{slot} = 4$

For SCS 15KHz:

$$L_{available_{PRS},i} = \frac{10}{10 * 1 * 14} * 4 * 1 * 4 = 1.14ms$$

For SCS=30KHz

$$L_{available_{PRS},i} = \frac{10}{20 * 1 * 14} * 4 * 1 * 4 = 0.571ms$$

$N_{sample}$= 4

$T_{last,i} = T_i + T_{available_PRS,i}$

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$

$T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$; where Tprs = 160 ms, and $T_{DRX}$ is 1280msec (depending on UE capabilities. Therefore, $T_{available_PRS,i} = 1280$

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 5.12s and 206.08s.

For SCS 30KHz The LPP time IE ranges between 5.12s and 103.6s.

The LPP time IE ranges between 6s and 207s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 207.3

The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.4.1.5-3 Table 16.4.1.5-4, Table 16.4.1.5-5 , Table 16.4.1.5-6 for reference.

**Table 16.4.1.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17***

| N' | $\lceil 4/N' \rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 16.4.1.5-4: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 15KHz***

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 16.4.1.5-5: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 30KHz***

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|

| | |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 16.4.1.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of correct events observed during repeated tests shall be at least 90%.

## 16.4.2 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 16.4.2.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] Clause 5.6.3.5 for single positioning frequency layer under AWGN propagation conditions in RRC_INACTIVE in FR1 standalone scenario.

### 16.4.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, supports DL-PRS measurement in RRC_INACTIVE state and reduced number of samples for PRS measurement in RRC_INACTIVE state.

### 16.4.2.3 Minimum conformance requirements

Same as in clause 16.4.1.3.

### 16.4.2.4 Test description

#### 16.4.2.4.1 Initial conditions

The supported test configurations in listed in Table 16.4.2.4-1.

**Table 16.4.2.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.4.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.4.2.5-1 and Table 16.4.2.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.4.2.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.4.2.4.2 Test procedure

Same as in clause 16.4.3.4.2.

### 16.4.2.4.3 Message contents

As defined in clause 16.4.1.4.3, with the following exceptions:

Table 16.4.1.4.3-1 replaces Table 16.4.2.4.3-6 , Table 16.4.2.4.3-2 replaces Table16.4.1.4.3-7 and Table 16.4.1.4.3-3 replaces Table 16.4.2.4.3-9.

**Table 16.4.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.3.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.3.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 &2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n1 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | TBD |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.4.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |

| | | | |
|---|---|---|---|
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.4.2.5 Test requirement

Table 16.4.2.5-1 and Table 16.4.2.5-2 define the primary level settings including test tolerances for the test

**Table 16.4.2.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |

| | | | | |
|---|---|---|---|---|
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX cycle length | s | 1, 2, 3 | 1.28 | |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 3 | |
| T2 | s | 1, 2, 3 | 5 | |

**Table 16.4.2.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | | | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| PRS configuration Note 5 | | 1 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 2 | PRS.1.2 FR1 | | PRS.1.2 FR1 | |
| | | 3 | PRS.2.2 FR1 | | PRS.2.2 FR1 | |
| PRS Resource slot offset | slots | 1, 2, 3 | 0 | | 0 | |
| PRS muting info | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0 | -Infinity | -6 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 3.08 | -Infinity | -0.63 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -94.92 | -Infinity | -98.63 |
| | | 2 | -Infinity | -94.92 | -Infinity | -98.63 |
| | | 3 | -Infinity | -91.91 | -Infinity | -95.62 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -98 | -98 | -Infinity | -98 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 2 | -98 | -98 | -Infinity | -98 |
| | | 3 | -95 | -95 | -Infinity | -95 |
| Io Note 3 | dBm/19.08 MHz | 1 | N/A | -61.05 | N/A | -61.05 |
| | dBm/19.08 MHz | 2 | | -61.05 | | -61.05 |
| | dBm/47.88 MHz | 3 | | -57.05 | | -57.05 |
| Propagation Condition | | 1, 2, 3 | AWGN | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels apply only for symbols where PRS is transmitted.

Note 4: OCNG shall be used such that all active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols, except for slots where PRS is transmitted by any of the cells.

Note 5: Void.

The UE shall perform and report the PRS-RSRP measurements for Cell 1 and Cell 2, within the time limit specified in clause 5.6.3.5, starting from the beginning of time interval T2.

NOTE: The actual overall delays measured in the test may be higher than the time duration above because of the uncertainty in acquiring the first available PRACH occasion to transition to RRC_CONNECTED state to report the measurements.

$$\mathrm{T}_{\mathrm{PRS-RSRP,i}} = \left( K_{\mathrm{carrier_PRS}} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,\mathrm{i}}}{N} \right\rceil * N_{sample} - 1 \right) * \mathrm{T}_{\mathrm{effect,i}} + \mathrm{T}_{\mathrm{last,i}}$$

$N_{RxBeam,i}$ = 1, $K_{\mathrm{carrier_PRS}}$=1

$N_{PRS,i}^{slot} = 4$

For SCS 15KHz:

$$L_{available_{PRS},i} = \frac{10}{10 * 1 * 14} * 4 * 1 * 4 = 1.14ms$$

For SCS=30KHz

$$L_{available_{PRS},i} = \frac{10}{20 * 1 * 14} * 4 * 1 * 4 = 0.571ms$$

$N_{sample}$= 1

$\mathrm{T}_{\mathrm{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$

$T_{\mathrm{effect,i}} = \left\lceil \frac{T_{\mathrm{i}}}{T_{available_PRS,\mathrm{i}}} \right\rceil * T_{available_PRS,\mathrm{i}}$

$T_{available_PRS,i} = LCM(T_{PRS,\mathrm{i}}, T_{DRX})$; where Tprs = 160 ms, and $T_{DRX}$ is 1280msec (depending on UE capabilities. Therefore, $T_{available_PRS,i} = 1280$

Where the remaining parameters depend on the UE capabilities.

For SCS 15KHz The LPP time IE ranges between 1.28s and 52.48s.

For SCS 30KHz The LPP time IE ranges between 1.28s and 26.88s.

The LPP time IE ranges between 2s and 53s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 53.3

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.4.2.5-3 Table 16.4.2.5-4, Table 16.4.2.5-5 , Table 16.4.2.5-6 for reference.

**Table 16.4.2.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17***

| N' | $\left\lceil \frac{4}{N'} \right\rceil$ |
|---|---|
| n1 | 4 |
| n2 | 2 |
| >=n4 | 1 |

**Table 16.4.2.5-4: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 15KHz***

| N | $\left\lceil \frac{1.143}{N} \right\rceil$ |
|---|---|
| nDot125 | 10 |
| nDot25 | 4 |
| nDot5 | 2 |
| >=n2 | 1 |

**Table 16.4.2.5-5: value of N based on *durationOfPRS-ProcessingSymbols-r17 for SCS 30KHz***

| N | $\left\lceil \frac{0.571}{N} \right\rceil$ |
|---|---|
| nDot125 | 5 |
| nDot25 | 2 |
| >=nDot1 | 1 |

**Table 16.4.2.5-6: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| durationOfPRS-ProcessingSymbolsInEveryTms | Teffect | Tlast |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6, C 2.5 and C4.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.4.3 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 16.4.3.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] Clause 5.6.3.5 for single positioning frequency layer under AWGN propagation conditions in RRC_INACTIVE in FR2 standalone scenario.

### 16.4.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supports DL-PRS measurement in RRC_INACTIVE state.

### 16.4.3.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 5.6.3.3) PRS-RSRP measurements, defined in TS 38.215 [4], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,\ total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max\left(T_{effect,i}\right)$$

Where:

- *i* is the index of positioning frequency layer,
- L is total number of positioning frequency layers,
- $T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left(K_{carrier_PRS_i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$ Where:

- $K_{carrier_PRS}$ is a scaling factor for PRS-based NR positioning measurements in RRC_INACTIVE. If the UE supports *parallelPRS-MeasRRC-Inactive-r17*, $K_{carrier_PRS}$ = 1. Otherwise,
- If Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$, $K_{carrier_PRS}$ equals to the sum of $K_{carrier}$ in 4.2.2.4 and one positioning layer.
- If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$, $K_{carrier_PRS}$ equals to the sum of $N_{layer}$ in 4.2.2.7 and one positioning layer.
- $N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping:
  - $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1, and if positioning frequency layer *i* is in FR2
    - $N_{RxBeam,i}$ equals to the value as UE reported in *supportedLowerRxBeamSweepingFactor-FR2* if the capability is reported by the UE for the band containing positioning frequency layer i, and LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-DL-AoD-RequestLocationInformation*.
    - $N_{RxBeam,i}$ equals to 8, otherwise.
- $L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only unmuted PRS resources that are not fully overlapped with other higher-priority DL signals/channels are considered.
- $N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,
- {N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSymbols-r17* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms-r17*in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],

- N' is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17* in clause 6.4.3 of TS 37.355 [34],

- $N_{sample}$ is the number of PRS-RSRP measurement samples and

  - $N_{sample}$ = 1, if UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [34], and the LMF indicates the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamplesrequestedDL-PRS-ProcessingSamples* [34], and the following conditions are met:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 2, if UE supports *supportedDL-PRS-ProcessingSamples-RRC-Inactive* [34], and the LMF indicates the UE to perform positioning measurements with reduced number of samples by *reducedDL-PRS-ProcessingSamples* [34], and the following conditions are not met:

    - PRS bandwidth is within the initial BWP and

    - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.

  - $N_{sample}$ = 4 otherwise

- $T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time,

  - $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,

  - $T_i$ corresponds to *durationOfPRS-ProcessingSymbolsInEveryTms-r17* in TS 37.355 [34],

  - $T_{available_PRS,i} = LCM(T_{PRS,i}, T_{DRX})$ is the least common multiple between $T_{PRS,i}$ and $T_{DRX}$,

  - $T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

  - $T_{DRX}$ is the DRX cycle length.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

- $T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

- $N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

## 16.4.3.4 Test description

### 16.4.3.4.1 Initial conditions

The supported test configurations in listed in Table 16.4.3.4-1.

**Table 16.4.3.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.4.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.4.3.5-1 and Table 16.4.3.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.4.3.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.4.3.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During T2 UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T = 50$ ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

The beginning of the time interval T2 shall be aligned with the first DRX cycle containing a DL PRS resource(s).

0. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 16.4.3.4.3-0.
1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 16.4.3.5-1 and Table 16.4.3.5-2. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
12. When T1 expires, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and switch the power setting from T1 to T2 as specified in Table 16.4.3.5-1 and Table 16.4.3.5-2.

13. Wait for the UE monitors for occurrence of the periodic event requested.

14. The UE shall transmit the UE transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 17.

15. The SS transmits an *RRCResume* message.

16. The UE transmits an *RRCResumeComplete* message. The time between the start of T2 and the UE send the *RRCResumeRequest* message should meet the requirements specified in TS 38.133 [50] Clause 5.6.3.5.

17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the NR-DL-AoD-ProvideLocationInformation IE within the response time (see clause 4.16.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message. The UE shall perform and report the RSTD measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1. If the UE transmits an NR-DL-AoD-ProvideLocationInformation IE including the nr-DL-PRS-RSRP-Result-r16 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the NR-DL-AoD-ProvideLocationInformation IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.

18. If the UE message at step 17 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

19. Repeat steps 2-18 in Tables 16.4.3.4-1 until the confidence level according to Annex D is achieved.

#### 16.4.3.4.3 Message contents

**Table 16.4.3.4.3-0: SIB1**

| Derivation Path: TS 38.508-1 [45], clause 4.6.1 with condition SDT |
|---|

**Table 16.4.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 16.4.3.4.3-2: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.3.4.3-3: CellGroupConfig (Table 16.4.3.4.3-2)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |

| | | | |
|---|---|---|---|
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.3.4.3-4: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 16.4.3.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 16.4.3.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.3.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.3.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.3.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.3.4.3-7: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.3.4.3-8: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 16.4.3.4.3-9 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 16.4.3.4.3-9: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| time | See 16.4.3.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.3.4.3-10: RRCRelease**

| |
|---|
| Derivation Path: TS 38.508-1 [45], Table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT |

**Table 16.4.3.4.3-11: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.4.3.5 Test requirement

Table 16.4.3.5-1 and Table 16.4.3.5-2 define the primary level settings including test tolerances for the test.

**Table 16.4.3.5-1: General test parameters for PRS RSRP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| CP length | | Config 1 | Normal | |
| DRX | | Config 1 | 0.64s | |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 83 | |

**Table 16.4.3.5-2: Cell-specific test parameters for PRS RSRP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption Note 7 | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.2 FR2 | | PRS.1.2 FR2 | |
| PRS muting configuration | | | Config 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz Note5 | | -99.1 | | -99.1 | |
| $N_{oc}$ Note2 | | dBm/SCS Note4 | Config 1 | -90.1 | | -90.1 | |
| SS-RSRP Note 3 | | dBm/SCS Note5 | Config 1 | -91 | -91 | -Infinity | -99 |
| PRS-RSRP Note 3 | | dBm/SCS Note5 | Config 1 | -Infinity | -92.1 | -Infinity | -100.1 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | Config 1 | -Infinity | -2.7 | -Infinity | -12.3 |
| PRS $\hat{E}_s/N_{oc}$ | | dB | Config 1 | -Infinity | -2 | -Infinity | -10 |

| | | | | |
|---|---|---|---|---|
| Io[Note3] | dBm/19 0.08 MHz Note5 | Config 1 | -55.69 | -55.69 |
| Propagation Condition | | Config 1 | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The PRS-RSRP measurement time fulfils the requirements specified in clause 16.4.3.3.

The UE shall perform and report the PRS RSRP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 5.6.3.5 starting from the beginning of time interval T2

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS-RSRP measurement for each correct event shall be within the PRS-RSRP reporting range specified in clause TS 38.133 clause 10.1.24.3.1.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period. The PRS-RSRP measurement period follows the equation:

$$\mathrm{T}_{\mathrm{PRS-RSRP,i}} = \left( K_{\mathrm{carrier_{PRS}}} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS,\mathrm{i}}}}{N} \right\rceil * N_{sample} - 1 \right) * \mathrm{T}_{\mathrm{effect,i}} + \mathrm{T}_{\mathrm{last,i}}$$

Where $K_{\mathrm{carrier_PRS}} = 1$, $N_{RxBeam,i}$ = 8, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,\mathrm{i}} = 0.035$, $N_{sample}$=4. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $\mathrm{T}_{\mathrm{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$\mathrm{T}_{\mathrm{available_PRS,i}} = \mathrm{LCM}(\mathrm{T}_{\mathrm{PRS,i}}, \mathrm{T}_{\mathrm{DRX}})$, where Tprs = 160 ms, and $\mathrm{T}_{\mathrm{DRX}}$= 640ms depending on UE capabilities. Therefore, $\mathrm{T}_{\mathrm{available_PRS,i}} = 640$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T}_{\mathrm{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 20.488s and 82.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 21s and 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 21.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.4.3.5-3, Table 16.4.3.5-4 and Table 16.4.3.5-5 for reference.

**Table 16.4.3.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 16.4.3.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >= nDot025 | 1 |

**Table 16.4.3.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.4.4 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 16.4.4.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement requirements specified in TS 38.133 [50] Clause 5.6.3.5 for single positioning frequency layer under AWGN propagation conditions in RRC_INACTIVE in FR2 standalone scenario.

### 16.4.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, supports DL-PRS measurement in RRC_INACTIVE state and reduced number of samples for PRS measurement in RRC_INACTIVE state.

### 16.4.4.3 Minimum conformance requirements

Same as in clause 16.4.3.3.

### 16.4.4.4 Test description

#### 16.4.4.4.1 Initial conditions

The supported test configurations in listed in Table 16.4.4.4-1.

**Table 16.4.4.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.4.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.4.4.5-1 and Table 16.4.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.4.4.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

#### 16.4.4.4.2 Test procedure

Same as in clause 16.4.3.4.2.

#### 16.4.4.4.3 Message contents

As defined in clause 16.4.3.4.3, with the following exceptions:

Table 16.4.4.4.3-1 replaces Table 16.4.3.4.3-6 and Table 16.4.4.4.3-2 replaces Table 16.4.3.4.3-9.

**Table 16.4.4.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.3.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.4.3.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.4.4.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.4.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 16.4.4.5 Test requirement

Table 16.4.4.5-1 and Table 16.4.4.5-2 define the primary level settings including test tolerances for the test.

**Table 16.4.4.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX cycle length | s | Config 1 | 0.64 | |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| T1 | s | Config 1 | 3 | |
| T2 | s | Config 1 | 22 | |

**Table 16.4.4.5-2: Cell specific test parameters**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |

| | | | | | | |
|---|---|---|---|---|---|---|
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | kHz | Config 1 | 120 | | 120 | |
| PRS configuration[Note8] | | Config 1 | PRS.1.2 FR2 | | PRS.1.2 FR2 | |
| PRS Resource slot offset | slots | Config 1 | 0 | | 0 | |
| PRS muting configuration | | Config 1 | ‘10’ | | ‘01’ | |
| EPRE ratio of PSS to SSS | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -103 | | -103 | |
| $N_{oc}$ Note2 | dBm/SCS Note4 | Config 1 | -94 | | -94 | |
| SS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -89 | -89 | -89 | -89 |
| PRS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -Infinity | -90.4 | -Infinity | -94.2 |
| PRS $\hat{E}_s/I_{ot}$ | dB | Config 1 | -Infinity | 0.3 | -Infinity | -5.6 |
| PRS $\hat{E}_s/N_{oc}$ | dB | Config 1 | -Infinity | 3.56 | -Infinity | -0.28 |
| Io[Note3] | dBm/190.08 MHz Note5 | Config 1 | N/A | -55.73 | N/A | -55.73 |
| Propagation Condition | | Config 1 | AWGN | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that all active cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols, except for slots where PRS is transmitted by any of the cells. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels apply only for symbols where PRS is transmitted. |
| Note 4: | PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 6: | As observed with 0 dBi gain antenna at the centre of the quiet zone |
| Note 7: | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |
| Note 8: | Void. |

The PRS-RSRP measurement time fulfils the requirements specified in clause 16.4.4.3.

The UE shall perform and report the PRS RSRP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 5.6.3.5 starting from the beginning of time interval T2

The rate of the correct events for each neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS-RSRP measurement for each correct event shall be within the PRS-RSRP reporting range specified in clause TS 38.133 clause 10.1.24.3.1.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the UE Rx-Tx time difference measurement period. The PRS-RSRP measurement period follows the equation:

$$\mathrm{T}_{\mathrm{PRS-RSRP,i}} = \left( K_{\mathrm{carrier_{PRS}}} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_{PRS},\mathrm{i}}}{N} \right\rceil * N_{sample} - 1 \right) * \mathrm{T}_{\mathrm{effect,i}} + \mathrm{T}_{\mathrm{last,i}}$$

Where $K_{\mathrm{carrier_PRS}} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,\mathrm{i}} = 0.035$, $N_{sample}$=1. N is the parameter *durationOfPRS-ProcessingSymbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive* from TS 37.355 [49], $\mathrm{T}_{\mathrm{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$\mathrm{T}_{\mathrm{available_PRS,i}} = \mathrm{LCM}(\mathrm{T}_{\mathrm{PRS,i}}, \mathrm{T}_{\mathrm{DRX}})$, where Tprs = 160 ms, and $\mathrm{T}_{\mathrm{DRX}}$= 640ms depending on UE capabilities. Therefore, $\mathrm{T}_{\mathrm{available_PRS,i}} = 640$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 1 - 1\right) * T_{effect} + \mathrm{T}_{\mathrm{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 21.12s. The value of the LPP time IE is rounded up to the next second. The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 22s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 22.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 16.4.4.5-3, Table 16.4.4.5-4 and Table 16.4.4.5-5 for reference.

**Table 16.4.4.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 16.4.4.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\lceil \frac{0.035}{N} \rceil$ |
|---|---|
| >= nDot025 | 1 |

**Table 16.4.4.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 16.5 NR PRS-RSRP measurement accuracy test case in RRC_INACTIVE

## 16.5.1 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 SA

### 16.5.1.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement accuracy requirements specified in TS 38.133 [50] Clause 10.1.24.2 for single positioning frequency layer under AWGN propagation conditions in RRC_INACTIVE in FR1 standalone scenario.

### 16.5.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supports DL-PRS measurement in RRC_INACTIVE state.

### 16.5.1.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 16.5.1.3-1 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE does not support positioning measurements with reduced number of samples, or LMF does not indicate UE to perform positioning measurements with reduced number of samples

**Table 16.5.1.3-1: PRS-RSRP absolute accuracy for FR1**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io [Note 7] range | | | | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | NR operating band groups [Note 8] | Minimum Io [Note 1] dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz [Note 6] | dBm/30kHz [Note 6] | dBm/60kHz [Note 6] | |
| ±3.5 | ±8 | ≥-3dB | ≥24 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±8.5 | ±13 | ≥-13dB | 24 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±6 | ±10.5 | | 52< BW≤ 104 | All | Note 4 | | | | |
| ±4.5 | ±9 | | BW >104 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 PRB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

The accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 16.5.1.3-2 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE does not support positioning measurements with reduced number of samples, or LMF does not indicate UE to perform positioning measurements with reduced number of samples

**Table 16.5.1.3-2: PRS-RSRP relative accuracy for FR1**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io [Note 7] range | | | | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | NR operating band groups [Note 8] | Minimum Io [Note 1] dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz [Note 6] | dBm/30kHz [Note 6] | dBm/60kHz [Note 6] | |
| ±3.5 | ±5.0 | ≥-3dB | ≥24 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±9.5 | ±11.0 | ≥-13dB | 24 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±6.5 | ±8.0 | | 52< BW≤ 104 | All | Note 4 | | | | |
| ±5.0 | ±6.5 | | BW >104 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

## 16.5.1.4 Test description

### 16.5.1.4.1 Initial conditions

The supported test configurations in listed in Table 16.5.1.4-1.

**Table 16.5.1.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | NR 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | NR 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | NR 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations in each supported band | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.5.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.5.1.5-1.
3. Propagation conditions are set according to clause 4.15.2.
4. Message contents are defined in clause 16.5.1.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.5.1.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration. The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state and all cells start to transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The NR-DL-AoD-RequestLocationInformation message and the DL-AoD assistance data as defined in clause 16.5.1.4.3 shall be provided to the UE during the set-up period. The beginning of the measurement period shall be aligned with the first DRX cycle containing a DL PRS resource(s).

1. Configure the Cell for RA-SDT with SIB1 configured according to TS 38.508-1 [45], clause 4.6.1 with condition SDT
2. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set parameters other than PRSconfiguration according to the Table 16.5.1.5-1. Propagation conditions are set according to clause 4.15.2.
4. Set-up period starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.

10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of measurement period, where ΔT = 50 ms.

11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.

12. When set-up period ends, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and set the PRS parameters as specified in Table 16.5.1.5-1.

13. Wait for the UE monitors for occurrence of the periodic event requested.

14. The UE shall transmit the UE transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the RRCResumeRequest message and then go to step 17.

15. The SS transmits an *RRCResume* message.

16. The UE transmits an *RRCResumeComplete* message.

17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.15.3).

18. If the UE message at step 17 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

19. The SS shall check the nr-DL-PRS-RSRP-Result-r16 value for Cell 2 in the *NR-DL-AoD-SignalMeasurementInformation* according to Table 16.5.1.5-3.

20. Repeat steps 2-19 in Table 16.5.1.4-1 until the confidence level according to Annex D is achieved.

21. Repeat step 2-20 for the other sub-test defined in Table 16.5.1.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 16.5.1.4.3 Message contents

As defined in clause 16.4.1.4.3, with the following exceptions:

Table 16.5.1.4.3-1 replaces Table 16.4.1.4.3-6 and Table 16.5.1.4.3-2 replaces Table 16.4.1.4.3-7.

**Table 16.5.1.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1&2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1 Config 1&2 |
| | 1 | 24 PRBs | Sub-test 1 Config 3 |
| | 21 | 104 PRBs | Subtest-2 Config 1&2 |

| | | | |
|---|---|---|---|
| | 28 | 132 PRBS | Subtest 2-Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.1.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.1.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.5.1.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | |
| n160-r16 | 10 | | |

| | | | |
|---|---|---|---|
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| | 1 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-test 1 |
| | 0 | | Sub-test 1 |
| n4-r16 | 1 | | Sub-test 2 |
| | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 16.5.1.5 Test requirement

Table 16.5.1.5-1 defines the primary level settings including test tolerances for the test.

**Table 16.5.1.5-1: PRS-RSRP test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Cell ID | | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | | freq1 | | freq1 | |
| Duplex mode | Config 1 | | FDD | | | |
| | Config 2,3 | | TDD | | | |
| TDD configuration | Config 1 | | Not Applicable | | | |
| | Config 2 | | TDDConf.1.1 | | | |
| | Config 3 | | TDDConf.2.1 | | | |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | | | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | | | |
| BWP BW | Config 1 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | | | |
| Downlink initial BWP configuration | | | DLBWP.0.1 | | | |
| Uplink initial BWP configuration | | | ULBWP.0.1 | | | |
| DRX Cycle | | ms | 1280 | | | |
| PDSCH Reference measurement channel | Config 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | Config 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | Config 3 | | SR2.1 TDD | | SR2.1 TDD | |
| RMSI CORESET Reference Channel | Config 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | Config 2 | | CR.1.1 TDD | | CR.1.1 TDD | |
| | Config 3 | | CR2.1 TDD | | CR2.1 TDD | |
| Control channel RMC | Config 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | Config 2 | | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | Config 3 | | CCR2.1 TDD | | CCR2.1 TDD | |
| PRS configuration | Config 1 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | PRS.2.3 FR1 | PRS.2.3 FR1 | PRS.2.4 FR1 | PRS.2.4 FR1 |
| PRS Resource slot offset (slot) | Config 1,2,3 | slot | 0 | 4 | 0 | 4 |
| SSB configuration | Config 1 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 |
| Time offset with Cell 1 | Config 1 | ms | - | 3 | - | 3 |
| | Config 2,3 | µs | - | 3 | - | 3 |
| SMTC configuration | Config 1 | | SMTC.2 | | | |
| | Config 2,3 | | SMTC.1 | | | |
| OCNG Patterns | | | OCNG pattern 1 | | | |
| PDSCH/PDCCH subcarrier spacing | Config 1,2 | kHz | 15 kHz | | | |
| | Config 3 | | 30 kHz | | | |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $N_{oc}$ Note2 | Config 1,2 | | dBm/15KhZ | -98 | | -98 | |
| | Config 3 | | | -98 | | -98 | |
| $N_{oc}$ Note2 | Config 1,2 | | dBm/SCS | -98 | | -98 | |
| | Config 3 | | | -95 | | -95 | |
| $\hat{E}_s/I_{ot}$ | | | dB | -2.41 | -12.12 | -2.41 | -12.12 |
| $\hat{E}_s/N_{oc}$ | | | dB | -2 | -10 | -2 | -10 |
| PRS-RSRP Note3 | Config 1, 2 | | dBm/SCS | -100 | -108 | -100 | -108 |
| | Config 3 | | | -97 | -105 | -97 | -105 |
| Io Note3 | Config 1,2 | | dBm/19.08MHz | -64.57 | | -64.57 | |
| | Config 3 | | dBm/47.88MHz | -60.58 | | -60.58 | |
| Propagation condition | | | | AWGN | | | |
| Antenna configuration | | | | 1x2 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Void.
Note 6: Void

**Table 16.5.1.5-2: PRS-RSRP accuracy requirements for the reported values**

| Test | Configuration | Report Mapping | Lowest reported value | Highest reported value |
|---|---|---|---|---|
| Sub-test 1 | Config 1& 2 | Absolute PRS RSRP accuracy for Cell 1 | 51 | 62 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 38 | 59 |
| | | Relative PRS RSRP accuracy for Cell 2 | 21 | 30 |
| | Config 3 | Absolute PRS RSRP accuracy for Cell 1 | 54 | 65 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 41 | 62 |
| | | Relative PRS RSRP accuracy for Cell 2 | 21 | 30 |
| Sub-test 2 | Config 1& 2 | Absolute PRS RSRP accuracy for Cell 1 | 51 | 62 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 41 | 56 |
| | | Relative PRS RSRP accuracy for Cell 2 | 24 | 30 |
| | Config 3 | Absolute PRS RSRP accuracy for Cell 1 | 54 | 65 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 45 | 58 |
| | | Relative PRS RSRP accuracy for Cell 2 | 25 | 30 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 16.5.2 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 with reduced sample number

### 16.5.2.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement accuracy requirements specified in TS 38.133 [50] Clause 10.1.24.2 for single positioning frequency layer under AWGN propagation conditions in RRC_INACTIVE in FR1 standalone scenario.

### 16.5.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supports DL-PRS measurement in RRC_INACTIVE state.

### 16.5.2.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 16.5.2.3-2 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

**Table 16.5.2.3-1: PRS-RSRP absolute accuracy for FR1 with reduced sample number**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io Note 6 range | | | | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | NR operating band groups Note 7 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | dBm/$BW_{Channel}$ |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 5 | dBm/30kHz Note 5 | dBm/60kHz Note 5 | |
| ±3.5 | ±8 | ≥0 | ≥48 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| ±8.5 | ±13 | ≥-6 | 48 ≤ BW ≤ 52 | All | Note 3 | | | | |
| ±6 | ±10.5 | | 52< BW≤ 104 | All | Note 3 | | | | |
| ±4.5 | ±9 | | BW >104 | All | Note 3 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 PRB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by $\Delta>0$, when applicable, as described in Sections B.3.2 and B.3.3.

NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 7: NR operating band groups are as defined in Section 3.5.2.

The relative PRS-RSRP accuracy is defined as accuracy of the difference between two PRS-RSRP measurements.

The relative PRS-RSRP accuracy requirements apply for the cases when PRS-RSRP is measured from PRS resources in the same PRS resource set in FR1.

The accuracy requirements for PRS-RSRP measurement for FR1 defined in Table 16.5.2.3-2 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band
- UE supports positioning measurements with reduced number of samples, and LMF indicates UE to perform positioning measurements with reduced number of samples
- AWGN channel

**Table 16.5.2.3-2: PRS-RSRP relative accuracy for FR1 with reduced sample number**

| Accuracy | | Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Io Note 6 range | | | | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS}/K_{comb}^{PRS}$) | NR operating band groups Note 7 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | | |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 5 | dBm/30kHz Note 5 | dBm/60kHz Note 5 | $dBm/BW_{Channel}$ |
| ±3.5 | ±5.0 | ≥0 | ≥48 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G, NR_TDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| ±9.5 | ±11.0 | ≥-6 | 48 ≤ BW ≤ 52 | All | Note 3 | | | | |
| ±6.5 | ±8.0 | | 52< BW≤ 104 | All | Note 3 | | | | |
| ±5.0 | ±6.5 | | BW >104 | All | Note 3 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: PRS bandwidth is as indicated in *prs-Bandwidth* in the OTDOA or DL-AoD assistance data defined in [34].
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 RB.
NOTE 4: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 6: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 7: NR operating band groups are as defined in Section 3.5.2.

## 16.5.2.4 Test description

### 16.5.2.4.1 Initial conditions

The supported test configurations in listed in Table 16.5.2.4-1.

**Table 16.5.2.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | NR 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | NR 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | NR 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations in each supported band | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.5.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 16.5.2.5-1 and Table 16.5.2.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.5.2.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.5.2.4.2 Test procedure

Same as in clause 16.5.1.4.2.

### 16.5.2.4.3 Message contents

As defined in clause 16.4.2.4.3, with the following exceptions:

Table 16.5.2.4.3-1 replaces Table 16.4.2.4.3-6 and Table 16.5.2.4.3-2 replaces Table 16.4.2.4.3-7.

**Table 16.5.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz1520 | | Config 1&2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | Sub-test 1 Config 1&2 |

| | | | |
|---|---|---|---|
| | 1 | 24 PRBs | Sub-test 1 Config 32 |
| | 21 | 104 PRBs | Subtest-2 Config 1&2 |
| | 28 | 132 PRBs | Subtest 2-Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.2.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 92 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 153 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.2.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.5.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | |
| n160-r16 | 10 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-test 1 |
| n4-r16 | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.5.2.5 Test requirement

**Table 16.5.2.5-1: PRS-RSRP test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | freq1 | | freq1 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Duplex mode | Config 1 | | FDD | | | |
| | Config 2,3 | | TDD | | | |
| TDD configuration | Config 1 | | Not Applicable | | | |
| | Config 2 | | TDDConf.1.1 | | | |
| | Config 3 | | TDDConf.2.1 | | | |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | | | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | | | |
| BWP BW | Config 1 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | | | |
| Downlink initial BWP configuration | | | DLBWP.0.1 | | | |
| Uplink initial BWP configuration | | | ULBWP.0.1 | | | |
| DRX Cycle | | s | 1.28 | | | |
| PDSCH Reference measurement channel | Config 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | Config 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | Config 3 | | SR2.1 TDD | | SR2.1 TDD | |
| RMSI CORESET Reference Channel | Config 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | Config 2 | | CR.1.1 TDD | | CR.1.1 TDD | |
| | Config 3 | | CR2.1 TDD | | CR2.1 TDD | |
| Control channel RMC | Config 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | Config 2 | | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | Config 3 | | CCR2.1 TDD | | CCR2.1 TDD | |
| PRS configuration | Config 1 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | PRS.2.3 FR1 | PRS.2.3 FR1 | PRS.2.4 FR1 | PRS.2.4 FR1 |
| PRS BW | Config 1 | | 52 PRBs | 52 PRBs | | |
| | Config 2 | | 52 PRBs | 52 PRBs | | |
| | Config 3 | | 48 PRBs | 48 PRBs | | |
| PRS Resource slot offset (slot) | Config 1,2,3 | slot | 0 | 4 | 0 | 4 |
| SSB configuration | Config 1 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 |
| Time offset with Cell 1 | Config 1 | ms | - | 3 | - | 3 |
| | Config 2,3 | µs | - | 3 | - | 3 |
| SMTC configuration | Config 1 | | SMTC.2 | | | |
| | Config 2,3 | | SMTC.1 | | | |
| OCNG Patterns | | | OCNG pattern 1 | | | |
| PDSCH/PDCCH subcarrier spacing | Config 1,2 | kHz | 15 kHz | | | |
| | Config 3 | | 30 kHz | | | |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| EPRE ratio of PRS to SSS | | | | | | |
| $N_{oc}$ Note2 | Config 1,2 | dBm/15Khz | -98 | | -98 | |
| | Config 3 | | | | | |
| $N_{oc}$ Note2 | Config 1,2 | dBm/SCS | -98 | | -98 | |
| | Config 3 | | -95 | | -95 | |
| $\hat{E}_s/I_{ot}$ | | dB | 0 | -6 | 0 | -6 |
| $\hat{E}_s/N_{oc}$ | | dB | 3.18 | -0.76 | 3.18 | -0.76 |
| PRS-RSRP Note3 | Config 1, 2 | dBm/SCS | -94.82 | -98.76 | -94.82 | -98.76 |
| | Config 3 | | -91.81 | -95.75 | -91.81 | -95.75 |
| Io Note3 | Config 1,2 | dBm/19.08MHz | -61.02 | | -61.02 | |
| | Config 3 | dBm/47.88MHz | -57.03 | | -57.03 | |
| Propagation condition | | | AWGN | | | |
| Antenna configuration | | | 1x2 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 16.5.2.5-2: PRS-RSRP accuracy requirements for the reported values**

| Test Configuration | Configuration | Report Mapping | Lowest reported value | Highest reported value |
|---|---|---|---|---|
| Sub-test 1 | Config 1& 2 | Absolute PRS RSRP accuracy for Cell 1 | 57 | 67 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 48 | 68 |
| | | Relative PRS RSRP accuracy for Cell 2 | 21 | 30 |
| | Config 3 | Absolute PRS RSRP accuracy for Cell 1 | 60 | 70 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 51 | 71 |
| | | Relative PRS RSRP accuracy for Cell 2 | 21 | 30 |
| Sub-test 2 | Config 1& 2 | Absolute PRS RSRP accuracy for Cell 1 | 57 | 67 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 50 | 65 |
| | | Relative PRS RSRP accuracy for Cell 2 | 24 | 30 |
| | Config 3 | Absolute PRS RSRP accuracy for Cell 1 | 60 | 70 |
| | | Relative PRS RSRP accuracy for Cell 1 | 27 | 30 |
| | | Absolute PRS RSRP accuracy for Cell 2 | 55 | 67 |
| | | Relative PRS RSRP accuracy for Cell 2 | 25 | 30 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 16.5.3 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 16.5.3.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement accuracy requirements specified in TS 38.133 [50] Clause 10.1.24.2 for single positioning frequency layer under AWGN propagation conditions in RRC_INACTIVE in FR2 standalone scenario.

### 16.5.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and supports DL-PRS measurement in RRC_INACTIVE state.

### 16.5.3.3 Minimum conformance requirements

Same as in clause 16.3.2.3.

### 16.5.3.4 Test description

#### 16.5.3.4.1 Initial conditions

The supported test configurations in listed in Table 16.5.3.4-1.

**Table 16.5.3.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.5.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.5.3.5-1 and Table 16.5.3.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.5.3.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

#### 16.5.3.4.2 Test procedure

The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration, PRS.1.3 FR2 and PRS.1.4 FR2. The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be

in RRC_CONNECTED state and shall not have any timing information of Cell 2. During the measurement period UE shall be in RRC_INACTIVE state and all cells start to transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The NR-DL-AoD-RequestLocationInformation message and the DL-AoD assistance data as defined in clause 16.5.3.4.3 shall be provided to the UE during the set-up period. The beginning of the measurement period shall be aligned with the first DRX cycle containing a DL PRS resource(s).

0. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 16.4.3.4.3-0.
1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set parameters other than PRSconfiguration according to the Table 16.5.3.5-1 and Table 16.5.3.5-2. Propagation conditions are set according to clause 4.16.2.
4. Set-up period starts.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of measurement period, where $\Delta T = 50$ ms.
11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
12. When set-up period ends, the SS shall transmit an RRCRelease message with SDT configuration to transit the UE to RRC_INACTIVE state and set the PRS parameters as specified in Table 16.5.3.5-1 and Table 16.5.3.5-2.
13. Wait for the UE monitors for occurrence of the periodic event requested.
14. The UE shall transmit the UE transmit an *RRCResumeRequest* message.
15. The SS transmits an *RRCResume* message.
16. The UE transmits an *RRCResumeComplete* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 19.
17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3).
18. If the UE message at step 17 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
19. The SS shall check the nr-DL-PRS-RSRP-Result-r16 value for Cell 2 in the *NR-DL-AoD-SignalMeasurementInformation* according to Table 16.5.3.5-3.
20. Repeat steps 2-19 in Table 16.5.3.4-1 until the confidence level according to Annex D is achieved.
21. Repeat step 2-20 for the other sub-test defined in Table 16.5.3.5-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

#### 16.5.3.4.3 Message contents

As defined in clause 16.4.3.4.3, with the following exceptions:

Table 16.5.3.4.3-1 replaces Table 16.4.3.4.3-6 and Table 16.5.3.4.3-2 replaces Table 16.4.3.4.3-7.

**Table 16.5.3.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.3.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.3.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.5.3.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | Sub-test 1 |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-test 1 |
| n4-r16 | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and |

| | | | |
|---|---|---|---|
| | | | Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 16.5.3.5 Test requirement

Table 16.5.3.5-1 and Table 16.5.3.5-2 define the primary level settings including the test tolerances for the test.

**Table 16.5.3.5-1: PRS-RSRP general test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| DRX cycle configuration | | 1.28s | - | 1.28s | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PRS configuration | | PRS.1.3 FR2 | PRS.1.3 FR2 | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS Resource slot offset | slot | 0 | 4 | 0 | 4 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |

| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
|---|---|

**Table 16.5.3.5-2: PRS-RSRP OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 7 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/15kHz Note4 | -99.1 | | Same as Test 1 | |
| $N_{oc}$ Note1 | dBm/SCS Note4 | -90.1 | | Same as Test 1 | |
| $\hat{E}_s / N_{oc}$ | dB | -2 | -10 | -2 | -10 |
| $E_S$ | dBm/SCS Note4 | - | - | - | - |
| PRS_RP Note2 | dBm/SCS | -92.1 | -100.1 | -92.1 | -100.1 |
| $\hat{E}_s / I_{ot\ BB}$ Note6 | dB | -2.68 | -12.30 | -2.68 | -12.30 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.69 | | -55.69 | |

Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 2: PRS_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: Void
Note 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 5: Void
Note 6: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The value of the LPP Time IE depends on the UE capabilities. The calculation procedure is the same as in clause 16.4.3.5 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.142}{N} \right\rceil * 4 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation.

**Table 16.5.3.5-3: PRS RSRP accuracy requirements for the reported values**

| Test Configuration | Report Mapping | Lowest reported value | Highest reported value |
|---|---|---|---|
| Sub-test 1 | Absolute PRS RSRP accuracy for Cell 1 | 59 | 94 |
| | Relative PRS RSRP accuracy for Cell 1 | 20 | 30 |
| | Absolute PRS RSRP accuracy for Cell 2 | 52 | 86 |
| | Relative PRS RSRP accuracy for Cell 2 | 20 | 30 |
| Sub-test 2 | Absolute PRS RSRP accuracy for Cell 1 | 61 | 93 |
| | Relative PRS RSRP accuracy for Cell 1 | 23 | 30 |
| | Absolute PRS RSRP accuracy for Cell 2 | 53 | 85 |
| | Relative PRS RSRP accuracy for Cell 2 | 23 | 30 |

The PRS RSRP measurement period fulfils the requirements specified in clause 4.16.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 16.5.4 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 with reduced sample number

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 16.5.4.1 Test purpose

The purpose of the test is to verify the PRS RSRP measurement accuracy requirements specified in TS 38.133 [50] Clause 10.1.24.2 for single positioning frequency layer with reduced sample number under AWGN propagation conditions in RRC_INACTIVE in FR2 standalone scenario.

### 16.5.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, supports DL-PRS measurement in RRC_INACTIVE state and reduced number of samples for PRS measurement in RRC_INACTIVE state.

### 16.5.4.3 Minimum conformance requirements

Same as in clause 16.3.4.3.

### 16.5.4.4 Test description

#### 16.5.4.4.1 Initial conditions

The supported test configurations in listed in Table 16.5.4.4-1.

**Table 16.5.4.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 16.5.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 16.5.4.5-1 and Table 16.5.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 16.5.4.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 16.5.4.4.2 Test procedure

Same as in clause 16.5.3.4.2.

### 16.5.4.4.3 Message contents

As defined in clause 16.5.3.4.3, with the following exceptions:

Table 16.5.4.4.3-1 replaces Table 16.4.3.4.3-6, Table 16.5.4.4.3-2 replaces Table 16.4.3.4.3-7 and Table 16.5.4.4.3-3 replaces Table 16.4.3.4.3-9.

**Table 16.5.4.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.4.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 16.5.4.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.5.4.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | Sub-test 1 |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-test 1 |
| n4-r16 | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Sub-test 1 Cell 1 and |

| | | | |
|---|---|---|---|
| | | | Sub-test 2 Cell 1 |
| | 4 | | Sub-test 1 Cell 2 and Sub-test 2 Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 16.5.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 16.5.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |

| | | | |
|---|---|---|---|
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 16.5.4.5 Test requirement

Table 16.5.4.5-1 and Table 16.5.4.5-2 define the primary level settings including the test tolerances for the test.

**Table 16.5.4.5-1: PRS-RSRP general test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| DRX cycle configuration | s | 0.64 | | | |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |

| | | | | | |
|---|---|---|---|---|---|
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PRS configuration | | PRS.1.3 FR2 | PRS.1.3 FR2 | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS BW | | 64 PRBs | 64 PRBs | | |
| PRS Resource slot offset | slot | 0 | 4 | 0 | 4 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| EPRE ratio of PRS to SSS | | | | | |
| Propagation conditions | | AWGN | AWGN | AWGN | AWGN |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | |

**Table 16.5.4.5-2: PRS-RSRP OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 7 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/15kHz Note4 | -103 | | Same as Test 1 | |
| $N_{oc}$ Note1 | dBm/SCS Note4 | -94 | | Same as Test 1 | |
| $\hat{E}_s / N_{oc}$ | dB | 3.56 | -0.28 | 3.56 | -0.28 |
| $E_s$ | dBm/SCS Note4 | - | - | - | - |
| PRS_RP Note2 | dBm/SCS | -90.4 | -94.2 | -90.4 | -94.2 |
| $\hat{E}_s / I_{ot\ BB}$ Note6 | dB | 0.35 | -5.64 | 0.35 | -5.64 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.73 | | -55.73 | |
| Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 2: PRS_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: Void<br>Note 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone<br>Note 5: Void | | | | | |

| | |
|---|---|
| Note 6: | Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. |
| Note 7: | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |

The value of the LPP Time IE depends on the UE capabilities. The calculation procedure is the same as in clause 16.4.4.5 Therefore, the value of the LPP Time IE is given by the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.142}{N} \right\rceil * 1 - 1\right) * T_{effect} + \mathrm{T}_{\mathrm{last}}$$

The resulting value shall be rounded up to the next integer and transmitted in the LPP-RequestLocationInformation.

**Table 16.5.4.5-3: PRS RSRP accuracy requirements for the reported values**

| Test Configuration | Report Mapping | Lowest reported value | Highest reported value |
|---|---|---|---|
| Sub-test 1 | Absolute PRS RSRP accuracy for Cell 1 | 66 | 76 |
| | Relative PRS RSRP accuracy for Cell 1 | 25 | 30 |
| | Absolute PRS RSRP accuracy for Cell 2 | 59 | 76 |
| | Relative PRS RSRP accuracy for Cell 2 | 20 | 30 |
| Sub-test 2 | Absolute PRS RSRP accuracy for Cell 1 | 66 | 76 |
| | Relative PRS RSRP accuracy for Cell 1 | 25 | 30 |
| | Absolute PRS RSRP accuracy for Cell 2 | 61 | 73 |
| | Relative PRS RSRP accuracy for Cell 2 | 23 | 30 |

The PRS RSRP measurement period fulfils the requirements specified in clause 4.16.3.

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 17 NR PRS-RSRPP measurement requirements

## 17.1 General

This clause defines the minimum performance requirements for NR UEs capable of PRS-RSRPP measurements.

## 17.2 NR PRS-RSRPP measurement period test cases in RRC_CONNECTED

### 17.2.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA

#### 17.2.1.1 Test purpose

The purpose of the test is to verify the PRS RSRPP measurement requirements specified in TS 38.133 [50] Clause 9.9.6.5 for single positioning frequency layer under AWGN propagation conditions in standalone scenario.

#### 17.2.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and maxDL-PRS-FirstPathRSRP-MeasPerTRP.

### 17.2.1.3 Minimum conformance requirements

When the physical layer receives *NR-DL-AoD-ProvideAssistanceData* message and *NR-DL-AoD-RequestLocationInformation* message from LMF via LPP [34], the UE shall be able to measure multiple (up to the UE capability specified in Clause 9.9.3.3) PRS-RSRP measurements, defined in TS 38.215 [4], from configured PRS resources for configured TRPs on configured positioning frequency layers, within $T_{PRS-RSRP,total}$ ms.

$$T_{PRS-RSRP,\ total} = \sum_{i=1}^{L} T_{PRS-RSRP,i} + (L-1) * \max(T_{effect,i})$$

where

*i* is the index of positioning frequency layer,

L is total number of positioning frequency layers,

$T_{effect,i}$ is the periodicity of the PRS-RSRP measurement in positioning frequency layer *i*.

$$T_{PRS-RSRP,i} = \left( CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

where

$CSSF_i$ is the carrier specific scaling factor for PRS-RSRP measurements specified in clause 9.1.5.2,

$K_{p,PRS,i}$ is a scaling factor for a positioning frequency layer to be measured within the associated measurement gap pattern, which is defined as $K_{p,PRS,i} = N_{total} / N_{available}$ for UE configured with concurrent measurement gap, and $K_{p,PRS,i} = 1$ for UE not configured with concurrent measurement gap.

For a window W of duration max($T_{PRS,i}$, MGRP_max), where MGRP_max is the maximum MGRP across all configured per-UE MG and per-FR MG within the same FR as the positioning frequency layer, and starting at the beginning of any associated gap occasions covering the PRS occasion:

$N_{total}$ is the total number of associated gap occasions covering PRS occasions within the window, including both dropped and non-dropped instances of the associated measurement gap within the window, and

$N_{available}$ is the number of non-dropped associated gap occasions covering PRS occasions within the window W, after further accounting for MG collisions by applying the selected gap collision rule

Requirements do not apply if $N_{available}$ =0.

$N_{RxBeam,i}$ is the scaling factor for Rx beam sweeping, and $N_{RxBeam,i}$=1 if positioning frequency layer *i* is in FR1 and if positioning frequency layer *i* is in FR2, $N_{RxBeam,i}$ is equal to the value reported by the UE in *supportedLowerRxBeamSweepingFactor-FR2* if the UE supports the capability for the band containing positioning frequency layer i, and the LMF indicates *lowerRxBeamSweepingFactor-FR2* in *NR-TDOA-RequestLocationInformation*. $N_{RxBeam,i}$ is equal to 8, otherwise.

$L_{available_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i to be measured during $T_{available_PRS,i}$, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26]. For calculation of $L_{available_PRS,i}$, only the PRS resources unmuted and fully or partially overlapped with MG are considered.

$N_{PRS,i}^{slot}$ is the maximum number of DL PRS resources of positioning frequency layer i configured in a slot,

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to *durationOfPRS-ProcessingSysmbols* in TS 37.355 [34] processed every T ms corresponding to *durationOfPRS-ProcessingSymbolsInEveryTms* in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to *supportedBandwidthPRS* in TS 37.355 [34],

N′ is UE capability for number of DL PRS resources that it can process in a slot as indicated by *maxNumOfDL-PRS-ResProcessedPerSlot* in clause 6.4.3 of TS 37.355 [34],

$N_{sample}$ is the number of PRS RSRP measurement samples, where

- $N_{sample}$ = 1 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and meets the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 2 if the UE supports *supportedDL-PRS-ProcessingSamples* [34], and the LMF requests the UE to perform positioning measurements with reduced number of samples, and does not meet the following conditions:
  - PRS bandwidth is within the active BWP and
  - Magnitude of difference between the serving cell's SS-RSRP and the neighbour cell's PRS-RSRP is within 6 dB.
- $N_{sample}$ = 4 otherwise.

$T_{last,i} = T_i + T_{available_PRS,i}$ is the measurement duration for the last PRS-RSRP sample, including the sampling time and processing time, if not all PRS resources to be measured are available in the same measurement gap occasion during $T_{available_PRS,i}$, otherwise $T_{last,i} = T_i + MGL$,

$T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$ is the periodicity of PRS-RSRP measurement in positioning frequency layer *i*,

$T_i$ corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34],

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$ is the least common multiple between $T_{PRS,i}$ and $MGRP_i$ ,

$T_{PRS,i}$ is the maximum PRS resource periodicity among all PRS resources in positioning frequency layer i,

$MGRP_i$ is the measurement gap repetition period in positioning frequency layer i.

If positioning frequency layer *i* has more than one DL PRS resource set with different PRS periodicities with muting, $T_{per}^{PRS\ with\ muting} = N_{muting} * T_{per}^{PRS}$, the least common multiple of $T_{per}^{PRS\ with\ muting}$ among the DL PRS resource sets is used to derive $T_{PRS,i}$, where:

$T_{per}^{PRS}$ is the periodicity of PRS resource sets given by the higher-layer parameter *DL-PRS-Periodicity*.

$N_{muting}$ is the scaling factor considering PRS resource muting. $N_{muting} = T_{muting}^{PRS} * L_{muting}$, where $T_{muting}^{PRS}$ is the muting repetition factor given by the higher-layer parameter *DL-PRS-MutingBitRepetitionFactor*, and $L_{muting}$ is the size of the bitmap $\{b^1\}$.

## 17.2.1.4 Test description

### 17.2.1.4.1 Initial conditions

The supported test configurations are listed in Table 17.2.1.4-1.

**Table 17.2.1.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.2.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.2.1.5-1 and Table 17.2.1.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.2.1.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.2.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. Both cells transmit PRS during T2.The *NR-DL-AoD-RequestLocationInformation* message and *NR-DL-AoD-ProvideAssistanceData* message as defined in TS 37.355 [49] shall be provided to the UE during T1. The last slot containing the two messages for the assistance data and location information request is denoted as #n.

The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources that is ΔT after slot #n, where ΔT = 50 ms is the maximum processing time of the assistance data and location information request.

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 17.2.1.5-1 and Table 17.2.1.5-2 as appropriate. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.
5. The SS shall send an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the NR-DL-AoD-ProvideCapabilities IE.
7. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-AoD-ProvideAssistanceData-r16 IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
8. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-AoD-RequestLocationInformation-r16 such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
9. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 17.2.1.5-2.
10. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the nr-DL-AoD-ProvideLocationInformation-r16 IE within the response time (see clause 4.16.3). The UE shall perform and report the PRS-RSRPP measurements for both Cell 1 and Cell 2. If the UE transmits an nr-DL-AoD-ProvideLocationInformation-r16 IE including the nr-DL-PRS-FirstPathRSRP-ResultDiff-r17 field for Cell 1 and Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the nr-DL-AoD-ProvideLocationInformation-r16IE with the nr-DL-PRS-FirstPathRSRP-ResultDiff-r17 fields included within the response time then the number of failure tests is increased by one.
11. If the UE message at step 10 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
12. Repeat step 2-11 until the confidence level according to Annex D is achieved.

17.2.1.4.3 Message contents

**Table 17.2.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.2.1.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.2.1.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.2.1.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.1.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.1.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 38 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.1.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.1.4.3-5: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.1.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.2.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.1.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 17.2.1.5 Test requirement

Table 17.2.1.5-1 and Table 17.2.1.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.2.1.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | NA | OFF |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 5 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 17.2.1.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |

| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
|---|---|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -88 | -88 | -Infinity | -88 |
| | | 2 | -88 | -88 | -Infinity | -88 |
| | | 3 | -85 | -85 | -Infinity | -85 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |
| | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.59 | | -60.59 |
| Propagation Condition | | 1, 2, 3 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d$ = 0.45 μs and $f_D$ = 5 Hz | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.2.1.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 9.9.6.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( CSSF_i * ceil(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

Where $CSSF_{PRS,i} = 1, K_{p,PRS,i} = 1,\ N_{RxBeam,i} = 1,\ N_{PRS,i}^{slot} = 2, L_{available_PRS,i} = 0.285, N_{sample} = 4$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.285}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 0.648s and 30.88s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 1s and 31s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 31.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.2.1.5-3, Table 17.2.1.5-4 and Table 17.2.1.5-5 for reference.

**Table 17.2.1.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| *maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.2.1.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\left\lceil \frac{0.285}{N} \right\rceil$ |
|---|---|
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table 17.2.1.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.2.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 17.2.2.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement meets the reduced sample measurement delay requirements specified in TS 38.133 [50] clause 9.9.6.5 in an environment with AWGN propagation conditions for reduced number of samples.

### 17.2.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples.

### 17.2.2.3 Minimum conformance requirements

Same as clause 17.2.1.3.

### 17.2.2.4 Test description

#### 17.2.2.4.1 Initial conditions

The supported test configurations are listed in Table 17.2.2.4-1.

**Table 17.2.2.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.2.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.2.2.5-1 and Table 17.2.2.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.2.2.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

#### 17.2.2.4.2 Test procedure

Same as clause 17.2.1.4.2.

#### 17.2.2.4.3 Message contents

**Table 17.2.2.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.2.2.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.2.2.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |

| | | | |
|---|---|---|---|
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.2.2.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.2.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 7 | 48 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.2.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 38 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.1.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.2.4.3-5: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.2.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |

| | | | |
|---|---|---|---|
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.2.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.2.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.2.2.5 Test requirement

**Table 17.2.2.5-1 and Table 17.2.2.5-2 define the primary level settings including the test tolerances for the test.Table 17.2.2.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| Measurement gap | | 1, 2, 3 | GP#24 or GP#0 Note 1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | NA | OFF |

| | | | | |
|---|---|---|---|---|
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 5 | |
| NOTE 1: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. | | | | |

**Table 17.2.2.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| | | | | | | |
| PRS muting info | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -102.6 | | | |
| | | 2 | -102.6 | | | |
| | | 3 | -99.6 | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0.38 | -Infinity | -5.62 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 2.98 | -Infinity | -0.87 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -99.62 | -Infinity | -103.47 |
| | | 2 | -Infinity | -99.62 | -Infinity | -103.47 |
| | | 3 | -Infinity | -96.62 | -Infinity | -100.47 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -100 | -100 | -Infinity | -104 |
| | | 2 | -100 | -100 | -Infinity | -104 |
| | | 3 | -97 | -97 | -Infinity | -101 |
| Io | dBm/19.08 MHz | 1 | N/A | -65.75 | N/A | -65.75 |
| | dBm/19.08 MHz | 2 | | -65.75 | | -65.75 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | dBm/47.88 MHz | 3 | | -61.77 | | -61.77 |
| Propagation Condition | | 1, 2, 3 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d$ = 0.45 µs and $f_D$ = 5 Hz | | | |
| Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | | | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | | |
| Note 3: SS-RSRP/PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | | |

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.2.2.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 9.9.6.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$\mathrm{T_{PRS-RSRP,i}} = \left(\mathrm{CSSF_i} * \mathrm{ceil}(\mathrm{K_{p,PRS,i}}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\mathrm{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,\mathrm{i}}}{N} \right\rceil * N_{sample} - 1\right) * \mathrm{T_{effect,i}} + \mathrm{T_{last}}$$

Where $\mathrm{CSSF_{PRS,i}} = 1, \mathrm{K_{p,PRS,i}} = 1,\ N_{RxBeam,i} = 1,\ N_{PRS,i}^{slot} = 2, L_{available_PRS,\mathrm{i}} = 0.285, N_{sample}$ =1. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $\mathrm{T_{last,i}} = T_{\mathrm{i}} + T_{available_PRS,\mathrm{i}}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#0) depending on UE capabilities. Therefore, $\mathrm{T_{available_PRS,i}} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.285}{N} \right\rceil * 1 - 1\right) * T_{effect} + \mathrm{T_{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 0.168s and 7.84s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 1s and 8s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 8.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.2.2.5-3, Table 17.2.2.5-4 and Table 17.2.2.5-5 for reference.

**Table 17.2.2.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.2.2.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| *durationOfPRS-ProcessingSysmbols* | $\lceil \frac{0.285}{N} \rceil$ |
|---|---|
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table 17.2.2.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.2.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA

### 17.2.3.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement meets the reporting delay requirements without measurement gaps specified in TS 38.133 [50] clause 9.9.6.6 in an environment with AWGN propagation conditions.

### 17.2.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP, supportedDL-PRS-ProcessingSamples and prs-Processing Window.

### 17.2.3.3 Minimum conformance requirements

Same as clause 17.2.1.3.

### 17.2.3.4 Test description

#### 17.2.3.4.1 Initial conditions

The supported test configurations are listed in Table 17.2.3.4-1.

**Table 17.2.3.4-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.2.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.2.3.5-1 and Table 17.2.3.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.2.3.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.2.3.4.2 Test procedure

The test consists of two sub-tests; Sub-test 1 is for Nsample = 4 and sub-test 2 is for Nsample = 1. Sub-test 2 is only conducted when UE also supporting supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall not have any timing information of Cell 2. All two cells transmit PRS during T2. The beginning of the time interval T2 shall be aligned with the beginning of the first PRS processing window instance containing the PRS resources.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

3. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.

2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 17.2.3.5-1 and Table 17.2.3.5-2. Propagation conditions are set according to clause 4.16.2.
4. T1 starts.
5. The SS shall transmit an LPP REQUEST CAPABILITIES message.
6. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
7. The SS shall transmit an RRCReconfiguration message with the PPW configuration.
8. The UE shall transmit RRCReconfigurationComplete message.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message $\Delta T$ ms before the start of T2, where $\Delta T$ = 50 ms.
11. The SS sends a Positioning Measurement Gap Activation command containing an ID to activate the associated measurement gap
12. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 17.2.3.5-2.
13. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the *NR-DL-AoD-ProvideLocationInformation* IE within the response time (see clause 4.16.3). The UE shall perform and report

the PRS-RSRPP measurements for Cell 2 with respect to the reference cell in the DL-TDOA assistance data, Cell 1. If the UE transmits an *NR-DL-AoD-ProvideLocationInformation* IE including the nr-DL-PRS-FirstPathRSRP-Result-r17 field for Cell 2 within the response time then the number of successful tests is increased by one. If the UE fails to report the *NR-DL-AoD-ProvideLocationInformation* IE with both the nr-DL-PRS-RSRP-Result-r16 fields included within the response time then the number of failure tests is increased by one.

14. If the UE message at step 13 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.

15. Repeat steps 2-14 in Table 17.2.3.4-1 until the confidence level according to Annex D is achieved.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 17.2.3.4.3 Message contents

**Table 17.2.3.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.2.3.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.2.3.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | OCTET STRING (CONTAINING CellGroupConfig) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.3.4.3-4: CellGroupConfig (RRCReconfiguration, Table 17.2.3.4.3-3)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 CHOICE { | The periodicity is 160ms and the starting slot offset is | | |

| | | | |
|---|---|---|---|
| | 11 ms for any SCS configuration | | |
| scs120 CHOICE { | | | |
| n1280 | 4 | | |
| } | | | |
| } | | | |
| length-r17 | 80 | | |
| type-r17 | type1A | | |
| priority-r17 | st1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.3.4.3-5: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.2.3.4.3-6 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.3.4.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | |
| | kHz30 | | Test Configuration 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Sub-test 1 Config 1 and Config 2 |
| | 28 | 132 PRBs | Sub-test 1 Config 3 |
| | 7 | 48 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.3.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | About 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 17 | About 0.58 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.3.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.3.4.3-7: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.3.4.3-8: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.2.3.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |

| | | | |
|---|---|---|---|
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | Sub-test 2 |
| | Not present | | Sub-test 1 |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.3.4.3-9: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.2.3.5 Test requirement

Table 17.2.3.5-1 and Table 17.2.3.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.2.3.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| PPW configuration | | 1,2,3 | PPW.1 | As defined in TS 38.133 [50] A.3.33 |
| CP length | | 1, 2, 3 | Normal | |
| DRX | | 1, 2, 3 | NA | OFF |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | - set to the UE reported capability for receive time difference threshold if the UE reported value is < 3µs<br>- 3µs otherwise | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 0 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | Same as time offset | |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | ms | 1, 2, 3 | 10 | In this test UE is configured to measure single PFL within the configured PPW. |

**Table 17.2.3.5-2: Cell specific test parameters T2**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| RMSI CORESET RMC configuration | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active DL BWP configuration | | 1, 2, 3 | DLBWP.1.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| | | | | | | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | -102.6 | -98 | -102.6 |
| | | 2 | -98 | -102.6 | -98 | -102.6 |
| | | 3 | -95 | -99.6 | -95 | -99.6 |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | -102.6 | -98 | -102.6 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -2.41 | 0.38 | -12.12 | -5.62 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -2 | 2.98 | -10 | -0.87 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -100 | -99.62 | -108 | -103.47 |
| | | 2 | -100 | -99.62 | -108 | -103.47 |
| | | 3 | -97 | -96.62 | -105 | -100.47 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -100 | -100 | -104 | -104 |
| | | 2 | -100 | -100 | -104 | -104 |
| | | 3 | -97 | -97 | -101 | -101 |
| Io | dBm/19.08 MHz | 1 | -64.57 | -65.75 | -64.57 | -65.75 |
| | dBm/19.08 MHz | 2 | -64.57 | -65.75 | -64.57 | -65.75 |
| | dBm/47.88 MHz | 3 | -60.59 | -61.77 | -60.59 | -61.77 |
| Propagation Condition | | 1, 2, 3 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d$ = 0.45 μs and $f_D$ = 5 Hz | | | |

Note 1: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.2.3.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 9.9.6.6 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The LPP response time calculation procedure could be divided into two situations based on whether the UE support ppw-durationOfPRS-Processing1-r17 or ppw-durationOfPRS-Processing2-r17.The PRS-RSRPP measurement period follows the equation:

$T_{\text{PRS-RSRP, total}} = \sum_{i=1}^{L} T_{\text{PRS-RSRP,i}} + (L-1) * \max\left(T_{\text{effect},i}\right)$ if any of the positioning frequency layers are in Case 1, or

$T_{\text{PRS-RSRP, total}} = \max\left(T_{\text{PRS-RSRP,i}} + T_{\text{uncertainty,i}}\right)$, if all the positioning frequency layers are in Case 2,

$$T_{\text{PRS-RSRP,i}} = \left(N_{RxBeam,i} * \left\lceil \frac{N_{PRS,\text{i}}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,\text{i}}}{N} \right\rceil * N_{sample} - 1\right) * T_{\text{effect,i}} + T_{\text{last}}$$

Where $N_{RxBeam,i} = 1$ $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,\text{i}} = 285$, $N_{sample}$=4 for sub-test 1, $N_{sample}$=1 for sub-test 2. N is the parameter *ppw-durationOfPRS-ProcessingSymbolsN* from TS 37.355 [49], N' is the parameter *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{\text{last,i}} = T_{\text{i}}$ +PPWL and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,\text{i}} = LCM\left(T_{PRS,\text{i}}, PPWRP_{\text{i}}\right)$; where Tprs = 160 ms, and the repetition periodicity $PPWRP_{\text{i}}$ =160ms. Therefore, $T_{\text{available_PRS,i}} = 160$

$T_i$ depends on the UE parameter *ppw-durationOfPRS-ProcessingSymbolsT* from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.285}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{\text{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 0.641s and 30.88s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 1s and 31s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 1.3s and 31.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.2.3.5-3, Table 17.2.3.5-4 and Table 17.2.3.5-5 for reference.

**Table 17.2.3.5-3: value of N' based on ppw-maxNumOfDL-PRS-ResProcessedPerSlot**

| *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.2.3.5-4: value of N based on *ppw-durationOfPRS-ProcessingSymbolsN***

| *ppw-durationOfPRS-ProcessingSymbolsN* | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| msDot125 | 3 |
| msDot25 | 2 |
| >=msDot5 | 1 |

**Table 17.2.3.5-5: value of $T_{effect}$ and $T_{last}$ based on ppw-durationOfPRS-ProcessingSymbolsT-r17**

| ppw-durationOfPRS-ProcessingSymbolsT-r17 | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| ms1 | 160 | 161 |
| ms2 | 160 | 162 |

| | | |
|---|---|---|
| ms4 | 160 | 164 |
| ms8 | 160 | 168 |
| ms16 | 160 | 176 |
| ms20 | 160 | 180 |
| ms30 | 160 | 190 |
| ms40 | 160 | 200 |
| ms80 | 160 | 240 |
| ms160 | 160 | 320 |
| ms320 | 320 | 480 |
| ms640 | 640 | 800 |
| ms1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.2.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.2.4.1 Test purpose

The purpose of the test is to verify the PRS RSRPP measurement requirements specified in TS 38.133 [50] Clause 9.9.6.5 for single positioning frequency layer in FR2 SA under AWGN propagation conditions in standalone scenario.

### 17.2.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples.

### 17.2.4.3 Minimum conformance requirements

Same as clause 17.2.1.3.

### 17.2.4.4 Test description

#### 17.2.4.4.1 Initial conditions

The supported test configuration is listed in Table 17.2.4.4-1.

**Table 17.2.4.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.2.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.2.4.5-1 and Table 17.2.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.2.4.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.2.4.4.2 Test procedure

Same as clause 17.2.1.4.2.

### 17.2.4.4.3 Message contents

**Table 17.2.4.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.2.4.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.2.4.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.2.4.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, |

| | | | |
|---|---|---|---|
| | | | i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.4.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.4.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.4.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.4.4.3-5: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 2 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.4.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.2.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |

| | | | |
|---|---|---|---|
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.4.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 17.2.4.5 Test requirement

Table 17.2.4.5-1 and Table 17.2.4.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.2.4.5-1: General test parameters for PRS RSRPP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | Config 1 | GP#13 or GP#24[Note1] | As specified in TS 38.133 [50] clause 9.1.2-1. |
| Measurement gap offset | | Config 1 | 39 | |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | | Config 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 17.2.4.5-2: Cell-specific test parameters for PRS RSRPP measurement reporting delay**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| AoA setup | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | Config 1 | Rough | | Rough | |
| TDD configuration | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.4 FR2 | | PRS.1.4 FR2 | |
| PRS muting configuration | | | Config 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | | | | | | | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | Config 1 | 0 | | 0 | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz Note5 | | -99.4 | | -99.4 | |
| $N_{oc}$ Note2 | | dBm/SCS Note4 | Config 1 | -90.4 | | -90.4 | |
| SS-RSRP Note 3 | | dBm/SCS Note5 | Config 1 | -90.4 | -90.4 | -Infinity | -90.4 |
| PRS-RSRP Note 3 | | dBm/SCS Note5 | Config 1 | -Infinity | -92.4 | -Infinity | -100.4 |
| PRS $\hat{E}_s/I_{ot}$ | | dB | Config 1 | -Infinity | -2.7 | -Infinity | -12.3 |
| PRS $\hat{E}_s/N_{oc}$ | | dB | Config 1 | -Infinity | -2 | -Infinity | -10 |
| Io Note3 | | dBm/19 0.08 MHz Note5 | Config 1 | -58.37 | -55.99 | -58.37 | -55.99 |
| Propagation Condition | | | Config 1 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d = 0.45$ µs and $f_D = 5$ Hz | | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone |
| Note 6: | As observed with 0 dBi gain antenna at the centre of the quiet zone |
| Note 7: | Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation |
| Note 8: | Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4. |

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.2.4.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 9.9.6.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

Where $CSSF_{PRS,i} = 1$, $K_{p,PRS,i} = 1$, $N_{RxBeam,i} = 8$, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.035$, $N_{sample} = 4$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 82.08s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.2.4.5-3, Table 17.2.4.5-4 and Table 17.2.4.5-5 for reference.

**Table 17.2.4.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |

| | |
|---|---|
| >=n2 | 1 |

**Table 17.2.4.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\lceil \frac{0.035}{N} \rceil$ |
|---|---|
| >= nDot025 | 1 |

**Table 17.2.4.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.2.5 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.2.5.1 Test purpose

The purpose of the test is to verify the PRS RSRPP measurement requirements for reduced number of samples specified in TS 38.133 [50] Clause 9.9.6.5 for single positioning frequency layer under AWGN propagation conditions in standalone scenario.

### 17.2.5.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 17.2.5.3 Minimum conformance requirements

Same as clause 17.2.2.3.

## 17.2.5.4 Test description

### 17.2.5.4.1 Initial conditions

The supported test configuration is listed in Table 17.2.5.4-1.

**Table 17.2.5.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.2.5.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.2.5.5-1 and Table 17.2.5.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.2.5.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.2.5.4.2 Test procedure

Same as clause 17.2.2.4.2.

### 17.2.5.4.3 Message contents

**Table 17.2.5.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.2.5.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.2.5.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.2.5.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.5.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.5.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.5.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.5.4.3-5: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 2 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.5.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |

| | | | |
|---|---|---|---|
| responseTime SEQUENCE { | | | |
| time | See 17.2.5.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.5.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.2.5.5 Test requirement

Table 17.2.5.5-1 and Table 17.2.5.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.2.5.5-1: General test parameters for PRS RSRPP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| Gap Pattern Id | | Config 1 | GP#13 or GP#24[Note1] | As specified in TS 38.133 [50] clause 9.1.2-1. |
| Measurement gap offset | | Config 1 | 39 | |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | | Config 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells |
| Expected RSTD | μs | Config 1 | 3 | |

| | | | | |
|---|---|---|---|---|
| Expected RSTD uncertainty | µs | Config 1 | 5 | |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |
| Note 1: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. | | | | |

**Table 17.2.5.5-2: Cell-specific test parameters for PRS RSRPP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.4 FR2 | | PRS.1.4 FR2 | |
| PRS muting configuration | | | Config 1 | ‘10’ | | ‘01’ | |
| EPRE ratio of PSS to SSS | | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz Note5 | | -106.45 | | -106.45 | |
| $N_{oc}$ Note2 | | dBm/SCS Note4 | Config 1 | -97.45 | | -97.45 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| SS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -97.45 | -95.22 | -Infinity | -99.18 |
| PRS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -Infinity | -90.3 | -Infinity | -92.1 |
| PRS $\hat{E}_s/I_{ot}$ | dB | Config 1 | -Infinity | 0.3 | -Infinity | -2.8 |
| PRS $\hat{E}_s/N_{oc}$ | dB | Config 1 | -Infinity | 7.08 | -Infinity | 5.27 |
| Io Note3 | dBm/19 0.08 MHz Note5 | Config 1 | -65.42 | -55.66 | -65.42 | -55.66 |
| Propagation Condition | | Config 1 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d$ = 0.45 μs and $f_D$ = 5 Hz | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation
Note 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor ΔMBP from TS 38.101-2 [55] Table 6.2.1.3-4.

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.2.5.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 9.9.6.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( CSSF_i * \text{ceil}(K_{p,PRS,i}) * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last}$$

Where $CSSF_{PRS,i} = 1, K_{p,PRS,i} = 1,\ N_{RxBeam,i} = 8,\ N_{PRS,i}^{slot} = 2, L_{available_PRS,i} = 0.035, N_{sample} = 1$. N is the parameter *durationOfPRS-ProcessingSysmbols* from TS 37.355 [49], N' is the parameter *maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM(T_{PRS,i}, MGRP_i)$; where Tprs = 160 ms, and MGRP is 80 (for GP#24) or 40 (for GP#13) depending on UE capabilities. Therefore, $T_{available_PRS,i} = 160$ ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 1 - 1\right) * T_{effect} + \mathrm{T}_{\mathrm{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 1.288s and 20.64s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 2s and 21s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 21.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.2.5.5-3, Table 17.2.5.5-4 and Table 17.2.5.5-5 for reference.

**Table 17.2.5.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot***

| ***maxNumOfDL-PRS-ResProcessedPerSlot*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.2.5.5-4: value of N based on *durationOfPRS-ProcessingSysmbols***

| ***durationOfPRS-ProcessingSysmbols*** | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >= nDot025 | 1 |

**Table 17.2.5.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| n8 | 160 | 168 |
| n16 | 160 | 176 |
| n20 | 160 | 180 |
| n30 | 160 | 190 |
| n40 | 160 | 200 |
| n80 | 160 | 240 |
| n160 | 160 | 320 |
| n320 | 320 | 480 |
| n640 | 640 | 800 |
| n1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.2.6 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.2.6.1 Test purpose

The purpose of the test is to verify the PRS RSRPP measurement outside MG requirements specified in TS 38.133 [50] Clause 9.9.6.6 for single positioning frequency layer under AWGN propagation conditions in FR2 in standalone scenario.

### 17.2.6.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and prs-Processing Window..

### 17.2.6.3 Minimum conformance requirements

Same as clause 17.2.3.3.

### 17.2.6.4 Test description

#### 17.2.6.4.1 Initial conditions

The supported test configuration is listed in Table 17.2.6.4-1.

**Table 17.2.6.4-1: Test Configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.2.6.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.2.6.5-1 and Table 17.2.6.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.2.6.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

#### 17.2.6.4.2 Test procedure

Same as clause 17.2.3.4.2.

#### 17.2.6.4.3 Message contents

**Table 17.2.6.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.2.6.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.2.6.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.2.6.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.6.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | Sub-test 1 |
| | 7 | 48 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.6.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | About 0 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 17 | About 0.58 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.2.6.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.6.4.3-5: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 |
|---|

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 2 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.6.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.2.6.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |

| | | | |
|---|---|---|---|
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | Sub-test 1 |
| | requested | | Sub-test 2 |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.2.6.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.2.6.5 Test requirement

Table 17.2.6.5-1 and Table 17.2.6.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.2.6.5-1: General test parameters for PRS RSRPP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| PPW configuration | | Config 1 | Table A.3.33-1: Reference PPW configuration | As defined in TS 38.133 [50] clause A.3.33 |

| | | | | |
|---|---|---|---|---|
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | | Config 1 | OFF | DRX is not used |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | - set to the UE reported capability for receive time difference threshold if the UE reported value is < 3µs<br>- 3µs otherwise | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 0 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | Same as time offset | |
| T1 | s | Config 1 | 5 | |
| T2 | ms | Config 1 | 10 | In this test UE is configured to measure single PFL within the configured PPW. |

**Table 17.2.6.5-2: Cell-specific test parameters during T2**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | Sub-test 1 | Sub-test 2 | Sub-test 1 | Sub-test 2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated DL BWP | | | DLBWP.1.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.4 FR2 | | PRS.1.4 FR2 | |
| PRS BW | | | | 48 PRBs for sub-test 2 | | 48 PRBs for sub-test 2 | |

| PRS muting configuration | | Config 1 | '10' | | '01' | |
|---|---|---|---|---|---|---|
| EPRE ratio of PSS to SSS | | Config 1 | 0 | | 0 | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -99.4 | -106.45 | -99.4 | -106.45 |
| $N_{oc}$ Note2 | dBm/SCS Note4 | Config 1 | -90.4 | -97.45 | -90.4 | -97.45 |
| SS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -92.4 | -95.22 | -100.4 | -99.18 |
| PRS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -92.4 | -90.3 | -100.4 | -92.1 |
| PRS $\hat{E}_s/I_{ot}$ | dB | Config 1 | -2.7 | 0.3 | -12.3 | -2.8 |
| PRS $\hat{E}_s/N_{oc}$ | dB | Config 1 | -2 | 7.08 | -10 | 5.27 |
| Io Note3 | dBm/190.08 MHz Note5 | Config 1 | -55.99 | -55.66 | -55.99 | -55.66 |
| Propagation Condition | | Config 1 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d = 0.45$ μs and $f_D = 5$ Hz | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone

Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

Note 8: Calculation of Es/Iot includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor ΔMBP from TS 38.101-2 [55] Table 6.2.1.3-4.

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.2.6.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 9.9.6.6 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the RSTD measurement period plus ΔT, where ΔT = 50ms. The LPP response time calculation procedure could be divided into two situations based on whether the UE support ppw-durationOfPRS-Processing1-r17 or ppw-durationOfPRS-Processing2-r17.The PRS-RSRPP measurement period follows the equation:

$T_{\text{PRS-RSRP, total}} = \sum_{i=1}^{L} T_{\text{PRS-RSRP,i}} + (L-1) * \max\left(T_{\text{effect},i}\right)$ if any of the positioning frequency layers are in Case 1, or

$T_{\text{PRS-RSRP, total}} = \max\left(T_{\text{PRS-RSRP,i}} + T_{\text{uncertainty,i}}\right)$, if all the positioning frequency layers are in Case 2,

$$T_{\text{PRS-RSRP,i}} = \left(N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{\text{effect,i}} + T_{\text{last}}$$

Where $N_{RxBeam,i}$ = 8, $N_{PRS,i}^{slot}$ = 2, $L_{available_PRS,i}$ = 0.035, $N_{sample}$=4 for sub-test 1, $N_{sample}$=1 for sub-test 2. N is the parameter *ppw-durationOfPRS-ProcessingSymbolsN* from TS 37.355 [49], N' is the parameter *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* from TS 37.355 [49], $T_{\text{last,i}} = T_i$ +PPWL and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$.

$T_{available_PRS,i} = LCM\left(T_{PRS,i}, PPWRP_i\right)$; where Tprs = 160 ms, and the repetition periodicity $PPWRP_i$ =160ms. Therefore, $T_{\text{available_PRS,i}} = 160$

$T_i$ depends on the UE parameter *ppw-durationOfPRS-ProcessingSymbolsT* from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{\text{last}}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.121s and 82.08s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.2.6.5-3, Table 17.2.6.5-4 and Table 17.2.6.5-5 for reference.

**Table 17.2.6.5-3: value of N' based on *ppw-maxNumOfDL-PRS-ResProcessedPerSlot***

| *ppw-maxNumOfDL-PRS-ResProcessedPerSlot* | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.2.6.5-4: value of N based on *ppw-durationOfPRS-ProcessingSymbolsN***

| *ppw-durationOfPRS-ProcessingSymbolsN* | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >= nDot125 | 1 |

**Table 17.2.6.5-5: value of $T_{effect}$ and $T_{last}$ based on ppw-durationOfPRS-ProcessingSymbolsT-r17**

| ppw-durationOfPRS-ProcessingSymbolsT-r17 | $T_{effect}$ | $T_{last}$ |
|---|---|---|

| | | |
|---|---|---|
| ms1 | 160 | 161 |
| ms2 | 160 | 162 |
| ms4 | 160 | 164 |
| ms8 | 160 | 168 |
| ms16 | 160 | 176 |
| ms20 | 160 | 180 |
| ms30 | 160 | 190 |
| ms40 | 160 | 200 |
| ms80 | 160 | 240 |
| ms160 | 160 | 320 |
| ms320 | 320 | 480 |
| ms640 | 640 | 800 |
| ms1280 | 1280 | 1440 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 17.3 NR PRS-RSRPP measurement accuracy test case in RRC_CONNECTED

## 17.3.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA

### 17.3.1.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.38.2 in an environment with two-tap channel conditions.

### 17.3.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and maxDL-PRS-FirstPathRSRP-MeasPerTRP.

### 17.3.1.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRPP measurement for FR1 defined in Table 10.1.38.2.1-1 and Table 10.1.38.2.1-3 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band

…

**Table 10.1.38.2.1-1: PRS-RSRPP absolute accuracy for FR1**

| **Accuracy** | | **Conditions** | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | **Io Note 7 range** | | | |
| **Normal condition** | **Extreme condition** | **PRS Ês/Iot** | **PRS BW** | **Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$)** | **NR operating band groups Note 8** | **Minimum Io Note 1 dBm / $SCS_{PRS}$** | | | **Maximum Io** |
| | | | | | | **dBm / $SCS_{PRS}$** | | | **dBm/$BW_{Channel}$** |
| **dB** | **dB** | **dB** | **PRB** | **-** | | **dBm/15kHz Note 6** | **dBm/30kHz Note 6** | **dBm/60kHz Note 6** | |
| ±4.1 | ±8.6 | ≥-3 | ≥24 | All | NR_FDD_FR1_A, | -127 | -124 | -121 | -50 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | NR_TDD_FR1_A, NR_SDL_FR1_A | | | | |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±5.8 | ±10.3 | ≥-13 | 24 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±4.9 | ±9.4 | | BW > 52 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the DL-TDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

## 17.3.1.4 Test description

### 17.3.1.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 17.3.1.4.1-1.

**Table 17.3.1.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.3.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.

2. The general test parameter settings are set up according to Table 17.3.1.5-1 and Table 17.3.1.5-2.

3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.3.1.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.3.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. Cell 1 and Cell 2 are both active during the complete test. The beginning of the measurement period shall be aligned with the first PRS positioning subframe of a positioning occasion in the reference cell. The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration.

The NR-DL-AoD-RequestLocationInformation message and NR-DL-AoD-ProvideAssistanceData message as defined in TS 37.355 shall be provided to the UE during T1. The last slot containing the two messages for the assistance data and location information request is denoted as #n. The beginning of the time interval T2 shall be aligned with the beginning of the first MG instance containing the PRS resources that is ΔT after slot #n, where ΔT = 50 ms is the maximum processing time of the assistance data and location information request

1. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release On, according to TS 38.508-1 [45] clause 4.5.
2. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
3. Set the parameters according to Table 17.3.1.5-1 as appropriate. Propagation conditions are set according to clause 4.16.2.
4. The SS shall send an LPP REQUEST CAPABILITIES message.
5. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the NR-DL-AoD-ProvideCapabilities IE.
6. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the nr-DL-AoD-ProvideAssistanceData-r16 IE. If the UE message at step 6 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
7. The SS shall send a LPP REQUEST LOCATION INFORMATION message, including the nr-DL-AoD-RequestLocationInformation-r16 such that the UE receives the message ΔT ms before the start of the measurement period, where ΔT = 50 ms.
8. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the nr-DL-AoD-ProvideLocationInformation-r16 IE.
9. If the UE message at step 8 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
10. The SS shall check the nr-DL-PRS-FirstPathRSRP-ResultDiff-r17 field for Cell 2 in the nr-DL-AoD-SignalMeasurementInformation-r16 according to Table 17.3.1.5-2.
11. Repeat step 2-10 until the confidence level according to Annex D is achieved.
12. Repeat step 1-11 for the other sub-test defined in Table 17.3.1.4-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 17.3.1.4.3 Message contents

**Table 17.3.1.4.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.3.1.4.3-2: LPP Request Capabilities**

| Information Element | Value/remark |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.3.1.4.3-3: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.3.1.4.3-4 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.1.4.3-4: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 8 | 24 PRBs | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | 21 | 104 PRBs | Sub-test 1-2 and Sub-test 2-2 |
| | 28 | 132 PRBs | Sub-test 3-2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.1.4.3-5 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.1.4.3-5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.1.4.3-5: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-test 2 |
| | n2 | | Sub-test 1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 2 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 2 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.1.4.3-6: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.3.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |

| | | | |
|---|---|---|---|
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.1.4.3-7: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 17.3.1.5 Test requirement

Table 17.3.1.5-1 defines the primary level settings including test tolerances for the test.

**Table 17.3.1.5-1: PRS-RSRPP test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | | freq1 | | freq1 | |
| Duplex mode | Config 1 | | FDD | | | |
| | Config 2,3 | | TDD | | | |
| TDD configuration | Config 1 | | Not Applicable | | | |
| | Config 2 | | TDDConf.1.1 | | | |
| | Config 3 | | TDDConf.2.1 | | | |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c} = 106$ | | | |
| | Config 2 | | 20: $N_{RB,c} = 106$ | | | |
| | Config 3 | | 50: $N_{RB,c} = 133$ | | | |
| BWP BW | Config 1 | | 20: $N_{RB,c} = 106$ | | | |
| | Config 2 | | 20: $N_{RB,c} = 106$ | | | |
| | Config 3 | | 50: $N_{RB,c} = 133$ | | | |
| Downlink initial BWP configuration | | | DLBWP.0.1 | | | |
| Downlink dedicated BWP configuration | | | DLBWP.1.1 | | | |
| Uplink initial BWP configuration | | | ULBWP.0.1 | | | |
| Uplink dedicated BWP configuration | | | ULBWP.1.1 | | | |
| TRS configuration | Config 1 | | TRS.1.1 FDD | NA | TRS.1.1 FDD | NA |
| | Config 2 | | TRS.1.1 TDD | NA | TRS.1.1 TDD | NA |
| | Config 3 | | TRS.1.2 TDD | NA | TRS.1.2 TDD | NA |
| DRX Cycle | | ms | Not Applicable | | | |
| Measurement gap | | | GP#24 or GP#0 Note 7 | | | |
| PDSCH Reference measurement channel | Config 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | Config 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | Config 3 | | SR2.1 TDD | | SR2.1 TDD | |
| RMSI CORESET Reference Channel | Config 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | Config 2 | | CR.1.1 TDD | | CR.1.1 TDD | |
| | Config 3 | | CR2.1 TDD | | CR2.1 TDD | |
| Control channel RMC | Config 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | Config 2 | | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | Config 3 | | CCR2.1 TDD | | CCR2.1 TDD | |
| PRS configuration | Config 1 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | PRS.2.3 FR1 | PRS.2.3 FR1 | PRS.2.4 FR1 | PRS.2.4 FR1 |
| PRS Resource slot offset (slot) | Config 1,2,3 | slot | 0 | 2 | 0 | 2 |
| SSB configuration | Config 1 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Time offset with Cell 1 | Config 1 | | ms | - | 3 | - | 3 |
| | Config 2,3 | | µs | - | 3 | - | 3 |
| SMTC configuration | Config 1 | | | SMTC.2 | | | |
| | Config 2,3 | | | SMTC.1 | | | |
| OCNG Patterns | | | | OCNG pattern 1 | | | |
| PDSCH/PDCCH subcarrier spacing | Config 1,2 | | kHz | 15 kHz | | | |
| | Config 3 | | | 30 kHz | | | |
| EPRE ratio of PSS to SSS | | | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| $N_{oc}$ Note2 | Config 1,2 | | dBm/15KhZ | -98 | | -98 | |
| | Config 3 | | | -98 | | -98 | |
| $N_{oc}$ Note2 | Config 1,2 | | dBm/SCS | -98 | | -98 | |
| | Config 3 | | | -95 | | -95 | |
| $\hat{E}_s/I_{ot}$ | | | dB | -2.41 | -12.12 | -2.41 | -12.12 |
| $\hat{E}_s/N_{oc}$ | | | dB | -2 | -10 | -2 | -10 |
| PRS-RSRP Note3 | Config 1, 2 | | dBm/SCS | -100 | -108 | -100 | -108 |
| | Config 3 | | dBm/SCS | -97 | -105 | -97 | -105 |
| Io Note3 | Config 1,2 | | dBm/19.08 MHz | -64.57 | | -64.57 | |
| | Config 3 | | dBm/47.88 MHz | -60.59 | | -60.59 | |
| Propagation condition | | | | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d = 0.45$ µs and $f_D = 5$ Hz | | | |
| Antenna configuration | | | | 1x2 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRPP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Void.

Note 6: Void.

Note 7: GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured.

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.2.1.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.3.1.5-2: PRS-RSRPP accuracy requirements for the reported values**

| Test Configuration | Subtest | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | Sub-test 1 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_40 | PRS_RSRPP_51 |
| | Sub-test 2 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_41 | PRS_RSRPP_50 |
| 2 | Sub-test 1 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_40 | PRS_RSRPP_51 |
| | Sub-test 2 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_41 | PRS_RSRPP_50 |
| 3 | Sub-test 1 | PRS_RSRPP_52 | PRS_RSRPP_61 | PRS_RSRPP_43 | PRS_RSRPP_54 |
| | Sub-test 2 | PRS_RSRPP_52 | PRS_RSRPP_61 | PRS_RSRPP_44 | PRS_RSRPP_53 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 17.3.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 17.3.2.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement with reduce number of samples meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.38.2 in an environment with two-tap channel conditions.

### 17.3.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 17.3.2.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRPP measurement for FR1 defined in Table 10.1.38.2.1-1 and Table 10.1.38.2.1-3 are valid under the following conditions:

- Conditions defined in 38.101-1 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band

…

**Table 10.1.38.2.1-3: PRS-RSRPP absolute accuracy for FR1 for reduced number of samples**

| Accuracy | | Conditions | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | | | |
| Normal condition | Extreme condition | PRS $\hat{E}s/Iot$ | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | NR operating band groups Note 8 | Minimum Io Note 1 dBm / $SCS_{PRS}$ | | Maximum Io |
| | | | | | | dBm / $SCS_{PRS}$ | | |
| dB | dB | dB | PRB | - | | dBm/15kHz Note 6 | dBm/30kHz Note 6 | dBm/60kHz Note 6 | dBm/$BW_{Channel}$ |
| ±3.9 | ±8.4 | ≥0 | ≥48 | All | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | -50 |
| | | | | | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | -50 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | NR_TDD_FR1_C | -126 | -123 | -120 | -50 |
| | | | | | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | -50 |
| | | | | | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | -50 |
| | | | | | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | -50 |
| | | | | | NR_FDD_FR1_G | -124 | -121 | -118 | -50 |
| | | | | | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | -50 |
| | | | | | Note 4 | | | | |
| | | | | | Note 4 | | | | |
| ±4.2 | ±8.7 | ≥-6 | 48 ≤ BW ≤ 52 | All | Note 4 | | | | |
| ±4.1 | ±8.6 | | BW >52 | All | Note 4 | | | | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the DL-TDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

## 17.3.2.4 Test description

### 17.3.2.4.1 Initial conditions

The test is defined with three possible Test Configurations. In the case that the UE supports more than one of these Test Configurations, then the UE is only required to be tested in one of the Test Configurations, chosen by the UE. The defined Test Configurations are specified in Table 17.3.2.4.1-1.

**Table 17.3.2.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.3.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.3.2.5-1 and Table 17.3.2.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.3.2.4.3.

5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.3.2.4.2 Test procedure

Same as defined in clause 17.3.1.4.2.

### 17.3.2.4.3 Message contents

As defined in clause 17.3.1.4.3, with the following exceptions:

Table 17.3.2.4.3-1 replaces Table 17.3.1.4.3-4, Table 17.3.2.4.3-2 replaces Table 17.3.1.4.3-5 and Table 17.3.2.4.3-3 replaces Table 17.3.1.4.3-6.

**Table 17.3.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 8 | 52 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.2.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 39 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.2.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 2 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.3.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.3.2.5 Test requirement

Table 17.3.2.5-1 defines the primary level settings including test tolerances for the test.

**Table 17.3.2.5-1: PRS-RSRPP test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| Cell ID | | | 489 | 0 |
| SSB ARFCN | | | freq1 | |
| Duplex mode | Config 1 | | FDD | |
| | Config 2,3 | | TDD | |
| TDD configuration | Config 1 | | Not Applicable | |
| | Config 2 | | TDDConf.1.1 | |
| | Config 3 | | TDDConf.2.1 | |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| BWP BW | Config 1 | | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| Downlink initial BWP configuration | | | DLBWP.0.1 | |
| Downlink dedicated BWP configuration | | | DLBWP.1.1 | |
| Uplink initial BWP configuration | | | ULBWP.0.1 | |
| Uplink dedicated BWP configuration | | | ULBWP.1.1 | |
| TRS configuration | Config 1 | | TRS.1.1 FDD | NA |
| | Config 2 | | TRS.1.1 TDD | NA |
| | Config 3 | | TRS.1.2 TDD | NA |
| DRX Cycle | | ms | Not Applicable | |

| | | | | |
|---|---|---|---|---|
| Measurement gap | | | GP#24 or GP#0 Note 5 | |
| PDSCH Reference measurement channel | Config 1 | | SR.1.1 FDD | - |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR2.1 TDD | |
| RMSI CORESET Reference Channel | Config 1 | | CR.1.1 FDD | - |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR2.1 TDD | |
| Control channel RMC | Config 1 | | CCR.1.1 FDD | - |
| | Config 2 | | CCR.1.1 TDD | |
| | Config 3 | | CCR2.1 TDD | |
| PRS configuration | Config 1 | | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | PRS.2.4 FR1 | PRS.2.4 FR1 |
| PRS bandwidth | Config 1,2,3 | | 52 PRBs | 52 PRBs |
| PRS Resource slot offset (slot) | Config 1,2,3 | slot | 0 | 2 |
| SSB configuration | Config 1 | | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | SSB.2 FR1 | SSB.2 FR1 |
| Time offset with Cell 1 | Config 1 | ms | - | 3 |
| | Config 2,3 | µs | - | 3 |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2,3 | | SMTC.1 | |
| OCNG Patterns | | | OCNG pattern 1 | |
| PDSCH/PDCCH subcarrier spacing | Config 1,2 | kHz | 15 kHz | |
| | Config 3 | | 30 kHz | |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | |
| $N_{oc}$ Note2 | Config 1,2 | dBm/15Khz | -98 | |
| | Config 3 | | | |
| $N_{oc}$ Note2 | Config 1,2 | dBm/SCS | -98 | |
| | Config 3 | | -95 | |
| $\hat{E}_s/I_{ot}$ | | dB | 0.32 | -5.64 |
| $\hat{E}_s/N_{oc}$ | | dB | 2.88 | -0.96 |
| PRS-RSRP Note3 | Config 1, 2 | dBm/SCS | -95.12 | -98.96 |
| | Config 3 | | -92.12 | -95.96 |
| SS-RSRP Note3 | Config 1, 2 | dBm/SCS | -95.77 | -99.73 |
| | Config 3 | | -92.77 | -96.73 |
| Io Note3 | Config 1,2 | dBm/19.08MHz | -61.22 | |
| | Config 3 | dBm/47.88MHz | -57.24 | |
| Propagation condition | | | Two-tap channel Note6 | |
| Antenna configuration | | | 1x2 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | |

| | |
|---|---|
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | GP#24 is configured if UE supports MG#24, otherwise GP#0 is configured. |
| Note 6: | The two-tap channel model is defined in 38.101-4 Annex B.2.4 (a = 1, $\tau_d$=0.45 μs and $f_D$=5 Hz). |

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.2.2.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.3.2.5-2: PRS-RSRPP accuracy requirements for the reported values**

| Test Configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|
| | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | PRS_RSRPP_54 | PRS_RSRPP_62 | PRS_RSRPP_50 | PRS_RSRPP_59 |
| 2 | PRS_RSRPP_54 | PRS_RSRPP_62 | PRS_RSRPP_50 | PRS_RSRPP_59 |
| 3 | PRS_RSRPP_57 | PRS_RSRPP_65 | PRS_RSRPP_53 | PRS_RSRPP_62 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 17.3.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.3.3.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.38.2 in an environment with two-tap channel conditions.

### 17.3.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and maxDL-PRS-FirstPathRSRP-MeasPerTRP.

### 17.3.3.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRPP measurement for FR2 defined in Table 10.1.38.2.1-2 and Table 10.1.38.2.1-4 are valid under the following conditions:

- Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.
- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band

…

**Table 10.1.38.2.1-2: PRS-RSRPP absolute accuracy for FR2**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Minimum Io Note 1 dBm / $SCS_{PRS}$ | Maximum Io |
| dB | dB | dB | PRB | - | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| | | | | | dBm/120kHz Note 6 / dBm/60kHz Note 6 | |
| ±6.0 | ±9.0 | ≥-3 | ≥24 | All | Same value as PRP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| | | | | | Note 4 | |
| | | | | | Note 4 | |
| ±7.3 | ±10.3 | ≥-13 | 24 ≤ BW ≤ 64 | All | Note 4 | |
| ±6.2 | ±9.2 | | BW >64 | All | Note 4 | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the DL-TDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 24 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by $\Delta>0$, when applicable, as described in Sections B.3.2 and B.3.3.
NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

## 17.3.3.4 Test description

### 17.3.3.4.1 Initial conditions

The supported test configurations in listed in Table 17.3.3.4.1-1.

**Table 17.3.3.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.3.3.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.3.3.5-1 and Table 17.3.3.5-2.
3. Propagation conditions are set according to clause 4.16.2.

4. Message contents are defined in clause 17.3.3.4.3.

5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.3.3.4.2 Test procedure

Same as defined in clause 17.3.1.4.2.

### 17.3.3.4.3 Message contents

As defined in clause 17.3.1.4.3, with the following exceptions:

Table 17.3.3.4.3-1 replaces Table 17.3.1.4.3-4 and Table 17.3.3.4.3-2 replaces Table 17.3.1.4.3-5.

**Table 17.3.3.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.3.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |

| | | | |
|---|---|---|---|
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.3.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.3.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-test 1 |
| n4-r16 | 1 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.3.3.5 Test requirement

Table 17.3.3.5-1 and Table 17.3.3.5-2 define the primary level settings including test tolerances for the test.

**Table 17.3.3.5-1: PRS-RSRPP general test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - | Not applicable | - |
| Measurement gap | | GP#13 or GP#24 Note2 | | | |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | μs | - | 3 | - | 3 |
| PRS configuration | | PRS.1.3 FR2 | PRS.1.3 FR2 | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS Resource slot offset | slot | 0 | 4 | 0 | 4 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d = 0.45$ μs and $f_D = 5$ Hz | | | |
| Antenna configuration | | 1x2 | 1x2 | 1x2 | 1x2 |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured. |

**Table 17.3.3.5-2: PRS-RSRPP OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 7 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/15kHz Note4 | -99.1 | | Same as Test 1 | |
| $N_{oc}$ Note1 | dBm/SCS Note4 | -90.1 | | Same as Test 1 | |
| $\hat{E}_s / N_{oc}$ | dB | -2 | -10 | -2 | -10 |
| Es | dBm/SCS Note4 | - | - | - | - |
| PRS_RP Note2 | dBm/SCS | -92.1 | -100.1 | -92.1 | -100.1 |
| $\hat{E}_s / I_{ot}$ BB Note6 | dB | -2.68 | -12.30 | -2.68 | -12.30 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.69 | | -55.69 | |

Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 2: PRS_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: Void
Note 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 5: Void
Note 6: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.2.4.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.3.3.5-3: PRS-RSRPP accuracy requirements for the reported values**

| Sub-test | Cell 1 | | Cell 2 | |
|---|---|---|---|---|
| | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | PRS_RSRPP_50 | PRS_RSRPP_83 | PRS_RSRPP_42 | PRS_RSRPP_75 |
| 2 | PRS_RSRPP_50 | PRS_RSRPP_83 | PRS_RSRPP_42 | PRS_RSRPP_75 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 17.3.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.3.4.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement with reduce number of samples meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.38.2 in an environment with two-tap channel conditions.

### 17.3.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning and maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples-RRC-CONNECTED.

### 17.3.4.3 Minimum conformance requirements

The absolute accuracy requirements for PRS-RSRPP measurement for FR2 defined in Table 10.1.38.2.1-2 and Table 10.1.38.2.1-4 are valid under the following conditions:

- Conditions defined in 38.101-2 Clause 7.3 for reference sensitivity are fulfilled.

- PRP $1,2|_{dBm}$ according to Annex B.2.14 for a corresponding Band

…

**Table 10.1.38.2.1-4: PRS-RSRPP absolute accuracy for FR2 for reduced number of samples**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | | | Io Note 7 range | |
| Normal condition | Extreme condition | PRS Ês/Iot | PRS BW | Repetition factor ($T_{rep}^{PRS} * L_{PRS} / K_{comb}^{PRS}$) | Minimum Io Note 1 dBm / $SCS_{PRS}$ | Maximum Io |
| dB | dB | dB | PRB | - | dBm / $SCS_{PRS}$ | dBm/$BW_{Channel}$ |
| | | | | | dBm/120kHz Note 6 / dBm/60kHz Note 6 | |
| ±5.9 | ±8.9 | ≥0 | ≥48 | All | Same value as PRP in Table B.2.14-2, according to UE Power class, operating band and angle of arrival | -50 |
| | | | | | Note 4 | |
| | | | | | Note 4 | |
| ±5.6 | ±8.6 | ≥-6 | 48 ≤ BW ≤ 64 | All | Note 4 | |
| ±5.4 | ±8.4 | | BW >64 | All | Note 4 | |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in an OFDM symbol.
NOTE 2: Void.
NOTE 3: PRS bandwidth is as indicated in *prs-Bandwidth* in the DL-TDOA or DL-AoD assistance data defined in [34].
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding requirement with the PRS bandwidth ≥ 48 RB.
NOTE 5: The serving cell, the reference cell, and the measured neighbour cell i are on the same carrier frequency.
NOTE 6: The condition level is increased by Δ>0, when applicable, as described in Sections B.3.2 and B.3.3.

NOTE 7: The Io is defined in PRS positioning subframes. The same Io range applies to PRS and non-PRS symbols. Io levels are different in PRS and non-PRS symbols within the same subframe.
NOTE 8: NR operating band groups are as defined in Section 3.5.2.

## 17.3.4.4 Test description

### 17.3.4.4.1 Initial conditions

The supported test configurations in listed in Table 17.3.4.4.1-1.

**Table 17.3.4.4.1-1: Test Configurations**

| Configuration | Description |
|---|---|
| 1 | 120 kHz SSB SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.3.4.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.3.4.5-1 and Table 17.3.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.3.4.4.3.
5. In the test there are two synchronous cells: Cell 1 and Cell 2. Cell 1 is the reference as well as the PCell. Cell 2 is the neighbour cell. All cells are on the same RF channel distributed in single positioning frequency layer.

### 17.3.4.4.2 Test procedure

Same as defined in clause 17.3.1.4.2.

### 17.3.4.4.3 Message contents

As defined in clause 17.3.1.4.3, with the following exceptions:

Table 17.3.4.4.3-1 replaces Table 17.3.1.4.3-4, Table 17.3.4.4.3-2 replaces Table 17.3.1.4.3-5 and Table 17.3.4.4.3-3 replaces Table 17.3.1.4.3-6.

**Table 17.3.4.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |

| | | | |
|---|---|---|---|
| dl-PRS-ResourceBandwidth-r16 | 7 | 48 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.4.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.3.4.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.4.4.3-2: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.3.4.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |

| | | | |
|---|---|---|---|
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.3.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 17.3.4.5 Test requirement

Table 17.3.4.5-1 and Table 17.3.4.5-2 define the primary level settings including test tolerances for the test.

**Table 17.3.4.5-1: PRS-RSRPP general test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 |
| SSB ARFCN | | freq1 | |
| Duplex mode | | TDD | |
| TDD configuration | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - |
| Downlink dedicated BWP configuration | | DLBWP.1.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - |
| Uplink dedicated BWP configuration | | ULBWP.1.1 | - |
| DRX cycle configuration | | Not applicable | - |
| Measurement gap | | GP#13 or GP#24 Note2 | |
| TRS configuration | | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | µs | - | 3 |
| PRS configuration | | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS bandwidth | | 48 PRBs | 48 PRBs |
| PRS Resource slot offset | slot | 0 | 4 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | |
| Propagation conditions | | Two-tap channel Note3 | |
| Antenna configuration | | 1x2 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: GP#24 is configured if UE supports MG#24, otherwise GP#13 is configured.
Note 3: The two-tap channel model is defined in 38.101-4 Annex B.2.4 (a = 1, $\tau_d$=0.45 µs and $f_D$=5 Hz).

**Table 17.3.4.5-2: PRS-RSRPP OTA related test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | |
| Assumption for UE beams Note 7 | | Rough | |
| $N_{oc}$ Note1 | dBm/15kHz Note4 | -103 | |
| $N_{oc}$ Note1 | dBm/SCS Note4 | -94 | |

| | | | |
|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | 3.55 | -0.29 |
| $E_s$ | dBm/SCS Note4 | - | - |
| PRS_RP Note2 | dBm/SCS | -90.42 | -94.26 |
| SS_RP Note2 | dBm/SCS | -86.77 | -90.73 |
| $\hat{E}_s / I_{ot}$ BB Note6 | dB | 0 | -6 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.74 | |

Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 2: PRS_RP, SS_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 3: Void

Note 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 5: Void

Note 6: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation.

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.2.4.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.3.4.5-3: PRS-RSRPP accuracy requirements for the reported values**

| Config | Cell 1 | | Cell 2 | |
|---|---|---|---|---|
| | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | PRS_RSRPP_52 | PRS_RSRPP_84 | PRS_RSRPP_48 | PRS_RSRPP_80 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

# 17.4 NR PRS-RSRPP measurement period test cases in RRC_INACTIVE

## 17.4.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA

### 17.4.1.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.5.5 in FR1 in standalone scenario when single positioning frequency layer is configured.

### 17.4.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and DL-PRS measurement in RRC_INACTIVE state.

### 17.4.1.3 Minimum conformance requirements

Same as defined in clause 16.4.3.3.

### 17.4.1.4 Test description

#### 17.4.1.4.1 Initial conditions

The test defines three possible test configurations; specified in Table 17.4.1.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 17.4.1.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.4.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.4.1.5-1 and Table 17.4.1.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.4.1.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

#### 17.4.1.4.2 Test procedure

The test consists of two consecutive time intervals, with duration of T1 and T2. During time duration T1, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During T2 UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the same numerology as the initial DL BWP.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during T1. The last TTI containing the two messages shall be provided to the UE $\Delta T$ ms before the start of T2, where $\Delta T = 50$ ms is the maximum processing time of the *DL-AoD assistance* data and location information request.

The beginning of the time interval T2 shall be aligned with the first DRX cycle containing a DL PRS resource(s).

3. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 17.4.1.4.3-1.
4. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.

3. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.

4. Set the parameters according to Table 17.4.1.5-1 and Table 17.4.1.5-2. Propagation conditions are set according to clause 4.16.2.
5. T1 starts.
6. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
7. The UE shall transmit RRCReconfigurationComplete message.
8. The SS shall transmit an LPP REQUEST CAPABILITIES message.
9. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
10. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 9 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
11. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of T2, where ΔT = 50 ms.
12. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
13. When T1 expires, the SS shall transmit an RRCRelease message including SDT configuration to transit the UE to RRC_INACTIVE state and switch the power setting from T1 to T2 as specified in Table 17.4.1.5-1 and Table 17.4.1.5-2.
14. Wait for the UE monitors for occurrence of the periodic event requested.
15. The UE shall transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 19.
16. The SS transmits an *RRCResume* message.
17. The UE transmits an *RRCResumeComplete* message. The time between the start of T2 and the UE send the *RRCResumeRequest* message should meet the requirements specified in TS 38.133 [50] Clause 5.6.3.5.
18. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message including the NR-DL-AoD-ProvideLocationInformation IE within the response time (see clause 4.16.3) plus the maximum RRC processing time 10ms between *RRCResume* message and *RRCResumeComplete* message.
19. The UE shall perform and report the PRS-RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1. If the UE transmits an NR-DL-AoD-ProvideLocationInformation IE including the nr-DL-PRS-FirstPathRSRP-ResultDiff-r17 field for Cell 2 field within the response time then the number of successful tests is increased by one. If the UE fails to report the NR-DL-AoD-ProvideLocationInformation IE with the nr-DL-PRS-FirstPathRSRP-ResultDiff-r17 field included within the response time then the number of failure tests is increased by one.
20. If the UE message at step 19 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
21. UE shall be released and moved to RRC_IDLE.
22. Repeat steps 2-21 in Tables 17.4.1.4-1 until the confidence level according to Annex D is achieved.

### 17.4.1.4.3 Message contents

**Table 17.4.1.4.3-1: SIB1**

| Derivation Path: TS 38.508-1 [45], clause 4.6.1 with condition SDT |
|---|

**Table 17.4.1.4.3-2: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 [44] clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 1 0 0 0 | DL-AoD | |

**Table 17.4.1.4.3-3: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [45], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.1.4.3-4: CellGroupConfig (Table 17.4.1.4.3-3)**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms1280 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.1.4.3-5: LPP Request Capabilities**

| **Information Element** | **Value/remark** |
|---|---|
| nr-DL-AoD-RequestCapabilities-r16 | TRUE |

**Table 17.4.1.4.3-6: LPP ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |

| | | | |
|---|---|---|---|
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData | Not present | | |
| otdoa-ProvideAssistanceData | Not present | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r15 | Not present | | |
| tbs-ProvideAssistanceData-r15 | Not present | | |
| wlan-ProvideAssistanceData-r15 | Not present | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 | Not present | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 17.4.1.4.3-7 | | |
| nr-SelectedDL-PRS-IndexList-r16 | Not present | | |
| nr-PositionCalculationAssistance-r16 SEQUENCE { | | | Depending on UE capabilities, i.e. support for UE-based DL-AoD |
| nr-TRP-LocationInfo-r16 | As defined in TS 37.571-5 [20] | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.1.4.3-7: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 21 | 104 PRBs | Config 1 and Config 2 |
| | 28 | 132 PRBs | Config 3 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.1.4.3-8 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.1.4.3-8 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.1.4.3-8: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.1.4.3-9: lcs-PeriodicTriggeredInvoke**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-PeriodicTriggeredInvokeArg::= SEQUENCE { | | | |
| referenceNumber | LCS-ReferenceNumber | | |
| h-gmlc-address | GSN-Address | | |
| qoS | Not present | | |
| reportingPLMNList | Not present | | |
| periodicLocation SEQUENCE { | | | |
| periodicLDRInfo SEQUENCE { | | | |
| reportingAmount | 5 | | |
| reportingInterval | 1 | | |
| reportingOptionMilliseconds | Not present | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| areaEventReporting | Not present | | |
| motionEventReporting | Not present | | |
| referenceNumberExt | LCS-ReferenceNumberExt | | |
| h-gmlc-callBackUri | Not present | | |
| supportedGADShapes | Not present | | |
| deferredRoutingIdentifier | Not present | | |
| reportingAccessTypes | 00000001 | NR | |
| multiplePositioningProtocolPDUs SIZE (1..maxNumLPPMsg)) OF PositioningProtocolPDU { | 1 entry | | |
| PositioningProtocolPDU[1] SEQUENCE { | | entry 1 | |
| PositioningProtocolPDU | Contains a LPP RequestLocationInformation message defined in Table 17.4.1.4.3-10 | | |
| } | | | |
| } | | | |
| controlPlane-CIoT-5GS-Optimisation | Not present | | |
| scheduledLocTime | Not present | | |
| eventReportAllowedArea | Not present | | |
| } | | | |

**Table 17.4.1.4.3-10: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.4.1.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if | |

| | | | |
|---|---|---|---|
| | | response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.1.4.3-11: RRCRelease**

| |
|---|
| Derivation Path: TS 38.508-1 [45], Table 4.6.1-16 with condition NR_RRC_INACTIVE and SDT |

**Table 17.4.1.4.3-12: LPP ProvideLocation Information**

| Derivation Path: 37.355 clause 6.5.12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | TRUE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | (0..255) | | |
| acknowledgement | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not present. | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | | |
| tbs-ProvideLocationInformation-r13 | Not present | | |
| wlan-ProvideLocationInformation-r13 | Not present | | |
| bt-ProvideLocationInformation-r13 | Not present | | |
| nr-ECID-ProvideLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Not present | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] SEQUENCE { | | entry 2 | |
| dl-PRS-ID-r16 | INTEGER (0..255) | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | | | |
| nr-ARFCN-r16 | | | |
| nr-DL-PRS-ResourceID-r16 | | | |
| nr-DL-PRS-ResourceSetID-r16 | | | |
| nr-TimeStamp-r16 | | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable. | | |
| nr-DL-PRS-RxBeamIndex-r16 | | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Present. Any value acceptable. | | |
| nr-los-nlos-Indicator-r17 | | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.4.1.5 Test requirement

Table 17.4.1.5-1 and Table 17.4.1.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.4.1.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |
| CP length | | 1, 2, 3 | Normal | |
| DRX | s | 1, 2, 3 | 1.28 | ON |
| Time offset between serving and neighbour cells | µs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | µs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | µs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 5 | |

**Table 17.4.1.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| PRS muting info | | 1, 2, 3 | '10' | | '01' | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98 | | | |
| | | 2 | -98 | | | |
| | | 3 | -95 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | -2.41 | -Infinity | -12.12 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | -2 | -Infinity | -10 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -100 | -Infinity | -108 |
| | | 2 | -Infinity | -100 | -Infinity | -108 |
| | | 3 | -Infinity | -97 | -Infinity | -105 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -88 | -88 | -Infinity | -88 |
| | | 2 | -88 | -88 | -Infinity | -88 |
| | | 3 | -85 | -85 | -Infinity | -85 |
| Io | dBm/19.08 MHz | 1 | N/A | -64.57 | N/A | -64.57 |
| | dBm/19.08 MHz | 2 | | -64.57 | | -64.57 |
| | dBm/47.88 MHz | 3 | | -60.59 | | -60.59 |
| Propagation Condition | | 1, 2, 3 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d$ = 0.45 μs and $f_D$ = 5 Hz | | | |

Note 1: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.4.1.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 5.6.3.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left(K_{carrier_PRS} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{RxBeam,i}$=1, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.285$, $N_{sample}$=4. N is the parameter durationOfPRS-ProcessingSymbols-r17 from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17 from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$. $T_{available_{PRS,i}} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 1280ms depending on UE capabilities. Therefore, $T_{available_{PRS,i}} = 1280$ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms-r17 from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.285}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 32s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 32s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and 32.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.4.1.5-3, Table 17.4.1.5-4 and Table 17.4.1.5-5 for reference.

**Table 17.4.1.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.4.1.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| ***durationOfPRS-ProcessingSymbols*** | $\left\lceil \frac{0.285}{N} \right\rceil$ |
|---|---|
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table 17.4.1.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |

| | | |
|---|---|---|
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.4.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 17.4.2.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.5.5 in FR1 in standalone scenario when single positioning frequency layer is configured and reduce sample measurements are requested by the LMF.

### 17.4.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples-RRC-Inactive.

### 17.4.2.3 Minimum conformance requirements

Same as defined in clause 16.4.3.3.

### 17.4.2.4 Test description

#### 17.4.2.4.1 Initial conditions

The test defines three possible test configurations; specified in Table 17.4.2.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 17.4.2.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.4.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.4.2.5-1 and Table 17.4.2.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.4.2.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

#### 17.4.2.4.2 Test procedure

Same as defined in clause 17.4.1.4.2.

#### 17.4.2.4.3 Message contents

As defined in clause 17.4.1.4.3, with the following exceptions:

Table 17.4.2.4.3-1 replaces Table 17.4.1.4.3-6 and Table 17.4.2.4.3-2 replaces Table 17.4.1.4.3-9.

**Table 17.4.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 7 | 48 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.1.4.3-7 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 µs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 µs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.1.4.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.2.4.3-2: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.4.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |

| | | | |
|---|---|---|---|
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.4.2.5 Test requirement

Table 17.4.2.5-1 and Table 17.4.2.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.4.2.5-1: General test parameters**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| Reference cell | | 1, 2, 3 | Cell 1 | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | 1, 2, 3 | Cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| RF Channel Number | | 1, 2, 3 | 1: Cell 1 and Cell 2 | |
| $BW_{channel}$ | MHz | 1 | 20: $N_{RB,c}$ = 106 | |
| | | 2 | 20: $N_{RB,c}$ = 106 | |
| | | 3 | 50: $N_{RB,c}$ = 133 | |
| SSB configuration | | 1 | SSB.1 FR1 | |
| | | 2 | SSB.1 FR1 | |
| | | 3 | SSB.2 FR1 | |
| SMTC configuration | | 1 | SMTC.2 | |
| | | 2 | SMTC.1 | |
| | | 3 | SMTC.1 | |

| | | | | |
|---|---|---|---|---|
| CP length | | 1, 2, 3 | Normal | |
| DRX | s | 1, 2, 3 | 1.28 | ON |
| Time offset between serving and neighbour cells | μs | 1, 2, 3 | 3 | Synchronous cells |
| Expected RSTD | μs | 1, 2, 3 | 3 | |
| Expected RSTD uncertainty | μs | 1, 2, 3 | 5 | |
| T1 | s | 1, 2, 3 | 2 | |
| T2 | s | 1, 2, 3 | 5 | |

**Table 17.4.2.5-2: Cell specific test parameters**

| Parameter | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|
| | | | T1 | T2 | T1 | T2 |
| TDD configuration | | 1 | N/A | | N/A | |
| | | 2 | TDDConf.1.1 | | TDDConf.1.1 | |
| | | 3 | TDDConf.2.1 | | TDDConf.2.1 | |
| PDSCH RMC configuration | | 1 | SR.1.1 FDD | | N/A | |
| | | 2 | SR.1.1 TDD | | | |
| | | 3 | SR.2.1 TDD | | | |
| RMSI CORESET RMC configuration | | 1 | CR.1.1 FDD | | N/A | |
| | | 2 | CR.1.1 TDD | | | |
| | | 3 | CR.2.1 TDD | | | |
| Dedicated CORESET RMC configuration | | 1 | CCR.1.1 FDD | | N/A | |
| | | 2 | CCR.1.1 TDD | | | |
| | | 3 | CCR.2.1 TDD | | | |
| OCNG Patterns | | 1, 2, 3 | OP.1 | | OP.1 | |
| TRS Configuration | | 1 | TRS.1.1 FDD | | N/A | |
| | | 2 | TRS.1.1 TDD | | | |
| | | 3 | TRS.1.2 TDD | | | |
| Initial BWP configuration | | 1, 2, 3 | DLBWP.0.1 ULBWP.0.1 | | N/A | |
| Active UL BWP configuration | | 1, 2, 3 | ULBWP.1.1 | | N/A | |
| PRS configuration | | 1 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 2 | PRS.1.4 FR1 | | PRS.1.4 FR1 | |
| | | 3 | PRS.2.4 FR1 | | PRS.2.4 FR1 | |
| PRS BW | | 1,2,3 | 48 PRBs | | 48PRBs | |
| PRS muting info | | 1, 2, 3 | ‘10’ | | ‘01’ | |
| $N_{oc}$ Note 2 | dBm/SCS | 1 | -98.6 | | | |
| | | 2 | -98.6 | | | |
| | | 3 | -95.6 | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | 1 | -98.6 | | | |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/I_{ot}$ | dB | 1 | -Infinity | 0.39 | -Infinity | -5.63 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS $\hat{E}_s/N_{oc}$ | dB | 1 | -Infinity | 2.98 | -Infinity | -0.88 |
| | | 2 | | | | |
| | | 3 | | | | |
| PRS-RSRP Note 3 | dBm/SCS kHz | 1 | -Infinity | -95.62 | -Infinity | -99.48 |
| | | 2 | -Infinity | -95.62 | -Infinity | -99.48 |
| | | 3 | -Infinity | -92.62 | -Infinity | -96.48 |
| SS-RSRP Note 3 | dBm/SCS kHz | 1 | -88 | -100 | -Infinity | -104 |
| | | 2 | -88 | -100 | -Infinity | -104 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 3 | -85 | -97 | -Infinity | -101 |
| Io | dBm/19.08 MHz | 1 | N/A | -61.75 | N/A | -61.75 |
| | dBm/19.08 MHz | 2 | | -61.75 | | -61.75 |
| | dBm/47.88 MHz | 3 | | -57.77 | | -57.77 |
| Propagation Condition | | 1, 2, 3 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d$ = 0.45 μs and $f_D$ = 5 Hz | | | |
| Note 1: The resources for uplink transmission are assigned after the end of time period T2 to UEs that do not support SDT for measurement reporting. | | | | | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | | |
| Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | | |

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.4.2.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 5.6.3.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126..

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left(K_{carrier_PRS} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{RxBeam,i}$=1, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.285$, $N_{sample}$=1. N is the parameter durationOfPRS-ProcessingSymbols-r17 from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17 from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$. $T_{available_{PRS,i}} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 1280ms depending on UE capabilities. Therefore, $T_{available_{PRS,i}} = 1280$ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms-r17 from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 1 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.285}{N} \right\rceil * 1 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 1.288s and 8.96s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 2s and 9s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 2.3s and 9.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.4.2.5-3, Table 17.4.2.5-4 and Table 17.4.2.5-5 for reference.

**Table 17.4.2.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.4.2.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| *durationOfPRS-ProcessingSymbols* | $\left\lceil \frac{0.285}{N} \right\rceil$ |
|---|---|
| nDot125 | 3 |
| nDot25 | 2 |
| >=nDot5 | 1 |

**Table 17.4.2.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| *durationOfPRS-ProcessingSymbolsInEveryTms* | $T_{effect}$ | $T_{last}$ |
|---|---|---|
| n8 | 1280 | 1288 |
| n16 | 1280 | 1296 |
| n20 | 1280 | 1300 |
| n30 | 1280 | 1310 |
| n40 | 1280 | 1320 |
| n80 | 1280 | 1360 |
| n160 | 1280 | 1440 |
| n320 | 1280 | 1600 |
| n640 | 1280 | 1920 |
| n1280 | 1280 | 2560 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.4.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.4.3.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 5.6.3.5 in FR2 in standalone scenario when single positioning frequency layer is configured.

### 17.4.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and DL-PRS measurement in RRC_INACTIVE state.

### 17.4.3.3 Minimum conformance requirements

Same as defined in clause 16.4.3.3.

### 17.4.3.4 Test description

#### 17.4.3.4.1 Initial conditions

The supported test configurations in listed in Table 17.4.3.4.1-1

**Table 17.4.3.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.4.3.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.4.3.5-1 and Table 17.4.3.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.4.3.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

### 17.4.3.4.2 Test procedure

Same as defined in clause 17.4.1.4.2.

### 17.4.3.4.3 Message contents

As defined in clause 17.4.1.4.3, with the following exceptions:

Table 17.4.3.4.3-1 replaces Table 17.4.1.4.3-3, Table 17.4.3.4.3-2 replaces Table 17.4.1.4.3-6 and Table 17.4.3.4.3-3 replaces Table 17.4.1.4.3-7.

**Table 17.4.3.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.3.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 27 | 128 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.3.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.3.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.3.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 17.4.3.5 Test requirement

Table 17.4.3.5-1 and Table 17.4.3.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.4.3.5-1: General test parameters for PRS RSRPP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |

| | | | | |
|---|---|---|---|---|
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | s | Config 1 | 0.64 | ON |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells. |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |

**Table 17.4.3.5-2: Cell-specific test parameters for PRS RSRPP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.4 FR2 | | PRS.1.4 FR2 | |
| PRS muting configuration | | | Config 1 | ‘10’ | | ‘01’ | |
| EPRE ratio of PSS to SSS | | | | | | | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of PDCCH DMRS to SSS | | Config 1 | 0 | | 0 | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -99.4 | | -99.4 | |
| $N_{oc}$ Note2 | dBm/SCS Note4 | Config 1 | -90.4 | | -90.4 | |
| SS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -89 | -91 | -Infinity | -99 |
| PRS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -Infinity | -92.4 | -Infinity | -100.4 |
| PRS $\hat{E}_s/I_{ot}$ | dB | Config 1 | -Infinity | -2.7 | -Infinity | -12.3 |
| PRS $\hat{E}_s/N_{oc}$ | dB | Config 1 | -Infinity | -2 | -Infinity | -10 |
| Io Note3 | dBm/19 0.08 MHz Note5 | Config 1 | -58.37 | -55.99 | -58.37 | -55.99 |
| Propagation Condition | | Config 1 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d = 0.45$ μs and $f_D = 5$ Hz | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols other than those in the slots with transmitted PRS.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone

Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.4.3.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 5.6.3.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left( K_{carrier_PRS} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1 \right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{RxBeam,i}$=8, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.035$, $N_{sample}$=4. N is the parameter durationOfPRS-ProcessingSymbols-r17 from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17 from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$. $T_{available_{PRS,i}} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 640ms depending on UE capabilities. Therefore, $T_{available_{PRS,i}} = 640$ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms-r17 from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 4 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 20.488s and 82.56s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 21s and 83s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 21.3s and 83.3s.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.4.3.5-3, Table 17.4.3.5-4 and Table 17.4.3.5-5 for reference.

**Table 17.4.3.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.4.3.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| ***durationOfPRS-ProcessingSymbols*** | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >=nDot125 | 1 |

**Table 17.4.3.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

## 17.4.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.4.4.1 Test purpose

The purpose of the test is to verify the PRS RSRPP measurement requirements specified in TS 38.133 [50] clause 5.6.5.5 for single positioning frequency layer under AWGN propagation conditions in FR2 standalone scenario for reduced number of samples

### 17.4.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples-RRC-Inactive.

### 17.4.4.3 Minimum conformance requirements

Same as defined in clause 16.4.3.3.

### 17.4.4.4 Test description

#### 17.4.4.4.1 Initial conditions

The supported test configurations in listed in Table 17.4.4.4.1-1

**Table 17.4.4.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.4.4.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.4.4.5-1 and Table 17.4.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.4.4.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

#### 17.4.4.4.2 Test procedure

Same as defined in clause 17.4.1.4.2.

### 17.4.4.4.3 Message contents

As defined in clause 17.4.1.4.3, with the following exceptions:

Table 17.4.4.4.3-1 replaces Table 17.4.1.4.3-3, Table 17.4.4.4.3-2 replaces Table 17.4.1.4.3-6, Table 17.4.4.4.3-3 replaces Table 17.4.1.4.3-7 and Table 17.4.4.4.3-4 replaces Table 17.4.1.4.3-9.

**Table 17.4.4.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.4.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 7 | 48 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.4.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.4.4.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.4.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 SEQUENCE { | | | |
| dl-prs-MutingBitRepetitionFactor-r16 | Not present | | |
| nr-option1-muting-r16 CHOICE { | | | |
| po2-r16 | 10 | | Cell 1 |
| | 01 | | Cell 2 |
| } | | | |
| } | | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 4 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.4.4.4.3-4: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.4.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if | |

| | | | |
|---|---|---|---|
| | | response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.4.4.5 Test requirement

Table 17.4.4.5-1 and Table 17.4.4.5-2 define the primary level settings including the test tolerances for the test.

**Table 17.4.4.5-1: General test parameters for PRS RSRPP measurement reporting delay**

| Parameter | Unit | Test configuration | Value | Comment |
|---|---|---|---|---|
| NR RF Channel Number | | Config 1 | 1: Cell 1 and Cell 2 | One TDD carrier frequency is used for the NR cells. |

| | | | | |
|---|---|---|---|---|
| Active cell | | Config 1 | NR cell 1 (PCell) | Cell 1 is the PCell and the DL-AoD reference cell in the positioning assistance data. |
| Neighbour cell | | Config 1 | NR cell 2 | Cell 2 is a neighbour cell in the positioning assistance data. |
| SMTC parameters | | Config 1 | SMTC.1 | As specified in TS 38.133 [50] clause A.3.11 |
| SSB parameters | | Config 1 | SSB.3 FR2 | As specified in TS 38.133 [50] clause A.3.10.2 |
| A3-Offset | dB | Config 1 | -6 | |
| Hysteresis | dB | Config 1 | 0 | |
| CP length | | Config 1 | Normal | |
| TimeToTrigger | s | Config 1 | 0 | |
| Filter coefficient | | Config 1 | 0 | L3 filtering is not used |
| DRX | s | Config 1 | 0.64 | ON |
| Time offset between serving and neighbour cells | | Config 1 | 3μs | Synchronous cells. |
| Expected RSTD | μs | Config 1 | 3 | |
| Expected RSTD uncertainty | μs | Config 1 | 5 | |
| T1 | s | Config 1 | 5 | |
| T2 | s | Config 1 | 7 | |

**Table 17.4.4.5-2: Cell-specific test parameters for PRS RSRPP measurement reporting delay**

| Parameter | | Unit | Test configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| | | | | T1 | T2 | T1 | T2 |
| AoA setup | | | Config 1 | Setup 1 as specified in TS 38.133 [50] clause A.3.15 | | | |
| Beam Assumption[Note 7] | | | Config 1 | Rough | | Rough | |
| TDD configuration | | | Config 1 | TDDConf.3.1 | | TDDConf.3.1 | |
| Duplex mode | | | Config 1 | TDD | | TDD | |
| $BW_{channel}$ | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP BW | | MHz | Config 1 | 200: $N_{RB,c}$ = 132 | | 200: $N_{RB,c}$ = 132 | |
| BWP configuration | Initial DL BWP | | Config 1 | DLBWP.0.1 | | N/A | |
| | Initial UL BWP | | | ULBWP.0.1 | | N/A | |
| | Dedicated UL BWP | | | ULBWP.1.1 | | N/A | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1) | | | Config 1 | OP.1 | | OP.1 | |
| PDSCH Reference measurement channel | | | Config 1 | SR.3.1 TDD | | - | |
| CORESET Reference Channel | | | Config 1 | CR.3.1 TDD | | - | |
| Dedicated CORESET RMC configuration | | | Config 1 | CCR.3.1 TDD | | - | |
| TRS configuration | | | Config 1 | TRS.2.1 TDD | | - | |
| PDSCH/PDCCH subcarrier spacing | | kHz | Config 1 | 120 | | 120 | |
| PRS configuration | | | Config 1 | PRS.1.4 FR2 | | PRS.1.4 FR2 | |
| PRS BW | | | Config 1 | 48 PRBs | | 48 PRBs | |
| PRS muting configuration | | | Config 1 | '10' | | '01' | |
| EPRE ratio of PSS to SSS | | | | | | | |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| EPRE ratio of PDCCH DMRS to SSS | | Config 1 | 0 | | 0 | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz Note5 | | -106.45 | | -106.45 | |
| $N_{oc}$ Note2 | dBm/SCS Note4 | Config 1 | -97.45 | | -97.45 | |
| SS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -89 | -86.77 | -Infinity | -90.73 |
| PRS-RSRP Note 3 | dBm/SCS Note5 | Config 1 | -Infinity | -90.34 | -Infinity | -92.15 |
| PRS $\hat{E}_s/I_{ot}$ | dB | Config 1 | -Infinity | 0.3 | -Infinity | -2.8 |
| PRS $\hat{E}_s/N_{oc}$ | dB | Config 1 | -Infinity | 7.08 | -Infinity | 5.27 |
| Io Note3 | dBm/19 0.08 MHz Note5 | Config 1 | -65.42 | -55.66 | -65.42 | -55.66 |
| Propagation Condition | | Config 1 | Two-tap channel defined in 38.101-4 Annex B.2.4, $a$ = 1, $\tau_d = 0.45$ µs and $f_D = 5$ Hz | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SS-RSRP/PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 6: As observed with 0 dBi gain antenna at the centre of the quiet zone

Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

Note 8: Calculation of Es/Iot$_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

The PRS-RSRPP measurement time fulfils the requirements specified in clause 17.4.4.3.

The UE shall perform and report the PRS RSRPP measurements for Cell 2 with respect to the reference cell in the DL-AoD assistance data, Cell 1, within the time duration specified in TS 38.133 [50] clause 5.6.3.5 starting from the beginning of time interval T2

The rate of the correct events for the neighbour cell observed during repeated tests shall be at least 90%, where the reported PRS RSRPP measurement for each correct event shall be within the PRS RSRPP reporting range specified in TS 38.133 Clause 10.1.38, i.e., between PRS RSRPP_0 and PRS RSRPP_126.

The response time is equal to the LPP time IE value plus the test tolerance. The LPP time IE value is derived from the PRS-RSRPP measurement period. The PRS-RSRPP measurement period follows the equation:

$$T_{PRS-RSRP,i} = \left(K_{carrier_PRS} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last,i}$$

Where $K_{carrier_PRS} = 1$, $N_{RxBeam,i}$=8, $N_{PRS,i}^{slot} = 2$, $L_{available_PRS,i} = 0.035$, $N_{sample}$=1. N is the parameter durationOfPRS-ProcessingSymbols-r17 from TS 37.355 [49], N' is the parameter maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17 from TS 37.355 [49], $T_{last,i} = T_i + T_{available_PRS,i}$ and $T_{effect,i} = \left\lceil \frac{T_i}{T_{available_PRS,i}} \right\rceil * T_{available_PRS,i}$. $T_{available_{PRS,i}} = LCM(T_{PRS,i}, T_{DRX})$, where Tprs = 160 ms, and $T_{DRX}$= 640ms depending on UE capabilities. Therefore, $T_{available_{PRS,i}} = 640$ms.

$T_i$ depends on the UE parameter durationOfPRS-ProcessingSymbolsInEveryTms-r17 from TS 37.355 [49]

Finally, it results in the following equation:

$$\left(1 * 8 * \left\lceil \frac{2}{N'} \right\rceil \left\lceil \frac{0.035}{N} \right\rceil * 1 - 1\right) * T_{effect} + T_{last}$$

Where the remaining parameters depend on the UE capabilities. The LPP time IE ranges between 5.128s and 17.28s. The value of the LPP time IE is rounded up to the next second (if the value is >128s, it should be rounded up to the next multiple of ten seconds). The result is transmitted in the response time IE in the LPP-RequestLocationInformation. The LPP time IE ranges between 6s and 18s.

The test tolerance for the response time is 300ms. Therefore, the response time ranges between 6.3s and ss.

The values of N', N and Ti and the effect in the response time equation are defined in Table 17.4.4.5-3, Table 17.4.4.5-4 and Table 17.4.4.5-5 for reference.

**Table 17.4.4.5-3: value of N' based on *maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive***

| ***maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive*** | $\left\lceil \frac{2}{N'} \right\rceil$ |
|---|---|
| n1 | 2 |
| >=n2 | 1 |

**Table 17.4.4.5-4: value of N based on *durationOfPRS-ProcessingSymbols***

| ***durationOfPRS-ProcessingSymbols*** | $\left\lceil \frac{0.035}{N} \right\rceil$ |
|---|---|
| >=nDot125 | 1 |

**Table 17.4.4.5-5: value of $T_{effect}$ and $T_{last}$ based on *durationOfPRS-ProcessingSymbolsInEveryTms***

| ***durationOfPRS-ProcessingSymbolsInEveryTms*** | **$T_{effect}$** | **$T_{last}$** |
|---|---|---|
| n8 | 640 | 648 |
| n16 | 640 | 656 |
| n20 | 640 | 660 |
| n30 | 640 | 670 |
| n40 | 640 | 680 |
| n80 | 640 | 720 |
| n160 | 640 | 800 |
| n320 | 640 | 960 |
| n640 | 640 | 1280 |
| n1280 | 1280 | 1920 |

The test tolerances are defined in clauses C.1.6 and C.2.5.

The rate of successful tests during repeated tests shall be at least 90% with a confidence level of 95%.

# 17.5 NR PRS-RSRPP measurement accuracy test case in RRC_INACTIVE

## 17.5.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA

### 17.5.1.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 10.1.38.2 in FR1 in standalone scenario when single positioning frequency layer is configured.

### 17.5.1.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and DL-PRS measurement in RRC_INACTIVE state.

### 17.5.1.3 Minimum conformance requirements

Same as defined in clause 17.3.1.3.

### 17.5.1.4 Test description

#### 17.5.1.4.1 Initial conditions

The test defines three possible test configurations; specified in Table 17.5.1.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 17.5.1.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |
| Note: The UE is only required to be tested in one of the supported test configurations. | |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.5.1.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.5.1.5-1.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.5.1.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

#### 17.5.1.4.2 Test procedure

The test consists of a set-up period and a measurement period. During the set-up period, the UE shall be in RRC_CONNECTED state and shall not have any timing information of Cell 2. During measurement period, the UE shall be in RRC_INACTIVE state and all cells transmit PRS resources within initial DL BWP of the UE and with the

same numerology as the initial DL BWP. The test consists of two sub-tests; the difference between the sub-tests is the PRS configuration.

The *NR-DL-AoD-ProvideAssistanceData* and *NR-DL-AoD-RequestLocationInformation* as defined in TS 37.355 [49] clause 6.5.11 shall be provided to the UE during set-up period. The beginning of the measurement period shall be aligned with the first DRX cycle containing a DL PRS resource(s).

5. Configure the Cell for RA-SDT with SIB1 configured according to the message content Table 17.4.1.4.3-1.
6. Ensure the UE is in State RRC_CONNECTED with generic procedure parameters Connectivity *NR*, Connected without release *On*, according to TS 38.508-1 [45] clause 4.5.
3. The SS shall send a RESET UE POSITIONING STORED INFORMATION message.
4. Set the parameters according to Table 17.5.1.5-1. Propagation conditions are set according to clause 4.16.2.
5. The SS shall transmit an RRCReconfiguration message with the DRX configuration.
6. The UE shall transmit RRCReconfigurationComplete message.
7. The SS shall transmit an LPP REQUEST CAPABILITIES message.
8. The UE shall transmit an LPP PROVIDE CAPABILITIES message indicating the DL-AoD capabilities supported by the UE in the *NR-DL-AoD-ProvideCapabilities* IE.
9. The SS shall send a LPP PROVIDE ASSISTANCE DATA message, including the *NR-DL-AoD-ProvideAssistanceData* IE. The position of neighbour Cell 2 is described in 3GPP TS 37.571-5 [20]. If the UE message at step 8 includes the ackRequested IE set to TRUE, then the SS shall send an acknowledgment in the LPP PROVIDE ASSISTANCE DATA message.
10. The SS shall transmit LCS Periodic-Triggered Invoke Request message with LPP REQUEST LOCATION INFORMATION embedded to start the periodic reporting procedure, including the *NR-DL-AoD-RequestLocationInformation* IE such that the UE receives the message ΔT ms before the start of measurement period, where ΔT = 50 ms.
11. The UE shall transmit an LCS Periodic-Triggered Invoke Return Result message.
12. When set-up period ends, the SS shall transmit an RRCRelease message including SDT configuration to transit the UE to RRC_INACTIVE state.
13. Wait for the UE monitors for occurrence of the periodic event requested.
14. The UE shall transmit an *RRCResumeRequest* message. If the UE also supporting SDT feature, the UE shall include LPP PROVIDE LOCATION INFORMATION message in the *RRCResumeRequest* message and then go to step 19.
15. The SS transmits an *RRCResume* message.
16. The UE transmits an *RRCResumeComplete* message.
17. The UE shall transmit a LPP PROVIDE LOCATION INFORMATION message, including the nr-DL-AoD-ProvideLocationInformation-r16 IE.
18. If the UE message at step 17 includes the *ackRequested* IE set to TRUE, the SS shall send a LPP acknowledgement message.
19. The SS shall check the nr-DL-PRS-FirstPathRSRP-ResultDiff-r17 field for Cell 2 in the nr-DL-AoD-SignalMeasurementInformation-r16 according to Table 17.5.1.5-2
20. UE shall be released and moved to RRC_IDLE.
21. Repeat steps 2-20 in Tables 17.5.1.4-1 until the confidence level according to Annex D is achieved.
22. Repeat step 1-21 for the other sub-test defined in Table 17.5.1.4-1 as appropriate.

If all (applicable) sub-tests pass, the whole test passes. If one (applicable) sub-test fails, the whole test fails.

### 17.5.1.4.3 Message contents

As defined in clause 17.4.1.4.3, with the following exceptions:

Table 17.5.1.4.3-1 replaces Table 17.4.1.4.3-6 and Table 17.5.1.4.3-2 replaces Table 17.4.1.4.3-7.

**Table 17.5.1.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 8 | 24 PRBs | Sub-test 1-1, Sub-test 2-1 and Sub-test 3-1 |
| | 21 | 104 PRBs | Sub-test 1-2 and Sub-test 2-2 |
| | 28 | 132 PRBs | Sub-test 3-2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.1.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |

| | | | |
|---|---|---|---|
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.1.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.1.4.3-2: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | Sub-test 2 |
| | n2 | | Sub-test 1 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |

| | | | |
|---|---|---|---|
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.5.1.5 Test requirement

Table 17.5.1.5-1 defines the primary level settings including test tolerances for the test.

**Table 17.5.1.5-1: PRS-RSRPP test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | | freq1 | | freq1 | |
| Duplex mode | Config 1 | | FDD | | | |
| | Config 2,3 | | TDD | | | |
| TDD configuration | Config 1 | | Not Applicable | | | |
| | Config 2 | | TDDConf.1.1 | | | |
| | Config 3 | | TDDConf.2.1 | | | |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | | | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | | | |
| BWP BW | Config 1 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | | | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | | | |
| Downlink initial BWP configuration | | | DLBWP.0.1 | | | |
| Uplink initial BWP configuration | | | ULBWP.0.1 | | | |
| TRS configuration | Config 1 | | TRS.1.1 FDD | NA | TRS.1.1 FDD | NA |
| | Config 2 | | TRS.1.1 TDD | NA | TRS.1.1 TDD | NA |
| | Config 3 | | TRS.1.2 TDD | NA | TRS.1.2 TDD | NA |
| DRX cycle length | Config 1,2,3 | ms | 1280 | | | |
| PDSCH Reference measurement channel | Config 1 | | SR.1.1 FDD | - | SR.1.1 FDD | - |
| | Config 2 | | SR.1.1 TDD | | SR.1.1 TDD | |
| | Config 3 | | SR2.1 TDD | | SR2.1 TDD | |
| RMSI CORESET Reference Channel | Config 1 | | CR.1.1 FDD | - | CR.1.1 FDD | - |
| | Config 2 | | CR.1.1 TDD | | CR.1.1 TDD | |
| | Config 3 | | CR2.1 TDD | | CR2.1 TDD | |
| Control channel RMC | Config 1 | | CCR.1.1 FDD | - | CCR.1.1 FDD | - |
| | Config 2 | | CCR.1.1 TDD | | CCR.1.1 TDD | |
| | Config 3 | | CCR2.1 TDD | | CCR2.1 TDD | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PRS configuration | Config 1 | | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | | PRS.1.3 FR1 | PRS.1.3 FR1 | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | | PRS.2.3 FR1 | PRS.2.3 FR1 | PRS.2.4 FR1 | PRS.2.4 FR1 |
| PRS Resource slot offset (slot) | Config 1,2,3 | | slot | 0 | 4 | 0 | 4 |
| SSB configuration | Config 1 | | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 | SSB.2 FR1 |
| Time offset with Cell 1 | Config 1 | | ms | - | 3 | - | 3 |
| | Config 2,3 | | µs | - | 3 | - | 3 |
| SMTC configuration | Config 1 | | | SMTC.2 | | | |
| | Config 2,3 | | | SMTC.1 | | | |
| OCNG Patterns | | | | OCNG pattern 1 | | | |
| PDSCH/PDCCH subcarrier spacing | Config 1,2 | | kHz | 15 kHz | | | |
| | Config 3 | | | 30 kHz | | | |
| EPRE ratio of PSS to SSS | | | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | | | | |
| $N_{oc}$ Note2 | Config 1,2 | | dBm/15KhZ | -98 | | -98 | |
| | Config 3 | | | -98 | | -98 | |
| $N_{oc}$ Note2 | Config 1,2 | | dBm/SCS | -98 | | -98 | |
| | Config 3 | | | -95 | | -95 | |
| $\hat{E}_s/I_{ot}$ | | | dB | -2.41 | -12.12 | -2.41 | -12.12 |
| $\hat{E}_s/N_{oc}$ | | | dB | -2 | -10 | -2 | -10 |
| PRS-RSRP Note3 | Config 1, 2 | | dBm/SCS | -100 | -108 | -100 | -108 |
| | Config 3 | | | -97 | -105 | -97 | -105 |
| Io Note3 | Config 1,2 | | dBm/19.08MHz | -64.57 | | -64.57 | |
| | Config 3 | | dBm/47.88MHz | -60.59 | | -60.59 | |
| Propagation condition | | | | Two-tap channel Note 7 | | | |
| Antenna configuration | | | | 1x2 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: PRS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Void.

Note 6: Void

Note 7: The two-tap channel model is defined in 38.101-4 Annex B.2.4 (a = 1, $\tau_d$=0.45 µs and $f_D$=5 Hz).

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.4.1.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.5.1.5-2: PRS-RSRPP accuracy requirements for the reported values**

| Test Configuration | Subtest | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | Sub-test 1 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_40 | PRS_RSRPP_51 |
| | Sub-test 2 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_41 | PRS_RSRPP_50 |
| 2 | Sub-test 1 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_40 | PRS_RSRPP_51 |
| | Sub-test 2 | PRS_RSRPP_49 | PRS_RSRPP_58 | PRS_RSRPP_41 | PRS_RSRPP_50 |
| 3 | Sub-test 1 | PRS_RSRPP_52 | PRS_RSRPP_61 | PRS_RSRPP_43 | PRS_RSRPP_54 |
| | Sub-test 2 | PRS_RSRPP_52 | PRS_RSRPP_61 | PRS_RSRPP_44 | PRS_RSRPP_53 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 17.5.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA

### 17.5.2.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement in RRC_INACTIVE state with reduce number of samples meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.38.2 in FR1 in standalone scenario when single positioning frequency layer is configured.

### 17.5.2.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples-RRC-Inactive.

### 17.5.2.3 Minimum conformance requirements

Same as defined in clause 17.3.2.3.

### 17.5.2.4 Test description

#### 17.5.2.4.1 Initial conditions

The test defines three possible test configurations; specified in Table 17.5.2.4-1. The UE is only required to be tested in one of the supported test configurations.

**Table 17.5.2.4-1: Test Configurations**

| Test Configuration | Description |
|---|---|
| 1 | 15 kHz SSB SCS, 20 MHz bandwidth, FDD duplex mode |
| 2 | 15 kHz SSB SCS, 20 MHz bandwidth, TDD duplex mode |
| 3 | 30 kHz SSB SCS, 50 MHz bandwidth, TDD duplex mode |

| Note: The UE is only required to be tested in one of the supported test configurations. |
|---|

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.5.2.4-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.13.
2. The general test parameter settings are set up according to Table 17.5.2.5-1.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.5.2.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR1.

### 17.5.2.4.2 Test procedure

Same as defined in clause 17.5.1.4.2.

### 17.5.2.4.3 Message contents

As defined in clause 17.4.1.4.3, with the following exceptions:

Table 17.5.2.4.3-1 replaces Table 17.4.1.4.3-6, Table 17.5.2.4.3-2 replaces Table 17.4.1.4.3-7 and Table 17.5.2.4.3-3 replaces Table 17.4.1.4.3-9.

**Table 17.5.2.4.3-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz15 | | Config 1 and Config 2 |
| | kHz30 | | Config 3 |
| dl-PRS-ResourceBandwidth-r16 | 8 | 52 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |

| | | | |
|---|---|---|---|
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.2.4.3-2 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.2.4.3-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.2.4.3-2: NR-DL-PRS-Info**

| Derivation Path: TS 37.355 [49] clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs15-r16 CHOICE { | | | Config 1 and Config 2 |
| n160-r16 | 10 | | |
| } | | | |
| scs30-r16 CHOICE { | | | Config 3 |
| n320-r16 | 20 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |

| | | | |
|---|---|---|---|
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.2.4.3-3: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.5.2.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded | |

| | | | |
|---|---|---|---|
| | | up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.5.2.5 Test requirement

Table 17.5.2.5-1 defines the primary level settings including test tolerances for the test.

**Table 17.5.2.5-1: PRS-RSRPP test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| Cell ID | | | 489 | 0 |
| SSB ARFCN | | | freq1 | |
| Duplex mode | Config 1 | | FDD | |
| | Config 2,3 | | TDD | |

| | | | | |
|---|---|---|---|---|
| TDD configuration | Config 1 | | Not Applicable | |
| | Config 2 | | TDDConf.1.1 | |
| | Config 3 | | TDDConf.2.1 | |
| $BW_{channel}$ | Config 1 | MHz | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| BWP BW | Config 1 | | 20: $N_{RB,c}$ = 106 | |
| | Config 2 | | 20: $N_{RB,c}$ = 106 | |
| | Config 3 | | 50: $N_{RB,c}$ = 133 | |
| Downlink initial BWP configuration | | | DLBWP.0.1 | |
| Uplink initial BWP configuration | | | ULBWP.0.1 | |
| TRS configuration | Config 1 | | TRS.1.1 FDD | NA |
| | Config 2 | | TRS.1.1 TDD | NA |
| | Config 3 | | TRS.1.2 TDD | NA |
| DRX Cycle length | Config 1,2,3 | ms | 1280 | |
| PDSCH Reference measurement channel | Config 1 | | SR.1.1 FDD | - |
| | Config 2 | | SR.1.1 TDD | |
| | Config 3 | | SR2.1 TDD | |
| RMSI CORESET Reference Channel | Config 1 | | CR.1.1 FDD | - |
| | Config 2 | | CR.1.1 TDD | |
| | Config 3 | | CR2.1 TDD | |
| Control channel RMC | Config 1 | | CCR.1.1 FDD | - |
| | Config 2 | | CCR.1.1 TDD | |
| | Config 3 | | CCR2.1 TDD | |
| PRS configuration | Config 1 | | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 2 | | PRS.1.4 FR1 | PRS.1.4 FR1 |
| | Config 3 | | PRS.2.4 FR1 | PRS.2.4 FR1 |
| PRS bandwidth | Config 1,2,3 | | 52 PRBs | 52 PRBs |
| PRS Resource slot offset (slot) | Config 1,2,3 | slot | 0 | 4 |
| SSB configuration | Config 1 | | SSB.1 FR1 | SSB.1 FR1 |
| | Config 2 | | SSB.1 FR1 | SSB.1 FR1 |
| | Config 3 | | SSB.2 FR1 | SSB.2 FR1 |
| Time offset with Cell 1 | Config 1 | ms | - | 3 |
| | Config 2,3 | µs | - | 3 |
| SMTC configuration | Config 1 | | SMTC.2 | |
| | Config 2,3 | | SMTC.1 | |
| OCNG Patterns | | | OCNG pattern 1 | |
| PDSCH/PDCCH subcarrier spacing | Config 1,2 | kHz | 15 kHz | |
| | Config 3 | | 30 kHz | |
| EPRE ratio of PSS to SSS | | dB | 0 | 0 |
| EPRE ratio of PBCH DMRS to SSS | | | | |
| EPRE ratio of PBCH to PBCH DMRS | | | | |
| EPRE ratio of PDCCH DMRS to SSS | | | | |
| EPRE ratio of PDCCH to PDCCH DMRS | | | | |
| EPRE ratio of PDSCH DMRS to SSS | | | | |
| EPRE ratio of PDSCH to PDSCH | | | | |
| EPRE ratio of OCNG DMRS to SSS(Note 1) | | | | |
| EPRE ratio of OCNG to OCNG DMRS (Note 1) | | | | |
| $N_{oc}$ Note2 | Config 1,2 | dBm/15Khz | -98 | |
| | Config 3 | | | |
| $N_{oc}$ Note2 | Config 1,2 | dBm/SCS | -98 | |
| | Config 3 | | -95 | |
| $\hat{E}_s/I_{ot}$ | | dB | 0.32 | -5.64 |
| $\hat{E}_s/N_{oc}$ | | dB | 2.88 | -0.96 |

| | | | | |
|---|---|---|---|---|
| PRS-RSRP Note3 | Config 1, 2 | dBm/SCS | -95.12 | -98.96 |
| | Config 3 | | -92.12 | -95.96 |
| SS-RSRP Note3 | Config 1, 2 | dBm/SCS | -95.77 | -99.73 |
| | Config 3 | | -92.77 | -96.73 |
| Io Note3 | Config 1,2 | dBm/19.08MHz | -61.22 | |
| | Config 3 | dBm/47.88MHz | -57.24 | |
| Propagation condition | | | Two-tap channel Note5 | |
| Antenna configuration | | | 1x2 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: PRS-RSRP, SS-RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: PRS-RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: The two-tap channel model is defined in 38.101-4 Annex B.2.4 (a = 1, $\tau_d$=0.45 μs and $f_D$=5 Hz).

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.4.2.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.5.2.5-2: PRS-RSRPP accuracy requirements for the reported values**

| Test Configuration | Cell 1 | | Cell 2 | |
|---|---|---|---|---|
| | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | PRS_RSRPP_54 | PRS_RSRPP_62 | PRS_RSRPP_50 | PRS_RSRPP_59 |
| 2 | PRS_RSRPP_54 | PRS_RSRPP_62 | PRS_RSRPP_50 | PRS_RSRPP_59 |
| 3 | PRS_RSRPP_57 | PRS_RSRPP_65 | PRS_RSRPP_53 | PRS_RSRPP_62 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 17.5.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.5.3.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement in RRC_INACTIVE state meets the requirements specified in TS 38.133 [50] clause 10.1.38.2 in FR2 in standalone scenario when single positioning frequency layer is configured.

### 17.5.3.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and DL-PRS measurement in RRC_INACTIVE state.

### 17.5.3.3 Minimum conformance requirements

Same as defined in clause 17.3.3.3.

### 17.5.3.4 Test description

#### 17.5.3.4.1 Initial conditions

The supported test configurations in listed in Table 17.5.3.4.1-1

**Table 17.5.3.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.5.3.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.5.3.5-1 and Table 17.5.3.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.5.3.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

#### 17.5.3.4.2 Test procedure

Same as defined in clause 17.5.1.4.2.

#### 17.5.3.4.3 Message contents

As defined in clause 17.4.1.4.3, with the following exceptions:

Table 17.5.3.4.3-1 replaces Table 17.4.1.4.3-3, Table 17.5.3.4.3-2 replaces Table 17.4.1.4.3-6 and Table 17.5.3.4.3-3 replaces Table 17.4.1.4.3-7.

**Table 17.5.3.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.3.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 3 | 32 PRBs | Sub-test 1 |
| | 27 | 128 PRBs | Sub-test 2 |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | Sub-test 1 |
| | n4 | | Sub-test 2 |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.3.4.3-3 | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.3.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.3.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | n2 | | Sub-test 1 |
| | Not present | | Sub-test 2 |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | Sub-test 1 |
| n4-r16 | 0 | | Sub-test 2 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 1 | | Sub-test 1 |
| n4-r16 | 1 | | Sub-test 2 |

| | | | |
|---|---|---|---|
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.5.3.5 Test requirement

Table 17.5.3.5-1 and Table 17.5.3.5-2 define the primary level settings including test tolerances for the test.

**Table 17.5.3.5-1: PRS-RSRPP general test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 | 489 | 0 |
| SSB ARFCN | | freq1 | | freq1 | |
| Duplex mode | | TDD | | TDD | |
| TDD configuration | | TDDConf.3.1 | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c} = 132$ | | 200: $N_{RB,c} = 132$ | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - | DLBWP.0.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - | ULBWP.0.1 | - |
| DRX cycle configuration | ms | 640 | | | |
| TRS configuration | | TRS.2.1 TDD | - | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - | CR.3.1 TDD | - |
| Control channel RMC | | CCR.3.1 TDD | - | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | µs | - | 3 | - | 3 |
| PRS configuration | | PRS.1.3 FR2 | PRS.1.3 FR2 | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS Resource slot offset | slot | 0 | 4 | 0 | 4 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | | | |
| Propagation conditions | | Two-tap channel Note 2 | | | |

| | | |
|---|---|---|
| Antenna configuration | | 1x2 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | |
| Note 2: The two-tap channel model is defined in 38.101-4 Annex B.2.4 (a = 1, $\tau_d$=0.45 μs and $f_D$=5 Hz). | | |

**Table17.5.3.5-2: PRS-RSRPP OTA related test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | | | |
| Assumption for UE beams Note 7 | | Rough | | Rough | |
| $N_{oc}$ Note1 | dBm/15kHz Note4 | -99.1 | | Same as Test 1 | |
| $N_{oc}$ Note1 | dBm/SCS Note4 | -90.1 | | Same as Test 1 | |
| $\hat{E}_s / N_{oc}$ | dB | -2 | -10 | -2 | -10 |
| $E_s$ | dBm/SCS Note4 | - | - | - | - |
| PRS_RP Note2 | dBm/SCS | -92.1 | -92.1 | -100.1 | -99 |
| $\hat{E}_s / I_{ot}$ BB Note6 | dB | -2.68 | -2.68 | -12.30 | -12.12 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.69 | | -55.69 | |

Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 2: PRS_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: Void
Note 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone
Note 5: Void
Note 6: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.
Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.4.3.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.5.3.5-3: PRS-RSRPP accuracy requirements for the reported values**

| Sub-test | Cell 1 | | Cell 2 | |
|---|---|---|---|---|
| | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | PRS_RSRPP_50 | PRS_RSRPP_83 | PRS_RSRPP_42 | PRS_RSRPP_75 |
| 2 | PRS_RSRPP_50 | PRS_RSRPP_83 | PRS_RSRPP_42 | PRS_RSRPP_75 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

## 17.5.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA

Editor's Note: This test case has been completed for the following configurations:

- Test frequency f ≤ 40.8 GHz

- UE PC3

This test case is incomplete for UE power classes other than PC3

This test case is incomplete for test frequencies > 40.8 GHz

### 17.5.4.1 Test purpose

The purpose of the test is to verify that the PRS-RSRPP measurement in RRC_INACTIVE state with reduce number of samples meets the accuracy requirements specified in TS 38.133 [50] clause 10.1.38.2 in FR2 in standalone scenario when single positioning frequency layer is configured.

### 17.5.4.2 Test applicability

This test applies to all types of NR UE release 17 onwards that supports DL-AoD positioning, maxDL-PRS-FirstPathRSRP-MeasPerTRP and supportedDL-PRS-ProcessingSamples-RRC-Inactive

### 17.5.4.3 Minimum conformance requirements

Same as defined in clause 17.3.4.3.

### 17.5.4.4 Test description

#### 17.5.4.4.1 Initial conditions

The supported test configurations in listed in Table 17.5.4.4.1-1

**Table 17.5.4.4.1-1: Supported test configurations**

| Config | Description |
|---|---|
| 1 | 120 kHz SSB and PRS SCS, 200 MHz bandwidth, TDD duplex mode |

Test Environment: Normal, as defined in TS 38.508-1 [45] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [45] clause 4.3.1.

Channel bandwidth to be tested: are specified in Table 17.5.4.4.1-1.

1. Connect the SS and AWGN noise sources to the UE antenna connector or antenna connectors as shown in Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 17.5.4.5-1 and Table 17.5.4.5-2.
3. Propagation conditions are set according to clause 4.16.2.
4. Message contents are defined in clause 17.5.4.4.3.
5. There are two cells in the test: PCell (Cell 1) and a neighbour cell (Cell 2). All cells are on the same RF channel in FR2.

#### 17.5.4.4.2 Test procedure

Same as defined in clause 17.5.1.4.2.

### 17.5.4.4.3 Message contents

As defined in clause 17.4.1.4.3, with the following exceptions:

Table 17.5.4.4.3-1 replaces Table 17.4.1.4.3-3, Table 17.5.4.4.3-2 replaces Table 17.4.1.4.3-6, Table 17.5.4.4.3-3 replaces Table 17.4.1.4.3-7 and Table 17.5.4.4.3-4 replaces Table 17.4.1.4.3-9.

**Table 17.5.4.4.3-1: CellGroupConfig**

| Derivation Path: 38.508-1 [45], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig SEQUENCE { | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| drx-onDurationTimer | ms6 | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | sl64 | | |
| drx-HARQ-RTT-TimerUL | sl64 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms640 | 0 | | |
| } | | | |
| shortDRX | not present | | |
| drx-SlotOffset | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.4.4.3-2: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | kHz120 | | |
| dl-PRS-ResourceBandwidth-r16 | 11 | 64 PRBs | |
| dl-PRS-StartPRB-r16 | 0 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n4 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | Cell 1 |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Cell 1 | | |
| nr-CellGlobalID-r16 | Cell 1 | | |

| | | | |
|---|---|---|---|
| nr-ARFCN-r16 | Cell 1 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.4.4.3-3 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Cell 2 |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Cell 2 | | |
| nr-CellGlobalID-r16 | Cell 2 | | |
| nr-ARFCN-r16 | Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 23 | About 3 μs | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 154 | About 5 μs | |
| nr-DL-PRS-Info-r16 | As specified in Table 17.5.4.4.3-3 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.4.4.3-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 CHOICE { | | | |
| scs120-r16 CHOICE { | | | |
| n1280-r16 | 80 | | |
| } | | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | s1 | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 27 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 2 entries | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 0 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |
| NR-DL-PRS-Resource-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-ResourceID-r16 | 1 | | |
| dl-PRS-SequenceID-r16 | 1 | | |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n4-r16 | 1 | | |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | Cell 1 |
| | 4 | | Cell 2 |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 17.5.4.4.3-4: LPP RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 1 | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | See 17.5.4.5 | Result of the response time calculation rounded up to the next second if response time <= 128s. Result of the response time calculation rounded up to the next multiple of ten seconds if response time > 128s. | |
| responseTimeEarlyFix-r12 | Not present | | Rel-12 onwards |
| unit-r15 | Not present | | |

| | | | |
|---|---|---|---|
| | ten-seconds | | Calculated response time >128s |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | Not present | | |
| otdoa-RequestLocationInformation | Not present | | |
| ecid-RequestLocationInformation | Not present | | |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | Not present | | |
| tbs-RequestLocationInformation-r13 | Not present | | |
| wlan-RequestLocationInformation-r13 | Not present | | |
| bt-RequestLocationInformation-r13 | Not present | | |
| nr-ECID-RequestLocationInformation-r16 | Not present | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Not present | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Present. Set depending on UE capabilities. | | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | | |
| reducedDL-PRS-ProcessingSamples-r17 | requested | | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | | |
| } | | | |
| multiMeasInSameReport-r17 | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 17.5.4.5 Test requirement

Table 17.5.4.5-1 and Table 17.5.4.5-2 define the primary level settings including test tolerances for the test.

**Table 17.5.4.5-1: PRS-RSRPP general test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |
| Cell ID | | 489 | 0 |
| SSB ARFCN | | freq1 | |
| Duplex mode | | TDD | |
| TDD configuration | | TDDConf.3.1 | |
| $BW_{channel}$ | MHz | 200: $N_{RB,c}$ = 132 | |
| Downlink initial BWP configuration | | DLBWP.0.1 | - |
| Uplink initial BWP configuration | | ULBWP.0.1 | - |
| DRX cycle configuration | ms | 640 | |
| TRS configuration | | TRS.2.1 TDD | - |
| TCI state | | TCI.State.0 | - |
| PDSCH Reference measurement channel | | SR.3.1 TDD | - |
| RMSI CORESET Reference Channel | | CR.3.1 TDD | - |

**Table**

| | | | |
|---|---|---|---|
| Control channel RMC | | CCR.3.1 TDD | - |
| OCNG Patterns | | OP.3 | OP.3 |
| SSB configuration | | SSB.3 FR2 | SSB.3 FR2 |
| SMTC configuration | | SMTC.1 | SMTC.1 |
| Time offset with Cell 1 | µs | - | 3 |
| PRS configuration | | PRS.1.4 FR2 | PRS.1.4 FR2 |
| PRS bandwidth | | 64 PRBs | 64 PRBs |
| PRS Resource slot offset | slot | 0 | 4 |
| PDSCH/PDCCH subcarrier spacing | kHz | 120 | 120 |
| EPRE ratio of PSS to SSS | dB | 0 | 0 |
| EPRE ratio of PBCH_DMRS to SSS | | | |
| EPRE ratio of PBCH to PBCH_DMRS | | | |
| EPRE ratio of PDCCH_DMRS to SSS | | | |
| EPRE ratio of PDCCH to PDCCH_DMRS | | | |
| EPRE ratio of PDSCH_DMRS to SSS | | | |
| EPRE ratio of PDSCH to PDSCH_DMRS | | | |
| EPRE ratio of OCNG DMRS to SSS Note 1 | | | |
| EPRE ratio of OCNG to OCNG DMRS Note 1 | | | |
| Propagation conditions | | Two-tap channel Note 2 | |
| Antenna configuration | | 1x2 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The two-tap channel model is defined in 38.101-4 Annex B.2.4 (a = 1, $\tau_d$=0.45 µs and $f_D$=5 Hz).

**17.5.4.5-2: PRS-RSRPP OTA related test parameters**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| | | Cell 1 | Cell 2 |
| Angle of arrival configuration | | Setup 1 according to TS 38.133 [50] clause A.3.15.1 | |
| Assumption for UE beams Note 7 | | Rough | |
| $N_{oc}$ Note1 | dBm/15kHz Note4 | -103 | |
| $N_{oc}$ Note1 | dBm/SCS Note4 | -94 | |
| $\hat{E}_s / N_{oc}$ | dB | 3.55 | -0.29 |
| $E_s$ | dBm/SCS Note4 | - | - |
| PRS_RP Note2 | dBm/SCS | -90.42 | -94.26 |
| SS_RP Note2 | dBm/SCS | -86.77 | -90.73 |
| $\hat{E}_s / I_{ot\ BB}$ Note6 | dB | 0 | -6 |
| Io Note2 | dBm/190.08 MHz Note4 | -55.74 | |

Note 1: Where used, interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 2: PRS_RP, SS_RP, Es/Iot and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 3: Void

Note 4: Equivalent power received by an antenna with 0 dBi gain at the centre of the quiet zone

Note 5: Void

Note 6: Calculation of $Es/Iot_{BB}$ includes the effect of UE internal noise up to the value assumed for the associated Refsens requirement in clause 7.3.2 of TS 38.101-2 [55], and an allowance of 1dB for UE multi-band relaxation factor $\Delta MB_P$ from TS 38.101-2 [55] Table 6.2.1.3-4.

Note 7: Information about types of UE beam is given in TS 38.133 [50] B.2.1.3, and does not limit UE implementation or test system implementation.

The value of the LPP Time IE depends on the UE capabilities. The calculation is as defined in clause 17.4.4.5. The ideal PRS-RSRPP is calculated by the following equation in the real tests:

$$\mathrm{RSRPP}_p \propto \left| \sum_k H_k \exp\left( j 2\pi D_p \frac{k}{N_{\mathrm{IFFT}}} \right) \right|^2$$

Where:

$H_k$ is the effective channel frequency response (over REs occupied by PRS) measured without receiver noise.

$D_p$ is the exact delay of the p-th path in the channel model.

**Table 17.5.4.5-3: PRS-RSRPP accuracy requirements for the reported values**

| Config | Cell 1 | | Cell 2 | |
|---|---|---|---|---|
| | Lowest reported value | Highest reported value | Lowest reported value | Highest reported value |
| 1 | PRS_RSRPP_52 | PRS_RSRPP_84 | PRS_RSRPP_48 | PRS_RSRPP_80 |

For the overall test to pass, the ratio of successful reported values in each sub-test shall be more than 90% with a confidence level of 95%.

# Annex A (informative): Connection Diagrams

Definition of Terms

**GNSS:** In this clause the term GNSS also includes the case where the only satellite system used is GPS.

**System Simulator or SS:** A device or system, which is capable of generating simulated NodeB and/or eNodeB signalling and analysing UE signalling responses on one RF channel, in order to create the required test environment for the UE under test. It will also include the following capabilities:

1. Control of the UE Tx output power through TPC commands
2. Measurement of signalling timing and delays.
3. Ability to simulate UTRAN and/or E-UTRAN and/or NR signalling.

**GNSS System Simulator or GSS:** A device or system, which is capable of generating simulated GNSS satellite transmissions in order to create the required test environment for the UE under test. It will also include the following capabilities:

1. Control of the output power of individual satellites and the simulation of atmospheric delays and multi-path.
2. Generation of appropriate assistance data to be transmitted to the UE via the SS.
3. Ability to synchronize with UTRAN and/or E-UTRAN and/or NR timing in the SS.

**MBS System Simulator or MSS:** A device or system, which is capable of generating simulated MBS transmissions in order to create the required test environment for the UE under test. It will also include the following capabilities:

1. Control of the output power of individual beacons and the simulation of delays and multi-path.
2. Generation of appropriate messaging to be transmitted to the UE via the SS.

**WLAN System Simulator or WSS:** A device or system, which is capable of generating simulated WLAN beacons in order to create the required test environment for the UE under test. It will also include the following capabilities:

1. Control of the output power of individual beacons and the simulation of delays and AWGN.

**BLE System Simulator or BSS:** A device or system, which is capable of generating simulated BLE advertising signals in order to create the required test environment for the UE under test. It will also include the following capabilities:

1. Control of the output power of individual BLE signals and the simulation of delays and AWGN.

**Test System:** A combination of devices brought together into a system for the purpose of making one or more measurements on a UE in accordance with the test case requirements. The following diagrams are all examples of Test Systems.

NOTE: The above terms are logical definitions to be used to describe the test methods used in the present document, in practice, real devices called "System Simulators" may also include additional measurement capabilities or may only support those features required for the test cases they are designed to perform.

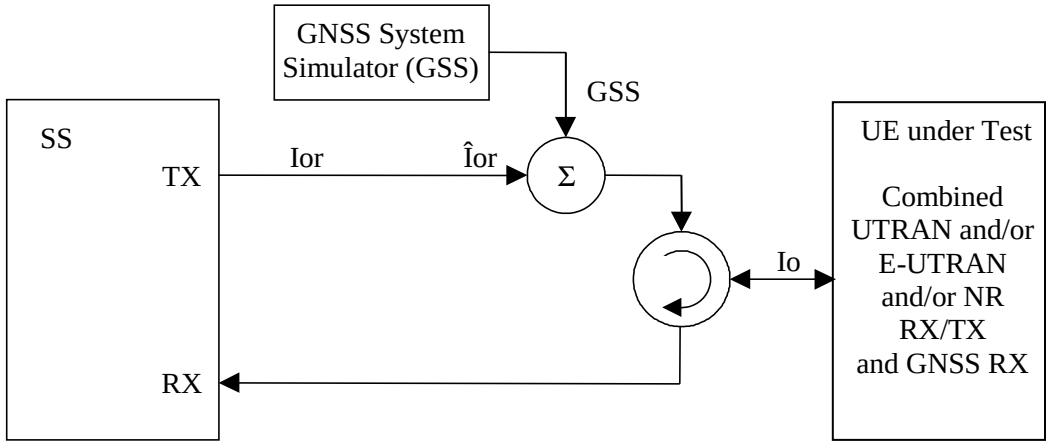



**Figure A.1: Connection for A-GNSS Minimum Performance requirements tests for UE with combined UTRAN and/or E-UTRAN and/or NR and GNSS antenna**

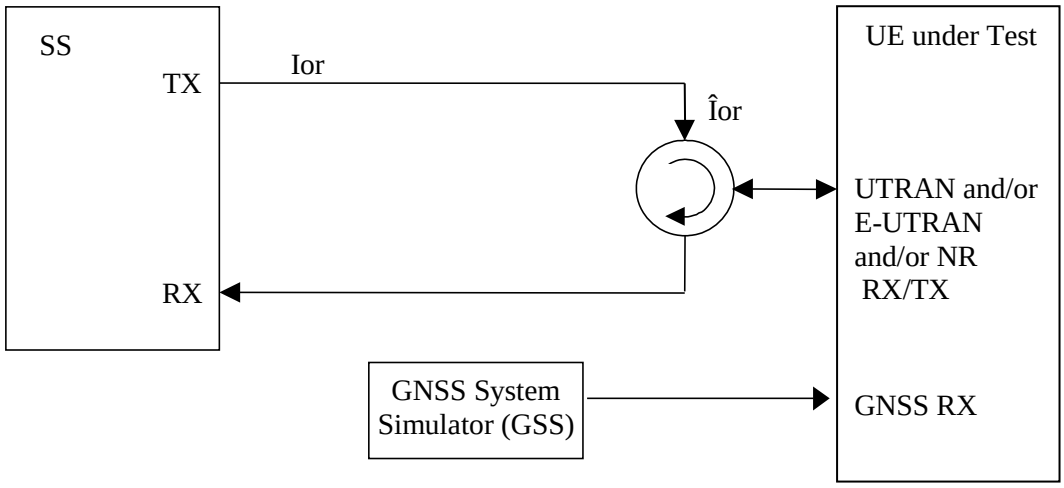



**Figure A.2: Connection for A-GNSS Minimum Performance requirements tests for UE with separate UTRAN and/or E-UTRAN and/or NR and GNSS antennas**

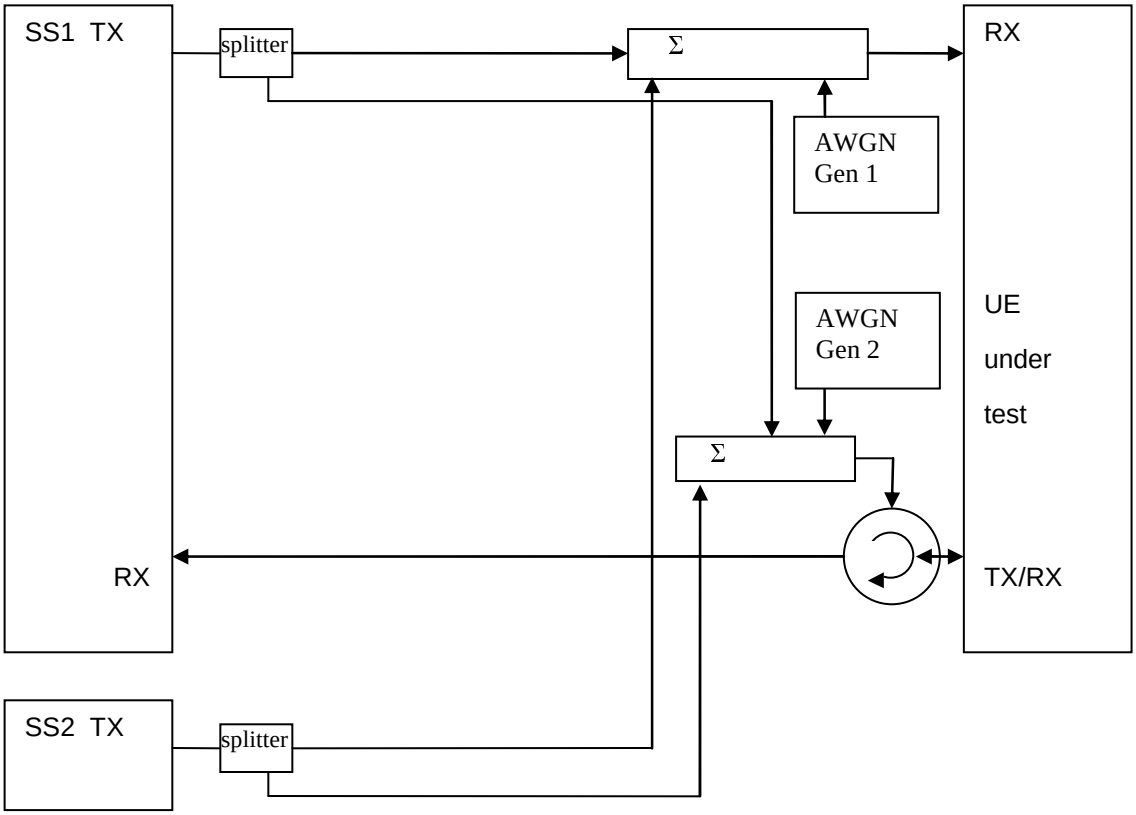



**Figure A.3: Connection for 2 cells OTDOA tests with static propagation**

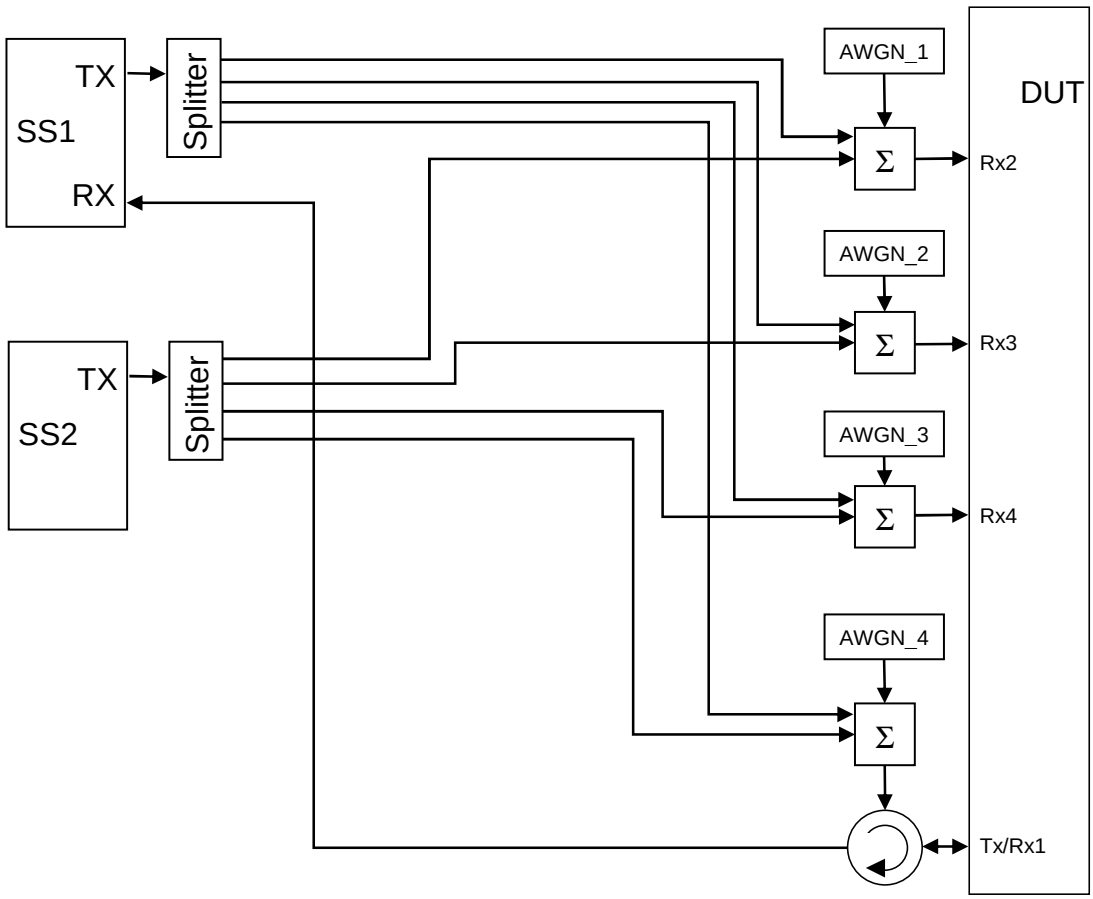



**Figure A.3a: Connection for 2 cells OTDOA tests with static propagation for 4Rx capable UE**

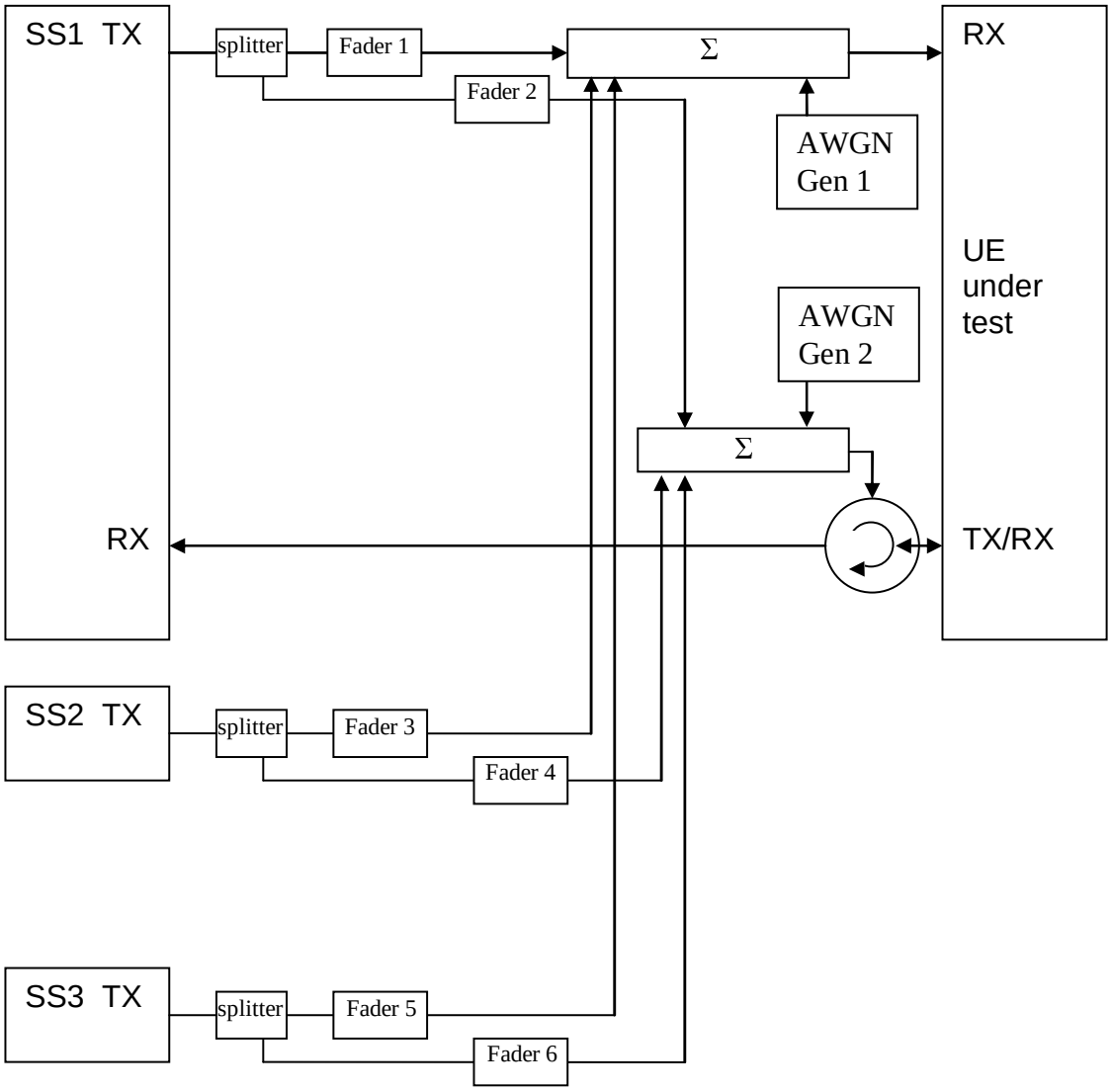



**Figure A.4: Connection for 3 cells OTDOA tests with multipath fading propagation conditions**

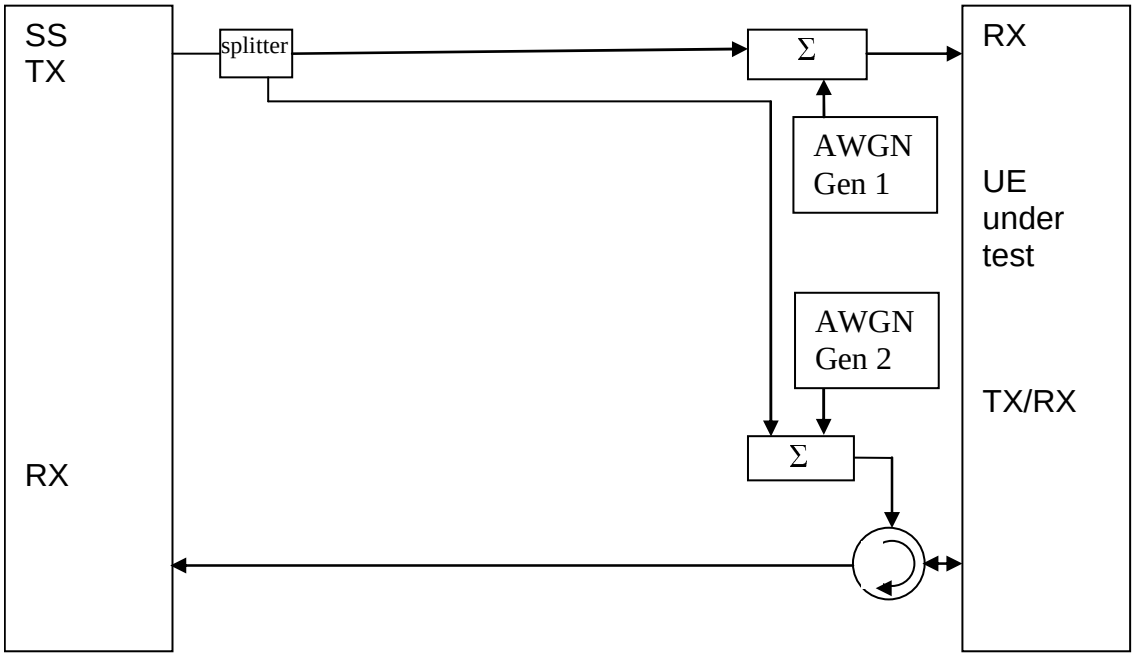



**Figure A.5: Connection for 1 cell ECID tests with static propagation conditions**

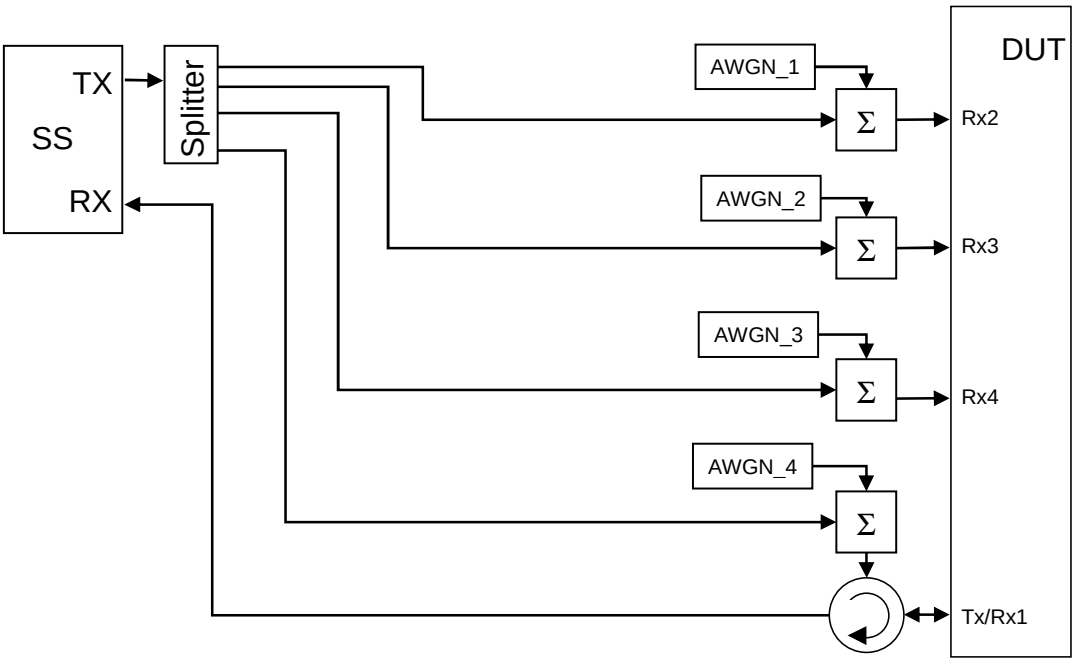



**Figure A.5a: Connection 1 cell ECID tests with static propagation for 4Rx capable UE**

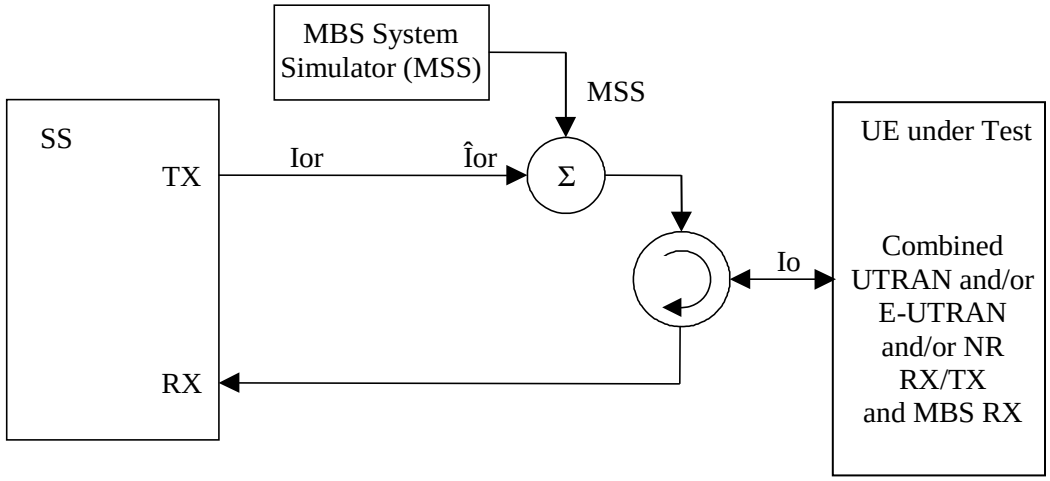



**Figure A.6: Connection for MBS Minimum Performance requirements tests for UE with combined UTRAN and/or E-UTRAN and/or NR and MBS antenna**

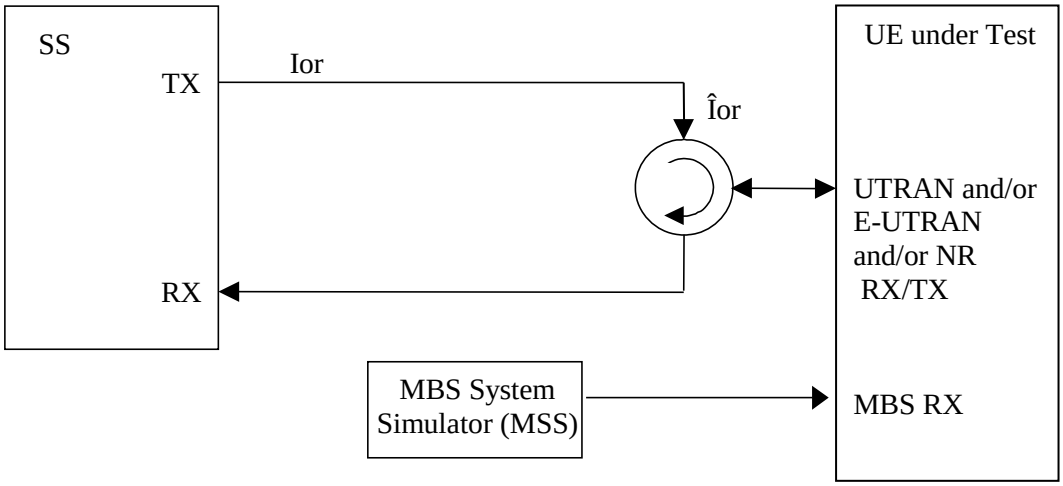



**Figure A.7: Connection for MBS Minimum Performance requirements tests for UE with separate UTRAN and/or E-UTRAN and/or NR and MBS antennas**

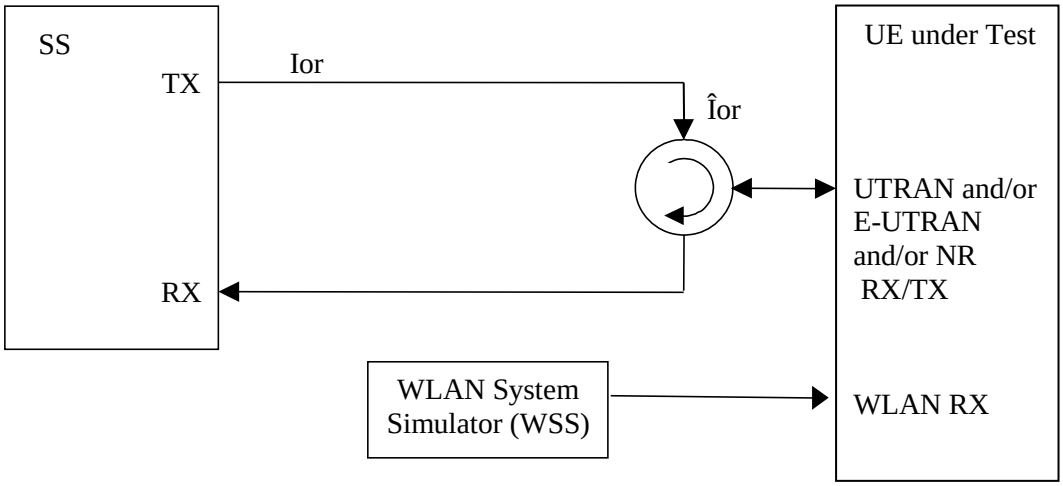



**Figure A.8: Connection for WLAN tests**
**for UE with separate UTRAN and/or E-UTRAN and/or NR and WLAN antennas**

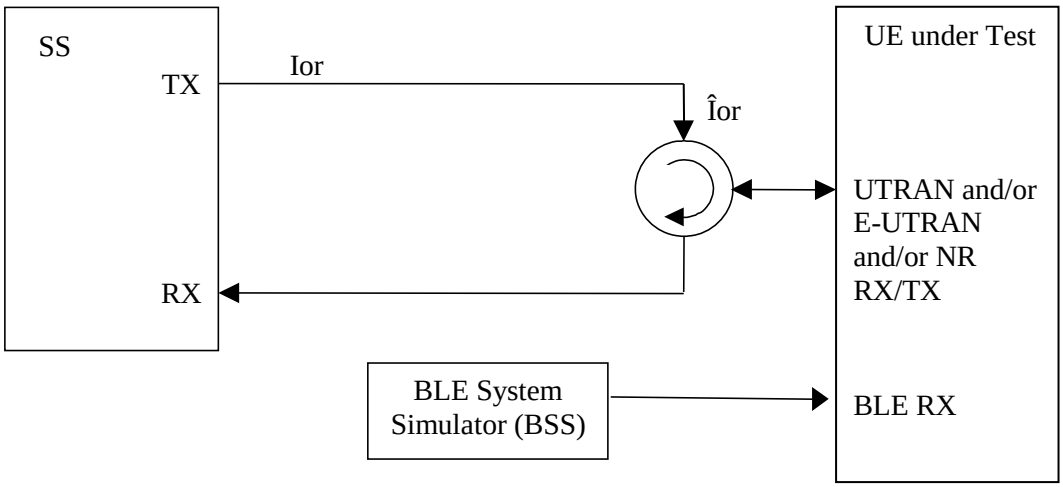



**Figure A.9: Connection for BLE tests**
**for UE with separate UTRAN and/or E-UTRAN and/or NR and BLE antennas**

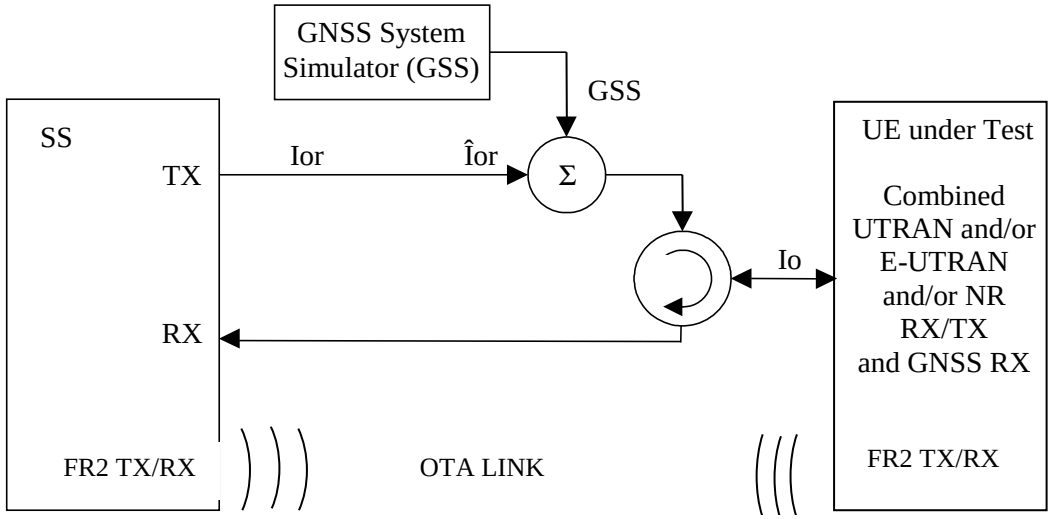



**Figure A.10: Connection for A-GNSS Minimum Performance requirements tests for UE with combined and/or E-UTRAN and/or NR FR1 and GNSS antenna and separate NR FR2 OTA connection**

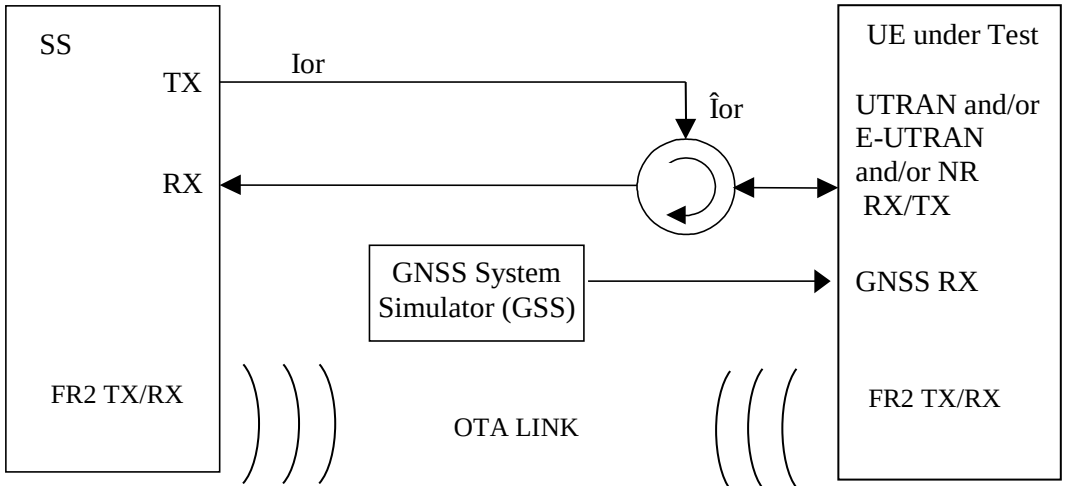



**Figure A.11: Connection for A-GNSS Minimum Performance requirements tests for UE with separate and/or E-UTRAN and/or NR FR1 and GNSS antennas and separate NR FR2 OTA connection**

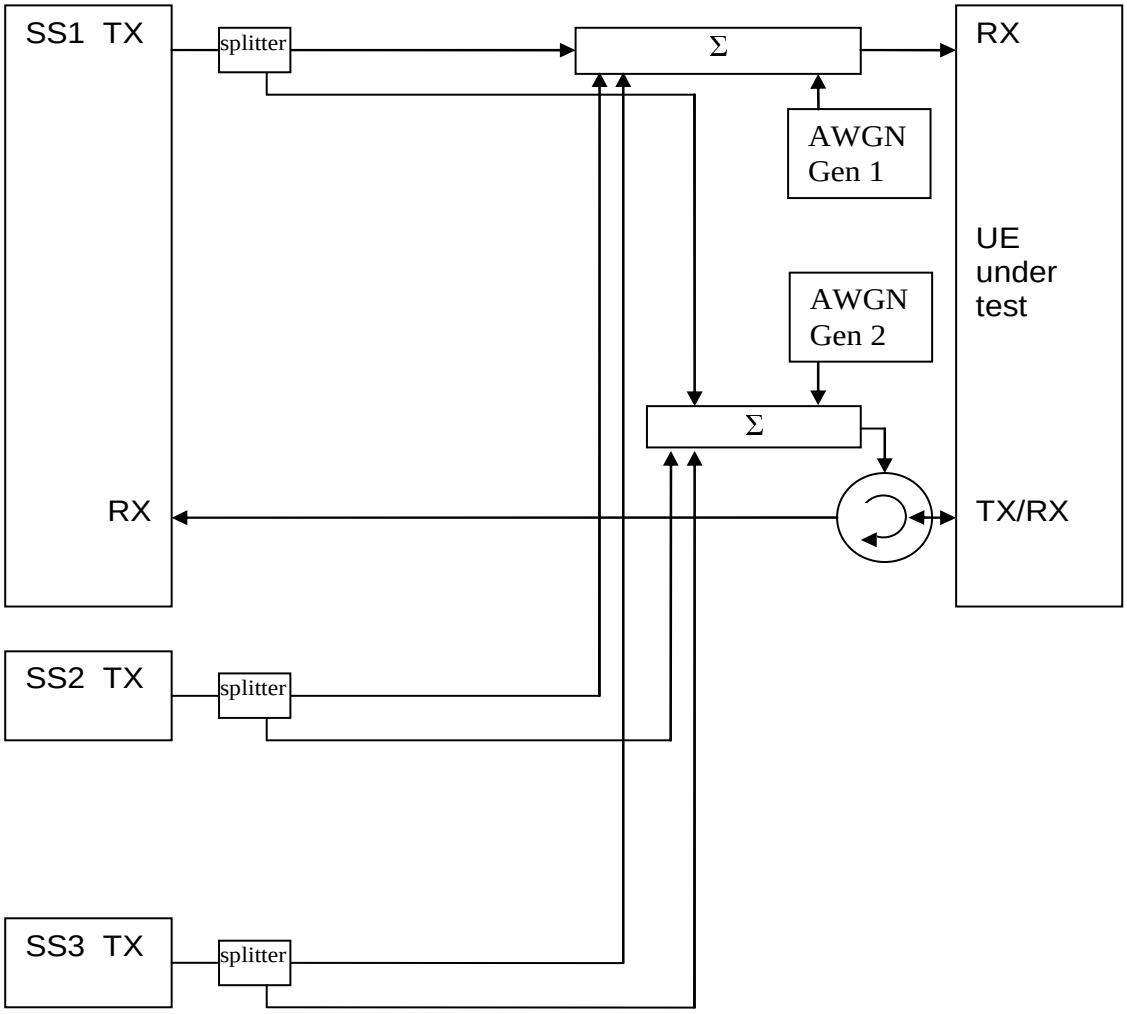



**Figure A.12: Connection for 3 cells DL-TDOA tests with static propagation conditions in NR FR1**

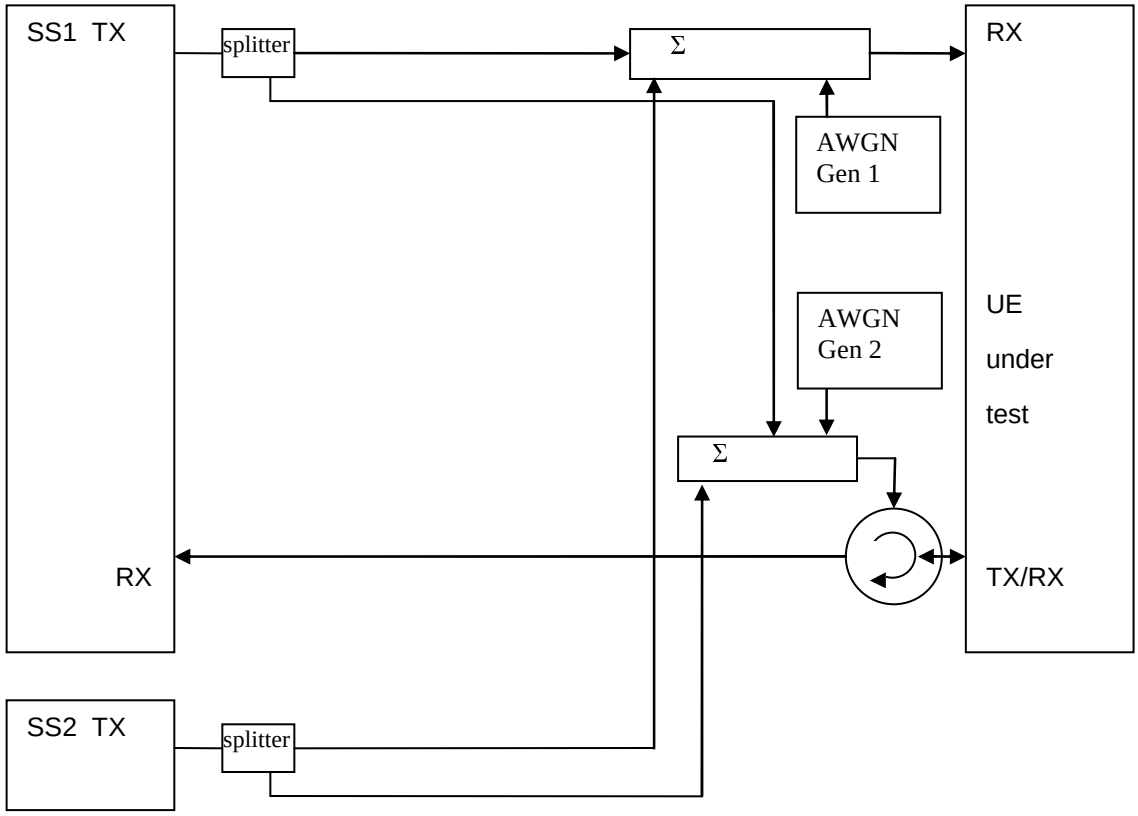



**Figure A.13: Connection for 2 cells DL-TDOA and/or Multi-RTT and/or DL-AoD tests with static propagation conditions in NR FR1**

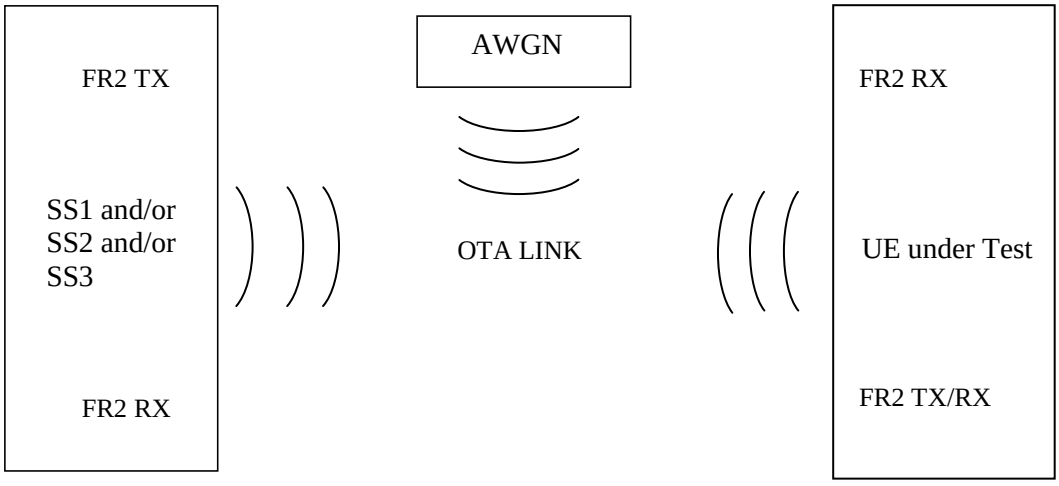



**Figure A.14: Connection for DL-TDOA and/or Multi-RTT and/or DL-AoD tests with static propagation conditions in NR FR2**

# Annex B (normative): Converting A-GNSS UE-assisted measurement reports into position estimates

## B.1 Introduction

In this clause the terms GNSS and A-GNSS also include the cases where the only satellite system used is GPS unless otherwise stated.

To convert the A-GNSS UE measurement reports in case of UE-assisted mode of A-GNSS into position errors, a transformation between the "measurement domain" (code-phases, etc.) into the "state" domain (position estimate) is necessary. Such a transformation procedure is outlined in the following clauses. The details can be found in [8-10] and [12-17].

## B.2 UTRAN UE measurement reports for A-GPS L1 C/A only

In case of UTRAN UE-assisted A-GPS L1 C/A only, the measurement parameters are contained in the RRC UE POSITIONING GPS MEASURED RESULTS IE (clause 10.3.7.93 in 3GPP TS 25.331 [30]). The measurement parameters required for calculating the UE position are:

1) Reference Time: The UE has two choices for the Reference Time:
   a) "UE GPS timing of cell frames";
   b) "GPS TOW msec".
2) Measurement Parameters: 1 to <maxSat>:
   a) "Satellite ID (SV PRN)";
   b) "Whole GPS chips";
   c) "Fractional GPS Chips";
   d) "Pseudorange RMS Error".

Additional information required at the system simulator:

1) "UE positioning GPS reference UE position" (clause 10.3.8.4c in 3GPP TS 25.331 [30]):
   Used for initial approximate receiver coordinates.
2) "UE positioning GPS navigation model" (clause 10.3.7.94 in 3GPP TS 25.331 [30]):
   Contains the GPS ephemeris and clock correction parameters as specified in [8]; used for calculating the satellite positions and clock corrections.
3) "UE positioning GPS ionospheric model" (clause 10.3.7.92 in 3GPP TS 25.331 [30]):
   Contains the ionospheric parameters which allow the single frequency user to utilize the ionospheric model as specified in [8] for computation of the ionospheric delay.

## B.3 UTRAN UE measurement reports for A-GNSS

In case of UTRAN UE-assisted A-GNSS, the measurement parameters are contained in the RRC UE POSITIONING GANSS MEASURED RESULTS IE (clause 10.3.7.93a in 3GPP TS 25.331 [30]). In case the UE provides also measurements on the GPS L1 C/A signal, the measurement parameters are contained in the RRC UE POSITIONING GPS MEASURED RESULTS IE (clause 10.3.7.93 in 3GPP TS 25.331 [30]). The measurement parameters required for calculating the UE position are:

1) Reference Time: The UE has two choices for the Reference Time:

a) "UE GANSS Timing of Cell Frames" and/or "UE GPS Timing of Cell Frames";

b) "GANSS TOD msec" and/or "GPS TOW msec" if GPS L1 C/A signal measurements are also provided.

NOTE: It is not expected that a UE will ever report both a GANSS TOD and a GPS TOW. However, if two time stamps are provided and they derive from different user times, be aware that no compensation is made for this difference and this could affect the location accuracy.

2) Measurement Parameters for each GANSS and GANSS Signal: 1 to <maxGANSSSat>:

a) "Satellite ID"; mapping according to table 10.3.7.88b in 3GPP TS 25.331 [30];

b) "GANSS Code Phase";

c) "GANSS Integer Code Phase";

d) "GANSS Integer Code Phase Extension";

e) "Code Phase RMS Error";

3) Additional Measurement Parameters in case of GPS L1 C/A signal measurements are also provided: 1 to <maxSat>:

a) "Satellite ID (SV PRN)";

b) "Whole GPS chips";

c) "Fractional GPS Chips";

d) "Pseudorange RMS Error".

Additional information required at the system simulator:

1) "UE Positioning GANSS Reference UE Position" or "UE Positioning GPS Reference UE Position" (clause 10.3.8.4c in 3GPP TS 25.331 [30]):
Used for initial approximate receiver coordinates.

2) "UE Positioning GANSS Navigation Model" and "UE Positioning GANSS Additional Navigation Models" (clauses 10.3.7.94a and 10.3.7.94b in 3GPP TS 25.331 [30]):
Contains the ephemeris and clock correction parameters as specified in the relevant ICD of each supported GANSS; used for calculating the satellite positions and clock corrections.

3) "UE Positioning GANSS Ionospheric Model" (clause 10.3.7.92a in 3GPP TS 25.331 [30]):
Contains the ionospheric parameters which allow the single frequency user to utilize the ionospheric model as specified in [15] for computation of the ionospheric delay.

4) "UE Positioning GANSS Additional Ionospheric Model" (clause 10.3.7.92b in 3GPP TS 25.331 [30]):
Contains the ionospheric parameters which allow the single frequency user to utilize the ionospheric model as specified in the relevant ICD of each supported GANSS [14], [37] for computation of the ionospheric delay.

5) "UE Positioning GANSS Time Model" (clause 10.3.7.97a in 3GPP TS 25.331 [30]):
Contains the GNSS-GNSS Time Offset for each supported GANSS. Note, that "UE Positioning GANSS Time Model" IE contains only the sub-ms part of the offset. Any potential integer seconds offset may be obtained from "UE Positioning GPS UTC Model" (clause 10.3.7.97 in 3GPP TS 25.331 [30]), "UE Positioning GANSS UTC Model" (clause 10.3.7.97c in 3GPP TS 25.331 [30]), or "UE Positioning GANSS Additional UTC Models" (clause 10.3.7.97d in 3GPP TS 25.331 [30]).

6) "UE Positioning GPS Navigation Model" (clause 10.3.7.94 in 3GPP TS 25.331 [30]):
Contains the GPS ephemeris and clock correction parameters as specified in [8]; used for calculating the GPS satellite positions and clock corrections in case of GPS L1 C/A signal measurements are the only GPS measurements provided in addition to GANSS measurements.

7) "UE Positioning GPS Ionospheric Model" (clause 10.3.7.92 in 3GPP TS 25.331 [30]):
Contains the ionospheric parameters which allow the single frequency user to utilize the ionospheric model as specified in [8] for computation of the ionospheric delay.

# B.4 E-UTRAN and NR UE measurement reports

In case of E-UTRAN and NR UE-assisted A-GNSS, the measurement parameters are contained in the LPP GNSS-SignalMeasurementInformation IE (clause 6.5.2.6 in 3GPP TS 36.355 [4] and 3GPP TS 37.355 [49]). The measurement parameters required for calculating the UE position are:

1) Reference Time: The UE has two choices for the Reference Time:
   a) "networkTime";
   b) "gnss-TOD-msec".
2) Measurement Parameters for each GNSS and GNSS signal: 1 to 64:
   a) " svID";
   b) "codePhase";
   c) "integerCodePhase";
   d) "codePhaseRMSError".

Additional information required at the system simulator:

1) "GNSS-ReferenceLocation" (clause 6.5.2.2 in 3GPP TS 36.355 [4] and 3GPP TS 37.355 [49]):
   Used for initial approximate receiver coordinates.
2) "GNSS-NavigationModel" (clause 6.5.2.2 in 3GPP TS 36.355 [4] and 3GPP TS 37.355 [49]):
   Contains the GNSS ephemeris and clock correction parameters as specified in the relevant ICD of each supported GNSS; used for calculating the satellite positions and clock corrections.
3) "GNSS-IonosphericModel" (clause 6.5.2.2 in 3GPP TS 36.355 [4] and 3GPP TS 37.355 [49]):
   Contains the ionospheric parameters which allow the single frequency user to utilize the ionospheric model as specified in the relevant ICD of each supported GNSS [8], [14], [15] and [37] for computation of the ionospheric delay.

# B.5 WLS position solution

The WLS position solution problem is concerned with the task of solving for four unknowns; $x_u$, $y_u$, $z_u$ the receiver coordinates in a suitable frame of reference (usually ECEF) and $b_u$ the receiver clock bias. It typically requires the following steps:

Step 1: Formation of pseudo-ranges

The observation of code phase reported by the UE for each satellite $SV_i$ is related to the pseudo-range/c modulo the "GNSS Code Phase Ambiguity" (UTRAN), or "gnss-CodePhaseAmbiguity" (E-UTRAN and NR), or modulo 1 ms (the length of the C/A code period) in case of GPS L1 C/A signal measurements. For the formation of pseudo-ranges, the integer number of milliseconds to be added to each code-phase measurement has to be determined first. Since 1 ms corresponds to a travelled distance of 300 km, the number of integer ms can be found with the help of reference location and satellite ephemeris. The distance between the reference location and each satellite $SV_i$ is calculated and the integer number of milliseconds to be added to the UE code phase measurements is obtained.

Step 2: Correction of pseudo-ranges for the GNSS-GNSS time offsets

In the case that the UE reports measurements for more than a single GNSS, the pseudo-ranges are corrected for the time offsets between the GNSSs relative to the selected reference time using the GNSS-GNSS time offsets available at the system simulator:

$$\rho_{GNSS_m,i} \equiv \rho_{GNSS_m,i} - c \cdot (t_{GNSS_k} - t_{GNSS_m}),$$

where $\rho_{GNSS_m,i}$ is the measured pseudo-range of satellite *i* of GNSS$_m$. The system time $t_{GNSS_k}$ of GNSS$_k$ is the reference time frame, and is the available GNSS-GNSS time offset, and *c* is the speed of light.

Step 3: Formation of weighting matrix

The UE reported "codePhaseRMSError" (E-UTRAN and NR) or "Code Phase RMS Error" and/or "Pseudorange RMS Error" (UTRAN) values are used to calculate the weighting matrix for the WLS algorithm [9]. According to 3GPP TS 25.331 [30] and 3GPP TS 36.355 [4] and 3GPP TS 37.355 [49], the encoding for this field is a 6 bit value that consists of a 3 bit mantissa, $X_i$ and a 3 bit exponent, $Y_i$ for each $SV_i$:

$$w_i = RMSError = 0.5 \times \left(1 + \frac{X_i}{8}\right) \times 2^{Y_i}$$

The weighting Matrix **W** is defined as a diagonal matrix containing the estimated variances calculated from the "codePhaseRMSError" (E-UTRAN and NR) or "Code Phase RMS Error" and/or "Pseudorange RMS Error" (UTRAN) values:

Step 4: WLS position solution

The WLS position solution is described in reference [9] and usually requires the following steps:

1) Computation of satellite locations at time of transmission using the ephemeris parameters and user algorithms defined in the relevant ICD of the particular GNSS. The satellite locations are transformed into WGS-84 reference frame, if needed.

2) Computation of clock correction parameters using the parameters and algorithms as defined in the relevant ICD of the particular GNSS.

3) Computation of atmospheric delay corrections using the parameters and algorithms defined in the relevant ICD of the particular GNSS for the ionospheric delay, and using the Gupta model in reference [10] p.121 equation (2) for the tropospheric delay. For GNSSs which do not natively provide ionospheric correction models (e.g., GLONASS), the ionospheric delay is determined using the available ionospheric model adapted to the particular GNSS frequency.

4) The WLS position solution starts with an initial estimate of the user state (position and clock offset). The Reference Location is used as initial position estimate. The following steps are required:

   a) Calculate geometric range (corrected for Earth rotation) between initial location estimate and each satellite included in the UE measurement report.

   b) Predict pseudo-ranges for each measurement including clock and atmospheric biases as calculated in 1) to 3) above and defined in the relevant ICD of the particular GNSS and [9].

   c) Calculate difference between predicted and measured pseudo-ranges $\Delta\boldsymbol{\rho}$

   d) Calculate the "Geometry Matrix" **G** as defined in [9]:

$$\mathbf{G} \equiv \begin{bmatrix} -\hat{\mathbf{1}}^T_{GNSS_1,1} & 1 \\ -\hat{\mathbf{1}}^T_{GNSS_1,2} & 1 \\ \vdots & \vdots \\ -\hat{\mathbf{1}}^T_{GNSS_1,n} & 1 \\ \vdots & \vdots \\ -\hat{\mathbf{1}}^T_{GNSS_m,1} & 1 \\ -\hat{\mathbf{1}}^T_{GNSS_m,2} & 1 \\ \vdots & \vdots \\ -\hat{\mathbf{1}}^T_{GNSS_m,l} & 1 \end{bmatrix} \text{ with } \hat{\mathbf{1}}_{GNSS_m,i} \equiv \frac{\mathbf{r}_{s_{GNSS_m,i}} - \hat{\mathbf{r}}_u}{\left|\mathbf{r}_{s_{GNSS_m,i}} - \hat{\mathbf{r}}_u\right|}$$ where $\mathbf{r}_{s_{GNSSm,i}}$ is the satellite position vector for $SV_i$ of $GNSS_m$ (calculated in 1) above), and $\hat{\mathbf{r}}_u$ is the estimate of the user location.

e) Calculate the WLS solution according to [9]:

$$\Delta\hat{\mathbf{x}} = \left(\mathbf{G}^T\mathbf{W}\mathbf{G}\right)^{-1}\mathbf{G}^T\mathbf{W}\Delta\boldsymbol{\rho}$$

f) Adding the $\Delta\hat{\mathbf{x}}$ to the initial state estimate gives an improved estimate of the state vector:

$$\hat{\mathbf{x}} \rightarrow \hat{\mathbf{x}} + \Delta\hat{\mathbf{x}}\,.$$

5) This new state vector $\hat{\mathbf{x}}$ can be used as new initial estimate and the procedure is repeated until the change in $\hat{\mathbf{x}}$ is sufficiently small.

Step 5: Transformation from Cartesian coordinate system to Geodetic coordinate system

The state vector $\hat{\mathbf{x}}$ calculated in Step 4 contains the UE position in ECEF Cartesian coordinates together with the UE receiver clock bias relative to the selected GNSS system time. Only the user position is of further interest. It is usually desirable to convert from ECEF coordinates $x_u$, $y_u$, $z_u$ to geodetic latitude φ, longitude λ and altitude $h$ on the WGS84 reference ellipsoid.

Step 6: Calculation of "2-D Position Errors"

The latitude φ / longitude λ obtained after Step 5 is used to calculate the 2-D position error.

# Annex C (normative): General test conditions and declarations

The requirements of this clause apply to all applicable tests in the present document.

In all the relevant clauses in this clause all 2D position error measurements shall be carried out according to the general rules for statistical testing in Annex D.

In this clause, the terms GNSS and A-GNSS also include the cases where the only satellite system used is GPS unless otherwise stated.

The test tolerances may not be valid for operating bands above 4200 MHz since some test system uncertainties are changed for frequencies above 4200 MHz. The test tolerances for bands above 4200 MHz are For Further Study [FFS].

## C.1 Acceptable uncertainty of Test System

The maximum acceptable uncertainty of the Test System is specified below for each test, where appropriate. The Test System shall enable the stimulus signals in the test case to be adjusted to within the specified range, and the equipment under test to be measured with an uncertainty not exceeding the specified values. All ranges and uncertainties are absolute values, and are valid for a confidence level of 95 %, unless otherwise stated.

A confidence level of 95 % is the measurement uncertainty tolerance interval for a specific measurement that contains 95 % of the performance of a population of test equipment.

It should be noted that the uncertainties in clause C.1 apply to the Test System operating into a nominal 50 ohm load and do not include system effects due to mismatch between the DUT and the Test System.

### C.1.1 Measurement of test environments

The measurement accuracy of the UE environmental test conditions, defined in Annex G or TS 36.508 [18] clause 4.1, shall be:

| | |
|---|---|
| Pressure | ±5 kPa |
| Temperature | ±2 degrees |
| Relative Humidity | ±5 % |
| DC Voltage | ±1.0 % |
| AC Voltage | ±1.5 % |
| Vibration | 10 % |
| Vibration frequency | 0.1 Hz |

The above values shall apply unless the test environment is otherwise controlled and the specification for the control of the test environment specifies the uncertainty for the parameter.

### C.1.2 A-GNSS Minimum Performance requirements

**Table C.1.1: Maximum Test System Uncertainty for A-GNSS Minimum Performance tests**

| Clause | Maximum Test System Uncertainty | | Derivation of Test System Uncertainty |
|---|---|---|---|
| 5.2.1, 6.2.1, 7.1.1, 13.2.1 Sensitivity Coarse Time Assistance | Coarse Time Assistance | ±200 ms | |
| | Absolute GNSS signal level | ±1 dB | |
| | Position error | ±0.05 m | Position error consists of ±0.05 m system uncertainty. The effect of position reporting resolution of approximately ±1.2 m (see note) is not included in the allowable test system uncertainty but is included in the Test Parameter Relaxations since this resolution limitation limits the reporting capability of the UE. For simplicity the combined Test Parameter Relaxation is given as ±1.3 m |

| Clause | Maximum Test System Uncertainty | | Derivation of Test System Uncertainty |
|---|---|---|---|
| | Response time | ± 300 ms | |
| 5.2.2, 6.2.2, 7.1.2, 13.2.2 Sensitivity Fine Time Assistance | Coarse Time Assistance | ±200 ms | |
| | Fine Time Assistance | ±1 us | |
| | Absolute GNSS signal level | ±1 dB | |
| | Position error | ±0.05 m | Position error as above |
| | Response time | ± 300 ms | |
| 5.3, 6.3, 7.2, 13.3 Nominal Accuracy | Coarse Time Assistance | ±200 ms | |
| | Absolute GNSS signal level | ±1 dB | |
| | Position error | ±0.05 m | Position error as above |
| | Response time | ± 300 ms | |
| 5.4, 6.4, 7.3, 13.4 Dynamic Range | Coarse Time Assistance | ±200 ms | |
| | Absolute GNSS signal level | ±1 dB | |
| | Relative GNSS signal level | ±0.2 dB | |
| | Position error | ±0.05 m | Position error as above |
| | Response time | ± 300 ms | |
| 5.5, 6.5, 7.4, 13.5 Multi-path scenario | Coarse Time Assistance | ±200 ms | |
| | Absolute GNSS signal level | ±1 dB | |
| | Relative GNSS signal level | ±0.2 dB | |
| | Position error | ±0.05 m | Position error as above |
| | Response time | ± 300 ms | |
| 5.6, 6.6, 7.5, 7.5A, 13.6, 13.7 Moving scenario and periodic update | Absolute GNSS signal level | ±1 dB | |
| | Position error | ±0.05 m | Position error as above |
| | Differential response time<br><br>± 100 ms | ± 100 ms | |

NOTE: For UE based mode the effect of position reporting resolution is given by:

$\sqrt{\left(\frac{90\times 2\times\pi\times R}{2E23\times 360}\right)^2+\left(\frac{360\times 2\times\pi\times R\times\cos\phi}{2E24\times 360}\right)^2}$ meters, where R is the radius of the earth and φ is the latitude of the location. For the GNSS scenarios defined in TS 37.571-5 [20] this equates to approximately.

Editor's note: this needs checking once the GNSS scenarios are agreed [TBD] m. For simplicity this is given as ±1.2 m.

For UE assisted mode it is assumed that the output from the WLS position solution calculation in Annex B is coded using the same position coding method as for UE based mode before being used to calculate position error. Therefore, the effect of reporting resolution will be the same as for UE based mode.

## C.1.3 ECID and OTDOA Measurement requirements

**Table C.1.3-1: Maximum Test System Uncertainty for ECID and OTDOA Measurement Requirements**

| Clause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|
| 8.1.1 E-UTRAN FDD UE Rx – Tx time difference case (Rel-9 to Rel-11) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB<br><br>±3Ts Uplink signal transmit timing relative to downlink | Note:<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [26] |

| | | |
|---|---|---|
| 8.1.1A E-UTRAN FDD UE Rx – Tx time difference case (Rel-12 onwards) | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.1B E-UTRAN FDD UE Rx – Tx time difference case for UE Category 1bis | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.2 E-UTRAN TDD UE Rx – Tx time difference case (Rel-9 to Rel-11) | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.2A E-UTRAN TDD UE Rx – Tx time difference case (Rel-12 onwards) | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.2B E-UTRAN TDD UE Rx – Tx time difference case for UE Category 1bis | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.3 E-UTRAN FDD UE Rx-Tx time difference under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>±3$T_S$ Uplink signal transmit timing relative to downlink | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN<br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [26] |
| 8.1.4 E-UTRAN TDD UE Rx-Tx time difference under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Same as 8.1.3 | Same as 8.1.3 |
| 8.1.5 E-UTRAN FDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>±3Ts Uplink signal transmit timing relative to downlink | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_3 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [26] |
| 8.1.6 E-UTRAN TDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 8.1.5 | Same as 8.1.5 |
| 8.1.7 E-UTRAN FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.8 E-UTRAN HD-FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.9 E-UTRAN TDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.1 | Same as 8.1.1 |

| | | |
|---|---|---|
| 9.1.1 FDD RSTD Measurement Reporting Delay | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3$ / $N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>Response Time = ± 300 ms | Note:<br>PRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN<br>PRS $\hat{E}s_3$ / $N_{oc}$ and $\hat{E}s_3$ / $N_{oc}$ are the ratios of cell 3 signal / AWGN<br><br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 9.1.1A FDD RSTD Measurement Reporting Delay for UE Category 1bis | Same as 9.1.1 | |
| 9.1.2 TDD RSTD Measurement Reporting Delay | Same as 9.1.1 | |
| 9.1.2A TDD RSTD Measurement Reporting Delay for UE Category 1bis | Same as 9.1.1 | |
| 9.1.3 FDD RSTD Measurement Accuracy | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = ± 1 Ts | Note:<br>PRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN |
| 9.1.3A FDD RSTD Measurement Accuracy for UE Category 1bis | Same as 9.1.3 | |
| 9.1.4 TDD RSTD Measurement Accuracy | Same as 9.1.3 | |
| 9.1.4A TDD RSTD Measurement Accuracy for UE Category 1bis | Same as 9.1.3 | |

| | | |
|---|---|---|
| 9.2.1 FDD-FDD inter-frequency RSTD measurement reporting delay | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>Response Time = ± 300 ms | Note:<br>PRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of cell 1 signal / AWGN for frequency 1<br>PRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of cell 2 signal / AWGN for frequency 2<br>PRS $\hat{E}s_3$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ are the ratios of cell 3 signal / AWGN for frequency 2<br><br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 9.2.1A FDD-FDD inter-frequency RSTD measurement reporting delay for UE Category 1bis | Same as 9.2.1 | |
| 9.2.2 TDD-TDD inter-frequency RSTD measurement reporting delay | Same as 9.2.1 | |
| 9.2.2A TDD-TDD inter-frequency RSTD measurement reporting delay for UE Category 1bis | Same as 9.2.1 | |
| 9.2.4 FDD-FDD inter frequency RSTD Accuracy | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = ± 2 Ts | Note:<br>PRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of cell 1 signal / AWGN for frequency 1<br>PRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of cell 2 signal / AWGN for frequency 2 |
| 9.2.4A FDD-FDD inter frequency RSTD Accuracy for UE Category 1bis | Same as 9.2.4 | |
| 9.2.5 TDD-TDD inter frequency RSTD Accuracy | Same as 9.2.4 | |
| 9.2.5A TDD-TDD inter frequency RSTD Accuracy for UE Category 1bis | Same as 9.2.4 | |
| 9.3.1.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.1.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.2.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.1.1 | Same as 9.1.1 |

| | | |
|---|---|---|
| 9.3.2.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.3.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.3.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.4.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.4.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.5.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.5.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.6.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.6.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.7.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.7.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.8.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.8.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.9.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.9.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.10.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.10.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.11.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.11.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.12.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.12.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.1.3 | Same as 9.1.3 |

| | | |
|---|---|---|
| 9.3.13 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.14 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.15 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.16 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.17 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.18 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.4.1.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.1.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.2.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.2.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.3.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.3.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.4.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.4.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.5.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.5.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.6.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.6.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.7.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.2.4 | Same as 9.2.4 |

| | | |
|---|---|---|
| 9.4.7.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.8.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.8.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.9.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.9.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.10.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.10.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.11.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.11.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.12.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.12.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.13 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.14 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.15 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.16 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.17 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.18 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.5.1 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = ± 1 Ts | Note:<br>NPRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of nCell 1 signal / AWGN<br>NPRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of nCell 2 signal / AWGN |

| | | |
|---|---|---|
| 9.5.2 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.5.1 | Same as 9.5.1 |
| 9.5.3 HD-FDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_3$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Response Time = ± 300 ms | Note:<br>NPRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of nCell 1 signal / AWGN<br>NPRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of nCell 2 signal / AWGN |
| 9.6.1 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = ± 2 Ts | Note:<br>NPRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of nCell 1 signal / AWGN for frequency 1<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of nCell 2 signal / AWGN for frequency 2 |
| 9.6.2 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.6.1 | Same as 9.6.1 |
| 9.6.3 HD-FDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Same as 9.5.3 | Same as 9.5.3 |
| 9.7.1 TDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = ± 1 Ts | Note:<br>NPRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of nCell 1 signal / AWGN<br>NPRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of nCell 2 signal / AWGN |
| 9.7.2 TDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.7.1 | Same as 9.7.1 |
| 9.7.3 TDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_3$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Response Time = ± 300 ms | Note:<br>NPRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of nCell 1 signal / AWGN<br>NPRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of nCell 2 signal / AWGN |

| | | |
|---|---|---|
| 9.8.1 TDD Inter-frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = ± 2 Ts | Note:<br>NPRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of nCell 1 signal / AWGN for frequency 1<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of nCell 2 signal / AWGN for frequency 2 |
| 9.8.2 TDD Inter-frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.8.1 | Same as 9.8.1 |
| 9.8.3 TDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Same as 9.7.3 | Same as 9.7.3 |
| 10.1 FDD RSTD Measurement Reporting Delay for Carrier Aggregation | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>Response Time = ± 300 ms | Note:<br>PRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of cell 1 signal / AWGN for frequency 1<br>PRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of cell 2 signal / AWGN for frequency 2<br>PRS $\hat{E}s_3$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ are the ratios of cell 3 signal / AWGN for frequency 2<br><br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 10.1A FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz | Same as 10.1 | Same as 10.1 |
| 10.1B FDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz +5 MHz Bandwidth | Same as 10.1 | Same as 10.1 |
| 10.1C FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.1 | Same as 10.1 |
| 10.2 TDD RSTD Measurement Reporting Delay for Carrier Aggregation | Same as 10.1 | Same as 10.1 |
| 10.2A TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz | Same as 10.1 | Same as 10.1 |
| 10.2B TDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz +5 MHz Bandwidth | Same as 10.1 | Same as 10.1 |

| | | |
|---|---|---|
| 10.2C TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.1 | Same as 10.1 |
| 10.2D TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz +10 MHz Bandwidth | Same as 10.1 | Same as 10.1 |
| 10.3 FDD RSTD Measurement Accuracy for Carrier Aggregation | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = ± 1 Ts | Note:<br>PRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of cell 1 signal / AWGN for frequency 1<br>PRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of cell 2 signal / AWGN for frequency 2<br>PRS $\hat{E}s_3$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ are the ratios of cell 3 signal / AWGN for frequency 2 |
| 10.3A FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-10 and Rel-11) | Same as 10.3 | |
| 10.3A_1 FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-12 onwards) | Same as 10.3 | Same as 10.3 |
| 10.3B FDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz+5 MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.3C FDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.4 TDD RSTD Measurement Accuracy for Carrier Aggregation | Same as 10.3 | |
| 10.4A TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-10 and Rel-11) | Same as 10.3 | |
| 10.4A_1 TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-12 onwards) | Same as 10.3 | Same as 10.3 |
| 10.4B TDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz+5 MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.4C TDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.4D TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz+10 MHz Bandwidth | Same as 10.3 | Same as 10.3 |

| | | |
|---|---|---|
| 10.5 FDD 3 DL CA RSTD Measurement Reporting Delay | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3$ / $N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_4$ / $N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_4$ / $N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br>Response Time = ± 300 ms | Note:<br>PRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of cell 1 signal / AWGN for frequency 1<br>PRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of cell 2 signal / AWGN for frequency 2<br>PRS $\hat{E}s_3$ / $N_{oc3}$ and $\hat{E}s_3$ / $N_{oc3}$ are the ratios of cell 3 signal / AWGN for frequency 3<br>PRS $\hat{E}s_4$ / $N_{oc3}$ and $\hat{E}s_4$ / $N_{oc3}$ are the ratios of cell 4 signal / AWGN for frequency 3<br><br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>PRS $\hat{E}s$ / $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 10.6 TDD 3 DL CA RSTD Measurement Reporting Delay | Same as 10.5 | Same as 10.5 |
| 10.7 FDD RSTD Measurement Accuracy for 3DL Carrier Aggregation | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3$ / $N_{oc3}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc3}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_4$ / $N_{oc3}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_4$ / $N_{oc3}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference (Intra-freq) = ± 1 Ts<br>Cell Timing Difference (Inter-freq) = ± 2 Ts | Note:<br>PRS $\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ are the ratios of cell 1 signal / AWGN for frequency 1<br>PRS $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$ are the ratios of cell 2 signal / AWGN for frequency 2<br>PRS $\hat{E}s_3$ / $N_{oc3}$ and $\hat{E}s_3$ / $N_{oc3}$ are the ratios of cell 3 signal / AWGN for frequency 3<br>PRS $\hat{E}s_4$ / $N_{oc3}$ and $\hat{E}s_4$ / $N_{oc3}$ are the ratios of cell 4 signal / AWGN for frequency 3 |
| 10.8 TDD RSTD Measurement Accuracy for 3DL Carrier Aggregation | Same as 10.7 | Same as 10.7 |
| In addition, the following Test System uncertainties and related constraints apply.<br>Any additional constraints are defined in the specific tests. | | |
| AWGN Bandwidth | | ≥ 1.08 MHz, 2.7 MHz, 4.5 MHz, 9 MHz, 13.5 MHz, 18 MHz;<br>$N_{RB}$ x 180kHz according to $BW_{Config}$ |
| AWGN absolute power uncertainty | | Test-specific |
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to average over $BW_{Config}$ | | ±2 dB |
| AWGN peak to average ratio | | ≥10 dB @0.001% |

| | |
|---|---|
| Signal-to noise ratio uncertainty | Test-specific |
| Fading profile power uncertainty | ±0.5 dB |
| Fading profile delay uncertainty, relative to frame timing | ±5 ns (excludes absolute errors related to baseband timing) |

## C.1.4 MBS Minimum Performance requirements

**Table C.1.4-1: Maximum Test System Uncertainty for MBS Minimum Performance tests**

| Clause | Maximum Test System Uncertainty | | Derivation of Test System Uncertainty |
|---|---|---|---|
| 11.1, 11.1A MBS Measurement Reporting Delay | Beacon power level | ±2 dB | |
| | Response time | ±300 ms | |
| 11.2, 11.2A MBS Sensitivity Measurement Accuracy | Beacon power level | ±2 dB | |
| | Code phase delay difference | ±5 ns | Code phase delay difference error value of +/- 5ns, being derived from 10% of the most stringent code phase delay measurement accuracy requirement |
| 11.3, 11.3A MBS Nominal Measurement Accuracy | Beacon power level | ±2 dB | |
| | Code phase delay | ±5 ns | Code phase delay error as above |
| 11.4, 11.4A MBS Dynamic Range Measurement Accuracy | Beacon power level | ±2 dB | |
| | Code phase delay | ±5 ns | Code phase delay error as above |
| 11.5, 11.5A MBS Measurement Accuracy in Multipath | Beacon power level | ±2 dB | |
| | Code phase delay | ±5 ns | Code phase delay error as above |
| Note: Code phase delay is equal to the propagation delay from the (simulated) beacon transmitter to the UE receive antenna based on the propagation distance in the test case. | | | |

## C.1.5 WLAN and BLE measurement requirements

**Table C.1.5-1: Maximum Test System Uncertainty for WLAN and BLE measurement tests**

| Clause | Maximum Test System Uncertainty | | Derivation of Test System Uncertainty |
|---|---|---|---|
| 12.1.1 WLAN AP Identification and reporting delay under nominal conditions | Response time | ±300 ms | |
| 12.1.2 WLAN AP Identification and reporting delay under dynamic range conditions | Response time | ±300 ms | |
| | AP Power Level Difference | ±1 dB | |
| 12.2.1 Bluetooth identification | Response time | ±300 ms | |

Note: AP Power Level Difference is the difference between the WLAN Received Power Level from the high power WLAN AP with respect to the low power WLAN APs.

## C.1.6 NR PRS-based measurement requirements

**Table C.1.6-1: Maximum Test System Uncertainty for NR PRS-based Measurement Requirements**

| Clause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|

| | | |
|---|---|---|
| 14.2.1 NR RSTD measurement period test case for single positioning frequency layer in FR1 SA | $N_{oc}$ ±1.5 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Response Time = 300ms | Note:<br>PRS $\hat{E}s_1 / N_{oc}$ and $\hat{E}s_1 / N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ are the ratios of cell 2 signal / AWGN<br>PRS $\hat{E}s_3 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ are the ratios of cell 3 signal / AWGN<br><br>Signal-to-noise ratio uncertainty ±0.3 dB |
| 14.2.2 NR RSTD measurement period test case for dual positioning frequency layers in FR1 SA | Same as 14.2.1 | |
| 14.2.3 NR RSTD measurement period test case for single positioning frequency layer in FR2 SA | $N_{oc}$ ±5.65 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_3 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Response Time = 300ms | Note:<br>PRS $\hat{E}s_1 / N_{oc}$ and $\hat{E}s_1 / N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ are the ratios of cell 2 signal / AWGN<br>PRS $\hat{E}s_3 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ are the ratios of cell 3 signal / AWGN<br><br>Signal-to-noise ratio uncertainty ±0.3 dB |
| 14.2.4 NR RSTD measurement period test case for dual positioning frequency layers in FR2 SA | Same as 14.2.3 | |
| 14.2.5 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 14.2.1 | |
| 14.2.6 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Same as 14.2.1 | |
| 14.2.7 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR1 SA | Same as 14.2.1 | |
| 14.2.8 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 14.2.3 | |
| 14.2.9 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Same as 14.2.3 | |
| 14.2.10 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR2 SA | Same as 14.2.3 | |
| 14.2.11 NR RSTD measurement reporting delay test case for PRS aggregation in FR1 | Same as 14.2.1 | Same as 14.2.1 |

| | | |
|---|---|---|
| 14.3.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA | $N_{oc}$ ±1.5 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = 32 Tc | Note:<br>PRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN |
| 14.3.2 NR RSTD measurement accuracy test case for dual positioning frequency layers in FR1 SA | Same as 14.3.1 | |
| 14.3.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA | $N_{oc}$ ±5.65 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = 32 Tc | Note:<br>PRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN |
| 14.3.4 NR RSTD measurement accuracy test case for dual positioning frequency layer in FR2 SA | Same as 14.3.3 | |
| 14.3.5 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 14.3.3 | |
| 14.3.6 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR1 SA | Same as 14.3.3 | |
| 14.3.7 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 14.3.3 | |
| 14.3.8 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR2 SA | Same as 14.3.3 | |
| 14.3.9 NR RSTD measurement accuracy test case for PRS aggregation in FR1 SA | Same as 14.3.1 | Same as 14.3.1 |
| 14.4.1 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR1 SA | Same as 14.3.3 | |
| 14.4.2 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 14.3.3 | |
| 14.4.3 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR2 SA | Same as 14.3.3 | |
| 14.4.4 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 14.3.3 | |

| | | |
|---|---|---|
| 14.5.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA | Same as 14.3.3 | |
| 14.5.2 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 14.3.3 | |
| 14.5.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA | Same as 14.3.3 | |
| 14.5.4 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 14.3.3 | |
| 15.2.1 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR1 SA | Noc ±1.5 dB averaged over BWConfig<br>PRS Ês1 / Noc ±0.3 dB averaged over BWConfig<br>Ês1 / Noc ±0.3 dB averaged over BWConfig<br>PRS Ês2 / Noc ±0.3 dB averaged over BWConfig<br>Ês2 / Noc ±0.3 dB averaged over BWConfig<br><br>Response Time = ± 300 ms | Note:<br>PRS Ês1 / Noc and Ês1 / Noc are the ratios of cell 1 signal / AWGN<br>PRS Ês2 / Noc and Ês2 / Noc are the ratios of cell 2 signal / AWGN<br><br>Tc = 1/(480000 x 4096), the basic timing unit defined in TS 38.211 [53] |
| 15.2.2 UE Rx-Tx time difference measurement period for dual positioning frequency layers in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.2.3 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR2 SA | Noc ±5.65 dB averaged over BWConfig<br>PRS Ês1 / Noc ±0.3 dB averaged over BWConfig<br>Ês1 / Noc ±0.3 dB averaged over BWConfig<br>PRS Ês2 / Noc ±0.3 dB averaged over BWConfig<br>Ês2 / Noc ±0.3 dB averaged over BWConfig<br><br>Response Time = ± 300 ms | PRS Ês1 / Noc and Ês1 / Noc are the ratios of cell 1 signal / AWGN<br>PRS Ês2 / Noc and Ês2 / Noc are the ratios of cell 2 signal / AWGN<br><br>Tc = 1/(480000 x 4096), the basic timing unit defined in TS 38.211 [53] |
| 15.2.4 UE Rx-Tx time difference measurement period for dual positioning frequency layers in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |
| 15.2.5 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.2.6 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.2.7 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with multiple RxTx TEGs in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.2.8 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |

| | | |
|---|---|---|
| 15.2.9 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |
| 15.2.10 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with RxTx TEG in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |
| 15.2.11 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.2.12 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |
| 15.2.13 UE Rx-Tx time difference measurement reporting delay test case in FR1 for Redcap UE without Rx FH | Same as 15.2.1 | Same as 15.2.1 |
| 15.2.14 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE without Rx FH | Same as 15.2.3 | Same as 15.2.3 |
| 15.2.16 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE with Rx FH | Same as 15.2.3 | Same as 15.2.3 |
| 15.3.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA | $N_{oc}$ ±1.5 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = 32 Tc | Note:<br>PRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN |
| 15.3.2 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA | $N_{oc}$ ±5.65 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = 32 Tc | Note:<br>PRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN |
| 15.3.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 15.3.1 | Same as 15.3.1 |
| 15.3.5 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 15.3.2 | Same as 15.3.2 |
| 15.3.6 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR2 SA | Same as 15.3.2 | Same as 15.3.2 |
| 15.3.7 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR1 SA | Same as 15.3.1 | Same as 15.3.1 |

| | | |
|---|---|---|
| 15.3.8 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR2 SA | Same as 15.3.2 | Same as 15.3.2 |
| 15.3.9 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH | Same as 15.3.1 | Same as 15.3.1 |
| 15.3.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH | Same as 15.3.2 | Same as 15.3.2 |
| 15.3.12 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH | Same as 15.3.2 | Same as 15.3.2 |
| 15.4.1 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.4.2 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.4.3 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |
| 15.4.4 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |
| 15.4.5 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR1 SA | Same as 15.2.1 | Same as 15.2.1 |
| 15.4.6 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR2 SA | Same as 15.2.3 | Same as 15.2.3 |
| 15.4.7 UE Rx-Tx time difference measurement reporting delay in FR1 for Redcap UE without Rx FH | Same as 15.2.1 | Same as 15.2.1 |
| 15.4.8 UE Rx-Tx time difference measurement reporting delay in FR2 for Redcap UE without Rx FH | Same as 15.2.3 | Same as 15.2.3 |
| 15.5.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA | Same as 15.3.1 | Same as 15.3.1 |
| 15.5.2 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR1 SA | Same as 15.3.1 | Same as 15.3.1 |
| 15.5.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA | $N_{oc}$ ±5.65 dB averaged over $BW_{Config}$<br>PRS $\hat{E}_{s_1} / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}_{s_1} / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}_{s_2} / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}_{s_2} / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = 33 Tc | Same as 15.3.2 |

| | | |
|---|---|---|
| 15.5.4 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR2 SA | $N_{oc}$ ±5.65 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>Cell Timing Difference = 33 Tc | Same as 15.3.2 |
| 15.5.7 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH | Same as 15.3.1 | Same as 15.3.1 |
| 15.5.8 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH | Same as 15.3.2 | Same as 15.3.2 |
| 15.5.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH | Same as 15.3.2 | Same as 15.3.2 |
| 16.2.1 PRS-RSRP measurement period test case for single positioning frequency layer in FR1 SA | Noc ±1.5 dB averaged over BWConfig<br>PRS Ês1 / Noc ±0.3 dB averaged over BWConfig<br>Ês1 / Noc ±0.3 dB averaged over BWConfig<br>PRS Ês2 / Noc ±0.3 dB averaged over BWConfig<br>Ês2 / Noc ±0.3 dB averaged over BWConfig | Note:<br>PRS Ês1 / Noc and Ês1 / Noc are the ratios of cell 1 signal / AWGN<br>PRS Ês2 / Noc and Ês2 / Noc are the ratios of cell 2 signal / AWGN<br><br>Signal-to-noise ratio uncertainty ±0.3 dB |
| 16.2.2 PRS-RSRP measurement period test case for dual positioning frequency layer in FR1 SA | Noc ±1.5 dB averaged over BWConfig<br>PRS Ês1 / Noc ±0.3 dB averaged over BWConfig<br>Ês1 / Noc ±0.3 dB averaged over BWConfig<br>PRS Ês2 / Noc2 ±0.3 dB averaged over BWConfig<br>Ês2 / Noc2 ±0.3 dB averaged over BWConfig | Note:<br>PRS Ês1 / Noc and Ês1 / Noc are the ratios of cell 1 signal / AWGN<br>PRS Ês2 / Noc2 and Ês2 / Noc2 are the ratios of cell 2 signal / AWGN<br><br>Signal-to-noise ratio uncertainty ±0.3 dB |
| 16.2.3 PRS-RSRP measurement period test case for single positioning frequency layer in FR2 SA | Noc ±5.65 dB averaged over BWConfig<br>PRS Ês1 / Noc ±0.3 dB averaged over BWConfig<br>Ês1 / Noc ±0.3 dB averaged over BWConfig<br>PRS Ês2 / Noc ±0.3 dB averaged over BWConfig<br>Ês2 / Noc ±0.3 dB averaged over BWConfig<br><br>Response Time = ± 300 ms | PRS Ês1 / Noc and Ês1 / Noc are the ratios of cell 1 signal / AWGN<br>PRS Ês2 / Noc and Ês2 / Noc are the ratios of cell 2 signal / AWGN<br><br>Tc = 1/(480000 x 4096), the basic timing unit defined in TS 38.211 [53] |
| 16.2.4 PRS-RSRP measurement period test case for dual positioning frequency layer in FR2 SA | Same as 16.2.3 | Same as 16.2.3 |
| 16.2.5 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR1 SA | Same as 16.2.1 | Same as 16.2.1 |
| 16.2.6 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Same as 16.2.1 | Same as 16.2.1 |
| 16.2.7 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR2 SA | Same as 16.2.3 | Same as 16.2.3 |
| 16.2.8 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Same as 16.2.3 | Same as 16.2.3 |

| | | |
|---|---|---|
| 16.3.1 PRS-RSRP measurement accuracy with PRS in FR1 | Noc ±1.5 dB averaged over BWConfig<br>PRS Ês1 / Noc ±0.3 dB averaged over BWConfig<br>Ês1 / Noc ±0.3 dB averaged over BWConfig<br>PRS Ês2 / Noc ±0.3 dB averaged over BWConfig<br>Ês2 / Noc ±0.3 dB averaged over BWConfig | Note:<br>PRS Ês1 / Noc and Ês1 / Noc are the ratios of cell 1 signal / AWGN<br>PRS Ês2 / Noc and Ês2 / Noc are the ratios of cell 2 signal / AWGN<br><br>Signal-to-noise ratio uncertainty ±0.3 dB |
| 16.3.2 PRS-RSRP measurement accuracy with PRS in FR2 | $N_{oc}$ ±5.65 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>PRS $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_1$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}s_2$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN |
| 16.3.3 PRS-RSRP measurement accuracy with PRS in FR1 with reduced sample number | Same as 16.3.1 | Same as 16.3.1 |
| 16.3.4 PRS-RSRP measurement accuracy with PRS in FR2 with reduced sample number | Same as 16.3.2 | Same as 16.3.2 |
| 16.4.1 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Same as 16.2.1 | Same as 16.2.1 |
| 16.4.2 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 16.2.1 | Same as 16.2.1 |
| 16.4.3 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR2 SA | Same as 16.2.3 | Same as 16.2.3 |
| 16.4.4 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 16.2.3 | Same as 16.2.3 |
| 16.5.1 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 SA | Same as 16.3.1 | Same as 16.3.1 |
| 16.5.2 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 with reduced sample number | Same as 16.3.1 | Same as 16.3.1 |
| 16.5.3 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 SA | Same as 16.3.2 | Same as 16.3.2 |
| 16.5.4 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 with reduced sample number | Same as 16.3.2 | Same as 16.3.2 |

| | | |
|---|---|---|
| 17.2.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Noc ±1.5 dB averaged over BWConfig<br>PRS $\hat{E}_{s1}$ / Noc ±0.3 dB averaged over BWConfig<br>$\hat{E}_{s1}$ / Noc ±0.3 dB averaged over BWConfig<br>PRS $\hat{E}_{s2}$ / Noc ±0.3 dB averaged over BWConfig<br>$\hat{E}_{s2}$ / Noc ±0.3 dB averaged over BWConfig | Note:<br>PRS $\hat{E}_{s1}$ / Noc and $\hat{E}_{s1}$ / Noc are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}_{s2}$ / Noc and $\hat{E}_{s2}$ / Noc are the ratios of cell 2 signal / AWGN<br><br>Signal-to-noise ratio uncertainty ±0.3 dB |
| 17.2.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.2.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.2.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA | $N_{oc}$ ±5.65 dB averaged over $BW_{Config}$<br>PRS $\hat{E}_{s1}$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}_{s1}$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>PRS $\hat{E}_{s2}$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}_{s2}$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>PRS $\hat{E}_{s1}$ / $N_{oc}$ and $\hat{E}_{s1}$ / $N_{oc}$ are the ratios of cell 1 signal / AWGN<br>PRS $\hat{E}_{s2}$ / $N_{oc}$ and $\hat{E}_{s2}$ / $N_{oc}$ are the ratios of cell 2 signal / AWGN |
| 17.2.5 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |
| 17.2.6 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |
| 17.3.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.3.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.3.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |
| 17.3.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |
| 17.4.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.4.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.4.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |

| | | |
|---|---|---|
| 17.4.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |
| 17.5.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.5.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 17.2.1 | Same as 17.2.1 |
| 17.5.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |
| 17.5.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 17.2.4 | Same as 17.2.4 |

# C.2 Test Parameter Relaxations (This clause is informative)

The Test Parameter Relaxations defined in this clause have been used to relax the Conformance requirement to derive the Test Requirements.

The Test Parameter Relaxations are derived from Test System uncertainties, regulatory requirements and criticality to system performance. As a result, the Test Parameter Relaxations may sometimes be set to zero.

The Test Parameter Relaxations should not be modified for any reason e.g. to take account of commonly known test system errors (such as mismatch, cable loss, etc.).

## C.2.1 A-GNSS Minimum Performance requirements

**Table C.2.1: Test Parameter Relaxations for A-GNSS Minimum Performance tests**

| Clause | Test Parameter Relaxation | |
|---|---|---|
| 5.2.1, 6.2.1, 7.1.1, 13.2.1 Sensitivity Coarse Time Assistance | Coarse Time Assistance | 200 ms |
| | Absolute GNSS signal level | 1 dB |
| | Position error | 1.3 m |
| | Response time | 300 ms |
| 5.2.2, 6.2.2, 7.1.2, 13.2.2 Sensitivity Fine Time Assistance | Coarse Time Assistance | 200 ms |
| | Fine Time Assistance | 1 us |
| | Absolute GNSS signal level | 1 dB |
| | Position error | 1.3 m |
| | Response time | 300 ms |
| 5.3, 6.3, 7.2, 13.3 Nominal Accuracy | Coarse Time Assistance | 200 ms |
| | Absolute GNSS signal level | 0 dB |
| | Position error | 1.3 m |
| | Response time | 300 ms |
| 5.4, 6.4, 7.3, 13.4 Dynamic Range | Coarse Time Assistance | 200 ms |
| | Absolute GNSS signal level | 1 dB |
| | Relative GNSS signal level | 0.2 dB |
| | Position error | 1.3 m |
| | Response time | 300 ms |
| 5.5, 6.5, 7.4, 13.5 Multi-path scenario | Coarse Time Assistance | 200 ms |
| | Absolute GNSS signal level | 0 dB |
| | Relative GNSS signal level | 0.2 dB |
| | Position error | 1.3 m |
| | Response time | 300 ms |

| | | |
|---|---|---|
| 5.6, 6.6, 7.5, 7.5A, 13.6, 13.7 Moving scenario and periodic update | Absolute GNSS signal level | 0 dB |
| | Position error | 1.3 m |
| | Differential Response Time | 100 ms |

## C.2.2 ECID and OTDOA Measurement requirements

**Table C.2.2: Test Parameter Relaxations for ECID and OTDOA Measurement requirements**

| Clause | Test Parameter Relaxation | |
|---|---|---|
| 8.1.1 E-UTRAN FDD UE Rx – Tx time difference case (Rel-9 to Rel-11) | Parameters<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.00dB<br>RxTx time difference value: | Test Tolerance<br>0dB<br>+0.3dB<br>±3Ts |
| 8.1.1A E-UTRAN FDD UE Rx – Tx time difference case (Rel-12 onwards) | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.1B E-UTRAN FDD UE Rx – Tx time difference case for UE Category 1bis | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.2 E-UTRAN TDD UE Rx – Tx time difference case (Rel-9 to Rel-11) | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.2A E-UTRAN TDD UE Rx – Tx time difference case (Rel-12 onwards) | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.2B E-UTRAN TDD UE Rx – Tx time difference case for UE Category 1bis | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.3 E-UTRAN FDD UE Rx-Tx time difference under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Parameters<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -3.00dB<br>$\hat{E}s_2 / N_{oc}$: +1.00dB | Test Tolerance<br>0dB<br>+0.3dB<br>0dB |
| 8.1.4 E-UTRAN TDD UE Rx-Tx time difference under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Same as 8.1.3 | Same as 8.1.3 |
| 8.1.5 E-UTRAN FDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Parameters<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -3.00dB<br>$\hat{E}s_2 / N_{oc}$: +3.00dB<br>$\hat{E}s_3 / N_{oc}$: +1.00dB | Test Tolerance<br>0dB<br>+0.4dB<br>0dB<br>0dB |
| 8.1.6 E-UTRAN TDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 8.1.5 | Same as 8.1.5 |
| 8.1.7 E-UTRAN FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.8 E-UTRAN HD-FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.9 E-UTRAN TDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.1 | Same as 8.1.1 |
| 9.1.1 FDD RSTD Measurement Reporting Delay | Response time | 300 ms |
| 9.1.1A FDD RSTD Measurement Reporting Delay for UE Category 1bis | Response time | 300 ms |
| 9.1.2 TDD RSTD Measurement Reporting Delay | Response time | 300 ms |
| 9.1.2A TDD RSTD Measurement Reporting Delay for UE Category 1bis | Response time | 300 ms |

| | | |
|---|---|---|
| 9.1.3 FDD RSTD Measurement Accuracy | For Test 2 and Test 4:<br>PRS $\hat{E}s_1$ / $N_{oc}$ averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ averaged over $BW_{Config}$<br><br>For all tests:<br>Cell Timing Difference | <br>+0.3 dB<br><br>+0.3 dB<br><br><br><br>± 1 Ts |
| 9.1.3A FDD RSTD Measurement Accuracy for UE Category 1bis | Same as 9.1.3 | Same as 9.1.3 |
| 9.1.4 TDD RSTD Measurement Accuracy | Same as 9.1.3 | Same as 9.1.3 |
| 9.1.4A TDD RSTD Measurement Accuracy for UE Category 1bis | Same as 9.1.3 | Same as 9.1.3 |
| 9.2.1 FDD-FDD inter-frequency RSTD measurement reporting delay | Response time | 300 ms |
| 9.2.1A FDD-FDD inter-frequency RSTD measurement reporting delay for UE Category 1bis | Response time | 300 ms |
| 9.2.2 TDD-TDD inter-frequency RSTD measurement reporting delay | Response time | 300 ms |
| 9.2.2A TDD-TDD inter-frequency RSTD measurement reporting delay for UE Category 1bis | Response time | 300 ms |
| 9.2.4 FDD-FDD inter frequency RSTD Accuracy | PRS $\hat{E}s_1$ / $N_{oc1}$ averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ averaged over $BW_{Config}$<br><br>Cell Timing Difference | +0.3 dB<br><br>+0.3 dB<br><br><br>± 2 Ts |
| 9.2.4A FDD-FDD inter frequency RSTD Accuracy for UE Category 1bis | Same as 9.2.4 | Same as 9.2.4 |
| 9.2.5 TDD-TDD inter frequency RSTD Accuracy | Same as 9.2.4 | Same as 9.2.4 |
| 9.2.5A TDD-TDD inter frequency RSTD Accuracy for UE Category 1bis | Same as 9.2.4 | Same as 9.2.4 |
| 9.3.1.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.1.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.2.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.2.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.3.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.3.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.4.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.4.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.5.1 HD-FDD intra-frequency RSTD Measurement Reporting | Same as 9.1.1 | Same as 9.1.1 |

| | | |
|---|---|---|
| Delay in CE Mode B for Category M1 | | |
| 9.3.5.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.6.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.6.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.1.1 | Same as 9.1.1 |
| 9.3.7.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.7.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.8.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.8.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.9.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.9.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.10.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.10.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.11.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.11.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.12.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.12.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.13 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.14 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.15 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.16 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.3.17 E-UTRAN HD-FDD intra-frequency RSTD measurement | Same as 9.1.3 | Same as 9.1.3 |

| | | |
|---|---|---|
| period test case in CE Mode B with longer PRS occasions | | |
| 9.3.18 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.1.3 | Same as 9.1.3 |
| 9.4.1.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.1.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.2.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.2.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.3.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.3.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.4.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.4.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.5.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.5.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.6.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.6.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Same as 9.2.1 | Same as 9.2.1 |
| 9.4.7.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.7.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.8.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.8.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.9.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.9.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.10.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.2.4 | Same as 9.2.4 |

| | | |
|---|---|---|
| 9.4.10.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.11.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.11.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.12.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.12.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.13 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.14 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.15 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.16 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.17 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.4.18 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Same as 9.2.4 | Same as 9.2.4 |
| 9.5.1 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | PRS $\hat{E}s_1$ / $N_{oc}$ averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc}$ averaged over $BW_{Config}$<br><br>Cell Timing Difference | <br>+0.3 dB<br><br>+0.3 dB<br><br>± 1 Ts |
| 9.5.2 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.5.1 | Same as 9.5.1 |
| 9.5.3 HD-FDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Response time | 300 ms |
| 9.6.1 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | PRS $\hat{E}s_1$ / $N_{oc1}$ averaged over $BW_{Config}$<br>PRS $\hat{E}s_2$ / $N_{oc2}$ averaged over $BW_{Config}$<br><br>Cell Timing Difference | +0.3 dB<br><br>+0.3 dB<br><br>± 2 Ts |
| 9.6.2 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.6.1 | Same as 9.6.1 |
| 9.6.3 HD-FDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Response time | 300 ms |

| | | |
|---|---|---|
| 9.7.1 TDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | PRS $\hat{E}_{s1}$ / $N_{oc}$ averaged over $BW_{Config}$<br>PRS $\hat{E}_{s2}$ / $N_{oc}$ averaged over $BW_{Config}$<br><br>Cell Timing Difference | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts |
| 9.7.2 TDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.7.1 | Same as 9.7.1 |
| 9.7.3 TDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Response time | 300 ms |
| 9.8.1 TDD Inter-frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | PRS $\hat{E}_{s1}$ / $N_{oc1}$ averaged over $BW_{Config}$<br>PRS $\hat{E}_{s2}$ / $N_{oc2}$ averaged over $BW_{Config}$<br><br>Cell Timing Difference | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts |
| 9.8.2 TDD Inter-frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | Same as 9.8.1 | Same as 9.8.1 |
| 9.8.3 TDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Response time | 300 ms |
| 10.1 FDD RSTD Measurement Reporting Delay for Carrier Aggregation | Response time | 300 ms |
| 10.1A FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz | Response time | 300 ms |
| 10.1B FDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz +5 MHz Bandwidth | Response time | 300 ms |
| 10.1C FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Response time | 300 ms |
| 10.2 TDD RSTD Measurement Reporting Delay for Carrier Aggregation | Response time | 300 ms |
| 10.2A TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz | Response time | 300 ms |
| 10.2B TDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz +5 MHz Bandwidth | Response time | 300 ms |
| 10.2C TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Response time | 300 ms |
| 10.2D TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz +10 MHz Bandwidth | Response time | 300 ms |
| 10.3 FDD RSTD Measurement Accuracy for Carrier Aggregation | PRS $\hat{E}_{s2}$ / $N_{oc2}$ averaged over $BW_{Config}$<br>PRS $\hat{E}_{s3}$ / $N_{oc2}$ averaged over $BW_{Config}$<br><br>Cell Timing Difference | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts |
| 10.3A FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-10 and Rel-11) | Same as 10.3 | Same as 10.3 |

| | | |
|---|---|---|
| 10.3A_1 FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-12 onwards) | Same as 10.3 | Same as 10.3 |
| 10.3B FDD RSTD Measurement Accuracy for Carrier Aggregation for 5MHz+5MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.3C FDD RSTD Measurement Accuracy for Carrier Aggregation for 10MHz+5MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.4 TDD RSTD Measurement Accuracy for Carrier Aggregation | Same as 10.3 | Same as 10.3 |
| 10.4A TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-10 and Rel-11) | Same as 10.3 | Same as 10.3 |
| 10.4A_1 TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-12 onwards) | Same as 10.3 | Same as 10.3 |
| 10.4B TDD RSTD Measurement Accuracy for Carrier Aggregation for 5MHz+5MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.4C TDD RSTD Measurement Accuracy for Carrier Aggregation for 10MHz+5MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.4D TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz+10 MHz Bandwidth | Same as 10.3 | Same as 10.3 |
| 10.5 FDD 3 DL CA RSTD Measurement Reporting Delay | Response time | 300 ms |
| 10.6 TDD 3 DL CA RSTD Measurement Reporting Delay | Response time | 300 ms |
| 10.7 FDD RSTD Measurement Accuracy for 3DL Carrier Aggregation | PRS $\hat{E}_{s3}$ / $N_{oc3}$ averaged over $BW_{Config}$<br>PRS $\hat{E}_{s4}$ / $N_{oc3}$ averaged over $BW_{Config}$<br>Cell Timing Difference (Intra-band)<br>Cell Timing Difference (Inter-band) | +0.3 dB<br>+0.3 dB<br>± 1 Ts<br>± 2 Ts |
| 10.8 TDD RSTD Measurement Accuracy for 3DL Carrier Aggregation | Same as 10.7 | Same as 10.7 |

## C.2.3 MBS Minimum Performance requirements

**Table C.2.3-1: Test Parameter Relaxations for MBS Minimum Performance tests**

| Clause | Test Parameter Relaxation | |
|---|---|---|
| 11.1,11.1A MBS Measurement Reporting Delay | Beacon power level | 0 dB (no relaxation) |
| | Response time | 300 ms |
| 11.2,11.2A MBS Sensitivity Measurement Accuracy | Beacon power level | 2 dB |
| | Code phase difference | 5 ns |
| 11.3,11.3A MBS Nominal Measurement Accuracy | Beacon power level | 0 dB (no relaxation) |
| | Code phase difference | 5 ns |
| 11.4,11.4A MBS Dynamic Range Measurement Accuracy | Beacon power level | 2 dB |
| | Code phase difference | 5 ns |
| 11.5,11.5A MBS Measurement Accuracy in Multipath | Beacon power level | 0 dB (no relaxation) |
| | Code phase difference | 5 ns |

## C.2.4 WLAN and BLE measurement requirements

**Table C.2.4-1: Test Parameter Relaxations for WLAN and BLE measurement tests**

| Clause | Test Parameter Relaxation | |
|---|---|---|
| 12.1.1 WLAN AP Identification and reporting delay under nominal conditions | Response time | 300 ms |
| 12.1.2 WLAN AP Identification and reporting delay under dynamic range conditions | Response time | 300 ms |
| | Low Power WLAN APs Received Power Level | 1 dB |
| 12.2.1 Bluetooth identification | Response time | 300 ms |

## C.2.5 NR PRS-based Measurement requirements

**Table C.2.5-1: Test Parameter Relaxations for NR PRS-based Measurement requirements**

| Clause | Test Parameter Relaxation | |
|---|---|---|
| 14.2.1 NR RSTD measurement period test case for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 14.2.2 NR RSTD measurement period test case for dual positioning frequency layers in FR1 SA | Response time | 300ms |
| 14.2.3 NR RSTD measurement period test case for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 14.2.4 NR RSTD measurement period test case for dual positioning frequency layers in FR2 SA | Response time | 300ms |
| 14.2.5 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response time | 300ms |
| 14.2.6 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Response time | 300ms |
| 14.2.7 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR1 SA | Response time | 300ms |
| 14.2.8 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response time | 300ms |
| 14.2.9 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Response time | 300ms |
| 14.2.10 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR2 SA | Response time | 300ms |
| 14.2.11 NR RSTD measurement reporting delay test case for PRS aggregation in FR1 | Response time | 300ms |
| 14.3.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA | Cell Timing Difference | 32 Tc |

| | | |
|---|---|---|
| 14.3.2 NR RSTD measurement accuracy test case for dual positioning frequency layers in FR1 SA | Cell Timing Difference | 32 Tc |
| 14.3.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA | Same as 14.3.1 | Same as 14.3.1 |
| 14.3.4 NR RSTD measurement accuracy test case for dual positioning frequency layer in FR2 SA | Same as 14.3.2 | Same as 14.3.2 |
| 14.3.5 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA | Cell Timing Difference | 32 Tc |
| 14.3.6 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR1 SA | Cell Timing Difference | 32 Tc |
| 14.3.7 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA | Cell Timing Difference | 32 Tc |
| 14.3.8 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR2 SA | Cell Timing Difference | 32 Tc |
| 14.3.9 NR RSTD measurement accuracy test case for PRS aggregation in FR1 SA | Cell Timing Difference | 32 Tc |
| 14.4.1 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 14.4.2 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response time | 300ms |
| 14.4.3 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 14.4.4 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response time | 300ms |
| 14.5.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA | Cell Timing Difference | 32 Tc |
| 14.5.2 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA | Cell Timing Difference | 32 Tc |
| 14.5.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA | Cell Timing Difference | 32 Tc |
| 14.5.4 NR RSTD measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA | Cell Timing Difference | 32 Tc |

| | | |
|---|---|---|
| 15.2.1 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 15.2.2 UE Rx-Tx time difference measurement period for dual positioning frequency layer in FR1 SA | Response time | 300ms |
| 15.2.3 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 15.2.4 UE Rx-Tx time difference measurement period for dual positioning frequency layer in FR2 SA | Response time | 300ms |
| 15.2.5 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response time | 300ms |
| 15.2.6 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Response time | 300ms |
| 15.2.7 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with multiple RxTx TEGs in FR1 SA | Response time | 300ms |
| 15.2.8 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response time | 300ms |
| 15.2.9 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Response time | 300ms |
| 15.2.10 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with RxTx TEG in FR2 SA | Response time | 300ms |
| 15.2.11 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR1 SA | Response time | 300ms |
| 15.2.12 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR2 SA | Response time | 300ms |
| 15.2.13 UE Rx-Tx time difference measurement reporting delay test case in FR1 for Redcap UE without Rx FH | Response time | 300ms |
| 15.2.14 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE without Rx FH | Response time | 300ms |
| 15.2.16 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE with Rx FH | Response time | 300ms |
| 15.3.1 UE Rx-Tx time difference measurement accuracy for single | Cell Timing Difference | 32 Tc |

| | | |
|---|---|---|
| positioning frequency layer in FR1 SA | | |
| 15.3.2 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA | Cell Timing Difference | 32 Tc |
| 15.3.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR1 SA | Cell Timing Difference | 32 Tc |
| 15.3.4 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR1 SA | Cell Timing Difference | 32 Tc |
| 15.3.5 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA | Cell Timing Difference | 32 Tc |
| 15.3.6 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR2 SA | Cell Timing Difference | 32 Tc |
| 15.3.7 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR1 SA | Cell Timing Difference | 32 Tc |
| 15.3.8 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR2 SA | Cell Timing Difference | 32 Tc |
| 15.3.9 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH | Cell Timing Difference | 32 Tc |
| 15.3.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH | Cell Timing Difference | 32 Tc |
| 15.3.12 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH | Cell Timing Difference | 32 Tc |
| 15.4.1 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 15.4.2 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response time | 300ms |
| 15.4.3 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 15.4.4 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response time | 300ms |
| 15.4.5 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR1 SA | Response time | 300ms |
| 15.4.6 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR2 SA | Response time | 300ms |
| 15.4.7 UE Rx-Tx time difference measurement reporting delay in FR1 for Redcap UE without Rx FH | Response time | 300ms |

| | | |
|---|---|---|
| 15.4.8 UE Rx-Tx time difference measurement reporting delay in FR2 for Redcap UE without Rx FH | Response time | 300ms |
| 15.5.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA | Cell Timing Difference | 32 Tc |
| 15.5.2 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR1 SA | Cell Timing Difference | 32 Tc |
| 15.5.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA | Cell Timing Difference | 33 Tc |
| 15.5.4 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR2 SA | Cell Timing Difference | 33 Tc |
| 15.5.7 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH | Cell Timing Difference | 33 Tc |
| 15.5.8 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH | Cell Timing Difference | 33 Tc |
| 15.5.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH | Cell Timing Difference | 33 Tc |
| 16.2.1 PRS-RSRP measurement period test case for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 16.2.2 PRS-RSRP measurement period test case for dual positioning frequency layer in FR1 SA | Response time | 300ms |
| 16.2.3 PRS-RSRP measurement period test case for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 16.2.4 PRS-RSRP measurement period test case for dual positioning frequency layer in FR2 SA | Response time | 300ms |
| 16.2.5 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR1 SA | Response time | 300ms |
| 16.2.6 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Response time | 300ms |
| 16.3.1 PRS-RSRP measurement accuracy with PRS in FR1 | All Test ConfigurationsTest 1:<br>Reported absolute PRP values<br>Reported relative PRP values<br><br>Test 2:<br>Reported absolute PRP values<br>Reported relative PRP values | Test Configuration 1 and 2Test 1:<br>Reported absolute PRP values: Cell 1 ±3.5, Cell 2: ±8.5.<br>Reported relative PRP values: Cell 1 ±3.5, Cell 2: ±9.5.<br><br>Test 2:<br>Reported absolute PRP values: Cell 1 ±3.5, Cell 2: ±6.<br>Reported relative PRP values: Cell 1 ±3.5, Cell 2: ±6.5.<br><br>Test Configuration 3<br>Test 1: |

| | | |
|---|---|---|
| | | Reported absolute PRP values: Cell 1 ±3.5, Cell 2: ±8.5.<br>Reported relative PRP values: Cell 1 ±3.5, Cell 2: ±9.5.<br><br>Test 2:<br>Reported absolute PRP values: Cell 1 ±3.5, Cell 2: ±4.5.<br>Reported relative PRP values: Cell 1 ±3.5, Cell 2: ±5. |
| 16.3.2 PRS-RSRP measurement accuracy with PRS in FR2 | Test 1:<br>Reported absolute PRP values<br>Reported relative PRP values<br><br>Test 2:<br>Reported absolute PRP values<br>Reported relative PRP values | Test 1:<br>Reported absolute PRP values: ±8.5<br>Reported relative PRP values: ±10<br><br>Test 2:<br>Reported absolute PRP values: ±6<br>Reported relative PRP values: ±7.5 |
| 16.3.3 PRS-RSRP measurement accuracy with PRS in FR1 with reduced sample number | Test 1:<br>Reported absolute PRP values<br>Reported relative PRP values<br><br>Test 2:<br>Reported absolute PRP values<br>Reported relative PRP values | Test 1:<br>Reported absolute PRP values: Cell 1 ±3.5, Cell 2: ±4.5.<br>Reported relative PRP values: Cell 1 ±3.5, Cell 2: ±6.5.<br><br>Test 2:<br>Reported absolute PRP values: Cell 1 ±3.5, Cell 2: ±4.5.<br>Reported relative PRP values: Cell 1 ±3.5, Cell 2: ±6.5 |
| 16.3.4 PRS-RSRP measurement accuracy with PRS in FR2 with reduced sample number | Test 1:<br>Reported absolute PRP values<br>Reported relative PRP values<br>Reported absolute PRP values<br>Reported relative PRP values | Test 1:<br>Reported absolute PRP values: ±8.5<br>Reported relative PRP values: ±10<br>Reported absolute PRP values: ±6<br>Reported relative PRP values: ±7.5 |
| 16.4.1 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 16.4.2 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response time | 300ms |
| 16.4.3 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 16.4.4 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response time | 300ms |
| 16.5.1 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 SA | Test 1:<br>Cell1 reported absolute PRP values<br>Cell 1 reported relative PRP values | Test 1:<br>Cell 1 reported absolute PRP values: ±3.5<br>Cell 1 reported relative PRP values: ±3.5 |

| | | |
|---|---|---|
| | Cell2 reported absolute PRP values<br>Cell 2 reported relative PRP values<br><br>Test 2:<br>Cell1 reported absolute PRP values<br>Cell 1 reported relative PRP values<br>Cell2 reported absolute PRP values<br>Cell 2 reported relative PRP values | Cell 2 reported absolute PRP values: ±8.5<br>Cell 2 reported relative PRP values: ±9.5<br><br>Test 2:<br>Cell 1 reported absolute PRP values: ±3.5<br>Cell 1 reported relative PRP values: ±3.5<br>Cell 2 reported absolute PRP values: ±6<br>Cell 2 reported relative PRP values: ±6.5 |
| 16.5.2 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 with reduced sample number | Same as 16.5.1 | Same as 16.5.1 |
| 16.5.3 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 SA | Test 1:<br>Cell1 reported absolute PRP values<br>Cell 1 reported relative PRP values<br>Cell2 reported absolute PRP values<br>Cell 2 reported relative PRP values<br><br>Test 2:<br>Cell1 reported absolute PRP values<br>Cell 1 reported relative PRP values<br>Cell2 reported absolute PRP values<br>Cell 2 reported relative PRP values | Test 1:<br>Cell 1 reported absolute PRP values: ±5<br>Cell 1 reported relative PRP values: ±5<br>Cell 2 reported absolute PRP values: ±8.5<br>Cell 2 reported relative PRP values: ±10<br><br>Test 2:<br>Cell 1 reported absolute PRP values: ±5<br>Cell 1 reported relative PRP values: ±5<br>Cell 2 reported absolute PRP values: ±6<br>Cell 2 reported relative PRP values: ±7.5 |
| 16.5.4 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 with reduced sample number | Same as 16.5.3 | Same as 16.5.3 |
| 17.2.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 17.2.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response time | 300ms |
| 17.2.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Response time | 300ms |
| 17.2.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 17.2.5 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response time | 300ms |
| 17.2.6 PRS-RSRPP measurement reporting delay test case for single | Response time | 300ms |

| | | |
|---|---|---|
| positioning frequency layer without measurement gap in FR2 SA | | |
| 17.3.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA | Test 1:<br>Cell1 reported absolute PRS-RSRPP values<br>Cell2 reported absolute PRS-RSRPP values<br><br>Test 2:<br>Cell1 reported absolute PRS-RSRPP values<br>Cell2 reported absolute PRS-RSRPP values | Test 1:<br>Cell 1 reported absolute PRS-RSRPP values: ±4.1<br>Cell 2 reported absolute PRS-RSRPP values: ±5.8<br><br>Test 2:<br>Cell 1 reported absolute PRS-RSRPP values: ±4.1<br>Cell 2 reported absolute PRS-RSRPP values: ±4.9 |
| 17.3.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA | Cell1 reported absolute PRS-RSRPP values<br>Cell2 reported absolute PRS-RSRPP values | Cell 1 reported absolute PRS-RSRPP values: ±3.9<br>Cell 2 reported absolute PRS-RSRPP values: ±4.2 |
| 17.3.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA | Test 1:<br>Cell1 reported absolute PRS-RSRPP values<br>Cell2 reported absolute PRS-RSRPP values<br><br>Test 2:<br>Cell1 reported absolute PRS-RSRPP values<br>Cell2 reported absolute PRS-RSRPP values | Test 1:<br>Cell 1 reported absolute PRS-RSRPP values: ±6<br>Cell 2 reported absolute PRS-RSRPP values: ±7.3<br><br>Test 2:<br>Cell 1 reported absolute PRS-RSRPP values: ±6<br>Cell 2 reported absolute PRS-RSRPP values: ±6.2 |
| 17.3.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA | Cell1 reported absolute PRS-RSRPP values<br>Cell2 reported absolute PRS-RSRPP values | Cell 1 reported absolute PRS-RSRPP values: ±5.9<br>Cell 2 reported absolute PRS-RSRPP values: ±5.6 |
| 17.4.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Response time | 300ms |
| 17.4.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response time | 300ms |
| 17.4.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response time | 300ms |
| 17.4.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response time | 300ms |
| 17.5.1 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR1 SA | Same as 17.3.1 | Same as 17.3.1 |
| 17.5.2 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA | Same as 17.3.2 | Same as 17.3.2 |
| 17.5.3 PRS-RSRPP measurement accuracy test case for single positioning frequency layer in FR2 SA | Same as 17.3.3 | Same as 17.3.3 |

| | | |
|---|---|---|
| 17.5.4 PRS-RSRPP measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA | Same as 17.3.4 | Same as 17.3.4 |

# C.3 Interpretation of measurement results

The measurement results returned by the Test System are compared - without any modification - against the Test Requirements as defined by the shared risk principle.

The Shared Risk principle is defined in TR 102 273-1-2 [7], clause 6.5.

The actual measurement uncertainty of the Test System for the measurement of each parameter shall be included in the test report.

The recorded value for the Test System uncertainty shall be, for each measurement, equal to or lower than the appropriate figure in clause C.1.

If the Test System for a test is known to have a measurement uncertainty greater than that specified in clause C.1, it is still permitted to use this apparatus provided that an adjustment is made value as follows.

Any additional uncertainty in the Test System over and above that specified in clause C.1 shall be used to tighten the Test Requirement - making the test harder to pass. (This may require modification of stimulus signals). This procedure will ensure that a Test System not compliant with clause C.1does not increase the chance of passing a device under test where that device would otherwise have failed the test if a Test System compliant with clause C.1 had been used.

# C.4 Derivation of Test Requirements (This clause is informative)

The Test Requirements have been calculated by relaxing the Conformance requirement of the core specification using the Test Parameter Relaxations defined in clause C.2. When the Test Parameter Relaxation is zero, the Test Requirement will be the same as the Conformance requirement. When the Test Parameter Relaxation is non-zero, the Test Requirements will differ from the Conformance requirement, and the formula used for this relaxation is given in table C.4.1, C.4.2, C.4.3 and C.4.4.

**Table C.4.1: Derivation of Test Requirements for A-GNSS Minimum Performance tests**

| Test | Conformance requirement in TS 25.171 [31] or TS 25.172 [19] or TS 36.171 [3] or 38.171 [43] | | Test Parameter Relaxation (TPR) | Test Requirement |
|---|---|---|---|---|
| 5.2.1, 6.2.1, 7.1.1, 13.2.1 Sensitivity Coarse Time Assistance | Coarse Time Assistance | ±2 s | 200 ms | Formulas: UL-TPR, LL+TPR: ±1.8 s |
| | Absolute GPS L1 C/A signal level (test 5.2.1, 7.1.1, 13.2.1, sub-test 1) | -142, -147 dBm | 1 dB | Level + TPR: -141, -146 dBm |
| | Absolute GNSS signal level (Galileo) | -142, -147 dBm | 1 dB | Level + TPR: -141, -146 dBm |
| | Absolute GNSS signal level (GPS) (test 6.2.1, 7.1.1, 13.2.1, sub-tests 4, 5, 8, and 10 to 13) | -142, -147 dBm | 1 dB | Level + TPR: -141, -146 dBm |
| | Absolute GNSS signal level (GLONASS) | -142, -147 dBm | 1 dB | Level + TPR: -141, -146 dBm |
| | Absolute GNSS signal level (BDS) | -136, -145 dBm | 1 dB | Level + TPR: -135, -144 dBm |
| | Position error | 100 m | 1.3 m | Error +TPR: 101.3 m |
| | Response time | 20 s | 300 ms | Time + TPR: 20.3 s |
| 5.2.2, 6.2.2, 7.1.2, 13.2.2 Sensitivity Fine Time Assistance | Coarse Time Assistance | ±2 s | 200 ms | Formulas: UL-TPR, LL+TPR: ±1.8 s |
| | Fine Time Assistance | ±10 us | 1 us | UL-TPR, LL+TPR: ±9 us |

| Test | Conformance requirement in TS 25.171 [31] or TS 25.172 [19] or TS 36.171 [3] or 38.171 [43] | | Test Parameter Relaxation (TPR) | Test Requirement |
|---|---|---|---|---|
| | Absolute GPS L1 C/A signal level (test 5.2.2, 7.1.2, 13.2.2, sub-test 1) | -147 dBm | 1 dB | Level + TPR: -146 dBm |
| | Absolute GNSS signal level (Galileo) | -147 dBm | 1 dB | Level + TPR: -146 dBm |
| | Absolute GNSS signal level (GPS) (test 6.2.2, 7.1.2, 13.2.2, sub-tests 4, 5, 8 and 10 to 13) | -147 dBm | 1 dB | Level + TPR: -146 dBm |
| | Absolute GNSS signal level (GLONASS) | -147 dBm | 1 dB | Level + TPR: -146 dBm |
| | Absolute GNSS signal level (BDS) | -147 dBm | 1 dB | Level + TPR: -146 dBm |
| | Position error | 100 m | 1.3 m | Error +TPR: 101.3 m |
| | Response time | 20 s | 300 ms | Time + TPR: 20.3 s |
| 5.3, 6.3, 7.2, 13.3 Nominal Accuracy | Coarse Time Assistance | ±2 s | 200 ms | Formulas: UL-TPR, LL+TPR: ±1.8 s |
| | Absolute GPS L1 C/A signal level (test 5.3, 7.2, 13.3, sub-test 1) | -130 dBm | 0 dB | Formulas: Level + TPR: -130 dBm |
| | Absolute GNSS signal level (Galileo) | -127 dBm | 0 dB | Level + TPR: -127 dBm |
| | Absolute GNSS signal level (GPS) (test 6.3, 7.2, 13.3, sub-tests 4, 5, 8 and 10 to 13) | -128.5 dBm | 0 dB | Level + TPR: -128.5 dBm |
| | Absolute GNSS signal level (GLONASS) | -131 dBm | 0 dB | Level + TPR: -131 dBm |
| | Absolute GNSS signal level (QZSS) | -128.5 dBm | 0 dB | Level + TPR: -128.5 dBm |
| | Absolute GNSS signal level (SBAS) | -131 dBm | 0 dB | Level + TPR: -131 dBm |
| | Absolute GNSS signal level (BDS) | -133 dBm | 0 dB | Level + TPR: -133 dBm |
| | Position error | 30 m | 1.3 m | Error +TPR: 31.3 m |
| | Response time | 20 s | 300 ms | Time + TPR: 20.3 s |
| 5.4, 6.4, 7.3, 13.4 Dynamic Range | Coarse Time Assistance | ±2 s | 200 ms | Formulas: UL-TPR, LL+TPR: ±1.8 s |
| | Absolute GPS L1 C/A signal level (test 5.4, 7.3, 13.4, sub-test 1) | -129 to -147 dBm | 1 dB | Level + TPR: each level +1 dBm |
| | Absolute GNSS signal level (Galileo) | -127.5 to -147 dBm | 1 dB | Level + TPR: each level +1 dBm |
| | Absolute GNSS signal level (GPS) (test 6.4, 7.3, 13.4, sub-tests 4, 5, 8 and 10 to 13) | -129 to -147 dBm | 1 dB | Level + TPR: each level +1 dBm |
| | Absolute GNSS signal level (GLONASS) | -131.5 to -147 dBm | 1 dB | Level + TPR: each level +1 dBm |
| | Absolute GNSS signal level (BDS) | -133.5 to -145 dBm | 1 dB | Level + TPR: each level +1 dBm |
| | Relative GPS L1 C/A signal level (test 5.4, 7.3, 13.4, sub-test 1) | 18 dB | 0.2 dB | Level - TPR: highest level -0.2 dB: -128.2 dBm |
| | Relative GNSS signal level (Galileo) | 19.5 dB | 0.2 dB | Level - TPR: highest level -0.2 dB: -126.7 dBm |
| | Relative GNSS signal level (GPS) (test 6.4, 7.3, 13.4, sub-tests 4, 5, 8 and 10 to 13) | 18 dB | 0.2 dB | Level - TPR: highest level -0.2 dB: -128.2 dBm |
| | Relative GNSS signal level (GLONASS) | 15.5 dB | 0.2 dB | Level - TPR: highest level -0.2 dB: -130.7 dBm |

| Test | Conformance requirement in TS 25.171 [31] or TS 25.172 [19] or TS 36.171 [3] or 38.171 [43] | | Test Parameter Relaxation (TPR) | Test Requirement |
|---|---|---|---|---|
| | Relative GNSS signal level (BDS) | 11.5 dB | 0.2 dB | Level - TPR: highest level -0.2 dB: -132.7 dBm |
| | Position error | 100 m | 1.3 m | Error +TPR: 101.3 m |
| | Response time | 20 s | 300 ms | Time + TPR: 20.3 s |
| 5.5, 6.5, 7.4, 13.5 Multi-path scenario | Coarse Time Assistance | ±2 s | 200 ms | Formulas: UL-TPR, LL+TPR: ±1.8 s |
| | Absolute GPS L1 C/A signal level (test 5.5, 7.4, 13.5, sub-test 1) | -130 dBm | 0 dB | Formulas: Level + TPR: -130 dBm |
| | Absolute GNSS signal level (Galileo) | -127 dBm | 0 dB | Level + TPR: -127 dBm |
| | Absolute GNSS signal level (GPS) (test 6.5, 7.4, 13.5, sub-tests 4, 5, 8 and 10 to 13) | -128.5 dBm | 0 dB | Level + TPR: -128.5 dBm |
| | Absolute GNSS signal level (GLONASS) | -131 dBm | 0 dB | Level + TPR: -131 dBm |
| | Absolute GNSS signal level (BDS) | -133 dBm | 0 dB | Level + TPR: -133 dBm |
| | Relative GPS L1 C/A signal level (test 5.5, 7.4, 13.5, sub-test 1)<br>-142, -147 dBm | 6 dB | 0.2 dB | Relative level + TPR: relative level + 0.2dB: 6.2 dB |
| | Relative GNSS signal level (all GNSSs) (test 6.5, 7.4, 13.5, sub-tests 4, 5, 8 and 10 to 13) | Y dB where "Y" is given in Table 4.2.2 | 0.2 dB | Relative level + TPR: relative level + 0.2dB: Y + 0.2 dB |
| | Position error | 100 m | 1.3 m | Error +TPR: 101.3 m |
| | Response time | 20 s | 300 ms | Time + TPR: 20.3 s |
| 5.6, 6.6, 7.5, 7.5A., 13.6, 13.7 Moving scenario and periodic update | Absolute GPS L1 C/A Signal level (test 5.6, 7.5, 7.5A., 13.6, 13.7, sub-test 1) | -130 dBm | 0 dB | Formulas: Level + TPR: -130 dBm |
| | Absolute GNSS signal level (Galileo) | -127 dBm | 0 dB | Level + TPR: -127 dBm |
| | Absolute GNSS signal level (GPS) (test 6.6, 7.5, 7.5A., 13.6, 13.7, sub-tests 4, 5, 8 and 10 to 13) | -128.5 dBm | 0 dB | Level + TPR: -128.5 dBm |
| | Absolute GNSS signal level (GLONASS) | -131 dBm | 0 dB | Level + TPR: -131 dBm |
| | Absolute GNSS signal level (BDS) | -133 dBm | 0 dB | Level + TPR: -133 dBm |
| | Position error | 100 m | 1.3 m | Error +TPR: 101.3 m |
| | Differential response time | 2s +/- 20 % | 100 ms | Time +TPR: 1.5 s and 2.5 s |

**Table C.4.2: Derivation of Test Requirements for ECID and OTDOA Measurement tests**

| Test | Minimum Requirement in TS 36.133 | Test Parameter Relaxation (TPR) | Test Requirement in TS 36.571-1 |
|---|---|---|---|
| 8.1.1 E-UTRAN FDD UE Rx - Tx time difference case (Rel-9 to Rel-11) | Test 1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.0dB<br>Reported RxTx time difference value: Measured value converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | Test 1:<br>0dB<br>0.3dB<br><br>Via mapping | Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s / N_{oc}$: -2.7.0dB<br>(Measured value from step 7 - 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +23) $T_s$ converted to RX- |

| | | | |
|---|---|---|---|
| | Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s$ / $N_{oc}$: -3.0dB<br>Reported RxTx time difference value: Measured value converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | Test 2:<br>0dB<br>0.3dB<br>Via mapping | TX_TIME_DIFFERENCE according to Table 4.6.3-1<br><br>Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>Measured value from step 7 -13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| 8.1.1A E-UTRAN FDD UE Rx – Tx time difference case (Rel-12 onwards) | Same as 8.1.1 | Same as 8.1.1 | Test 1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s$ / $N_{oc}$: -2.7.0dB<br>(Measured value from step 7 - 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br><br>Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>Measured value from step 7 -10) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +10) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| 8.1.1B E-UTRAN FDD UE Rx – Tx time difference case for UE Category 1bis | Same as 8.1.1 | Same as 8.1.1 | Same as 8.1.1A |
| 8.1.2 E-UTRAN TDD UE Rx - Tx time difference case (Rel-9 to Rel-11) | Same as 8.1.1 except use Table 4.6.3-2 | Same as 8.1.1 | Same as 8.1.1 except use Table 4.6.3-2 |
| 8.1.2A E-UTRAN TDD UE Rx – Tx time difference case (Rel-12 onwards) | Same as 8.1.1 except use Table 4.6.3-2 | Same as 8.1.1 | Same as 8.1.1A except use Table 4.6.3-2 |
| 8.1.2B E-UTRAN TDD UE Rx – Tx time difference case for UE Category 1bis | Same as 8.1.1 except use Table 4.6.3-2 | Same as 8.1.1 | Same as 8.1.1A except use Table 4.6.3-2 |
| 8.1.3 E-UTRAN FDD UE Rx-Tx time difference under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Test 1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -3.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.00dB<br>Reported RxTx time difference value: Measured value converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | Test 1:<br>0dB<br>0.3dB<br>0dB<br>Via mapping | Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.70dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.00dB<br>Measured value from step 7 -13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |

| | | | |
|---|---|---|---|
| | Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -3.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.00dB<br>Reported RxTx time difference value: Measured value converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | Test 2:<br>0dB<br>0.3dB<br>0dB<br>Via mapping | Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.70dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.00dB<br>Measured value from step 7 -13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| 8.1.4 E-UTRAN TDD UE Rx-Tx time difference under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Same as 8.1.3 except use Table 4.6.3-2 | Same as 8.1.3 | Same as 8.1.3 except use Table 4.6.3-2 |
| 8.1.5 E-UTRAN FDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Test 1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -3.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.00dB<br>$\hat{E}s_3$ / $N_{oc}$: +1.00dB<br>Reported RxTx time difference value: Measured value converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br><br>Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -3.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.00dB<br>$\hat{E}s_3$ / $N_{oc}$: +1.00dB<br>Reported RxTx time difference value: Measured value converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | Test 1:<br>0dB<br>0.4dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0.4dB<br>0dB<br>0dB<br>Via mapping | Test 1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.60dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.00dB<br>$\hat{E}s_3$ / $N_{oc}$: +1.00dB<br>Measured value from step 7 -13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br><br>Test 2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.60dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.00dB<br>$\hat{E}s_3$ / $N_{oc}$: +1.00dB<br>Measured value from step 7 -13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| 8.1.6 E-UTRAN TDD UE Rx–Tx time difference under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 8.1.5 except use Table 4.6.3-2 | Same as 8.1.5 | Same as 8.1.5 except use Table 4.6.3-2 |

| | | | |
|---|---|---|---|
| 8.1.7 E-UTRAN FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | $N_{oc}$: -98dBm/15kHz<br>$\hat{E}s$ / $N_{oc}$: -3.0dB<br>Reported RxTx time difference value: Measured value converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 | 0dB<br>0.3dB<br><br>Via mapping | $N_{oc}$: -98dBm/15kHz<br>$\hat{E}s$ / $N_{oc}$: -2.7.0dB<br>Category M1: (Measured value from step 7 - 23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +23) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>Category M2: (Measured value from step 7 -13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1<br>To<br>(Measured value from step 7 +13) $T_s$ converted to RX-TX_TIME_DIFFERENCE according to Table 4.6.3-1 |
| 8.1.8 E-UTRAN HD-FDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.7 | Same as 8.1.7 | Same as 8.1.7 |
| 8.1.9 E-UTRAN TDD UE Rx-Tx time difference case for Category M1/M2 UE in CEModeA | Same as 8.1.7 except use Table 4.6.3-2 | Same as 8.1.7 | Same as 8.1.7 except use Table 4.6.3-2 |
| 9.1.1 FDD RSTD Measurement Reporting Delay | Response Time = 3 s | 300 ms | Time + TPR: 3.3 s |
| 9.1.1A FDD RSTD Measurement Reporting Delay for UE Category 1bis | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.1.2 TDD RSTD Measurement Reporting Delay | Response Time = 3 s | 300 ms | Time + TPR: 3.3 s |
| 9.1.2A TDD RSTD Measurement Reporting Delay for UE Category 1bis | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.1.3 FDD RSTD Measurement Accuracy | For Test 2 and Test 4:<br>PRS $\hat{E}s_1$ / $N_{oc}$ = -6dB<br>PRS $\hat{E}s_2$ / $N_{oc}$ = -13dB<br><br>Cell timing difference:<br>Test 1 and 2 = ±15 Ts<br>Test 3 and 4 = ±5 Ts | <br>+0.3 dB<br>+0.3 dB<br><br>± 1 Ts | <br>Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR,<br>±16 Ts<br>±6 Ts |
| 9.1.3A FDD RSTD Measurement Accuracy for UE Category 1bis | Same as 9.1.3 | Same as 9.1.3 | Same as 9.1.3 |
| 9.1.4 TDD RSTD Measurement Accuracy | Same as 9.1.3 | Same as 9.1.3 | Same as 9.1.3 |
| 9.1.4A TDD RSTD Measurement Accuracy for UE Category 1bis | Same as 9.1.3 | Same as 9.1.3 | Same as 9.1.3 |
| 9.2.1 FDD-FDD inter-frequency RSTD measurement reporting delay | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.2.1A FDD-FDD inter-frequency RSTD measurement reporting delay for UE Category 1bis | Response Time = 11 s | 300 ms | Time + TPR: 11.3 s |
| 9.2.2 TDD-TDD inter-frequency RSTD measurement reporting delay | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |

| | | | |
|---|---|---|---|
| 9.2.2A TDD-TDD inter-frequency RSTD measurement reporting delay for UE Category 1bis | Response Time = 11 s | 300 ms | Time + TPR: 11.3 s |
| 9.2.4 FDD-FDD inter frequency RSTD Accuracy | PRS $\hat{E}s_1$ / $N_{oc1}$ = -6dB<br>PRS $\hat{E}s_2$ / $N_{oc2}$ = -13dB<br><br>Cell timing difference:<br>Test 1 = ±21 Ts<br>Test 2 = ±9 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR,<br>±23 Ts<br>±11 Ts |
| 9.2.4A FDD-FDD inter frequency RSTD Accuracy for UE Category 1bis | Same as 9.2.4 | Same as 9.2.4 | Same as 9.2.4 |
| 9.2.5 TDD-TDD inter frequency RSTD Accuracy | Same as 9.2.4 | Same as 9.2.4 | Same as 9.2.4 |
| 9.2.5A TDD-TDD inter frequency RSTD Accuracy for UE Category 1bis | Same as 9.2.4 | Same as 9.2.4 | Same as 9.2.4 |
| 9.3.1.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.3.1.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.3.2.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.3.2.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.3.3.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.3.3.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.3.4.1 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Response Time = 13 s | 300 ms | Time + TPR: 13.3 s |
| 9.3.4.2 FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Test 1: Response Time 13 s<br>Test 2: Response Time 6 s | 300 ms | Test 1: Time + TPR: 13.3 s<br>Test 2: Time + TPR: 6.3 s |
| 9.3.5.1 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Response Time = 13 s | 300 ms | Time + TPR: 13.3 s |
| 9.3.5.2 HD-FDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Test 1: Response Time 13 s<br>Test 2: Response Time 6 s | 300 ms | Test 1: Time + TPR: 13.3 s<br>Test 2: Time + TPR: 6.3 s |
| 9.3.6.1 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Response Time = 13 s | 300 ms | Time + TPR: 13.3 s |

| | | | |
|---|---|---|---|
| 9.3.6.2 TDD intra-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Test 1: Response Time 13 s<br>Test 2: Response Time 6 s | 300 ms | Test 1: Time + TPR: 13.3 s<br>Test 2: Time + TPR: 6.3 s |
| 9.3.7.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | PRS $\hat{E}_{s1}$ / $N_{oc1}$ = -6 dB<br>PRS $\hat{E}_{s2}$ / $N_{oc2}$ = -13 dB<br><br>Cell timing difference = ±15 Ts | +0.3 dB<br>+0.3 dB<br><br>±1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±16 Ts |
| 9.3.7.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | PRS $\hat{E}_{s1}$ / $N_{oc1}$ = -6 dB<br>PRS $\hat{E}_{s2}$ / $N_{oc2}$ = -13 dB<br><br>Cell timing difference= ±6 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±7 Ts |
| 9.3.8.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.3.7.1 | Same as 9.3.7.1 | Same as 9.3.7.1 |
| 9.3.8.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.3.7.2 | Same as 9.3.7.2 | Same as 9.3.7.2 |
| 9.3.9.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.3.7.1 | Same as 9.3.7.1 | Same as 9.3.7.1 |
| 9.3.9.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.3.7.2 | Same as 9.3.7.2 | Same as 9.3.7.2 |
| 9.3.10.1 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | PRS $\hat{E}_{s1}$ / $N_{oc1}$ = -15dB<br>PRS $\hat{E}_{s2}$ / $N_{oc2}$ = -15dB<br><br>Cell timing difference= ±15 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -14.7 dB<br><br>Timing difference ± TPR, ±16 Ts |
| 9.3.10.2 FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | PRS $\hat{E}_{s1}$ / $N_{oc1}$ = -15 dB<br>PRS $\hat{E}_{s2}$ / $N_{oc2}$ = -15 dB<br><br>Cell timing difference= ±6 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -14.7 dB<br><br>Timing difference ± TPR, ±7 Ts |
| 9.3.11.1 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.3.10.1 | Same as 9.3.10.1 | Same as 9.3.10.1 |
| 9.3.11.2 HD-FDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.3.10.2 | Same as 9.3.10.2 | Same as 9.3.10.2 |
| 9.3.12.1 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.3.10.1 | Same as 9.3.10.1 | Same as 9.3.10.1 |
| 9.3.12.2 TDD intra-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.3.10.2 | Same as 9.3.10.2 | Same as 9.3.10.2 |
| 9.3.13 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Response Time = 3 s | 300 ms | Time + TPR: 3.3 s |
| 9.3.14 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Response Time = 3 s | 300 ms | Time + TPR: 3.3 s |
| 9.3.15 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Response Time = 3 s | 300 ms | Time + TPR: 3.3 s |

| | | | |
|---|---|---|---|
| 9.3.16 E-UTRAN FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Response Time = 3 s | 300 ms | Time + TPR: 3.3 s |
| 9.3.17 E-UTRAN HD-FDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Response Time = 3 s | 300 ms | Time + TPR: 3.3 s |
| 9.3.18 E-UTRAN TDD intra-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.4.1.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Response Time = 16 s | 300 ms | Time + TPR: 16.3 s |
| 9.4.1.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Test 1: Response Time 16 s<br>Test 2: Response Time 11 s | 300 ms | Test 1: Time + TPR: 16.3 s<br>Test 2: Time + TPR: 11.3 s |
| 9.4.2.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Response Time = 16 s | 300 ms | Time + TPR: 16.3 s |
| 9.4.2.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Test 1: Response Time 16 s<br>Test 2: Response Time 11 s | 300 ms | Test 1: Time + TPR: 16.3 s<br>Test 2: Time + TPR: 11.3 s |
| 9.4.3.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M1 | Response Time = 16 s | 300 ms | Time + TPR: 16.3 s |
| 9.4.3.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode A for Category M2 | Test 1: Response Time 16 s<br>Test 2: Response Time 11 s | 300 ms | Test 1: Time + TPR: 16.3 s<br>Test 2: Time + TPR: 11.3 s |
| 9.4.4.1 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Response Time = 42 s | 300 ms | Time + TPR: 42.3 s |
| 9.4.4.2 FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Test 1: Response Time 42 s<br>Test 2: Response Time 11 s | 300 ms | Test 1: Time + TPR: 42.3 s<br>Test 2: Time + TPR: 11.3 s |
| 9.4.5.1 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Response Time = 42 s | 300 ms | Time + TPR: 42.3 s |
| 9.4.5.2 HD-FDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Test 1: Response Time 42 s<br>Test 2: Response Time 11 s | 300 ms | Test 1: Time + TPR: 42.3 s<br>Test 2: Time + TPR: 11.3 s |
| 9.4.6.1 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M1 | Response Time = 42 s | 300 ms | Time + TPR: 42.3 s |
| 9.4.6.2 TDD inter-frequency RSTD Measurement Reporting Delay in CE Mode B for Category M2 | Test 1: Response Time 42 s<br>Test 2: Response Time 11 s | 300 ms | Test 1: Time + TPR: 42.3 s<br>Test 2: Time + TPR: 11.3 s |

| | | | |
|---|---|---|---|
| 9.4.7.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | PRS $\hat{E}s_1$ / $N_{oc1}$ = -6 dB<br>PRS $\hat{E}s_2$ / $N_{oc2}$ = -13 dB<br><br>Cell timing difference= ±21 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±23 Ts |
| 9.4.7.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | PRS $\hat{E}s_1$ / $N_{oc1}$ = -6 dB<br>PRS $\hat{E}s_2$ / $N_{oc2}$ = -13 dB<br><br>Cell timing difference= ±10 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±12 Ts |
| 9.4.8.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.4.7.1 | Same as 9.4.7.1 | Same as 9.4.7.1 |
| 9.4.8.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.4.7.2 | Same as 9.4.7.2 | Same as 9.4.7.2 |
| 9.4.9.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M1 | Same as 9.4.7.1 | Same as 9.4.7.1 | Same as 9.4.7.1 |
| 9.4.9.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode A for Category M2 | Same as 9.4.7.2 | Same as 9.4.7.2 | Same as 9.4.7.2 |
| 9.4.10.1 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | PRS $\hat{E}s_1$ / $N_{oc1}$ = -15 dB<br>PRS $\hat{E}s_2$ / $N_{oc2}$ = -15 dB<br><br>Cell timing difference= ±21 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -14.7 dB<br><br>Timing difference ± TPR, ±23 Ts |
| 9.4.10.2 FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | PRS $\hat{E}s_1$ / $N_{oc1}$ = -15 dB<br>PRS $\hat{E}s_2$ / $N_{oc2}$ = -[13] dB<br><br>Cell timing difference= ±10 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -[12.7] dB<br><br>Timing difference ± TPR, ±12 Ts |
| 9.4.11.1 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.4.10.1 | Same as 9.4.10.1 | Same as 9.4.10.1 |
| 9.4.11.2 HD-FDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.4.10.2 | Same as 9.4.10.2 | Same as 9.4.10.2 |
| 9.4.12.1 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M1 | Same as 9.4.10.1 | Same as 9.4.10.1 | Same as 9.4.10.1 |
| 9.4.12.2 TDD inter-frequency RSTD Measurement Accuracy in CE Mode B for Category M2 | Same as 9.4.10.2 | Same as 9.4.10.2 | Same as 9.4.10.2 |
| 9.4.13 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.4.14 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.4.15 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode A with longer PRS occasions | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.4.16 E-UTRAN FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |

| | | | |
|---|---|---|---|
| 9.4.17 E-UTRAN HD-FDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Response Time = 6 s | 300 ms | Time + TPR: 6.3 s |
| 9.4.18 E-UTRAN TDD inter-frequency RSTD measurement period test case in CE Mode B with longer PRS occasions | Response Time = 11 s | 300 ms | Time + TPR: 11.3 s |
| 9.5.1 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | NPRS $\hat{E}s_1$ / $N_{oc}$ = -6dB<br>NPRS $\hat{E}s_2$ / $N_{oc}$ = -13dB<br><br>Cell timing difference= ±20 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±21 Ts |
| 9.5.2 HD-FDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | NPRS $\hat{E}s_1$ / $N_{oc}$ = -15dB<br>NPRS $\hat{E}s_2$ / $N_{oc}$ = -15dB<br><br>Cell timing difference= ±32 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -14.7 dB<br><br>Timing difference ± TPR, ±33 Ts |
| 9.5.3 HD-FDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Response Time = 78 s | 300 ms | Time + TPR: 78.3 s |
| 9.6.1 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | NPRS $\hat{E}s_1$ / $N_{oc1}$ = -6dB<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ = -13dB<br><br>Cell timing difference= ±28 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±30 Ts |
| 9.6.2 HD-FDD Inter-Frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | NPRS $\hat{E}s_1$ / $N_{oc1}$ = -15dB<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ = -15dB<br><br>Cell timing difference= ±40 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -14.7 dB<br><br>Timing difference ± TPR, ±42 Ts |
| 9.6.3 HD-FDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Same as 9.5.3 | Same as 9.5.3 | Same as 9.5.3 |
| 9.7.1 TDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | NPRS $\hat{E}s_1$ / $N_{oc}$ = -6dB<br>NPRS $\hat{E}s_2$ / $N_{oc}$ = -13dB<br><br>Cell timing difference= ±20 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±21 Ts |
| 9.7.2 TDD Intra frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | NPRS $\hat{E}s_1$ / $N_{oc}$ = -15dB<br>NPRS $\hat{E}s_2$ / $N_{oc}$ = -15dB<br><br>Cell timing difference= ±32 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -14.7 dB<br><br>Timing difference ± TPR, ±33 Ts |
| 9.7.3 TDD Intra frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Response Time = 79 s | 300 ms | Time + TPR: 79.3 s |
| 9.8.1 TDD Inter-frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in normal coverage | NPRS $\hat{E}s_1$ / $N_{oc1}$ = -6dB<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ = -13dB<br><br>Cell timing difference= ±28 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±30 Ts |
| 9.8.2 TDD Inter-frequency RSTD Measurement Accuracy for NB-IOT Inband Mode in enhanced coverage | NPRS $\hat{E}s_1$ / $N_{oc1}$ = -15dB<br>NPRS $\hat{E}s_2$ / $N_{oc2}$ = -15dB<br><br>Cell timing difference= ±40 Ts | +0.3 dB<br>+0.3 dB<br><br>± 2 Ts | Level + TPR, -14.7 dB<br>Level + TPR, -14.7 dB<br><br>Timing difference ± TPR, ±42 Ts |
| 9.8.3 TDD Inter frequency RSTD Measurement Reporting Delay for NB-IOT Inband Mode in enhanced coverage | Response Time = 90 s | 300 ms | Time + TPR: 90.3 s |

| | | | |
|---|---|---|---|
| 10.1 FDD RSTD Measurement Reporting Delay for Carrier Aggregation | Test 1:<br>Response Time = 3 s<br><br>Test 2:<br>Response Time = 6 s | <br>300 ms<br><br><br>300 ms | <br>Time + TPR: 3.3 s<br><br><br>Time + TPR: 6.3 s |
| 10.1A FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.1B FDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz +5 MHz Bandwidth | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.1C FDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.2 TDD RSTD Measurement Reporting Delay for Carrier Aggregation | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.2A TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.2B TDD RSTD Measurement Reporting Delay Carrier Aggregation for 5 MHz +5 MHz Bandwidth | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.2C TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.2D TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20 MHz +10 MHz Bandwidth | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.3 FDD RSTD Measurement Accuracy for Carrier Aggregation | PRS $\hat{E}s_2 / N_{oc2}$ = -6dB<br>PRS $\hat{E}s_3 / N_{oc2}$ = -13dB<br><br>Cell timing difference= ±5 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±6 Ts |
| 10.3A FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-10 and Rel-11) | Same as 10.3 | Same as 10.3 | Same as 10.3 |
| 10.3A_1 FDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-12 onwards) | PRS $\hat{E}s_2 / N_{oc2}$ = -6dB<br>PRS $\hat{E}s_3 / N_{oc2}$ = -13dB<br><br>Cell timing difference= ±4 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±5 Ts |
| 10.3B FDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz+5 MHz Bandwidth | PRS $\hat{E}s_2 / N_{oc2}$ = -6dB<br>PRS $\hat{E}s_3 / N_{oc2}$ = -13dB<br><br>Cell timing difference= ±6 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±7 Ts |
| 10.3C FDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.3B | Same as 10.3B | Same as 10.3B |
| 10.4 TDD RSTD Measurement Accuracy for Carrier Aggregation | Same as 10.3 | Same as 10.3 | Same as 10.3 |

| | | | |
|---|---|---|---|
| 10.4A TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-10 and Rel-11) | Same as 10.3 | Same as 10.3 | Same as 10.4 |
| 10.4A_1 TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz (Rel-12 onwards) | PRS $\hat{E}_{S2}$ / $N_{oc2}$ = -6dB<br>PRS $\hat{E}_{S3}$ / $N_{oc2}$ = -13dB<br><br>Cell timing difference= ±4 Ts | +0.3 dB<br>+0.3 dB<br><br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR, ±5 Ts |
| 10.4B TDD RSTD Measurement Accuracy for Carrier Aggregation for 5 MHz+5 MHz Bandwidth | Same as 10.3B | Same as 10.3B | Same as 10.3B |
| 10.4C TDD RSTD Measurement Accuracy for Carrier Aggregation for 10 MHz+5 MHz Bandwidth | Same as 10.3B | Same as 10.3B | Same as 10.3B |
| 10.4D TDD RSTD Measurement Accuracy for Carrier Aggregation for 20 MHz+10 MHz Bandwidth | Same as 10.3 | Same as 10.3 | Same as 10.3 |
| 10.5 FDD 3 DL CA RSTD Measurement Reporting Delay | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.6 TDD 3 DL CA RSTD Measurement Reporting Delay | Same as 10.1 | Same as 10.1 | Same as 10.1 |
| 10.7 FDD RSTD Measurement Accuracy for 3DL Carrier Aggregation | PRS $\hat{E}_{S3}$ / $N_{oc3}$ = -6dB<br>PRS $\hat{E}_{S4}$ / $N_{oc3}$ = -13dB<br><br>Cell timing difference=<br>Cell 1: ±10 Ts, ±9 Ts, ±8 Ts<br>Cell 2: ±10 Ts, ±9 Ts, ±8 Ts<br>Cell 4: ±6 Ts, ±5 Ts, ±4 Ts | +0.3 dB<br>+0.3 dB<br><br><br>± 2 Ts<br>± 2 Ts<br>± 1 Ts | Level + TPR, -5.7 dB<br>Level + TPR, -12.7 dB<br><br>Timing difference ± TPR,<br>Cell 1: ±12 Ts, ±11 Ts, ±10 Ts<br>Cell 2: ±12 Ts, ±11 Ts, ±10 Ts<br>Cell 4: ±7 Ts, ±6 Ts, ±5 Ts |
| 10.8 TDD RSTD Measurement Accuracy for 3DL Carrier Aggregation | Same as 10.7 | Same as 10.7 | Same as 10.7 |

**Table C.4.3: Derivation of Test Requirements for MBS Minimum Performance tests**

| Test | Conformance requirements in 11.1.3, 11.2.3, 11.3.3, 11.4.3 and 11.5.3 | | Test Parameter Relaxation (TPR) | Test Requirement |
|---|---|---|---|---|
| 11.1, 11.1A MBS Measurement Reporting Delay | Beacon power level | -30 dBm | 0 dB | Level+TPR: -30 dBm |
| | Response time | 12 seconds | 300 ms | Time+TPR: 12.3 s |
| 11.2 MBS Sensitivity Measurement Accuracy | Beacon power level | -130 dBm | 2 dB | Level+TPR: -128 dBm |
| | Code phase difference | $2.35 \times 10^{-4}$ ms | 5 ns | Error+TPR: $2.40 \times 10^{-4}$ ms |
| 11.2A MBS Sensitivity Measurement Accuracy | Beacon power level | -130 dBm | 2 dB | Level+TPR: -128 dBm |
| | Code phase difference | $2.35 \times 10^{-4}$ ms | 5 ns | Error+TPR: $2.40 \times 10^{-4}$ ms |
| | Code phase difference | $9.3 \times 10^{-5}$ ms | 5 ns | Error+TPR: $9.8 \times 10^{-5}$ ms |
| 11.3 MBS Nominal Measurement Accuracy | Beacon power level | -30 dBm | 0 dB | Level+TPR: -30 dBm |
| | Code phase difference | $7.1 \times 10^{-5}$ ms | 5 ns | Error+TPR: $7.6 \times 10^{-5}$ ms |
| 11.3A MBS Nominal Measurement Accuracy | Beacon power level | -30 dBm | 0 dB | Level+TPR: -30 dBm |
| | TB1 Code phase difference | $7.1 \times 10^{-5}$ ms | 5 ns | Error+TPR: $7.6 \times 10^{-5}$ ms |
| | TB2 Code phase difference | $2.8 \times 10^{-5}$ ms | 5 ns | Error+TPR: $3.3 \times 10^{-5}$ ms |
| 11.4 MBS Dynamic Range Measurement Accuracy | High Power Beacon power level | -30 dBm | 0 dB | Level+TPR: -30 dBm |
| | Code phase difference | $7.1 \times 10^{-5}$ ms | 5 ns | Error+TPR: $7.6 \times 10^{-5}$ ms |
| | Low Power Beacon power level | -130 dBm | 2 dB | Level+TPR: -128 dBm |
| | Code phase difference | $2.35 \times 10^{-4}$ ms | 5 ns | Error+TPR: $2.40 \times 10^{-4}$ ms |
| | High Power Beacon power level | -30 dBm | 0 dB | Level+TPR: -30 dBm |

| | | | | |
|---|---|---|---|---|
| 11.4A MBS Dynamic Range Measurement Accuracy | TB1 High Power code phase difference | $7.1 \times 10^{-5}$ ms | 5 ns | Error+TPR: $7.6 \times 10^{-5}$ ms |
| | TB2 High Power code phase difference | $2.8 \times 10^{-5}$ ms | 5 ns | Error+TPR: $3.3 \times 10^{-5}$ ms |
| | Low Power Beacon power level | -130 dBm | 2 dB | Level+TPR: -128 dBm |
| | TB1 Low Power code phase difference | $2.35 \times 10^{-4}$ ms | 5 ns | Error+TPR: $2.40 \times 10^{-4}$ ms |
| | TB2 Low Power code phase difference | $9.3 \times 10^{-5}$ ms | 5 ns | Error+TPR: $9.8 \times 10^{-5}$ ms |
| 11.5, 11.5A MBS Measurement Accuracy in Multipath | Beacon power level | -30 dBm | 0 dB | Level+TPR: -30 dBm |
| | Code phase difference | $2.35 \times 10^{-4}$ ms | 5 ns | Error+TPR: $2.40 \times 10^{-4}$ ms |

**Table C.4.4: Derivation of Test Requirements for WLAN and BLE measurement tests**

| Test | Conformance requirements in 12.1.1, 12.1.2 and 12.2.1 | | Test Parameter Relaxation (TPR) | Test Requirement |
|---|---|---|---|---|
| 12.1.1 WLAN AP Identification and reporting delay under nominal conditions | Response time | 20.85 seconds | 300 ms | Time+TPR: 21.15 s |
| 12.1.2 WLAN AP Identification and reporting delay under dynamic range conditions | Response time | 20.85 seconds | 300 ms | Time+TPR: 21.15 s |
| | Low Power WLAN APs Received Power Level | WLAN 2.4 GHz band: -74 dBm WLAN 5 GHz band: -79 dBm | 1 dB | Power+TPR: WLAN 2.4 GHz band: -73 dBm WLAN 5 GHz band: -78 dBm |
| 12.2.1 Bluetooth identification | Response time | 10.85 seconds | 300 ms | Time+TPR: 11.15 s |

**Table C.4.5: Derivation of Test Requirements for NR RSTD Measurement tests**

| Test | Minimum Requirement in TS 38.133 | Test Parameter Relaxation (TPR) | Test Requirement in TS 37.571-1 |
|---|---|---|---|
| 14.2.1 NR RSTD measurement period test case for single positioning frequency layer in FR1 SA | Response Time between 1 s and 210 s depending on UE capabilities | 300 ms | Time + TPR: between 1.3 s and 210.3 s depending on UE capabilities |
| 14.2.2 NR RSTD measurement period test case for dual positioning frequency layers in FR1 SA | Response Time between 2 s and 104 s depending on UE capabilities | 300 ms | Time + TPR: between 2.3 s and 104.3 s depending on UE capabilities |
| 14.2.3 NR RSTD measurement period test case for single positioning frequency layer in FR2 SA | Response Time between 6 s and 250 s depending on UE capabilities | 300 ms | Time + TPR: between 6.3 s and 250.3 s depending on UE capabilities |
| 14.2.4 NR RSTD measurement period test case for dual positioning frequency layers in FR2 SA | Response Time between 11 s and 210 s depending on UE capabilities | 300 ms | Time + TPR: between 11.3 s and 210.3 s depending on UE capabilities |
| 14.2.5 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response Time between 1 s and 28 s depending on UE capabilities | 300 ms | Time + TPR: between 1.3 s and 28.3 s depending on UE capabilities |
| 14.2.6 NR RSTD measurement reporting delay test case for single positioning frequency layer | Response Time between 1 s and 14 s depending on UE capabilities | 300 ms | Time + TPR: between 1.3 s and 14.3 s depending on UE capabilities |

| | | | |
|---|---|---|---|
| without measurement gap in FR1 SA | | | |
| 14.2.7 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR1 SA | Response Time between 1 s and 870 s depending on UE capabilities | 300 ms | Time + TPR: between 1.3 s and 870.3 s depending on UE capabilities |
| 14.2.8 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response Time between 2 s and 31 s depending on UE capabilities | 300 ms | Time + TPR: between 2.3 s and 31.3 s depending on UE capabilities |
| 14.2.9 NR RSTD measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Response Time between 5 s and 25 s depending on UE capabilities | 300 ms | Time + TPR: between 5.3 s and 25.3 s depending on UE capabilities |
| 14.2.10 NR RSTD measurement reporting delay test case for single positioning frequency layer with Rx TEG in FR2 SA | Response Time between 11 s and 500 s depending on UE capabilities | 300 ms | Time + TPR: between 11.3 s and 500.3 s depending on UE capabilities |
| 14.2.11 NR RSTD measurement reporting delay test case for PRS aggregation in FR1 | Response Time between 3 s and 84 s depending on UE capabilities | 300 ms | Time + TPR: between 3.3 s and 84.3 s depending on UE capabilities |
| 14.3.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA | Cell Timing Difference = 3μs | 32 Tc | Time + TPR: 3μs ± 32 Tc |
| 14.3.2 NR RSTD measurement accuracy test case for dual positioning frequency layers in FR1 SA | Cell Timing Difference = 3μs | 32 Tc | Time + TPR: 3μs ± 32 Tc |
| 14.3.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA | Test 1:<br>Cell Timing Difference = 3μs± 88 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 50 Tc | 32 Tc | Test 1: Time + TPR: 3μs ± 120 Tc<br>Test 2: Time + TPR: 3μs ± 82 Tc |
| 14.3.4 NR RSTD measurement accuracy test case for dual positioning frequency layer in FR2 SA | Test 1:<br>Cell Timing Difference = 3μs± 328 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 278 Tc | 32 Tc | Test 1: Time + TPR: 3μs ± 360 Tc<br>Test 2: Time + TPR: 3μs ± 310 Tc |
| 14.3.5 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR1 SA | Test Configuration 1 and 2<br>Test 1:<br>Cell Timing Difference = 3μs±202 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 110 Tc<br><br>Test Configuration 3<br>Test 1:<br>Cell Timing Difference = 3μs±116 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 72 Tc | 32 Tc | Test Configuration 1 and 2<br>Test 1: Time + TPR: 3μs ± 234 Tc<br>Test 2: Time + TPR: 3μs ± 142 Tc<br><br>Test Configuration 3<br>Test 1: Time + TPR: 3μs ± 148 Tc<br>Test 2: Time + TPR: 3μs ± 104 Tc |
| 14.3.6 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR1 SA | Test Configuration 1 and 2<br>Test 1:<br>Cell Timing Difference = 3μs±284 Tc<br>Test 2: | 32 Tc | Test Configuration 1 and 2<br>Test 1: Time + TPR: 3μs ± 316 Tc<br>Test 2: Time + TPR: 3μs ± 142 Tc |

| | | | |
|---|---|---|---|
| | Cell Timing Difference = 3μs± 110 Tc<br><br>Test Configuration 3<br>Test 1:<br>Cell Timing Difference = 3μs±179 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 72 Tc | | Test Configuration 3<br>Test 1: Time + TPR: 3μs ± 211 Tc<br>Test 2: Time + TPR: 3μs ± 104 Tc |
| 14.3.7 NR RSTD measurement accuracy test case for single positioning frequency layer with reduced number of samples in FR2 SA | Test 1:<br>Cell Timing Difference = 3μs± 61 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 50 Tc | 32 Tc | Test 1: Time + TPR: 3μs ± 93 Tc<br>Test 2: Time + TPR: 3μs ± 82 Tc |
| 14.3.8 NR RSTD measurement accuracy test case for single positioning frequency layer with Rx TEG in FR2 SA | Test 1:<br>Cell Timing Difference = 3μs±88 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 50 Tc | 32 Tc | Test 1: Time + TPR: 3μs ± 120 Tc<br>Test 2: Time + TPR: 3μs ± 82 Tc |
| 14.3.9 NR RSTD measurement accuracy test case for PRS aggregation in FR1 SA | Test Configuration 1 and 2:<br>Cell Timing Difference = 3μs ± 344Tc<br>Test Configuration 3:<br>Cell Timing Difference = 3μs ± 299Tc | 32 Tc | Test Configuration 1 and 2:<br>Time + TPR: 3μs ± 376Tc<br><br>Test Configuration 3:<br>Time + TPR: 3μs ± 331Tc |
| 14.4.1 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR1 SA | Response Time<br>between 6 s and 109 s depending on UE capabilities | 300 ms | Time + TPR: between 6.3 s and 109.3 s depending on UE capabilities |
| 14.4.2 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response Time<br>between 2 s and 29 s depending on UE capabilities | 300 ms | Time + TPR: between 2.3 s and 29.3 s depending on UE capabilities |
| 14.4.3 NR RSTD measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response Time<br>between 21 s and 124 s depending on UE capabilities | 300 ms | Time + TPR: between 21.3 s and 124.3 s depending on UE capabilities |
| 14.4.4 NR RSTD measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response Time<br>between 6 s and 32 s depending on UE capabilities | 300 ms | Time + TPR: between 6.3 s and 32.3 s depending on UE capabilities |
| 14.5.1 NR RSTD measurement accuracy test case for single positioning frequency layer in FR1 SA | Test Configuration 1 and 2<br>Test 1:<br>Cell Timing Difference = 3μs±284 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 110 Tc<br><br>Test Configuration 3<br>Test 1:<br>Cell Timing Difference = 3μs±179 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 72 Tc | 32 Tc | Test Configuration 1 and 2<br>Test 1: Time + TPR: 3μs ± 316 Tc<br>Test 2: Time + TPR: 3μs ± 142 Tc<br><br>Test Configuration 3<br>Test 1: Time + TPR: 3μs ± 211 Tc<br>Test 2: Time + TPR: 3μs ± 104 Tc |
| 14.5.2 NR RSTD measurement accuracy for single positioning frequency | Test Configuration 1 and 2<br>Test 1: | 32 Tc | Test Configuration 1 and 2<br>Test 1: Time + TPR: 3μs ± 234 Tc<br>Test 2: Time + TPR: 3μs ± 142 Tc |

| | | | |
|---|---|---|---|
| layer with reduced number of samples in FR1 SA | Cell Timing Difference = 3μs±202 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 110 Tc<br><br>Test Configuration 3<br>Test 1:<br>Cell Timing Difference = 3μs±116 Tc<br>Test 2:<br>Cell Timing Difference = 3μs± 72 Tc | | <br><br><br><br><br>Test Configuration 3<br>Test 1: Time + TPR: 3μs ± 148 Tc<br>Test 2: Time + TPR: 3μs ± 104 Tc |
| 14.5.3 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA | Cell Timing Difference = 3μs | 32 Tc | Time + TPR: 3μs ± 32 Tc |
| 14.5.4 NR RSTD measurement accuracy test case for single positioning frequency layer in FR2 SA | Sub-test 1:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: -1.44dB<br>PRS Ês2 / Noc: -3.65dB<br><br>Cell Timing Difference = 3μs± 93 Tc<br><br><br>Sub-test 2:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: -1.44dB<br>PRS Ês2 / Noc: -3.65dB<br><br>Cell Timing Difference = 3μs± 82 Tc | -2.6dB<br>0.88dB<br>0.93dB<br><br>Via mapping<br><br><br>-2.6dB<br>0.88dB<br>0.93dB<br>Via mapping | Noc: -100.6dBm/15kHz<br>PRS Ês1 / Noc: -0.56dB<br>PRS Ês2 / Noc: -2.72dB<br><br>Test 1: Time + TPR: 3μs ± 125 Tc<br><br><br>Noc: -100.6dBm/15kHz<br>PRS Ês1 / Noc: -0.56dB<br>PRS Ês2 / Noc: -2.72dB<br><br>Test 1: Time + TPR: 3μs ± 114 Tc |

**Table C.4.6: Derivation of Test Requirements for UE Rx–Tx time difference Measurement tests**

| Test | Minimum Requirement in TS 38.133 | Test Parameter Relaxation (TPR) | Test Requirement in TS 37.571-1 |
|---|---|---|---|
| 15.2.1 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR1 SA | Response Time between 1 s and 103 s depending on UE capabilities | 300ms | Time + TPR: between 1.3 s and 103.3 s depending on UE capabilities |
| 15.2.2 UE Rx-Tx time difference measurement period for dual positioning frequency layers in FR1 SA | Response Time between 2.3 s and 53.3 s depending on UE capabilities | 300ms | Time + TPR: between 2.3 s and 53.3 s depending on UE capabilities |
| 15.2.3 UE Rx-Tx time difference measurement period for single positioning frequency layer in FR2 SA | Response Time between 6 s and 170 s depending on UE capabilities | 300ms | Time + TPR: between 6.3 s and 170.3 s depending on UE capabilities |
| 15.2.4 UE Rx-Tx time difference measurement period for dual positioning frequency layers in FR2 SA | Response Time between 11 s and 84 s depending on UE capabilities | 300ms | Time + TPR: between 11.3 s and 84.3 s depending on UE capabilities |
| 15.2.5 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response Time between 1 s and 52 s depending on UE capabilities | 300ms | Time + TPR: between 1.3 s and 52.3 s depending on UE capabilities |
| 15.2.6 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency | Response Time between 1 s and 205 s depending on UE capabilities | 300ms | Time + TPR: between 1.3 s and 205.3 s depending on UE capabilities |

| | | | |
|---|---|---|---|
| layer without measurement gap in FR1 SA | | | |
| 15.2.7 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with multiple RxTx TEGs in FR1 SA | Response Time<br>between 1 s and 164 s depending on UE capabilities | 300ms | Time + TPR: between 1.3 s and 164.3 s depending on UE capabilities |
| 15.2.8 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response Time<br>between 2 s and 21 s depending on UE capabilities | 300ms | Time + TPR: between 2.3 s and 21.3 s depending on UE capabilities |
| 15.2.9 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Response Time<br>Sub-test 1: between 3 s and 21 s depending on UE capabilities<br>Sub-test 2: between 6 s and 83 s depending on UE capabilities | 300ms | Sub-test 1: Time + TPR: between 3.3 s and 21.3 s depending on UE capabilities<br>Sub-test 2: Time + TPR: between 6.3 s and 83.3 s depending on UE capabilities |
| 15.2.10 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with RxTx TEG in FR2 SA | Response Time<br>between 11 s and 170 s depending on UE capabilities | 300ms | Time + TPR: between 11.3 s and 170.3 s depending on UE capabilities |
| 15.2.11 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR1 SA | Response Time<br>between 3 s and 42 s depending on UE capabilities | 300ms | Time + TPR: between 3.3 s and 42.3 s depending on UE capabilities |
| 15.2.12 UE Rx-Tx time difference measurement reporting delay test case for PRS aggregation in FR2 SA | Response Time<br>between 42 s and 170 s depending on UE capabilities | 300ms | Time + TPR: between 42.3 s and 170.3 s depending on UE capabilities |
| 15.2.13 UE Rx-Tx time difference measurement reporting delay test case in FR1 for Redcap UE without Rx FH | Response Time<br>between 2 s and 83 s depending on UE capabilities | 300ms | Time + TPR: between 2.3 s and 83.3 s depending on UE capabilities |
| 15.2.14 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE without Rx FH | Response Time<br>between 11 s and 42 s depending on UE capabilities | 300ms | Time + TPR: between 11.3 s and 42.3 s depending on UE capabilities |
| 15.2.16 UE Rx-Tx time difference measurement reporting delay test case in FR2 for Redcap UE with Rx FH | Response Time<br>between 11 s and 330 s depending on UE capabilities | 300ms | Time + TPR: between 11.3 s and 330.3 s depending on UE capabilities |
| 15.3.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA | Config 1 and Config 2:<br>Sub-test 1: Cell Timing Difference = 3μs± 238 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 86 Tc<br>Config 3:<br>Sub-test 1: Cell Timing Difference = 3μs± 137 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 39 Tc | 32 Tc | Config 1 and Config 2:<br>Sub-test 1: Cell Timing Difference = 3μs± 270 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 118 Tc<br>Config 3:<br>Sub-test 1: Cell Timing Difference = 3μs± 169 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 71 Tc |
| 15.3.2 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA | Sub-test 1: Cell Timing Difference = 3μs± 44 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 24 Tc | 32 Tc | Sub-test 1: Cell Timing Difference = 3μs± 76 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 56 Tc |
| 15.3.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer | Config 1 and Config 2:<br>Sub-test 1: Cell Timing Difference = 3μs± 139 Tc | 32 Tc | Config 1 and Config 2:<br>Sub-test 1: Cell Timing Difference = 3μs± 171 Tc |

| | | | |
|---|---|---|---|
| with reduced number of samples in FR1 SA | Sub-test 2: Cell Timing Difference = 3μs± 155 Tc<br>Config 3:<br>Sub-test 1: Cell Timing Difference = 3μs± 86 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 95 Tc | | Sub-test 2: Cell Timing Difference = 3μs± 187 Tc<br>Config 3:<br>Sub-test 1: Cell Timing Difference = 3μs± 118 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 127 Tc |
| 15.3.4 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR1 SA | Config 1 and Config 2:<br>Sub-test 1: Cell Timing Difference = 3μs± 292 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 98 Tc<br>Config 3:<br>Sub-test 1: Cell Timing Difference = 3μs± 155 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 48 Tc | 32 Tc | Config 1 and Config 2:<br>Sub-test 1: Cell Timing Difference = 3μs± 324 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 130 Tc<br>Config 3:<br>Sub-test 1: Cell Timing Difference = 3μs± 187 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 80 Tc |
| 15.3.5 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with reduced number of samples in FR2 SA | Cell Timing Difference = 3μs± 33 Tc | 32 Tc | Time + TPR: 3μs ± 65 Tc |
| 15.3.6 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer with RxTx TEG in FR2 SA | Sub-test 1: Cell Timing Difference = 3μs± 56 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 26 Tc | 32 Tc | Sub-test 1: Cell Timing Difference = 3μs± 88 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 58 Tc |
| 15.3.7 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR1 SA | Config 1 and Config 2:<br>Cell 1: Cell Timing Difference = 3μs± 38 Tc<br>Cell 2: Cell Timing Difference = 3μs± 40 Tc<br>Config 3:<br>Cell 1: Cell Timing Difference = 3μs± 30 Tc<br>Cell 2: Cell Timing Difference = 3μs± 32 Tc | 32 Tc | Config 1 and Config 2:<br>Cell 1: Cell Timing Difference = 3μs± 70 Tc<br>Cell 2: Cell Timing Difference = 3μs± 72 Tc<br>Config 3:<br>Cell 1: Cell Timing Difference = 3μs± 62 Tc<br>Cell 2: Cell Timing Difference = 3μs± 64 Tc |
| 15.3.8 UE Rx-Tx time difference measurement accuracy test case for PRS aggregation in FR2 SA | Config 1:<br>Cell 1: Cell Timing Difference = 3μs± 22 Tc<br>Cell 2: Cell Timing Difference = 3μs± 22 Tc | 32 Tc | Config 1:<br>Cell 1: Cell Timing Difference = 3μs± 54 Tc<br>Cell 2: Cell Timing Difference = 3μs± 54 Tc |
| 15.3.9 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH | Config 1&2&4:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 238 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 249 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 262 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 258 Tc<br>Config 3:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 137 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 129 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 138 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 135 Tc | 32 Tc | Config 1&2&4:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 270 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 281 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 293 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 290 TcConfig 3:<br>Config 3:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 169 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 161 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 170 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 167 Tc |
| 15.3.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH | Config 1:<br>Cell 1: Cell Timing Difference = 3μs±31 Tc<br>Cell 2: Cell Timing Difference = 3μs± 33 Tc | 32 Tc | Config 1:<br>Cell 1: Cell Timing Difference = 3μs± 63 Tc<br>Cell 2: Cell Timing Difference = 3μs± 65 Tc |

| | | | |
|---|---|---|---|
| 15.3.12 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH | Config 1:<br>Cell 1: Cell Timing Difference = 3μs±31 Tc<br>Cell 2: Cell Timing Difference = 3μs± 51 Tc | 32 Tc | Config 1:<br>Cell 1: Cell Timing Difference = 3μs± 63 Tc<br>Cell 2: Cell Timing Difference = 3μs± 83 Tc |
| 15.4.3 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response Time<br>between 21 s and 83 s depending on UE capabilities | 300 ms | Time + TPR: between 21.3 s and 83.3 s depending on UE capabilities |
| 15.4.4 UE Rx-Tx time difference measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response Time<br>between 6 s and 22 s depending on UE capabilities | 300 ms | Time + TPR: between 6.3 s and 22.3 s depending on UE capabilities |
| 15.4.5 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR1 SA | Response Time<br>Config 1&2:<br>between 6 s and 86 s depending on UE capabilities<br>Config 3:<br>between 12 s and 45 s depending on UE capabilities | 300 ms | Config 1&2:<br><br>Time + TPR: between 6.3 s and 86.3 s depending on UE capabilities<br>Config 3:<br>Time + TPR: between 12.3 s and 45.3 s |
| 15.4.6 UE Rx-Tx time difference measurement reporting delay for PRS aggregation in FR2 SA | Response Time<br>between 21 s and 83 s depending on UE capabilities | 300 ms | Time + TPR: between 21.3 s and 83.3 s depending on UE capabilities |
| 15.4.7 UE Rx-Tx time difference measurement reporting delay in FR1 for Redcap UE without Rx FH | Response Time<br>Config 1&2:<br>between 6 s and 84 s depending on UE capabilities<br>Config 3:<br>between 6 s and 43 s depending on UE capabilities | 300 ms | Config 1&2:<br><br>Time + TPR: between 6.3 s and 84.3 s depending on UE capabilities<br>Config 3:<br>Time + TPR: between 6.3 s and 43.3 s |
| 15.4.8 UE Rx-Tx time difference measurement reporting delay in FR2 for Redcap UE without Rx FH | Response Time<br>between 21 s and 83 s depending on UE capabilities | 300ms | Time + TPR: between 21.3 s and 83.3 s depending on UE capabilities |
| 15.5.1 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR1 SA | Config 1 and Config 2:<br>Sub-test 1: Cell 1 Timing Difference = 3μs± 238 Tc<br>Sub-test 2: Cell 1 Timing Difference = 3μs± 86 Tc<br>Sub-test 1: Cell 2 Timing Difference = 3μs± 261 Tc<br>Sub-test 2: Cell 2 Timing Difference = 3μs± 93 Tc<br><br>Config 3:<br>Sub-test 1: Cell 1 Timing Difference = 3μs± 137 Tc<br>Sub-test 2: Cell 1 Timing Difference = 3μs± 39 Tc<br>Sub-test 1: Cell 2 Timing Difference = 3μs± 138 Tc<br>Sub-test 2: Cell 2 Timing Difference = 3μs± 40 Tc | 32 Tc | Config 1 and Config 2:<br>Sub-test 1: Cell 1 Timing Difference = 3μs± 270 Tc<br>Sub-test 2: Cell 1 Timing Difference = 3μs± 118 Tc<br>Sub-test 1: Cell 2 Timing Difference = 3μs± 293 Tc<br>Sub-test 2: Cell 2 Timing Difference = 3μs± 125 Tc<br><br>Config 3:<br>Sub-test 1: Cell 1 Timing Difference = 3μs± 169 Tc<br>Sub-test 2: Cell 1 Timing Difference = 3μs± 71 Tc<br>Sub-test 1: Cell 2 Timing Difference = 3μs± 170 Tc<br>Sub-test 2: Cell 2 Timing Difference = 3μs± 72 Tc |
| 15.5.3 UE Rx-Tx time difference measurement accuracy for single positioning frequency layer in FR2 SA | Sub-test 1: Cell Timing Difference = 3μs± 44 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 24 Tc | 33 Tc | Sub-test 1: Cell Timing Difference = 3μs± 77 Tc<br>Sub-test 2: Cell Timing Difference = 3μs± 57 Tc |
| 15.5.4 UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR2 SA | Cell Timing Difference = 3μs± 33 Tc | 33 Tc | Cell Timing Difference = 3μs± 66 Tc |

| | | | |
|---|---|---|---|
| 15.5.7 UE Rx-Tx time difference measurement accuracy test case in FR1 for Redcap UE without Rx FH | Config 1&2&4:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 238 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 249 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 262 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 258 Tc<br>Config 3:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 137 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 129 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 138 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 135 Tc | 32 Tc | Config 1&2&4:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 270 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 281 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 293 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 290 TcConfig 3:<br>Config 3:<br>Cell 1 for 2RX redcap UE: Cell Timing Difference = 3μs± 169 Tc<br>Cell 1 for 1RX redcap UE: Cell Timing Difference = 3μs± 161 Tc<br>Cell 2 for 2RX redcap UE: Cell Timing Difference = 3μs± 170 Tc<br>Cell 2 for 1RX redcap UE: Cell Timing Difference = 3μs± 167 Tc |
| 15.5.8 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE without Rx FH | Config 1:<br>Cell 1: Cell Timing Difference = 3μs±31 Tc<br>Cell 2: Cell Timing Difference = 3μs± 33 Tc | 32 Tc | Config 1:<br>Cell 1: Cell Timing Difference = 3μs± 63 Tc<br>Cell 2: Cell Timing Difference = 3μs± 65 Tc |
| 15.5.10 UE Rx-Tx time difference measurement accuracy test case in FR2 for Redcap UE with Rx FH | Config 1:<br>Cell 1: Cell Timing Difference = 3μs±31 Tc<br>Cell 2: Cell Timing Difference = 3μs± 51 Tc | 32 Tc | Config 1:<br>Cell 1: Cell Timing Difference = 3μs± 63 Tc<br>Cell 2: Cell Timing Difference = 3μs± 83 Tc |

**Table C.4.7: Derivation of Test Requirements for PRS-RSRP Measurement tests**

| Test | Minimum Requirement in TS 38.133 | Test Parameter Relaxation (TPR) | Test Requirement in TS 37.571-1 |
|---|---|---|---|
| 16.2.1 PRS-RSRP measurement period test case for single positioning frequency layer in FR1 SA | Response Time between 1 s and 210 s depending on UE capabilities | 300 ms | Time + TPR: between 1.3 s and 210.3 s depending on UE capabilities |
| 16.2.2 PRS-RSRP measurement period test case for dual positioning frequency layer in FR1 SA | Response Time between 2 s and 104 s depending on UE capabilities | 300 ms | Time + TPR: between 2.3 s and 104.3 s depending on UE capabilities |
| 16.2.3 PRS-RSRP measurement period test case for single positioning frequency layer in FR2 SA | Response Time between 6 s and 170 s depending on UE capabilities | 300ms | Time + TPR: between 6.3 s and 170.3 s depending on UE capabilities |
| 16.2.4 PRS-RSRP measurement period test case for dual positioning frequency layer in FR2 SA | Response Time between 11 s and 84 s depending on UE capabilities | 300ms | Time + TPR: between 11.3 s and 84.3 s depending on UE capabilities |
| 16.2.5 PRS-RSRP measurement reporting delay test case for single positioning frequency with reduced number of samples in FR1 SA | Response Time between 1 s and 52 s depending on UE capabilities | 300ms | Time + TPR: between 1.3 s and 52.3 s depending on UE capabilities |
| 16.2.6 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Response Time between 1 s and 7 s depending on UE capabilities | 300ms | Time + TPR: between 1.3 s and 205.3 s depending on UE capabilities |
| 16.2.7 PRS-RSRP measurement reporting | Response Time | 300ms | Time + TPR: between 2.3 s and 21.3 s depending on UE capabilities |

| | | | |
|---|---|---|---|
| delay test case for single positioning frequency with reduced number of samples in FR2 SA | between 2 s and 21 s depending on UE capabilities | | |
| 16.2.8 PRS-RSRP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR2 SA | Response Time<br>Sub-test 1: between 2 s and 21 s depending on UE capabilities<br>Sub-test 2: between 6 s and 83 s depending on UE capabilities | 300ms | Sub-test 1: Time + TPR: between 2.3 s and 21.3 s depending on UE capabilities<br>Sub-test 2: Time + TPR: between 6.3 s and 83.3 s depending on UE capabilities |
| 16.3.1 PRS-RSRP measurement accuracy with PRS in FR1 | Absolute power, log: -98 dBm/15Hz<br>PRS Es/Noc Cell#1: -2<br>PRS Es/Noc Cell#2: -10<br><br>Test Configuration 1 and 2<br>Test 1<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±9.5<br><br>Test2<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±6<br>Reported relative PRP2 values: ±6.5<br><br>Test Configuration 3<br>Test 1<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±9.5<br><br>Test2<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±4.5<br>Reported relative PRP2 values: ±5 | 0<br>0<br>0<br>Via mapping | Absolute power, log: -98 dBm/15Hz<br>PRS Es/Noc Cell#1: -2<br>PRS Es/Noc Cell#2: -10<br><br>Test Configuration 1 and 2<br>Test 1<br>Reported absolute PRP1 : PRS_RSRP_51 to PRS_RSRP_62<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_38 to PRS_RSRP_59<br>Reported relative PRP2 : DIFFRSRP_21 to DIFFRSRP_30<br><br>Test 2<br>Reported absolute PRP1 : PRS_RSRP_51 to PRS_RSRP_62<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_41 to PRS_RSRP_56<br>Reported relative PRP2 : DIFFRSRP_24 to DIFFRSRP_30<br><br>Test Configuration 3<br>Test 1<br>Reported absolute PRP1 : PRS_RSRP_54 to PRS_RSRP_65<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_41 to PRS_RSRP_62<br>Reported relative PRP2 : DIFFRSRP_21 to DIFFRSRP_30<br><br>Test 2<br>Reported absolute PRP1 : PRS_RSRP_54 to PRS_RSRP_65<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_45 to PRS_RSRP_58<br>Reported relative PRP2 : DIFFRSRP_25 to DIFFRSRP_30 |
| 16.3.2 PRS-RSRP measurement accuracy with PRS in FR2 | Test 1:<br>Reported absolute PRP1 values: ±8.5<br>Reported relative PRP1 values: ±10<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±10<br><br>Test 2: | Via mapping | Test 1:<br>Reported absolute PRP1 : PRS_RSRP_54 to PRS_RSRP_85<br>Reported relative PRP1 : DIFFRSRP_22 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_46 to PRS_RSRP_77<br>Reported relative PRP2 : DIFFRSRP_20 to DIFFRSRP_30<br><br>Test 2: |

| | | | |
|---|---|---|---|
| | Reported absolute PRP1 values: ±6<br>Reported relative PRP1 values: ±7.5<br>Reported absolute PRP2 values: ±6<br>Reported relative PRP2 values: ±7.5 | | Reported absolute PRP1 : PRS_RSRP_54 to PRS_RSRP_85<br>Reported relative PRP1 : DIFFRSRP_24 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_46 to PRS_RSRP_77<br>Reported relative PRP2 : DIFFRSRP_23 to DIFFRSRP_30 |
| 16.3.3 PRS-RSRP measurement accuracy with PRS in FR1 with reduced sample number | Absolute power, log: -98 dBm/15Hz<br>PRS Es/Noc Cell#1: 2.23<br>PRS Es/Noc Cell#2: -1.73<br>Test 1<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±6<br>Reported relative PRP2 values: ±6.5<br><br>Test2<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±4.5<br>Reported relative PRP2 values: ±5 | 0<br>0.69<br>0.80<br>Via mapping | Absolute power, log: -98 dBm/15Hz<br>PRS Es/Noc Cell#1: 2.92<br>PRS Es/Noc Cell#2: -0.93<br>Test 1<br>Reported absolute PRP1 : PRS_RSRP_56 to PRS_RSRP_66<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_50 to PRS_RSRP_65<br>Reported relative PRP2 : DIFFRSRP_24 to DIFFRSRP_30<br><br>Test 2<br>Reported absolute PRP1 : PRS_RSRP_59 to PRS_RSRP_69<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_55 to PRS_RSRP_67<br>Reported relative PRP2 : DIFFRSRP_25 to DIFFRSRP_30 |
| 16.3.4 PRS-RSRP measurement accuracy with PRS in FR2 with reduced sample number | Reported absolute PRP1 values: ±5<br>Reported relative PRP1 values: ±5<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±10 | Via mapping | Reported absolute PRP1 : PRS_RSRP_66 to PRS_RSRP_76<br>Reported relative PRP1 : DIFFRSRP_25 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_61 to PRS_RSRP_73<br>Reported relative PRP2 : DIFFRSRP_23 to DIFFRSRP_30 |
| 16.4.1 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Response Time between 6 s and 53 s depending on UE capabilities | 300ms | Time + TPR: between 6.3 s and 207.3 s depending on UE capabilities |
| 16.4.2 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Response Time between 2 s and 15 s depending on UE capabilities | 300ms | Time + TPR: between 2.3 s and 53.3 s depending on UE capabilities |
| 16.4.3 PRS-RSRP measurement reporting delay test case for single positioning frequency layer in FR2 SA | Response Time between 21 s and 83 s depending on UE capabilities | 300ms | Time + TPR: between 21.3 s and 83.3 s depending on UE capabilities |
| 16.4.4 PRS-RSRP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Response Time between 6 s and 22 s depending on UE capabilities | 300ms | Time + TPR: between 6.3 s and 22.3 s depending on UE capabilities |
| 16.5.1 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 SA | Test Configuration 1 and 2<br>Test 1:<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5 | Via mapping | Test Configuration 1 and 2<br>Test 1:<br>Reported absolute PRP1 : PRS_RSRP_51 to PRS_RSRP_62<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30 |

| | | | |
|---|---|---|---|
| | Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±9.5<br><br>Test 2:<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±6<br>Reported relative PRP2 values: ±6.5<br><br>Test Configuration 3<br>Test 1:<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±9.5<br><br>Test 2:<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±4.5<br>Reported relative PRP2 values: ±5 | | Reported absolute PRP2 : PRS_RSRP_38 to PRS_RSRP_59<br>Reported relative PRP2 : DIFFRSRP_21 to DIFFRSRP_30<br><br>Test 2:<br>Reported absolute PRP1 : PRS_RSRP_51 to PRS_RSRP_62<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_41 to PRS_RSRP_56<br>Reported relative PRP2 : DIFFRSRP_24 to DIFFRSRP_30<br><br>Test Configuration 3<br>Test 1:<br>Reported absolute PRP1:PRS_RSRP_54 to PRS_RSRP_65<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_41 to PRS_RSRP_62<br>Reported relative PRP2 : DIFFRSRP_21 to DIFFRSRP_30<br><br>Test 2:<br>Reported absolute PRP1 : PRS_RSRP_54 to PRS_RSRP_65<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_45 to PRS_RSRP_58<br>Reported relative PRP2 : DIFFRSRP_25 to DIFFRSRP_30 |
| 16.5.2 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR1 with reduced sample number | Test Configuration 1 and 2<br>Test 1:<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±9.5<br><br>Test 2:<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±6<br>Reported relative PRP2 values: ±6.5<br><br>Test Configuration 3<br>Test 1:<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±8.5 | Via mapping | Test Configuration 1 and 2<br>Test 1:<br>Reported absolute PRP1 : PRS_RSRP_57 to PRS_RSRP_67<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_48 to PRS_RSRP_68<br>Reported relative PRP2 : DIFFRSRP_21 to DIFFRSRP_30<br><br>Test 2:<br>Reported absolute PRP1 : PRS_RSRP_57 to PRS_RSRP_67<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 : PRS_RSRP_50 to PRS_RSRP_65<br>Reported relative PRP2 : DIFFRSRP_24 to DIFFRSRP_30<br><br>Test Configuration 3<br>Test 1:<br>Reported absolute PRP1 : PRS_RSRP_60 to PRS_RSRP_70<br>Reported relative PRP1 : DIFFRSRP_27 to DIFFRSRP_30 |

| | | | |
|---|---|---|---|
| | Reported relative PRP2 values: ±9.5<br><br>Test 2：<br>Reported absolute PRP1 values: ±3.5<br>Reported relative PRP1 values: ±3.5<br>Reported absolute PRP2 values: ±4.5<br>Reported relative PRP2 values: ±5 | | Reported absolute PRP2 ：<br>PRS_RSRP_51 to PRS_RSRP_71<br>Reported relative PRP2 ：<br>DIFFRSRP_21 to DIFFRSRP_30<br><br>Test 2：<br>Reported absolute PRP1 ：<br>PRS_RSRP_60 to PRS_RSRP_70<br>Reported relative PRP1 ：<br>DIFFRSRP_27 to DIFFRSRP_30<br>Reported absolute PRP2 ：<br>PRS_RSRP_55 to PRS_RSRP_67<br>Reported relative PRP2 ：<br>DIFFRSRP_25 to DIFFRSRP_30 |
| 16.5.3 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 SA | Test 1：<br>Reported absolute PRP1 values: ±5<br>Reported relative PRP1 values: ±5<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±10<br><br>Test 2：<br>Reported absolute PRP1 values: ±5<br>Reported relative PRP1 values: ±5<br>Reported absolute PRP2 values: ±6<br>Reported relative PRP2 values: ±7.5 | Via mapping | Test 1：<br>Reported absolute PRP1 ：<br>PRS_RSRP_59 to PRS_RSRP_94<br>Reported relative PRP1 ：<br>DIFFRSRP_20 to DIFFRSRP_30<br>Reported absolute PRP2 ：<br>PRS_RSRP_52 to PRS_RSRP_86<br>Reported relative PRP2 ：<br>DIFFRSRP_20 to DIFFRSRP_30<br><br>Test 2：<br>Reported absolute PRP1 ：<br>PRS_RSRP_61 to PRS_RSRP_93<br>Reported relative PRP1 ：<br>DIFFRSRP_23 to DIFFRSRP_30<br>Reported absolute PRP2 ：<br>PRS_RSRP_53 to PRS_RSRP_85<br>Reported relative PRP2 ：<br>DIFFRSRP_23 to DIFFRSRP_30 |
| 16.5.4 PRS-RSRP measurement accuracy test case for single positioning frequency layer with PRS in FR2 with reduced sample number | Test 1：<br>Reported absolute PRP1 values: ±5<br>Reported relative PRP1 values: ±5<br>Reported absolute PRP2 values: ±8.5<br>Reported relative PRP2 values: ±10<br><br>Test 2：<br>Reported absolute PRP1 values: ±5<br>Reported relative PRP1 values: ±5<br>Reported absolute PRP2 values: ±6<br>Reported relative PRP2 values: ±7.5 | Via mapping | Test 1：<br>Reported absolute PRP1 ：<br>PRS_RSRP_66 to PRS_RSRP_76<br>Reported relative PRP1 ：<br>DIFFRSRP_25 to DIFFRSRP_30<br>Reported absolute PRP2 ：<br>PRS_RSRP_59 to PRS_RSRP_76<br>Reported relative PRP2 ：<br>DIFFRSRP_20 to DIFFRSRP_30<br><br>Test 2：<br>Reported absolute PRP1 ：<br>PRS_RSRP_66 to PRS_RSRP_76<br>Reported relative PRP1 ：<br>DIFFRSRP_25 to DIFFRSRP_30<br>Reported absolute PRP2 ：<br>PRS_RSRP_61 to PRS_RSRP_73<br>Reported relative PRP2 ：<br>DIFFRSRP_23 to DIFFRSRP_30 |

**Table C.4.8: Derivation of Test Requirements for PRS-RSRPP Measurement tests**

| Test | Minimum Requirement in TS 38.133 | Test Parameter Relaxation (TPR) | Test Requirement in TS 37.571-1 |
|---|---|---|---|
| 17.2.1 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR1 SA | Config 1and 2:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB | Config 1and 2:<br>0dB<br>0dB<br>0dB | Config 1and 2:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB |

| | | | |
|---|---|---|---|
| | Config 3:<br>Noc: -95dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Response Time<br>between 1 s and 31 s depending on UE capabilities | Config 3:<br>0dB<br>0dB<br>0dB<br><br>300 ms | Config 3<br>Noc: -95dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Time + TPR: between 1.3 s and 31.3 s depending on UE capabilities |
| 17.2.2 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR1 SA | Config 1and 2:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: 2.23dB<br>PRS Ês2 / Noc: 1.72dB<br><br>Config 3:<br>Noc: -95dBm/15kHz<br>PRS Ês1 / Noc: 2.23dB<br>PRS Ês2 / Noc: 1.72dB<br><br>Response Time<br>between 1 s and 8 s depending on UE capabilities | Config 1and 2:<br>-4.6dB<br>0.75dB<br>0.85dB<br><br>Config 3:<br>-4.6 dB<br>0.75dB<br>0.85dB<br><br>300 ms | Config 1and 2:<br>Noc: -102.6dBm/15kHz<br>PRS Ês1 / Noc: 2.98dB<br>PRS Ês2 / Noc: -0.87dB<br><br>Config 3<br>Noc: -99.6dBm/15kHz<br>PRS Ês1 / Noc: 2.98dB<br>PRS Ês2 / Noc: -0.87dB<br><br>Time + TPR: between 1.3 s and 8.3 s depending on UE capabilities |
| 17.2.3 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer without measurement gap in FR1 SA | Config 1and 2 sub-test 1:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Config 3 sub-test 1:<br>Noc: -95dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Config 1and 2 sub-test 2:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: 2.23dB<br>PRS Ês2 / Noc: 1.72dB<br><br>Config 3 sub-test 2:<br>Noc: -95dBm/15kHz<br>PRS Ês1 / Noc: 2.23dB<br>PRS Ês2 / Noc: 1.72dB<br><br>Response Time<br>Sub-test 1: between 1 s and 31 s depending on UE capabilities<br>Sub-test 2: between 1 s and 8 s depending on UE capabilities | Config 1and 2 sub-test 1:<br>0dB<br>0dB<br>0dB<br><br>Config 3 sub-test 1:<br>0dB<br>0dB<br>0dB<br><br>Config 1and 2 sub-test 2:<br>-4.6dB<br>0.75dB<br>0.85dB<br><br>Config 3 sub-test 2:<br>-4.6 dB<br>0.75dB<br>0.85dB<br><br>300ms | Config 1and 2 sub-test 1:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Config 3 sub-test 1:<br>Noc: -95dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Config 1and 2 sub-test 2:<br>Noc: -102.6dBm/15kHz<br>PRS Ês1 / Noc: 2.98dB<br>PRS Ês2 / Noc: -0.87dB<br><br>Config 3 sub-test 2:<br>Noc: -99.6dBm/15kHz<br>PRS Ês1 / Noc: 2.98dB<br>PRS Ês2 / Noc: -0.87dB<br><br>Sub-test 1: Time + TPR: between 1.3 s and 31.3 s depending on UE capabilities<br>Sub-test 2: Time + TPR: between 1.3 s and8.3 s depending on UE capabilities |
| 17.2.4 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer in FR2 SA | Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Response Time<br>between 6 s and 83 s depending on UE capabilities | -1.4dB<br>0dB<br>0dB<br><br>300 ms | Noc: -99.4dBm/15kHz<br>PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Time + TPR: between 6.3 s and 83.3 s depending on UE capabilities |
| 17.2.5 PRS-RSRPP measurement reporting delay test case for single positioning frequency layer with reduced number of samples in FR2 SA | Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: 2.23dB<br>PRS Ês2 / Noc: -1.73dB<br><br>Response Time<br>between 2 s and 21 s depending on UE capabilities | -8.45dB<br>4.85dB<br>7dB<br><br>300 ms | Noc: -106.45dBm/15kHz<br>PRS Ês1 / Noc: 7.08dB<br>PRS Ês2 / Noc: 5.27dB<br><br>Time + TPR: between 2.3 s and 21.3 s depending on UE capabilities |
| 17.2.6 PRS-RSRPP measurement reporting | Sub-test 1:<br>Noc: -98dBm/15kHz | Sub-test 1:<br>-1.4dB | Sub-test 1:<br>Noc: -99.4dBm/15kHz |

| | | | |
|---|---|---|---|
| delay test case for single positioning frequency layer without measurement gap in FR2 SA | PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Sub-test 2:<br>Noc: -98dBm/15kHz<br>PRS Ês1 / Noc: 2.23dB<br>PRS Ês2 / Noc: -1.73dB<br><br>Response Time<br>Sub-test 1: between 6 s and 83 s depending on UE capabilities<br>Sub-test 2: between 2 s and 21 s depending on UE capabilities | 0dB<br>0dB<br><br>Sub-test 2:<br>-8.45dB<br>4.85dB<br>7dB<br><br>300ms | PRS Ês1 / Noc: -2.00dB<br>PRS Ês2 / Noc: -10.00dB<br><br>Sub-test 2:<br>Noc: -106.45dBm/15kHz<br>PRS Ês1 / Noc: 7.08dB<br>PRS Ês2 / Noc: 5.27dB<br><br>Sub-test 1: Time + TPR: between 6.3 s and 83.3 s depending on UE capabilities<br>Sub-test 2: Time + TPR: between 2.3 s and 21.3 s depending on UE capabilities |

# Annex D (normative): Rules for statistical testing

## D.1 Test Method

In this clause the terms GNSS and A-GNSS also include the cases where the only satellite system used is GPS unless otherwise stated.

Each test is performed in the following manner:

a) Setup the required test conditions.

b) Start each repetition after having applied the message 'RESET UE POSITIONING STORED INFORMATION'. This ensures that each result is independent from the previous one.

c) Make the required measurement a repeated number of times. The results, measured, are simplified to:

good result, if the measured result is ≤ limit.

bad result, if the measured result is > limit

For the relevant A-GNSS test cases measure the 2D position and Time to First Fix (TTFF) a repeated number of times. Measure the 2D position and Time to First Fix (if applicable) repeated times. Start each repetition after having applied the message 'RESET UE POSITIONING STORED INFORMATION'. This ensures that each result is independent from the previous one. The results, measured, are simplified to:

good result, if the 2D position and TTFF are ≤ limit.

bad result, if the 2D position or TTFF or both are > limit

d) Record the number of results (ns) and the number of bad results (ne)

e) Stop the test at a pass or a fail event.

f) Once the test is stopped, decide according to the pass fail decision rules (D.4.2)

## D.2 Error Ratio (ER)

The Error Ratio (ER) is defined as the ratio of bad results (ne) to all results (ns).

(1-ER is the success ratio)

## D.3 Test Design

A statistical test is characterised by:

Test-time, Selectivity and Confidence level

### D.3.1 Confidence level

The outcome of a statistical test is a decision. This decision may be correct or in-correct. The Confidence Level CL describes the probability that the decision is a correct one. The complement is the wrong decision probability (risk) D = 1-CL

### D.3.2 Introduction: Supplier Risk versus Customer Risk

There are two targets of decision:

a) A measurement on the pass-limit shows, that the DUT has the specified quality or is better with probability CL (CL e.g.95%) This shall lead to a “pass decision”

The pass-limit is on the good side of the specified DUT-quality. A more stringent CL (CL e.g.99%) shifts the pass-limit further into the good direction. Given that the quality of the DUTs is distributed, a greater CL passes less and better DUTs.

A measurement on the bad side of the pass-limit is simply “not pass” (undecided)

aa) Complementary:

A measurement on the fail-limit shows, that the DUT is worse than the specified quality with probability CL.

The fail-limit is on the bad side of the specified DUT-quality. A more stringent CL shifts the fail-limit further into the bad direction. Given that the quality of the DUTs is distributed, a greater CL fails less and worse DUTs.

A measurement on the good side of the fail-limit is simply “not fail”.

b) A DUT, known to have the specified quality, shall be measured and decided pass with probability CL. This leads to the pass limit.

For CL e.g. 95%, the pass limit is on the bad side of the specified DUT-quality. CL e.g.99% shifts the pass-limit further into the bad direction. Given that the DUT-quality is distributed, a greater CL passes more and worse DUTs.

bb) A DUT, known to be an ($\varepsilon \rightarrow 0$) beyond the specified quality, shall be measured and decided fail with probability CL.

For CL e.g.95%, the fail limit is on the good side of the specified DUT-quality.

Note the different sense for CL in (a), (aa) versus (b), (bb).

NOTE: For constant CL in all 4 bullets, (a) is equivalent to (bb) and (aa) is equivalent to (b).

## D.3.3 Supplier Risk versus Customer Risk

The table below summarizes the different targets of decision.

**Table D.3.3: Equivalent statements**

| | Equivalent statements, using different cause-to-effect-directions, and assuming CL = constant >0.5 | |
|---|---|---|
| cause-to-effect-directions | Known measurement result → estimation of the DUT’s quality | Known DUT’s quality → estimation of the measurement’s outcome |
| Supplier Risk | **A measurement on the pass-limit shows, that the DUT has the specified quality or is better (a)** | A DUT, known to have an ($\varepsilon \rightarrow 0$) beyond the specified DUT-quality, shall be measured and decided fail (bb) |
| Customer Risk | A measurement on the fail-limit shall shows, that the DUT is worse than the specified quality (aa) | **A DUT, known to have the specified quality, shall be measured and decided pass (b)** |

NOTE: The bold text shows the obvious interpretation of Supplier Risk and Customer Risk. The same statements can be based on other DUT-quality-definitions.

## D.3.4 Introduction: Standard test versus early decision concept

In standard statistical tests, a certain number of results (ns) is predefined in advance of the test. After ns results the number of bad results (ne) is counted and the error ratio (ER) is calculated as ne/ns.

Applying statistical theory, a decision limit can be designed, against which the calculated ER is compared to derive the decision. Such a limit is one decision point and is characterised by:

- D: the wrong decision probability (a predefined parameter)
- ns: the number of results (a fixed predefined parameter)
- ne: the number of bad results (the limit based on just ns)

In the formula for the limit, D and ns are parameters and ne is the variable. In the standard test ns and D are constant. The property of such a test is: It discriminates between two states only, depending on the test design:

- pass (with CL) / undecided (undecided in the sense: finally undecided)
- fail (with CL) / undecided (undecided in the sense: finally undecided)
- pass (with CL) / fail (with CL) (however against two limits).

In contrast to the standard statistical tests, the early decision concept predefines a set of (ne, ns) co-ordinates, representing the limit-curve for decision. After each result a preliminary ER is calculated and compared against the limit-curve. After each result one may make the decision or not (undecided for later decision). The parameters and variables in the limit-curve for the early decision concept have a similar but not equal meaning:

- D: the wrong decision probability (a predefined parameter)
- ns: the number of results (a variable parameter)
- ne: the number of bad results (the limit. It varies together with ns)

To avoid a “final undecided” in the standard test, a second limit must be introduced and the single decision co-ordinate (ne, ns) needs a high ne, leading to a fixed (high) test time. In the early decision concept, having the same selectivity and the same confidence level an “undecided” does not need to be avoided, as it can be decided later. A perfect DUT will hit the decision coordinate (ne, ns) with ne=0. This test time is short.

## D.3.5 Standard test versus early decision concept

For Supplier Risk:
The wrong decision probability D in the standard test is the probability, to decide a DUT in-correctly in the single decision point. In the early decision concept there is a probability of in-correct decisions d at each point of the limit-curve. The sum of all those wrong decision probabilities accumulate to D. Hence $d<D$

For Customer Risk:
The correct decision probability CL in the standard test is the probability, to decide a DUT correctly in the single decision point. In the early decision concept there is a probability of correct decisions cl at each point of the limit-curve. The sum of all those correct decision probabilities accumulate to CL. Hence $cl<CL$ or $d>D$

## D.3.6 Selectivity

There is no statistical test which can discriminate between a limit-DUT-quality and a DUT-quality which is an ($\varepsilon \rightarrow 0$) apart from the limit in finite time and confidence level $CL>1/2$. Either the test discriminates against one limit with the results pass (with CL)/undecided or fail (with CL)/undecided, or the test ends in a result pass (with CL)/fail (with CL) but this requires a second limit.

For $CL>0.5$, a (measurement-result = specified-DUT-quality), generates undecided in test “supplier risk against pass limit” (a in clause D.3.2) and also in the equivalent test against the fail limit (aa in clause D.3.2)

For $CL>0.5$, a DUT, known to be on the limit, will be decided pass for the test “customer risk against pass limit” (b in clause D.3.2) and also in the equivalent test against fail limit (bb in clause D.3.2).

This overlap or undecided area is not a fault or a contradiction, however it can be avoided by introducing a Bad or a Good DUT quality according to:

- Bad DUT quality: specified DUT-quality * M ($M>1$)
- Good DUT quality: specified DUT-quality * m ($m<1$)

Using e.g. $M>1$ and CL=95% the test for different DUT qualities yield different pass probabilities:

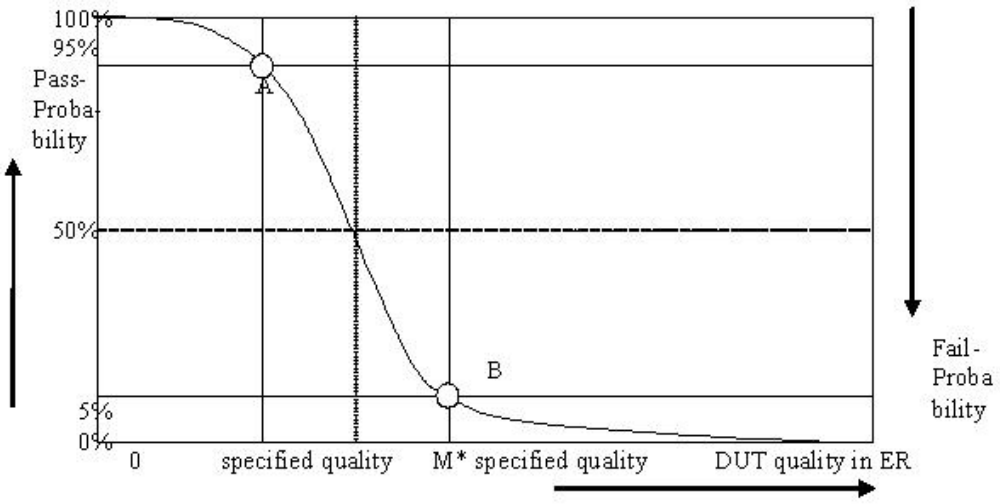



**Figure D.3.6: Pass probability versus DUT quality**

## D.3.7 Design of the test

The test is defined according to the following design principles:

1. The early decision concept is applied.
2. A second limit is introduced: Bad DUT factor M>1
3. To decide the test pass:

   Supplier risk is applied based on the Bad DUT quality

To decide the test fail

   Customer Risk is applied based on the specified DUT quality

The A-GNSS test cases are defined using the following parameters:

1. Specified DUT quality: ER = 0.05
2. Bad DUT quality: M=1.5 (selectivity)
3. Confidence level CL = 95% (for specified DUT and Bad DUT-quality)

The ECID and OTDOA test cases are defined using the following parameters:

1. Specified DUT quality: ER = 0.1
2. Bad DUT quality: M=1.5 (selectivity)
3. Confidence level CL = 95% (for specified DUT and Bad DUT-quality)

This has the following consequences:

a) A measurement on the fail limit is connected with 2 equivalent statements:

| | |
|---|---|
| A measurement on the fail-limit shows, that the DUT is worse than the specified DUT-quality | A DUT, known to have the specified quality, shall be measured and decided pass |

A measurement on the pass limit is connected with the complementary statements:

| | |
|---|---|
| A measurement on the pass limit shows, that the DUT is better than the Bad DUT-quality. | A DUT, known to have the Bad DUT quality, shall be measured and decided fail |

The left column is used to decide the measurement.

The right column is used to verify the design of the test by simulation.

The simulation is based on the two fulcrums A and B only in Figure D.3.6. There is freedom to shape the remainder of the function.

b) Test time

1. The minimum and maximum test time is fixed.
2. The average test time is a function of the DUT's quality.
3. The individual test time is not predictable (except ideal DUT).

c) The number of decision co-ordinates (ne, ns) in the early decision concept is responsible for the selectivity of the test and the maximum test time. Having fixed the number of decision co-ordinates there is still freedom to select the individual decision co-ordinates in many combinations, all leading to the same confidence level.

# D.4 Pass fail decisions

## D.4.1 Numerical definition of the pass fail limits for A-GNSS test cases

**Table D.4.1: Pass/fail limits for A-GNSS test cases**

| ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 77 | NA | 43 | 855 | 576 | 86 | 1525 | 1297 | 129 | 2173 | 2050 |
| 1 | 106 | NA | 44 | 871 | 592 | 87 | 1540 | 1314 | 130 | 2188 | 2067 |
| 2 | 131 | NA | 45 | 887 | 608 | 88 | 1556 | 1331 | 131 | 2203 | 2085 |
| 3 | 154 | NA | 46 | 903 | 625 | 89 | 1571 | 1349 | 132 | 2218 | 2103 |
| 4 | 176 | NA | 47 | 919 | 641 | 90 | 1586 | 1366 | 133 | 2233 | 2121 |
| 5 | 197 | NA | 48 | 935 | 657 | 91 | 1601 | 1383 | 134 | 2248 | 2139 |
| 6 | 218 | 42 | 49 | 951 | 674 | 92 | 1617 | 1401 | 135 | 2263 | 2156 |
| 7 | 238 | 52 | 50 | 967 | 690 | 93 | 1632 | 1418 | 136 | 2277 | 2174 |
| 8 | 257 | 64 | 51 | 982 | 706 | 94 | 1647 | 1435 | 137 | 2292 | 2192 |
| 9 | 277 | 75 | 52 | 998 | 723 | 95 | 1662 | 1453 | 138 | 2307 | 2210 |
| 10 | 295 | 87 | 53 | 1014 | 739 | 96 | 1677 | 1470 | 139 | 2322 | 2227 |
| 11 | 314 | 100 | 54 | 1030 | 756 | 97 | 1692 | 1487 | 140 | 2337 | 2245 |
| 12 | 333 | 112 | 55 | 1046 | 772 | 98 | 1708 | 1505 | 141 | 2352 | 2263 |
| 13 | 351 | 125 | 56 | 1061 | 789 | 99 | 1723 | 1522 | 142 | 2367 | 2281 |
| 14 | 369 | 139 | 57 | 1077 | 805 | 100 | 1738 | 1540 | 143 | 2381 | 2299 |
| 15 | 387 | 152 | 58 | 1093 | 822 | 101 | 1753 | 1557 | 144 | 2396 | 2317 |
| 16 | 405 | 166 | 59 | 1108 | 839 | 102 | 1768 | 1574 | 145 | 2411 | 2335 |
| 17 | 422 | 180 | 60 | 1124 | 855 | 103 | 1783 | 1592 | 146 | 2426 | 2352 |
| 18 | 440 | 194 | 61 | 1140 | 872 | 104 | 1798 | 1609 | 147 | 2441 | 2370 |
| 19 | 457 | 208 | 62 | 1155 | 889 | 105 | 1813 | 1627 | 148 | 2456 | 2388 |
| 20 | 474 | 222 | 63 | 1171 | 906 | 106 | 1828 | 1644 | 149 | 2470 | 2406 |
| 21 | 492 | 237 | 64 | 1186 | 922 | 107 | 1844 | 1662 | 150 | 2485 | 2424 |
| 22 | 509 | 251 | 65 | 1202 | 939 | 108 | 1859 | 1679 | 151 | 2500 | 2442 |
| 23 | 526 | 266 | 66 | 1217 | 956 | 109 | 1874 | 1697 | 152 | 2515 | 2460 |
| 24 | 543 | 281 | 67 | 1233 | 973 | 110 | 1889 | 1714 | 153 | 2530 | 2478 |
| 25 | 560 | 295 | 68 | 1248 | 990 | 111 | 1904 | 1732 | 154 | 2544 | 2496 |
| 26 | 577 | 310 | 69 | 1264 | 1007 | 112 | 1919 | 1750 | 155 | 2559 | 2513 |
| 27 | 593 | 325 | 70 | 1279 | 1024 | 113 | 1934 | 1767 | 156 | 2574 | 2531 |
| 28 | 610 | 341 | 71 | 1295 | 1040 | 114 | 1949 | 1785 | 157 | 2589 | 2549 |
| 29 | 627 | 356 | 72 | 1310 | 1057 | 115 | 1964 | 1802 | 158 | 2603 | 2567 |
| 30 | 643 | 371 | 73 | 1326 | 1074 | 116 | 1979 | 1820 | 159 | 2618 | 2585 |
| 31 | 660 | 387 | 74 | 1341 | 1091 | 117 | 1994 | 1838 | 160 | 2633 | 2603 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 32 | 676 | 402 | 75 | 1357 | 1108 | 118 | 2009 | 1855 | 161 | 2648 | 2621 |
| 33 | 693 | 418 | 76 | 1372 | 1126 | 119 | 2024 | 1873 | 162 | 2662 | 2639 |
| 34 | 709 | 433 | 77 | 1387 | 1143 | 120 | 2039 | 1890 | 163 | 2677 | 2657 |
| 35 | 725 | 449 | 78 | 1403 | 1160 | 121 | 2054 | 1908 | 164 | 2692 | 2675 |
| 36 | 742 | 465 | 79 | 1418 | 1177 | 122 | 2069 | 1926 | 165 | 2707 | 2693 |
| 37 | 758 | 480 | 80 | 1433 | 1194 | 123 | 2084 | 1943 | 166 | 2721 | 2711 |
| 38 | 774 | 496 | 81 | 1449 | 1211 | 124 | 2099 | 1961 | 167 | 2736 | 2729 |
| 39 | 790 | 512 | 82 | 1464 | 1228 | 125 | 2114 | 1979 | 168 | 2751 | 2747 |
| 40 | 807 | 528 | 83 | 1479 | 1245 | 126 | 2128 | 1997 | 169 | 2765 | NA |
| 41 | 823 | 544 | 84 | 1495 | 1263 | 127 | 2143 | 2014 | | | |
| 42 | 839 | 560 | 85 | 1510 | 1280 | 128 | 2158 | 2032 | | | |

NOTE: The first column is the number of bad results (ne)
The second column is the number of results for the pass limit ($ns_p$)
The third column is the number of results for the fail limit ($ns_f$)

## D.4.2 Pass fail decision rules for A-GNSS test cases

Having observed 0 bad results, pass the test at ≥77 results, otherwise continue

Having observed 1 bad result, pass the test at ≥106 results, otherwise continue

Having observed 2 bad results, pass the test at ≥131 results, otherwise continue

etc. until

Having observed 6 bad results, pass the test at ≥218 results, fail the test at ≤ 42 results, otherwise continue

Having observed 7 bad results, pass the test at ≥238 results, fail the test at ≤ 52 results, otherwise continue

etc. until

Having observed 168 bad results, pass the test at ≥2751 results, fail the test at ≤2747 results, otherwise continue

Having observed 169 bad results, pass the test at ≥2765 results, otherwise fail

NOTE: an ideal DUT passes after 77 results. The maximum test time is 2765 results.

## D.4.3 Numerical definition of the pass fail limits for ECID, OTDOA, MBS, WLAN and BLE test cases

**Table D.4.3: Pass/fail limits for ECID, OTDOA, MBS, WLAN and BLE test cases**

| ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 33 | NA | 43 | 408 | 283 | 86 | 737 | 644 | 129 | 1056 | 1021 |
| 1 | 46 | NA | 44 | 416 | 291 | 87 | 745 | 653 | 130 | 1064 | 1030 |
| 2 | 58 | 2 | 45 | 424 | 299 | 88 | 752 | 661 | 131 | 1071 | 1039 |
| 3 | 69 | 5 | 46 | 432 | 307 | 89 | 760 | 670 | 132 | 1078 | 1048 |
| 4 | 79 | 8 | 47 | 440 | 315 | 90 | 767 | 679 | 133 | 1086 | 1057 |
| 5 | 89 | 12 | 48 | 447 | 324 | 91 | 775 | 687 | 134 | 1093 | 1066 |
| 6 | 99 | 17 | 49 | 455 | 332 | 92 | 782 | 696 | 135 | 1100 | 1074 |
| 7 | 109 | 22 | 50 | 463 | 340 | 93 | 790 | 705 | 136 | 1108 | 1083 |
| 8 | 118 | 27 | 51 | 471 | 348 | 94 | 797 | 713 | 137 | 1115 | 1092 |
| 9 | 127 | 33 | 52 | 478 | 356 | 95 | 804 | 722 | 138 | 1122 | 1101 |
| 10 | 136 | 39 | 53 | 486 | 365 | 96 | 812 | 731 | 139 | 1130 | 1110 |
| 11 | 145 | 45 | 54 | 494 | 373 | 97 | 819 | 739 | 140 | 1137 | 1119 |
| 12 | 154 | 51 | 55 | 502 | 381 | 98 | 827 | 748 | 141 | 1144 | 1128 |
| 13 | 163 | 58 | 56 | 509 | 389 | 99 | 834 | 757 | 142 | 1152 | 1137 |
| 14 | 172 | 64 | 57 | 517 | 398 | 100 | 842 | 766 | 143 | 1159 | 1147 |
| 15 | 180 | 71 | 58 | 525 | 406 | 101 | 849 | 774 | 144 | 1166 | 1155 |
| 16 | 189 | 78 | 59 | 532 | 414 | 102 | 857 | 783 | 145 | 1174 | 1164 |
| 17 | 197 | 85 | 60 | 540 | 423 | 103 | 864 | 792 | 146 | 1181 | 1173 |
| 18 | 206 | 92 | 61 | 548 | 431 | 104 | 871 | 801 | 147 | NA | 1182 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 19 | 214 | 99 | 62 | 555 | 440 | 105 | 879 | 809 | 148 | | |
| 20 | 223 | 106 | 63 | 563 | 448 | 106 | 886 | 818 | 149 | | |
| 21 | 231 | 113 | 64 | 571 | 456 | 107 | 894 | 827 | 150 | | |
| 22 | 239 | 120 | 65 | 578 | 465 | 108 | 901 | 836 | 151 | | |
| 23 | 248 | 128 | 66 | 586 | 473 | 109 | 909 | 844 | 152 | | |
| 24 | 256 | 135 | 67 | 594 | 482 | 110 | 916 | 853 | 153 | | |
| 25 | 264 | 142 | 68 | 601 | 490 | 111 | 923 | 862 | 154 | | |
| 26 | 272 | 150 | 69 | 609 | 499 | 112 | 931 | 871 | 155 | | |
| 27 | 281 | 157 | 70 | 616 | 507 | 113 | 938 | 880 | 156 | | |
| 28 | 289 | 165 | 71 | 624 | 516 | 114 | 946 | 888 | 157 | | |
| 29 | 297 | 173 | 72 | 632 | 524 | 115 | 953 | 897 | 158 | | |
| 30 | 305 | 180 | 73 | 639 | 533 | 116 | 960 | 906 | 159 | | |
| 31 | 313 | 188 | 74 | 647 | 541 | 117 | 968 | 915 | 160 | | |
| 32 | 321 | 196 | 75 | 654 | 550 | 118 | 975 | 924 | 161 | | |
| 33 | 329 | 204 | 76 | 662 | 558 | 119 | 983 | 933 | 162 | | |
| 34 | 337 | 211 | 77 | 669 | 567 | 120 | 990 | 941 | 163 | | |
| 35 | 345 | 219 | 78 | 677 | 575 | 121 | 997 | 950 | 164 | | |
| 36 | 353 | 227 | 79 | 684 | 584 | 122 | 1005 | 959 | 165 | | |
| 37 | 361 | 235 | 80 | 692 | 592 | 123 | 1012 | 968 | 166 | | |
| 38 | 369 | 243 | 81 | 700 | 601 | 124 | 1019 | 977 | 167 | | |
| 39 | 377 | 251 | 82 | 707 | 610 | 125 | 1027 | 986 | 168 | | |
| 40 | 385 | 259 | 83 | 715 | 618 | 126 | 1034 | 994 | 169 | | |
| 41 | 393 | 267 | 84 | 722 | 627 | 127 | 1042 | 1003 | | | |
| 42 | 400 | 275 | 85 | 730 | 635 | 128 | 1049 | 1012 | | | |

The first column is the number of errors (ne = number of exceeded delays or number of wrong reports)

The second column is the number of samples for the pass limit ($ns_p$ , ns=Number of samples= number of successes + number of exceedings or number of reports)

The third column is the number of samples for the fail limit ($ns_f$)

## D.4.4 Pass fail decision rules for ECID, OTDOA, MBS, WLAN and BLE test cases

Having observed 0 errors, pass the test at 33+ samples, otherwise continue

Having observed 1 error, pass the test at 46+ samples, otherwise continue

Having observed 2 errors, pass the test at 58+ samples, fail the test at 2 samples, otherwise continue

Having observed 146 errors, pass the test at 1181+ samples, fail the test at 1173- samples, otherwise continue

Having observed 147 errors, fail the test at 1182- samples,

Where x+ means: x or more, x- means x or less

NOTE: an ideal DUT passes after 33 samples. The maximum test time is 1181 samples.

## D.4.5 Background information to the pass fail limits

There is freedom to design the decision co-ordinates (ne, ns).

The binomial distribution and its inverse is used to design the pass and fail limits. Note that this method is not unique and that other methods exist.

$$fail(ne, d_f) := \frac{ne}{(ne + qnbinom(d_f, ne, ER))}$$

$$pass(ne, cl_p, M) := \frac{ne}{(ne + qnbinom(cl_p, ne, ER \cdot M))}$$

Where

fail(..) is the error ratio for the fail limit

pass(..) is the error ratio for the pass limit

ER is the specified error ratio e.g. 0.05

ne is the number of bad results. This is the variable in both equations

M is the Bad DUT factor M=1.5

$d_f$ is the wrong decision probability of a single (ne, ns) co-ordinate for the fail limit.
It is found by simulation to be $d_f = 0.004$

$cl_p$ is the confidence level of a single (ne, ns) co-ordinate for the pass limit.
It is found by simulation to be $cl_p = 0.9975$

qnbinom(..): The inverse cumulative function of the negative binomial distribution

The simulation works as follows:

A large population of limit DUTs with true ER = 0.05 is decided against the pass and fail limits.

$cl_p$ and $d_f$ are tuned such that CL (95%) of the population passes and D (5%) of the population fails.

A population of Bad DUTs with true ER = M*0.05 is decided against the same pass and fail limits.

$cl_p$ and $d_f$ are tuned such that CL (95%) of the population fails and D (5%) of the population passes.

This procedure and the relationship to the measurement is justified in clause D.3.7. The number of DUTs decreases during the simulation, as the decided DUTs leave the population. That number decreases with an approximately exponential characteristic. After 169 bad results all DUTs of the population are decided.

NOTE: The exponential decrease of the population is an optimal design goal for the decision co-ordinates (ne, ns), which can be achieved with other formulas or methods as well.

# Annex E (normative): Conditions for ECID and OTDOA requirements

## E.1 Conditions for E-CID UE Rx – Tx time difference Measurements

This clause defines the E-UTRAN RSRP applicable for ECID UE Rx-Tx time difference Measurements for a corresponding operating band

The conditions for E-UTRAN ECID UE Rx-Tx time difference measurements are as defined in Table E.1-1.

**Table E.1-1: Conditions for ECID UE Rx-Tx time difference measurements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum RSRP Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -127 |
| | FDD_B | -126.5 |
| | FDD_C, TDD_C | -126 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 2 |
| | FDD_G | -124 |
| | FDD_H | -123.5 |
| | FDD_N | -120.5 |
| NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3 of TS 36.133 [23]. | | |
| NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 3: E-UTRA operating band groups are as defined in clause 4.4.2. | | |

### E.1.1 Conditions for E-CID UE Rx – Tx time difference by UE Category M1/M2

This clause defines the E-UTRAN RSRP applicable for ECID UE Rx-Tx time difference Measurements for a corresponding operating band for UE Category M1 and M2.

The conditions for CE mode A intra-frequency E-UTRAN FDD, HD-FDD and TDD measurements are defined in Table E.1.1-1.

**Table B.2.14-1: E-UTRAN ECID UE Rx-Tx time difference measurements for FDD, HD-FDD and TDD for CE mode A**

| Parameter | E-UTRA operating band groups Note 3 | Minimum RSRP Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD-M1_A, TDD-M1_A | -127 |
| | FDD-M1_D | -125.5 |
| | FDD-M1_E, TDD-M1_E | -125 |
| | FDD-M1_F | -124.5 Note 2 |
| | FDD-M1_G | -124 |
| | FDD-M1_N | -120.5 |
| NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3 of TS 36.133 [23]. | | |
| NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 3: E-UTRA operating band groups are as defined in clause 4.4.2. | | |

# E.2 Conditions for OTDOA intra-frequency RSTD Measurements

This clause defines the E-UTRAN intra-frequency PRP_1,2 applicable for OTDOA intra-frequency RSTD measurements for a corresponding operating band.

The conditions for E-UTRAN OTDOA intra-frequency RSTD measurements are as defined in Table E.2-1

**Table E.2-1: Conditions for OTDOA intra-frequency RSTD measurements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum PRP1,2 Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -127 |
| | FDD_B | -126.5 |
| | FDD_C, TDD_C | -126 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 2 |
| | FDD_G | -124 |
| | FDD_H | -123.5 |
| | FDD_N | -120.5 |
| NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3 of TS 36.133[23]. | | |
| NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 3: E-UTRA operating band groups are as defined in clause 4.4.2. | | |

## E.2.1 Conditions for OTDOA intra-frequency RSTD Measurements by UE Category M1 and M2

This clause defines the E-UTRAN intra-frequency PRP1,2 applicable for a corresponding operating band.

The conditions for CE mode A intra-frequency E-UTRAN HD-FDD, FDD and TDD measurements are defined in Table E.2.1-1.

The conditions for CE mode B for intra-frequency E-UTRAN HD-FDD, FDD and TDD measurements are defined in Table E.2.1-2.

**Table E.2.1-1: E-UTRAN intra-frequency measurements for HD-FDD, FDD and TDD for CE mode A**

| Parameter | E-UTRA operating band groups Note 2 | Minimum PRP1,2 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -127 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 1 |
| | FDD_G | -124 |
| | FDD_N | -120.5 |
| NOTE 1: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 2: E-UTRA operating band groups are as defined in Section 4.4.2. | | |

**Table E.2.1-2: E-UTRAN intra-frequency measurements for HD-FDD, FDD and TDD for CE mode B**

| Parameter | E-UTRA operating band groups Note 2 | Minimum PRP1,2 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -136 |
| | FDD_D | -134.5 |
| | FDD_E, TDD_E | -134 |
| | FDD_F | -133.5 Note 1 |

| | | |
|---|---|---|
| | FDD_G | -133 |
| | FDD_N | -129.5 |
| NOTE 1: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 2: E-UTRA operating band groups are as defined in Section 4.4.2. | | |

# E.3 Conditions for OTDOA inter-frequency RSTD Measurements

This clause defines the E-UTRAN inter-frequency PRP_1,2 applicable for OTDOA Inter-frequency RSTD measurements for a corresponding operating band.

The conditions for E-UTRAN OTDOA inter-frequency RSTD measurements are as defined in Table E.2-1.

## E.3.1 Conditions for OTDOA inter-frequency RSTD Measurements by UE Category M1 and M2

This clause defines the E-UTRAN inter-frequency PRP1,2 applicable for a corresponding operating band.

The conditions for CE mode A inter-frequency E-UTRAN HD-FDD, FDD and TDD measurements are defined in Table E.3.1-1.

The conditions for CE mode B for inter-frequency E-UTRAN HD-FDD, FDD and TDD measurements are defined in Table E.3.1-2.

**Table E.3.1-1: E-UTRAN inter-frequency measurements for HD-FDD, FDD and TDD for CE mode A**

| **Parameter** | **E-UTRA operating band groups** Note 2 | **Minimum PRP1,2** |
|---|---|---|
| | | **dBm/15kHz** |
| **Conditions** | FDD_A, TDD_A | -127 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 1 |
| | FDD_G | -124 |
| | FDD_N | -120.5 |
| NOTE 1: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 2: E-UTRA operating band groups are as defined in Section 4.4.2. | | |

**Table E.3.1-2: E-UTRAN inter-frequency measurements for HD-FDD, FDD and TDD for CE mode B**

| **Parameter** | **E-UTRA operating band groups** Note 2 | **Minimum PRP1,2** |
|---|---|---|
| | | **dBm/15kHz** |
| **Conditions** | FDD_A, TDD_A | -136 |
| | FDD_D | -134.5 |
| | FDD_E, TDD_E | -134 |
| | FDD_F | -133.5 Note 1 |
| | FDD_G | -133 |
| | FDD_N | -129.5 |
| NOTE 1: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 2: E-UTRA operating band groups are as defined in Section 4.4.2. | | |

# E.4 Conditions for UE Rx-Tx Time Difference Measurement under Time Domain Measurement Resource Restriction with CRS Assistance Information

This clause defines the E-UTRAN RSRP applicable for UE Rx-Tx Time Difference Measurement under Time Domain Measurement Resource Restriction with CRS Assistance Information for a corresponding operating band.

The conditions for UE Rx-Tx time difference measurements, when time domain measurement resource restriction pattern and CRS assistance information are provided, are as defined in Table E.1-1.

# E.5 Conditions for NB-IOT OTDOA intra-frequency RSTD Measurements

This clause defines the NB-IoT intra-frequency PRP1,2 applicable for a corresponding operating band.

The conditions for intra-frequency RSTD measurements in normal coverage are defined in Table E.5-1.

The conditions for intra-frequency RSTD measurements in enhanced coverage are defined in Table E.5-2.

**Table E.5-1: NB-IoT intra-frequency RSTD measurements for HD-FDD in normal coverage**

| Parameter | E-UTRA operating band groups Note 1 | Minimum PRP1,2 Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | NFDD_G | -129 |
| NOTE 1: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1 | | |

**Table E.5-2: NB-IoT intra-frequency RSTD measurements for HD-FDD in enhanced coverage**

| Parameter | E-UTRA operating band groups Note 1 | Minimum PRP1,2 Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | NFDD_G | -135 |
| NOTE 1: E-UTRA operating band groups for NB-IoT are as defined in Section 4.11.1 | | |

# E.6 Conditions for NB-IOT OTDOA inter-frequency RSTD Measurements

This clause defines the NB-IoT intra-frequency PRP1,2 applicable for a corresponding operating band.

The conditions for inter-frequency RSTD measurements in normal coverage are defined in Table E.5-1.

The conditions for inter-frequency RSTD measurements in enhanced coverage are defined in Table E.5-2.

# Annex F (normative): UTRAN Generic procedures

## F.1 General

This normative annex specifies the set up and release procedure that shall be used for each UTRAN test case.

In this clause the terms GNSS and A-GNSS also include the cases where the only satellite system used is GPS unless otherwise stated.

## F.2 UTRAN connection set up

### F.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters. The default system information, as specified in clause 6.1 of TS 34.108 [28], is broadcast with the exceptions of SIB15, SIB15.1, SIB15.2 and SIB15.3 which are not broadcast.

User Equipment:

- The UE shall be operated in Normal Propagation Conditions as specified in clause 5.2.1 of TS 34.108 [28].
- The UE is in state "MM idle" state with valid TMSI and CKSN.
- The UE is in state "PMM idle" with valid P-TMSI.

### F.2.2 Procedures

CS Domain

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | <-- | | PAGING TYPE1 (PCCH) | Paging (CS domain, TMSI) |
| 3 | --> | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | <-- | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | --> | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | --> | | PAGING RESPONSE | RR |
| 7 | <-- | | AUTHENTICATION REQUEST | MM |
| 8 | --> | | AUTHENTICATION RESPONSE | MM |
| 9 | <-- | | SECURITY MODE COMMAND | RRC |
| 10 | --> | | SECURITY MODE COMPLETE | RRC |

PS Domain

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | PAGING TYPE1 (PCCH) | Paging (PS domain, PMSI or IMSI) |
| 2 | --> | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | <-- | | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | --> | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC (Transport Channel: DCH or FACH) |
| 5 | --> | | SERVICE REQUEST | GMM |
| 6 | <-- | | AUTHENTICATION REQUEST | GMM |
| 7 | --> | | AUTHENTICATION RESPONSE | GMM |
| 8 | <-- | | SECURITY MODE COMMAND | RRC |

| | | | |
|---|---|---|---|
| 9 | --> | SECURITY MODE COMPLETE | RRC |

## F.2.3 Specific message contents

The default message contents specified in clause 9.1 of TS 34.108 [28] will be used for the Moving Scenario and Periodic Update test. For all Minimum Performance TTFF Tests the default message contents specified in clause 9.1 of TS 34.108 [28] will be used with the following exception.

Contents of PAGING TYPE1:

| Information Element | Value/remark |
|---|---|
| Paging Cause | Terminating High Priority Signalling |

Contents of RRC CONNECTION SETUP:

For A-GNSS performance testing in CELL_DCH state: The RRC Connection Setup is defined in clause 9.1.1 of TS 34.108 [28] "Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH)".

For A-GNSS performance testing in CELL_FACH state: The RRC Connection Setup is defined in clause 9.1.1 of TS 34.108 [28] "Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH)".

Contents of RRC CONNECTION SETUP COMPLETE:

| Information Element | Value/remark |
|---|---|
| UE radio access capability<br>- UE positioning capability | Defines the A-GNSS mode the UE supports (UE-based, UE-assisted, or both). UE shall be tested for all modes it supports. |

# F.3 UTRAN connection release

## F.3.1 Procedure

**Table F.3.1: UTRAN connection release procedure**

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | RRC CONNECTION RELEASE | RRC |
| 2 | --> | | RRC CONNECTION RELEASE COMPLETE | RRC |

## F.3.2 Specific message contents

The default message contents specified in clause 9.1 of TS 34.108 [28] are used.

# Annex G (normative): Environmental conditions

## G.1 General

This normative annex specifies the environmental requirements of the UE. Within these limits the requirements of the present documents shall be fulfilled.

## G.2 Environmental requirements

The requirements in this clause apply to all types of UE(s).

### G.2.1 Temperature

The UE shall fulfil all the requirements in the full temperature range of:

**Table G.2.1.1**

| | |
|---|---|
| +15°C to +35°C | for normal conditions (with relative humidity up to 75 %) |

### G.2.2 Voltage

The UE shall fulfil all the requirements in the full voltage range, i.e. the voltage range between the extreme voltages.

The manufacturer shall declare the lower and higher extreme voltages and the approximate shutdown voltage. For the equipment that can be operated from one or more of the power sources listed below, the lower extreme voltage shall not be higher, and the higher extreme voltage shall not be lower than that specified below.

**Table G.2.2.1**

| Power source | Normal conditions voltage |
|---|---|
| AC mains | nominal |
| Regulated lead acid battery | 1.1 × nominal |
| Non regulated batteries:<br>- Leclanché / lithium<br>- Mercury/nickel and cadmium | <br>Nominal<br>Nominal |

# Annex H (informative): MBS Beacon parameters

## H.1 General

This informative annex consolidates a description of the MBS beacon parameters that are specified in the MBS ICD [38]. They are provided here for reference.

## H.2 Beacon parameters

This is a summary of the MBS beacon parameters to be used for MBS testing:

**Beacon PN Code:** 1023 chip length for TB1. Actual PN codes are listed it the MBS ICD [38]

**MBS Beacon Configuration:** TB1 (2.046 MHz, contains data) [38]

**MBS Packet Type:** Type 2 [38]

**MBS Transmitter ID (TxID):** Field used to signal a unique ID that identifies each transmitter within one major deployment area, such as within North America. Range: [0, $2^{15}$-1] [38]

**Slot Index:** This is the physical time slot within a MBS transmission period, in which a transmitter is transmitting. Each slot is 100 ms in duration and a MBS transmission period is 1 sec long. [38]

**All other fields:** Set to the min value (bit value equal to 0) for testing [38]

# Annex I (normative): Conditions for NR PRS-based measurements

## I.1 General

This clause defines the following conditions for NR PRS-based measurements and corresponding procedures performed based on PRS: PRP and PRS Ês/Iot, applicable for a corresponding operating band.

## I.2 General

The conditions are defined in Table I.2-1 for FR1 NR cells.

The conditions are defined in Table I.2-2 for FR2 NR cells.

Editor's notes for Table I.2-1and Table I.2-2:

- The Table is not complete, FFS for PRS-RSRP and UE Rx-Tx.

**Table I.2-1: Conditions for NR PRS-based measurements in FR1**

| Parameter | NR operating band groups Note1 | Minimum PRP1,2 | | | PRS Ês/Iot for RSTD measurement reference cell PRS resource |
|---|---|---|---|---|---|
| | | dBm / $SCS_{PRS}$ | | | dB |
| | | $SCS_{PRS}$ = 15 kHz | $SCS_{PRS}$ = 30 kHz | $SCS_{PRS}$ = 60 kHz | |
| Conditions | NR_FDD_FR1_A, NR_TDD_FR1_A, NR_SDL_FR1_A | -127 | -124 | -121 | ≥ -6<br>≥ -13 |
| | NR_FDD_FR1_B | -126.5 | -123.5 | -120.5 | |
| | NR_TDD_FR1_C | -126 | -123 | -120 | |
| | NR_FDD_FR1_D, NR_TDD_FR1_D | -125.5 | -122.5 | -119.5 | |
| | NR_FDD_FR1_E, NR_TDD_FR1_E | -125 | -122 | -119 | |
| | NR_FDD_FR1_F | -124.5 | -121.5 | -118.5 | |
| | NR_FDD_FR1_G | -124 | -121 | -118 | |
| | NR_FDD_FR1_H | -123.5 | -120.5 | -117.5 | |
| NOTE 1: NR operating band groups are defined in TS 38.101-1 [54] clause 3.5.2. | | | | | |

**Table I.2-2: Conditions for NR PRS-based measurements in FR2**

| Parameter | Angle of arrival | NR operating bands | Minimum PRP1,2 Note 2, Note 3 | | | | | PRS Ês/Iot for RSTD measurement reference cell PRS resource |
|---|---|---|---|---|---|---|---|---|
| | | | dBm / $SCS_{PRS}$ | | | | | dB |
| | | | $SCS_{PRS}$ = 120 kHz | | | | $SCS_{PRS}$ = 60 kHz | |
| | | | UE power class | | | | UE power class | |
| | | | 1 | 2 | 3 | 4 | 1, 2, 3, 4 | |
| Conditions | Rx Beam Peak | n257 | $-128.3+Y_1$ | -113.8 | -112.1 | $-127.8+Y_4$ | (Value for $SCS_{PRS}$ = 120 kHz) - 3dB | ≥ -6<br>≥ -13 |
| | | n258 | $-128.3+Y_1$ | -113.8 | -112.1 | $-127.8+Y_4$ | | |
| | | n259 | | | -108.5 | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | n260 | -125.3+$Y_1$ | | -109.5 | -125.8+$Y_4$ | | |
| | | n261 | -128.3+$Y_1$ | -113.8 | -112.1 | -127.8+$Y_4$ | | |
| | Spherical coverage Note 1 | n257 | -120.3+$Z_1$ | -102.8 | -101.2 | -118.8+$Z_4$ | (Value for $SCS_{PRS}$ = 120 kHz) - 3dB | ≥ -6<br>≥ -13 |
| | | n258 | -120.3+$Z_1$ | -102.8 | -101.2 | -118.8+$Z_4$ | | |
| | | n259 | | | -95.7 | | | |
| | | n260 | -117.3+$Z_1$ | | -96.9 | -113.8+$Z_4$ | | |
| | | n261 | -120.3+$Z_1$ | -102.8 | -101.2 | -118.8+$Z_4$ | | |

NOTE 1: Values based on EIS spherical coverage as defined in clause 7.3.4 of TS 38.101-2 [55]. Side condition applies for directions in which EIS spherical coverage requirement is met.
NOTE 2: Values specified at the Reference point to give minimum PRS Ês/Iot, with no applied noise.
NOTE 3: For UEs that support multiple FR2 bands, Rx Beam Peak values are increased by $\Delta MB_{P,n}$ and spherical coverage values are increased by $\Delta MB_{S,n}$, the UE multi-band relaxation factor in dB specified in clause 6.2.1 of TS 38.101-2 [55].

# Annex J (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2010-08 | RAN5#48 | R5-104316 | | | Initial draft TS 36.571-1 created | | 0.0.0 |
| 2010-11 | RAN5#49 | R5-106613 | | | V1.0.0 created for presentation to RAN Plenary | 0.0.0 | 1.0.0 |
| 2011-02 | RAN5#50 | R5-110124 | | | Various values and corrections added | 1.0.0 | 1.1.0 |
| 2011-08 | RAN5#52 | R5-113133 | | | Text changes from R5-112139, R5-112386, R5-112837, R5-112838, R5-112839 added | 1.1.0 | 1.2.0 |
| 2011-08 | RAN5#53 | | | | Text changes from R5-113135, R5-113150, R5-114066, R5-113587 added | 1.2.0 | - |
| 2011-11 | RAN5#53 | R5-115206 | | | Initial draft TS 37.571-1 created from TS 36.571-1, TS 34.171 and TS 34.172 | - | 1.0.0 |
| 2011-11 | RAN5#53 | R5-115207 | | | V2.0.0 created for presentation to RAN Plenary with additions from R5-115246, R5-115247, R5-115248, R5-115809 | 1.0.0 | 2.0.0 |
| 2011-12 | RAN#54 | - | - | - | Moved to Rel-9 with editorial changes only | 2.0.0 | 9.0.0 |
| 2012-03 | RAN#55 | R5-120087 | 0001 | - | Modify OTDOA connection diagrams | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120089 | 0002 | - | OTDOA parameter corrections | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120414 | 0003 | - | Adding ECID test cases to Annexes in TS 37.571-1 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120822 | 0004 | - | Correct A-GNSS signalling | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120823 | 0005 | - | ECID procedure modifications | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120893 | 0006 | - | OTDOA procedure updates | 9.0.0 | 9.1.0 |
| 2012-06 | RAN#56 | R5-121126 | 0007 | - | Update to Figure 9.1.1.3-1 | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | R5-121127 | 0008 | - | Clarification to notes in tests 9.1.3 & 9.1.4 | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | R5-121128 | 0009 | - | Clarifications to frequencies and bandwidths to be used | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | R5-121129 | 0010 | - | Setting responseTime in ECID test cases | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | R5-121130 | 0011 | - | Modifications to signalling used in OTDOA test cases | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | R5-121908 | 0012 | - | Adding operating band 26 to TS 37.571-1 | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | - | - | - | Added missing contents from R5-121126, R5-121127, R5-121128 | 9.2.0 | 9.2.1 |
| 2012-06 | RAN#56 | - | - | - | Upgraded to v10.0.0 with no change. | 9.2.1 | 10.0.0 |
| 2012-09 | RAN#57 | R5-123066 | 0013 | - | Correction to RSTD Measurement Accuracy Tests 9.1.3 and 9.1.4 | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123913 | 0014 | - | Addition of RRM Test Case 9.8.4 TDD inter-frequency RSTD Accuracy | 10.0.0 | 10.1.0 |
| 2012-12 | RAN#58 | R5-125136 | 0015 | - | Corrections to references | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125188 | 0016 | - | Correction to LPP Message Content for GNSS Moving Scenario Test | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125806 | 0018 | - | New test case 10.1 FDD RSTD Measurement Reporting Delay for Carrier Aggregation | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125807 | 0019 | - | New test case 10.2 TDD RSTD Measurement Reporting Delay for Carrier Aggregation | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125808 | 0020 | - | New test case 10.3 FDD RSTD Measurement Accuracy for Carrier Aggregation | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125809 | 0021 | - | New test case 10.4.TDD RSTD Measurement Accuracy for Carrier Aggregation | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125831 | 0022 | - | Adding bands 28 and 44 to TS 37.571-1 | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125847 | 0023 | - | Corrections to procedures for RSTD tests | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125848 | 0024 | - | Correction of OCNG Patterns for UE Rx - Tx Time Difference Test Cases | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-125916 | 0025 | - | Add editor's note for value of Iprs for test case 9.1.4 | 10.1.0 | 10.2.0 |
| 2012-12 | RAN#58 | R5-124120 | 0026 | - | New common text for test cases 10.1 - 10.4 for RSTD for Carrier Aggregation | 10.1.0 | 10.2.0 |
| 2013-03 | RAN#59 | R5-130959 | 0027 | | LBS Perf: Corrections to TCs 8.1.1 and 8.1.2 | 10.2.0 | 10.3.0 |
| 2013-06 | RAN#60 | R5-131097 | 0028 | - | Removal of Note 1 from OTDOA parameter tables | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131176 | 0029 | - | Clarification to RSTD Delay Test procedures | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131943 | 0030 | - | New Test Case for FDD-FDD inter-frequency RSTD Accuracy | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131944 | 0031 | - | New Test Case for FDD-FDD inter-frequency RSTD measurement reporting delay | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131945 | 0032 | - | OTDOA test case alignment with RAN 4 | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131946 | 0033 | - | Corrections to ECID and OTDOA tests<br>Note: same contents as R5-131945 was submitted by accident. | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131947 | 0034 | - | RSTD test parameter updates | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131993 | 0035 | - | Test Description for TDD inter-frequency accuracy test case | 10.3.0 | 10.4.0 |
| 2013-06 | RAN#60 | R5-131994 | 0036 | - | New test case for TDD inter-frequency RSTD reporting delay 9.2.2 | 10.3.0 | 10.4.0 |
| 2013-09 | RAN#61 | R5-133173 | 0037 | - | Tidy up of Table 9.2.1.4.1-1 | 10.4.0 | 10.5.0 |
| 2013-09 | RAN#61 | R5-133174 | 0038 | - | Corrections to ECID and OTDOA tests | 10.4.0 | 10.5.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2013-09 | RAN#61 | R5-133375 | 0039 | - | Uncertainties and Test Tolerances for RSTD test cases 9.1.1 and 9.1.2 | 10.4.0 | 10.5.0 |
| 2013-09 | RAN#61 | R5-133378 | 0040 | - | Uncertainties and Test Tolerances for RSTD test cases 9.1.3 and 9.1.4 | 10.4.0 | 10.5.0 |
| 2013-09 | RAN#61 | R5-133848 | 0041 | - | LBS Perf: Uncertainties and test tolerances for TCs 8.1.1 and 8.1.2 | 10.4.0 | 10.5.0 |
| 2013-09 | RAN#61 | R5-133885 | 0042 | - | LBS Perf: Revision of test procedure for TC-s 8.1.1-2 | 10.4.0 | 10.5.0 |
| 2013-12 | RAN#62 | R5-134200 | 0043 | - | Updates to ECID and RSTD tests following RAN 4 updates | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134202 | 0044 | - | Addition of Capability exchange in ECID and RSTD tests | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134205 | 0045 | - | Addition of Applicabilities for 9.2.1 - 9.2.5 | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134849 | 0046 | - | Addition of missing acknowledgements in ECID tests | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134850 | 0047 | - | Corrections to references for OCNG and RMC | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134899 | 0048 | - | Introduction 8.1.3 E-UTRAN FDD UE Rx-Tx time difference (feICIC) | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134970 | 0049 | - | Introduction 8.1.4 E-UTRAN TDD UE Rx-Tx time difference (feICIC) | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134979 | 0050 | - | Addition of new tests 10.1a, 10.2a, 10.3a and 10.4a for 20MHz CA | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-134980 | 0051 | - | LBS Perf: Corrections to RSTD reporting tests | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-135016 | 0052 | - | Uncertainties and Test Tolerances for RSTD test cases 9.2.1 and 9.2.2 | 10.5.0 | 10.6.0 |
| 2013-12 | RAN#62 | R5-135018 | 0053 | - | Uncertainties and Test Tolerances for RSTD test cases 9.2.4 and 9.2.5 | 10.5.0 | 10.6.0 |
| 2014-03 | RAN#63 | R5-140107 | 0054 | - | Corrections to PRS_RA in RSTD tests | 10.6.0 | 10.7.0 |
| 2014-03 | RAN#63 | R5-140278 | 0055 | - | Addition of E-UTRA band groups | 10.6.0 | 10.7.0 |
| 2014-03 | RAN#63 | R5-140308 | 0056 | - | LBS RF: Aperiodic CQI configuration for 1.4 MHz bandwidth subtests | 10.6.0 | 10.7.0 |
| 2014-03 | RAN#63 | R5-141033 | 0057 | - | RSTD test case updates | 10.6.0 | 10.7.0 |
| 2014-03 | RAN#63 | R5-140875 | 0058 | - | Additions to TC 8.1.6 E-UTRAN TDD UE Rx-Tx time difference (feICIC) | 10.7.0 | 11.0.0 |
| 2014-03 | RAN#63 | R5-141010 | 0059 | - | Additions to TC 8.1.5 E-UTRAN FDD UE Rx-Tx time difference (feICIC) | 10.7.0 | 11.0.0 |
| 2014-06 | RAN#64 | R5-142098 | 0060 | - | Corrections for OCNG patterns defined in RSTD Tables | 11.0.0 | 11.1.0 |
| 2014-06 | RAN#64 | R5-142211 | 0061 | - | Additions to feICIC UE Rx-Tx test cases in Annex C | 11.0.0 | 11.1.0 |
| 2014-06 | RAN#64 | R5-142302 | 0062 | - | LBS RF: Aperiodic CQI configuration for 1.4 MHz bandwidth tests | 11.0.0 | 11.1.0 |
| 2014-06 | RAN#64 | R5-143109 | 0063 | - | Additions to TC 8.1.6 E-UTRAN TDD UE Rx-Tx time difference (feICIC) | 11.0.0 | 11.1.0 |
| 2014-06 | RAN#64 | R5-143111 | 0064 | - | Additions to FDD interruption requirements for SCell | 11.0.0 | 11.1.0 |
| 2014-06 | RAN#64 | R5-143180 | 0065 | - | Additions to TC 8.1.5 E-UTRAN FDD UE Rx-Tx time difference (feICIC) | 11.0.0 | 11.1.0 |
| 2014-06 | RAN#64 | R5-143211 | 0066 | - | LBS RF: Update of RSTD tests | 11.0.0 | 11.1.0 |
| 2014-09 | RAN#65 | R5-144080 | 0068 | - | Corrections to RSTD Measurement Reporting Delay for Carrier Aggregation tests | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144125 | 0069 | - | Corrections to Physical Cell Id (PCI) Configuration Conditions in UE Rx-Tx time difference feICIC | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144193 | 0072 | - | Cell-specific test parameters for E-UTRAN | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144215 | 0081 | - | Clarification to RSTD Reporting Delay tests | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144409 | 0083 | - | Update to initial conditions and measurement procedure in for UTRA A-GPS and A-GNSS tests | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144553 | 0084 | - | Removal of editors note in TC 9.1.4 | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144620 | 0085 | - | Updates OTDOA Neighbour Cell Info List | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144763 | 0086 | - | Correction for RSTD Measurement Accuracy in CA requirements in RRM | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144789 | 0070 | - | Uncertainties and Test Tolerances for RSTD test cases 10.1, 10.1A, 10.2 and 10.2A | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144790 | 0071 | - | Uncertainties and Test Tolerances for RSTD test cases 10.3, 10.3A, 10.4 and 10.4A | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144791 | 0082 | - | Updates to Annex E of TS 37.571-1 | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144792 | 0087 | - | RSTD tests RAN 4 alignment | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144865 | 0088 | - | Editor's note to PRS levels with fading | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144870 | 0089 | - | Corrections to Note 3 for RSTD CA tests | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144914 | 0090 | - | Corrections to TDD RSTD Measurement Reporting Delay for Carrier Aggregation | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144915 | 0091 | - | RSTD CA Measurement Accuracy connection diagrams | 11.1.0 | 11.2.0 |
| 2014-09 | RAN#65 | R5-144203 | 0073 | - | Addition of new TC 10.1B FDD RSTD Measurement Reporting Delay CA for 5+5MHz | 11.2.0 | 12.0.0 |
| 2014-09 | RAN#65 | R5-144204 | 0074 | - | Addition of new TC 10.1C FDD RSTD Measurement Reporting Delay CA for 10+5MHz | 11.2.0 | 12.0.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2014-09 | RAN#65 | R5-144205 | 0075 | - | Addition of new TC 10.2B TDD RSTD Measurement Reporting Delay CA for 5+5MHz | 11.2.0 | 12.0.0 |
| 2014-09 | RAN#65 | R5-144206 | 0076 | - | Addition of new TC 10.2C TDD RSTD Measurement Reporting Delay CA for 10+5MHz | 11.2.0 | 12.0.0 |
| 2014-09 | RAN#65 | R5-144207 | 0077 | - | Addition of new TC 10.3B FDD RSTD Measurement Accuracy CA for 5+5MHz | 11.2.0 | 12.0.0 |
| 2014-09 | RAN#65 | R5-144208 | 0078 | - | Addition of new TC 10.3C FDD RSTD Measurement Accuracy CA for 10+5MHz | 11.2.0 | 12.0.0 |
| 2014-09 | RAN#65 | R5-144209 | 0079 | - | Addition of new TC 10.4B TDD RSTD Measurement Accuracy CA for 5+5MHz | 11.2.0 | 12.0.0 |
| 2014-09 | RAN#65 | R5-144210 | 0080 | - | Addition of new TC 10.4C TDD RSTD Measurement Accuracy CA for 10+5MHz | 11.2.0 | 12.0.0 |
| 2014-12 | RAN#66 | R5-145133 | 0092 | - | Update Galileo ICD reference | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145211 | 0093 | - | Io value incorrectly calculated | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145212 | 0094 | - | Editorial Note clarification | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145213 | 0095 | - | Alignment of Es/Nos value | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145214 | 0096 | - | Duplicated Io values listed in RSTD tables | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145215 | 0097 | - | Correction to References in Specification | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145254 | 0098 | - | LBS Perf: Corrections to measurement gap configuration | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145490 | 0099 | - | Test Tolerances for TC 8.1.5 E-UTRAN FDD UE Rx-Tx time difference (feICIC) | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145491 | 0100 | - | Test Tolerances for TC 8.1.6 E-UTRAN TDD UE Rx-Tx time difference (feICIC) | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145492 | 0101 | - | Uncertainties and Test Tolerances to Annex C for feICIC UE Rx-Tx test cases | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145502 | 0102 | - | Correction to periodicity of ABS pattern in UE RX-TX time difference for feICIC | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145503 | 0103 | - | Introduction of BDS testing in Annex C of 37.571 | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145510 | 0104 | - | Correction to Annex E notes and tables | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145836 | 0105 | - | Changes to RSTD tests to align with RAN 4 | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145843 | 0106 | - | Introduction of feICIC applicability statement for UE Rx-TX Time Difference test cases | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145864 | 0107 | - | Introduction of content for BDS and UTRA TDD UE in section 1-3 of TS 37.571-1 | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145865 | 0108 | - | Introduction of content for BDS in section 4 in TS 37.571-1 | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145870 | 0109 | - | Corrections to measurement procedures for UTRA A-GPS and A-GNSS tests | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145871 | 0110 | - | Correction to UE Rx-Tx Time difference tests | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145920 | 0111 | - | Introduction of test cases for BDS and UTRA TDD UE in section 6 of TS 37.571-1 | 12.0.0 | 12.1.0 |
| 2014-12 | RAN#66 | R5-145930 | 0112 | - | Introduction of test cases for BDS in section 7 of TS 37.571-1 | 12.0.0 | 12.1.0 |
| 2015-03 | RAN#67 | R5-150051 | 0113 | - | Updates to RSTD values and terminology following changes in RAN 4 | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150074 | 0114 | - | Remove incorrect note from CA RSTD accuracy tests | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150082 | 0115 | - | Uncertainties and Test Tolerances for RSTD test cases 10.1, 10.1A, 10.1B, 10.1C, 10.2, 10.2A, 10.2B, and 10.2C | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150085 | 0116 | - | Uncertainties and Test Tolerances for RSTD test cases 10.3B, 10.3C, 10.4B, and 10.4C | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150107 | 0117 | - | Corrections to table headings in CA RSTD tests | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150378 | 0118 | - | Very minor corrections to references for feICIC test cases, 8.1.5 and 8.1.6 | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150609 | 0119 | - | Abbreviation Corrections for BDS in 37.571-1 | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150833 | 0120 | - | Addition of BDS ICD reference | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150834 | 0121 | - | Corrections to BDS Test Requirements for Minimum Performance tests | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150835 | 0122 | - | The Io Unit Parameter is Incorrect | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150836 | 0123 | - | Inconsistent Text Referenced | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150837 | 0124 | - | Missing Abbreviations in Specification | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150888 | 0125 | - | New TC: TDD RSTD Measurement Accuracy for Carrier Aggregation for 20MHz+10MHz bandwidth | 12.1.0 | 12.2.0 |
| 2015-03 | RAN#67 | R5-150911 | 0126 | - | New TC: TDD RSTD Measurement Reporting Delay for Carrier Aggregation for 20MHz +10MHz Bandwidth | 12.1.0 | 12.2.0 |
| 2015-06 | RAN#68 | R5-151070 | 0128 | - | Delete "FFS" from ECID test conditions | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151083 | 0129 | - | Uncertainties and Test Tolerances for RSTD test case 10.2D | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151085 | 0130 | - | Uncertainties and Test Tolerances for RSTD test case 10.4D | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151086 | 0131 | - | RSTD accuracy changes for Rel-12 | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151088 | 0132 | - | Formatting error in Parameter Sensitivity Coarse Tables | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151089 | 0133 | - | Incorrect Expected RSTD value in Table 9.2.5.4.1-1 | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151162 | 0135 | - | Correction of the TPR of Absolute GNSS signal level for Dynamic Range | 12.2.0 | 12.3.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment TS 36.571-1** | **Old** | **New** |
| 2015-06 | RAN#68 | R5-151331 | 0136 | - | Addition of band 32 to 37.571-1 | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151335 | 0137 | - | Corrections to message contents for feICIC TCs in 37.571-1 | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151912 | 0134 | 1 | LPP responseTime update and correction | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151913 | 0138 | 1 | Introduction of new test case 8.1.3 to 37.571-1 | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-151914 | 0139 | 1 | Introduction of new test case 8.1.4 to 37.571-1 | 12.2.0 | 12.3.0 |
| 2015-06 | RAN#68 | R5-152014 | 0127 | 1 | Add TDD to A-GNSS testing | 12.2.0 | 12.3.0 |
| 2015-09 | RAN#69 | R5-153253 | 0140 | - | CA LBS: Clarification of PHICH configuration | 12.3.0 | 12.4.0 |
| 2015-09 | RAN#69 | R5-153661 | 0144 | - | Update of Galileo OS SIS ICD reference | 12.3.0 | 12.4.0 |
| 2015-09 | RAN#69 | R5-153863 | 0143 | 1 | Update of feICIC Test cases 8.1.5 and 8.1.6 | 12.3.0 | 12.4.0 |
| 2015-09 | RAN#69 | R5-153864 | 0141 | 1 | Update of eICIC Test case 8.1.3 | 12.3.0 | 12.4.0 |
| 2015-09 | RAN#69 | R5-153865 | 0142 | 1 | Update of eICIC Test case 8.1.4 | 12.3.0 | 12.4.0 |
| 2015-09 | RAN#69 | - | - | - | update of the "non-specific references" in section 2 according to the approved R5-153582 and an action point on ETSI MCC | 12.3.0 | 12.4.0 |
| 2015-12 | RAN#70 | R5-155018 | 0145 | - | Uncertainties and Test Tolerances for RSTD Test Cases 10.3A_1 and 10.4A_1 | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-155035 | 0146 | - | Incorrect Table Note referenced in LPP Request Table | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-155063 | 0149 | - | Incorrect references in TDD test cases | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-155064 | 0150 | - | Reference [2] has no explanation | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-155065 | 0151 | - | Remove square brackets from RSTD tests | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-155066 | 0152 | - | Incorrect Section number referenced | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-155081 | 0153 | - | Editorial changes to correct Section and Table references | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-155875 | 0154 | 1 | Two new 3 DL CA RSTD Measurement Reporting Delay test cases | 12.4.0 | 12.5.0 |
| 2015-12 | RAN#70 | R5-156111 | 0155 | 1 | Two new 3 DL CA RSTD Measurement Accuracy test cases | 12.4.0 | 12.5.0 |
| 2016-03 | RAN#71 | R5-160041 | 0156 | - | Correction to Cells in OTDOA 3DL RSTD Measurement | 12.5.0 | 12.6.0 |
| 2016-03 | RAN#71 | R5-160900 | 0164 | - | Add Cell values in RSTD Table for 3DL RSTD | 12.5.0 | 12.6.0 |
| 2016-03 | RAN#71 | R5-160909 | 0157 | 1 | Correction of Cell Time offset in RSTD CA | 12.5.0 | 12.6.0 |
| 2016-03 | RAN#71 | R5-161016 | 0161 | 1 | Add Cell values in OTDOA table for 3DL RSTD Measurement Reporting Delay | 12.5.0 | 12.6.0 |
| 2016-03 | RAN#71 | R5-161053 | 0158 | 1 | Correction to Trstd values in 3DL RSTD Measurement Accuracy test cases | 12.5.0 | 12.6.0 |
| 2016-03 | RAN#71 | R5-161054 | 0160 | 1 | Addition of antenna diagram Figure for 3DL CA test cases | 12.5.0 | 12.6.0 |
| 2016-06 | RAN#72 | R5-162114 | 0165 | - | Uncertainties and Test Tolerances for RSTD Test Cases 10.5 and 10.6 | 12.6.0 | 12.7.0 |
| 2016-06 | RAN#72 | R5-162116 | 0166 | - | Uncertainties and Test Tolerances for RSTD Test Cases 10.7 and 10.8 | 12.6.0 | 12.7.0 |
| 2016-06 | RAN#72 | R5-163116 | 0167 | 1 | Uncertainties and Test tolerances for TS 37.571-1 Test cases 8.1.3 and 8.1.4 | 12.6.0 | 12.7.0 |
| 2016-06 | RAN#72 | R5-162970 | 0168 | 1 | Add missing LTE FDD TDD bands to E-UTRA Band Groups | 12.7.0 | 13.0.0 |
| 2016-06 | RAN#72 | R5-162971 | 0169 | 1 | Add missing LTE FDD band to Annex E | 12.7.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165350 | 0179 | - | Incorrect FDD Band reference noted for Band 32 | 13.0.0 | 13.1.0 |
| 2016-09 | RAN#73 | R5-165360 | 0181 | - | Correct editorial changes in Annex C of 37.571-1 | 13.0.0 | 13.1.0 |
| 2016-09 | RAN#73 | R5-166125 | 0173 | 1 | Updates to the UE Rx – Tx Time Difference tests for Rel-12 onwards | 13.0.0 | 13.1.0 |
| 2016-09 | RAN#73 | R5-166126 | 0178 | 1 | Add missing references to GPS and Galileo and A-GPS and A-Galileo | 13.0.0 | 13.1.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment TS 36.571-1** | **Old** | **New** |
| 2016-09 | RAN#73 | R5-166127 | 0180 | 1 | Add Derivation of Test Requirements for test cases 8.1.5 and 8.1.6 | 13.0.0 | 13.1.0 |
| 2016-09 | RAN#73 | R5-166168 | 0182 | 1 | Addition of performance test specification for Indoor Positioning Enhancements (MBS) | 13.0.0 | 13.1.0 |
| 2016-09 | RAN#73 | R5-166179 | 0171 | 1 | Addition of editor's notes for TDD UE Rx-TX tests | 13.0.0 | 13.1.0 |
| 2016-09 | RAN#73 | R5-166181 | 0174 | 1 | Unification of Channel BW testing requirements for OTDOA 3 DL CA test cases | 13.0.0 | 13.1.0 |
| 2016-09 | RAN#73 | R5-166283 | 0183 | - | Change of Reference Channel for 1.4 MHz RSTD tests | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168060 | 0185 | - | Change references to Reference Channel for RSTD tests | 13.1.0 | 13.2.0 |
| 2016-12 | RAN#74 | R5-168063 | 0186 | - | Change of applicability of UE Rx-Tx tests for TDD | 13.1.0 | 13.2.0 |
| 2016-12 | RAN#74 | R5-168147 | 0187 | - | Corrections for errors in 37.571-1 | 13.1.0 | 13.2.0 |
| 2016-12 | RAN#74 | R5-169602 | 0184 | 1 | Addition of test tolerances to the performance test specification for Indoor Positioning Enhancements (MBS) | 13.1.0 | 13.2.0 |
| 2016-12 | RAN#74 | R5-169611 | 0190 | 1 | Clarification of MBS beacon code phase delay | 13.1.0 | 13.2.0 |
| 2016-12 | RAN#74 | R5-169663 | 0188 | 1 | Editorial correction on OTDOA TC10.6 | 13.1.0 | 13.2.0 |
| 2016-12 | RAN#74 | R5-168396 | 0189 | - | Band 70 addition to band groups in 37.571-1 | 13.2.0 | 14.0.0 |
| 2017-01 | RAN#74 | - | - | - | correction of floating point of R5-169602 in Table 11.4.5-3 | 14.0.0 | 14.0.1 |
| 2017-03 | RAN#75 | R5-171304 | 0195 | - | Removal of square brackets in the performance test specification for Indoor Positioning Enhancements (MBS) | 14.0.1 | 14.1.0 |
| 2017-03 | RAN#75 | R5-171902 | 0191 | 1 | Update TS 37.571-1 with Addition of LTE Band 48 | 14.0.1 | 14.1.0 |
| 2017-03 | RAN#75 | R5-171904 | 0192 | 1 | Clarification on DRX for Single Mode OTDOA Measurement Reporting Delay Test Cases | 14.0.1 | 14.1.0 |
| 2017-03 | RAN#75 | R5-171905 | 0193 | 1 | Clarification on DRX for 2CC OTDOA Measurement Reporting Delay Test Cases | 14.0.1 | 14.1.0 |
| 2017-03 | RAN#75 | R5-171906 | 0194 | 1 | Clarification on DRX for 3CC OTDOA Measurement Reporting Delay Test Cases | 14.0.1 | 14.1.0 |
| 2017-06 | RAN#76 | R5-172179 | 0197 | - | Add Minimum Performance Sub-tests for 3 GNSS | 14.1.0 | 14.2.0 |
| 2017-06 | RAN#76 | R5-172623 | 0198 | - | Introduction of periodical reporting capability for GNSS | 14.1.0 | 14.2.0 |
| 2017-06 | RAN#76 | R5-173364 | 0200 | 1 | Introduction of MBS Assistance Data Measurement Test Cases | 14.1.0 | 14.2.0 |
| 2017-06 | RAN#76 | R5-173414 | 0201 | 1 | Addition of ACKs in step 5 of test procedures | 14.1.0 | 14.2.0 |
| 2017-09 | RAN#77 | R5-173569 | 0202 | - | Correction of PRS Subframe Offset for TC 10.5 and 10.6 | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-173570 | 0203 | - | Correction of SRS-Bandwidth for ECID TC 8.1.3 and 8.1.4 | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-173571 | 0204 | - | Correction of SRS-Bandwidth for ECID TC 8.1.5 and 8.1.6 | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-173572 | 0205 | - | Correction of message contents for ECID (Editorial Change) | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-173576 | 0209 | - | WLAN and BLE Annex D updates (Editorial Change) | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-173863 | 0214 | - | Editorial change to clarify the MBS test cases applicability | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-174053 | 0215 | - | Update Statement Concerning Test System Uncertainties for Operating Bands Above 3 GHz | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-175116 | 0206 | 1 | New Abbreviations and References for WLAN and BLE (Editorial Change) | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-175117 | 0207 | 1 | WLAN test conditions | 14.2.0 | 14.3.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment TS 36.571-1** | **Old** | **New** |
| 2017-09 | RAN#77 | R5-175118 | 0208 | 1 | WLAN and BLE Connection Diagrams | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-175119 | 0212 | 1 | BLE test conditions | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-175186 | 0210 | 1 | New WLAN AP Identification in Nominal Accuracy Test | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-175187 | 0211 | 1 | New WLAN AP Identification in Dynamic Range Test | 14.2.0 | 14.3.0 |
| 2017-09 | RAN#77 | R5-175188 | 0213 | 1 | New BLE Reporting Test | 14.2.0 | 14.3.0 |
| 2017-12 | RAN#78 | R5-176110 | 0219 | - | Add 4Rx support for OTDOA/ECID tests – Tests | 14.3.0 | 14.4.0 |
| 2017-12 | RAN#78 | R5-176111 | 0220 | - | Add 4Rx support for OTDOA/ECID tests – Connection Diagrams | 14.3.0 | 14.4.0 |
| 2017-12 | RAN#78 | R5-177118 | 0222 | 1 | Adapt LTE A-GNSS test cases for BL/CE devices | 14.3.0 | 14.4.0 |
| 2017-12 | RAN#78 | R5-177411 | 0216 | 1 | Complete WLAN and BLE test cases | 14.3.0 | 14.4.0 |
| 2017-12 | RAN#78 | R5-177413 | 0217 | 1 | Editorial - Alignment of 2CC 5MHz OTDOA Nprs with core spec | 14.3.0 | 14.4.0 |
| 2017-12 | RAN#78 | R5-177414 | 0218 | 1 | Add 4Rx support for OTDOA/ECID tests – Common Sections | 14.3.0 | 14.4.0 |
| 2017-12 | RAN#78 | R5-177415 | 0221 | 1 | Add release information for sub-tests of test case 7.5. | 14.3.0 | 14.4.0 |
| 2017-12 | RAN#78 | R5-176791 | 0223 | - | Band 72 addition to band groups in 37.571-1 | 14.4.0 | 15.0.0 |
| 2017-12 | RAN#78 | R5-176816 | 0224 | - | Band 71 addition to band groups in 37.571-1 | 14.4.0 | 15.0.0 |
| 2018-03 | RAN#79 | R5-180295 | 0228 | - | New OTDOA Cat1bis TC 9.1.3A and 9.1.4A | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180296 | 0229 | - | New OTDOA Cat1bis TC 9.2.4A and 9.2.5A | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180297 | 0230 | - | Annex C OTDOA Cat1bis tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180298 | 0231 | - | UE Category M1/M2 General Sections for OTDOA | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180299 | 0232 | - | New OTDOA Cat M1/M2 reporting delay normal mode tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180300 | 0233 | - | New OTDOA Cat M1/M2 reporting delay enhanced mode tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180301 | 0234 | - | New OTDOA Cat M1/M2 reporting accuracy normal mode tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180303 | 0236 | - | New OTDOA Cat M1/M2 inter-freq reporting delay normal mode tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180304 | 0237 | - | New OTDOA Cat M1/M2 inter-freq reporting delay enhanced mode tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180307 | 0240 | - | New ECID Cat M1/M2 tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180308 | 0241 | - | Annex C OTDOA and ECID Cat M1/M2 | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180311 | 0244 | - | NB-IOT Annex E | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180325 | 0245 | - | Band 68 addition to band groups in 37.571-1 | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180352 | 0246 | - | [Editorial] Correct normative reference for minimum conformance requirements | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180583 | 0247 | - | feMTC Annex E | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180584 | 0248 | - | 4Rx support for OTDOA 2CC | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-180585 | 0249 | - | 4Rx support for OTDOA 3CC | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181360 | 0250 | 2 | WLAN core specification updates | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181584 | 0235 | 1 | New OTDOA Cat M1/M2 reporting accuracy enhanced mode tests | 15.0.0 | 15.1.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment TS 36.571-1** | **Old** | **New** |
| 2018-03 | RAN#79 | R5-181585 | 0238 | 1 | New OTDOA Cat M1/M2 inter-freq reporting accuracy normal mode tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181586 | 0239 | 1 | New OTDOA Cat M1/M2 inter-freq reporting accuracy enhanced mode tests | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181599 | 0226 | 1 | New OTDOA Cat1bis TC 9.1.1A and 9.1.2A | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181600 | 0227 | 1 | New OTDOA Cat1bis TC 9.2.1A and 9.2.2A | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181605 | 0242 | 1 | NB-IOT General Sections | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181606 | 0243 | 1 | NB-IOT OTDOA Test Cases | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181613 | 0251 | 1 | Addition of the Band 74 information into TS 37.571-1 | 15.0.0 | 15.1.0 |
| 2018-06 | RAN#80 | R5-182218 | 0256 | - | Common clauses updates for new NB-IOT OTDOA tests | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-182219 | 0257 | - | Annex C updates for NB-IOT OTDOA | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-182279 | 0258 | - | New ECID Cat1bis tests | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-182280 | 0259 | - | New ECID Cat1bis tests - Annexes | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-182382 | 0261 | - | Corrections to WLAN dynamic range test case | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-182384 | 0263 | - | Corrections to maximum response time for WLAN test cases | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183252 | 0252 | 1 | Completion of OTDOA NB-IOT TC 9.5.1 and 9.5.2 | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183253 | 0253 | 1 | Completion of OTDOA NB-IOT TC 9.6.1 and 9.6.2 | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183261 | 0264 | 1 | Clarifications for RSSI reporting in WLAN and BLE test cases | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183844 | 0254 | 1 | New OTDOA NB-IOT TC 9.5.3 | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183845 | 0255 | 1 | New OTDOA NB-IOT TC 9.6.3 | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183847 | 0260 | 1 | Removing editor note from A-GNSS min perf test cases for Cat M1 | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183848 | 0262 | 1 | Clarifications and additions to EUTRAN, WLAN and BLE test frequencies and bandwidths | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183849 | 0265 | 1 | Corrections to WLAN and BLE applicabilities | 15.1.0 | 15.2.0 |
| 2018-09 | RAN#81 | R5-184041 | 0266 | - | Clarifications and corrections to Bluetooth identification test | 15.2.0 | 15.3.0 |
| 2018-09 | RAN#81 | R5-184189 | 0271 | - | Band groups added to specification | 15.2.0 | 15.3.0 |
| 2018-09 | RAN#81 | R5-185416 | 0268 | 1 | Correction to nrs-CRS-PowerOffset-r13 for NB-IOT OTDOA tests | 15.2.0 | 15.3.0 |
| 2018-09 | RAN#81 | R5-185417 | 0269 | 1 | NB-IOT OTDOA reporting delay test cases not testable | 15.2.0 | 15.3.0 |
| 2018-09 | RAN#81 | R5-185419 | 0270 | 1 | Changes to eMTC OTDOA tests | 15.2.0 | 15.3.0 |
| 2018-09 | RAN#81 | R5-185552 | 0267 | 1 | Applicability of tests for types and Categories of UE | 15.2.0 | 15.3.0 |
| 2018-12 | RAN#82 | R5-186489 | 0272 | - | Resubmission of CR 0269 | 15.3.0 | 15.4.0 |
| 2018-12 | RAN#82 | R5-186615 | 0274 | - | Clarification of the meaning of A-GPS | 15.3.0 | 15.4.0 |
| 2018-12 | RAN#82 | R5-186616 | 0275 | - | Addition of two missing triple-GNSS test cases | 15.3.0 | 15.4.0 |
| 2018-12 | RAN#82 | R5-186617 | 0276 | - | Updates to Table 4B.2-1 | 15.3.0 | 15.4.0 |
| 2018-12 | RAN#82 | R5-187983 | 0277 | 1 | Editorial Changes for TS 37.571-1 | 15.3.0 | 15.4.0 |
| 2019-03 | RAN#83 | R5-191607 | 0281 | - | Editorial Changes for TS 37.571-1 | 15.4.0 | 15.5.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2019-03 | RAN#83 | R5-192506 | 0278 | 1 | Addition of general NR information | 15.4.0 | 15.5.0 |
| 2019-03 | RAN#83 | R5-192646 | 0282 | 1 | Editorial updates for RSTD NB-IOT tests | 15.4.0 | 15.5.0 |
| 2019-03 | RAN#83 | R5-192851 | 0279 | 1 | Corrections to RSTD reporting accuracy NB-IOT tests | 15.4.0 | 15.5.0 |
| 2019-03 | RAN#83 | R5-192642 | 0280 | 1 | Band 53 introduction in TS 37-571-1 | 15.5.0 | 16.0.0 |
| 2019-06 | RAN#84 | R5-194099 | 0283 | - | Addition of A-GNSS tests for NR | 16.0.0 | 16.1.0 |
| 2019-06 | RAN#84 | R5-194100 | 0284 | - | Updates to Table 4B.2-1 for WLAN and BLE for NR | 16.0.0 | 16.1.0 |
| 2019-06 | RAN#84 | R5-194260 | 0285 | - | 38.509 reference update | 16.0.0 | 16.1.0 |
| 2019-06 | RAN#84 | R5-195009 | 0286 | 1 | MBS Tests for NR | 16.0.0 | 16.1.0 |
| 2019-06 | RAN#84 | R5-195450 | 0287 | 2 | Alignment 9.5.3 and 9.6.3 with core spec | 16.0.0 | 16.1.0 |
| 2019-09 | RAN#85 | R5-196604 | 0292 | - | Alignment of 9.5.3 and 9.6.3 to core spec | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197356 | 0294 | - | Missing Bands in Tables | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197357 | 0295 | - | Corrections to GNSS Multi-path Tests | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197358 | 0296 | - | Correct RSTD Test Parameter Values | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197381 | 0288 | 1 | Corrections to RSTD tests for NB-IoT | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197471 | 0289 | 1 | Alignment of NR terminology | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197473 | 0290 | 2 | Add default conditions for FR2 positioning tests for performance | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197477 | 0291 | 1 | Clarify LPP transport for min perf positioning | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197478 | 0293 | 1 | Correct MBS Applicability Table and Editorial Updates | 16.1.0 | 16.2.0 |
| 2019-12 | RAN#86 | R5-198063 | 0297 | - | Addition of User Plane testing of A-GNSS to clause 7 for LTE | 16.2.0 | 16.3.0 |
| 2019-12 | RAN#86 | R5-198817 | 0302 | - | Editorial Correction to Multi-Path Scenario Test Parameter Table | 16.2.0 | 16.3.0 |
| 2019-12 | RAN#86 | R5-198821 | 0303 | - | Add Reference to RRM Specifications for Frequency Band Groups | 16.2.0 | 16.3.0 |
| 2019-12 | RAN#86 | R5-198822 | 0304 | - | Editorial Corrections for RSTD Test Cases | 16.2.0 | 16.3.0 |
| 2019-12 | RAN#86 | R5-199090 | 0300 | 1 | Update to min perf positioning tests | 16.2.0 | 16.3.0 |
| 2020-03 | RAN#87 | R5-201013 | 0305 | 1 | Editorial changes to TS 37.571-X titles to remove references to individual RATs | 16.3.0 | 16.4.0 |
| 2020-06 | RAN#88 | R5-202803 | 0308 | 1 | Update of the Note 1 in the Power level and satellite allocation tables for the Sensitivity Coarse time assistance requirements and tests for E-UTRA and NR. | 16.4.0 | 16.5.0 |
| 2020-09 | RAN#89 | R5-204979 | 0309 | 1 | Clarification on frequency band test conditions for A-GNSS | 16.5.0 | 16.6.0 |
| 2020-12 | RAN#90 | R5-206774 | 0311 | 1 | Addition of BDS B1C Signal test contents in TS 37.571-1 | 16.6.0 | 16.7.0 |
| 2020-12 | RAN#90 | R5-206783 | 0310 | 1 | Definition of A-GNSS Sensitivity testing conditions for EN-DC | 16.6.0 | 16.7.0 |
| 2021-03 | RAN#91 | R5-210757 | 0318 | - | Update to Annex C for TDD NB-IOT RSTD measurement test cases | 16.7.0 | 16.8.0 |
| 2021-03 | RAN#91 | R5-211796 | 0312 | 1 | Addition of TDD NB-IOT RSTD measurement test case 9.7.1 | 16.7.0 | 16.8.0 |
| 2021-03 | RAN#91 | R5-211797 | 0313 | 1 | Addition of TDD NB-IOT RSTD measurement test case 9.7.2 | 16.7.0 | 16.8.0 |
| 2021-03 | RAN#91 | R5-211798 | 0314 | 1 | Addition of TDD NB-IOT RSTD measurement test case 9.7.3 | 16.7.0 | 16.8.0 |
| 2021-03 | RAN#91 | R5-211799 | 0315 | 1 | Addition of TDD NB-IOT RSTD measurement test case 9.8.1 | 16.7.0 | 16.8.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment TS 36.571-1** | **Old** | **New** |
| 2021-03 | RAN#91 | R5-211800 | 0316 | 1 | Addition of TDD NB-IOT RSTD measurement test case 9.8.2 | 16.7.0 | 16.8.0 |
| 2021-03 | RAN#91 | R5-211801 | 0317 | 1 | Addition of TDD NB-IOT RSTD measurement test case 9.8.3 | 16.7.0 | 16.8.0 |
| 2021-06 | RAN#92 | R5-212200 | 0320 | - | Addition of Derivation of Test Requirements for NB-IOT and CA RSTD tests | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212201 | 0321 | - | Corrections to RSTD accuracy requirements values for UE Category M1 and M2 test cases | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212202 | 0322 | - | Corrections to RSTD accuracy requirements values for NB-IOT test cases | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212234 | 0323 | - | Addition of Derivation of Test Requirements for RSTD and Category M1 and M2 RSTD tests | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212235 | 0324 | - | Correction to Derivation of Test Requirements for UE Rx-Tx tests for Category M1 and M2 UEs | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212237 | 0325 | - | Updates to Correction to Derivation of Test Requirements for UE Rx-Tx tests for Category M1 and M2 UEs | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212238 | 0326 | - | Addition of Test Parameter Relaxations for UE Rx - Tx time difference cases | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212242 | 0327 | - | Clarifications to FDD NB-IoT RSTD test case applicabilities and other corrections | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212244 | 0328 | - | Corrections to RSTD test cases for Category M1 and M2 UEs | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212247 | 0329 | - | Updates to requirements for RSTD test cases for Category M1 and M2 UEs | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-212397 | 0330 | - | Update Release 15 and onwards references for TS 36.355 to TS 37.355 | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-213134 | 0332 | - | Add OTDOA feMTC test case 9.3.14 | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-213135 | 0333 | - | Add OTDOA feMTC test case 9.3.15 | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-213138 | 0336 | - | Add OTDOA feMTC test case 9.3.18 | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-213139 | 0337 | - | OTDOA feMTC test cases - Annexes | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-213992 | 0331 | 1 | Add OTDOA feMTC test case 9.3.13 | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-213993 | 0334 | 1 | Add OTDOA feMTC test case 9.3.16 | 16.8.0 | 16.9.0 |
| 2021-06 | RAN#92 | R5-213994 | 0335 | 1 | Add OTDOA feMTC test case 9.3.17 | 16.8.0 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215973 | 0338 | 1 | Addition of Multi-RTT, DI-AoD and DL-TDOA positioning method test conditions | 16.9.0 | 16.10.0 |
| 2021-09 | RAN#93 | R5-215974 | 0340 | 1 | Addition of conditions for NR PRS-based measurements and connection diagrams | 16.9.0 | 16.10.0 |
| 2021-12 | RAN#94 | R5-217133 | 0346 | - | Update number of satellites for multi-GNSS for LTE tests | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-217134 | 0347 | - | Update number of satellites for multi-GNSS for NR tests | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-217183 | 0355 | - | EN-DC frequency bands for testing of A-GNSS sensitivity requirements | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-218283 | 0348 | 1 | Add OTDOA feMTC test case 9.4.13 | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-218284 | 0349 | 1 | Add OTDOA feMTC test case 9.4.14 | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-218285 | 0350 | 1 | Add OTDOA feMTC test case 9.4.15 | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-218286 | 0351 | 1 | Add OTDOA feMTC test case 9.4.16 | 16.10.0 | 16.11.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2021-12 | RAN#94 | R5-218287 | 0352 | 1 | Add OTDOA feMTC test case 9.4.17 | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-218288 | 0353 | 1 | Add OTDOA feMTC test case 9.4.18 | 16.10.0 | 16.11.0 |
| 2021-12 | RAN#94 | R5-218289 | 0354 | 1 | OTDOA feMTC test cases - Annexes | 16.10.0 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220518 | 0356 | - | Addition of test cases for UE Rx-Tx time difference measurement period | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-220534 | 0358 | - | Update test applicability to allow for support of limited GNSS combinations for 5G tests | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-220535 | 0359 | - | Update test applicability to allow for support of limited GNSS combinations | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221201 | 0360 | - | Update for the signal conditions for FR2 test cases | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221202 | 0361 | - | New NR RSTD test case 14.2.1 | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221203 | 0362 | - | New NR RSTD test case 14.2.2 | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221204 | 0363 | - | New NR RSTD test case 14.3.1 | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221205 | 0364 | - | New NR RSTD test case 14.3.2 | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221206 | 0365 | - | New NR PRS-RSRP test case 16.2.1 | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221207 | 0366 | - | New NR PRS-RSRP test case 16.2.2 | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221208 | 0367 | - | New NR PRS-RSRP test case 16.3.1 | 16.11.0 | 16.12.0 |
| 2022-03 | RAN#95 | R5-221887 | 0357 | 1 | Addition of UE Rx-Tx time difference measurement test uncertainties and test parameter relaxations | 16.11.0 | 16.12.0 |
| 2022-06 | RAN#96 | R5-222596 | 0368 | - | Addition of measurement period requirements in Multi-RTT test conditions, DL-TDOA test conditions and DL-AoD test conditions | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-222597 | 0369 | - | Correction of NR RSTD test cases 14.2.1, 14.2.2, 14.3.1 and 14.3.2 | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-222598 | 0370 | - | Addition of SRS configuration in UE Rx-Tx time difference measurement period test cases | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-222599 | 0371 | - | Addition of new RSTD accuracy test case 14.2.3 | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-222600 | 0372 | - | Addition of new RSTD accuracy test case 14.2.4 | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-222601 | 0373 | - | Addition of new RSTD accuracy test case 14.3.3 | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-222603 | 0375 | - | Addition of new UE Rx-TX time difference accuracy test case 15.3.1 | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-222604 | 0376 | - | Addition of new UE Rx-TX time difference accuracy test case 15.3.2 | 16.12.0 | 16.13.0 |
| 2022-06 | RAN#96 | R5-223748 | 0374 | 1 | Addition of new RSTD accuracy test case 14.3.4 | 16.12.0 | 16.13.0 |
| 2022-09 | RAN#97 | R5-223972 | 0379 | - | Annexes for TC 14.2.1 and 14.3.1 | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-224401 | 0381 | - | Correction of UE Rx-Tx time difference accuracy test cases | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-224403 | 0383 | - | Addition of new PRS RSRP measurement period test case 16.2.3 | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-224404 | 0384 | - | Addition of new PRS RSRP measurement period test case 16.2.4 | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-224478 | 0386 | - | Core spec alignment 9.6.3 | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-225698 | 0380 | 1 | Correction to UE Rx-TX time difference measurement period test cases 15.2.1 and 15.2.2 | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-225740 | 0382 | 1 | Correction of NR RSTD test cases | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-225741 | 0385 | 1 | Addition of new PRS RSRP measurement accuracy test case 16.3.2 | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-225877 | 0377 | 1 | Complete TC 14.2.1 including TT analysis results | 16.13.0 | 16.14.0 |
| 2022-09 | RAN#97 | R5-225878 | 0378 | 1 | Complete TC 14.3.1 including TT analysis results | 16.13.0 | 16.14.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2022-12 | RAN#98 | R5-225944 | 0388 | | Addition of NR PRS-based measurement requirements for Multi-RTT test cases | 16.14.0 | 16.15.0 |
| 2022-12 | RAN#98 | R5-226460 | 0389 | | Addition of accuracy requiremets for RSTD and PRS-RSRP | 16.14.0 | 16.15.0 |
| 2022-12 | RAN#98 | R5-226489 | 0394 | | Update on Annexes for RSTD TT results | 16.14.0 | 16.15.0 |
| 2022-12 | RAN#98 | R5-227843 | 0387 | 1 | Correction to UE Rx-Tx time difference test cases 15.2.3, 15.2.4, 15.3.1 and 15.3.2 | 16.14.0 | 16.15.0 |
| 2022-12 | RAN#98 | R5-228005 | 0390 | 1 | Complete TC 14.2.1 with TT analysis results | 16.14.0 | 16.15.0 |
| 2022-12 | RAN#98 | R5-228006 | 0391 | 1 | Complete TC 14.3.1 with TT analysis results | 16.14.0 | 16.15.0 |
| 2022-12 | RAN#98 | R5-228007 | 0392 | 1 | Update TC 14.2.2 with TT analysis results | 16.14.0 | 16.15.0 |
| 2022-12 | RAN#98 | R5-228008 | 0393 | 1 | Update TC 14.3.2 with TT analysis results | 16.14.0 | 16.15.0 |
| 2023-03 | RAN#99 | R5-230337 | 0404 | - | Addition of accuracy requirements for UE Rx-Tx time difference | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-230927 | 0405 | - | Update TC 14.3.2 with TT analysis results | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231749 | 0395 | 1 | Correction to RSTD test case 14.2.3 | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231750 | 0396 | 1 | Correction to RSTD test case 14.2.4 | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231751 | 0397 | 1 | Correction to RSTD test case 14.3.3 | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231752 | 0398 | 1 | Correction to RSTD test case 14.3.4 | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231753 | 0399 | 1 | Correction to PRS-RSRP test case 16.2.3 | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231754 | 0400 | 1 | Correction to PRS-RSRP test case 16.2.4 | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231755 | 0402 | 1 | Addition of NR PRS-based measurement requirements for NR RSTD and PRS-RSRP test cases | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231798 | 0406 | 1 | Update minimum conformance requirements for dual PFL for TC 14.3.2 | 16.15.0 | 16.16.0 |
| 2023-03 | RAN#99 | R5-231799 | 0403 | 1 | Introduction of BDS B2a and B3I signal test contents in TS 37.571-1 | 16.16.0 | 17.0.0 |
| 2023-06 | RAN#100 | R5-232079 | 0407 | - | Correction to PRS-RSRP test cases 16.3.2 | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-232080 | 0408 | - | Addition of NR PRS-based measurement requirements for PRS-RSRP accuracy test case | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-232438 | 0416 | - | Annex C updated for PRS-RSRP TT results | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-232710 | 0417 | - | Completion 16.3.1 with TT analysis results | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-233651 | 0410 | 1 | Completion 16.2.1 with TT analysis results | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-233652 | 0411 | 1 | Completion 16.2.2 with TT analysis results | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-233680 | 0413 | 1 | Corrections to A-GNSS minimum performance test scenarios | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-233681 | 0414 | 1 | Corrections to A-GNSS minimum performance test scenarios chapter 13 | 17.0.0 | 17.1.0 |
| 2023-09 | RAN#101 | R5-234205 | 0418 | - | Addition of new NR RSTD reporting delay test case 14.2.8 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234206 | 0419 | - | Addition of new NR RSTD reporting delay test case 14.2.9 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234207 | 0420 | - | Addition of new NR RSTD reporting delay test case 14.2.10 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234208 | 0421 | - | Addition of new NR RSTD measurement accuracy test case 14.3.7 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234209 | 0422 | - | Addition of new NR RSTD measurement accuracy test case 14.3.8 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234210 | 0423 | - | Addition of new NR UE Rx-Tx time difference reporting delay test case 15.2.8 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234211 | 0424 | - | Addition of new NR UE Rx-Tx time difference reporting delay test case 15.2.9 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234212 | 0425 | - | Addition of new NR UE Rx-Tx time difference reporting delay test case 15.2.10 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234213 | 0426 | - | Addition of new NR UE Rx-Tx time difference measurement accuracy test case 15.3.5 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234214 | 0427 | - | Addition of new NR UE Rx-Tx time difference measurement accuracy test case 15.3.6 | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-235786 | 0428 | 1 | Addition of missed TT analysis results for 16.3.1 in Annex C.4.7 | 17.1.0 | 17.2.0 |
| 2023-12 | RAN#102 | R5-236010 | 0430 | | Update TC 14.2.8 with TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236011 | 0431 | | Update TC 14.2.9 with TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236012 | 0432 | | Update TC 14.2.10 with TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236014 | 0434 | | Update TC 14.3.8 with TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236015 | 0435 | | Update TC 15.3.5 with TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236017 | 0437 | | Addition of NR PRS-based measurement requirements in annex C to include TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236111 | 0438 | | Addition of new NR PRS-RSRP reporting delay test case 16.2.7 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236112 | 0439 | | Addition of new NR PRS-RSRP reporting delay test case 16.2.8 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236113 | 0440 | | Addition of new NR PRS-RSRP accuracy test case 16.3.4 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236114 | 0441 | | Addition of new NR PRS-RSRPP reporting delay test case 17.2.1 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236115 | 0442 | | Addition of new NR PRS-RSRPP reporting delay test case 17.2.2 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236116 | 0443 | | Addition of new NR PRS-RSRPP reporting delay test case 17.2.3 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236117 | 0444 | | Addition of new NR PRS-RSRPP reporting delay test case 17.2.4 | 17.2.0 | 17.3.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2023-12 | RAN#102 | R5-236118 | 0445 | | Addition of new NR PRS-RSRPP reporting delay test case 17.2.5 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236119 | 0446 | | Addition of new NR PRS-RSRPP reporting delay test case 17.2.6 | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-236259 | 0447 | | Update positioning test condition to include Rel-17 NR performance requirements | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-237794 | 0433 | 1 | Update TC 14.3.7 with TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-237795 | 0436 | 1 | Update TC 15.3.6 with TT analysis results | 17.2.0 | 17.3.0 |
| 2023-12 | RAN#102 | R5-237794, R5-237795 | - | - | implementation of missing R5-237794, R5-237795 | 17.3.0 | 17.3.1 |
| 2024-03 | RAN#103 | R5-240103 | 0450 | - | Update TC 15.2.10 with TT analysis results | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240104 | 0451 | - | Update TC 16.2.7 with TT analysis results | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240105 | 0452 | - | Update TC 16.2.8 with TT analysis results | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240107 | 0454 | - | Addition of NR PRS-based measurement requirements in annex C to include TT analysis results | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240108 | 0455 | - | Addition of new RRC_INACTIVE NR RSTD reporting delay test case 14.4.3 | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240109 | 0456 | - | Addition of new NR RSTD reporting delay test case 14.4.4 in RRC_INACTIVE state | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240110 | 0457 | - | Addition of new NR RSTD accuracy test case 14.5.3 in RRC_INACTIVE state | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240112 | 0459 | - | Addition of new NR PRS-RSRP reporting delay test case 16.4.3 in RRC_INACTIVE state | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240113 | 0460 | - | Addition of new NR PRS-RSRP reporting delay test case 16.4.4 in RRC_INACTIVE state | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240114 | 0461 | - | Addition of new NR PRS-RSRP accuracy test case 16.5.3 in RRC_INACTIVE state | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240115 | 0462 | - | Addition of new NR PRS-RSRP accuracy test case 16.5.4 in RRC_INACTIVE state | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240116 | 0463 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.3.1 | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240117 | 0464 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.3.2 | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240118 | 0465 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.3.3 | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240119 | 0466 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.3.4 | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-240120 | 0467 | - | Addition of measurement period requirements in RRC_INACTIVE state for NR positioning methods | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241897 | 0448 | 1 | Update TC 15.2.8 with TT analysis results | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241898 | 0449 | 1 | Update TC 15.2.9 with TT analysis results | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241899 | 0453 | 1 | Update TC 16.3.4 with TT analysis results | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241911 | 0458 | 1 | Addition of new NR RSTD accuracy test case 14.5.4 in RRC_INACTIVE state | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241912 | 0468 | 1 | Correction to PRS-RSRPP positioning test cases | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241913 | 0469 | 1 | Correction to PRS-RSRP positioning test cases | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241914 | 0470 | 1 | Correction to NR RSTD positioning test cases | 17.3.1 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241915 | 0471 | 1 | Correction to NR UE Rx-Tx time difference positioning test cases | 17.3.1 | 17.4.0 |
| 2024-06 | RAN#104 | R5-242201 | 0483 | - | Update TC 15.2.2 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242208 | 0490 | - | Update TC 15.3.2 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242209 | 0491 | - | Update TC 15.3.5 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242210 | 0492 | - | Update TC 15.3.6 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242211 | 0493 | - | Update TC 16.2.3 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242212 | 0494 | - | Update TC 16.2.4 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242213 | 0495 | - | Update TC 16.2.7 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242214 | 0496 | - | Update TC 16.2.8 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242215 | 0497 | - | Update TC 16.3.2 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242216 | 0498 | - | Update TC 16.3.4 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242471 | 0500 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.4.1 in RRC_INACTIVE state | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242472 | 0501 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.4.2 in RRC_INACTIVE state | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242473 | 0502 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.4.3 in RRC_INACTIVE state | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242474 | 0503 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.4.4 in RRC_INACTIVE state | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242475 | 0504 | - | Addition of new NR PRS-RSRPP measurement period test case 17.4.1 in RRC_INACTIVE | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242476 | 0505 | - | Addition of new NR PRS-RSRPP measurement period test case 17.4.2 in RRC_INACTIVE | 17.4.0 | 17.5.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2024-06 | RAN#104 | R5-242477 | 0506 | - | Addition of new NR PRS-RSRPP measurement period test case 17.4.3 in RRC_INACTIVE | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242478 | 0507 | - | Addition of new NR PRS-RSRPP measurement period test case 17.4.4 in RRC_INACTIVE | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242479 | 0508 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.5.1 in RRC_INACTIVE | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242480 | 0509 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.5.2 in RRC_INACTIVE | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242481 | 0510 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.5.3 in RRC_INACTIVE | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-242482 | 0511 | - | Addition of new NR PRS-RSRPP measurement accuracy test case 17.5.4 in RRC_INACTIVE | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243760 | 0475 | 1 | Update TC 14.2.8 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243761 | 0476 | 1 | Update TC 14.2.9 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243762 | 0477 | 1 | Update TC 14.2.10 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243763 | 0480 | 1 | Update TC 14.3.7 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243764 | 0481 | 1 | Update TC 14.3.8 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243765 | 0486 | 1 | Update TC 15.2.8 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243766 | 0487 | 1 | Update TC 15.2.9 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243767 | 0488 | 1 | Update TC 15.2.10 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243768 | 0499 | 1 | Update NR PRS-based measurement requirements in annex based on the updated TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243784 | 0473 | 1 | Update TC 14.2.3 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243785 | 0474 | 1 | Update TC 14.2.4 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243786 | 0478 | 1 | Update TC 14.3.3 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243787 | 0479 | 1 | Update TC 14.3.4 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243788 | 0482 | 1 | Update TC 15.2.1 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243789 | 0484 | 1 | Update TC 15.2.3 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243790 | 0485 | 1 | Update TC 15.2.4 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243791 | 0489 | 1 | Update TC 15.3.1 with TT analysis results | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243803 | 0472 | 1 | Addition of Test Case 14.2.6 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243804 | 0512 | 1 | Addition of Test Case: UE Rx-Tx time difference measurement accuracy with reduced number of samples in FR2 SA | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243805 | 0513 | 1 | Addition of UE Rx-Tx time difference measurements in RRC_INACTIVE test case | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243857 | 0514 | 1 | Addition of Test Case 14.2.5 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243858 | 0515 | 1 | Addition of Test Case 14.2.7 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243859 | 0516 | 1 | Addition of Test Case 14.3.5 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243860 | 0517 | 1 | Addition of Test Case 14.3.6 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243861 | 0518 | 1 | Addition of Test Case 14.4.1 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243862 | 0519 | 1 | Addition of Test Case 14.4.2 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243863 | 0520 | 1 | Addition of Test Case 14.5.1 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243864 | 0521 | 1 | Addition of Test Case 14.5.2 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243865 | 0522 | 1 | Addition of Test Case 15.2.5 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243866 | 0523 | 1 | Addition of Test Case 15.2.6 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243867 | 0524 | 1 | Addition of Test Case 15.2.7 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243868 | 0525 | 1 | Addition of Test Case 15.3.3 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243869 | 0526 | 1 | Addition of Test Case 15.3.4 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243870 | 0527 | 1 | Addition of Test Case 16.2.5 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243871 | 0528 | 1 | Addition of Test Case 16.2.6 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243872 | 0529 | 1 | Addition of Test Case 16.3.3 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243873 | 0530 | 1 | Addition of Test Case 16.4.1 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243874 | 0531 | 1 | Addition of Test Case 16.4.2 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243875 | 0532 | 1 | Addition of Test Case 16.5.1 | 17.4.0 | 17.5.0 |
| 2024-06 | RAN#104 | R5-243876 | 0533 | 1 | Addition of Test Case 16.5.2 | 17.4.0 | 17.5.0 |
| 2024-09 | RAN#105 | R5-244062 | 0534 | - | Update TC 14.4.3 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244063 | 0535 | - | Update TC 14.4.4 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244064 | 0536 | - | Update TC 14.5.3 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244066 | 0538 | - | Update TC 15.4.3 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244067 | 0539 | - | Update TC 15.4.4 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244068 | 0540 | - | Update TC 16.4.3 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244069 | 0541 | - | Update TC 16.4.4 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244070 | 0542 | - | Update TC 16.5.3 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244071 | 0543 | - | Update TC 16.5.4 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244354 | 0551 | - | Editorial correction on TS 37.571-1 test case titles | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-244546 | 0552 | - | Completion of UE Rx-Tx time difference measurement accuracy in RRC_INACTIVE test case including TTs | 17.5.0 | 17.6.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2024-09 | RAN#105 | R5-244547 | 0553 | - | Completion of UE Rx-Tx time difference measurement accuracy with reduced number of samples in RRC_INACTIVE test case including TTs | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245234 | 0554 | - | Addition of test case 15.5.2 for UE Rx-Tx time difference measurements | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245236 | 0556 | - | Corrections to NR RSTD measurement reporting delay test case 14.2.5 | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245237 | 0557 | - | Corrections to NR RSTD measurement accuracy test case 14.3.5 | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245238 | 0558 | - | Corrections to NR RSTD measurement accuracy test case 14.3.6 | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245239 | 0559 | - | Corrections to PRS-RSRP measurement accuracy test case 16.5.1 | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245240 | 0560 | - | Corrections to PRS-RSRP measurement accuracy test case 16.5.2 | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245317 | 0561 | - | Addition of test case 15.5.1 for UE Rx-Tx time difference measurement accuracy | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245936 | 0544 | 1 | Update TC 17.2.1 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245937 | 0545 | 1 | Update TC 17.2.2 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245938 | 0546 | 1 | Update TC 17.2.3 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245939 | 0547 | 1 | Update TC 17.2.4 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245940 | 0548 | 1 | Update TC 17.2.5 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245941 | 0549 | 1 | Update TC 17.2.6 with TT analysis results | 17.5.0 | 17.6.0 |
| 2024-09 | RAN#105 | R5-245942 | 0550 | 1 | Update NR PRS-based measurement requirements in annex based on the updated TT analysis results | 17.5.0 | 17.6.0 |
| 2024-12 | RAN#106 | R5-246119 | 0563 | - | Correction of RSTD measurement reporting delay test case 14.2.5 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246140 | 0564 | - | Update TC 14.5.4 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246144 | 0568 | - | Update TC 17.3.2 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246145 | 0569 | - | Update TC 17.3.3 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246147 | 0571 | - | Update TC 17.4.1 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246148 | 0572 | - | Update TC 17.4.2 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246149 | 0573 | - | Update TC 17.4.3 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246150 | 0574 | - | Update TC 17.4.4 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-246152 | 0576 | - | Update TC 17.5.2 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247313 | 0580 | - | Correction of RSTD RSTD measurement period test case 14.2.1 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247314 | 0581 | - | Correction of RSTD measurement reporting delay test case 14.2.2 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247315 | 0582 | - | Correction of RSTD measurement accuracy test case 14.3.1 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247316 | 0583 | - | Correction of RSTD measurement accuracy test case 14.3.2 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247323 | 0584 | - | Correction of RSTD measurement reporting delay test case 14.2.6 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247324 | 0585 | - | Correction of RSTD measurement reporting delay test case 14.2.7 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247325 | 0586 | - | Correction of RSTD measurement accuracy test case 14.3.5 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247326 | 0587 | - | Correction of RSTD measurement accuracy test case 14.3.6 including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247888 | 0565 | 1 | Update TC 15.4.1 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247889 | 0566 | 1 | Update TC 15.4.2 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247890 | 0567 | 1 | Update TC 17.3.1 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247891 | 0575 | 1 | Update TC 17.5.1 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247892 | 0577 | 1 | Update TC 17.5.3 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247893 | 0579 | 1 | Update NR PRS-based measurement requirements in annex based on the updated TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247894 | 0588 | 1 | Correction of RSTD measurement accuracy test case 14.5.1 in RRC_INACTIVE including TT | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247895 | 0589 | 1 | Correction of RSTD measurement accuracy test case 14.5.2 including TT with RRC_INACTIVE | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247981 | 0570 | 1 | Update TC 17.3.4 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | R5-247982 | 0578 | 1 | Update TC 17.5.4 with TT analysis results | 17.6.0 | 17.7.0 |
| 2024-12 | RAN#106 | - | - | - | Administrative release upgrade to match the release of TS 37.571-2 which was upgraded at RAN#106 to Rel-18 due to Rel-18 relevant CRs | 17.7.0 | 18.0.0 |
| 2025-03 | RAN#107 | R5-250887 | 0594 | - | Correction of PRS-RSRP measurement accuracy test case 16.3.1 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-250908 | 0602 | - | Correction of test case 16.2.5 including TT | 18.0.0 | 18.1.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2025-03 | RAN#107 | R5-250911 | 0605 | - | Correction of test case 16.4.1 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-250912 | 0606 | - | Correction of test case 16.4.2 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-250915 | 0609 | - | Message content corrections for NR RSTD measurement period test cases 14.2.X | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-250916 | 0610 | - | Alignment to Rel16 RSTD, PRS-RSRP and Rx-Tx time difference measurement test cases | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251443 | 0611 | 1 | Correction to R16 RSTD accuracy test parameters for NR RSTD measurement accuracy test cases | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251444 | 0612 | 1 | Correction to R17 RSTD accuracy test parameters for NR RSTD measurement accuracy test cases | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251624 | 0590 | 1 | Correction of NR RSTD measurement period test case 14.2.1 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251625 | 0591 | 1 | Correction of NR RSTD measurement period test case 14.2.2 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251626 | 0592 | 1 | Correction of PRS-RSRP measurement period test case 16.2.1 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251627 | 0593 | 1 | Correction of PRS-RSRP measurement period test case 16.2.2 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251628 | 0603 | 1 | Correction of test case 16.2.6 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251629 | 0604 | 1 | Correction of test case 16.3.3 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251630 | 0607 | 1 | Correction of test case 16.5.1 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251631 | 0608 | 1 | Correction of test case 16.5.2 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251709 | 0595 | 1 | Annex C update with new TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251710 | 0596 | 1 | Correction of test case 14.2.7 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251711 | 0597 | 1 | Correction of RSTD reporting delay test case 14.4.1 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251712 | 0598 | 1 | Correction of RSTD reporting test case 14.4.2 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251713 | 0599 | 1 | Correction of UE Rx-Tx time difference measurement test case 15.2.5 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251714 | 0600 | 1 | Correction of UE Rx-Tx time difference measurement test case 15.2.6 including TT | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251715 | 0601 | 1 | Correction of UE tx-rx measurement period test case 15.2.7 including TT | 18.0.0 | 18.1.0 |
| 2025-06 | RAN#108 | R5-252279 | 0613 | - | Specification quality improvement for 37.571-1 clause 4 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252280 | 0614 | - | Specification quality improvement for 37.571-1 clause 14 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252281 | 0615 | - | Specification quality improvement for 37.571-1 clause 15 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252282 | 0616 | - | Specification quality improvement for 37.571-1 clause 16 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252283 | 0617 | - | Specification quality improvement for 37.571-1 clause 17 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252773 | 0618 | - | Positioning spec quality improvement draft CR clauses 05_0902 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252774 | 0619 | - | Positioning spec quality improvement draft CR clauses 0903_0904 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252775 | 0620 | - | Positioning spec quality improvement draft CR clauses 0905_13 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252893 | 0625 | - | Annex C update with new TT for Rel 17 test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252903 | 0626 | - | Message content update for R16 RSTD measurement accuracy test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252904 | 0627 | - | Message content update for R17 RSTD measurement accuracy test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252905 | 0628 | - | Correction for R16 FR1 UE Rx-Tx time difference measurement test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252906 | 0629 | - | Correction for R17 FR1 UE Rx-Tx time difference measurement test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252907 | 0630 | - | Message content corrections for Rel-16 NR PRS-RSRP measurement period test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252908 | 0631 | - | Core Spec alignment for number of cells to be provided in the assistance data for R16 test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252909 | 0632 | - | Core Spec alignment for number of cells to be provided in the assistance data for R17 test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252910 | 0633 | - | Message content update for R16 PRS RSRP and Rx-Tx accuracy test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252911 | 0634 | - | Message content update for R17 PRS RSRP and Rx-Tx accuracy test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252912 | 0635 | - | Correction of test case 14.2.7 according to existing TT | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252913 | 0636 | - | Correction of test case 16.5.2 according to existing TT | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252914 | 0637 | - | Positioning Spec quality improvement draft CR for Annexes | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252915 | 0638 | - | Positioning Spec quality improvement draft CR for clauses 1 and 2 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253573 | 0621 | 1 | Correction of test case 15.3.3 including TT | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253574 | 0622 | 1 | Correction of test case 15.3.4 including TT | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253575 | 0623 | 1 | Correction of test case 15.5.1 including TT | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253576 | 0624 | 1 | Correction of test case 15.5.2 including TT | 18.1.0 | 18.2.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2025-09 | RAN#109 | R5-253849 | 0639 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.11 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253850 | 0640 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.12 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253851 | 0641 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.13 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253852 | 0642 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.14 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253853 | 0643 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.15 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253855 | 0645 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.7 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253856 | 0646 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.8 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253857 | 0647 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.9 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253858 | 0648 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.10 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253859 | 0649 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.11 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253860 | 0650 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.12 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253861 | 0651 | - | Addition of release 18 Multi-RTT test conditions | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254155 | 0652 | - | Core Spec alignment for PRS Resource slot offset of PRS-RSRPP test cases | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254442 | 0653 | - | Adding new test case NR RSTD measurement reporting delay for PRS aggregation in FR1 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254443 | 0654 | - | Introducing new test case NR RSTD measurement reporting delay for PRS aggregation in FR2 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254444 | 0655 | - | Adding new NR RSTD measurement accuracy test case for PRS aggregation in FR1 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254446 | 0657 | - | Introduce RSTD measurement requirements based on PRS aggregation | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254715 | 0658 | - | Addition of new NR RSTD measurement reporting delay test cases for RedCap UE | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-255332 | 0644 | 1 | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.16 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-255333 | 0656 | 1 | Introducing new NR RSTD measurement accuracy test case for PRS aggregation in FR2 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-255334 | 0659 | 1 | Addition of new NR RSTD measurement accuracy test cases for RedCap UE | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253849 | 0639 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.11 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253850 | 0640 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.12 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253851 | 0641 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.13 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253852 | 0642 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.14 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253853 | 0643 | - | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.15 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-255332 | 0644 | 1 | Addition of new UE Rx-Tx time difference reporting delay test case 15.2.16 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253855 | 0645 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.7 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253856 | 0646 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.8 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253857 | 0647 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.9 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253858 | 0648 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.10 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253859 | 0649 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.11 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253860 | 0650 | - | Addition of new UE Rx-Tx time difference accuracy test case 15.3.12 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-253861 | 0651 | - | Addition of release 18 Multi-RTT test conditions | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254155 | 0652 | - | Core Spec alignment for PRS Resource slot offset of PRS-RSRPP test cases | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254442 | 0653 | - | Adding new test case NR RSTD measurement reporting delay for PRS aggregation in FR1 | 18.2.0 | 18.3.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment TS 36.571-1 | Old | New |
| 2025-09 | RAN#109 | R5-254443 | 0654 | - | Introducing new test case NR RSTD measurement reporting delay for PRS aggregation in FR2 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254444 | 0655 | - | Adding new NR RSTD measurement accuracy test case for PRS aggregation in FR1 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-255333 | 0656 | 1 | Introducing new NR RSTD measurement accuracy test case for PRS aggregation in FR2 | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254446 | 0657 | - | Introduce RSTD measurement requirements based on PRS aggregation | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254715 | 0658 | - | Addition of new NR RSTD measurement reporting delay test cases for RedCap UE | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-254715 | 0659 | 1 | Addition of new NR RSTD measurement accuracy test cases for RedCap UE | 18.2.0 | 18.3.0 |
| 2025-12 | RAN#110 | R5-255704 | 0660 | - | Addition of new UE Rx-Tx time difference measurement test cases in RRC_INACTIVE state | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255713 | 0661 | - | Correction of UE Rx-Tx time difference measurement period test case 15.2.1 including TT | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255714 | 0662 | - | Correction of UE Rx-Tx time difference measurement period test case 15.2.2 including TT | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255715 | 0663 | - | Correction of UE Rx-Tx time difference measurement period test case 15.4.1 including TT | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255717 | 0664 | - | Correction to assistance data for measurement period test cases | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255827 | 0665 | - | Introducing new connection diagrams for 4 or 6 cells testing in FR1 and FR2 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255828 | 0666 | - | Adding new FR1 RSTD measurement reporting delay test case for PRS aggregation | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255829 | 0667 | - | Adding new FR2 RSTD measurement reporting delay test case for PRS aggregation | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255830 | 0668 | - | Adding new FR1 RSTD measurement accuracy test case for PRS aggregation | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-255831 | 0669 | - | Adding new FR2 RSTD measurement accuracy test case for PRS aggregation | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256423 | 0670 | - | Correction of NR RSTD measurement accuracy in FR1 for Redcap UE without Rx FH test case 14.3.11 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256424 | 0671 | - | Correction of NR RSTD measurement accuracy in FR2 for Redcap UE without Rx FH test case 14.3.12 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256425 | 0672 | - | Correction of NR RSTD measurement accuracy in FR1 for Redcap UE with Rx FH test case 14.3.13 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256426 | 0673 | - | Correction of NR RSTD measurement accuracy in FR2 for Redcap UE with Rx FH test case 14.3.14 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256427 | 0674 | - | Correction of NR RSTD measurement reporting delay test case in FR1 for Redcap UE without Rx FH test case 14.2.13 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256428 | 0675 | - | Correction of NR RSTD measurement reporting delay test case in FR2 for Redcap UE without Rx FH test case 14.2.14 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256429 | 0676 | - | Correction of NR RSTD measurement reporting delay test case in FR1 for Redcap UE with Rx FH test case 14.2.15 | 18.3.0 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256430 | 0677 | - | Correction of NR RSTD measurement reporting delay test case in FR2 for Redcap UE with Rx FH test case 14.2.16 | 18.3.0 | 18.4.0 |
| 2026-03 | RAN#111 | R5-260022 | 0681 | - | Update TC 15.3.7 with TT analysis results | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260023 | 0682 | - | Update TC 15.3.8 with TT analysis results | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260053 | 0685 | - | Updating test requirement with TT analysis results for FR1 test case 14.3.9 | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260203 | 0687 | - | Addition of UE Rx-Tx time difference reporting delay test case for normal UE and Redcap UE in RRC INACTIVE state | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260204 | 0688 | - | Addition of UE Rx-Tx time difference accuracy test case for normal UE and Redcap UE in RRC INACTIVE state | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260795 | 0689 | - | Adding new test cases for PRS-RSRP measurement delay for Redcap UE without Rx FH | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260796 | 0690 | - | Adding new test cases for PRS-RSRP measurement delay for Redcap UE with Rx FH | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260797 | 0691 | - | Adding new FR1 test case for NR RSTD measurement delay for Redcap UE without Rx FH in RRC INACTIVE state | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260927 | 0693 | - | Correction of test case 16.3.1 TT analysis | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260930 | 0694 | - | Corrections of UE Rx-Tx time difference measurement period FR1 test cases | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260931 | 0695 | - | Correction of LPP Time calculation for Rel 16 FR1 measurement accuracy test cases | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-261569 | 0692 | 1 | Adding new FR2 test case for NR RSTD measurement delay for Redcap UE without Rx FH in RRC INACTIVE state | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-261659 | 0678 | 1 | Update TC 15.2.11 with TT analysis results | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-261660 | 0679 | 1 | Update TC 15.2.12 with TT analysis results | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-261661 | 0680 | 1 | Update TC 15.2.13 with TT analysis results | 18.4.0 | 18.5.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment TS 36.571-1** | **Old** | **New** |
| 2026-03 | RAN#111 | R5-261662 | 0683 | 1 | Annex C correction for Rel-18 positioning test cases including TT | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-261663 | 0684 | 1 | Updating test requirement with TT analysis results for FR1 test case 14.2.11 | 18.4.0 | 18.5.0 |
| 2026-03 | RAN#111 | R5-260194 | 0686 | - | Introduction of BDS B2b signal information in TS 37.571-1 | 18.5.0 | 19.0.0 |
| 2026-06 | RAN#112 | R5-261749 | 0698 | | Update TC 15.2.16 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-261750 | 0699 | | Update TC 15.4.8 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-261751 | 0700 | | Update TC 15.3.9 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-261752 | 0701 | | Update TC 15.5.7 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-261753 | 0702 | | Update TC 15.3.10 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-261755 | 0704 | | Update TC 15.5.8 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-261757 | 0706 | | Update TC 15.4.5 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-261759 | 0708 | | Update TC 15.4.7 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-262056 | 0710 | | Updating message contents for FR1 RSTD measurement reporting delay test case 14.4.7 | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-262057 | 0711 | | Updating message contents for FR2 RSTD measurement reporting delay test case 14.4.8 | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-263232 | 0707 | 1 | Update TC 15.4.6 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-263265 | 0712 | 1 | Updating message contents for PRS-RSRP measurement delay test cases for Redcap UE without Rx FH | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-263266 | 0713 | 1 | Updating message contents for PRS-RSRP measurement delay test cases for Redcap UE with Rx FH | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-263383 | 0697 | 1 | Update TC 15.2.14 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-263384 | 0703 | 1 | Update TC 15.3.12 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-263385 | 0705 | 1 | Update TC 15.5.10 with TT analysis results | 19.0.0 | 19.1.0 |
| 2026-06 | RAN#112 | R5-263386 | 0709 | 1 | Anne`x C correction for Rel-18 positioning test cases including TT | 19.0.0 | 19.1.0 |

# History

| Version | Date | Status |
|---|---|---|
| V19.0.0 | April 2026 | Publication |
| V19.1.0 | September 2026 | Publication |
| | | |
| | | |
| | | |